The four-dimensional Yang–Mills construction

Foundations and the existence argument

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Status. This is an incomplete draft of the existence argument. Local results below have proofs from their displayed hypotheses. The construction estimates and the quantitative inputs to the phase, symmetry and ultraviolet arguments still require proofs before the final existence statement can be asserted on the basis of this article. The conditional reconstruction is developed below, including locality and the spectral condition. Outstanding inputs are stated explicitly. All prose remains a draft pending independent human mathematical and source-correspondence review.

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1 The construction problem and the order of its limits

The aim is to construct a relativistic field theory from finite-dimensional Wilson integrals. There are two limiting operations. First the lattice spacing tends to zero while the physical period stays fixed. Then the period tends to infinity. The observables and bare action coefficients must be chosen coherently before either operation: changing the family in response to a requested sequence of periods would give a weaker statement.

We assume familiarity with finite-dimensional Lie groups, probability Haar measure on compact groups, Lebesgue integration and dominated convergence, and elementary Hilbert-space theory. The specific compactness, quotient and operator arguments used below are supplied where they enter the construction.

Fix a closed subgroup $G \subset U(r_G)$, with its specified faithful matrix representation. Its real Lie algebra is

$$\mathfrak{g} = \{X \in M_{r_G}(\mathbb{C}) : \exp(tX) \in G \text{ for every } t \in \mathbb{R}\}.$$

We assume that $\mathfrak{g}$ is nonabelian and simple: its only real subspaces invariant under every $\operatorname{ad} X$, $X \in \mathfrak{g}$, are 0 and $\mathfrak{g}$. Coordinates on $\mathbb{R}^4$ are written $x = (\boldsymbol{x}, t)$, with time last. The Minkowski form is $t^2 - |\boldsymbol{x}|^2$, and the closed forward cone is $\overline{V}_+ = \{(\boldsymbol{p}, E) : |\boldsymbol{p}| \le E\}$.

The lattice family consists of strictly increasing positive integer sequences P_p and D_N, a spacing $a_N = D_N^{-1}$, real coefficients $\beta_{N,p}$, and local gauge-invariant insertion functions $O_{N,p,v}(x;U)$ indexed by tokens v. The periodic lattice has $P_p D_N$ sites along each axis and physical period P_p. Section 2 describes its Wilson probability measure and moments. Each insertion is continuous in U. For each token a single finite integer-offset stencil controls the insertion at every spacing, period and placement. The running condition is

$$\exists \sigma > 0 \quad \forall \varepsilon > 0 \quad \exists N_0 \quad \forall N \geq N_0 \quad \forall p : \qquad \left| \frac{\beta_{N,p}}{\log(a_N^{-1})} - \sigma \right| < \varepsilon. \tag{1}$$

In particular, the choice of N_0 is uniform in the period index.

Let $\mathrm{Conf}_n(\mathbb{R}^4)$ be the set of ordered n-tuples of pairwise distinct points. A separated test is a smooth complex function with compact support in this set, extended by zero outside it. For insertion tokens $\boldsymbol{v} = (v_1, \ldots, v_n)$, write $M_{N,p}(\boldsymbol{v}; \boldsymbol{x})$ for their Wilson moment and put

$$W_{N,p}(\boldsymbol{v}; f) = a_N^{4n} \sum_{\boldsymbol{x} \in \Lambda_{N,p}^n} f(a_N \boldsymbol{x}) M_{N,p}(\boldsymbol{v}; \boldsymbol{x}). \tag{2}$$

Here the coordinates of $\boldsymbol{x}$ are chosen in the centered fundamental cell. A period is suitable for f if its support lies strictly inside that physical cell in every coordinate.

The limiting data have several related components. For every integer grade $d \geq 0$ there is a finite-dimensional complex space $\mathcal{F}_d$. It has a complex-linear injection into a common field-label space; every label is represented at some grade. Tokens map onto the disjoint union $\coprod_d \mathcal{F}_d$. A compact metric space X, a probability measure ν of full support, and a selected character $\chi_* \in X$ describe the phases. For graded fields $\boldsymbol{A}$, let $S_{\boldsymbol{A}}^{\chi}(\boldsymbol{x})$ and $S_{\boldsymbol{A}}^{\mathrm{mix}}(\boldsymbol{x})$ denote the phase and mixed separated correlations. They are smooth on configuration space, permutation symmetric, and continuous in χ, with

$$S_{\boldsymbol{A}}^{\mathrm{mix}}(\boldsymbol{x}) = \int_X S_{\boldsymbol{A}}^{\chi}(\boldsymbol{x}) \, d\nu(\chi). \tag{3}$$

The mixed correlations are the Wilson limits. The selected correlations are the Euclidean correlations to be continued into the reconstructed theory. These two roles should be kept distinct even when the phase space has only one point.

Definition 1.1 (Reconstructed selected theory)**.** *The selected theory consists of a complex Hilbert space $\mathcal{H}$, a normalized vacuum Ω, a common dense invariant domain $\mathcal{D}$ containing Ω, and complex-linear fields $\Phi(A, f) : \mathcal{D} \to \mathcal{D}$, linear separately in the label and test, where $f \in \mathcal{S}(\mathbb{R}^4)$. For fixed A, ψ, the map $f \mapsto \Phi(A, f)\psi$ is continuous in the Schwartz topology. The domain is the span of field words applied to the vacuum. There is a conjugate-linear label adjoint $A \mapsto A^*$ with*

$$\langle \Phi(A, f)\psi, \eta \rangle = \langle \psi, \Phi(A^*, \overline{f})\eta \rangle.$$

A strongly continuous unitary representation of the proper orthochronous Poincaré group preserves the domain and the vacuum, and transforms fields with the specified action on labels and the ordinary affine pullback of tests. Its translation spectral measure is supported in $\overline{V}_+$, and every translation-invariant vector is a scalar multiple of Ω. Fields commute on $\mathcal{D}$ when their test supports are spacelike separated. Every nonzero label acts nontrivially for some test, and at least one smeared field is nonzero.

For fixed spatial positions the selected correlations have a holomorphic continuation $F_{\boldsymbol{A}}(\boldsymbol{z}; \boldsymbol{x})$ on $\operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n$. At strictly ordered Euclidean times its value at $z_i = it_i$ is $S_{\boldsymbol{A}}^{\chi_*}((\boldsymbol{x}_i, t_i)_i)$. For every strictly increasing real height vector $\boldsymbol{h}$ and every tuple of Schwartz tests, its integrals on $z_i = t_i + i\epsilon h_i$ converge as $\epsilon \downarrow 0$ to the corresponding vacuum expectation of the smeared fields.

The existence target is one regular Wilson family such that, for every strictly increasing period-index sequence $p(k)$, a strictly increasing $q(k)$ and one hierarchy with the preceding properties can be found. For each separated test there are row limits $w_p(\boldsymbol{v}; f)$ satisfying

$$\lim_{N\to\infty} W_{N,p}(\boldsymbol{v}; f) = w_p(\boldsymbol{v}; f) \qquad \text{at each suitable fixed } p, \tag{4}$$

$$\lim_{k\to\infty} w_{p(q(k))}(\boldsymbol{v}; f) = \int f(\boldsymbol{x}) S^{\mathrm{mix}}_{\boldsymbol{A}}(\boldsymbol{x})\, d\boldsymbol{x}, \tag{5}$$

where A_i is the class of v_i. Suitability holds eventually along the selected period sequence. The same q and hierarchy work for all arities, tokens and tests. The hierarchy may depend on the requested sequence; the Wilson family may not.

The full endpoint also specifies a common operator-product expansion and a short-distance trace test, described in Section 204. These are additional obligations on the same hierarchy. The target concerns a theory of the constructed gauge-invariant observables; identifying a particular label with a separately prescribed physical observable requires its own construction argument.

2 Finite lattice gauge theory

The starting point is a probability measure on finitely many group-valued links. Compactness makes this measure elementary to define; the difficulty of the continuum problem is to control its observables uniformly as the number of links grows. We first fix the action and its normalization.

2.1 The periodic lattice and its probability law

Fix a closed subgroup $G \subset U(r_G)$ in a specified faithful unitary representation, with nonabelian simple real Lie algebra as in Section 1. The group is compact. For a positive integer ℓ, let

$$\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4, \qquad E_\ell = \Lambda_\ell \times \{0,1,2,3\}.$$

A link configuration is a function $U : E_\ell \to G$. The four coordinates are Euclidean; this finite model does not distinguish a time direction in its action. An oriented link in the negative direction is represented by the inverse of the corresponding positively oriented link.

For $x \in \Lambda_\ell$ and distinct axes i, j, the positively oriented plaquette holonomy based at x is

$$U_{ij}(x) = U_i(x) U_j(x + e_i) U_i(x + e_j)^{-1} U_j(x)^{-1}.$$

Count each pair $i < j$ once. The action is

$$S_\ell(U) = \sum_{x\in\Lambda_\ell} \sum_{0\le i<j<4} \left(1 - \frac{\operatorname{Re}\operatorname{tr} U_{ij}(x)}{r_G}\right).$$

Each summand is real and nonnegative: a unitary matrix has each diagonal entry of modulus at most one, so $\operatorname{Re}\operatorname{tr} V \le r_G$. The convention has no extra factor of $1/2$ over faces. This action is dimensionless. The bare coefficient β multiplies it in the Boltzmann weight; it is not the blocking-kernel penalty used later in the RG construction.

Let H_G be probability Haar measure on G, and let

$$\rho_\ell = \bigotimes_{(x,i)\in E_\ell} H_G, \quad Z_\ell(\beta) = \int e^{-\beta S_\ell(U)}\, d\rho_\ell(U), \quad d\mu_{\ell,\beta}(U) = Z_\ell(\beta)^{-1} e^{-\beta S_\ell(U)} d\rho_\ell(U).$$

The partition function is finite and strictly positive. Indeed the link space is a finite product of compact groups, hence compact, and S_ℓ is continuous; therefore $e^{-\beta S_\ell}$ is bounded and integrable. It is strictly positive at every configuration, so its integral against the probability measure ρ_ℓ is positive. Integrating the displayed density gives $Z_\ell(\beta)/Z_\ell(\beta) = 1$, so $\mu_{\ell,\beta}$ is a probability measure. This argument works for every real β for which the displayed definition is used; the construction theorem later selects its own coefficient sequence.

Proposition 2.1 (Gauge invariance of the finite action and law)**.** *For any map $g : \Lambda_\ell \to G$, transform links by*

$$(g \cdot U)_i(x) = g(x)U_i(x)g(x + e_i)^{-1}.$$

Then $S_\ell(g \cdot U) = S_\ell(U)$ and $\mu_{\ell,\beta}$ is invariant under $U \mapsto g \cdot U$.

Proof. Substitution into the four factors of $U_{ij}(x)$ cancels the group elements at the three intermediate vertices. The remaining factors give

$$(g \cdot U)_{ij}(x) = g(x)U_{ij}(x)g(x)^{-1}.$$

Trace is invariant under conjugation, since for matrices A, B of matching finite size $\mathrm{tr}(AB) = \mathrm{tr}(BA)$; hence every plaquette summand and their finite sum are unchanged.

For each link (x, i), the transformed value is obtained from $U_i(x)$ by left and right multiplication by fixed group elements. Left and right translations preserve Haar measure. The full transformation acts separately on the coordinates of the finite product E_ℓ, so the product Haar measure ρ_ℓ is preserved. Since the density $Z_\ell(\beta)^{-1}e^{-\beta S_\ell(U)}$ is invariant, the weighted measure is preserved as well. This proves invariance of the law, not gauge invariance of every observable. An insertion must separately satisfy the relevant invariance condition. □

2.2 Scales, observables, and the ordered limit

For mesh index N and physical-period index p, the family uses strictly increasing positive-integer sequences D_N and P_p. Set

$$a_N = D_N^{-1}, \qquad \ell_{N,p} = P_p D_N = P_p/a_N, \qquad \Lambda_{N,p} = (\mathbb{Z}/\ell_{N,p}\mathbb{Z})^4.$$

Thus $\ell_{N,p}$ is a site count per axis, while P_p is the physical linear period. A site is embedded in Euclidean spacetime by multiplying the symmetric least-absolute residue in each coordinate by a_N. A token v specifies an insertion $O_{N,p}(v, z; U)$. We require the insertion to be continuous and gauge invariant, with the fixed-stencil locality specified in Section 1.

The finite m-point moment is the integral of the product of its m insertions under $\mu_{\ell_{N,p},\beta_{N,p}}$. Smearing uses a lattice sum weighted by a_N^{4m}. Continuity and compactness of the finite configuration space give integrability of each finite product. The convergence interface first demands, at each fixed suitable physical volume, a full $N \to \infty$ mesh limit of these smeared moments, and only then a volume limit along a specified strictly increasing subsequence. It is not an arbitrary joint or diagonal limit. The support of a test function must eventually fit inside the period box and its arguments must remain separated as specified in Section 1.

The next task is to control these moments as the number of links tends to infinity. The required estimates must concern the same bare coefficients and local insertions at all cutoffs and periods. The extraction argument in the next section isolates a local modulus of continuity and a local size bound; the renormalization construction must supply those estimates with the indicated order of quantifiers.

3 From strip receipts to an order-two atlas

The local flat-sector estimate requires an order-two atlas for the two response kernels. This section proves the quantitative extraction of that atlas from the exponentially localized bed kernels and the supplied tower cross-kernel. It isolates what the construction has to provide: the strip receipt for the tower kernel and its first differences. The receipt-to-atlas argument itself is proved here.

3.1 Cover and installed kernels

Write $\mathcal{C} = \mathbb{Z}^4$ with $|x|_1 = \sum_{a=1}^4 |x_a|$. For an odd positive integer L, let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$, let π_L be reduction modulo L, and let ℓ_L denote the centered representative. There is a unique $\rho_L(x) \in \mathcal{C}$ such that

$$x = \ell_L(\pi_L x) + L\rho_L(x).$$

For $x, y \in \mathcal{C}$, put

$$d_L(x,y) = L^{-1}|x - Ly|_1.$$

The centered representative satisfies $|\ell_L(r)|_1 \le 2L$, hence, for every $z \in \mathcal{C}$,

$$d_L(x,y) \le |y - \rho_L(x)|_1 + 2 \le |y - \rho_L(x)|_1 + 4. \tag{6}$$

We retain the harmless margin 4 used in the finite-scale estimates.

For finite type sets F, G, a cross-stencil is a family $T(r,i,j,z) \in \mathbb{R}$ indexed by $r \in \Lambda_L$, $i \in F$, $j \in G$, and $z \in \mathcal{C}$. Its width-α strip cost is the per-label bound

$$\sum_{z \in \mathcal{C}} e^{\alpha|z|_1} |T(r,i,j,z)| \le C.$$

For each coordinate direction a, define its residue difference by

$$(\nabla_a^L T)(r,i,j,z) = T(r + \pi_L(e_a), i, j, z - \epsilon_a(r)) - T(r,i,j,z), \qquad \epsilon_a(r) = \begin{cases} e_a, & 2\ell_L(r)_a = L-1, \\ 0, & \text{otherwise.} \end{cases}$$

Here L is odd, so the boundary condition in the displayed case is equivalently $2\ell_L(r)_a = L-1$; this is the centered-lift carry convention. The installed cover kernel is

$$\widehat{T}((x,i),(y,j)) = T(\pi_L x, i, j, y - \rho_L x). \tag{7}$$

Its forward difference is

$$(\partial_a \widehat{T})((x,i),(y,j)) = \widehat{T}((x+e_a,i),(y,j)) - \widehat{T}((x,i),(y,j)).$$

The centered-lift identity gives the exact relation

$$\partial_a \widehat{T} = \widehat{\nabla_a^L T}. \tag{8}$$

Indeed, if $x = \ell_L(r) + Lv$, then $x + e_a = \ell_L(r + \pi_L(e_a)) + L(v + \epsilon_a(r))$; substituting this identity in (7) produces the displayed difference formula.

Lemma 3.1 (Installation of strip receipts)**.** *Suppose $0 \le \sigma \le \alpha$. If T has strip cost C/L^2 at width α and every $\nabla_a^L T$ has strip cost C/L^3 at that width, then for every fixed pair of types i, j,*

$$\sum_y e^{\sigma d_L(x,y)} |\widehat{T}((x,i),(y,j))| \le e^{4\sigma} C L^{-2},$$
$$\sum_x e^{\sigma d_L(x,y)} |\widehat{T}((x,i),(y,j))| \le e^{4\sigma} C L^{2},$$
$$\sum_y e^{\sigma d_L(x,y)} |\partial_a \widehat{T}((x,i),(y,j))| \le e^{4\sigma} C L^{-3}.$$

Thus $\widehat{T}$ has the order-two atlas bounds with cost $e^{4\sigma}C$: row L^{-2}, column L^2, and first-difference row L^{-3}. These are per-label bounds; summing over a finite type set adds its cardinality explicitly.

Proof. Fix x and put $r = \pi_L x$, $v = \rho_L x$. In the row sum, the substitution $z = y - v$ is a bijection of $\mathcal{C}$. By (6),

$$\sum_y e^{\sigma d_L(x,y)} |\widehat{T}((x,i),(y,j))| \le e^{2\sigma} \sum_z e^{\sigma |z|_1} |T(r,i,j,z)| \le e^{2\sigma} C L^{-2} \le e^{4\sigma} C L^{-2}.$$

The last inequality uses $\sigma \ge 0$.

For the column, decompose every x uniquely as $x = \ell_L(r) + Lv$, where $(r, v) \in \Lambda_L \times \mathcal{C}$. At fixed r, the installed coefficient is $T(r, i, j, y - v)$ and

$$d_L(\ell_L(r) + Lv, y) \le |v - y|_1 + 2,$$

again because $|\ell_L(r)|_1 / L \le 2$. Thus the sum over v is at most $e^{2\sigma} C L^{-2}$ for each r. There are $|\Lambda_L| = L^4$ residues, so the column sum is at most $e^{2\sigma} C L^2 \le e^{4\sigma} C L^2$. Finally apply the row argument to each $\nabla_a^L T$, use (8), and replace the cost C/L^2 by C/L^3. □

3.2 Composition with translation-invariant bed kernels

A translation-invariant row stencil I on a finite type set F has coefficients $I(i, b, u)$ for $u \in \mathcal{C}$. Its strip cost C_I at width α means

$$\sum_{u \in \mathcal{C}} e^{\alpha |u|_1} |I(i,b,u)| \le C_I \quad \text{for every } i, b \in F.$$

It acts on the cover by convolution. Let J be a second such stencil, on a finite type set G, with strip cost C_J. Given a cross kernel K from $\mathcal{C} \times F$ to $\mathcal{C} \times G$, form its sandwich

$$S((x,i),(y,j)) = \sum_{u,v \in \mathcal{C}} \sum_{b \in F,\, c \in G} I(i,b,u) K((x+u,b),(y+v,c)) J(c,j,-v).$$

Assume the sums below are absolutely convergent; this follows, in each application here, from the weighted bounds just stated and Tonelli's theorem.

Lemma 3.2 (Sandwich atlas estimate)**.** *Suppose I, J have nonnegative strip costs C_I, C_J at width α, and K has per-label atlas bounds of order two with cost Q at width σ. If $0 \le \sigma \le \alpha$ and $L \ge 1$, then S has per-label order-two atlas bounds at width σ with cost*

$$|F|\,|G|\,C_I C_J Q.$$

The first input difference of S is obtained by replacing K with its first input difference in the same formula; in particular its row cost is at most $|F|\,|G| C_I C_J Q / L^3$.

Proof. For $x, y, u, v \in \mathcal{C}$, the triangle inequality gives

$$d_L(x, y) \le d_L(x + u, y + v) + |u|_1/L + |v|_1.$$

Consequently the exponential weight at (x, y) is bounded by the central kernel weight multiplied by $e^{\alpha|u|_1} e^{\alpha|v|_1}$, since $0 \le \sigma/L \le \alpha$ and $0 \le \sigma \le \alpha$. Sum absolute values first in y, v, c. For each fixed u, b, the kernel row bound, summed over the $|G|$ possible intermediate output types c, is $|G|QL^{-2}$. For each resulting intermediate output, the stencil J contributes at most C_J. Summing in u, b against I costs at most $|F|C_I$. The row bound is therefore

$$|F|C_I \cdot |G|QL^{-2} \cdot C_J.$$

For the column, fix the external input type i and sum first in x, u, b. The stencil I contributes C_I; the kernel column bound, summed over the $|F|$ possible input types, is $|F|QL^2$; and summing the remaining displacement and type of J contributes $|G|C_J$. This gives $|F||G|C_I C_J QL^2$.

Subtract the formulas for $S((x + e_a, i), (y, j))$ and $S((x, i), (y, j))$ term by term with the same summation variables. Since I, J do not depend on their torus positions, this replaces $K((x + u, b), (y + v, c))$ by

$$K((x + u + e_a, b), (y + v, c)) - K((x + u, b), (y + v, c)).$$

Repeating the row estimate with the difference-row cost Q/L^3 proves the last assertion. Absolute convergence follows from the same calculation with all exponential weights replaced by 1; this licenses the subtraction. □

Theorem 3.3 (Response atlas from primitive strip receipts)**.** *Let F be the common finite type set of the fine inverse stencil and the coarse metric stencil, and let T be the tower cross-kernel. Suppose the two bed stencils have nonnegative strip costs C_f, C_c at width α, and $C_T \ge 0$, while T has base strip cost C_T/L^2 and each first residue-difference strip cost C_T/L^3 at that width. For $0 \le \sigma \le \alpha$, let $\widehat{T}$ be the installed cross-kernel and let*

$$K = I_f \, \widehat{T} \, I_c$$

denote its convolution sandwich with the fine inverse and coarse metric stencils. Then K has the order-two atlas bounds at width σ with cost

$$Q = |F|^2 e^{4\sigma} C_f C_c C_T. \tag{9}$$

In particular its per-label row, column and first-difference row bounds are Q/L^2, QL^2 and Q/L^3, respectively.

Proof. Lemma 3.1 gives the order-two atlas for $\widehat{T}$ with cost $e^{4\sigma} C_T$. Apply Lemma 3.2 with I_f and I_c; the two intermediate type sums each contribute one factor $|F|$. This yields exactly (9) and all three asserted estimates. □

Remark 3.4 (What this estimate does not supply). The theorem starts with the weighted cross-kernel receipts for T and its first differences. The RG construction must still prove these receipts, uniformly in the period, from its chosen cumulative kernels and defect factorization. At the public construction seam inspected here, this quantitative tower receipt is an input to the construction record; the local atlas calculation cannot replace it. Thus the theorem closes the passage from receipts to the hypotheses of the flat-sector propagation, but not the construction of the receipts themselves.

4 One-step propagation of a neutral curvature representative

This section gives the local mathematical content of the one-step flat-sector update. It concerns an order-zero quadratic precision difference; it does not assert convergence of the full Wilson theory. The inputs are exact one-step Schur data and weighted kernel estimates. All cost sums below are absolute weighted sums in $[0,\infty]$; signed kernel sums are used only after their absolute convergence has been established.

4.1 Spaces and estimates

Let $\mathcal{C}=\mathbb{Z}^4$ be the cover with Laurent distance $\operatorname{sep}(x,y)=\sum_{a=1}^4|x_a-y_a|$. For a positive integer scale L, set

$$\delta_L(x,y)=L^{-1}\operatorname{sep}(x,Ly),\qquad \operatorname{bill}(a,t)=|a|e^t.$$

For finite label sets I,J, a kernel K is a real function on $(\mathcal{C}\times I)\times(\mathcal{C}\times J)$. All three atlas bounds below are per fixed finite label: for every cover point x, input label i, and output label j,

$$\sum_{y\in\mathcal{C}}\operatorname{bill}\big(K((x,i),(y,j)),\omega\delta_L(x,y)\big)\le bL^{-2}, \tag{10}$$

$$\sum_{x\in\mathcal{C}}\operatorname{bill}\big(K((x,i),(y,j)),\omega\delta_L(x,y)\big)\le bL^{2},\quad (y,j)\text{ fixed}, \tag{11}$$

$$\sum_{y\in\mathcal{C}}\operatorname{bill}\big(K((x+e_a,i),(y,j))-K((x,i),(y,j)),\omega\delta_L(x,y)\big)\le bL^{-3}. \tag{12}$$

In the column line, i and (y,j) are fixed and the sum is over the cover coordinate of the input. These are the order-two atlas costs. Summing a row bound over all $|J|$ output labels or a column bound over all $|I|$ input labels contributes the corresponding explicit cardinality factor; neither line above already includes that factor. Here and below an estimate stated at width ω also holds at every smaller nonnegative width.

A rank-one packet P has coefficients $P((x,i),(v,j))$, with both positions in $\mathcal{C}$ and both types in I. Define its folded cost by

$$\|P\|_{\tau,\mathrm{fold}}:=\sup_{(x,i)}\sum_{(v,j)\in\mathcal{C}\times I}|P((x,i),(v,j))|e^{\tau\operatorname{sep}(x,v)}.$$

It is neutral when $\sum_{v\in\mathcal{C}}P((x,i),(v,j))=0$ for every x,i,j. It is diagonally stationary when translating both cover positions by the same unit vector leaves each coefficient unchanged.

Let T,U map fine cover-label spaces to coarse cover-label spaces, using the same finite type set I, $|I|=m$. The push is

$$(P\star(T,U))((y,k),(z,l))=\sum_{(x,i)\in\mathcal{C}\times I}T((x,i),(y,k))\sum_{(v,j)\in\mathcal{C}\times I}P((x,i),(v,j))U((v,j),(z,l)). \tag{13}$$

Its folded cost is defined by the same formula as for P, with fine labels replaced by coarse labels and τ replaced by σ.

4.2 Centering and the kernel row bound

Lemma 4.1 (Centering bound from an order-two atlas)**.** *Suppose U has the per-label difference bound* (12) *with cost p and width ω. For $x,v\in\mathcal{C}$ and types $j,l\in I$, set*

$$U_x(v,j;z,l)=U((v,j),(z,l))-U((x,j),(z,l)).$$

If $\tau > 0$, $0 \le \sigma \le \omega$, and $2\sigma \le L\tau$, then for every fixed output type l,

$$\sum_{z\in\mathcal{C}} |U_x(v,j;z,l)|e^{\sigma\delta_L(x,z)} \le \frac{2p}{\tau L^3} e^{\tau\,\mathrm{sep}(x,v)}. \tag{14}$$

Proof. Let $a = \mathrm{sep}(x,v)$. Choose a coordinate path $x = v_0, v_1, \dots, v_a = v$ with each increment a signed unit vector. The cover distance triangle inequality gives

$$\delta_L(x,z) \le \mathrm{sep}(x,v_k)/L + \delta_L(v_k,z) \le a/L + \delta_L(v_k,z).$$

For each path edge, the per-label difference row bound (12) (the negative edge has the same absolute value after translating its base point) therefore bounds its weighted row at center x by $pL^{-3}e^{\sigma a/L}$. Telescoping the coefficient difference and summing the a edge bounds yields

$$\sum_z |U((v,j),(z,l)) - U((x,j),(z,l))|e^{\sigma\delta_L(x,z)} \le pL^{-3}ae^{\sigma a/L}.$$

Since $0 \le \sigma/L \le \tau/2$,

$$ae^{\sigma a/L} \le ae^{\tau a/2} \le (2/\tau)e^{\tau a},$$

where the last inequality is $u \le e^u$ for $u = \tau a/2 \ge 0$. This proves the claim, including $a = 0$. □

Remark 4.2. The path bound is for one fixed output label. Summing it over all m output labels gives the aggregate outgoing-row bound used below. No cancellation has yet been used. Neutrality is used next to replace U by this centered kernel inside the convolution.

4.3 Weighted push estimate

Lemma 4.3 (Weighted convolution)**.** *Let P be a neutral rank-one packet with $\|P\|_{\tau,\mathrm{fold}} \le r$, where $r \ge 0$. Suppose T has the order-two atlas row and column bounds with cost $q \ge 0$, and U has the order-two atlas bounds with cost $p \ge 0$, both at width ω. Assume explicitly $\tau > 0$, $0 \le \sigma \le \omega$, and $2\sigma \le L\tau$. Then*

$$\|P \star (T,U)\|_{\sigma,\mathrm{fold}} \le \frac{m^2 q(2/\tau)p}{L}\, r.$$

Proof. Before using neutrality, the uncentered absolute sum is already finite. For a fixed output (y,k), first sum over the outgoing cover coordinate and type: its per-label row budget is pL^{-2}, so summing the m output types costs mpL^{-2}. The packet factor then sums to at most r, since its nonnegative exponential weight is at least one. Finally the root column has per-input- type budget qL^2 and summing the m input types costs mqL^2. Thus

$$\sum_{x,i,v,j,z,l} |T((x,i),(y,k))P((x,i),(v,j))U((v,j),(z,l))| \le (mqL^2)r(mpL^{-2}) = m^2qpr.$$

This finite majorant justifies Fubini and all subsequent changes of order, including the neutrality replacement. For fixed (x,i) and output type l, neutrality gives

$$\sum_{v,j} P((x,i),(v,j))U((v,j),(z,l)) = \sum_{v,j} P((x,i),(v,j))[U((v,j),(z,l)) - U((x,j),(z,l))].$$

Indeed, for each j, the subtracted term is independent of v and its coefficient sum is zero. The folded bound makes these coefficient sums absolutely convergent; the Atlas pointwise bounds and

the centered-row bound make the products summable. Thus this replacement and the following uses of Tonelli are legitimate.

Fix an output market (y,k). For output (z,l), the cover geometry gives

$$\operatorname{sep}(y,z) \le \delta_L(x,y) + \delta_L(x,z).$$

After taking absolute values in the push and summing in (z,l), the root factor is weighted by $\sigma\delta_L(x,y)$ and the outgoing factor by $\sigma\delta_L(x,z)$. For every fixed pair of input and output types of U, Lemma 4.1 bounds the outgoing cover sum by $2p(\tau L^3)^{-1}e^{\tau \operatorname{sep}(x,v)}$. Summing over its m output types gives the factor m. The exponential is exactly the packet weight on the leg v. Summing the packet over (v,j) therefore costs at most r for each fixed (x,i).

The remaining root sum is over (x,i). For the fixed output (y,k), the column bound in (11) is per fixed input type i; summing over the m possible i costs mqL^2. Consequently the whole nonnegative sum is bounded by

$$(mqL^2)\left(m\frac{2p}{\tau L^3}\right)r = \frac{m^2q(2/\tau)p}{L}r.$$

This bound is finite and proves both absolute convergence of the defining push and the asserted folded estimate. More formally, the argument applies first to finite subsets of each countable cover sum; the uniform right-hand side bounds their increasing limit, so monotone convergence gives the full sums and justifies every rearrangement. □

Remark 4.4. The estimate is a neutral-packet estimate: the extra factor L^{-1} comes from centering U and using its first-difference row cost L^{-3}, while the coarse-root column cost contributes L^2. Without neutrality, the centering identity is unavailable and the stated contraction does not follow.

4.4 Periodization and matching of the pushed representative

For each positive integer n, let $\Lambda_n = (\mathbb{Z}/n\mathbb{Z})^4$, let $\pi_n : \mathcal{C} \to \Lambda_n$ be coordinatewise reduction, and let $\ell_n : \Lambda_n \to \mathcal{C}$ choose the centered representative with $\pi_n\ell_n$ the identity. The fibre of π_n over y is exactly $\{\ell_n(y) + nr : r \in \mathcal{C}\}$: if $\pi_n z = y$, then each coordinate of $z - \ell_n(y)$ is divisible by n, with unique quotient. Fix positive periods n_f, n_c, n when using a composition. For a kernel K and output side n, define its displayed finite matrix by

$$(D_nK)((x,i),(y,j)) := \sum_{r\in\mathcal{C}} K((x,i),(\ell_n(y)+nr,j)).$$

When the input side is n_f and output side is n_c, K matches a matrix A if

$$(D_{n_c}K)((x,i),(y,j)) = A((\pi_{n_f}(x),i),(y,j)) \quad \text{for every } (x,i)\in\mathcal{C}\times I, (y,j)\in\Lambda_{n_c}\times J. \tag{15}$$

Thus matching requires the same value for every cover representative of the input torus point, not just its chosen lift. A kernel is city-covariant at ratio L if

$$K((x+Lu,i),(y+u,j)) = K((x,i),(y,j)) \quad (u,x,y\in\mathcal{C}). \tag{16}$$

For kernels K and U with an intermediate finite type set J, define their cover composition by

$$(K\diamond U)((x,i),(z,k)) := \sum_{v\in\mathcal{C}}\sum_{j\in J} K((x,i),(v,j))U((v,j),(z,k)).$$

Suppose K matches A from side n_f to side n_c, and U matches B from side n_c to side n. Assume for each fixed external input and output that the total double absolute sum over the intermediate and final cover coordinates and intermediate labels is finite. Then

$$(D_n(K \diamond U))((x,i),(y,k)) = \sum_{w \in \Lambda_{n_c}} \sum_{j \in J} A((\pi_{n_f}(x),i),(w,j))B((w,j),(y,k)). \tag{17}$$

Indeed, replace the output fibre in D_n by its explicit coset, interchange absolutely convergent sums, and first sum in the final cover lift:

$$\begin{aligned}(D_n(K \diamond U))((x,i),(y,k)) &= \sum_{v,j} K((x,i),(v,j)) \sum_{r \in \mathcal{C}} U((v,j),(\ell_n(y)+nr,k)) \\ &= \sum_{v,j} K((x,i),(v,j))B((\pi_{n_c}(v),j),(y,k)) \\ &= \sum_{w,j} A((\pi_{n_f}(x),i),(w,j))B((w,j),(y,k)).\end{aligned}$$

The second equality is the matching identity for U. For the third, partition $\mathcal{C}$ into the disjoint fibres $\pi_{n_c}^{-1}(w)$ and use the matching identity for K on each fibre. This proves the exact finite-matrix product, including the sum over intermediate types.

The transpose also has an exact matching rule. Suppose $n_f = Ln_c$ and T is city-covariant at ratio L and matches M from side n_f to side n_c. Then $\operatorname{swap} T$, defined by $(\operatorname{swap} T)((x,j),(y,i)) = T((y,i),(x,j))$, matches M^{T} from side n_c to side n_f. For fixed $x \in \mathcal{C}$ and $y \in \Lambda_{n_f}$, its display is

$$\begin{aligned}(D_{n_f} \operatorname{swap} T)((x,j),(y,i)) &= \sum_{v \in \mathcal{C}} T((\ell_{n_f}(y)+n_f v,i),(x,j)) \\ &= \sum_{v \in \mathcal{C}} T((\ell_{n_f}(y),i),(x-n_c v,j)) \\ &= \sum_{r \in \mathcal{C}} T((\ell_{n_f}(y),i),(\ell_{n_c}(\pi_{n_c}x)+n_c r,j)) \\ &= (D_{n_c}T)((\ell_{n_f}(y),i),(\pi_{n_c}x,j)) = M((y,i),(\pi_{n_c}x,j)).\end{aligned}$$

The second line uses city covariance with shift $u = n_c v$. In the third line use $x = \ell_{n_c}(\pi_{n_c}x) + n_c \operatorname{rent}_{n_c}(x)$ and the bijective reindexing $r = \operatorname{rent}_{n_c}(x) - v$ of $\mathcal{C}$. The last line is the matching identity and equals the $((\pi_{n_c}x,j),(y,i))$ entry of M^{T}. Thus no unproved periodization rule is needed.

For a rank-one packet, define its two-point kernel by $P^\sharp((x,i),(v,j)) = P((x,i),(v,j))$. Direct substitution in the push definition gives the cover-kernel identity

$$(P \star (T,U))^\sharp = (\operatorname{swap} T) \diamond (P^\sharp \diamond U). \tag{18}$$

For completeness, the absolute sum in the two compositions is finite. The row atlas estimate implies $|U((v,j),(z,l))| \le pL^{-2}$, and the folded bound implies $\sum_{v,j} |P((x,i),(v,j))| \le r$. Hence $\sum_{v,j} |P((x,i),(v,j))U((v,j),(z,l))| \le pL^{-2}r$. The outer sum against $\operatorname{swap} T$ is finite by the column atlas bound, summed over its m input types. These same estimates justify Fubini. If $P^\sharp$ matches A, and T, U match M, N, respectively, apply (17) first to $P^\sharp$ and U, then to $\operatorname{swap} T$ and that composition. The transpose calculation above supplies the match M^{T} for $\operatorname{swap} T$. The resulting displayed coarse matrix is exactly $M^{\mathsf{T}}AN$.

4.5 Exact propagation of the quadratic representative

Let $E_n^1 = \mathbb{R}^{\Lambda_n \times I_1}$ and $E_n^2 = \mathbb{R}^{\Lambda_n \times I_2}$ be the one-form and curvature coordinate spaces; $d_n : E_n^1 \to E_n^2$. The precision matrices Q_H, Q_Y act on $E_{n_f}^1$, and $Q_{H'}, Q_{Y'}$ act on $E_{n_c}^1$. The curvature representative matrix A acts on $E_{n_f}^2$, whose type set is the same I used for the packet legs. The mean matrices $M_A, N_A : E_{n_c}^1 \to E_{n_f}^1$, and the matched carrier matrices $C_Y, C_H : E_{n_c}^2 \to E_{n_f}^2$.

Here are the exact identities used below and their derivations. The two one-step precision inputs share a blocking (coarse-averaging) map $B : E_{n_f}^1 \to E_{n_c}^1$ and penalty coefficient κ. For the two histories they say

$$Q_H M_A = B^{\mathsf{T}} Q_{H'}, \quad BM_A = I - \kappa Q_{H'}, \qquad Q_Y N_A = B^{\mathsf{T}} Q_{Y'}, \quad BN_A = I - \kappa Q_{Y'}.$$

The floor data make all four precision matrices symmetric. Transposing the first identity for Y gives $N_A^{\mathsf{T}} Q_Y = Q_{Y'} B$, while transposing $BN_A = I - \kappa Q_{Y'}$ gives $N_A^{\mathsf{T}} B^{\mathsf{T}} = I - \kappa Q_{Y'}$. Therefore

$$\begin{aligned} N_A^{\mathsf{T}}(Q_Y - Q_H) M_A &= Q_{Y'} BM_A - N_A^{\mathsf{T}} B^{\mathsf{T}} Q_{H'} \\ &= Q_{Y'}(I - \kappa Q_{H'}) - (I - \kappa Q_{Y'}) Q_{H'} \\ &= Q_{Y'} - Q_{H'}. \end{aligned}$$

This proves the mixed Schur identity

$$Q_{Y'} - Q_{H'} = N_A^{\mathsf{T}}(Q_Y - Q_H) M_A. \tag{19}$$

For either history, write G, G' for fine/coarse metric matrices, D, D' for the diagonal matrices, R for the installed transfer on curvature coordinates, and Z for its mean on one-form coordinates. Its carrier is $C = G^{-1} R G'$. The floor equalities on exact forms and the diagonal tower identity are respectively $G d_f = D d_f$, $G' d_c = D' d_c$, and $R D' d_c = D d_f Z$. The floor inverse identity is $G^{-1} G = I$. Thus, with all domains as above,

$$C d_c = G^{-1} R G' d_c = G^{-1} R D' d_c = G^{-1} D d_f Z = G^{-1} G d_f Z = d_f Z.$$

Applying this calculation to the Y history with $Z = N_A$ and the H history with $Z = M_A$ proves the inherited identities

$$C_Y d_c = d_f N_A, \qquad C_H d_c = d_f M_A. \tag{20}$$

Proposition 4.5 (One-step flat-sector propagation)**.** *Let P be a rank-one packet satisfying neutrality, diagonal stationarity, and $\|P\|_{\tau,\mathrm{fold}} \le r$. Suppose the fine two-point kernel associated with P matches a matrix A, which represents the curvature difference, and*

$$d_f^{\mathsf{T}} A d_f = Q_Y - Q_H.$$

Suppose T, U are matched, on the coarse torus, by their carrier matrices $M = C_Y$ and $N = C_H$, respectively. They satisfy the order-two atlas bounds with nonnegative costs q, p and width ω used in Lemma 4.3, and are city-covariant at ratio L. The fine and coarse torus periods satisfy $n_f = L n_c$. Assume $\tau > 0$, $r \ge 0$, $0 \le \sigma \le \omega$, and $2\sigma \le L\tau$. Also assume the exact Schur and inherited identities (19) *and* (20)*. Finally suppose U has the travel property that each fixed pair of finite labels has a row sum independent of the cover point. Then the pushed packet $P' = P \star (T, U)$ has the following properties:*

1. *P' is neutral;*

2. *P' is diagonally stationary;*

3. *its folded cost satisfies*

$$\|P'\|_{\sigma,\mathrm{fold}} \le \frac{m^2 q(2/\tau)p}{L}\, r;$$

4. *P' is matched by $M^{\mathsf{T}}AN$, and*

$$d_c^{\mathsf{T}}(M^{\mathsf{T}}AN)d_c = Q_{Y'} - Q_{H'}.$$

Proof. The folded estimate is Lemma 4.3. For neutrality, expand the pushed constant-column sum, interchange the sums using the same absolute weighted bounds, and first sum in the output cover coordinate of U. Its value depends only on the finite labels and is independent of the intermediate cover point by the travel assumption. The remaining cover sum is then the neutral column sum of P, hence is zero. For diagonal stationarity, translate the outer summation variable and then the inner one. The two covariance identities for T and U, together with the simultaneous stationarity of P, leave each summand unchanged. Absolute convergence just established permits these translations in the sums.

By the exact reindexing argument in (17) and the transpose calculation following it, the pushed packet is matched by $M^{\mathsf{T}}AN$; the two applications have absolutely convergent sums by the weighted estimates. Finally, using (20),

$$\begin{aligned} d_c^{\mathsf{T}}M^{\mathsf{T}}ANd_c &= (Md_c)^{\mathsf{T}}A(Nd_c) \\ &= (d_fN_A)^{\mathsf{T}}A(d_fM_A) \\ &= N_A^{\mathsf{T}}(d_f^{\mathsf{T}}Ad_f)M_A \\ &= N_A^{\mathsf{T}}(Q_Y - Q_H)M_A \\ &= Q_{Y'} - Q_{H'}, \end{aligned}$$

where the final equality is (19). This proves the propagated curvature identity. □

Remark 4.6 (Inputs still required from the RG construction). The proposition proves the one-step packet algebra and weighted estimate conditional on the two actual order-two atlas bounds, exact tower matching, quadratic floors, and travel condition. Those hypotheses are mathematical inputs from the selected construction and must be checked there. In particular, this section does not prove that a given construction supplies these bounds, nor does it upgrade the order-zero quadratic statement to the full hierarchy of Wilson observables.

5 A scalar decay estimate for cutoff errors

The following elementary lemma converts a one-step contraction with a polynomially controlled source into a power of the ratio of scales. In an RG application, the difficult step is to prove that the difference of two quadratic precisions has a positive coefficient majorant satisfying the hypotheses below. The lemma here does not prove that RG estimate.

Proposition 5.1 (Contraction with a polynomially controlled source)**.** *Let $M \in \mathbb{N}$, and let*

$$t_0 < t_1 < \cdots < t_M$$

be real numbers whose successive gaps satisfy

$$\Delta_j := t_{j+1} - t_j \geq \delta > 0 \qquad (0 \leq j < M).$$

Fix $s \in \mathbb{R}$, $Z \geq 0$, $d \in \mathbb{N}$, $c_0, \theta > 0$, *and* $A \geq 0$. *Put* $T = t_M$, *choose*

$$\mu = \min\{c_0, \theta/2\}, \qquad \lambda = \theta - \mu, \qquad R = \max\left\{2, \frac{2(d+1)}{\lambda}\right\}, \qquad y_j = R + Z + |t_j - s|,$$

and set $q = \exp(-\lambda\delta/2)$ *and* $C = A/(1-q)$. *Suppose* $f_0, \ldots, f_M$ *and* $g_0, \ldots, g_{M-1}$ *are nonnegative real numbers such that, for* $0 \leq j < M$,

$$f_j \leq e^{-\theta\Delta_j} f_{j+1} + g_j, \qquad e^{c_0(T-t_j)} g_j \leq A y_j^d, \tag{21}$$

and at the terminal index

$$f_M \leq A y_M^d. \tag{22}$$

Then, for every $j \in \{0, \ldots, M\}$,

$$f_j \leq C y_j^d e^{-\mu(T-t_j)}. \tag{23}$$

If positive scales are defined by $a_j = e^{-t_j}$, *the exponential factor in* (23) *equals* $(a_M/a_j)^\mu$.

Proof. The constants in the conclusion depend on $A, d, c_0, \theta, \delta$ but not on s or Z. The choices give $0 < \mu \leq c_0$, $\mu \leq \theta/2$, and hence $\lambda = \theta - \mu \geq \theta/2 > 0$. Thus $0 < q < 1$ and $C \geq A$. Introduce the tilted errors

$$F_j = e^{\mu(T-t_j)} f_j, \qquad G_j = e^{\mu(T-t_j)} g_j.$$

Since $t_j \leq T$ and $\mu \leq c_0$, the source bound in (21) implies

$$0 \leq G_j \leq A y_j^d. \tag{24}$$

Multiplying the recurrence by $e^{\mu(T-t_j)}$ gives

$$F_j \leq e^{-\lambda\Delta_j} F_{j+1} + G_j. \tag{25}$$

We first control the change in the polynomial weight. The triangle inequality gives

$$y_{j+1} = R + Z + |t_{j+1} - s| \leq y_j + \Delta_j.$$

Because $y_j \geq R \geq 2(d+1)/\lambda$, we have $d/y_j \leq \lambda/2$. Using $1 + x \leq e^x$ for $x \geq 0$, we obtain

$$\begin{aligned} e^{-\lambda\Delta_j} y_{j+1}^d &\leq y_j^d e^{-\lambda\Delta_j} \left(1 + \frac{\Delta_j}{y_j}\right)^d \\ &\leq y_j^d e^{-(\lambda - d/y_j)\Delta_j} \\ &\leq y_j^d e^{-\lambda\Delta_j/2} \leq q y_j^d. \end{aligned} \tag{26}$$

For $d = 0$, the same inequalities hold with the polynomial factor equal to one.

We now argue backward from M. At the endpoint, $F_M = f_M \leq A y_M^d \leq C y_M^d$. Suppose $F_{j+1} \leq C y_{j+1}^d$. Equations (25), (24), and (26) imply

$$F_j \leq C q y_j^d + A y_j^d = (Cq + A) y_j^d = C y_j^d,$$

where the last equality follows from $C = A/(1-q)$. Induction proves this bound for every j. Undoing the tilt gives

$$f_j = e^{-\mu(T-t_j)} F_j \leq C y_j^d e^{-\mu(T-t_j)},$$

as required. Finally, $e^{-(T-t_j)} = e^{-t_M + t_j} = a_M/a_j$, so the exponential factor is the stated scale ratio to the power μ. □

Remark 5.2 (Scope of the recurrence input). The proposition is a complete scalar argument once the nonnegative recurrence, its polynomially bounded tilted source, and its endpoint estimate are known. In the Yang–Mills construction, those data must be obtained from the scale-by-scale propagation of a neutral curvature representative for the difference of two quadratic RG precisions. Establishing that propagation requires the exact Schur-complement, locality, and weighted-kernel hypotheses of the RG step. This proposition supplies none of them. It therefore proves a power-rate consequence of that RG input, not the RG input itself or convergence of the public Wilson moments.

6 Weighted strip bounds for the tower kernel

The atlas estimate in Section 3 needs a Laurent strip bound for the tower cross-kernel and for its first residue differences. The construction supplies these bounds from a stronger momentum-space estimate: each entry of the alias-dependent wire symbol is holomorphic and bounded by a summable polynomial weight in the alias. This section proves the conversion, including the scale powers and the centered carry in the difference. The entrywise wire estimate is then stated in its actual form, and its source proof is cross-referenced in the audit record.

6.1 From an analytic cube to a weighted Fourier strip

For $p = (p_1, \dots, p_4) \in \mathbb{C}^4$, let

$$\mathcal{Q}_\eta = \{p : |\operatorname{Im} p_a| < \eta, \quad |\operatorname{Re} p_a| < \pi + \eta \text{ for all } a\}.$$

Suppose f is holomorphic on $\mathcal{Q}_\eta$, $|f(p)| \le A$ there, and for every real p with $|p_a| < \pi + \eta$ and integer vector n for which $|p_a + 2\pi n_a| < \pi + \eta$ in each coordinate, $f(p + 2\pi n) = f(p)$. Define u on the horizontal strip $|\operatorname{Im} p_a| < \eta$ by translating each real coordinate into $[-\pi, \pi]$ before evaluating f. The overlap condition makes this definition independent of the representative on the cell boundaries. Moreover u is holomorphic: on each open cell it is a translate of f; on a shared boundary the two holomorphic expressions agree on a real open box, hence agree on the complex overlap by applying the one-variable identity theorem successively in the four coordinates. The same argument at a meeting of several cells gives a single analytic continuation. The bound $|u| \le A$ follows from the bound on f.

Let

$$\widehat{u}(n) = (2\pi)^{-4} \int_{[-\pi,\pi]^4} u(x) e^{-in\cdot x}\, dx, \qquad n \in \mathbb{Z}^4.$$

For $h, b > 0$ with $h + b < \eta$, shifting the ath contour by $-i(h+b)\operatorname{sgn}(n_a)$ gives

$$|\widehat{u}(n)| \le A e^{-(h+b)|n|_1}. \tag{27}$$

To justify the shift, apply Cauchy's theorem on the rectangle with horizontal sides at imaginary parts 0 and $-(h+b)\operatorname{sgn}(n_a)$. The two vertical integrals cancel because both u and $e^{-in_a p_a}$ are 2π-periodic in the real direction. Repeat in each coordinate; the integrand remains holomorphic and bounded throughout the product of the four rectangles. On the shifted contour the exponential factor has modulus $e^{-(h+b)|n|_1}$, proving the estimate.

The geometric series is bounded explicitly. Since $e^{-b} \le (1+b)^{-1}$,

$$\sum_{m\in\mathbb{Z}} e^{-b|m|} = \frac{1 + e^{-b}}{1 - e^{-b}} \le \frac{2(1+b)}{b}.$$

Taking products over the four coordinates and using (27) yields

$$\sum_{n\in\mathbb{Z}^4} e^{h|n|_1}|\widehat{u}(n)| \le A\left(\frac{2(1+b)}{b}\right)^4. \tag{28}$$

If real coefficients are needed, replace $\widehat{u}(n)$ by its real part; the estimate is unchanged. If in addition $u(-x)=\overline{u(x)}$ for real x, the Fourier coefficients are real, since conjugating the integral and changing x to $-x$ gives $\widehat{\overline{u(n)}}=\widehat{u}(n)$.

Lemma 6.1 (Uniform strip extraction)**.** *Let $\mathcal{I}$ be any index set, $w_i\ge 0$, and let f_i be a family of symbols satisfying the preceding hypotheses with common $\eta>0$ and $|f_i|\le Aw_i$. If $0<h,b$ and $h+b<\eta$, then their real Fourier coefficient kernels $K_i(n)=\operatorname{Re}\widehat{u}_i(n)$ satisfy*

$$\sum_{n\in\mathbb{Z}^4} e^{h|n|_1}|K_i(n)| \le A\left(\frac{2(1+b)}{b}\right)^4 w_i \quad (i\in\mathcal{I}).$$

The width and the prefactor are independent of the index.

Proof. Apply (28) to f_i/w_i when $w_i>0$. When $w_i=0$, the bound forces $f_i=0$, so the result is immediate. Taking real parts can only decrease absolute values. □

6.2 The finite-residue average and its scale powers

For a real cross-kernel $K(r,i,j,z)$, the strip bound of width σ and cost A means

$$\sum_{z\in\mathbb{Z}^4} e^{\sigma|z|_1}|K(r,i,j,z)| \le A \quad \text{for every fixed } r,i,j.$$

The estimate is per fixed pair of finite labels; summing over labels later contributes their cardinalities.

For an odd side L, let $\Lambda_L=(\mathbb{Z}/L\mathbb{Z})^4$ and choose the centered lift $\ell_L(r)$. Write the alias momentum as

$$q_l(p)=\frac{p+2\pi\ell_L(l)}{L}, \qquad l\in\Lambda_L.$$

For a word $u=(a_1,\dots,a_s)$ in the four coordinate directions, put

$$\chi_u(q)=\prod_{v=1}^{s}(e^{iq_{a_v}}-1), \qquad \chi_\varnothing=1.$$

The rescaled row symbol made from a thread $F(P,p,l)$ is

$$\mathcal{S}_{e,u}(P,r,p)=L^{-e}\sum_{l\in\Lambda_L} e^{iq_l(p)\cdot\ell_L(r)}\chi_u(q_l(p))F(P,p,l). \tag{29}$$

Here e is a fixed nonnegative integer; the tower two-form uses $e=2$. For each type pair, let $K_{e,u}(P,r,i,j,z)$ be the real part of the normalized Fourier coefficient at z of the corresponding entry of $\mathcal{S}_{e,u}(P,r,\cdot)$.

Define

$$\Gamma_L(l)=\prod_{a=1}^{4}(1+|\ell_L(l)_a|), \qquad w_j(l)=\Gamma_L(l)^{-j}.$$

For the centered representatives,

$$\sum_{l\in\Lambda_L}\Gamma_L(l)^{-2}\le 3^4=81. \tag{30}$$

Indeed, each coordinate lift is a distinct integer, so the left side is at most $\prod_{a=1}^4\sum_{m\in\mathbb{Z}}(1+|m|)^{-2}$. The one-dimensional sum is at most $1+2\int_1^\infty x^{-2}\,dx=3$. Also, with $g_L(l)=1+\max_a|\ell_L(l)_a|$, we have $g_L(l)\le\Gamma_L(l)$ and $\Gamma_L(l)\ge 1$.

Assume every scalar matrix entry of $F(P,p,l)$ is holomorphic on a fixed complex cube $\mathcal{Q}_\eta$ and bounded there by $A_0\Gamma_L(l)^{-j}$, uniformly in P,L,l. Assume also the overlap law needed to glue the residue-symbol coefficients across reciprocal shifts. For $p\in\mathcal{Q}_\eta$, the alias satisfies

$$|\operatorname{Im} q_l(p)_a|\le\eta/L,\qquad |q_l(p)_a|\le\frac{C_\eta(1+|\ell_L(l)_a|)}{L},$$

where one may take $C_\eta=3\pi+2\eta$ for $L\ge 1$. The elementary inequality $|e^{iz}-1|\le|z|e^{|\operatorname{Im} z|}$ therefore gives

$$|\chi_u(q_l(p))|\le C_\eta^s e^{s\eta}\,L^{-s}g_L(l)^s.$$

Moreover $|e^{iq_l(p)\cdot\ell_L(r)}|\le e^{2\eta}$ because each $|\ell_L(r)_a|\le(L-1)/2$. Hence, whenever $j\ge s+2$,

$$\begin{aligned}|\mathcal{S}_{e,u}(P,r,p)|&\le e^{2\eta+s\eta}C_\eta^sA_0L^{-(e+s)}\sum_{l\in\Lambda_L}g_L(l)^s\Gamma_L(l)^{-j}\\&\le 81e^{2\eta+s\eta}C_\eta^sA_0L^{-(e+s)}.\end{aligned} \tag{31}$$

The second line follows from $g_L(l)^s\le\Gamma_L(l)^s$ and $\Gamma_L(l)^{s-j}\le\Gamma_L(l)^{-2}$.

Applying Lemma 6.1 to the family indexed by (P,r) gives a common width $\sigma>0$ and a cost $C_{e,s}L^{-(e+s)}$, where

$$C_{e,s}=81e^{2\eta+s\eta}C_\eta^sA_0\left(\frac{2(1+b)}{b}\right)^4$$

for any fixed $b>0$ with $2b<\eta$. Thus each row of the cross-kernel obtained by Fourier coefficient extraction has strip cost $C_{e,s}/L^{e+s}$, uniformly in P, the torus residue, and the period.

Lemma 6.2 (Residue differences are the true cover differences)**.** *For a coefficient row K on $\Lambda_L\times\mathcal{C}$, define*

$$(\nabla_a^LK)(r,i,j,z)=K(r+\pi_L(e_a),i,j,z-c_a(r))-K(r,i,j,z),$$

where $c_a(r)$ is the centered-lift carry characterized in the proof below. For a word $u=(a_1,\dots,a_s)$ set $K_{e,\emptyset}=K_{e,0}$ and $K_{e,(a,u)}=\nabla_a^LK_{e,u}$. If $K_{e,u}$ is the real Fourier coefficient row of $\mathcal{S}_{e,u}$, then $K_{e,(a,u)}=\nabla_a^LK_{e,u}$. Consequently the order-s difference row has width cost $C_{e,s}L^{-(e+s)}$.

Proof. The centered lift has a unique carry $c_a(r)\in\mathbb{Z}^4$ such that

$$\ell_L(r+\pi_L(e_a))+Lc_a(r)=\ell_L(r)+e_a.$$

For odd L, $c_a(r)$ is e_a precisely at the upper centered boundary in coordinate a, and is zero otherwise. In the symbol (29), multiplication by the new factor $e^{iq_l(p)_a}-1$ gives

$$\mathcal{S}_{e,(a,u)}(P,r,p)=e^{ip\cdot c_a(r)}\mathcal{S}_{e,u}(P,r+\pi_L(e_a),p)-\mathcal{S}_{e,u}(P,r,p).$$

To check the phase, use $Lq_l(p)=p+2\pi\ell_L(l)$ and $e^{2\pi i\ell_L(l)\cdot c_a(r)}=1$; the change from $\ell_L(r)$ to $\ell_L(r+\pi_L(e_a))$ contributes exactly $e^{iq_l(p)_a}e^{-ip\cdot c_a(r)}$. This is the Fourier transform of the displacement reindexing by $c_a(r)$ in the definition of ∇_a^L. Fourier coefficients are unique for these absolutely summable rows, so the coefficient identity follows. Repeating this one-step argument proves the formula for every word. □

6.3 The actual wire receipt

The preceding estimates apply to the construction's actual dual cross-stencil $F(P,p,l)$. Its suppliers give the following exact data. Let $j,t\ge 4$, $k>0$, let $\mathcal{I}$ be any index set, choose a flight P_i of stages (j,t) and a site $l_i\in\Lambda_{L_i}$ for each $i\in\mathcal{I}$. There are common $\eta,A>0$ such that every scalar matrix entry of $F(P_i,p,l_i)$ is holomorphic on $\mathcal{Q}_\eta$ and bounded there by $A\Gamma_{L_i}(l_i)^{-j}$, uniformly in the indices and the period. The thread also obeys the precise reciprocal-shift law

$$F(P,p+2\pi x,l)=F(P,p,l+\operatorname{wrap}_L(x)),\tag{32}$$

whenever both momenta lie in the supplier's real enlarged cube. Here $\operatorname{wrap}_L(x)$ is the residue characterized by $\ell_L(l+\operatorname{wrap}_L(x))\equiv\ell_L(l)+x \pmod{L\mathbb{Z}^4}$. The formal supplier proves these statements for every odd side $L\ge 3$, every flight with the two stage indices at least 4, and every $k>0$. Their factor-by-factor proof is recorded in Section 7; the present section proves the quantitative consequences of those exact supplier estimates.

Here is the branch algebra in mathematical notation. Let $m_P(p)=\operatorname{area}_L(p)^j$ be the patch mass factor (distinct from the cumulative smooth factor of the train), $b_v(p)$ the vector of four bulb factors for a train v, and $\beta_v(p)$ its bend factor. For a vector g indexed by the four coordinate directions, let $\mathcal{M}_r(g)$ be the diagonal matrix whose entry on an r-face is the product of the g_a over its directions. Let

$$\mathcal{T}_2(x)_{ff}=\sum_{a\in f}x_a,\qquad \operatorname{yen}(g)(p)=g(p)g(-p).$$

For $v\in\mathbb{C}^4$, let $\mathsf{W}_r(v)$ denote the map of exterior multiplication $\alpha\mapsto v\wedge\alpha$ from r-faces to $(r+1)$-faces. Put

$$\Sigma_S^r(p)=\mathsf{W}_r\big((S(e^{ip_a}-1))_{a=1}^4\big),\quad \mathcal{P}_v(p)=\beta_v(p)^{-1}\mathcal{M}_2(b_v(p)^{-1}),\mathcal{V}_v(p)=\beta_v(p)\mathcal{M}_2(b_v(p)).$$

Write $q_l=(p+2\pi\ell_L(l))/L$, including q_0 at the zero residue. Write $r_L(q)_a=\operatorname{rail}_L(q_a)$. The r-flux and latch matrices are

$$\Phi_{P,r}(p,l)=\big[m_P(-q_l)\mathcal{M}_r(r_L(-q_l))\big]^{\mathsf{T}},\qquad \Lambda_{P,r}(p)=m_P(q_0)^{-1}\mathcal{M}_r(r_L(q_0)^{-1}).$$

The nonzero-residue port vector and scalar are

$$D_{P,l,a}(p)=\frac{b_{P_{\text{wing}}}(q_l)_a}{\operatorname{blow}_{P_{\text{wing}}.mass,L}(p,l)},\qquad R_{P,l}(p)=\operatorname{blow}_{P_{\text{wing}}.mass,L}(p,l)^2\beta_{P_{\text{wing}}}(q_l).$$

The land divisor is $\rho_P(p)=\operatorname{divisor}(P_{\text{land}}.mass)(p)$, and the tail vector is

$$\operatorname{tail}_{P,a}(p)=k^{-1}+\sum_{l\in\Lambda_L,\,l\neq 0}\operatorname{yen}(m_P(\cdot)r_L(\cdot)_a)\,(q_l)D_{P,l,a}(p).$$

Further set

$$h_P(p)=\Lambda_{P,2}(p)\Sigma_1^1(p),\quad f_P(p)=\frac{\operatorname{yen}(m_P)(q_0)\,\beta_{P_{\text{land}}}(p)}{\beta_{P_{\text{wing}}}(q_0)},\quad M_P(p)=\mathcal{M}_1(b_{P_{\text{land}}}(p)),$$

and, with $x_a(p)=\operatorname{tail}_{P,a}(p)/b_{P_{\text{land}}}(p)_a$,

$$B_P(p)=\mathcal{P}_{P_{\text{wing}}}(q_0)\big(\mathcal{T}_2(x(p))-\rho_P(p)\mathcal{M}_2(x(p))\big),$$

$$N_P(p)=\rho_P(p)^2I-\rho_P(p)\mathcal{P}_{P_{\text{land}}}(p)\Sigma_1^1(p)M_P(p)\Sigma_1^1(-p)^{\mathsf{T}}.$$

For a nonzero alias l, let $D_{P,l}$ be the vector with components $D_{P,l,a}$ above and set $Y_{P,l}(p) = \mathrm{yen}(m_P)(q_l)\beta_{P_{\mathrm{land}}}(p)/R_{P,l}(p)$. The dual stencil is exactly

$$W_{P,0}(p) = f_P(p)\Phi_{P,2}(p,0) + \big(B_P(p)h_P(p)M_P(p) - \mathcal{P}_{P_{\mathrm{wing}}}(q_0)h_P(p)\mathcal{M}_1(\mathrm{tail}_P(p))\big)\Sigma_1^1(-p)^{\mathsf{T}}, \quad (33)$$

$$W_{P,l}(p) = \mathcal{P}_{P_{\mathrm{wing}}}(q_l)\Sigma_L^1(q_l)\mathcal{M}_1(D_{P,l}(p))\Phi_{P,1}(p,l)\Sigma_1^1(-p)^{\mathsf{T}} + Y_{P,l}(p)\Phi_{P,2}(p,l)N_P(p), \quad l \neq 0. \quad (34)$$

These identities follow by expanding the definitions of the two branches and the listed auxiliary matrices; they are not additional hypotheses.

The response-space notation in those formulas is as follows. For a train v and side S, set $\mathrm{fieldD}_{S,v}(p) = \mathcal{M}_1(S^{-2}\mathscr{D}_v(p))$, where $\mathscr{D}_v$ is the vector of level responses. The hinge matrix is $\mathrm{hinge}_{P,r}(q) = m_P(q)\mathcal{M}_r(r_L(q))$; the flux is $\Phi_{P,r}(p,l) = \mathrm{hinge}_{P,r}(-q_l)^{\mathsf{T}}$; and $\mathrm{mirage}_{1,1}(p) = \Sigma_1^1(-p)^{\mathsf{T}}$.

Here is the exact reduction behind the branch calculations. For a train v of mass side M, direction a, and momentum p, the level response is

$$\mathscr{D}_{v,a}(p) = \mathrm{feed}_{v,a}\big((\mathrm{divisor}_1)^{-1}\big)(p) + k^{-1}\,\mathrm{noise}_{v,a}(p),$$
$$\mathrm{feed}_{v,a}(f)(p) = M^{-2}\sum_{r\in\Lambda_M} D_{v,a}(q_r^{(M)}(p))D_{v,a}(-q_r^{(M)}(p))f(q_r^{(M)}(p)).$$

The feed is the normalized finite alias sum of the two-sided drive $D_{v,a}(q)D_{v,a}(-q)$ times its input, and the noise is the sum of the constant-input feeds along the finite nozzle history. Every train drive is reciprocal-periodic. The reciprocal divisor input is periodic because its chip factors are invariant under integer reciprocal shifts. A finite alias sum of a periodic summand is periodic: a reciprocal shift permutes its alias sites. More explicitly, for a hop of side L from a train of mass side M, the mixed-radix alias bijection $\Lambda_{LM} \to \Lambda_L \times \Lambda_M$ is determined by $\ell_{LM}(r) \equiv \ell_L(l) + L\ell_M(v) \pmod{LM}$, with the unique centered representatives differing by a multiple of LM absorbed into a reciprocal shift. This is a bijection because every residue modulo LM has a unique decomposition into a residue modulo L and then a residue modulo M after division by L. Thus $q_r^{(LM)}(p) = q_v^{(M)}(q_l^{(L)}(p))$ modulo $2\pi\mathbb{Z}^4$. Reindexing this bijection in the normalized feed sum and using $(LM)^{-2} = L^{-2}M^{-2}$ gives

$$\mathrm{feed}_{\mathrm{hop}}(f) = \mathrm{slide}_L(\mathrm{feed}_v(f)), \qquad D_{\mathrm{hop},a}(z) = \mathrm{pullLine}_{P,a}(Mz)D_{v,a}(z).$$

The nozzle list has a top entry with constant drive, contributing 1; each earlier entry is the lifted nozzle list of the predecessor train. Reindexing the same aliases gives $\mathrm{Noise}_{\mathrm{hop}} = 1 + \mathrm{slide}_L\,\mathrm{Noise}_v$. Consequently $\mathscr{D}_{\mathrm{hop},a} = \mathrm{slide}_{L,a}\,\mathscr{D}_{v,a} + k^{-1}$, the response feedback used below. Induction on the finite nozzle length proves reciprocal periodicity of every $\mathscr{D}_{v,a}$. The level scalar

$$\rho_v(p) = \sum_a \frac{J_a(p)}{\mathscr{D}_{v,a}(p)}, \qquad J_a(p) = (e^{ip_a} - 1)(e^{-ip_a} - 1),$$

is periodic because each J_a and each $\mathscr{D}_{v,a}$ is periodic. Off $p = 0$, the source recovery identity is $\rho_v = \mathrm{divisor}_M(p)^2\beta_v(p)$ and the level response identity is $\mathscr{D}_{v,a} = b_{v,a}/\,\mathrm{divisor}_M(p)$. Substituting these identities in the definitions of pulse and vault gives

$$\mathrm{pulse}_v(p) = \rho_v(p)^{-1}\mathcal{M}_2(\mathscr{D}_v(p)^{-1}), \qquad \mathrm{vault}_v(p) = \rho_v(p)\mathcal{M}_2(\mathscr{D}_v(p)).$$

Thus pulses are reciprocal-periodic on every nonzero overlap. A nonzero reciprocal shift cannot have both endpoints in the small real cube while one endpoint is zero; the elementary overlap separation inequality proves this. These are the level-response and pulse identities used below, not periodicity of the unscaled divisor $\mathrm{divisor}_M$ (which is not periodic when $M > 1$).

For clarity, the branch algebra is verified here. All products below are matrix products in the indicated face spaces. At a nonzero alias write $q = q_l$, $s = \rho_P(p)$, $D = \mathcal{M}_1(D_{P,l})$, $D' = \mathcal{M}_1(b_{P_{\mathrm{land}}}(p))$, and abbreviate the two order-one signals and the order-one/order-two fluxes by their displayed symbols. The port definitions and the divisor recovery identity give

$$\mathrm{chainY}_P(p,q) = Y_{P,l}(p)\,\rho_P(p)^2, \qquad \mathrm{portD}_{P,l,a}(p) = L^{-2}\mathscr{D}_{P_{\mathrm{wing}},a}(q_l).$$

The first equality follows by substituting $R_{P,l} = \mathrm{blow}^2_{P_{\mathrm{wing}}.mass,L}\,\beta_{P_{\mathrm{wing}}}(q)$ into $Y_{P,l}$ and then using $\rho_P^2\beta_{P_{\mathrm{land}}}/\beta_{P_{\mathrm{wing}}}$ for the ratio of the two bend factors; the second is the definition of the port vector after cancellation of its blow-up denominator. Also

$$N_P = s^2 I - s\,\mathcal{P}_{P_{\mathrm{land}}}(p)\Sigma_1^1(p)D'\Sigma_1^1(-p)^{\mathsf{T}}, \qquad \mathrm{notch}_P = s^2(I - \mathrm{cloud}_P).$$

For the first identity use $s\,\mathrm{fieldD}_{1,P_{\mathrm{land}}} = \mathcal{M}_1(b_{P_{\mathrm{land}}})$ in the definition of the cloud; the second follows from the mouth identity and the definition of the notch. Substituting these identities in the nonzero branch, and using $Y_{P,l}s^2 = \mathrm{chainY}_P$, gives exactly

$$\mathcal{P}_{P_{\mathrm{wing}}}(q)\Sigma_L^1(q)D\Phi_{P,1}\Sigma_1^1(-p)^{\mathsf{T}} + \mathrm{chainY}_P(p,q)\Phi_{P,2}(p,l)(I - \mathrm{cloud}_P(p)) = W_{P,l}(p).$$

This is the stake branch identity, including its order of matrix factors.

The heel is, by definition,

$$\mathrm{heel}_{P,a}(p) = \mathrm{yen}(\mathrm{pullLine}_{P,a})(q_0)b_{P_{\mathrm{wing}},a}(q_0).$$

The relevant one-step identity for this response is

$$\mathscr{D}_{P_{\mathrm{land}},a}(p) = \mathrm{slide}_{P,a}\,\mathscr{D}_{P_{\mathrm{wing}},a}(p) + k^{-1}, \qquad (\mathrm{slide}_{P,a}\,g)(p) = L^{-2}\sum_{l\in\Lambda_L}\mathrm{yen}(\mathrm{pullLine}_{P,a})(q_l)g(q_l).$$

Use the port-to-response equality $\mathrm{portD}_{P,l,a} = L^{-2}\mathscr{D}_{P_{\mathrm{wing}},a}(q_l)$ in this finite sum. The zero-residue summand is $\mathrm{heel}_{P,a}(p)/s$, by the zero-port identity $\mathrm{blow}_{P_{\mathrm{wing}}.mass,L}(p,0) = s$; the remaining residues sum to the above-alias tail. Adding k^{-1} gives $\mathscr{D}_{P_{\mathrm{land}},a} = \mathrm{heel}_{P,a}/s + \mathrm{tail}_{P,a}$. Since off the origin $\mathscr{D}_{P_{\mathrm{land}},a} = b_{P_{\mathrm{land}},a}/s$, multiplication by s proves the bulb network identity

$$b_{P_{\mathrm{land}},a}(p) = \mathrm{heel}_{P,a}(p) + s\,\mathrm{tail}_{P,a}(p).$$

For the zero branch write $q = q_0$, $s = \rho_P(p)$, H and H' for the wing and land pulses at q and p, $d = \Sigma_1^1(p)$, $\delta = \Sigma_L^1(q)$, $v = \Phi_{P,2}(p,0)$, $u = \Phi_{P,1}(p,0)$, and $m = \mathrm{mirage}_{1,1}(p)$. Let $f_1 = \mathrm{hinge}_{P,1}(q)$ and $f_2 = \mathrm{hinge}_{P,2}(q)$, $\lambda = \Lambda_{P,2}(p)$, $R = \mathcal{M}_1(\mathrm{tail}_P(p))$, $B = B_P(p)$, $D = \mathrm{fieldD}_{L,P_{\mathrm{wing}}}(q)$, $D' = \mathrm{fieldD}_{1,P_{\mathrm{land}}}(p)$, and $y = \mathrm{foot}_P(p)$. These symbols will be used in the brace, one-form, and cancellation identities below. For the brace relation, abbreviate the land and wing bulb vectors at their respective arguments by f and g, the two patch masses by $c_0 = m_P(q_0)$ and $d_0 = m_P(-q_0)$, and the corresponding rails by $E_a = \mathrm{rail}_L(q_{0,a})$ and $F_a = \mathrm{rail}_L(-q_{0,a})$. Put $Z_a = \mathrm{heel}_{P,a}(p)/f_a$ and $x_a = \mathrm{tail}_{P,a}(p)/f_a$. The definitions of heel and the port-to-response formula give, coordinate by coordinate,

$$g_a^{-1}Z_a = c_0 d_0\,F_a f_a^{-1}E_a.$$

Inserting this in the diagonal two-form pulse and multiplying by the land pulse, two-form flux, and two-form hinge gives $y\,vH'f_2 = H\mathcal{M}_2(Z)$. The bulb network identity gives $Z_a = 1 - sx_a$. On a two-face $\{a,b\}$, direct multiplication yields

$$1 - (1 - sx_a)(1 - sx_b) = s(x_a + x_b - sx_a x_b).$$

Since $\mathcal{T}_2(x)$ has diagonal entry $x_a + x_b$ and $\mathcal{M}_2(x)$ has entry $x_a x_b$, this is the matrix identity $I - \mathcal{M}_2(1 - sx) = s(\mathcal{T}_2(x) - s\mathcal{M}_2(x))$. Multiplying by $H = \mathcal{P}_{P_{\mathrm{wing}}}(q_0)$ proves $sB = H - yvH'f_2$, the brace relation used in the cancellation.

For the one-form flux, each matrix entry of the hinge-flux product is diagonal in the coordinate direction. At the zero alias it equals

$$\big(f_1 Du\big)_{aa} = \mathrm{yen}(m_P)(q_0)\,\mathrm{rail}_L(q_{0,a})\,\mathrm{rail}_L(-q_{0,a})\,\mathrm{fieldD}_{L,P_{\mathrm{wing}}}(q_0)_a.$$

Because $\mathrm{fieldD}_{L,P_{\mathrm{wing}}}(q_0)_a = b_{P_{\mathrm{wing}},a}(q_0)/s$, the right side is $\mathrm{heel}_{P,a}(p)/s$. The scalar toe-feed identity above says this equals $\mathscr{D}_{P_{\mathrm{land}},a}(p) - \mathrm{tail}_{P,a}(p)$, proving $f_1 Du = D' - R$ entry by entry. The equality $\mathrm{chainY}_P(p, q_0) = \mathrm{foot}_P(p)$ follows as follows. Divisor recovery gives $\rho_{P_{\mathrm{land}}} = s^2 \beta_{P_{\mathrm{land}}}$ and $\rho_{P_{\mathrm{wing}}} = d_w^2 \beta_{P_{\mathrm{wing}}}$, where $d_w = \mathrm{divisor}(P_{\mathrm{wing}}.mass)(q_0)$. The zero-port identity gives $s = L^2 d_w$. Since $\mathrm{scaleR}_{1,P_{\mathrm{land}}} = \rho_{P_{\mathrm{land}}}$ and $\mathrm{scaleR}_{L,P_{\mathrm{wing}}} = L^4 \rho_{P_{\mathrm{wing}}}$, their ratio is $\beta_{P_{\mathrm{land}}}/\beta_{P_{\mathrm{wing}}}$. Multiplying by $\mathrm{yen}(m_P)(q_0)$ gives both the chain and foot expressions.

The bank and land-response identities are

$$\mathrm{mouth}_P = \mathcal{M}_1(b_{P_{\mathrm{land}}}), \qquad s\,\mathrm{fieldD}_{1,P_{\mathrm{land}}} = \mathcal{M}_1(b_{P_{\mathrm{land}}}).$$

Thus $\mathrm{mouth}_P = sD'$. The latch inverse, hinge transport, and the order-one flux calculation give the identities

$$\begin{aligned} f_2\lambda = I, \qquad \lambda f_2 = I, \\ f_2\delta = df_1, \\ f_1 Du = D' - R, \\ sB = H - y\,vH'f_2, \qquad \mathrm{mouth}_P = sD'. \end{aligned}$$

The hinge-transport line follows entrywise from the alias relation $L\,\mathrm{rail}_L(q_{0,a})\,\mathrm{chip}(q_{0,a}) = \mathrm{chip}(p_a)$, inserted into the definitions of the hinge and signal. The third line is the one-form identity obtained by multiplying the flux by its hinge; the fourth is the brace identity after the two-form flux calculation. The mouth identity was derived just above. Also $\mathrm{chainY}_P(p, q) = y$. These identities imply, with no commutation of noncommuting factors,

$$\begin{aligned} H\delta Du\,m = H\lambda df_1 Du\,m = H\lambda d(D' - R)m, \\ B\lambda d\,\mathrm{mouth}_P\; m - H\lambda dR\,m = H\lambda dD'm - yvH'dD'm - H\lambda dR\,m. \end{aligned}$$

For the first equality use $\delta = \lambda df_1$ (which follows from $f_2\delta = df_1$ and $f_2\lambda = I$). For the second use $\mathrm{mouth}_P = sD'$ and $sB = H - yvH'f_2$, then cancel the adjacent $f_2\lambda$. Adding yv to the second line yields $H\lambda d(D' - R)m + yv(I - H'dD'm)$, and its first term is $H\delta Du\,m$. This is precisely the zero-alias script formula and proves the copper branch identity. Each displayed identity is an equality of the finite matrices defined above; the component chapter supplies the scalar divisor, rail, and bend inputs on the common analytic cube.

The reciprocal-shift law used in the Fourier argument also follows from the actual wire symbol, rather than being an independent periodicity assumption. For $x \in \mathbb{Z}^4$ and a residue l, the centered alias arithmetic gives a cover vector y such that

$$q_l(p + 2\pi x) = q_{l+\mathrm{wrap}_L(x)}(p) + 2\pi y.$$

Every factor of the script is invariant under these reciprocal shifts. The level responses are periodic by the finite-feed and nozzle induction above; hence their diagonal matrices and the scalar ρ_v are

periodic. The pulses are periodic on nonzero overlaps by their displayed level-source formula. The signals and the jewel factors are periodic because they are formed from $e^{ip_a} - 1$ and its opposite-momentum partner. The patch mass is periodic by the plank identity. The scalar factor is exactly

$$\mathrm{chainY}_P(p,q) = \mathrm{yen}(m_P)(q)\,\frac{\rho_{P_{\mathrm{land}}}(p)}{L^4\rho_{P_{\mathrm{wing}}}(q)}.$$

It is periodic by the patch-mass and level-response identities. The other factors in scriptT are level responses, pulses, signals, hinges, or fluxes already shown periodic. Substitution in its definition therefore yields $\mathrm{scriptT}_P(p+2\pi x, q+2\pi y) = \mathrm{scriptT}_P(p,q)$. For nonzero external momentum, the copper and stake branch identities just proved identify the wire with this script; the zero momentum endpoint follows directly from the definitions. Combining this script invariance with the residue permutation $l \mapsto l + \mathrm{wrap}_L(x)$ gives (32). More explicitly, when both external momenta lie in the real enlarged cube and $x \neq 0$, both are nonzero; the alias exchange above supplies the compensating fine reciprocal vector, and the shift identity for the script removes it. When $x = 0$ the claim is identity. This is the reciprocal-shift content of the wire matching argument; the companion audit records its exact source declarations. The real enlarged cube has coordinate radius $\pi + \mathrm{zone}$, where $\mathrm{zone} = \pi/4$ is the padding. Since $\pi + \mathrm{zone} < 2\pi$, no nonzero reciprocal vector can connect zero momentum to another point of this cube; this is the separation used above.

The product estimates use the following weights, all uniform on a common complex cube and independent of L:

factor	weight in the alias label l
mass factor $m_P(q_l)$	$\Gamma_L(l)^{-j}$
rail factors and reciprocal rails at the zero alias	1
dual multipliers, bend ratios at zero alias, correction and tail terms	1
scaled signal $\Sigma^1_L(q_l)$	$g_L(l)$
port mask $\mathcal{M}_1(D_{P,l})$	$g_L(l)^{-2}$
flux $\Phi_{P,r}(p,l)$	$\Gamma_L(l)^{-j}$
levy and notch at nonzero alias	1

The component suppliers prove these bounds by the estimates listed in Section 7. For the zero branch every product has total weight 1. In the first nonzero product the total weight is at most $g_L(l)g_L(l)^{-2}\Gamma_L(l)^{-j} \le \Gamma_L(l)^{-j}$; the second has weight $\Gamma_L(l)^{-j}$. For matrix products, apply $|(AB)_{ij}| \le \sum_r |A_{ir}||B_{rj}|$; every finite intermediate type sum contributes a fixed cardinality. Taking the larger of the two branch constants and the smaller of their widths yields the entrywise weighted analytic estimate stated above. The branch multiplication and the weights it must deliver are explicit here; Section 7 derives the component estimates used in that multiplication.

Theorem 6.3 (Wire symbol yields the tower's order-e strip receipt)**.** *Under $j \ge 4$, odd L, and the entrywise weighted analytic bound together with the reciprocal-shift overlap identity* (32)*, there are $\sigma > 0$ and $C_{e,j} > 0$, independent of the flight, residue, and period, such that for every word u of length s with $s \le j-2$, the cross-kernel row $K_{e,u}$ has strip width σ and cost $C_{e,j}/L^{e+s}$ for every flight, residue, and fixed type pair. In particular $e = 2$ gives the tower component T the base cost $C_{2,j}/L^2$ and each first-difference cost $C_{2,j}/L^3$ required by Lemma 3.1.*

Proof. For a fixed word u of length s, the finite-residue estimate (31) gives a common analytic bound with factor $L^{-(e+s)}$. To check the overlap, let $x \in \mathbb{Z}^4$ and let $l' = l + \mathrm{wrap}_L(x)$. By the defining congruence for the wrap, there is $n \in \mathbb{Z}^4$ such that

$$q_l(p + 2\pi x) = q_{l'}(p) + 2\pi n.$$

The chip factors and the exponential displacement factor are unchanged by this integral reciprocal shift, and the supplier law (32) changes $F(P, p + 2\pi x, l)$ into $F(P, p, l')$. The map $l \mapsto l'$ is a permutation of the finite residue set, so reindexing the finite sum proves equality of the two row symbols on the real overlap. Holomorphic uniqueness extends the equality to the overlapping complex domain used for coefficient extraction. Lemma 6.1 therefore gives the strip cost $C_{e,s}/L^{e+s}$. There are only finitely many words in the four coordinate directions of length at most $j-2$. The width σ is the same for every word; take the maximum of the finitely many costs $C_{e,s}$ over all words of length at most $j-2$ to obtain one cost $C_{e,j}$. Finally Lemma 6.2, iterated along u, identifies the word's Fourier coefficient row with the iterated cover difference. This is exactly the stated receipt. □

Remark 6.4 (Remaining local analytic inputs). This proof discharges the passage from a uniform alias-weighted analytic bound to all required tower strip receipts, with explicit L^{-2-s} scaling and the carry-corrected differences. Its mathematical dependence on the actual dual-kernel producer is restricted to the entrywise weighted analytic estimate and the component inequalities listed in the audit. In particular, the finite-residue geometry, scale powers, geometric coefficient summation, overlap gluing, and finite uniformization are proved here. The selected response atlas then follows from Theorem 3.3.

7 Component estimates for the dual wire symbol

This section supplies the factor bounds used in the dual-stencil calculation of Section 6. All constants below may depend on the fixed smoothing stages, the positive coupling parameter, and the four-dimensional cell conventions, but not on the odd side length or the alias. Write

$$\Gamma_L(l) = \prod_{a=1}^{4} (1 + |\ell_L(l)_a|), \qquad g_L(l) = 1 + \max_a |\ell_L(l)_a|.$$

The centered representative satisfies $g_L(l) \le \Gamma_L(l)$. For $z \in \mathbb{C}$ define

$$\mathrm{rail}_L(z) = \frac{1}{L} \sum_{v=0}^{L-1} e^{ivz}, \qquad \mathrm{plank}_L(z) = e^{-i\lfloor L/2 \rfloor z}\, \mathrm{rail}_L(z).$$

For $p \in \mathbb{C}^4$ set

$$\mathrm{area}_L(p) = \prod_{a=1}^{4} \mathrm{plank}_L(p_a).$$

Set zone $= \pi/4$ and $R_0 = 3\pi/2$. Since $(e^{iz} - 1)/z$ extends continuously and has no zero on the closed disk $|z| \le R_0$, compactness gives constants $0 < \ell \le h$ such that

$$\ell|z| \le |e^{iz} - 1| \le h|z| \qquad (|z| \le R_0).$$

Put $p_0 = \min(1, \ell/h)$ and $c_0 = hR_0/p_0$. These are the uniform rail/plank lower and principal-plank constants used below. In the divisor estimates write low $= \ell$ and high $= h$; for real four-vectors, $|p|$ denotes the Euclidean norm. At each stage j, a train is built by finite recursion. The top train top_j has mass side 1 and extra factor 1. A hop has output stage j, coarse dilation an odd integer $L \ge 3$, and a predecessor train of stage $J \in \{j, j+1\}$; if the predecessor has mass side M, the hop

has mass side LM and extra factor $\mathrm{area}_{LM}(p)^{J-j}$ times the predecessor's extra factor. Thus every train is a finite history, and no bound on its number of hops is assumed. For a side M, set

$$s_M(p)_a = M\big(e^{ip_a/M} - 1\big), \qquad d_M(p) = \sum_{a=1}^{4} s_M(p)_a s_M(-p)_a.$$

Thus d_M is the divisor, and for real p it is the sum of the four squared seed magnitudes. A train v of stage j and mass side $M = v.mass$ has cumulative filter $S_v(p) = \mathrm{area}_M(p)^j v.extra(p)$; this is distinct from the patch mass factor $m_P(p) = \mathrm{area}_{P.L}(p)^{P.J}$ used in the wire branches. A flight with indices $j, t \geq 4$ consists of an odd coarse side $L \geq 3$, a wing train of stage j, and indices $t \leq j \leq t+1$; its land train is the hop of that wing with output stage t. Its patch mass is the separate factor $m_P = \mathrm{area}_L^j$. For a train of side M and direction a, define its drive, lid, and halo by

$$D_{v,a}(q) = S_v(q)\,\mathrm{rail}_M(q_a), \quad \ell_{v,a}(p) = D_{v,a}(p/M)D_{v,a}(-p/M), \quad h_v(p) = S_v(p/M)S_v(-p/M).$$

For an alias side L, write $q_l^{(L)}(p) = (p + 2\pi\ell_L(l))/L$. For a train of side M, set $\mathrm{blow}_{M,L}(p,l) = L^2 d_M(q_l^{(L)}(p))$. The lift bound $|\ell_L(l)_a| \leq (L-1)/2$ gives $q_l^{(L)}(\mathcal{Q}_\eta) \subset \mathcal{Q}_\eta$. The feed with constant input one, the high-alias term, the noise drain, bulb, and bend are

$$\begin{aligned}
F_{v,a}(p) &= M^{-2} \sum_{l \in \Lambda_M} D_{v,a}(q_l^{(M)}(p)) D_{v,a}(-q_l^{(M)}(p)),\\
H_{v,a}(p) &= \sum_{l \neq 0} \frac{D_{v,a}(q_l^{(M)}(p)) D_{v,a}(-q_l^{(M)}(p))}{M^2 d_1(q_l^{(M)}(p))},\\
G_{v,a}(p) &= H_{v,a}(p) + k^{-1} N_{v,a}(p),\\
b_{v,a}(p) &= \ell_{v,a}(p) + d_M(p) G_{v,a}(p),\\
\beta_v(p) &= h_v(p)^{-1} - \sum_{a=1}^{4} \frac{J_a(p) G_{v,a}(p)}{\ell_{v,a}(p) b_{v,a}(p)}, \qquad J_a(p) = (e^{ip_a} - 1)(e^{-ip_a} - 1).
\end{aligned}$$

The noise is defined from a list: the top train has empty list; a hop of coarse side L has as its list one top train of side one followed by the images of every entry of the predecessor's list under that same hop. For a list $\mathcal{N}$, its noise at direction a is the sum, over $u \in \mathcal{N}$, of the feed

$$\begin{aligned}
\mathrm{feed}_{u,a}(p) &= \frac{1}{u.mass^2} \sum_{l \in \Lambda_{u.mass}} D_{u,a}(q_l^{(u.mass)}(p)) D_{u,a}(-q_l^{(u.mass)}(p)),\\
N_{v,a}(p) &= \sum_{u \in \mathcal{N}(v)} \mathrm{feed}_{u,a}(p).
\end{aligned}$$

In the high-alias term, the divisor side is the unit side, as in the source definition of the high part of the drain; the port blow-up below uses the indicated train side M.

7.1 The one-dimensional alias estimate

Lemma 7.1 (Rail and box decay)**.** *There is a constant B such that, for every odd $L \geq 1$, every $p \in \mathcal{Q}_\eta$ with η in a fixed sufficiently small cube, and every centered residue l,*

$$|\,\mathrm{rail}_L(q_l^{(L)}(p)_a)|, |\,\mathrm{plank}_L(q_l^{(L)}(p)_a)| \leq B(1 + |\ell_L(l)_a|)^{-1}.$$

At the zero residue the reciprocal rail is also uniformly bounded on a common smaller complex cube.

Proof. If $\ell_L(l)_a = 0$, the imaginary part of $Lq_l^{(L)}(p)_a = p_a$ is bounded by η, and the normalized finite geometric sum defining the rail has modulus at most $\exp(\eta)$; multiplication by the centered phase defining the plank costs at most another fixed exponential factor. If $\ell_L(l)_a \neq 0$, the centered-lift estimate for aliases gives

$$(\pi/4)(1 + |\ell_L(l)_a|) \leq L|q_l^{(L)}(p)_a| \leq C(1 + |\ell_L(l)_a|).$$

The lower bound uses that $|\operatorname{Re} p_a| < \pi + \eta$ and the lift is a nonzero integer; the upper bound is the triangle inequality. On the fixed disk containing these momenta, the function $(e^{iz} - 1)/z$ is bounded above and bounded away from zero, with its value at zero defined by continuity. The identity

$$L \operatorname{rail}_L(z)(e^{iz} - 1) = e^{iLz} - 1$$

then bounds the rail by a constant divided by $L|e^{iz} - 1|$, hence by the stated inverse lift weight. The plank differs from the rail by a centered phase whose modulus is bounded on the same complex cube. At the zero alias, the real-frequency rail has a uniform positive lower bound on the contracted disk. The following quantitative continuation promotes it to a common complex cube, uniformly in the side. Suppose a holomorphic function has modulus at most Cw on a cube of width η, with $w > 0$, and modulus at least cw on its real slice of width θ. At a complex point p, let $y = \operatorname{Re} p$. The one-variable Schwarz estimate on a ball centered at y gives

$$|f(p) - f(y)| \leq \frac{4Cw}{\eta} \|p - y\|_\infty$$

provided $\|p - y\|_\infty \leq \eta/4$ and the ball lies in the original cube. Thus on the smaller cube of width $r = \min(\eta/4, \theta, c\eta/(8C))$ one has $|f(p)| \geq (c/2)w$. In particular the reciprocal is holomorphic there and bounded by $2/(cw)$. Applying this with $w = 1$ to the zero-alias rail proves the asserted uniform reciprocal bound. □

Lemma 7.2 (Principal plank estimate with unit coefficient)**.** *If $M \leq L$, z is real, and $L|z| \leq R_0$, then*

$$|\operatorname{plank}_M(z)| \geq \exp\bigl(-c_0 M/L\bigr).$$

Proof. The centered-sum formula is

$$\operatorname{plank}_M(z) = \frac{1}{M} \sum_{v=0}^{M-1} e^{i(v - \lfloor M/2 \rfloor)z}.$$

Each centered integer has modulus at most M. The chip upper bound therefore gives $|\operatorname{plank}_M(z) - 1| \leq hM|z| \leq hR_0 x$, where $x = M/L$. The rail-chip quotient and the two-sided chip comparison also give $|\operatorname{plank}_M(z)| \geq p_0$ because $M|z| \leq R_0$ and $|z| \leq R_0$. Consequently $1 \leq |\operatorname{plank}_M(z)| + hR_0 x$ by the triangle inequality. Since $e^{c_0 x} \geq 1 + c_0 x$ and $|\operatorname{plank}_M(z)|/p_0 \geq 1$, it follows that $e^{c_0 x}|\operatorname{plank}_M(z)| \geq |\operatorname{plank}_M(z)| + hR_0 x \geq 1$. Rearranging proves the claimed estimate. The coefficient c_0 is fixed independently of the number of train steps; this is why its exponential loss can be summed by the geometric side-ratio argument. □

7.2 Mass, signals, and nonzero-alias denominators

Lemma 7.3 (Mass and flux weights)**.** *For a flight P with patch side L and stage j, its patch mass factor $m_P(q) = \operatorname{area}_L(q)^j$ obeys, on a common complex cube,*

$$|m_P(q_l^{(L)}(p))| \leq C_m^j \Gamma_L(l)^{-j}.$$

The r-face flux has entrywise bound $C_{m,r}\Gamma_L(l)^{-j}$ for each fixed r used in the wire formula. At the zero alias, the latch factor $m_P(q_0)^{-1}$ and reciprocal rails are bounded independently of L on a common smaller complex cube.

Proof. The area is the product of the four one-dimensional plank factors. Apply Lemma 7.1 in each coordinate and raise the result to the jth power. The product of the four coordinate weights is exactly $\Gamma_L(l)^{-j}$. A flux entry is this mass evaluated at $-q_l$ times the corresponding rail factors at $-q_l$, followed by transpose; the rail factors have uniformly bounded modulus, so the same weight remains. At zero alias, the patch mass obeys

$$|m_P(q_0)| \geq \exp(-4j\, c_0)$$

on the real cube: apply the principal one-dimensional plank lower bound with scale ratio one in each coordinate. Its upper analytic-cube bound and this real lower bound give a uniform reciprocal on a smaller complex cube by the quantitative continuation estimate above. The zero-alias rails are handled by their real lower bound and the same continuation. □

Lemma 7.4 (Scaled signal weight)**.** *For every face degree r, the scaled signal has entrywise bound $C_s g_L(l)$, independent of r. In particular the one-form signal costs at most $C_s g_L(l)$.*

Proof. The signal is exterior multiplication by the vector with entries $L(e^{iq_l^{(L)}(p)_a} - 1)$. Its matrix entry from an r-face to an $(r+1)$-face is zero unless the latter face is obtained by adjoining one coordinate; in that case it is, up to sign, the corresponding single vector entry. The alias upper bound and $|e^{iz} - 1| \leq |z|e^{|\operatorname{Im} z|}$ bound this entry by $C(1 + |\ell_L(l)_a|) \leq C g_L(l)$. Thus the cost is one grade factor for every r. □

Lemma 7.5 (Cumulative train, noise, bulb, and bend bounds)**.** *Fix a stage $j \geq 4$ and $k > 0$. Uniformly over every finite train history of that stage and its mass side M, there is a common complex cube on which for every p in the cube and every alias l, the cumulative filter $S_v(q_l^{(M)}(p))$ and directional drive $D_{v,a}(q_l^{(M)}(p))$ are bounded by $C\Gamma_M(l)^{-j}$. At the zero alias the real lower bound $|S_v(p/M)| \geq c > 0$ holds uniformly on the fixed real cube. On a possibly smaller common complex cube, the noise drain, bulb, bend, and all reciprocals used in the pulse and vault are bounded. On its real slice there are uniform positive lower bounds for each directional lid, bulb, and bend. For each nonzero alias l, the directional drive therefore has the stated bound $C\Gamma_M(l)^{-j}$, evaluated at the aliased momentum.*

Proof. For a train v of stage J and side M, write $S_v(p) = \operatorname{area}_M(p)^J v_{\mathrm{extra}}(p)$. A hop with output stage j and predecessor of stage $J \in \{j, j+1\}$ has side LM and satisfies

$$S_{\mathrm{hop}}(p) = \operatorname{area}_L(Mp)^J S_v(p), \qquad v_{\mathrm{extra,hop}}(p) = \operatorname{area}_{LM}(p)^{J-j} v_{\mathrm{extra}}(p).$$

The first identity follows by multiplying the four box averages and using their dilation composition law. For the upper bound, if p lies in the strip of width zone/L_0 and $M \leq L_0$, each box factor has modulus at most $e^{\mathrm{zone}M/L_0}$, hence the fourfold product at most $e^{4\mathrm{zone}M/L_0}$. Induction over histories proves $|v_{\mathrm{extra}}(p)| \leq e^{8\mathrm{zone}M/L_0}$. At a hop with $J = j$, the extra is unchanged and the mass increases. If $J = j+1$, the added box factor contributes exponent $4\mathrm{zone}LM/L_0$; the predecessor's extra is bounded by exponent $(8/3)\mathrm{zone}LM/L_0$ since $L \geq 3$. Their sum is at most $8\mathrm{zone}LM/L_0$.

For the lower bound, the principal one-dimensional estimate with coefficient one states that if $M \leq L_0$, z is real, and $L_0|z| \leq R_0$, then $|\operatorname{plank}_M(z)| \geq e^{-c_0 M/L_0}$. Four coordinates give the same estimate with exponent multiplied by four. The same induction proves $|v_{\mathrm{extra}}(p/L_0)| \geq e^{-8c_0 M/L_0}$.

For $J = j$, the extra is unchanged and the mass increases. For $J = j+1$, the new box costs exponent $4c_0LM/L_0$, and the old extra costs at most $(8/3)c_0LM/L_0$; their sum is bounded by $8c_0LM/L_0$. This is the geometric mass-ratio estimate that keeps the bound independent of the history length.

For $M = 1$ the only alias is zero and the top train has $S_v = D_{v,a} = 1$. For $M > 1$, the alias estimate for each plank gives $|\operatorname{area}_M(q_l^{(M)}(p))| \le B^4\Gamma_M(l)^{-1}$, while the upper extra estimate at $L_0 = M$ costs at most $e^{8\mathrm{zone}}$. Consequently, for the fixed output stage j, $|S_v(q_l^{(M)}(p))| \le (B^4)^j e^{8\mathrm{zone}}\Gamma_M(l)^{-j}$. At zero alias the lower estimate with $L_0 = M$ gives $|S_v(p/M)| \ge \exp(-4j\,c_0 - 8\,c_0)$. The zero-root rail is at least p_0 and at most a uniform constant. Thus every directional drive and its two-sided lid have uniform upper and positive lower bounds there. At any alias, the train smooth upper bound is $C\Gamma_M(l)^{-j}$ and the rail is bounded, so the drive has that same weight.

The divisor $d_M(p)$ satisfies, for real p in the fixed disk,

$$\mathrm{low}^2|p|^2 \le d_M(p) \le \mathrm{high}^2|p|^2.$$

Indeed each seed magnitude lies between $\mathrm{low}|p_a|$ and $\mathrm{high}|p_a|$, and the divisor is the sum of their squares. The single-feed noise term uses the constant function one and equals

$$M^{-2}\sum_{l\in\Lambda_M} D_{v,a}(q_l^{(M)}(p))D_{v,a}(-q_l^{(M)}(p)).$$

The alias drive bound gives a summand at most $C\Gamma_M(l)^{-2j} \le C\Gamma_M(l)^{-2}$, whose sum is at most $81C$; the retained factor M^{-2} therefore bounds the feed by $81CM^{-2}$. The source noise list is empty for a top train and, for a hop of dilation L, consists of one top train followed by the hop applied to every entry of the previous list. Its total price T obeys $T_{\mathrm{top}} = 0$ and $T_{\mathrm{hop}} = 1 + L^{-2}T_{\mathrm{old}}$. As $L \ge 3$, induction gives $T \le 1 + 2/9 < 2$. Summing the feed estimates at these actual inverse-square side prices bounds the full accumulated noise uniformly.

The high-alias drain term is a finite sum of two-sided drive products times the reciprocal blow-up $[M^2 d_1(q_l^{(M)}(p))]^{-1}$. For nonzero l, choose a coordinate attaining $\max_a |\ell_M(l)_a|$. The centered-alias lower bound in that coordinate and the seed lower bound give $M^2 d_1(q_l^{(M)}(p)) \ge c\,g_M(l)^2$ on the real cube. Its complex upper bound has the same $g_M(l)^2$ weight. The quantitative continuation argument above therefore bounds its reciprocal by $Cg_M(l)^{-2}$ on a common smaller cube. At $M = 1$ the high-alias sum is empty. For $M > 1$, the remaining drive product has weight $\Gamma_M(l)^{-2j}$, so this sum is bounded by the exponent-two residue sum. Thus the full drain is uniformly bounded. It is nonnegative on the real slice.

The bulb is the directional lid plus $d_M(p)$ times the drain. The real lower bounds for the lid and nonnegativity of both latter factors give a positive lower bound for the bulb; the complex upper bound follows from the already established bounds. The halo is $S_v(p/M)S_v(-p/M)$ and has a positive real lower bound by the same cumulative-filter estimate. The exact bend identity follows as follows: the rail-chip relation gives $J_a(p)h_v(p) = s_M(p)_a s_M(-p)_a \ell_{v,a}(p)$, where $h_v = S_v(p/M)S_v(-p/M)$ and $\ell_{v,a}$ is the directional lid. Therefore $d_M(p)/h_v(p) = \sum_a J_a(p)/\ell_{v,a}(p)$. Combining this with $b_{v,a} = \ell_{v,a} + d_M G_{v,a}$ yields

$$d_M(p)\,\beta_v(p) = \sum_{a=1}^{4} \frac{J_a(p)}{b_v(p)_a}, \qquad J_a(p) = (e^{ip_a} - 1)(e^{-ip_a} - 1).$$

Indeed, each summand on the right is $J_a/\ell_{v,a} - d_M J_a G_{v,a}/(\ell_{v,a}b_{v,a})$, which is exactly the corresponding term in $d_M\beta_v$. On the real cube, $J_a \ge 0$, $\sum_a J_a \ge \mathrm{low}^2|p|^2$, and $d_M(p) \le \mathrm{high}^2|p|^2$. If $|b_v(p)_a| \le C_b$, the identity gives $\beta_v(p) \ge \mathrm{low}^2/(C_b\mathrm{high}^2)$ for $p \ne 0$; at zero $\beta_v(0) = 1$. The bend's

complex-cube upper bound follows from its defining formula: first shrink the cube using the positive real lower bounds for the halo and lid, so their reciprocals are bounded. Next the bulb has a positive real lower bound and a complex upper bound, so shrink again to bound its reciprocal. With these factors, the displayed bend formula gives a complex upper bound. The real identity then gives a positive lower bound for the bend; shrink once more to bound its reciprocal for the pulse and vault. This proves all assertions uniformly over train histories. □

Lemma 7.6 (Port denominator and levy weights)**.** *For every nonzero alias, the port vector has entrywise weight $C_D g_L(l)^{-2}$, the reciprocal port normalizer has weight $C_R g_L(l)^{-4}$, and the levy scalar is uniformly bounded when $j \geq 4$.*

Proof. The bulb and reciprocal bend bounds are those of Lemma 7.5. The blow-up is $L^2 d_M(q_l^{(L)}(p))$, where M is the wing train's side. Choose a coordinate a_* attaining $\max_a |\ell_L(l)_a|$; since $l \neq 0$, its lift is nonzero. The centered alias lower bound gives $L|q_l^{(L)}(p)_{a_*}| \geq c(1 + |\ell_L(l)_{a_*}|) = c g_L(l)$ on the real cube. The seed lower bound $|M \operatorname{chip}(q_l^{(L)}(p)_{a_*}/M)| \geq \operatorname{low}|q_l^{(L)}(p)_{a_*}|$ then implies $L^2 d_M(q_l^{(L)}(p)) \geq c' g_L(l)^2$. The analytic-cube upper bound for the blow-up has the same weight $g_L(l)^2$, so the quantitative continuation argument gives its reciprocal at cost $C g_L(l)^{-2}$ on a common smaller cube. The port vector is the bounded bulb times this reciprocal. The reciprocal port normalizer is the reciprocal of blow-up squared times the bend; it therefore costs $C g_L(l)^{-4}$. Finally, the levy contains two patch mass factors, of total weight $\Gamma_L(l)^{-2j}$, times the bounded land bend and this reciprocal port normalizer. Since both weights are at most one, the levy is uniformly bounded. □

7.3 Uniform tails and regular zero-alias factors

Lemma 7.7 (Uniform tail and regular factors)**.** *The tail vector, the correction matrix, the notch matrix, and every zero-alias dual multiplier in the dual-stencil formula are bounded on a common complex cube, uniformly in L and the train. The same is true of the bulb and bend factors and their reciprocals wherever they occur in the wire formula.*

Proof. The train, bulb, bend, and reciprocal bounds are Lemma 7.5; the nonzero port bounds are Lemma 7.6. For each nonzero alias, a tail summand is the product of two patch masses, two bounded rail factors, and a port vector. It is bounded by $C\Gamma_L(l)^{-2j} g_L(l)^{-2} \leq C\Gamma_L(l)^{-2}$. Thus the sum is uniformly bounded, since

$$\sum_{l \in \Lambda_L} \Gamma_L(l)^{-2} \leq \left(\sum_{n \in \mathbb{Z}} (1 + |n|)^{-2} \right)^4 \leq 81.$$

The extra constant k^{-1} is finite. The divisor, scaled signals, and all zero-alias ratios in the correction and notch are bounded by the preceding train lemma. Their formulas contain only finitely many sums and products of these quantities, so the triangle inequality and the finite-dimensional matrix product bound complete the uniform estimate. □

7.4 Closing the two wire branches

Proposition 7.8 (Entrywise weighted wire bound)**.** *For fixed stages $j, t \geq 4$ and $k > 0$, each scalar entry of the dual stencil is holomorphic on one common complex cube and satisfies*

$$|W_{P,0}(p)| \leq C_0, \qquad |W_{P,l}(p)| \leq C_1 \Gamma_L(l)^{-j} \quad (l \neq 0),$$

uniformly in odd $L \geq 3$. Hence the unified bound is $C\Gamma_L(l)^{-j}$.

Proof. Use the zero and nonzero branch formulas in Section 6. In the zero branch, Lemma 7.3 and Lemma 7.7 show that every factor has weight one. In the first nonzero branch product, the scaled signal, port vector, and flux contribute respectively $g_L(l)$, $g_L(l)^{-2}$, and $\Gamma_L(l)^{-j}$. Their product is at most $\Gamma_L(l)^{-j}$. The second nonzero product has levy weight one and flux weight $\Gamma_L(l)^{-j}$. Finite intermediate matrix sums multiply the constant by only the fixed dimension of the face spaces. The branch constants and widths can be replaced by their maximum and minimum, respectively. This proves the assertion. □

8 The exact identities of the original transport kernels

We now identify the kernels estimated in the preceding sections with the transports used in the quadratic construction. This also accounts for the harmonic mode. Throughout this section $k > 0$, the block side L is odd and at least three, and the adjacent train orders j, t satisfy $4 \le t \le j \le t+1$.

8.1 Covariance recursion and exterior algebra

Fix a nonzero real coarse momentum p in the common real cube containing the Brillouin cube, and write $q_l = (p + 2\pi l)/L$ for its centered aliases. In the following formulas the fine exterior symbol is rescaled: $\delta_r = L d_r(q_l)$, whereas $d_r = d_r(p)$ is the coarse symbol. Stars denote Hermitian adjoints at these real momenta. Let H_l, H be the fine and coarse two-form pulse symbols. Let D_l be the diagonal matrix with entries $L^{-2} D_{f,a}(q_l)$, and D the diagonal matrix with entries $D_{c,a}(p)$, where $D_{v,a}$ is the propagated one-form covariance from Section 7. Set

$$x = d_0, \quad x_l = \delta_0, \quad R = d_1^* H d_1, \quad R_l = \delta_1^* H_l \delta_1.$$

All diagonal entries of D, D_l are positive. Put

$$\rho = x^* D^{-1} x, \quad \pi = \rho^{-1} x^* D^{-1}, \quad c = \rho^{-1} D^{-1} x.$$

The corresponding fine objects have subscript l. The pulse formula is $H = \rho^{-1} \bigwedge^2 D^{-1}$; with the stated scaling, $H_l = \rho_l^{-1} \bigwedge^2 D_l^{-1}$ as well. Consequently

$$R = D^{-1} - \rho^{-1} D^{-1} x x^* D^{-1}, \qquad DR = I - x\pi, \qquad RD = I - c x^*. \tag{35}$$

For completeness, if $D^{-1} = \operatorname{diag}(f_a)$, the diagonal entry of $d_1^*(\bigwedge^2 D^{-1}) d_1$ at a is $f_a \sum_{b \ne a} f_b |x_b|^2$, and its off-diagonal entry at (a, b) is $-f_a f_b x_a \overline{x_b}$. Division by $\rho = \sum_b f_b |x_b|^2$ proves the first equality; multiplication by D proves the other two. In particular $Rx = 0$ and $x^* R = 0$.

Let m_l be the patch mass filter at q_l and define

$$F_{r,l} = m_l \bigwedge^r \operatorname{diag}(\operatorname{rail}_L(q_{l,a})).$$

The geometric-series identity $L \operatorname{rail}_L(q_{l,a})(e^{i q_{l,a}} - 1) = e^{i p_a} - 1$ gives

$$F_{r+1,l} \delta_r = d_r F_{r,l}. \tag{36}$$

The covariance recursion is

$$\sum_l F_{1,l} D_l F_{1,l}^* = D - k^{-1} I. \tag{37}$$

Here the sum is a sum over all L^4 aliases, with no averaging factor. To verify the normalization and the recursion, the feed of a train of total side M is

$$M^{-2} \sum_{u \in \Lambda_M} |\operatorname{drive}_{v,a}(q_u)|^2 f(q_u).$$

On appending a block, its drive is the old drive times the new patch filter evaluated at the old rescaled momentum. Decomposing a residue modulo LM into its residues modulo L and M therefore turns this feed into the successive two feeds: the factors are $(LM)^{-2} = L^{-2}M^{-2}$, and the summands agree by periodicity. The noise-prefix list of the new train consists of its new unit prefix and the old prefixes with the block appended. Thus its noise feed is one plus the patch feed of the old noise. Applying these two facts to the free covariance and to each earlier independent noise gives $D_{c,a} = k^{-1} + \operatorname{slide}_a D_{f,a}$. The diagonal entries of (37) are precisely this equality. All off-diagonal entries vanish.

8.2 The two punctured transport formulas

Define scalars

$$a = \sum_l |m_l|^2 \operatorname{tr} D_l, \qquad e_l = \overline{m_l} \operatorname{tr} D_l / a, \qquad y_l = |m_l|^2 \rho / \rho_l.$$

The principal mass filter is nonzero on the real cube. Since each D_l is positive, its contribution shows $a > 0$. In particular

$$\sum_l e_l m_l = 1. \tag{38}$$

The source constructs the following rescaled columns:

$$\mathcal{A}_l = D_l F_{1,l}^* R + e_l x_l \pi, \tag{39}$$

$$\mathcal{T}_l = H_l \delta_1 D_l F_{1,l}^* d_1^* + y_l F_{2,l}^* (I - C), \qquad C = H d_1 D d_1^*. \tag{40}$$

The complement $I - C$ is part of the two-form transport; it cannot be discarded merely because it vanishes on weighted exact forms. Indeed (35) and $d_1 x = 0$ imply

$$RDd_1^* = d_1^*, \qquad d_1 DR = d_1, \qquad d_1^*(I - C) = 0, \qquad (I - C)Hd_1 = 0.$$

Also, the adjoint of (36) gives $x_l^* F_{1,l}^* d_1^* = 0$. These identities prove

$$R_l \mathcal{A}_l = F_{1,l}^* R, \qquad \delta_1^* \mathcal{T}_l = F_{1,l}^* d_1^*, \qquad \mathcal{T}_l H d_1 = H_l \delta_1 \mathcal{A}_l. \tag{41}$$

Here are the cancellations explicitly. The second term in $R_l \mathcal{A}_l$ vanishes because $R_l x_l = 0$; in the first, $R_l D_l = I - c_l x_l^*$ and $x_l^* F_{1,l}^* R = 0$. For the second identity the first term is $R_l D_l F_{1,l}^* d_1^* = F_{1,l}^* d_1^*$, while the second is $y_l F_{1,l}^* d_1^* (I - C) = 0$. For the third identity the complement term vanishes by $(I - C)Hd_1 = 0$, and the remaining term is $H_l \delta_1 D_l F_{1,l}^* R$. The other side equals this because $\delta_1 x_l = 0$.

Finally (36) at degree zero, (38), and (37) give

$$\begin{aligned} \sum_l F_{1,l} \mathcal{A}_l &= (D - k^{-1} I) R + x\pi \\ &= I - k^{-1} R. \end{aligned} \tag{42}$$

This is the identity that incorporates the new Gaussian noise.

8.3 Removing the puncture without changing the kernels

We check that these formulas are the analytic columns whose bounds were proved earlier. In this continuation every real-momentum adjoint is written as transposition after reflection $p \mapsto -p$; for example, $|m(p)|^2$ continues as $m(p)m(-p)$. For an alias-indexed factor the reflection is $q_l \mapsto -q_l$, equivalently $(p, l) \mapsto (-p, -l)$. Thus none of the complex-momentum formulas below uses literal complex conjugation of its variable. Let s be the coarse divisor, b_a the coarse bulb and $b_{l,a}$ the fine bulb. Write $q_0 = p/L$ and $u_a = \mathrm{tail}_a$, including the current noise. Splitting the covariance recursion into its principal alias and the other aliases gives

$$b_a = |m_0 \operatorname{rail}_L(q_{0,a})|^2 b_{0,a} + s u_a. \tag{43}$$

For $p \neq 0$ this follows by multiplying $D_a = |m_0 \operatorname{rail}_L(q_{0,a})|^2 b_{0,a}/s + u_a$ by s. Both sides of (43) are continuous, so it holds at zero as well.

For the one-form column put $B_l = D_l F_{1,l}^* D^{-1}$ and define $r_{l,a}$ by $B_l x = \operatorname{diag}(r_{l,a}) x_l$. Entrywise,

$$r_{l,a} = \overline{m_l} (D_l)_{aa} |\operatorname{rail}_L(q_{l,a})|^2 / D_{aa}.$$

Using (35) in (39) gives

$$\mathcal{A}_l = B_l + \operatorname{diag}(e_l - r_{l,a}) x_l \pi.$$

Now $D^{-1} = s \operatorname{diag}(b_a^{-1})$ and $\pi = s^{-1}\sigma$, where $\sigma = (\mathrm{bend}_c)^{-1} x^* \operatorname{diag}(b_a^{-1})$ is analytic. For a nonprincipal alias, D_l has the regular port formula; thus $e_l - r_{l,a} = s g_{l,a}$ with

$$g_{l,a} = \overline{m_l} \left(\frac{\operatorname{tr} D_l}{sa} - \frac{(D_l)_{aa} |\operatorname{rail}_L(q_{l,a})|^2}{b_a} \right).$$

For the principal alias let h be the sum of the nonprincipal terms in a. Then $sa = |m_0|^2 \sum_a b_{0,a} + sh$, and (43) instead gives the regular formula

$$g_{0,a} = m_0^{-1} \left(\frac{u_a}{b_a} - \frac{h}{sa} \right).$$

These are exactly the principal and nonprincipal gain formulas. The denominators b_a, sa, and m_0 have the common analytic lower bounds established in the component estimates. Consequently

$$\mathcal{A}_l = B_l + \operatorname{diag}(g_{l,a}) x_l \sigma$$

is the already estimated analytic column, including at the origin.

For the two-form column, at a nonprincipal alias write $\lambda_l = |m_l|^2 \, \mathrm{bend}_c \,/\, \mathrm{portR}_l$. Then $y_l = s^2 \lambda_l$ and

$$s^2 (I - C) = s^2 I - s H d_1 \operatorname{diag}(b_a) d_1^*.$$

Substitution in (40) is exactly the regular nonprincipal wire formula. At the principal alias set $J = F_{2,0}^{-1}$, $U = \operatorname{diag}(u_a)$, $M = \operatorname{diag}(b_a)$, and $z_a = u_a/b_a$. The inverse exists on the common cube. The exterior chain rule and the principal covariance split give

$$\delta_1 = J d_1 F_{1,0}, \qquad F_{1,0} D_0 F_{1,0}^* = D - U.$$

The cancellation needed for the apparent remaining pole is

$$s\mathcal{B} = H_0 - y_0 F_{2,0}^* H F_{2,0}, \qquad \mathcal{B} = H_0 \left(\operatorname{diag}_{\{a,b\}}(z_a + z_b) - s \bigwedge^2 \operatorname{diag}(z_a) \right). \tag{44}$$

To verify it, put $Z_a = |m_0 \operatorname{rail}_L(q_{0,a})|^2 b_{0,a}/b_a = 1 - sz_a$ by (43). The pulse formula shows $y_0 F_{2,0}^* H F_{2,0} = H_0 \bigwedge^2 \operatorname{diag}(Z_a)$: on a face $\{a, b\}$ both sides have the same product of its two diagonal entries and the same scalar bend ratio. Finally $1 - (1 - sz_a)(1 - sz_b) = s(z_a + z_b - sz_a z_b)$ proves (44) entrywise.

Using $M = sD$, the principal expression therefore becomes

$$\mathcal{T}_0 = y_0 F_{2,0}^* + (\mathcal{B} J d_1 M - H_0 J d_1 U) d_1^*. \tag{45}$$

Indeed its second term expands to $H_0 J d_1 (D - U) d_1^* - y_0 F_{2,0}^* H d_1 D d_1^*$, which is (40) after the two chain identities. All factors in (45) have the analytic bounds already proved. This verifies agreement with the principal wire as well as the nonprincipal one.

At $p = 0$ the principal analytic columns have values $\mathcal{A}_0 = I$ and $\mathcal{T}_0 = I$: the bulbs, bends, rails, and principal filter have value one and the exterior symbols vanish. For $l \neq 0$ the patch mass filter vanishes at $q_l(0)$. Indeed some nonzero alias coordinate has a nontrivial Lth root of unity, and the geometric-series rail vanishes there; the patch mass contains a positive power of its squared modulus. The nonprincipal one-form column is zero because both its leading term and gain contain that filter. The two-form column is zero because its first term contains $d_1(0)^* = 0$ and its other term contains the mass filter. Thus (41) and (42) also hold at $p = 0$.

8.4 Returning to physical counting measure

The installed one-form and two-form columns at alias l are $L^{-1}\mathcal{A}_l$ and $L^{-2}\mathcal{T}_l$. The block mean Q is the alias row with columns $LF_{1,l}$. In physical counting measure the adjoint alias column of Q is $L^{-3}F_{1,l}^*$: the factor L is multiplied by the coarse-to-fine volume ratio L^{-4}. Since $\delta_r = L d_r(q_l)$, the second and third identities in (41) become respectively

$$d_{1,f}^* T = Q^* d_{1,c}^*, \qquad T H_c d_{1,c} = H_f d_{1,f} A_0.$$

The alias sum becomes $QA_0 = I - k^{-1} d_{1,c}^* H_c d_{1,c}$. Fourier diagonalization on each finite torus now gives the real matrix identities

$$T^{\mathsf{T}} d_{1,f} = d_{1,c} Q, \qquad T H_c d_{1,c} = H_f d_{1,f} A_0, \qquad Q A_0 = I - k^{-1} d_{1,c}^{\mathsf{T}} H_c d_{1,c}. \tag{46}$$

This argument includes every finite period, including the one-site period, and uses the Fourier coefficients of the original analytic columns. No new choice of transport has been made.

9 Positive Laurent rows and the actual quadratic beds

The quadratic construction needs positive factors with exponentially decaying coefficients, uniformly in the finite period. Taking a separate matrix square root on each torus would not supply such coefficients. We construct one analytic symbol and then recover its Laurent row. This is also how the construction preserves its zero-frequency conventions.

9.1 A uniform analytic inverse and square root

Let I be a fixed finite index set of cardinality r. A real Laurent row is a collection $A_{ij}(z)$, $i, j \in I$, $z \in \mathbb{Z}^4$. Suppose, for some $\eta, C > 0$,

$$\sup_{i,j} \sum_{z \in \mathbb{Z}^4} e^{\eta |z|_1} |A_{ij}(z)| \leq C. \tag{47}$$

Its symbol

$$\widehat{A}(p) = \sum_z A(z) e^{ip\cdot z}$$

is periodic in each real coordinate and holomorphic whenever $\max_a |\operatorname{Im} p_a| < \eta$. If the index set is empty all the statements below are vacuous, so assume $r \geq 1$. On every smaller strip of width σ, termwise differentiation is justified by

$$|z|_1 e^{\sigma|z|_1} \leq (\eta - \sigma)^{-1} e^{\eta|z|_1}.$$

Thus each entry has derivative norm at most $C/(\eta - \sigma)$, for the maximum norm on momenta. This also proves local uniform convergence of the differentiated series. In operator norm the corresponding Lipschitz bound is at most $rC/(\eta - \sigma)$, since a matrix with entries bounded by a has operator norm at most ra.

Suppose in addition that for every real momentum p,

$$vI \leq \widehat{A}(p) \leq MI, \qquad 0 < v \leq M, \tag{48}$$

as Hermitian forms. These bounds are uniform over the whole real torus. Let $\sigma = \eta/2$, $D = rC/(\eta - \sigma)$, and choose

$$0 < \tau \leq \min\{\sigma, v/(2D)\}.$$

For a complex momentum in this strip, put $q = \operatorname{Re} p$. The segment joining q to p lies in the strip of width σ, and therefore

$$\|\widehat{A}(p) - \widehat{A}(q)\| \leq v/2.$$

Diagonalizing the Hermitian matrix at q gives

$$\|M^{-1}\widehat{A}(q) - I\| \leq 1 - v/M.$$

Hence throughout the smaller strip

$$\|M^{-1}\widehat{A}(p) - I\| \leq 1 - v/(2M) =: \rho < 1. \tag{49}$$

All constants depend only on the original uniform bounds and the fixed index set.

Put $Z(p) = M^{-1}\widehat{A}(p) - I$. The normally convergent series

$$\widehat{F}(p) = \sqrt{M} \sum_{n \geq 0} \binom{1/2}{n} Z(p)^n, \qquad \widehat{B}(p) = M^{-1} \sum_{n \geq 0} (-Z(p))^n$$

define holomorphic periodic symbols on this strip. Normal convergence follows from $\rho < 1$ and absolute convergence of the scalar power series on every smaller closed disk. Cauchy multiplication gives $\widehat{B}\widehat{A} = I$ and $\widehat{F}^2 = \widehat{A}$. On the real torus diagonalization gives eigenvalues λ^{-1} and $\sqrt{\lambda}$, respectively, where $v \leq \lambda \leq M$. Consequently

$$M^{-1}I \leq \widehat{B} \leq v^{-1}I, \qquad \sqrt{v}I \leq \widehat{F} \leq \sqrt{M}I. \tag{50}$$

In particular these are the positive inverse and positive square root, with no choice of eigenvectors as the momentum varies.

We verify that they are symbols of real exponentially summable rows. Either symbol is bounded on a still smaller strip, say of width $\tau_0 > 0$, by a constant K. Its Fourier coefficient is the integral over $[-\pi, \pi]^4$ against $e^{-ip\cdot z}$, divided by $(2\pi)^4$. Shift the ath contour to imaginary part $-\tau_1 \operatorname{sign}(z_a)$,

where $0 < \tau_1 < \tau_0$. Cauchy's theorem applies one coordinate at a time; the vertical sides cancel by periodicity. The coefficient is therefore bounded by $Ke^{-\tau_1|z|_1}$. For $0 < e < \tau_1$,

$$\sum_z e^{e|z|_1}|F_{ij}(z)| \le K\left(\sum_{n\in\mathbb{Z}} e^{-(\tau_1-e)|n|}\right)^4 < \infty,$$

and likewise for B. The original real row satisfies $\widehat{A}(-p) = \overline{\widehat{A}(p)}$ on real momenta. Both power series have real coefficients, so their symbols have the same symmetry. Changing p to $-p$ in the Fourier integral shows that every recovered coefficient is real. Absolute Fourier convergence identifies the recovered symbols with the displayed analytic functions on the real torus.

On a finite torus, periodize these same rows. The discrete Fourier transform diagonalizes their convolution matrices, with symbols evaluated at the finite Fourier frequencies. Thus on every period n, simultaneously,

$$B_nA_n = I, \qquad F_n^{\mathsf{T}} = F_n, \qquad F_n^{\mathsf{T}}F_n = A_n, \qquad \sqrt{v}I \le F_n \le \sqrt{M}I. \tag{51}$$

The same transform transfers the inverse bounds in (50). This includes the zero frequency and the one-site torus. The construction has used a fixed number of form labels; no bound involving the growing number of lattice sites has entered the analytic inverse or root estimate.

9.2 The amendment of the two-form precision

Fix a smoothing stage $j \ge 4$ and the positive kernel parameter k. For every finite train at this stage, let H and B be the pulse and dual-pulse rows constructed in Sections 6 and 7. Their real symbols are diagonal, positive and reciprocal. Their uniform strip bounds therefore also give uniform positive lower bounds: if a diagonal entry of each row is at most C and their product is one, each is at least C^{-1}. The preceding Fourier argument yields $HB = BH = I$ on every period, with uniform spectral bounds.

Let d_1, d_2 be the exterior differences from one-forms to two-forms and from two-forms to three-forms; thus $d_2d_1 = 0$. For a fixed amendment coefficient $l \ge 0$, define

$$X = d_2B, \qquad A = B + lX^{\mathsf{T}}X, \qquad G = A^{-1}, \qquad R = I + ld_2Bd_2^{\mathsf{T}}, \qquad D = R^{-1}. \tag{52}$$

These formulas denote compositions of Laurent rows and hence also their matrices on every finite period. Products and sums preserve exponential summability, after taking a common strip width. On real frequencies $A \ge B > 0$ and $R \ge I$, because $B > 0$. The analytic inverse construction supplies G, D with uniform strip and spectral bounds. The constants may depend on j, k, l, but are independent of the train and finite period.

Two identities explain the choice of this amendment. First, $XHd_1 = d_2d_1 = 0$, so

$$AHd_1 = d_1, \qquad Gd_1 = Hd_1.$$

Thus the potential precision is unchanged. Second, expansion of (52) gives $RX = d_2A$, since B is symmetric. Multiplication by D on the left and G on the right therefore gives

$$XG = Dd_2. \tag{53}$$

Both G and D are symmetric. At zero momentum $d_2 = 0$ and $B = I$, so $G(0) = I$; the harmonic convention is retained.

Choose a uniform lower spectral bound $b > 0$ for G, and allow any direct precision $0 < p \le b/2$. Then $G - pI \ge (b/2)I$. It has a uniform strip bound, including the subtracted constant row, so the analytic square-root construction gives a row F with

$$F_n \ge 0, \qquad F_n^{\mathsf{T}} F_n = G_n - pI$$

for every finite period, with uniform strip bounds. The full root matrix is

$$J_p = \begin{bmatrix} \sqrt{p} I \\ F_n \end{bmatrix}, \qquad J_p^{\mathsf{T}} J_p = G_n.$$

Together with H, B, G, A this is the actual quadratic bed. Its inverse, symmetry, positivity and exterior-intertwining properties have all been verified above. In the top train the pulse is the identity, so its diagonal bed row is the identity as required. The admissible upper bound for p and all row bounds are selected before any lattice volume or train is chosen.

10 Construction of the finite precision tower

This section explains the finite-dimensional algebra that turns the selected quadratic beds and their curvature factor into an exact Wilson tower. It also records how the beds and towers are produced on the actual schedule. All matrices act on finite real torus spaces; there is no limiting or spectral argument in the tower identities.

10.1 The exact finite-torus block identity

Let $\mathcal{E}_f^r$ and $\mathcal{E}_c^r$ be the real coordinate spaces of r-form labels on the fine and coarse tori. The first coboundaries are $D_f : \mathcal{E}_f^1 \to \mathcal{E}_f^2$ and $D_c : \mathcal{E}_c^1 \to \mathcal{E}_c^2$; the next coarse coboundary is $d_2 : \mathcal{E}_c^2 \to \mathcal{E}_c^3$, with $d_2 D_c = 0$. Let $J : \mathcal{E}_f^2 \to \mathcal{Y}_f$ and $J' : \mathcal{E}_c^2 \to \mathcal{Y}_c$ be the full precision-factor maps for the fine and coarse beds, respectively, and write

$$W = J^{\mathsf{T}} J, \qquad W' = J'^{\mathsf{T}} J', \qquad R = W^{-1}, \qquad R' = (W')^{-1}.$$

Assume W, W' are symmetric and the displayed inverses exist. Let T be the installed two-form transport $T : \mathcal{E}_c^2 \to \mathcal{E}_f^2$, and let $A_0 : \mathcal{E}_c^1 \to \mathcal{E}_f^1$ be the selected one-form transport. The conditional mean $\overline{T} : \mathcal{E}_f^1 \to \mathcal{E}_c^1$ and these transports satisfy

$$T^{\mathsf{T}} D_f = D_c \overline{T}, \qquad T W' D_c = W D_f A_0, \tag{54}$$

where W and R act on $\mathcal{E}_f^2$, while W' and R' act on $\mathcal{E}_c^2$. Put $s = \sqrt{k}$, with $k > 0$.

Define the curvature remainder by

$$Z = R' - T^{\mathsf{T}} R T - k^{-1} D_c D_c^{\mathsf{T}}. \tag{55}$$

Suppose this remainder has the curvature factorization

$$W' Z W' = d_2^{\mathsf{T}} P^{\mathsf{T}} P d_2. \tag{56}$$

Let $P : \mathcal{E}_c^3 \to \mathcal{E}_c^3$ be a real matrix with the curvature factorization above. Define the coarse projection and its curvature image by

$$\Pi = J' R' J'^{\mathsf{T}}, \qquad X = P d_2 R' J'^{\mathsf{T}}.$$

The three blocks in the tower are

$$M_F = J R T J'^{\mathsf{T}}, \qquad M_N = s^{-1} D_c^{\mathsf{T}} J'^{\mathsf{T}}, \qquad M_O = \begin{bmatrix} I - \Pi \\ X \end{bmatrix}. \tag{57}$$

Proposition 10.1 (Exact block completion)**.** *Assume the inverses and factorization above exist and the two transport identities* (54) *hold. Then the blocks in* (57) *satisfy*

$$M_F^{\mathsf{T}} M_F + M_N^{\mathsf{T}} M_N + M_O^{\mathsf{T}} M_O = I, \tag{58}$$

$$s M_N^{\mathsf{T}} = J' D_c, \tag{59}$$

$$M_F J' D_c = J D_f A_0, \tag{60}$$

$$M_N J' D_c = s^{-1} D_c^{\mathsf{T}} W' D_c, \tag{61}$$

$$M_O J' D_c = 0, \tag{62}$$

$$M_F^{\mathsf{T}} J D_f + M_N^{\mathsf{T}} (-s\,\overline{T}) = 0. \tag{63}$$

The three blocks have common domain $\mathcal{Y}_c$ and respective ranges $\mathcal{Y}_f$, $\mathcal{E}_c^1$, and $\mathcal{Y}_c \oplus \mathcal{E}_c^3$.

Proof. The inverse of a symmetric invertible real matrix is symmetric. Indeed, from $WR = I$ and $W^{\mathsf{T}} = W$ we obtain $R^{\mathsf{T}} W = I$, hence $R^{\mathsf{T}} = R^{\mathsf{T}}(WR) = (R^{\mathsf{T}} W)R = R$; the same argument applies to R'.

The matrix Π is the orthogonal projection onto the range of J'. First,

$$\Pi^{\mathsf{T}} = J' R' J'^{\mathsf{T}} = \Pi.$$

Second,

$$\Pi^2 = J' R' (J'^{\mathsf{T}} J') R' J'^{\mathsf{T}} = J' R' W' R' J'^{\mathsf{T}} = \Pi, \qquad \Pi J' = J' R' W' = J'.$$

For the curvature block, substitute (56):

$$\begin{aligned} X^{\mathsf{T}} X &= J' R' d_2^{\mathsf{T}} P^{\mathsf{T}} P d_2 R' J'^{\mathsf{T}} \\ &= J' R' (W' Z W') R' J'^{\mathsf{T}} = J' Z J'^{\mathsf{T}}. \end{aligned}$$

The two middle factors cancel because $R'W' = W'R' = I$. The other two Gram terms are

$$M_F^{\mathsf{T}} M_F = J' T^{\mathsf{T}} R T J'^{\mathsf{T}}, \qquad M_N^{\mathsf{T}} M_N = J' (k^{-1} D_c D_c^{\mathsf{T}}) J'^{\mathsf{T}}.$$

By (55), $T^{\mathsf{T}} R T + k^{-1} D_c D_c^{\mathsf{T}} + Z = R'$. Consequently the sum of the three block Gram matrices is

$$J' R' J'^{\mathsf{T}} + (I - \Pi)^{\mathsf{T}} (I - \Pi) = \Pi + I - \Pi = I,$$

using $\Pi^{\mathsf{T}} = \Pi$ and $\Pi^2 = \Pi$. This proves (58) as an identity on the full coarse augmented row space $\mathcal{Y}_c$; no restriction to the range of J' is made.

The remaining formulas are direct consequences of the block definitions and (54). Since $s^2 = k$, $s M_N^{\mathsf{T}} = J' D_c$. For the fine transport,

$$M_F J' D_c = J R T (J'^{\mathsf{T}} J') D_c = J R (T W' D_c) = J R (W D_f A_0) = J D_f A_0.$$

For the normal block,

$$M_N J' D_c = s^{-1} D_c^{\mathsf{T}} W' D_c.$$

For the image block, its first component is $(I - \Pi) J' D_c = 0$, since $\Pi J' = J'$. Its second component is

$$X J' D_c = P d_2 R' J'^{\mathsf{T}} J' D_c = P d_2 R' W' D_c = P d_2 D_c = 0,$$

using $R'W' = I$ and $d_2 D_c = 0$. This proves (62).

Finally, the transpose of M_F gives

$$M_F^{\mathsf{T}} J D_f = J' T^{\mathsf{T}} R (J^{\mathsf{T}} J) D_f = J' T^{\mathsf{T}} D_f.$$

By $T^{\mathsf{T}} D_f = D_c \overline{T}$, this is $J' D_c \overline{T}$. Meanwhile $M_N^{\mathsf{T}} (s\overline{T}) = J' D_c \overline{T}$. The two terms therefore cancel with the sign in (63). This proves the orthogonality identity. □

10.2 The selected schedule, its beds, and the tower estimates

The schedule input is a family of priced bed and flight receipts. At each integer scale $j \geq 4$, for every allowed direct precision $p_{\mathrm{direct}} \leq \gamma_j$ and every train of order j, the bottom receipt supplies a bed whose four precision rows are fixed as above, whose five rows have the stated weighted strip costs, whose floors hold on every torus side, and whose real precision is sworn positive. It also makes the top bed's diagonal the identity. For every admissible adjacent pair of orders j, t (with $t = j$ or $t + 1 = j$), the flight receipt supplies donated unmodified primal/two-form transports with the order-$j - 2$ difference bounds, together with a curvature factor of the stated strip cost and positive lower quadratic bound. The block-length credit requires the separately selected lower bound on L before its quadratic floor is used. These receipts are obtained by the one-flight frontier and the uniform bed pricing; they are construction inputs here, not conclusions of the block algebra.

Fix the actual level schedule A and a terminal cutoff N. Its train at level N is the constant top train. Descending one level, use the flight whose wing is the train already constructed at level $i + 1$ and whose land is the train at level i. Thus, for every $i < N$, the level-i train is the landing train of the flight whose wing is the already defined train at level $i + 1$. This uses the actual adjacent block at level i; no second schedule is selected.

The bed at each (N, i) is first constructed for the corresponding train. The source bed proposition fixes its four precision rows to the train's pulse, dual pulse, amended precision and amended inverse. It constructs the remaining root row as a square factor of the positive remainder. The finite-volume root matrix is the vertical stack

$$J_p = \begin{bmatrix} \sqrt{p_{\mathrm{direct}}} I \\ F \end{bmatrix}, \qquad J_p^{\mathsf{T}} J_p = p_{\mathrm{direct}} I + F^{\mathsf{T}} F.$$

This identity follows by splitting the row index of the product into its two disjoint blocks. The lower root factor is obtained from the finite matrix square-factor receipt, so the equality holds at every torus side and includes the zero modes. The floor receipt then gives symmetry, inverse identities and the uniform quadratic form bounds required of a bed.

For a step $i < N$, the flight's donated primal and two-form transports are the source towers' A_0 and T. The tangent identities give

$$T^{\mathsf{T}} D_f = D_c \overline{T}, \qquad T W' D_c = W D_f A_0,$$

where W, W' are the actual adjacent bed metrics and the overline is the flight's prescribed conditional mean. The flight's rebase receipt supplies the curvature factorization of the remainder Z in (55). Therefore the preceding block proposition applies to these actual adjacent beds, not just to an abstract pair of positive matrices.

The source tower-price theorem also supplies explicit uniform strip costs. Here is the finite matrix bookkeeping behind that assertion. Let α be the common strip width after narrowing the two bed widths and donated width; write Ω, Ω' for the two bed costs, V for the curvature factor cost, L for the odd block side, X for the maximum difference order, $k > 0$ for the penalty, and $n_r = |\mathrm{TypeFace}(r)|$ for the number of r-form labels. Put

$$\begin{aligned}
A &= \max\{\|\sqrt{p_{\mathrm{direct}}}\|, \Omega\}, & B &= \max\{\|\sqrt{q_{\mathrm{direct}}}\|, \Omega'\}, & & \\
C_A &= n_2 A \Omega, & C_B &= n_2 B \Omega', & C_L &= n_2 C_A \Omega, \quad C_F = n_2 C_L B, \\
C_N &= k^{-1/2} n_2 D_1 B, & C_{\mathrm{proj}} &= n_2 C_B B, & C_{\mathrm{up}} &= n_3 V D_2, \quad C_X = n_2 C_{\mathrm{up}} \Omega', \\
C_O &= \max\{1 + C_{\mathrm{proj}}, n_2 C_X B\}, & M_0 &= \max\{\Omega, C_F, C_N, C_O\}, & &
\end{aligned}$$

where

$$D_r = \sum_{a\in \mathrm{TypeFace}(r+1)} \sum_{b\in \mathrm{TypeFace}(r)} \sum_{z\in \mathrm{supp}(\delta_r)} e^{\alpha|z|_1}\|(\delta_r)_{ab}(z)\|, \qquad r = 1, 2,$$

is the finite stencil cost of the rth coboundary, where for a cover displacement z we set $|z|_1 = \sum_{j=1}^{4} |z_j|$. For ordinary cover convolution, the basic weighted row bound can be checked entrywise. If $C_{ij} = \sum_b A_{ib} * B_{bj}$ is the routed cover convolution and the weighted absolute sums of each entry of A and B are at most a_{ib} and b_{bj}, then

$$\sum_z e^{\alpha|z|_1}|C_{ij}(z)| \le \sum_b a_{ib}b_{bj}.$$

Indeed, write each convolution coefficient as a sum over $x+y=z$, apply the triangle inequality, use $|x+y|_1 \le |x|_1+|y|_1$, and factor the two nonnegative sums over x and y. Summing the intermediate label b then gives the finite cardinality factor in the strip estimate. Each n_2 factor below is this count for a two-form intermediate label; n_3 counts the three-form labels in the curvature factor. A finite coboundary stencil contributes D_r. The fine-to-coarse composition uses a mixed version of this estimate. Fix a fine residue r and a fine cover displacement a. There is a unique coarse displacement $\kappa(r,a)$ and a new fine residue $r' = \mathrm{wrap}(\mathrm{lift}\, r + a)$ such that

$$\mathrm{lift}\, r + a = \mathrm{lift}\, r' + L\kappa(r,a).$$

Coordinatewise, put $u = \mathrm{lift}\, r$, $v = \mathrm{lift}\, r'$, and $n = L$. Both representatives lie in the centered interval, so $|u-v| \le n-1$. The displayed identity yields $n|\kappa_i| \le |a_i|+n-1$. If $|\kappa_i| > |a_i|$, integrality gives $|\kappa_i| \ge |a_i|+1$, hence $n|\kappa_i| \ge n|a_i|+n \ge |a_i|+n$, a contradiction. Thus $|\kappa(r,a)|_1 \le |a|_1$. In the mixed composition, each a selects exactly the single residue r' above, and its coarse shift is $\kappa(r,a)$. Apply the routed-convolution estimate at this fixed residue; summing the finite intermediate form label costs n_2, with no sum over all possible residues. This is the $C_L = n_2 C_A \Omega$ bound, uniform in L.

The fine factor JR has cost C_A. Its mixed composition with T has cost C_L, and the final composition with J'^{T} has cost C_F; the normal block has cost C_N; the projection and curvature rows have costs C_{proj}, C_X, respectively. Stacking the projection complement and curvature rows gives C_O. These are precisely the products in the displayed formulas; the explicit factor L^{-2} in the donated two-form strip is retained separately. The resulting tower has

$$M_N, M_O \text{ strip-cost at most } M_0, \qquad M_F \text{ strip-cost at most } M_0 L^{-2},$$

and its rth iterated primal and two-form differences have costs at most $M_0 L^{-(1+r)}$ and $M_0 L^{-(2+r)}$ for $0 \le r \le X$. These latter bounds are inherited unchanged from the donated tower: the block construction retains its A_0, T rows, whose input receipts already include every iterated difference bound through order X.

For completeness, the construction has two distinct stages. First, the schedule supplies a bed for every (N,i) and the recursively defined flight supplies its one-step data. The bed and hop receipts together with the exact block proposition above give a horizon with row, floor, crossing, and roll data. Second, the finite tower-price construction narrows widths and increases costs, while retaining the original A_0, T rows; the new rows M_F, M_N, M_O are the block completion just proved. The construction theorem performs this on the actual level object and retains the same bed family and the actual primal and two-form transports. In particular its retained-horizon equality is equality of the bed families; its tower equality preserves the A_0, T rows used by the schedule.

The source contract's remaining numerical receipts are explicit: the chosen price functions bound the bed strip widths/costs, floor constants, iterated tower costs, and the one-step crossing and roll

identities at all N, i and finite torus sides. The top bed is the unit diagonal because the top train is constant. These are exactly the fields of the horizon contract; they are not consequences of the block identity alone. In the public construction the level schedule and these receipts are attached to the preselected run, as stated next.

10.3 Actual path data and the public binding

Path data enter through the selected history of that run. Fix a finite torus, a plaquette face b, and a one-form variation a. For a path slot with coefficient c_{bi}, cover offset y_{bi}, and plaquette label p_{bi}, its physical face is the translate $b + y_{bi}$ modulo the torus. Define the path linear response by the absolutely convergent sum $\ell_{\mathrm{path}}(b, a) = \sum_i c_{bi}\, d_1 a(b + y_{bi}, p_{bi})$. The attachment identity says that, after summing all path coefficients with fixed (p, y), one obtains the corresponding cover entry of the root row. The weighted row bound implies absolute summability of the path coefficients: its exponential weight is at least one. It also implies absolute summability after multiplying by the bounded face curl of a. Therefore we may first group the path sum by physical face, and then sum over the finite set of torus faces. Let F_W be the actual roots block in the selected bed factorization, acting on periodic two-form fields. Periodizing the cover entries by the attachment identity gives, for each color component q,

$$(\ell_{\mathrm{path}}(b, a))_q = \big[F_W((d_1 a)_q)\big](b),$$

where F_W is the actual root matrix on the torus and $d_1 a$ is the finite-torus coboundary. The right side is a finite matrix-vector product. This is the entrywise path-to-root identity used in the physical quadratic-form computation; the absolute bound justifies the only infinite regrouping.

For the selected RG history, the construction receipt gives the attachment identity for every cutoff N, level $i \le N$, and period label p, between the bed $H_{N,i}$ and the path at that same history, cutoff, level, and period. The public adapter carries the original levels and retained horizon into the component theorem; it does not choose another schedule. Thus the path-to-root identity above applies to the same bed rows that enter the tower calculation. The weighted row bound is an additional priced path input: it is what supplies absolute summability, and is not concluded from attachment alone.

The matrix identities proved here discharge the tower side of the selected construction. The path attachment and absolute bounds are separate receipts of the selected path history, and are kept explicit in the source crosswalk rather than inferred from the matrix bounds alone.

11 The curvature factor and the retained horizon

The positive beds of Section 9 must be joined by one common family of transports. The remaining issue is the positivity of the curvature defect. The construction first writes that defect as a double exterior derivative, then adds a fixed positive amendment before taking its square root.

11.1 An exponentially summable exterior lift

We give the row-level lifting argument, because a cycle on a single finite torus would not suffice. Write $q_a(p) = e^{ip_a} - 1$ and $d_r(p) = q(p) \wedge \cdot$ on r-forms. Let $A(p)$ be the symbol of an exponentially summable row from $(r+1)$-forms to a fixed coordinate space, where $r + 1 > 0$, and suppose $A(p) d_r(p) = 0$ for every real p.

For one coordinate a, let $A_0(p) = A(p_1, \dots, 0, \dots, p_4)$. The difference $A - A_0$ is divisible by q_a as a Laurent row. To see this coefficientwise, replace each monomial in that coordinate by

$$\frac{e^{inp_a} - 1}{e^{ip_a} - 1} = \begin{cases} \sum_{h=0}^{n-1} e^{ihp_a}, & n > 0, \\ 0, & n = 0, \\ -\sum_{h=n}^{-1} e^{ihp_a}, & n < 0. \end{cases}$$

These are finite identities, including at $p_a = 0$ by continuity. If the original row has weighted norm at width η, the quotient has weighted norm at width $\eta/2$ bounded by $2/\eta$ times the original one. Each monomial produces $|n|$ terms of no greater absolute displacement, and

$$|n| e^{(\eta/2)|z|_1} \le (2/\eta) e^{\eta|z|_1} \quad (|n| \le |z|_1).$$

This proves both convergence and the claimed bound.

Let E_a denote exterior multiplication by the ath basis vector, and $\iota_a = E_a^{\mathsf{T}}$ the corresponding contraction. The exterior identity $d_r \iota_a + \iota_a d_{r+1} = q_a I$ follows on every coordinate face by distinguishing whether that face contains a; the two possible orders of adjoining another coordinate have opposite signs. Set

$$R_a = \frac{A - A_0}{q_a} \iota_a, \qquad S_a = A_0(I - E_a \iota_a).$$

Then

$$R_a d_{r+1} + S_a = A. \tag{64}$$

For $q_a \neq 0$, multiply this identity by q_a and use $A d_r = 0$ and $A_0 d_r = q_a A_0 E_a$; the latter follows from the cycle identity at $p_a = 0$. This proves it there, and continuity proves it also at $q_a = 0$. Multiplication on the right by d_r shows that S_a remains a cycle. The row S_a has zero columns on faces containing a, and on the other columns it simply evaluates p_a at zero.

Apply this operation successively in the four coordinate directions. Previously vanished columns remain zero. Every positive-degree face contains some coordinate, so the final residual is zero. Summing the four successive R_a gives a row $\mathcal{L}(A)$ with

$$\mathcal{L}(A) d_{r+1} = A.$$

There are only four operations. The width is reduced a fixed number of times and the cost is multiplied by a fixed constant; any scalar scale factor in the initial bound is retained. Transposition gives the left-handed lift. All these identities hold on every finite period by Fourier diagonalization. Conversely, an exponentially summable row identity valid on all finite periods holds for all real symbols: the finite Fourier frequencies are dense and the symbols are continuous.

We shall also need a two-sided version. Suppose a two-form row E satisfies $E d_1 = 0$ and $d_1^{\mathsf{T}} E = 0$ on every period. Let W be a left lift, so $d_2^{\mathsf{T}} W = E$, and put $V = W d_1$. Then $d_2^{\mathsf{T}} V = 0$. Lift once more on the left to obtain T with $d_3^{\mathsf{T}} T = V$. The row $T d_0$ is annihilated on the left by d_3^{T}. A further left lift would take values in five-forms, whose coordinate space is zero in dimension four. Hence $T d_0 = 0$. The right lift therefore gives $X d_1 = T$. Consequently

$$W' = W - d_3^{\mathsf{T}} X \quad \text{satisfies} \quad W' d_1 = 0, \qquad d_2^{\mathsf{T}} W' = E.$$

A final right lift C with $C d_2 = W'$ gives

$$d_2^{\mathsf{T}} C d_2 = E. \tag{65}$$

If E is symmetric, replace C by $(C + C^{\mathsf{T}})/2$. The identity and exponential row bounds are preserved, and the new C is symmetric.

11.2 The unamended defect and its scale factor

For one flight let H_c, B_c and H_f, B_f be the reciprocal pulse and dual-pulse rows on the coarse and fine lattices. Let T be its two-form transport and A_0 its one-form transport. The conditional mean is denoted by Q. The exact unamended transport identities are

$$T^{\mathsf{T}} d_{1,f} = d_{1,c} Q, \qquad T H_c d_{1,c} = H_f d_{1,f} A_0, \qquad Q A_0 = I - k^{-1} d_{1,c}^{\mathsf{T}} H_c d_{1,c}. \tag{66}$$

These identities were proved for the original kernel in Section 8; the amendment does not alter them on exact forms, by (53) and $G d_1 = H d_1$. Define

$$Z_0 = B_c - T^{\mathsf{T}} B_f T - k^{-1} d_{1,c} d_{1,c}^{\mathsf{T}}, \qquad E = H_c Z_0 H_c.$$

The matrices are symmetric. Moreover

$$T^{\mathsf{T}} B_f T H_c d_{1,c} = T^{\mathsf{T}} d_{1,f} A_0 = d_{1,c}\big(I - k^{-1} d_{1,c}^{\mathsf{T}} H_c d_{1,c}\big).$$

Substituting this in $Z_0 H_c d_{1,c}$ gives zero. Thus $E d_{1,c} = 0$, and symmetry gives the left cycle identity. The two-sided lift yields a symmetric row C such that

$$Z_0 = (d_{2,c} B_c)^{\mathsf{T}} C (d_{2,c} B_c). \tag{67}$$

Indeed multiply (65) by B_c on both sides and use $B_c H_c = H_c B_c = I$.

We check uniformity in the block side L. The dual transport and the row $B_f T$ have entrywise strip cost $O(L^{-2})$, including their required cover differences, by the wire estimates of Sections 6 and 7. The Gram product $T^{\mathsf{T}} B_f T$ sums over L^4 fine residues and a fixed number of form labels. Its entrywise cost is consequently $O(L^4 L^{-2} L^{-2}) = O(1)$. The other rows in E have uniform costs. All lifts use fixed exterior index sets on the coarse side, so the resulting cost of C is uniform in L too.

There is a second lift with a stronger scale gain. Put

$$Y = d_{2,f} B_f T H_c.$$

The second identity in (66) gives $Y d_{1,c} = d_{2,f} d_{1,f} A_0 = 0$. Its right lift is a row S with $S d_{2,c} = Y$. One fine exterior difference is a signed sum of at most four cover differences of $B_f T$; the first difference estimate supplies one additional L^{-1}. Thus each entry of Y, and then of S, has strip cost $O(L^{-3})$. Multiplying the lift identity by B_c gives

$$S d_{2,c} B_c = d_{2,f} B_f T. \tag{68}$$

The row $S^{\mathsf{T}} S$ now has cost $O(L^{-2})$: its summation has L^4 residues, while both factors cost L^{-3}. Retaining this count is essential.

11.3 Making the curvature row positive

Choose amendment coefficients $m, l \geq 0$ for the fine and coarse beds. With $D_c = (I + l d_{2,c} B_c d_{2,c}^{\mathsf{T}})^{-1}$, set

$$R_0 = C + lI - m S^{\mathsf{T}} S, \qquad R = D_c^{\mathsf{T}} R_0 D_c.$$

The uniform bounds just proved give constants $C_0, C_1 > 0$, independent of the train and L, with

$$|\langle u, Cu\rangle| \leq C_0 \|u\|^2, \qquad \langle u, S^{\mathsf{T}} S u\rangle \leq C_1 L^{-2} \|u\|^2.$$

Choose $l \geq C_0 + 2$ and $L^2 \geq mC_1$. Then

$$\langle u, R_0 u \rangle \geq \|u\|^2.$$

This is positivity on the entire three-form space. No projection onto a momentum-dependent range is needed. The row $D_c^{-1} = I + l d_{2,c} B_c d_{2,c}^{\mathsf{T}}$ has uniformly bounded operator norm, say at most K. Therefore

$$\langle u, Ru \rangle \geq \|D_c u\|^2 \geq K^{-2} \|u\|^2.$$

The products defining R have a uniform exponential row bound. Section 9 supplies its positive square-root row P, uniformly over all flights of these stages.

It remains to identify this factor with the actual amended curvature defect. Write

$$A_c = B_c + l(d_{2,c}B_c)^{\mathsf{T}}(d_{2,c}B_c), \qquad A_f = B_f + m(d_{2,f}B_f)^{\mathsf{T}}(d_{2,f}B_f), \qquad G_c = A_c^{-1}.$$

Equations (67) and (68), followed by expansion, give

$$A_c - T^{\mathsf{T}} A_f T - k^{-1} d_{1,c} d_{1,c}^{\mathsf{T}} = (d_{2,c}B_c)^{\mathsf{T}} R_0 (d_{2,c}B_c).$$

Using $(d_{2,c}B_c)G_c = D_c d_{2,c}$ from (53), multiply by G_c on both sides to obtain

$$G_c\big(A_c - T^{\mathsf{T}} A_f T - k^{-1} d_{1,c} d_{1,c}^{\mathsf{T}}\big) G_c = d_{2,c}^{\mathsf{T}} P^{\mathsf{T}} P d_{2,c}. \tag{69}$$

This is precisely the curvature factorization in Section 10, with the actual amended beds. All formulas hold on every finite period, since they were constructed from the same Laurent rows.

11.4 Choosing the prices before the scale sequence

For each adjacent pair of stages j, t, where $t = j$ or $t = j - 1$, let $a_{j,t} > 0$ be the lower requirement on the coarse amendment coefficient just obtained. Select

$$l_i = \max(a_{i,i}, a_{i+1,i}).$$

Then every permitted transition has $l_t \geq a_{j,t}$ and $l_j \geq 0$. Its required block-size floor is finite because the fine amendment $m = l_j$ is now fixed. Combine the two permitted flight floors at stage j by their maximum. Combine bed and flight strip widths and positive lower bounds by minima, and upper bounds by maxima. All widths and lower bounds stay positive, since each combination is finite.

Choose the stage's direct precision below its positive bed ceiling. Choose its block length above the flight floor. These are requirements on the original stage profiles and pre-length floors; they are imposed before the common schedule of Section 201. They hold for both within-stage steps and a change to the next stage, however many times either occurs.

At a terminal cutoff N, start with the top train and descend along those very block lengths. At each level use its positive bed from Section 9; at each step use (69) and the exact block completion of Section 10. The latter's strip calculation supplies the prescribed iterated difference bounds for its unchanged A_0, T rows and the displayed strip bounds for its remaining blocks. This changes neither the bed rows nor the original one-form and two-form transports A_0, T. It only constructs the remaining three blocks from their common positive curvature factor. Taking finite minima and maxima of the quoted prices gives a single family of bounds valid for every N, every intermediate level and every finite period. This proves the bed and curvature suppliers used by the horizon construction, using the original pulse transport identities proved in (46). The attachment and bounds of the nonlinear path remain the separate construction step.

12 Signed lattice symmetry of the quadratic precision

The path attachment uses the quadratic precision of the selected bed. We verify its invariance under signed permutations of the coordinate axes, including the change of anchor caused by reversing a bond. This is a property of the bilinear form on one-cochains; the curvature metric need not satisfy a stronger symmetry away from exact curvatures.

Let s be a signed permutation of the four axes, acting on sites and momenta by the same orthogonal integer matrix. A positive bond (x, a) becomes the directed bond from sx to $sx + se_a$. If $se_a = -e_b$, its value in positive-bond coordinates is minus the value at $(sx - e_b, b)$. This defines the pullback R_s on real one-cochains. For a canonical plaquette $p = (a, b)$, $a < b$, let $r_s(x, p)$ be its image with its two axes reordered increasingly and its lower vertex shifted for each reversed axis. Its orientation sign is

$$\epsilon_s(p) = \epsilon_s(a)\epsilon_s(b)\,\mathrm{sgn}(s_{\mathrm{perm}}(b) - s_{\mathrm{perm}}(a)),$$

where $\epsilon_s(a) \in \{-1, 1\}$ is the sign of the image axis. In the permutation subscripts, only the unsigned permutation is meant. Equivalently the last factor is 1 if the two image axes are in increasing order and -1 otherwise. The map r_s is a permutation of canonical plaquettes on every finite torus and $\epsilon_s(p)^2 = 1$.

Let d denote the discrete exterior derivative. The boundary of the image of an oriented plaquette is the image of its boundary, with the same orientation sign, so

$$(dR_s a)(x, p) = \epsilon_s(p)(da)(r_s(x, p)). \tag{70}$$

One can verify the identity directly by substituting $(da)(x, a, b) = a(x, a) + a(x + e_a, b) - a(x + e_b, a) - a(x, b)$. Reversing an axis replaces each affected bond by its negative at the shifted lower endpoint; exchanging the two axes reverses the whole expression. These two operations generate all the signs and permutations, so the formula holds also across the torus seam and for even periods.

Assume the blocking penalty is positive and the kernel level is at least four, as in the retained construction. Let P be the unamended curvature row from Section 9. Its Fourier symbol is diagonal in the plaquette axes. For $q \in [-\pi, \pi]^4 \setminus \{0\}$, its entry at $p = (a, b)$ is

$$\widehat{P}_{pp}(q) = \rho(q)^{-1} D_a(q)^{-1} D_b(q)^{-1}, \qquad \rho(q) = \sum_c \frac{4\sin^2(q_c/2)}{D_c(q)}.$$

Here D_c are the positive direction denominators of the selected finite tower. Their defining alias sums satisfy $D_{s_{\mathrm{perm}}(c)}(sq) = D_c(q)$: a signed permutation bijects the alias indices, preserves their scalar product weights, and permutes the single distinguished axis. The identity also follows term by term through the finite tower recursion. The sine squares are unchanged by sign and permuted with the same axes. Consequently $\rho(sq) = \rho(q)$ and

$$\widehat{P}_{pp}(q) = \widehat{P}_{p',p'}(sq), \qquad p' = sp$$

with the image axes canonically ordered. At $q = 0$ the unamended symbol is the identity matrix, so the same identity holds there too. Off-diagonal entries vanish.

The Laurent rows are absolutely summable. Uniqueness of Fourier coefficients and the change of variable $q \mapsto sq$ therefore give

$$P_{pp}(z) = P_{p'p'}(sz), \qquad P_{pq} = 0 \quad (p \neq q).$$

Let P_n denote the periodized row on a torus of side n. The lower vertex of an image plaquette has the form $sx + t_p$, where t_p depends only on its axes. For two plaquettes with the same axes,

this shift cancels from their separation. Reindexing the absolutely convergent periodization sum by $z \mapsto sz$ thus proves

$$P_n((x,p),(y,q))\epsilon_s(p)\epsilon_s(q) = P_n(r_s(x,p), r_s(y,q)). \tag{71}$$

For unequal axes both sides are zero; for equal axes the two signs multiply to one. This cancellation is why the anchor shifts cannot be ignored before passing to the bilinear form.

Using (70) and (71), expand the finite double sum:

$$\begin{aligned}\langle dR_s a, P_n dR_s b\rangle &= \sum_{u,v} \epsilon_s(u)\epsilon_s(v)(da)(r_s u)P_n(u,v)(db)(r_s v)\\ &= \sum_{u,v} (da)(r_s u)P_n(r_s u, r_s v)(db)(r_s v)\\ &= \langle da, P_n db\rangle.\end{aligned}$$

The last equality reindexes by the permutation r_s.

The positive amendment of Section 11 vanishes on exact curvatures: if M is the amended metric, then $d^* M_n d = d^* P_n d$. Hence its quadratic bilinear form satisfies

$$\langle R_s a, d^* M_n dR_s b\rangle = \langle a, d^* M_n db\rangle.$$

For a Lie-algebra-valued field, expand in an orthonormal colour basis. The signed lattice pullback acts separately in every colour, and the gauge quadratic form is the sum of the real bilinear forms just considered. Summing their equalities proves the required invariance for every pair of such fields and every period. The positive square root used to represent the amended metric is not needed in this symmetry argument.

13 Finite-scale propagation of the neutral curvature representative

This section records the finite-scale construction that supplies the scalar recurrence used in Section 5. It starts from a terminal difference of two quadratic precisions, propagates a representative of that difference modulo linear-curvature null directions, and controls its cost at every earlier level. The one-step operation is the local Schur and matching argument of Section 4; the present section supplies its actual finite-history inputs and closes the resulting recurrence. No limiting argument is used here.

13.1 Data on a finite interval of levels

For each x in an index set X, fix one schedule, a measuring level k_x, and finite indices $i_0(x) \le N_x \le n_x$. In formulas for one fixed x, suppress its subscript and write

$$J_i = J(i), \qquad L_i = |\text{block at step } i|, \qquad t_i = \log a_i^{-1}, \qquad \Omega_i = \Omega(J_i), \qquad \sigma_i = w(J_i),$$

where a_i is the positive length at level i. The schedule identities are

$$t_{i+1} - t_i = \log L_i, \qquad L_i \ge 3.$$

Let $\mathcal{F}$ be the finite set of face labels and put $m = |\mathcal{F}|$. For a horizon metric H at a torus side c, let D_c be the discrete exterior derivative from one-cochains to two-cochains and write its quadratic precision as

$$\mathscr{Q}(H,c) = D_c^{\mathsf{T}} H_c D_c,$$

where H_c is the finite matrix obtained from the metric by periodization. The packets below represent differences of these forms by periodizing a cover curvature matrix between the two copies of D_c. The scale allotment for adjacent levels and its one-step null correction are

$$\mathcal{A}_i = m^2 e^4 \Omega_{i+1}^2 \Omega_i, \qquad \nu_i(u) = \frac{2m^2 \mathcal{A}_i^2}{u L_i} \quad (u > 0). \tag{72}$$

These quantities are nonnegative. Indeed, the row bounds on each quadratic precision imply $\Omega_i \geq 0$. The expression $\mathcal{A}_i$ is the allotment entering the two order-two response-kernel estimates. The factor m^2 in ν_i counts the two finite face labels; the factor L_i^{-1} and the factor $2/u$ are the centered null-packet correction.

Assume the following finite data.

1. For every member x of an index set X, there is a horizon of quadratic precisions $H_x(N, i)$, with row bounds at widths $\alpha(J_i)$ and costs Ω_i. These bounds include the diagonal, crossing, and torus-floor estimates used in the one-step lemma. For every $i_0(x) \leq i < N_x$, the actual Wilson tower precision at (N_x, i) equals the order-two precision of the flight from level $i+1$ to level i.

2. There is a positive width curve w, with $w(k) \leq \alpha(k)$, such that for every $i < N$,

$$\sigma_i \leq \min\{1, \alpha(J_i), \alpha(J_{i+1})\}, \qquad 2\sigma_i \leq L_i \sigma_{i+1}, \qquad \nu_i(\sigma_{i+1}) \leq \exp\bigl(-\theta(t_{i+1} - t_i)\bigr) \tag{73}$$

   for one fixed $\theta > 0$.

3. The terminal allowance $T_{x,N} = 2m\Omega_N$ has a uniform polynomial age bound

$$T_{x,N} \leq A_0 \, \mathfrak{a}_x(N)^d, \qquad \mathfrak{a}_x(i) = 2 + Z_x + |t_i - t_{k_x}|, \tag{74}$$

   with $Z_x \geq 0$, integer $d \geq 0$, and $A_0 \geq 0$ independent of x, N, n.

The first two items are the complete inputs represented by the plan and funding conditions below. The first inequality in (73) keeps all response kernels inside both adjacent analytic widths. The second is the block-folding requirement. The third pays the exact one-step null-packet cost.

13.2 Production of the funded width and top allowance

For a map price X, put

$$\omega_j = \min\{1, \alpha_X(j), \alpha_X(j+1)\}, \qquad A_X(j,u) = m^2 e^4 \Omega_X(j)^2 \Omega_X(u),$$

$$b_X(j,u) = m A_X(j,u) \left(m \frac{2}{\omega_j} A_X(j,u) \right) = \frac{2m^2 A_X(j,u)^2}{\omega_j}.$$

$$E_X(j) = \left\lceil \frac{2}{\omega_j} + b_X(j,j)^2 + b_X(j,j-1)^2 \right\rceil.$$

These are the source's entrance ticket and two adjacent row prices. The schedule satisfies $J_{i+1} = J_i$ or $J_i + 1$. Suppose the selected floors give $E_X(J_{i+1}) \leq L_i$ and the delay receipt gives $\Omega_X(J_i) \leq \log \widehat{\beta}_i$. Then the world width is $w(j) = \omega_j$ and $\theta = 1/2$.

The width is positive, at most one, and no larger than α_X. If the stage rises, its bridge width is exactly ω_{J_i}; if the stage stays fixed, the bridge is $\min\{1, \alpha_X(J_i)\} \geq \omega_{J_i}$. The entrance summand $2/\omega_{J_{i+1}}$ implies $L_i \omega_{J_{i+1}} \geq 2$, while $w(J_i) \leq 1$, proving $2w(J_i) \leq L_i w(J_{i+1})$.

Since J_i is either J_{i+1} or its predecessor, $b_X(J_{i+1}, J_i)^2 \le L_i$. The quote, by the definitions of allotment and neutral coin, is exactly

$$\mathrm{quote}_i(w(J_{i+1})) = \frac{2m^2 A_X(J_{i+1}, J_i)^2}{w(J_{i+1})L_i} = \frac{b_X(J_{i+1}, J_i)}{L_i} \le L_i^{-1/2} = e^{-(t_{i+1}-t_i)/2}.$$

This proves the paid condition and fixes its exponent from the actual entrance-price formula.

For the top allowance let $s_G > 0$ be the gauge slope constant and choose $Z_x = 3 + \log \widehat{\beta}_{k_x}$. The selected spacing hypothesis includes $\widehat{\beta}_{k_x} > 1$; therefore $Z_x = 3 + \log \widehat{\beta}_{k_x} \ge 0$. The schedule's delayed-scale inequality and $\log r \le r - 1$ imply $\log \widehat{\beta}_i \le \log \widehat{\beta}_{k_x} + s_G|t_i - t_{k_x}|$. Consequently

$$\Omega_X(J_i) \le (1 + s_G)(2 + Z_x + |t_i - t_{k_x}|).$$

As $T_{x,N} = 2m\Omega_X(J_N)$, the top cap holds with degree one and constant $2m(1 + s_G)$. Thus the full funding conclusion follows from the selected integer entrance floor and the post-length delay, rather than being an independent large-scale assumption.

13.3 Production of the horizon rows and exact tower identity

The selected public route supplies a horizon and an exact carrier identity, but their local row construction is a separate producer obligation. The source route retains finite-period rows, adds a Gram correction preserving the curl action, takes square roots on a common smaller strip, and checks the resulting block identities including harmonic modes. This paragraph records that route and does not treat those operations as proved here. The exact mathematical input consumed below is: for every x, N, i, a positive horizon metric with the stated diagonal, crossing, floor and row bounds, and an identity of its tower curvature with the order-two flight from $i + 1$ to i on every required finite torus. Producing these horizons and verifying the Gram, root, and harmonic identities remains an open construction step in this chapter. The native component recipe retains the same schedule when binding these receipts to the selected run.

13.4 The selected run's nonlinear one-step supplier

At the selected adjacent levels, let L be the block side, β the lower scale, B_0, B_1 the lower and upper budget radii, μ the shape strength, and Q the menu target. The shipment tariff is

$$N\ \mathrm{siteFactor}(B_0)\exp((Q.\mathrm{relative} + 1)\,\mathrm{preCost}(Q.\mathrm{halo} + 5))\,a \le \mathrm{ribbon}(Q, L, \beta), \tag{75}$$

where N is the occupancy count and a is the available shipment amount. The second-row allotment is

$$\mathfrak{s}(\mu, L, \beta) = 2\exp\Big(-\frac{\mu}{16}\Delta_{G,L}(\beta)^2\Big), \tag{76}$$

with $\Delta_{G,L}$ the fine-scale monitor.

The selected response produces ungrouped terms $A_i(U)$ and $D_i(U)$, with support indices $s(i)$ and $p(i)$, respectively. Let $a_i, d_i \ge 0$ be summable majorants for their weighted prices, and let S_i, T_i be their support sets. The precise paid inputs are

$$\mathrm{price}_r(A_i(U)) \le a_i, \qquad \sum_{i:z\in S_i} a_i e^{(b+1/2)\,\mathrm{coverNumber}(B_0,S_i)} \le 1, \tag{77}$$

$$\mathrm{price}_r(D_i(U)) \le d_i, \qquad \sum_{i:z\in T_i} d_i e^{(b+1)\,\mathrm{coverNumber}(B_0,T_i)} \le \mathfrak{s}(\mu, L, \beta). \tag{78}$$

Every A_i has zero quiet coefficient and is supported in the assigned support $S_i = s(i)$. Every D_i is supported in the full box support $T_i = p(i).1$ and in the prescribed threshold shell. Thus the cover weights in the ledgers are the supports used by the resulting families themselves. The selected response receipts also give, for every box p and configuration U,

$$\sum_{i:p(i)=p} [D_i(U)]_{\varnothing} = \mathrm{profile}_p(U), \tag{79}$$

$$\sum_{i:p(i)=p} [D_i(U)]_j = \sum_l \mathrm{owl}(q_l)_j(p, U), \tag{80}$$

and the ungrouped expenditure identity

$$\mathcal{D}(\mathcal{E})(U) = e^{E(U)}\,\mathrm{pass}\left(\sum_i A_i(U)\right) * \mathrm{field}\left(\Big(\sum_{i:p(i)=p} D_i(U)\Big)_p, \mathrm{flags}(U)\right). \tag{81}$$

The field coefficient identity is for each box p and singleton mode j; higher coefficients are the full products in the field polynomial.

Theorem 13.1 (Paid menu assembly and shipment conversion)**.** *Assume the selected history provides the ungrouped paid terms and exact identities above, at lower exponent b and upper exponent $e_1 \geq 100$. Assume the menu offer has exponent $e_1 - 4$ and allowance a, and that the birth estimate, foam, and radius hypotheses stated in the proof hold. Then the grouped families*

$$F_s(U) = \sum_{i:s(i)=s} A_i(U), \qquad K_p(U) = \sum_{i:p(i)=p} D_i(U)$$

have the same rooted budgets as the ungrouped families, retain the support and shell conditions, and satisfy the per-box quiet and singleton identities and the push identity in (79)– (81)*. The selected door, tuning, and slot-separation conditions are retained. The producers of the paid terms are treated separately in Section 14; this theorem proves their tariff conversion and assembly.*

Proof. For menu i, write n_i^{birth} for the cover number of its birth region at radius B_0 and n_i^{crest} for the cover number of its crest at radius B_1. The selected birth estimate is

$$(Q.\mathrm{relative} + 1)n_i^{\mathrm{birth}} \leq (Q.\mathrm{relative} + 1)\,\mathrm{preCost}(Q.\mathrm{halo} + 5) + \frac{5}{4}\varepsilon_Q n_i^{\mathrm{crest}}.$$

Here $\varepsilon_Q > 0$. Let w_i be the menu wage, equal to its intrinsic profit times its total menu fee, and set $f_i = w_i \exp(2\varepsilon_Q n_i^{\mathrm{crest}})$. Foam at scale L, $B_1 \leq 2B_0$, $Q.\mathrm{relative} \geq 4$, and the profile hypotheses give $4\varepsilon_Q < e_1$. Hence

$$\frac{13}{4}\varepsilon_Q < \frac{13}{16}e_1 \leq e_1 - 4,$$

where the last inequality follows from $e_1 \geq 100$. Adding the wage weight to the birth estimate gives

$$e^{(Q.\mathrm{relative}+1)n_i^{\mathrm{birth}}} f_i \leq e^{(Q.\mathrm{relative}+1)\,\mathrm{preCost}(Q.\mathrm{halo}+5)} w_i e^{(e_1-4)n_i^{\mathrm{crest}}}.$$

For every named marker v, the offer gives $\sum_{i:v\in A_i} w_i e^{(e_1-4)n_i^{\mathrm{crest}}} \leq a$, where A_i is the set of available named origins for menu i. Choose, for each positive wage, a named origin in the menu mode whose parent image lies in that menu's birth region. The birth estimate and the displayed $13\varepsilon_Q/4$ margin transfer the offer to a rooted budget on birth regions, with exponent $Q.\mathrm{relative} + 1$ and budget

$c = \exp((Q.\text{relative} + 1)\,\text{preCost}(Q.\text{halo} + 5))a$. Apply the rooted projection estimate: if the origin family has occupancy at most N at each parent site and every positive-wage mode has one such origin in its birth support, then lowering the rooted exponent by one gives

$$\sum_{i:z\in\text{birth}_i} f_i e^{Q.\text{relative coverNumber}(B_0,\text{birth}_i)} \le N\,\text{siteFactor}(B_0)c \le \text{ribbon}(Q,L,\beta). \tag{82}$$

The final inequality is exactly the purchased tariff (75); the tariff is an input from the schedule. Zero-wage terms contribute nothing.

Now group terms by their support and box indices. The first ledger already uses exactly the supports $S_i = s(i)$, and the second uses exactly the box supports $T_i = p(i).1$. By the triangle inequality, $\text{price}_r(F_s(U)) \le \sum_{i:s(i)=s} a_i$ and $\text{price}_r(K_p(U)) \le \sum_{i:p(i)=p} d_i$. Summing these bounds over fibers at a fixed site gives exactly the two budgets in (77)– (78). Tonelli's theorem applies because the majorants are nonnegative and summable. Absolute convergence allows each finite marker coefficient to be grouped termwise, proving the per-box quiet and singleton identities.

The pass is a finite coefficient operation and commutes with the absolutely convergent grouping of A_i. The field operation is a finite sum of finite products on the marker algebra and likewise commutes with grouping the D_i by box. Substituting both regroupings into the ungrouped expenditure identity gives the stated push identity, including the exponential scalar and the pass of $\sum_s F_s$. All support conditions pass to the unions within each fiber. This proves the assembly without assuming the grouped conclusion as an input. □

13.5 The terminal representative

Lemma 13.2 (Terminal neutral representative)**.** *Let H_x satisfy the row and floor bounds at level N, and suppose $0 < \sigma_N \le \alpha(J_N)$. There is a summable, translation-covariant packet B_N which is neutral, has folded cost at most $T_{x,N}$ at width σ_N, and represents*

$$\mathscr{Q}(H_x(n,N),c) - \mathscr{Q}(H_x(N,N),c)$$

as a linear-curvature quadratic form on every torus side c occurring at level N.

Proof. Set $Y = H_x(n,N)$ and $Z = H_x(N,N)$, and define the packet by subtracting their saddle kernels entrywise. The row bounds give the two floor inequalities with the common parameter Ω_N and width σ_N. For each metric row R, its saddle kernel is the plain translation-invariant kernel evaluated on a constant output position. A strip bound of width σ_N and cost Ω_N makes this saddle packet's folded norm at most $m\Omega_N$: sum first over the cover position and use the strip bound for each of the m output labels. The triangle inequality therefore bounds the difference packet by $m(\Omega_N + \Omega_N) = T_{x,N}$. Translation covariance follows from translation invariance of each plain kernel.

For neutrality, fix the root position and output face label and sum the packet over the cover. Absolute convergence follows from the folded bound. Each saddle sum is the zero-frequency row sum of its metric. The floor condition pins both metric zero modes to the same identity matrix; hence their row sums agree and the difference sum is zero. Finally, on each torus side c, periodizing the packet gives the difference of the two metric matrices. Each floor identifies its metric with the quadratic form obtained from its full map, so subtracting the identities represents the precision difference by the exterior derivative of the difference of those finite curvature matrices. This proves the claim without inverting the curl precision; the identity holds on the tested curvature form including its harmonic zero mode. □

13.6 One backward step and the actual kernel inputs

Lemma 13.3 (One-step predecessor)**.** *Fix $i < N$ and assume a packet B_{i+1} represents the difference $\mathscr{Q}(H_x(n, i+1), c) - \mathscr{Q}(H_x(N, i+1), c)$ on every required torus side. Suppose the two order-two response kernels for the histories n and N have common exponential width*

$$\omega_i = \min\{1, \alpha(J_i), \alpha(J_{i+1})\}$$

and cost $\mathcal{A}_i$, satisfy the city and matching identities with the corresponding tower carriers, and the response kernel used for the coarser history is a travelling kernel. Denote the finer-history response by T_i and the coarser-history response by U_i. If B_{i+1} is neutral and translation covariant and has folded cost $p \geq 0$ at width $u = \sigma_{i+1}$, then the composed packet

$$B_i = B_{i+1} \star (T_i, U_i)$$

is neutral and translation covariant, represents the quadratic-precision difference at level i on every required torus side, and has folded cost at width σ_i at most $\nu_i(u)p$.

Proof. The two kernels are the responses obtained by composing the inverse precision at the fine level $i+1$, the installed Wilson tower from $i+1$ to i, and the metric at the coarse level i, once for cutoff n and once for cutoff N. The deep order-two tower estimate and the two adjacent precision row estimates give their order-two atlas bounds. The priced atlas initially contains $e^{4\omega_i}\Omega_{i+1}$ in its high-side entry; since $0 \leq \omega_i \leq 1$, enlarging this to $e^4\Omega_{i+1}$ gives exactly the allotment $\mathcal{A}_i$ in (72). The crossing identities for the actual towers give the required matches with their periodized carrier maps. The response composition has the city law because each factor has its own translation law and these laws add under convolution. The widths are ω_i for both responses.

For the remaining travel condition, the Wilson tower's zero-alias column is the toll-L_i identity, namely $L_i^{-2}I$. Periodicity then gives a position-independent column sum for the installed tower; this scalar normalization is sufficient for the travel product. The fine cross-row bound and the inverse-precision row bound make the two preceding convolutions absolutely summable; absolute convergence permits changing their order. The sum of a convolution of two travelling kernels is the product of their position-independent row sums, so the full coarser response is travelling. These are precisely the finite matching, city, and travel hypotheses in Section 4; applying its proved Schur and weighted-packet estimate gives the asserted identities and cost $\nu_i(u)p$. The width, positivity, and folding assumptions in (73) are exactly that lemma's remaining hypotheses. □

13.7 Backward induction and the scalar rate

Theorem 13.4 (Uniform finite-interval flat-sector bound)**.** *Suppose the terminal and one-step hypotheses above hold for every $x \in X$ and every finite interval $N \leq n$. There are nonnegative constants A_1 and an integer d_1, and an exponent $c_1 > 0$, independent of x, N, n, and packets B_i for all $i \in [i_0(x), N_x]$, such that each B_i is neutral and translation covariant, represents $\mathscr{Q}(H_x(n,i), c) - \mathscr{Q}(H_x(N,i), c)$ on all required torus sides, and has folded cost*

$$\|B_i\|_{\sigma_i,\mathrm{fold}} \leq A_1\, \mathfrak{a}_x(i)^{d_1} \left(\frac{a_N}{a_i}\right)^{c_1}.$$

Proof. Define the nonnegative allowance recursively by

$$C_N = T_{x,N}, \qquad C_i = C_{i+1} \exp\bigl(-\theta(t_{i+1} - t_i)\bigr) \quad (i < N).$$

At $i = N$, the terminal lemma constructs B_N with cost at most C_N. If B_{i+1} has cost at most C_{i+1}, the one-step lemma and the funding inequality give

$$\|B_i\|_{\sigma_i,\mathrm{fold}} \le \nu_i(\sigma_{i+1})C_{i+1} \le C_{i+1}\exp\bigl(-\theta(t_{i+1}-t_i)\bigr) = C_i.$$

Neutrality, covariance, and the tested curvature identity are inherited from the one-step lemma. Backward induction therefore constructs all packets and proves their costs are bounded by C_i.

For each fixed x, the schedule times have gaps $\Delta_i = t_{i+1} - t_i = \log L_i \ge \log 3$. The scalar estimate of Section 5, applied on this finite interval with terminal value C_N, zero interior source, source decay exponent $c_0 = \theta$, and the age $R + Z_x + |t_i - t_{k_x}|$, converts the terminal polynomial bound (74) and the exact recurrence for C_i into

$$C_i \le A_1\bigl(2 + Z_x + |t_i - t_{k_x}|\bigr)^{d_1} e^{-c_1(t_N - t_i)}.$$

The scalar estimate gives $c_1 = \theta/2$ and $d_1 = d$. The fixed R in that estimate is absorbed into the age because $2 + Z_x + |t_i - t_{k_x}| \ge 2$. The constants are independent of x and of the interval length. Since $e^{-(t_N - t_i)} = a_N/a_i$, this is the claimed bound. □

Remark 13.5 (Role in the selected construction). This finite-scale producer is the one used by the flat supply in the selected construction. The public route first selects the Wilson run with treasury and period funding and retains its original sheet, quote, and post-length payments. Its native component choice includes the flat entrance supplier. Realization applies that supplier to the very same selected schedule and horizon. The flat entrance supplier obtains an admissible spacing schedule and invokes the finite-history supplier whose meshing step uses the world construction proved above. This chapter discharges that row-level producer conditional on the stated funding, response-kernel, tower-matching, and cost hypotheses. Their origin in the complete public construction remains tracked in the accompanying audit; the present theorem alone is not the final continuum or moment limit.

14 Menu fees and the two response budgets

This section expands the two quantitative suppliers consumed by the selected one-step construction. Work at one selected adjacent pair of levels and retain its schedule, menu family, tuned representation, and activity model. All sums below use the selected countable menu or field index sets; no new selection of menus or rates is made.

14.1 The paid fragrance response

Let i be a fragrance label. Its order is n_i and its spatial labels are $u_{i,1}, \dots, u_{i,n_i}$. Write c_i for the corresponding Ursell coefficient and $t_{i,\ell}$ for the selected tuned component, with ℓ running over the distinguished base label and the n_i spatial labels. The fee majorant is

$$\ell_i = \frac{|c_i|}{(n_i+1)!} \prod_{\ell \in \{0,1,\dots,n_i\}} t_{i,\ell}. \tag{83}$$

A switching observable is paid at cost c when its coefficient series is controlled by a nonnegative majorant with total sum at most c, uniformly for all allowed interpolation parameters and configurations. The trained fragrance response is $s_i(U,t,F) = Y_i(\mathrm{color}(U,t,F))$; the paid observable is

$$\tau_i(U,t,F) = s_i(U,t,F) - s_i(U,0,\varnothing),$$

where F is a finite set of selected menu indices. The product formula for s_i writes it as its Ursell coefficient divided by $(n_i + 1)!$ times the product of the component votes. The product has $n_i + 1$ factors: one distinguished base position and the n_i spatial positions. The switching payment for a product is as follows. For a finite marker set I, a score family $g_t(A)$ represents

$$f_t(F) = \sum_{A \subseteq F} t^{|A|} g_t(A), \qquad 0 \le t \le 1.$$

A payment certificate supplies a nonnegative majorant $b(A)$, independent of t and the configuration, with $|g_t(A)| \le b(A)$, $\sum_A b(A) \le c$, and convergence of each score as $t \downarrow 0$. If g_t and h_t represent two paid observables, the score for their product is

$$(g_t \star_t h_t)(C) = \sum_{A \cup B = C} t^{|A \cap B|} g_t(A) h_t(B).$$

Expanding the product and regrouping by $C = A \cup B$ gives the original weight, since $|A| + |B| = |A \cup B| + |A \cap B|$. The coefficient bound at $t = 1$ is the same convolution of the two nonnegative majorants; its total sum is the product of their sums. There are finitely many scores, so continuity at $t = 0$ is preserved. Iterating over the $n_i + 1$ component votes and multiplying by the scalar coefficient gives the fee ℓ_i in (83). The zero-menu term is paid at the same cost; the triangle inequality for the two score majorants therefore pays τ_i at cost $2\ell_i$.

The selected forest estimate gives a ready cluster bound at exponent $a + e$, with rooted budget P_0 and degree ratio $0 \le \rho < 1$. Here a is the selected earned exponent and e the selected allowance; in the one-step data, $a + e = \text{earn} + w = r_4$. Fix a site z and mark each underlying component whose full support contains z. The zero-depth cluster-tail estimate sums all connected clusters containing at least one marked component. Its geometric series is $\sum_{m \ge 0} \rho^m = (1 - \rho)^{-1}$, so the total cluster charge at z is at most $P_0/(1 - \rho)$. For each such cluster, the connected-coefficient cover inequality bounds its Ursell coefficient times the support currency by the fee majorant of its union support. The resulting fragrance fee ledger is exactly

$$\sum_{i : z \in R_i} \ell_i \exp(r_4 \, \text{coverNumber}(B, R_i)) \le \frac{P_0}{1 - \rho}, \tag{84}$$

where R_i is the full fragrance support and B is the selected lower radius. Here P_0 is the pledged first-row allotment, and ρ is the upward sail loss. This is the zero-depth specialization of the geometric cluster-tail estimate. No extra exponent margin is used. The coefficient sequence is summable with an explicit finite-volume factor. The selected conditions give $r_4 \ge 0$. For nonnegative fees on nonempty supports, sum the rooted bound over all sites in the finite lower torus. Each fee contributes at at least one site, with its exponential factor at least one. There are $|\Lambda_c|$ sites, so

$$\sum_i \ell_i \le |\Lambda_c| \frac{P_0}{1 - \rho}.$$

This is the source's nonempty-support counting argument. Its finite-volume factor is needed for summability; it is not absorbed into the rooted budget. The rooted bound prices supports, while this summability justifies the countable expansion.

The subtraction cost is $2\ell_i$. Multiplying the rooted fee sequence by two scales its rooted sum and budget by two. The selected stage identities identify the resulting exponent and budget as

$$r_4 = e_{\text{earn}} + w, \qquad d_4 = P_0 (1 - \rho)^{-1},$$

so the response fee ledger has exponent r_4 and budget $2d_4$. Its zero coefficient at the empty marker mode follows from the ways/color identity: when the marker set is empty, every way has the same value as at interpolation parameter zero, so the trained response equals its subtracted zero-menu value. Nonemptiness of each menu mode is used separately in the nilpotence calculation below. The exact locality statement is a noninterference implication. For each menu label i, marker site u in its crest image, and activity label j, if u is outside the support of the jth forced activity term, erasing switch i leaves that activity term unchanged. Consequently a term can depend on switch i only when its support contains the entire selected crest image. The implication passes to the absolutely convergent sum, with no cancellation argument.

For the combined observable, the selected cash identity first holds as an identity of switched scalar values, for every $t \in [0,1]$, finite marker set F, and configuration U:

$$e^{E(U)} e^{\sum_i \tau_i(U,t,F)} \operatorname{gas}_{\mathcal{A}}\big(K(t,F)\big)(U) = \operatorname{push}(\operatorname{buy}(t,F))(U). \tag{85}$$

Here $K(t,F)$ is the final response code, and $\mathcal{A}$ is the same activity family before and after the push. To verify the identity, let L_0, L_1 be the logarithm sums of the entrance and branched responses and let G_0, G_1 be their gas polynomials. The switching calculation gives $G_1 = G_0 e^{L_0}$ and $\operatorname{gas}_{\mathcal{A}}(K) = G_1 e^{L_1}$. The base purchase identity is $\operatorname{push}(\operatorname{buy}) = C(U)M(U)G_0(U)$, where C is the selected overhead and M the homestead multiplier. The cash assembly gives $M = e^{\sum_i s_i(U,t,F)}$ and the zero-menu multiplier is $e^{t_0} = \mathsf{P}_{\mathrm{vac}}(U)$. Finally, the selected exponent identity is

$$e^{E(U)} e^{L_0+L_1} = C(U)\mathsf{P}_{\mathrm{vac}}(U).$$

Multiplying the switching identities and substituting this exponent identity reduces the left side of (85) to $C(U)M(U)G_0(U)$, which is the base purchase. The paid fees are summable, so the logarithm series converges absolutely and each switched scalar product is well-defined. For $t > 0$, the triangular evaluation map $g \mapsto (f(F) = \sum_{A \subseteq F} t^{|A|} g(A))$ is injective: induction on $|F|$ recovers $g(F)$ after the proper-subset terms are known. Thus the switched scalar identity determines equal score families for every $t > 0$; letting $t \downarrow 0$ under the uniform summable score bounds gives equal limiting scores. To obtain the nilpotent polynomial identity, for a family of scores $g_t(A)$ and square-zero elements z_i, take $\sum_A (\lim_{t \downarrow 0} g_t(A)) \prod_{i \in A} z_i$. Addition is preserved by linearity. Multiplication is preserved because the score convolution has weight $t^{|A \cap B|}$, whose limit is zero whenever the same marker occurs in both factors; the corresponding product then vanishes by $z_i^2 = 0$. The exponential follows by applying this product rule to its power series. For the countable logarithm sum, the uniform summable fee majorants permit dominated convergence coefficient by coefficient and then summation over markers. This establishes the coefficientwise closed-response identity used in the first paid response certificate; scalar equality alone would not suffice.

The full response certificate also includes a paid gas response and a locality statement. These are distinct obligations: the rooted fragrance ledger above does not imply the gas budget. The source proves the first by combining the selected hard and soft budgets into a rate price, then harvesting it through the fused response. The locality statement follows from the explicit crest support of each menu contribution.

14.2 Pricing the second response row

At level m, let P be the lower budget point, Ξ the selected parameter profile, T_0 the shelved activity, T_1 the branched activity, and α the selected kinetic parameter. Write S, σ, C for the clip, width and path count in Ξ, and $H = \operatorname{credit}_G(\Xi)\delta_E(\beta)$. The time profile and budget point satisfy the selected matching relation. The ordinary field request is

$$f = \operatorname{softNeed}_G(P, S, \sigma, C, \alpha, H) \tag{86}$$

By definition this is $f = \mathrm{softReset}_G(P,H) + \mathrm{hitNeed}(P,\alpha,H)\,\mathrm{lowHard}(P,S,\sigma,C)$. Write $R = \mathrm{raiser}(P)$ for the raised point. It changes only the support exponent, so its radius agrees with the base radius: $B_+ = B = P.B$. Retain the same selected profile Ξ for its clip, width, path count, and credit. The raised final demand is the base-point price which pays a vapor receipt at R:

$$f_+ = F(H) + N(P,\alpha,H)Q(P,S,\sigma,C), \tag{87}$$

Let $N(P,\alpha,H) = \mathrm{hitNeed}(P,\alpha,H)$ and define the selected reset and soft-column prices by

$$\begin{aligned} F(H) &= \mathcal{P}(1, 4b+2, 0, 1/2, \mathrm{resetLoss}(G,P), H, e+2, 81),\\ Q(P,S,\sigma,C) &= \mathcal{P}(1, b, 0, \sigma/8, 1, d_{\mathrm{sig}} L_{\mathrm{low}}(S,C), e+2, C_{\mathrm{low}}), \end{aligned}$$

where $b = P^B$, $e = P.\mathrm{supportExponent}$, d_{sig} is the number of signed directions, and C_{low} is the fixed low-column count. The footprint price used here is the explicit function

$$\mathcal{P}(P,W,k,\epsilon,C,\gamma,a,v) = \frac{e^{av}\gamma C\,P\ \mathrm{ballCost}(W)\,\epsilon^{-(4+k)}(4+k)!e^{\epsilon}}{1-e^{-\epsilon}}.$$

Thus $f_+ = F(H)+N(P,\alpha,H)Q(P,S,\sigma,C)$ is the reset plus hit-weighted soft-column cost. These are the two terms in the enlarged- support vapor estimate at the raised point R; in general $f_+ \neq f$. The estimate is used with the profile's clip, width and path count, and its credit $H = \mathrm{credit}_G(\Xi)\delta_E(\beta)$. The matching relation supplies equality of power, order and support exponent between P and Ξ. The estimate requires the selected local table, bounds, width and decent-row conditions, the room condition, and a coarse torus side length greater than 20. In the source both footprint prices use the base budget's support exponent plus two; raising the budget changes the target receipt, not the argument at which the numerical price is evaluated. There are two distinct applications of the arrival certificate in the selected proof. The precursor is the purchased entrance response, of cost $z_0 = 2P_Q\exp(-E_0)$, where P_Q is the selected pledge and E_0 its marking exponent. The gain inequality gives $2P_Q \le 1$, so $0 \le z_0 \le \exp(-E_0)$. For the ordinary path, the column-tail receipt gives $0 \le p \le \beta_F^{-1/8}$; the first arrival clause prices the precursor and the finite-product harvesting estimate transfers it to the branched activity at stems. For the raised path, the first-row column receipt is instead $0 \le p_3 \le \exp(\chi_0)\beta_F^{-1/8}$, with χ_0 denoting the selected cover ticket. This is the soft allowance paired with its raised precursor. The final vapor estimate is a separate local estimate and has a different budget, stated below.

For the final row, the second arrival clause is used with the exact same budget, clip, width, path count, and schedule identity. Its hypotheses are $|\alpha| \le \alpha_{\max}$, equality of the current parameters with the selected profile, and $f \ge 0$. It returns a rate price κ for the pair of goals

$$\mathrm{straw} = 2\ \mathrm{lauge}(L,\mu/4,\beta,1/4), \qquad \mathrm{stems} = 2\ \mathrm{lauge}(L,\mu/4,\beta,1/2).$$

By the definition of the exponential taper,

$$\mathrm{stems} = 2\ \mathrm{lauge}(L,\mu/8,\beta,1), \qquad \mathrm{straw} = 2\ \mathrm{lauge}(L,\mu/8,\beta,1/2).$$

This exact factor of two is kept in the payment: no volume factor is removed from the row budget.

The rate certificate has hard goal stems, soft goal f, and target toll straw. Its two numerical conditions are

$$(2B+1)^4\big(e^{\kappa}\,\mathrm{stems} + 2f\big) \le 1, \tag{88}$$

$$\mathrm{Quote}(\kappa, e^{\kappa}\,\mathrm{stems} + 2f) \le \mathrm{straw}\,. \tag{89}$$

Here B is the integer budget radius. The rate also requires $\kappa \geq 0$, $0 \leq$ stems, and $0 \leq f \leq 1$. The first inequality pays the fourth-degree local inflation and the soft reserve; the second is the selected financial price. The arrival certificate supplies all these fields. To pass a paid precursor to the widened activity family, use the selected injection u from old activity labels to the new drawer labels. Put the old code at label $u(i)$ and set it to zero at every label outside the image. The widening identity says the forced field at $u(i)$ is exactly the old forced field at i, and all forced fields outside the image vanish. Thus the score majorant is extended by zero with the same total price; the physical supports at retained labels are unchanged, so the rooted hit receipt is transported without any site map. The payment's support-exponent allowance is the one supplied by the widening step.

Set $b =$ stems and $w = f$. The rate inflation is $\Gamma = e^{\kappa} b + 2w$. By definition, $\mathrm{Quote}(\kappa, \Gamma) = e^{-\kappa}(e^{\Gamma} - 1)$. The capital and quote inequalities above therefore give exactly the hypotheses and site output of the hard/soft fusion lemma in Section 91:

$$(2B + 1)^4 \Gamma \leq 1, \qquad e^{-\kappa}(e^{\Gamma} - 1) \leq \text{straw}\,.$$

That section proves the gas identity (477) by a weight-preserving bijection of compatible families. It also proves the priced site estimate (480) from $h_B \Gamma \leq 1$. In the selected construction, the widening identity transports the paid hard response, and the finite-product harvesting estimate applies these results to the hard precursor at stems and the soft logarithmic response at f. The rate return inequality is exactly $e^{-\kappa}(e^{\Gamma} - 1) \leq$ straw, so (480) gives the purchased site budget. All overlap cases are accounted for by the compatible-family bijection, including eligibility and exact flag coverage.

For the ordinary path, the second arrival clause prices the request (86) directly at the straw row, using the selected profile equality, exact schedule identity, and within-step bound on $|\alpha|$. For the raised path, the enlarged-support local vapor estimate pays f_+ from (87); the cover-weighted allowance p_3 belongs to its earlier precursor receipt, not to this final vapor bound. The final vapor receipt is attached to the raised budget, while its second rate clause prices at the base budget point with the selected profile's clip, width, paths and credit. In particular, with raised rate κ_+ and the common radius $B_+ = B$ it requires

$$(2B_+ + 1)^4 (e^{\kappa_+} \text{ stems} + 2f_+) \leq 1, \qquad e^{-\kappa_+}(e^{e^{\kappa_+} \text{ stems} + 2f_+} - 1) \leq \text{straw}\,.$$

The proof first widens the support and then harvests the finite product. In both cases the payment is transported to the result activity by the finalization identity. The ordinary paid-response certificate uses the first path; a separate raised branch of the construction uses the second path. Neither path follows from the other by changing only the exponent.

14.3 Constructing the per-box response from the paid rows

For the per-box construction, let z_i be the normalized polynomial supported on the nonempty marker mode S_i, with coefficient a_i/ϕ_i, where a_i is the menu charge and $\phi_i = \mathrm{cost}_r(S_i)|a_i|$. Let $stamp_i$ be the polynomial on that mode with coefficient a_i. Then

$$[z_i]_{\varnothing} = 0, \qquad z_i^2 = 0, \qquad \mathrm{price}_r(z_i) \leq 1, \qquad \phi_i z_i = stamp_i. \tag{90}$$

If $\phi_i = 0$, positivity of the cost on a nonempty mode forces $a_i = 0$, and these statements hold with both polynomials zero. If $\phi_i > 0$, they follow directly from the definition of the weighted price and the fact that the mode is not disjoint from itself.

For each activity label j, configuration U, and $t \in [0,1]$, write $g_{j,t}(A,U)$ for the switching score of the selected code. The paid certificate gives a nonnegative summable majorant $b_j(A)$ with $|g_{j,t}(A,U)| \le b_j(A)$, and each score has a limit as $t \downarrow 0$. Define the activity polynomial

$$F_j(U) = \sum_A \Big(\lim_{t\downarrow 0} g_{j,t}(A,U) \Big) \prod_{i\in A} z_i.$$

Its weighted price is bounded by $\sum_A b_j(A)$: submultiplicativity of weighted price gives

$$\operatorname{price}_r\Big(\prod_{i\in A} z_i\Big) \le 1.$$

The triangle inequality then gives the bound first for finite partial sums and then for their absolutely convergent limit. A product of two normalized menu factors either vanishes by a repeated marker or has support of size at least two. Therefore the coefficient of a singleton marker in F_j comes only from singleton score sets $A = \{i\}$.

The exact local jet receipt, proved in Section 15, is

$$\lim_{t\downarrow 0} g_{j,t}(\{i\},U) = \phi_i D_{i,j}(U), \tag{91}$$

where $D_{i,j}$ is the derivative of the selected one-menu final response at activity label j and configuration U. Using it in the coefficient formula above and then using $\phi_i z_i = \mathit{stamp}_i$ gives

$$[F_j(U)]_{\{v\}} = \sum_i [D_{i,j}(U)\, \mathit{stamp}_i]_{\{v\}},$$

which is the quoted singleton coefficient. The series converges absolutely: the door-volume estimate bounds the hinge coefficients by the menu's weighted load, and the menu offer has a summable fee-load ledger.

Group activity polynomials by the box label $p = \operatorname{barcode}(j)$:

$$K_p(U) = \sum_{j:\, \operatorname{barcode}(j)=p} F_j(U).$$

Each fiber is countable. The paid coefficient majorants and their rooted hit ledger imply absolute convergence on each fiber and preserve its price when the terms are summed. The quiet coefficient of K_p is the activity profile on p: at the empty marker set the interpolated code is its vacuum code, so its score reconstructs that value. Thus the profile term is retained, while the singleton calculation above concerns only nonquiet coefficients. Higher coefficients of K_p contain all compatible products and are not sums of singleton responses.

Write $\mathcal{L}_t(A,U)$ for the combined score of the full fragrance response; its evaluation satisfies $\sum_{A\subseteq F} t^{|A|}\mathcal{L}_t(A,U) = \sum_i \tau_i(U,t,F)$. For any such score family $q_t(A)$, define its limiting marker polynomial by

$$\mathsf{Pass}(q) = \sum_A \Big(\lim_{t\downarrow 0} q_t(A) \Big) \prod_{i\in A} z_i.$$

For a support mode s, define $\mathsf{Field}((K_p), s)$ by the finite sum

$$\mathsf{Field}((K_p), s) = \sum_{\mathcal{C}:\, \bigsqcup_{p\in\mathcal{C}} Q_p = s} \prod_{p\in\mathcal{C}} K_p,$$

where a box is a pair $p = (P_p, Q_p)$ consisting of its full support P_p and assigned flag $Q_p \subseteq P_p$. In every family in the sum the full supports P_p are pairwise disjoint and the flags have union s; a nonzero box term has nonempty assigned flag. This is the support pattern condition: disjoint flags alone would permit two interacting polymers to overlap and would not reproduce the gas. Grouping by barcode preserves the condition. Indeed, two nonzero compatible activity terms cannot have the same barcode, because that barcode has a nonempty full support. Thus each barcode occurs at most once in a compatible family; its full support is unchanged by grouping, and its assigned flag remains a subset of that support. Expanding the products after this bijective grouping gives exactly the original full-support-disjoint families and their flag union. Absolute summability justifies the countable regrouping. Because the full logarithm score has zero quiet coefficient, its marker polynomial is nilpotent on the finite marker algebra. Its truncated exponential is therefore a finite polynomial. The full pushed expression is

$$\mathrm{push}(\mathrm{Expense})(U) = e^{E(U)} \exp\big(\mathsf{Pass}(\mathcal{L}(U))\big) * \mathsf{Field}\big((K_p(U))_p, \mathrm{flags}(U)\big). \tag{92}$$

For a finite ticket truncation, expand the purchase as its finite ticket sum and push each ticket term using linearity of the kernel integral; the lawful input makes each term integrable. The score-to-marker map preserves products because repeated switches vanish. Its exponential is the finite nilpotent exponential, and it preserves the countable fragrance sum by the summable majorants. Expanding the field pattern then gives (92); the absolute price bounds pass to the countable limit. The quiet profile, singleton coefficient, support, shell, measurability, and gauge conditions are exactly the receipts needed for the per-box response. The hinge identity (91) and the selected geometric price receipts are upstream inputs; the next section proves the hinge identity itself.

14.4 What these estimates produce

The two row estimates have separate roles. The fragrance estimate gives a summable family of paid logarithmic terms, with rooted budget $2P_0/(1-\rho)$ at exponent r_4. The field estimate gives a paid activity response at the straw budget (76). Together with the selected crest locality and the coefficientwise cash identity (85), they supply the paid-response inputs to the per-box assembly in Theorem 13.1. All numerical dependence on L, β, μ, the budget radius, and the menu profile remains visible in the displayed prices.

15 The single-menu derivative supplies the field jet

Fix one selected step and its finite menu family indexed by $i \in I$. Let M_i be the leaf attached to menu i, let $r_i \geq 0$ be its tuning weight, and let ϕ_i be its charged fee. The selected menu profile has $r_i = \phi_i$. For each activity label j and configuration U, let $K(t, F)_j(U)$ be the final response code with switch parameter t and selected menu set $F \subseteq I$. The activity payment supplies a score family $g_{j,U,t}(A)$ satisfying

$$K(t,F)_j(U) = \sum_{A \subseteq F} t^{|A|} g_{j,U,t}(A), \qquad |g_{j,U,t}(A)| \leq b_{j,U}(A), \qquad \sum_{A \subseteq I} b_{j,U}(A) < \infty, \tag{93}$$

with each score converging as $t \downarrow 0$. The exact selected interpolation identities are

$$K(t, \varnothing)_j(U) = P_i(0)_j(U), \qquad K(t, \{i\})_j(U) = P_i(r_i t)_j(U), \tag{94}$$

where $P_i(\theta)$ is the final response produced from the single leaf M_i at amplitude θ. These are not approximations: with no active switch the selected color is the vacuum color, and with just switch i its color is the single-leaf color at amplitude $r_i t$.

The path P_i is differentiable at zero. Before the last branch, its color has the form $c + \theta a$ for fixed c, a, so every activity entry is affine in θ. Each selected fragrance entry is a finite product of such entries, raised to fixed integer powers and divided by a fixed factorial. If $f_1, \dots, f_N$ are differentiable at zero, then

$$\left(\prod_{k=1}^{N} f_k\right)'(0) = \sum_{k=1}^{N} f_k'(0) \prod_{\ell \neq k} f_\ell(0),$$

by repeated use of the product rule. The final response at a fixed activity coordinate is itself a product over a finite set of labels; branches outside the selected address range are identically zero, and branches inside it are these finite products. Thus every coordinate $P_i(\theta)_j(U)$ is differentiable at zero. Denote its derivative by

$$D_{i,j}(U) = \frac{d}{d\theta} P_i(\theta)_j(U)\Big|_{\theta=0}.$$

This is exactly the hinge of the selected gauge door, since that door is defined by differentiating the same one-leaf final response.

For completeness, the score extraction from (93) is triangular. For $t > 0$ and any $F \subseteq I$,

$$g_{j,U,t}(F) = t^{-|F|} \left(K(t,F)_j(U) - \sum_{A \subsetneq F} t^{|A|} g_{j,U,t}(A) \right).$$

Induction on $|F|$ therefore shows that all scores are uniquely determined by the switched values. The payment model also gives convergence to the zero-parameter score; the uniform summable majorant in (93) permits passage to that limit.

Take $F = \{i\}$. From (93),

$$K(t,\{i\})_j(U) = g_{j,U,t}(\varnothing) + t\, g_{j,U,t}(\{i\}).$$

At $F = \varnothing$ the same representation gives $K(t,\varnothing)_j(U) = g_{j,U,t}(\varnothing)$. Subtract these equations and use (94):

$$t\, g_{j,U,t}(\{i\}) = P_i(r_i t)_j(U) - P_i(0)_j(U).$$

Dividing by t and letting $t \downarrow 0$, differentiability at zero implies

$$\lim_{t \downarrow 0} g_{j,U,t}(\{i\}) = r_i D_{i,j}(U) = \phi_i D_{i,j}(U). \tag{95}$$

This is the exact menu-to-field jet identity required in Section 14.3. It supplies the fee-weighted hinge for each menu separately; the subsequent finite marker-algebra calculation and absolutely convergent barcode sum produce the per-box singleton coefficient.

16 Scalar prices for a nonlinear step

This section proves the scalar scale estimates for the shipment, probe, purchase, lodging, and credit fields of the finite-menu tariff. The allowance and gain estimates come from a separate arrival receipt on the same selected run; they are not consequences of these scalar inequalities. All limits below are as the scale parameter x tends to infinity, with the block length fixed.

Fix a block side L, an integer occupancy bound N, a positive strength μ, a positive relative weight λ, an exponent $\rho \in (-1, 0)$, and a fixed cost $C > 0$. Write

$$b(x) = x + c_L, \qquad R(x) = \lceil (\log x)^2 \rceil, \qquad M(x) = \exp\big((\log\log b(x))^2\big),$$

where c_L is fixed once L is fixed. Set

$$f(x) = b(x)^{-1}, \qquad y(x) = b(x)^{-2},$$

$$k(x) = \big(N\mathfrak{s}(R(b(x))) + 4\big)y(x), \qquad z(x) = 2\exp\big(-\mu M(x)^2/16\big) + N\mathfrak{s}(R(x))y(x).$$

Here

$$\mathfrak{s}(r) = e\,\big(2(16r + 9)\big)^4, \qquad \mathcal{R}(x) = b(x)^\rho e^{\lambda C}.$$

Lemma 16.1 (Rate calculus and decaying costs)**.** *For a function u, say that it has rate a if, for every $\delta > 0$, eventually $|u(x)| \le b(x)^{a+\delta}$. Products add rates, sums of finitely many terms keep the larger rate, and multiplication by a fixed constant does not change a rate. Moreover, if u has rate $a < 0$, then $u(x) \to 0$.*

The functions $R(x)$ and every fixed polynomial in $R(x)$ have rate zero. The same is true of $\mathfrak{s}(R(x))$ and $\mathfrak{s}(R(b(x)))$. If $\mu > 0$, then the first summand in $z(x)$ has rate less than every real number; in particular it has rate -2. The fixed shift yields

$$R(b(x)) \le 2R(x) \quad \textit{eventually.} \tag{96}$$

Consequently k and z have rate -2, as does

$$\big(2R(x) + 1\big)^4 z(x).$$

Proof. For products, apply the two rate bounds with exponent slack $\delta/2$ and use

$$b(x)^{a+\delta/2} b(x)^{c+\delta/2} = b(x)^{a+c+\delta}.$$

For sums, use the same slack and the fact that $b(x)^{\delta/2}$ eventually exceeds the fixed number of summands. Fixed constants are absorbed by any positive slack. Since $b(x) \to \infty$, a negative rate implies convergence to zero by choosing an intermediate exponent still below zero.

Since $b(x) = x + c_L$, the ratio $(\log b(x))^2/(\log x)^2$ tends to one. Thus, eventually, $(\log b(x))^2 \le (3/2)(\log x)^2$ and $(\log x)^2 \ge 2$. Therefore

$$R(b(x)) \le (\log b(x))^2 + 1 \le (3/2)(\log x)^2 + 1 \le 2(\log x)^2 \le 2R(x),$$

as claimed. Also, the definition $R(x) = \lceil (\log x)^2 \rceil$ and the fixed-shift relation show that $R(x)$ is bounded by a constant times $(\log b(x))^2$ for large x. Applying this same comparison with argument $b(x)$ controls $R(b(x))$. Every fixed power of $\log b(x)$ is smaller than $b(x)^\delta$ for every $\delta > 0$; hence R and its fixed polynomials have rate zero. The site factor is a fixed quartic polynomial in its argument, so it also has rate zero, including after the comparison (96).

Finally, $M(x)^2 = \exp(2(\log\log b(x))^2)$. For every $A > 0$,

$$\frac{\mu}{16} M(x)^2 \ge A \log b(x)$$

for all sufficiently large x, since with $t = \log b(x)$ the logarithm of the ratio $M(x)^2/t$ is $2(\log t)^2 - \log t \to \infty$. Taking A as large as desired proves the claimed decay of the Gaussian summand. The remaining rate statements follow by the product and sum rules just proved. □

Proposition 16.2 (Eventually purchased tariff)**.** *Assume $\rho \in (-1,0)$, $\lambda > 0$, and $\mu > 0$. Then, for all sufficiently large x, the following inequalities hold simultaneously:*

$$0 \le f(x), \qquad 0 < y(x) \le 1, \tag{97}$$

$$N\mathfrak{s}(R(x))e^{(\lambda+1)C} f(x) \le \mathcal{R}(x), \tag{98}$$

$$k(x) + e^{k(x)}k(x) \le f(x), \tag{99}$$

$$(2R(x)+1)^4 z(x) \le \tfrac{1}{2}, \qquad e^{z(x)} - 1 \le 1. \tag{100}$$

The lodging and credit cost is $z(x)$, whose Gaussian term is

$$2\exp(-\mu M(x)^2/16).$$

Here the fine monitor equals $M(x)$. Thus these are exactly the nonnegativity, probe, shipment, purchase, lodging, and credit conditions required by the local tariff.

Proof. For x large, $b(x) > 1$, so $f = b^{-1} > 0$ and $0 < y = b^{-2} \le 1$. The polynomial factor has rate zero, while f has rate -1. Thus

$$N\mathfrak{s}(R(x))e^{(\lambda+1)C} f(x)$$

has rate -1. Choose q_0 strictly between -1 and ρ. Eventually the left side is at most $b(x)^{q_0}$, which is at most $b(x)^{\rho}$ for large x. Multiplying the fixed factor $e^{\lambda C}$ on the right gives

$$N\mathfrak{s}(R(x))e^{(\lambda+1)C} f(x) \le b(x)^{\rho} e^{\lambda C},$$

as claimed.

By the rate-calculus lemma, k has rate -2. In particular, eventually $0 \le k \le 1$ and $k + ek \le b^{-1}$: the latter follows by applying the rate--2 estimate at exponent -1. Since $e^k \le e$ for $k \le 1$, this proves (99). The product $(2R+1)^4 z$ has rate -2, so it is eventually at most $1/2$. Also $z \to 0$ and $z \ge 0$; hence eventually $z \le \log 2$, which gives $e^z - 1 \le 1$. This proves all the claims. □

Remark 16.3 (Placement in the selected construction). On the selected run, the Gaussian pressure estimate supplies this tariff. The arrival receipt supplies local activity, allowance, and gain bounds; the reset receipt supplies separate rate payments for soft response. The delay is enlarged to satisfy all finitely many transitions at each stage. The finite-menu estimate uses arrival and reset as distinct inputs: neither is a consequence of the tariff. Its shipment budget is the tariff inequality proved above, combined with the predecessor's paid activity bounds.

17 The paid menu response on the selected history

The centered-field argument requires the menu response and the ordinary transition to belong to one history. We now identify their parameters and prices. The analytic inputs are the funded path and Gaussian data used in the preceding one-step estimates; this step makes no new choice of a background transition or of a door transporting insertions.

Fix a finite history, adjacent scales $m+1, m$, and one point of its interpolating path. Denote the fine representation by R_f, the result by R_c, and the corresponding local coordinate change and insertion transport by $\mathcal{D}$. The construction of the unmarked result fixes all three. Use the same tuned representation, forced array and derivative of the one-leaf response when constructing the menu terms of Sections 14 and 15. Their quiet identity is therefore the profile of R_c, and their singleton identity uses the hinge of precisely $\mathcal{D}$.

Write $b_f, b_c \ge 100$ for the support exponents, N for the finite occupancy bound, and $u \ge 0$ for the menu shipment allowance. A menu offer is measured at exponent $b_f - 4$ and allowance u. Its marker weights are positive, its modes nonempty, and the parent images of its named sites lie in its crests. The shipment inequality of Section 16, together with the birth-region conversion in Theorem 13.1, gives the rooted majorants required for the paid component construction. This is uniform over every finite menu satisfying the offer bound.

Let a be the earned exponent and $e > 0$ the additional reserve, so $a \ge b_c + 3$. Let $P_0, P_1 \ge 0$ be the ordinary and flagged component budgets, $0 \le \rho \le 1/2$ the cluster ratio, and $E_0 \ge 0$ the marked exponential cost. The field-price calculation gives

$$4e^{E_0} P_1 + 4\frac{P_0}{1-\rho} \le V_Q \le \tfrac{1}{2}. \tag{101}$$

Here the first inequality follows from the same component majorants used in the cluster sums. To make its numerical content explicit, the total pledge is $V_Q = 10(P_0 + E')$, with $P_1 = E' e^{-E_0}$, $E' \ge 0$. Since $(1-\rho)^{-1} \le 2$, its left side is at most $4E' + 8P_0 \le 10(E' + P_0)$. The upper bound $V_Q \le 1/2$ is purchased by increasing the entry coupling: both P_0 and E' have the fixed negative upper exponent of the component estimates. This is the same arrival threshold used to pay their degree prices, so all inequalities hold together.

The logarithmic response has the rooted bound $2P_0/(1-\rho)$ at exponent $a + e$, by (84). In particular it is at most 1, indeed at most $1/4$, by (101). Replacing its exponent by $b_c + 1/2$ only decreases each nonnegative summand, since $b_c + 1/2 \le a + e$ and all cover numbers are nonnegative. The flagged response has the separate rooted bound

$$t = 2\exp\{-\mu \Delta_{G,L}(\beta)^2/16\}$$

at exponent $b_c + 1$. This is the second, larger-reserve response from Section 14, paid by the two successive absorptions of Section 200. It is not deduced from the logarithmic bound.

The paid marker substitution and grouping now yield ordinary and flagged families with exactly the ledgers (77) and (78). Their supports, threshold shells and quiet and singleton identities are those of Theorem 13.1. The full push identity retains the exponential of the ordinary polynomial and the complete flagged gas. In particular no linearity of that full expression in the marker families is used. The derivative construction fixes the tuning law, and the retained regular and forced slots provide the input labels used by the offer. Thus the response is available for the same door $\mathcal{D}$ and the same pair R_f, R_c.

17.1 Identifying the centered transfer allowances

Let $\widehat{\beta}_m$ be the selected running coupling and L the block side at this step. Their exact relation is

$$\widehat{\beta}_{m+1} = \beta_F(\widehat{\beta}_m, L) = \widehat{\beta}_m + s_G \log L.$$

The fine and coarse radii are correspondingly $B_f = B(\widehat{\beta}_{m+1})$ and $B_c = B(\widehat{\beta}_m)$. Take the offer allowance and the single-insertion price to be

$$u = \widehat{\beta}_{m+1}^{-1}, \qquad \delta = \widehat{\beta}_{m+1}^{-2}.$$

These are the tariff's station and probe prices; they are evaluated at the same fine coupling. The tariff proved in Section 16 says precisely that $u \ge 0$, $0 < \delta \le 1$, and

$$T + e^T T \le u, \qquad T = N\mathfrak{s}(B_f)\delta + 4\delta,$$
$$(2B_c + 1)^4 (t + N\mathfrak{s}(B_c)\delta) \le \tfrac{1}{2}, \qquad e^{t + N\mathfrak{s}(B_c)\delta} - 1 \le 1.$$

The exponent margins are $b_f - 4$ for the input offer, $b_c + 1/2$ for the ordinary output and $b_c + 1$ for its flagged output. These are exactly the margins in the local served-transfer proof of Section 57. The finite wing and its singleton coefficients are transported by $\mathcal{D}$, whose soundness is the exact one-leaf transition already retained above.

Finally, the point of the interpolating path represents the fine history entry at $(N, m+1)$, and its result represents the coarse entry at (N, m). Substituting these two equalities in the formulas just proved changes no map or coefficient. It identifies the door with the history's chosen adjacent-scale door. Hence the paid menu response and the three displayed payment inequalities are the ones required in Section 19. The remaining construction of the funded nonlinear path is independent of this handoff; conditional on that funded path and its retained slot and representation data, there is no additional finite-menu or tariff hypothesis here.

18 Centered cell budgets imply finite-cutoff control

This section proves the quantitative step used before taking any continuum limit. The estimates are conditional on a centered cell bound along one fixed finite-volume trajectory. They do not produce that bound from the renormalisation-group flow; the latter is a separate and substantially stronger input recorded in the blueprint. The point of the argument here is to show in detail how a bound on one observable centered in one cell controls arbitrary separated tuples, and how the same bound controls changes of lattice positions.

Write a_N for the lattice spacing at mesh index N, and $\ell_{N,p}$ for the number of sites in one coordinate direction of the period-p torus. Thus its physical linear period is $P_p = a_N \ell_{N,p}$, independent of N when the schedule is embedded. A configuration of q marked points is a point of $E_q = (\mathbb{R}^4)^q$. Let Conf_q be the set of configurations with no coincident points, and let Gen_q consist of configurations for which each coordinate projection is injective on the marked points. For period p, let Cyc_p be the set where every pairwise coordinate difference avoids $P_p\mathbb{Z}$, and let NoHit_p be the set where each pair is nonperiodic in at least one coordinate. Thus $\mathrm{Gen}_q \subset \mathrm{Conf}_q$, while NoHit_p permits coordinatewise ties. We use the sup norm on each spatial coordinate when discussing separations. A tuple is *cyclically separated at scale r* if it lies in Cyc_p and, for each pair and coordinate, its coordinate difference has distance at least r from every multiple of P_p. It is *separated in some coordinate at scale r* if for each pair at least one coordinate has this property. No condition is imposed on tuples with coincident marked points when we assert a compact-set bound only away from collisions.

For a local observable O (that is, one of the tokens in the fixed observable register of the finite-volume family) and a lattice site x, let $O_N(x)$ be its normalized finite-volume readout. For a finite list $\mathbf{O} = (O_0, \dots, O_{q-1})$ and sites $\mathbf{x} = (x_0, \dots, x_{q-1})$, write

$$F_{N,p,\mathbf{O}}(\mathbf{x}) = \mathbb{E}_{N,p}\left[\prod_{i=0}^{q-1} O_{iN}(x_i)\right].$$

The same notation is used for the lifted lattice function on E_q. The list is allowed to contain any fixed local observables, including reflected or conjugated observables. A centered cell of physical width comparable to a_m is the centered link tiling at depth $m \le N$, with its first chamber centered at the chosen site.

Definition 18.1 (Centered budget)**.** *For an observable O, a nonnegative number A, and indices $m \le N$, a centered budget $\mathsf{CB}(O; A, m, N, p)$ means: (i) the depth of O is represented at volume N; (ii) at every torus site x, the readout $O_N(x)$ belongs to the interior of the chamber-positive class for the centered tiling of depth m based at x; and (iii) its centered root norm is at most A. The root*

norm is the Mth root of the absolute normalized expectation of the product of the observable copied, with the prescribed reflected/conjugated orientation, to all M cells of that tiling. The positive cone and the copies are those of the finite-volume reflection-positive transfer structure.

For a centered tiling $\mathcal{T}$ with M cells, write

$$\mathcal{R}_{\mathcal{T}}(F) := \left| \mathbb{E}_{N,p} \left[\prod_{C \in \mathrm{Cell}(\mathcal{T})} F^{(C)} \right] \right|^{1/M},$$

where $F^{(C)}$ is the prescribed translated copy of F, reflected and conjugated on alternating chambers. This notation is used only when the copies are chamber-positive and the normalized expectation is defined.

The quantitative assumption has two distinct orders of quantifiers. For every fixed observable O and period index p, there is a tail of depths m such that for each such m one can choose $A = A(O, p, m) \geq 0$ and a mesh tail on which $\mathsf{CB}(O; A, m, N, p)$ holds. In addition, for each O there is a tail of depths m such that one can choose $A = A(O, m) \geq 0$ for which there is an outer period-index tail and, for each period in it, an inner mesh-index tail on which the budget holds. The constants may depend on the displayed fixed data. This is the precise centered-budget hypothesis used in the next result.

Lemma 18.2 (Finite families share one budget)**.** *Suppose each observable in a finite list has the fixed-period centered-budget property (respectively, the expanding-period property). Then, after increasing the depth tail, there is one nonnegative budget for the whole list and one common eventuality on which all the individual centered budgets hold. The same is true for any finite list of the observables obtained by reflecting or conjugating a member of the original list.*

Proof. For each member O_i, choose its depth threshold m_i, and let $m_* = \max_i m_i$. Fix any $m \geq m_*$. At this particular depth, the hypotheses give a possibly depth-dependent budget $A_i(m)$ and an eventuality for O_i. Take $A(m) = \max_i A_i(m)$ and intersect the finitely many mesh tails (or the finitely many ordered (N, p) tails). The result is a common eventuality and a single budget for the entire list at this same depth m. Since this works for every $m \geq m_*$, it proves the asserted eventual-in-depth statement. No monotonicity of the centered budgets as depth changes is used or claimed. There are finitely many reflected orientations of each fixed observable (four coordinate reflections and the choice of conjugation), so they can be included in the same finite list. This argument does not claim a bound uniform over all observables or over an uncountable test class. □

Theorem 18.3 (Centered budgets give local moduli and compact bounds)**.** *Assume the finite-volume Wilson family has the following properties, as well as the centered-budget property above. Its fixed schedule satisfies $a_N \to 0$, $P_p \to \infty$ along the period schedule, and the nested lattice scales and centered tilings have the stated integer alignment; the finite-volume side is divisible into the $4s$ equal chambers used by the centered transfer word. Each registered observable has a fixed finite stencil, its readout is continuous and covariant under translations and coordinate reflections, and conjugation is implemented by the reflected token. Every coordinate cut has a positive seam and the associated continuous transfer representation: a compact metric slice space X, a finite Borel measure μ, a continuous positive-semidefinite heat kernel T, and continuous kernels B_i for local slab observables. If a cyclic word consists of bricks B_i of common width w and nonnegative integer free gaps H_i, with $qw + \sum_i H_i = L$, the unnormalized finite-volume expectation is the cyclic kernel integral*

$$\mathrm{Cl}_\mu(K_1, \ldots, K_r) := \int_{X^r} \prod_{j=1}^{r} K_j(x_j, x_{j+1})\, d\mu^r(x_1, \ldots, x_r), \qquad x_{r+1} = x_1,$$

for the collapsed cyclic list of q brick kernels and the individual unit heat steps, which has $r = q + \sum_i H_i$ kernel factors. The physical tour length is $L = qw + \sum_i H_i$. Division by the L-step partition trace $Z = \mathrm{Cl}_\mu(T, \dots, T) > 0$ gives the normalized Wilson expectation. The finite-volume representation identifies a change of one marked position by l steps with replacing two free gaps H_i, H_j by $H_i - l, H_j + l$. Finally, for every cut tiling $\mathcal{T}$, finite index set I, injection $c : I \to \mathrm{Cell}(\mathcal{T})$, and chamber-positive functions F_i, the planted inequality

$$\left| \mathbb{E}_{N,p} \left[\prod_{i \in I} F_i^{(c(i))} \right] \right| \le \prod_{i \in I} \mathcal{R}_{\mathcal{T}}(F_i)$$

holds. These are the precise locality, transfer, symmetry, reality, seam, and cell-packing properties used below. Then the following four conclusions hold.

1. *For each fixed period p, each compact set $K \subset \mathrm{Cyc}_p$, each finite list $\mathbf{O}$, and every $\varepsilon > 0$, there exist $\delta > 0$ and a mesh threshold N_0 such that, for $N \ge N_0$, any two lattice tuples $\mathbf{x}, \mathbf{y}$ whose physical embeddings lie in K and satisfy $|\mathbf{x} - \mathbf{y}| < \delta$ obey*

$$|F_{N,p,\mathbf{O}}(\mathbf{x}) - F_{N,p,\mathbf{O}}(\mathbf{y})| < \varepsilon.$$

2. *For each compact $K \subset \mathrm{Gen}_q$, each finite list $\mathbf{O}$, and every $\varepsilon > 0$, there are $\delta > 0$ and an outer period-index threshold such that for every period index in that tail the same mesh modulus holds eventually in the inner mesh index. The mesh threshold may depend on the period, while δ does not.*

3. *For each compact $K \subset \mathrm{Conf}_q$ and each finite list $\mathbf{O}$, there are $D < \infty$ and an outer period-index threshold such that for every period index in that tail the same finite-volume functions have absolute value at most D eventually in the inner mesh index on lattice tuples embedded in K.*

4. *For each fixed period p, each compact $K \subset \mathrm{NoHit}_p$, and each finite list $\mathbf{O}$, there is $D_p < \infty$ such that $|F_{N,p,\mathbf{O}}(\mathbf{x})| \le D_p$ for all lattice tuples embedded in K, eventually in the mesh limit. This hypothesis permits coordinatewise ties and applies even when the points are not separated in every coordinate.*

The first modulus may depend on the fixed period; the second modulus is uniform in the outer period tail. The third bound has one constant in that tail, with a mesh threshold that may depend on the period. The fourth bound is only for a fixed period. No assertion extends a bound through point collisions.

Proof. We first derive the directional estimate. Fix a compact set K in the fixed-period cyclic patch. Since that patch is open, there is a compact neighborhood K' of K still contained in it. Compactness and continuity of the finitely many periodic coordinate-distance functions give a number $\rho > 0$ such that every tuple in K' has, for every pair of marks and every axis, distance at least 4ρ from the period lattice. Choose a centered-budget depth m with $a_m < \rho$, and choose one common budget A for the finite list consisting of the requested observables and their axis-reflected/conjugated versions. For sufficiently fine mesh N, the stencil fit conditions hold and the mesh is small enough for the following integer choices.

Move one marked site by $l \ge 0$ lattice steps in an axis α, where $l a_N \le \rho$. For a negative displacement, reverse the endpoints and apply the same estimate along the reversed segment; the enlarged compact neighborhood keeps both paths admissible. Let $R = \rho / a_N$ be the corresponding

radius in lattice units. The coordinate gaps in the cyclic order exceed the common brick width plus $2R$: the compact separation gives at least $4\rho/a_N$ lattice steps, whereas the brick has width at most $2a_m/a_N < 2R$. Subtracting that width leaves free gaps greater than $2R$. The move changes two free gaps from (H_0, H_{q-1}) to $(H_0 - l, H_{q-1} + l)$; all other gaps remain fixed, and total tour length is unchanged.

The centered root bound supplies the column estimate needed for this tour. Cut a marked observable at the transfer plane through its centered cell. Locality identifies it with a continuous one-dimensional slab observable, represented by a brick of width $w = 2a_m/a_N$ in lattice units. The reflected cut gives a second slab function, obtained from the reflected token; both are in the positive cone and have centered root test at most A. The centered tiling has $4s$ cells along that axis. Put the $2s$ alternating original and reflected slab functions in the distinct even-indexed cells; their spacing is two cells, and the planted inequality applies to this injection. The planted inequality bounds the absolute expectation of their product by A^{2s}. Translation and reflection identify that product with the s-fold reflected column word of the brick. Thus the normalized column moment to power $\theta = 1/(2s)$ is at most A.

We now justify the transfer estimate for the continuous slice kernels; they are not finite matrices. Fix a compact slice space X, finite measure μ, continuous kernels $T, B_0, \dots, B_{q-1}$, and the positive integer word length $L = 2sw$. Choose finite Borel partitions $\{C_u^{(j)}\}_u$ of X whose maximum cell diameter tends to zero, and choose a tag $\xi_u^{(j)} \in C_u^{(j)}$ in each nonempty cell. Form the sampled matrix

$$(K_j)_{uv} = \mu(C_u^{(j)})^{1/2} K(\xi_u^{(j)}, \xi_v^{(j)}) \mu(C_v^{(j)})^{1/2}.$$

If T is a positive-semidefinite kernel, T_j is a positive-semidefinite matrix: it is the Gram matrix of the tagged kernel, conjugated by the diagonal matrix of square-root cell masses. For any fixed cyclic list $K_1, \dots, K_r$, multiplication and taking the matrix trace gives the exact finite sum

$$\operatorname{Tr}((K_1)_j \cdots (K_r)_j) = \sum_{u_1,\dots,u_r} \prod_{t=1}^{r} \mu(C_{u_t}^{(j)}) K_t(\xi_{u_t}^{(j)}, \xi_{u_{t+1}}^{(j)}), \qquad u_{r+1} = u_1.$$

The integrand $\prod_t K_t(x_t, x_{t+1})$ is continuous on compact X^r. Since the cell diameters tend to zero, this tagged sum converges to its integral, the cyclic kernel integral $\mathrm{Cl}_\mu(K_1, \dots, K_r)$. The same quadrature sequence works for the partition word, each shifted tour word, and each fixed replicated column word $((B_i)^* B_i)^s$, since these are finitely many continuous cyclic integrands. In particular, if $Z = \mathrm{Cl}_\mu(T, \dots, T) > 0$ uses L copies, then $Z_j = \operatorname{Tr}(T_j^L) \to Z$ and $Z_j > 0$ eventually. Define the adjoint kernel by $B_i^*(x, y) = \overline{B_i(y, x)}$. The column quantities $f_{i,j}$ converge to

$$f_i = \left|\mathrm{Cl}_\mu(B_i^*, B_i, \dots, B_i^*, B_i)\right|^{1/(2s)},$$

and the centered root bound just proved says $f_i \le AZ^{1/(2s)}$.

For completeness, the finite-matrix estimate applied to these sampled matrices is an immediate Schatten–Hölder bound. For a finite matrix A, write $\|A\|_r = (\operatorname{Tr}|A|^r)^{1/r}$ for $1 \le r < \infty$ and use the operator norm for $r = \infty$. The two-factor Schatten Hölder inequality is $\|AB\|_r \le \|A\|_u \|B\|_v$ when $1/r = 1/u + 1/v \le 1$. It follows from Tropp's generalized unitarily invariant norm Hölder inequality (*Matrix Analysis*, Theorem 6.32): take the unitarily invariant norm to be the Schatten r norm and its conjugate powers to be u/r and v/r. The endpoint cases follow by continuity of finite-dimensional Schatten norms as an exponent tends to infinity; when both tend to infinity this is the operator norm inequality. Iterating this two-factor statement with partial sums of reciprocal exponents, and finishing with $|\operatorname{Tr} A| \le \|A\|_1$, gives $|\operatorname{Tr}(A_1 \cdots A_k)| \le \prod_j \|A_j\|_{r_j}$ whenever $\sum_j r_j^{-1} = 1$. Put

$Z_j = \mathrm{Tr}(T_j^L)$ and $\theta = 1/(2s)$. For each insertion $B_{i,j}$ use exponent $2s$, so its Schatten norm is

$$f_{i,j} = \left(\mathrm{Tr}\left((B_{i,j}^* B_{i,j})^s\right)\right)^{1/(2s)}.$$

For a heat factor $T_j^{h_i}$ use exponent $L/\operatorname{Re} h_i$; if $\operatorname{Re} h_i = 0$, use ∞. The spectral theorem gives $\|T_j^{h_i}\|_{L/\operatorname{Re} h_i} = Z_j^{\operatorname{Re} h_i/L}$, and $\|T_j^{h_i}\|_\infty \le 1$ on the imaginary boundary after the zero-eigenvalue convention. The reciprocal exponents sum to one because $q/(2s) + \sum_i \operatorname{Re} h_i/L = 1$. Hence

$$\left|\mathrm{Tr} \prod_i B_{i,j} T_j^{h_i}\right| \le \left(\prod_i f_{i,j}\right) Z_j^{1-q\theta}.$$

Now take $h_0 = H_0 - z$, $h_{q-1} = H_{q-1} + z$, and $h_i = H_i$ otherwise. Their sum is constant; each real part is nonnegative for $|z| \le 2R$ because every free gap exceeds $2R$. The matrix powers are entire in z by spectral calculus, with the zero eigenspace assigned the constant-zero value. Thus this trace is an entire function $\Phi_j(z)$ satisfying

$$|\Phi_j(z)| \le \left(\prod_i f_{i,j}\right) Z_j^{1-q\theta}.$$

For any real $t \in [0, l]$, the radius-R circle around t is inside $|z| \le 2R$, so Cauchy's estimate gives $|\Phi_j'(t)| \le (\prod_i f_{i,j}) Z_j^{1-q\theta}/R$. Integrate from 0 to l. Taking $j \to \infty$ in the resulting endpoint inequality is justified by the cyclic quadrature convergence above. It gives the continuous-kernel estimate

$$|F_{N,p,\mathbf{O}}(\mathbf{x} + l a_N e_{k,\alpha}) - F_{N,p,\mathbf{O}}(\mathbf{x})| \le \frac{l}{R} \frac{(\prod_i f_i) Z^{1-q\theta}}{Z} \le \frac{A^q}{\rho}\, l a_N.$$

This is the required one-coordinate bound. It derives the continuous transfer estimate from finite sampled matrices and explicitly passes the traces, denominator, and all column norms to the limit; no finite-dimensional representation of the Wilson measure is assumed.

To compare arbitrary lattice tuples whose embeddings lie in K, change their $4q$ scalar coordinates one at a time. Intermediate tuples can lie outside K; by first choosing K' as above and then taking δ smaller than its neighborhood margin, every intermediate tuple remains in K'. The same A, ρ work at all stages. The sum of the $4q$ coordinate displacements is at most $4q$ times the sup distance, so

$$|F_{N,p,\mathbf{O}}(\mathbf{x}) - F_{N,p,\mathbf{O}}(\mathbf{y})| \le 4q\, \frac{A^q}{\rho}\, |\mathbf{x} - \mathbf{y}|.$$

Choosing $\delta < \varepsilon\rho/(4qA^q)$, with any sufficiently small neighborhood margin if $A = 0$, proves the fixed-period modulus. For $q = 0, 1$ there is no separated pair and the zero-/one-point correlation is constant under position changes by the empty-word identity or lattice translation covariance.

For the expanding-period modulus, let K be compact in the generic set, where each coordinate projection separates distinct points. Choose a compact neighborhood still inside that open set. Since its coordinate differences are bounded and the physical period tends to infinity, for every sufficiently large period all pairs are uniformly separated from every periodic image; the same positive ρ works for all those periods. Choose the centered-budget depth m with $a_m < \rho$. The expanding clause gives a common budget A eventually in period, then eventually in mesh for each such period. The preceding coordinate telescope uses the same ρ, A, δ for all these periods, with the inner mesh threshold allowed to depend on the period. This is exactly the expanding modulus. Genericity, rather than absence of full point collisions alone, is needed here because all four coordinate gaps are used by the directional telescope.

We next prove both compact-bound assertions. The cases $q = 0, 1$ are immediate: the empty correlation is 1, while for one observable the planted inequality with one occupied chamber bounds its expectation by its centered root norm, at most A; translation covariance makes this uniform in its position. Hence assume $q \geq 2$. Let K be compact with no full collisions. Define for each pair $i \neq j$

$$d_{ij}(x) = \max_{\alpha=1,\dots,4} |x_i^\alpha - x_j^\alpha|.$$

Its minimum over the finite set of pairs and compact K is a positive number. For sufficiently large periods, every coordinate difference has absolute value less than half the physical period; hence its distance to the period lattice is its absolute value. It follows that there is $\rho_0 > 0$ such that every pair in every tuple of K is separated in at least one torus coordinate by ρ_0, uniformly on a period tail. The same conclusion for a fixed period holds under the weaker NoHit hypothesis: use instead the continuous positive function

$$d_{ij,p}(x) = \max_{\alpha=1,\dots,4} \operatorname{dist}(x_i^\alpha - x_j^\alpha, P_p\mathbb{Z}).$$

NoHit says precisely that $d_{ij,p}(x) > 0$ for every pair; compactness and finiteness of the pair set give $\min_{x\in K, i\neq j} d_{ij,p}(x) > 0$. This maximum over coordinates is essential: taking the minimum over coordinates would incorrectly rule out coordinatewise ties.

Here is the packing argument, with the two scales distinguished. Choose a coarse depth m with $a_m < \rho_0$. Then choose a finer depth $k \geq m$ in the relevant centered-budget tail so that the budget at depth k is available and

$$6qa_k < a_m.$$

For expanding bounds use the expanding centered-budget clause; for a fixed-period bound use the fixed-p clause. After fixing k, take the mesh index large enough that every fixed stencil fits inside a depth-k chamber. The finite torus placement lemma supplies a translate of the coarse tiling X of depth m such that all q marked sites lie in its margin of three fine-cell spans. The proof is a boundary count in each coordinate: one point forbids at most $6a_k$ of the coarse translation interval, so all q forbid at most $6qa_k < a_m$; hence an allowed translate exists. Let D_i be the coarse cell of X containing x_i. These cells are distinct, since two sites in the same cell differ by less than $a_m < \rho_0$ in every coordinate, contradicting the some-coordinate separation.

The centered budgets are at the fine depth k, whereas the tuple will be placed in a coarse tiling X of depth m. First choose the home site o of X on the fine center grid and in chamber zero of X. Match the entire coarse translated-copy orbit of the observable at o, one copy for each cell of X, injectively to cells of the fine centered tiling at o. The aligned scale divisibility supplies the matching; the required reflected and conjugated orientations are in the finite family for which the common budget was chosen. The fine planted inequality bounds the absolute expectation of the whole coarse copy product by $A^{|\mathrm{Cell}(X)|}$. By the definition of the root norm, the root norm of the observable at o relative to the coarse tiling X is at most A. This is the full replicated-orbit argument; bounding just the original tuple's q entries would not prove a coarse root estimate.

For each marked point x_i, let D_i be its assigned coarse cell. The tiling's pointed-site map chooses y_i in the home chamber (the chamber based at cell zero), and reflection of the margin shows that y_i has a margin of three fine-cell spans from the home-chamber boundary. The coarse cell orientation determines a reflected or conjugated token at y_i; signed-coordinate covariance identifies its copy in D_i with the original observable at x_i. The finite list of all such oriented tokens was included in the common budget. We next move the coarse home point from o to any such y_i. In each of four coordinates, bracket y_i by two fine-center-grid points separated by at most one fine span d. The sixteen corner points lie in the margin of two spans. The fixed-stencil fit gives chamber interiority

at every intermediate point on these four coordinate segments. For a fixed axis, let u_j be the root test at successive integer translates of the full coarse copied product, and let M_X be the number of cells. Translate the entire copied product by one lattice step. In each chamber the translated copy becomes one of the two neighboring translates, with the sign determined by that chamber's orientation. Reflection positivity applied to this translated product bounds its moment by the planted product of the root tests of those neighboring translates. Pairing cells by the involution that reverses the orientation along this axis gives exactly $M_X/2$ copies of each neighbor. Thus

$$u_j^{M_X} \le u_{j-1}^{M_X/2} u_{j+1}^{M_X/2},$$

and raising both sides to the power $2/M_X$ gives $u_j^2 \le u_{j-1}u_{j+1}$. If an interior value exceeds both endpoint bounds, choose a maximum u_j on the finite interval; it is positive. Both neighbors are at most u_j, so the inequality forces both to equal u_j. Repeating this at the adjacent interior maxima propagates that value to an endpoint, contradicting the endpoint bound. This proves the finite-interval maximum principle, including possible zeros elsewhere. Repeatedly applying this maximum principle along the four segments bounds the root at y_i by the maximum of the sixteen corner roots, all at most A. This proves the coarse root estimate at each needed marked home point, with chamber interiority there; covariance transfers that bound to the coarse insertion associated with x_i.

Now apply the planted inequality in the coarse tiling X to the injection $i \mapsto D_i$. It bounds the tuple expectation by the product of the q coarse root tests, hence by A^q. All choices were uniform over tuples in K, so this is the expanding compact bound; for fixed p it gives the fixed-period compact NoHit bound. The former uses compactness away from full collisions and the period tail; the latter uses only NoHit at that fixed period and permits coordinate ties. Neither gives control at a torus-coincidence configuration, nor smoothness through a collision diagonal. □

Remark 18.4 (What remains upstream). The theorem assumes the centered budget. The exact upstream estimate needed is: for each grade d, norm bound $R > 0$, and exponent slack $\eta > 0$, there is $A \ge 0$ such that, for every depth m beyond the reference depth for grade d, eventually in volume index N, uniformly for every period index p and every label v of grade d with $\|v\| \le R$, the centered root norm is at most $Aa_m^{-d-\eta}$, with representation and chamber-interiority. The same volume tail works for all p; restricting either to a fixed period or to the cofinal period schedule gives the two budget regimes used above. No proof of this Wilson estimate is contained here. In particular, the finite-volume cell-packing and transfer arguments above cannot supply its decay in a_m, nor can they supply the required uniformity in labels of bounded norm. This is the next analytic input to establish from the renormalisation-group trajectory.

19 The centered estimate on the selected Wilson history

Fix the family selected for a compact simple gauge group and positive penalty by the construction theorem. The compactness argument in Section 18.3 needs a uniform root bound for its centered local observables. We derive it from the paid single-source history selected with that family. The essential finite-volume input is a marked-copy estimate: placing one copy in each cell turns the RG bound for a finite census into a bound on the normalized root moment. The scale schedule and the factorial-column estimate then make its cost at most the desired power of the mesh.

Theorem 19.1 (Uniform centered power bound from the single-source receipts)**.** *Assume the selected same-history construction supplies the single-source quantitative fields listed below, with one*

common budget for all labels in each bounded grade ball, together with its width and interiority receipts. For each grade d, each $R > 0$, and each $\varepsilon > 0$, there is a finite $A \geq 0$ such that for every $m \geq m_{\mathrm{ref}}(d)$, for all sufficiently large volumes N, uniformly in every period p, every label v of grade d with $\|v\| \leq R$, and every site x, the centered cell at depth m and site x contains the observable in its positive chamber and

$$\mathcal{R}_{m,N,p}(O_{N,p}(v,x)) \leq A a_m^{-d-\varepsilon}.$$

The volume threshold may depend on m, but not on p, v, or x.

The argument below proves this conditional estimate through the actual single-source supplier used by the public construction. The upstream production of the uniform fields is tracked separately and is not folded into this theorem's hypotheses.

Proof. **19.1 The finite-volume marked-copy estimate**

Fix a grade d, a label-norm radius $R > 0$, a target depth m, and an opened depth $k \geq m$. The finite-dimensional grade-d label space has compact closed R-ball, and its coefficient-size function is continuous. Thus

$$g_{d,R} = \max\left(1,\ \sup_{\|v\| \leq R} \mathrm{coeffSize}_d(v)\right) < \infty.$$

Use this coefficient-size radius in the local payment and width estimates. Write a_i for the lattice spacing, $\sigma_i = \log(a_i^{-1})$, and

$$P(d,R,k) = a_k^d\, W(d, g_{d,R}, k), \qquad E(d,R;m,k) = \left(\frac{a_m}{a_k}\right)^4 t_k + 6 - \log P(d,R,k),$$

where $W(d,R,k)$ is the width allocated to the grade-d coefficient ball at depth k, and t_k is the vacuum treasury per site at that depth. All these quantities are positive where they occur in logarithms. Let M_m be the number of cells in the centered tiling of depth m. For an observable copy U_j, write $c_k(U_j)$ for the local production cost assigned by the selected history; the paid-ticket inequality is $P(d,R,k)c_k(U_j) \leq \mathrm{fly}_k$.

We now give the finite-census estimate used in the proof. Let V be the cells of the centered depth-m tiling, so $|V| = M_m$. For each $j \in V$, let U_j be the transported observable placed at that cell. Write $\mathcal{C}_V$ for the finite squarefree marker algebra: its basis elements are subsets $I \subseteq V$, multiplication is disjoint union when the subsets are disjoint and zero otherwise, and

$$\mathrm{price}_r(F) = \sum_{I \subseteq V} |F_I| \prod_{j \in I} r_j, \qquad \mathrm{toll}_r(I) = \prod_{j \in I} r_j$$

for positive weights r_j. The mode-zero coefficient is the quiet coefficient. The polynomial pass is the finite exponential series through degree $|V|$; it satisfies $\mathrm{price}_r(\mathrm{pass}\, F) \leq \exp(\mathrm{price}_r(F))$, by submultiplicativity and the ordinary exponential-series bound.

The single-source receipt at each intermediate scale represents the propagated density in the form

$$X(U) = e^{E(U)}\, \mathrm{pass}\left(\mathrm{Mach}(F(U)) + \sum_S H_S(U)\right) \mathrm{field}\big(\overline{K}(U) + \mathrm{Mach}(G(U)) + K(U)\big)\,.$$

Here $\mathrm{Mach}(z) = \sum_{j \in V} \mathrm{send}_j(z_j)$ places a scalar source in the marker for copy j; each local unit H_S and forced-box source K_S has marker degree at least two whenever it is nonzero. The field is

the sum over admissible families of pairwise disjoint boxes whose FLAG sets have union equal to the global flag set, with each term the product of its box factors. The vacuum terms are $\overline{K}$ and the vacuum part of the box factors. This is the representation supplied by the source's served-slab receipt.

For positive r_j, the support budgets give four contributions to the price. The scalar observables contribute at most $|V|x$ by the first part of the vault bound. The box-marked observables contribute at most $|V|y$ by its second part. The local unit family and the forced-box family each contribute at most $|V|f$ by their towed support budgets. Here is the origin-counting step in full. For an active unit $H_S(U) \neq 0$, choose a nonzero marker coefficient. The degree condition gives a marker j in that coefficient; the support-placement condition for that unit says its site o_j lies in S. Thus every active support contains a named origin. For a forced-box term the same argument gives o_j in its box. If $a_S \geq 0$ is the envelope supplied by the towed bound, its weighted coefficient price $q_S(U)$ satisfies $q_S(U) \leq a_S$ and $\sum_{S:o_j\in S} a_S \exp(e\,\mathrm{cover}(S)) \leq f$ at every origin. Only active supports need be charged; a zero polynomial has price zero even if its envelope is positive. Because $e \geq 0$, the exponential is at least one. Summing the envelope bounds over origins therefore gives $\sum_{S\,\mathrm{active}} q_S(U) \leq |V|f$. Finally, the unmarked vacuum activities contribute at most su_0, where s is the number of lattice sites and u_0 is the per-site vacuum allowance. Thus the total price inside the finite pass is at most $|V|(x+f)$, and the total price of the box factors is at most $su_0 + |V|(y+f)$.

For the field, sum first over admissible box families. Dropping the disjointness restriction only adds nonnegative terms, so its price is at most the product over boxes of one plus the price of that box factor. The inequality $1 + t \leq e^t$ bounds this by the exponential of the sum of box prices. The same $1 + t \leq e^t$ argument for the finite polynomial pass, together with submultiplicativity of price, now yields the pointwise estimate

$$\mathrm{price}_r(X(U)) \leq \exp(E(U) + |V|(x + y + 2f) + su_0)\,. \tag{102}$$

The lawfulness estimate bounds the scalar action by scalar + s exCost, and the pointwise vacuum price contributes $s\kappa$. Because the prior is a probability measure and $|X_I(U)|\,\mathrm{toll}_r(I) \leq \mathrm{price}_r(X(U))$, integrating (102) gives

$$\left|\int X_I\, d\mu\right| \mathrm{toll}_r(I) \leq \exp(\mathrm{scalar} + s(\mathrm{exCost} + \kappa) + |V|(x + y + 2f))\,.$$

Assume the action parameters satisfy $0 < \rho \leq \rho_0 \leq 1$, where ρ_0 is the flag-vanishing threshold, and $\beta \geq 0$, the Wilson action is lawful, and its representation is pointwise nonnegative. The product prior is the normalized product Haar probability μ on edge variables. The positive partition function has the explicit lower bound

$$Z \geq \exp(\mathrm{scalar} - s\,[\mathrm{exCost} + \beta\,\mathrm{baseBudget}(\rho) + 4d_G \log(16/\rho)])\,.$$

Here $d_G = \dim_{\mathbb{R}} G.\mathrm{Mat}$ is the dimension of the ambient real matrix space used by the gauge representation. For completeness, the Haar small-ball estimate used here follows from a packing argument. Put $D = d_G$ and use the matrix norm defining the radius. Choose a maximal finite family with pairwise distances strictly greater than $r\gamma$. Its open balls of radius $r\gamma/2$ are disjoint and lie in the ambient ball of radius 2γ. Comparing Euclidean volumes gives $M(r/4)^D \leq 1$. Maximality says the closed balls of radius $r\gamma$ cover the group. By Haar invariance, each such ball has the same Haar measure as the identity ball, so $1 \leq M\,\mu(B(e, r\gamma))$ and hence $\mu(B(e, r\gamma)) \geq (r/4)^D$. With γ the source color scale and $r = \rho/4$, each edge ball has probability at least $(\rho/16)^{d_G}$. Restrict every oriented edge to this ball: a four-edge plaquette path differs from the identity by at most the sum

of its four edge deviations, hence its chord is at most ρ and the flag term vanishes. The loop action is bounded by the stated loop cost, and nonnegativity of β bounds the remaining local action by β times the base budget. Independence under the normalized product prior gives event probability at least $(\rho/16)^{4sd_G}$. On that event the exponential density is at least the scalar exponential times the exponential of minus the displayed action and vacuum costs. Integrating this nonnegative density over the event proves the bound. Dividing the numerator estimate by this denominator gives the integrated mode estimate

$$\frac{\left|\int X_I \, d\mu\right|}{Z} \operatorname{toll}_r(I) \le \exp(s\,[2\ \mathrm{exCost} + \kappa + \beta\ \mathrm{baseBudget} + 4d_G \log(16/\rho)] + |V|(x+y+2f))\,, \tag{103}$$

where $Z > 0$ is the Wilson partition function, ρ is the small-ball radius below the action threshold, and d_G is the real dimension of the ambient gauge-matrix space. This is a finite-volume estimate; no infinite-volume measure is involved.

Now propagate the terminal birth polynomial b_N backwards through the finite-volume transitions: $b_i = T_i b_{i+1}$ for $i = N-1, \dots, k$. Start from the actual terminal birth, which is integrable because it is a finite polynomial in continuous observables on compact configuration space. At each step the transition's finite mass bound sends an integrable function to an integrable function; the kernel mass identity preserves its integral against the corresponding normalized prior. The mode extraction commutes with this transition because it is a continuous linear coefficient map on the finite-dimensional polynomial space: applying it before or after integration gives the same scalar integral. Induction therefore preserves integrability, each coefficient identity, and the same partition normalization at every scale. The transition equation and terminal birth identity consequently give

$$\frac{\int (b_k)_I \, d\mu_k}{Z_{N,p}} = \mathbb{E}_{N,p} \prod_{j \in I} U_j.$$

The intermediate height, locality, and terminal Fourier statements are the exact hypotheses used to identify the propagated coefficient with the served-slab census; the conversion from these fields to the displayed squarefree price is set out below. The weighted local-payment inequality is $r_j c_k(U_j) \le \mathrm{fly}_k$. Separation controls the cover cost of each multi-marker support, while the occupancy bound controls the number of named origins per site. These are the geometric and payment hypotheses needed for the four support-price bounds above.

At the opened depth, the weight march returns its initial value, so $r_j = P(d, R, k)$ for all j. The local scalar bounds are $x = 3\,\mathrm{fly}_k$ and $y = \mathrm{fly}_k$; the action estimate has $f = 1$ and $\mathrm{fly}_k \le 1$. Thus $x + y + 2f = 4\,\mathrm{fly}_k + 2 \le 6$. The finite-volume treasury identity bounds the extensive bracket in (103) by $M_m (a_m/a_k)^4 t_k$. Taking $I = V$ therefore gives

$$\left| \mathbb{E}_{N,p} \prod_{j \in V} U_j \right| P(d, R, k)^{M_m} \le \exp\left(M_m \left[\left(\frac{a_m}{a_k} \right)^4 t_k + 6 \right] \right). \tag{104}$$

This is the marked-copy inequality with all terms traced: the $6M_m$ is the sum of the $3\,\mathrm{fly}_k + \mathrm{fly}_k + 2$ scalar, forced, and action charges; the treasury is the cost of all $M_m (a_m/a_k)^4$ scale-k cells.

By definition, the centered root test is X^{1/M_m}, where $X = |\mathbb{E}_{N,p} \prod_j U_j|$. If $X = 0$ the desired estimate is immediate. Otherwise divide (104) by $P^{M_m} > 0$ to get

$$X \le \left(\frac{\exp((a_m/a_k)^4 t_k + 6)}{P(d, R, k)} \right)^{M_m}.$$

The map $x \mapsto x^{1/M_m}$ is increasing on $[0,\infty)$ and $M_m \geq 1$, so taking its value on the two sides yields

$$\mathcal{R}_{m,N,p}(O_{N,p}(v,x)) \leq \frac{\exp((a_m/a_k)^4 t_k + 6)}{P(d,R,k)} = \exp(E(d,R;m,k)), \tag{105}$$

for $\|v\| \leq R$, once the centered tiling has interiority at the chosen site. Translation covariance transfers the estimate from the origin to any site; the finitely many signed-coordinate copies have the same coefficient and payment bounds. In the source proof, the marked-copy inequality is obtained by applying its finite-census RG estimate to the full census, then applying the root-test inequality. No continuum or limiting operation is used in this step.

19.2 The width bill and the choice of opened depth

Set $n_d = 2\,\mathrm{winter}(d)$ and, at scale i, write c_i for the column count, ν_i for the factorial order, $\widehat{\beta}_i$ for the hatted coupling, and

$$W_i = \frac{\widehat{\beta}_i^{-n_d/2}}{g_{d,R}\, c_i \prod_{j<i} c_{j+1}}, \qquad L_i = \frac{W_i}{\nu_i!}.$$

These are the actual width, envelope, and paid last-column quantities in the factorial-column estimate. The accumulated column-log estimate $F_i = o(\sigma_i)$ is proved by the scale-increment telescoping in Section 53.3. The hatted coupling has the exact schedule form $\widehat{\beta}_i = \bar{\beta} + c_G \sigma_i$ with $c_G > 0$. Thus for all sufficiently large i there is a fixed $C > 0$ such that $1 \leq \widehat{\beta}_i \leq C\sigma_i$, and

$$0 \leq \frac{\log \widehat{\beta}_i}{\sigma_i} \leq \frac{\log(C\sigma_i)}{\sigma_i} \longrightarrow 0.$$

This proves the logarithmic coupling subpower directly from the affine reference schedule; the actual finite-volume coefficient's running limit is not used in this width step. The paid factorial condition $\nu_i! \leq c_i$ and the width identity give, for every $z > 0$, eventually

$$5 - \log L_i \leq \frac{z}{2}\sigma_i.$$

The exact logarithmic expression is

$$5 - \log L_i = 5 + \log g_{d,R} + \frac{n_d}{2} \log \widehat{\beta}_i + \log c_i + \sum_{j<i} \log c_{j+1} + \log(\nu_i!).$$

For $i \geq 1$, the sum $F_i = \sum_{j<i} \log c_{j+1}$ contains $\log c_i$; since $\nu_i! \leq c_i$, it also dominates $\log(\nu_i!)$. Thus the last three terms are bounded by $3F_i$. The fixed prefactor and coupling-log terms are $o(\sigma_i)$ by the estimates just given. This is exactly the I04 inequality with slack $z/2$, and makes the finite-prefix handling explicit. Since $\nu_i! \geq 1$, one has $L_i \leq W_i$, and therefore $5 - \log W_i \leq z\sigma_i/2$. For all sufficiently large i, also $1 \leq z\sigma_i/2$. Hence

$$6 - \log W_i \leq z\sigma_i.$$

The physical palette width is $P(d,R,i) = a_i^d W_i$, so $-d \log a_i = d\sigma_i$ yields the exact bill

$$6 - \log P(d,R,i) \leq (d+z)\sigma_i. \tag{106}$$

This also explains the constants: the last-column estimate is on $5 - \log L_i$; the extra 1 is absorbed into $z\sigma_i/2$, and the physical grade factor contributes exactly $d\sigma_i$. No additional arbitrary width slack is hidden in (106).

Let $k(m)$ be the depth opened by the selected schedule for target depth m. The schedule gives $k(m) \geq m$ and $k(m) \to \infty$. Given $\eta > 0$, choose

$$0 < z \leq \min\left\{\frac{\eta}{4}, \frac{\eta}{4(d+\eta)}\right\}.$$

The schedule's pay bound and the vacuum-treasury estimate give, eventually in m,

$$w^4 q_{k(m)} \leq z\sigma_{k(m)}, \qquad t_{k(m)} \leq q_{k(m)}, \qquad w = \frac{a_m}{a_{k(m)}} \geq 1,$$

where $q_k \geq 1$ is the paid fiat at depth k. Also $\sigma_{k(m)} - \sigma_m = \log w \leq w^4 q_{k(m)}$. Combining these facts with (106) yields

$$E(d, R; m, k(m)) \leq (d + 2z)\sigma_{k(m)} \leq (d+\eta)(1-z)\sigma_{k(m)} \leq (d+\eta)\sigma_m.$$

The middle inequality follows from the two restrictions on z. Hence, for any $0 < \eta < \varepsilon$, the opened-depth estimate is at most $a_m^{-d-\eta}$ for every sufficiently large m.

For the finitely many smaller depths, choose an opened $k(m) \geq m$ and use the finite estimate (105). The maximum of 1 and the finitely many numbers

$$\exp(E(d, R; m, k(m)))\, a_m^{d+\varepsilon}$$

is a single finite constant A. It is independent of the period, volume, site, and label within the specified coefficient ball. For late depths, $a_m^{-d-\eta} \leq a_m^{-d-\varepsilon}$ because $a_m < 1$ and $\eta < \varepsilon$. This proves the power bound at all depths at or beyond the grade reference. The centered-cell interiority and the common eventual volume threshold are supplied by the selected schedule's interior-clearance receipt.

19.3 The centered supplier on the public construction route

Write $S_B = e\{2(16B+9)\}^4$ for the packing factor of Section 57.

The public construction uses a single-source per-label supplier, followed by a downward induction across the scale transitions. We spell out the quantitative propagation and cost checks; the local transfer argument itself is supplied in Section 57. The public route does not call the separate multisource-lane construction. The finite-menu nonlinear response remains an explicit selected-history input to this conditional proof. At the adjacent step it must provide ordinary and flagged polynomial ledgers for finite nonquiet offers supported in their crests, under the history's regularity, arrival, response-rate, slot, and shipment-tariff conditions. The ordinary ledger is paid at exponent equal to the lower profile exponent plus $1/2$ and site budget 1; the flagged ledger is paid at exponent plus 1 and site budget $2\exp(-\mu\Delta_{G,L}(\beta)^2/16)$. Its push identity fixes the quiet coefficient and the specified singleton coefficients for each box only; no arbitrary nonquiet coefficients are assumed. The offer threshold in this application is the upper support exponent minus 4, with allowance fixed by that step's tariff. The conditional response record also retains propagation, tuning, slot separation, and the finite-wing price data used below. A separate producer argument is required to derive this record uniformly from the public run; the source path and its current review status are recorded in the evidence file. The adjacent-scale finite-wing input is produced from the selected history: its nested realization, tariff, climbing-door data, and support-exponent bounds are passed through the one-step response construction. At step i the two support exponents are at least 100, the price shrinkage $d_i = \delta_i$ lies in $(0, 1]$, and the one-step total price t_i and local-response allowance y_i obey

the exact inequalities displayed below. These are the finite-wing response and one-step payment data required by the local-transfer step. The tariff and compatibility data are carried on the same chosen history; their producer is identified in the evidence file. At a step i, the resulting one-step conditions are $\nu_i, \nu_{i+1} \geq 100$, $0 < d_i \leq 1$, and, with $T_i = r_i\, S_{B_{i+1}} d_i + 4d_i$, $T_i + e^{T_i} T_i \leq t_i$ and $(2B_i + 1)^4 (y_i + r_i\, S_{B_i} d_i) \leq \frac{1}{2}$, $e^{y_i + r_i\, S_{B_i} d_i} - 1 \leq 1$. These are the one-step payment inequalities, with t_i the total response price and y_i the local-response allowance. The climbing witness and tariff are used by the one-step response lemma to produce the finite wing; their nonlinear estimate is a separately tracked input, not inferred from scale geometry.

For occupancy and separation, write B_i for the integer separation scale, $o_i(j)$ for the designated site, r_i for the occupancy bound, and $W_i(j) = \phi_{i-1} \cdots \phi_k w(j)$ for the marched weight. The scale fields give $0 < \phi_i, \delta_i$, $1 \leq \delta_i \phi_i e^{B_i/128}$, $B_{i+1}^2 \leq \ell_i B_i^2$, and $8 \leq B_i$, where ℓ_i is the integer ratio count. The migration identity is $o_{i+1} = \mathrm{embed}_i \circ o_i$, and r_i is nondecreasing. Occupancy means that each site has at most r_i names mapped to it. The injective embedding maps each fine fiber into the coarse fiber over its parent. Since the occupancy bounds are nondecreasing, both the old fiber and its embedded fine fiber have multiplicity at most r_i at step i: the old occupancy is bounded by r_i, while the fine occupancy is bounded by the old bound and hence also by r_i. Iterating this comparison propagates occupancy from the opened scale through all finer scales. For a mode with at least two names, distinct origins have coarse separation at least B_i^2. The scale comparison gives fine separation at least ℓ_i times that coarse separation. The split inequality $B_{i+1}^2 \leq \ell_i B_i^2$ therefore propagates the required separation forward.

At a given scale, take any mode I of cardinality at least two and any support X containing its origin image. The following tour argument proves the required geometric estimate. Choose a minimizing cover of X by $D = \mathrm{cover}(B_i, X)$ balls of radius B_i. In the source connected-halo cover, the centers form a connected graph: two centers are adjacent when their radius-B_i halos meet or contain sites at max-distance at most one. Thus adjacent centers are at max-distance at most $2B_i + 1$. Take a spanning tree. Join each tree edge by a unit max-distance path, advancing coordinates simultaneously so its length is at most $2B_i + 1$, and traverse each such path out and back. This gives a walk visiting every center with length at most $2(2B_i + 1)(D - 1) \leq D(4B_i + 2)$. Enumerate its unit steps as $R(0), \ldots, R(N)$; successive sites, including intermediate sites that need not be cover centers, are at distance at most one. For each origin $o_i(j)$, choose a covering center and a time t_j at which the walk visits it; that center is within distance B_i of the origin. If $j \neq j'$ are in the mode, their origins are separated by at least B_i^2. The triangle inequality and the unit-speed walk give, after ordering $t_j \leq t_{j'}$, $t_j + (B_i^2 - 2B_i) \leq t_{j'}$. Therefore the buckets of width $B_i^2 - 2B_i$ containing these times are distinct. Since each time is at most N,

$$(|I| - 1)(B_i^2 - 2B_i) \leq N.$$

For $B_i \geq 8$, $B_i^2 \leq 4(|I| - 1)(B_i^2 - 2B_i)/|I|$ when $|I| \geq 2$. Combining with the tour length and $4B_i + 2 \leq 16B_i$ yields $B_i |I| \leq 64D$. The support X can be any support containing the origin image, so in particular

$$B_i |I| \leq 64\ \mathrm{cover}(B_i, X).$$

Set $a_j = W_i(j)$ and $v_j = \delta_i W_{i+1}(j)$. Since $W_{i+1} = \phi_i W_i$, the march and the inequality $1 \leq \delta_i \phi_i e^{B_i/128}$ give $a_j \leq v_j e^{B_i/128}$, with both weights positive. Multiplying over I and applying the preceding geometric estimate yields

$$\prod_{j \in I} a_j \leq \prod_{j \in I} v_j\ \exp\{\mathrm{cover}(B_i, X)/2\}.$$

This is the support-price comparison with cover exponent $1/2$ and multiplier one. Its definition only asks about modes of cardinality at least two, so no singleton or empty-mode case is needed.

Together with propagated occupancy, site migration, and positive marched weights, this gives the site/separation package data used downstream. For each label v, let A_i be the scale-dependent grade dilation, $\tau_i(v)$ its coefficient in the same-history representation, and Γ_i the paid column count. Define the paid bond $\Lambda_i(v) = \|A_i\|\,\mathrm{coeffSize}(\tau_i(v))$. The representation supplies the coefficient cone estimate $q_i(v) \le \Gamma_i \mathrm{coeffSize}(\tau_i(v))$ at each scale, while its dilation step gives $\Lambda_{i+1}(v) \le \ell_i^d \Gamma_{i+1}\Lambda_i(v)$. The paid scale data give, at every i, $0 < \phi_i \le 1$, $\phi_i \ell_i^d \Gamma_{i+1}\zeta_i \le \zeta_{i+1}$, $\zeta_i\Gamma_i \le 1$, $\psi_i \ge 0$, and $2\,\mathrm{orbitCost}(\psi_i + 2\Gamma_{i+1})\zeta_{i+1} \le 1$. The opened-scale payment is $w_k\Lambda_k(v) \le \zeta_k$. The marched weights satisfy $w_{i+1} = \phi_i w_i$. Multiplying $\Lambda_{i+1}(v) \le \ell_i^d\Gamma_{i+1}\Lambda_i(v)$ by w_{i+1} and using the displayed price inequality gives $w_{i+1}\Lambda_{i+1}(v) \le \zeta_{i+1}$ from $w_i\Lambda_i(v) \le \zeta_i$; induction proves the bound at all scales. The one-step estimate charges the observable by $b = w_{i+1}$, and $b \le w_i$, so $b\|A_i\|q_i(v) \le b\Lambda_i(v)\Gamma_i \le w_i\Lambda_i(v)\Gamma_i \le \zeta_i\Gamma_i \le 1$. For the finite-wing increment, put $b = w_{i+1}$ and $r = w_i$. The march has $b \le r$; the adjacent paid bound and the column budget give exactly

$$2b\|A_{i+1}\|\,\mathrm{orbitCost}\,\mathrm{coeffSize}(\tau_{i+1}(v))(\psi_i + 2\Gamma_{i+1}) \le 1.$$

These are the two numeric inequalities passed to the scale-step construction. The same-history compatibility bound is a finite set of tales, one for each coefficient in the frame; each tale relates the fine exponential entry and forced profile to the coarser forced profile, with the gauge-door carrying the same local coordinate change. Together with the gauge-door transport law, it yields a finite family of insertions and the two bounds $2w_{i+1}s \le 1$ and $w_{i+1}t \le 1$. The charge estimate above is the first term in that tale budget; the displayed orbit-cost inequality pays the remaining coefficient increment and column contribution.

Here is the per-scale profile construction. Reorient the coefficient by the selected motion and gauge move, and denote the resulting audited coefficient by $Q_i(v)$. Let $F_i(v)$ be its exponential-entry field and let $g_i(v)$ be the sequence of marked activities obtained by applying the grade dilation A_i to its forced activities. For clarity, the coefficient coin is

$$\chi_i(v;U) = (R_i.\mathrm{eval}(U))F_i(v;U) + e^{E_i(U)}\sum_j N_{i,j}(g_i(v);U),$$

where E_i is the scalar action exponent. The audition identities identify this coin with the oriented density and identify the marked profile with the motion-transported forced profile. The remnant estimate gives a summable dominating function for $Q_i(v)$.forced, hence for $g_i(v)$. Reorientation preserves the normalized prior because it is a composition of a lattice motion and a gauge change; therefore the transformed ordinary field remains integrable. At the terminal level the same identity is the birth coefficient equation $R_N.\mathrm{eval}(U)b_N(v;U) = \chi_N(v;U)$. This proves the profile and coefficient identities used at every scale, rather than only a formal compatibility of names.

At terminal scale, the calibrated response construction uses the actual attainment, the place bound

$$\|S(\mathrm{axis}_j, 0, U)\| \le t,$$

cover number at most one, and bank inequality $e^{\nu_N+2}\mathfrak{c}_N(t)\zeta_N \le u_N$ to produce the response with $w_N a_N \le u_N$ and the prescribed forced hook. Here $\mathfrak{c}_N(t)$ is the full coefficient cost at section budget t. The terminal response's hook agrees with the profile identity above, so the exact terminal birth coefficient is recovered.

Define the pitch π_i by $\pi_i = \nu_i$ for $i < N$ and $\pi_N = \nu_N - 3$. Then $\nu_{i+1} - 4 \le \pi_{i+1}$ at every transition, including the terminal step. Now descend from scale N. The terminal birth identity gives each mode coefficient as the manifest. To check terminal servedness quantitatively, put $c_N = \mathrm{forcedAllowance}_N$ and $C_N = c_N + r_N S_{B_N} u_N$. The terminal receipt gives

$$((2B_N+1)^4)C_N \le 3, \qquad e^{C_N} - 1 \le 1, \qquad \nu_N - 3 \ge 0.$$

The terminal response at label v has price $a_N(v) \le \|A_N\| e^{\nu_N+2} \mathfrak{c}_N(t_N) \mathrm{coeffSize}(\mathrm{core}(\tau_N(v)))$. Since core deletion does not increase coefficient size, the paid bond gives $w_N \|A_N\| \mathrm{coeffSize}(\mathrm{core}(\tau_N(v))) \le \zeta_N$. Hence $w_N a_N(v) \le e^{\nu_N+2} \mathfrak{c}_N(t_N) \zeta_N \le u_N$, by the bank budget. Its toll estimate converts this into the required weighted section bound; the forced response is dominated. More explicitly, the terminal bank price requires $t_N \ge 1$ and

$$e^{\nu_N+2}\, \mathfrak{c}_N(t_N)\, \zeta_N \le u_N,$$

and gives the response price estimate stated above. Combining this toll control with the terminal birth expansion, Fourier hook, and the two displayed smallness conditions gives the served representation with exponent $\nu_N - 3$, exactly the terminal pitch.

For the induction step, assume integrability and the manifest identity at scale $i+1$. Applying the finite transition kernel preserves integrability by its finite mass bound; applying a mode coefficient before or after integration agrees because coefficient extraction is a continuous linear map on the finite-dimensional polynomial space. Thus the pushed coefficient at scale i is the transition applied to the scale-$i+1$ coefficient, and the exact push identity identifies it with the scale-i manifest. The one-step geometry supplies positive old and new weights, occupancy at both the old and embedded site families with multiplicity at most r_i, the parent-site identity, and the support-price comparison. The transported representation is dominated, and its finite-wing response satisfies $2w_{i+1}s \le 1$ and $w_{i+1}t \le 1$. The one-step payment data are

$$0 < d_i \le 1, \quad \nu_i, \nu_{i+1} \ge 100, \quad T_i + e^{T_i} T_i \le t_i, \quad (2B_i+1)^4 (y_i + r_i\, S_{B_i} d_i) \le \tfrac{1}{2}, \quad e^{y_i + r_i\, S_{B_i} d_i} - 1 \le 1,$$

where $T_i = r_i S_{B_{i+1}} d_i + 4 d_i$ is the prepared insertion price and t_i its response budget. The local-transfer theorem of Section 57 applies with these values; it yields servedness at scale i. Finite downward induction proves the served-slab statement for every $m \in [k, N]$. The remaining section-compatibility and response-construction receipts used to obtain these displayed one-step inputs are named explicitly in the evidence file; they are part of the conditional input, not consequences of the transfer lemma.

The remaining public-construction obligation is to produce these inputs uniformly on one selected history. The run selector's continuation and terminal full-volume supplier provide levels, doors, weights, and budgets; the same-history section supplier provides adjacent-scale compatibility, place, and bank bounds; and the terminal supplier provides the birth polynomial, transition identities, and Fourier hook. The source trace and exact upstream declaration chain are in the evidence file. Where that chain forwards a receipt rather than proving the uniform bounded-label inequalities from earlier data, that is recorded as an open production obligation. The present derivation is conditional on those explicitly listed quantitative fields, and does not infer them merely from a receipt's name.

The dependency chain is therefore

$$\begin{aligned} \text{funded selected run} &\longrightarrow \text{paid same-history receipts} \longrightarrow (104) \\ &\longrightarrow (105) \longrightarrow \text{uniform centered power bound} \longrightarrow \text{continuum extraction.} \end{aligned}$$

The compact-bound and local-modulus arguments of Section 18.3 use that final centered estimate. The separate multisource-lane route, whose offered lanes already package terminal births and raw adjacent-scale bounds, is not called along this public dependency chain. In particular, the normalized overlay-density price calculation in the optional multisource draft is not an input here. □

20 Producing the selected finite-volume transfer data

The centered estimate uses a finite-volume representation at every period and volume. On the public construction route these data are produced on the same selected history: the run chooses one terminal voice, then reconstructs compatible voices at all earlier scales. This section isolates that producer and explains the two algebraic steps that turn the reconstruction into transfer data. The tariff and one-step interpolation estimates remain the quantitative inputs of the selected history; the proof below does not replace them by an unpriced compactness argument.

20.1 One-step response and its finite-dimensional assembly

Fix a volume, period, and adjacent scales $m, m+1$. Let F be the finite-dimensional complex coefficient space of the chosen grade and let $s \geq 1$ be the number of paid columns at scale $m+1$. Write $\mu_F(u) = \sum_a |u_a|$ for its coefficient ℓ^1 norm, where $|\cdot|$ is the ordinary complex modulus, and let $\mathrm{core}(u)$ erase the vacuum coordinate. The scale dilation vowel_m multiplies coordinate a by $\ell_m^{d-\deg(a)}$, so

$$\mu_F(u) \leq \mu_F(\mathrm{vowel}_m\, u) \leq \ell_m^d \mu_F(u).$$

A voice at scale $m+1$ consists of a density family f_v, a nonnegative charge, and an index isomorphism $\iota : F \to F$. The family is complex-linear in v and its motion covariance is semilinear according to the independent Boolean conjugation flag in Motion = Signed × Bool; it is not determined by coordinate orientation. For each v, the selected heard certificate supplies the format and remainder bounds; the associated finite-volume integrability lemma gives $f_v \in L^1$ of the normalized prior. We do not use continuity of the densities. The selected transition defines the Wilson push

$$(\mathsf{P}_m f)(U) = \ell_m^d \int f(V)\, K_m(U, dV),$$

where $K_m(U, dV)$ is the one-step Wilson transition density against normalized Haar measure and ℓ_m is the block side. Thus P_m includes the physical block-volume factor ℓ_m^d; it is not a probability-preserving Markov operator. The selected transition package also supplies slice integrability: for every output U, the transition-weighted section of a prior-integrable input is integrable, and the resulting pushed function is measurable. This is the conditional-kernel conclusion attached to the lawful transition, not a consequence of finite total mass alone. The transition commutes with the selected signed motions. These properties make the push well-defined and linear, and for each signed motion g,

$$g \cdot (\mathsf{P}_m f_v) = \mathsf{P}_m f_{g \cdot v}.$$

Indeed, both identities follow first for finite linear combinations from linearity of the integral; the motion identity follows by the change of variables preserving the Wilson transition density. These identities are exact, not limiting statements.

The selected ticket's one-column estimates have three exact forms. For each nonzero frame axis e_a, it supplies a coarse coefficient whose density is the pushed fine density, whose orbit-cost-weighted lead error is at most $\eta \,\|\mathrm{core}(e_a)\|_1$, whose activity residual price is at most $C \,\|\mathrm{vowel}_m(\mathrm{core}(e_a))\|_1$, and whose finite gate-crossing certificate costs t. For each regular supported term of mass u, both coupon prices are ηu and the gate-crossing cost is $2u$. For each forced term of mass u on a marked wing site, both coupon prices are $\eta u/\kappa$ and the gate-crossing cost is $2u$, where $\kappa = \sqrt{\mathrm{forcedAllowance}} > 0$. Each such coupon includes a format bound, the exact density or activity identity, an orbit-cost lead bound, a finite response expansion, and its paid gate-crossing estimate

through the selected door. These are the exact hypotheses used below; we do not infer their numerical bounds from compactness.

On the public route these coupons are produced together for the selected adjacent history. A held spring supplies endpoints $x < y$, an interval parameter z, and a finite local realization $K(z)$ whose endpoint fields are the two selected Wilson fields. The gate is the composed local comparison across those endpoints; the same spring gives the climb witness. For a grade d with $d+3$ no larger than the coarse clock, the ticket's wing contains the origin-supported principal coordinate modes and every marked keeper site. It is finite because the frame has finitely many axes and the terminal keeper has finite support.

Write E_{m+1} for the support exponent at the fine level, ϑ_{m+1} for its profile title, and b_m for the block side. The raw coupon prices are $\eta_m = \mathrm{rate}(b_m)$ for the lead and $C_m = E(J_m)$ for the residual, where J_m is the fine clock and E is the selected schedule. The exact total gate-crossing allowance is

$$t_m = 2e^{E_{m+1}}\vartheta_{m+1}.$$

The selected clock interpolation chooses one heading for the whole wing. Its three coupon families are exactly the axis, regular-activity, and forced-activity cases displayed above. In the raw ticket, the axis lead price is $\eta_m \mu_F(\mathrm{core}\, e_a)$, the residual price is $C_m \mu_F(\mathrm{vowel}_m(\mathrm{core}\, e_a))$, and the crossing bill is t_m. A regular support of mass u has equal lead and residual price $\eta_m u$ and bill $2u$; a marked forced support of mass u has equal price $\eta_m u/\kappa$ and bill $2u$. These are the three fields of the finite coupon certificate. The origin unit is supplied by its exact unit coupon.

The coupon-production step is conditional on its field estimates. Assume the selected spring supplies the two endpoint fields and a sound local section; assume the heading forecast and cap satisfy the two-sided map bound, the principal-field estimate has the displayed axis prices, each supported regular field has the displayed regular prices, and each slot-fitted forced field at a marked keeper site has the displayed forced prices. Then one heading and one ticket for the whole finite wing are obtained by applying those three estimates to their respective terms and collecting the resulting certificates. The heading forecast comes from the calibrated flight matrix, and the selected schedule-price bound supplies the cap's triangular-price estimate. The axis, regular, and forced estimates are separate hypotheses; no one of them follows from another. This is the precise conditional finite coupon package used here. It leaves the derivation of those analytic field estimates and the selected schedule-price data to their upstream producers.

More precisely, the source's finite interpolation lemma has this conditional interface: when the spring endpoints, slot table, motion/rate data, schedule-price bound, and keeper are those of the selected history, and $d+3 \le J_m$, it returns one heading and a ticket for every finite wing containing the origin principal modes. The ticket has axis residual price C_m, lead price η_m, and crossing allowance t_m, with the regular and forced prices just listed. The heading and all prices are common to that whole wing. Substituting the wing formed by the origin modes together with the keeper's marked sites gives the ticket used by the public run. The interface and its three analytic coupon estimates are the precise source inputs here; this conditional assembly does not reprove their field-level derivation.

The ticket prices C_m, η_m are then checked against the one-step reserves. The purchase inequalities are

$$\mu_F(T_m v) \le C_m \mu_F(v), \qquad \mu_F(T_m^{-1} v) \le C_m \mu_F(v).$$

The same selected clock purchase pays the ticket through

$$\begin{aligned} 1 &\le C_m, & 0 &\le \eta_m, \\ \eta_m(1+s)\,\mathrm{orbitCost}(F) &\le \mathrm{barunit}(b_m), \\ (C_m + \eta_m s)\,\mathrm{orbitCost}(F) &\le W_{m+1}. \end{aligned}$$

where s is the number of paid columns and W_{m+1} is the available fine-level column reserve. The one-column theorem combines the raw coupon with these purchases: after all s columns its residual and lead allowances are $c_m = C_m + \eta_m s$ and $\epsilon_m = \eta_m(1+s)$, and its crossing bill is $t_m + 2s$. These inequalities are the actual interpolation and funding checks consumed by the one-column construction; the proof does not choose a new adjacent pair or reset the history.

The reason the ticket pays these three kinds of term is the local decomposition of the marked field. For a fine coefficient P, first apply the local rack map: it preserves the coefficient coordinate and exponential entry, while its keeper supplies the forced-profile and remainder bounds. The density is then

$$(\text{exponential entry}) \cdot (\text{local regular part}) + \sum_j (\text{forced activity at } j) \cdot (\text{dressed field at } j).$$

The regular summands use the regular coupons. A forced summand with zero coefficient vanishes. For a nonzero forced coefficient, its support is marked and belongs to the selected finite wing; its absolute mass uses the forced coupon after division by κ. The origin term uses the principal axis coupon. The source's remnant and exponential envelopes dominate both countable activity series absolutely; Tonelli's theorem applied to the resulting absolute majorant proves integrability and permits summing the densities term by term. The coupon-summation identity adds the costs and errors, so the three displayed prices give the one-column response and preserve the exact pushed density.

The local selected-history ticket is a finite set of marked insertions with a linear cap. It is paid with nonnegative allowances C, η, t and yields an endpoint response on every coefficient of an incoming voice. In the normalization used by the source, the response allowances after accounting for all s columns are

$$c = C + \eta s, \qquad \epsilon = \eta(1+s), \qquad t + 2s.$$

For each frame coordinate e_a, the one-column estimate gives a marked polynomial Z_a for $j_a = \iota^{-1}(e_a)$. Applying a selected motion g gives $Z_{g,a}$ with density $g \cdot (\mathsf{P}_m f_{j_a}) = \mathsf{P}_m f_{\iota^{-1}(g \cdot e_a)}$, lead error at most $\epsilon\, \mu_F(\operatorname{core} e_a)$, residual activity at most cD_a, and charged cost at most $t + 2s$, where

$$D_a = \mu_F(\operatorname{vowel}_m(\operatorname{core} e_a)) .$$

Let $\mathcal{G} = \text{Motion} = \text{Signed} \times \text{Bool}$ be the finite source motion set. Write $g = (\pi, \epsilon)$, where π is a signed coordinate permutation and $\epsilon \in \{0,1\}$ is the independent conjugation flag. Let ρ_g be its (possibly semilinear) action on F and let σ_g be its scalar action, identity for $\epsilon = 0$ and complex conjugation for $\epsilon = 1$. Then the actual averaging chips are

$$\chi(v; g, a) = \frac{1}{|\mathcal{G}|}\, \sigma_g\big((\rho_g^{-1} \iota v)_a\big) , \qquad (g, a) \in \mathcal{G} \times \{1, \dots, q\},$$

where $q = \dim_{\mathbb{C}} F$. Applying the inverse motion, taking its ath coordinate, and then applying the matching scalar action makes $v \mapsto \chi(v; g, a)$ complex-linear, including when $\epsilon = 1$. Averaging the reindexed coordinate expansions gives

$$\iota(v) = \sum_{g,a} \chi(v; g, a)\, \rho_g e_a, \qquad v = \sum_{g,a} \chi(v; g, a)\, \iota^{-1}(\rho_g e_a).$$

For any selected h, reindexing $g = hq'$ gives

$$\chi(\rho_h v; hg, a) = \sigma_h(\chi(v; g, a)),$$

so the assembled response has the required semilinear motion covariance. Thus conjugation comes from the independent flag, not from whether the signed coordinate permutation reverses orientation. The action preserves the grade and commutes with the core, but it need not permute the coordinate axes. Define the finite frame orbit price by

$$\operatorname{orbitCost}(F) := 1 + \sum_{g\in\mathcal{G}} \sum_{a=1}^{q} \mu_F(\rho_g e_a).$$

For any $x = \sum_a x_a e_a$, axis expansion and the triangle inequality give

$$\mu_F(\rho_g x) \le \sum_a |x_a| \mu_F(\rho_g e_a) \le \operatorname{orbitCost}(F)\mu_F(x).$$

Apply this bound to ρ_g^{-1} and to $x = \operatorname{core}(\iota v)$ or $x = \operatorname{vowel}_m(\operatorname{core}(\iota v))$. The scalar actions preserve modulus, and both core and vowel commute with the motion. Averaging over g in the displayed chip formula gives the two estimates

$$\sum_{(g,a)} |\chi(v; g, a)|\, \mu_F(\operatorname{core} e_a) \le \operatorname{orbitCost}(F)\mu_F(\operatorname{core}(\iota v)),$$

$$\sum_{(g,a)} |\chi(v; g, a)| D_a \le \operatorname{orbitCost}(F)\mu_F(\operatorname{vowel}_m(\operatorname{core}(\iota v)))\,.$$

They are the finite tally inequalities obtained by summing the absolute chip weights against the axis and dilated-axis costs. Therefore the weighted sums of lead errors and residual activities are at most $\epsilon \operatorname{orbitCost}(F)\mu_F(\operatorname{core}(\iota v))$ and $c \operatorname{orbitCost}(F)\mu_F(\operatorname{vowel}_m(\operatorname{core}(\iota v)))$, respectively; the corresponding finite tally of gate-crossing bills is the purchased wing allowance. These are the exact finite sums paid by the selected column and orbit-cost budgets. The estimates hold after each signed motion because the marked ticket is closed under those motions and the Wilson transition is motion covariant. Here is the finite assembly step. Let $\iota : F \to F$ be the voice's coefficient-index isomorphism, and let $\mathcal{J}$ be the finite tally of pairs (g, a) consisting of a selected signed motion g and a coordinate index a. Put e_a for the coordinate vector. Let $j_a = \iota^{-1}(e_a)$ and $v_{g,a} = g{\cdot}j_a = \iota^{-1}(g{\cdot}e_a)$; the equality uses equivariance of the index map. The frame interpolation coefficients $\chi(v; g, a)$ are the coefficients of the finite frame expansion of $\iota(v)$, so

$$\iota(v) = \sum_{(g,a)\in\mathcal{J}} \chi(v; g, a)(g \cdot e_a), \qquad v = \sum_{(g,a)\in\mathcal{J}} \chi(v; g, a) v_{g,a}.$$

For each coordinate a, first take the one-column response at j_a, then apply the selected signed motion g to it. The resulting moved response $Z_{g,a}$ has fine density $g \cdot f_{j_a}$ and, by covariance of the transition, coarse density $g \cdot (\mathsf{P}_m f_{j_a}) = \mathsf{P}_m f_{v_{g,a}}$. Its scalar lead is one term in the reconstructed coordinate, and its forced profile is the corresponding moved profile. Assemble them by the finite sum

$$Z(v) = \sum_{(g,a)\in\mathcal{J}} \chi(v; g, a) Z_{g,a}.$$

The assembled scalar lead and forced profile are the same finite weighted sums of their one-column counterparts. By the exact one-column density identity, linearity of the push, and the second frame identity,

$$\begin{aligned} \operatorname{density}(Z(v)) &= \sum_{(g,a)} \chi(v; g, a) g \cdot (\mathsf{P}_m f_{j_a}) \\ &= \mathsf{P}_m \Big(\sum_{(g,a)} \chi(v; g, a) f_{v_{g,a}} \Big) = \mathsf{P}_m f_v. \end{aligned}$$

All coefficient and residual bounds are summed with $|\chi(v;g,a)|$. The frame interpolation inequalities are proved by the finite frame identities: the lead error is bounded by ϵ orbitCost$(F)\|\operatorname{core}(\iota v)\|_1$, and the residual by c orbitCost$(F)\|\operatorname{vowel}_m(\operatorname{core}(\iota v))\|_1$. The first is the triangle inequality followed by the frame orbit-cost bound; the second follows by summing the coordinate bounds and the vowel size inequality. The paid ticket bounds the total gate-crossing costs by the corresponding finite-wing budget. Thus no infinite series or exchange of an infinite sum with the transition integral occurs. The one-step direction estimates give the cone bounds; here is the numerical argument. Put $L = b_m$, let J_f and J_c be the fine and coarse clocks with $J_f \le J_c + 1$, and set $C = W.K(J_f)$, $D_f = W.K(J_f + 1)$, $D_c = W.K(J_c + 1)$. Let s_f be the fine column count, s_c the coarse column count, and $e = \operatorname{barunit}(L)$. Write u and v for the coarse and fine frame coordinates, r and t for their charges, and $y = \operatorname{vowel}_L(v)$. The direction certificate provides a coefficient isomorphism B such that

$$\|Bx\|_1,\ \|B^{-1}x\|_1 \le C\|x\|_1, \qquad B(\operatorname{core} x) = \operatorname{core}(Bx),$$

$$\|u - By\|_1 \le e(\|\operatorname{core} v\|_1 + t), \qquad r \le C\|\operatorname{core} y\|_1 + et, \qquad t \le s_f\|\operatorname{core} v\|_1.$$

The selected price growth gives $1 \le C \le D_f$ and, by the slow-step condition, $C \le D_c$. The bank certificate says

$$100000 D_f^6 (1 + s_f)^2 \le \operatorname{puffs}(L), \qquad e = \operatorname{puffs}(L)^{-3}.$$

In particular $\operatorname{puffs}(L) \ge 1$ and $2C(1 + s_f) \le \operatorname{puffs}(L)$, so $2Ce(1 + s_f) \le \operatorname{puffs}(L)^{-2} \le 1$ and $e \ge 0$. The fine room certificate gives $64D_f^2 \le s_f$, and the coarse room certificate gives $64D_c^2 \le s_c$. Hence $2C \le s_f$ and $4C^2 \le s_c$. Since $\|\operatorname{core} y\|_1 = \|\operatorname{vowel}_L(\operatorname{core} v)\|_1 \ge \|\operatorname{core} v\|_1$, the incoming charge bound implies

$$2Ce(\|\operatorname{core} v\|_1 + t) \le 2Ce(1 + s_f)\|\operatorname{core} v\|_1 \le \|\operatorname{core} y\|_1.$$

Consequently $2C\|u - By\|_1 \le \|\operatorname{core} y\|_1$. Put $\delta = \|u - By\|_1$. Then $C\delta \le \|y\|_1/2$, and the inverse norm bound gives

$$\|y\|_1 = \|B^{-1}(By)\|_1 \le C\|By\|_1 \le C\|u\|_1 + C\delta \le C\|u\|_1 + \tfrac{1}{2}\|y\|_1,$$

so $\|y\|_1 \le 2C\|u\|_1$. In the other direction,

$$\|u\|_1 \le \|By\|_1 + \delta \le C\|y\|_1 + \frac{1}{2C}\|y\|_1 \le 2C\|y\|_1,$$

since $C \ge 1$. Apply the same absorption argument to the core vectors; the core projection contracts the ℓ^1 norm and commutes with B. It yields

$$\|\operatorname{core} y\|_1 \le 2C\|\operatorname{core} u\|_1.$$

It follows, using $\|v\|_1 \le \|y\|_1 \le L^d\|v\|_1$, that

$$\|v\|_1 \le s_f\|u\|_1, \qquad \|u\|_1 \le s_f L^d\|v\|_1.$$

Also $et \le C\|\operatorname{core} y\|_1$, by the displayed bank estimate, so

$$r \le 2C\|\operatorname{core} y\|_1 \le 4C^2\|\operatorname{core} u\|_1 \le s_c\|\operatorname{core} u\|_1.$$

These are exactly the charge, fine-to-coarse, and coarse-to-fine bounds used below. In particular, the two norm inequalities are full coefficient norms; the core norm occurs only in the charge estimate.

The response defines a linear map $Q_m : F \to F$ by its coarse reference coordinate. The first cone inequality gives $\|\iota(v)\|_1 \le s_f\|Q_m v\|_1$. If $Q_m v = 0$, then $\iota(v) = 0$; since ι is an isomorphism,

$v = 0$. Thus Q_m is injective and, since F is finite dimensional, bijective. The one-step output stores this isomorphism as the new voice index. The density family is not precomposed by its inverse. Across several scales, the full-norm cone inequality is iterated from the reference to the terminal frame; the exact recurrence and terminal isomorphism give the single global transfer normalization used below.

Lemma 20.1 (Finite backward reconstruction)**.** *Let $t \le N$. Suppose a selected finite-volume history has (i) a terminal voice which agrees with the calibrated top coefficient frame and satisfies the terminal reflection condition; (ii) a finite-mass, motion-covariant Wilson push between each adjacent pair of scales; (iii) at every $m \in [t, N)$ a selected door, a sound local section, and a finite paid ticket whose one-step response satisfies the column estimates and cap conditions described above; and (iv) the room and bank bounds needed to pay the cumulative responses. Then there is a sequence of voices V_m, $t \le m \le N$, such that the terminal voice is the selected one, every adjacent density satisfies*

$$\operatorname{density}(V_m(v)) = \mathsf{P}_m \operatorname{density}(V_{m+1}(v)),$$

and each scale has the selected reality, motion, and coefficient-cone bounds. The pushed coefficient maps define a linear transfer system, normalized by the inverse of its reference coefficient map.

Proof. Use downward induction. At $m = N$, take the supplied terminal voice. Suppose V_{m+1} has been constructed and has the required integrability, reality, and motion properties. Apply the one-step response assembly above to its paid ticket. The density identity follows from the exact finite linear assembly. The local cap gives the response and residual bounds at scale m; the paid room and bank inequalities keep the accumulated expenditure within its assigned budget. The terminal reflection condition propagates because the Wilson push is invariant under reflection and the ticket is real on the selected real subspace. The coefficient-cone estimate propagates by the one-step paid inequality. The one-step construction packages its own invertible index map with its density family as a voice; we do not alter the family by precomposing with the inverse. This completes one induction step. There are only $N - t$ steps, so finite downward induction constructs every V_m and all the claimed identities. Finally, if the reference readout of a coefficient v is zero, the full-norm cone inequality propagates zero from scale t to scale N. The terminal voice index is an isomorphism, so $v = 0$. The reference map is injective and hence bijective on F; its inverse gives the one global normalization in the conclusion. □

20.2 The transfer map and its normalization

For the unnormalized music just constructed, write $x_m(v)$ for the coefficient index of its voice at scale m. The full-norm cone estimates at each adjacent pair have the form

$$\|x_{m+1}(v)\|_1 \le s_{m+1}\|x_m(v)\|_1, \qquad \|x_m(v)\|_1 \le s_{m+1}\ell_m^d\|x_{m+1}(v)\|_1,$$

where s_{m+1} is the paid column count and ℓ_m the block side. Let $Q_t(v) = x_t(v)$ be the reference readout. If $Q_t(v) = 0$, the first inequality propagates zero successively through every scale up to N. At the terminal scale x_N is an isomorphism, so $v = 0$. Thus Q_t is injective, and finite dimensionality makes it bijective. This is the sole global normalization. The transfer data record the unnormalized music entries and set their terminal coordinate map to Q_t^{-1}. When viewed on external labels w, every scale uses the same reparametrization $V_m(Q_t^{-1}w)$; no scale gets an independently fitted inverse. The selected public supplier applies this construction for every grade and every pair (N, p) beyond the grade reference. Its first finitary step is the selected endpoint: the terminal frame is obtained from the calibrated section and the keeper chosen by the nested history. The endpoint is on top

because the selected Wilson trajectory represents the terminal field and the paid terminal receipt gives the exact coefficient identities. Its reflection condition follows from the real section and the terminal keeper. The selected climb provides the door and the finite ticket at each adjacent pair. The ticket's allowance is the scale's tariff, and the paid room and bank bounds control the total cost. Applying Lemma 20.1 and then the reference normalization gives the transfer data required by the centered-volume calculation.

The resulting ordinary mathematical object is a music of compatible voices with a selected terminal frame, the adjacent finite-wing estimates, and a normalized transfer map. In the center argument, the separate terminal birth polynomial is the finite product of the named observables. It is continuous on compact configuration space and hence integrable. The backward transition identity identifies its propagated coefficients with the voice at each intermediate scale. This is the only integral passage needed for the finite-volume marked-copy argument; all later bounds concern normalized finite-volume expectations.

20.3 Scope of the producer statement

The preceding induction proves the finite reconstruction from its exact selected-history ticket and terminal inputs. In the public run, the schedule is chosen from the purchased trajectory so that the grade reference lies before every requested terminal level. The selected climb and terminal receipt are retained on this same trajectory; they are not assembled by selecting a new history for each period. The consumer then uses the resulting transfer system to form the terminal birth and the root estimate.

The quantitative production of the ticket's nonlinear coefficient estimates is a distinct finite-volume step. The selected public construction supplies it through the field interpolation, mode-slot, and shipment records; this chapter only states the numerical inequalities that the one-step theorem consumes. Likewise, uniformity over every coefficient in a bounded grade ball must be established from those records before the conditional centered theorem becomes unconditional. The evidence file records the exact source chain and distinguishes the finite backward assembly proved here from these upstream quantitative estimates.

21 Producing the occupied, migrating, separated sites

The mode-moment receipt needs a site assignment at its opened scale: it must have bounded occupancy, agree with its descendants under every selected coarse-graining map, and keep distinct names in every nontrivial mode far apart. On the public root-test route, one finite copy of the opened lattice supplies all three properties. This section proves that geometric production from the selected tab-opening schedule.

Let $m \le k \le N$ be levels in one selected history and fix a period p. Put $V = \Lambda_m$ for the finite site set at level m. Write a_j for the physical lattice spacing at level j, $L_j = a_j/a_{j+1} \in \mathbb{N}$ for the exact adjacent ratio, and B_j for the schedule's covering radius. Write n_j for the integer side of the periodic lattice Λ_j, so an aligned embedding with ratio $q_j = a_m/a_j$ has $n_j = q_j n_m$. The selected tab schedule gives

$$L_j \ge 3, \qquad B_{j+1}^2 \le 2B_j^2, \qquad B_k^2 \le a_m/a_k$$

when the root level m is opened at k with parameter one. At every level $j \ge m$, define

$$o_j(x) = \mathrm{embed}_{m\to j}(x), \qquad x \in V,$$

using the canonical aligned lattice embedding, and put $o_j(x) = 0$ below m. Let B_j be the schedule's separation radius at level j; for the selected Wilson history this is also the radius of the local configuration there.

Lemma 21.1 (Opened-copy geometry)**.** *Assume the selected family identifies the represented field radius with B_j, and the schedule bounds just stated hold. Then the opened-copy assignment has the following properties for $k \le j \le N$:*

1. *$o_j : V \to \Lambda_j$ is injective, and hence has occupancy at most one.*
2. *For each $k \le j < N$, the parent of $o_{j+1}(x)$ is $o_j(x)$.*
3. *For distinct $x, y \in V$, $\operatorname{sep}(o_j(x), o_j(y)) \ge B_j^2$.*
4. *At the terminal level, the root copy map sends each $x \in V$ to $o_N(x)$.*

Consequently, if an occupancy schedule r_j obeys $1 \le r_j$, the assignment is migrating, has occupancy at most r_k at scale k, and satisfies the separated-origin condition for every census mode whose distinct names are drawn from V.

Proof. The opening condition gives $B_k^2 \le a_m/a_k$. We propagate this to every later j. If $B_j^2 \le a_m/a_j$, then

$$B_{j+1}^2 \le 2B_j^2 \le 2\frac{a_m}{a_j} \le L_j \frac{a_m}{a_j} = \frac{a_m}{a_{j+1}},$$

because $L_j \ge 3$. Thus $a_m/a_j \ge B_j^2$ for all $j \ge k$. The level-m to level-j ratio $q_j = a_m/a_j$ is an integer, by multiplication of the adjacent integer ratios.

For sites on the periodic lattice Λ_m, define separation as the maximum over coordinates of the shortest periodic distance, namely

$$\operatorname{sep}(x, y) = \max_{\alpha} \min_{r \in \mathbb{Z}} |\widetilde{x}_\alpha - \widetilde{y}_\alpha + r n_m|.$$

Here $\widetilde{x}, \widetilde{y}$ are any integer lifts of the periodic sites; the minimum does not depend on the chosen lifts. Choose in each coordinate a centered integer representative of the difference. Under the aligned ratio embedding of factor q_j, each such representative is multiplied by q_j and remains centered modulo the new periodic side $n_j = q_j n_m$, including the half-side tie case. Hence

$$\operatorname{sep}(o_j(x), o_j(y)) = q_j \operatorname{sep}(x, y).$$

Distinct x, y in the level-m lattice have integer separation at least one. Therefore

$$\operatorname{sep}(o_j(x), o_j(y)) = q_j \operatorname{sep}(x, y) \ge q_j \ge B_j^2.$$

In particular they are distinct; this proves injectivity and occupancy at most one. The exact ratio-composition identity for canonical embeddings identifies the parent of $o_{j+1}(x)$ with $o_j(x)$. By the definition of the opened root cell, its terminal copy map is precisely the canonical embedding o_N. Finally, $1 \le r_k$ upgrades occupancy one to the requested bound, while the separation estimate is exactly the required bound for every pair of distinct names in any nontrivial mode. □

For the selected root test, take its finite root set as V. Lemma 21.1 gives the migration, occupancy, and separated-origin inputs used by the finite-volume moment receipt. It gives these inputs for the same selected period, stop level, and schedule; it does not choose new sites independently at each scale. The estimate does not supply the weights or the response budget, which remain separate selected-history inputs.

22 Propagating the selected terminal birth to a moment bound

This section records the finite-volume terminal receipt used on the public selected-history route. It starts from the calibrated family, its same-history multiscale data, and the birth polynomial made from a finite list of observables. The result is a mode-by-mode normalized moment estimate. It is a conditional receipt: it uses the selected-history service and migration data stated below, rather than deriving those upstream construction inputs.

Let $\mathcal{V}$ be a finite marker set and let B be its census. For each $a \in \mathcal{V}$, choose a grade label s_a, a terminal site $o_N(a)$, and a signed coordinate motion g_a. Let O_a be the corresponding local cylinder observable, and define its terminal translate by

$$T_a(U) = g_a \cdot O_a\big(\mathrm{pull}(o_N(a), U)\big).$$

Assume the selected family makes each T_a continuous on the compact terminal configuration space. Write R_N for the selected terminal representative and set

$$\mathcal{B}_N(U) = R_N(U)\ \mathrm{Pass}\left(\sum_{a\in\mathcal{V}} e_{\{a\}} T_a(U)\right),$$

where the marker algebra has square-free multiplication: $e_S e_{S'} = e_{S\cup S'}$ for disjoint subsets and 0 otherwise. The polynomial pass is the finite exponential in this nilpotent algebra. For every mode $I \subseteq \mathcal{V}$, its coefficient satisfies

$$(\mathcal{B}_N(U))_I = R_N(U) \prod_{a\in I} T_a(U).$$

The coefficient for the empty mode is R_N. This is the selected terminal birth identity. For this selected terminal representative, R_N is the bare Wilson density. Write

$$Z_{N,p} := \int R_N(U)\, d\mu_N(U) > 0, \qquad \omega_{N,p}(F) := Z_{N,p}^{-1} \int R_N(U) F(U)\, d\mu_N(U).$$

Theorem 22.1 (Selected-history birth moment bound)**.** *Fix a selected family, volume N, period p, grade d, and reference scale k with $C.\mathrm{ref}(d) \le k < N$. Suppose the family satisfies its calibration conditions and the same-history finite-volume receipt supplies the terminal birth, its backward Wilson transitions, the scale-k service resources, and the positive vacuum treasury. Assume also the geometric inputs at k: positive weights w_a, site assignments $o_k(a)$, an occupancy bound, the migration identity for the site assignments, and the separated-origin condition for every nontrivial mode. Suppose further that*

$$w_a\, \mathfrak{m}_a \le \mathfrak{f}_k \le 1 \quad (a \in \mathcal{V}),$$

where $\mathfrak{m}_a$ is the paid coefficient size of label s_a at scale k and $\mathfrak{f}_k$ is the scale's small price. Require the service receipt to give measurable mode densities H_I, a common scalar σ, and real costs A_k, D_k such that its representative is nonnegative, its support exponent is nonnegative, and its reserve is $f = 1$, with

$$w_I |H_I(U)| \le \exp(\sigma + A_k), \quad Z_{N,p} \ge \exp(\sigma - D_k), \quad A_k + D_k \le |\Lambda_k| \mathcal{T}_k + |\mathcal{V}|(x + y + 2f),$$

where $w_I = \prod_{a\in I} w_a$, $x = 3\mathfrak{f}_k$, and $y = \mathfrak{f}_k$. The same-history receipt identifies H_I with the scale-k coefficient of the backward birth. Then for every census mode I,

$$\left|\omega_{N,p}\left(\prod_{a\in I} T_a\right)\right| \prod_{a\in I} w_a \le \exp\left(|\Lambda_k|\, \mathcal{T}_k + 6|\mathcal{V}|\right),$$

where $\omega_{N,p}$ is the normalized finite-volume Wilson expectation, Λ_k is the scale-k lattice, and $\mathcal{T}_k$ is the finite-volume vacuum treasury cost. The conclusion is conditional on the listed service, geometry, and paid-coefficient hypotheses.

Proof. First, $\mathcal{B}_N$ is integrable. The selected terminal representative is continuous, each T_a is continuous by assumption, and the finite marker polynomial is a finite sum of products of these functions. Hence every coefficient of $\mathcal{B}_N$ is continuous. The terminal configuration space is compact, so each coefficient is integrable against the normalized product-Haar prior.

For $k \le i \le N$, let $\mathcal{B}_i$ be the coefficient polynomial obtained by the selected backward transition. It satisfies

$$\mathcal{B}_i = \mathrm{Push}_i(\mathcal{B}_{i+1}), \qquad \mathcal{B}_N \text{ as above},$$

where the fine-to-coarse kernel is normalized in its output variable:

$$\int k_i(U,V)\, d\mu_i(V) = 1 \qquad \text{for each fine configuration } U.$$

The density push is therefore

$$(\mathrm{Push}_i\, F)(V) = \int k_i(U,V) F(U)\, d\mu_{i+1}(U),$$

where U is fine and V is coarse. For an integrable coefficient F, slice integrability and Fubini give the precise conservation identity

$$\int (\mathrm{Push}_i\, F)(V)\, d\mu_i(V) = \int F(U) \left(\int k_i(U,V)\, d\mu_i(V) \right) d\mu_{i+1}(U) = \int F(U)\, d\mu_{i+1}(U).$$

For this terminal birth, R_N is the bare Wilson density, so $Z_{N,p} = \int R_N\, d\mu_N$. For the selected Wilson transition the density is bounded on the finite configuration space, and the source slice-integrability property ensures that each coefficient of the product integrand is integrable on every fixed slice. Fubini therefore applies at each step. Since coefficient extraction is a continuous linear map on the finite-dimensional marker algebra, it commutes with this finite integral. Inductively, for each mode I,

$$\int (\mathcal{B}_k(U))_I\, d\mu_k(U) = \int (\mathcal{B}_N(U))_I\, d\mu_N(U) = Z_{N,p}\, \omega_{N,p} \left(\prod_{a \in I} T_a \right),$$

where $Z_{N,p} > 0$ is the finite-volume Wilson partition function. The second equality uses the terminal birth identity and the definition of normalized expectation. This establishes the exact finite-volume transfer of the moment; no continuum limit or interchange of an infinite sum is involved.

At scale k, the same-history receipt yields a served marked representation of $\mathcal{B}_k$. Its finite response estimate gives at marker a a coefficient bound by three times the paid coefficient size and a total activity bound by that paid size. Multiplying by w_a and using $w_a \mathfrak{m}_a \le \mathfrak{f}_k$ gives the coefficient and activity bills $x = 3\mathfrak{f}_k$ and $y = \mathfrak{f}_k$. The factor three is the local coefficient bound from the response estimate; the activity bill is its unmultiplied support charge. The separated-origin condition and occupancy estimate are what make this finite response estimate valid for a whole mode, and migration transports those facts through the backward scales. Since $\mathfrak{f}_k \le 1$, these satisfy $x + y + 2 \le 6$.

Here are the precise conditional price assumptions used at the scale-k service step. The marked representation is lawful, its representative is pointwise nonnegative, its support exponent is nonnegative, and its served reserve is $f = 1$. For each mode I, the local density estimate is

$$w_I |H_I(U)| \le \exp(\sigma + A_k), \qquad w_I = \prod_{a \in I} w_a,$$

with $x = 3\mathfrak{f}_k$ and $y = \mathfrak{f}_k$. The vacuum denominator bound is

$$Z_{N,p} \geq \exp(\sigma - D_k),$$

and the receipts give

$$A_k + D_k \leq |\Lambda_k|\mathcal{T}_k + |\mathcal{V}|(x + y + 2f), \qquad f = 1.$$

The reserve contributes the term $2f$; the coefficient $+2$ is not paid merely by the marker fee. Integrating the pointwise inequality yields

$$w_I \left| \int H_I \, d\mu_k \right| \leq \exp(\sigma + A_k).$$

Divide by the positive denominator lower bound. The scalar σ cancels:

$$w_I \frac{|\int H_I \, d\mu_k|}{Z_{N,p}} \leq \exp(A_k + D_k) \leq \exp\big(|\Lambda_k|\mathcal{T}_k + 6|\mathcal{V}|\big),$$

using $x + y + 2f = 4\mathfrak{f}_k + 2 \leq 6$. Equation (22) identifies this normalized coefficient integral with the desired Wilson moment and proves the result. □

The terminal product, continuity, and exact backward integration have been proved above. The service certificate, its local coefficient envelope, the origin-counting response, and the finite-volume vacuum lower bound remain explicit inputs to this receipt. Their source correspondence and the exact producer boundary are recorded in the evidence ledger.

23 Large entrance scales for the selected Geyser

This section proves the finite-volume scale estimates used by the selected Geyser receipt. It isolates the elementary large-parameter estimates and then explains how their separate eventual thresholds are made into one monotone delay sequence. All scale, profile, and gauge data below are the actual fixed data of the selected run.

Write

$$B(x) = \lceil (\log x)^2 \rceil, \qquad h(x) = e^{-\frac{1}{2}\log x}, \qquad f_d(x) = x^{-(\deg F_d + 4)}, \qquad s(x) = \tfrac{1}{4}h(x),$$

where F_d is the finite frame at grade d. For a fixed period j, write L_j for its side, $b_j = c_G \log L_j$, $y_j(x) = x + b_j$, and $p_j = Q(j)$. The stage quantities are the column count C_j, support exponent E_j, and the finite camera bound A_d; each is fixed while $x \to \infty$.

Lemma 23.1 (Uniform eventual scale estimates for one transition)**.** *Fix a period j, two adjacent stages i and j with* $\mathrm{Step}(j, i)$*, and a finite grade range $0 \leq d \leq i$. For each grade let $C_i \geq 1$ and $C_j \geq 1$ be the corresponding column counts, $E_i, E_j \geq 0$ the support exponents, $A_d \geq 1$ its camera count, $a_d \geq 1$ its motion orbit cost, and $m_d = \deg F_d$ its frame degree. Let $L_j \geq 1$ be the integer side and assume the finite title cost $T_j \geq 0$. Put*

$$r_j = 2e^{E_j} T_j, \qquad T_j = \text{the finite title cost at stage } j.$$

Then there is x_0 so that every $x \geq x_0$ satisfies

$$8 \leq B(x), \qquad y_j(x)^3 \leq e^{B(x)/128}, \tag{107}$$

$$(2B(x) + 1)^4 \, (\mathrm{stain}(x) + p_j \, \mathrm{siteFactor}(B(x))s(x)) \leq 3,$$

$$e^{\mathrm{stain}(x) + p_j \, \mathrm{siteFactor}(B(x))s(x)} - 1 \leq 1, \tag{108}$$

and, for every $d \le i$,

$$f_d(x)C_i \le 1, \tag{109}$$
$$2a_d\big(r_j + 2C_j\big) f_d(y_j(x)) \le 1, \tag{110}$$
$$e^{E_i+2}A_d x^{m_d} f_d(x) \le s(x), \tag{111}$$
$$y_j(x)^{-1} \le 1, \qquad y_j(x)^{-1} L_j^d C_j f_d(x) \le f_d(y_j(x)), \tag{112}$$

as well as $\nu_d \le B(x)$, where ν_d is the minimum stencil side at grade d.

Proof. We check each bound from its large-x mechanism. Since $B(x) \to \infty$, eventually $B(x) \ge 8$ and $B(x) \ge \nu_d$ for all the finitely many grades. The first polynomial estimate follows from the same elementary comparison: for any fixed b and integer q, eventually $x + b \le 2x$, $C2^q \le x$, and $(q+1)\log x \le (\log x)^2/128$. Thus

$$C(x+b)^q \le C2^q x^q \le x^{q+1} = e^{(q+1)\log x} \le e^{B(x)/128}.$$

Apply this with $C = 1, q = 3, b = b_j$ to get (107).

For (108), set $H = h(x)$. We verify the reserves directly. The hat is $H = x^{-1/2}$, so $H \le 1/4$ eventually. From the definition,

$$\mathrm{smallK}(x) = \exp\big(-\tfrac{1}{2}\exp\big(\tfrac{1}{5}(\log\log x)^2\big)\big).$$

Writing $y = \log\log x$, for $y \ge 10$ we have $y^2/5 - y \ge 10 > \log 6$, hence $\exp(y^2/5) \ge 6e^y = 6\log x$. Comparing exponents yields $\mathrm{smallK}(x) \le H^6$.

For the contact bound, put $t = \frac{1}{2}\log x$, so $H = e^{-t}$. For $t \ge 1$, $B(x) = \lceil 4t^2 \rceil \le 5t^2$, and therefore

$$2B(x) + 1 \le 11t^2, \qquad \mathrm{siteFactor}(B(x)) = e\big(2(16B(x)+9)\big)^4 \le e(178t^2)^4.$$

Thus the contact expression is at most $C_p t^{16} e^{-t}$ for a constant C_p depending only on the fixed period. For any fixed integer $q \ge 0$, the exponential series gives $e^t \ge t^{q+1}/(q+1)!$, and so $Ct^q e^{-t} \le C(q+1)!/t \to 0$. Taking $q = 16$ proves the contact reserve. Consequently, for large x,

$$0 < H \le \tfrac{1}{4}, \qquad \mathrm{smallK}(x) \le H^6, \qquad (1 + p_j\,\mathrm{siteFactor}(B(x)))(2B(x)+1)^4 H \le 1.$$

The stain is $\mathrm{smallK}(x)^2$. Squaring the small-K bound gives $\mathrm{stain}(x) \le H^{12} \le H$ because $0 < H \le 1$. Also $s(x) = H/4 \le H$. Therefore

$$\mathrm{stain}(x) + p_j\,\mathrm{siteFactor}(B(x))s(x) \le (1 + p_j\,\mathrm{siteFactor}(B(x)))H,$$

which proves the product bound by the third reserve estimate. Since $(2B+1)^4 \ge 2$ once $B \ge 8$, the exponent in that display is at most $1/2$. It follows from $1/2 < \log 2$ that its exponential minus one is at most $e^{1/2} - 1 < 1$.

For the remaining five estimates, fix $d \le i$ and put $m = \deg F_d + 4$, so $m \ge 1$, and $D = L_j^d C_j$. All constants are fixed, while $x \to \infty$ and $y = x + b_j \to \infty$. Eventually

$$x \ge \max\{1, C_i, 4e^{E_i+2}A_d\}, \quad y \ge \max\{1, 2a_d(r_j + 2C_j), D2^m\}, \quad y \le 2x.$$

Now $f_d(x) = x^{-m}$, so $C_i x^{-m} \le 1$ and

$$2a_d(r_j + 2C_j)y^{-m} \le 1.$$

For the camera term, put $m = m_d + 4 = \deg F_d + 4$; then

$$e^{E_i+2} A_d x^{\deg F_d} f_d(x) = \frac{e^{E_i+2} A_d}{x^4} \le \frac{1}{4\sqrt{x}} = s(x),$$

using $4e^{E_i+2} A_d \le x \le x^{7/2}$ for $x \ge 1$ (the weaker power inequality suffices). Also $y^{-1} \le 1$.

Finally, $D2^m \le y$ and $y \le 2x$ imply

$$D y^m \le y x^m.$$

Indeed $y^m \le (2x)^m$ and $D2^m \le y$. Dividing by the positive quantity $y^{m+1} x^m$ yields

$$y^{-1} D x^{-m} \le y^{-m},$$

which is the last inequality. There are finitely many $d \le i$, so the finitely many thresholds just chosen have a common maximum. This proves the lemma. □

Theorem 23.2 (One monotone delay for all selected transitions)**.** *Fix the selected schedule, profile, gauge, period sides, and an initial delay sequence z_j. There is a coordinatewise larger delay sequence v_j, nondecreasing in j, such that for every later delay sequence $u_j \ge v_j$, every j, i, x with* $\mathrm{Step}(j,i)$ *and $u_j \le x$ satisfies* (107)–(112) *for the actual profile, as well as the frame-fit inequality $\nu_d \le B(x)$ for every $d \le i$.*

Proof. For a fixed pair (j,i) with $\mathrm{Step}(j,i)$, apply Lemma 23.1 to its finite range $d \le i$. It provides a threshold $t_{j,i}$ above which all its conclusions hold. A fixed j has only two possible stages, $i = j$ and (when $j > 0$) $i = j - 1$, because the source relation is $i \le j \le i + 1$. Set t_j to the maximum of the available $t_{j,i}$, and enlarge it to include z_j. Define

$$v_j = \max_{0 \le \ell \le j} t_\ell.$$

Then v_j is nondecreasing and $z_j \le v_j$. Whenever $v_j \le x$, we have $t_j \le x$, so the required estimates hold for either possible stage. If $u \ge v$ coordinatewise and $u_j \le x$, then $v_j \le x$, giving the same conclusions. This is the finite-prefix maximum construction used to collect the eventual estimates; no uniformity over unbounded profile ranges is presumed at a fixed scale. □

Use in the Geyser producer. The theorem gives the exact quantitative service inequalities used by the selected finite producer: the cubic mint bound, stain/contact reserve, five interpolation bounds, and finite stencil fit. In the source these are combined with the selected history, opening, menu, and Freight inputs in the Geyser constructor. This section proves the large-entrance estimates and their common monotone threshold; it does not prove the separate selected nonlinear menu response or the paid purchase that supplies the fixed profile data.

Proposition 23.3 (From the scale inequalities to the Geyser service fields)**.** *Let x_m be the selected hatted scale, L_m its block side, and $B_m = B(x_m)$. Let E_m be its support exponent and T_m its finite title cost. Suppose the selected schedule and delay u satisfy*

$$x_{m+1} = x_m + b_{J(m+1)}, \qquad x_m \ge 1, \qquad B_{m+1}^2 \le 2B_m^2, \qquad L_m \ge 2, \qquad u_{J(m+1)} \le x_m,$$

with $u_j \ge v_j$ as in Theorem 23.2. Assume $\mathrm{threshold}_m > 0$ *and* $\mathrm{threshold}_m^{-1} \le x_m$*. Let $d \le J(k)$, $k \le m$, and set*

$$\zeta_m = f_d(x_m), \quad \phi_m = x_{m+1}^{-1}, \quad \delta_m = x_{m+1}^{-2}, \quad \psi_m = 2e^{E_{m+1}} T_{m+1}, \quad \eta_m = s(x_m).$$

Then the scale estimates give, at every $m \ge k$, the local coin, room, decay, and norm-cost inequalities in the definitions of Fiesta*,* Cherry*, and* Banks*, with price* cameras(d)*. They also give the Brine water, mint, split, and support inequalities. If the local-section bounds are taken from the source camera estimate, namely* $\|\mathrm{places}_d(i,0,U)\| \le \mathrm{cameras}(d)$*, its place bound is uniform in m; its footprint cover is at most one once $\nu_d \le B_m$.*

Proof. Apply Theorem 23.2 at $j = J(m+1)$, $i = J(m)$, and $x = x_m$. The selected clock identities give $\mathrm{Step}(j,i)$ and identify the next scale with $y_j(x_m) = x_{m+1}$. The delay condition places x_m beyond the collected threshold. Since $d \le J(k) \le J(m)$, its creek inequalities apply to grade d.

Write F_d for the grade frame and $m_d = \deg F_d$. The first creek inequality is

$$x_m^{-(m_d+4)}\,\mathrm{columns}_m \le 1,$$

which is the Fiesta coin inequality. Its second is

$$2\,\mathrm{orbitCost}(F_d)(\psi_m + 2\,\mathrm{columns}_{m+1})x_{m+1}^{-(m_d+4)} \le 1,$$

the Fiesta room inequality. The title cost is nonnegative, so the remaining Fiesta sign condition holds.

The shifted reciprocal estimate $x_{m+1}^{-1} \le 1$ is the Cherry level inequality. The final creek estimate reads

$$x_{m+1}^{-1}L_m^d\mathrm{columns}_{m+1}x_m^{-(m_d+4)} \le x_{m+1}^{-(m_d+4)};$$

this is exactly the Cherry cost inequality with $\phi_m = x_{m+1}^{-1}$ and the adjacent marker prices ζ_m, ζ_{m+1}.

For Banks, the cost price is at least one by the camera bound. Its candle cost is

$$e^{E_m+2}\,\mathrm{cameras}(d)\,\mathrm{threshold}_m^{-m_d}\,x_m^{-(m_d+4)}.$$

The threshold comparison and $\mathrm{threshold}_m^{-1} \le x_m$ bound this by

$$e^{E_m+2}\,\mathrm{cameras}(d)\,x_m^{m_d}x_m^{-(m_d+4)} \le s(x_m) = \eta_m,$$

using the third creek inequality. This verifies the Banks candle bound. The separate source camera estimate gives the stated uniform place bound; the footprint estimate follows from the fitted stencil radius and the one-site cover lemma.

For Brine, positivity is immediate because $x_{m+1} > 0$. Its mint inequality follows from the second scale estimate:

$$\delta_m\phi_m e^{B_m/128} = x_{m+1}^{-3}e^{B_m/128} \ge 1.$$

The contact inequality supplies support $B_m \ge 8$. Finally,

$$B_{m+1}^2 \le 2B_m^2 \le L_mB_m^2,$$

so the scale split holds. These are exactly the four Brine clauses used in the Geyser-to-Ponds conversion. □

24 The service receipt on the selected public run

The finite-volume moment estimate uses one service certificate at each period, reference scale, and terminal volume. The public route builds these certificates from a single funded run and its same-history multiscale record. This section follows that finite producer and spells out how its fields are used; it does not assume that a new history can be selected for each volume.

Theorem 24.1 (Same-history service certificates)**.** *Fix the selected run and a campaign album on its trail, with its one family and gauge doors. For each grade d, let k_d denote its selected reference scale. Assume the funded record on this trail supplies (i) a single transition history with retention clauses at every level and a terminal endowment; (ii) the scale room and bank budgets, initial Brine inequalities, and a nondecreasing occupancy schedule ρ_i; (iii) for every grade d and $k_d \le k \le N$, a common height bound X and the activity, paid-cost, nested-support, footprint, and local-section estimates stated below; (iv) for each such grade and terminal volume, a calibrated finite music with source ledgers, adjacent door estimates, and its terminal forced-profile identity; and (v) gradewise thresholds, positive scale prices with adjacent growth, and finite local tone tickets on those same doors. Then for every period p, reference level $k \ge 1$, and terminal volume N, these records form the service certificate used by the public centered root-test argument. All fields refer to the same selected history.*

Proof. The selected run fixes one trail T, one family C, transfer stories, and gauge doors at every adjacent scale; the purchase record supplies the multiscale ledgers. The funded-history producer is applied to exactly this tuple. Its output contains a retained history, the tab schedule, a terminal keeper, marked portfolios, purchases, and weak specifications. The history's cost field contains the numerical room and bank estimates, the Brine estimates, monotone occupancy, gradewise height data, and terminal music records. All are attached to the same selected run.

Fix p, k, N. The service certificate is assembled from those records as follows.

First, restrict the retained history to levels $i \ge k$. At each adjacent scale in this range, the selected spring and purchased tariff give a local comparison. The endpoint support exponents are at least 100, so the finite interpolation constructor returns both its local gate ticket and the duty estimate. These are exactly the clauses needed for the retained-history field; the terminal endowment is the terminal clause of the same history.

Next, use the room and bank estimates from the same clock. Restrict the initial Brine inequalities to start at k: water, mint, split, and support bounds are the original inequalities with their scale index re-based. The occupancy schedule remains monotone under this restriction.

Finally, for every grade d with reference level at most k, the height record at (d, k, N, p) supplies a common constant X and the four data needed downstream: the activity and paid-cost bounds at all levels above k, the nested-support bound, the volume-wide footprint cover estimate, and the uniform bound on the local-section insertions. The terminal music record supplies the other part: its finite music begins with the calibrated factory entries, earns the source ledger at the grade reference, pays each adjacent transition through the same doors, and satisfies the terminal forced-profile identity. Joining these witnesses gives, for each such grade, the complete terminal service record. Since all these clauses are universally quantified over periods, grades, and terminal volumes in the funded record, their assembly is uniform over (p, k, N) and introduces no new history.

The same-history per-label construction applies this service record to a finite list of observables. Its multiscale propagation gives a served marked representation at the reference scale. For a finite set V of labels, let w_i be the positive marker weights, $X(U)$ the represented marked polynomial, A the local Wilson representative, $\sigma = \mathrm{scalar}(A)$ its common scalar, and $\mathcal{T}_k$ the vacuum cost per site. Write $\mathrm{toll}_w(I) = \prod_{i \in I} w_i$ and $\mathrm{price}_w(Y) = \sum_I |Y_I| \, \mathrm{toll}_w(I)$ for its mode toll and polynomial price. Suppose the terminal data satisfy the hypotheses of the selected root-test estimate: $k \le N$, the local section is sound, the representation is selected, weights are positive, the paid prices obey $w_i \, \mathrm{paid}_i \le \zeta_i(k)$, and the label sites have the stated migration, occupancy, and separation bounds. If $a = \mathfrak{f}_k \le 1$ is the paid label price, the finite response estimate gives the vault budgets $x = 3a$ and $y = a$. The local representative is lawful, has nonnegative coupling and pointwise nonnegative evaluation, and its support exponent is nonnegative. Choose $0 < s \le \min\{1, \theta\}$ below its positive

action threshold θ. Its served representation has reserve $f = 1$ and integrable mode coefficients, as furnished by the finite transition receipt. The pointwise price estimate is

$$\mathrm{price}_w(X(U)) \le \exp\big(\mathrm{scalar}(A) + |\Lambda_k|\big(\mathrm{exCost}(A) + \kappa\big) + |V|(x + y + 2)\big)\,.$$

Indeed this is the source fine-price formula with $x = 3a$, $y = a$, and reserve $f = 1$; the two extra units are charged by that reserve. The denominator lower bound is

$$\int \mathrm{eval}(A)\,d\mu \ge \exp\big(\mathrm{scalar}(A) - |\Lambda_k|\big(\mathrm{exCost}(A) + \beta\ \mathrm{baseBudget} + 4d_G \log(16/s)\big)\big)\,,$$

where $\beta \ge 0$ and d_G is the real dimension of the ambient matrix representation space. The same vacuum treasury receipt bounds the per-site bill

$$2\,\mathrm{exCost}(A) + \kappa + \beta\ \mathrm{baseBudget} + 4d_G \log(16/s) \le \mathcal{T}_k.$$

The common scalar cancels. Integrating a fixed mode coefficient, using its positive toll, and dividing by this denominator gives the finite local bound

$$\frac{\big|\int X_I\,d\mu\big|}{\int \mathrm{eval}(A)\,d\mu}\,\mathrm{toll}_w(I) \le \exp\Big\{|\Lambda_k|\big(2\,\mathrm{exCost}(A) + \kappa + \beta\ \mathrm{baseBudget} + 4d_G \log(16/s)\big) + |V|(4a + 2)\Big\}.$$

This follows directly from the displayed numerator and denominator: $|X_I|\,\mathrm{toll}_w(I) \le \mathrm{price}_w(X)$ pointwise, so $\big|\int X_I\big|\,\mathrm{toll}_w(I) \le \int \mathrm{price}_w(X)$; the prior is normalized, and $x + y + 2 = 4a + 2 \le 6$. The finite marked-coefficient estimate is therefore at most $\exp(|\Lambda_k|\mathcal{T}_k + 6|V|)$. In the public root test, let M be the number of replicated labels at terminal scale m, each with common weight $w > 0$, and put $q = (a_m/a_k)^4$, the four-dimensional volume ratio. The centered spacer estimate applies to this moment bound. The aligned side-count identity is $|\mathrm{Site}(\Lambda_k)| = Mq$. Thus the finite-volume exponent is $M(q\mathcal{T}_k + 6)$. Taking the normalized Mth root and dividing by w gives the centered root-test estimate

$$\exp(q\mathcal{T}_k + 6)/w.$$

This is the marked-copy spacer estimate of Section 103. The treasury contribution survives with factor q in the root exponent; it is not canceled by the finite-volume denominator. Finite backward Wilson integration preserves each mode integral, as proved in Section 22.

The group of source fields that supplies the service record, same-history propagation, and finite local price is the actual selected funded package; the displayed computations show how its clauses enter the bound without replacing them by unpriced per-volume assumptions. □

The upstream source boundary remains the construction of the funded record itself: its inputs are the selected campaign purchase, transition tariff, delayed scale choices, same-trail transfer data, climbing doors, and Freight records. The ordinary proof unit records the exact public caller chain and which quantitative inputs are furnished there.

25 Convergence of correlations along the full mesh sequence

We now pass from convergence of the local data at each fixed level to convergence of the finite-volume correlations themselves. The two limits are different: a terminal correlation contains interactions

involving several marked insertions, whereas the fixed-level convergence theorem controls the single-insertion coefficients and their forced profiles. The missing estimate makes the effect of the former interactions small at a sufficiently fine intermediate level, uniformly in the terminal cutoff.

Fix a period p, a finite set V of marked insertions, their grades d_i, coefficient vectors z_i, and one of the modes S of their finite marker algebra. Throughout the argument the history, the local fields, and the transfer maps belong to the one selected family. At cutoff N, let $R_{N,m}$ be its representation at level $m \le N$. Its product-Haar prior is denoted by $\mu_{m,p}$, and the positive terminal Wilson partition function by $Z_{N,p}$. The transferred polynomial density $X_{N,m}$ satisfies the exact identities

$$X_{N,m} = \mathcal{T}_{N,m} X_{N,m+1}, \qquad \int X_{N,m}\, d\mu_{m,p} = \int X_{N,m+1}\, d\mu_{m+1,p}.$$

Here $\mathcal{T}_{N,m}$ is the actual gauge-covariant transfer at the coupling of $R_{N,m+1}$; it is applied coefficient by coefficient. The integral of the coefficient of mode S, divided by $Z_{N,p}$, is the correlation under consideration. In particular all normalizations in this section use the same partition function.

25.1 The fixed-level data and their uniform bounds

At level m write $\Phi_{N,m,i}(U)$ for the single-insertion term in the exponent and $\omega_{N,m,i}(Q,U)$ for its forced profile, where Q records a support together with its exact set of flagged sites. Both include the field's scaling factor, its translated origin, and its prescribed motion. These are the actual transferred fields of Section 169.

For every fixed m above the entry levels of the finitely many grades, the fixed-level comparison supplies uniform Cauchy convergence in U of each $\Phi_{N,m,i}$ and each $\omega_{N,m,i}(Q,\cdot)$, and eventual uniform boundedness. This assertion uses both parts of that comparison. The finite-dimensional principal coefficient vectors are Cauchy. The captured remainder vectors are Cauchy in the local norm which controls their evaluation uniformly in U. Evaluation, translation, and the fixed motion preserve these bounds. The local masks are the same at this fixed level for every sufficiently large N, so multiplying by a mask preserves uniform Cauchy convergence even if the mask is discontinuous. The selected paths agree with the stationary paths of the common horizon; their larger-width convergence supplies the path part of the vacuum comparison. The fixed-level activity comparison supplies its remaining part.

Here and below finite initial segments cause no difficulty. A Cauchy sequence of coefficient vectors is bounded, including its finite initial segment. Since the coefficient price is a continuous seminorm times the fixed scaling factor, there is a number A_i such that the actual paid coefficient of insertion i at a fixed root r is at most A_i for every cutoff. Choose

$$w_i = \frac{\zeta_{d_i}(r)}{|A_i| + 1} > 0, \tag{113}$$

where $\zeta_d(r) > 0$ is the available single-insertion allowance. Then w_i times that paid coefficient is at most $\zeta_{d_i}(r)$. The root r is chosen after the finite list of grades, so these allowances are positive there.

For later use, at any fixed level m and for any fixed positive weights v_i, there exist $x_m, y_m \ge 0$, independent of all sufficiently large N, such that

$$v_i |\Phi_{N,m,i}(U)| \le x_m, \qquad \sum_Q v_i |\omega_{N,m,i}(Q,U)| \le y_m \quad (i \in V). \tag{114}$$

Indeed, take eventual bounds a_i and $b_{i,Q}$ for these functions and put $x_m = \sum_i v_i |a_i|$ and $y_m = \sum_{i,Q} v_i |b_{i,Q}|$. There are only finitely many supports and flag inventories on the fixed torus. Taking

the maximum of the finitely many entry thresholds proves both inequalities simultaneously. Measurability follows from the continuous local functions, measurable masks and absolutely summable forced series in their construction.

25.2 Separating the higher marker terms

Write $\mathsf{M}(\Phi) = \sum_i \Phi_i e_{\{i\}}$ for the single-marker polynomial. With the level and cutoff suppressed, the transferred density has the exact form

$$\begin{aligned}\mathcal{R}(H,K)(U) = e^{R.\mathrm{exponent}(U)}\,\mathrm{Pass}\Bigg(\mathsf{M}(\Phi(U)) + \sum_P H_P(U)\Bigg) \\ \times\, \mathcal{F}\big(\overline{R}_Q(U) + \mathsf{M}(\omega(Q,U)) + K_Q(U); \mathrm{flags}(U)\big)\,.\end{aligned} \tag{115}$$

The finite nilpotent exponential Pass was defined in Section 19. In this formula $\overline{R}_Q$ is the vacuum forced profile. For an inventory F, $\mathcal{F}(A;F)$ is the sum of $\prod_{Q\in\mathcal{Q}} A_Q$ over families $\mathcal{Q}$ with disjoint supports whose flag inventories have union F. This is a finite sum on the fixed torus. Every nonzero mode in H_P or K_Q contains at least two markers. Its marker origins lie in its support. The functions H_P vanish off the healthy set of P; the functions K_Q vanish unless the flag inventory is exactly that specified by Q. Both are local, gauge-invariant and measurable.

For positive marker weights v_i, use the weighted coefficient price of Section 19. The higher terms have nonnegative majorants a_P and b_Q bounding this price uniformly in the field. Their rooted support budgets, with weight $e^{e_m \operatorname{cov}_{B_m}(P)}$, are at most one, for both H and K. Here e_m is the level's support exponent, B_m its support scale and cov_{B_m} the cover number already used in those budgets.

We explain why these statements concern the actual density. The centered terminal construction gives (115) at level N with exponent $e_N - 3$. Its price uses

$$c_N = \kappa_N + r_N\,\mathrm{siteFactor}(B_N)u_N, \qquad (2B_N+1)^4 c_N \le 3, \qquad e^{c_N} - 1 \le 1,$$

where r_N is the allowed number of origins per site, κ_N the vacuum forced allowance and u_N the terminal insertion cost. These are the terminal estimates in the centered birth construction. At a step $m+1 \to m$, the purchased finite-menu estimate accepts support exponent $e_{m+1} - 4$ and produces the full exponent e_m. The incoming exponent is sufficient, including at $m+1 = N$, because $e_N - 4 \le e_N - 3$. The one-insertion transfer identities identify its single-marker coefficients with the actual $\Phi_{N,m,i}$ and $\omega_{N,m,i}$; they are not replaced by estimates. The local transfer ledger and the occupancy budget then give precisely the higher-term majorants above. Integrability propagates through the same transfer. Backward induction proves the assertion at every $m < N$.

The hypotheses for this induction are the funded finite-menu service of Section 17 and the centered service of Section 24. Their single-insertion component is obtained as follows. At the root use (113), and propagate weights by $v_i(m+1) = \varphi_m v_i(m)$. The centered scale estimates give

$$0 < \varphi_m, \quad 0 < \delta_m, \quad 1 \le \delta_m \varphi_m e^{B_m/128}, \quad B_{m+1}^2 \le L_m B_m^2, \quad B_m \ge 8, \quad e_m \ge 4.$$

The actual single-insertion transfer estimate and its coefficient price recursion propagate the root inequality to every later level. The terminal local section has cover number at most one and its purchased coefficient bound. The coefficient generators in that estimate therefore give the terminal insertion, the exact single-insertion recursion, and its local finite transfer ledgers. Summing their forced coefficients by support and exact flag inventory returns the original $\omega_{N,m,i}$: this is the defining profile identity, not a change of the fields. The remaining applicability conditions are the funded common service and the convergence comparisons just described; their simultaneous production is required in the common-family assembly.

25.3 Removing the higher terms at a fine intermediate level

Suppose distinct origins at level j are separated by at least B_j^2 in the lattice metric. A support of a nonzero higher term contains at least two such origins. The cover estimate of Section 19 gives

$$B_j \le 64 \operatorname{cov}_{B_j}(P).$$

Put $S_j = e^{B_j/64}$. For $|z| \le S_j$, replace H, K by zH, zK. The support and flag conditions are unchanged. The price majorant of a nonzero term is multiplied by $|z|$, and

$$|z| e^{(e_j-2)\operatorname{cov}_{B_j}(P)} \le e^{\operatorname{cov}_{B_j}(P)} e^{(e_j-2)\operatorname{cov}_{B_j}(P)} \le e^{e_j \operatorname{cov}_{B_j}(P)}. \tag{116}$$

Zero terms retain the zero majorant. Thus the inflated density has the same cost one at exponent $e_j - 2$. Its single-insertion terms are unchanged. Transport it from j down to a fixed root $r < j$. The first transfer accepts it since $e_j - 4 \le e_j - 2$; subsequent steps again have the full exponent. The preceding induction therefore applies, with the same root single-insertion terms and weights.

For clarity, separation also supplies the weight reservation needed in this transport. If a mode contains $q \ge 2$ distinct origins, the stronger cover estimate is $B_m q \le 64 \operatorname{cov}_{B_m}(P)$. Changing each marker weight by at most $e^{B_m/128}$ multiplies its toll by at most $e^{\operatorname{cov}_{B_m}(P)/2}$. This pays the half-unit reservation in the finite-menu transfer. The identities between parent origins preserve occupancy, and $B_{m+1}^2 \le L_m B_m^2$ preserves separation under the exact lattice embeddings. These observations verify the geometric conditions at every step of the transport.

Let $X_{N,j} = \mathcal{R}(H, K)$ be the actual density at $j < N$, and set

$$F_N(z) = \frac{1}{Z_{N,p}} \int \big(\mathcal{R}(zH, zK)(U)\big)_S \, d\mu_{j,p}(U).$$

This is holomorphic in z. Indeed, the marker algebra is nilpotent and the inventory sum finite, so the expression is a polynomial in z with measurable coefficients. The price bounds give integrable coefficient majorants on every smaller disk. At the root, let x_r, y_r be the eventual bounds (114) with weights w_i. The normalized density estimate proved in Section 19 gives

$$|F_N(z)| \le C_r := \frac{\exp\{|\Lambda_{r,p}|\tau_r + |V|(x_r + y_r + 2)\}}{\operatorname{toll}_w(S)} \qquad (|z| < S_j). \tag{117}$$

To apply it, transfer the inflated density to r. Its integral and its vacuum normalization remain exactly those above. The support budgets at r are one, and its single-insertion data have the root bounds x_r, y_r. This proves (117) with a constant independent of j and of all sufficiently large N. The bounds at j needed for integrability may depend on j; they do not enter C_r.

If $S_j > 2$, Cauchy's estimate bounds $|F_N'(z)|$ by $2C_r/S_j$ for $|z| \le 1$: the closed disk of radius $S_j/2$ about such a point lies in $|z| < S_j$. Integrating the derivative on $[0, 1]$ therefore gives

$$\left| \frac{1}{Z_{N,p}} \int (X_{N,j})_S \, d\mu_{j,p} - \frac{1}{Z_{N,p}} \int (\mathcal{R}(0,0))_S \, d\mu_{j,p} \right| \le 2C_r e^{-B_j/64}. \tag{118}$$

The first term is the original terminal correlation. Only the terms with at least two markers were erased in the second term.

25.4 Cauchy convergence of the erased expression

Fix j. Remove the scalar normalization from the representation and put

$$D_{N,j}(U) = e^{-R_{N,j}.\mathrm{scalar}}\mathcal{R}(0,0)(U), \qquad \ell_{N,j} = e^{-R_{N,j}.\mathrm{scalar}}Z_{N,p}.$$

The second term in (118) is exactly $\int (D_{N,j})_S \, d\mu_{j,p}/\ell_{N,j}$; the exponential scalar cancels. The fixed-level vacuum estimate gives a number $a_{j,p} > 0$ with $\ell_{N,j} \geq a_{j,p}$ eventually, and the fixed-level convergence theorem gives Cauchy convergence of $\ell_{N,j}$.

The functions $(D_{N,j})_S$ are uniformly Cauchy and eventually uniformly bounded. To see this directly from (115), combine the fixed-level vacuum, activity, single-insertion and forced-profile comparisons. There are finitely many coordinates on the fixed torus. Finite products, sums and the nilpotent exponential are continuous functions of those coordinates, uniformly continuous on the compact closures of their bounded ranges. Fixed masks can be kept in every term, as noted earlier. These operations consequently preserve uniform Cauchy convergence. As the prior is a probability measure,

$$\left| \int (D_{N,j})_S \, d\mu_{j,p} - \int (D_{M,j})_S \, d\mu_{j,p} \right| \leq \sup_U |(D_{N,j}(U) - D_{M,j}(U))_S| \longrightarrow 0.$$

The numerator is Cauchy and the denominator has a positive limit, so their ratio is Cauchy as well.

Now $B_j \to \infty$. Given $\varepsilon > 0$, first choose a fixed $j > r$ with $S_j > 2$ and $2C_r e^{-B_j/64} < \varepsilon/4$. At this fixed j the erased expressions differ by less than $\varepsilon/4$ for sufficiently large cutoffs $N, M > j$. Equation (118) and the triangle inequality bound the difference of the original correlations by $3\varepsilon/4 < \varepsilon$. This proves Cauchy convergence along the full cutoff sequence, without choosing an additional subsequence.

25.5 From nested-grid points to all distinct configurations

It remains to supply the separated origins used above and to pass to general points. Call a physical point ancestral if it lies on the mesh at some level. Integer block lengths imply that it lies on every finer mesh, and its addresses obey the exact embedding identity. Ancestral configurations are dense: round each coordinate to the nearest mesh point and use $a_m \to 0$.

For a fixed ancestral configuration of distinct points on the period p torus, let $\eta > 0$ be its minimum pairwise physical separation. For zero or one insertion take any $\eta > 0$ instead. Its lattice separation at level m is at least η/a_m. The chosen schedule satisfies $a_m B_m^2 \to 0$. Here is the quantitative reason for this limit. Its support-scale estimate eventually gives $B_{m+1}^2 \leq 2B_m^2$, whereas the block lengths satisfy $L_m \geq 3$ and $a_{m+1} = a_m/L_m$. Thus $a_{m+1}B_{m+1}^2 \leq (2/3)a_m B_m^2$ on a tail; iteration proves the limit. Choose a common root after all entry levels and after $a_m B_m^2 \leq \eta$ holds on that tail. The origins are then separated by B_m^2 at every later level. They are distinct at the root, so occupancy there is one, within the purchased occupancy schedule. This verifies all geometric conditions for the preceding Cauchy argument.

Finally apply the all-period finite-difference bounds of Section 28. They give local boundedness and an eventual common modulus on NoHit_p, the open set of configurations distinct modulo the physical period. The rounded lattice readouts need not themselves be continuous; their mesh errors tend to zero. The explicit local bound is $C(|x-y| + 2da_N)$ for a fixed finite coordinate dimension d, so one first chooses $|x-y|$ small and then the mesh tail. Cauchy convergence on the dense ancestral set in this domain extends to locally uniform convergence there: cover a compact set by finitely many neighborhoods with a common equicontinuity modulus, choose an ancestral point in each, and use the triangle inequality between each point and its chosen center. The corresponding higher

difference bounds give the smooth limiting correlations by the argument of Section 27. Restriction to the ordered Euclidean configuration domain gives the stated position-space ultraviolet limit.

In applying this result to the selected construction, the cutoff parameter is N itself and every branch is the history chosen by the same shooting construction. The alignment identities identify its paths with the common horizon; the selected transfer maps and their based local sections identify all of the Φ and ω above. The common centered service supplies the root allowances and their positive tails, the transfer ledgers, and the scale estimates. These identifications are part of the common-family assembly: a service on a different history would not establish the result here.

26 Ordered limits and Euclidean properties

The lattice spacing is sent to zero at fixed volume before the volume is sent to infinity. A subsequence of pairs (N_k, p_k) tending to infinity in both coordinates need not give either iterated limit. We record the compactness argument that produces one common pair of subsequences for all finite correlation signatures, and then explain which Euclidean properties pass to the resulting table. The construction is conditional on uniform finite-cutoff estimates stated below; their derivation from the Wilson trajectory is a separate input.

Fix one Wilson family $\mathcal{C}$. Write a_N for its mesh, P_p for its physical linear period, and $\ell_{N,p}$ for the number of sites in that period. For an n-tuple of field labels v, let $U_{N,p,v}$ denote the finite-lattice correlation, namely the finite expectation of the chosen insertions. We assume it is multilinear in each label. The factor a_N^{4n} appears when these values are smeared by a test function. Let Cyc_p be the open set of configurations for which no two distinct insertions have coordinate differences equal to an integer multiple of the physical period P_p in any coordinate. Write NoHit_p for configurations in which each distinct pair has at least one coordinate difference that is not an integer multiple of P_p. This excludes torus-coincident pairs but permits coordinatewise ties. Let Gen be the set for which every coordinate projection separates distinct insertions. Let Conf be the set of configurations with no coincident points. Thus Gen $\subset$ Conf; Gen also excludes noncoincident coordinate-projection ties, while Conf permits them. The table S below is the limiting Euclidean correlation table.

Here are the two uniform estimates used for extraction. In each statement "eventually" has its ordinary meaning along the indicated strictly increasing sequence.

Definition 26.1 (Uniform bounds and moduli)**.** *Fix an ordered pair $f = (f_{\mathrm{mesh}}, f_{\mathrm{per}})$ of strictly increasing sequences.*

1. *A local mesh modulus means that for every fixed period p, arity n, label tuple v, compact $K \subset \mathrm{Cyc}_p$, and $\varepsilon > 0$, there are $\delta > 0$ and j_0 such that, for $j \geq j_0$ and lattice configurations $x, y \in K$ at mesh $a_{f_{\mathrm{mesh}}(j)}$,*

$$|U_{f_{\mathrm{mesh}}(j),p,v}(x) - U_{f_{\mathrm{mesh}}(j),p,v}(y)| < \varepsilon \quad \text{whenever } |x - y| < \delta.$$

   The modulus may depend on p, n, v, K, ε.

2. *An expanding-volume modulus means that for every n, v, compact $K \subset \mathrm{Gen}$, and $\varepsilon > 0$, there are $\delta > 0$ and thresholds k_0 and $j_0(k)$ such that for $k \geq k_0$, $j \geq j_0(k)$, and lattice configurations $x, y \in K$ at mesh $a_{f_{\mathrm{mesh}}(j)}$,*

$$|U_{f_{\mathrm{mesh}}(j),f_{\mathrm{per}}(k),v}(x) - U_{f_{\mathrm{mesh}}(j),f_{\mathrm{per}}(k),v}(y)| < \varepsilon \quad \text{whenever } |x - y| < \delta.$$

   The mesh threshold may depend on the fixed volume index k.

3. *A compact norm bound means both: for every fixed p, n, v and compact $K \subset \mathrm{NoHit}_p$, the lattice values $|U_{f_{\mathrm{mesh}}(j),p,v}(x)|$ are bounded by one constant for all $x \in K$ and all sufficiently large j; and for every n, v and compact $K \subset \mathrm{Conf}$ they are bounded by one constant for all sufficiently large pairs in the ordered sense $\forall k \geq k_0\ \forall j \geq j_0(k)$.*

In the ordered-volume clause, the outer threshold is the period threshold and the inner mesh threshold may depend on that period. In particular these assumptions do not assert a single diagonal limit or a modulus uniform over all periods.

Proposition 26.2 (Common ordered extraction)**.** *Fix a countable algebraic basis for the local field-label spaces. Assume the local mesh modulus, expanding-volume modulus, and compact norm bound of Definition 26.1 for every finite tuple from this basis, for the same family $\mathcal{C}$ and initial order f. Then there are strictly increasing subsequences t, u of the mesh and volume indices. Put $\rho_k = f_{\mathrm{per}}(u(k))$. There are intermediate continuous functions $J_{p,v}$ on each Cyc_p and a table S continuous on* Gen*, with the following properties:*

1. *for each fixed p, n, v, the address-cell interpolants along the mesh subsequence $f_{\mathrm{mesh}} \circ t$ converge to $J_{p,v}$ uniformly on every compact subset of Cyc_p;*
2. *for each n, v, $J_{\rho_k,v}$ converges to S uniformly on every compact subset of* Gen *as $k \to \infty$;*
3. *for each separated test function suitable at fixed p, the finite-lattice smeared correlations converge first along the mesh subsequence to the integral against $J_{p,v}$, and then along the volume subsequence to the integral against S (extended by zero off* Gen*, which does not change this integral).*

All signature limits use the same t, u. The mesh subsequence is chosen independently of the requested test support. The volume limit is only asserted along $f_{\mathrm{per}} \circ u$, and the mesh limit remains an ordinary limit along the full mesh subsequence at each eligible fixed period.

Proof. We give the compactness argument, including the common-subsequence step. For each mesh index j and point x, let $Q_j(x)$ be the cell representative selected by the lattice address map, so that $|Q_j(x) - x| \leq c_n a_{f_{\mathrm{mesh}}(j)}$ for a constant c_n depending only on the arity. Define the position-space function $\widetilde{U}_j(x) = U_{f_{\mathrm{mesh}}(j),p,v}(Q_j(x))$. This function is defined even when x is not a lattice point. Let K be a compact subset of an open patch and choose a compact neighborhood K' still inside that patch. For $x, y \in K$ with $|x - y| < \delta/2$, the representatives $Q_j(x), Q_j(y)$ belong to K' for all sufficiently large j, and

$$|Q_j(x) - Q_j(y)| \leq |x - y| + 2c_n a_{f_{\mathrm{mesh}}(j)} < \delta.$$

The lattice modulus on K' therefore makes $\widetilde{U}_j$ asymptotically equicontinuous on K; the rounded functions need not themselves be continuous. The compact norm bound on K' makes their values bounded there for all sufficiently large j.

The basis-signature set and periods are countable. Choose a countable dense subset D of each patch. The values $\widetilde{U}_j(d)$, over all dense points and signatures for the fixed period, lie eventually in bounded closed disks in $\mathbb{C}$. A diagonal subsequence makes all these scalar sequences converge. We check that this gives compact-uniform convergence, not just convergence on D. Let K be compact in the patch and $\varepsilon > 0$. Choose a compact neighborhood K' inside the patch. Use the common modulus on K' to choose a radius δ for error $\varepsilon/3$. Finitely many balls of radius $\delta/3$ centered at points of $D \cap K'$ cover K. At these finitely many centers the selected subsequence is Cauchy. For any

$x \in K$, choose one such center d; for all large indices i, i', the two modulus errors $|\widetilde{U}_i(x) - \widetilde{U}_i(d)|$ and $|\widetilde{U}_{i'}(x) - \widetilde{U}_{i'}(d)|$ are each less than $\varepsilon/3$, while the center values differ by less than $\varepsilon/3$. The triangle inequality proves the subsequence is uniformly Cauchy on K. Completeness of $\mathbb{C}$ gives a uniform limit on K; limits agree on overlaps, and the same modulus estimate proves the resulting function is continuous. Exhausting the patch by compact sets gives one continuous limit there and uniform convergence on every compact subset. Diagonalize over the countable collection of signatures and a compact exhaustion of each patch to obtain one mesh subsequence t and all the intermediate functions $J_{p,v}$.

Now consider the sequence of mesh-limit functions $J_{f_{\mathrm{per}}(k),v}$ on Gen. For a compact $K \subset \mathrm{Gen}$ choose a compact neighborhood $K' \subset \mathrm{Gen}$ with K in its interior. For all sufficiently large k, K' lies inside the fundamental box; since $K' \subset \mathrm{Gen}$, this gives $K' \subset \mathrm{Cyc}_{f_{\mathrm{per}}(k)}$. If $x, y \in K$ and $|x - y| < \delta/2$, their representatives at mesh $a_{f_{\mathrm{mesh}}(j)}$ lie in K' and have mutual distance less than δ for all sufficiently large j. Apply the expanding-volume modulus at the fixed outer index k, then pass $j \to \infty$ using the already established mesh convergence on K'. This proves asymptotic equicontinuity of the functions $J_{f_{\mathrm{per}}(k),v}$ on K. Similarly, the ordered compact norm bound and mesh convergence give a uniform bound for these functions on K for all sufficiently large k.

Choose countable dense subsets in each generic configuration space and repeat the closed-disk diagonal argument, now with k as the sequence index. The finite-cover argument proves compact-uniform convergence on every compact subset of Gen and gives a continuous limit S. Diagonalizing over the countable signature index makes the same volume subsequence u work for all signatures. Use arity together with finite tuples from a countable algebraic basis as the countable signature index. Every arbitrary finite tuple of labels is a finite linear combination of such basis tuples, and finite lattice moments are multilinear, so finite linearity extends these limits to arbitrary labels without any further subsequence. This is a diagonal argument over the countable basis index, not over the uncountably many arbitrary test functions.

It remains to pass from pointwise limits to smeared moments. Let φ be a separated test, meaning its support K is compact and contained in Conf, and fix an eligible period p. Assume p is eligible for this test, so K lies strictly inside the comparison cell. Choose a compact neighborhood K_1 of K whose closure stays inside the comparison cell. For all fine meshes, every cell representative of a point in K lies in K_1. The cell boundaries in this bounded region lie in finitely many affine hyperplanes and have Lebesgue measure zero. The complement of the cyclic patch is a countable union of affine hyperplanes and therefore has Lebesgue measure zero. At almost every point in K, the cell representatives tend to that point and eventually belong to the cyclic patch. Extend $J_{p,v}$ by zero off that patch. The mesh convergence gives almost-everywhere convergence of the cell density to $\varphi J_{p,v}$. Since K is compactly contained in the fundamental cell and in Conf, it lies in the fixed-period no-hit set. Choose a compact neighborhood K_2 of K contained in that open set. For all sufficiently fine meshes the rounded samples of points in K lie in K_2. The fixed-period compact norm bound on K_2 bounds the finite-lattice moments for all sufficiently fine meshes. If $\varphi(Q_N(x)) \neq 0$, the cell representative $Q_N(x)$ belongs to K and hence to K_2. Therefore the absolute value of the piecewise constant cell density is at most $D\|\varphi\|_\infty$ on the whole fundamental box and is zero outside the union of cells meeting the support. The indicator of the bounded fundamental box times this constant is integrable. The exact cell identity writes the smeared sum as the integral of this density: each cell has volume a_N^{4n}. The strict cell margin keeps all samples inside the no-wrap region. At points outside the closed support K, the cell density is eventually zero because $Q_N(x) \to x$; outside the fixed box it is identically zero. Dominated convergence proves the fixed-period smeared limit.

For the volume limit, choose compact neighborhoods $K_0 \subset K_1 \subset \mathrm{Conf}$ of the support K. For sufficiently fine meshes, every cell representative of a point in K_0 lies in K_1. The ordered compact norm bound on K_1 gives one constant for these lattice values once the period index is sufficiently

large and then the mesh threshold is met. For all sufficiently large period indices, K_1 lies inside the fundamental cell. At generic points $x \in K$, the no-wrap equivalence identifies the cyclic patch with Gen, and the mesh evaluation lemma passes the lattice bound to $J_{f_{\mathrm{per}}(k),v}(x)$. The set $K \setminus \mathrm{Gen}$ is contained in finitely many coordinate hyperplanes and has Lebesgue measure zero. Therefore these intermediate functions are bounded by the same constant almost everywhere on K. Extend S by zero off Gen. The volume-point convergence holds at each $x \in K \cap \mathrm{Gen}$, hence almost everywhere on K. Dominated convergence gives the second integral limit for every separated test, including supports that meet noncoincident coordinate-projection ties. Here Gen supplies a full-measure set for the integral passage, while the larger set Conf contains the entire test support. □

The two compactness stages use distinct moduli: one at each fixed period and one in the ordered volume limit. The argument preserves the order of limits: no estimate uniform in an unbounded volume threshold is substituted for the permitted threshold $j_0(k)$.

Proposition 26.3 (Passage of Euclidean identities and positivity)**.** *Let S be a table obtained with the ordered smeared limits of Proposition 26.2. Assume in addition that:*

1. *finite-lattice correlations satisfy exact permutation and lattice translation identities, signed coordinate symmetries, conjugation, and normalization at admissible grid points; labels transform accordingly;*
2. *the ordered point limits exist on the generic domain and S is continuous there; assume also that for each signature it extends to a smooth function on* Conf *agreeing with S on* Gen*;*
3. *for each finite list of positive-time smeared observables, the exact finite-lattice pairing identity expresses each reflected product as a finite linear combination of separated smeared correlations covered by the preceding extraction, with coefficients independent of the lattice indices; the same combination defines the limiting smeared pairing. The finite-lattice smeared reflection form is nonnegative for all sufficiently late indices in the ordered sense.*

Then the corresponding permutation, translation, signed coordinate-permutation, and conjugation identities hold for S on the noncoincident domain, and the smeared reflection form of S is nonnegative for every finite list of admissible positive-time test observables. If, in addition, S satisfies the separate cap bound required by the Euclidean hierarchy interface, these conclusions together with normalization give its Euclidean positivity and symmetry clauses.

Proof. For permutation and conjugation, fix the signature and a generic configuration and approximate it by admissible lattice configurations. At each sufficiently fine mesh the finite-lattice change of variables gives the exact identity; the point limits along the common ordered filter and uniqueness of limits in $\mathbb{C}$ give the identity for S. A signed coordinate permutation preserves the lattice, so the same argument applies after transforming the grid points and the insertion labels. For translation by an arbitrary vector $z \in E_4$, choose lattice translations z_j converging to z as the mesh tends to zero. The finite trajectory's anchoring identity gives equality between the two grid evaluations separated by z_j. The grid points converge to the desired translated configurations, and continuity of the limiting correlation gives the translation identity. The generic set is dense in the noncoincident domain; if an identity has first been proved only there, continuity of both sides extends it throughout that domain.

For positivity, fix m positive-time smeared observables $A_1, \dots, A_m$ and coefficients $c_1, \dots, c_m \in \mathbb{C}$. The pairing assumption and the ordered convergence of each separated entry show that every

entry in the finite reflection matrix converges to its limiting smeared pairing. At each sufficiently late admissible lattice pair, reflection positivity gives

$$Q_{N,p} := \sum_{i,j=1}^{m} \overline{c_i} c_j \, \langle \theta A_i \, A_j \rangle_{N,p} \;\geq\; 0.$$

There are only m^2 terms. Since every matrix entry has the same ordered limit, finite-sum continuity gives

$$Q_{N,p} \longrightarrow \sum_{i,j} \overline{c_i} c_j \langle \theta A_i A_j \rangle_S.$$

This convergence holds along the same ordered filter. The nonnegative real half-line is closed, so the limit is nonnegative. Since the list and coefficients were arbitrary, the limiting smeared reflection matrix is positive semidefinite. Pointwise reflection positivity for point insertions then follows from the separate bump-approximation argument in Lemma 32.1; that step uses continuity away from coincident points and is not a consequence of generic point convergence alone.

These limit arguments establish only the clauses just named. The extraction gives continuity on Gen; passing identities to Conf requires the separate smoothness hypothesis there. Smoothness on Conf includes noncoincident coordinate-projection ties but excludes actual coincident points. Field quotients and Wightman reconstruction do not follow from a subsequence argument; they are separate hypotheses or results in the hierarchy and reconstruction units. □

Remark 26.4 (Scope of the extraction). The local estimates are not a proof that the Wilson family has a convergent subsequence by themselves. One must first establish the exact norm and oscillation bounds for that family, uniformly over the compact sets and with the specified fixed-period or ordered-volume quantifiers. Also, the extracted first surface is only continuous on generic patches. The later proof of smoothness on the full noncoincident set (including noncoincident coordinate-projection ties), followed by the full Euclidean hierarchy, has its own proof obligations.

27 Smooth limits from lattice differences

The extraction argument initially gives continuity only on generic configuration patches. A stronger family of difference bounds removes coordinate ties and identifies every derivative of the limit. This argument also explains the additional input needed for convergence along the whole mesh sequence: pointwise Cauchy convergence on a dense set, without another subsequence selection.

We state the analytic result on an arbitrary open set $U \subset \mathbb{R}^d$. In the fixed-period application, $d = 4n$, U is the set of configurations with no coincident points modulo the period, and the functions are periodic lifts of lattice correlations. Thus U includes configurations with coordinate ties and can cross a fundamental-domain seam. Write $\Gamma_N = a_N \mathbb{Z}^d$, where $a_N > 0$ and $a_N \to 0$. For functions $F_N : \Gamma_N \to \mathbb{C}$, define

$$\Delta_{N,j} F_N(x) = \frac{F_N(x + a_N e_j) - F_N(x)}{a_N}, \qquad F_{N,\alpha} = \Delta_{N,\alpha_1} \cdots \Delta_{N,\alpha_k} F_N$$

for a finite list α of coordinates, including the empty list. The order convention is that prefixing a coordinate applies one further difference to the previous function. Choose cell representatives $Q_N : \mathbb{R}^d \to \Gamma_N$ by rounding each coordinate, so $|Q_N x - x|_1 \leq d a_N$ and $Q_N x = x$ on Γ_N.

Theorem 27.1 (Full-sequence smooth convergence)**.** *Assume that for every compact $K \subset U$ and every finite list α, there are $C_{K,\alpha} < \infty$ and $N_{K,\alpha}$ such that*

$$|F_{N,\alpha}(x)| \le C_{K,\alpha} \quad (N \ge N_{K,\alpha},\ x \in \Gamma_N \cap K). \tag{119}$$

Suppose also that there is a dense subset $D \subset U$ for which $F_N(Q_N x)$ is Cauchy for every $x \in D$. Then there is a smooth function $f : U \to \mathbb{C}$ such that, for every α,

$$F_{N,\alpha}(Q_N x) \longrightarrow \partial^\alpha f(x) \quad \textit{uniformly on every compact subset of } U. \tag{120}$$

No further subsequence is taken. In particular, whenever $N_k \to \infty$ and $x_k \in \Gamma_{N_k}$ tends to $x \in U$, $F_{N_k,\alpha}(x_k) \to \partial^\alpha f(x)$.

Proof. We give the details in the ℓ^1 norm; equivalence of norms in fixed dimension gives the same local convergence in the Euclidean norm. All estimates below are made in a compact neighborhood strictly inside U, and finitely many thresholds in (119) can be replaced by their maximum.

Grid Lipschitz and Taylor estimates. Fix α, and abbreviate $G_N = F_{N,\alpha}$. On a slightly enlarged compact set, bound all its first differences by B_1 and all its second differences by B_2. Join sufficiently nearby grid points x, y by moving one coordinate at a time, always monotonically toward y. The path has total length $R = |y - x|_1$ and stays in the coordinate box between x, y. Each edge changes G_N by at most $a_N B_1$, including a negative edge by using the forward difference at its lower endpoint. Telescoping gives

$$|G_N(y) - G_N(x)| \le B_1 |y - x|_1. \tag{121}$$

Apply the same argument to each $\Delta_{N,j} G_N$. At every base point z of an edge of the path, $|\Delta_{N,j} G_N(z) - \Delta_{N,j} G_N(x)| \le B_2 R$. Subtract the linear function whose coordinate coefficients are $\Delta_{N,j} G_N(x)$ from the edge increments, and telescope again:

$$\left| G_N(y) - G_N(x) - \sum_j (y_j - x_j) \Delta_{N,j} G_N(x) \right| \le B_2 |y - x|_1^2. \tag{122}$$

The slightly enlarged compact set ensures that all shifted points used by these differences remain inside the region of the bounds. This is possible because only finitely many shifts occur and $a_N \to 0$.

Extension from dense pointwise convergence. Equation (121) and the rounding bound imply on any compact patch, for sufficiently nearby x, y,

$$|G_N(Q_N x) - G_N(Q_N y)| \le B_1 (|x - y|_1 + 2 d a_N).$$

Thus the rounded functions are asymptotically equicontinuous; they need not themselves be continuous. For $\alpha = \varnothing$, this and the Cauchy property on D imply uniform Cauchy convergence on each compact set. To see it explicitly, cover that compact set by finitely many sufficiently small balls with centers in D, lying in a common compact neighborhood in U. At the centers, choose one common Cauchy threshold. Choose the mesh still finer so that the two oscillation errors, from a point to its center for each of the two mesh functions, are small. The triangle inequality then bounds their difference uniformly on the whole compact set. Completeness of $\mathbb{C}$ gives a uniform limit. The displayed modulus makes the limit continuous. Limits on overlapping compact neighborhoods agree by pointwise uniqueness, giving a single continuous f on U.

This reasoning also proves a useful moving-point assertion: if the rounded G_N converge locally uniformly to a continuous g, and grid points $y_N \to y \in U$, then $G_N(y_N) \to g(y)$. Indeed, eventually

the points lie in one compact neighborhood; use $Q_N y_N = y_N$, uniform convergence there, and continuity of g. The same statement holds after any indices $N_k \to \infty$.

Cauchy convergence of the next differences. Suppose now the rounded $G_N = F_{N,\alpha}$ converge locally uniformly to g. Fix $x \in U$ and a coordinate j. Take a small $h > 0$ such that a compact neighborhood of the segment $[x, x + he_j]$ lies in U. Put $h_N = a_N\lceil h/a_N\rceil$, $x_N = Q_N x$, and $y_N = x_N + h_N e_j$. Then $h \le h_N < h + a_N$, so the moving-point assertion implies convergence of

$$V_N = \frac{G_N(y_N) - G_N(x_N)}{h_N} \longrightarrow \frac{g(x + he_j) - g(x)}{h}.$$

The Taylor estimate gives

$$|\Delta_{N,j} G_N(x_N) - V_N| \le B_2 h_N.$$

Here B_2 can be fixed on one compact neighborhood for every sufficiently small h. Given $\epsilon > 0$, first choose h so that $4B_2 h < \epsilon/2$, then choose N large so that $h_N \le 2h$ and V_N is Cauchy with error less than $\epsilon/2$. The triangle inequality proves that $\Delta_{N,j} G_N(Q_N x)$ is Cauchy. Bounds for its first differences give asymptotic equicontinuity by the previous argument. It therefore converges locally uniformly to a continuous function g_j.

Identification of derivatives and induction. Apply (122) with $Q_N x, Q_N y$, for fixed nearby x, y, and pass to the limit using the just-established convergence of all first differences. This gives

$$|g(y) - g(x) - \sum_j (y_j - x_j) g_j(x)| \le B_2 |y - x|_1^2.$$

The bound proves Fréchet differentiability, with $Dg(x)[u] = \sum_j u_j g_j(x)$. Since the g_j are continuous, g is C^1. Repeat this argument for every finite list α. Induction gives locally uniform limits for all iterated differences and identifies them with the corresponding derivatives of f. Their continuity proves $f \in C^k$ for every finite k, hence smoothness. The case $d = 0$ is simply convergence of one scalar sequence; there are no derivative steps. This completes the proof. □

Corollary 27.2 (Other fixed difference stencils)**.** *The same limits hold for any fixed finite composition of forward, backward or centered coordinate differences, provided every difference is divided by its full physical interval length.*

Proof. For fixed integers $b, c \ge 0$ with $b + c > 0$, telescope:

$$\frac{G_N(x + ca_N e_j) - G_N(x - ba_N e_j)}{(b + c)a_N} = \frac{1}{b + c} \sum_{r=0}^{b+c-1} \Delta_{N,j} G_N(x + (r - b)a_N e_j).$$

Each of the finitely many sample points approaches the same point, so the moving-point conclusion in Theorem 27.1 gives the same derivative for each summand. Translation averages commute with coordinate differences. A finite composition is therefore a finite average of shifted iterated unit forward differences, and has exactly the stated limiting derivative. Uniform convergence on compacts follows by using one slightly enlarged compact neighborhood for all the finitely many shifts. No assertion is made when the stencil sizes vary with N. □

The theorem is applied separately for each fixed period, label tuple and finite derivative request. Its derivative bounds are supplied by the full-precision lattice estimates, and its dense Cauchy input by the renormalization-group comparison of different cutoffs. Neither input follows from a subsequential limit alone. Once they are supplied, the fixed-period smooth kernel and all its locally uniform difference limits follow without further mesh extraction. The fixed-period smeared limits then follow by the cell-integral dominated-convergence argument of Proposition 26.2, applied to each fixed stencil.

28 Buying the full-mesh finite-difference bounds

We prove the finite-volume estimate needed before any continuum extraction. The important order is to pay for the bounded projected coefficient first, and only then use the exact Wilson difference identity. The result is uniform over every period at all sufficiently fine meshes. The proof has two parts: a finite dimensional absorption estimate for one coefficient row, and a packing argument which plants finitely many such rows into one large periodic box.

For a period p, mesh N, field list $v = (v_1, \dots, v_n)$, and finite list l of slot-and-coordinate directions, write $K_{N,p;v,l}(x)$ for the finite Wilson expectation of the product of the corresponding iterated periodic difference insertions at the positions $x = (x_1, \dots, x_n)$. The normalization is $\Delta_{N,\alpha}F(x) = a_N^{-1}\big(F(x + a_N e_\alpha) - F(x)\big)$. Thus the theorem below concerns the unscaled derivative kernel; the exact one-step identity used in its proof is $F(x + a_N e_\alpha) - F(x) = a_N \Delta_{N,\alpha}F(x)$.

Lemma 28.1 (Absorption of the projected coefficient)**.** *Let I be a directed set of mesh and period indices, and let X_i be a finite-dimensional normed coefficient space. Suppose that for each i there are a projection $P_i : X_i \to X_i$, a linear map $L_i : X_i \to Y_i$, and a constant $Z > 0$ such that*

$$\|P_i z\| \le Z\|L_i(P_i z)\|. \tag{1}$$

Let $x_i \in X_i$, put $X_i^0 = P_i x_i$, and set $\lambda_i = (1 + \|X_i^0\|)^{-1}$. Assume the following estimates hold eventually in I:

$$\|L_i x_i\| \le c, \tag{2}$$

$$\|L_i(\lambda_i(x_i - X_i^0))\| \le \frac{1}{2Z}. \tag{3}$$

Then eventually

$$\|X_i^0\| \le 2Zc + 1. \tag{5}$$

Proof. Write $R_i = x_i - X_i^0$. By linearity and (2),

$$\|X_i^0\| \le Z\|L_i X_i^0\| \le Zc + Z\|L_i R_i\|.$$

Since $\lambda_i = (1 + \|X_i^0\|)^{-1}$, (3) implies $Z\|L_i R_i\| \le (1 + \|X_i^0\|)/2$. Therefore

$$\|X_i^0\| \le Zc + \tfrac{1}{2}(1 + \|X_i^0\|),$$

and rearranging gives (5). All estimates were eventual, so their finite set of thresholds can be replaced by their maximum. □

Proposition 28.2 (One coefficient row is bounded at every period)**.** *Let $C \subseteq \prod_i X_i$ be one class of coefficient row families satisfying the following normalized-cost conditions. Suppose that the full family $x = (x_i)$ belongs to C, that C is closed under subtraction and pointwise multiplication by scalar families whose absolute values are at most one, and that every uniformly norm-bounded row family belongs to C. Suppose also that the detector estimates*

1. *$\|P_i z\| \le Z\|L_i(P_i z)\|$ for all $z \in X_i$ with a scale-independent $Z > 0$;*
2. *for the primary paid trial there are assay values $b_i \in Y_i$ with $\|b_i\| \le v$ and $\|L_i x_i - b_i\| \to 0$;*
3. *for every row family $z \in C$ satisfying $P_i z_i = 0$ for every i, one has $\|L_i z_i\| \to 0$.*

Then, eventually, uniformly over all indices in the directed mesh-period system,

$$\|P_i x_i\| \le 2Z(v+1)+1.$$

The uniformity means that the estimates in (2) and (3) hold beyond one mesh threshold for every period, as in the source full-mesh receipts.

Proof. From (2), eventually $\|L_i x_i\| \le v+1 =: c$. Put $X_i^0 = P_i x_i$ and

$$\lambda_i = (1+\|X_i^0\|)^{-1}, \qquad z_i = \lambda_i(x_i - X_i^0).$$

The scalar family $\lambda = (\lambda_i)_i$ satisfies $|\lambda_i| \le 1$, so $\lambda x \in C$. The family λX^0 is uniformly bounded by one, so it belongs to C as well. Therefore $z = \lambda(x - X^0) \in C$ by subtraction, and $P_i z_i = 0$ because $P_i(x_i - P_i x_i) = 0$. By (3), for all sufficiently fine meshes $\|L_i z_i\| \le (2Z)^{-1}$. Since $L_i z_i = \lambda_i L_i(x_i - X_i^0)$, estimate (1) gives

$$\|X_i^0\| \le Z\|L_i X_i^0\| \le Zc + Z\|L_i(x_i - X_i^0)\| \le Zc + \frac{1+\|X_i^0\|}{2}.$$

Rearranging proves the claim. The common mesh threshold is the maximum of the thresholds in (2) and (3). □

The cap alone bounds a projected row; the source separately uses that bound to pay the difference trial. Here are the precise closure steps. Let x be the coefficient row for the requested insertion and let F be its primary paid trial. The exact equation gives an owed row $e = \mathrm{upper}(x) - \mathrm{hops}(F)$, with a root-test cost u tending to zero. Applying the cap to (x, F, e) bounds Px uniformly. Thus $y = Px$ is a uniformly bounded row family and its primary insertion is paid by the bounded-coefficient receipt. The row $x - y$ lies in the normalized cost class by the class closure, and $P(x-y) = 0$. The sweep theorem therefore makes $\mathrm{upper}(x-y)$ owed with a cost tending to zero. The source identity is

$$\mathrm{hops}(F) = \mathrm{upper}(y) + \mathrm{upper}(x-y) - e.$$

It follows from $\mathrm{upper}(x) = \mathrm{upper}(y) + \mathrm{upper}(x-y)$ and the defining equation for e. The paid budget is closed under finite addition and subtraction: once each vanishing cost is below the unit allowance at a sufficiently fine mesh, the primary paid budget together with the two unit owed budgets yields a finite paid budget for the requested difference trial. This is exactly the projection-and-remainder argument; it does not require the unsplit equation residual to be bounded by a quantity that depends on $\|Px\|$.

Theorem 28.3 (Full-mesh bounds for all finite differences)**.** *Fix a family for which the preceding coefficient-row estimate and the paid-remainder decomposition just described apply to every finite list of fields and difference directions. Assume that bounded primary coefficient rows have paid root-test budgets, that the exact periodic one-step difference identity holds, and that translated insertions satisfy the product root-test inequality in distinct interior cells. Then for each finite field list v, difference list l, and $r > 0$, there are $B < \infty$ and a mesh threshold N_0 such that for every $N \ge N_0$, every period p, and every periodic position tuple x such that any two distinct slot indices have sites separated by at least r in physical units (even if the corresponding insertions agree),*

$$|K_{N,p;v,l}(x)| \le B. \tag{7}$$

The threshold is independent of p; the bound is independent of both N and p.

Proof. First fund one row. For a field label w, the constant coefficient row $x_i = w$ has bounded normalized cost. The assumed primary receipt and the preceding cap-and-remainder argument give a paid budget for each finite difference list applied to that insertion, eventually at every mesh and every period. For a general finite list $v = (v_1, \dots, v_n)$ and positional derivative list, make one trial for each slot by applying to v_j the subsequence of directions assigned to that slot. The exact one-step identity is the periodic discrete difference formula

$$F_j(x + a_N e_\alpha) - F_j(x) = a_N \, \delta_{a_N e_\alpha} F_j(x),$$

with the normalization of the difference operator. Iterating this identity and the exact product measure definition gives the desired multi-difference kernel, including wrapped coordinates. The list is finite. Choose a common centered scale beyond the finitely many thresholds in these paid records; at that scale all its root-test budgets hold eventually in mesh, uniformly over periods.

We now explain the two-scale tiling argument. Choose a tiling scale ℓ and a finer recentering scale k so that the physical tiling span a_ℓ is below r and

$$6n a_k < a_\ell.$$

First choose ℓ with $a_\ell < r$, then use $a_k \to 0$ to choose k with the displayed margin. At a fine mesh N beyond both scales, let t be the integer span of scale k and let T be a periodic tiling whose side is the integer span of scale ℓ. The aligned-scale identities give $a_N t = a_k$ and $a_N T.\text{side} = a_\ell$. Thus $n(2(3t)) < T.\text{side}$. The tiling placement lemma, applied with margin $3t$, gives one translated tiling and an assignment of the n insertion sites to cells, each site lying at least $3t$ from the cell boundary. This is uniform in the period: alignment gives the same physical side a_ℓ for every period index, once N is sufficiently fine.

The cell assignment is injective for a geometric reason. If two sites were assigned to the same cell, their periodic maximum-coordinate separation would be at most the integer tiling side $T.\text{side}$ of that cell. Multiplying by a_N would give physical separation at most $a_N T.\text{side} = a_\ell < r$, contradicting the assumption. The three-span margin is exactly what the centering lemma needs to transport each local insertion from the centered tile to its assigned cell.

For each occupied cell, recenter its insertion to the distinguished center cell. The Wilson law is invariant under translations and the signed coordinate symmetries used by the lattice. The centering lemma therefore transports both the interior support condition and the same root-test budget to the occupied cell. No statistical independence is asserted: all factors are evaluated under the same Wilson law. Instead, the planted-cell inequality applies to the translated copies in distinct cells and gives

$$\left| \mathbb{E} \prod_{j=1}^{n} F_j(x_j) \right| \le \prod_{j=1}^{n} b_j,$$

where b_j is the paid root-test bound for the jth trial. Their finite product is a constant independent of mesh and period. This proves (7). □

Remark 28.4 (What this estimate supplies). The statement is the full-mesh separated-kernel estimate: each finite difference order and each positive separation have one eventual mesh threshold valid at all periods. The bounds at fixed period also control compact families of separated positions and their coordinate ties. The estimate does not itself give convergence of smeared correlations; that is the separate same-history convergence estimate. With both inputs, the positional theorem follows by the interpolation, compactness, and dominated-convergence argument in Section 26.

29 From one-field purchases to the public comparison grid

This section isolates the quantitative steps on the selected finite-volume route to the comparison grid. It proves the finite-frame price estimate, the radial decay once the trajectory bounds hold, and the realification and four-term assembly steps. The local approximation and separated-pair receipts are substantive upstream inputs; they are recorded explicitly at the end rather than treated as consequences of the assembly.

Write a_j for the physical spacing at depth j, c_j for the number of columns there, and σ_j for its running scale. For a coefficient L of grade d, let $\|L\|_1$ be its coefficient norm in the finite frame. The price appearing in the construction is

$$K_j(L) = 2\exp(\ell_j)\|L\|_1,$$

where ℓ_j is the accumulated column logarithm. The rooted price of a block of side at depth m, compared with a fine observable at depth j, is

$$S_{m,j} = \exp\left(\left(\frac{a_m}{a_j}\right)^4 \mathcal{T}_j + 5\right),$$

with $\mathcal{T}_j$ the vacuum treasury. This is the four-dimensional marked-copy estimate proved in Section 103.

Lemma 29.1 (Finite-frame recovery from approximate coordinate atoms)**.** *Let V be a finite-dimensional complex coefficient space with coordinate norm $\|\cdot\|_1$ and basis $(e_s)_{s\in I}$. Suppose a family $(y_s)_{s\in I}$ satisfies $\|e_s - y_s\|_1 \le 1/4$ for every s. Define $Av = \sum_{s\in I} v_s y_s$. Then A is invertible and $\|A^{-1}z\|_1 \le 2\|z\|_1$ for all $z \in V$.*

Proof. For $v = \sum_s v_s e_s$, the triangle inequality gives

$$\|v - Av\|_1 = \left\|\sum_s v_s(e_s - y_s)\right\|_1 \le \sum_s |v_s|\|e_s - y_s\|_1 \le \tfrac{1}{4}\|v\|_1.$$

Consequently $\|Av\|_1 \ge 3\|v\|_1/4$. Thus A is injective and, by finite dimensionality, surjective. The displayed inequality gives $\|A^{-1}z\|_1 \le 4\|z\|_1/3 \le 2\|z\|_1$. □

Lemma 29.2 (Linear assembly preserves the local price)**.** *Fix a finite coefficient frame and a compact local region W. Suppose the complex basis approximants Z_s satisfy*

$$\sup_x |Z_s(x)| \le A_0, \qquad Z_s \text{ is supported in } W, \qquad \|Z_s - f_s\|_{\mathrm{root},m} \le \epsilon_0 \quad (s \in I),$$

where f_s are the finite-volume insertions corresponding to the frame columns. Let $\mathcal{C}$ denote the centered chamber-interior test class; assume $Z_s, f_s, Z_s - f_s \in \mathcal{C}$ and, for finite combinations,

$$\rho\left(\sum_s t_s H_s\right) \le \sum_s |t_s|\rho(H_s) \qquad (H_s \in \mathcal{C}),$$

where ρ is the rooted test. If $L = \sum_s t_s f_s$ with $\sum_s |t_s| \le K$, then

$$Z = \sum_s t_s Z_s$$

is supported in W, belongs to $\mathcal{C}$, satisfies $\sup_x |Z(x)| \le A_0 K$, and

$$\|Z - L\|_{\mathrm{root},m} \le K\epsilon_0.$$

Proof. Support is preserved under finite linear combinations. For every x, $|Z(x)| \le \sum_s |t_s||Z_s(x)| \le A_0 K$. Also $Z - L = \sum_s t_s(Z_s - f_s)$, so subadditivity and homogeneity give the root bound. The same closure assumptions give $Z - L \in \mathcal{C}$. No independence assumption is used: all rooted tests are evaluated in the same finite-volume Wilson law. □

Proposition 29.3 (A finite-depth real approximation receipt)**.** *Fix grade d, a star-real coefficient L, a block depth m, a station $J \le N$, and a fine depth $k < N$ with $C.ref(d) \le m \le k$. Assume the following selected-history data at these depths. The rooted quantity below is the source root test for the same finite-volume law and centered chamber; we assume its triangle inequality, homogeneity, and invariance under complex conjugation, as in the selected root-test receipts.*

1. *each coordinate coefficient has a complex local approximation in the window of radius $R_d + 4$, with supremum bound A_0 and rooted error at most $S_{J,m}/w$;*
2. *the finite-volume insertion columns span the requested insertion L, with coefficient sum at most $K_k(L)$, and the resulting complex sum Z lies in the centered chamber interior; that interior is preserved by the real-part projection used below;*
3. *moving each local approximation from the fine centered box back to the block of depth m preserves its support inside the block and its rooted error bound;*
4. *the scales fit: the local window embeds in the block, the station box is open at depth m, and*

$$w \,\|\, \mathrm{alto}_{m,d} \,\|\, \frac{4(1+c_k)}{\mathrm{gain}(k,m)} \le \frac{\zeta_{d,m}}{2}, \qquad w > 0, \quad \zeta_{d,m} > 0.$$

Then there is a real field B_N supported in the centered block, uniformly bounded by $A_0 K_k(L)$, whose rooted distance from the insertion L is at most

$$K_k(L)\,\frac{S_{J,m}}{w}.$$

The same bound holds at every period once the station and scale inequalities hold there.

Proof. Use the representation and coefficient bound in item 2. Transport the complex coordinate approximants to the common block and form their complex linear combination Z. The finite-sum lemma gives $|Z| \le A_0 K_k(L)$ and rooted error at most $K_k(L) S_{J,m}/w$. The exact selected-family reality identity says that the target insertion U is real when L is star-real. Put $B = \mathrm{Re}\, Z$. Then $|B| \le |Z|$, and $\mathrm{paint}(B)$ has the same local support. The selected realification lemma, namely chamber-interior compatibility for averaging a field with its conjugate, gives its chamber interior. Since $U = \overline{U}$,

$$\mathrm{paint}(B) - U = \tfrac{1}{2}\big((Z-U) + \overline{(Z-U)}\big).$$

Conjugation preserves the rooted test, and its triangle inequality and homogeneity therefore give the same rooted error bound for B. This is the realification argument of the selected route, not an assumption that the complex coordinate approximants are themselves real. The support condition and scale inequality ensure that all summands are legal tests in one block. Uniformity in period follows from the common station and aligned physical block in the hypotheses. □

Lemma 29.4 (Replacing a finite local word)**.** *Let $\mathcal{I}$ be a finite index set, $\mathcal{T}$ a finite tiling, and τ_a its copy maps, indexed by the cells $a \in \mathrm{Cell}(\mathcal{T})$. Assume each copy map is additive. Let μ be a probability law and ρ the same nonnegative rooted test on all functions considered. Let $a : \mathcal{I} \to \mathrm{Cell}(\mathcal{T})$ be*

injective. Suppose X_i, Y_i lie in a linear local test class that is closed under cell copies, and suppose all finite products below are integrable.

$$\rho(X_i) \le M, \qquad \rho(Y_i) \le M, \qquad \rho(Y_i - X_i) \le \delta \quad (i \in \mathcal{I}),$$

where $M, \delta \ge 0$. Assume the planted-product estimate

$$\left| \int \prod_{i \in \mathcal{I}} \tau_{a(i)}(Z_i)\, d\mu \right| \le \prod_{i \in \mathcal{I}} \rho(Z_i)$$

for every family in this local test class. If $|\mathcal{I}| = n \ge 1$, then

$$\left| \int \prod_{i \in \mathcal{I}} \tau_{a(i)}(Y_i)\, d\mu - \int \prod_{i \in \mathcal{I}} \tau_{a(i)}(X_i)\, d\mu \right| \le n\delta M^{n-1}.$$

For $n = 0$ the two empty products agree and the difference is zero.

Proof. Order the finite index set as $i_1, \dots, i_n$, and define $Z^{(r)}$ to use Y_{i_s} for $s \le r$ and X_{i_s} for $s > r$. Then the difference of the two products is the sum, for $r = 1, \dots, n$, of the product with $\tau_{a(i_r)}(Y_{i_r} - X_{i_r})$ in position r and the hybrid factors in all other positions. The copy operation is additive, including its reflected cases, so this is an exact telescoping identity. Each hybrid family remains in the local test class. The planted-product estimate bounds the absolute value of the rth integral by δM^{n-1}: the changed factor costs at most δ and each of the other $n-1$ factors costs at most M. Summing these n bounds proves the claim. □

Remark 29.5 (The selected pair receipt). This is the quantitative replacement step used to turn centered single-word receipts into whole-word product receipts. Its local inputs are: for each copy and each mesh/period in the fused late set, the exact insertion X_i and real prepared insertion Y_i lie in the centered chamber-interior test class, both rooted costs are at most a common M, their difference costs at most $\mathrm{costHint}(k)$, and the prepared field has a finite supremum bound. The finite-dimensional spin frame is absorbed by summing its centered and purchase budgets. For an injective list of slots, the measure's planted inequality is applied to each telescoping term above. Thus the word-product discrepancy is bounded by $n\ \mathrm{costHint}(k) M^{n-1}$, uniformly over the finite grid positions; concatenating two finite lists gives the four pair receipts used in the reflection-difference assembly. The selected proof still needs the origin receipts that feed the centered single-word estimates, but the many-factor replacement estimate itself is proved here.

Lemma 29.6 (The radial choice makes the coefficient price affordable)**.** *Fix d, m, J and $\varepsilon > 0$. Suppose the scale schedule satisfies*

$$\sigma_j \longrightarrow +\infty, \qquad \ell_j = o(\sigma_j), \qquad \log(1 + c_j) = o(\sigma_j), \qquad \mathrm{gain}(j,m)^{-1} \le C_m e^{-\gamma(\sigma_j - \sigma_m)}$$

for some $\gamma > 0$. Assume $\| \mathrm{alto}_{m,d} \| > 0$, $\zeta_{d,m} > 0$, and $\mathrm{gain}(j,m) > 0$. Let w_j be the positive allowance

$$w_j = \frac{\zeta_{d,m}\, \mathrm{gain}(j,m)}{8 \| \mathrm{alto}_{m,d} \| (1 + c_j)}.$$

If $\|L\|_1$ and $\zeta_{d,m}^{-1}$ are fixed, then for all sufficiently large j,

$$K_j(L)\, \frac{S_{J,m}}{w_j} \le \varepsilon.$$

Proof. For fixed d, m, J, L, all factors except $\exp(\ell_j)$, $1+c_j$, and $\mathrm{gain}(j,m)^{-1}$ are constant in j; the remaining treasury ratio in $S_{J,m}$ is fixed at the chosen m, J. By the first two hypotheses, for any $\eta > 0$ both $\exp(\ell_j)$ and $1+c_j$ are eventually at most $\exp(\eta\sigma_j)$. Choose $\eta < \gamma/3$. The gain factor is bounded by a constant times $e^{-\gamma\sigma_j}$, after absorbing $e^{\gamma\sigma_m}$ into that constant. The product is thus bounded by a constant times $e^{-(\gamma-2\eta)\sigma_j}$, which tends to zero. This gives the claimed threshold. The source obtains the first two subpower estimates from the accumulated column logarithm and column-count bounds; the gain decay is the scale-flight estimate. □

Theorem 29.7 (Conditional assembly of the public comparison grid)**.** *Fix a finite list of star-real tokens v_i satisfying the source plainness condition, and a generic grid of pairwise separated positions x_i. Write H_i and R_i for the exact unreflected and reflected whole-word insertions at common volume, period, and translation; reflection acts on both labels and positions. For each approximation index k, suppose there are real measurable approximants $F_{k,N,p,t,i}$ and $G_{k,N,p,t,i}$ to the unreflected and reflected insertions. Assume:*

1. *$A(k) \geq 0$, $\delta(k) \geq 0$, and $\delta(k) \to 0$. For each fixed k, t, i, eventually in the fused mesh/period filter f.*fuse*, $F_{k,N,p,t,i}$ and $G_{k,N,p,t,i}$ are measurable and bounded in absolute value by the common reserve $A(k)$.*

2. *For each k and each $i < j$, there is a translation threshold such that for every t beyond it, eventually in f.*late *in (N,p), the four approximating products and their target products are integrable and their integrals differ by at most $\delta(k)$:*

$$\int F_iF_j \leftrightarrow \int \mathrm{Re}(H_iH_j), \qquad \int F_iG_j \leftrightarrow \int \mathrm{Re}(H_iR_j),$$
$$\int G_iF_j \leftrightarrow \int \mathrm{Re}(R_iH_j), \qquad \int G_iG_j \leftrightarrow \int \mathrm{Re}(R_iR_j).$$

   The arrows mean that the absolute difference of the two real numbers is at most $\delta(k)$.

*Then the real plots $P_{k,N,p,t,i} = F_{k,N,p,t,i} - G_{k,N,p,t,i}$, with reserve $2A(k)$ and error schedule $4\delta(k)$, define a covering satisfying the exact Works quantifiers: controls hold eventually in f.*fuse *for each fixed k, t, i; and for every k and $i < j$, its pair product differs from the bill*

$$\int \mathrm{Re}\big[(H_i - R_i)(H_j - R_j)\big]$$

*by at most $4\delta(k)$ eventually in f.*late *for every sufficiently large translation t. Thus k is the approximation index whose error tends to zero; it is not the extraction index. This is a conditional finite assembly statement, not a proof of the four local pair receipts.*

Proof. Put $P_i = F_i - G_i$. The bound $|P_i| \leq |F_i| + |G_i| \leq 2A(k)$ proves the reserve assertion. The identity

$$P_iP_j = F_iF_j - F_iG_j - G_iF_j + G_iG_j$$

is an exact identity of real functions. The target bill has the same four-term expansion after taking real parts of the exact complex products $H_iH_j, H_iR_j, R_iH_j, R_iR_j$. By the four assumed receipts and the triangle inequality, the difference of the two integrals is at most $4\delta(k)$. The common late set exists because there are finitely many controls and four receipts per pair. The translation threshold is chosen first, followed by the mesh/period late set. This is precisely the order of quantifiers in Works. □

Remark 29.8 (Selected-route boundary). The public endpoint reaches this construction through the selected run's history records, then uses the finite-dimensional insertion recovery, a finite real sum, a radial choice of the intermediate depth, and the four-term reflection expansion. The arguments above prove the coefficient recovery, finite-sum estimates, radial decay under the displayed scale bounds, and the final four-term assembly. The selected-history receipts that remain inputs are: the real local coordinate approximations, their rooted one-field errors, the back-transport support and price estimate, and the separated pair integral estimates. These are the substantive content of the history lemmas producing the local approximants and their served pair bounds; this section does not treat their names alone as proofs. The public theorem also separately requires same-history full-mesh convergence for positioned correlations. This chapter supplies the real grid estimate on its own hypotheses and does not identify it with that convergence theorem.

30 The selected one-scale linear interpolation

We next expand the finite construction that turns the one-column coupons into one coherent coarser voice. It is essential that the coupons be combined before the coefficient is chosen: choosing separately for each label would not give a single linear voice, and the next scale's cone estimate would not apply. Let $\mathcal{T}_m$ denote the actual fine-to-coarse density map, including its physical block-volume factor. It is integration against the selected transition kernel, so its linearity on the displayed integrable columns follows from the probability-kernel identity.

Let F be the finite coefficient frame, V a voice at the fine scale, and write I for its invertible coefficient readout. Assume the heard identities hold: the played coefficient has x-coordinate $I(v)$, its density is a complex-linear function of v, the density obeys the stated motion covariance, and its charge is bounded by $s\|\operatorname{core} I(v)\|_1$, where s is the fine column count. For the profile conclusions below assume the coefficient law is sound under local reorientation, the forced profiles transform covariantly, and the local evaluation is nonzero on every configuration. Density covariance alone does not imply covariance of the exponential entry. Let L be the block scale and T the selected heading. Set

$$D(z) = \|\operatorname{vowel}_L(\operatorname{core} z)\|_1, \quad c = C + \eta s, \quad \epsilon = \eta(1+s), \quad R_F = \operatorname{orbitCost}(F).$$

The frame has basis $(e_j)_{j\in J}$ and finite motion set $\mathcal{M} = \text{Signed} \times \{0,1\}$; the second coordinate is the independent conjugation flag. Write $\sigma_g(c) = c$ or $\bar{c}$ according to that flag, and gz for the corresponding coefficient action. The exact repainting coefficient is

$$\chi_{g,j}(z) = |\mathcal{M}|^{-1}\sigma_g((g^{-1}z)_j), \qquad R_F = 1 + \sum_{g\in\mathcal{M}}\sum_{j\in J}\|g(e_j)\|_1.$$

It is complex-linear: applying g^{-1} to cz contributes $\sigma_{g^{-1}}(c)$, and the outer σ_g changes it back to c. For fixed g, coordinate expansion gives

$$\sum_j \chi_{g,j}(z)g(e_j) = |\mathcal{M}|^{-1}z;$$

summing over g proves exact reconstruction. The weighted bounds used below are stronger than the unweighted reconstruction bound. For $h(z) = \|\operatorname{core} z\|_1$ or $h(z) = \|\operatorname{vowel}_L(\operatorname{core} z)\|_1$, coordinate expansion gives

$$\sum_{g,j}|\chi_{g,j}(z)|h(e_j) = |\mathcal{M}|^{-1}\sum_g h(g^{-1}z).$$

For each fixed motion, the weighted coordinate sum is exactly $h(g^{-1}z)$, by expanding in the frame basis; the core and grade weights commute with motion. The finite patch-cost estimate gives $h(g^{-1}z) \le R_F h(z)$, and therefore yields

$$\sum_{g,j} |\chi_{g,j}(z)| h(e_j) \le R_F h(z). \tag{2}$$

This weighted identity also explains why one must include the motion cost: a motion need not preserve the frame's ℓ^1 norm. The same coefficient formula with $h(z) = \|z\|_1$ gives the unweighted budget used for finite-wing additions:

$$\sum_{g,j} |\chi_{g,j}(z)| \le R_F \|z\|_1,$$

by the same finite patch-cost estimate. Assume each grade weight in vowel_L lies in $[1, L^d]$; this is the grading convention used below. The selected purchase inequalities are

$$C \ge 1, \quad \eta \ge 0, \quad \eta(1+s)R_F \le \mathrm{barunit}(L), \quad (C + \eta s)R_F \le K_{J(m+1)}. \tag{1}$$

For each basis vector e_j, set $v_j = I^{-1}e_j$. The fine played column $V(v_j)$ is formatted and integrable, and its remainder is bounded by its charge. The selected one-column purchase record has three clauses: the principal insertion when $j \ne 0$, every regular remainder atom with its support and supremum bounds, and every forced atom with its support-mass bound. The forced clause applies only to nonzero addresses; the keeper condition places every such address in the funded wing. The unit coordinate uses the exact vacuum coupon. The density has an absolutely convergent principal-plus-regular-plus-forced expansion: the regular and forced majorants are dominated by the coefficient envelope and remainder bound. For each component, the corresponding purchase clause gives a coupon. The coupon-merger theorem sums their densities and profiles termwise; Tonelli applies to the nonnegative majorants, and each scalar cost sum is bounded by the given purchase budget. This gives the exact basis coupon with lead $T(L)e_j$. Before simplification its three budgets are

$$\eta(\|\operatorname{core} e_j\|_1 + V.\operatorname{charge}(v_j)), \quad CD(e_j) + \eta V.\operatorname{charge}(v_j), \quad t + 2V.\operatorname{charge}(v_j).$$

Each paid column receipt controls its lead after every motion: the transported coefficient $X_{g,j}$ satisfies $\|X_{g,j} - T(L)(ge_j)\|_1 \le \epsilon \|\operatorname{core} e_j\|_1$ for all g, j. This is the all-motion clause in the one-column payment; no norm-isometry of a motion is used. The heard charge estimate gives $V.\operatorname{charge}(v_j) \le s\|\operatorname{core} e_j\|_1$; $\|\operatorname{core} e_j\|_1 \le 1$ and $D(e_j) \ge \|\operatorname{core} e_j\|_1$. Hence the budgets are at most $\epsilon\|\operatorname{core} e_j\|_1$, $cD(e_j)$, and $t + 2s$. The marked coefficient/profile payment is proved in Section 95. The forced component is paid there after the forced allowance is at most one. The selected primary and regular-card comparisons remain explicit analytic inputs; the one-scale local transition estimate of Section 57 does not itself show that the selected history satisfies those two purchase clauses. Every component here is evaluated under one background law and one transition kernel.

Lemma 30.1 (The paid one-step cone used in the interpolation)**.** *Let $K \ge 1$, $s \ge 1$, $L \ge 1$, and put $q = L^{1/8}$, $e = q^{-3}$. Suppose*

$$100000D^6(1+s)^2 \le q, \quad K \le D, \quad 64D^2 \le c_-, \quad 64D^2 \le c_+.$$

Let B be an invertible linear map with both B and B^{-1} of ℓ^1 norm at most K, commuting with deletion of the unit coordinate. For vectors u, v set $y = \mathrm{vowel}_L(v)$ and assume

$$\|u - By\|_1 \le e(\|\operatorname{core} v\|_1 + t), \qquad r \le K\|\operatorname{core} y\|_1 + et, \qquad 0 \le t \le s\|\operatorname{core} v\|_1.$$

Then

$$r \le c_- \| \operatorname{core} u\|_1, \qquad \|v\|_1 \le c_+ \|u\|_1, \qquad \|u\|_1 \le c_+ L^d \|v\|_1.$$

Proof. The bank inequality, $K \le D$, and $K, D \ge 1$ imply

$$2Ke(1+s) = \frac{2K(1+s)}{q^3} \le 1.$$

Indeed the assumed lower bound on q is at least $2K(1+s)$, and $q \ge 1$. The error bound and $t \le s\| \operatorname{core} v\|_1$ therefore give

$$2K\|u - By\|_1 \le \| \operatorname{core} v\|_1 \le \| \operatorname{core} y\|_1,$$

where the second inequality is the coordinatewise fact that all grade-weights in vowel_L are at least one. The elementary perturbation estimate for B now gives

$$\|y\|_1 \le 2K\|u\|_1, \qquad \|u\|_1 \le 2K\|y\|_1.$$

Apply the same estimate after deleting the unit coordinate. The map B commutes with that deletion and deletion is contractive, so the same error bound holds for $\operatorname{core} u - B(\operatorname{core} y)$. Thus

$$\| \operatorname{core} y\|_1 \le 2K\| \operatorname{core} u\|_1.$$

Also

$$et \le e(1+s)\| \operatorname{core} v\|_1 \le \| \operatorname{core} y\|_1,$$

since $2Ke(1+s) \le 1$ and $2K \ge 1$. Hence

$$r \le 2K\| \operatorname{core} y\|_1 \le 4K^2\| \operatorname{core} u\|_1 \le c_-\| \operatorname{core} u\|_1.$$

Finally, the grade weights also satisfy $\|y\|_1 \le L^d\|v\|_1$. Consequently

$$\|v\|_1 \le \|y\|_1 \le 2K\|u\|_1 \le c_+\|u\|_1, \quad \|u\|_1 \le 2K\|y\|_1 \le 2KL^d\|v\|_1 \le c_+L^d\|v\|_1.$$

The two room assumptions imply the needed $4K^2 \le c_-$ and $2K \le c_+$. □

Proposition 30.2 (Common linear interpolation of the purchased columns)**.** *Assume $s \ge 1$, $L \ge 1$, and $t \ge 0$. Write the heading as $T(L) = B \circ \operatorname{vowel}_L$, where B commutes with core deletion and $\|B\|_1, \|B^{-1}\|_1 \le D_0 := K_{J(m+1)}$. Assume the bank and room bounds*

$$100000D_0^6(1+s)^2 \le L^{1/8}, \qquad 64D_0^2 \le c_-, \quad 64D_0^2 \le c_+,$$

with $\operatorname{barunit}(L) = L^{-3/8}$ *and the purchase inequalities* (1)*. These are exactly the scalar hypotheses of Lemma 30.1, with $K = D = D_0$. The basis coupons have their stated exact density/profile identities and a common gauge door; their wings are the motion images of the common funded wing. There is a coarser voice Q with:*

1. *every $Q(v)$ is formatted and has remainder bound $a(v) = cR_FD(Iv)$ at the coarse forced allowance;*
2. *its readout $I_Q(v) = Q(v).x$ is a linear isomorphism;*
3. *at the centered anchor, the exact transition identity $Q(v).\text{density} = \mathcal{T}_m(V(v).\text{density})$ holds;*

4. *its lead discrepancy satisfies*

$$\|Q(v).x - T(L)I(v)\|_1 \le \epsilon R_F \|\operatorname{core} I(v)\|_1;$$

5. *the finite-wing witness for the exponential entry and forced profile of $V(v)$ has budgets*

$$R_F\|I(v)\|_1(t+2s) \quad \textit{and} \quad a(v),$$

respectively.

The coefficient, regular, and forced fields are complex-linear in v. The density and readout obey motion covariance, while the forced profile and exponential entry obey the corresponding covariance after reindexing their address dictionaries; the conjugation flag is independent of orientation.

Proof. For each basis coordinate put $v_j = I^{-1}e_j$. Its played coefficient has x-coordinate e_j by the heard identity. Let Z_j be the coefficient from its elementary coupon. For each motion g, the transition slice has a move taking the coarse origin to the required fine anchor. Reorient Z_j by that move and, on the conjugating branch, apply complex conjugation to all coefficient fields. The centered transport identities give a coefficient $Z_{g,j}$ with

$$Z_{g,j}.x = g(Z_j.x), \qquad Z_{g,j}.\text{density} = g(\mathcal{T}_m(V(v_j).\text{density}))\,,$$

with the same remainder and two Tale budgets. The remainder assertion follows from the turned tab estimate and the coupon's future bound; the finite wing is carried to its image under the move. Soundness of the transition door gives the common transport, and the side condition $2 \le n_m$ makes every move admissible on the selected schedule. This explains the coordinate, format/remainder, density, and wing transport used below.

Index these transported coefficients by the finite set

$$\mathcal{I} = \{(g,j) : g \text{ a signed motion},\ j \text{ a frame coordinate}\}.$$

For a vector v, assign weight

$$w_v(g,j) = \chi_{g,j}(Iv).$$

The finite orbit coefficients are the averaging/rebasing coefficients; their identity is the finite reconstruction formula

$$Iv = \sum_{(g,j)\in\mathcal{I}} w_v(g,j)\, g(e_j). \tag{2}$$

Define $Q(v)$ by summing the transported coefficients with these weights, using the disjoint-address merge that retains their support and label data. Equation (2) makes the operation complex-linear in v. Since the index set is finite, there is no limiting argument in this assembly.

The x-coordinate is the weighted sum of the transported column coordinates. Its leading part is reconstructed by the finite motion average, since $T(L)$ intertwines the motion action. For the error use the paid all-motion estimate and (2) with $h(z) = \|\operatorname{core} z\|_1$:

$$\begin{aligned}
\|Q(v).x - T(L)Iv\|_1 &\le \sum_{g,j} |\chi_{g,j}(Iv)|\, \|X_{g,j} - T(L)(ge_j)\|_1 \\
&\le \epsilon \sum_{g,j} |\chi_{g,j}(Iv)| \|\operatorname{core} e_j\|_1 \\
&\le \epsilon R_F \|\operatorname{core} Iv\|_1.
\end{aligned}$$

Each transported remainder has price $cD(e_j)$. Summing its absolute coefficient-weighted price and using (2) with $h(z) = \| \operatorname{vowel}_L(\operatorname{core} z)\|_1$ gives

$$\sum_{g,j} |\chi_{g,j}(Iv)| cD(e_j) \le cR_F D(Iv).$$

The finite coefficient merge preserves formatting and adds these remainder budgets, proving item 1.

For the density, linearity of the transition integral is justified by the integrability of each played column. Motion covariance of the heard density identifies the transported coupon density with the corresponding motion of the transitioned basis density. Sum with the weights in (2): the result is $\mathcal{T}$ applied to the density of $V(v)$. The regular and forced address arrays are finite coefficient sums, hence complex-linear in v; their address dictionaries are reindexed by motion and need not themselves be invariant. The heard density is motion-covariant by hypothesis. The forced profile covariance is an explicit assumption. For the exponential entry, use the coefficient identity $\operatorname{density}(U) = \operatorname{eval}(U) \operatorname{exponentialEntry}(U) + \operatorname{forcedDress}(U)$. The profile covariance and lawful dressing identity give covariance of the last term; subtract it from the density covariance identity and divide by the nonzero local evaluation. This proves the needed exponential-entry covariance. Thus the readout and the two profile outputs obey the finite motion reindexing identities, without asserting invariance of raw address arrays. Every summand is supported on the same common stencil, so the output support is independent of v. The unit identity follows because the core of the unit coordinate vanishes, so the lead error is zero, and the heading preserves the unit; the scale map preserves the unit normalization.

For each v, put $u = Q(v).x$ and $y = \operatorname{vowel}_L(Iv)$. The heading map gives the two-sided map bound with $K = K_{J(m+1)}$; (1) bounds the leading error by $\operatorname{barunit}(L)\| \operatorname{core}(Iv)\|_1$ and the new charge $a(v) = cR_F D(Iv)$ by $K\| \operatorname{core} y\|_1$. Heard gives $V.\operatorname{charge}(v) \le s\| \operatorname{core}(Iv)\|_1$. Apply Lemma 30.1 with the room and bank bounds. It yields the new charge cone and the forward/backward coordinate estimates. In particular, if $Q(v).x = 0$, its estimate $\|Iv\|_1 \le c_+\|Q(v).x\|_1$ gives $Iv = 0$, and therefore $v = 0$. Thus the linear coordinate map of the new voice is injective and, in finite dimension, is an isomorphism.

Finally, transport each finite-wing witness to its motion address, multiply it by $w_v(g, j)$, and take their finite union. The finite repainting identity and motion covariance show that the exponential entries and forced profiles sum to those of $V(v)$. The first budget is bounded by

$$\sum_{(g,j)} |w_v(g, j)|(t + 2s) \le R_F\|Iv\|_1(t + 2s),$$

while the second is the already proved remainder-price sum $a(v)$. The finite-wing addition rule therefore gives the required Tale witness. This proves item 5 and completes the common-pencil construction. □

The proposition is the finite assembly used by the selected one-scale transition. The source status of its three purchase clauses is not uniform. The forced clause is supplied by the large-projection estimate after the forced allowance is at most one and the keeper contains every nonzero slot. The regular clause still requires the regular-card estimate for the small-support, vanishing-jet principal term. The main clause for each nonunit basis coordinate still requires the selected primary-card and budget-card comparison. Neither missing clause follows from the finite pencil. Once the exact component receipts are supplied, their absolutely convergent density decomposition is merged by the dominated-series argument above, and the finite-wing merger sums the two budgets using the same motion coefficients. No compactness or diagonal extraction is involved.

31 Same-run Cauchy input for the full mesh

We record the actual Cauchy route used to upgrade the positioned limit to the whole mesh. This is a same-run statement. Fix a selected run and its campaign, Winter witness H, and Pillars witness W, together with the chosen growth function f_i and the selected history tables C_i. For a fixed period p, word length n, list of local labels v, and physical tuple x which is both non-hit and an ancestor point, write

$$o_j(x) = \mathrm{wrap}_{a_j}\big(\mathrm{address}_{a_j}(x)\big), \qquad U_i(x) = C_i.\,\mathrm{numeric}\big(f_i, p, v, o_{f_i}(x)\big).$$

The selected source route proves that $(U_i(x))_i$ is Cauchy from the growth, price, transfer, and geometric receipts of one fixed run. This subsection translates the Cauchy implication and the analytic interpolation step; the upstream proof that the selected run supplies all the receipts is recorded separately below.

The Cauchy step uses one quantitative insertion comparison. We state its exact logical form so the upstream estimate is visible. Fix a finite label set A, a reference rung k, a later rung j, and a terminal mesh index i. Let U_i be the full-volume pilot and $V_{j,i}$ its fixed-rung interpolant. Suppose $F_0 < \infty$ is independent of j, $G_j \to \infty$, and, for every sufficiently large fixed j, the same-history receipts give

$$|U_i - V_{j,i}| \le \frac{F_0}{G_j/2} \quad \text{for all sufficiently large } i, \tag{3}$$

and $(V_{j,i})_i$ is Cauchy at each such fixed j. A sufficient exact condition for this fixed-rung Cauchyness is that its finite mode integrand is a polynomial in finitely many measurable coefficient fields that converge pointwise, these mode integrands have a common integrable dominator, and their positive partition normalizers converge to a positive limit. Dominated convergence then gives convergence of the numerator integrals, and division by the normalizers gives convergence of $V_{j,i}$. Under these hypotheses $(U_i)_i$ is Cauchy: given $\varepsilon > 0$, first choose j so that $F_0/(G_j/2) < \varepsilon/4$, then take the maximum of the threshold in (3) and the fixed-j Cauchy threshold for error $\varepsilon/4$. For later i, i',

$$|U_i - U_{i'}| \le |U_i - V_{j,i}| + |V_{j,i} - V_{j,i'}| + |V_{j,i'} - U_{i'}| < 3\varepsilon/4 < \varepsilon.$$

Lemma 31.1 (Holomorphic insertion estimate)**.** *Let $S > 2$ and let Φ be holomorphic on $B(0, S)$. If $|\Phi(z)| \le F_0$ for all $|z| < S$, then*

$$|\Phi(1) - \Phi(0)| \le F_0/(S/2). \tag{4}$$

Proof. Put $r = S/2$. For every w with $|w| \le 1$, the closed disk $\overline{B}(w, r)$ lies in $B(0, S)$ since $1 + S/2 < S$. The Cauchy integral formula on its boundary gives $|\Phi'(w)| \le F_0/r$. Integrating the derivative on the real segment from 0 to 1 proves (4). □

Lemma 31.2 (Uniform confluence supplies the fixed-rung vault)**.** *Let I be a finite set, let $\mathcal{B}$ be a finite set, and let $\mathcal{X}$ be a nonempty set. Suppose $F_{n,i} : \mathcal{X} \to \mathbb{C}$ and $G_{n,i,b} : \mathcal{X} \to \mathbb{C}$ have uniform eventual bounds: for each $i \in I$ there is $c_i \ge 0$ such that eventually $\sup_{x\in\mathcal{X}} |F_{n,i}(x)| \le c_i$, and for each i, b there is $d_{i,b} \ge 0$ such that eventually $\sup_{x\in\mathcal{X}} |G_{n,i,b}(x)| \le d_{i,b}$. For any weights $w_i > 0$, there are finite x_0, y_0 such that, eventually in n,*

$$w_i|F_{n,i}(x)| \le x_0, \qquad \sum_{b\in\mathcal{B}} w_i|G_{n,i,b}(x)| \le y_0 \quad (i \in I,\ x \in \mathcal{X}),$$

where one may take

$$x_0 = \sum_{i\in I} w_i c_i, \qquad y_0 = \sum_{i\in I}\sum_{b\in\mathcal{B}} w_i d_{i,b}.$$

Proof. There are only finitely many bounds. Intersect their eventual tails to get one index n_0 after which all hold at once. For each fixed i, positivity of the weights and nonnegativity of every summand give $w_i|F_{n,i}(x)| \le w_i c_i \le x_0$. Likewise $\sum_b w_i|G_{n,i,b}(x)| \le \sum_b w_i d_{i,b} \le y_0$. This is precisely the two-part vault bound used for the scalar and box-marked fields. In the selected construction, the source confluence statement gives an eventual uniform bound over configurations for each fixed rung and each field. Since the label set and the box set of a fixed finite torus are finite, their thresholds can be intersected once. No compactness or extraction is needed. □

The fixed-rung vaults in this argument are produced by the same history, not by an additional choice of subsequence. At a fixed rung j, the retained history interpolation gives uniform confluence of each scalar insertion and each box-marked insertion. Applying Lemma 31.2 to the finite label set, with weights w_i, gives the constants t, v in the later-rung vault. Applying it at the opened rung k gives x, y in the opened-rung vault. The constants may depend on k, j, the finite labels, and the weights, but the opened-rung constants used in the Cauchy comparison are held fixed as j varies. The positive weights are obtained from the paid allowances B_i by $w_i = \zeta_i/(|B_i| + 1)$; indeed $w_i|B_i| \le \zeta_i$ and $w_i > 0$ when $\zeta_i > 0$.

In the public same-run application, these are actual receipts of the selected history. Its branch-stability estimate gives uniform confluence at every fixed rung for the scalar and box-marked fields, and its credit estimate gives the finite constants B_i. The preceding finite sum therefore supplies the two vaults required at the opened rung and at each later rung. The history's scale profile supplies the positive weights and their march relation. Its selected position map supplies the remaining geometry: ancestor-point separation gives injectivity at the chosen rung, hence occupancy one, and continues to give the required Drake inequalities at later rungs. These properties are intersected with the eventual pond, positive-reserve, and delay tails before fixing the rung; thus the estimates are simultaneous for the finite label family and hold along the original volume sequence.

To apply Lemma 31.1, let $P_z(U)$ be the selected census-mode coefficient of the restored field at rung j, with its principal and forced field families multiplied by z, and put

$$\Phi(z) = Z_{N,p}^{-1} \int P_z(U)\, d\mu_j(U).$$

The finite restored polynomial depends polynomially on z. The quantitative majorant establishing integrability of its coefficients is given below.

Here is the selected-field verification of its uniform bound, conditional on the served-slab and uniform price receipts at rung j. Write B_j for its Drake separation scale and E_j for the common support exponent. Assume $B_j \ge 8$, $E_j \ge 2$, and $S_j = e^{B_j/64} > 2$. The last inequality is a separate late-scale condition; it does not follow from $B_j \ge 8$ alone. For a nonzero polynomial component on a support X, each nonzero mode has at least two markers, and the support-placement condition puts every named origin of those markers in X. Drake separation says distinct origins of that mode are at max-distance at least B_j^2. The connected-halo tour estimate in Section 19 therefore gives

$$B_j\,|I| \le 64\,\mathrm{cover}(B_j, X), \qquad \mathrm{cover}(B_j, X) \ge B_j/64,$$

for its marker set I. Thus $|z| < e^{B_j/64} \le e^{\mathrm{cover}(B_j,X)}$. Scaling the component by z multiplies its envelope by $|z|$. Its original weighted support bill contains the factor $e^{E_j\,\mathrm{cover}(B_j,X)}$; since $|z| \le e^{2\,\mathrm{cover}(B_j,X)}$, the same bill pays the scaled envelope at exponent $E_j - 2$. This argument applies to every active support in both the principal and forced families; zero components cost nothing. The correction families are scaled, while the exponential-entry and marked activity carriers are fixed. The support, gauge, and minimum-degree conditions remain true under scalar multiplication; the envelopes in the support ledger scale by $|z|$. Hence the same served decomposition with those scaled

correction families witnesses servedness of the restored field at exponent $E_j - 2$. This does not assert that the restore operation itself is linear. For integrability, the vault bounds the finitely many scalar and marked fields and the served ledger bounds the finitely many correction coefficients. The selected lawfulness upper bound for the action and the vacuum-activity envelope give the pointwise exponential majorant proved in Section 102; all coefficient fields are measurable. Expanding the finite restored polynomial therefore gives a measurable, integrable majorant for each coefficient of P_z, uniform for z in the disk. The prior is a probability measure. Thus the integral defining Φ is holomorphic on $B(0, S_j)$.

To pass to the opened rung, the served witness at the lowered exponent is propagated by the downward servedness induction: at each intervening scale the retained matchup and wake data give the transition hypotheses, and the exact push identity fixes each marked coefficient. The induction's integrability step follows from the transition's finite probability mass. This is the served-transfer result of Section 19, with its displayed one-step response and payment hypotheses; Fubini supplies the mode-integral identity below, but by itself does not imply servedness. The backward transition from rung j to k is finite, and at each transition, if $K_m(U, V)$ denotes its kernel from finer configuration U to coarser configuration V, its normalization is $\int K_m(U, V)\, d\mu_m(V) = 1$ for every U. For each integrable coefficient q, the push identity is

$$q_m(V) = \int K_m(U, V) q_{m+1}(U)\, d\mu_{m+1}(U).$$

Moreover, nonnegativity and normalization give

$$\begin{aligned} &\int\int K_m(U, V)|q_{m+1}(U)|\, d\mu_{m+1}(U) d\mu_m(V) \\ &\quad = \int |q_{m+1}(U)|\, d\mu_{m+1}(U) < \infty. \end{aligned}$$

Fubini therefore proves

$$\int q_m(V)\, d\mu_m(V) = \int q_{m+1}(U) \left(\int K_m(U, V)\, d\mu_m(V) \right) d\mu_{m+1}(U) = \int q_{m+1}(U)\, d\mu_{m+1}(U).$$

This proves the mode-integral preservation at each transition. The opened-rung vaults and finite-volume mode estimate of Section 103 then give exactly

$$|\int P_z\, d\mu_j|\ \mathrm{toll}_w(A) \le Z_{N,p} \exp(|\mathrm{Site}_k|\mathcal{T}_k + |A|(x + y + 2)),$$

$$|\Phi(z)| \le F_0 := \frac{\exp(|\mathrm{Site}_k|\mathcal{T}_k + |A|(x + y + 2))}{\mathrm{toll}_w(A)}.$$

All weights, the opened rung, and its vault budgets are fixed as j varies, so F_0 is independent of j. This checks the bound hypothesis of the holomorphic estimate from the explicit scaling, backward transfer, and integrated-mode receipts.

In the public pilot comparison, the finite transition recurrence identifies $U_i = \Phi(1)$ and $V_{j,i} = \Phi(0)$, with the same terminal law and normalization. The terminal birth identity supplies the value at $z = 1$; the later-rung restoration with its correction families set to zero supplies the value at $z = 0$. The opened rung k is used to price this value after transfer. The transition preserves each mode integral, so this identification introduces no change of partition function. Thus (4) is precisely (3), not an additional compactness input.

There is an exact finite identity connecting this pilot observable to the ordinary finite-volume Wilson observable. For the census mode containing all n labels, all copy switches are the identity. The product of the local insertions is therefore exactly the finite-volume numeric observable, at the same address tuple $o_{f_i}(x)$. Thus Cauchyness of the pilot sequence is Cauchyness of $U_i(x)$ itself; this identity introduces no error term and no subsequence.

To pass from ancestor points to the full positional function, use that ancestor points are dense. The full-mesh local Lipschitz receipt on a compact subset of the non-hit set, together with $a_i \to 0$, gives asymptotic uniform equicontinuity there: if the continuum points are within $\delta/2$, their lattice addresses are within δ once $2c_n a_i < \delta/2$. For compact $K \subset U = \mathrm{NoHit}$, choose a compact neighborhood $K \Subset K\prime \Subset U$ and a common asymptotic modulus on $K\prime$. A finite $\delta/3$-net for K can be chosen with centers in the dense ancestor set intersected with U. Pointwise Cauchyness at these finitely many centers gives one threshold. For any two later indices and any $x \in K$, compare their values at x to the value at a net center: the two modulus errors are each below $\varepsilon/3$, and the center values differ by less than $\varepsilon/3$. Hence the functions are uniformly Cauchy on K; completeness gives a uniform limit, and the same modulus makes it continuous. The generic finite-difference bounds in Section 28 provide the required bounds and local Lipschitz receipts from the same run. The application is conditional on the following same-run receipts: uniform branch confluence and a paid credit for each fixed finite label set; the pond, migration, occupancy, and Drake data needed to construct the transition fields; a served witness at each chosen later rung with its displayed support-price bounds; the finite downward service-transfer hypotheses; and a fixed-rung Cauchy sequence. It also uses the full-mesh local Lipschitz bound on each compact subset. Given these receipts, (3) is the quantitative insertion estimate just proved, and the pointwise Cauchy and compact-uniform limits required by the full-mesh positioned construction follow for every period, length, and label word. Together with the same-run finite-difference bounds in Section 28, these receipts yield the smooth positioned correlation along the full mesh, without further extraction. For source correspondence, the branch-stability result at a fixed rung gives confluence of the scalar insertion and of every box-marked insertion; the credit result gives the paid amounts. The finite-sum lemma above converts those uniform bounds into the two vault inequalities. At the opened rung the positive weights are fixed, and at a later rung the marched weights are used with later-rung confluence. The pond package is passed from the selected run's scale-field package, and its history recurrence builds the per-label transition fields used by the served-slab step. These source fields are mapped in the evidence record; the mathematical production of them is not claimed here. The detailed local service propagation and its explicit nonlinear response input are stated in Section 19. The alternative detector route through trained rows and null-row sweeps is not called by this public construction.

32 Reflection positivity and the Euclidean Hilbert space

Reflection positivity is the mechanism that turns Euclidean correlations into an inner product. A second ingredient is needed to make positive-time translation contractive: a bound on correlations as the two halves move apart. We give the complete argument, keeping this bound explicit.

A word $w = ((x_1, v_1), \dots, (x_n, v_n))$ records finitely many distinct insertion points and their labels. Write $S(w)$ for its correlation and $\varnothing$ for the empty word. Fix a coordinate axis and call its coordinate t. Let $\mathcal{W}_+$ consist of words whose points have strictly positive t, including the empty word. Reflection θ reverses the order of a word, reflects the points across $t = 0$, and applies the reflected conjugate to each label. Assume $\theta^2 = 1$. For $v, w \in \mathcal{W}_+$, the concatenation $(\theta v)w$ has no coincident points: its two parts lie in opposite open half-spaces.

The inputs used in this section are normalization $S(\varnothing) = 1$, translation invariance, the reflection relation $S(\theta w) = \overline{S(w)}$, permutation symmetry on distinct points, and the following two conditions:

$$\sum_{i,j=1}^{m} \overline{c_i} c_j S((\theta w_i) w_j) \geq 0 \quad (w_i \in \mathcal{W}_+,\ c_i \in \mathbb{C}), \tag{123}$$

$$x \longmapsto S((\theta v)(w + x)) \text{ is continuous and bounded on } \{x : t(x) \geq 0\} \quad (v, w \in \mathcal{W}_+). \tag{124}$$

The positivity in (123) means that the displayed complex number is real and nonnegative.

Lemma 32.1 (From smeared to point reflection positivity)**.** *Suppose the correlations are continuous at every configuration of distinct points. Assume reflection positivity for finite sums of smooth compactly supported tests on positive-time configurations, with arbitrary fixed label tuples for the separate summands. Then* (123) *holds.*

Proof. Fix positive words $w_1, \ldots, w_m$ and coefficients $c_1, \ldots, c_m$. For each nonempty word choose a small configuration neighborhood in which its points stay distinct and strictly in the positive half-space. Place a nonnegative smooth bump of integral one in that neighborhood, centered at the word's positions, with support radius ϵ. Such bumps are obtained by translation and rescaling of $\exp(-1/(1 - |x|^2))$ on the unit ball, zero outside, followed by division by its positive integral. The empty word has a zero-dimensional configuration space; use the constant test one there and counting measure of total mass one.

For each pair i, j, the concatenation of a reflected configuration near w_i with a configuration near w_j has distinct points. Within either half this follows from the choice of neighborhood; across halves it follows from the strict time separation. Thus its correlation kernel is continuous near the pair of centers, even if some spatial coordinates agree. Write K_{ij}^{ϵ} for its integral against the two bumps. Nonnegativity and unit integrals give

$$|K_{ij}^{\epsilon} - S((\theta w_i) w_j)| \leq \sup \big| K_{ij}(x, y) - K_{ij}(x_i, y_j) \big| \longrightarrow 0,$$

where the supremum is over the shrinking product support. The kernels are bounded on these compact supports, so the integrals are well-defined. Smeared reflection positivity says $\sum_{i,j} \overline{c_i} c_j K_{ij}^{\epsilon} \geq 0$. There are finitely many terms, and the nonnegative real half-line is closed in $\mathbb{C}$. Passing to the limit proves (123). □

This argument needs continuity at all distinct-point configurations. Continuity only where every coordinate is pairwise different would not suffice: a word and its reflection share spatial coordinates, including in the diagonal entries of the reflection matrix.

Lemma 32.2 (A sufficient separated-correlation bound)**.** *Suppose the correlations are smooth at distinct points and, for each fixed finite list of labels, there are $C \geq 0$ and $A \geq 0$ such that*

$$|S_{\boldsymbol{v}}(\boldsymbol{x})| \leq C \rho^{-A} \quad \textit{whenever } 0 < \rho \leq 1 \textit{ and } |x_i - x_j|_\infty \geq \rho \ (i \neq j).$$

Then (124) *holds.*

Proof. Fix v, w. The distances between points within either word are unchanged by translation. For a reflected point from v and a point of $w + x$, their time separation is the sum of their original positive times plus $t(x)$. It is bounded below by a positive number independent of x when $t(x) \geq 0$. There are only finitely many pairs. Choose $\rho \leq 1$ smaller than all their positive lower bounds; if there are no pairs, take $\rho = 1$. The assumed estimate gives the required uniform bound. The configuration depends continuously on x and always remains in the distinct-point set, so smoothness gives continuity. □

Proposition 32.3 (Hilbert space and vacuum)**.** *There is a Hilbert space $\mathcal{H}_E$ with vectors $[w]$, $w \in \mathcal{W}_+$, whose linear span is dense and such that*

$$\langle [v], [w] \rangle = S((\theta v)w).$$

The vector $\Omega_E = [\varnothing]$ has norm one.

Proof. On the free complex vector space of finitely supported coefficients on $\mathcal{W}_+$, define

$$B(c, d) = \sum_{v,w} \overline{c_v} d_w S((\theta v)w).$$

Reflection, $\theta^2 = 1$, and permutation symmetry show $B(d, c) = \overline{B(c, d)}$; positivity follows from (123). For completeness, Cauchy–Schwarz follows by applying positivity to $c + \lambda d$: if $B(d, d) > 0$, minimize the resulting quadratic expression in λ; if $B(d, d) = 0$, varying the magnitude and phase of λ forces $B(c, d) = 0$. Thus $|B(c, d)|^2 \le B(c, c)B(d, d)$. Consequently the null vectors $\mathcal{N} = \{c : B(c, c) = 0\}$ are orthogonal to every vector and form a subspace. The quotient by $\mathcal{N}$ is an inner-product space. Complete it and map each word to the class of its basis vector. This gives the stated inner product and density. Finally $\|\Omega_E\|^2 = S(\varnothing) = 1$. □

Let $\tau_s w$ translate every point of w by s in the positive time direction, where $s \ge 0$. Define for a finite formal sum c the vector $T_s c = \sum_w c_w [\tau_s w]$. At this point T_s is defined on formal sums; descent to the null quotient will follow from its norm bound. Translation invariance gives

$$\langle T_s c, T_t d \rangle = \langle T_0 c, T_{s+t} d \rangle. \tag{125}$$

Indeed, reflecting the translation by s shifts the first word by $-s$; translating the whole concatenation by s gives the right side. Sesquilinearity extends the identity from words to finite sums.

Lemma 32.4 (The contraction estimate)**.** *For every finite sum c and every $s \ge 0$, $\|T_s c\| \le \|T_0 c\|$.*

Proof. For a single word, (125) gives $\|[\tau_s w]\|^2 = S((\theta w)(\tau_{2s} w))$. Condition (124) bounds this expression uniformly in s. The triangle inequality therefore bounds $\|T_s c\|$ uniformly for every finite sum c. Put $M = \sup_{s \ge 0} \|T_s c\| < \infty$ and $a = \|T_0 c\|$. By (125) and Cauchy–Schwarz,

$$\|T_s c\|^2 = |\langle T_0 c, T_{2s} c \rangle| \le aM.$$

Taking the supremum gives $M^2 \le aM$. Since $M \ge 0$, either $M = 0$, or division by M gives $M \le a$. Both cases prove the result. □

Theorem 32.5 (Euclidean translations)**.** *Positive time translations extend uniquely to a strongly continuous semigroup E_s, $s \ge 0$, of self-adjoint contractions on $\mathcal{H}_E$. Spatial translations extend to a strongly continuous unitary representation U_y, $y \in \mathbb{R}^3$, commuting with E_s. Both leave Ω_E fixed.*

Proof. The contraction estimate sends null vectors to null vectors. It therefore defines a contraction on the quotient, and then uniquely on its completion. On word vectors, $E_0 = 1$ and $E_s E_t = E_{s+t}$; continuity of the bounded operators extends these identities from their dense span to the whole Hilbert space. Equation (125) with one time equal to zero says that E_s is symmetric. A bounded symmetric operator defined everywhere is self-adjoint: its adjoint has the same matrix elements against all vectors, hence is the same operator.

For a spatial shift y, reflection commutes with the shift. Translation invariance gives $\langle [v + y], [w + y] \rangle = \langle [v], [w] \rangle$. Thus the shift descends to an isometry on the quotient and its completion.

Its inverse is the shift by $-y$, so it is unitary. The group law and commutation with E_s follow on words from commutation of ordinary translations, and extend by density. The empty word is unchanged by any translation, proving the vacuum assertions.

It remains to prove strong continuity. For a word w, the square of $\|E_s w - E_t w\|$, with word brackets understood, is the sum of the two diagonal inner products minus twice the real part of the cross term. By (125) these are values of the continuous function $u \mapsto S((\theta w)(\tau_u w))$ at $2s, 2t, s+t$. Their combination tends to zero as $s \to t$. The identical norm-square argument and (124) give continuity of spatial translations on word vectors. Finite linear combinations remain continuous. For an arbitrary ψ, choose a finite word sum v with $\|\psi - v\| < \varepsilon/3$. For either family V_z, its norm bound gives

$$\|V_z\psi - V_{z_0}\psi\| \le 2\|\psi - v\| + \|V_z v - V_{z_0} v\|.$$

The last term is smaller than $\varepsilon/3$ near z_0, proving strong continuity. □

This theorem establishes Euclidean Hilbert-space dynamics from the displayed correlation hypotheses. Relativistic reconstruction also requires field operators, rotations, locality, boundary distributions and a selected phase with a unique invariant vacuum. These do not follow merely by naming the Hilbert space. One possible literature route is the corrected reconstruction theorem of Osterwalder–Schrader [1, §IV.1]; applying that version requires checking its linear growth condition as well as its covariance, positivity, symmetry and clustering assumptions. That comparison is still outstanding here, particularly the passage from bounds for each fixed label tuple to the required growth control and the treatment of the selected phase.

33 The algebra of time-invariant observables

The Hilbert space constructed from an unselected correlation table need not have a one-dimensional vacuum space. We first identify the whole subspace left fixed by time evolution. Compressed slab observables on this subspace will then distinguish its phases. Throughout this section the hypotheses of Theorem 32.5 remain in force. We write $\mathcal{H} = \mathcal{H}_E$ and $\Omega = \Omega_E$. In this section e_0 denotes the chosen time axis and x_0 its coordinate; this local notation does not change the earlier coordinate ordering.

Proposition 33.1 (The large-time projection)**.** *The subspace*

$$\mathcal{H}_0 = \{v \in \mathcal{H} : E_t v = v \text{ for every } t \ge 0\}$$

is closed. If P_0 denotes its orthogonal projection, then $E_t v \to P_0 v$ in norm as $t \to \infty$, for every $v \in \mathcal{H}$.

Proof. Each $E_t - I$ is continuous, so its kernel is closed; their intersection is $\mathcal{H}_0$. Fix v. Contractivity and the semigroup law show that $a(t) = \|E_t v\|^2$ decreases to its nonnegative infimum ℓ. Self-adjointness gives, for $s \le t$,

$$\langle E_s v, E_t v\rangle = \langle v, E_{s+t} v\rangle = \|E_{(s+t)/2} v\|^2.$$

The scalar on the right is real. Consequently

$$\|E_s v - E_t v\|^2 = a(s) + a(t) - 2a((s+t)/2) \le a(s) - a(t) \le a(s) - \ell.$$

For large s, t this tends to zero, after exchanging s, t if necessary. Completeness gives a norm limit v_0. For fixed $r \ge 0$, continuity of E_r and $E_r E_t = E_{r+t}$ give $E_r v_0 = v_0$. For $z \in \mathcal{H}_0$, self-adjointness gives $\langle E_t v, z\rangle = \langle v, E_t z\rangle = \langle v, z\rangle$. Passing to the limit shows $v - v_0 \perp \mathcal{H}_0$. These two properties characterize the orthogonal projection, so $v_0 = P_0 v$. □

We next explain the analytic estimate needed to insert an entire word as a bounded operator. A word w lies in a slab of width $d > 0$ if all its points are distinct and have time coordinates in $(0, d)$. For a positive word v, put

$$I_w v = w\,\tau_d v, \qquad w^\sharp = \tau_d \theta w.$$

Both are positive words, and $w^\sharp$ lies in the same slab. Let I_w also denote the linear map on finite formal word sums. Translation invariance and reflection give

$$\langle [I_w c], [b] \rangle = \langle [c], [I_{w^\sharp} b] \rangle. \tag{126}$$

Indeed, translating the configuration $\theta(w\tau_d v)\,u = (\tau_{-d}\theta v)(\theta w)u$ by d gives $(\theta v)(\tau_d \theta w)(\tau_d u)$, the configuration in the right-hand pairing. This proves the identity on words and then on finite sums.

Lemma 33.2 (From repeated insertion to a bounded operator)**.** *Fix a slab word w and $D > 0$. Suppose that for every positive word v there is $B_v \geq 0$ such that*

$$\big\| [(I_{w^\sharp} I_w)^n v] \big\|^2 \leq B_v D^{4n} \quad (n = 0, 1, 2, \ldots). \tag{127}$$

There is a unique bounded operator $A_d(w)$ on $\mathcal{H}$ with $A_d(w)[v] = [w\tau_d v]$; it satisfies $\|A_d(w)\| \leq D$.

Proof. Write $T = I_{w^\sharp} I_w$ on formal sums and $Xc = [c]$. The triangle inequality and (127) imply, for $c = \sum_v c_v v$,

$$\|XT^n c\| \leq C_c D^{2n}, \qquad C_c = \sum_v |c_v| \sqrt{B_v} < \infty.$$

Equation (126) shows that T is symmetric for the possibly degenerate form induced by X. Repeatedly moving T across the inner product therefore gives $\|XT^n c\|^2 = \langle Xc, XT^{2n} c \rangle$. Put $b_n = D^{-2n} \|XT^n c\|$ and $M = \sup_{n \geq 0} b_n < \infty$. Cauchy–Schwarz gives $b_n^2 \leq \|Xc\| b_{2n} \leq \|Xc\| M$. Taking the supremum yields $M^2 \leq \|Xc\| M$, hence $M \leq \|Xc\|$. Finally,

$$\|XI_w c\|^2 = |\langle Xc, XTc \rangle| \leq \|Xc\| \|XTc\| \leq D^2 \|Xc\|^2.$$

Thus insertion respects the null space and induces a bounded operator on the quotient. Its unique extension to the completion has the asserted norm and action. Uniqueness follows from density of the word span. □

The estimate in (127) concerns arbitrarily long repeated words; the constants B_v may depend on the exterior word. Bounds for each fixed correlation order alone do not supply this estimate. Its production from the lattice construction remains an input at this stage. Assume henceforth that it holds, with some D, for every slab word. The resulting operators obey

$$\begin{aligned} A_d(\varnothing) &= E_d, \\ A_{d+e}(w\tau_d v) &= A_d(w) A_e(v), \\ A_d(w)^* &= A_d(w^\sharp), \\ A_{d+r}(w) &= A_d(w) E_r, \qquad A_{r+d}(\tau_r w) = E_r A_d(w). \end{aligned} \tag{128}$$

Here each word lies in its indicated slab and $r \geq 0$. The first, second and last two identities follow by applying both sides to a positive word and concatenating and translating its entries. The adjoint identity follows from (126). All extend from the dense word span by boundedness. Spatial translation also gives $U_y A_d(w) = A_d(w + y) U_y$, by the same calculation on words.

33.1 Spatial invariance of the ground space

Assume now full orthogonal covariance of the correlation table: there is an action on quotient field labels such that, for every $K \in O(4)$ and every distinct-point word w, the jointly rotated word Kw satisfies $S(Kw) = S(w)$. Rotation includes this label action in what follows. To evaluate a rotated word, project each raw label to its field class, rotate that class, and choose a raw representative. Correlations are independent of that choice. No group action on the chosen raw representatives is assumed. This is stronger than covariance under signed coordinate permutations.

Lemma 33.3. *Every $v \in \mathcal{H}_0$ is fixed by all spatial translations.*

Proof. For two distinct-point words p, q, set $F_{p,q}(x) = S(p(q + x))$ whenever the two clusters are disjoint. Translate θp and q separately far enough in positive time to obtain positive words v, w. If the translation amounts are r, s, then, whenever $x_0 \geq r + s$, translation invariance gives

$$F_{p,q}(x) = \langle [v], U_{x_\perp} E_{x_0 - r - s}[w] \rangle, \qquad x_\perp = x - x_0 e_0. \tag{129}$$

The time separation also ensures that the two clusters are disjoint. For any fixed $h \in \mathbb{R}$ and any path $x(t)$ with $x_0(t) \to \infty$, it follows that

$$F_{p,q}(x(t) + he_0) - F_{p,q}(x(t)) \longrightarrow 0. \tag{130}$$

Indeed, the difference of the two heat-evolved vectors in (129) tends to zero by Proposition 33.1; the spatial unitary has norm one, even when $x_\perp(t)$ varies without bound.

If $b_0 > 0$, choose an orthogonal map K taking b to $|b|e_0$. Orthogonality implies $(Ke_0)_0 = b_0/|b| > 0$. Rotate both clusters and apply (130) along the path tKe_0, whose time coordinate tends to infinity. Covariance gives

$$F_{p,q}(te_0 + b) - F_{p,q}(te_0) \longrightarrow 0.$$

For a spatial y, take $b = y + e_0$ and subtract (130) along $x(t) = te_0 + y$, with $h = 1$. We obtain $F_{p,q}(te_0 + y) - F_{p,q}(te_0) \to 0$. Apply this with $p = \theta v, q = w$ for positive words. The two correlations are $\langle [v], U_y E_t[w] \rangle$ and $\langle [v], E_t[w] \rangle$. Their limits show $\langle [v], (U_y - I)P_0[w] \rangle = 0$. Density of word vectors first gives $(U_y - I)P_0[w] = 0$ and then $(U_y - I)P_0 = 0$. Since P_0 is the identity on $\mathcal{H}_0$, the conclusion follows. □

33.2 Compression and separation of clusters

For a slab word define $M(w) = P_0 A_d(w)|_{\mathcal{H}_0}$. The width is immaterial: increasing it appends an E_r, which is the identity on $\mathcal{H}_0$; two widths can be compared with their sum. Likewise a positive time shift inserts E_r on the left, and $P_0 E_r = P_0$. Spatial shifts give the same operator by Lemma 33.3 and spatial covariance. For a negative time shift, reverse the comparison. Every finite distinct-point word can be translated into a slab. Thus $M(w)$ is well-defined for every such word, independent of both the placement and width, and

$$M(w + b) = M(w), \qquad \langle \Omega, M(w)\Omega \rangle = S(w). \tag{131}$$

The latter follows from $A_d(w)\Omega = [w]$ after a slab placement, $P_0\Omega = \Omega$, and translation invariance.

For a positive word also put $X(w) = A_d(w)|_{\mathcal{H}_0}$, as a map into $\mathcal{H}$. Width independence follows as above, and

$$X(w)\Omega = [w], \qquad M(w)\Omega = P_0[w], \qquad M((\theta v)w) = X(v)^* X(w).$$

For the last identity, place the reflected word into the slab of width d, append the word w in a subsequent slab of width e, and use the concatenation and adjoint identities in (128). Compression then gives exactly the displayed product. In particular $M(\theta v) = M(v)^*$. This also shows that the generators are closed under adjoints, after translation into slabs.

Lemma 33.4 (Separated-cluster limit)**.** *Let $w_1, \dots, w_m$ be words with distinct points. Suppose $b_1(n), \dots, b_m(n) \in \mathbb{R}^4$ satisfy $(b_{j+1}(n) - b_j(n))_0 \to \infty$ for $j < m$. Then*

$$S((w_1 + b_1(n)) \cdots (w_m + b_m(n))) \longrightarrow \langle \Omega, M(w_1) \cdots M(w_m)\Omega \rangle.$$

The concatenated words have distinct points for all sufficiently large n.

Proof. Fixed translations of individual words, compensated in their b_j, preserve the assertion and the gap condition. Make all words positive. We prove a stronger vector limit by induction from the last cluster: after translating the concatenation so that its first cluster has its original position, its vector tends to $X(w_1)M(w_2) \cdots M(w_m)\Omega$. For one cluster this is $X(w_1)\Omega = [w_1]$.

Choose a slab width d for w_1. Write the displacement of the second cluster relative to the first as $g_n e_0 + y_n$, with y_n spatial. Eventually $g_n > d$. If z_n is the positive word vector of the remaining clusters relative to the second, the whole vector is $A_d(w_1)U_{y_n}E_{g_n-d}z_n$. By induction $z_n \to z = X(w_2)M(w_3) \cdots M(w_m)\Omega$. The estimates

$$\|U_{y_n}E_{g_n-d}z_n - P_0 z\| \le \|z_n - z\| + \|E_{g_n-d}z - P_0 z\| \longrightarrow 0$$

use contraction, unitarity and $U_{y_n}P_0 z = P_0 z$. Boundedness of $A_d(w_1)$ proves the vector limit. The same induction shows positivity and distinctness eventually, since the successive time gaps tend to infinity. Pairing the vector limit with Ω, and using translation invariance and $P_0\Omega = \Omega$, proves the scalar assertion. The empty list has correlation one and needs no induction. □

Proposition 33.5 (Commutativity)**.** *The norm-closed unital star algebra $\mathcal{A} \subset B(\mathcal{H}_0)$ generated by the $M(w)$ is commutative, and Ω is cyclic and separating for $\mathcal{A}$.*

Proof. Write $F(w_1, \dots, w_m) = \langle \Omega, \prod_j M(w_j)\Omega \rangle$, in the indicated order. If $b_0 > 0$ and $(Kb)_0 > 0$, place the clusters at jnb, rotate the configuration by K, and use Lemma 33.4 before and after rotation. Covariance gives $F(w_1, \dots, w_m) = F(Kw_1, \dots, Kw_m)$.

To exchange the middle two of four words, use one spatial axis e_1 and displace the four clusters by

$$0, \quad n(e_0 + 2e_1), \quad n(2e_0 + e_1), \quad 3n(e_0 + e_1).$$

They are ordered by time as written. Interchange e_0, e_1 by an orthogonal map K. The resulting time order is first, third, second, fourth. Permutation symmetry of distinct-point correlations and the cluster limit give $F(x, y, w, z) = F(Kx, Kw, Ky, Kz)$. Use the preceding rotation identity with $b = e_0 + e_1$, fixed by K, to conclude $F(x, y, w, z) = F(x, w, y, z)$.

The vectors $M(v)\Omega = P_0[v]$, for positive words v, have dense linear span in $\mathcal{H}_0$: approximate any vector in $\mathcal{H}_0$ by word sums in $\mathcal{H}$ and apply P_0. For two generators, their commutator has zero matrix elements between these dense vectors, since

$$\langle M(u)\Omega, [M(w), M(z)]M(v)\Omega \rangle = F(\theta u, w, z, v) - F(\theta u, z, w, v) = 0.$$

The commutator is bounded, hence is zero. The generators and their adjoints therefore commute; polynomial approximation and continuity of multiplication show that their norm closure is commutative. The same density proves cyclicity. If $A\Omega = 0$, then $AB\Omega = BA\Omega = 0$ for every $B \in \mathcal{A}$. Cyclicity and boundedness give $A = 0$, which is separation. □

This proves commutativity from the displayed slab and covariance inputs. The next construction must establish norm separability of $\mathcal{A}$, represent the vacuum state on its compact character space, and show that each resulting whole-word correlation table retains the required regularity, null spaces and field identities. Those conclusions are not part of Proposition 33.5.

34 Regularity in operator norm

To pass from scalar correlation identities to the character construction, we need regularity in operator norm. Weak continuity of matrix elements alone would not provide it. The additional ingredient is a local uniform bound on the operators representing the differentiated insertions.

Lemma 34.1 (Operator remainder estimate)**.** *Let U be an open subset of a finite-dimensional real normed space E, let H be a complex Hilbert space, and let $\mathcal{D}$ be a dense linear subspace. Suppose $A : U \to B(H)$ and $B : U \to \mathcal{L}_{\mathbb{R}}(E, B(H))$ satisfy*

$$D\langle u, A(\cdot)v\rangle(z)[h] = \langle u, B(z)[h]v\rangle \quad (u, v \in \mathcal{D},\ z \in U,\ h \in E).$$

If the segment from x to y lies in U and $\|B(z) - B_0\| \le C$ along the segment, with $C \ge 0$, then

$$\|A(y) - A(x) - B_0[y - x]\| \le C\|y - x\|. \tag{132}$$

Consequently, local boundedness of B implies local Lipschitz continuity of A; continuity of B implies $DA = B$ in operator norm.

Proof. Fix $u, v \in \mathcal{D}$. Along the segment, subtract the linear function $z \mapsto \langle u, B_0[z]v\rangle$. The derivative of the resulting complex-valued function has norm at most $C\|u\|\|v\|$. The scalar mean-value inequality yields

$$|\langle u, (A(y) - A(x) - B_0[y - x])v\rangle| \le C\|u\|\|v\|\|y - x\|.$$

This inequality also follows directly by composing with the segment and then with every real-linear functional of norm one on $\mathbb{C}$, using the real mean-value theorem, and taking the supremum. Density extends the inequality to all $u, v \in H$. For any bounded operator T, the supremum of $|\langle u, Tv\rangle|$ over unit vectors u, v is $\|T\|$: Cauchy–Schwarz gives one inequality and, when $Tv \neq 0$, the choice $u = Tv/\|Tv\|$ gives the other. This proves (132).

On a sufficiently small ball where $\|B\| \le C$, use $B_0 = 0$ to obtain the Lipschitz bound. If B is continuous at x, then for any $\epsilon > 0$ choose such a ball with $\|B(z) - B(x)\| \le \epsilon$. Taking $B_0 = B(x)$ proves that the remainder is at most $\epsilon\|y - x\|$, which is precisely Fréchet differentiability with derivative $B(x)$. □

We apply the lemma to the slab operators of Section 33. For fixed width $d > 0$, arity n, and labels $\boldsymbol{v} = (v_1, \dots, v_n)$, write $A_{\boldsymbol{v}}(x) = A_d((x_1, v_1) \cdots (x_n, v_n))$ on

$$U_{d,n} = \{x : x_i \neq x_j\ (i \neq j), \quad 0 < (x_i)_0 < d\}.$$

This is open, since finitely many strict inequalities describe the time conditions and finitely many diagonals have been excluded.

The two quantitative and differential inputs are as follows. First, for each grade bound D and label norm bound R, there are $C > 0$ and $K \in \mathbb{N}$ such that every slab word of length n, with these label bounds, satisfies

$$\|A_d(w)\| \le C^n (n/r)^{Kn} \tag{133}$$

whenever $0 < r \le 1$, its points have mutual sup-norm distances at least r, and each time coordinate lies in $[r, d-r]$. The constants are independent of n, d and the positions. For the empty word the right side is interpreted as one, in agreement with $A_d(\varnothing) = E_d$. Second, for each raw label v of grade d' and each coordinate j, choose a raw label $\delta_j v$ of grade $d'+1$ representing its field derivative. Assume that the smooth scalar correlations obey the exact slot identity

$$\partial_{(x_i)_j} S_{\boldsymbol{v}}(x) = S_{\boldsymbol{v}[i \leftarrow \delta_j v_i]}(x) \tag{134}$$

at distinct points, also with any finite list of exterior insertions. The choice of raw representative need not itself be linear. For a phase using the original field quotient, the identity includes the assertion that the chosen representatives have the indicated values in that phase.

Proposition 34.2 (Smooth slab operators)**.** *Under* (133) *and* (134)*, $A_{\boldsymbol{v}} : U_{d,n} \to B(\mathcal{H})$ is smooth in operator norm, and*

$$\partial_{(x_i)_j} A_{\boldsymbol{v}}(x) = A_{\boldsymbol{v}[i \leftarrow \delta_j v_i]}(x).$$

Proof. Fix a point $x \in U_{d,n}$. Choose $0 < r \le 1$ below its finitely many positive time margins and pairwise distances. In a neighborhood of x, the same margins and distances are at least $r/2$. For every fixed label tuple, (133) therefore gives a local operator bound. The same holds for every one-slot differentiated tuple, since there are finitely many labels and their grades and norms have finite maxima.

Define the real-linear derivative candidate

$$B_{\boldsymbol{v}}(x)[h] = \sum_{i=1}^{n} \sum_{j=0}^{3} h_{i,j} A_{\boldsymbol{v}[i \leftarrow \delta_j v_i]}(x).$$

The finite collection of local bounds gives a local bound on $\|B_{\boldsymbol{v}}(x)\|$. For positive words p, q,

$$\langle [p], A_{\boldsymbol{v}}(x)[q] \rangle = S((\theta p)\, w(x, \boldsymbol{v})\, \tau_d q).$$

The exterior points lie respectively below zero and above d, so all configurations remain distinct near x. Differentiating by (134) gives exactly the matrix elements of $B_{\boldsymbol{v}}$. Sesquilinearity extends this statement to finite word sums, a dense linear subspace. The remainder lemma now proves norm continuity of $A_{\boldsymbol{v}}$, for every tuple.

In particular each differentiated tuple has norm-continuous operators. Thus the finite sum defining $B_{\boldsymbol{v}}$ is norm-continuous. A second application of the remainder lemma proves $DA_{\boldsymbol{v}} = B_{\boldsymbol{v}}$. Inductively, if all the tuple maps are C^k, then each $B_{\boldsymbol{v}}$ is C^k as a finite sum of fixed coordinate functionals times such maps. The derivative formula makes all tuple maps C^{k+1}. Starting from continuity proves smoothness of every finite order, with the stated coordinate derivatives. □

Corollary 34.3 (Norm separability of the compression algebra)**.** *If the scalar correlations are multilinear in the finite-dimensional raw label spaces, the hypotheses of Proposition 34.2 imply that $\mathcal{A}$ is separable in operator norm.*

Proof. Fix a finite basis in every raw label space. At fixed positions, the slab operator is multilinear in the labels: pairing the purported linear identity with any two word vectors reduces it to scalar multilinearity; density then proves equality of bounded operators. Every compressed word operator is therefore a finite linear combination of ones with basis labels.

Place a word in positive time and choose a positive rational width greater than all its time coordinates. Approximate its distinct positions by rational positions inside this slab. Operator-norm continuity from Proposition 34.2, followed by the bounded compression $A \mapsto P_0 A|_{\mathcal{H}_0}$, proves

norm convergence of the compressed operators. There are countably many rational widths, rational configurations, arities and finite basis-label tuples. Their compressed operators thus generate a norm-dense subalgebra. Star polynomials in this countable set with coefficients in $\mathbb{Q} + i\mathbb{Q}$ form a countable set dense in the whole norm-closed algebra: approximate the finitely many generators and coefficients in any given polynomial, and then use the definition of the closure. This is the required norm separability. □

The first analytic input, the estimate uniform in the length of the word, is stronger than the separated-correlation bound used to construct time translations. The second input is an exact differential identity. The argument above supplies operator smoothness and norm separability from these two inputs; it leaves their derivation from the lattice fields to the construction chapters.

35 Characters and selection of a vacuum

The preceding section constructs a commutative algebra on the whole ground space. Its characters provide a useful way of separating the vacuum state into components. The argument below also explains why a one-dimensional ground space is an almost-everywhere conclusion, rather than an assumption about the unselected table.

Proposition 35.1 (The representing probability)**.** *Suppose the algebra $\mathcal{A}$ of Proposition 33.5 is separable in operator norm. Its space Ξ of unital complex star characters, with the topology of pointwise convergence on $\mathcal{A}$, is compact and metrizable. There is a probability measure μ on Ξ, positive on every nonempty open set, such that*

$$\int_{\Xi} \chi(A)\, d\mu(\chi) = \langle \Omega, A\Omega \rangle \quad (A \in \mathcal{A}). \tag{135}$$

Proof. The algebra is a nonzero unital commutative C^*-algebra: it is a norm-closed star subalgebra of $B(\mathcal{H}_0)$, and $\|\Omega\| = 1$. The commutative Gelfand theorem identifies it isometrically, as a star algebra, with $C(\Xi)$, by $A \mapsto \widehat{A}$, where $\widehat{A}(\chi) = \chi(A)$ [2, Theorem 2.13]. Its character space is compact since the algebra is unital.

Choose a countable norm-dense sequence (A_j) in $\mathcal{A}$. The continuous map $\chi \mapsto (\chi(A_j))_j$ into $\mathbb{C}^{\mathbb{N}}$ is injective: equality on the dense sequence implies equality everywhere by continuity of the characters. A continuous injection of a compact space into a Hausdorff space is a homeomorphism onto its image (images of closed sets are compact and hence closed). The countable product is metrizable, for example by $\sum_j 2^{-j-1} \min(1, |z_j - z'_j|)$, so Ξ is metrizable.

Transfer the functional $A \mapsto \langle \Omega, A\Omega \rangle$ to $C(\Xi)$. It is positive: a nonnegative continuous function is $\overline{g}g$ for its continuous nonnegative square root g, and its inverse Gelfand transform has the form B^*B. The functional takes the value $\|B\Omega\|^2 \geq 0$ there. The Riesz representation theorem supplies a positive Borel measure representing it [3, Chapter 6, Theorem 32]. Compactness gives $C_c(\Xi) = C(\Xi)$, so that theorem applies directly. At the constant function one it gives $\mu(\Xi) = \|\Omega\|^2 = 1$.

To prove full support, suppose a nonempty open set O has zero measure. Choose $x \in O$ and $r > 0$ with $B(x, r) \subset O$. The function $f(y) = \max(0, r - \operatorname{dist}(x, y))$ is continuous, nonzero at x, and zero outside O. Let A be its inverse Gelfand transform. Then (135) applied to A^*A gives $\|A\Omega\|^2 = \int |f|^2\, d\mu = 0$. Separation of Ω forces $A = 0$, contradicting $f(x) = r$. Thus every such open set has positive measure. □

For $\chi \in \Xi$, define the whole-word values

$$S_\chi(w) = \chi(M(w)) \quad \text{for distinct-point words } w. \tag{136}$$

It is essential to apply the character to the operator for the entire word. The compressed operator $M(w_1w_2)$ need not equal $M(w_1)M(w_2)$ at finite separation. Equations (135) and (131) give the barycenter identity

$$S(w) = \int_{\Xi} S_\chi(w)\, d\mu(\chi). \tag{137}$$

For a fixed word the integrand is continuous, and hence integrable.

Some properties of S_χ already follow at this level. Normalization and translation invariance follow from $M(\varnothing) = I$ and (131). It is also reflection positive. For positive words v_i and coefficients c_i, the Gram identity in the preceding section gives

$$\sum_{i,j} \overline{c_i} c_j M((\theta v_i)v_j) = \left(\sum_i c_i X(v_i)\right)^* \left(\sum_j c_j X(v_j)\right) \geq 0 \quad \text{in } B(\mathcal{H}_0).$$

The operator on the left lies in $\mathcal{A}$. Positivity is the same in a unital C^*-subalgebra as in the ambient operator algebra (spectral permanence and continuous functional calculus give its nonnegative square root in the subalgebra; see [2, Theorems 2.17 and 2.19]). Characters take positive elements to nonnegative numbers. Applying χ proves the desired Gram positivity. Regularity and the remaining covariance and field identities still need their own operator estimates; the barycenter formula alone does not establish them.

Lemma 35.2 (A common countable collection of words)**.** *Suppose the raw label spaces are finite-dimensional complex vector spaces indexed by the nonnegative integers. Suppose that each S_χ is multilinear in its labels and continuous on the distinct-point configuration space, and has the Hilbert realization of Proposition 32.3. There is one countable collection of positive words whose vectors span a dense subspace in every $\mathcal{H}_\chi$.*

Proof. Choose a finite basis in every raw label space. Take all positive words with rational coordinates and with basis-vector labels, for every arity and every finite list of label grades; include the empty word. There are countably many choices. Every word is a finite linear combination of words with basis labels, by expanding each label in its basis. This identity holds for the Hilbert-space vectors: pair the difference with any word vector and use multilinearity of the correlation in the positive-word labels. All such inner products vanish, so density forces the difference to be zero. For a fixed basis-labelled positive word, approximate its positions by rational positions, remaining within the open set of positive distinct-point configurations. Its vectors converge in every $\mathcal{H}_\chi$: the squared norm of their difference is the sum of two diagonal Gram values minus twice the real part of the cross Gram value, and continuity makes this tend to zero. The reflected and positive halves remain strictly separated throughout this limit. The full word span is dense by construction; thus the selected countable span is dense as well. The positions and bases were chosen without reference to χ, so the collection is common to every phase. □

Theorem 35.3 (A one-dimensional ground space in almost every phase)**.** *Assume every S_χ has the Euclidean Hilbert-space dynamics of Theorem 32.5, with unit vacuum Ω_χ. Assume there is a common countable collection of positive words whose vectors span densely in every phase Hilbert space. Then, for μ-almost every χ,*

$$\mathcal{H}_{\chi,0} = \mathbb{C}\Omega_\chi.$$

In particular there exists a phase with this property.

Proof. Fix a positive word v. For integers $n \geq 0$, set

$$f_n(\chi) = \|E_{\chi,n}[v]_\chi\|^2 = S_\chi((\theta\tau_n v)(\tau_n v)).$$

Each is a continuous function of χ, by (136). Contractivity gives $0 \leq f_n \leq f_0$; the latter is integrable on the compact probability space. The large-time projection theorem gives $f_n(\chi) \to \|P_{\chi,0}[v]_\chi\|^2$. Dominated convergence and the barycenter identity imply

$$\int_\Xi \|P_{\chi,0}[v]_\chi\|^2 \, d\mu = \lim_{n\to\infty} \|E_n[v]\|^2 = \|P_0[v]\|^2. \tag{138}$$

All limit functions here are measurable as pointwise limits of the continuous f_n.

Apply (135) to $M(v)^*M(v)$. Since characters preserve products and adjoints and $M(v)\Omega = P_0[v]$,

$$\int_\Xi |S_\chi(v)|^2 \, d\mu = \|P_0[v]\|^2.$$

Moreover $\langle \Omega_\chi, P_{\chi,0}[v]_\chi \rangle = S_\chi(v)$, because the projection fixes the vacuum. Orthogonal decomposition along the unit vector Ω_χ gives

$$\|P_{\chi,0}[v]_\chi - S_\chi(v)\Omega_\chi\|^2 = \|P_{\chi,0}[v]_\chi\|^2 - |S_\chi(v)|^2.$$

The nonnegative right side is integrable and, by the two preceding identities, has integral zero. It is therefore zero almost everywhere.

Apply this conclusion to the common countable collection and intersect its full-measure sets. For every χ in the intersection, the projection of its dense word span lies in $\mathbb{C}\Omega_\chi$. This one-dimensional subspace is closed. Continuity of the projection therefore implies $\operatorname{ran} P_{\chi,0} \subset \mathbb{C}\Omega_\chi$. The reverse inclusion holds because the vacuum is fixed. Finally a set of measure one in a probability space is nonempty. □

This selection argument requires the phase Hilbert spaces and their contraction semigroups before it can be applied. It does not construct them from a probability decomposition alone. Retention of the raw-label null spaces, field derivatives, stress tensor and relativistic reconstruction remain additional obligations for the chosen phase.

36 Regularity and bounded insertions in the component theories

Several hypotheses in the selection theorem can now be supplied from the unselected operator estimates. Assume in this section that (133) and (134) hold for the unselected theory, and that its correlations are multilinear and permutation symmetric. By Corollary 34.3, the representing probability of Proposition 35.1 is then available.

Proposition 36.1 (Smoothness and caps in every character)**.** *The whole-word functions S_χ are multilinear in the raw labels, permutation symmetric, and smooth on distinct-point configurations. They obey the slot derivative identity* (134)*. For every grade bound D and label norm bound R, the constants C, K in* (133) *also give*

$$|S_\chi(w)| \leq C^n (n/r)^{Kn} \tag{139}$$

for all χ, all words with these label bounds, and $0 < r \leq 1$ with pairwise sup-norm separations at least r. For every fixed label tuple and derivative order q, there are $C_q \geq 0$ and $A_q \in \mathbb{N}$, independent of χ and the positions, such that

$$\|D^q S_{\chi,\boldsymbol{v}}(x)\| \leq C_q r^{-A_q}$$

on the same separated configurations.

Proof. Multilinearity of slab operators was proved in the norm-separability argument. Their permutation symmetry follows by the same dense matrix-element test, using scalar permutation symmetry. Compression and the linear character preserve both identities. For local smoothness, fix a distinct configuration, translate it into a slab, and keep the translation and width fixed in a small neighborhood. Proposition 34.2 gives smoothness of the operator-valued map and its slot derivative formula. Compression is bounded linear, and the character is bounded linear on $\mathcal{A}$; their composition gives the same statements for S_χ. Equivalently one may extend the character to the ambient operator space by Hahn–Banach before composing; the extension agrees on every compressed word and its derivatives.

To obtain (139), translate a given word into a slab with one unit of padding at each time boundary. Translation leaves all pairwise separations unchanged, and the boundary padding is at least r. Contractivity of characters and of orthogonal compression gives

$$|S_\chi(w)| = |\chi(M(w))| \le \|M(w)\| \le \|A_d(w)\| \le C^n (n/r)^{Kn}.$$

For the empty word use normalization. For nonzero arity, iterate the slot derivative formula q times. There are finitely many resulting label tuples, one for each sequence of q coordinate directions. Their grades and raw-label norms have finite maxima, so apply (139) with these maxima. Expand the real q-linear derivative in the standard coordinate basis of $\mathbb{R}^{4n}$. On unit arguments every coordinate coefficient has absolute value at most one, so its operator norm is bounded by the sum of the absolute values of its at most $(4n)^q$ entries. This yields the claimed constants and the integer exponent $A_q = Kn$ for the enlarged label bounds. At arity zero the function is constant one, with all positive-order derivatives zero. □

Corollary 36.2 (Phase Hilbert spaces and selection)**.** *Every character table has the Hilbert space and strongly continuous contraction semigroup of Section 32. The common countable family in Lemma 35.2 exists. Consequently Theorem 35.3 applies, giving a phase whose time-invariant subspace is exactly its vacuum line.*

Proof. Normalization, translation invariance and time reflection positivity were proved after (136). The adjoint identity $M(\theta w) = M(w)^*$ gives $S_\chi(\theta w) = \overline{S_\chi(w)}$. For a word in arbitrary position, this operator identity follows by translating it into a slab, using its opposite word there and the slab adjoint identity, and then using placement independence of M. Proposition 36.1 supplies permutation symmetry, continuity and the separated zeroth-order bound. Lemma 32.2 gives the boundedness under separating translations required in (124). All hypotheses of the Hilbert-space and contraction construction are therefore met for each S_χ. Multilinearity, finite-dimensional raw-label spaces and continuity give the common countable dense span by Lemma 35.2. The selection theorem now applies. □

Lemma 36.3 (Smeared reflection positivity)**.** *For any correlation table with continuous distinct-point correlations and point reflection positivity, the reflection form is nonnegative on smooth compactly supported positive-time configuration tests.*

Proof. For one fixed label tuple, the map $x \mapsto [w(x)]$ into the GNS space is norm-continuous on positive-time distinct-point configurations. Indeed its squared norm difference is a finite combination of continuous Gram entries, as in Lemma 35.2. On the compact support of a test φ, it is bounded and its image is a separable compact set. Hence the compactly supported continuous vector function $\varphi(x)[w(x)]$ is Bochner integrable. Denote its integral by Ψ_φ. For two such tests, continuity of the inner product and Fubini give

$$\langle \Psi_\varphi, \Psi_\psi \rangle = \iint \overline{\varphi(x)} \psi(y) S((\theta w(x)) w(y))\, dx\, dy.$$

The scalar integrand is absolutely integrable: on the product of the two compact supports the two reflected halves have a positive time gap, each half has internally distinct points, and the continuous kernel is bounded. Different arities and labels cause no change to this argument; each pair has its own compact product support. The empty word uses the one-point zero-dimensional configuration space. A finite combination of tests has reflection form $\| \sum_i c_i \Psi_{\varphi_i} \|^2 \geq 0$. □

Proposition 36.4 (Slab bounds survive character selection)**.** *Let $A_{\chi,d}(w)$ denote insertion in the Hilbert space of a character table. Such an operator exists for every slab word, and*

$$\|A_{\chi,d}(w)\| \leq \|A_d(w)\|.$$

In particular the uniform bound (133) *holds in every phase with the same constants. Its slab operators are smooth in operator norm and obey the same slot derivative identities.*

Proof. Let $a = A_d(w)$ in the unselected Hilbert space and put $c = \|a\|$. For a positive word v, let $v_n = (I_{w^\sharp} I_w)^n v$. The maps $X(v) : \mathcal{H}_0 \to \mathcal{H}$ from Section 33 satisfy

$$X(I_w v) = aX(v), \qquad X(v_n) = (a^* a)^n X(v).$$

The first equality is the slab concatenation law restricted to $\mathcal{H}_0$; the second follows by induction and the adjoint law. Therefore $\|X(v_n)\| \leq c^{2n} \|X(v)\|$. For any $D > c$, the character Gram identity gives

$$\|[v_n]_\chi\|^2 = \chi\big(M((\theta v_n) v_n)\big) = \chi\big(X(v_n)^* X(v_n)\big) \leq \|X(v)\|^2 D^{4n}.$$

Here $X(v_n)^* X(v_n) \in \mathcal{A}$ by its equality with the compressed whole-word operator, and the character bound applies in that algebra. Lemma 33.2 produces a phase insertion operator with norm at most D. These operators are all identical: their action on every positive word is prescribed, and the word span is dense. Taking the infimum over $D > c$ proves the bound c, also when $c = 0$. The unselected uniform estimate therefore transfers. The character slot identities were proved in Proposition 36.1; applying Proposition 34.2 inside the phase Hilbert space proves its operator smoothness and derivative formulas. □

These results construct the time-axis Euclidean component theories from the unselected slab and differential estimates. Full orthogonal covariance of each character, and ordinary conjugation separately from reflected conjugation, still require the reflection-transport argument. Equality of null spaces requires the detector inputs in Section 38.

37 Ordinary reflections and covariance of individual phases

Reflected conjugation is built into the Hilbert inner product. Ordinary reflection, which acts complex-linearly on labels, is a different operation. Proving that it fixes each compressed observable is the extra step needed to obtain orthogonal covariance in each character separately. We denote ordinary time reflection by ρ; thus θ also conjugates labels and reverses the word, whereas ρ does neither. Permutation symmetry makes the reversal immaterial to M.

Lemma 37.1 (Positivity from bounded real approximations)**.** *Suppose real numbers $b_t(i,j)$, for integers $i < j$, satisfy $b_t(i,j) \to a$ as $t \to \infty$, with the same limit a for every pair. Suppose there are $\epsilon_k \geq 0$ tending to zero and finite $R_k \geq 0$ with the following property. For every k and finite m, for all sufficiently large t there exist simultaneous real random variables $X_1, \ldots, X_m$, possibly on cutoff-dependent probability spaces, such that eventually in the ordered cutoff limit*

$$|X_i| \leq R_k, \qquad \mathbb{E}(X_i X_j) \leq b_{\mathrm{cut},t}(i,j) + \epsilon_k \quad (i < j),$$

where $b_{\mathrm{cut},t}(i,j) \to b_t(i,j)$ *in that cutoff limit. Then* $a \geq 0$.

Proof. The variables are bounded and measurable, so their products are integrable. Nonnegativity of a square gives

$$0 \leq \mathbb{E}\left(\sum_{i=1}^{m} X_i\right)^2 \leq mR_k^2 + 2\sum_{i<j}\big(b_{\mathrm{cut},t}(i,j) + \epsilon_k\big).$$

For fixed k, m, t, all eventual conditions can be imposed together: only finitely many variables and pairs occur. Remove the cutoff in the specified order, then send $t \to \infty$, obtaining $0 \leq mR_k^2 + m(m-1)(a+\epsilon_k)$. For $m \geq 2$, divide by $m(m-1)$ and send $m \to \infty$. This yields $a + \epsilon_k \geq 0$. Now send $k \to \infty$. The order of these limits matters: R_k may diverge as k grows. □

The lattice construction applies the lemma to a real-labelled word w at an internally generic grid configuration, minus its ordinary reflection. Translate several copies along one vector with strictly positive coordinates, spacing the i-th copy by it times that vector. For fixed distinct copies and large t, their combined configurations are generic in every coordinate. The covariance to be approximated is the four-term expression

$$S(w_i w_j) - S(w_i(\rho w)_j) - S((\rho w)_i w_j) + S((\rho w)_i(\rho w)_j),$$

or its real part. By Lemma 33.4, its large-separation limit is

$$a = \mathrm{Re}\langle \Omega, (M(w) - M(\rho w))^2 \Omega\rangle. \tag{140}$$

The input required here is the existence of the bounded real variables in Lemma 37.1, with errors independent of the number and offsets of copies. Ordinary positivity of a Wilson probability measure supplies the nonnegative square once these simultaneous approximations are available; it does not supply the approximations or their uniform error bounds.

Proposition 37.2 (Ordinary reflection fixes compressed words)**.** *Assume the bounded real approximation property just described holds on a dense set of internally generic grid configurations for each tuple of real basis labels. Assume the norm-regularity hypotheses of Section 34 and that each raw label space has a basis fixed by conjugation. Then* $M(\rho w) = M(w)$ *for every distinct-point word.*

Proof. First take a real basis-labelled word at an admitted configuration. Since its labels are fixed by conjugation, permutation symmetry and the adjoint identity give $M(\rho w) = M(\theta w) = M(w)^*$. Thus $D = M(w) - M(\rho w)$ satisfies $D^* = -D$. The previous lemma and (140) imply

$$0 \leq \mathrm{Re}\langle \Omega, D^2 \Omega\rangle = -\|D\Omega\|^2.$$

Hence $D\Omega = 0$, and separation of Ω for $\mathcal{A}$ gives $D = 0$. Norm continuity of compressed words and their reflected words extends the identity from the dense admitted configurations to all distinct configurations for these basis labels. Both M and ordinary reflection are complex-multilinear in the labels. Expanding an arbitrary tuple in the real basis therefore extends the equality to every tuple of complex labels. □

We make explicit one other algebraic point used below. Define a null raw label to be one whose insertion gives zero correlation with every finite list of exterior insertions at distinct points. A slab operator containing a null label is zero: pair it with two arbitrary word vectors, use the null-label property on the resulting whole correlation, and use density. Consequently $M(w)$ depends only on the quotient classes of its labels. Assume that the orthogonal action on those classes is an actual group action and agrees, for signed coordinate transformations, with the raw signed-label action after projection. These are the precise compatibilities used when composing rotations below.

Proposition 37.3 (Orthogonal covariance in each character)**.** *Under these compatibilities and Proposition 37.2, $M(Kw) = M(w)$ for every $K \in O(4)$ and distinct-point word w. Thus $S_\chi(Kw) = S_\chi(w)$ for every character.*

Proof. Write $F(p, w) = \langle \Omega, M(p)M(w)\Omega \rangle$. We first show simultaneous invariance

$$F(Kp, Kw) = F(p, w) \qquad (K \in O(4)). \tag{141}$$

The cluster argument in Proposition 33.5 already proves this whenever there exists a vector b with $b_0 > 0$ and $(Kb)_0 > 0$. Put $a = (Ke_0)_0$. If $a > -1$, choose $b = e_0 + K^{-1}e_0$; orthogonality gives $b_0 = (Kb)_0 = 1 + a > 0$. If $a = -1$, then $(\rho K e_0)_0 = 1$, so the argument applies to ρK. Individual invariance under ρ removes the extra reflection from each factor and again gives (141).

For any $J \in O(4)$, rotate a pair backwards by J^{-1}, apply ordinary reflection to its second factor alone, and rotate forwards by J. Equation (141) gives

$$F(p, w) = F(J^{-1}p, J^{-1}w) = F(J^{-1}p, \rho J^{-1}w) = F(p, J\rho J^{-1}w).$$

There is no ambiguity from raw representatives, by quotient independence. Pair values separate compressed operators: if $F(p, w) = F(p, z)$ for every p, replace p by θv to obtain

$$\langle M(v)\Omega, (M(w) - M(z))\Omega \rangle = 0.$$

The linear span of $M(v)\Omega$ is dense in $\mathcal{H}_0$, so $(M(w) - M(z))\Omega = 0$; separation gives $M(w) = M(z)$. Thus M is individually invariant under every conjugate $J\rho J^{-1}$.

These conjugates are all hyperplane reflections: map e_0 to any chosen unit normal by an orthogonal map. For completeness, hyperplane reflections generate $O(d)$. Given an orthogonal map K, if $Ke_1 \neq e_1$, reflection in the hyperplane normal to $Ke_1 - e_1$ sends Ke_1 to e_1; this follows immediately from the reflection formula and $|Ke_1| = |e_1| = 1$. Composing with this reflection gives a map fixing e_1. It preserves $e_1^\perp$, and induction on dimension writes its restriction as a product of reflections there, extended to fix e_1. If it already fixes e_1, start with that restriction. Dimension zero is trivial. Invariance under products follows from the group action on quotient labels. This proves individual invariance under every orthogonal map, and evaluation by a character gives the asserted phase covariance. □

Ordinary conjugation now follows as well. Since θw is a permutation of $\rho\overline{w}$,

$$S_\chi(\overline{w}) = S_\chi(\rho\overline{w}) = S_\chi(\theta w) = \overline{S_\chi(w)}.$$

Signed coordinate covariance follows from the compatibility of the quotient action with the raw signed action. Transporting time-axis reflection positivity by signed permutations gives reflection positivity about every coordinate axis: choose a signed permutation taking that axis to the time axis, rotate each positive word and its labels, and use covariance and the conjugation compatibility in the Gram matrix. Lemma 36.3 then supplies the corresponding smeared positivity. The unproved quantitative input in this chapter is the bounded real approximation property on arbitrarily fine generic grids; its uniformity over finitely many replicated copies is indispensable.

38 Keeping the same field space in each phase

A field label may disappear when one passes to a component of a mixture. For the selected phase to represent the intended fields, one must rule out additional null labels. The following argument explains the role of the finite detectors and the uniform moment estimates used for this purpose.

Lemma 38.1 (Moment bounds at every character)**.** *Let $A \in \mathcal{A}$, let $t \geq 0$, and suppose*

$$\langle \Omega, (A^*A)^n \Omega \rangle \leq t^{2n} \qquad (n = 0, 1, 2, \ldots).$$

Then $|\chi(A)| \leq t$ for every $\chi \in \Xi$.

Proof. The representing measure gives $\int |\chi(A)|^{2n}\, d\mu \leq t^{2n}$. If $|\chi_0(A)| > t$, choose p strictly between these numbers. The nonempty open set $O = \{\chi : |\chi(A)| > p\}$ has positive measure c by full support. Hence $cp^{2n} \leq \int |\chi(A)|^{2n}\, d\mu \leq t^{2n}$. Since $p > 0$, division gives $c \leq (t/p)^{2n} \to 0$, a contradiction. This also covers $t = 0$, when the right side already vanishes for positive n. □

Here is the elementary two-point calculation to which these moment estimates are applied. At a fixed scale k, let a, b denote two insertions at distinct points and let $j_a, j_b, \kappa \in \mathbb{C}$ be the prescribed comparison one-point values and connected two-point value. Form the elements

$$D_a = M(a) - j_a I, \qquad D_b = M(b) - j_b I,$$
$$D_{ab} = M(ab) - j_a M(b) - j_b M(a) + (j_a j_b - \kappa) I.$$

Assume the preceding lemma's moment bounds hold for each of these three elements with the same number $\epsilon_k \geq 0$. It follows for every character that

$$|S_\chi(a) - j_a| \leq \epsilon_k, \tag{142}$$
$$|S_\chi(ab) - (\kappa + j_a j_b)| \leq \epsilon_k (1 + |j_a| + |j_b|). \tag{143}$$

Indeed, evaluating D_{ab} and rearranging gives

$$S_\chi(ab) - (\kappa + j_a j_b) = \chi(D_{ab}) + j_a (S_\chi(b) - j_b) + j_b (S_\chi(a) - j_a),$$

so the triangle inequality proves (143). If $\epsilon_k \to 0$ and the prescribed one-point values converge, these errors tend to zero in every phase, with a bound independent of the character. This conclusion requires moments of the full corrected operators D_{ab}, not only pointwise estimates for the unselected two-point correlations.

Proposition 38.2 (Finite detectors prevent an additional null space)**.** *Let E be a complex vector space of raw labels, and let $N \subset N_\chi$ be subspaces. Write $\pi : E \to E/N = V$ and $\pi_\chi : E \to E/N_\chi = V_\chi$. Let W be a finite-dimensional complex normed space. Suppose there are linear maps $J_k : W \to E$, linear tests $\ell_{k,i} : V_\chi \to \mathbb{C}$, $1 \leq i \leq r$, and an injective linear map $F : W \to \mathbb{C}^r$ such that:*

1. *πJ_k is surjective for all sufficiently large k;*
2. *for every vector of a fixed basis of W,*

$$F_k w := \big(\ell_{k,i}(\pi_\chi J_k w)\big)_{i=1}^r \longrightarrow Fw.$$

Then $N_\chi = N$.

Proof. Convergence on a finite basis implies convergence in operator norm. To see this, let $e_1, \ldots, e_s$ be that basis with continuous coordinate functionals e_j^*. Then

$$\|(F_k - F)w\| \leq \sum_{j=1}^s |e_j^*(w)|\, \|(F_k - F)e_j\| \leq \|w\| \sum_{j=1}^s \|e_j^*\|\, \|(F_k - F)e_j\|.$$

The finite sum tends to zero. If $W \neq \{0\}$, compactness of its unit sphere and injectivity of F give $c = \min_{\|w\|=1} \|Fw\| > 0$. Choose k late enough that $\|F_k - F\| < c/2$ and πJ_k is onto. Then $\|F_k w\| \geq (c/2)\|w\|$, so F_k is injective. If $W = \{0\}$, surjectivity already gives $V = 0$, and the result is immediate.

Now take $e \in N_\chi$. By surjectivity choose $w \in W$ with $\pi J_k w = \pi e$. Thus $J_k w - e \in N \subset N_\chi$, and since $e \in N_\chi$, also $J_k w \in N_\chi$. All tests vanish on $\pi_\chi J_k w = 0$; hence $F_k w = 0$. Injectivity forces $w = 0$, so $\pi e = \pi J_k 0 = 0$, or $e \in N$. This proves $N_\chi \subset N$; the reverse inclusion was assumed. □

In the application at label degree d, W is the comparison field space, J_k consists of its corrected labels at the chosen scales, and the tests are one- or two-point correlations with fixed exterior labels. Estimates (142)–(143), combined with continuity at the limiting distinct test configurations, give convergence on a basis. The comparison tests separate W, so their joint map F is injective. The first hypothesis of the proposition is the eventual spanning statement for the corrected labels in the unselected quotient. Establishing these detector moment bounds, the separating comparison tests and eventual spanning from the lattice construction is still required. The proposition proves exactly how those inputs imply that every phase has the same null space; it does not infer their existence from finite dimensionality alone.

39 Complex heat time and real-time translations

The complex heat extension used in the boundary argument can be constructed from the real Euclidean semigroup alone. This also fixes the sign of the real-time translation group. Let $(E_t)_{t\geq 0}$ be a strongly continuous semigroup of self-adjoint contractions on a complex Hilbert space, with $E_0 = I$. Each E_t is positive because $E_t = E_{t/2}^2$.

We use continuous functional calculus for a bounded self-adjoint operator: continuous scalar functions on its spectrum act as a unital isometric star homomorphism, with the usual composition rule. This is the self-adjoint instance of the continuous calculus in [2, Theorem 2.19]. All uses below are for a positive contraction and scalar functions continuous at zero; an imaginary power at zero is never inserted into this calculus.

Proposition 39.1. *There is a holomorphic operator-valued function $E(z)$ on $\operatorname{Re} z > 0$ such that*

$$\|E(z)\| \leq 1, \quad E(z+w) = E(z)E(w), \quad E(z)^* = E(\overline{z}), \quad E(t) = E_t \quad (t > 0).$$

Each E_t, $t > 0$, is injective and has dense range. There is a strongly continuous unitary group $V(t)$, $t \in \mathbb{R}$, characterized by

$$V(t)E(z) = E(z)V(t) = E(z - it) \qquad (\operatorname{Re} z > 0). \tag{144}$$

Every bounded operator commuting with all E_t commutes with $E(z)$ and $V(t)$. If $E_t\Omega = \Omega$ for all t, then $E(z)\Omega = V(t)\Omega = \Omega$.

Proof. Put $A = E_1$. Its spectrum is contained in $[0, 1]$: positivity excludes negative spectrum and the contraction bound excludes values above one. For $\operatorname{Re} z > 0$, define

$$f_z(\lambda) = \begin{cases} \exp(z \log \lambda), & \lambda > 0, \\ 0, & \lambda = 0, \end{cases} \qquad E(z) = f_z(A).$$

This scalar function is continuous at zero and has magnitude at most one. The norm bound, product law, and adjoint identity follow from the calculus. Holomorphy holds in the uniform norm on the

scalar functions, hence in operator norm. Indeed, on a compact neighborhood of any such z choose $\delta > 0$ with $\operatorname{Re} z \geq 2\delta$. The functions $\lambda^\delta |\log \lambda|^j$ are bounded on $(0, 1]$ and tend to zero at zero for each fixed j. Taylor's formula for the scalar exponential then gives, uniformly in $\lambda \in [0, 1]$ and for sufficiently small h,

$$|f_{z+h}(\lambda) - f_z(\lambda) - h\lambda^z \log \lambda| \leq C_\delta |h|^2.$$

At zero the derivative is defined to be zero. This proves complex differentiability into the continuous-function space.

For a positive integer n, put $B = E_{1/n}$. The semigroup law gives $B^n = A$. Functional calculus on the positive operator B yields $(B^n)^{1/n} = B$, because $(\lambda^n)^{1/n} = \lambda$ for $\lambda \geq 0$. The composition rule thus gives $E(p/n) = A^{p/n} = B^p = E_{p/n}$ for positive integers p, n. Approximate any positive real t by positive rationals. Operator-norm continuity of $E(z)$ and strong continuity of the original semigroup show, after application to each vector, that $E(t) = E_t$.

If $E_t x = 0$, then $\|E_{t/2} x\|^2 = \langle x, E_t x \rangle = 0$. Iteration gives $E_{t/2^n} x = 0$ for every n, and strong continuity at zero implies $x = 0$. Self-adjointness then gives dense range, since the orthogonal complement of the range is the kernel of the adjoint. For real t and $r > 0$, the product and adjoint identities imply

$$E(r - it)^* E(r - it) = E(2r) = E_r^* E_r, \qquad \|E(r - it)x\| = \|E_r x\|.$$

Define an isometry initially on the dense range of A by $V(t)Ax = E(1 - it)x$. Injectivity of A makes the definition unambiguous; the displayed norm identity extends it uniquely to an isometry of the whole Hilbert space.

For $\operatorname{Re} z > 0$, both equalities in (144) may be checked on Ax. For example,

$$V(t)E(z)Ax = V(t)AE(z)x = E(1 - it)E(z)x = E(z - it)Ax.$$

The other order is identical. Thus these are bounded-operator identities. In particular, on Ax,

$$V(t)V(s)Ax = V(t)E(1 - is)x = E(1 - i(s + t))x = V(t + s)Ax.$$

Density proves the group law. Since $V(0) = I$, each isometry has inverse $V(-t)$ and is unitary. For Ax, the orbit is continuous by holomorphy of $E(1 - it)$; approximation by the dense range of A, with the unitary norm bound on both sides, proves strong continuity for every vector.

Commutation with A implies commutation with every continuous function of A: approximate such a function uniformly on its real spectrum by polynomials. Thus a bounded operator commuting with all E_t commutes with $E(z)$. Checking on Ax then gives commutation with $V(t)$. Finally $A\Omega = \Omega$ implies $f(A)\Omega = f(1)\Omega$, first for polynomials and then for continuous f. Apply this to f_z and then the definition of $V(t)$ to obtain the vacuum assertions. □

Suppose now that spatial translations $U(x)$ form a strongly continuous unitary representation commuting with the heat semigroup and fixing Ω_E. The proposition defines a strongly continuous unitary representation of all real translations,

$$T_E(a^0, \mathbf{a}) = V(a^0)U(\mathbf{a}).$$

The group law follows from commutation. Joint strong continuity follows by adding and subtracting $V(a^0)U(\mathbf{b})\xi$ and using the unitary norms. Moreover $E(\varepsilon - it)\xi = V(t)E_\varepsilon \xi \to V(t)\xi$ as $\varepsilon \downarrow 0$, which justifies the boundary notation $E(-it)$ used in the heat-vector chapter. No operator-norm continuity at the boundary is required.

Lemma 39.2. *The vectors fixed by every $V(t)$ are exactly the vectors fixed by every E_s. Consequently, if the selected heat ground space is $\mathbb{C}\Omega_E$, the same is true of the subspace fixed by all real translations.*

Proof. A heat-fixed vector satisfies $A\xi = \xi$, so functional calculus and the definition of V give $V(t)\xi = \xi$. Conversely suppose $V(t)\xi = \xi$. The holomorphic vector function $g(z) = E(z)\xi$ satisfies $g(z - it) = E(z)V(t)\xi = g(z)$. Its derivative in the imaginary direction is zero; complex differentiability therefore makes its full derivative zero. It is constant on the connected right half-plane. Since $E_s\xi \to \xi$ as $s \downarrow 0$, this constant is ξ. A vector invariant under all translations is, in particular, invariant under $V(t)$, and hence belongs to the heat ground space. The vacuum itself is translation-invariant, which proves the last assertion. □

Combining this lemma with the isometric translation intertwiner in Proposition 46.5 proves uniqueness of the invariant vacuum in the reconstructed field Hilbert space: an invariant ξ maps to $c\Omega_E = V(c\Omega)$, and injectivity of the intertwiner gives $\xi = c\Omega$. Here the intertwiner denoted V in that proposition is a different map from the one-parameter group used in this section.

40 Analytic time continuation and joint boundary values

We next construct the tempered boundary distribution used in the field reconstruction argument. The crucial bound is a power of the smallest imaginary-time gap. No bound uniform in the number of insertions is needed in this section.

40.1 Centered insertions and continuation

Work in the selected Euclidean Hilbert space with heat contractions E_t, spatial translations $U(x)$, and vacuum Ω_E. In this section we use the complex heat extension $E(z)$ on $\operatorname{Re} z > 0$: it is holomorphic in operator norm, satisfies the semigroup law and $\|E(z)\| \le 1$, agrees with E_t for positive real z, and commutes with $U(x)$. These properties were established in Proposition 39.1.

For a finite-level label v, define $B_p(v)$, $p > 0$, to be the slab operator of width $2p$ with its single insertion at spatial position zero and time p. All exterior matrix elements are precisely the corresponding Euclidean correlations. Multilinearity of those correlations, tested against the dense span of Euclidean words on both sides, proves linearity in v. The same argument, using compatibility of the finite-level inclusions in every correlation, proves that these maps agree under inclusion. They therefore define one linear map $B_p : E \to \mathcal{B}(\mathcal{H}_E)$ on the union of the field spaces. This use of all exterior words is what makes the union map well-defined.

The single-insertion case of the all-copy slab bound gives, for each fixed $v \in E$, constants $C_v > 0$ and an integer $a_v \ge 0$ such that

$$\|B_p(v)\| \le C_v p^{-a_v}, \qquad 0 < p < 1. \tag{145}$$

Indeed, choose a finite level representing v and a bound for the norm of its chosen raw representative. The centered slab has both boundary clearances p, so the estimate with one insertion applies with buffer p. The padding identities for slab operators give

$$B_{p+u}(v) = E_u B_p(v) E_u \qquad (p > 0,\ u \ge 0). \tag{146}$$

Both statements retain the constants of the all-copy estimate; no uniform bound over all labels in the union is asserted.

For times $z_i = t_i + iy_i$ with $y_1 < \cdots < y_n$, choose $0 < p < 1$ with $4p < y_j - y_i$ whenever $i < j$. Write

$$R_p(z, x) = E(-iz - 2p)U(x).$$

Define

$$F_n(z, x; v) = \langle \Omega_E, B_p(v_1) R_p(z_2 - z_1, x_2 - x_1) B_p(v_2) \cdots R_p(z_n - z_{n-1}, x_n - x_{n-1}) B_p(v_n) \Omega_E \rangle. \quad (147)$$

For $n = 1$ the intervening factors are absent; for $n = 0$ the value is one. All arguments of E have positive real part.

Lemma 40.1. *The value* (147) *is independent of the admissible width p. It is multilinear in the labels, jointly continuous in (z, x), and holomorphic in the complex time variables on the strictly ordered tube. At purely imaginary times it equals the Euclidean correlation of the insertions at (x_i, y_i). For each fixed label tuple there are $C \geq 0$ and $m \in \mathbb{N}$ such that, for $0 < k \leq 1$,*

$$y_j - y_i \geq k \ (i < j) \quad \Longrightarrow \quad |F_n(t + iy, x; v)| \leq C k^{-m} \quad (148)$$

uniformly in all real times t and spatial positions x.

Proof. Compare widths p and $q = p + u$. For $u > 0$, the semigroup law and spatial commutation imply

$$R_p(z, x) = E_u R_q(z, x) E_u.$$

Substituting (146) for every B_q in the product at width q merges the two adjacent E_u factors around each R_q into R_p. The exterior E_u factors disappear against the invariant vacuum. This proves equality; $u = 0$ is immediate, and any two widths can be ordered. The admissibility inequalities are strict, so the same fixed width works on a neighborhood of every point of the tube. There the product is holomorphic in all time variables. It is jointly continuous in the spatial variables as a vector matrix element: strong continuity of $U(x)$, local norm continuity of the other factors, and their local uniform bounds prove this successively from the rightmost vector. No operator-norm continuity of spatial translations is required. Multilinearity follows from that of each B_p.

At imaginary times each intervening factor is the ordinary positive heat translation through the unused gap. Starting with the rightmost insertion, the slab insertion rule and the time and space translation rules show that the suffix train after a preceding reference (X, Y) is the Euclidean word vector with suffix positions $(x_i - X, y_i - Y - p)$. For the first suffix point its preceding heat duration is $y_i - Y - 2p > 0$: the centered slab places that insertion at height p, and translation through this duration places it at $y_i - Y - p$. The remaining suffix points follow the same translation, proving the induction invariant. Pairing the resulting train with the initial centered insertion gives the complete word shifted by $(-x_1, -y_1 + p)$. Pairing with the vacuum and using translation invariance removes this last shift and gives the asserted Euclidean correlation.

All intervening factors have norm at most one. Thus the absolute value is at most $\prod_i \|B_p(v_i)\|$, since $\|\Omega_E\| = 1$. When the gaps are at least k, choose $p = k/8$. Equation (145) gives

$$m = \sum_i a_{v_i}, \qquad C = \prod_i (C_{v_i} 8^{a_{v_i}}),$$

and proves (148). The empty product and sum also cover $n = 0$. □

40.2 Differentiating tested functions

Fix the arity and labels, and write $X = (t_1, x_1, \dots, t_n, x_n)$ for real positions. For strictly ordered heights y and a joint Schwartz test f, put

$$\psi_y(f) = \int F_n(t + iy, x; v) f(X)\, dX.$$

For a real height vector a, let ∂_a denote differentiation of the real time coordinates in direction a, and put $D_a = -i\partial_a$ on tests.

Lemma 40.2. *The integral defining ψ_y is absolutely convergent, depends continuously on y in the open ordered cone, and satisfies*

$$\frac{d}{ds}\psi_{y+sa}(f) = \psi_{y+sa}(D_a f)$$

whenever the heights remain strictly ordered.

Proof. Near any strictly ordered y, all gaps have a common positive lower bound. Equation (148) therefore supplies a bound uniform in X and in the nearby heights. Multiplication by the integrable function $|f|$ proves integrability and, by dominated convergence, height continuity. For the derivative, restrict the holomorphic function to the complex line which changes all complex time coordinates by ζa. A fixed disc about zero stays in the tube for every real configuration X and every height in a sufficiently small neighborhood. The same gap bound holds throughout that disc. Cauchy's formula therefore bounds the first complex derivative by a constant independent of X. The Cauchy–Riemann equations give the pointwise height derivative $i\partial_a F$. This uniform bound times $|f|$ justifies differentiation under the integral.

To move the derivative onto the test, use smooth compactly supported cutoffs $\chi_R(X) = \chi(X/R)$ equal to one on the radius-R ball and supported on the radius-$2R$ ball. Integration by parts for the real time direction gives

$$\int i(\partial_a F) f \chi_R = -\int iF(\partial_a f)\chi_R - \int iFf\partial_a\chi_R.$$

The last integral has magnitude at most a fixed constant times $R^{-1}\|f\|_{L^1}$, hence tends to zero. The other two integrals converge by the uniform bounds for F and $\partial_a F$ and the integrability of $f, \partial_a f$. This proves the identity, including its sign. □

40.3 An exact Taylor formula at the edge

Let $r = (0, 1, \dots, n-1)$ and $D = D_r$. If y is only weakly ordered, then $y + sr$ has all distinct-index gaps at least s for $s > 0$; for arities zero and one the conditions on pairs are empty. Choose m, C from (148) for the fixed labels. Define

$$\widetilde{\psi}_y(f) = \sum_{j=0}^{m} \frac{(-1)^j}{j!}\psi_{y+r}(D^j f) - \int_0^1 \frac{(-s)^m}{m!}\psi_{y+sr}(D^{m+1} f)\, ds. \tag{149}$$

This formula uses values strictly inside the tube even when some heights of y coincide.

Theorem 40.3 (Joint tempered boundary)**.** *Formula* (149) *defines a continuous linear functional of f on the joint Schwartz space, continuously in every weakly ordered y. It agrees with $\psi_y(f)$ when y is strictly ordered. In particular*

$$W_n(v)(f) := \widetilde{\psi}_0(f)$$

is a tempered distribution and

$$\psi_y(f) \longrightarrow W_n(v)(f) \quad \text{as } y \to 0 \text{ through all strictly ordered heights.}$$

The limit is multilinear in the labels. No restriction on the ratios of height gaps is imposed.

Proof. First take strictly ordered y and set

$$P(s) = \sum_{j=0}^{m} \frac{(-s)^j}{j!} \psi_{y+sr}(D^j f).$$

By Lemma 40.2, differentiating the finite sum cancels successive terms, leaving

$$P'(s) = \frac{(-s)^m}{m!} \psi_{y+sr}(D^{m+1} f).$$

Integrating from zero to one and using $P(0) = \psi_y(f)$ proves the asserted identity with (149).

For weakly ordered y and $0 < s \leq 1$, the gap estimate gives

$$\left| \frac{(-s)^m}{m!} \psi_{y+sr}(D^{m+1} f) \right| \leq \frac{C}{m!} \|D^{m+1} f\|_{L^1}.$$

Thus the remainder is absolutely integrable uniformly in y. For each $s > 0$ its integrand is continuous in y on the closed ordered cone, by Lemma 40.2. Dominated convergence proves continuity of the remainder, and the finite sum in (149) is continuous for the same reason. The formula also gives the uniform estimate

$$|\widetilde{\psi}_y(f)| \leq C \sum_{j=0}^{m} \frac{\|D^j f\|_{L^1}}{j!} + \frac{C}{m!} \|D^{m+1} f\|_{L^1}.$$

These are continuous seminorms of f in the Schwartz topology. Explicitly, in real dimension $d = 4n$, each L^1 norm is bounded by $\int (1 + |X|)^{-d-1} dX$ times the corresponding weighted supremum norm; D^j is a finite sum of ordinary derivatives. This proves continuity as a test functional.

The cone of strictly ordered heights accumulates at every weakly ordered height, by the path $y + sr$. Hence its limits determine the functional uniquely. Linearity in f and multilinearity in every label follow by passing those identities for the interior integrals to the common boundary limit. This also shows that the answer is independent of the chosen power bound m, C, although those bounds may depend on the label tuple. Finally, continuity of (149) at zero and its agreement with the interior integrals give the full asserted limit. □

The theorem constructs boundary values for joint tests on all real positions, not just factorized tests and not just pointwise limits. To obtain the heat vectors needed for positivity, one must also identify their Gram matrix with these same continued functions at mixed positive and negative heights. That identification is the next step.

41 The first spectral support restriction

The joint boundary theorem also controls the Fourier support of the continued distributions. We use the radian convention

$$\widehat{f}(X) = \int e^{-iX \cdot p} f(p)\, dp, \qquad \widehat{W}(f) = W(\widehat{f}).$$

Thus the inverse test transform has factor $(2\pi)^{-4n}$. The dot product here is the Euclidean pairing of the absolute coordinates, including each real time coordinate; no relative-coordinate reduction has been made. For a height tuple y, write $d(y) = (y_1 e_0, \ldots, y_n e_0)$.

Lemma 41.1 (Exponential testing identity)**.** *For strictly ordered y and every smooth compactly supported frequency test f,*

$$\widehat{W}_n(v)(f) = \psi_y\big(\, e^{d(y)\cdot p}\widehat{f}(p)\,\big). \tag{150}$$

Proof. Set $f_s(p) = e^{sd(y)\cdot p}f(p)$ and $g_s = \widehat{f}_s$. All f_s have the same compact support. Differentiating their explicit formula, including all spatial derivatives, shows that $s \mapsto f_s$ is differentiable in the Schwartz topology with derivative $(d(y)\cdot p)f_s$. To see the topology directly, every weighted derivative is supported in that fixed compact set; Taylor's formula in s, uniformly on it for each derivative, proves convergence in each Schwartz seminorm. Continuity of the Fourier transform then gives

$$g_s' = \widehat{(d(y)\cdot p)f_s} = -D_y g_s.$$

The sign follows by differentiating the kernel $e^{-iX\cdot p}$ and using $D_y = -i\partial_y$.

For $s > 0$, differentiate $\psi_{sy}(g_s)$. The varying-test step is legitimate with the uniform seminorm bound proved in Theorem 40.3: write a difference quotient as the sum of the change in the test at the new height and the change in height with the old test fixed. The first part converges to $\psi_{sy}(g_s')$, because the test quotient converges in Schwartz space and the family of functionals has a common continuous seminorm bound. Lemma 40.2 handles the second part. Their sum is $\psi_{sy}(-D_y g_s) + \psi_{sy}(D_y g_s) = 0$. The function is therefore constant on $(0,\infty)$. As $s \downarrow 0$, $g_s \to \widehat{f}$ in Schwartz space, and the same uniform bound plus the joint boundary limit gives $\psi_{sy}(g_s) \to W_n(v)(\widehat{f})$. Comparing this with its value at $s = 1$ proves the formula. □

Proposition 41.2. *For every label tuple, the distribution $\widehat{W}_n(v)$ is supported in*

$$H_n = \left\{ p : \sum_{i=1}^{n} y_i p_i^0 \geq 0 \text{ for every } y_1 \leq \cdots \leq y_n \right\}.$$

Equivalently,

$$\sum_{i=1}^{n} p_i^0 = 0, \qquad \sum_{i=j}^{n} p_i^0 \geq 0 \quad (2 \leq j \leq n).$$

Proof. Fix a weakly ordered y and a smooth compactly supported f whose support lies in the open set $d(y)\cdot p < -1$. Put $y(t) = ty + r$, where $r = (0, 1, \ldots, n-1)$ and $t \geq 0$. These heights are strictly ordered, with all gaps at least one. The functions

$$h_t(p) = e^{d(y(t))\cdot p} f(p)$$

tend to zero in Schwartz space. Indeed, on the fixed compact support every derivative is a finite sum of a fixed smooth coefficient times a power of t times $e^{td(y)\cdot p}$, and the latter is bounded by e^{-t}. All polynomially weighted derivative suprema therefore tend to zero. Equation (150) and the gap bound at $k = 1$ imply

$$|\widehat{W}_n(v)(f)| = |\psi_{y(t)}(\widehat{h}_t)| \leq C\|\widehat{h}_t\|_{L^1} \longrightarrow 0.$$

Thus the distribution vanishes on that open set. If a point has $d(y)\cdot p < 0$, rescale y by a positive constant so that its value is less than -1. A neighborhood of the point then has the same strict inequality. This proves support in H_n.

To identify H_n, the constant height tuples 1 and -1 give zero total time momentum. The weakly increasing step tuple equal to zero before j and one from j onward gives nonnegativity of each suffix sum. Conversely, for any weakly ordered y,

$$\sum_i y_i p_i^0 = y_1 \sum_i p_i^0 + \sum_{j=2}^{n} (y_j - y_{j-1}) \sum_{i=j}^{n} p_i^0 \geq 0.$$

This proves the equivalence, with the empty and single-slot conventions included. □

Corollary 41.3 (Conditional Lorentz improvement)**.** *If the boundary distributions are covariant under the continuous finite-dimensional label representation of the proper orthochronous Lorentz group, then the support of every $\widehat{W}_n(v)$ lies in*

$$\left\{ p : \sum_i p_i = 0, \quad \sum_{i=j}^{n} p_i \in \overline{V}_+ \ (2 \le j \le n) \right\}, \qquad \overline{V}_+ = \{p : p^0 \ge |\mathbf{p}|\}.$$

Proof. Fourier transformation of the covariance identity changes momenta by the dual Lorentz action: this follows by the linear change of variables in the test transform, whose Jacobian has absolute value one. The transformed labels are again an allowed tuple, so Proposition 41.2 applies after every such dual Lorentz transformation. The inverse-transpose action ranges over the same group: for the Minkowski metric matrix η, $\Lambda^{-T} = \eta\Lambda\eta$, which preserves the future cone, and this operation is an involution. In particular every transformed suffix momentum has nonnegative time coordinate.

A real vector q has this property exactly when $q \in \overline{V}_+$. One implication follows from preservation of the future cone. For the other, if $q^0 < 0$ the identity transformation suffices. If $0 \le q^0 < |\mathbf{q}|$, choose a boost in the direction of $\mathbf{q}$ with speed β strictly between $q^0/|\mathbf{q}|$ and one; the transformed time coordinate $(q^0 - \beta|\mathbf{q}|)/\sqrt{1-\beta^2}$ is negative, a contradiction. This proves the suffix assertion. The total momentum has zero time coordinate in every frame. The identity frame makes its time coordinate zero, and the same boost argument forces its spatial part to vanish. □

The Lorentz covariance hypothesis in this last corollary is still to be proved from the infinitesimal Euclidean rotation identities. The conclusion here is support of the distributions in momentum variables; constructing the projection-valued momentum measure of the Hilbert-space representation is a separate step.

42 Compatibility of the Lorentz action across field levels

The continuation argument uses one Lorentz action on the union of the field spaces. Constructing an action separately at every finite level would not be enough: the changes of basis used to compare with the free theory need not commute with the inclusions. The compatibility is instead proved from the Euclidean action and the infinitesimal boost relation.

Write E_d for the interacting finite-dimensional complex field space and $\iota_{de} : E_d \to E_e$, $d \le e$, for its injective inclusions. Let E_d^0 be the corresponding reference spaces. The following input isolates the finite-level comparison produced by the calibration argument.

Input 42.1 (Calibrated finite-level actions)**.** *For each d there is a complex-linear isomorphism $Q_d : E_d^0 \to E_d$. The reference spaces carry continuous representations R_d^0 of $O(4)$ and T_d^0 of the proper orthochronous Lorentz group. For each spatial coordinate a, their generators satisfy*

$$\frac{d}{dt} T_d^0(B_a(t))\Big|_{t=0} = -iL_{d,a}^0,$$

where $L_{d,a}^0$ is the generator of the Euclidean time–a plane rotation, with the conventions of Section 43. On spatial rotations the two reference actions agree. Define $R_d(g) = Q_d R_d^0(g) Q_d^{-1}$ and let $L_{d,a}$ be its indicated Euclidean generator. These actions satisfy

$$R_e(g)\iota_{de} = \iota_{de} R_d(g), \qquad L_{d,a} Q_d = Q_d L_{d,a}^0.$$

There are antilinear involutions J_d, J_d^0, compatible with inclusions, such that $J_d Q_d = Q_d J_d^0$ and J_d^0 commutes with T_d^0. The inclusions form a directed system and the field union is its algebraic direct limit.

The comparison maps Q_d are the same maps that define R_d; a fresh choice at the Lorentz stage would not justify the displayed generator identity. Production of these maps and the reference actions is separate from the compatibility proof below.

Proposition 42.2. *Under Input 42.1, the finite-level representations*

$$T_d(\Lambda) = Q_d T_d^0(\Lambda) Q_d^{-1}$$

commute with all inclusions. They therefore induce one representation T on the field union, continuous on each finite level. Its spatial action is the Euclidean one, its boost generator on E_d is $-iL_{d,a}$, and it commutes with the induced involution J.

Proof. Conjugation immediately gives the representation laws and continuity on each level. Differentiation at zero, using the fixed linear map Q_d, gives the boost generator $K_{d,a} = -iL_{d,a}$. Differentiating the Euclidean inclusion law along its plane rotation gives

$$L_{e,a}\iota_{de} = \iota_{de} L_{d,a}, \qquad K_{e,a}\iota_{de} = \iota_{de} K_{d,a}.$$

For $v \in E_d$, the two curves

$$f(t) = T_e(B_a(t))\iota_{de} v, \qquad g(t) = \iota_{de} T_d(B_a(t)) v$$

have the same initial value and solve $h' = K_{e,a} h$. The derivative at arbitrary t follows from the group law by translating the parameter in the derivative at zero; for g use the generator intertwining just proved. Uniqueness of this constant-coefficient equation can be seen directly: the derivative of $e^{-tK_{e,a}}(f(t) - g(t))$ is zero, and its value at zero is zero. Hence the inclusion commutes with every coordinate boost.

It commutes with each spatial rotation because the transported spatial action is R_d and the Euclidean inclusion law already holds. For fixed d, e, the transformations commuting with ι_{de} form a subgroup. Products follow by two substitutions. For inverses, if $T_e(\Lambda)\iota_{de} = \iota_{de} T_d(\Lambda)$, multiply on the left and right by the respective inverse transformations. Spatial rotations and coordinate boosts generate the full proper orthochronous group, by the explicit reduction of Λe_0 in the proof of Theorem 43.3. Thus every transformation commutes with the inclusions. Notice that this argument has not asserted any compatibility of the maps Q_d themselves.

Define the action on a union element represented by $v \in E_d$ to be represented by $T_d(\Lambda)v$. If another representative lies in E_e, pass to a common higher level; the inclusion compatibility proves the same resulting class. This proves well-definedness. Linearity and the group laws hold after putting finitely many representatives in a common level, and the inverse action is that of Λ^{-1}. The asserted spatial and derivative identities follow in the same way. Finally

$$J_d T_d(\Lambda) Q_d v = Q_d J_d^0 T_d^0(\Lambda) v = Q_d T_d^0(\Lambda) J_d^0 v = T_d(\Lambda) J_d Q_d v.$$

Surjectivity of Q_d and compatibility of the involutions prove commutation with J on the union. □

43 From Euclidean rotations to Lorentz covariance

This section proves covariance of the boundary distributions from the Euclidean rotation identities, retaining the same field labels. The finite-dimensional Lorentz action on those labels must first be available. We state its required compatibility explicitly rather than identify it with an arbitrary representation having the same dimension.

On each finite level let R be the Euclidean rotation representation, and write L_a for its infinitesimal generator in the plane consisting of time and spatial coordinate a. Our convention for the coordinate generator is

$$\mathcal{R}_a = \sum_i \left(x_i^a \partial_{y_i} - y_i \partial_{x_i^a} \right).$$

Assume R acts compatibly under the inclusions of field levels, and the smooth Euclidean correlation $S(v; X)$ obeys simultaneous rotation covariance. Differentiating it gives

$$\mathcal{R}_a S(v; X) = -\sum_i S(v_1, \dots, L_a v_i, \dots, v_n; X). \tag{151}$$

Indeed, covariance says $S(v; R_t X) = S(R_{-t} v; X)$; the ordinary chain rule on positions and the finite multilinear expansion on labels give the displayed formula. This is applied only at distinct Euclidean positions.

Suppose also that the proper orthochronous Lorentz group has a representation ρ on the field union preserving every finite level, continuous there, compatible with the inclusions, and agreeing with R on spatial rotations. For the coordinate boost $B_a(s)$, with coordinate infinitesimal generator $x^a \partial_t + t \partial_{x^a}$, require

$$\frac{d}{ds} \rho(B_a(s)) v \Big|_{s=0} = -i L_a v. \tag{152}$$

Proposition 42.2 supplies this action from the calibrated finite-level actions. Their construction remains an explicit earlier input. Set $K_a = -iL_a$.

Proposition 43.1 (The tested boost identity)**.** *For strictly ordered y and any joint Schwartz test f,*

$$\sum_i \psi_y(v_1, \dots, K_a v_i, \dots, v_n; f) = \psi_y \left(v; \sum_i [x_i^a \partial_{t_i} + (t_i + i y_i) \partial_{x_i^a}] f \right). \tag{153}$$

Consequently the boundary distributions satisfy

$$\sum_i W_n(v_1, \dots, K_a v_i, \dots, v_n)(f) = W_n(v)(\mathcal{B}_a f), \qquad \mathcal{B}_a = \sum_i (x_i^a \partial_{t_i} + t_i \partial_{x_i^a}). \tag{154}$$

Proof. First suppose f has compact support C. For a complex parameter c put $z_i(c) = i y_i + c t_i$ and

$$F_c(v; X) = F_n(z(c), x; v), \qquad T_c f = \sum_i [x_i^a \partial_{t_i} + c z_i(c) \partial_{x_i^a}] f.$$

Let D consist of parameters for which $z(c)$ has strictly ordered imaginary parts for every $X \in C$. This is an open convex set containing the real axis: its constraints are affine strict inequalities in $\operatorname{Im} c$, and compactness of C makes their margins uniform near each admissible parameter. Consider the integral

$$H(c) = \int_C \left[F_c(v; X) T_c f(X) - c \sum_i F_c(v_1, \dots, K_a v_i, \dots, v_n; X) f(X) \right] dX.$$

It is holomorphic on D. In fact the integrand is jointly continuous on $D \times C$ and holomorphic in c for every X; on each compact subdisc, Cauchy's formula gives uniformly bounded derivatives, so differentiation under this finite-volume integral is justified.

At $c = i\sigma$, put $Y_i = y_i + \sigma t_i$. All these heights are strictly ordered on a neighborhood of C, and $F_{i\sigma}(v; X) = S(v; (x, Y))$. The real vector field

$$A_\sigma = \sum_i \left[x_i^a \partial_{t_i} - \sigma(y_i + \sigma t_i) \partial_{x_i^a} \right]$$

is divergence free. Under the map $(t, x) \mapsto (Y, x)$ it becomes $\sigma \mathcal{R}_a$. Thus (151) gives

$$A_\sigma F_{i\sigma}(v; X) = -\sigma \sum_i S(v_1, \ldots, L_a v_i, \ldots, v_n; (x, Y)) = -i\sigma \sum_i F_{i\sigma}(v_1, \ldots, K_a v_i, \ldots, v_n; X).$$

Also $T_{i\sigma} f = A_\sigma f$. The integrand of $H(i\sigma)$ is therefore $A_\sigma(F_{i\sigma}(v; X) f(X))$, whose integral is zero. The product is smooth near C and zero off C, so its extension by zero is a smooth compactly supported function; no regularity of the continued function at positions outside that neighborhood is being assumed. The identity theorem on the connected set D now gives $H(1) = 0$, which is (153) for compactly supported tests.

To pass to Schwartz tests, choose a smooth cutoff $\chi(X/R)$ equal to one near zero. Then $\chi(X/R) f \to f$ in every Schwartz seminorm: on the outside region use one extra polynomial decay weight, and for each positive-order derivative of the cutoff use its factor R^{-1} or a higher inverse power and the same tail estimate. At fixed strictly ordered y, every ψ_y is continuous on Schwartz space, and all coordinate multiplications and derivatives in (153) are continuous there. The compact-test identity consequently extends to f.

Finally take the full ordered-height limit $y \to 0$. The test on the right is $\mathcal{B}_a f + \sum_i i y_i \partial_{x_i^a} f$. Each fixed-test pairing $\psi_y(v; \partial_{x_i^a} f)$ has a finite boundary limit, so multiplication by y_i makes the correction tend to zero. All other terms have their specified joint boundary limits, proving (154). □

Lemma 43.2 (Affine pullback of tests)**.** *The action $P_g f(X) = f(g^{-1}X)$ of the Poincaré group, acting on all positions at once, is continuous in the Schwartz topology. For coordinate boosts,*

$$\frac{d}{ds} P_{B_a(s)} f \Big|_{s=0} = -\mathcal{B}_a f$$

in that topology.

Proof. On a compact neighborhood of a fixed invertible affine map, its linear part and inverse have uniformly bounded norms and its translation is bounded. Hence $1 + |X|$ and $1 + |g^{-1}X|$ are comparable uniformly there. The chain rule expresses every derivative of $f \circ g^{-1}$ as a finite sum of derivatives of f at $g^{-1}X$ times products of matrix entries. Each weighted supremum seminorm is therefore bounded by finitely many Schwartz seminorms of f, uniformly on that neighborhood.

For continuity, split the supremum into a fixed ball and its complement. On the ball, uniform continuity of each derivative and continuous variation of the matrix coefficients give convergence. On the complement, the comparison of weights and one extra decay seminorm make the tail uniformly small. This proves continuity in every seminorm. For the boost derivative, the pointwise chain rule at zero gives $-\mathcal{B}_a f$. To justify the same derivative in Schwartz space, use Taylor's formula along the smooth curve of invertible matrices $B_a(-s)$. Its second derivative acts on f as a finite sum of first and second derivatives composed with $B_a(-s)$, multiplied by coefficients polynomial in X of degree at most two. The preceding weight comparison bounds every seminorm of this second derivative uniformly for s in a small compact interval. The integral Taylor remainder is therefore $O(s^2)$ in each seminorm, proving the stated derivative. □

Theorem 43.3. *Under the Euclidean covariance and compatible label-action hypotheses above, the boundary distributions satisfy*

$$W_n(\rho(\Lambda)v_1, \dots, \rho(\Lambda)v_n)(P_\Lambda f) = W_n(v)(f)$$

for every proper orthochronous Lorentz transformation Λ*. They are also invariant under common spacetime translations.*

Proof. For a fixed coordinate boost set

$$A(s) = W_n(\rho(B_a(s))v_1, \dots, \rho(B_a(s))v_n)(P_{B_a(s)}f).$$

Put the finitely many labels in one common finite level. Expanding them in a finite basis expresses $A(s)$ as a finite sum of smooth scalar coefficients times fixed-label distributions applied to the moving test. The label coefficients are smooth along the boost: its group law and the derivative at zero give the linear differential equation $\rho(B_a(s))' = \rho(B_a(s))K_a$ on each finite level. Thus Lemma 43.2 and (152) justify the ordinary product rule at zero. Its label terms are the left side of (154), and its test term is the negative of the right side. Hence $A'(0) = 0$. The representation laws for labels and pullbacks give the same calculation at any s, using the labels and test already moved by $B_a(s)$. Therefore $A'(s) = 0$ everywhere and $A(s) = A(0)$.

Spatial rotations and common spatial translations are inherited by the strict-time continuation from their Euclidean identities. To check this without an assumption of holomorphy in spatial coordinates, fix all real positions, compare the two scalar continued expressions, and replace each time by $iy_i + ct_i$. Both sides are holomorphic on an open strip about the real axis and agree for $c = i\sigma$ near zero by Euclidean covariance. The identity theorem gives equality at $c = 1$. For a common real-time translation b, compare times $iy_i + c(t_i + b)$ on one side with $iy_i + ct_i$ on the other. Their gaps agree, and at $c = i\sigma$ they differ by the common Euclidean time translation σb. The same proof applies. Changing variables in the joint test integrals, with unit absolute Jacobian, and taking boundary limits gives spatial-rotation covariance and translation invariance of W.

Finally these spatial rotations and coordinate boosts generate the proper orthochronous Lorentz group. Here is a direct verification. Given Λ, its image $q = \Lambda e_0$ is future timelike with Minkowski norm squared one. A boost in coordinate a with rapidity $-\operatorname{artanh}(q^a/q^0)$ makes that spatial component zero; $|q^a| < q^0$ ensures the parameter exists. It leaves the other spatial components unchanged. Three such boosts produce a vector with zero spatial part and positive time component, necessarily e_0. Their product B therefore makes $B\Lambda$ fix e_0. Preservation of the Minkowski form then shows that $B\Lambda$ preserves its spatial orthogonal complement and is orthogonal on it; determinant one makes its spatial part a rotation. Thus Λ is a product of coordinate boosts and a spatial rotation. Covariance is closed under products and inverses by the representation laws, proving the assertion for every Λ. □

Together with Corollary 41.3, this gives the full forward suffix-cone support of the boundary distributions, conditional only on the specified compatible Lorentz action and the previously stated Euclidean construction inputs. It does not yet prove spacelike commutativity.

44 Continuation to the full spacetime tube

The ordered complex-time functions constructed earlier are holomorphic in the time variables. Locality needs a larger domain: the spatial variables must also become complex. We construct this continuation from the spectral support of the boundary distributions. A smooth cutoff near that

support makes the Fourier–Laplace expression an actual Schwartz test. Keeping the cutoff also permits limits in which some imaginary gaps lie on the boundary of the forward cone.

Fix n and set $D = 4n$. All coordinates in this section are absolute coordinates in $\mathbb{R}^D$, rather than relative coordinates. Write

$$K_n = \left\{ p : \sum_i p_i = 0, \quad \sum_{i=j}^{n} p_i \in \overline{V}_+ \ (2 \le j \le n) \right\},$$
$$C_n = \{ y : y_{i+1} - y_i \in \overline{V}_+ \ (1 \le i < n) \},$$
$$C_n^\circ = \{ y : y_{i+1} - y_i \in V_+ \ (1 \le i < n) \}, \qquad V_+ = \{ q : q^0 > |\mathbf{q}| \}.$$

The superscript on C_n° denotes the strict-gap set, which is open in the full space. For $n = 0$ or 1 the gap conditions are vacuous. The tube is $\mathcal{T}_n = \mathbb{R}^D + iC_n^\circ$. By Corollary 41.3 and Theorem 43.3, the distribution $\widehat{W}_n(v)$ has support in K_n, under the hypotheses stated there.

Lemma 44.1 (Dual positivity and a strict margin)**.** *If $p \in K_n$ and $y \in C_n$, then $p \cdot y \ge 0$. For each $y \in C_n^\circ$ there are $c > 0$ and a neighborhood N of y such that*

$$p \cdot u \ge c|p| \qquad (p \in K_n, \ u \in N).$$

Proof. Put $q_j = \sum_{i=j}^{n} p_i$. The zero total momentum and telescoping give

$$p \cdot y = \sum_{j=2}^{n} q_j \cdot (y_j - y_{j-1}).$$

The Euclidean pairing of two forward-cone vectors is nonnegative: $a \cdot b = a^0 b^0 + \mathbf{a} \cdot \mathbf{b} \ge a^0 b^0 - |\mathbf{a}||\mathbf{b}| \ge 0$. This proves the first assertion. Choose $r > 0$ with $B(y, r) \subseteq C_n^\circ$, and put $c = r/2$. For $u \in B(y, r/2)$ and $p \ne 0$, the point $u - cp/|p|$ belongs to $B(y, r)$ and hence to C_n. Applying the first assertion to this point gives $p \cdot u - c|p| \ge 0$. The case $p = 0$ is immediate. This argument also covers the vacuous gap cases, where $K_n = \{0\}$. □

44.1 A smooth cutoff and closed imaginary shifts

Let $K_n^{(a)} = \{ p : \operatorname{dist}(p, K_n) < a \}$. Choose a nonnegative smooth bump ρ, supported in $B(0, 1/2)$, with integral one, and define

$$\chi = \rho * \mathbf{1}_{K_n^{(2)}}.$$

Then $0 \le \chi \le 1$, $\chi = 1$ on $K_n^{(1)}$, and $\operatorname{supp} \chi \subseteq K_n^{(3)}$, with support understood as closed support. Indeed, for $p \in K_n^{(1)}$ every point $p - q$ with $q \in \operatorname{supp} \rho$ has distance less than $3/2$ from K_n; the convolution therefore integrates the constant one. Nonzero values require distance less than $5/2$ from K_n, and their closure still has distance at most $5/2 < 3$. Every derivative is globally bounded, since

$$\partial^\alpha \chi = (\partial^\alpha \rho) * \mathbf{1}_{K_n^{(2)}}, \qquad \|\partial^\alpha \chi\|_\infty \le \|\partial^\alpha \rho\|_1.$$

Differentiating the compactly supported convolution proves smoothness: difference quotients of the smooth factor converge in L^1, so their convolutions with the bounded indicator converge uniformly. In dimension zero simply take $\chi = 1$.

For $y \in C_n$ set $a_y(p) = e^{-y \cdot p} \chi(p)$. This is a smooth multiplier of Schwartz space. More precisely, for every R and multi-index α there is a finite constant $A_{R,\alpha}$ such that

$$\sup_p |\partial^\alpha a_y(p)| \le A_{R,\alpha} \qquad (y \in C_n, \ |y| \le R). \tag{155}$$

On the closed support of χ choose $q \in K_n$ with $|p-q| < 3$. Dual positivity gives $-y \cdot p \le 3|y|$. Each derivative of the exponential adds a product of components of y, while derivatives of χ are bounded. Leibniz's rule proves (155); outside that support all derivatives of a_y vanish.

For each $f \in \mathscr{S}$, the map $y \mapsto a_y f$ is continuous from C_n to $\mathscr{S}$. On a compact frequency ball this follows uniformly for every derivative from the explicit smooth formula. Outside a large ball, the uniform derivative bounds and Leibniz's rule bound a seminorm of weight k by a constant times $(1+|p|)^{-1}$ times finitely many seminorms of f of weight $k+1$. The tail is therefore uniformly small. Combining the compact and tail estimates proves convergence in every Schwartz seminorm, including at closed-cone boundary shifts.

For a fixed label tuple abbreviate $w = \widehat{W}_n(v)$ and define a tempered distribution W_y by

$$W_y(f) = w(a_y \check{f}), \qquad \check{f}(p) = (2\pi)^{-D} \int e^{ip\cdot x} f(x)\, dx. \tag{156}$$

The preceding continuity shows that $W_y(f)$ is continuous in $y \in C_n$. Moreover $W_0 = W_n(v)$. To see this, multiplication by χ leaves w unchanged because $\chi = 1$ on a neighborhood of its support. The support assertion applies to Schwartz tests as well as compact tests: multiply a test vanishing near the support by compact cutoffs, apply distributional support, and pass to the Schwartz limit. Fourier inversion now gives $w(\check{f}) = W_n(v)(f)$.

44.2 The holomorphic Fourier–Laplace expression

For $z = x + iy \in \mathcal{T}_n$ define

$$L_z(p) = e^{ip\cdot z}\chi(p), \qquad H_n(v;z) = (2\pi)^{-D} w(L_z).$$

Here $p \cdot z$ is the complex bilinear extension of the Euclidean pairing, with no conjugation. The test L_z is in $\mathscr{S}$. In fact, the margin lemma at y and the distance-three support property give, on $\operatorname{supp}\chi$,

$$-y \cdot p \le 3(|y| + c) - c|p|.$$

To obtain this, choose $q \in K_n$ with $|p-q| < 3$, use $y \cdot q \ge c|q|$, and $|q| \ge |p| - 3$. Frequency derivatives of L_z only introduce bounded derivatives of χ and powers of the fixed vector z. They consequently have exponential decay. The same estimates hold in a sufficiently small neighborhood of z, with a common positive decay rate.

Proposition 44.2. *The map $z \mapsto L_z$ is complex differentiable into Schwartz space, with derivative $h \mapsto i(p\cdot h)L_z(p)$. Consequently $H_n(v;z)$ is holomorphic jointly in all D complex coordinates. For each fixed $y \in C_n^\circ$ it grows at most polynomially in x, and for every $f \in \mathscr{S}$,*

$$W_y(f) = \int H_n(v; x+iy) f(x)\, dx, \tag{157}$$

with an absolutely convergent integral.

Proof. The difference from the proposed derivative is

$$L_{z+h} - L_z - i(p\cdot h)L_z = L_z\big(e^{ip\cdot h} - 1 - ip\cdot h\big).$$

For $|h| \le 1$, every fixed-order frequency derivative of the factor in parentheses has absolute value at most $C_\alpha |h|^2 (1+|p|)^2 e^{|h||p|}$. At derivative order zero use $|e^u - 1 - u| \le |u|^2 e^{|u|}$; at order one use $|e^u - 1| \le |u| e^{|u|}$; at higher orders differentiate the exponential, which contributes at least two factors

of h. These inequalities follow directly from its power series. Choose $|h| < c/2$. Leibniz's rule and the exponential decay of L_z show that the remainder is $O(|h|^2)$ in each Schwartz seminorm. The derivative map is continuous from the finite-dimensional parameter space into $\mathscr{S}$, since multiplication by each frequency coordinate is continuous. Composition with the continuous linear functional w proves complex differentiability of H_n; in finite complex dimension this gives holomorphy.

For fixed y, put $g(p) = a_y(p) = L_{iy}(p)$. Modulation becomes translation under Fourier transformation:

$$\widehat{L_{x+iy}}(u) = \widehat{g}(u - x).$$

For $s_{k,j}(h) = \sup_u |u|^k \|D^j h(u)\|$, the triangle inequality gives the explicit translation bound

$$s_{k,j}(h(\,\cdot - x)) \le 2^k \big(s_{k,j}(h) + |x|^k s_{0,j}(h)\big).$$

A continuous linear functional on Schwartz space is bounded by finitely many such seminorms. Fourier inversion and the displayed bound therefore bound $|w(L_{x+iy})|$ by a polynomial in $|x|$. Translation is continuous in Schwartz space: its derivatives converge uniformly on compact sets, and the same bound with one extra decay weight controls the tails. Thus the integrand in (157) is continuous and absolutely integrable for every Schwartz f.

First let f be compactly supported. Fourier inversion, pointwise in p, gives

$$a_y(p)\check{f}(p) = (2\pi)^{-D} \int f(x) L_{x+iy}(p)\, dx.$$

This integral can be passed through w in the Schwartz topology. For an explicit justification take Riemann sums in a box containing the support of f. The integrand is continuous into Schwartz space and has compact parameter support. For each seminorm the error is bounded by the box volume times its uniform modulus of continuity in that seminorm. The sums therefore converge in every seminorm to the displayed function; continuity of w gives (157). For general f, use $f_R = \kappa(x/R)f$ with $0 \le \kappa \le 1$ and $\kappa = 1$ near zero. Then $f_R \to f$ in Schwartz space, while polynomial growth gives an integrable majorant for $|H_n(v; x + iy) f_R(x)|$ independent of R. Continuity on the left and dominated convergence on the right finish the proof. □

Proposition 44.3 (Identification with the ordered-time function)**.** *For strictly increasing scalar heights $y_1 < \cdots < y_n$,*

$$H_n(v; X + id(y)) = F_n(t + iy, x; v), \qquad X = ((t_1, x_1), \ldots, (t_n, x_n)).$$

For any $Y \in C_n$, the integrals of $H_n(v; X + iY')f(X)$ converge to $W_Y(f)$ as $Y' \to Y$ through C_n°.

Proof. At fixed strict scalar heights, ψ_y is a tempered distribution by the uniform real-position bound on F_n. Let h be a compactly supported smooth frequency test. Since $\chi = 1$ near K_n and w is supported there, the exponential testing identity gives

$$w(a_{d(y)}h) = w(e^{-d(y)\cdot p}h) = \psi_y(\widehat{h}).$$

Both sides are continuous in h on Schwartz space. Compact-test density therefore extends this equality to every $h \in \mathscr{S}$. Apply inverse Fourier transformation: $W_{d(y)} = \psi_y$. Equation (157) now identifies the integrals of the two proposed continuous functions against every Schwartz test. They agree pointwise: a nonzero difference would have, after a constant phase change, strictly positive real part on a ball and would be detected by a nonnegative smooth bump there. The final boundary statement follows from (157) and continuity of $W_{Y'}(f)$ on C_n. □

Proposition 44.4 (Lorentz covariance in the tube)**.** *For every proper orthochronous Lorentz matrix* Λ*,*

$$H_n(v; z) = H_n(\rho(\Lambda)v; \Lambda z) \qquad (z \in \mathcal{T}_n).$$

The action is simultaneous on all slots.

Proof. The matrix Λ preserves strict forward gaps, so Λz is in the tube. Its dual $A = \Lambda^{-T}$ preserves K_n, because it is again proper orthochronous and preserves total momentum and suffix cones. The Fourier form of Theorem 43.3, obtained by the change of variables of determinant one, is

$$\widehat{W}_n(\rho(\Lambda)v)(h \circ A^{-1}) = \widehat{W}_n(v)(h).$$

The two Schwartz functions L_z and $L_{\Lambda z} \circ A$ agree near K_n. Indeed their exponential factors agree because $(Ap) \cdot (\Lambda z) = p \cdot z$. At each $p \in K_n$, both cutoffs are one on the neighborhood $B(p, 1) \cap A^{-1}B(Ap, 1)$. Their difference therefore vanishes near the support of w. Apply the Fourier covariance identity to $h = L_{\Lambda z} \circ A$, discard this vanishing difference, and multiply by $(2\pi)^{-D}$. This is exactly the claimed equality. □

45 The heat-vector bridge

We now supply the independent-height Gram limit in Input 46.1 from the joint boundary theorem. Throughout this section the complex heat and slab properties used in Section 40 are retained. The involution J is the one induced on the field union by Euclidean time reflection and conjugation. This identification is essential in the Gram formula below.

For $0 < y_1 < \cdots < y_n$, real positions $X_i = (t_i, x_i)$, and an admissible width p satisfying $4p < y_1$ and $4p < y_j - y_i$ for $i < j$, define

$$\begin{aligned} Q_{p,y}(X; v) &= E(-i(t_1 + iy_1) - p)U(x_1)B_p(v_1) \\ &\cdot R_p(t_2 - t_1 + i(y_2 - y_1), x_2 - x_1)B_p(v_2) \cdots R_p(t_n - t_{n-1} + i(y_n - y_{n-1}), x_n - x_{n-1})B_p(v_n)\Omega_E. \end{aligned} \tag{158}$$

The empty tuple gives Ω_E. The subtraction in the first complex heat time is p, whereas that between two insertions is $2p$. Each complex heat factor is a contraction, so

$$\|Q_{p,y}(X; v)\| \le \prod_i \|B_p(v_i)\|. \tag{159}$$

For fixed p, y, v this vector is continuous in X, by the same successive strong-continuity argument used for (147). Consequently $f(X)Q_{p,y}(X; v)$ is Bochner integrable for every joint Schwartz test f. To justify strong measurability without a global separability assumption on the Hilbert space, its continuous image of the separable position space is separable. The bound (159) then gives the integrable majorant $|f(X)| \prod_i \|B_p(v_i)\|$.

Lemma 45.1 (Mixed Gram identity)**.** *For positive strictly ordered height tuples* y, z*, separately admissible widths* p, q*, positions* X, Y *and labels* v, u*, put*

$$\widehat{y} = (-y_n, \ldots, -y_1, z_1, \ldots, z_m), \quad \widehat{X} = (X_n, \ldots, X_1, Y_1, \ldots, Y_m), \quad \widehat{v} = (Jv_n, \ldots, Jv_1, u_1, \ldots, u_m).$$

Then

$$\langle Q_{p,y}(X; v), Q_{q,z}(Y; u)\rangle = F_{n+m}(\widehat{t} + i\widehat{y}, \widehat{x}; \widehat{v}). \tag{160}$$

Here $\widehat{t}, \widehat{x}$ *are the real time and spatial parts of* $\widehat{X}$*. The real positions in the first tuple are reversed in order; their real time coordinates are not negated.*

Proof. When all real time coordinates are zero, the vector in (158) is the Euclidean word vector with insertions (x_i, y_i) and labels v_i. This follows directly by induction from the right: a centered insertion adds its point at height p, the preceding positive heat interval supplies the missing distance, and the spatial translation changes the common reference point. More explicitly, with a preceding reference (X_0, Y_0) and preceding half-width q_0, the partial vector has points $(x_i - X_0, y_i - Y_0 - q_0)$. The recursion uses first heat time $y_1 - Y_0 - q_0 - p$; after insertion it produces the same formula with the first point included. Taking $q_0 = 0$ and $(X_0, Y_0) = 0$ gives the stated word. Pairing the two Euclidean words reflects the first tuple in time and applies J to its labels. Permutation symmetry allows its reversed order, and the imaginary-time identity of Lemma 40.1 proves (160) in this case. This argument does not require equal widths p, q.

For arbitrary fixed real positions, introduce one complex parameter c. In the first vector replace its times by $iy_i + ct_i$, and in the second by $iz_j + cs_j$. Call the resulting holomorphic vector functions $A(c)$ and $B(c)$. Since all clearance inequalities are strict and there are only finitely many of them, there is a $\delta > 0$ such that these vectors are well-defined on $|\operatorname{Im} c| < \delta$. The same strip works for $A(\overline{c})$, after decreasing δ if necessary. The scalar function

$$c \longmapsto \langle A(\overline{c}), B(c) \rangle$$

is holomorphic: expand the difference quotient and use antilinearity in the first argument, which cancels the conjugation of its increment. Its proposed right side is F at times $i\widehat{y} + c\widehat{t}$, with the fixed combined spatial tuple. This too is holomorphic on a sufficiently small horizontal strip; its heights are the concatenation of the reversed negatives of $y_i - (\operatorname{Im} c)t_i$ and the tuple $z_j + (\operatorname{Im} c)s_j$, and remain strictly ordered there.

At $c = i\tau$ near zero, both vectors have purely imaginary times, with these modified positive height tuples. The already proved Euclidean identity therefore equates the two scalar functions. The identity theorem on the connected strip extends equality to $c = 1$. This is the desired identity with arbitrary real positions. Empty tuples cause no difficulty: their vector is the fixed vacuum and the same argument applies. □

For a word $b = ((v_1, f_1), \dots, (v_n, f_n))$, choose

$$y(t) = (t, 2t, \dots, nt), \qquad p(t) = \tfrac{1}{8} \min(t, 1), \quad t > 0,$$

and define

$$h_t(b) = \int \Big(\prod_i f_i(X_i) \Big) Q_{p(t), y(t)}(X; v)\, dX.$$

The clearance conditions hold for every $t > 0$, and the fixed-height bound above proves integrability. No uniform pointwise bound as $t \downarrow 0$ is asserted or needed.

Proposition 45.2. *These vectors satisfy* (161) *for the boundary functional of Theorem 40.3, and* $h_t(1) = \Omega_E$. *They also intertwine real translations of all tests with the Euclidean real-time translation representation.*

Proof. For words b, c of lengths n, m, use Lemma 45.1 inside the scalar product of their Bochner integrals. The integrand is absolutely integrable on the product space: its absolute value is bounded by the product of the two integrable scalar test magnitudes times the two fixed-height bounds (159). Fubini and the coordinate permutation $(X, Y) \mapsto (X_n, \dots, X_1, Y_1, \dots, Y_m)$ therefore give

$$\langle h_s(b), h_t(c) \rangle = \psi_{(-ns, \dots, -s, t, \dots, mt)}(f_{b^* c}),$$

with labels $(Jv_n, \dots, Jv_1, u_1, \dots, u_m)$ and joint product test f_{b^*c}, whose first factors are the reversed conjugates of the tests in b. The displayed heights are strictly ordered for every $s, t > 0$ and approach zero as the two parameters independently approach zero. Theorem 40.3 gives the limit $w(b^*c) = K(b, c)$. The zero-dimensional integral for the empty word is the vacuum itself.

For translations write $T_E(a) = E(-ia^0)U(\mathbf{a})$, where the imaginary-axis values of E denote its strongly continuous unitary boundary values. These boundary values and their law with the interior extension are part of the complex heat extension used here. Translating all real positions by a leaves all differences in (158) unchanged; its first factor changes by the commuting factor $T_E(a)$. Hence

$$Q_{p,y}(X + a; v) = T_E(a)Q_{p,y}(X; v).$$

For the empty word the same assertion follows from translation invariance of the vacuum. Substitute $X \mapsto X + a$ in the integral with translated tests $f_i(\cdot - a)$ and move the bounded operator $T_E(a)$ outside the Bochner integral. This proves the translation identity for every $t > 0$. □

Thus the heat-vector limit required for positivity and for the isometric embedding of Proposition 46.5 follows from the slab bounds and complex heat extension. Lorentz covariance, spacelike locality, and retention of nonzero labels at the boundary are distinct remaining steps; they have not been inferred from the Gram identity.

46 From boundary values to fields on a common domain

The Euclidean Hilbert space and the relativistic field Hilbert space have related, but different, definitions. The bridge between them is a limit of vectors with small positive imaginary times. This section proves the bridge and the ensuing operator construction. The preceding two sections construct the analytic boundary values and verify their heat-vector bridge from the stated slab hypotheses. Positivity of the boundary values will now follow from that bridge.

Write E for the algebraic union of the finite-dimensional field spaces and $\mathscr{S} = \mathscr{S}(\mathbb{R}^4; \mathbb{C})$ for the Schwartz space. A letter is a pair $a = (v, f) \in E \times \mathscr{S}$, and a word is a finite ordered list of letters. Let $J : E \to E$ be an antilinear involution, put $a^* = (Jv, \overline{f})$, and define the star of a word by reversing its order and starring each letter. The empty word is denoted by 1. Concatenation is written without an intervening symbol.

Suppose $W_n(v_1, \dots, v_n)$ is a tempered distribution on $(\mathbb{R}^4)^n$, multilinear in the labels. Its value on a product test function defines $w(a_1 \cdots a_n)$. Set

$$K(b, c) = w(b^*c).$$

The joint distribution, rather than merely this restricted family of tests, will be needed in constructing the boundary values. For the operator argument below the following properties specify exactly what is used.

Input 46.1 (Boundary data)**.** *The functional w is multilinear separately in every label and every test function, and $w(1) = 1$. For fixed words b, c and labels v, u, the function*

$$(f, g) \longmapsto K((v, f)b, (u, g)c)$$

is jointly continuous in the Schwartz topology. There are vectors $h_t(b)$, $t > 0$, in the selected Euclidean Hilbert space $\mathcal{H}_E$, with unit vacuum Ω_E, such that

$$\lim_{s,t\downarrow 0} \langle h_s(b), h_t(c) \rangle = K(b, c), \tag{161}$$

$$\lim_{t\downarrow 0} h_t(1) = \Omega_E.$$

The two positive parameters in the first limit vary independently.

Here and below the inner product is antilinear in its first argument. In the application $h_t(b)$ is the integral of a product of bounded slab insertions and intervening complex-time translations, at strictly increasing heights $(t, 2t, \dots, nt)$. Thus the integrability of this vector-valued integral and the joint limit (161) require proof. A scalar boundary limit along one fixed ray does not by itself give this input.

Lemma 46.2 (Removing the imaginary heights)**.** *Under Input 46.1, each $h_t(b)$ has a norm limit $q(b) \in \mathcal{H}_E$, and*

$$\langle q(b), q(c)\rangle = K(b,c), \qquad q(1) = \Omega_E.$$

In particular K is a positive semidefinite kernel.

Proof. Fix b. Expanding a squared norm gives

$$\|h_s(b) - h_t(b)\|^2 = \langle h_s(b), h_s(b)\rangle + \langle h_t(b), h_t(b)\rangle - \langle h_s(b), h_t(b)\rangle - \langle h_t(b), h_s(b)\rangle.$$

All four terms tend to $K(b,b)$, including the diagonal terms, by the joint limit. The expression therefore tends to zero as $s, t \downarrow 0$. Completeness gives the norm limit. Continuity of the inner product gives the asserted Gram identities and the specified empty-word limit gives the vacuum. For any finitely many coefficients c_i and words b_i,

$$\sum_{i,j} \overline{c_i} K(b_i, b_j) c_j = \Big\| \sum_i c_i q(b_i) \Big\|^2 \geq 0.$$

□

Let $\mathcal{P}$ be the vector space of formal finite complex linear combinations of words. Extend K sesquilinearly to $\mathcal{P}$ and let $\mathcal{N} = \{p : K(p,p) = 0\}$. Cauchy–Schwarz for a positive semidefinite form follows by applying positivity to $p + zq$ and minimizing over z (the case $K(q,q) = 0$ follows by allowing $|z|$ to grow). Consequently $\mathcal{N}$ is the radical of the form. The quotient $\mathcal{D} = \mathcal{P}/\mathcal{N}$ is an inner-product space; let $\mathcal{H}$ be its completion and $[b] \in \mathcal{D}$ the class of a word.

Proposition 46.3 (The common invariant domain)**.** *There is a bilinear map*

$$(v, f) \longmapsto \Phi(v, f) : \mathcal{D} \to \mathcal{D}, \qquad \Phi(v, f)[b] = [(v, f)b].$$

The vacuum $\Omega = [1]$ has norm one, the domain $\mathcal{D}$ is dense and spanned by its word vectors, and

$$\begin{aligned} \langle \Phi(v,f)\xi, \eta\rangle &= \langle \xi, \Phi(Jv, \overline{f})\eta\rangle, \\ \langle \Omega, \Phi(a_1)\cdots\Phi(a_n)\Omega\rangle &= w(a_1 \cdots a_n). \end{aligned} \tag{162}$$

For each v and $\xi \in \mathcal{D}$, the map $f \mapsto \Phi(v,f)\xi$ is continuous from $\mathscr{S}$ to $\mathcal{H}$. As densely defined operators on $\mathcal{H}$, $\Phi(Jv, \overline{f}) \subseteq \Phi(v,f)^*$; in particular every field operator is closable.

Proof. Let L_a denote left concatenation on formal sums. Word reversal gives the exact identity

$$K(L_a p, q) = K(p, L_{a^*} q).$$

If $p \in \mathcal{N}$, the right side vanishes for every q. Taking $q = L_a p$ proves $K(L_a p, L_a p) = 0$. Thus L_a preserves $\mathcal{N}$ and descends to $\mathcal{D}$. This is the necessary null-space argument; positivity without the concatenation identity would not suffice. The same identity proves (162). Multilinearity of w shows

that every matrix element against a word of the defect in a proposed linearity identity is zero. Since words span $\mathcal{D}$, each such defect is zero. This proves bilinearity of Φ even though no multilinearity relations were imposed on the formal word space initially.

The vacuum norm, cyclicity and moment identity follow directly from the construction. To prove continuity first take $\xi = [b]$ and write $\eta_f = [(v, f)b]$. At any f_0,

$$\|\eta_f - \eta_{f_0}\|^2 = K((v, f)b, (v, f)b) + K((v, f_0)b, (v, f_0)b) - 2\,\mathrm{Re}\,K((v, f_0)b, (v, f)b),$$

which tends to zero by the assumed joint continuity. Finite linear combinations give the result for every $\xi \in \mathcal{D}$. Finally (162), with $\mathcal{D}$ dense, is precisely the criterion for the asserted adjoint inclusion. The adjoint domain contains the dense space $\mathcal{D}$; hence the original operator is closable. Equivalently, if $\xi_n \to 0$ and $\Phi(v, f)\xi_n \to \eta$, pairing with each vector of $\mathcal{D}$ gives $\eta = 0$, which directly proves closability. □

Proposition 46.4 (Locality and retention of labels)**.** *Suppose, whenever f, g have mutually spacelike supports, that*

$$w(b(v, f)(u, g)c) = w(b(u, g)(v, f)c)$$

for all exterior words b, c. Then $[\Phi(v, f), \Phi(u, g)] = 0$ on $\mathcal{D}$. Suppose also that a label v whose insertion has zero expectation against all exterior words and all tests must be zero. Then for every $v \neq 0$ there is an f with $\Phi(v, f) \neq 0$.

Proof. Pair the commutator applied to $[c]$ with $[b]$. Its value is the difference of the two assumed expectations, with left exterior word b^*, and so is zero. Density and linearity prove the operator identity on $\mathcal{D}$. If all $\Phi(v, f)$ vanish, then for any b, c, f,

$$w(b(v, f)c) = \langle [b^*], \Phi(v, f)[c]\rangle = 0,$$

because $(b^*)^* = b$. The detection assumption now gives $v = 0$. □

46.1 The group action and its Euclidean embedding

Let G be the Poincaré group with proper orthochronous Lorentz part. Suppose its action on labels and pullback action on test functions induce an action $b \mapsto gb$ on words, preserving the empty word. Assume

$$K(gb, gc) = K(b, c), \qquad g \longmapsto K(b, gc) \text{ is continuous} \tag{163}$$

for all words. These are properties of the continued distributions and the label action on each finite level, to be established before this construction is applied.

Proposition 46.5. *There is a strongly continuous unitary representation U of G on $\mathcal{H}$ with $U(g)[b] = [gb]$. It fixes Ω, preserves $\mathcal{D}$, and satisfies field covariance there. There is also a linear isometry $V : \mathcal{H} \to \mathcal{H}_E$ with $V[b] = q(b)$ and $V\Omega = \Omega_E$. If, for every translation a, the heat vectors obey*

$$h_t(ab) = U_E(a)h_t(b)$$

for all sufficiently small $t > 0$, then $VU(a) = U_E(a)V$.

Proof. Extend the action on words linearly. Gram invariance makes it an isometry of the quotient; the inverse action makes its extension to the completion unitary. Multiplication and identity laws hold on words, hence on the dense span, hence on the completion. For a word b, Gram invariance and (163) give

$$\|[gb] - [g_0 b]\|^2 = 2K(b, b) - 2\,\mathrm{Re}\,K(g_0 b, gb) \longrightarrow 0.$$

Thus orbit maps are continuous first on finite word sums and then on every vector: approximate that vector by a finite sum, and use the isometry at both g and g_0 to bound the approximation errors uniformly. The vacuum and domain statements follow on words. The equality $g(ab) = (ga)(gb)$ gives covariance on their span.

The map $\sum c_i[b_i] \mapsto \sum c_i q(b_i)$ preserves squared norms by Lemma 46.2, so is well-defined and extends isometrically. Its vacuum identity is already known. Taking the norm limit in the stated translation identity for heat vectors proves intertwining on every word; boundedness then extends it to $\mathcal{H}$. □

These arguments construct all fields on one common invariant domain. They still require the continued distributions, their locality and label-detection properties, the continuous group action, and the independent-height limit. The forward spectral condition and uniqueness of the invariant vacuum require additional facts about the Euclidean translation representation and are not consequences of the formal word construction alone.

47 Joint tests and survival of field labels

Two details complete the common-domain reconstruction, apart from locality and the momentum projection measure. First, the joint tempered distributions supply the test-function continuity used in the word construction. Second, a label detectable by Euclidean correlations remains detectable after continuation to real time. The latter uses the full-tube construction and its identification with the ordered-time function; no pointwise boundary values of the fields are required.

Lemma 47.1. *The finite linear span of functions $\prod_{i=1}^{n} f_i(X_i)$, $f_i \in \mathscr{S}(\mathbb{R}^4)$, is dense in $\mathscr{S}((\mathbb{R}^4)^n)$. The product map from the n test spaces into the joint test space is continuous multilinear.*

Proof. Every joint derivative of a product separates into the corresponding single-slot derivatives. Also $1 + |X| \le \prod_i (1 + |X_i|)$ up to a fixed constant depending on n. Thus each weighted derivative supremum of the product is bounded by a constant times a product of single-slot Schwartz seminorms. This proves continuous multilinearity.

For density, compactly supported smooth joint tests are already dense by the cutoff argument in Proposition 43.1. Fix such a test f, and choose single-slot smooth compact cutoffs χ_i with product $\chi(X) = \prod_i \chi_i(X_i)$ equal to one on $\operatorname{supp} f$. Let η_R be a smooth compact frequency cutoff approaching one. Fourier inversion on Schwartz space gives

$$f_R = \chi\, \mathcal{F}^{-1}(\eta_R \widehat{f}) \longrightarrow \chi f = f$$

in Schwartz space. Here the Fourier transform and its inverse are continuous inverse maps on that space, with the convention of Section 41; see [4, Theorem 1.3, p.22]. Multiplication by the fixed compact cutoff is continuous by the Leibniz rule.

For fixed R, put $g = \eta_R \widehat{f}$. Fourier inversion writes

$$f_R(X) = (2\pi)^{-4n} \int_{\operatorname{supp} g} g(p) \prod_i \left[\chi_i(X_i) e^{ip_i \cdot X_i}\right], dp.$$

This integral is a limit in Schwartz space of finite sums of its displayed product tests. To verify that assertion explicitly, enclose the compact frequency support in a box and use ordinary rectangular Riemann sums. All summands and the integral have the fixed compact position support $\operatorname{supp} \chi$. On the product of that support with the frequency box, every position derivative of the integrand

is continuous, hence uniformly continuous. The error in each weighted derivative supremum is bounded by the frequency-box volume times its uniform modulus of continuity at the partition diameter, times the maximum position weight on the compact support. It tends to zero. Thus the same sequence of partitions converges in every Schwartz seminorm. Each Riemann sum belongs to the stated span, so does f_R in its closure, and taking $R \to \infty$ proves density. For $n = 0$ the joint space is one-dimensional and the empty product one already spans it. □

Corollary 47.2. *For the jointly tempered distributions constructed in Theorem 40.3, the word functional is separately linear in every label and test. Its mixed Gram expressions*

$$(f, g) \longmapsto K((v, f)b, (u, g)c)$$

are jointly continuous. Thus these parts of Input 46.1 follow from the construction.

Proof. Multilinearity follows by composing the label-multilinear distribution with the multilinear product test. In the Gram expression the first variable occurs as $\overline{f}$ in a fixed slot and the second as g in another fixed slot; all other factors and labels are fixed. Conjugation is continuous on Schwartz space. Lemma 47.1 and continuity of the fixed-label distribution therefore give the asserted joint continuity. The empty-word value is one because the zero-insertion continuation is constantly one. □

Lemma 47.3 (A zero boundary has zero strict continuation)**.** *Fix an ordered label tuple v. If $W_n(v) = 0$ as a joint tempered distribution, then $F_n(t + iy, x; v) = 0$ for every strictly ordered height tuple y and every real position tuple (t, x).*

Proof. The full-tube function is defined by $H_n(v; z) = (2\pi)^{-4n}\widehat{W}_n(v)(L_z)$. It is therefore identically zero if $W_n(v) = 0$. Proposition 44.3 identifies its restriction at $z = X + id(y)$ with the ordered-time function for every strictly ordered y. This proves the assertion. □

Let T denote the selected Euclidean hierarchy. At each finite raw label level, its null space consists of labels whose insertion vanishes in every distinct-point correlation with arbitrary exterior labels. Assume the selected and original null spaces agree at every level, as established by the detector argument when its moment and rank hypotheses are supplied. Assume also the finite-level quotient embeddings into the union are injective and every union label has a finite-level representative.

Proposition 47.4 (Boundary detection)**.** *If a union label v satisfies*

$$w(b(v, f)c) = 0$$

for every pair of exterior words b, c and every single-slot Schwartz test f, then $v = 0$. Consequently every nonzero label defines a nonzero smeared field operator in Proposition 46.3.

Proof. Take any ordered tuple of labels containing v, at any slot. The assumption says that its joint distribution vanishes on every product test: group the factors before the distinguished slot into b, and those after it into c. Lemma 47.1 and tempered continuity therefore make that entire joint distribution zero. By Lemma 47.3, its strict-time continuation is zero. At real times zero, the pure-imaginary identity gives zero for the corresponding selected Euclidean correlation whenever the Euclidean times are strictly increasing.

For arbitrary positions with pairwise distinct time coordinates, sort them by time. The label tuple is permuted with the positions and still contains v, so the preceding zero-distribution argument

applies to the sorted tuple. Euclidean permutation symmetry transfers its zero value back to the original ordering. Such position tuples are dense in the full configuration space of distinct points: perturb the finitely many time coordinates by distinct arbitrarily small amounts, avoiding the finitely many equality hyperplanes. Openness of the distinct-point configuration space keeps all points distinct during a sufficiently small perturbation. Continuity of the Euclidean correlations extends the zero value to every configuration, including those with time ties.

Represent v by a finite-level raw label. The preceding conclusion, with arbitrary raw exterior labels projected into the union, is exactly the all-exterior null criterion for that label in T. Equality of selected and original null spaces makes its original quotient class zero. Its union image is v, so $v = 0$. This proves the detection premise of Proposition 46.4; that proposition then supplies the nonzero smeared operator for every nonzero label. □

This argument closes the boundary-detection step conditional on the already specified equality of Euclidean null spaces. The estimate and finite-rank inputs establishing that equality remain part of the quantitative phase construction.

48 Positive energy and the joint momentum measure

The Fourier support of the vacuum distributions and the spectrum of the translation operators are different assertions. We now prove the latter. The essential analytic fact is elementary: a positive finite measure whose Fourier transform extends boundedly to the upper half-plane cannot charge negative frequencies. Positivity is what makes a Gaussian probe detect such a charge without cancellation.

We use the spectral theorem for strongly continuous unitary representations of locally compact abelian groups, in the form of the SNAG theorem [5, Theorem D.3.1, p.397]. Applied to a strongly continuous unitary representation T of $\mathbb{R}^4$, it supplies a unique regular projection-valued measure P on $\mathbb{R}^4$ with

$$T(a) = \int e^{ip\cdot a}\, dP(p), \qquad P(\mathbb{R}^4) = I. \tag{164}$$

Here we identify the character indexed by p with $a \mapsto e^{ip\cdot a}$; the pairing is Euclidean, as in Section 41. The hypotheses hold for the translation representation constructed in Proposition 46.5, and for its Euclidean counterpart in Section 39. In particular, continuity is strong operator continuity, not operator-norm continuity. For each vector v, $\mu_v(B) = \langle v, P(B)v\rangle$ is a positive regular finite measure, of mass $\|v\|^2$, with characteristic function $\langle v, T(a)v\rangle$.

Lemma 48.1 (A bounded upper-half-plane extension)**.** *Let μ be a positive finite Borel measure on $\mathbb{R}^d$, and let $\ell : \mathbb{R}^d \to \mathbb{R}$ be continuous. Suppose a function F is continuous on $\operatorname{Im} z \geq 0$, holomorphic on $\operatorname{Im} z > 0$, bounded there by C, and satisfies*

$$F(t) = \int e^{it\ell(p)}\, d\mu(p) \qquad (t \in \mathbb{R}).$$

Then $\operatorname{supp}\mu \subseteq \{p : \ell(p) \geq 0\}$.

Proof. Fix $\varepsilon > 0$, $b \in \mathbb{R}$ and $s \geq 0$. Multiply $F(z)$ by $\exp(-ibz - \varepsilon z^2)$ and integrate around the rectangle with corners $-R, R, R+is, -R+is$. Cauchy's theorem applies: the function is continuous on the rectangle and holomorphic in its interior. One can also apply the theorem above height $\delta > 0$ and let $\delta \downarrow 0$; continuity on each compact rectangle justifies that limit. On either vertical edge the magnitude is at most $C\exp(-\varepsilon R^2 + |b|s + \varepsilon s^2)$. The vertical integrals tend to zero. The horizontal

integrals converge absolutely, since at height u the common majorant is $C\exp(-\varepsilon t^2 + bu + \varepsilon u^2)$. Consequently the integrals along heights zero and s agree, and

$$\left|\int_{\mathbb{R}} F(t)e^{-ibt-\varepsilon t^2}\,dt\right| \le C\sqrt{\frac{\pi}{\varepsilon}}\,e^{bs+\varepsilon s^2}. \tag{165}$$

Fubini applies to the boundary integral because μ is finite and $e^{-\varepsilon t^2}$ is integrable. The Gaussian transform gives

$$\int_{\mathbb{R}} e^{it(\ell(p)-b)-\varepsilon t^2}\,dt = \sqrt{\frac{\pi}{\varepsilon}}\,e^{-(\ell(p)-b)^2/(4\varepsilon)}.$$

For completeness, if $G(a) = \int e^{iat-\varepsilon t^2}\,dt$, differentiation under the integral and integration by parts give $G'(a) = -aG(a)/(2\varepsilon)$. The value $G(0) = \sqrt{\pi/\varepsilon}$ follows by squaring the Gaussian integral and using polar coordinates. Solving this differential equation proves the formula with its stated normalization.

Suppose $p_0 \in \operatorname{supp}\mu$ and $b = \ell(p_0) < 0$. In (165) choose $s = -b/(2\varepsilon)$ and cancel the positive factor $\sqrt{\pi/\varepsilon}$. Positivity of the resulting integral yields

$$\int e^{-(\ell(p)-b)^2/(4\varepsilon)}\,d\mu(p) \le Ce^{-b^2/(4\varepsilon)}.$$

The open neighborhood $A = \{p : (\ell(p) - b)^2 < b^2/4\}$ of p_0 has mass $m > 0$. Its contribution to the left side is at least $me^{-b^2/(16\varepsilon)}$. Hence $m \le Ce^{-3b^2/(16\varepsilon)}$ for every $\varepsilon > 0$. Letting $\varepsilon \downarrow 0$ is a contradiction. □

Apply the lemma first on the Euclidean Hilbert space. Write $T_E(t, \mathbf{x}) = V(t)U_E(\mathbf{x})$ for its joint translations, where V is the unitary time group of Proposition 39.1. For $v \in \mathcal{H}_E$ define

$$F_v(t + is) = \langle v, V(t)E_s v\rangle \qquad (s \ge 0).$$

This function is continuous on the closed upper half-plane by strong continuity and the uniform operator bounds. On its interior it equals $\langle v, E(-iz)v\rangle$, so is holomorphic and bounded by $\|v\|^2$. Its boundary is the characteristic function of the time coordinate of the joint measure μ_v^E. The lemma gives

$$\operatorname{supp}\mu_v^E \subseteq \{p : p^0 \ge 0\}. \tag{166}$$

Spatial covariance is not used in this step.

Theorem 48.2 (Forward spectrum and the unique vacuum)**.** *Suppose the word construction has a strongly continuous Poincaré representation U with proper orthochronous Lorentz part, and an isometric translation intertwiner $V : \mathcal{H} \to \mathcal{H}_E$ as in Proposition 46.5. Then its joint momentum measure satisfies*

$$P(\mathbb{R}^4 \setminus \overline{V}_+) = 0, \qquad \overline{V}_+ = \{p : p^0 \ge |\mathbf{p}|\}.$$

If the heat-fixed subspace of $\mathcal{H}_E$ is $\mathbb{C}\Omega_E$, then the vectors fixed by all translations on $\mathcal{H}$ are exactly $\mathbb{C}\Omega$.

Proof. The intertwining identity and isometry give

$$\langle v, U(a)v\rangle = \langle Vv, T_E(a)Vv\rangle.$$

Thus μ_v and μ_{Vv}^E have equal characteristic functions and are equal measures. Here is the measure-uniqueness step explicitly. For any Schwartz function f, Fourier inversion and Fubini express its

integral against a finite measure in terms of that characteristic function; Fubini follows from integrability of $\widehat{f}$. The two measures therefore agree on all smooth compactly supported tests. Continuous compactly supported tests are uniformly approximated, in a fixed compact neighborhood of their supports, by smooth mollifications. They too have equal integrals. Uniqueness for regular measures then gives equality, by the Riesz representation theorem. In particular every μ_v obeys (166).

For a Lorentz matrix Λ, covariance of translations implies

$$\mu_{U(\Lambda)v} = (\Lambda^{-T})_* \mu_v.$$

Indeed the characteristic function of the left side at a equals $\langle v, U(\Lambda^{-1}a)v\rangle$, which is that of the right side. The preceding uniqueness argument applies again. A homeomorphism carries the support of a measure to the support of its pushforward: every neighborhood of the image has a preimage neighborhood of positive measure, and the converse follows by its inverse. It follows that, for any $p \in \operatorname{supp} \mu_v$,

$$(\Lambda^{-T}p)^0 \geq 0 \quad \text{for every proper orthochronous } \Lambda.$$

The dual matrices $\Lambda^{-T} = \eta\Lambda\eta$, with $\eta = \operatorname{diag}(1,-1,-1,-1)$, range over the same group. The identity gives $p^0 \geq 0$. If $|\mathbf{p}| > p^0$, take a boost in the direction $\mathbf{p}/|\mathbf{p}|$ with positive rapidity r such that $\tanh r > p^0/|\mathbf{p}|$. Its time coordinate is $p^0 \cosh r - |\mathbf{p}| \sinh r < 0$, a contradiction. Thus every μ_v is supported in $\overline{V}_+$. The complement of the support of a finite Borel measure on $\mathbb{R}^4$ has measure zero: it is a union of a countable collection of zero-measure basis neighborhoods. Hence $\mu_v(\overline{V}_+^c) = 0$ for every v. Since $P(B)$ is an orthogonal projection, $\|P(B)v\|^2 = \mu_v(B)$, which proves the asserted operator identity.

Finally let v be translation invariant. Its image Vv is fixed by all Euclidean real-time translations. The fixed-vector equivalence proved in Section 39 makes it heat fixed. Therefore $Vv = c\Omega_E = V(c\Omega)$; injectivity of the isometry gives $v = c\Omega$. The converse follows from vacuum invariance. This proves uniqueness under the stated heat hypothesis, and makes no spectral-gap assertion. □

49 Spacelike exchange and locality

We prove that exchanging two adjacent spacelike-separated insertions does not change the boundary distribution. The argument keeps all the other insertions present. It first makes the two tube functions analytic near a partial boundary with one tied imaginary-time gap. Euclidean symmetry identifies their germs when the real times vanish. Analytic continuation along a path then identifies them at arbitrary spacelike configurations. Finally their partial boundary distributions agree, and the remaining imaginary heights can be removed.

Fix adjacent slots $i, j = i+1$ and let σ exchange them. It acts on tuples of labels, real points and complex points by the same permutation. Put

$$O_{ij} = \{X : X_j - X_i \text{ is spacelike}\} = \{X : |X_j^0 - X_i^0| < |\mathbf{X}_j - \mathbf{X}_i|\}.$$

Choose scalar heights y with $y_i = y_j$ and with every other ordered pair strictly increasing. Then $d(y) \in C_n$ and $\sigma d(y) = d(y)$. The point $X + id(y)$ lies in the closures of both $\mathcal{T}_n$ and $\sigma^{-1}\mathcal{T}_n$: strictly separate the tied heights in the appropriate order by arbitrarily small amounts. The functions to be compared are

$$z \longmapsto H_n(v; z), \qquad z \longmapsto H_n(v \circ \sigma; \sigma z).$$

49.1 Small imaginary boosts enter both tubes

For a spatial unit vector e, write, in time-first coordinates,

$$C_e(t,\mathbf{x}) = (e\cdot\mathbf{x}, te), \qquad P_e(t,\mathbf{x}) = (t,(e\cdot\mathbf{x})e).$$

The boost and its entire complex extension are $B_e(z) = I + \sinh z\, C_e + (\cosh z - 1)P_e$. For a spacelike vector a, the choice $e = \mathbf{a}/|\mathbf{a}|$ gives $C_e a = (|\mathbf{a}|, a^0 e) \in V_+$.

Lemma 49.1. *For $X \in O_{ij}$ and $q = X + id(y)$ there are e and $\tau > 0$ such that $B_e(i\tau)q \in \mathcal{T}_n$ and $\sigma B_e(-i\tau)q \in \mathcal{T}_n$. Moreover, for all z sufficiently close to q in $\mathcal{T}_n$, $B_e(ib)z \in \mathcal{T}_n$ for every $0 \le b \le \tau$. For all z sufficiently close to q in $\sigma^{-1}\mathcal{T}_n$, $\sigma B_e(-ib)z \in \mathcal{T}_n$ throughout the same interval.*

Proof. Take e from the spacelike difference $X_j - X_i$ as above. For any adjacent difference $z_{k+1} - z_k$, write its real and imaginary parts as x and Y. The imaginary part after the positive imaginary boost is

$$Y + \sin b\, C_e x + (\cos b - 1)P_e Y. \tag{167}$$

At q, every gap other than (i,j) has $Y \in V_+$. By openness and continuity those finitely many transformed gaps remain in V_+ for (b,z) in a product neighborhood of $(0,q)$. At the distinguished gap $Y = 0$ and $C_e x \in V_+$. For small b rewrite (167) as

$$Y + \sin b\left(C_e x - \frac{\sin b}{1+\cos b} P_e Y\right).$$

The bracket is continuous near $(0,q)$ and lies in V_+ there, after shrinking the neighborhood. For $0 < b < \pi$, the sine is positive. If $Y \in \overline{V}_+$, the sum is strictly in V_+: the spatial triangle inequality shows that a closed forward-cone vector plus a positive multiple of a strict one is strict.

For the negative boost followed by the exchange, the original tied gap is reversed and the sine changes sign. Its inward vector is again $C_e(X_j - X_i)$. All other imaginary gaps at q remain strictly forward, since the tied heights agree and are the only tie. The same neighborhood argument applies. Take a common product neighborhood for both orientations and all gaps, and choose $\tau > 0$ inside its positive parameter interval. At q the closed-gap condition gives the two strict endpoints. At nearby interior points it gives strictness for $b > 0$, while $b = 0$ follows from membership in the relevant original tube. This proves the asserted uniform path property, including its endpoint at zero. □

49.2 Holomorphic germs across the tied gap

The labels in any finite tuple lie in one common finite-dimensional level. On that level the continuous representation $r \mapsto \rho(B_e(r))$ has an entire extension. Here is the elementary argument, which uses operator-norm continuity only in this finite-dimensional space. For a continuous one-parameter group $R(r)$ in its matrix algebra, put $A(t) = \int_0^t R(s)\,ds$. Then $A(t)/t \to I$ as $t \to 0$, so $A(a)$ is invertible for a sufficiently small nonzero a. The group law gives

$$R(t)A(a) = A(t+a) - A(t).$$

The right side is continuously differentiable. Thus R is differentiable; differentiating the group law gives $R'(t) = R(t)K$, $K = R'(0)$. The derivative of $R(t)e^{-tK}$ is zero and its value at zero is I. Therefore $R(t) = e^{tK}$, and $R(z) = e^{zK}$ is the required entire extension. This construction is compatible with the finite-level inclusions, since the real representations are compatible and the identity theorem extends their equality to complex parameters.

For a point q and parameters supplied by Lemma 49.1, define near q

$$A_q(z) = H_n(R(i\tau)v; B_e(i\tau)z),$$
$$B_q(z) = H_n(R(-i\tau)(v \circ \sigma); \sigma B_e(-i\tau)z).$$

They are holomorphic in a neighborhood of q, because their arguments at q lie strictly in the tube. They extend the respective original tube functions. To prove this for A_q, fix a nearby $z \in \mathcal{T}_n$ for which the entire path stays in the tube. The function of one complex parameter

$$\zeta \longmapsto H_n(R(\zeta)v; B_e(\zeta)z)$$

is holomorphic on an open neighborhood of the segment $i[0, \tau]$. Moving labels cause no difficulty: expansion in a basis of the common finite level makes the expression a finite sum of holomorphic coefficients times the fixed-label holomorphic tube functions. For real ζ near zero the expression equals $H_n(v; z)$ by tube covariance. The identity theorem on the connected component containing the segment makes it constant along that segment. Its value at $i\tau$ proves the assertion. The same proof with the negative segment and the exchange gives the assertion for B_q.

Such a germ is uniquely determined by its original tube function. Indeed, two candidate germs are holomorphic on a common ball about q and agree on the intersection of that ball with the relevant open tube. This intersection contains a nonempty open set because q is in the tube's closure. The identity theorem on the ball makes the germs equal.

Lemma 49.2 (Equality on the Euclidean slice)**.** *If every real time coordinate of X is zero, then the germs A_q and B_q coincide at $q = X + id(y)$.*

Proof. Let M be the real-linear Wick map that sends Euclidean time to imaginary time and leaves spatial coordinates real. Put $p = ((y_k, \mathbf{X}_k))_{k=1}^n$, so $Mp = q$. This is a distinct-point Euclidean configuration. All pairs other than (i, j) have different time coordinates, and the distinguished pair has different spatial coordinates because its real separation is spacelike. The Euclidean correlation S_v is consequently smooth near p.

On the open side where the Euclidean times are strictly ordered, $A_q \circ M = S_v$ near p, by the tube restriction and pure-imaginary identification. On the open side where their exchanged tuple is strictly ordered, $B_q \circ M = S_v$, using Euclidean permutation symmetry as well. The point p is in the closure of each side. Differentiate the respective identities on their open sides. For each order, continuity of the derivatives up to p shows that the two real derivatives at p both equal the corresponding derivative of S_v.

The image of M spans the full complex coordinate space over $\mathbb{C}$: each spatial basis vector is already in the image, and each time basis vector is $-i$ times an image vector. By the chain rule the real derivatives just compared are the restrictions of the complex multilinear derivatives of A_q and B_q to this image. Complex multilinearity and spanning make all their full complex derivatives equal. Their convergent local power series therefore agree. This proves equality of germs, rather than merely equality of values on the Euclidean slice. □

The equality propagates to every $X \in O_{ij}$. Keep its spatial coordinates fixed and multiply all its real time coordinates by $s \in [0, 1]$. The distinguished separation remains spacelike throughout. Along $q(s) = ((sX_k^0, \mathbf{X}_k))_k + id(y)$ both germs exist, and both underlying tubes have $q(s)$ in their closures. Germ equality is locally constant in s. To verify this, choose a common ball where the two germs at $q(s_0)$ are holomorphic and extend their original tube functions. For s near s_0, uniqueness of germs identifies the germs at $q(s)$ with these same functions. The identity theorem on the ball says that equality near either point is equivalent to equality near the other. Thus the set of parameters with equal germs and its complement are both open in $[0, 1]$. Connectedness and Lemma 49.2 at $s = 0$ give equality at $s = 1$.

49.3 From germs to boundary distributions

Let $W_{d(y),v}$ denote the partial boundary distribution (156). Near a given $X \in O_{ij}$ it is represented by the germ A_q at the partial height:

$$W_{d(y),v}(f) = \int A_q(Z + id(y))f(Z)\,dZ \tag{168}$$

for compactly supported smooth f in a sufficiently small ball about X. Indeed, choose a smaller product neighborhood of $(X, d(y))$ on which the germ is bounded and agrees with the tube function at interior heights. For $Y \to d(y)$ through C_n°, its integrals against f converge to the displayed integral by dominated convergence, using a constant times $|f|$ as majorant. The tube boundary theorem identifies the same limit with $W_{d(y),v}(f)$. Applying this argument to the exchanged tube, and changing variables by the coordinate permutation of absolute Jacobian one, gives the analogous formula for $W_{\sigma d(y),v\circ\sigma}(f \circ \sigma)$ with germ B_q. Since these germs agree, the two distributions agree locally on O_{ij}.

They agree on every Schwartz test whose closed support is contained in O_{ij}. For compact support, cover it by finitely many of the local balls and apply a smooth partition of unity. For arbitrary Schwartz support contained in this open set, multiply by compact cutoffs approaching one in the Schwartz topology; their supports remain in O_{ij} and continuity passes the equality to the limit.

Theorem 49.3 (Adjacent spacelike exchange)**.** *For every joint Schwartz test f supported in O_{ij},*

$$W_n(v)(f) = W_n(v \circ \sigma)(f \circ \sigma).$$

Consequently the reconstructed field operators commute on their common domain whenever their test functions have mutually spacelike supports.

Proof. There is a sequence of tied profiles of the kind used above tending to zero. For example, with slots numbered $1, \dots, n$, take $y_k^{(m)} = a_k/(m+1)$ where $a_k = k$ except $a_j = i$. All other ordered pairs have positive gaps, and the exchanged profile is unchanged. The preceding partial-distribution equality holds for each m. Continuity on the closed shift set, and $W_0 = W_n(v)$, give the asserted equality by letting $m \to \infty$.

For the operator conclusion take any two exterior words and place the two test functions f_i, f_j between them. The closed support of their joint product test is contained in the inverse image of $\operatorname{supp} f_i \times \operatorname{supp} f_j$ under the two coordinate projections: wherever either factor vanishes on a neighborhood, so does the product. Mutually spacelike supports therefore put that joint support in O_{ij}. The distributional identity exchanges exactly these two letters and retains all spectators. Proposition 46.4 now proves that the commutator vanishes on $\mathcal{D}$. □

50 One field prescription for every truncation degree

An operator-product expansion must use the same coefficients when its truncation degree changes. Choosing a new basis independently at each finite degree would not ensure this. We construct a single compatible system of fields and finite correlation tests, one degree at a time. The tests are fixed before choosing the cluster geometry or its spectators.

Let E_n be the finite-dimensional complex space of physical fields through degree n. Denote its injective inclusion maps by $\iota_{mn} : E_m \to E_n$, $m \le n$, with $\iota_{nn} = I$ and $\iota_{nk}\iota_{mn} = \iota_{mk}$. These are

the physical filtration maps, so inserting a field and its image in a higher degree gives the same separated correlations.

An exterior test λ consists of finitely many fixed spectator labels and a separated configuration including a distinguished first position. Inserting $v \in E_n$ at that first position and evaluating the mixed Euclidean correlation defines a linear functional $\lambda_n(v)$. The physical field quotient has been defined by vanishing of all such correlations. Thus these functionals separate E_n:

$$[\lambda_n(v) = 0 \text{ for every exterior test } \lambda] \implies v = 0. \tag{169}$$

Their compatibility with filtration is $\lambda_n \iota_{mn} = \lambda_m$. For a finite linear combination $b = \sum_{\lambda \in F} b_\lambda \lambda$ use the notation $b_n = \sum_{\lambda \in F} b_\lambda \lambda_n$. The same finite combination then defines a test on every level.

Lemma 50.1. *Every linear functional on E_n is b_n for some finite combination b of exterior tests.*

Proof. Let V be the span of the exterior functionals inside E_n^*. If V were proper, a basis of V would consist of fewer than $\dim E_n$ functionals. The linear map from E_n to their coordinate values would then have a nonzero kernel, by rank–nullity. A nonzero vector in that kernel would be annihilated by all of V, hence by every exterior test, contrary to (169). Therefore $V = E_n^*$. Membership in a span means a finite linear combination, as required. □

Put $E_{-1} = 0$ and $d_n = \dim(E_n / \iota_{n-1,n} E_{n-1})$, where the denominator is zero when $n = 0$. A channel is a pair $\alpha = (n, j)$ with $1 \le j \le d_n$; its degree is $|\alpha| = n$. We seek fields $u_\alpha \in E_{|\alpha|}$ and finite exterior tests b_α such that, for every n,

$$b_{\alpha,n}(\iota_{|\beta|,n} u_\beta) = \delta_{\alpha\beta} \quad (|\alpha|, |\beta| \le n), \tag{170}$$

and

$$v = \sum_{|\alpha| \le n} b_{\alpha,n}(v) \iota_{|\alpha|,n} u_\alpha \qquad (v \in E_n). \tag{171}$$

The sums contain finitely many channels. No assertion of a convergent infinite expansion is involved.

Proposition 50.2. *There is one choice of all the fields u_α and tests b_α satisfying* (170) *and* (171) *simultaneously for every n. Moreover $b_{\alpha,m} = 0$ whenever $m < |\alpha|$.*

Proof. At degree zero choose a basis of E_0 and represent each coordinate functional by a finite exterior test, using Lemma 50.1. The two identities are then the usual basis identities.

Suppose the fields and tests through degree n have been chosen. Write $R = \iota_{n,n+1}$, let $q : E_{n+1} \to E_{n+1}/R(E_n)$ be the quotient map, and set

$$Bv = \sum_{|\alpha| \le n} b_{\alpha,n+1}(v) \iota_{|\alpha|,n} u_\alpha.$$

Test compatibility and (171) give $BR = I_{E_n}$. Choose a linear right inverse S of q; it exists by lifting a finite basis of the quotient. Correct this arbitrary lift by putting

$$F = S - RBS.$$

Then $qF = I$ and $BF = 0$. The old tests vanish on the new image F. Indeed (170) gives $b_{\alpha,n} B = b_{\alpha,n+1}$ for every old channel α, so $b_{\alpha,n+1} F = b_{\alpha,n} BF = 0$.

Every $v \in E_{n+1}$ has the exact decomposition

$$v = Fqv + RBv. \tag{172}$$

To prove it, $q(v - Fqv) = 0$ implies $v - Fqv = Ra$ for some $a \in E_n$. Applying B and using $BF = 0$, $BR = I$ gives $a = Bv$. Now choose a basis (e_j) of the quotient. The new fields are Fe_j. For each new channel choose a finite exterior test representing $e_j^* q$ on E_{n+1}. Leave every old field and every old test unchanged. The new tests annihilate $R(E_n)$, are dual to the new fields because $qF = I$, and the old tests annihilate the new fields as just proved. Old fields and tests retain their mutual duality by compatibility. This proves all four old/new cases of (170). Expanding the two terms in (172) proves (171) at degree $n + 1$.

Iterate. A channel and its finite test, once chosen, never change. Their value in the resulting infinite list is the one fixed at their birth degree. Any identity at degree n involves only the finitely many channels already fixed by stage n, so it remains valid in the final list. Finally take $m < |\alpha|$ and expand $v \in E_m$ using (171). Lift that identity to degree $|\alpha|$ and apply b_α. Biorthogonality makes every term zero, because all its channels have degree at most m. Compatibility then gives $b_{\alpha,m}(v) = 0$. □

The subtraction RBS is the reason this construction works at all degrees. It places every new quotient lift in the exact kernels of all old tests, while retaining the old tests as actual finite correlation evaluations.

50.1 Testing a product at the cluster center

For each exterior test λ, translate its distinguished position to zero. Write its other positions as $z_{\lambda,j}$ and its spectator labels as $w_{\lambda,j}$. These positions are pairwise distinct and none is zero. For a cluster of fields v_i at x_i and center o, define

$$\mathcal{R}_\lambda(\boldsymbol{x}, \boldsymbol{v}; o) = S_{\mathrm{mix}}\big((v_i, x_i - o)_i, (w_{\lambda,j}, z_{\lambda,j})_j\big)$$

when the combined configuration is separated, and assign value zero otherwise. This convention gives a function everywhere; every asymptotic use below is in its separated region. Define

$$c_\alpha^{\mathrm{raw}}(\boldsymbol{x}, \boldsymbol{v}; o) = \sum_{\lambda \in \operatorname{supp} b_\alpha} b_{\alpha,\lambda} \mathcal{R}_\lambda(\boldsymbol{x}, \boldsymbol{v}; o), \tag{173}$$

and the finite field-valued expression

$$H_n(\boldsymbol{x}, \boldsymbol{v}; o) = \sum_{|\alpha| \le n} c_\alpha^{\mathrm{raw}}(\boldsymbol{x}, \boldsymbol{v}; o) \iota_{|\alpha|,n} u_\alpha. \tag{174}$$

Each coefficient is fixed independently of the truncation degree, the phase, or any later choice of exterior spectators. It is multilinear in the cluster labels on its separated region. The separation condition depends only on positions, so the zero extension is multilinear as well. Permutation symmetry of the correlations gives simultaneous permutation symmetry in cluster positions and labels. Simultaneous translation of the cluster and center leaves their differences unchanged.

For a single field $v \in E_m$ placed at o, translation invariance and the definition of the exterior test give

$$c_\alpha^{\mathrm{raw}}((o), (v); o) = b_{\alpha,m}(v).$$

All the detector configurations in this evaluation are separated. Hence, for $m \le n$, (171) implies

$$H_n((o), (v); o) = \iota_{mn} v. \tag{175}$$

The same testing operation is compatible with replacing any input field by its filtration image, since separated correlations have that compatibility.

For every fixed finite list of coefficients there is a common positive detector clearance. To see this, take the minimum of the finitely many nonzero lengths $|z_{\lambda,j}|$ occurring in their finite test supports; if there are no spectator positions, there is no such restriction. If $|x_i - o|$ is less than half that minimum, no cluster point can meet a detector position. Distinctness among cluster points and among detector points supplies the remaining separation conditions. The same argument works after any orthogonal rotation of the relative cluster, because rotations preserve $|x_i - o|$. Smoothness of the mixed correlations on separated configurations makes each such finite coefficient collection smooth on this common region. This is a finite-degree statement; it gives no uniform clearance for infinitely many degrees.

The raw expressions above do not yet have the required rotational and reality properties, because the finite tests were chosen by linear algebra. Averaging the field-valued expression over $O(4)$ and symmetrizing conjugation supplies those properties. The subsequent homogeneous projection must be made using these averaged tests themselves, so that its lower-degree components agree exactly. These operations and their quantitative error estimates are the next steps of the OPE construction.

51 Covariant homogeneous OPE coefficients

The finite tests of Section 50 are chosen for exact compatibility, without imposing rotational symmetry. We first average their field-valued output, retaining both the input and output rotations. We then construct homogeneous complements using these averaged tests. This order ensures that the final coefficients have compatible finite truncations as well as the required symmetries.

Write $R_n(K)$ for the continuous $O(4)$ action on E_n and J_n for complex conjugation of fields. Assume, as supplied by the field structure, that these actions commute with the inclusions, that J_n is an antilinear involution, and that $J_n R_n(K) = R_n(K) J_n$. Haar measure dK on $O(4)$ is normalized to have mass one.

51.1 Averaging on a common separated region

For a fixed degree n, let $\mathcal{U}_n$ be the set of cluster configurations $(\boldsymbol{x}, o)$ for which every exterior test occurring through degree n remains separated from the relative cluster after every orthogonal rotation. This is an open set. Indeed, for a fixed finite test, separation is an open condition in the configuration and the rotation. If it holds for every rotation at a configuration, compactness of $O(4)$ supplies finitely many rotation neighborhoods and a common configuration neighborhood on which it still holds. Intersect over the finitely many tests. The set is invariant under simultaneous translations, rotations and permutations of the cluster. It contains every sufficiently small separated cluster around its center, by the clearance argument of the preceding section.

On $\mathcal{U}_n$ define

$$A_n(\boldsymbol{x}, \boldsymbol{v}; o) = \int_{O(4)} R_n(K) H_n(K^{-1}\boldsymbol{x}, R(K^{-1})\boldsymbol{v}; K^{-1}o)\, dK, \tag{176}$$

where the action on each input label uses its own degree. Set $A_n = 0$ outside $\mathcal{U}_n$. The integrand is continuous in K, takes values in a finite-dimensional space, and is therefore integrable on the compact group. Multilinearity passes through the integral; the zero extension preserves it because the domain condition is independent of labels.

The map A_n is smooth in positions on $\mathcal{U}_n$. Here one must not assume differentiability of the representation in K to justify position derivatives. Expand the input labels and output in finite bases. Each matrix entry of $R(K)$ is continuous and bounded on the compact group. The remaining

factors are derivatives in positions of separated correlations evaluated at rotated coordinates. On a sufficiently small closed configuration neighborhood times $O(4)$, all finitely many separations have a positive minimum. The correlation derivatives of each fixed order are therefore continuous and bounded on this compact set. The chain rule contributes only bounded entries of the rotation matrix. Thus each position derivative has a uniform integrable bound. Differentiation under the integral, repeated for every finite order, proves smoothness. The same finite-basis expansion justifies smooth variation of labels when needed.

Left invariance of Haar measure gives rotation covariance. For a fixed $L \in O(4)$ substitute $K = LH$ in $A_n(L\boldsymbol{x}, R(L)\boldsymbol{v}; Lo)$. Since $(LH)^{-1}L = H^{-1}$, the integrand becomes $R_n(L)$ times the integrand in (176). Hence

$$A_n(L\boldsymbol{x}, R(L)\boldsymbol{v}; Lo) = R_n(L)A_n(\boldsymbol{x}, \boldsymbol{v}; o).$$

This also holds outside the domain by its invariance. Translation and permutation identities follow directly from the corresponding identities for H_n inside the integral. For a single field $v \in E_q$ at its center, with $q \le n$, (175) shows that the integrand is the constant $R_n(K)\iota_{qn}R_q(K^{-1})v = \iota_{qn}v$. Therefore $A_n((o),(v);o) = \iota_{qn}v$.

Now define the real-symmetric readout

$$C_n(\boldsymbol{x}, \boldsymbol{v}; o) = \tfrac{1}{2}\big(A_n(\boldsymbol{x}, \boldsymbol{v}; o) + J_n A_n(\boldsymbol{x}, J\boldsymbol{v}; o)\big). \tag{177}$$

It is still complex multilinear: an input scalar is conjugated by the input J and conjugated back by the output J_n. It is smooth on $\mathcal{U}_n$, has the same translation, permutation and rotation identities, and satisfies

$$C_n(\boldsymbol{x}, J\boldsymbol{v}; o) = J_n C_n(\boldsymbol{x}, \boldsymbol{v}; o), \qquad C_n((o),(v);o) = \iota_{qn}v \quad (q \le n).$$

The first identity uses $J^2 = I$ and exchanges the two summands. The second follows from the corresponding identity for A_n and compatibility of J with inclusions.

51.2 Compatible projections onto new degrees

For every pair $m, q \ge 0$, define a linear map

$$L_{mq} : E_q \to E_m, \qquad L_{mq}v = C_m((0),(v);0).$$

The centered single-field configuration always belongs to $\mathcal{U}_m$. The preceding identities give

$$L_{mq} = \iota_{qm} \ (q \le m), \qquad L_{mq}\iota_{rq} = L_{mr} \ (r \le q). \tag{178}$$

For the second equality, every exterior correlation used in the raw tests is unchanged under the physical inclusion. This equality passes through rotation averaging, since the inclusions commute with rotations, and through conjugation averaging for the same reason. It holds even when $m < r$; no extension of a low-degree inverse is being assumed. The maps L_{mq} intertwine rotations and conjugation as well.

Define recursively

$$K_m = I_{E_m} - \sum_{i<m} \iota_{im}K_i L_{im}, \qquad P_{mq} = K_m L_{mq}, \qquad V_m = \bigcap_{i<m} \ker P_{im}.$$

The empty sum makes $K_0 = I$. These are finite sums of fixed linear maps. From (178), for $q \le m$,

$$P_{mq}v = \iota_{qm}v - \sum_{i<m} \iota_{im}P_{iq}v, \qquad \sum_{i\le m} \iota_{im}P_{iq}v = \iota_{qm}v. \tag{179}$$

Also $P_{mq}\iota_{rq} = P_{mr}$ whenever $r \le q$. If $q < m$, apply the second identity at degree $m-1$ and then include in E_m. Its sum is the one subtracted in the first identity, so

$$P_{mq} = 0 \quad (q < m). \tag{180}$$

For $v \in V_m$, the recursion with $q = m$ gives $P_{mm}v = v$.

Lemma 51.1. *The range of K_m is contained in V_m. If $v \in V_j$, then $P_{mj}v = \delta_{mj}v$, interpreting the right side as zero when $m \neq j$. Consequently*

$$E_m = \bigoplus_{j \le m} \iota_{jm} V_j.$$

Proof. We prove the range assertion by induction on m. Fix $i < m$ and apply P_{im} to the defining formula for $K_m x$. Compatibility gives

$$P_{im}K_m x = P_{im}x - \sum_{j<m} P_{ij}(K_j L_{jm} x).$$

By the induction hypothesis $K_j L_{jm} x \in V_j$. If $i < j$, its image under P_{ij} is zero by the definition of V_j; if $i > j$, it is zero by (180). At $i = j$, the identity on V_i proved above makes the term $K_i L_{im} x = P_{im}x$. The sum therefore cancels the first term. This proves $K_m x \in V_m$. The asserted action on V_j follows from the same three cases.

Take $q = m$ in (179). Each $P_{jm}x = K_j L_{jm} x$ belongs to V_j, so this identity proves spanning. If $\sum_{j \le m} \iota_{jm} v_j = 0$ with $v_j \in V_j$, apply P_{km} and use inclusion compatibility. All terms vanish except v_k. Thus every $v_k = 0$, proving directness. □

Induction in the formula for K_m shows that it commutes with rotations and conjugation: the inclusions and L_{im} intertwine them, and the previous K_i do so by induction. It follows that V_m is invariant under both operations, since all its defining tests intertwine them. No claim that the original arbitrary quotient lifts were covariant was needed.

Define the homogeneous output at degree m by

$$F_m(\boldsymbol{x}, \boldsymbol{v}; o) = K_m C_m(\boldsymbol{x}, \boldsymbol{v}; o) \in V_m.$$

It is multilinear, has the same symmetries as C_m, and is smooth on $\mathcal{U}_m$. At a single centered field $v \in E_q$ it equals $P_{mq}v$. In particular the finite packet

$$\mathcal{P}_D(\boldsymbol{x}, \boldsymbol{v}; o) = \sum_{m \le D} \iota_{mD} F_m(\boldsymbol{x}, \boldsymbol{v}; o)$$

satisfies the exact compatibility

$$P_{jD}\mathcal{P}_D = F_j \qquad (j \le D), \tag{181}$$

by Lemma 51.1. On one centered field of degree $q \le D$, the packet equals $\iota_{qD}v$ by (179).

51.3 Real channel bases and scalar coefficients

The conjugation J_m restricts to an antilinear involution of V_m. Its fixed real vector space has a real basis $(e_{m,\alpha})$; this is also a complex basis of V_m. Indeed every $v \in V_m$ decomposes as

$$v = \frac{v + J_m v}{2} + i\frac{v - J_m v}{2i},$$

and both displayed components are fixed by J_m. This proves complex spanning. If $\sum_\alpha z_\alpha e_{m,\alpha} = 0$, applying J_m and then adding and subtracting the two equations shows separately that the real and imaginary coefficient sums vanish. Real linear independence forces every $z_\alpha = 0$.

Choose these bases once for every m and write

$$F_m(\boldsymbol{x}, \boldsymbol{v}; o) = \sum_\alpha c_{\boldsymbol{v}}^{m,\alpha}(\boldsymbol{x}; o) e_{m,\alpha}. \tag{182}$$

The coordinate functionals are continuous because the spaces are finite dimensional. Thus the coefficients inherit smoothness, multilinearity, translation and permutation symmetry; reality gives conjugation of their values under conjugation of all input labels. Rotation covariance holds for the field-valued output, or equivalently through the finite matrix of R_m on this basis. The individual basis vectors need not be rotation invariant.

Let $\ell_{\chi,m,o,\boldsymbol{y},\boldsymbol{w}}$ be the linear functional obtained by inserting an E_m field at o alongside separated spectators, in any phase χ or in the mixed state. Physical inclusion compatibility gives $\ell_{\chi,D}\iota_{mD} = \ell_{\chi,m}$. Therefore evaluation of the packet is exactly

$$\ell_{\chi,D}(\mathcal{P}_D) = \sum_{m \le D} \sum_\alpha c_{\boldsymbol{v}}^{m,\alpha}(\boldsymbol{x}; o) S_\chi(e_{m,\alpha}, \boldsymbol{w}; o, \boldsymbol{y}).$$

This identifies the final scalar coefficients with one fixed covariant field prescription, independent of phase, spectators and truncation. It does not yet estimate the difference from the original cluster correlation. That quantitative difference is the next proof obligation.

52 Centered bounds, phase prices, and matched discrepancies

The raw OPE estimates must hold in every character phase, including phases of zero measure for the original mixture. An estimate of the mixed-state expectation alone does not give this. We prove the required transfer by bounding all even moments of a compressed whole-word operator. The cutoff argument keeps a matched discrepancy as a single observable throughout; it never replaces its small price by separate prices for its two terms.

We use the finite-volume planted inequality and the change-of-cell root estimate established in Section 18. In the latter, a centered observable whose root norm is at most B can be moved, with the corresponding reflection or conjugation, into a coarser cell at a margin of three of its original cell spans, still with root norm at most B. The proof uses the entire reflected copy orbit and coordinatewise log convexity. These statements apply to any observable satisfying the centered interior and root conditions, including a difference of two observables.

52.1 A mixed two-scale chessboard estimate

Let W_0, W be aligned link tilings of a finite torus, with the side of W_0 dividing that of W. Write $\mathcal{R}_W$ for the root norm. Suppose fine-group observables F_j have centered prices $A_j \ge 0$ at a span dividing the side of W_0, and their positions u_j lie in the required three-span margins of W_0. Suppose further that

$$\operatorname{dist}_\infty(u_j, u_{j'}) > \operatorname{side}(W_0) \quad (j \ne j').$$

For one fixed cell C of W, collect the fine-group observables lying in C. Their product is a copy in C of an interior observable Z_C, and

$$\mathcal{R}_W(Z_C) \le \prod_{u_j \in C} A_j. \tag{183}$$

Indeed pull each factor back to the home cell of W, with the reflection and conjugation determined by C. Now copy this product into every cell of W. The copied positions still lie in the corresponding fine-tiling margins, since the tilings are aligned and their sides divide. They occupy distinct cells of W_0. Points copied into different cells of W cannot share a fine cell; within one coarse cell, reflection preserves distances, and the displayed separation prevents sharing a fine cell. The planted inequality on W_0 therefore bounds the full copied-product expectation by

$$\left(\prod_{u_j\in C} A_j\right)^{\#\mathrm{Cell}(W)}.$$

Taking the defining root proves (183). Interiority of the home product follows by multiplying interior factors.

Let a second group G_i have centered prices $B_i \geq 0$ at spans dividing the side of W, with positions v_i in the corresponding margins of W. Assume their mutual distances, and their distances to all u_j, exceed side(W). The G_i then occupy distinct coarse cells, and none of those cells contains an F_j. Apply the planted inequality on W to the individual G_i and the products Z_C. This gives

$$\left|\mathbb{E}\left[\prod_i G_i(\cdot + v_i)\prod_j F_j(\cdot + u_j)\right]\right| \leq \prod_i B_i \prod_j A_j. \tag{184}$$

Grouping by cells loses no factors: the disjoint groups of indices $\{j : u_j \in C\}$ partition the fine-group list. Empty groups contribute one.

Here are sufficient geometric conditions for placing both groups. Use depths $b \leq f \leq l \leq n$ and $f \leq q$. The coarse tiling is at depth b, the fine tiling at depth l, the centered prices for the coarse group at depth q, and those for the fine group at depth n. If the group sizes are I_0, J_0 and

$$10I_0 a_f < a_b, \qquad 6J_0 a_n < a_l, \tag{185}$$

there is a common origin for both tilings placing all coarse-group points at margin $3a_q$ and all fine-group points at margin $3a_n$. To prove this, first choose the coarse origin coordinate by coordinate. For each point, origins giving distance less than $5a_f$ from a cell boundary occupy an interval of length at most $10a_f$ in the circle of possible origins modulo a_b. The first inequality leaves an allowed origin. Next choose the fine origin modulo a_l avoiding the intervals of total length at most $6J_0 a_n$ forbidden by the fine-group points. It can be reached from the chosen coarse origin by a shift of coordinate size at most $a_l \leq a_f$. This shift decreases the coarse margins by at most $2a_f$ even with the endpoint conventions for link cells, leaving $3a_f \geq 3a_q$. On the finite lattice, if a cell side is B and the desired integer margin is r, a point forbids the stripe residues $0, \ldots, r-1$ and $B-r, \ldots, B-1$: at most $2r$ residues of the origin modulo B. Summing over the points leaves a permitted origin whenever their total count is less than B. Multiplying (185) by a_N^{-1} gives precisely these strict inequalities, first with margin $5a_f/a_N$, then with margin $3a_n/a_N$. To attain the permitted fine origin, choose a common lattice shift with $0 \leq s_\mu < a_l/a_N$ in each coordinate. The coarse $5a_f/a_N$ margin loses at most $2a_f/a_N$, leaving the required three spans. Nested mesh divisibility makes all sides and margins integers.

A Euclidean separation greater than $2a_b$ implies a maximum-coordinate separation greater than a_b in four dimensions; likewise for a_l. For fixed physical grid points these inequalities persist on the periodic lattice for all sufficiently large periods: their finitely many coordinate differences then do not wrap. They persist at all finer cutoffs by the exact mesh embedding. Thus Euclidean separation at $2a_b$ within and across the coarse group, and at $2a_l$ within the fine group, suffices for (184) on the ordered cutoff tail.

52.2 Even replicas and the norm in the ground algebra

Let Z be a finite linear combination of products of local fields, at fixed coordinate-generic grid configurations. Write $A \in \mathcal{A}$ for the same linear combination of their compressed whole-word operators. The cutoff version $Z_{N,p}$ may use labels converging in their fixed finite-dimensional label spaces along a nontrivial refinement of the ordered cutoff limit.

Suppose each cutoff $Z_{N,p}$ factors as the product of two groups as in (184), with the same centered price bounds and geometry. The statement includes a group consisting of a single matched discrepancy. Set $C = \prod_i B_i \prod_j A_j$. For each $m \geq 1$, translate $2m$ copies of $Z_{N,p}$ by the even coarse-cell displacements

$$d_j(t) = 2tj\, a_b(1,1,1,1), \qquad t \in \mathbb{N}, \quad 0 \leq j < 2m,$$

conjugating the copies with even j. These translations preserve all tilings' margins, because they are multiples of the coarse cell side, and this side is a multiple of the fine side. For fixed m, copies at distinct indices become separated in every coordinate as $t \to \infty$. For each sufficiently large fixed t, take the period large enough to avoid wrapping, and then the cutoff fine enough to use all the centered prices. Applying (184) to the replicated lists gives

$$\left| \mathbb{E}_{N,p} \prod_{j=0}^{2m-1} \begin{cases} \overline{Z_{N,p}(\cdot + d_j(t))}, & j \text{ even}, \\ Z_{N,p}(\cdot + d_j(t)), & j \text{ odd} \end{cases} \right| \leq C^{2m}. \tag{186}$$

Only the two original arities occur in (185). Once an origin works for one copy, it works for all its even-cell translates. The period and the translation threshold may depend on m; the bound C does not.

Expand the finite polynomial in (186) and first pass to the ordered cutoff limit at fixed t. For sufficiently large t each expanded word is coordinate-generic, so the point-limit theorem applies. Converging labels cause no extra difficulty: in a fixed basis a multilinear correlation is a finite sum of basis correlations times products of label coordinates. The basis correlations converge and the coordinates converge, proving convergence of each term. This also works on any finer nontrivial cutoff filter. It does not assume that the moving labels have separate limits in the two cutoff indices.

Next let $t \to \infty$. The separated-cluster limit of Lemma 33.4, together with conjugation becoming adjoint under ground compression, identifies the limit as

$$\langle \Omega, (A^* A)^m \Omega \rangle.$$

For a linear combination this follows by expanding the ordered product; its alternating factors are exactly $A^*, A, \ldots, A^*, A$. Thus

$$\int_{\Xi} |\chi(A)|^{2m}\, d\mu(\chi) \leq C^{2m} \quad (m \geq 1).$$

Full support of μ now gives $|\chi(A)| \leq C$ for every character. If one value exceeded C, continuity would give an open set of positive measure on which it exceeded some $C' > C$. Then $\mu(O)(C')^{2m} \leq C^{2m}$ for every m, a contradiction. The same argument covers $C = 0$. The isometric Gelfand transform consequently gives

$$\|A\| \leq C. \tag{187}$$

Both every character evaluation and the mixed-state functional have norm one, so the same bound holds for their evaluations of A.

52.3 Removing coordinate ties

To extend (187) to arbitrary separated configurations, first approximate all their points by the generic grids of (848). Strict Euclidean separations persist. For each fixed character, its correlation is continuous on separated configurations, so the generic bound passes to the limit. Taking the supremum over characters then proves the ground-algebra norm bound at the original configuration. This uses scalar continuity in each phase, not norm continuity of the ground-algebra-valued correlation.

For a matched discrepancy whose cluster is already fixed and generic, only the spectator positions are approximated. They must remain coordinate-generic relative to both that cluster and the origin. Here is the required extension of the grid construction. Let W be the finite set consisting of the cluster points and the origin, J_0 the number of spectators, and $K = (J_0 + 1)(|W| + 1)$. In each coordinate choose for spectator j a residue from the block

$$\{j(|W| + 1), \dots, j(|W| + 1) + |W|\}$$

which avoids the residues $\lfloor w_\mu/h \rfloor \bmod K$ for $w \in W$. There are $|W|+1$ candidates and at most $|W|$ forbidden residues. The blocks are disjoint, so the chosen residues distinguish all spectators. Adding the chosen residue to $K\lfloor y_{j,\mu}/(hK) \rfloor$ gives a grid point within hK in each coordinate, and it cannot equal any w_μ. Thus these spectator grids converge and retain all required coordinate inequalities. If the new grid is finer than the fixed cluster grid, reexpress the cluster points on it by the integer mesh ratio; their physical positions and cutoff observables are unchanged. Apply the preceding replica argument on that common grid, then take the spectator approximation limit in each phase. This proves the complete-discrepancy bound at arbitrary separated spectator configurations.

52.4 Engineering rates from centered prices

We now use the centered estimate in its full quantifier order: for a fixed degree d, bounded set of labels and every $\epsilon > 0$, one constant controls the centered root price at all depths after the reference depth, with an eventual cutoff tail uniform in period and those labels. In particular a fixed field of degree d_i has a price

$$B_i(k) = (1 + C_i)a_k^{-d_i-\epsilon} + 1$$

for a suitable $C_i \geq 0$. If $s_{k_n}/t_n \to u \geq 0$, this positive price has logarithmic rate $(d_i + \epsilon)u$. For a finite group the product therefore has rate $(\sum_i d_i + \epsilon \# I)u$.

More generally, suppose two groups have mutual separations at upper scales $0 \leq a \leq b$, and cross separation at scale a. This means that for every positive slack the corresponding distances eventually exceed $e^{-(a+\delta)t_n}$ or $e^{-(b+\delta)t_n}$. Choose mesh depths of logarithmic rates

$$a + \delta/3, \quad a + \delta/2, \quad a + \delta, \quad b + 3\delta/4, \quad b + \delta$$

for the coarse placement, buffer, coarse-group price, fine placement, and fine-group price respectively. The inequalities of the rates imply all depth orderings and (185) eventually. The separation assumptions imply the required strict gaps at the placement meshes. The bounds just proved yield ground-algebra upper rate

$$(w_1 + \epsilon I_0)(a + \delta) + (w_2 + \epsilon J_0)(b + \delta).$$

Let $\epsilon, \delta \downarrow 0$ by allocating half of any desired rate slack to these choices and half to the eventual estimate. The result is the upper rate $w_1 a + w_2 b$, uniformly along every sequence of phases, because the ground-algebra norm controls them all.

For the OPE, take the fixed-clearance spectators as the first group ($a = 0$) and the scale-x cluster as the second ($b = x$). Their cross clearance is fixed for all sufficiently large indices. This proves the rough rate $xw(p)$ used in Section 175. Taking every group at scale zero proves rate zero for the fixed single-field spectator insertion in Input 176.1.

Finally suppose a cutoff matched discrepancy is centered and priced by A_n at a depth of logarithmic rate $b > 0$, and its coefficient label converges along a refinement of the same ordered cutoff limit. Assume A_n has upper rate γ. Choose, for any small $0 < \delta < b$, placement, buffer, fine-placement and spectator-price depths of rates $\delta/4, \delta/2, 3\delta/4, \delta$. The centered discrepancy price at rate b fits this geometry. Equation (187) and the spectator approximation give upper rate at most $\gamma + (w_2 + \epsilon J_0)\delta$ for the entire discrepancy, in every phase and at arbitrary fixed-clearance spectator positions. Sending $\delta \downarrow 0$ proves rate γ. Thus the finite-cutoff match supplies exactly the complete-discrepancy part of Input 175.1, once its price and convergent label are constructed. Neither this transfer nor the even-moment argument creates that finite-cutoff match.

53 Triangular transport and the physical filtration

The distance estimate in (850) concerns the physical filtration. A bare coefficient of low engineering degree need not stay in the corresponding coordinate plane under renormalization. We first estimate its transported plane, then identify limits of that plane by their correlations. The two parts use different information: triangular estimates for the first, and the comparison of local observables for the second.

53.1 Expansion of the low coordinates

Fix a degree d, its finite coefficient space F, and an integer $0 \le e < d$. Each coordinate has an engineering weight in $\{0, \dots, d\}$. Write $\mu(v) = \sum_\alpha |v_\alpha|$ and let P retain the coordinates of weight at most e. For a block ratio $L \ge 1$, let

$$(D_L v)_\alpha = L^{d - \deg \alpha} v_\alpha.$$

Thus P commutes with D_L and

$$\mu(PD_L v) \ge L^{d-e}\mu(Pv), \qquad \mu((1-P)D_L v) \le L^{d-e-1}\mu(v). \tag{188}$$

Both inequalities follow by summing the coordinate inequalities.

Here is the exact one-step input needed from the calibrated construction. At consecutive levels $i+1, i$, the readouts of the same bare vector are a, b. There is an invertible linear map B preserving $\ker P$, with

$$\mu(Bv),\ \mu(B^{-1}v) \le K\mu(v), \qquad \mu(b - BD_L a) \le \epsilon\mu(a).$$

The last estimate follows in the construction from the stronger bound with the nonconstant-coordinate part of a on the right. Its numerical conditions are

$$K \ge 1, \quad s_0, s \ge 64K^2, \quad s \le L, \quad \epsilon \ge 0, \quad 2K\epsilon(1+s) \le 1. \tag{189}$$

Here $s_0 = \zeta_i$ and $s = \zeta_{i+1}$ are the column allowances. These triangular estimates concern the actual readouts, including their nonlinear corrections. Their production is still a construction input.

Since B maps the finite-dimensional space $\ker P$ injectively into itself, it maps it onto itself. Its inverse therefore also preserves $\ker P$. Applying this observation to $Bv - PBv$ gives

$$\mu(Pv) = \mu(PB^{-1}PBv) \le K\mu(PBv). \tag{190}$$

Suppose now that $\mu(a) \le s\mu(Pa)$, and set $Y = \mu(PD_L a)$. By (188) and $s \le L$,

$$\mu(D_L a) \le Y + L^{d-e-1} s\mu(Pa) \le 2Y, \qquad \mu(a) \le sY.$$

The error $E = \mu(b - BD_L a)$ consequently satisfies $2KE \le Y$. On the other hand, (190) gives $Y \le K(\mu(Pb) + E)$, so $Y \le 2K\mu(Pb)$. It follows that

$$\mu(b) \le 3KY \le 6K^2\mu(Pb) \le s_0\mu(Pb), \qquad L^{d-e}\mu(Pa) \le s\mu(Pb). \tag{191}$$

If instead $Pa = 0$, (188), $\epsilon \le 1$, and $\mu(a) \le \mu(D_L a)$ give

$$\mu(b) \le (K+1)\mu(D_L a) \le sL^{d-e-1}\mu(a). \tag{192}$$

The lost power of L distinguishes the transverse estimate from the expansion of the low coordinates.

53.2 Subtracting the transported low plane

Fix a finest level N, a period, and a reference level $U \le N$. Let $T_k : F \to F$ be the linear readout at level k, for $U \le k \le N$. Thus $T_N = 1$ and $T_U = K_N$, where K_N is the invertible map to reference coefficients. For each adjacent pair assume the preceding one-step estimates, with $a_k = L_k a_{k+1}$ for the meshes. Define

$$J_k = \prod_{i<k} \zeta_{i+1}, \qquad H_k = \prod_{i<k} (1+\zeta_i)\zeta_{i+1}.$$

All these factors are at least one. If $Pv = v$, backward induction in (191) gives

$$\mu(T_k v) \le \zeta_k \mu(PT_k v), \qquad J_k (a_k/a_N)^{d-e}\mu(v) \le J_N \mu(PT_k v). \tag{193}$$

For the second inequality, multiply the one-step lower expansion by J_i, use $J_{i+1} = \zeta_{i+1}J_i$, and telescope the mesh ratios. The terminal equality $T_N v = v$ starts both inductions.

In particular PT_k restricted to $\operatorname{ran} P$ is injective. Its domain and codomain have equal finite dimension, so it is onto. For any $y \in F$ choose $v \in \operatorname{ran} P$ with $PT_k v = PT_k y$. The first bound in (193) shows

$$PT_k(y - v) = 0, \qquad \mu(T_k(y-v)) \le (1+\zeta_k)\mu(T_k y). \tag{194}$$

This cancellation must be repeated after each step: the fixed coordinate complement is not asserted to be invariant under the exact readouts. If $PT_{i+1}y = 0$, first apply (192), then (194) at level i. The cost is $(1+\zeta_i)\zeta_{i+1}L_i^{d-e-1}$. Induction therefore gives, whenever $U \le k \le j \le N$ and $PT_j y = 0$, a $v \in \operatorname{ran} P$ such that

$$PT_k(y - v) = 0, \qquad H_k \mu(T_k(y-v)) \le H_j (a_k/a_j)^{d-e-1}\mu(T_j y).$$

At an induction step add the newly subtracted low vector to the old one; their sum remains in $\operatorname{ran} P$. The recurrence for H_k and the identity $a_i = L_i a_{i+1}$ give the displayed product.

Cancel first at level k, then transport to U, and discard the factor $H_U \ge 1$. For every reference vector x this yields a vector y in the transported plane $K_N \operatorname{ran} P$ with

$$\|x - y\| \le F_{k,d,e}\, q_k(x), \qquad F_{k,d,e} = H_k(1+\zeta_k)(a_U/a_k)^{d-e-1}, \tag{195}$$

where $q_k(x) = \mu(T_k K_N^{-1} x)$, and the reference norm is chosen so that $\|z\| \le \mu(z)$. Indeed apply the preceding result to $K_N^{-1}x$ and use $T_U = K_N$. For comparison, the calibrated bound $q_i(x) \le \zeta_{i+1}L_i^d q_{i+1}(x)$ gives by forward induction

$$\|x\| \le J_k (a_U/a_k)^d q_k(x). \tag{196}$$

At U use $q_U(x) = \mu(x)$ and $J_U \ge 1$; the recurrence for J and the mesh identity give each subsequent step.

53.3 Subpower calibration and smeared discrepancies

The cumulative column factor used above is subpower for a precise reason. Put $s_i = \log(1/a_i) \geq 0$ and $C_N = \sum_{i<N} \log \zeta_{i+1} = \log J_N$. The schedule provides numbers $h_i \to \infty$ and the bound

$$0 \leq \log \zeta_{i+1} \leq h_i^{-2} \log L_i, \qquad s_{i+1} - s_i = \log L_i > 0.$$

The first inequality follows also when the scheduled logarithmic cost contains other nonnegative charges, since it then dominates $\log \zeta_{i+1}$. Given $\epsilon > 0$, choose k so that $h_i^{-2} \leq \epsilon/2$ for $i \geq k$. Telescoping gives

$$C_N \leq C_k + \tfrac{\epsilon}{2}(s_N - s_k) \leq C_k + \tfrac{\epsilon}{2} s_N.$$

Eventually $C_k \leq \epsilon s_N/2$, because $s_N \to \infty$. Thus $C_N = o(s_N)$. In particular J_N has logarithmic rate zero; so does ζ_N, since $1 \leq \zeta_N \leq J_N$ for $N \geq 1$.

The calibrated reverse estimate is

$$\mu(T_{i+1} K_N^{-1} x) \leq \zeta_{i+1} \mu(T_i K_N^{-1} x).$$

Starting at $T_U K_N^{-1} x = x$ and enlarging the product to the full prefix gives $\mu(T_i K_N^{-1} x) \leq J_i \mu(x)$. At $i = N$, therefore,

$$\|K_N^{-1} x\| \leq J_N \mu(x) \leq J_N \, \dim(F) \, \|x\|,$$

using the coordinate supremum norm for the last inequality. For every fixed radius R and $\epsilon > 0$, both J_N and $\dim(F)R$ are eventually bounded by $e^{\epsilon s_N/2}$. Consequently

$$\sup_p \sup_{\|x\| \leq R} \|K_{N,p}^{-1} x\| \leq e^{\epsilon s_N} \quad \text{eventually in } N. \tag{197}$$

The same estimate applies to any bounded family along the ordered cutoff filter. The calibration inequalities are uniform in the period; no new period extraction is needed.

We will also use the following passage from separated lattice bounds to smeared tests. Let a trial $Z_{N,p}$ replace one distinguished field in an n-point correlation, and suppose that, for each fixed spectator list and every $\delta > 0$, its correlation has absolute value at most $A_\delta |r_{N,p}|$ whenever all physical pairwise separations are at least δ. Assume $r_{N,p} \to 0$ along the ordered cutoff filter. For every smooth test φ compactly supported in the separated configuration space, the corresponding lattice sum, with the factor a_N^{4n}, tends to zero.

Indeed the test support K is compact and has positive distance from every pairwise diagonal. Choose a smaller clearance $\delta > 0$. For sufficiently large periods, its lattice representatives have that clearance also in the periodic metric: a bounded set does not meet a periodic translate of itself near a diagonal. Express the weighted lattice sum exactly as the integral of its piecewise constant function on the mesh cells in the fundamental region. A nonzero cell value has its corner in K. The cell diameter tends to zero, so every such cell lies in a fixed compact enlargement B of K. The absolute value of the step function is at most

$$\mathbf{1}_B \, \|\varphi\|_\infty A_\delta |r_{N,p}|.$$

Its integral is bounded by $|B| \|\varphi\|_\infty A_\delta |r_{N,p}|$, which tends to zero. This is a norm estimate on the same ordered filter, not an interchange of mesh and volume limits. Coordinate ties do not cause a problem: the test support is separated in the Euclidean sense, and one separating coordinate for each pair suffices to control its periodic separation.

53.4 Identifying the limiting plane by tests

We spell out the finite-dimensional limit argument, since convergence of the moving planes themselves is neither needed nor asserted. Let V be a finite-dimensional complex normed space, p_c projections converging in operator norm to a projection p, and $f_{\theta,c}$ linear functionals converging in norm to f_θ, one for each test θ. Assume

$$\ker p = \bigcap_\theta \ker f_\theta. \tag{198}$$

The index c tends along any nontrivial filter. Suppose vectors u_c satisfy, for every θ,

$$f_{\theta,c}(u_c) \longrightarrow M_\theta, \qquad \frac{f_{\theta,c}(u_c - p_c u_c)}{1 + \|p_c u_c\|} \longrightarrow 0. \tag{199}$$

Then there is $u \in \operatorname{ran} p$ with $f_\theta(u) = M_\theta$ for all θ.

To prove this, put $b_c = (1 + \|p_c u_c\|)^{-1}$ and $z_c = b_c p_c u_c$. The pairs (b_c, z_c) lie in the compact set $[0,1] \times \overline{B}_1$. Along a finer nontrivial filter they converge to (b, z), with $b + \|z\| = 1$. Since $p_c z_c = z_c$, convergence of the projectors gives $pz = z$. Equations (199) imply $f_\theta(z) = bM_\theta$. If $b = 0$, (198) forces $pz = 0$, hence $z = 0$, contradicting $b + \|z\| = 1$. Thus $b > 0$ and $u = b^{-1}z$ proves the claim. There is only one compact extraction; the limits for every test pass to its refinement. No extraction indexed by the tests is used.

Here is how the required hypotheses arise for the transported plane. We make explicit the remaining comparison input. A local trial at degree e is an observable with fixed finite support whose product with a_N^e has subpower group supremum and subpower bounds for the prescribed left derivatives, uniformly in its lattice origin. Here the derivative order, spatial weight power and direction threshold are the purchased cutoff-dependent parameters: the bound holds simultaneously for every ordered left derivative of length at most that cutoff's purchased order, with directions satisfying its weighted threshold. Bounds only for each fixed derivative order would not suffice. The local comparison construction supplies degree-e labels u_c with subpower bare norm, whose difference from that trial has a centered price tending to zero. It also supplies the projections $p_c \to p$ of (198). Labels in $\ker p_c$ with subpower bare norm have vanishing tested correlations. Bounded reference labels have subpower bare norm by (197). The local comparison and projection assertions are substantive inputs still to be proved from the construction. All statements use the same ordered cutoff filter.

For clarity, these inputs imply the second limit in (199) without assuming u_c bounded. Multiplication by $b_c = (1+\|p_c u_c\|)^{-1} \le 1$ preserves subpower bare norm. The vectors $b_c p_c u_c$ are bounded, so also have subpower bare norm. Their difference $b_c(u_c - p_c u_c)$ consequently has subpower bare norm and belongs to $\ker p_c$, because $p_c^2 = p_c$. The vanishing-test assertion gives exactly the normalized residual limit. The first limit follows by comparing the trial with u_c: the centered estimate gives a separated correlation bound by a constant times the price, and the step-integral estimate just proved sends this bound to zero in every smeared correlation. Linearity here is legitimate at finite cutoff, since the trials are bounded measurable functions and the remaining fields are continuous on the compact configuration space.

It remains to check the degree of a bounded transported low vector $v_c \in K_N \operatorname{ran} P$. Write $x_c = K_N^{-1} v_c$. Taking $k = U$ in (193) gives

$$a_N^{-(d-e)} \|x_c\| \le a_U^{-(d-e)} J_N \mu(v_c).$$

Section 53.3 gives zero logarithmic rate for J_N. Thus the bare coefficients of $w_c = a_N^{-(d-e)} v_c$ have subpower norm. The calibrated local-section bounds give subpower prices for $a_N^d \mathcal{O}_d(w_c)$. By

linearity,

$$a_N^d \mathcal{O}_d(w_c) = a_N^e \mathcal{O}_d(v_c).$$

The same literal observable is therefore an admissible degree-e trial, with its original finite support. If $v_c \to v$ along a refinement, its tested correlations converge to those of the reference label v: convergence of linear tests on a fixed basis is convergence in operator norm. Apply the preceding comparison and compact argument to obtain a degree-e label having all the same smeared correlations. The smooth correlation kernels consequently agree: their difference integrates to zero against every compactly supported smooth test on the separated configuration domain, hence vanishes there. By the physical null-space definition, the class of v belongs to E_e.

Only boundedness and low-plane membership on the refined filter are needed. To apply comparison inputs stated on the original filter, replace v_c by zero whenever either $\|v_c\| > \|v\| + 1$ or low-plane membership fails. This modified family is bounded and in the plane on the original filter, and equals v_c eventually on the refinement. It has the same limit and the same limiting correlations there.

53.5 The distance estimate after extraction

Suppose reference labels $v_c \to v$ on a refinement of the ordered cutoff filter and $q_k(v_c) \le V$ eventually, with k fixed. Choose $y_c \in K_N \operatorname{ran} P$ by (195). They are eventually bounded, because $\|y_c\| \le \|v_c\| + F_{k,d,e}V$. Extract a further nontrivial filter on which $y_c \to y$. The preceding argument puts $[y]$ in E_e, and continuity of the norm gives

$$[y] \in E_e, \qquad \|v - y\| \le F_{k,d,e}V. \tag{200}$$

The whole-vector bound (196) passes to the limit directly. Both estimates hold for the same limit v; only the auxiliary approximant y requires a further extraction.

Finally these finite-scale factors have the advertised rates. By Section 53.3, $\log J_k = o(s_k)$ and $\log \zeta_k = o(s_k)$, where $s_k = \log(1/a_k)$. The column allowances also satisfy $\zeta_k \ge 2$. In fact

$$H_k \zeta_k \le J_k^3 \zeta_0.$$

This holds at $k = 0$. For the induction step use $1 + \zeta_k \le \zeta_k \zeta_{k+1}$ and the recurrences for H, J:

$$H_{k+1}\zeta_{k+1} \le H_k \zeta_k \zeta_{k+1}^3 \le J_k^3 \zeta_0 \zeta_{k+1}^3 = J_{k+1}^3 \zeta_0.$$

Thus H_k also has zero logarithmic rate. If $k_n \to \infty$, $t_n \to +\infty$, and $s_{k_n}/t_n \to a$, then $F_{k_n,d,e}$ has rate $(d - e - 1)a$, while the factor in (196) has rate da. Taking $d = q$ and $e = m - 1$ proves precisely the factors in (849)–(850), once a cutoff match with quota V_n and convergent reference labels has been produced. This argument preserves the exact reference labels and physical filtration; it does not require convergence of the individual renormalization matrices.

54 The weighted support estimate for grouping

The loss in (841) pays for two effects. Moving several marks to a common representative enlarges the supports carrying them. Expanding the resulting product then creates connected unions of these enlarged supports. We prove the two counting estimates that control these operations. They apply to distinct activity labels; no claim about a logarithmic expansion with repeated labels is needed.

54.1 Connected unions with distinct labels

Let X_i be finite supports in a four-dimensional periodic lattice, indexed by an arbitrary set, and let $c_M(X)$ be their covering number at a positive integer block size M. We use its elementary properties

$$|X| \le (2M+1)^4 c_M(X), \qquad c_M(X \cup Y) \le c_M(X) + c_M(Y) \quad \text{if } X \cap Y \neq \varnothing.$$

Let $f_i \ge 0$, $a \ge 0$, $\gamma \ge 0$, and suppose that, for every site x,

$$\sum_{i : x \in X_i} f_i e^{(a+v)c_M(X_i)} \le \gamma, \qquad (2M+1)^4 \gamma \le v. \tag{201}$$

All sums of nonnegative terms have their supremum-of-finite-sums meaning. A nonempty finite label set A is connected when its intersection graph, with an edge for overlapping supports, is connected. Then

$$\sum_{\substack{A \text{ finite, connected} \\ x \in \bigcup_{i \in A} X_i}} e^{a c_M(\bigcup_{i \in A} X_i)} \prod_{i \in A} f_i \le e^{\gamma} - 1. \tag{202}$$

We prove this by a finite recursion. For a finite set of allowed labels D and a support Y, let $\mathcal{P}(D, Y)$ consist of all subsets of D whose every connected component has a support meeting Y; the empty subset is included. Set

$$F(D, Y) = \sum_{A \in \mathcal{P}(D,Y)} \prod_{i \in A} p_i, \qquad p_i = f_i e^{a c_M(X_i)}.$$

The following inequality retains each label at most once:

$$F(D, Y) \le \prod_{\substack{i \in D \\ X_i \cap Y \neq \varnothing}} \bigl(1 + p_i F(D \setminus \{i\}, X_i)\bigr). \tag{203}$$

To see it, choose for each connected component of A one label whose support meets Y. Different components choose different labels. Deleting the chosen label leaves a set every one of whose connected components meets its support: follow a path to the deleted vertex in the old component. Record that remaining set at the chosen label, and record the empty option at every unchosen label. The union of the recorded sets with their chosen labels recovers A, so the encoding is injective. Its weight is exactly $\prod_{i \in A} p_i$. Expanding the product on the right of (203) permits additional choices, possibly sharing labels between different options; all their weights are nonnegative. This proves the inequality without introducing factorial or ordering losses.

Strong induction on $|D|$ gives

$$F(D, Y) \le e^{|Y|\gamma} \le e^{v c_M(Y)}.$$

Indeed insert the induction bound in (203) and use $1 + r \le e^r$:

$$F(D, Y) \le \exp\left(\sum_{\substack{i \in D \\ X_i \cap Y \neq \varnothing}} f_i e^{(a+v)c_M(X_i)} \right) \le e^{|Y|\gamma}.$$

The last inequality sums (201) over sites of Y; a support meeting Y is counted at least once. The cardinality bound and (201) prove the second inequality, which closes the induction. Now take

$Y = \{x\}$ and remove the empty subset, whose weight is one. For a connected A, order its vertices so that each new support meets the preceding union; a spanning tree supplies such an order. Repeated covering subadditivity gives $c_M(\bigcup_{i\in A} X_i) \le \sum_{i\in A} c_M(X_i)$. Thus each term in (202) is bounded by its corresponding product of p_i. This proves the desired estimate for every finite family of connected sets: take D to contain all their labels. Taking the supremum over finite families proves the full assertion and its absolute summability.

54.2 Paying for the new representative sites

Let a finite set of old marks I be partitioned into groups V, with a chosen old mark in each group and a positive integer bound h on the group size. Write r_i for the old site, c_v for its group's representative site, and $\pi : I \to V$ for the partition. Assume that at most h old marks belong to a group and at most ℓ old sites occur at any one lattice site. The same occupancy bound holds for the representatives, since their selected old marks are distinct. Suppose

$$c_M(\{r_i, c_{\pi(i)}\}) \le k.$$

An activity indexed by α has a mark set $A_\alpha \subseteq I$ and support X_α containing every r_i for $i \in A_\alpha$. Enlarge it to

$$\widetilde{X}_\alpha = X_\alpha \cup \{c_{\pi(i)} : i \in A_\alpha\}.$$

Adding the pairs one at a time gives

$$c_M(\widetilde{X}_\alpha) \le c_M(X_\alpha) + k|A_\alpha|, \tag{204}$$

because each pair meets the original support at r_i. For positive mark weights w_i and $0 < \eta \le 1$, put

$$\widetilde{w}_i = (\eta/h) w_i e^{-\sigma k}, \qquad \sigma \ge 0.$$

For every nonnegative coefficient majorant q_α,

$$\left(\prod_{i\in A_\alpha} \widetilde{w}_i\right) q_\alpha e^{\sigma c_M(\widetilde{X}_\alpha)} \le (\eta/h)^{|A_\alpha|} \left(\prod_{i\in A_\alpha} w_i\right) q_\alpha e^{\sigma c_M(X_\alpha)}. \tag{205}$$

This follows immediately from (204); the exponential price paid by each mark cancels its added connecting cover.

We need a local summation version of this observation. Suppose $b, g \ge 0$ and nonnegative quantities F_α have the bound $\sum_{\alpha: y\in X_\alpha} F_\alpha \le g$ at every site. Suppose new weighted quantities $W_\alpha \ge 0$ satisfy $W_\alpha \le bF_\alpha$ for all α, and the stronger $W_\alpha \le (b/h)F_\alpha$ whenever $A_\alpha \ne \varnothing$. Then

$$\sum_{\alpha: y\in \widetilde{X}_\alpha} W_\alpha \le (1+\ell) bg. \tag{206}$$

The old-support contribution is at most bg. If y is a new site of $\widetilde{X}_\alpha$, some group v has $c_v = y$ and some $i \in A_\alpha \cap \pi^{-1}(v)$ has $r_i \in X_\alpha$. Bound this contribution by summing over all such v and all its old marks. For fixed i the sum is at most bg/h; there are at most h marks in that group, and at most ℓ groups at y. This proves (206). Activities with no marks have no added representative sites and are accounted for entirely in the old-support contribution. This is why the factor $1/h$ is needed only for activities carrying marks.

54.3 The resulting local majorant

Take nonnegative budget constants T, A, B. There are three kinds of entries in the regrouping expansion. The first are singleton coefficients u_i, with $w_i|u_i| \le T$. Their local sum, including the support weight, is at most $\ell T e^{\sigma}$, because $c_M(\{r_i\}) \le 1$ and the occupancy is at most ℓ. The second are the nonempty marked coefficients of the ordinary activity, whose local weighted sum is at most A. Together these have bound

$$g_0 = \ell T e^{\sigma} + A.$$

Both carry a nonempty mark set. Equation (205) bounds their new weight by η/h times the old weight. Applying (206) with $b = \eta$ gives $(1+\ell)\eta g_0$. Each entry is split into its healthy and unhealthy support cases by indicator functions. Each piece is bounded by the same coefficient majorant, so their combined bound is twice this.

The third kind comprises the activity on flagged supports, including its zero-mark part. Let its original local weighted sum be at most B. For these entries (205) gives a factor at most one in general, and at most $1/h$ for a nonempty mark set. Thus (206) with $b = 1$ gives $(1 + \ell)B$. The complete local majorant is therefore

$$\Gamma = 2(1 + \ell)\eta(\ell T e^{\sigma} + A) + (1 + \ell)B. \tag{207}$$

This bound is independent of the total number of cells and groups.

Apply (202) with support exponent $\sigma = \nu + v$, provided $(2M + 1)^4\Gamma \le v$. Every connected set of distinct entries then has local summed price at most $e^{\Gamma} - 1$ with exponent ν. For a component whose mark set is a union of complete groups, let $W_v = \prod_{i\in\pi^{-1}(v)} \widetilde{w}_i$. The product of its old mark weights is exactly the product of these W_v over its new group marks. Distinct constituent entries have disjoint marks when their squarefree marker product is nonzero; there is no multiplicity correction in this equality. Each constituent-label set must yield at most one output entry across the ordinary and flagged kinds; its support and prescribed flag set are fixed by its constituents. This uniqueness is proved in Section 55. Grouping the connected terms by their resulting support and group mark set, the triangle inequality preserves the local bound $e^{\Gamma} - 1$. Restrictions to the healthy or prescribed-flag events only reduce the coefficient majorants. Thus both the ordinary and flagged connected arrays have this same bound.

For (841), the old weights are δW_i, $\eta = \delta$, $\sigma = \nu + 1/2$, and the maximum group size is h. The new weight of an old mark is precisely $\delta^2 W_i e^{-(\nu+1/2)k}/h$. Their product over a group is the numerator in (841). The purchased smallness conditions are $(2M + 1)^4\Gamma \le 1/2$ and $e^{\Gamma} - 1 \le 1$. They provide the claimed unit bound for both connected arrays before the subsequent factor $1/8$ is reserved for their local presentation. This proves the quantitative support and coefficient estimates used by regrouping. The exact polynomial decomposition, its gauge/locality properties and the local projection estimate are separate steps; these counting bounds do not by themselves establish those identities.

55 The exact regrouping identity

The estimates of Section 54 concern an exact reorganization of the marked density. We now prove the identity. In particular the coefficient of the empty mark set, which determines the vacuum activity, is preserved.

55.1 Squarefree marks and complete groups

For a finite mark set I, use the commutative algebra with basis e_A, $A \subseteq I$, and multiplication

$$e_A e_B = \begin{cases} e_{A \cup B}, & A \cap B = \varnothing, \\ 0, & \text{otherwise.} \end{cases}$$

Its unit is $e_\varnothing$. The ideal spanned by the nonempty A is nilpotent, so its exponential is a finite sum. In particular, for finitely many nonempty A_r,

$$\exp\left(\sum_r z_r e_{A_r}\right) = \prod_r (1 + z_r e_{A_r}),$$

since each individual summand squares to zero and all commute. Repeated use of a mark thus vanishes automatically.

Let $\pi : I \to V$ be the partition from the previous section, with nonempty groups $I_v = \pi^{-1}(v)$. Define a linear map $\mathcal{R}$ to the analogous algebra on V by

$$\mathcal{R} e_A = \begin{cases} e_B, & A = \bigcup_{v \in B} I_v \text{ for some } B \subseteq V, \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently, the coefficient at B after applying $\mathcal{R}$ is the old coefficient at $\pi^{-1}(B)$. This map is not asserted to preserve arbitrary products. It does preserve a product of monomials whose mark sets are individually unions of complete groups. If these sets are disjoint, the assertion follows from their unions; if they overlap, both products vanish. We will apply multiplicativity only after the connected-component decomposition has made every component complete in this sense.

55.2 Ordinary and flagged entries

Fix a lattice configuration and let F be its set of flags. Take finite families of ordinary and flagged entries. An ordinary entry r has a nonempty old mark set A_r, a support X_r and a coefficient z_r. A flagged entry r has, in addition, a nonempty prescribed flag set $Z_r \subseteq X_r$, and its coefficient vanishes unless $X_r \cap F = Z_r$. Its mark set may be empty. The flagged part of the density is the sum over selections of flagged entries with pairwise disjoint supports and prescribed flag sets whose union is F. Multiply this sum by the exponential of the ordinary entries. The finite exponential formula shows that the product is a sum over sets S of distinct ordinary and flagged entries, with weight

$$\prod_{r \in S} z_r e_{A_r},$$

subject to the flagged-support and flag-inventory conditions. Terms with overlapping old mark sets vanish.

Enlarge each support as in (204), writing $\widetilde{X}_r$ for the result. Split each ordinary entry into two entries with the same marks, multiplied respectively by $\mathbf{1}_{\widetilde{X}_r \cap F = \varnothing}$ and its complement. Call the second kind *dirty*; all flagged entries are dirty too. The sum of the two pieces is the original entry, so this split leaves the density unchanged. At most one piece is nonzero at any fixed configuration. Every enlarged support contains the representative c_v of each group mentioned by its old marks. A flagged entry with no marks is not enlarged.

We describe a canonical partition of a selected set S. First join entries which mention a common group. Call a component of this first graph hot if it contains a dirty entry. Next join hot components

whenever their enlarged supports intersect. The resulting components are the components used below. Equivalently each entry always carries its group names as graph coordinates, and carries its enlarged support as additional graph coordinates precisely when its first component is hot. Connected components are then taken in this single intersection graph. Distinct components have disjoint group-name sets, and distinct hot components have disjoint enlarged supports. Every component is connected by ordinary support overlaps: a common group supplies its representative site as a common point.

This partition is unchanged by restricting to one of its components. Indeed a path using group names never leaves that component, so its hot status can be determined within it. Conversely, a family of internally connected components with disjoint group-name sets and pairwise disjoint hot supports is exactly the component partition of its union. No group-name path or hot-support edge can cross between members of the family. These two observations prove the uniqueness needed for the reorganization.

55.3 Reconstructing the selected density

Consider a nonzero old term which survives $\mathcal{R}$. Its total mark set is a union of complete groups. Every one of its components also has this property. For if one mark in a group occurs in that component, every other selected mark of the same group is joined to it through their common group name. Since the total selection contains the whole group, the component contains the whole group.

A cold component contains only healthy ordinary entries, so its coefficient is supported on configurations with no flags in its combined enlarged support. It has a nonempty group mark set. A hot component necessarily contains a flagged entry. To prove this last assertion, choose a dirty ordinary entry if no flagged entry has yet been found. Its enlarged support meets F. The global flag-inventory condition supplies a selected flagged entry whose prescribed flag set contains that point. Both entries are hot and their supports intersect, so they are in the same component, a contradiction.

For each possible cold component, make a new ordinary entry: its support is the union of its enlarged supports, its coefficient is the product of its constituent coefficients, and its new marks are its group names. For each possible component containing a flagged entry, make a new flagged entry with that same combined support, flag set Z equal to the union of its old prescribed flag sets, and coefficient

$$\mathbf{1}_{\widetilde{X}\cap F=Z}\prod_{r\in S} z_r.$$

Only internally connected components with disjoint old mark sets, complete group marks and pairwise disjoint old flagged supports are included. The new flag set is nonempty.

For the components of a nonzero old selection, the displayed indicator is one. Any flag in the enlarged support of a hot component belongs to a selected flagged entry somewhere in the old term; intersection of the two hot supports puts that entry in this same component. The converse inclusion follows from the old flag conditions. Distinct hot supports are disjoint, and their prescribed flag sets still partition F. Distinct components have disjoint new mark sets. The old term therefore gives an allowed new term with exactly the same coefficient after $\mathcal{R}$.

Conversely, take a nonzero term in the new ordinary-exponential and flagged-selection expansion. Its component entries have disjoint new mark sets and pairwise disjoint hot supports. Their inventories of old labels are disjoint. A shared old label carrying a mark would force a shared group name; a shared label without marks must be a flagged entry and would force intersecting hot supports. Thus no old label is counted twice. Within each new component the old flagged supports are disjoint; between components they lie inside disjoint hot supports. Their old flag sets have union

F. The old mark sets are disjoint, since they are disjoint within components and complete disjoint groups between components. Hence their union gives an allowed old selection. The uniqueness of the component partition proves that this construction inverts the preceding one.

The products of coefficients agree in both directions. On complete group components, $\mathcal{R}$ preserves their monomial product by the first subsection. Summing the resulting weight-preserving bijection proves the exact regrouping identity. All sums here are finite: the torus, mark sets, supports and their admissible patterns are finite. No convergence or rearrangement assumption enters this algebraic step. The connected-array estimates of Section 54 apply to these very component entries. The ordinary and flagged kinds cannot represent the same constituent set, since one is cold and the other contains a flagged entry; thus summing their bounds does not hide an additional multiplicity.

55.4 Vacuum, locality, and covariance

A new entry with no group marks must be a single old zero-mark flagged entry. In fact no ordinary entry has empty marks. All entries of such a component are therefore flagged entries with no marks, whose supports were not enlarged. Their old supports are pairwise disjoint, so no support edge connects two different entries; and there are no group-name edges. Connectedness forces a singleton. Its new support, flag set and coefficient coincide with the old ones; the indicator is already implied by the old flag condition. Thus the zero-mark coefficient of every flagged support is unchanged. The ordinary array has no zero-mark coefficient. The same vacuum activity is retained by the regrouping.

The construction also preserves the required local dependence. Each coefficient is a finite product of original coefficients, with indicators determined by the flags in the combined support. It therefore depends only on the original field and flag data in that support. Products, finite sums and these measurable indicators preserve measurability. If the original coefficients and flags are gauge invariant, so are the new coefficients. Healthy ordinary components vanish when their support meets the flags; flagged components vanish unless the intersection with the flags is their specified flag set. These are the support conditions used in the local representations. They do not assert smoothness across a flag boundary.

Finally translations, signed coordinate changes and the accompanying conjugation preserve the construction whenever they preserve the original coefficient data. They relabel entries bijectively, preserve group-name incidence, health and flag inventory, and send support intersections to support intersections. Hence they send the canonical components to the corresponding components after transformation. Reindexing the finite sums proves covariance of the new arrays, including every coefficient whose marks lie in the copied group family. This keeps the replication identities through repeated regrouping. To obtain the local normalized-field representation used in (841), one must in addition extract its singleton coefficients and apply the purchased local analytic projection bounds; those estimates are not consequences of this finite algebraic identity.

56 Recovering the local fields after regrouping

The preceding regrouping gives a polynomial density with controlled coefficients. To continue the scale induction, its linear coefficients must again be presented as local fields, with the remaining terms assigned to interactions involving at least two marks. We give this passage explicitly. It uses no Taylor approximation: the separation by the number of marks is exact.

Fix a finite periodic lattice, a finite mark set V, and positive weights w_v. Write e_A for the squarefree monomial belonging to an allowed mark set A, as in the preceding section. Ordinary

polynomial activities F_X are indexed by supports X; flagged activities $K_{X,Z}$ are indexed by pairs with $\varnothing \neq Z \subseteq X$. Their coefficients are measurable, gauge invariant, and depend only on the field and flag data in X. A nonzero coefficient with mark set A requires $\{o_v : v \in A\} \subseteq X$. Ordinary activities have zero constant coefficient and vanish unless X is healthy. Flagged activities vanish unless the flags in X are exactly Z.

Put $\nu = \nu_R$, $M = M_R$ for the support parameters of the current representation. Suppose the ordinary and flagged arrays have the weighted coefficient bounds from Section 54, with local hit bounds a and b, respectively. In concrete terms there are nonnegative coefficient majorants $t_{X,A}$ and $s_{(X,Z),A}$, valid for every field configuration, such that for every site x

$$\sum_{X \ni x} e^{\nu c_M(X)} \sum_A t_{X,A} \prod_{v \in A} w_v \leq a, \tag{208}$$
$$\sum_{(X,Z): x \in X} e^{\nu c_M(X)} \sum_A s_{(X,Z),A} \prod_{v \in A} w_v \leq b.$$

The underlying lattice is finite, so the support indices in these expressions are finite. The activity dictionary used to represent them may nevertheless be countable.

We use the following precise property of that dictionary. Every nonempty support has an unused ordinary slot with that support and zero background coefficient. Every admissible flagged pair has a flagged slot with that pair, eligible whenever the specified flag condition holds. A real measurable gauge-invariant function h in such a slot is an allowed elementary insertion when it has the required local dependence and vanishes off the slot's eligibility set. If $|h| \leq t$, its cost is

$$e^{\nu c_M(X)} t.$$

This is a property of the available representation, not a consequence of the support count. Its production in the evolving activity dictionary remains an input to this local argument.

56.1 The price of a singleton

Fix a mark v. In the first inequality of (208), take $x = o_v$ and retain only the coefficient of $e_{\{v\}}$. Coefficients on supports not containing o_v vanish. Dividing by $w_v > 0$ gives majorants $t_X \geq 0$ with

$$|[e_{\{v\}}] F_X(U)| \leq t_X, \qquad \sum_X e^{\nu c_M(X)} t_X \leq a/w_v.$$

The same argument for the flagged array gives majorants $s_{X,Z}$ with weighted sum at most b/w_v. There is no factor $|V|$ in these bounds: only one coordinate of the price at o_v was retained.

For an ordinary singleton function f supported in X, use the unused ordinary slot and write $f = \operatorname{Re} f + \mathrm{i} \operatorname{Im} f$. Both real functions have bound t_X, retain local dependence, measurability and gauge invariance, and vanish outside the healthy set. They are therefore two elementary insertions, with coefficients 1 and i, of total cost at most $2e^{\nu c_M(X)} t_X$. They contribute f to the ordinary field and zero to its flagged profile. The same construction in the flagged slot contributes zero to the ordinary field and f at the specified flagged pair, at cost at most $2e^{\nu c_M(X)} s_{X,Z}$. Zero functions require no slot.

Consequently, for arbitrary constants $u_v \in \mathbb{C}$, the pair

$$g_v(U) = u_v + \sum_X [e_{\{v\}}] F_X(U), \qquad q_v(X, Z; U) = [e_{\{v\}}] K_{X,Z}(U) \tag{209}$$

has a finite elementary representation anchored at o_v, with constant part u_v and cost at most

$$2a/w_v + 2b/w_v. \tag{210}$$

Indeed concatenate the finite lists just constructed. Adding lists adds their costs, and multiplication by a scalar multiplies cost by its modulus. The constant u_v is recorded separately and costs nothing in this representation; its later affordability is a separate condition.

Each flagged elementary insertion is carried by its chosen index in the countable activity dictionary. Adding the finitely many such indexed functions gives a family $k_{v,j}(U)$ with a summable uniform majorant. Summing it over all indices with flagged pair (X, Z) gives exactly $q_v(X, Z; U)$. Thus a finite local representation also provides the dominated indexed family required by the transfer formula. The estimates concern the local representation and do not require a canonically chosen index for each singleton.

56.2 Exact separation of the higher terms

Let $k_j^0(U)$ be the real background flagged activities and let $E(U)$ be the ordinary background exponent. For a fixed configuration U, denote its flag set by $\mathcal{Z}(U)$. For an array L indexed by flagged pairs set

$$\mathcal{G}_U(L) = \sum_{\substack{\mathcal{P} \text{ a set of flagged pairs} \\ \text{supports pairwise disjoint, } \bigcup_{(X,Z)\in\mathcal{P}} Z = \mathcal{Z}(U)}} \prod_{(X,Z)\in\mathcal{P}} L_{X,Z}.$$

The current density is

$$D(U) = e^{E(U)} \exp\left(\sum_v u_v e_{\{v\}} + \sum_X F_X(U)\right) \mathcal{G}_U(K(U)). \tag{211}$$

Assume the constant coefficient of $K_{X,Z}$ is the sum of k_j^0 over dictionary indices with pair (X, Z); assume also the background activities are absolutely summable. These are the vacuum consistency and validity conditions retained in the preceding regrouping.

Let $F_X^{\geq 2}$ and $K_{X,Z}^{\geq 2}$ denote the arrays obtained by discarding coefficients with fewer than two marks. Then, coefficient by coefficient,

$$\sum_v u_v e_{\{v\}} + \sum_X F_X = \sum_v g_v e_{\{v\}} + \sum_X F_X^{\geq 2},$$
$$K_{X,Z} = \sum_{j:\,(X_j,Z_j)=(X,Z)} k_j^0 + \sum_v q_v(X, Z) e_{\{v\}} + K_{X,Z}^{\geq 2}.$$

Substitution in (211) proves the desired local representation of the same density, with fields g_v, flagged profiles q_v, and interactions of order at least two. Discarding coefficients cannot increase a sum of nonnegative coefficient majorants. Hence the two higher arrays retain hit bounds a and b. Their support, eligibility and invariance properties are retained as well, since each remaining coefficient is unchanged.

For completeness, this representation has the expected linear density. Put $r(U) = e^{E(U)}\mathcal{G}_U(k^0)$, the unmarked density. Expanding a product to first order gives

$$[e_{\{v\}}]D(U) = r(U)g_v(U) + e^{E(U)} \sum_{S \text{ admissible}} \sum_{j\in S} k_{v,j}(U) \prod_{i\in S\setminus\{j\}} k_i^0(U). \tag{212}$$

Here admissibility means that the indexed supports are disjoint and their flags cover $\mathcal{Z}(U)$. The exponential contributes its singleton coefficient g_v, while the flagged product contributes exactly one differentiated factor. Replacing sums over flagged pairs by sums over dictionary indices is legitimate: if $B_0 = \sum_j |k_j^0(U)|$ and $B_v = \sum_j |k_{v,j}(U)|$, the absolute sum of all differentiated products, even without admissibility, is bounded by $B_v \prod_j (1 + |k_j^0(U)|) \le B_v e^{B_0}$. The corresponding vacuum sum is bounded by e^{B_0}. Finite expansions followed by absolute convergence establish the identity.

Equation (212) also explains why the ordinary singleton is determined by its density and flagged profile whenever $r(U) \neq 0$: subtract the flagged term and divide by $r(U)$. This is the uniqueness used when a transfer step first identifies the singleton density rather than the ordinary field itself.

56.3 Returning to the scale induction

Apply this construction to the regrouped arrays of the preceding section. If both hit bounds are at most Z, each grouped field has cost at most $4Z/w_v$, and the higher interactions have hit bound Z. The regrouping creates no separately recorded linear constant, so here $u_v = 0$. In particular, when $Z = 1$, reducing the mark weight from w_v to $w'_v = w_v/8$ leaves

$$\text{singleton cost} \le \frac{4}{w_v} = \frac{1}{2w'_v}.$$

The coefficient prices of the higher interactions only decrease under this reduction. This proves the representation and the singleton budget needed after a grouping operation, conditional on the stated dictionary and background-validity properties.

Assume the input arrays and constants have the covariance inherited from the preceding regrouping. Translations and coordinate symmetries then preserve the ordinary singletons by reindexing the finite support sum in (209); conjugating symmetries also conjugate its scalar coefficients. The flagged profiles transform in the same way. No equivariance of the auxiliary choice of dictionary indices is needed, because only their summed profiles occur in the representation and in (212). This argument applies to any finite number of slots in each copied cell. The remaining analytic step is to replace these local fields by degree-bounded coefficient fields while controlling the error under transfer to the next scale. The singleton extraction alone does not provide that projection estimate.

57 Transporting a local representation through one scale

We now explain the common transport argument used by the centered estimate and the local OPE construction. Its analytic input is a bound for integrating finitely many elementary insertions against one fixed background transition. The argument below turns that bound into a transition for arbitrary local fields represented by absolutely summable elementary insertions. It also restores the prescribed singleton profiles after integration. The same background transition is used throughout.

57.1 From an anchored estimate to a site estimate

The first estimate accounts for summing over all named sources. Write

$$S_M = e\{2(16M + 9)\}^4.$$

Suppose at most N marks are anchored at any one site. Let X_i be supports, A_i their nonempty sets of marks, and $f_i \ge 0$. Every i with $f_i > 0$ is assumed to have some $v \in A_i$ with $o_v \in X_i$. If

$$\sum_{i : v \in A_i} e^{(a+1)c_M(X_i)} f_i \le c \quad \text{for every } v,$$

where $c \geq 0$, then

$$\sum_{i:x\in X_i} e^{ac_M(X_i)} f_i \leq NS_M c \quad \text{for every } x. \tag{213}$$

Indeed, the defining connected-cover bound gives, for $x, y \in X$,

$$\operatorname{dist}(x,y) \leq (4M+2)c_M(X) + 2M.$$

Hence a term on the left of (213) is at most

$$\sum_{v\in A_i} e^{1-\operatorname{dist}(x,o_v)/(4M+2)} e^{(a+1)c_M(X_i)} f_i.$$

Retain the anchored v to verify this inequality; all other terms are nonnegative. Sum first over i, use the anchored bound, and then the occupancy bound. It remains to check

$$\sum_y e^{-\operatorname{dist}(x,y)/(4M+2)} \leq \{2(16M+9)\}^4.$$

Lift $y - x$ to its minimal integer representative. These lifts are distinct, and their ℓ^1 norm is at most four times their periodic maximum distance. With $b = 1/(4(4M+2))$, the sum is therefore bounded by

$$\sum_{z\in\mathbb{Z}^4} e^{-b|z|_1} = \left(1 + \frac{2}{e^b - 1}\right)^4 \leq \{2(1+b)/b\}^4 = \{2(16M+9)\}^4.$$

This proves the estimate, uniformly in the period. The argument also proves summability for countable activity families by bounding every finite subsum.

57.2 Preparing a finite family of insertions

Consider a fine representation R and a coarse representation E. Let $\mathcal{T}$ be their block-integration operator. For a fine elementary insertion M, write $\mathcal{Q}(M)$ for its quoted coarse flagged profile. We assume this map is real linear under absolutely summable combinations in a fixed slot, with a uniform bound by the elementary insertion cost. These are properties of the fixed transition; their analytic construction is not being asserted here.

For each mark v, suppose we have an anchored expansion, using only finitely many distinct slots but possibly countably many functions in each slot,

$$g_v - u_v = \sum_j z_{vj}\ \operatorname{ord}(M_{vj}), \qquad q_v = \sum_j z_{vj}\ \operatorname{flag}(M_{vj}),$$
$$q'_v = \sum_j z_{vj}\ \mathcal{Q}(M_{vj}) + \mathcal{H}_E(\psi_v).$$

Here $\sum_j |z_{vj}| \operatorname{cost}(M_{vj}) \leq y_v$, and $\psi_v(X,U)$ is a coarse response anchored at the parent of o_v. It has local dependence, is measurable and gauge invariant, and vanishes when X is healthy. Its cost hypothesis is

$$\sum_X e^{(p+1)c_{M_E}(X)} w_v \sup_U |\psi_v(X,U)| \leq W. \tag{214}$$

An available majorant can replace the displayed supremum. The operator $\mathcal{H}_E$ will be described below. The constant u_v is outside the priced sum.

Compress the countable sum separately in each slot. Its real and imaginary parts converge uniformly, since their absolute values are bounded by the summable insertion weights. The sums are real measurable local gauge-invariant functions in that same slot. Their bounds are the respective sums of absolute coefficient weights. Thus two real insertions, multiplied by 1 and i, replace the whole complex series. Both the old profiles and their quoted profiles are preserved, by the assumed continuity and linearity. The resulting finite list for mark v has weighted cost at most $2w_v y_v$. If $2w_v y_v \le d_0$ and the fine-site occupancy is at most N_f, (213) bounds its site price at exponent a by $N_f S_{M_R} d_0$, provided $a+1 \le \nu_R$.

Suppose the higher ordinary and flagged arrays in the fine local representation each have site price at most q at exponent a. The real/imaginary construction of Section 56, now applied to every coefficient of order at least two, represents them by finite elementary insertions at total site price at most $4q$. The unused-slot property is used here. Combining the two lists gives a finite list with price

$$T = N_f S_{M_R} d_0 + 4q. \tag{215}$$

Its ordinary sum is $\sum_v (g_v - u_v) e_{\{v\}} + H$, and its flagged sum is $\sum_v q_v e_{\{v\}} + K$, where H, K are the higher arrays. All marks of each insertion lie in its support.

The elementary integration input takes a family of linear slot insertions. To turn the exponential insertions in the last display into that form, adjoin, at each ordinary slot, the product of every compatible set of at least two members of the list. Compatibility means that their mark sets are disjoint and their union is an allowed monomial. Keep all original members. The added product uses the same slot and support, and its price is the product of the constituent prices. No product is added to a flagged slot.

This operation has price at most

$$T + e^T T. \tag{216}$$

To verify it, let $r_j \ge 0$ be the product of the marker toll, the modulus of the complex coefficient, and the elementary supremum-norm weight, excluding the support exponential $e^{a c_M(X)}$. For one ordinary slot put $s = \sum_j r_j$. Because every member has at least one mark in that support and $a \ge 0$, the site bound at any point of the support implies $s \le T$. The sum of the product prices of all nonempty subsets is at most

$$\prod_j (1 + r_j) - 1 \le e^s - 1 \le s e^s \le e^T s.$$

The added compatible subsets of size at least two form a subfamily. Multiplying by the support exponential and summing over distinct slots meeting a given site gives at most $e^T T$. Adding the original price proves (216).

The density identity is exact. For each ordinary slot, the sum of all the compatible products is its squarefree exponential minus one: expand $\prod_j (1 + m_j) = \exp(\sum_j m_j)$, where each nonempty monomial m_j has square zero. Terms with incompatible marks vanish in the algebra. Thus replacing the exponential at that slot by one plus the expanded list changes no coefficient. Flagged slots have not changed. Denote the resulting elementary density by $\mathcal{E}$. The fine density X satisfies

$$X = \exp\left(\sum_v u_v e_{\{v\}} \right) \mathcal{E}. \tag{217}$$

Every added insertion has at least two marks, so the sum of quoted singleton profiles is still $q'_v - \mathcal{H}_E(\psi_v)$.

57.3 Absorbing the coarse response

We state explicitly the nonlinear integration input used at this point. For every prepared finite list of price at most M_*, the operator $\mathcal{T}$ produces ordinary and flagged arrays $\widehat{F}, \widehat{K}$, with the support and eligibility properties of Section 56, and

$$\mathcal{T}\mathcal{E} = e^{E_E} \exp\left(\sum_X \widehat{F}_X\right) \mathcal{G}(\widehat{K}). \tag{218}$$

The ordinary array has no constant term and site price at most k at exponent $t + \eta$; the flagged array has site price at most u at exponent p. Its vacuum coefficient is the coarse background profile, and its singleton coefficient is precisely the sum of the quoted singleton insertions. This estimate is required uniformly in the finite list and period, subject to the specified parent-site occupancy. Its production is a separate analytic obligation.

Assume $T + e^T T \le M_*$. Require also that every prepared elementary density is integrable under the fine prior and in each transition slice. The lawful background representation and its transition-kernel property supply these integrability conditions in the application. Integration is componentwise on the finite-dimensional polynomial algebra, so its continuous linear maps commute with an integrable transition. In particular the constant polynomial exponential in (217) can be pulled out. We obtain the same representation as (218), with exponent $P = \sum_v u_v e_{\{v\}} + \sum_X \widehat{F}_X$.

Set $J(U) = \sum_{v,X} \psi_v(X, U) e_{\{v\}}$. We next move J from the ordinary exponent into the flagged array, without changing the density. In a term of

$$\mathcal{G}(\widehat{K}) \exp(J),$$

regard each selected flagged pair and each selected response insertion (v, X) as a vertex, and join vertices whose supports overlap. Every nonzero response insertion meets a flag, because it vanishes on healthy supports. That flag is covered by a selected flagged pair in the gas term. Hence every connected component contains a flagged vertex. Within a component the flagged supports are disjoint; give the component the union support and the union of their flag sets. Define $\overline{K}$ as the sum of these component products with that output pair, retaining its required flag indicator. Every flag in a response support is covered by an old flagged vertex meeting that support, hence by a vertex in the same component. Thus the component flag set is exactly the set of flags in its union support.

Connected-component decomposition is a bijection on nonzero terms. In the forward direction, different components have disjoint supports and their flag sets cover all flags. In the reverse direction, the constituent vertices of components with disjoint output supports cannot coincide or acquire a new overlap edge. Their union recovers the original selection. The products of their coefficients agree. Thus

$$\mathcal{G}(\overline{K}) = \mathcal{G}(\widehat{K}) \exp(J), \qquad e^P \mathcal{G}(\widehat{K}) = e^{P-J} \mathcal{G}(\overline{K}). \tag{219}$$

All these expansions are finite in support and mark indices.

With no response vertex a connected component is a single old flagged vertex, since old flagged supports are disjoint. This proves preservation of the vacuum. With exactly one response vertex (v, X), every flagged vertex in its component meets X: there are no edges between distinct flagged vertices, so any path to the response must be a direct edge. Let $\Pi_E(X; Y, Z; U)$ be the sum of products of the background flagged activities over these nonempty, pairwise-disjoint collections meeting X, with union support Y after adjoining X and union flag set Z, including the output flag indicator. Then

$$\mathcal{H}_E(\psi_v)(Y, Z; U) = \sum_X \psi_v(X, U) \Pi_E(X; Y, Z; U).$$

This defines the operator in the transition expansion above. Two or more response vertices have degree at least two and cannot contribute a singleton. Therefore

$$[e_{\{v\}}]\overline{K} = [e_{\{v\}}]\widehat{K} + \mathcal{H}_E(\psi_v) = q'_v.$$

Replacing the background profiles in Π_E by their countable indexed sums is justified by absolute convergence, bounded by the product of $1 + |k_j^0|$, as in (212).

The response insertions have site price at most $N_c S_{M_E} W$ at exponent p, by (214) and (213), where N_c is the parent-site occupancy. Put

$$A = u + N_c S_{M_E} W.$$

If

$$t + \eta \geq 0, \qquad (2M_E + 1)^4 A \leq p - (t + \eta), \tag{220}$$

the connected-support estimate (202) bounds the site price of $\overline{K}$ at exponent $t + \eta$ by $e^A - 1$. Its coefficients are products of old coefficient majorants, and the map from a constituent set to its output pair is unique; thus the counting lemma applies without a multiplicity factor. The ordinary higher coefficients retain price k, since J is linear. Local dependence, gauge invariance, measurability and the flag condition follow from finite products and support-local flag indicators.

Finally suppose the singleton coefficient of the actual integrated density has already been identified as the linear density of a prescribed pair (g'_v, k'_v), with k'_v dominated and profile q'_v. Assume the coarse unmarked density is nonzero. The uniqueness following (212) identifies the ordinary singleton of $P - J$ with g'_v. Removing all coefficients of degree zero and one now yields a local representation with these prescribed fields, with ordinary price k and flagged price $e^A - 1$ at exponent $t + \eta$.

57.4 Changing the mark weights

The final operation pays for changing weights. Suppose $w_v, w'_v > 0$ and for every allowed monomial B with $|B| \geq 2$ supported in X,

$$\prod_{v \in B} w'_v \leq a_* \prod_{v \in B} w_v \, e^{\eta c_{M_E}(X)}, \qquad a_* \geq 0. \tag{221}$$

Multiply this inequality by each coefficient majorant and sum. It changes an exponent-$t + \eta$ price C into an exponent-t price at most $a_* C$. Hence the transported representation has both higher-array prices bounded by g whenever

$$a_* k \leq g, \qquad a_*(e^A - 1) \leq g.$$

Here is the specialization used in the scale induction. Start with weights $I_v > 0$, higher-array price 1, and exponent at least $\nu_R - 4$. Choose $0 < d \leq 1$ and first reduce the weights to $w_v = dI_v$. A monomial with at least two marks acquires the factor $d^{|B|} \leq d$, so the higher-array price becomes $q = d$. Suppose the singleton transition costs obey $2I_v y_v \leq 1$ and $I_v r_v \leq 1$, with response cost r_v measured at the full coarse exponent plus two. After multiplying by w_v, the fine singleton budget is $d_0 = d$ and the response budget is $W = d$ at $p = \nu_E + 1$.

Assume $\nu_R, \nu_E \geq 100$, and write N for a bound on both occupancies. The two numerical conditions are

$$\begin{aligned} &T = NS_{M_R} d + 4d, && T + e^T T \leq M_*, && \\ &A = u + NS_{M_E} d, && (2M_E + 1)^4 A \leq \tfrac{1}{2}, && e^A - 1 \leq 1. \end{aligned} \tag{222}$$

Use the nonlinear input with fine exponent $\nu_R - 4$, ordinary output exponent $\nu_E + 1/2$ and price 1, and flagged output exponent $\nu_E + 1$ and price u. The fine anchored estimate requires only $\nu_R - 3 \le \nu_R$, and all exponents used for support counting are nonnegative. Equation (222) therefore verifies (220) with $t = \nu_E$ and $\eta = 1/2$. Any new weights satisfying (221) with $a_* = 1$ give a coarse local representation at exponent ν_E and price 1. This proves the one-scale propagation from the stated elementary transition estimate and singleton transition expansions. It does not replace the proof of that analytic estimate or the construction of those expansions.

58 Local jet extraction and exact masked splitting

A transferred local field is not yet a coefficient field. Its low-order jet has to be extracted, and subtracting that jet changes the support of its healthy-set indicator. The difference of indicators is retained as an explicit flagged response. We give the extraction and this exact splitting before applying it to the one-step integral.

Fix a finite lattice, an anchor o, a degree d, and a coefficient frame V_d for the gauge-invariant curvature-jet polynomials of weighted degree at most d. A curvature derivative of order j has weight $j + 2$. Write $\mu(v) = \sum_\alpha |v_\alpha|$, e_0 for the unit coordinate, $v^\circ = v - v_0 e_0$, and D_t for the diagonal map multiplying a coordinate with r curvature factors by t^r. The budget consists of a threshold $0 < t \le 1$, derivative order $q \ge d$, spatial power $p \ge d$, positive support exponent u, and integer covering radius $B \ge 1$.

A regular field on a support X is a measurable, gauge-invariant function depending only on its links, whose exponential-coordinate germ is C^q at every configuration healthy on X. Its ordered left derivatives exist through order q, including at the threshold boundary. Write $\|f\|_{o,X} \le c$ when $c \ge 0$ and all these derivatives of orders at most q have modulus at most c, for direction lists whose every entry satisfies

$$|A(e)| \le t(1 + \operatorname{dist}(e_-, o))^p.$$

In particular the zeroth-order estimate bounds f on the healthy set. No derivatives of a healthy-set indicator occur in this definition.

58.1 Reading the unmasked jet

Let $H_o(J, h)$ be the local test holonomy for a curvature jet J, and write $h \cdot J$ for weighted scaling, with component $h^{j+2} J_j$. Its defining local identities are

$$H_o(J, 0) = 1, \qquad H_o(J, h) = H_o(h \cdot J, 1), \qquad H_o(gJ, h) = g\, H_o(J, h)\, g^{-1}.$$

Here only the components of weights at most d are retained on the right. The finite-dimensional map $Z \mapsto H_o(Z, 1)$ is analytic near $Z = 0$ in the genuine group exponential chart. These properties of the chosen test holonomy, and the existence of the local polynomial section below, are the geometric inputs used in this section.

Here a support X is nonwrapping at o precisely when $10 \operatorname{dist}(x, o) < n$ for every $x \in X$, where n is the lattice side. For such a support, define $\Pi_{o,X} f$ by requiring that for every J, the functions

$$h \longmapsto f(H_o(J, h)), \qquad h \longmapsto (\Pi_{o,X} f)(h \cdot J)$$

have the same derivatives at zero through order d. For a wrapping support the convention is $\Pi_{o,X} f = f(1)\, 1$.

We first prove existence in the nonwrapping case. In the finite space of truncated jets put $F(Z) = f(H_o(Z,1))$. The identity configuration is healthy, and the local logarithm of $H_o(Z,1)$ is analytic and vanishes at zero. Thus F is C^d near zero. Its Taylor polynomial

$$T(Z) = \sum_{r=0}^{d} \frac{1}{r!} D^r F(0)[Z, \dots, Z]$$

is a polynomial in the real coordinates of the curvature jets. Delete its monomials of weighted degree greater than d, obtaining P. The ordinary Taylor polynomial has the same derivatives as F through order d. Composing with the polynomial curve $h \cdot J$ preserves that equality: in the chain rule for a derivative of order $k \le d$, only derivatives of F of orders at most k enter. A deleted monomial becomes a multiple of h^r with $r > d$, so contributes nothing through order d. Hence P has exactly the required test jets.

Gauge invariance of f and covariance of H give the same test jets for $P(gJ)$ and $P(J)$. Their difference has weighted degree at most d. For each fixed J, all its coefficients as a polynomial in h vanish, since its derivatives at zero vanish. Taking $h = 1$ proves $P(gJ) = P(J)$. Thus P belongs to the gauge-invariant label space. The same argument applied to two candidates proves uniqueness. Subtraction and scalar multiplication of the test identities now show that $\Pi_{o,X}$ is linear on regular fields with a fixed support. The wrapping convention has these properties as well.

For later use set

$$v(f) = D_t \operatorname{coord}(\Pi_{o,X} f).$$

The uniform estimate for this map follows from a precise local bound on test holonomies. Namely, for every fixed J, d and $r \le d$, suppose the logarithmic test curve $Z(h) = \log H_o(tJ, h)$ satisfies

$$|Z^{(r)}(0)(e)| \le a_r(J,d)\, t(1 + \operatorname{dist}(e_-, o))^{\max(r-1,0)}. \tag{223}$$

The constants are independent of the period, anchor, support and threshold. We prove this estimate for the actual radial test below; first we explain precisely how it is used.

Here is why it gives a constant $R_d \ge 1$ such that

$$\mu(v(f)) \le R_d c. \tag{224}$$

First bound a mixed exponential-coordinate derivative of order k on directions $w_1, \dots, w_k$ bounded by fixed multiples $a_1, \dots, a_k$ of the allowed direction weight. Restrict the symmetric kth Fréchet derivative at zero to their span, using the map $Lx = \sum_i x_i w_i$ from $\mathbb{R}^k$. Its diagonal at x is bounded by $(1 + \sum_i |x_i| a_i)^k c$: normalize Lx by this positive factor and use the repeated same-direction left derivative along $\exp(sLx)$. Such a repeated derivative equals the derivative of $f(\exp(sLx))$; the one-parameter group identity proves this by induction. Only the Fréchet derivative in this finite-dimensional chart is being treated as symmetric.

Diagonal evaluations separate symmetric tensors on $\mathbb{R}^k$. A finite number of them therefore bound any fixed mixed evaluation. For completeness, if separating linear functionals ℓ_i on a finite-dimensional space obey $|\ell_i A| \le b_i c$, but no uniform norm bound follows, choose A_n, c_n with $\|A_n\| > (n+1)c_n$. A subsequence of $A_n/\|A_n\|$ converges on the unit sphere. Every ℓ_i vanishes on its limit, a contradiction. Apply this argument to the symmetric tensors just described. The resulting mixed bound depends on k and the a_i, not on the lattice.

The finite-order chain rule for $f(\exp Z(h))$, together with (223) and $p \ge d$, now bounds each fixed jet probe by $C_{J,k} c$. These probes, for $k \le d$, separate V_d: a weighted polynomial all of whose test derivatives vanish is zero, by the uniqueness argument above. Apply the same finite-dimensional argument to V_d to obtain (224). The weights are exactly D_t, since evaluating a polynomial at tJ multiplies a monomial with r curvature factors by t^r. For a wrapping support only $f(1)$ remains, and $|f(1)| \le c$.

58.2 Uniform bounds for the logarithmic test links

We establish (223) by the same rescaling used to construct the test. A truncated radial connection is a finite sum of homogeneous connection tensors J_j of degree $j+1$. On a positive link starting at a covering-space point x in direction e_i, its coefficient at test scale h and amplitude t is

$$\sum_{j=0}^{d-2} t h^{j+2} J_j(e_i)[x+se_i,\dots,x+se_i], \qquad 0 \le s \le 1.$$

An empty sum is allowed. The link is the endpoint of the right-multiplication equation $U' = UA$, $U(0) = 1$, with this Lie-algebra-valued coefficient. The local logarithm is taken only near $h = 0$.

We need an elementary parameter regularity fact for this equation. On continuous matrix paths with the supremum norm define $T_A V(s) = \int_0^s V(r)A(r)\,dr$. Its norm is at most $\|A\|_\infty$. If A depends linearly on a finite-dimensional parameter z, then T_A is a continuous linear function of z. For z near zero, $\|T_A\| < 1$, and the integral equation has the unique solution

$$U = (I - T_A)^{-1}1 = \sum_{k=0}^{\infty} T_A^k 1.$$

The convergent operator power series makes this an analytic function of z. The given link solution satisfies the same integral equation, so agrees with it. Endpoint evaluation and the analytic local group logarithm therefore give an analytic logarithmic endpoint map $E(z)$ near zero, with $E(0) = 0$. This argument applies to the finite-dimensional space of all polynomial-in-s connection profiles of the indicated degrees, not only to profiles coming from a particular link.

Put $R = 1 + \operatorname{dist}(e_-, o)$, $r = R^{-1}$ and $y = r\,\operatorname{lift}(e_- - o)$. The centered lift has each coordinate at most R in modulus, so $|y|_2 \le 2$, while $0 < r \le 1$. Consider the polynomial profile

$$P(y,r,a,h)(s) = \sum_{j=0}^{d-2} a h^{j+2} J_j(e_i)[y+rse_i,\dots,y+rse_i].$$

For all parameters (y, r, a) the profile is zero at $h = 0$. Consequently

$$Q_k(y,r,a) = \frac{d^k}{dh^k} E(P(y,r,a,h))\Big|_{h=0}$$

is smooth in (y, r, a): near each point it is obtained by composing a polynomial map with the analytic germ E, then differentiating a fixed finite number of times. Also $Q_k(y, r, 0) = 0$. On the compact set $|y|_2 \le 2$, $0 \le r, a \le 1$, its derivative with respect to a is bounded. Integrating that derivative from zero to a gives $|Q_k(y,r,a)| \le C_{J,d,k,i} a$. This compactness is in the fixed profile-parameter space and has no dimension dependence on the lattice.

Homogeneity of J_j shows that the actual link at scale hr and amplitude t has profile $P(y, r, tr, h)$: both coefficients are $t h^{j+2} r^{j+2} J_j(e_i)[x+se_i,\dots,x+se_i]$. Uniqueness of the link equation identifies their endpoints. Differentiating this identity at zero gives

$$r^k |Z^{(k)}(0)(e)| \le C_{J,d,k,i} t r.$$

For $k \ge 1$ this is precisely $|Z^{(k)}(0)(e)| \le C_{J,d,k,i} t R^{k-1}$. For $k = 0$ the logarithm is zero. Taking the maximum over the four directions proves (223). The argument uses only derivatives at zero; it asserts no common analytic radius for links arbitrarily far from the anchor.

58.3 The support correction

Choose a linear local section of the labels, supported on a fixed stencil Y containing o. Its placed field $P_t(v)$ represents $D_t^{-1}v$, has the prescribed jet, and satisfies

$$\|P_t(v)\|_{o,Y} \le w\mu(v), \qquad w \ge 0. \tag{225}$$

The stencil is assumed to fit without wrapping. The section sends e_0 to the constant one. These are hypotheses about the chosen section; they do not follow from the abstract existence of Π. Let χ_X be the indicator that X is healthy, and put

$$p_t(v) = v_0 + \chi_Y P_t(v^\circ), \qquad Z = X \cup Y.$$

Then $|p_t(v)| \le (1+w)\mu(v)$ everywhere, and $p_t(v) = P_t(v)$ when Y is healthy. Define

$$g = f - P_t(v(f)), \tag{226}$$

$$r = \chi_X f - p_t(v(f)) - \chi_Z g. \tag{227}$$

These definitions give the exact, pointwise identity

$$\chi_X f = p_t(v(f)) + \chi_Z g + r. \tag{228}$$

The field g is regular on Z. It has zero identity value and, when Z is nonwrapping, zero test jets through order d. Indeed Y is nonwrapping, so Z is nonwrapping exactly when X is; subtract the two equal jets in that case. In the wrapping case $v(f) = f(1)e_0$, and the same subtraction gives the zero identity value.

The response r is measurable, gauge invariant, and determined by the links and final flags in Z. It vanishes whenever Z is healthy, by (228) and $p_t(v) = P_t(v)$ there. Put $A_d = 1 + (1+w)R_d$. Then

$$\|g\|_{o,Z} \le A_d c, \qquad |r(U)| \le A_d c. \tag{229}$$

The first estimate is the triangle inequality using (224)–(225). For the second, $r = 0$ on the healthy set of Z. On its complement $\chi_Z = 0$, so $|r| \le |\chi_X f| + |p_t(v(f))| \le c + (1+w)R_d c$. This proves the global bound without differentiating any mask.

58.4 Putting the response into the existing gas

Let K_j be the original flagged activities, with supports P_j and nonempty flag inventories $I_j \subseteq P_j$. An admissible finite family has disjoint supports and its inventories cover exactly the final flags. Let $\mathcal{G}(U)$ be its gas sum. For each response r_i supported on Z_i and each nonempty pairwise disjoint family q of old activities meeting Z_i, assume a distinct available address with support and inventory

$$Z_i \cup \bigcup_{j \in q} P_j, \qquad \bigcup_{j \in q} I_j.$$

The address's additional eligibility predicate must hold whenever its support has exactly this flag inventory. No zero value of its old activity is required. Insert at this address the coefficient

$$r_i(U) \prod_{j \in q} K_j(U)$$

when eligible, and zero otherwise. Denote the resulting source array by $\mathcal{K}(r)$. This is a linear operation in r, with addresses fixed before their values are evaluated.

Its exact marked-gas sum is

$$\mathcal{G}^{[1]}(\mathcal{K}(r))(U) = \Big(\sum_i r_i(U)\Big)\mathcal{G}(U). \tag{230}$$

To prove this, take a nonzero term on the right, consisting of i and an admissible old family b. Because r_i vanishes on healthy Z_i, some final flag lies in Z_i. Exact inventory coverage implies that the subfamily $q = \{j \in b : P_j \cap Z_i \neq \varnothing\}$ is nonempty. Replace precisely this subfamily by the designated new address. Every remaining support is disjoint from the new support; the inventories still cover the same flags. This produces an admissible pointed family on the left, with identical value. Conversely, expand the distinguished new address into its encoded family q. The other supports are disjoint from its total support, so q is exactly the subfamily meeting Z_i. This recovers i, b uniquely. Thus there is a value-preserving bijection of nonzero terms. Absolute summability, established next, permits reindexing.

Let $C_B(X)$ be the covering number used in the local norm and set

$$m(X,a) = e^{uC_B(X)}a, \qquad m_+(X,a) = e^{(u+2)C_B(X)}a.$$

Suppose $|K_j| \le b_j$, $b_j \ge 0$, and $\sum_{j:x\in P_j} m(P_j, b_j) \le v$ at every site x. For one response with $|r_i| \le a_i$, the inserted array has total mass at most

$$m(Z_i, a_i)\big(e^{|Z_i|v} - 1\big). \tag{231}$$

Indeed the total mass of old activities meeting Z_i is at most $|Z_i|v$, by summing their site bounds over Z_i. The mass of the union is at most the product of the masses, since each old support meets Z_i and their covering numbers are subadditive along such unions. Summing products over nonempty finite subsets is bounded by $e^{\sum b} - 1$; dropping the disjointness restriction only increases this nonnegative sum. This proves (231). If the sum of the right sides of (231) over i is finite, summing these bounds proves absolute domination of the inserted array and hence of its pointed gas series. The reserved-mass hypothesis below will establish this finiteness. The old gas series is absolutely convergent under the same site bound on the finite lattice. This justifies (230).

58.5 The two-unit reserve in the support exponent

Let T bound $C_B(Y)$ and put $h_B = (2B+1)^4$. The elementary covering estimates used here are

$$C_B(X \cup Y) \le C_B(X) + T \quad (o \in X \cap Y), \qquad |X| \le h_B C_B(X).$$

Set $F_d = e^{(u+2)T}A_d$. Then

$$m_+(X \cup Y, A_d c) \le F_d m_+(X, c). \tag{232}$$

This follows immediately by inserting the first covering inequality in the exponential. If $h_B v \le 1$, then for $a \ge 0$,

$$m(X,a)(e^{|X|v} - 1) \le h_B v\, m_+(X,a). \tag{233}$$

For $D = C_B(X)$, put $q_0 = h_B v$. We have $|X|v \le Dq_0 \le D$. Using $e^x - 1 \le xe^x$ for $x \ge 0$ and $D \le e^D$ gives $e^{|X|v} - 1 \le q_0 D e^D \le q_0 e^{2D}$, as required.

Consider now a raw expansion with seed v_0, regular fields f_i on anchored supports X_i, and an existing forced source k. Suppose $\|f_i\|_{o,X_i} \le c_i$ and

$$\sum_i m_+(X_i, c_i) \le a, \qquad \mathrm{mass}(k) \le \kappa b, \qquad \kappa = \sqrt{v} > 0, \quad h_B\kappa \le 1.$$

All constants a, b are nonnegative. Form $v_i = v(f_i)$, g_i, r_i as above, and define the coefficient field by

$$v = v_0 + \sum_i v_i, \qquad \text{regular terms } g_i \text{ on } X_i \cup Y, \qquad \text{forced array } k + \mathcal{K}(r). \tag{234}$$

The coordinate sum converges absolutely by (224) and $\sum_i c_i \le a$. Its change is bounded by $R_d a$. The regular remainder mass is at most $F_d a$. Furthermore $h_B v = h_B \kappa^2 \le \kappa \le 1$, so (233) and (232) bound the added forced mass by $\kappa F_d a$. Thus the normalized total remainder bound is

$$2F_d a + b. \tag{235}$$

The original density is preserved exactly: sum (228), use linearity of p_t, and then (230) for the response terms. Multiplication by the common vacuum exponential preserves this identity. Every sum is absolutely dominated by the bounds just proved. In particular this construction uses the original vacuum activities and the original threshold predicates throughout.

Finally, collect the responses with the same support: $\psi(Z) = \sum_{i: X_i \cup Y = Z} r_i$. Their majorants obey

$$\sum_Z m_+ \left(Z, \sum_{i: X_i \cup Y = Z} A_d c_i \right) \le F_d a.$$

Consequently ψ is an anchored, gauge-invariant response vanishing on healthy supports, with the stated reserved mass. Its insertion profile is exactly the profile of $\mathcal{K}(r)$: grouping the absolutely convergent terms by their encoded support and inventory proves the equality. These are the response and profile identities needed by the marked transport of Section 57. The raw expansion and its bounds still have to be produced by the one-step integral; this section proves the projection, mask correction, and cost conversion once that expansion is given.

59 Constructing the local polynomial sections

We now construct the fixed section and its price used in Section 58. The construction uses a finite collection of based, transported plaquettes. The finite collection and all constants are chosen before the lattice period and threshold.

59.1 A finite sampler for radial jets

Let $\mathcal{J}_d$ be the finite-dimensional real space of radial connection jets $A = (A_0, \dots, A_{d-2})$. Here A_j is a homogeneous one-form of polynomial degree $j+1$ satisfying the radial identity $\sum_i x_i A_{j,i}(x) = 0$. It is this radial description of curvature jets that underlies the label space. If $d < 2$, the space is zero. For a lattice path from the origin to x and an oriented plaquette at x, let $s(U)$ be the Lie-algebra curvature of that plaquette, transported back to the origin. Its value is zero at the identity. Define the sampler $S_s(A)$ by evaluating s on the test holonomy of the truncated radial connection A at scale one.

The endpoint argument in Section 58 shows that S_s is analytic near zero. Differentiating the link integral equation at the zero connection gives the line integral of A. At the identity, the derivative of a product is the sum of its factor derivatives, and inversion changes their signs. The derivative of transported plaquette curvature is consequently

$$DS_s(0)A = \oint_{\partial \square_{x,ij}} \sum_{l=0}^{d-2} A_l.$$

Transport along the based path contributes nothing to this derivative: it conjugates a curvature whose value at zero is zero. The Lie projection of the linearized plaquette is its Lie-algebra value.

These linear samplers jointly separate $\mathcal{J}_d$. To see this, for a homogeneous A_l write

$$N_l(x,h) = \int_0^1 \int_0^1 \partial_i A_{l,j}(x + hse_j + hre_i)\, dr\, ds - \int_0^1 \int_0^1 \partial_j A_{l,i}(x + hse_i + hre_j)\, dr\, ds.$$

The fundamental theorem of calculus identifies the unit-square circulation with $N_l(x,1)$. Also $N_l(hx,h) = h^l N_l(x,1)$ and $N_l(x,0) = \partial_i A_{l,j}(x) - \partial_j A_{l,i}(x)$. Suppose all sampled circulations vanish. If k is the largest remaining degree, multiply the identity at an integer point z by h^k and use homogeneity to get

$$\sum_{l=0}^{k} h^{k-l} N_l(hz,h) = 0.$$

For any real x, choose integer z_h with $hz_h \to x$ as $h \downarrow 0$. Continuity gives $N_k(x,0) = 0$. Thus $dA_k = 0$. The radial identity and Euler's identity give, componentwise,

$$\sum_i x_i(\partial_i A_{k,j} - \partial_j A_{k,i}) = (k+2)A_{k,j}.$$

Indeed the first term is $(k+1)A_{k,j}$, and differentiation of $\sum_i x_i A_{k,i} = 0$ makes the second term $-A_{k,j}$. Hence $A_k = 0$. Descending induction proves $A = 0$.

A finite subcollection already separates. Start with the whole space as the common kernel. If that kernel is nonzero, one of the samplers is nonzero on a vector in it; adding that sampler strictly reduces the kernel dimension. After finitely many steps the kernel is zero. Fix the resulting slots $s_1, \dots, s_m$ and the analytic map $S = (S_{s_1}, \dots, S_{s_m})$ into the panel of m Lie-algebra values. Its derivative $D = DS(0)$ is injective.

59.2 Equivariant recovery and polynomial substitution

The panel has its invariant Euclidean inner product. Constant gauge transformations act orthogonally there, preserve $\operatorname{ran} D$, and intertwine D with their action on $\mathcal{J}_d$. Let π be orthogonal projection onto $\operatorname{ran} D$ and put $C = D^{-1}\pi$. The range and its orthogonal complement are invariant, so C is equivariant and $CD = I$. The analytic inverse function theorem applied to CS at zero gives an analytic local inverse g. Set $F = gC$. Then

$$F(0) = 0, \qquad F(S(A)) = A$$

near zero. For each fixed gauge transformation, equivariance of CS and uniqueness of its local inverse imply equivariance of F as germs. This assertion needs no common invariant neighborhood for all transformations simultaneously.

Let $T_d F$ be the Taylor polynomial of this one recovery map through ordinary degree d. It is equivariant: differentiate the germ identity $F(Rx) = R'F(x)$ at zero, and use linearity of R, R' in each Taylor coefficient. It has zero constant term. For each real homogeneous basis label p_α define a polynomial in panel coordinates by

$$P_\alpha(x) = p_\alpha(T_d F(x)).$$

It is gauge invariant and real valued on real panels, and $P_0 = 1$. If p_α has r_α curvature factors, every monomial of P_α has ordinary panel degree at least r_α: each substituted coordinate has zero

constant term, so each of the r_α factors contributes at least one panel coordinate. No deletion of higher-degree terms after substitution is necessary.

These polynomials read the correct jets. Along the radial curve $A(h) = h \cdot J$, put $W(h) = S(A(h))$. It is smooth and $W(0) = 0$. The Taylor polynomial $T_d F$ and F have identical derivatives through order d at zero, so their compositions with W do too. Since $F(W(h)) = A(h)$ near zero, composition with the polynomial p_α gives

$$P_\alpha\big(s_1(H(J,h)), \dots, s_m(H(J,h))\big) \quad \text{and} \quad p_\alpha(h \cdot J)$$

the same derivatives through order d. Linearly extending the P_α therefore gives a local section T_0 of every label in V_d at the level of test jets.

59.3 Lattice symmetry and a common stencil

Let $\mathcal{H}$ be the finite group of signed coordinate permutations. For its action ρ on labels, define

$$T(v)(U) = \frac{1}{|\mathcal{H}|} \sum_{s \in \mathcal{H}} T_0(\rho(s^{-1})v)(\mathrm{pull}_s\, U). \tag{236}$$

The test holonomy transforms by the inverse tensor action. This can be checked directly in the link equation. Positive coordinate changes transport its coefficient. For a reversed link, if $u' = uA$, then $u(1)^{-1}u(1-h)$ solves the reversed equation with coefficient $-A(1-h)$ and starts at the identity. Uniqueness identifies its endpoint with the inverse link. Constant conjugation likewise conjugates the solution of the equation. These identities prove the stated tensor and gauge transformation laws of the sampler.

It follows that each summand of (236) has label v. Their average does also. Reindexing the finite group proves the signed-permutation covariance $T(v)(\mathrm{pull}_s\, U) = T(\rho(s)v)(U)$. Its unit is one, and it commutes with ordinary conjugation because the basis labels and the initial panel polynomials are real. Each transported plaquette transforms into another transported plaquette, possibly with a minus sign from reversing its orientation. Take the finite union of the slots in all these terms and enclose their paths and plaquettes in a centered box Y.

The average remains polynomial in this common finite list of slots. Replacing a slot by a signed slot changes no monomial degree. Moreover $\rho(s)$ preserves the number of curvature factors in a homogeneous label. Thus the polynomial representing the column $T(e_\alpha)$ still has no terms of panel degree below r_α. Each slot is itself a polynomial in the positive-link matrix entries and their conjugates: use matrix multiplication for the path and plaquette, conjugate transpose for inverse links, and the real-linear Lie projection of $(M - M^*)/2$ for curvature. A real-linear matrix functional is a linear combination of real and imaginary entry parts, each a linear combination of entries and their conjugates. Therefore T is a genuine finite polynomial cylinder on Y.

59.4 Threshold prices at every period

Fix derivative order q and spatial power p. For any period, anchor o and allowed direction A, a link from the fixed stencil satisfies $|A(e)| \le C_Y t$, since its torus distance from o is at most the size of its chosen covering-space position. An ordered left derivative of a positive-link matrix for the list $(A_1, \dots, A_k)$ is exactly $\mathrm{col}(A_k(e)) \cdots \mathrm{col}(A_1(e))U_e$, where col is the Lie-algebra matrix representation. The inverse link is its conjugate transpose. Consequently all derivatives of order $k \le q$ of any fixed path matrix are bounded by Ct^k, uniformly in the period and t. The product rule has finitely many terms, and compactness of the matrix group bounds the undifferentiated factors. Smoothness holds globally in exponential coordinates.

The same derivative bound holds for each transported plaquette slot. At order zero a stronger estimate comes from health: on a configuration healthy on the projected stencil, the plaquette curvature has norm at most t, and parallel transport preserves that norm. At every positive order k, the previous bound is at most Ct because $0 < t \le 1$. Thus each slot and each of its tested derivatives through order q has norm at most Ct on that healthy set. This also holds when the stencil wraps: the path identities and the bound on torus distance still apply.

Apply the ordered product rule to a monomial with r slot factors. Each term assigns the derivatives to those factors in their original orders. Every factor, including those receiving no derivative, contributes a bound $C_i t$. The monomial is therefore bounded by Ct^r through order q. Finite summation and $t \le 1$ give

$$\|T(e_\alpha)\|_{o,Y} \le C_{d,q,p} t^{r_\alpha}. \tag{237}$$

This proves both regularity and the exact factor-count threshold power of the constructed section. Multiplying its column coefficients by t^{-r_α} and using the triangle inequality yields (225) with $w = C_{d,q,p}$.

All choices are independent of the period and threshold. They can also be priced before choosing a scale: for an integer D, choose one bound for each triple $d, q, p \le D$ and set

$$w_D = 1 + \sum_{d,q,p \le D} \max(0, C_{d,q,p}).$$

It bounds every requested price in this finite range. The stencil fits whenever the side exceeds $\max_{x \in Y,i} 20(|x_i| + 1)$. These are fixed local requirements for the later calendar to satisfy. The construction here supplies the section and its uniform price; it does not choose that calendar.

60 Polynomial moments and decay of irrelevant insertions

The subtraction in Section 58 removes the low-order holonomy tests of an insertion. We now explain why this improves its coarse-to-fine estimate. The argument has two parts: a qualitative implication from holonomy tests to polynomial moments, and a quantitative decomposition of the actual interpolating field. Both use the same centered lattice lift as the local subtraction.

60.1 From holonomy tests to polynomial moments

Let X be a nonwrapping support about o on a torus of side at least two. Let f be local on X, gauge invariant, and C^d in logarithmic coordinates near the identity. Suppose that every holonomy test through degree d vanishes:

$$\frac{d^j}{dt^j} f(H_o(J,t))\Big|_{t=0} = 0, \qquad 0 \le j \le d, \quad \text{every } J. \tag{238}$$

Tests with a larger available cutoff are allowed; only degrees through d enter the argument. For $a \ge 1$ let $\mathcal{M}_a$ be the space of link fields screened to the links with both endpoints in X, whose value on a link in direction i is a polynomial of degree at most $a - 1$ in $x = \mathrm{lift}(e_- - o)$. We claim that

$$D^m(f \circ \exp)(0)[v_1, \dots, v_m] = 0 \quad \text{if } v_i|_X \in \mathcal{M}_{a_i}, \quad a_i \ge 1, \quad \sum_i a_i \le d. \tag{239}$$

The restriction means screening the link field, not restricting its support before applying the coarse-to-fine map.

Here are the details of the change of coordinates behind this claim. Introduce a site gauge parameter q and the finite list J of radial connection coefficients of weights $2, 3, \ldots$. For each link set

$$\Psi(q, J)_e = \log\bigl(e^{q(e_-)} H_o(J, 1)_e e^{-q(e_+)}\bigr)$$

on the links tested by X, and set it to zero elsewhere. This is an analytic germ with $\Psi(0) = 0$. The endpoint map is analytic by the integral-equation argument in Section 58. Locality and gauge invariance give, as germs,

$$(f \circ \exp) \circ \Psi(q, J) = f(H_o(J, 1)). \tag{240}$$

The identity uses only tested links and a finite common chart.

Give a direction (q, J) filtration degree a when q is polynomial of degree at most a on X and the components of J of weight greater than a vanish. Denote this space by $\mathcal{K}_a$. The following two properties hold:

$$D\Psi(0)\mathcal{K}_a = \mathcal{M}_a, \qquad 1 \le a \le d, \tag{241}$$

$$D^r\Psi(0)[u_1, \ldots, u_r] \in \mathcal{M}_{\sum_i a_i}, \qquad u_i \in \mathcal{K}_{a_i},\ r \ge 1. \tag{242}$$

We prove them, since the nonlinear coordinate change matters.

For the first property the linearized link is

$$D\Psi(0)(q, J)_{(x,i)} = q(x) - q(x + e_i) + \int_0^1 A_J(x + se_i)(e_i)\, ds.$$

Nonwrapping and side at least two identify the centered lift at the endpoint with $x + e_i$. Every constant one-form is the gradient of a linear site polynomial, so it occurs with $J = 0$. For a homogeneous polynomial one-form Q of degree $k \ge 1$, put

$$r(x) = \frac{1}{k+1} \sum_i x_i Q_i(x), \qquad K_i = Q_i - \partial_i r.$$

Euler's identity gives $\sum_i x_i K_i(x) = 0$. Thus K is precisely an admissible radial connection coefficient of weight $k + 1$. Taking $q = -r$ and that coefficient gives the link average of Q. The difference between this average and $Q_i(x)$ has degree at most $k - 1$: expand $Q_i(x + se_i)$ as a polynomial in s, and observe that its constant term is $Q_i(x)$. Induction on the degree supplies the difference by lower-weight parameters. Splitting a general polynomial into homogeneous parts proves (241).

For (242), write the link germ in variables

$$q(x), \qquad q(x + e_i) - q(x), \qquad A_J(x + se_i).$$

For a direction of degree a, the first expression has degree at most a in x, and each of the other two has degree at most $a - 1$. When both of the latter expressions vanish, the link germ is identically zero, since $e^{q(x)} 1 e^{-q(x)} = 1$. Expand any multilinear derivative according to these three components in each argument. The term using only the first components vanishes. Every remaining term contains a component with a one-degree loss; its degree is at most $\sum_i a_i - 1$. Screening completes (242).

Equation (238) implies that all mixed derivatives of the right side of (240) with total filtration weight at most d vanish. To verify this precisely, take its Taylor polynomial through ordinary degree d in the finite coordinates J. Substitute the weighted scaling $J_a \mapsto t^a J_a$. The coefficient of every t^j, $j \le d$, is zero for all J, by (238). Each such coefficient is a polynomial, hence all its monomial

coefficients vanish. Polarization gives the asserted mixed derivatives. Gauge-coordinate derivatives vanish because the right side is independent of q.

Apply the chain rule to the left side of (240). Induct on the number m of arguments in (239), lifting each v_i to $u_i \in \mathcal{K}_{a_i}$ by (241). The partition into singletons contributes exactly the desired derivative of $f \circ \exp$. Every other partition has fewer than m blocks; by (242), its arguments have the block weights, whose sum is still $\sum_i a_i$. The induction hypothesis kills all these terms. The whole derivative of the composite is zero, so the singleton term is zero too. The case $m = 0$ is $f(1) = 0$. Finally, locality makes $f \circ \exp$ equal to its composition with the linear screening map. Differentiating that identity allows unscreened directions whose restrictions have the stated polynomial form. This proves (239).

60.2 A discrete Taylor decomposition on the cover

Write $|x|_1 = \sum_{i=1}^{4} |x_i|$ for $x \in \mathbb{Z}^4$ and $\Delta_i v(x) = v(x + e_i) - v(x)$. For a word w in the four axes let Δ_w be the corresponding iterated difference. Suppose $d, p \in \mathbb{N}$, $L \geq 1$, $p \geq d$, $M \geq 0$, $\theta > 0$, and a colour-valued cover field v satisfies

$$\|\Delta_w v(x)\| \leq M\theta L^{-1-|w|}(1 + |x|_1/L)^p, \qquad |w| \leq d. \tag{243}$$

There are polynomial pieces P_j, $0 \leq j < d$, and a remainder R_d such that

$$v(x) = \sum_{j<d} P_j(x) + R_d(x), \qquad \begin{cases} \deg P_j \leq j, \\ \|P_j(x)\| \leq C_{d,p} M\theta L^{-j-1}(1 + |x|_1)^p, \\ \|R_d(x)\| \leq C_{d,p} M\theta L^{-d-1}(1 + |x|_1)^p. \end{cases} \tag{244}$$

The constant precedes the choice of L and of the period.

For completeness use the total-degree Newton polynomials

$$P_j(x) = \sum_{|\alpha|=j} \binom{x}{\alpha} \Delta^\alpha v(0), \qquad \binom{x}{\alpha} = \prod_{i=1}^{4} \frac{x_i(x_i - 1) \cdots (x_i - \alpha_i + 1)}{\alpha_i!}.$$

These formulas are polynomial identities for real x, so allow negative lattice offsets. The identity $\binom{t+1}{a} - \binom{t}{a} = \binom{t}{a-1}$ implies

$$\Delta_i P_{j+1}(v) = P_j(\Delta_i v), \qquad P_j(v)(0) = 0 \ (j > 0).$$

They are also the polynomials obtained by the recursion

$$P_{j+1}(v)(x) = (j + 1)^{-1} \sum_i x_i P_j(\Delta_i v)(x - e_i).$$

To check this, expand the falling factorials and use $\sum_i \alpha_i = j + 1$. Their coefficients give

$$\|P_j(x)\| \leq c_j M\theta L^{-j-1}(1 + |x|_1)^j, \qquad c_j = [4(j + 1)]^j.$$

Indeed each falling-factorial factor is bounded by $(1 + |x|_1 + j)^j$, while summing the reciprocal factorials at fixed degree contributes $4^j/j! \leq 4^j$.

If $|x|_1 \leq L$, all differences of order d in the ball of radius L are at most $2^p M\theta L^{-d-1}$. The remainder $R_d = v - \sum_{j<d} P_j$ is zero at zero for $d > 0$, and $\Delta_i R_d(v) = R_{d-1}(\Delta_i v)$. Induction on d, followed by summation along a shortest signed coordinate path from zero to x, gives

$$\|R_d(x)\| \leq 2^p M\theta L^{-d-1}(1 + |x|_1)^d.$$

For a negative step the forward difference is evaluated at the new endpoint, which still lies in the same ball. Thus this summation does not assume positive offsets. Since $p \ge d$, it is bounded by the remainder in (244). If $|x|_1 > L$, then

$$(1 + |x|_1/L)^p \le 2^p L^{-p}(1 + |x|_1)^p$$

and, for $j < d$, $L^{d-j} \le (1 + |x|_1)^{d-j}$. Apply these inequalities to (243) with the empty word and to the piece bounds, then use $R_d = v - \sum_{j<d} P_j$. This gives (244) with $C_{d,p} = 2^p + \sum_{j<d} c_j$. When $d = 0$ only the remainder is present, and the original bound suffices.

Evaluate these pieces at the actual centered lift on the torus. The lift satisfies $|\operatorname{lift}(y - o)|_1 \le 4 \operatorname{dist}(y, o)$. Multiplying the constant by 4^p therefore gives a global decomposition into d polynomial pieces and a last remainder. The screened jth piece belongs to $\mathcal{M}_{j+1}$. Nothing is required of the polynomial behavior of the last remainder, including across the lift boundary.

60.3 Applying the decomposition to the actual kernel

The coarse-to-fine linear field is the installed A_0 kernel, with the physical normalization of Section 8. Its iterated spatial differences obey the strip estimates

$$\|\Delta_w A_0\|_\alpha \le A L^{-1-|w|}, \qquad |w| \le d.$$

These are the A_0 difference bounds in Section 11; no such assertion about all the other tower blocks is needed here. If a coarse input obeys $\|u(y)\| \le \theta(1 + \operatorname{dist}(y, o_c))^p$, where o_c is the parent of the chosen fine anchor o, use the installed row at $o + \operatorname{wrap}(x)$. Its summation addresses are the parent of this site plus $\operatorname{wrap}(z)$, where z is the cover displacement in that residue row. The parent map can cross a block boundary; its centered residue changes at the same time. The coordinate division with remainder gives

$$1 + \operatorname{dist}\big(\operatorname{parent}(o + \operatorname{wrap}(x)) + \operatorname{wrap}(z), o_c\big) \le 3(1 + |x|_1/L)(1 + |z|_1).$$

Indeed in each coordinate the parent displacement is the integer quotient of x plus a residue correction of absolute value at most one; periodic distance can only decrease this bound. Taking the maximum of the coordinate bounds and then adding the cover shift gives the displayed factor three. Taking absolute values in the convolution now costs $3^p(1 + |x|_1/L)^p$ times the strip row sum with the additional weight $(1 + |z|_1)^p$. The inequality $(1 + t)^p \le C_{p,\alpha} e^{\alpha t}$ absorbs that weight into the available strip estimate. Sum also over the fixed form types. Differences can be taken on the row before summation by absolute convergence; the installed odd-block carry identifies them with the cover forward differences. We obtain (243), with M depending only on A, α, p and the fixed component count. The preceding decomposition therefore applies to this actual field, uniformly in the lattice size and block factor.

60.4 Quantitative cancellation and displacement

Let $\|\cdot\|_P$ denote the direction norm

$$\|v\|_P = \sup_e \frac{\|v_e\|}{\theta(1 + \operatorname{dist}(o, e_-))^p}.$$

Say that a direction admits a graded decomposition at scale $0 \le \xi \le 1$ and cost $C \ge 1$ if

$$v = \sum_{a=1}^{d+1} v_a, \qquad \|v_a\|_P \le C\xi^a, \qquad v_a|_X \in \mathcal{M}_a \quad (a \le d).$$

The preceding construction gives this property with $\xi = L^{-1}$. Suppose the regular derivative bound of $F = f \circ \exp$ is

$$|D^m F(b)[v_1, \dots, v_m]| \le Q\tau \prod_i \|v_i\|_P, \qquad m \le n, \tag{245}$$

on the logarithmic region used below, with $n \ge d+1$. Here Q is the fixed conversion constant from the local regularity norm, and τ is its price. The chart and health conditions in that norm are retained in (245).

At zero, expand each argument into its $d+1$ pieces. All terms of total weight at most d vanish by (239). Every remaining term has a factor at most $C^m \xi^{d+1}$. There are $(d+1)^m$ terms. For $m \le n$ this proves

$$|D^m F(0)[v_1, \dots, v_m]| \le Q\tau((d+1)C)^n \xi^{d+1} =: S.$$

Now let the displacement b have the same decomposition, and suppose that the segment tb, $0 \le t \le 1$, remains in the healthy logarithmic region, with the tested links and direction widths bounded by one as required by the regular norm. At order n, (245) bounds the derivative on this segment by S, since each argument has norm at most $(d+1)C\xi$ and $n \ge d+1$. If the order-$m+1$ derivative is bounded there by $(n-m)S$, the fundamental theorem of calculus applied to $t \mapsto D^m F(tb)[v_1, \dots, v_m]$ bounds its change from zero by $(n-m)S$. Its value at zero is at most S. Descending induction gives the uniform bound $(n+1)S$ at every order through n. This proves the attenuation after displacement; it does not require differentiating the health indicator.

61 The elementary one-step projection estimate

We connect the local splitting to the elementary insertion estimate used in Section 170. The analytic input is now the localized expansion of a real inserted field under one fixed nonlinear block integral. We state the part of that expansion used here and prove the remaining conversion and its constants.

Let $\mathcal{T}$ be the fine-to-coarse density transfer with block ratio L, and let f be one real elementary insertion with anchored fine support X, majorant c, and mass $m = m_{\mathrm{fine}}(X, c) \ge 0$. Use the same fixed coarse background, threshold and activity alphabet as the unmarked transition. A localized expansion has real regular terms h_i on anchored coarse supports X_i and a real flagged source array k such that

$$\mathcal{T} f = \rho_{\mathrm{coarse}} \sum_i \chi_{X_i} h_i + e^{E_{\mathrm{coarse}}} \mathcal{G}^{[1]}(k). \tag{246}$$

All regular and marked series must be absolutely dominated, and the original insertion must be integrable under the fine prior and every transition slice. These conditions are part of the analytic expansion. They justify the displayed identity as an identity of actual integrals. In addition, require the marked identity: the raw forced profile of k is the transferred singleton profile of the insertion in its fixed ordinary or flagged input slot, with the same background and activity alphabet. This profile identity is a separate part of the expansion hypothesis.

Split the localized terms into the distinguished principal term, when present, and the remaining terms. The required bounds have the following form, in the coarse derivative norm and reserved support mass:

$$\sum_{i \ mnonprincipal} m_+(X_i, \|h_i\|) \le M, \tag{247}$$

$$m_+(X_*, \|h_*\|) \le V\, m/g, \tag{248}$$

$$\mathrm{mass}(k) \le Ds. \tag{249}$$

Here $g > 0$ is the normalization gain, $V \geq 0$ a fixed-order derivative price, $D \geq 0$ the insertion debt, and $s \geq 0$ the flagged-response price. For a flagged elementary insertion there is no principal term. For a regular elementary insertion there is just one, at the fixed principal location. The indices and supports are selected before forming complex linear combinations.

61.1 The principal-term reserve

The principal estimate (248) follows from three more primitive bounds used by the local integral expansion. Write $D_X = C_B(X_*)$, let $\omega \geq 0$ be its localized majorant, and suppose

$$\|h_*\| \leq D\omega\,\Phi(X_*), \qquad \omega e^{(a+e)D_X} \leq 1, \qquad D \leq m/g,$$

where $e \geq 0$ and $a \geq u + 3$. Its derivative price has the form $\Phi(X_*) \leq V e^{D_X/2}$. The latter is the polynomial-to-exponential bound on the support size: for a fixed integer $r \geq 0$ one can take $C_r = \sup_{z \geq 0} z^r e^{-z/2} < \infty$ and bound z^r by $C_r e^{z/2}$. The additional fixed factor in V is the spatial weight of the derivative directions at the given coupling. Consequently

$$e^{(u+2)D_X}\|h_*\| \leq DV\omega e^{(u+5/2)D_X} \leq DV\omega e^{(a+e)D_X} \leq Vm/g.$$

If $D_X \geq B$, the same calculation leaves a factor $e^{-B/2}$, because $a + e - (u + 5/2) \geq 1/2$. This is the stronger principal estimate used when its support is large. The local derivative estimate and the weighted bound on ω remain substantive inputs from the nonlinear expansion.

61.2 Complex fields and the raw expansion

To treat a complex insertion, take two real elementary insertions with the same support, type and majorant, and combine them as $f = f_1 + if_2$. Their masses are the same m; their principal support and index agree. For $z \in \mathbb{C}$, define

$$H_i = z(h_i^{(1)} + ih_i^{(2)}), \qquad K = z(k^{(1)} + ik^{(2)}).$$

The sum of the two real majorants times $|z|$ dominates each complex term. Thus the nonprincipal reserved mass is at most $2|z|M$, the principal reserved mass is at most $2|z|Vm/g$, and $\mathrm{mass}(K) \leq 2|z|Ds$. The raw seed is zero. Pad the common countable list by zero terms on the singleton anchor support if a natural-number indexing is needed; neither the represented density nor its cost changes.

Linearity of masks, marked-gas insertion and $\mathcal{T}$, with absolute convergence of the series, gives the exact raw density $z\mathcal{T}(f_1 + if_2)$. This is the reason the same supports and principal indices were retained. Applying the splitting of Section 58 to these raw terms preserves the identity. In the elementary application take $z = L^d$ and $f_2 = 0$ with the same majorant as f_1. The bound with the factor two remains valid and avoids making the chosen dictionary depend on whether one component happens to vanish.

61.3 The projection allowance

Let F_d be the support-join and jet-extraction factor from (232), let $C_{\mathrm{orb}} \geq 1$ be the finite coefficient-action price, and let $C_{\mathrm{turn}} \geq 1$ be the price for transporting the local remainder norm under the required lattice motions. Put

$$H_d = 1 + 2F_d(1 + C_{\mathrm{orb}} + C_{\mathrm{turn}}), \qquad \eta_0 = \mathrm{rate}(L)/H_d.$$

Assume the already chosen coupling satisfies

$$2L^dM \le \eta_0 m, \qquad 2\kappa^{-1}L^dDs \le \eta_0 m, \qquad (2B+1)^4\kappa \le 1, \tag{250}$$

where $\kappa = \sqrt{\text{coarse forced allowance}} > 0$. The first two inequalities are required uniformly over insertions. They are paid by the same coupling delay as the chosen nonlinear transition; choosing a different history here is not allowed.

Set

$$E = 2L^dV/g, \qquad e_0 = (\eta_0 + |E|)m.$$

Then $e_0 \ge \eta_0 m \ge 0$. The principal bill is at most e_0, and the nonprincipal bill is at most $\eta_0 m$. The total raw regular reserved mass is therefore at most $\eta_0 m + e_0 \le 2e_0$. Write $p_0 = F_d(\eta_0 m + e_0) \le 2F_d e_0$ and $b_0 = 2\kappa^{-1}L^dDs \le e_0$. The local splitting gives a coefficient change at most p_0, a regular remainder of mass at most p_0, a normalized forced remainder of mass at most $b_0 + p_0$, and a response of reserved mass at most p_0. To bound the coefficient change by p_0, use $R_d \le F_d$, which follows from $w \ge 0$ and the definition of F_d.

It follows directly that

$$C_{\mathrm{orb}}\mu(q) \le C_{\mathrm{orb}}p_0 \le H_d e_0, \qquad C_{\mathrm{turn}}p_0 + (b_0 + p_0) \le H_d e_0.$$

The response cost is also at most $H_d e_0$. Therefore any Q_e satisfying

$$Q_e \ge (\eta_0 + |2L^dV/g|)H_d \tag{251}$$

gives a coefficient field with coefficient and remainder bounds $Q_e m$ and exact density $L^d\mathcal{T}f$. This is the numerical projection bound in (821).

The marked transition is retained as well. The original real insertion occupies its fixed ordinary or flagged slot with cost m; the real/imaginary convention just used gives the available bound $2m$. Its raw output forced profile is the transferred singleton profile. Splitting adds exactly the hook of the response ψ constructed in Section 58, of cost at most $Q_e m$. Thus the transition identity has fine cost $2m$, coarse response cost $Q_e m$, and multiplier L^d, precisely as required in Section 170. This last assertion uses the marked profile hypothesis stated alongside (246); the scalar density identity alone would not suffice.

The analytic work remaining at this point is explicit: construct the common localized expansion with its singleton identity, prove (247) and (249), and fund (250) on the selected history. The finite projection, support correction, principal reserve conversion, and passage to the elementary allowance have now been proved from those inputs.

62 Connected clusters with repeated labels

Section 54 counted connected sets of distinct activity labels. A logarithmic expansion is different: its vertices are numbered occurrences, and the same activity label may occur several times. We now prove the bound for this expansion, including its factorials and the identity with the real logarithm of the gas. No positivity of the gas is assumed in obtaining the identity.

Let Λ be a finite periodic lattice. Each label i has a nonempty support X_i, and write $c(X) = C_B(X)$ and $h_B = (2B+1)^4$. We use $|X| \le h_B c(X)$ and covering subadditivity for intersecting supports. The incompatibility relation R is symmetric and reflexive, and

$$R(i,j) \quad \Longrightarrow \quad X_i \cap X_j \ne \varnothing.$$

The converse is not required. Let $w_i \geq 0$, $a \geq 0$, $\gamma > 0$, $\delta > 0$, and suppose

$$\sup_{x \in \Lambda} \sum_{i : x \in X_i} w_i e^{(a+\delta)c(X_i)} \leq \gamma, \qquad q_0 = \frac{4h_B \gamma}{\delta} \leq q < 1. \tag{252}$$

In particular $q > 0$.

For an ordered tuple $z = (i_0, \dots, i_m)$, let $T(z)$ be the graph on $\{0, \dots, m\}$ with an edge $\{r, s\}$ when $r \neq s$ and $R(i_r, i_s)$. Define its connected coefficient and majorant by

$$u(z) = \sum_{\substack{G \subseteq T(z) \\ G \text{ connected}}} (-1)^{|E(G)|}, \qquad W(z) = \frac{|u(z)|}{(m+1)!} \prod_{r=0}^{m} w_{i_r}.$$

For $m = 0$ the graph with no edges is connected and $u(z) = 1$. Repeated labels are allowed in every tuple sum below.

62.1 Cancellation before taking absolute values

For any graph T on $\{0, \dots, m\}$,

$$\left| \sum_{\substack{G \subseteq T \\ G \text{ connected}}} (-1)^{|E(G)|} \right| \leq \#\{\text{spanning trees of } T\}. \tag{253}$$

Here is the cancellation argument. Fix a total order of the vertices and root at zero. For each connected G, let $d(v)$ be its distance from zero, and choose the least neighbor $p(v)$ at distance $d(v) - 1$ as the parent of each $v \neq 0$. The resulting parent edges form a spanning tree P. They determine d by their path lengths.

For a fixed tree P, define $U(P) \subseteq T$ to contain the tree edges and all further edges satisfying these two rules: their levels differ by at most one, and, if they join v at level r to w at level $r - 1$, then $w \geq p(v)$ in the chosen order. Precisely the graphs with

$$P \subseteq G \subseteq U(P)$$

have this breadth-first parent tree. One implication follows from shortest-path distance and the least-parent rule. Conversely, the tree paths show that a graph in this interval reaches every vertex in at most $d(v)$ steps. The level restriction implies that no path from zero reaches it in fewer steps, so the distances are exactly d. The second rule makes $p(v)$ its least neighbor at the preceding level. This proves the other implication and partitions all connected graphs into these intervals.

The alternating sum over one interval is $(-1)^m (1-1)^{|U(P) \setminus P|}$, hence has absolute value at most one. Equivalently, if there is a free edge, toggling it pairs terms with opposite signs. Summing over P proves (253). Taking absolute values separately for all connected graphs would lose this cancellation and would not give the required estimate.

62.2 The weighted forest estimate

A parent map $p : \{1, \dots, m\} \to \{0, \dots, m\}$ is called rooted if iterating it always reaches the external root zero. Place a fixed support Y at zero and a support X_{i_v} at vertex $v \neq 0$. Only assignments in which each child support meets its parent support are retained. We claim

$$\frac{1}{m!} \sum_{p \text{ rooted}} \sum_{i_1, \dots, i_m} \mathbf{1}_{\{\text{all parent edges overlap}\}} \prod_{v=1}^{m} w_{i_v} e^{ac(X_{i_v})} \leq e^{\delta c(Y)} q^m. \tag{254}$$

The labels in this sum need not be distinct.

Put $t = h_B\gamma/\delta > 0$, $F(Y) = \max(\delta c(Y), 1)$, and introduce the nonnegative kernel

$$K(Y,i) = \frac{\mathbf{1}_{\{X_i \cap Y \neq \varnothing\}} w_i e^{(a+\delta)c(X_i)}}{tF(Y)}.$$

The site bound in (252) implies

$$\sum_i \mathbf{1}_{\{X_i \cap Y \neq \varnothing\}} w_i e^{(a+\delta)c(X_i)} \leq |Y|\gamma \leq h_B\gamma c(Y) \leq tF(Y).$$

Thus every row of K sums to at most one. For a fixed rooted parent map, the sum over all label assignments of the product of its edge kernels is at most one: sum the label at a leaf first, apply its row bound, and repeat until no vertex remains. For infinite alphabets one first restricts to a finite alphabet; the row bounds still hold, and nonnegative convergence removes the restriction.

Let d_v be the number of children of vertex v, including the external root. The exponential series gives

$$F(Z)^k \leq k! e^{\delta c(Z)} \quad (k \geq 0).$$

Indeed if $\delta c(Z) \geq 1$ this is one term in the exponential series; otherwise use $F(Z) = 1$ and $k! e^{\delta c(Z)} \geq 1$. Replacing each overlap weight with its kernel expression and collecting the factors at parents gives

$$\begin{aligned}\prod_{v=1}^{m} &\mathbf{1}_{\{X_{i_v} \text{ meets parent}\}} w_{i_v} e^{ac(X_{i_v})} \\ &\leq t^m e^{\delta c(Y)} \left(\prod_{v=0}^{m} d_v! \right) \prod_{v=1}^{m} K(X_{i_{p(v)}}, i_v),\end{aligned}$$

where $X_{i_0} = Y$. To check the reserve cancellation, before the inequality there is a factor $e^{-\delta c(X_{i_v})}$ at every nonroot vertex. Applying the bound on F^{d_v} supplies exactly one inverse factor there, leaving only $e^{\delta c(Y)}$ at the external root.

It remains to sum the child factorials. Summing over all parent maps, including those which are not rooted, can only increase the answer. For fixed child counts $d_0 + \cdots + d_m = m$ there are $m!/\prod_v d_v!$ maps. Therefore

$$\sum_{p:\{1,\ldots,m\}\to\{0,\ldots,m\}} \prod_{v=0}^{m} d_v! = m! \binom{2m}{m} \leq m! 4^m.$$

The binomial coefficient counts weak compositions of m into $m+1$ parts. The inequality follows by comparing it with the sum of all binomial coefficients of order $2m$. After division by $m!$, the kernel bound and this count prove (254), since $(4t)^m = q_0^m \leq q^m$. This also proves the case $m = 0$, whose left side is one.

A distinguished root may carry a different nonnegative weight v_i. If $\sum_i v_i e^{(a+\delta)c(X_i)} \leq b$, multiply (254) with $Y = X_{i_0}$ by $v_{i_0} e^{ac(X_{i_0})}$ and sum i_0. The result is the rooted bound bq^m, with denominator $m!$ for the other vertices.

62.3 A root anywhere in the tuple

Let Q be any property of an activity label and assume

$$\sum_{i:Q(i)} w_i e^{(a+\delta)c(X_i)} \le b.$$

Apply (253) to each occurrence graph and the rooted forest bound with $v_i = \mathbf{1}_{Q(i)} w_i$ at a specified vertex. This gives

$$\sum_{z} \frac{|u(z)|}{m!} \mathbf{1}_{Q(i_r)} \prod_{s=0}^{m} w_{i_s} e^{ac(X_{i_s})} \le b q^m$$

for every root position r. Relabeling occurrence positions preserves both the connected coefficient and the tuple sum, so the same bound holds for all r. If a tuple contains at least one label satisfying Q, its cluster majorant, with denominator $(m+1)!$, is bounded by $1/(m+1)$ times the sum of the displayed rooted integrands over all r. Summing yields

$$\sum_{z:\,\exists r\, Q(i_r)} \frac{|u(z)|}{(m+1)!} \prod_{r=0}^{m} w_{i_r} e^{ac(X_{i_r})} \le b q^m. \tag{255}$$

This is the normalization that pays for choosing a root; no additional factor $m+1$ remains.

If $u(z) \neq 0$, its occurrence graph contains a connected spanning subgraph, hence a spanning tree. Incompatibility edges imply support overlap. Adding supports along this tree gives

$$c\left(\bigcup_{r=0}^{m} X_{i_r}\right) \le \sum_{r=0}^{m} c(X_{i_r}).$$

As $a \ge 0$, the exponential weight of the actual union is therefore bounded by the product weight in (255). Take $Q(i)$ to mean $x \in X_i$ and $b = \gamma$. The union contains x exactly when some label has this property. Finally sum over $m \ge z$ to obtain

$$\sum_{m \ge z} \sum_{i_0,\dots,i_m} \mathbf{1}_{\{x \in \bigcup_r X_{i_r}\}} W(i_0,\dots,i_m) e^{ac(\bigcup_r X_{i_r})} \le \frac{\gamma q^z}{1-q}. \tag{256}$$

Thus the connected estimate (498) follows from a vertex site bound and the explicit smallness inequality in (252): use the exponent $a+e$ there in place of a, the cluster vertex price t in place of γ, and the tail parameter γ of that section in place of q. The reserve δ pays for tree branching and is distinct from the reserve e used later for spatial derivative weights.

62.4 The exact logarithm identity

Now let $v_i \in \mathbb{R}$ be possibly signed activities and apply the preceding estimates to $w_i = |v_i|$. Define the gas and connected sum

$$Z = \sum_{A \text{ finite,compatible}} \prod_{i \in A} v_i, \qquad \mathcal{L} = \sum_{m \ge 0} \frac{1}{(m+1)!} \sum_{i_0,\dots,i_m} u(i_0,\dots,i_m) \prod_{r=0}^{m} v_{i_r}.$$

A finite set is compatible if it contains no incompatible distinct labels. The empty set contributes one. We will prove

$$Z = e^{\mathcal{L}} > 0, \qquad \log Z = \mathcal{L}. \tag{257}$$

First both series converge absolutely. Nonempty supports and (252) imply $\sum_i |v_i| \le |\Lambda|\gamma$, so the gas is dominated by $\prod_i (1 + |v_i|) \le e^{|\Lambda|\gamma}$. For the logarithm, use (255) with Q always true and $b = |\Lambda|\gamma$, and drop the support weights. Its absolute sum at order m is at most $|\Lambda|\gamma q^m$; hence the total is at most $|\Lambda|\gamma/(1-q)$.

Begin with a finite alphabet D. Let $P(t)$ be its gas polynomial, with each activity replaced by tv_i, and let

$$L(t) = \sum_{m \ge 0} \ell_m t^{m+1}, \qquad \ell_m = \frac{1}{(m+1)!} \sum_{i_0,\dots,i_m \in D} u(i_0,\dots,i_m) \prod_r v_{i_r}.$$

The bound $|\ell_m| \le |\Lambda|\gamma q^m$ shows that L is differentiable for $|t| < q^{-1}$ by termwise differentiation. Indeed on $|t| \le u < q^{-1}$ the derivative terms are bounded by the summable series $|\Lambda|\gamma(m+1)(qu)^m$. This interval contains $[0, 1]$.

We verify the coefficient identity $P' = L'P$, rather than assuming a formal logarithm can be evaluated. On a tuple of n occurrences, the sum of $(-1)^{|E(G)|}$ over all subgraphs of its incompatibility graph is $(1-1)^{|E(T)|}$. It is one if the graph has no edge and zero otherwise. Because R is reflexive, a tuple with repeated labels has an edge between equal occurrences and contributes zero. Thus summing this all-graph weight over tuples, multiplying by their activities, and dividing by $n!$ gives exactly the coefficient p_n of $P(t)$: each compatible n-element set has $n!$ enumerations.

For $n = m + 1$, distinguish occurrence zero in an arbitrary graph. Its connected component consists of zero and some k of the other m vertices. Choose those vertices in $\binom{m}{k}$ ways. A graph is then uniquely specified by a connected graph on this component and an arbitrary graph on its complement, with no cross edges. Edge signs and label products factor. Summing the choices and dividing by $m!$ gives the exact recurrence

$$(m+1)p_{m+1} = \sum_{k=0}^{m} (k+1)\ell_k p_{m-k}.$$

Absolute convergence of L' and the finiteness of P allow its coefficients to be multiplied and summed, proving $P' = L'P$ on the interval. Hence

$$\frac{d}{dt}\{P(t)e^{-L(t)}\} = 0.$$

At zero, $P(0) = 1$ and $L(0) = 0$. The mean value theorem implies $P(t)e^{-L(t)} = 1$ throughout the interval, and evaluation at $t = 1$ proves (257) for the finite alphabet.

Finally let finite alphabets increase to all labels. Every gas term and every occurrence tuple is eventually retained. The absolute majorants already proved are independent of this restriction. Dominated convergence therefore gives $P_D(1) \to Z$ and $L_D(1) \to \mathcal{L}$. Continuity of the exponential yields $Z = e^{\mathcal{L}}$, and its strict positivity gives the claimed real logarithm. For an arbitrary index set the same argument uses the directed family of all finite subsets; alternatively absolute summability leaves only countably many nonzero terms. This completes the proof, with no assumption of positivity or a choice of a complex logarithm branch.

Applied to the uniformly dominated tilted activities of Section 94, this result constructs their positive normalization and logarithmic cluster series once the vertex site bound (252) is established. The joint bounds proved there then justify differentiating it. Constructing the actual block-integral activities and proving their vertex bounds remain separate analytic tasks.

63 Prices for packets and marked connected components

The vertex bound used in Section 62 combines two populations: mandatory defect packets and optional local activities. We prove the counting and reserve estimates for this combination, then show how a single insertion is added to its actual alphabet. The analytic bound on the weight of one packet remains a separate input.

63.1 Counting physical defect packets

Fix a block ratio $L \geq 3$ and nonnegative integers ℓ, B. A seed is a fine plaquette, a coarse oriented bond, or a coarse site. These are distinct types even when they have the same location. Each seed has a coarse anchor and a finite set of points within max-distance one of its anchor. For a plaquette these points are the coarse parents of its touched fine sites; for a bond they are its two endpoints; for a site it is the site itself. For the plaquette, the parent-distance estimate is $L \operatorname{dist}(\operatorname{parent}(x), \operatorname{parent}(y)) \leq L + 2$. Since the distance is integral and $L \geq 3$, it is at most one. The halo of radius ℓB around these points is its support, contained in the ball of radius $\ell B + 1$ about the anchor. A packet is a nonempty finite set of seeds whose halo intersection graph is connected. Its support is the union of these halos, and its size n is the number of typed seeds.

The number of seeds with anchors in a set Y is at most $T_L|Y|$, where

$$T_L = L^4 \#\{\text{plaquette axes}\} + 4 + 1.$$

The fine site is determined by its coarse parent and one of L^4 residues, which proves the plaquette count; the two other counts are immediate. Consequently at most

$$H = (2(\ell B + 1) + 1)^4 T_L$$

seed halos can contain a given site. If two halos overlap, their anchors have distance at most $2(\ell B + 1)$, so each seed has at most

$$D = (4(\ell B + 1) + 1)^4 T_L \geq 1$$

possible neighboring seeds. These bounds remain valid when balls wrap around the periodic lattice, since wrapping only decreases the number of distinct sites.

There are at most HD^{2n} packets of size n containing a fixed site in their support. To prove this, choose a seed whose halo contains that site; there are at most H choices. A connected n-seed packet has a spanning tree. Traversing every tree edge out and back gives a walk of length $2(n-1)$, starting at the chosen seed, whose visited set is exactly the packet. There are at most $D^{2(n-1)}$ such walks. Different visited sets cannot come from the same walk, so this counts packets without an injectivity assumption on their placements. Enlarging the bound to HD^{2n} also covers $n = 1$ uniformly. There is no packet with $n = 0$.

A single halo has B-cover number at most $s_\ell = (2\ell + 3)^4$. For $B \geq 1$ it lies in the ball of radius $(\ell + 1)B$, tiled by B-balls with centers on the B-spaced grid; there are at most $(2\ell + 3)^4$ centers, forming a connected grid. For $B = 0$ use at most $3^4 \leq s_\ell$ singleton centers. Covering subadditivity along a spanning tree therefore bounds a packet's cover number by ns_ℓ.

Suppose its nonnegative weight obeys the analytic estimate

$$p_A \leq C \exp((s - g)n(A)), \qquad C \geq 0, \tag{258}$$

where s is the expense per seed and g the gain per seed. For a support exponent $h \geq 0$, put

$$q_p = D^2 \exp(s - g + hs_\ell).$$

If $q_p \le r_p < 1$, multiplication by the support weight, the packet count, and a geometric sum give

$$\sum_{\substack{A:\, x\in X_A\\ n(A)\ge n_0}} p_A e^{hc(X_A)} \le HC\frac{r_p^{\max(1,n_0)}}{1-r_p}. \tag{259}$$

Indeed the summand at fixed size n is at most HCq_p^n. The lower limit is $\max(1,n_0)$ because packets are nonempty. This explains the single-seed extraction in the packet price. If a packet has several analytic variants, the hypothesis (258) must bound their *sum* at that packet. It cannot be applied separately to every variant for free. A variant with smaller support has no larger support price and is handled by the same fiberwise summation.

63.2 Reserving the cost of a whole component

Let $f_i \ge 0$ be weights on a combined alphabet, with supports X_i, extra nonnegative costs d_i, and nonnegative marks e_i. Define

$$\widetilde{f}_i = f_i \exp(2\varepsilon d_i + e_i), \qquad D(A) = \sum_{i\in A}(c(X_i)+d_i), \qquad E(A) = \sum_{i\in A} e_i,$$

where $\varepsilon > 0$. Assume the vertex condition (252) holds for $\widetilde{f}$ with support exponent $a+2\varepsilon$, site price γ, tree reserve δ and tail ratio q. We will use the same tree bound for distinct connected sets.

For each connected finite set A containing a root label i, choose a spanning tree rooted at i. If $|A|-1 = m$, the $m!$ enumerations of its other labels give $m!$ different numbered trees. Their weights in (254) each contain $1/m!$. Thus the sum over distinct sets and their chosen trees is bounded by the full numbered tree sum. Choose a root whose support contains x, or bound by the sum over all such roots. The rooted bound established in Section 62, followed by the geometric sum in m, proves

$$\sum_{\substack{A \text{ connected},\ x\in\cup_{i\in A}X_i\\ |A|-1\ge m}} \prod_{i\in A} \widetilde{f}_i e^{(a+2\varepsilon)c(X_i)} \le \frac{\gamma q^m}{1-q}. \tag{260}$$

No logarithmic coefficient is present in this distinct-set estimate.

Let $k, m \ge 0$ be integers and restrict the sets further by $D(A) \ge T$ and $E(A) \ge U$, where T, U are real. The elementary polynomial estimate is

$$(1+D)^k \le P_{\varepsilon,k} e^{\varepsilon D}, \qquad P_{\varepsilon,k} = \varepsilon^{-k} k! e^{\varepsilon} \quad (D \ge 0).$$

Since $D \ge T$ and $E \ge U$,

$$e^{\varepsilon D} \le e^{-\varepsilon T - U} e^{2\varepsilon D + E}.$$

For connected A we also have $c(\cup_i X_i) \le \sum_i c(X_i)$. Multiplying these inequalities, and then using (260), proves

$$\sum_A \left(\prod_{i\in A} f_i\right)(1+D(A))^k e^{ac(\cup_{i\in A}X_i)} \le P_{\varepsilon,k} e^{-\varepsilon T - U}\frac{\gamma q^m}{1-q}. \tag{261}$$

The sum on the left has all the preceding connectedness, site, cardinality, cost and mark restrictions. The two reserves have separate uses: one pays the polynomial, and the other leaves the explicit tail $e^{-\varepsilon T}$.

63.3 Combining the two vertex populations

For an optional label take its original weight w_i, extra cost $d_i \ge 0$ and no mark. Assume its site estimate with the full head exponent $h = a + 2\varepsilon + \delta$ is

$$\sup_x \sum_{i:x\in X_i} w_i e^{2\varepsilon d_i} e^{hc(X_i)} \le \Gamma_o.$$

For a packet of size n, set extra cost bn and mark μn, where $b, \mu \ge 0$. Its modified weight is $p_A e^{(2\varepsilon b+\mu)n}$. Thus (259) applies with seed expense replaced by $s + 2\varepsilon b + \mu$, leaving the analytic packet bound (258) unchanged. If the resulting packet site price is Γ_p, the combined alphabet has site price at most $\Gamma_p + \Gamma_o$. Choose $\gamma > 0$ with

$$\Gamma_p + \Gamma_o \le \gamma, \qquad 4h_B\gamma/\delta \le q < 1.$$

This is exactly the vertex condition needed for (261).

If the allowed component contains at least r seeds in total, then $E(A) \ge \mu r$. Hence the right-hand side becomes

$$P_{\varepsilon,k} e^{-\varepsilon T - \mu r} \frac{\gamma q^m}{1-q}. \tag{262}$$

This bound controls simultaneously excess label count m, polynomial order k, minimum seed count r, and minimum total cost T. In particular, $r = 1$ may be used only for components known to contain a mandatory packet. The support of an output may be smaller than the component support. Summing all output majorants over each component fiber first, a bound by C_0 times its component price gives C_0 times (262), by support monotonicity and nonnegative summation. This is the required rule for summing analytic variants without losing their multiplicities.

63.4 Adding a normalized elementary insertion

The insertion changes one optional label already present in the alphabet. Its fine support has cover number c_f, support exponent ν_f, and original scalar majorant $w \ge 0$. Let $g > 0$ denote the normalization gain and set

$$j = g e^{-(\nu_f - \varepsilon)c_f}, \qquad D = \frac{w e^{\nu_f c_f} e^{-\varepsilon c_f}}{g}.$$

Then $Dj = w$ exactly. Thus D is the insertion debt and j the normalized vertex weight, not two independent prices.

Let X_* be this label's actual coarse footprint, and suppose its projection geometry gives

$$hc(X_*) \le hP + \varepsilon c_f, \qquad \nu_f \ge 4\varepsilon, \tag{263}$$

where $h = a + 2\varepsilon + \delta$ is the head exponent and P the fixed halo constant. Its extra cost is c_f. Therefore

$$j e^{2\varepsilon c_f} e^{hc(X_*)} \le g \exp\{hP + (4\varepsilon - \nu_f)c_f\} \le g e^{hP} =: R_*.$$

At a fixed site there is just this one modified label. Replacing its old nonnegative weight by j thus increases the optional site bound by at most R_*. This follows even if j is smaller than the old weight, by bounding the new array by the old array plus the single new entry. Set the new optional price and total vertex price to

$$\Gamma_o' = \Gamma_o + R_*, \qquad \gamma' = \gamma + R_*.$$

The packet estimates are unchanged. Once $4h_B\gamma'/\delta \le q' < 1$, the preceding mixed-component bound applies to the modified alphabet with these prices. Retaining only components containing the marked label decreases each nonnegative sum. This produces the marked-molecule estimate used for the order-zero tail in Section 94.

For the actual projected footprint the sufficient geometric condition is $h(10P/L) \le \varepsilon$, together with the comparison of fine and coarse cover radii. Indeed the projected cover estimate is $c(X_*) \le P + (10P/L)c_f$, which gives (263). The fixed block ratio is chosen large enough for this condition before the coupling is increased. If g and γ have negative upper exponent $q_* < 0$ in the fine coupling, then $R_* = ge^{hP}$ and γ' have the same upper exponent, with L and the halo constants fixed. The factor h_B has upper exponent zero. The degree price therefore has upper exponent q_*, and is eventually bounded by the strictly larger power $H^{q_*/2}$, which tends to zero. This proves affordability of the augmented tree price under precisely these rate and geometric inputs.

The combinatorial part of the mixed and marked estimates is now proved. The remaining analytic input is the packet weight bound (258), including all its variants, together with the optional activity site bound and the quoted projection and rate estimates on the actual selected block step.

64 Simultaneous Gaussian warnings and integrated activities

The expansion contains factors which are small only after integration. A warning factor, for example, need not be small on a configuration with a large fluctuation. What is small is the probability that all warnings in a given term are present. Their fluctuations are correlated, and several warnings may be witnessed by the same Gaussian coordinate. We first account for both effects explicitly.

64.1 A bound for correlated coordinates

Let I be a finite coordinate set, and let Z be a centred real Gaussian vector with covariance matrix A. Suppose

$$A \ge 0, \qquad v^{\mathsf T}Av \le M\sum_{i\in I} v_i^2 \quad (v \in \mathbb{R}^I), \qquad M \ge 0. \tag{264}$$

The covariance may be singular. For $J \subseteq I$, $t \in \mathbb{R}$ and $s \ge 0$,

$$\mathbb{P}\{|Z_i| \ge t \text{ for every } i \in J\} \le 2^{|J|}\exp\{(-st + Ms^2/2)|J|\}. \tag{265}$$

Indeed, for each choice of signs $\epsilon_i \in \{-1, 1\}$ on J, put $v_i = \epsilon_i$ on J and $v_i = 0$ elsewhere. On the corresponding orthant event, $v \cdot Z \ge t|J|$. The Gaussian exponential identity gives

$$\mathbb{E}e^{sv\cdot Z} = e^{s^2 v^{\mathsf T}Av/2} \le e^{Ms^2|J|/2}.$$

For completeness, write $Z = A^{1/2}G$ with G a vector of independent standard real Gaussians. In each coordinate, completing the square in $(2\pi)^{-1/2}\int e^{hu-u^2/2}\,du$ gives $e^{h^2/2}$; multiplication proves the identity, including zero eigenvalues of A. Markov's inequality applied to $e^{sv\cdot Z}$ bounds one orthant by $e^{(-st+Ms^2/2)|J|}$, and the union of the $2^{|J|}$ orthants contains the event in (265). This also proves the formula when J is empty.

Let Λ be the coarse periodic lattice, with its lattice distance. Write

$$b_D = (2D+1)^4.$$

Every ball of radius D has at most b_D sites, including on small periods where some representatives coincide. Assign each Gaussian coordinate a site $g(i) \in \Lambda$, with at most p coordinates at any site.

Warning addresses form a finite set K, with site map $o : K \to \Lambda$ and at most m addresses at any site. For a set $X \subseteq K$, define

$$\mathcal{C}(X; t, D) = \{z : \ \forall a \in X \ \exists i \in I, \ d(o(a), g(i)) \le D, \quad |z_i| \ge t\}.$$

These are measurable sets, being finite intersections of finite unions of closed coordinate inequalities.

Proposition 64.1. *If $H \ge 0$ and*

$$M/2 + H b_D m \le t, \tag{266}$$

then

$$\mathbb{P}\{Z \in \mathcal{C}(X; t, D)\} \le (2 b_D p \, e^{-H})^{|X|}. \tag{267}$$

Proof. For each warning in X, choose a witnessing coordinate within distance D. There are at most $(b_D p)^{|X|}$ such assignments. For one assignment, let J be the set of distinct coordinates chosen. Then $|J| \le |X|$. Conversely, any fixed chosen coordinate can witness at most $b_D m$ addresses, since all such addresses lie within distance D of its site. Thus $|X| \le b_D m |J|$. Applying (265) with $s = 1$, the event belonging to this assignment has probability at most

$$2^{|J|} e^{(-t + M/2)|J|} \le 2^{|X|} e^{-H b_D m |J|} \le 2^{|X|} e^{-H|X|}.$$

The union bound over assignments proves the assertion. If a warning has no possible coordinate, its event is empty; the empty set of warnings has probability one. Thus no nonemptiness exception is needed. □

64.2 The scales used for warnings

Fix the block length L. Let $h = h(x) = \widehat{\beta}_f$ be the fine reference coupling, so that $h = x + c_L$ for a constant c_L depending on the fixed group and block length. For sufficiently large x, put

$$B = \lceil (\log x)^2 \rceil, \qquad P = \exp\{(\log \log h)^2\}, \qquad d = 1 + B + (\log h)^2, \qquad \tau = P/\sqrt{h}. \tag{268}$$

The actual fine coupling satisfies $h/2 \le \beta \le 2h$. The fluctuation is written $\mu + \beta^{-1/2} \operatorname{erect}(Z)$; μ is its mean and erect identifies colour coordinates with bond variables. For a fixed $\xi > 0$, the coordinate warning threshold is

$$t = \xi \tau \sqrt{\beta} \ge \xi P/2.$$

The weaker constant $1/2$ is convenient and follows from $\sqrt{\beta} \ge \sqrt{h}/2$.

For the actual warning alphabet, its density per coarse site is bounded by the fixed integer

$$m = L^4 \#\{\text{plaquette axes}\} + 2\#\{\text{axes}\} + 1 + L^4 + L^4 \#\{\text{axes}\}.$$

The five first types are fine plaquette masks, coarse bond masks, coarse bond likelihoods, coarse site likelihoods, and fine site contractions; the last type is a fine bond cutoff. The coordinate density p is the fixed bound for the tubular colour coordinates. Both bounds count distinct addresses, including different types at the same site.

Suppose that the physical covariance satisfies (264) with a constant M independent of the period and of x. Set $D = 2B$, $\varepsilon = \xi/4$, and

$$E(x) = 2 b_{2B} p \exp\{-\varepsilon \sqrt{P}\}. \tag{269}$$

Then, for all sufficiently large x, uniformly in the period and in X,

$$\mathbb{P}\{Z \in \mathcal{C}(X; t, 2B)\} \le E(x)^{|X|}. \tag{270}$$

Here is the scale comparison needed to justify this assertion. First $d \le 7(\log h)^2$ eventually, since $h = x + c_L$, $\log x \le 2 \log h$ eventually, and $B < (\log x)^2 + 1$. For every fixed $a > 0$, integer $r \ge 0$ and constant $C \ge 0$,

$$C d^r \le P^a \quad \text{eventually.} \tag{271}$$

Taking logarithms, it suffices to compare a constant plus $2r \log\log h$ with $a(\log\log h)^2$; the latter dominates. Since $b_{2B} m$ is bounded by a fixed multiple of d^4, (271) implies $b_{2B} m \le \sqrt{P}$ and $M/2 \le \varepsilon P$ eventually. Consequently

$$M/2 + \varepsilon \sqrt{P}\, b_{2B} m \le 2\varepsilon P = \xi P/2 \le t.$$

Proposition 64.1, with $H = \varepsilon\sqrt{P}$, gives (270).

In fact E decays faster than every negative power of h. To see this, write $y = \log\log h$. Then $\log h = e^y$ and $\sqrt{P} = e^{y^2/2}$. For each $N > 0$,

$$\varepsilon e^{y^2/2} - N e^y - \log(2p b_{2B}) \longrightarrow +\infty,$$

because $\log b_{2B} = O(y)$ and $e^{y^2/2}/e^y \to \infty$. Thus $E(x) \le h^{-N}$ eventually. The thresholds may depend on L, ξ, M, p, m, N; they do not depend on the lattice period.

64.3 Why a nonzero warning forces an excursion

We spell out the deterministic step used to connect the preceding probability estimate with the actual factors. Let q be the number of colour coordinates on one bond. On the event that all coordinates within distance B of a site obey $|Z_i| \le \xi\tau\sqrt{\beta}$,

$$\|\beta^{-1/2} \operatorname{erect}(Z)_b\| \le q\xi\tau, \qquad \|\mu_b + \beta^{-1/2} \operatorname{erect}(Z)_b\| \le (C + q\xi)\tau$$

for the bonds in question, provided $\|\mu_b\| \le C\tau$ there. The first inequality follows by the triangle inequality in an orthonormal colour basis; it does not require a sharp Euclidean bound. Choose $q\xi \le 1$ and an outer cutoff plateau $a \ge C + 1$. The bond cutoff is then identically one throughout this smaller event.

The remaining masks use the corresponding fine and predictor estimates. In the source construction these estimates supply, outside the enlarged seed region, a local error bound $\delta\tau$ for each fine or predictor reading, together with the guide bounds $A\tau/L^2$ and $A\tau$, respectively. The local error has three parts: the mean error, the change under the controlled fluctuation, and the nonlinear remainder. Each is bounded by $\delta\tau/3$. If the positive quadratic constants of the fine action and predictor penalty are c_f, c_p, the resulting bounds are

$$\text{fine reading} \le \frac{2}{\sqrt{c_f}}(A/L^2 + \delta)\tau, \qquad \text{predictor reading} \le \frac{2}{\sqrt{c_p}}(A + \delta)\tau. \tag{272}$$

Choose $\delta = \sqrt{c_f}/32$, then L large enough that $A/L^2 \le \delta$, and finally $W > (2/\sqrt{c_p})(A+\delta)$. The first coefficient in (272) is at most $1/8 < 1/4$, and the second is strictly less than W. The corresponding fine and predictor masks therefore equal one; the local likelihood factors also equal one by the same local plateau estimates. Taking ξ smaller if necessary bounds the linear fluctuation error by $\delta\tau/3$; the two remaining errors are eventually below $\delta\tau/3$ by their strictly smaller scale estimates.

A warning factor is either a bounded cutoff times one of these factors minus one, or a bond cutoff minus one. Factors inside their compulsory seed region are defined to be zero. Hence, under the stated local estimates, a nonzero warning at an address implies a coordinate excursion within distance $2B$ of its assigned site. The local estimates producing (272), and the uniform physical covariance bound, remain explicit analytic inputs here. The probability and integration arguments do not assume or replace their production.

64.4 Integration before multiplication of prices

Fix a tree sector. Its integration variables are a Haar variable Y and the single Gaussian vector Z above. Their reference measure $\nu(dY)\gamma_A(dZ)$ is a product of probability measures. All factors in a term are evaluated at the same (Y, Z).

A term is indexed by a finite set S of distinct vertices of four types: connected packets, forced activities, exponential logarithmic corrections, and warnings. Assume these factors are measurable and satisfy the following uniform pointwise bounds:

vertex i	$\lvert f_i(Y,Z)\rvert \le c_i$
packet with n_i seeds	$C_0 e^{(s-g)n_i}$
forced activity i	v_i
logarithmic correction i	w_i
warning i	T

Here all displayed prices are nonnegative. The price w_i bounds the actual factor $e^{\ell_i} - 1$ as well as the logarithmic quantity when that additional estimate is used elsewhere. A disconnected packet is assigned factor and price zero. Write $X(S)$ for the distinct warning addresses in S. Suppose each individual nonzero warning forces its coordinate excursion, as established under the local inputs above. Thus for every finite S, a nonzero product forces $Z \in \mathcal{C}(X(S); t, 2B)$. Then

$$\int \left| \prod_{i\in S} f_i(Y,Z) \right| d\nu\, d\gamma_A \le \prod_{i\in S} \widetilde{c}_i, \qquad \widetilde{c}_i = \begin{cases} E(x)T, & i \text{ a warning}, \\ c_i, & \text{otherwise}. \end{cases} \tag{273}$$

Indeed the integrand is bounded by $\mathbf{1}_{\mathcal{C}(X(S);t,2B)}(Z) \prod_{i\in S} c_i$. It is integrable, being measurable and uniformly bounded on a probability space. Fubini's theorem and (270) give a bound $E(x)^{|X(S)|} \prod c_i$, which is exactly the right-hand side of (273). In particular, no product of marginal warning probabilities has been used.

The same calculation covers one marked insertion. Let its distinguished vertex be i_* and let its measurable factor F_* satisfy $|F_*| \le w_*$. Multiply it by a cutoff taking values in $[0, 1]$. Define the marked term to be zero unless $i_* \in S$, and otherwise replace f_{i_*} by this cutoff times F_*. Applying (273) to $S \setminus \{i_*\}$ gives

$$\int |\text{marked term}|\, d\nu\, d\gamma_A \le \mathbf{1}_{\{i_*\in S\}} w_* \prod_{i\in S\setminus\{i_*\}} \widetilde{c}_i. \tag{274}$$

If the removed vertex was a warning, its probability price is removed as well. The remaining warnings alone are used in the event bound. This is the marked vertex price required for the connected sums of Section 63.

For the actual factors one may take $T = 1 + \#\{\text{frame tags}\} + \#\{\text{point tags}\}$. Indeed the unmodified mask or likelihood lies in $[0, T]$, the enforcing cutoff lies in $[0, 1]$, and $T \ge 1$, so its cutoff times the difference from one has absolute value at most T. A bond cutoff minus one is bounded by one. Thus the small warning price in (273) is supplied by the proved Gaussian estimate. The packet, forced, and logarithmic pointwise bounds in the table are separate analytic obligations; (273) is the precise passage from those bounds to integrated vertex products.

65 Gauge comparison and the geometry of large defects

We now prove the geometric estimates used in the seed energy bound. The essential analytic point is a first-power estimate for the actual coarse predictor. It must hold on both branches of its definition

and for every output with positive weight. Small-field uniqueness alone would not give the estimate needed here.

Write $r(g) = D^{-1}\|g - 1\|$ for the normalized matrix chord. Thus $r \le 2$, $r(g^{-1}) = r(g)$, r is conjugation invariant, and $r(gh) \le r(g) + r(h)$. For a finite lattice box X, let

$$E_X(U) = \sum_{\substack{(x,a)\\ x,x+e_a\in X}} r(U_{x,a}), \qquad F_X(U) = \sum_{\substack{(x,p)\\ \text{all corners in } X}} r(U_p).$$

The sums include positive bonds and one orientation for each plaquette. All boxes in the first part of the argument lie in the covering lattice $\mathbb{Z}^4$; periodic multiplicities will be restored at the end.

65.1 An axial gauge with a first-power bound

For the box of radius R, with side $q = 2R + 1$, there is a coordinate path gauge H such that

$$E_X(H \cdot U) \le 16 q F_X(U). \tag{275}$$

Here the gauge at x is parallel transport from a suitable base point to x, moving along the coordinates in a prescribed order. We prove more generally the bound $4kqF_X$ on a k-dimensional coordinate slab. It is zero on a one-point slab. Suppose the assertion holds in k dimensions and add a coordinate j. Choose a slice perpendicular to j with least plaquette sum; its sum is at most F_X/q, since the plaquettes in different slices are distinct. Choose the base point there by induction, and extend the paths by moving last along j. All j-bonds in the resulting gauge are the identity.

For a transverse bond, two neighbouring slice values differ in chord by at most the chord of the plaquette between them. Indeed, with the two vertical bonds equal to 1, that plaquette is ab^{-1}, and $|r(a) - r(b)| \le r(ab^{-1})$. Summing over the transverse directions costs at most four times the plaquette sum between the slices. Telescoping in j shows that the bond sums of any two slices differ by at most $4F_X$. The total bond sum is consequently at most

$$q\left(4kq\frac{F_X}{q} + 4F_X\right) = 4(k+1)qF_X.$$

This proves (275), with the required coordinate path form of the gauge. In particular the fallback rule which permits axial gauges with bound $K_{\mathrm{ax}}qF_X$ has an eligible choice whenever $K_{\mathrm{ax}} \ge 16$.

We also need a path counting estimate. If $N = |X|$, then for any fixed coordinate order,

$$\sum_{x,y\in X} r(U_{x\to y}) \le 4NqE_X(U). \tag{276}$$

For each coordinate segment use the sum of the bond chords on its coordinate line. During that segment the other coordinates are taken from either x or y, according to whether they have already been changed. Exchanging these coordinate slots between the two endpoints is a bijection of $X \times X$. Thus the mixed point is uniformly distributed when the endpoints are uniformly distributed. Averaging the sum over its coordinate line gives $qE_{X,a}/N$, where $E_{X,a}$ is the bond sum in direction a. Bounding each of the at most four segments by qE_X/N and multiplying by N^2 proves (276). Reversed segments have the same bound because the chord is invariant under inversion.

65.2 Synchronizing minimizing frames

Let I and A be nonempty finite sets, and suppose $W_{ij\alpha} \in G$, with $W_{ii\alpha} = 1$ and $W_{ji\bar{\alpha}} = W_{ij\alpha}^{-1}$ for a permutation $\alpha \mapsto \bar{\alpha}$ of A. A minimizing frame $h : I \to G$ minimizes

$$\mathcal{E}(h) = \operatorname*{av}_{i,j,\alpha} \tfrac{1}{2} r(h_i W_{ij\alpha} h_j^{-1})^2.$$

Such a minimum exists by compactness. Put

$$b_i = \operatorname*{av}_{j,\alpha} r(W_{ij\alpha}), \qquad b = \operatorname*{av}_i b_i.$$

There is a constant C_G, independent of both finite sets and of W, such that some $c \in G$ satisfies

$$\operatorname*{av}_i r(h_i c^{-1}) \le C_G b. \tag{277}$$

Here are the details which improve the square-root estimate to this first-power bound. Comparison with the constant frame gives $\mathcal{E}(h) \le b$, since $r^2/2 \le r$. The triangle inequality and $(u+v)^2 \le 2u^2 + 2v^2$ therefore give

$$\operatorname*{av}_{i,j} \|h_i - h_j\|^2 \le 8D^2 b. \tag{278}$$

Write $m = \operatorname{av}_j h_j$ for the matrix centroid and $m_i = \operatorname{av}_{j,\alpha} h_j W_{ij\alpha}^{-1}$ for its local replacement. Then

$$\|m_i - m\| \le D b_i, \qquad h_i \text{ is a nearest group element to } m_i. \tag{279}$$

The norm bound follows by averaging matrix differences. For the second assertion, change only h_i in the minimizing energy. The two off-diagonal parts have equal costs by the reversal identity. The diagonal term in the actual energy remains zero. Freezing its second endpoint would add the nonnegative term $r(k h_i^{-1})^2/2$, which is zero at $k = h_i$; hence h_i also minimizes the full frozen-neighbour cost. Expanding the squared Frobenius norms shows that this is exactly nearest-point minimization for m_i.

The local tubular projection of the group has derivative the orthogonal tangent projection at 1. On a sufficiently small ambient ball its derivative has norm at most 2. Shrinking the ball also ensures that every nearest point is this local projection. Thus choose $\delta > 0$ so that two matrices in the δ-ball have nearest points at distance at most twice their distance. This also follows directly from the first-order nearest-point estimate used for the group data. Set

$$\sigma = \frac{(\delta/2)^2}{8D^2}, \qquad a = \frac{\delta}{2D}.$$

If $b < \sigma$, choose c nearest to m and rotate the frame by c^{-1}. In the rotated coordinates 1 is nearest to its centroid. Choose k whose mean squared distance to the frame is at most the double average in (278). Jensen's inequality and the nearest-point property give

$$\|m - 1\|^2 \le \|m - h_k\|^2 \le \operatorname*{av}_j \|h_j - h_k\|^2 < (\delta/2)^2.$$

For $b_i < a$, (279) puts m_i in the δ-ball and gives $D r(h_i) \le 2D b_i$. For $b_i \ge a$ use $r(h_i) \le 2 \le 2b_i/a$. Averaging proves the desired inequality in this case. If $b \ge \sigma$, simply use the bound $2 \le 2b/\sigma$. Thus (277) holds with $C_G = \max\{2, 2/a, 2/\sigma\}$. Left and right rotations give the same chord, by conjugation invariance.

Apply this to the coordinate path transports on a patch X of side q. The path orders supply A, with reversed orders supplying the involution above. If f is the actual minimizing frame and H is any comparison gauge, then $h_i = f_i H_i^{-1}$ minimizes the path cost for $H \cdot U$. This follows by substituting into every path product; normalizing a competitor at the patch centre is harmless because a common left rotation preserves the cost. Equations (276) and (277) yield

$$\sum_{i \in X} r(c^{-1} f_i H_i^{-1}) \le 4 C_G q E_X(H \cdot U).$$

For an active fallback frame, eligibility instead gives $E_X(f \cdot U) \le K_{\mathrm{ax}} q F_X(U)$. The pure-gauge discrepancy $d_i = f_i H_i^{-1}$ satisfies

$$E_X(d \cdot 1) \le E_X(f \cdot U) + E_X(H \cdot U).$$

This is the bondwise triangle inequality, with conjugation removing $H \cdot U$. Its path transport from i to j is $d_i d_j^{-1}$. Average (276) over j and select one j whose sum is no larger than the average. Taking $c = d_j$ proves the same conclusion with the added axial term. Both branches therefore satisfy, for $A_G = \max\{4C_G, 4\}$,

$$\sum_{i \in X} r(c^{-1} f_i H_i^{-1}) \le A_G q \{E_X(H \cdot U) + K_{\mathrm{ax}} q F_X(U)\}. \tag{280}$$

Only positivity of the selected frame weight is required; the fallback assertion would be false for an ineligible tag.

65.3 The actual coarse predictor

Fix the smoothing order $J \ge 4$ and the kernel constants. The odd block side L may vary. The patch has radius $(J+3)L$, and the macro box has radius $(J+7)L$. Fix a coarse site x, lift the field around Lx to this macro box, and write P_x for its plaquette chord sum. All translated patches at coarse offsets in the radius-two box lie inside the macro box. Their connecting straight paths of length L lie there as well.

By (275) choose a macro gauge H with $E_{\mathrm{macro}}(H \cdot U) \le D_0 L P_x$, where D_0 depends only on J. Applying (280) on each translated patch gives constants $c_y \in G$ with

$$\sum_i r(c_y^{-1} f_{x+y}(i) H_{i+Ly}^{-1}) \le C_0 L^2 P_x. \tag{281}$$

Indeed each patch side is at most a constant times L, and its bond and plaquette sums are bounded by the macro sums. The constants C_0, D_0 are fixed before choosing L.

The patch weights w_i are the distribution of the sum of J independent uniform points in the small box of side L. They obey

$$w_i \le L^{-4}. \tag{282}$$

Condition on all but the first point. For a prescribed sum there is at most one possible first point, among L^4 equiprobable choices. This proves (282), including coincidences among different tuples of points.

For a coarse bond from $x + y$ to $x + y + e_a$, let Y_i be its connector in the two selected frames. Gauging the middle straight path by H expresses $c_y^{-1} Y_i c_{y+e_a}$ as the product of the two frame errors

and that path transport. At each of its L steps, translation of i is injective; hence the sum of the path chords is at most $LE_{\mathrm{macro}}(H \cdot U)$. Equations (281) and (282) give

$$\sum_i w_i r(c_y^{-1} Y_i c_{y+e_a}) \le (2C_0 + D_0) L^{-2} P_x. \tag{283}$$

The predictor Q either minimizes the weighted squared chord to the connectors, or selects a connector when that minimum is at least the fixed positive cutoff ζ. For every group element b its output satisfies

$$r(Qb^{-1}) \le 2 \max\{1, \zeta^{-1}\} \sum_i w_i r(Y_i b^{-1}).$$

On the minimizing branch, Q is nearest to the weighted matrix centroid; comparison with b and two triangle inequalities prove the factor 2. On the other branch, $\zeta \le \sum_i w_i r(Y_i b^{-1})^2/2 \le \sum_i w_i r(Y_i b^{-1})$, while $r(Qb^{-1}) \le 2$. This proves the claim for every output tag. Using $b = c_y c_{y+e_a}^{-1}$ in (283) bounds each gauged coarse bond by $C_1 L^{-2} P_x$. The four corner gauges cancel around a plaquette, so

$$r(Q_p) \le C L^{-2} P_x, \qquad r(V_p) \le \sum_{e \in \partial p} r(V_e Q_e^{-1}) + C L^{-2} P_x, \quad C = 4C_1. \tag{284}$$

The second inequality follows by replacing the four factors of the plaquette one at a time, conjugating the intervening products. No smallness assumption on U was used. The hypothesis is that the frames at the radius-two coarse offsets have positive weight.

65.4 Periodic multiplicity and the large-defect witness

On a periodic lattice, let $m(x, p)$ count the lifts of the fine plaquette p wholly contained in the macro box centred at Lx. Then, with $r_p = r(U_p)$,

$$P_x = \sum_p m(x, p) r_p. \tag{285}$$

This equality counts lifts, not merely distinct periodic plaquettes. If the fine period is greater than $2(J + 7)L$, different anchors in one macro box cannot wrap to the same point, and $m(x, p) \le 1$. Furthermore, if $m(x, p) > 0$, the coarse parent of p lies at periodic sup distance at most $R = J + 8$ from x: the fine anchor is within $(J+7)L$ of Lx, its distance to its block centre is at most $L/2$, and distances between embedded coarse sites are L times their coarse distances. Consequently, under the same nonwrapping condition,

$$\sum_x m(x, p) \le (2R + 1)^4. \tag{286}$$

The periodic ball bound holds even when the ball wraps, since reducing an integer box modulo the period is surjective onto that ball.

A seed collection consists of fine plaquettes $\mathcal{F}$, coarse predictor bonds $\mathcal{I}$ and coarse flags $\mathcal{G}$. Its size is $n = |\mathcal{F}| + |\mathcal{I}| + |\mathcal{G}|$. Associate to it the coarse parents of the fine plaquette corners, the endpoints of the predictor bonds and the flag sites. Their union has size at most $4n$. A radius-b neighbourhood of this union therefore contains at most

$$v(b)n, \qquad v(b) = 4(2b + 1)^4 N_p,$$

coarse plaquette addresses, where N_p is the number of plaquette orientations. For $B \geq 1$, $v(2B) \leq v(2)B^4$. This proves the outer-region bound used in Section 71.

Let t be the fine threshold, T the very large threshold, and W the predictor cutoff parameter. Retain the seeds in $\mathcal{F}$ with $r_p > T$ and those in $\mathcal{I}$ with $r(V_e Q_e^{-1}) > T$, and discard the flags. Call this collection the very large seeds, of size d_A. The local sector conditions say that, in the radius-$5B$ seed neighbourhood, unlisted fine chords are at most $t/2$, unlisted predictor chords at most $2Wt$, and all selected frame weights are positive.

Suppose

$$R \leq \lfloor 4B/5 \rfloor, \quad t/2 \leq T, \quad 2Wt \leq T, \quad \big(4 + CL^{-2}(2(J+7)L+1)^4 N_p\big)\, T \leq \delta. \tag{287}$$

Then every coarse plaquette of chord greater than δ whose anchor lies in the radius-$2B$ seed neighbourhood is within distance R of a very large seed. To prove this, suppose no such seed exists. Every fine plaquette counted by P_x has parent within R of x. If listed, its chord is at most T, or it would be a witness; if unlisted, its chord is at most $t/2 \leq T$. The same reasoning applies to the four predictor discrepancies, using $2Wt \leq T$. All these addresses and the required radius-two frames lie in the radius-$5B$ neighbourhood, since x starts within $2B$ and the added radii are at most $3B$. Thus (284) applies and gives $r(V_p) \leq \delta$, a contradiction.

It follows that these large coarse plaquettes belong to the hit region, defined using the radius-$\lfloor 4B/5 \rfloor$ neighbourhood of the very large seeds, and their number is at most $v(R)d_A$. The whole hit region has at most $v(2)B^4 d_A$ plaquettes. A coarse flag is, by definition, a corner of a coarse plaquette with chord above the coarse threshold s; each such plaquette has at most four corners. Hence

$$|\mathcal{G}| \leq 4\#\{p \text{ in the outer region} : r(V_p) > s\}.$$

Splitting these plaquettes at δ gives precisely the flag count and the $K_0 d_A$ large-plaquette count used in the energy allocation.

All constants in these geometric counts can be chosen before L and W: take, for example, $K_0 = v(R) + v(2) + R + (2R+1)^4$. For each fixed L, W, the conditions (287) hold eventually along the coupling parameter. Indeed $B \to \infty$, $T = h^{-49/100} \to 0$, and $t/T = P(h)h^{-1/100} \to 0$, by the threshold estimates already proved. The allowed coarse periods are at least $100(1+B)$, which also ensures the nonwrapping condition for the macro box. Thus the geometric constants are independent of both the volume and the subsequent choice of block side. The guide bounds and the hit and unhit energy comparison estimates are separate analytic inputs; this argument establishes the geometric part of their allocation.

66 Kernel sums and the bounded guide fields

The energy argument uses kernels after parallel transport, truncation and periodization. We derive their row and column bounds from the cover estimates, and then bound the clipped guide fields. The inputs are the tower strip bounds and the exponentially summable path expansions at the two adjacent stages. The passage to the physical kernels does not require a small-field assumption.

66.1 Counting types before periodization

Let $K_{ab}(z)$ be a scalar cover kernel, with finite type sets $\mathcal{A}, \mathcal{B}$, satisfying, for every fixed pair of types,

$$\sum_{z \in \mathbb{Z}^4} |K_{ab}(z)| e^{\eta|z|} \leq C, \qquad \eta \geq 0.$$

A physical entry from (x,a) to (y,b) is a sum of selected terms $K_{ab}(z)\,\mathrm{Ad}(U_{\gamma_{x,z}})$ with $y = x + \mathrm{wrap}(z)$. Adjoint transport is an isometry of the real colour space, including along winding paths, so the operator norm of each term is at most $|K_{ab}(z)|$. Absolute convergence therefore permits the following rearrangements:

$$\sup_{x,a}\sum_{y,b}\|F_{(x,a),(y,b)}\| \le |\mathcal{B}|C, \qquad \sup_{y,b}\sum_{x,a}\|F_{(x,a),(y,b)}\| \le |\mathcal{A}|C. \tag{288}$$

For the first, fix z,b: exactly one y can contribute. For the second, fix z,a: exactly one $x = y - \mathrm{wrap}(z)$ can contribute. Dropping the truncation only increases the absolute sum. Thus different lifts of one periodic displacement are all paid.

The wide kernel Φ has fine doubled plaquette rows and coarse doubled plaquette columns. In coarse coordinates a fine type includes its residue in the block. There are L^4 residues. Its fixed-type strip cost is C/L^2, so (288) gives, after absorbing only fixed orientation counts,

	row sum	column sum
Φ	a/L^2	aL^2
Φ'	a	a
N	a	a

(289)

with a fixed before L. Here Φ' has predictor-bond rows and coarse doubled-plaquette columns; N has auxiliary coarse rows and the same columns. Their fixed-type strip costs are bounded independently of L. The tower estimates provide precisely these three strip inputs; no residue factor is present in the narrow two kernels. In particular the column bound for Φ cannot be replaced by its row bound.

All three kernels are cut at path length at most $R_0 = \lfloor B/100000 \rfloor$ in fine units. The endpoint of the wide path beginning at a fine site of residue r and ending at coarse displacement z is the embedded coarse point at parent plus z: this is the path from r to Lz on the cover, reduced modulo the fine period. The parent changes by at most one coarse step on every fine bond. For the narrow paths the ordinary endpoint bound applies. Thus every nonzero entry has anchor separation at most $B/50$ in coarse units. This is an incidence assertion for these kernels, and does not bound all auxiliary reads of a larger operation.

66.2 Absolute path expansions and their transposes

At one stage write the path expansion of a plaquette field in the form

$$\ell_p(U) = \sum_{i\ge 0} c_{a,i}\ \mathrm{Ad}(U_{\gamma_{p,i}})F(U_{p+s_{a,i}}), \qquad p = (x,a).$$

The label i includes the target plaquette orientation and the transport path. Its length $\ell_{a,i}$ bounds the distance from x to the target anchor. The stage input is

$$\sum_i |c_{a,i}|e^{\sigma\ell_{a,i}} \le P, \qquad \sigma > 0, \quad P > 0, \tag{290}$$

uniformly in a, with clipping relaxation radius $r_0 > 0$ satisfying $\sigma r_0 \ge 1$. Both stages have such bounds, with constants fixed before the step side L. We retain the full label set when different labels land at the same periodic plaquette.

Keep terms with $d\ell_{a,i} \le R_0$, where $d = 1$ on the fine stage and $d = L$ on the coarse stage. Let ℓ_p^{cut} be their sum and let

$$k_{pu} = \sum_{\substack{i: d\ell_{a,i}\le R_0 \\ p+s_{a,i}=u}} |c_{a,i}|.$$

Absolute convergence follows from (290). The curvature bound $\|F(U_u)\| \le r(U_u)$ gives

$$\|\ell_p^{\mathrm{cut}}(U)\| \le \sum_u k_{pu} r(U_u), \quad \sum_u k_{pu} \le P, \quad \sum_p k_{pu} \le N_p P. \tag{291}$$

For the column bound, fix an input orientation a and a label i. Translation by its offset is a permutation of the periodic sites, so at most one anchor can land at the specified u. Sum first over labels and then over the N_p possible input orientations. This argument also explains why merging repeated targets before taking absolute values would not justify the bound.

A retained site satisfies $r(U_u) \le t \exp\{d(p,u)/r_0\}$ for every target u. Since $d(p,u) \le \ell_{a,i}$ and $1/r_0 \le \sigma$, (290) implies

$$\|\ell_p^{\mathrm{cut}}(U)\| \le Pt. \tag{292}$$

The direct component is $\sqrt{a_0} F(U_p)$, where $a_0 > 0$ is the stage quadratic coefficient. Its clipped version multiplies it by $\min\{1, t/r(U_p)\}$, with multiplier 1 at zero. Its norm is at most $\sqrt{a_0} t$. The second component is ℓ_p^{cut} at retained sites and zero at discarded sites. Thus the two-component clipped field v satisfies the pointwise bound

$$\|v_{p,\varepsilon}\| \le A_0 t, \qquad A_0 = (N_p + 1)P + \sqrt{a_0}. \tag{293}$$

This same A_0 bounds both row and column sums in (291).

The selected and discarded rows have the locality needed for the energy proof. For a cut path, the parent separation is at most its length plus one, hence at most $R_0 + 1$. A witness to discarding at lattice distance k obeys $te^{k/r_0} < r(U_u) \le 2$. For $t = P(h)/\sqrt{h} \ge h^{-1/2}$ this implies $k < r_0 \log(2\sqrt{h})$. With $h = x + c_L$, c_L fixed, this is at most $r_0 \log x$ eventually. Since $B = \lceil (\log x)^2 \rceil$, both $R_0 + 1$ and $k + 1$ are at most $B/50$ eventually. The parent map increases distances by at most one when compared with fine lattice distance; therefore these bounds give the asserted coarse anchor locality too. All tails are uniform in the period.

Finally let $s = t(x)$ be the coarse threshold and $t = t(x + c_L)$ the fine threshold. The shift c_L is nonnegative: its group factor is a positive multiple of a sum of squares of adjoint matrix entries, and $\log L \ge 0$. For large x, $s \le 2t$: the numerator $P(x) = \exp\{(\log\log x)^2\}$ is increasing once $\log x \ge 1$, and $\sqrt{x + c_L} \le 2\sqrt{x}$ for $x \ge c_L$. Choose A larger than the two stage constants A_0. Then (293), applied on the coarse stage, and (289) imply

$$\|v_j\| \le dt, \qquad \|(\Phi v)_i\| \le dL^{-2} t, \qquad \|(\Phi' v)_b\| \le dt, \qquad d = 2A(1 + a). \tag{294}$$

Indeed a matrix with operator-norm row sum r sends a field bounded by u to a field bounded by ru. These are exactly the guide bounds used for the fine and predictor energy rows in Section 71, with constants fixed before L, W. The derivation uses the stated tower and path-expansion inputs; the construction of the path expansions remains a separate input to be proved below.

67 Paying for the affected comparison terms

The hit region consists of coarse plaquettes near the very large seeds. Its size is controlled by Section 65. We now bound the comparison terms on this region, including the negative coarse energy of large plaquettes. The decisive distinction is between a small contribution at every tested row and a summable excess. The former may have constants depending on L, W; the latter must retain constants chosen before L.

Keep the notation s, t for coarse and fine thresholds, with $s \le 2t$. Let a, d be the kernel and guide constants in (289) and (294). Let D be the predictor-curvature constant in (284), and let

N bound the macro column count and the hit-region size. These constants are fixed before L, W. Write $g = \sqrt{p}$, where $p > 0$ is the predictor penalty, and

$$u = \sqrt{a_f} + d, \qquad v_0 = \sqrt{a_c} + d.$$

Here a_f, a_c are the direct quadratic coefficients. All kernels and fields below are real; a star denotes the physical Hilbert-space adjoint, after transport and truncation.

67.1 A small part and a summable excess

For a matrix F_{ij} with operator-norm row bound r and column bound c, suppose $\|z_i\| \le b + e_i$ on the inputs of the tested columns $j \in S$, with $b, e_i \ge 0$. Then

$$\sum_{j \in S} \|(F^* z)_j\| \le |S| cb + r \sum_i e_i. \tag{295}$$

Indeed $\|(F^* z)_j\| \le \sum_i \|F_{ij}\| \|z_i\|$ because an operator and its adjoint have the same norm. The constant part is bounded using the column sum. In the excess part exchange the two finite sums and enlarge S to all columns; the row sum then gives $r \sum_i e_i$. Inputs outside the nonzero stencil need no estimate.

For the unclipped two-component plaquette field X, define the excess

$$E_p^\sharp = \sqrt{a_0} E_p + \sum_q k_{pq} E_q,$$

where k is the nonnegative path matrix of (291). If $r(U_q) \le b + E_q$ on the local stencil, those bounds give, for either component ε,

$$\|X_{p,\varepsilon}\| \le (\sqrt{a_0} + d) b + E_p^\sharp, \qquad \sum_{p,\varepsilon} E_p^\sharp \le 2(\sqrt{a_0} + d) \sum_p E_p. \tag{296}$$

For the direct component use the curvature bound. For the path component use the row bound for the constant part. For the total excess use the column bound. The factor 2 is the actual number of components, even though a sharper componentwise estimate is possible.

Freeze the following excess arrays:

$$E_p = \mathbf{1}_{\{p \in \mathcal{F},\ r(U_p) > t\}} r(U_p), \qquad F_e = \mathbf{1}_{\{e \in \mathcal{I}\}} r(V_e Q_e^{-1}), \qquad \mathcal{B} = L^{-2} \sum_p E_p + \sum_e F_e.$$

The threshold in E is t, and F includes every listed predictor bond. The larger threshold T is used only to count the hit region. The sector conditions give, throughout the radius-$5B$ neighbourhood,

$$r(U_p) \le t + E_p, \qquad r(V_e Q_e^{-1}) \le 2Wt + F_e. \tag{297}$$

For a listed fine plaquette outside the excess set its chord is at most t; an unlisted one is at most $t/2$. Listed predictor bonds are included entirely in F, while unlisted ones are at most $2Wt$.

Let $M_L = (2(J+7)L + 1)^4 N_p$ bound the number of lifted plaquettes in a macro box. Put

$$r_* = 8W + DL^{-2} M_L, \qquad H_p = \sum_{e \in \partial p} F_e + DL^{-2} \sum_q m(x_p, q) E_q, \qquad H_* = 4N_p + DN_p N.$$

Equations (284) and (297) imply

$$r(V_p) \le r_* t + H_p \quad \text{in the radius-}3B\text{ neighbourhood}, \qquad \sum_p H_p \le H_* \mathcal{B}. \tag{298}$$

For the local assertion, the four boundary bonds contribute $8Wt$ and their excesses; the macro row contributes $DL^{-2}M_L t$ and its excess. All input anchors remain within radius $5B$ once $B \ge N+1$, and the required nearby frames have positive weight. For the total assertion, each of the four boundary-bond maps, for a fixed plaquette orientation, is injective on the periodic lattice; thus its sum is at most $\sum_e F_e$. The macro column sum is at most N for each fine plaquette, and summing over coarse plaquette orientations contributes N_p. Consequently

$$\sum_p H_p \le 4N_p \sum_e F_e + DN_p N L^{-2} \sum_q E_q \le H_* \mathcal{B}.$$

In contrast to r_*, the coefficient H_* is independent of L, W.

67.2 The two comparison rows

Let S be the hit plaquettes, $m = |S|$, and $\widetilde{S} = S \times \{0,1\}$, so $|\widetilde{S}| = 2m$. Put $Z_e = gF(V_e Q_e^{-1})$ for the predictor curvature field. The three norm totals entering the linear comparison satisfy

$$\sum_{j \in \widetilde{S}} \|(\Phi^* X_f)_j\| \le 2m\, aL^2 ut + 2au\mathcal{B}, \tag{299}$$

$$\sum_{j \in \widetilde{S}} \|((\Phi')^* Z)_j\| \le 2m\, ag(2Wt) + ag\mathcal{B}, \tag{300}$$

$$\sum_{j \in \widetilde{S}} \|(X_c)_j\| \le 2m\, v_0 r_* t + 2v_0 H_* \mathcal{B}. \tag{301}$$

For (299), use (296) with $b = t$ and then (295) with row a/L^2 and column aL^2. The total excess is at most $2u \sum E$, giving $2auL^{-2} \sum E \le 2au\mathcal{B}$. The two successive stencils extend the tested radius-$2B$ region first to radius $3B$ and then to radius $5B$, so the assumed local envelope is sufficient. For (300), use $\|Z_e\| \le g(2Wt + F_e)$ and the row and column bounds a for Φ'. For (301), apply (296) to (298) and sum its excess. Again the stencil lies within the region where that inequality is valid.

The linear comparison row is

$$L_p = \sum_{\varepsilon=0}^{1} \langle v_{p,\varepsilon}, (\Phi^* X_f + (\Phi')^* Z - X_c)_{p,\varepsilon} \rangle.$$

By $\|v_j\| \le dt$ and Cauchy–Schwarz, the sum over S is bounded below by $-dt$ times the sum of the three right sides above. The quadratic defect row is

$$D_p = \frac{1}{2} \sum_{\varepsilon=0}^{1} \Big\langle v_j, \big(\{1 - \Phi^*\Phi - (\Phi')^*\Phi' - N^*N\}v\big)_j \Big\rangle, \qquad j = (p,\varepsilon).$$

There is no assertion here that the bracket is zero at a curved field. From (294) and the column bounds,

$$\|\Phi^*\Phi v\|_\infty \le adt, \quad \|(\Phi')^*\Phi' v\|_\infty \le adt, \quad \|N^*N v\|_\infty \le a^2 dt.$$

In the first inequality the factor L^2 in the column cancels the L^{-2} in the guide bound. Thus

$$D_p \geq -d_0 t^2, \qquad d_0 = d(d + ad + ad + a^2 d). \tag{302}$$

The factor $1/2$ cancels the two component slots.

Let $\mathcal{C}_2$ be the coarse plaquettes in the outer region with chord greater than δ. By Section 65, $\mathcal{C}_2 \subset S$. Therefore

$$\sum_{p \in \mathcal{C}_2} r(V_p) \leq m r_* t + H_* \mathcal{B}.$$

Writing λ_c for the coarse linear coefficient and using $s \leq 2t$, combine this with (299)–(302) to obtain

$$\sum_{p \in S} (L_p + D_p) \geq \lambda_c s \sum_{p \in \mathcal{C}_2} r(V_p) - C_1 t \mathcal{B} - Z_0 t^2 m, \tag{303}$$

where the explicit nonnegative constants are

$$C_1 = d(2au + ag + 2v_0 H_*) + 2\lambda_c H_*,$$
$$Z_0 = 2d\{aL^2 u + 2agW + v_0 r_*\} + d_0 + 2\lambda_c r_*.$$

This is just addition of the bounds; in particular C_1 is chosen before L, W, while Z_0 may depend on them. The geometric estimate $m \leq N B^4 d_A$ now changes the last term into $-Z t^2 B^4 d_A$, with $Z = N Z_0 + 1$. Since $\mathcal{B} = L^{-2} \sum_{\mathcal{S}} r(U_p) + \sum_{\mathcal{I}} r(V_e Q_e^{-1})$, (303) is the hit-region input used in Section 71. Its proof retains the two distinct thresholds and all transposed-kernel multiplicities. The unhit-region comparison still requires the local flat-field approximation.

68 A common small gauge away from very large seeds

The remaining energy comparison needs a gauge in which both the fine and coarse fields are small near an unhit plaquette. The fine field is given on its actual block-tree slice: its tree bonds equal 1. We use that tree, including paths crossing a periodic seam.

68.1 Comparing one fine bond with the coarse replica

In each block the rooted forest gives a path from a fine site to its block centre, of length at most $4L$. On the tree slice its transport is 1, since every step is a tree bond or its inverse. For a fine bond $e = (x, a)$, concatenate the reverse path from x to its root, the bond itself, and the path from $x + e_a$ to its root. This gives a walk w between the two roots with

$$|w| \leq 8L + 1, \qquad U_w = U_e.$$

Lift this entire walk to the covering lattice starting at zero, with origin the first root. Its length fits in the macro radius $(J + 7)L$ because $J \geq 4$. Its endpoint z lies in that macro box. For fine period at least $100(J + 7)L$, wrapping is injective on the macro box. Hence $z = 0$ if the two roots coincide, and $z = L e_a$ if the bond crosses to the next block. The latter is true even at a periodic seam: the periodic endpoints agree with the two stated cover endpoints, and injectivity identifies their lifts.

Use the macro gauge H of Section 65 at the first coarse root i, and write P_i for its macro plaquette sum. If its bond sum is at most $D_0 L P_i$, the chord of the transport of w is at most its length times that bond sum. This deliberately allows repeated steps and gives

$$r(H_0 U_e H_z^{-1}) \leq (8L + 1) D_0 L P_i. \tag{304}$$

The normalized local frames take value 1 at their own centres. Thus the centre term of (281) gives

$$r(H_{Ly}c_y) \le C_0 L^2 P_i \quad (y = 0, e_a).$$

Together with the gauged predictor bond bound this yields

$$r(H_0 Q_{i,a} H_{Le_a}^{-1}) \le (2C_0 L^2 + C_1 L^{-2}) P_i.$$

Indeed insert $c_0 c_0^{-1}$ and $c_{e_a} c_{e_a}^{-1}$ and apply the triangle inequality to the two centre errors and the predictor term.

The coarse replica $\overline{V}$ is 1 on internal fine bonds and $V_{i,a}$ on crossing bonds. If e is internal, (304) has $z = 0$ and conjugation removes H_0. If it crosses, compare the last two displayed bounds in the same macro gauge and then compare $Q_{i,a}$ with $V_{i,a}$. Thus, with

$$D_L = (8L + 1) D_0 L + 2C_0 L^2 + C_1 L^{-2},$$

we obtain the shell comparison

$$r(U_e \overline{V}_e^{-1}) \le D_L P_i + \begin{cases} 0, & e \text{ internal to a block}, \\ r(V_{i,a} Q_{i,a}^{-1}), & e \text{ crossing from } i. \end{cases} \tag{305}$$

The constant is allowed to depend on the fixed block side. The only frame hypothesis is the same positive-weight condition in the macro comparison. No replacement of the prescribed forest was made.

68.2 Constructing the gauge in an unhit neighbourhood

Fix an outer plaquette with anchor o which is not in the hit region. Let T be the very large threshold. Throughout distance $\lfloor 4B/5 \rfloor$ from o, all fine plaquette chords and all predictor discrepancies are at most T. For listed defects this follows from being unhit: any value above T would place a very large seed in that neighbourhood. For unlisted defects it follows from the sector bounds $t/2 \le T$ and $2Wt \le T$.

Write $M = (J + 7)L$ and $K_L = (2M + 1)^4 N_p$. For B sufficiently large after fixing L, W, every macro box whose coarse centre lies within $r = \lfloor B/2 \rfloor$ of o is contained, in coarse parent distance, in the above neighbourhood. Its macro sum is at most $K_L T$, with lifted multiplicities included. The necessary frames are active there as well. By (284), every coarse plaquette with anchor within r has chord at most

$$F_L T, \qquad F_L = 4 + C L^{-2} K_L.$$

The allowed coarse period is at least $100(1 + B)$, so the cover box of radius r is nonwrapping. Apply the axial gauge construction (275) on this box. Its plaquette sum is at most $(2r + 1)^4 N_p F_L T$, so the total gauged bond sum is at most $16(2r + 1)^5 N_p F_L T$. Descend the cover gauge by the centred lift $y \mapsto \mathrm{lift}(y - o)$. For a bond whose anchor is within $\lfloor 2B/5 \rfloor$ of o, both endpoints lie in the box; the descended gauge therefore agrees with the cover gauge at both endpoints. Each such coarse bond has chord at most

$$H_L B^5 T, \qquad H_L = 16\, 3^5 N_p F_L. \tag{306}$$

This follows by bounding one nonnegative summand by the total and using $2r + 1 \le 3B$.

Call the resulting coarse gauge z and use $z \circ \mathrm{parent}$ on the fine lattice. Under this block-constant gauge, the replica of V becomes the replica of $z \cdot V$: odd L ensures the prescribed parent changes across a positive fine bond either not at all or by the corresponding positive coarse step. Thus its

bond chord is either zero or bounded by (306). On the other hand, the chord of $U_e\overline{V}_e^{-1}$ is unchanged by the simultaneous gauge, since that product is conjugated at the initial endpoint. Equation (305) consequently gives

$$r((z\circ\mathrm{parent})\cdot U)_e \le (D_L K_L+1)T + H_L B^5 T.$$

With $C_L = H_L + D_L K_L + 1$ and $B \ge 1$, both fine and coarse fields therefore satisfy the common bound

$$r((z\circ\mathrm{parent})\cdot U)_e \le h(x), \qquad r(z\cdot V)_b \le h(x), \qquad h(x) = C_L B^{10} T, \tag{307}$$

on the radius-$\lfloor 2B/5\rfloor$ neighbourhood of o. The fine bound uses parent distance for the anchor. All choices, including the gauge, refer to the same original fields, tree and tags. The threshold for x may depend on L, W and the fixed construction parameters, but not on the torus volume or on the chosen unhit plaquette. This proves the common small-gauge input needed for the local comparison.

69 The minimizing branches and the predictor derivative

We establish the branch properties used in the local comparison. The cutoffs for uniqueness can be chosen using only the group, before choosing a patch. Analyticity near the identity is then a local finite-dimensional statement, and its derivative includes the root normalization explicitly.

69.1 Uniform uniqueness of minimizing frames

Use the finite path family $W_{ij\alpha}$ and minimizing energy from Section 65. Regard group elements as matrices, and let Π be the orthogonal projection onto the Lie algebra. The analytic nearest-point projection π near 1 has $D\pi(1) = \Pi$. By continuity of its derivative, for a sufficiently small convex ball and all A, B in that ball,

$$\|\pi(A) - \pi(B) - \Pi(A-B)\| \le \tfrac{1}{4}\|A-B\|. \tag{308}$$

This follows by integrating $D\pi - \Pi$ along the segment. Choose $\delta > 0$ for such a ball, and choose $0 < \varepsilon \le \min\{\delta/4, 1/4\}$. Suppose

$$\|W_{ij\alpha} - 1\| \le \varepsilon \quad \text{for every } i, j, \alpha.$$

We claim that any two minimizing frames differ by a common left multiplication, independently of the number of points or paths.

Comparison with the constant frame and the squared triangle inequality give

$$\operatorname*{av}_{i,j} \|h_i - h_j\|^2 \le 4\varepsilon^2.$$

For clarity, the mean squared transported discrepancy $\|h_i W_{ij\alpha} - h_j\|^2$ is at most ε^2 by minimality. Each untransported discrepancy squared is at most twice this term plus $2\varepsilon^2$; averaging gives the claim. Rotate each minimizing frame separately so that 1 is nearest to its centroid m. Choosing a point with mean squared distance no larger than the double average and applying Jensen gives $\|m-1\| \le 2\varepsilon$. Its feedback matrix $m_i = \mathrm{av}_{j,\alpha}\, h_j W_{ij\alpha}^{-1}$ satisfies $\|m_i - m\| \le \varepsilon$, so $\|m_i - 1\| \le 3\varepsilon < \delta$. The self-consistency proved in (279) therefore identifies $h_i = \pi(m_i)$.

Let h, k be the two rotated minimizers and set $v = \max_i \|h_i - k_i\|$. Their feedback differences D_i satisfy

$$\|D_i\| \le v, \qquad \|D_i - (m_h - m_k)\| \le v\varepsilon.$$

The first follows by averaging and unitary invariance. For the second, subtract the centroid difference to obtain the average of $(h_j - k_j)(W_{ij\alpha}^{-1} - 1)$ and use the matrix product bound. Since 1 is nearest to each centroid, $\Pi(m_h - 1) = \Pi(m_k - 1) = 0$: differentiate squared distance along $\exp(tX)$ for each tangent vector X. Thus $\|\Pi D_i\| \le v\varepsilon$. At an index attaining the maximum, (308) yields

$$v \le v\varepsilon + \tfrac{1}{4}v \le \tfrac{1}{2}v,$$

so $v = 0$. Undoing the rotations proves uniqueness up to a common left multiplier. If both frames are normalized at their centre, that multiplier is 1.

The actual frame branch tests that every coordinate path transport, after one of its rooted axial gauges, has chord less than a fixed cutoff. Take this cutoff at most ε/D. Choose one of those axial gauges; the resulting W satisfies the preceding matrix bound. Changing variables by that gauge in the minimizing energy proves uniqueness of the normalized frame on the entire tested branch. These cutoffs do not depend on the patch size or smoothing order. At the identity all tested chords vanish, so this branch is an open neighbourhood of the identity field.

For the predictor branch, let $w_i \ge 0$, $\sum_i w_i = 1$, and let $m = \sum_i w_i Y_i$. Expanding the squares shows that minimizing $\sum_i w_i r(Y_i u^{-1})^2/2$ is equivalent to choosing a nearest group element to m. If the weighted cost at some u is less than ζ, Jensen gives

$$\|m - u\|^2 \le \sum_i w_i \|Y_i - u\|^2 < 2D^2\zeta.$$

The group has a uniform nearest-point tube, obtained from the local tube by left translation. Choose ζ so that $\sqrt{2D^2\zeta}$ is smaller than its radius. Then the predictor minimum is unique on its tested branch, uniformly over all finite weight distributions. Finally the fallback frame rule has an eligible choice by (275) with axial factor 16. Together these facts construct the required branch constants for any fixed positive predictor penalty.

69.2 Analyticity of the normalized frame

Fix a patch and a finite torus for the moment. Write a for the fine logarithmic bond field and $h_i(a)$ for the actual frame matrices. At $a = 0$ they are all 1. The chosen minimizer is continuous on the uniqueness branch. To verify this without assuming regularity of a choice function, consider a convergent sequence of input fields. Compactness of the finite product of groups gives a convergent subsequence of any sequence of minimizers. Continuity of the energy and the centre constraint makes every limit a minimizer at the limiting field. Uniqueness makes that limit the selected one. A sequence staying outside a fixed neighbourhood of that value would contradict this argument, which proves continuity.

Let o be the patch centre and let $W_{ij\alpha}(a)$ be its actual coordinate path transports. Define an ambient matrix-valued system

$$\mathcal{F}_i(a, H) = \begin{cases} H_o - 1, & i = o, \\ H_i - \pi\big(\operatorname{av}_{j,\alpha} H_j W_{ij\alpha}(a)^{-1}\big), & i \ne o. \end{cases}$$

It is analytic near $(0, \mathbf{1})$. Continuity of the actual frame and its feedback, together with self-consistency, shows that $\mathcal{F}(a, h(a)) = 0$ near zero. The derivative in the ambient frame variable is

$$(\mathcal{D}v)_o = v_o, \qquad (\mathcal{D}v)_i = v_i - \Pi\bar{v} \quad (i \ne o), \qquad \bar{v} = \operatorname*{av}_j v_j.$$

It is invertible. Indeed if $\mathcal{D}v = 0$, put $A = \Pi\bar{v}$. Then $v_o = 0$ and $v_i = A$ for $i \neq o$. If the patch has n points, $\bar{v} = (n-1)A/n$, so applying Π gives $A = (n-1)A/n$ and $A = 0$. The derivative is an injective endomorphism of a finite -dimensional space and is therefore bijective. The analytic implicit function theorem gives a unique analytic local solution. Continuity places the actual frame in that local solution neighbourhood, so $h(a)$ is analytic. This argument works in ambient matrices; it does not require choosing coordinates for a complement at every frame.

69.3 The root correction in the derivative

For a Lie-algebra-valued bond perturbation a, write $\int_\gamma a$ for its signed sum along a path. Define

$$\tau_y(i;a) = -\operatorname*{av}_{j,\alpha} \int_{i\to j,\alpha} a,$$

where the paths are in the patch centred at the coarse site y. Differentiate the feedback equation at zero. If $\dot{h}_i$ is the matrix derivative and $\overline{\dot{h}}$ its uniform average, differentiation of the inverse transport gives

$$\dot{h}_i = \Pi\overline{\dot{h}} + \operatorname{col}(\tau_y(i;a)).$$

The feedback identity holds also at the centre. Since the normalized frame has $h_o(a) = 1$, its centre derivative is zero. Subtraction therefore yields

$$\dot{h}_i = \operatorname{col}\{\tau_y(i;a) - \tau_y(o;a)\}. \tag{309}$$

In particular the frame derivative uses uniform averages over patch points and path orders, before the separate connector weights enter.

For a coarse bond $e = (y, b)$, the connector at patch point i consists of its frame at y, the length-L fine path, and the inverse frame at $y + e_b$. Its colour derivative is consequently

$$\tau_y(i;a) - \tau_y(o;a) + \int_{y+i\to y+i+Le_b} a - \tau_{y+e_b}(i;a) + \tau_{y+e_b}(o;a).$$

At zero all connectors equal 1. Nearby the predictor is on its minimizing branch and equals π of their weighted matrix centroid. This follows from the weighted nearest-point identity above; the branch remains open since its minimum cost is at most the continuous cost of comparison with 1, which vanishes at zero. Thus the predictor is analytic near zero. The derivative of π is Π, which fixes the above colour derivatives, so the predictor derivative is their weighted sum.

Define the bond averaging map and the site correction by

$$(\mathcal{A}a)_{y,b} = \sum_i w_i \int_{y+i\to y+i+Le_b} a,$$
$$(\mathcal{R}a)_y = \tau_y(o;a) - \sum_i w_i \tau_y(i;a).$$

The weighted connector derivative is exactly

$$\mathcal{Q}a = \mathcal{A}a + d_c^{(0)}\mathcal{R}a. \tag{310}$$

The sign is determined by the subtraction of the frame at the final endpoint: the remaining root terms are $(\mathcal{R}a)_{y+e_b} - (\mathcal{R}a)_y$. For a fine gradient $a = d_f^{(0)}v$, path sums telescope, giving

$$(\mathcal{R}d_f^{(0)}v)_y = v(Ly) - \sum_i w_i v(Ly+i), \qquad \mathcal{Q}d_f^{(0)}v = d_c^{(0)}\{v(Ly)\}_y.$$

This checks the root normalization directly. Equations (309) and (310) prove the analytic predictor law and derivative used in Section 70. Its transplantation argument then supplies a uniform fixed-patch Taylor radius for all periods.

The covariance needed there follows from the same definitions. A gauge change turns a frame into $f_i' = g_o f_i g_i^{-1}$. On a minimizing branch it preserves the energy and centre normalization, so uniqueness identifies this with the selected frame; on a fallback branch the identity follows by telescoping its rooted paths. Connector endpoint factors then reduce to $g_{Ly} Y_i g_{L(y+e_b)}^{-1}$. The weighted cost is invariant under this change of variables. Uniqueness identifies the minimizing predictor, and direct substitution handles a selected connector on the other branch. The original tags are retained in a gauge change. These identities give the block-constant covariance used in the unhit comparison.

70 The comparison error on the unhit region

We compare the actual cut kernels with the full linear kernels at the identity. The linear cancellation uses the derivative of the actual rooted predictor. Its gradient correction disappears only after the coarse curl is applied. Throughout this section L and the patch are fixed; constants may depend on them but are uniform in the period, the truncation length, and the selected tags.

70.1 The full linear identities

Let B_f, B_c be the two stacked curvature maps: their first components are $\sqrt{a_f}$ and $\sqrt{a_c}$ times the identity, and their second components are the full path-root kernels. Path coefficients are aligned with these root kernels by summing all labels with the same displacement and target orientation. Absolute convergence in (290) permits periodization and grouping. Consequently the linearizations of the two plaquette fields are

$$X_f^0 = B_f d_f a, \qquad X_c^0 = B_c d_c b,$$

where a, b are fine and coarse logarithmic bond fields and d_f, d_c are the discrete exterior derivatives from bonds to plaquettes. Let F_0, I_0, N_0 be the full real kernels corresponding to Φ, Φ', N at the identity. The flat stack identities needed here are

$$\begin{aligned} F_0^* F_0 + I_0^* I_0 + N_0^* N_0 &= 1, \\ \sqrt{p}\, I_0^* &= B_c d_c, \\ F_0^* B_f d_f &= \sqrt{p}\, I_0^* \mathcal{A}. \end{aligned} \tag{311}$$

Here $\mathcal{A}$ is the unrooted linear averaging map. These are identities of full finite matrices in physical counting coordinates, for the same step and the same two stages. We state their exact role because none of them is asserted for the transported cut kernels. Their production, together with the aligned path expansion, is the flat-stack input to this comparison.

The derivative $\mathcal{Q}$ of the actual predictor has the form $\mathcal{Q} = \mathcal{A} + d_c^{(0)} \mathcal{R}$, where $d_c^{(0)}$ sends site fields to bond gradients. Thus, since $d_c d_c^{(0)} = 0$,

$$\sqrt{p}\, I_0^* \mathcal{Q} = B_c d_c \mathcal{A}.$$

The discrete identity $d_c d_c^{(0)} = 0$ follows by cancellation of the four endpoint differences around each plaquette. With $Z^0 = \sqrt{p}\,(b - \mathcal{Q} a)$, the last two identities in (311) therefore give

$$F_0^* X_f^0 + I_0^* Z^0 - X_c^0 = 0. \tag{312}$$

The first identity in (311) also gives $(1 - F_0^* F_0 - I_0^* I_0 - N_0^* N_0) w = 0$ for every doubled coarse field w. No closedness assumption on w or removal of harmonic modes is used.

70.2 Uniform Taylor bounds for local reads

For a real logarithmic field a, put $U_e = \exp(a_e)$ and use its supremum norm. On a fixed small group chart,

$$r(\exp v) \le C\|v\|, \qquad \|\log g\| \le Cr(g), \qquad \exp(\log g) = g.$$

These follow by integrating the first derivative of the analytic exponential and local logarithm on smaller closed balls. Likewise, Taylor's formula for the product of the four exponentials around a plaquette gives

$$\|F(U_p) - (da)_p\| \le C\|a\|_\infty^2, \qquad \|(da)_p\| \le C\|a\|_\infty. \tag{313}$$

Indeed differentiating the product at zero gives the signed sum of the four logarithms; the derivative of the curvature coordinate is the identity on the Lie algebra. Bound the second derivative on the fixed four-slot ball and integrate the remainder. Repeated periodic bonds simply repeat coordinates in that four-slot map and do not change the constant.

For the actual predictor we use its local analytic law at the identity, for each allowed tag, with derivative $\mathcal{Q}$ above. The law is part of the branch construction: it includes uniqueness of the normalized minimizing frame on the branch where that frame is used. It implies the uniform estimate

$$\|Z - Z^0\|_\infty \le Ch^2, \qquad \|Z^0\|_\infty \le Ch, \quad h = \max\{\|a\|_\infty, \|b\|_\infty\}, \tag{314}$$

with a radius independent of the torus. Here is why finite-dimensional Taylor estimates give that uniformity. A predictor bond reads only two frame patches, their coordinate paths, and their length-L connectors. This is a fixed finite set of cover bond anchors. Choose one reference torus large enough that these anchors have their actual cover lifts and its two coarse endpoints are distinct. Transplant the local fine logarithms to it by reading the original periodic field at each corresponding cover anchor. Transplant the coarse logarithm at the tested bond as a constant field, and copy the actual two frame tags and connector tag. The transplanted sup norm does not increase. All path products, frame scores, selected frames and connectors agree. On a minimizing branch, equality of scores gives equality of the selected normalized frame by its uniqueness; on the other branch the chosen axial paths agree directly.

There are only finitely many tags and tested directions on this reference torus. Take the minimum of their positive Taylor radii and the maximum of their Taylor bounds. Equality of the transplanted reads along every ray $v \mapsto v(a, b)$ gives equality of their derivatives at zero as well, so these bounds prove (314) on every torus, including short ones. This proves the uniformization; the stated analytic branch law and identification of its derivative are required inputs.

70.3 Truncation and transport errors

For cutoff R in fine units define

$$\omega(R, h) = (1 + R)h + e^{-\tau R}, \qquad \tau > 0.$$

The kernel strip width here is strictly positive, $\eta > 0$. The value of τ can be decreased a finite number of times below. If all logarithms have norm at most h in the chart, every link has chord at most Ch, and a path of length k has chord at most Ckh by the product triangle inequality. On the colour space,

$$\|\operatorname{Ad}(g) - 1\| \le 2Dr(g). \tag{315}$$

Here D is the fixed colour normalization. To see this, write $gXg^* - X = (g - 1)Xg^* + X(g^* - 1)$, use the Frobenius product bound and unitary invariance, then divide by the colour normalization.

Consider a transported kernel term with cover displacement z and physical path length $k \le D_L(1+|z|)$. This bound holds for the actual paths: a narrow path has fine length $L|z|$, while the wide path from residue r to Lz has length at most $L|z|+4L$. If $k \le R$, (315) bounds the difference from the scalar flat term by $CRh|K_{ab}(z)|$. If $k > R$, the flat term is omitted, and

$$1 \le e^{\eta} e^{\eta|z|} e^{-\eta R/D_L}.$$

Paying both cases with the weighted strip sum and using the same row and column counting as (288) proves

$$\|F\|_{\mathrm{row}}, \|F\|_{\mathrm{col}}, \|F_0\|_{\mathrm{row}}, \|F_0\|_{\mathrm{col}} \le C, \qquad \|F-F_0\|_{\mathrm{row}}, \|F-F_0\|_{\mathrm{col}} \le C\omega(R,h). \tag{316}$$

This holds for each of the three kernels, after increasing C and choosing $\tau \le \eta/D_L$. The type counts, including residues, are absorbed after L is fixed. Every cover term is estimated before periodization, so winding paths cause no exception.

The same argument applies to the two plaquette fields. For a retained path term, first use (313), then (315) on its transport. Its error is at most $C(1+R)h^2$ times its coefficient. For an omitted term use its full linear value, bounded by Ch, and $1 \le e^{\sigma\ell} e^{-\sigma R/d}$ when $d\ell > R$. The weighted absolute path sum therefore gives

$$\|X^0\|_\infty \le Ch, \qquad \|X - X^0\|_\infty \le Ch\omega(R,h) \tag{317}$$

for both stages, with $\tau \le \min\{\sigma_f, \sigma_c/L\}$. Absolute convergence justifies subtracting and summing the path series, even if the set of labels below the length cutoff is infinite. Equation (314) has the same form since $h \le \omega(R,h)$.

If F, F_0 satisfy (316) and z, z^0 satisfy (317), then

$$F^* z - F_0^* z^0 = F^*(z - z^0) + (F - F_0)^* z^0$$

has sup norm at most $2C^2 h\omega(R,h)$. For an arbitrary bounded field w, the same decomposition gives

$$\|(F^*F - F_0^*F_0)w\|_\infty \le 2C^2 \|w\|_\infty \omega(R,h).$$

Insert these bounds in the two full linear cancellations. After increasing C, the actual defect vectors satisfy

$$\|\Phi^* X_f + (\Phi')^* Z - X_c\|_\infty \le Ch\omega(R,h), \qquad \|(1 - \Phi^*\Phi - (\Phi')^*\Phi' - N^*N)w\|_\infty \le C\|w\|_\infty \omega(R,h). \tag{318}$$

The argument keeps w arbitrary; it requires no regularity of the clipping indicators that will later define w.

70.4 Restriction to a neighbourhood and summation

A cut kernel entry has endpoints within distance R in coarse anchor units and reads only the transport path of that length. The cut path field reads through radius $R+4$. A predictor bond reads through a fixed patch radius R_{pred}. Thus the first defect at anchor o depends only on fine and coarse bonds through radius $3R+4+R_{\mathrm{pred}}$, using parent distance for fine bonds. For example, the outer transpose moves the input anchor by at most R, its kernel path adds at most R, and the input field adds $R+4$ or the predictor radius. The second defect, as an operator applied to a fixed w, reads its kernels through radius $2R$. Equality of these reads gives equality term by term in each cut series and finite matrix sum.

Suppose a coarse gauge z makes the fine and coarse bond chords at most h on this neighbourhood. Take the local logarithms there and set them to zero outside. Their global sup norms are at most Ah and their exponentials agree with the gauged fields on all required reads. Apply (318) to these extensions. Replacing h by Ah changes only the constant, since for $A \geq 1$, $\omega(R, Ah) \leq A\omega(R, h)$.

To remove the gauge, every transported plaquette term transforms by $\mathrm{Ad}(z_o)$ at its anchor; the endpoint gauge factors cancel. The same endpoint calculation for kernels gives

$$F'_{ij} = \mathrm{Ad}(z_i) F_{ij}\, \mathrm{Ad}(z_j)^{-1},$$

with the parent anchor on fine rows. Taking physical adjoints preserves this identity, because adjoint transport is orthogonal. The actual predictor is covariant with the same tags under the block-constant fine gauge. Hence both defect vectors transform by $\mathrm{Ad}(z_o)$, provided the free field w is transformed at its anchors. Its norm is unchanged. Thus the local version of (318) holds for the original fields and for any bounded w. In particular we may use the original clipped field v without changing its indicators in the extension argument.

Take $R = \lfloor B/100000 \rfloor$ and $h = C_L B^{10} T$ from (307). For large B, $3R + 4 + R_{\mathrm{pred}} \leq \lfloor 2B/5 \rfloor$, so this estimate applies to every unhit outer plaquette. As $\|v\|_\infty \leq dt$, its two comparison rows are bounded below by minus

$$\mathcal{L} = 2dt\, Ch\omega(R, h) + dt\, Cdt\, \omega(R, h). \tag{319}$$

There are two components in the linear row, while the factor $1/2$ in the quadratic row cancels its two components.

For completeness, the required delay is stronger than $h \to 0$. Write $H = x + c_L$, so $T = H^{-49/100}$ and $t = P(H)H^{-1/2} \geq H^{-1/2}$. Any fixed power of $B = \lceil (\log x)^2 \rceil$ is smaller than any positive power of H eventually. Consequently, for every fixed integer n,

$$B^n h \longrightarrow 0, \qquad B^n \frac{h^2}{t} \longrightarrow 0,$$

the latter because $T^2/t \leq H^{-48/100}$. Also $e^{-\tau R} \leq H^{-1}$ eventually: R grows quadratically in $\log x$, whereas $\log H$ grows linearly. It follows that

$$B^n \omega(R, h) \longrightarrow 0, \qquad B^n \frac{h\omega(R, h)}{t} \longrightarrow 0.$$

Expanding ω proves the second assertion from the h^2/t bound and $e^{-\tau R}/t \leq H^{-1/2}$. The outer region has at most $NB^4 n$ plaquettes for n seeds. Given $\epsilon > 0$, choose the coupling threshold so that $NB^4 \mathcal{L} \leq \epsilon t^2$. Summing (319) gives

$$\sum_{p \text{ outer, unhit}} (L_p + D_p) \geq -\epsilon t^2 n. \tag{320}$$

This is the unhit-region estimate used in the energy allocation. Its remaining construction inputs are the full flat stack and aligned path expansions, and the analytic law and derivative of the actual predictor. The estimates, localization, common-gauge use and delay from those inputs have all been given above.

71 The energy gain assigned to defect seeds

The exponential gain in Section 84 comes from a lower bound on the energy assigned to a finite set of defects. Fine plaquette defects, predictor defects and coarse flags require different arguments. In particular, some coarse terms can be negative. The proof retains a reserve from the very large fine and predictor defects to pay for these terms. We give the local algebra and the final allocation, keeping the geometric and local comparison inputs explicit.

71.1 A quartic bound for the curvature defect

Use the real Frobenius inner product in the defining unitary matrix representation of the group. Write $\mathfrak{g}$ for its tangent space at the identity and Π for orthogonal projection onto $\mathfrak{g}$. Let $D > 0$ be the colour normalization, so that

$$\|U - 1\| = D\, r(U), \qquad \|\operatorname{col}(v)\| = D\|v\|.$$

Here $r(U)$ is the group chord and col is the isometric identification up to the factor D. The curvature coordinate $F(U)$ is obtained by projecting $(U - U^*)/2$ onto $\mathfrak{g}$. Since the Hermitian part of $U - 1$ is orthogonal, for the real Frobenius inner product, to every skew-Hermitian matrix,

$$\operatorname{col}(F(U)) = \Pi(U - 1).$$

To check the orthogonality, if $H^* = H$ and $X^* = -X$, cyclicity of the trace shows that $\operatorname{Re}\operatorname{tr}(HX^*)$ is its own negative. Pythagoras therefore gives the exact identity

$$D^2\{r(U)^2 - \|F(U)\|^2\} = \|(1 - \Pi)(U - 1)\|^2. \tag{321}$$

In particular $\|F(U)\| \le r(U)$ for every group element.

The local tubular projection in the group data is analytic, equals the identity on the group near 1, and has derivative Π at 1. Taylor's formula on a sufficiently small ambient ball consequently gives

$$\|(1 - \Pi)(U - 1)\| \le C\|U - 1\|^2$$

for nearby group elements. Explicitly, bound the second derivative of the projection on a smaller closed ball and integrate its second-order Taylor remainder along the segment from 1 to U. This proves the inequality with one constant $C > 0$. Combining it with (321) gives constants $K, \delta > 0$, depending only on the group data, such that

$$r^2 - Kr^4 \le \|F(U)\|^2 \le r^2, \qquad \tfrac{3}{4}r^2 \le \|F(U)\|^2 \quad \text{when } r = r(U) \le \delta. \tag{322}$$

One can take $K = C^2D^2$ and decrease δ until $K\delta^2 \le 1/4$. Thus the nonnegative defect $\mathfrak{f}(U) = (r(U)^2 - \|F(U)\|^2)/2$ is at most $Kr(U)^4/2$ in that neighbourhood.

This proves the group estimate needed for the coarse energy row. Let $a_c > 0$ be its quadratic coefficient and $\lambda_c > 0$ its linear coefficient, with $\lambda_c \le a_c/8$. At coarse threshold $s \ge 0$, the clipping function and the radial attenuation are

$$q_s(r) = \begin{cases} a_c r^2/2, & r \le s, \\ \lambda_c s r, & r > s, \end{cases} \qquad \alpha_s(r) = \begin{cases} 1, & r = 0, \\ \min\{1, s/r\}, & r > 0. \end{cases}$$

Cancellation of the other coarse quadratic components gives the row

$$C_s(U) = a_c(\alpha_s - \alpha_s^2/2)\|F(U)\|^2 - q_s(r(U)). \tag{323}$$

For $0 \le s \le \delta$ this has three useful bounds:

$$\begin{aligned} &r \le s: && C_s(U) \ge -B_0 s^4, \quad B_0 = a_c K/2, \\ &s < r \le \delta: && C_s(U) \ge (a_c/4) s r, \\ &r > \delta: && C_s(U) \ge -\lambda_c s r. \end{aligned} \tag{324}$$

For the first, $\alpha_s = 1$ and use the quartic defect. In the other two ranges $\alpha_s = s/r \in [0, 1]$, so $\alpha_s - \alpha_s^2/2 \ge s/(2r)$. In the middle range this gives $C_s \ge (3a_c/8 - \lambda_c) s r \ge a_c s r/4$; in the last range drop the nonnegative curvature term. These arguments include $s = 0$.

71.2 The fine row and the discarded quadratic component

Let the fine threshold be $t > 0$. Write $a_f, \lambda_f > 0$ for the fine quadratic and linear coefficients, and let r_i be the fine plaquette chords. The direct part of its energy row is

$$D_i = q_t(r_i) + \tfrac{1}{2}\|v_i\|^2 - \langle v_i, y_i\rangle, \qquad \|y_i\| \le \sqrt{a_f}\, r_i,$$

where q_t uses coefficients a_f, λ_f. The second part is

$$H_i = \mathbf{1}_{\{i \text{ retained}\}} \tfrac{1}{2}\|u_i\|^2 + \tfrac{1}{2}\|v_i'\|^2 - \langle v_i', u_i\rangle.$$

Their sum is the fine row. The guide bounds are $\|v_i\|, \|v_i'\| \le \epsilon t$, with $\epsilon = C/L^2$. The guide constant $C \ge 0$, the coefficients $a_f, \lambda_f > 0$, and the fine clipping radius $R > 0$ are fixed before L is chosen.

Let $\mathcal{F}$ be the listed fine defects and $\mathcal{S} = \{i \in \mathcal{F} : r_i > t\}$ their large part. The local sector conditions ensure $r_i \ge t/4$ on $\mathcal{F}$ and $r_i \le t/2$ on unlisted plaquettes in the relevant neighbourhood. For $r_i \le t$, Cauchy–Schwarz gives

$$D_i \ge \tfrac{1}{2}(\sqrt{a_f}\, r_i - \|v_i\|)^2 \ge 0.$$

If additionally $i \in \mathcal{F} \setminus \mathcal{S}$ and $\epsilon \le \sqrt{a_f}/8$, the last square is at least $a_f t^2/128$. For $i \in \mathcal{S}$, the linear part of the clipping and $\epsilon\sqrt{a_f} \le \lambda_f/4$ give $D_i \ge (3\lambda_f/4)t r_i$. Summing over the fine rows in the $2B$ neighbourhood of the seeds therefore gives

$$\sum_i D_i \ge \frac{3\lambda_f}{4} t \sum_{i \in \mathcal{S}} r_i + \frac{a_f}{128} t^2 |\mathcal{F} \setminus \mathcal{S}|. \tag{325}$$

The second component is nonnegative on retained rows, since there it is $\|u_i - v_i'\|^2/2$. It is not necessarily nonnegative on discarded rows. We count these rows on the original fine lattice. A discarded row at i has a witness j with

$$t \exp\{d(i,j)/R\} < r_j.$$

The finite-range locality conditions ensure that each such witness belongs to $\mathcal{S}$. For $y \ge 1$, the number of fine plaquette addresses within distance $R \log y$ of a point is at most

$$C_R y, \qquad C_R = (8R+1)^4 \#\{\text{plaquette axes}\}.$$

Indeed $1 + (\log y)/4 \le y^{1/4}$ implies $2R \log y + 1 \le (8R+1) y^{1/4}$, and the lattice ball has at most $(2\lfloor R \log y \rfloor + 1)^4$ sites. A union bound over witnesses therefore gives, for the discarded set $\mathcal{D}$,

$$t|\mathcal{D}| \le C_R \sum_{j \in \mathcal{S}} r_j. \tag{326}$$

There is no factor L^4 in this count.

For the shortened path field u_i, use the nonnegative finite-range coefficient matrix k_{ij}, whose row and column sums are at most C and which satisfies $\|u_i\| \le \sum_j k_{ij} r_j$. Locality and the sector conditions imply $r_j \le t$ for a contributing j outside $\mathcal{S}$. Thus

$$\|u_i\| \le Ct + \sum_{j \in \mathcal{S}} k_{ij} r_j, \qquad \sum_{i \in \mathcal{D}} \|u_i\| \le C(C_R + 1) \sum_{j \in \mathcal{S}} r_j.$$

The second inequality uses (326) for the constant part and the column bound for the large part. Since $H_i \ge -\epsilon t\|u_i\|$ on discarded rows, choosing L so large that $\epsilon C(C_R+1) \le \lambda_f/4$ and combining with (325) proves

$$F_A := \sum_i (D_i + H_i) \ge \frac{\lambda_f}{2} t \sum_{i\in\mathcal{S}} r_i + d_f t^2 |\mathcal{F}\setminus\mathcal{S}|, \qquad d_f = a_f/128 > 0. \tag{327}$$

The geometric row and column estimates and the guide bound used here are explicit inputs; the discarded-row argument shows precisely how they enter the energy estimate.

For predictor defects the argument is simpler. Let $p > 0$ be the predictor penalty and z_i its bond chord. A predictor energy row is at least $(\sqrt{p}\, z_i - \|w_i\|)^2/2$, with $\|w_i\| \le Ct$. On each listed predictor defect $i \in \mathcal{I}$, the sector condition is $z_i \ge Wt$. Choose W after L so that $C \le \sqrt{p}\, W/2$. The unlisted rows are nonnegative, and hence

$$I_A \ge \frac{p}{8} \sum_{i\in\mathcal{I}} z_i^2. \tag{328}$$

71.3 The reserve for very large defects

Let $T \ge t$ be a larger threshold and put

$$\mathcal{F}_T = \{i \in \mathcal{F} : r_i > T\}, \quad \mathcal{I}_T = \{i \in \mathcal{I} : z_i > T\}, \quad d_A = |\mathcal{F}_T| + |\mathcal{I}_T|.$$

The use of T here does not change the seed set. Let $m \ge 0$ satisfy

$$m \le \lambda_f/8, \qquad m \le d_f, \qquad m \le p/32.$$

If $c_F \le \lambda_f/4$, then (327) implies

$$F_A \ge mt^2|\mathcal{F}| + mtT|\mathcal{F}_T| + c_F t \sum_{i\in\mathcal{S}} r_i. \tag{329}$$

To check this, both $t|\mathcal{S}|$ and $T|\mathcal{F}_T|$ are at most $\sum_{\mathcal{S}} r_i$. Their two charges and the c_F charge use at most $(2m + c_F)t\sum_{\mathcal{S}} r_i \le (\lambda_f/2)t\sum_{\mathcal{S}} r_i$. The remaining fine seeds are paid by $d_f t^2$.

Similarly suppose $W \ge 1$ and $C_1 \le pW/16$. Then (328) implies

$$I_A \ge mt^2|\mathcal{I}| + mtT|\mathcal{I}_T| + C_1 t \sum_{i\in\mathcal{I}} z_i. \tag{330}$$

Indeed $z_i \ge Wt \ge t$ on every listed bond, so $t^2|\mathcal{I}| \le \sum z_i^2$ and $tT|\mathcal{I}_T| \le \sum z_i^2$. Also $C_1 t z_i \le (p/16)z_i^2$. The two first charges use at most $2m\sum z_i^2 \le (p/16)\sum z_i^2$, leaving the required linear charge within the total $p/8$.

71.4 Coarse flags and the affected region

Let $\mathcal{O}$ be the coarse plaquette rows in the $2B$ neighbourhood of all seeds, and let $\mathcal{H} \subseteq \mathcal{O}$ be the rows affected by a very large defect. Split the coarse plaquettes into

$$\mathcal{C}_0 = \{r_i \le s\}, \quad \mathcal{C}_1 = \{s < r_i \le \delta\}, \quad \mathcal{C}_2 = \{r_i > \delta\}, \qquad i \in \mathcal{O}.$$

The geometry supplies a constant K_0, fixed before L, W, such that

$$\mathcal{C}_2 \subseteq \mathcal{H}, \quad |\mathcal{C}_2| \le K_0 d_A, \quad n_g \le 4(|\mathcal{C}_1| + K_0 d_A), \quad |\mathcal{O}| \le K_0 B^4 n, \tag{331}$$

where n_g is the number of coarse flag seeds and $n = |\mathcal{F}| + |\mathcal{I}| + n_g$ is the total number of seeds. These are the precise geometric counting inputs used below. For $t \le 2s$ and $m \le a_c/64$, the middle bound in (324) gives at least $4mt^2$ on each row of $\mathcal{C}_1$. Summing all three ranges yields

$$C_A \ge 4mt^2|\mathcal{C}_1| - \lambda_c s \sum_{i\in\mathcal{C}_2} r_i - B_0 s^4 |\mathcal{O}|. \tag{332}$$

The remaining two comparison rows, denoted together by L_A, are split over $\mathcal{H}$ and its complement. Their local estimates have the form

$$L_{\mathcal{H}} \ge \lambda_c s \sum_{i\in\mathcal{C}_2} r_i - \frac{C_1}{L^2} t \sum_{i\in\mathcal{S}} r_i - C_1 t \sum_{i\in\mathcal{I}} z_i - Z t^2 B^4 d_A, \tag{333}$$

$$L_{\mathcal{O}\setminus\mathcal{H}} \ge -\eta t^2 n. \tag{334}$$

The constant C_1 is fixed before L, W, whereas Z may depend on both. The second estimate is available for every fixed $\eta > 0$ by taking x sufficiently large after the other parameters. The row estimates producing these inequalities remain inputs here. The positive normalization row, denoted N_A, satisfies $N_A \ge 0$. The total assigned energy is exactly $F_A + I_A + C_A + N_A + L_{\mathcal{H}} + L_{\mathcal{O}\setminus\mathcal{H}}$.

Choose L large enough for the fine estimate and $C_1/L^2 \le \lambda_f/4$, and W large enough for the predictor estimate, $W \ge 1$ and $C_1 \le pW/16$. Choose

$$0 < m \le \min\{\lambda_f/8, d_f, p/32, a_c/64\}, \qquad \eta = m/4.$$

Suppose finally that

$$(4mK_0 + ZB^4)t^2 \le mtT, \qquad B_0 s^4 K_0 B^4 \le (m/4)t^2. \tag{335}$$

Adding (329)–(334) cancels the three explicit linear expenses. The first inequality in (335) leaves $4mK_0t^2 d_A$ from the very-large-defect reserve, and the second pays the small coarse rows. Thus the assigned energy is at least

$$mt^2(|\mathcal{F}| + |\mathcal{I}|) + 4mt^2(|\mathcal{C}_1| + K_0 d_A) - \tfrac{m}{4}t^2 n - \tfrac{m}{4}t^2 n \ge \tfrac{m}{2}t^2 n.$$

The last step is the flag count in (331). This is the required positive gain per seed, with $\mu = m/2$.

71.5 The order of choices and orientation average

For the actual thresholds put

$$t = \frac{P(h)}{\sqrt{h}}, \qquad s = \frac{P(x)}{\sqrt{x}}, \qquad T = h^{-49/100}, \qquad h = x + c_L.$$

As $x \to \infty$ at fixed L,

$$s \to 0, \quad t/s \to 1, \quad B^r\, t/T \to 0, \quad B^r s \to 0 \quad \text{for every fixed } r \ge 0.$$

For the ratio t/s, subtract the logarithms: $\log t = (\log\log h)^2 - \frac{1}{2}\log h$ and the corresponding formula at x. Their difference tends to zero under a fixed shift, by the mean value theorem, since the derivative of $(\log\log x)^2 - \frac{1}{2}\log x$ tends to zero. The other statements follow from B and P growing slower than every positive power, using $-1/2 + 49/100 = -1/100$ for t/T. It follows that

$0 < s \le \delta$, $t \le 2s$, $t \le T$, and both inequalities (335) hold eventually. For the second, divide by t^2 and use $s/t \to 1$ and $B^4 s^2 \to 0$. All choices have therefore been made in the order

$$\text{fixed group and row data ; } L \text{ large ; } W \text{ large ; } \mu > 0 \text{ ; } x \text{ large,}$$

with no dependence on the lattice period in the last threshold.

The packet construction uses an average of the assigned energies over signed coordinate transformations. Here the block side is odd, so the parent map commutes with the signed coordinate action. We use the patch's signed covariance laws for the predictor and the local likelihoods. Turn the tree, seed set and tag array together with the fields. Tree edges remain equal to the group identity; the predictor and likelihood transform covariantly; distances, chords and the number of seeds are preserved. The local sector conditions, including the strict positivity of the retained site likelihoods, therefore hold in every transformed channel. Applying the same energy bound there and averaging with normalized weights yields $Q_A^{\mathrm{core}} \ge \mu t^2 n$. This supplies the core input to (425) once the geometric, guide and comparison inputs specified in this section have been established. The average is of energies on transformed copies of the same data; there is no independent choice of trees or tags between channels.

72 The local annular remainder

This section proves the annular estimate used in Section 84. The comparison is made before Gaussian integration, on the actual fine configuration, coarse configuration and predictor tags. In particular, removing a remote cutoff does not permit us to replace the predictor by a newly chosen one. The argument needs only the local conditions stated below.

Fix the block length L, the predictor patch, the positive direct coefficients a_f, a_c, and the predictor penalty $p > 0$. Retain the exponent convention of Section 84, with large parameter $h = h(x)$. Put $t = P(h)/\sqrt{h}$, let s be the coarse threshold, and let B and R be the shell and truncation radii. We use $s \le 2t$, $R = \lfloor B/100000 \rfloor$, and the eventual geometric inequalities

$$100(1 + B) \le c, \qquad 10000(\rho + R_{\mathrm{macro}} + L + 10) \le B,$$

where c is the coarse period and ρ the predictor reach. Both B and R have upper exponent zero, whereas $e^{-\sigma R}$ has every negative upper exponent for fixed $\sigma > 0$. These properties follow from $B \asymp (\log x)^2$. All thresholds in this section are chosen after the fixed local data; all estimates are uniform in c.

The incoming linear data are the path and kernel bounds of Section 66, the flat identities of (311), and the additional orthogonality

$$N_0 B_c d_c = 0. \tag{336}$$

The last identity is the null-row part of the same flat decomposition. Its construction, like that of the flat identities, belongs to the flat-stack input. We will identify exactly where it is used.

72.1 A common local chart

Call a coarse patch clean when no coarse flag occurs there. On a clean patch every plaquette chord is at most s. The axial gauge argument in Section 68, applied on a box of radius $d+1$ with $d \le 32B$, gives a gauge in which the bond chords on the smaller box are bounded by

$$u_0 = 16\{2(32B + 1) + 1\}^5 N_{\mathrm{plaq}}(2t).$$

Thus u_0 has upper exponent $-1/2$. For large x these bonds are in the logarithm chart. Take their logarithms and set all other bond logarithms to zero. This gives a full array b with $\|b\|_\infty \le Cu_0$ whose exponential agrees with the gauged coarse configuration on the required patch. No assertion about the coarse configuration outside that patch is needed.

Write $V^\uparrow$ for the block replica of V. Suppose, on a patch about o, the fine configuration and a colour array u satisfy

$$U_e = \exp(u_e) V_e^\uparrow, \qquad \|u_e\| \le w(x), \tag{337}$$

where $w \ge 0$ has upper exponent $-1/2$. Under the block-constant lift of the coarse gauge, u_e is conjugated at its initial point; its norm is unchanged. Since a replica bond is either the identity or one coarse bond, the chord triangle inequality gives

$$r(U_e) \le Cw + Cu_0.$$

Consequently both fields have small bonds in this same gauge, with a bound $H(x) \ge w(x)$ of upper exponent $-1/2$. The coarse logarithm extension also has norm at most H after enlarging H.

For any prescribed $\delta > 0$, all predictor reads on a sufficiently small such patch lie in the minimizing branch and have chord at most δ. Here is the uniformity argument. At the identity configuration the normalized minimizing frames and the best connector are the identity. Their continuity was proved in Section 69. The branch inequalities are strict there. The finitely many frame tags, connector tags and axes therefore have a common neighbourhood on which these inequalities hold and the predictor chord is less than δ. In the product topology on cover configurations, this neighbourhood contains a cylinder restricting finitely many bonds. Enclose these bonds in one fixed patch and choose one positive chord radius for all of them. Every torus configuration whose lifted patch satisfies these restrictions belongs to the same neighbourhood, independently of its period. For large x the patch fits within $B/2$ and its bonds satisfy the restrictions. In particular, the normalized tag likelihoods are exactly one there: each tag has probability the reciprocal of the number of tags in the minimizing branch. The four-link triangle inequality also makes every touched fine plaquette chord at most δ. These statements hold for the original tags, including tags retained after remote masks were erased.

72.2 Affine approximation at the actual coarse field

Let $\mathcal{C}(U, V)$ be the row field consisting of the two fine curvature rows and the predictor row:

$$\mathcal{C}(U, V) = \big(X_f(U), \sqrt{p}\, F(VQ(U)^{-1})\big).$$

Here X_f includes the truncated path row and the direct curvature row, and Q is the actual predictor. Each row norm is the physical colour norm at one address; an array norm is its supremum over addresses. No whole-volume Euclidean norm is used in these estimates. Its full linearization is

$$\mathcal{C}^0(a, b) = \big(B_f d_f a, \sqrt{p}\,(b - \mathcal{Q}a)\big).$$

The chart estimates of Section 70 give, for $\|a\|, \|b\| \le H$,

$$\|\mathcal{C}^0(a, b)\|_\infty \le CH, \qquad \|\mathcal{C}(e^a, e^b) - \mathcal{C}^0(a, b)\|_\infty \le CH\{(1 + R)H + e^{-\sigma R}\}. \tag{338}$$

The predictor term has a quadratic error; the path row has the displayed additional truncation error. The right side has upper exponent -1.

Define $A_\infty a = \mathcal{C}^0(a, 0)$, and let A_R be its truncated version. Absolute exponential summability of the path coefficients and the four-bond curl estimate give

$$\|A_\infty a\|_\infty + \|A_R a\|_\infty \le C\|a\|_\infty, \qquad \|(A_\infty - A_R)a\|_\infty \le C\|a\|_\infty e^{-\sigma R}. \tag{339}$$

Indeed, each face curl has norm at most $4\|a\|_\infty$. For a discarded path of length $\ell > R$, insert $e^{-\sigma R}e^{\sigma\ell} \ge 1$ into its absolute coefficient bound and sum. The predictor part is local and is not truncated.

Let A_R^V be the actual transported row: input colours are parallel transported along the prescribed coarse routes before the same cut linear row is applied. If the coarse bonds along these routes have chord at most h_0, and the routes have length at most d, then

$$\|(A_R^V - A_R)v\| \le 2CD_{\mathrm{col}}\, dh_0\, \|v\|_\infty. \tag{340}$$

The route product has chord at most dh_0 by the triangle inequality. The bound $\|\operatorname{Ad}(g) - 1\| \le 2D_{\mathrm{col}}r(g)$ then bounds each changed input, and (339) bounds the resulting row. Only inputs within distance $R + \rho + 4$ occur; $4(R + \rho + 4) \le d$ ensures the routes remain in the controlled patch. One may set all other input differences to zero to apply the supremum-norm bound.

We claim that, under (337) on a clean patch of radius $B/2$, there are nonnegative functions M, E, of upper exponents $-1/2$ and -1 respectively, such that at its central row

$$\|\mathcal{C}(U, V)\| \le M, \qquad \|\mathcal{C}(U, V) - \mathcal{C}(V^\uparrow, V) - A_R^V u\| \le E. \tag{341}$$

To prove this, work in the common gauge above and restrict to the smaller radius $d = \lfloor B/4 \rfloor$. Extend u by zero outside this patch, obtaining v with $\|v\|_\infty \le H$. Let a be the linear replica of the coarse logarithm extension b; then $\|a\|_\infty \le H$ and $e^a = (e^b)^\uparrow$.

The local product logarithm satisfies

$$e^{w_e} = e^{v_e}e^{a_e}, \qquad \|w - (v + a)\|_\infty \le JH^2, \qquad \|w\|_\infty \le J'H.$$

For completeness, $(v, a) \mapsto \log(e^v e^a)$ is analytic near zero, vanishes at zero and has derivative $(v, a) \mapsto v + a$. Integrate its second derivative on a smaller closed ball to obtain the quadratic remainder. The first-derivative bound gives the last inequality, bond by bond, with constants independent of the number of bonds. Locality identifies the actual central rows with those at (e^w, e^b) and (e^a, e^b).

Put $z = \mathcal{C}^0(w, b)$ and $z_0 = \mathcal{C}^0(a, b)$. Linearity gives the exact decomposition

$$\begin{aligned}\mathcal{C}(U, V) - \mathcal{C}(V^\uparrow, V) - A_R^V u = {} & \mathcal{C}(U, V) - z + A_\infty(w - v - a) \\ & + z_0 - \mathcal{C}(V^\uparrow, V) + (A_\infty - A_R)v + (A_R - A_R^V)v.\end{aligned}$$

The equality uses locality to replace u by v in the transported row. Equations (338)– (340) bound these five terms by

$$2E_0 + CJH^2 + CHe^{-\sigma R} + C'BH^2,$$

where E_0 has upper exponent -1. This proves the error bound in (341); the bound on the row itself follows from its linear bound plus E_0. Gauge covariance of all three actual rows and invariance of the colour norm return the estimate to the original gauge.

72.3 Subtracting the selected squares

Let g be the coarse guide row and assume the already established guide bound $\|g\| \le At$. Define the actual and affine residuals

$$v = \mathcal{C}(U, V) - g, \qquad v_0 = \mathcal{C}(V^{\uparrow}, V) - g + A_R^V u.$$

They satisfy $\|v\| \le m = M + At$ and $\|v - v_0\| \le E$. The elementary identity for a difference of squares implies

$$\left| \frac{\|v\|^2}{2} - \frac{\|v_0\|^2}{2} \right| \le Z := \frac{E(2m + E)}{2}. \tag{342}$$

Indeed $|\|v\| - \|v_0\|| \le E$ and $\|v_0\| \le m + E$. Hence Z has upper exponent $-3/2$.

Write $\operatorname{flaw}(q) = (r(q)^2 - \|F(q)\|^2)/2$. The curvature argument in Section 71 gives

$$0 \le \operatorname{flaw}(q) \le K r(q)^4/2, \qquad \|F(q)\|^2 \ge 3r(q)^2/4$$

for small $r(q)$. The branch and plaquette bounds above put the required arguments in this neighbourhood. Since the appropriate direct component of $\mathcal{C}$ has norm at most M,

$$r(q_f) \le 2M/\sqrt{a_f}, \qquad r(q_I) \le 2M/\sqrt{p}.$$

Thus their flaw contributions are bounded by

$$X = a_f (K/2)(2M/\sqrt{a_f})^4, \qquad Y = p(K/2)(2M/\sqrt{p})^4,$$

both of upper exponent -2.

On a kept fine row below its threshold the fine energy is exactly

$$a_f \operatorname{flaw}(q_f) + \frac{1}{2} \sum_{j=0}^{1} \|v_{f,j}\|^2.$$

Expand the squares to verify this: their cross terms are the guide pairings, and their direct curvature square together with a_f flaw restores $a_f r(q_f)^2/2$. The predictor energy has, without a kept condition, the exact form

$$p \operatorname{flaw}(q_I) + \|v_I\|^2/2.$$

It follows from (342) that subtracting the two affine fine squares costs at most $X + 2Z$ in absolute value, and subtracting the predictor square costs at most $Y + Z$. One common nonnegative bound is $X + Y + 2Z$, of upper exponent $-3/2$. At an unselected fine or predictor address no square is subtracted. Its corresponding displayed energy is nonnegative. This gives the required one-sided bound there, without asserting an absolute bound on an unselected row.

72.4 The remaining coarse rows

Three further terms occur in the assigned energy. First, on a coarse plaquette with $r \le s$, the coarse counterterm is

$$\frac{a_c}{2}\{\|F(q)\|^2 - r(q)^2\}.$$

It is nonpositive and its absolute value is at most $a_c K s^4/2$, of upper exponent -2.

Second, consider a guide kernel $K(V)$ at one central row. On the clean patch its clipped coarse field agrees with the ordinary truncated curvature field at every address touched by this row: the

curvature is below threshold, and the exponential kept test is automatic outside the smaller radius where it needs checking. Kernel locality and the local agreement $V = e^b$ then give the exact central-row equality

$$[K(V)\,\mathrm{vision}(V)]_i = [K(e^b)X_c(e^b)]_i.$$

This equality is taken before any supremum norm is used. Add and subtract $K(e^b)X_c^0(b)$ to obtain

$$\big\|[K(V)\,\mathrm{vision}(V) - K_0X_c^0(b)]_i\big\| \le C\|X_c(e^b) - X_c^0(b)\|_\infty + C\omega(R,H)\|X_c^0(b)\|_\infty,$$

where $\omega(R,H) = (1+R)H + e^{-\tau R}$. The row bounds in (316) justify this inequality. The globally small extension b permits both supremum estimates on the right; no global bound on the original V is required. The right side has upper exponent -1. For the null row $K = N$, its full linear comparator vanishes by (336). The null energy, half the square of this row, consequently has upper exponent -2. This is the only extra flat identity used here.

Finally, the linear and quadratic mismatch rows are precisely the gaps estimated in (318). The common local gauge and locality argument used there apply to the present grafted configuration, with smallness bound H of upper exponent $-1/2$. Pairing the linear gap with a guide of size At gives at most $At\,CH\omega(R,H)$ for each of the two fine-row channels. Pairing the quadratic gap with that guide gives at most $At\,CAt\,\omega(R,H)/2$ per channel. Hence their combined absolute cost is at most

$$2At\,CH\omega(R,H) + (At)^2C\omega(R,H),$$

of upper exponent $-3/2$. Adding all row costs gives one nonnegative function $r(x)$ of upper exponent $-3/2$.

72.5 The protected annulus of an actual packet

We verify the local hypotheses on the packet itself. Its fine configuration is formed in a fixed tree slice. On active free bonds it is $\exp(u_e)V_e^{\uparrow}$; on frozen free bonds it is the fixed slice value Y_e; on tree bonds it is the identity. Let $K > 0$ be a fixed shell constant and define

$$\ell_K(q) = \chi\big((r(q)/(Kt))^2\big)\log q,$$

where the smooth cutoff equals one for $r(q) \le Kt$ and vanishes for $r(q) \ge 2Kt$. The local logarithm estimate gives $\|\ell_K(q)\| \le 2\gamma Kt$ everywhere. This assertion also holds outside the chart because the cutoff is zero there; choose x so that the closed ball $r(q) \le 2Kt$ lies in the logarithm chart. The resulting map is smooth wherever its group argument varies smoothly, by the same two local cases.

Define the actual colour signal $\widetilde{u}$ to be u on active bonds, $\ell_K(Y_e(V_e^{\uparrow})^{-1})$ on frozen free bonds, and zero on tree bonds. On a weak interior patch the fine plaquettes have chord at most $t/2$, the predictor mismatch at most $2Wt$, and the selected frames have positive likelihood. The tree comparison (305) then gives

$$r(U_e(V_e^{\uparrow})^{-1}) \le Kt, \qquad K = C\,N_{\mathrm{macro}}/2 + 2W,$$

where N_{macro} counts the fine plaquettes in the fixed macro patch. The frozen cutoff is therefore one and its logarithm exponentiates to the original mismatch. On tree bonds both U_e and $V_e^{\uparrow}$ are the identity. Thus on this patch

$$U_e = \exp(\widetilde{u}_e)V_e^{\uparrow}.$$

If the active variables there obey $\|u_e\| \le w(x)$ and $2\gamma Kt \le w(x)$, the signal obeys the same bound on all bonds. This proves (337) for the actual signal.

Let A be the packet's seed set, and consider an energy address whose coarse anchor belongs to its $200B$ neighbourhood but not its $2B$ neighbourhood. Retain the weak fine, predictor and coarse flag conditions through $300B$, and the bounds on active variables through $304B$. For a bond within B of this anchor, its $B/4$ interior patch lies within the $300B$ neighbourhood and outside the seeds. This follows from the triangle inequality: the outer distance is at most $201B + B/4 < 300B$, whereas entering the seeds would put the anchor within $B + B/4 < 2B$ of them. The retained conditions therefore give the weak interior just used. The active-variable condition is available since $201B < 304B$. The same argument supplies a clean coarse patch at the anchor.

Signed changes of orientation preserve these conclusions, allowing one additional bond of radius for reversed bonds. Norms are unchanged by the associated conjugations. Finally, the fine kept test holds on this annulus: a plaquette whose test is nontrivial has distance at most $B/25$ by the exponential threshold bound; the weak interior makes its chord at most $t/2 \le te^{\mathrm{dist}\,/r_0}$. At all other plaquettes the right side is at least the universal chord bound 2. Thus the actual fine energy equals the square expression used above, including at threshold equality.

At every selected address the extracted affine square is the square of the actual signal row with the same tags, orientations and coarse field. At every unselected address it is zero. We have therefore proved, on the protected annulus,

$$\text{actual assigned energy at the address } \ge -r(x). \tag{343}$$

Here $r \ge 0$ has upper exponent $-3/2$. Summing over its at most CB^4 addresses per seed and averaging over the signed channels gives exactly the annular input of (425). This proves the value estimate; derivative bounds for the density, covariance and chart factors remain separate inputs to the packet jet estimate.

73 The Gaussian quadratic form in a sector

The quadratic reference must remain nondegenerate after the large-field bonds have been frozen. A lower bound for a flat Hessian alone would not establish this. We prove the lower bound for the actual transported, selected rows, using a separate local gauge around each active bond. All norms in the quadratic inequalities below are physical counting norms: squared colour norms are summed over bonds without dividing by the volume.

Fix the tree, seed set, coarse field and original predictor tags of Section 72. Let $\mathcal{H}$ be the active free bonds and let $P_{\mathcal{H}}$ denote restriction to them, extended by zero on all other bonds. The seeds include exactly the actual coarse flags. An active bond has both coarse endpoints outside the $10B$ seed neighbourhood. A row is selected when its anchor is outside the $6B$ seed neighbourhood. These are fixed selections for this quadratic form.

Let q be the number of signed channels and write $\lambda = q^{-1/2}$. For a channel s and a fine or predictor row a, set

$$D_{s,a}u = \mathbf{1}_{\{(s,a)\ mselected\}} A^V_{s,a} P_{\mathcal{H}} u.$$

The map $A^V_{s,a}$ is the actual transported cut row, including the signed displacement of input colours. In orthonormal colour coordinates, define M by

$$(Mu)_{s,a} = \lambda D_{s,a} u.$$

Here and below a colour array and its scalar coordinate vector are identified. Define the completed precision matrix

$$A = M^*M + P_{\mathcal{H}^c}. \tag{344}$$

Every column of M indexed by an inactive bond is zero. Consequently A acts as the identity on the inactive coordinates and has no cross terms between them and the active coordinates. The completion supplies no energy on the active subspace.

If b is the vector of selected affine rows at zero fluctuation, including the same factor λ, linearity of the transported rows gives

$$\mathcal{Q}(u) = \tfrac{1}{2}\|b + Mu\|_2^2.$$

This is exactly the orientation average of the extracted squares: $\lambda^2 = 1/q$ supplies its normalization. In particular,

$$\mathcal{Q}(v + u) - \mathcal{Q}(v) = \langle u, M^*b + M^*Mv\rangle + \tfrac{1}{2}\langle u, M^*Mu\rangle.$$

The same fine field, tags and orientation choices are used in every term. We have not averaged independently chosen Gaussian matrices.

73.1 A local linear inequality from the tree comparison

Let w be a fine colour array which vanishes on the fixed tree. The tree comparison of Section 68, with coarse configuration equal to the identity, implies

$$\|w_e\| \le K \sum_{z\in\mathcal{B}} \sum_{\alpha} \|(d_f w)_{\bar{z}+Lo,\alpha}\| + \|(\mathcal{Q}w)_{o,a}\|, \qquad o = \mathrm{parent}(e), \tag{345}$$

where a is the axis of e, $\mathcal{B}$ is the fixed macro box, and $\mathcal{Q}$ is the derivative of the actual predictor. The box sum may count periodic images more than once; this only enlarges its nonnegative right side.

Here is the passage to the linear inequality. Apply the nonlinear comparison to $U_e(\varepsilon) = \exp(\varepsilon w_e)$. It remains in the same tree slice since w vanishes on the tree. For sufficiently small ε all required frame likelihoods are positive: at zero every frame is in the minimizing branch, and the finitely many conditions are open. Divide the comparison by $|\varepsilon|$ and let $\varepsilon \to 0$ through nonzero real values. If $g(0) = 1$ and its matrix derivative is the colour matrix of v, then

$$\lim_{\varepsilon\to 0} \frac{r(g(\varepsilon))}{|\varepsilon|} = \|v\|.$$

This follows directly from differentiability and the physical normalization of the chord. The derivatives of the fine links, plaquettes and predictor are respectively w, $d_f w$ and $\mathcal{Q}w$, the last by Section 69. Passing through the finite box sum proves (345). The limiting neighbourhood can depend on the finite torus; the resulting constant K, inherited from the nonlinear tree inequality, does not.

73.2 A lower bound for the actual selected rows

For an anchor o, define

$$J_o(u)^2 = \sum_{\mathrm{dist}(o,\mathrm{parent}(e))\le 3B} \|(P_{\mathcal{H}}u)_e\|^2,$$

$$R_o(u)^2 = \sum_{\mathrm{dist}(o,\mathrm{anchor}(s,a))\le R_{\mathrm{macro}}+2} \|D_{s,a}u\|^2.$$

The square roots are nonnegative. An active bond at o is more than $10B$ from the seeds. Since the macro radius is much smaller than B, every row with anchor within $R_{\mathrm{macro}} + 2$ of o is selected

in the identity signed channel. Also the coarse patch of radius $2B+1$ is clean, because the coarse flags belong to the seeds. The local axial construction gives a gauge g with

$$r((g\cdot V)_e)\le h_0(x)\quad\text{on radius } 2B,$$

where $h_0\ge 0$ has upper exponent $-1/2$.

Apply (345) to $w=\mathrm{Ad}(g)P_{\mathcal{H}}u$. This array still vanishes on the tree. For every required row, the transport estimate (340) and gauge covariance give

$$\|(A_Rw)_a\|\le R_o(u)+\varepsilon(x)J_o(u),\qquad \varepsilon(x)=C\,2D_{\mathrm{col}}(B+1)h_0(x). \tag{346}$$

Indeed the actual row in the chosen gauge has the same norm as its selected original row and is at most $R_o(u)$. Every input used in the comparison lies within $3B$ of o, so its norm is at most $J_o(u)$. All comparison routes lie inside the radius $2B$ patch. The identity signed channel suffices for this lower bound; the sum defining R_o may include all the other channels.

The direct fine component of A_Rw is $\sqrt{a_f}\,d_fw$, and its predictor component is $-\sqrt{p}\,\mathcal{Q}w$. Writing $N_{\mathrm{macro}}=|\mathcal{B}|N_{\mathrm{plaq}}$, insert (346) into (345). We obtain

$$\|(P_{\mathcal{H}}u)_e\|\le ZR_o(u)+E(x)J_o(u),\quad Z=KN_{\mathrm{macro}}/\sqrt{a_f}+1/\sqrt{p},\quad E=Z\varepsilon. \tag{347}$$

This also holds on inactive bonds because its left side is zero. Z is fixed after L; E has upper exponent $-1/2$.

There are L^4N_{axis} fine positive bonds per coarse site. Let

$$S_L(d)=(2d+1)^4L^4N_{\mathrm{axis}}.$$

For every nonnegative family f_a anchored on coarse sites, changing the order of the two finite sums shows

$$\sum_e\sum_{\mathrm{dist}(\mathrm{parent}(e),\mathrm{anchor}(a))\le d} f_a\le S_L(d)\sum_a f_a.$$

For fixed a, at most $(2d+1)^4$ coarse sites can occur, each with at most L^4N_{axis} bonds. This remains valid when the neighbourhood wraps around the torus: wrapping reduces the number of distinct sites.

Square (347), using $(r+s)^2\le 2r^2+2s^2$, and sum over bonds. The counting bound gives

$$\|P_{\mathcal{H}}u\|_2^2\le 2Z^2S_L(R_{\mathrm{macro}}+2)\sum_{s,a}\|D_{s,a}u\|^2+2E(x)^2S_L(3B)\|P_{\mathcal{H}}u\|_2^2.$$

The last coefficient has upper exponent -1 and is eventually at most $1/2$. Move it to the left to get

$$\|P_{\mathcal{H}}u\|_2^2\le a_*\sum_{s,a}\|D_{s,a}u\|^2,\qquad a_*=1+4Z^2S_L(R_{\mathrm{macro}}+2).$$

The added one ensures a strictly positive constant in every case. Since $\|Mu\|_2^2=\lambda^2\sum\|D_{s,a}u\|^2$, the orthogonal active/inactive decomposition proves

$$\langle u,Au\rangle\ge\kappa\|u\|_2^2,\qquad \kappa=\min\{\lambda^2/a_*,1\}>0. \tag{348}$$

This includes an empty active set, in which case $A=1$. No globally small gauge for V has been constructed or assumed.

73.3 Upper bounds with spatial weights

We also need a bound independent of the period on the matrix entries and their weighted sums. The cut row A_R vanishes on an impulse at a bond farther than $R+\rho+4$ from its anchor. Together with (339), this implies

$$e^{\sigma d}\|(A_R\delta_e v)_a\| \le 2Ce^{\sigma(\rho+6)}\|v\|, \qquad d = \operatorname{dist}(\operatorname{anchor}(a), \operatorname{parent}(e)).$$

To see this, if $d \le \rho+5$ use the uniform cut-row bound. Otherwise take $r = d-(\rho+5)$, so that $A_r\delta_e v = 0$. If $R < r$ locality gives zero as well. For $R \ge r$, compare both A_R and A_r with A_∞; the two tails are at most $C\|v\|e^{-\sigma r}$ each. This proves the stated bound.

Signed displacement and parallel transport send a single impulse to a single impulse with unchanged colour norm. The coarse anchor may move by one under reversal of a bond. Thus the preceding constant becomes $2Ce^{\sigma(\rho+7)}$ for the actual oriented row. Selection and restriction to active bonds only set entries to zero. In scalar coordinates,

$$e^{\sigma \operatorname{dist}(i,j)}|M_{ij}| \le b_* = \lambda\, 2Ce^{\sigma(\rho+7)}.$$

The anchors of input coordinates have multiplicity at most $m = L^4 N_{\mathrm{axis}} q_{\mathrm{col}}$, where q_{col} is the colour dimension. Output-coordinate anchors have multiplicity at most

$$n = q\,(2L^4 N_{\mathrm{plaq}} + N_{\mathrm{axis}})q_{\mathrm{col}}.$$

For $\eta > 0$ put $P_0(\eta) = \sum_{k\ge 0}(2k+1)^4 e^{-\eta k} < \infty$. Grouping coordinates by their distance from an anchor gives, for $0 \le r \le \sigma/2$,

$$\sup_i \sum_j e^{r\operatorname{dist}(i,j)}|M_{ij}| \le b_* m P_0(\sigma/2), \qquad \sup_j \sum_i e^{r\operatorname{dist}(i,j)}|M_{ij}| \le b_* n P_0(\sigma/2).$$

Choose one constant C_* larger than these two bounds and one. The triangle inequality for anchor distances proves the product rule for such weighted row and column sums. Therefore

$$\sup_i \sum_j e^{r\operatorname{dist}(i,j)}|A_{ij}| \le C_*^2 + 1. \tag{349}$$

The extra one is the inactive diagonal. Symmetry gives the same column bound. The usual finite Cauchy–Schwarz argument gives $\|A\|_{2\to 2} \le C_*^2 + 1$: bound $|(Au)_i|^2$ by the absolute row sum times $\sum_j |A_{ij}||u_j|^2$ and then sum over i. Also $M_{ij} = 0$ beyond distance $B/4$ by its stencil, so $A_{ij} = 0$ beyond $B/2$.

Combining (348) and (349) proves a strictly positive, uniform spectral interval for the actual completed precision matrix. In particular its inverse exists and satisfies $0 < A^{-1} \le \kappa^{-1}1$. For $\beta > 0$, the centered finite-dimensional Gaussian with covariance $\beta^{-1}A^{-1}$ is consequently well defined, including when there are no active coordinates. Restricting to the active subspace gives the physical Gaussian; the inactive identity coordinates are an algebraic completion. Spatial locality of the inverse, comparison of local Gaussian normalizations, and control of the affine mean require the additional estimates developed at their respective steps.

74 Local inverses and Gaussian normalization

The Gaussian covariance used in the expansion is assembled from local principal inverses. It is not the full inverse of the precision matrix. We prove its positivity, estimate the resulting error, and

express the normalization through local terms and convergent correction series. This follows the real interpolation from the identity used in the construction.

Let I be a finite coordinate set, with anchors a_i on a coarse four-dimensional torus. Assume at most m coordinates have any given anchor. For a real matrix T define

$$\|T\|_\eta = \max\left\{ \sup_i \sum_j e^{\eta\,\mathrm{dist}(a_i,a_j)}|T_{ij}|, \sup_j \sum_i e^{\eta\,\mathrm{dist}(a_i,a_j)}|T_{ij}| \right\}.$$

On an empty coordinate set both suprema are zero. The triangle inequality for distance and an interchange of finite sums give $\|ST\|_\eta \le \|S\|_\eta\|T\|_\eta$ for $\eta \ge 0$. Transposition preserves this norm, and decreasing η cannot increase it. It dominates the Euclidean operator norm by the row-and-column Cauchy–Schwarz estimate used above.

Let $A = A^*$ and suppose

$$A \ge r1, \qquad \|A\|_\eta \le Q, \qquad 0 < r \le 1, \quad Q \ge 1, \quad \eta > 0. \tag{350}$$

Section 73 supplies these constants for the actual completed sector precision, after fixing L and then taking the large parameter sufficiently large. Every principal submatrix has the same lower bound and weighted estimate: extend a test vector by zero for the first assertion, and restrict the nonnegative row and column sums for the second.

74.1 Exponential decay of the inverse

Set

$$\theta = \min\{1, r/(2Q)\}, \qquad \gamma = \theta\eta, \qquad c_0 = \min\{1, r/2\}.$$

For every $u \ge 0$, the matrix $A + u1$ is invertible and

$$e^{\gamma\,\mathrm{dist}(a_i,a_j)}|(A+u1)^{-1}_{ij}| \le \{c_0(1+u)\}^{-1}. \tag{351}$$

We give the weighted argument, which is uniform in the number of coordinates. Fix j and put $s_i = e^{\gamma\,\mathrm{dist}(a_i,a_j)}$. For $0 \le \theta \le 1$ and $|v| \le \theta D$ with $D \ge 0$, convexity of the exponential gives

$$|e^v - 1| \le \theta e^D.$$

For positive v, use $e^{\theta D} \le 1 + \theta(e^D - 1)$; for negative v, use $1 - e^v \le e^{-v} - 1$. The reverse triangle inequality for distances now implies

$$|s_i/s_k - 1| \le \theta e^{\eta\,\mathrm{dist}(a_i,a_k)}.$$

Conjugate $A + u1$ by the positive diagonal matrix with entries s_i. Its difference F from $A + u1$ has entries $(A + u1)_{ik}(s_i/s_k - 1)$. The diagonal contribution is zero, so both absolute row and column sums of F are at most $\theta Q \le r/2$. Thus $|\langle w, Fw\rangle| \le (r/2)\|w\|_2^2$.

Let $y = (A + u1)^{-1}e_j$ and $w_i = s_i y_i$. The conjugated equation is $(A + u1 + F)w = e_j$, since $s_j = 1$. It follows that

$$(r/2 + u)\|w\|_2^2 \le w_j \le \|w\|_2.$$

If $w \ne 0$, division gives $\|w\|_2 \le (r/2 + u)^{-1} \le \{c_0(1 + u)\}^{-1}$; the case $w = 0$ is immediate. Each $|w_i|$ is bounded by this norm, proving (351). Positive definiteness gave invertibility before this computation. The same proof applies to every principal submatrix.

Put $k = \gamma/3$ and

$$S = c_0^{-1} m P_0(\gamma - 2k), \qquad P_0(t) = \sum_{n \geq 0} (2n+1)^4 e^{-tn}.$$

Summing (351) at $u = 0$ with weight $e^{2k\,\mathrm{dist}}$ gives $\|A^{-1}\|_{2k} \leq S$. The summation uses the anchor multiplicity m and the same distance-shell count as in Section 73.

74.2 Principal windows and their error

For each coordinate j, choose a window W_j containing every coordinate within distance B of a_j and no coordinate farther than $B + 2$. Define P by its columns:

$$P_{ij} = \begin{cases} (A_{W_j}^{-1})_{ij}, & i \in W_j, \\ 0, & i \notin W_j. \end{cases}$$

Indices inside a window mean the corresponding principal coordinates. All these inverses exist by (350). Applying (351) separately to each column yields the same pointwise exponential bound for P, hence $\|P\|_{2k} \leq S$. This also controls column sums even though the windows vary with the column.

Let $E = AP - 1$. If $i \in W_j$, the principal inverse equation gives $E_{ij} = 0$: the column $P_{\cdot j}$ is zero outside W_j, so the full product at i is exactly the principal product. In particular $E_{ij} = 0$ when the anchor distance is at most B. The product bound gives $\|E\|_{2k} \leq QS + 1$. On its remaining entries the weight can be reduced from $2k$ to k at cost e^{-kB}. Therefore

$$\|E\|_k \leq \epsilon_B := e^{-kB}(QS + 1), \qquad \|P - A^{-1}\|_k \leq S\epsilon_B. \tag{352}$$

The second inequality uses the exact identity $P - A^{-1} = A^{-1}E$. Notice the order of the factors.

The physical covariance is the symmetric matrix

$$C = \tfrac{1}{2}(P + P^*). \tag{353}$$

Symmetry of A^{-1} and (352) give

$$\|C\|_k \leq S, \qquad \|C - A^{-1}\|_k \leq \delta_B := S\epsilon_B, \qquad \|AC - 1\|_k \leq Q\delta_B.$$

The upper bound $A \leq Q1$ implies $A^{-1} \geq Q^{-1}1$. It can also be checked directly: for $w = A^{-1}v$ and $x = Q^{-1}v$, positivity of $\langle w - x, A(w - x)\rangle$ and $\langle x, Ax\rangle \leq Q\|x\|^2$ give $\langle v, A^{-1}v\rangle \geq Q^{-1}\|v\|^2$. Consequently, once $\delta_B \leq 1/(2Q)$,

$$C \geq (2Q)^{-1}1. \tag{354}$$

The actual bond windows are colour independent: a coordinate belongs to W_j when some coarse endpoint of its bond is within B of some endpoint of the bond of j. Each endpoint is within one of its bond's parent anchor. This proves the required B and $B + 2$ bounds and shows that complete colour blocks occur together. The matrix C has range at most $B + 2$. Its closeness to A^{-1} is a conclusion of the preceding estimates, not a replacement of one matrix by the other.

For the sector precision, inactive rows and columns of A are those of the identity. This property passes to every principal inverse, hence to P and C. Also M^*b vanishes on inactive coordinates, so the chosen local mean $-P^*M^*b$ vanishes there. These observations justify restricting the local Gaussian to the active coordinates without changing its physical entries.

74.3 Ordered inverse series

If a matrix E satisfies $\|E\|_k \le \epsilon < 1$, then

$$(1+E)^{-1} = \sum_{n=0}^{\infty} (-E)^n.$$

Indeed the partial geometric identity has remainder $(-E)^N$, whose norm tends to zero; the series converges absolutely in the finite matrix space. For $E = AP - 1$ this gives

$$A^{-1} = \sum_{n \ge 0} P(-E)^n. \tag{355}$$

For $D = AC - 1$ the corresponding identity is

$$C^{-1} = \sum_{n \ge 0} (-D)^n A, \qquad C^{-1} - A = \sum_{n \ge 0} (-D)^{n+1} A. \tag{356}$$

Multiply these formulas by A on the left and by C on the right, respectively, to verify them. No commutation of P, C and A is asserted. Since $B \asymp (\log x)^2$ and $k > 0$ is fixed after L, ϵ_B and δ_B have every negative upper exponent. Both convergence conditions can thus be met on one sufficiently late tail, uniformly over sectors and periods.

74.4 Logarithms as local terms and correction series

For $0 \le s \le 1$ put $A_s = 1 + s(A-1)$. The hypotheses (350) hold for A_s with the same r, Q, η, since $r \le 1 \le Q$. Let P_s be its window inverse and $E_s = A_s P_s - 1$. All bounds above are uniform in s. Define the local diagonal terms and their correction terms by

$$\ell_i = \int_0^1 [(A-1)P_s]_{ii}\, ds, \qquad \ell_{i,n} = \int_0^1 [(A-1)P_s(-E_s)^n]_{ii}\, ds.$$

Thus $\ell_i = \ell_{i,0}$. We have the exact identity

$$\log\det A = \sum_i \ell_i + \sum_i \sum_{n \ge 1} \ell_{i,n}. \tag{357}$$

To justify it, the determinant's linear term at the identity is the trace, as follows by expanding the finite determinant by permutations. Factoring

$$A_{s+t} = (1 + t(A-1)A_s^{-1})A_s$$

therefore gives $\frac{d}{ds} \log\det A_s = \operatorname{tr}((A-1)A_s^{-1})$. All A_s are positive definite. Their inverses and principal inverses vary continuously, by the adjugate formula and the nonvanishing determinant. Integrate the derivative from zero to one and substitute (355). Its diagonal terms are bounded, uniformly in s, by $(Q+1)S\epsilon_B^n$. This summable geometric bound justifies termwise integration for each i. The coordinate sum is finite. In particular,

$$|\ell_i| \le (Q+1)S, \qquad \sum_{n \ge 1} |\ell_{i,n}| \le \frac{(Q+1)S\epsilon_B}{1-\epsilon_B}.$$

These are per-coordinate bounds. Summing them over the entire torus introduces its coordinate count; it does not yield a per-seed estimate without the further support assignment.

There is a second determinant correction for the covariance C. With $D = AC - 1$ and $\|D\|_k < 1$, the path $1 + sD$ is invertible for $s \in [0,1]$ by its Neumann series. Its real determinant starts at one and never vanishes, so it remains positive. The same determinant derivative and geometric domination give

$$\log\det A + \log\det C = \sum_i \sum_{n\geq 0} \int_0^1 [D(-sD)^n]_{ii}\, ds. \tag{358}$$

Here $1 + D = AC$ and both A, C have positive determinant, so multiplicativity of the determinant justifies the sum of logarithms. Although D need not be symmetric, no spectral logarithm of D is used. The integral of its nth term is $(-1)^n (D^{n+1})_{ii}/(n+1)$, with absolute value at most $\|D\|_k^{n+1}/(n+1)$.

74.5 Local comparison of normalization terms

For a physical bond v, group its colour coordinates and set $J_v(A) = \sum_{i:\, i \text{ lies over } v} \ell_i$. The term ℓ_i reads only the principal window W_i:

$$[(A-1)P_s]_{ii} = [(A_{W_i} - 1)(1 + s(A_{W_i} - 1))^{-1}]_{ii}.$$

This follows because column i of P_s vanishes outside W_i. Suppose A and Z both satisfy (350), use the same windows, and differ entrywise by at most e on $W_i \times W_i$. The window has at most $\Delta = m(2B+5)^4$ coordinates. Thus the unweighted row and column norms of $A_{W_i} - Z_{W_i}$ are at most Δe. The same bound holds for the difference of their interpolants. The inverse identity, with its order retained, is

$$(Z_s)_{W_i}^{-1} - (A_s)_{W_i}^{-1} = (Z_s)_{W_i}^{-1}((A_s)_{W_i} - (Z_s)_{W_i})(A_s)_{W_i}^{-1}.$$

Each inverse has row-and-column norm at most S. Splitting the difference of the two local products into a difference of first factors and a difference of inverses bounds its norm by

$$\{S + (Q+1)S^2\}\Delta e.$$

Integration and the colour sum give

$$|J_v(A) - J_v(Z)| \leq q_{\mathrm{col}}\{S + (Q+1)S^2\}\Delta e, \tag{359}$$

provided the entrywise hypothesis holds on the window for each colour over v.

Finally J_v is invariant under bondwise orthogonal changes of colour frame. The windows contain whole colour blocks, so such a change restricts to an orthogonal matrix on every window and commutes with the projection onto the colours over v. Principal interpolants and their inverses are conjugated by this matrix. The sum of diagonal entries over v is the trace after that projection; cyclicity of the finite trace proves invariance before integration. This is why one compares the whole colour trace, rather than assigning independent gauge-invariant meanings to its individual diagonal terms.

75 The Gaussian mean away from seeds

The local covariance is accompanied by a local mean. Its defining formula alone does not make the affine energy small: a uniform bound on the inverse would only preserve the size of the original source. The additional gain comes from the stationary linear response. We first prove its cancellation, and then compare it with the actual mean on one common clean neighbourhood.

75.1 The stationary response cancels every linear row

Fix the step, forest and full flat kernels of Section 70. Write $\mathcal{U}_F b$ for the fine stationary field in tree coordinates with coarse input b. It is the tree-straightened installed linear response. Its relative field is

$$v_F(b) = \mathcal{U}_F b - \overline{b},$$

where $\overline{b}$ is the linear coarse replica. On every forest bond this difference is zero. We use the actual diagonalization identity for this response: if K_f, K_c are the fine and coarse quadratic forms and $\mathcal{Q}$ is the rooted linear predictor,

$$K_c = \mathcal{U}_F^* K_f \mathcal{U}_F + p(1 - \mathcal{Q}\mathcal{U}_F)^*(1 - \mathcal{Q}\mathcal{U}_F). \tag{360}$$

These are the full physical forms, including the prescribed root correction. This is an incoming flat construction identity; it does not follow merely from positivity of the forms.

Let b_s and a_s be the pullbacks, by one signed lattice symmetry, of b and $\mathcal{U}_F b$. Set

$$w = B_c d_c b_s, \qquad u = B_f d_f a_s, \qquad z = \sqrt{p}\,(b_s - \mathcal{Q} a_s), \qquad U = F_0 w, \quad V = I_0 w.$$

All norms in this paragraph are counting Euclidean norms, summed over their physical sites and colours. The two curvature squares are the corresponding quadratic forms. Indeed expand each colour component in the curvature factorization $K = d^* B^* B d$; the predictor square is exactly the last term in (360). Thus that identity gives

$$\|u\|_2^2 + \|z\|_2^2 = \|w\|_2^2. \tag{361}$$

The same equality holds after the signed pullback: the two full quadratic forms are invariant, and the nonlinear predictor energy is invariant with the transported tags. Apply that last invariance to $\exp(tb)$ and $\exp(t\mathcal{U}_F b)$, divide by t^2 and let $t \to 0$. The derivative proved in Section 69 gives the stated predictor square. This also explains why the rooted response need not itself commute with the chosen symmetry.

The full stack identities give

$$N_0 w = 0, \qquad F_0^* U + I_0^* V = w, \qquad F_0^* u + I_0^* z = w.$$

The first is the extra identity $N_0 B_c d_c = 0$; the second follows from the stack isometry and the first, and the third is (312). Taking inner products with w gives

$$\|U\|_2^2 + \|V\|_2^2 = \|w\|_2^2, \qquad \langle U, u\rangle + \langle V, z\rangle = \|w\|_2^2.$$

Together with (361), these imply $\|U - u\|_2^2 + \|V - z\|_2^2 = 0$. Every summand is nonnegative, so $U = u$ and $V = z$ at every address. In the notation of affine energy rows, this says

$$C_0(\overline{b_s}, b_s) - G_0(b_s) + \mathcal{A}_\infty(v_F(b))_s = 0. \tag{362}$$

Here C_0 is the linear fine-curvature/predictor pair, G_0 the linear guide, and $\mathcal{A}_\infty$ the full fluctuation row; the subscript s includes the signed action on bonds. Linearity of C_0 in its fine input and commutation of the replica with the signed action give the displayed equation. In particular the cancellation holds for each row, not just after summing the energy.

We will also need a volume-independent bound

$$\|v_F(b)\|_\infty \le C_{\mathrm{resp}} \|b\|_\infty. \tag{363}$$

Here is the quantitative tree estimate behind this bound. Put $r_0 = (J+3)L$ and let the installed cross kernel have weighted row price C/L and width $\eta \geq 0$. At fine sites within tL of a fixed origin, moving the exponential weight costs at most $e^{\eta t}$. Consequently its action on a scalar coarse field bounded by that weight is at most

$$A/L, \qquad A = e^{\eta(12(J+3)+5)} e^{\eta}\, 4 \max(C, 1).$$

A tree hike has length at most $4L$. Each coordinate route in the root patch has length at most $8r_0$; the root correction is the difference of two averages of such path integrals. It is therefore bounded by $16 r_0 A/L$. Combining the hike and root correction gives $C_1 = (4 + 16(J+3))A$ for the straightening potential. Taking its bond gradient costs $2C_1$, so the extracted response costs at most $A + 2C_1$. Subtracting the replica costs another e^{η} in the weighted estimate. For a field of sup norm one all its colour coordinates satisfy the weighted premise, since the weight is at least one. Summing the absolute colour coordinates bounds the physical colour norm. Thus one may take

$$C_{\mathrm{resp}} = \max\{1, q \max(0, A + 2C_1 + e^{\eta})\},$$

where q is the colour dimension. Homogeneity proves (363) for arbitrary b. All tree and path lengths were bounded before periodization, so the estimate is uniform in the period and in the forest.

75.2 A common logarithmic comparison field

Retain the upper-exponent convention of Section 84. Away from the seed set the coarse field V is clean: its plaquette chords are bounded by the small threshold. If the ball of radius $d+1$ about o is clean, $d \leq 32B$, the axial construction gives a gauge with bond chords, on the radius-d ball, at most

$$16\{2(32B+1)+1\}^5 N_p\, 2\tau_c.$$

Here τ_c is the coarse small threshold and has upper exponent $-1/2$. Apply the local logarithm there and extend the logarithmic field by zero elsewhere. We obtain a single global field b and a coarse gauge g such that

$$g \cdot V = \exp b \text{ on the radius-}d \text{ ball}, \qquad \|b\|_{\infty} \leq w(x), \tag{364}$$

where $w \geq 0$ has upper exponent $-1/2$. The period and chart-radius conditions hold at one threshold depending only on the fixed step parameters. The zero extension is used for the full linear kernels; no smallness of V outside the ball is asserted.

Consider a row whose anchor is at distance greater than $20B$ from all seeds, and suppose (364) holds on its radius-$2B$ ball. In the original gauge use the candidate

$$v = \mathrm{Ad}(g^{-1} \circ \mathrm{parent}) v_F(b).$$

It vanishes on tree bonds and has norm at most $W = C_{\mathrm{resp}} w$. Within the stencil of this remote row all non-tree bonds are active. Thus the sector's affine signal agrees there with v, including its tree zeros. Gauge covariance transfers its actual affine row to the same b and $v_F(b)$.

Let $C(V)$ be the fine-curvature/predictor pair at the coarse replica, $G(V)$ the guide, and $\mathcal{A}_R^V$ the transported cut fluctuation row. Their residual is $C(V) - G(V) + \mathcal{A}_R^V v_F(b)$ in this common gauge. Subtract (362). The four errors are

$$\begin{aligned} &C(V) - C_0(\bar{b}, b), \qquad -(G(V) - G_0(b)), \\ &(\mathcal{A}_R^V - \mathcal{A}_R) v_F(b), \qquad (\mathcal{A}_R - \mathcal{A}_{\infty}) v_F(b). \end{aligned}$$

The first two have upper exponent -1, by the local Taylor and guide comparisons of Sections 70 and 72: their bounds are fixed multiples of $w\{(1+R)w + e^{-\sigma R}\}$. The clean condition makes the guide's clipping inactive on precisely the read stencil; it is removed before the comparison with the globally extended logarithm. For the third error, all coarse transport bonds have chord at most Cw on the common hull. The adjoint estimate and the absolute path price give a bound $C(B+1)wW$. The fourth is at most $CWe^{-\sigma R}$. The cutoff $R = \lfloor B/100000 \rfloor$ has an exponentially small tail of every prescribed negative upper exponent. We have therefore proved, for one $a(x) \geq 0$ of upper exponent -1,

$$\|\mathcal{R}_j(v)\| \leq a(x) \tag{365}$$

on every row whose comparison hull is included in the common clean ball. Here $\mathcal{R}_j$ is the actual unscaled affine row, with its original orientation and tags. One field b is used for all rows in the ball, so these estimates can be inserted into a matrix identity without changing the candidate from row to row.

75.3 Comparing the candidate with the chosen local mean

Use counting coordinates and the matrices of Sections 73 and 74. Write M for the selected row matrix, $A = M^*M + \Pi_{\text{inactive}}$, $\ell = M^*r(0)$ for the linear source, and

$$r(v) = Mv + r(0), \qquad P = \text{the padded principal inverse}, \qquad m = -P^*\ell, \quad E = AP - 1.$$

The row $r_j(v)$ is $s\mathcal{R}_j(v)$ when selected, and zero otherwise; $s > 0$ is the fixed signed-channel normalization. The chosen physical mean μ is obtained from m by reassembling its colour coordinates. We have the exact identity

$$m - v = E^*v - P^*(Av + \ell), \qquad Av + \ell = M^*r(v) + \Pi_{\text{inactive}}v. \tag{366}$$

For the first equality use $E^* = P^*A - 1$ and cancel P^*Av. The second follows directly from the definition of A and ℓ.

Let S bound both unweighted row and column sums of P, let ε_B bound those of E, and let Y bound those of M. These constants are supplied by the weighted estimates already proved; ε_B has every negative upper exponent. If the normal residual $Av + \ell$ is bounded by b_0 on the principal window attached to coordinate i, (366) gives

$$|m_i - v_i| \leq Sb_0 + \varepsilon_B\|v\|_\infty. \tag{367}$$

Indeed the ith row of P^* is supported on that very window, and the column bounds give both estimates. This uses P^* in the mean; replacing it by the symmetrized covariance would change the identity.

Fix a target row at anchor o farther than $30B$ from every seed. Choose (364) on the radius-$8B$ ball. For each coordinate within B of o, its principal window lies within $3B$ of o, since the window radius is at most $B+2$. At such window coordinates the inactive term vanishes: a non-tree bond is active there, and a tree bond has candidate value zero. A nonzero entry of M^* reads a row within $B/4$, so all its affine residuals obey (365), using subballs of the same radius-$8B$ comparison. Hence $b_0 = Ysa$, and

$$|m_i - v_i| \leq U(x) := SYs\,a(x) + \varepsilon_B W(x) \quad \text{when } d(o,i) \leq B.$$

The function U has upper exponent -1.

The target row is selected and reads only coordinates within $B/4$. Its difference on μ and v is linear, so the row bound on M and the passage from colour coordinates to the colour norm imply

$$\|\mathcal{R}_o(\mu) - \mathcal{R}_o(v)\| \le qs^{-1}YU.$$

Combining this with (365) proves

$$\|\mathcal{R}_o(\mu)\| \le a(x) + qs^{-1}YU(x), \qquad d(o, \text{seeds}) > 30B. \tag{368}$$

The right-hand side has upper exponent -1, uniformly in the period, forest, tags and all sectors satisfying the fixed construction conditions. This is the required extra order of smallness of the actual affine mean. The argument uses the flat stationary response and its diagonal identity; it makes no assertion about derivatives of the mean or a nonlinear global stationary solution.

76 Producing the warning plateaus

We prove the deterministic input to the warning estimate of Section 64. Fix the patch and the positive fine-action and predictor-penalty constants c_f, c_p. We retain the funded path bounds, the clean-sector precision bounds and the stationary response of Sections 73 and 75. Let $t = \text{fiery}(x)$, $B = \text{radius}(x)$ and $q = \dim \mathfrak{g}$. We use δ_0 for a fixed small error allowance, to distinguish it from the field scale t.

The guide estimate supplies a constant $A > 0$, chosen before L, such that the fine guide has norm at most At/L^2 and the predictor guide at most At. Choose

$$\delta_0 = \frac{\sqrt{c_f}}{32}, \qquad \frac{A}{L^2} \le \delta_0, \qquad W > \frac{2}{\sqrt{c_p}}(A + \delta_0). \tag{369}$$

For example, any integer lower bound strictly larger than A/δ_0 suffices for L, since $L \ge 1$. We will choose the remaining constants after L and W, and finally one lower bound for x. None of these choices depends on the lattice period or on the seed allocation.

76.1 A uniform bound for the affine mean

First we need $\|\mu_e\| \le Ct$ with a fixed post-L constant C. A bound with upper exponent $-1/2$ alone would not imply this stronger comparison with the particular scale t.

On a clean ball of fixed radius n, the axial gauge construction of Section 65 bounds every coarse bond chord by $K_n t$, where one may take

$$K_n = 32(2(n+1)+1)^5 \#\{\text{plaquette axes}\}.$$

The number n is fixed before x. For a fine plaquette in the coarse replica, use $n = 4$: each of its four links has chord at most $K_4 t$ in that gauge, so its curvature norm is at most $4K_4 t$. Every path term retained by the fine cutoff leaves the required radius-five margin at its endpoint inside the clean B-ball: its endpoint separation plus five is at most B. Apply the same radius-four argument separately at that endpoint. Unitary transport preserves its norm; summing the absolute path coefficients bounds the entire fine base stroke by

$$4K_4(\sqrt{c_f} + P_f)t,$$

where P_f is the funded absolute fine-path price. This is a sum of individual fixed-radius estimates; no gauge on the whole growing B-ball is required for this bound.

For a predictor base stroke, take n equal to the fixed predictor read radius. The logarithm chart extends the locally gauged coarse field to a global logarithmic field of norm at most $K'_n t$, while preserving the actual links on that stencil. Its linear replica has no larger norm. The fixed-patch Taylor estimate from Section 69 bounds the predictor by $J(\|w\| + \|w\|^2)$ for the pair of logarithmic fields w. Eventually $\|w\| \le K'_n t \le 1$, so this is at most $2JK'_n t$. Locality and gauge covariance identify it with the original predictor base stroke for every original tag choice. We have therefore proved a common bound $A_1 t$ for both base-stroke types on clean balls.

Let $s > 0$ be the fixed row normalization, N_1 the row operator bound, and γ, K the shell-access constants. The saved frozen logarithmic field is bounded by $2\gamma K t$. A selected affine row at zero is the base stroke minus its guide plus the row applied to this saved field. Unselected rows vanish. Hence all normalized affine rows have norm at most $b_1 t$, where

$$b_1 = s(A_1 + A + 2N_1\gamma K).$$

Let Y bound the absolute column sums of the row matrix, and let P_0 bound both row and column sums of the unsymmetrized local inverse P. The affine linear coefficient ℓ and the mean $m = -P^T\ell$ then satisfy coordinatewise

$$|\ell_i| \le Y b_1 t, \qquad |m_i| \le P_0 Y b_1 t, \qquad \|\mu_e\| \le q P_0 Y b_1 t.$$

Each inequality is the elementary bound $|\sum_j M_{ij} v_j| \le (\sum_j |M_{ij}|) \sup_j |v_j|$; the last sums the q colour coordinates. Choose once and for all

$$C = 1 + \max\{0, b_1, Y b_1, q P_0 Y b_1\} > 0. \tag{370}$$

The same algebra gives the residual identity $Am + \ell = -A(P - A^{-1})^T \ell$. Its entries are bounded by a fixed multiple of $\mathrm{distPay}(B)t$, and hence have every prescribed negative upper exponent by the local inverse tail estimate. The mean and its residual thus come from the same affine system.

76.2 The genuine local minimizing branches

We also need exact likelihood values, before restoring other masks. Fix any $\varepsilon > 0$. There exist a radius $r > 0$ and a finite read distance d, depending on the patch and ε, such that if all fine bond chords within distance d of a coarse site are at most r, then its site likelihood equals one, every outgoing bond likelihood equals one, and the predictor bonds have chord at most ε.

Here is the uniformity argument. On the covering lattice, the identity configuration lies in the open unique-minimizer branches for both the frame and the connector. Their values there are the identity. The continuity and uniqueness proved in Section 69 imply a neighbourhood on which these branches remain selected and each predictor chord is below ε. Intersect these neighbourhoods over the finitely many axes and frame-tag pairs. A product-topology neighbourhood contains one obtained by restricting finitely many covering bonds to chord balls of a common positive radius r. Enclose their anchors in a ball of finite read distance d.

Pulling a periodic configuration to the cover preserves all these finite reads, even when some periodic bonds coincide. Thus the same r, d work for every period and every tag array. On a minimizing branch the chosen tag is uniform: its mass is the reciprocal of the number of tags. The physical likelihood multiplies by that number, so its value is exactly one. This proves both likelihood identities, not merely their convergence to one.

Suppose now that the coarse field is clean on the B-ball about o and that, on its half-ball, the actual fine field is $U = \exp(v)\,\mathrm{replica}\,V$ with $\|v_e\| \le w(x)$, where w has upper exponent $-1/2$. The local axial gauge and logarithm extension put both coarse and fine chords below a common bound

$W_1(x)$ of upper exponent $-1/2$. Indeed the coarse logarithm has that bound, the exponential is locally Lipschitz, and

$$\operatorname{chord}(\exp(v_e)\operatorname{replica} V_e) \le \operatorname{chord}(\exp(v_e)) + \operatorname{chord}(\operatorname{replica} V_e).$$

Choose x so that $B \ge 2d+12$ and $W_1(x) \le \min\{r, \varepsilon/4\}$, applying the preceding branch result with target $\varepsilon/2$. The predictor/coarse relative chord is at most $\varepsilon/2 + W_1(x) \le \varepsilon$. Every fine plaquette chord is at most $4W_1(x) \le \varepsilon$. Gauge covariance transfers these statements and the likelihood identities to the original fields.

76.3 The three errors on a small fluctuation event

Let o have distance greater than $30B$ from the seeds. The exact flag condition implies that the coarse field is clean on its needed hull. For $\beta > 0$, assume

$$|z_i| \le \xi t\sqrt{\beta} \quad \text{for every coordinate whose anchor lies within } B \text{ of } o. \tag{371}$$

With $v = \mu + \beta^{-1/2} z$, interpreted in the colour basis, (370) gives

$$\|v_e - \mu_e\| \le q\xi t, \qquad \|v_e\| \le (C + q\xi)t.$$

Away from the seeds the actual fine field agrees with this increment of the coarse replica. Thus the small-chart branch result just proved applies to it, with $w = (C + q\xi)t$.

For a fine or predictor row based at o, write $B_p(U, V)$ for its actual stroke minus its guide, and $R_p(v)$ for the transported affine row with the same coarse field, tags and saved variables. There is a nonnegative error $H(x)$ of upper exponent -1 such that

$$\|B_p(U, V) - R_p(v)\| \le H(x). \tag{372}$$

To detail the estimate, use the preceding common local gauge and extend its two logarithmic fields by zero outside the read hull. Let h bound their norms; it has upper exponent $-1/2$. The local product logarithm differs from the sum by at most Jh^2. Taylor expansion of each genuine fine/predictor stroke at zero leaves a remainder E_0 of upper exponent -1. Subtract the corresponding Taylor formula at the replica. The five resulting errors are bounded by

$$2E_0 + N_1 Jh^2 + N_1 h e^{-\sigma R} + 2N_1 c_G B A_2 h^2. \tag{373}$$

Here $R = \lfloor B/100000 \rfloor$ is the retained path cutoff, and $\sigma > 0$ is the funded path decay exponent. They are, respectively, the two Taylor remainders, the product-log error acted on by the linear row, the cutoff tail of that row, and the change from flat to coarse-field transport. For the last term, each transport path has length at most B, each coarse chord is at most $A_2 h$, and the adjoint perturbation bound is $2c_G$ times the path chord; multiply by the input norm h and the row price N_1. The retained read radius is at most $B/4$, so truncating the extended fields does not alter the actual row. Since B has upper exponent zero and $e^{-\sigma R}$ has every negative exponent, every term in (373) has upper exponent -1. This proves (372) with the original tags.

The affine row matrix has absolute row price Y after normalization by s. Therefore (371) implies

$$\|R_p(v) - R_p(\mu)\| \le q s^{-1} Y \xi t. \tag{374}$$

The local mean comparison in (368) gives a nonnegative $E(x)$ of upper exponent -1 with $\|R_p(\mu)\| \le E(x)$. Choose

$$\xi = \min\left\{(q+1)^{-1}, \frac{\delta_0}{3qs^{-1}|Y| + 1}\right\} > 0. \tag{375}$$

Then $q\xi \le 1$ and $qs^{-1}Y\xi \le \delta_0/3$. An upper-exponent-$-1$ function is eventually at most $\delta_0 t/3$: multiplying it by $\widehat{\beta}_f^{1/2}$ gives a quantity tending to zero, while $t = \widehat{\beta}_f^{-1/2}\,\mathrm{fineMonitor} \ge \widehat{\beta}_f^{-1/2}$. Hence $H, E \le \delta_0 t/3$ eventually. The triangle inequality proves

$$\|B_p(U,V)\| \le \delta_0 t. \tag{376}$$

Choose the earlier small-chart target ε below the fixed curvature-comparison radius. In that neighbourhood, $\mathrm{chord}(g) \le 2\|\,\mathrm{curvature}(g)\|$. Adding the guide bounds to (376), and retaining the square-root penalty normalizations, gives

$$\mathrm{toll}(U,p) \le \frac{2}{\sqrt{c_f}}(A/L^2 + \delta_0)t < t/4, \qquad \mathrm{wire}(Q(U),V,(o,a)) \le \frac{2}{\sqrt{c_p}}(A+\delta_0)t < Wt.$$

The strict inequalities follow from (369); the first coefficient is at most $1/8$. Both physical likelihoods are already exactly one by the local minimizing-branch argument.

76.4 The cutoff margin and the final order of choices

Set $a = \max\{C+1, \gamma K\}$. The same event has $\|v_e\| \le (C+q\xi)t \le at$, so the broad bond cutoff is one. For derivatives, use the event with $\xi/2$. It gives the strict margin $C + q\xi/2 < a$, while the fine and predictor inequalities above are strict as well. Continuity makes these masks constant on a neighbourhood of the base point. This is exactly the plateau required in Section 87; no derivative is taken through a chosen gauge or a varying seed assignment.

The order is now explicit: choose the guide bound A and error allowance before L; choose L, then W; obtain the fixed shell and precision constants, then C, a, ξ; finally enlarge the x threshold for the finite-read branches, the two order--1 errors, the chart radius and the inverse-series bounds. This supplies the previously displayed safety input on every funded, stamped sector. Funding, the stationary response and the guide bounds retain their separate construction hypotheses.

77 The actual Haar density in exponential coordinates

The chart factor in the packet is the density of probability Haar measure. We construct it from the given local exponential and logarithm, keeping its scalar normalization. Write $\mathfrak{g}$ for the physical colour space, with its counting Lebesgue measure, and exp for the group exponential. The local logarithm is inverse to exp near zero. After shrinking the radius, exp is a homeomorphism from a ball D about zero onto an open neighbourhood U of the identity. To check openness, if b is in a sufficiently small open set O, continuity of the local logarithm gives a neighbourhood of $\exp b$ whose logarithms lie in O; the local right-inverse identity places that neighbourhood in $\exp O$.

For a group element u, where the expression stays in the chart set

$$T_u(b) = \log(u \exp b), \qquad H(b) = DT_{(\exp b)^{-1}}(b), \qquad J(b) = \det H(b).$$

The map $(b,c) \mapsto \log((\exp b)^{-1}\exp c)$ is analytic near every point (b,b): its matrix argument is the identity there, where the specified logarithm is analytic. Differentiating in c and restricting to the diagonal shows that H, and hence J, is analytic. At zero, T_1 agrees with the identity on a neighbourhood, so $H(0) = 1$ and $J(0) = 1$. Shrink D so that $1/2 < J(b) < 2$ throughout D.

77.1 The overlap identity and normalization

If $b, T_u(b) \in D$, put $y = T_u(b)$. Locally at b,

$$T_{(\exp b)^{-1}} = T_{(\exp y)^{-1}} \circ T_u.$$

Indeed $\exp y = u \exp b$, and the intermediate exponential and logarithm cancel on this overlap. Differentiate at b and take determinants to obtain

$$J(T_u(b)) \det DT_u(b) = J(b). \tag{377}$$

Both densities are positive, so the transition determinant is positive as well.

Let $\nu = \exp_*(1_D J(b)\, db)$. It is a nonzero finite measure, supported on U: positivity gives nonzero mass, and the upper bound $J < 2$ on the bounded ball gives finiteness. For a measurable $A \subset U$, change variables by T_u on $\{b \in D : u \exp b \in A\}$. This map is injective there and has image $\{y \in D : \exp y \in A,\ u^{-1} \exp y \in U\}$; its inverse is $T_{u^{-1}}$. Equation (377) therefore gives

$$\nu(u^{-1}A) = \nu(A \cap uU). \tag{378}$$

All sets lie in open chart domains, so the ordinary finite-dimensional change-of-variables formula applies also to their measurable subsets.

Let h be probability Haar measure on the compact group. Integrate (378) over u with respect to h. Tonelli's theorem and right invariance give

$$\int \nu(u^{-1}A)\, dh(u) = \int\int 1_A(ux)\, dh(u)\, d\nu(x) = \nu(G)h(A).$$

On the other side, inversion and right translation preserve h, so

$$\int \nu(A \cap uU)\, dh(u) = \int 1_A(x) \int 1_U(u^{-1}x)\, dh(u)\, d\nu(x) = h(U)\nu(A).$$

The open nonempty set U has positive Haar measure. Define $c_{\mathrm{chart}} = h(U)/\nu(G) > 0$. Then, for every measurable $A \subset U$, $h(A) = c_{\mathrm{chart}}\nu(A)$. Since ν is supported on U, this is the exact identity of measures

$$\exp_*\big(1_D c_{\mathrm{chart}} J(b)\, db\big) = h|_U. \tag{379}$$

This argument constructs both the analytic density and its scalar normalization from the local chart and the given probability Haar measure. It makes no assertion that additive Lebesgue measure is a Haar measure on the group.

77.2 The protected chart correction

Fix $a > 0$ and use the prescribed small scale $t = P(h)/\sqrt{h} > 0$ from (268). It has upper exponent $-1/2$, and $t^{-1} \le \sqrt{h}$ because $P(h) \ge 1$; thus its reciprocal has upper exponent $1/2$. Let $\chi : [0, \infty) \to [0, 1]$ be the smooth cutoff, equal to one on $[0, 1]$ and zero on $[4, \infty)$. For a fluctuation bond u_i set

$$\gamma_a(u_i) = \chi\left(\frac{\|u_i\|^2}{a^2 t^2}\right), \qquad j_i(u) = \gamma_{2a}(u_i) \log J(u_i).$$

Outside the chart this last expression can use any specified measurable extension of J and the total real logarithm convention; the cutoff makes its value zero there once $4at$ is smaller than the chart radius. Where $\gamma_a(u_i) \neq 0$, we have $\|u_i\| \le 2at$, and hence $\gamma_{2a}(u_i) = 1$. Positivity of J then proves

$$e^{j_i(u)} = J(u_i) \quad \text{where } \gamma_a(u_i) \neq 0. \tag{380}$$

Thus the doubled cutoff preserves the true Haar factor on the support of the narrower protected integral.

Since $\log J$ is analytic near zero and vanishes there, the fundamental theorem of calculus on a smaller closed ball gives $|\log J(v)| \le C\|v\|$. If $\gamma_{2a}(u_i)$ is nonzero, $\|u_i\| \le 4at$; if it is zero, $j_i = 0$. Therefore, globally in u,

$$|j_i(u)| \le 4aCt. \tag{381}$$

In particular the bound $|j_i| \le 1$ used in the packet estimate holds at one sufficiently large coupling threshold, uniformly in the period and in the number of chart bonds.

We record the differentiated version with its exact incoming premise. Suppose a variation $u_i(y)$ is C^k near y_0, and its increment $u_i(y) - u_i(y_0)$ has derivatives through order k bounded by $v(x)$, where v has upper exponent $-1/2$. This is the compensated variation supplied by the Gaussian change of variables; its production is separate from the chart calculation here. If $\gamma_b(u_i(y))$ is zero on a neighbourhood of y_0, all its jets vanish. Otherwise continuity gives $\|u_i(y_0)\| \le 2bt$. Thus every derivative, including order zero, of u_i at y_0 is bounded by $v + 2bt$. For $g = u_i/(bt)$ these bounds are at most

$$K_b = (bt)^{-1}(v + 2bt),$$

which has upper exponent zero. The derivatives of $\|g\|^2$ are bounded by a fixed polynomial in K_b; this follows directly by the product rule for the inner product. Derivatives of χ are bounded on $[0, 4]$, and its value is at most one. The chain rule, indexed by partitions of the derivative arguments, therefore gives

$$\|D^r \gamma_b(u_i(y_0))\| \le L_b(x)^r, \qquad 0 \le r \le k, \tag{382}$$

for an $L_b \ge 1$ of upper exponent zero. The zero-order inequality uses the range of χ; a uniform amplitude from the positive-order bounds can be absorbed into L_b.

Apply this to the broader cutoff $b = 2a$. If its germ vanishes, so do all jets of j_i. Otherwise $\|u_i(y_0)\| \le 4at$, and all jets of u_i are bounded by $V = v + 4at$, which tends to zero. All derivatives of $\log J$ through order k are bounded on a fixed smaller chart ball. For positive derivative order, each partition term in the chain rule contains at least one derivative of u_i; since $V \le 1$, its product is at most V. At order zero use $|\log J(u_i(y_0))| \le CV$. Hence

$$\|D^r(\log J \circ u_i)(y_0)\| \le C_k V \quad (0 \le r \le k).$$

Leibniz's formula with (382) yields

$$\|D^r j_i(y_0)\| \le C_k V(1 + L_{2a})^k \quad (0 \le r \le k). \tag{383}$$

The common right-hand side has upper exponent $-1/2$. The argument covers points on the mask boundary and arbitrary Gaussian inputs: smallness is inferred from the nonvanishing germ of the protected factor, not assumed for the Gaussian field globally.

78 Local normalization and its derivatives

The local scalar correction is a difference between the actual sector and its flat reference. The reference uses the same forest, tags, cutoff radius and windows; its coarse field is the identity and its seed sets are empty. Write A° for its completed precision matrix. With the local colour trace $J_e(A)$ of Section 74, the correction at a physical bond is

$$c_e = \frac{1}{2}\{J_e(A^\circ) - J_e(A)\}. \tag{384}$$

We first establish its scalar bounds and then differentiate the actual window formula. A bound on the full determinant would not provide these local estimates.

78.1 Uniform and remote scalar bounds

Let $\|A\|_\eta, \|A^\circ\|_\eta \le Q$ and let S be the uniform row and column bound for every principal inverse along $A_s = 1 + s(A-1)$ and the corresponding flat interpolation, $0 \le s \le 1$. The local formula is

$$J_e(A) = \sum_{\alpha=1}^{q} \int_0^1 [(A-1)P_s]_{(e,\alpha),(e,\alpha)}\, ds.$$

Thus $|J_e(A)| \le q(Q+1)S$, and the same holds for A°. By (384),

$$|c_e| \le q(Q+1)S. \tag{385}$$

Both precisions meet the hypotheses of the Gram estimate: the flat reference has no flags, and gauge transformations preserve the flags of the actual sector. Hence the same post-L constants can be used for the actual, flat and gauge-transformed sectors.

If the anchor of e is farther than $30B$ from the seeds, choose a single gauge making the coarse bond chords at most $Cw(x)$ on the radius-$6B$ ball, as in (364). Here w has upper exponent $-1/2$. For any coordinate in the window of (e,α), its anchor lies within $B+2$ of that of e. On these coordinates the inactive diagonal of the gauged sector agrees with the flat reference: exactly the tree bonds are inactive. If an entry of the selected row matrix meets such a coordinate, either its row anchor lies within $B/4$ or both entries are zero. In the former case that row is selected in both sectors. The gauge transport comparison, applied to the same unit coordinate input, gives

$$|M'_{pi} - M^\circ_{pi}| \le d(x), \qquad d(x) = C_1(B+1)w(x).$$

The same estimate holds for the other tested coordinate. All transport paths lie in the common $6B$ hull. Expanding the two products in the Gram entry gives

$$|M'_{pi}M'_{pj} - M^\circ_{pi}M^\circ_{pj}| \le d(x)|M'_{pj}| + d(x)|M^\circ_{pi}|.$$

Sum over p and use the column bound Y for both row matrices:

$$|A'_{ij} - A^\circ_{ij}| \le 2Y d(x) =: \delta(x) \quad \text{on each tested principal window.} \tag{386}$$

No count of all rows enters this bound.

The window contains at most $\Delta_B = m(2B+5)^4$ coordinates. The local resolvent comparison of Section 74 now yields

$$|J_e(A') - J_e(A^\circ)| \le q\{S + (Q+1)S^2\}\Delta_B\delta(x).$$

The complete colour trace is gauge invariant; the endpoint windows are invariant under colour rotations. Therefore $J_e(A') = J_e(A)$, and

$$|c_e| \le q\{S + (Q+1)S^2\}\Delta_B\delta(x) =: a_0(x), \qquad d(e,\text{seeds}) > 30B. \tag{387}$$

We dropped a harmless factor $1/2$. Since B has upper exponent zero, a_0 has upper exponent $-1/2$.

78.2 Matrix derivative bounds with spatial weights

For a C^k matrix-valued map $A(y)$ near y_0, a convenient quantitative statement, for $f, v, \gamma \geq 0$, is the existence of a nonnegative matrix H such that

$$\|D^r A_{ij}(y_0)\| \leq H_{ij} v^r \quad (0 \leq r \leq k), \qquad \|H\|_\gamma \leq f. \tag{388}$$

The norm on the right is the maximum of the weighted row and column sums. Thus it controls a common entrywise majorant of all jets, not just the norm of each derivative separately. If A and B have bounds (f, v) and (g, z) with $g, z \geq 0$, Leibniz's rule gives the envelope HK for their product with bounds $(fg, v + z)$: the binomial sum is $(v + z)^r$, and the triangle inequality for distance proves $\|HK\|_\gamma \leq \|H\|_\gamma \|K\|_\gamma$. Addition, transpose and restriction to a principal set preserve the corresponding bounds. These operations introduce no volume factor.

Suppose $A(y_0)$ is invertible and its inverse has weighted price g. Set $F = A(y_0)^{-1}$ and $V(y) = 1 - FA(y)$. Then $V(y_0) = 0$, and, wherever $A(y)$ is invertible,

$$A(y)^{-1} = \sum_{r=0}^{k} V(y)^r F + V(y)^{k+1} A(y)^{-1}. \tag{389}$$

Multiplying the finite geometric identity on the right by $A(y)^{-1}$ proves this without commuting any factors. Invertibility holds on a neighbourhood by continuity of the determinant. The inverse is C^k there by the adjugate formula. Every derivative of order at most k of V^{k+1} vanishes at y_0: in each Leibniz term at least one of the $k + 1$ factors is undifferentiated and equals zero. The remainder in (389) consequently has zero jets through order k.

The envelope for V has bounds $(1 + fg, v)$. Applying the product rules to the finite sum proves an inverse envelope with bounds

$$T_k(f, g) = \sum_{r=0}^{k} (1 + fg)^r g, \qquad kv. \tag{390}$$

Only the inverse at the base point was priced. A uniform inverse bound on an entire neighbourhood was not assumed.

Apply this argument separately in every prescribed principal window. Each restriction of $A(y)$ retains the envelope (f, v), and each base inverse has weighted price S. Its (i, j) entry has majorant $T_k(f, S) e^{-\gamma d(i,j)}$. Padding it by zero outside the window gives a common envelope for $P(y)$, because window membership is fixed. Both its row and column sums are bounded by

$$F_B = \Delta_B T_k(f, S), \qquad \text{with derivative scale } kv. \tag{391}$$

For the row bound, even though different columns use different windows, any nonzero entry has $d(i, j) \leq B + 2$, and there are at most Δ_B such j. The same count proves the column bound.

The error $A(y)P(y) - 1$ vanishes, as a germ, whenever $d(i, j) \leq B$. Indeed each such coordinate belongs to the window, and its principal inverse solves the corresponding row equation for all nearby y. Erasing these zero jets and decreasing the spatial weight from γ to $\gamma/2$ gives an envelope with price

$$e^{-\gamma B/2}(fF_B + 1), \qquad \text{derivative scale } v + kv. \tag{392}$$

This is the differentiated version of the boundary error. Multiplying by the full inverse envelope also prices $P - A^{-1}$; taking the symmetric part and then multiplying by A prices the covariance defect $A(P + P^*)/2 - 1$. For fixed k all these prices are a polynomial in B times $e^{-\gamma B/2}$, hence have every prescribed negative upper exponent. The derivative scales are fixed multiples of v.

78.3 The actual sector matrix supplies the premise

Freeze the seeds, tree, tags, windows and active sets while varying the coarse field $U(y)$. Suppose every link matrix has jets

$$\|D^r U_e(y_0)\|_{\mathrm{F}} \le R_G v^r, \qquad R_G = \sqrt{\dim(\text{matrix representation})}, \quad 0 \le r \le k.$$

The same bound holds for inverses by taking the adjoint, for signed pullbacks, and for replicas. Constant gauge transformations preserve it by unitary invariance of the Frobenius norm.

A product along a walk of length n has amplitude R_G and derivative scale $nR_G v$. To verify the absence of an exponential amplitude, split its first factor as $U(y_0) + (U(y) - U(y_0))$. Multiplication by $U(y_0)$ is an isometry; in every positive-order term from the difference, its bound $R_G v^r$ is at most $(R_G v)^r$. The binomial formula therefore adds $R_G v$ to the derivative scale while preserving the amplitude of the remaining product. Induction proves the claim, including inverse letters and repeated physical links.

For the selected sector row matrix, a single coordinate input is transported from its physical bond to the row anchor. The coordinate path has length at most $4B$. A reversed signed bond adds its replica adjoint, whose derivative scale is $2v$. The two sides of the transported adjoint add twice the path scale. Thus the transported input has fixed amplitude and derivative scale

$$z = 8BR_G v + 2v. \tag{393}$$

The remaining cut flat row is a fixed linear map. The weighted impulse estimate from Section 73, before periodization, bounds its operator norm by $Ce^{-\eta d(p,i)}$. The extra one-step shift from a reversed bond is absorbed into this fixed C. Hence

$$\|D^r M_{pi}(y_0)\| \le Ce^{-\eta d(p,i)} z^r.$$

Inactive columns, unselected rows and entries outside the cutoff are identically zero throughout the variation; their indicators are not differentiated.

For $0 < \gamma < \eta$, sum the displayed envelope using the bounded number of row and column coordinates per coarse site. The shell sum $\sum_{n\ge 0}(2n+1)^4 e^{-(\eta-\gamma)n}$ is finite. It follows that M has an envelope (C_γ, z), with C_γ independent of the volume and of B. The identity $A = M^*M + \Pi_{\text{inactive}}$ therefore supplies (388) with

$$f = C_\gamma^2 + 1, \qquad v_A = 2z.$$

The inactive projection is constant. If the incoming link scale v has upper exponent $-1/2$, so do z and v_A.

78.4 Differentiating the local logarithmic trace

Choose γ also below the retained inverse width. The glide $A_s(y) = 1 + s(A(y) - 1)$ has envelope $(f + 2, v_A)$ uniformly for $0 \le s \le 1$. Its principal inverses at y_0 satisfy the same uniform price S as before. Multiplication by $A(y) - 1$ and (391) consequently give, for every local diagonal entry, the amplitude

$$(f+1)\Delta_B T_k(f+2, S)$$

and derivative scale $(k+1)v_A$, uniformly in s.

To justify differentiation under the integral, note that the principal determinants are nonzero at every (y_0, s) on the compact interval. There are finitely many windows. Joint continuity gives one neighbourhood of y_0 on which all these determinants remain nonzero for all $s \in [0, 1]$. The

adjugate formula makes the integrand jointly C^k there. On a smaller closed parameter ball times the interval, all derivatives are bounded by compactness. Difference quotients and the fundamental theorem of calculus, followed by dominated convergence, now permit differentiation under the integral, successively through order k. For this argument the variation space is finite dimensional, as it is for each finite-volume field variation. The local bounds at y_0 are unaffected by the auxiliary compactness bound. Summing the q colour entries proves

$$\|D^r J_e(A(y_0))\| \le q(f+1)\Delta_B T_k(f+2,S)\{(k+1)v_A\}^r, \quad 0 \le r \le k. \tag{394}$$

The flat-reference term in (384) is constant in y. Combining its scalar bound with (394), and enlarging the harmless factor $1/2$, gives an envelope for $c_e(y)$ with amplitude $b(x)$ of upper exponent zero and derivative scale $z_1(x)$ of upper exponent $-1/2$. It follows that all its jets are bounded by $b(x)(1+z_1(x))^k$, again of upper exponent zero. At a remote bond we can do better: its value is at most $a_0(x)$ by (387), while for positive orders, once $0 \le z_1 \le 1$,

$$\|D^r c_e(y_0)\| \le b(x) z_1(x)^r \le b(x) z_1(x).$$

Thus the common bound $a_0 + bz_1$, of upper exponent $-1/2$, controls all jets through order k at remote bonds. This obtains small normalization jets without differentiating a chosen local gauge. The needed link variation and its coherent Gaussian compensation remain distinct inputs to the eventual packet variation.

79 Varying the coarse field while holding the fine field fixed

Differentiating a packet directly in its coarse field would also change the fine configuration represented by its fluctuation variable. The resulting derivatives would meet the fine masks and branch choices. We instead make the actual compensated change of variables. It is a global bijection on the fluctuation space near the chosen coarse configuration, and agrees with the required logarithmic adjustment where the protected integrand can be nonzero.

79.1 A globally defined adjustment of one logarithm

Choose fixed $d, A > 0$ so small that

$$\log(\exp u) = u \quad (\|u\| \le 2d)$$

and the matrix logarithm is analytic at $\exp(u)M$ whenever $\|u\| \le 2d$ and $\|M-1\| \le A$. Also require $\exp\log(\exp(u)g) = \exp(u)g$ for $\|u\| \le d$ and $\|g-1\| \le A$. Such constants exist by continuity of multiplication and the exponential at $(0,1)$, after choosing a smaller product neighbourhood inside the given logarithm domain. The constants are chosen before the coupling threshold and before the particular cutoff.

Using the same smooth cutoff as before, define

$$\psi(M,u) = \chi(\|u\|^2/d^2)\{\log(\exp(u)M) - u\}, \qquad F_M(u) = u + \psi(M,u).$$

Here group elements are represented by their unitary matrices in the logarithm expression. The correction is zero for $\|u\| \ge 2d$. On the closed ball it is smooth in (M,u); outside the ball it vanishes locally, so these formulas define a smooth map everywhere for M sufficiently near 1. At $M = 1$ the correction and all its u derivatives vanish.

Compactness of $\|u\| \le 2d$ now supplies one neighbourhood of $M = 1$ on which, for every u,

$$\|\psi(M,u)\| < d/2, \qquad \|D_u\psi(M,u)\| < 1/2. \tag{395}$$

Indeed these inequalities hold at $(1,u)$, and their continuous left-hand sides are uniformly small on that compact set; off it they are zero. Integration of the derivative on line segments shows that $\psi(M,\cdot)$ is 1/2-Lipschitz. Thus F_M is injective, since $F_M(u) = F_M(v)$ implies $\|u-v\| \le \|u-v\|/2$. For every z, iteration of $u \mapsto z - \psi(M,u)$ is a contraction on the complete colour space. Its iterates are Cauchy because successive distances form a geometric series; their limit is its unique fixed point and solves $F_M(u) = z$. Hence F_M is surjective as well.

Its derivative is $1 + D_u\psi$, which is invertible by the convergent geometric series. The determinant

$$p(M,u) = \det(1 + D_u\psi(M,u))$$

is positive for all u after a further uniform restriction on M: it equals one at $M = 1$, is continuous on the compact u ball, and is one outside it. Thus F_M is a smooth bijection with positive Jacobian. Its local smooth inverses agree by injectivity, giving a global smooth inverse.

For an actual group element g near 1 and $\|u\| \le d$, the cutoff equals one. Consequently

$$\exp(F_g(u)) = \exp(u)g. \tag{396}$$

The map is global, but this exact group identity is used only within that smaller ball.

79.2 The coherent coarse and fluctuation variables

Fix a sector with coarse field V_0, and keep its seeds, forest, tags, frozen fine bonds and active sets fixed. In the affine source replace the clipped coarse guide by its unmasked path-field formula; let $\widetilde{\ell}(V)$ be the resulting linear source and put

$$\nu(V) = -P(V)^*\widetilde{\ell}(V)$$

in physical colour coordinates. This specifies the extension used in the change of variables. At a valid sector, every selected row has a clean neighbourhood. The clipping is then inactive at every read of that row, as proved in the guide comparison. Its unmasked source equals the original source, and hence $\nu(V_0) = \mu$. The same statement holds at any other valid sector. Away from valid sectors the unmasked expression is retained as its definition.

Let $g_\beta = \beta^{-1/2} > 0$ and identify a counting Gaussian coordinate z with its colour vector on each fine bond. The extended fluctuation field is

$$u_0 = \nu(V_0) + g_\beta z.$$

For a nearby coarse field V, define separately at each fine bond

$$M_i(V) = \overline{V}_{0,i}\overline{V}_i^{-1}, \quad u_i(V) = F_{M_i(V)}(u_{0,i}), \quad z_i(V) = g_\beta^{-1}\{u_i(V) - \nu_i(V)\}. \tag{397}$$

The replica is the prescribed one. It follows identically that $\nu(V) + g_\beta z(V) = u(V)$, and at $V = V_0$ the map is the identity on z.

The fine field on an active bond is $\exp(u_i)\overline{V}_i$; on an inactive bond it is the fixed original fine field. Suppose $8at \le d/2$ and the varied active fluctuation satisfies $\|u_i(V)\| \le 8at$. By (395),

$$\|u_{0,i}\| \le \|u_i(V)\| + \|\psi(M_i(V), u_{0,i})\| < d.$$

Equation (396) therefore gives

$$\exp(u_i(V))\overline{V}_i = \exp(u_{0,i})\overline{V}_{0,i}. \tag{398}$$

Inactive bonds agree by definition. On the protected support all fine reads are consequently fixed, including their masks, likelihoods and selected branch tags. The larger $8at$ allowance is deliberate: at a point where the broader cutoff germ is not identically zero, continuity first gives $\|u_i(V_0)\| \le 4at$, then gives the strict $8at$ bound on a neighbourhood. A finite set of protected bonds has one common such neighbourhood. No condition is imposed on the unprotected Gaussian coordinates.

79.3 Uniform derivative bounds for the compensation

Fix the derivative order k. On $\{1\} \times \{u : \|u\| \le 2d\}$, all derivatives in M of ψ and p through order k have fixed bounds, by smoothness and compactness. Outside this set their u-sections are respectively zero and one as functions of M. Hence these bounds hold for every $u \in \mathfrak{g}$, with constants independent of the Gaussian coordinate. Moreover $\psi(1,u) = 0$ and $p(1,u) = 1$.

Suppose the coarse variation $V(y)$ has the link derivative bounds of Section 78, with $V(y_0) = V_0$ and scale $v(x)$ of upper exponent $-1/2$. Inversion, replication and multiplication by the fixed unitary replica show that $M_i(V(y))$ has the same bound. In a chain-rule term of positive order $r \le k$, the product of derivatives of M_i has total scale v^r; its amplitude is bounded by a constant depending only on k and the representation. There are finitely many partitions of r. Thus, for every $u_{0,i}$,

$$\|D_y^r \psi(M_i(V(y_0)), u_{0,i})\| \le C_k v^r, \qquad \|D_y^r p(M_i(V(y_0)), u_{0,i})\| \le C_k' v^r \quad (1 \le r \le k).$$

Their zero-order values are zero and one. Since $v \to 0$, enlarge constants and obtain

$$\|D_y^r (u_i(y) - u_{0,i})|_{y=y_0}\| \le s(x), \qquad \|D_y^r p(M_i(V(y)), u_{0,i})|_{y=y_0}\| \le D(x)^r \quad (0 \le r \le k), \tag{399}$$

where $s \ge 0$ has upper exponent $-1/2$, and $D \ge 0$ has upper exponent zero. For the first bound use $C_k v$ for positive orders and zero at order zero; for the second take $D = \max(1, C_k')v$, increasing the fixed constant if necessary. For $r \ge 1$, $\max(1, C_k')^r \ge C_k'$. At order zero the second bound is equality, with the convention $D^0 = 1$. These are precisely the compensated increment bounds used in Section 77. They are uniform over all Gaussian samples; they do not require a bound on $z_i(y)$ or on the derivatives of the mean extension.

The incoming link-variation premise itself holds for the prescribed scaled exponential variations. For example, let $V_i(y) = \exp(\epsilon L_i y) V_{0,i}$, where the linear maps L_i have uniformly bounded norm and y stays in a fixed bounded set. The derivatives of the matrix exponential are uniformly bounded on the resulting fixed colour ball. The linear chain rule contributes ϵ^r at order r. Right multiplication by $V_{0,i}$ is an isometry. Enlarging one fixed constant therefore gives the link bound with $v = C_k \epsilon$. In particular the required scale holds when ϵ has upper exponent $-1/2$, as for the prescribed inverse-square-root coupling normalization. Finite collections of marked directions are handled by taking the maximum of their norms.

79.4 The exact finite-dimensional Jacobian

For a fixed nearby V, the map in (397) acts separately on the colour coordinate of every fine bond. On a finite set P of bonds it is conjugate, by the fixed coordinate regrouping, to the product of maps

$$z_i \longmapsto g_\beta^{-1}\{F_{M_i(V)}(\nu_i(V_0) + g_\beta z_i) - \nu_i(V)\}.$$

Each is a bijection. Its derivative is exactly $DF_{M_i(V)}(u_{0,i})$: the two scalar factors g_β^{-1} and g_β cancel. The derivative of the product is block diagonal, so its determinant is

$$\prod_{i\in P} p(M_i(V), u_{0,i}) > 0. \tag{400}$$

A fixed regrouping of counting coordinates conjugates the derivative and does not change its determinant. This remains true if the regrouped spaces are temporarily given different finite-dimensional norms. For the empty set the map is the identity of the zero-dimensional space and the product is one.

The ordinary change-of-variables formula therefore gives, for every nonnegative measurable test f on these coordinates (and hence for every integrable test),

$$\int f(z')\,dz' = \int f(z(V)) \prod_{i\in P} p(M_i(V), u_{0,i})\,dz.$$

Together with (398) and (399), this supplies the coherent protected variation and its exact Jacobian. Differentiating a Gaussian density under this substitution still requires its own covariance and mean score bounds; those do not follow from this Jacobian calculation.

80 Derivatives of the mean and of the local action

We now prove the differentiated estimates needed for the logarithm of a packet. The pointwise comparison with a stationary response is not differentiated: its gauge was chosen from the base configuration. Instead we use that one fixed gauge to express nearby configurations in logarithmic coordinates, estimate the smooth unmasked expressions there, and finally restore the original gauge. This distinction keeps the argument valid at the boundary of a cutoff.

Fix a derivative order j, a patch, the branch constants, the forest, tags and sector indicators. Assume the uniform path bounds, clean-sector conditions and positive precision bounds of Sections 73–78. All these choices precede the large-x limit. A coarse variation $V(y)$ is measured by its matrix derivatives on each bond:

$$\|D^r \rho(V_e)(y_0)\| \le R_G v(x)^r \quad (0 \le r \le j), \qquad v \ge 0,$$

where R_G is the fixed representation bound and v has upper exponent $-1/2$. We use the derivative notation $\mathcal{B}_j$ of Section 84; the same definition applies to colour-valued and operator-valued functions.

80.1 Uniform bounds for the actual predictor branch

The analytic branch of Section 69 gives bounds of every fixed finite order on a neighbourhood of the identity. We need that neighbourhood to be independent of the torus size. Here is the finite-read argument that ensures this.

For a fixed patch and direction, take the union of the sites visited by all frame paths, their translates to the neighbouring coarse root, and all the straight connectors of length L. This is a finite subset of the covering lattice. Choose a reference torus large enough that lifting after wrapping fixes every site in this union, and with distinct neighbouring coarse roots. There are only finitely many tag choices and output bonds on this reference torus. For each, analyticity gives a positive logarithmic radius and a bound on derivatives through order k. Take the minimum of the radii and one plus the sum of the bounds.

For a bond on any other torus, pull its logarithmic coordinates to the reference torus by translation and wrapping. This linear map has norm at most one in the supremum norm, even when some input coordinates coincide. Assign the two reference frame tags the actual tags at the two endpoints, and assign the actual predictor tag to the reference bond. Every read path has the same transport. Thus the two frame score functions coincide; uniqueness on the minimizing branch, and the explicit axial formula on its complement, identify the frames. The connector and predictor then coincide as well. Composition with the norm-one linear map transfers the derivative bound without a dimension factor.

Consequently, for each k there are $r, C > 0$, independent of the period, such that the actual predictor on logarithmic fields of norm at most r belongs to $\mathcal{B}_k(C, 1)$ on each output bond. Shrink r once more to use the logarithm and exponential charts. The chain rule shows that a near-identity group variation with derivative scale v has predictor derivative scale $C_k v$. This statement includes the original tags and the branch tests.

In particular, on a fixed ball in pairs (b, u) of coarse and fine logarithmic fields, the four fields

$$V = \exp b, \qquad U = \exp u \operatorname{replica} V, \qquad Q(U), \qquad Q(\operatorname{replica} V)$$

have uniform derivatives of every prescribed finite order. Indeed both coordinate projections have norm one; differentiation of the exponential on a fixed compact chart, replication, and the unitary product rule give uniform bounds for V, U. Their chords are bounded by a fixed multiple of $\max(\|b\|, \|u\|)$, so the predictor estimate just proved applies after shrinking the ball.

Let $f(b, u)$ be one unmasked affine residual at zero fluctuation, namely the fine or predictor base stroke minus its transported coarse guide, with coarse field $\exp b$. It does not depend on u, but retaining the pair space allows the same chart for all local actions. The preceding bounds and the absolute path prices imply

$$f \in \mathcal{B}_k(A(x), 1; (b, u)) \quad \text{for } \|(b, u)\| \le r, \qquad f(0, 0) = 0, \tag{401}$$

where A has upper exponent zero. To see the rate assertion, each cut path has length at most a fixed multiple of $R + 1$ and bounded absolute coefficient sum. Transport adds a derivative scale $C(R+1)$; a guide sums over at most $C(R+1)^4$ inputs with bounded coefficients. At fixed order, the product rule therefore costs only a polynomial in $R + 1$. The predictor bounds are fixed constants. Here $R = \lfloor B/100000 \rfloor$ has upper exponent zero. At the identity all curvature strokes vanish, the predictor equals the identity, and the guides are linear in zero coarse strokes, which proves the last equality in (401).

80.2 Recovering small derivatives by flat interpolation

We give the interpolation step explicitly because its constants are independent of the dimension of the logarithmic field space. Let $f : E \to F$ be C^k near the closed ball of radius B_0 in a finite-dimensional normed space, with $k \ge 3$. Suppose

$$f \in \mathcal{B}_k(A, 1; z) \quad (\|z\| \le B_0), \qquad \|f(z)\| \le u \quad (\|z\| \le 3t), \quad 0 < t \le 1, \quad 3t \le B_0,$$

where $A, u \ge 0$. Put

$$W = 1000(u/t^2 + AB_0), \qquad Y = 1000u/t + 3WB_0,$$
$$H = u + YB_0 + Yt + Wt^2 + At^3.$$

Then, throughout the larger ball,

$$f \in \mathcal{B}_k(H, t^{-1}). \tag{402}$$

First, if $\|Df(z)-Df(a)\| \le ed$ and $\|f(z)\| \le u$ on $\overline{B}(a,d)$, integration along a segment of length d in any unit direction gives

$$\|Df(a)\| \le 2u/d + ed.$$

Write $H_0 = \|D^2 f(0)\|$. The third derivative bound gives $\|D^2 f(z)\| \le H_0 + A\|z\|$. Apply the preceding inequality on balls of radius $t/4$ centred in $\overline{B}(0,2t)$. These balls lie in $\overline{B}(0,3t)$, and hence

$$\|Df(z)\| \le A_1 := 8u/t + (H_0 + 3At)t/4 \quad (\|z\| \le 2t).$$

Apply the same inequality to Df on the radius-$2t$ ball, using the third derivative bound to control the variation of $D^2 f$. It yields $H_0 \le A_1/t + 2At$, and therefore

$$H_0 \le \tfrac{32}{3}u/t^2 + \tfrac{11}{3}At \le 50(u/t^2 + At).$$

Integration out to radius B_0 now gives $\|D^2 f\| \le W$, $\|Df\| \le Y$, and $\|f\| \le u + YB_0$. For orders $r \ge 3$, the original bound gives $t^r\|D^r f\| \le At^3$. Each of the bounds for orders zero, one, two and at least three is a summand of H, proving (402).

Apply this with $k = j+3$ and $t = h^{-1/2}$. This t is the flat probing radius, not the larger fluctuation threshold $\tau = P(h)h^{-1/2}$. In a clean ball, choose one gauge from the base field $V(y_0)$ and keep it fixed. The small-log construction and the smooth local logarithm give a logarithmic variation $b(y)$, zero outside the required ball, with

$$b \in \mathcal{B}_j(F(x), 1; y_0), \qquad F \text{ of upper exponent } -1/2.$$

For positive derivatives this follows by composing the bounded derivatives of the logarithm with the bond derivative scale v; for the value it is the small-log bound at the base field. Finitely many strict chart inequalities give a common neighbourhood on which exponentiating b agrees with the gauge-transformed field on the whole read stencil. No derivative of the chosen gauge is taken.

Set $B_0 = F + 3t$, which eventually lies in the fixed ball of (401). Since $f(0) = 0$, integration of its first derivative gives $\|f\| \le 3At$ on the radius-$3t$ ball. Thus take $u = 3At$ above. The upper exponents of W, Y, H are, respectively, $1/2, 0, -1/2$: each term follows from the displayed formulas, B_0, t of upper exponent $-1/2$, and t^{-1} of upper exponent $1/2$.

The chain rule for $f(b(y), 0)$ costs at most a fixed factorial times H and a fixed-order polynomial in $1 + 2F(1 + t^{-1})$. To verify this directly, in a term indexed by a partition into m blocks, the outer derivative contributes Ht^{-m} and each inner derivative contributes at most $2F$; there are at most $j!$ partitions. The displayed polynomial has upper exponent zero. Thus every derivative through order j of the residual has upper exponent $-1/2$. Locality identifies this expression with the actual unmasked residual in the fixed gauge; gauge covariance restores the original field by a fixed isometry.

80.3 The source and the mean

The selected affine source also contains frozen cut logarithms. They obey the same upper exponent $-1/2$ derivative bound. For completeness, write such a logarithm as

$$\chi\big(\mathrm{chord}(U)^2/Z^2\big)\,\log U, \qquad Z = K\tau,$$

with fixed $K > 0$. If the base chord is strictly larger than $2Z$, the function is locally zero. Otherwise normalize the matrix displacement by $R_G Z$. Its value is at most two and its positive derivatives have size bounded by constants times v/Z after enlarging the scale. Since Z^{-1} has upper exponent $1/2$, the derivatives of the cutoff have upper exponent zero. On the fixed logarithm chart, the value

of $\log U$ is at most CZ; its positive derivatives are at most $C_j v$ after $v \le 1$. The product rule proves the assertion, including the closed boundary $\operatorname{chord}(U) = 2Z$.

On a selected row, the zero-fluctuation affine source is the fixed row scale times the sum of the unmasked residual and the transported frozen logarithm. The latter transport reads only a radius-B neighbourhood and has bounded absolute row price. Its derivative scale is bounded by $C(B+1)v + 1$, of upper exponent zero. Combining the preceding estimates gives

$$r(0) \in \mathcal{B}_j(F_0(x), 1) \quad \text{coordinatewise}, \qquad F_0 \text{ of upper exponent } -1/2. \tag{403}$$

On unselected rows the function is identically zero. The selection is held fixed throughout this differentiation.

Use the same counting-coordinate matrices M, A, P as before, and put $\ell = M^* r(0)$ and $\nu = -P^*\ell$. Weighted row and column derivative bounds for M, together with (403), show that each coordinate of ℓ has derivatives of upper exponent $-1/2$. The padded inverse derivative bounds have upper exponent zero, so the same is true of ν. Reassembling a colour vector costs at most its fixed colour dimension. On the clean selected sector ν equals the actual mean μ, by the local clipping identity already proved.

There is a stronger estimate for the normal residual. Since A is symmetric, on a neighbourhood where it is invertible,

$$A\nu + \ell = -A(P - A^{-1})^*\ell. \tag{404}$$

This follows by substituting $\nu = -P^*\ell$ and using $(A^{-1})^* = A^{-1}$. Every prescribed finite derivative of $P - A^{-1}$ has every prescribed negative upper exponent, by Section 78. The weighted product rule therefore proves the same assertion for (404). For $0 < \beta \le 2h$, multiplication by $\sqrt{\beta}$ loses at most upper exponent $1/2$, so the normalized linear Gaussian score $-\sqrt{\beta}(A\nu + \ell)$ still has any prescribed upper exponent. This proves the score bound without replacing the local mean by the exact global inverse mean.

80.4 Local action derivatives with a held fine field

We also require a cruder bound that holds for every fixed fine configuration U. Keep U and its actual predictor $Q(U)$ fixed while varying the coarse field. Since $v \le 1$ eventually, all coarse matrix derivatives may be bounded with scale one.

The cut curvature strokes and predictor-noise stroke have a common constant amplitude C_0 and derivative scale $D_0 = C(R+1)$. For curvature strokes this follows by differentiating their cut path sums, whose absolute coefficients are bounded; the unextended plaquette alternative is a product of four unitary bonds. For the noise stroke use the product $VQ(U)^{-1}$ and the bounded curvature map. A transported guide adds a finite input sum and transport: its amplitude is at most $C(R+1)^4 C_0$ and its derivative scale at most $D_0 + C(R+1)$. A second application of the transported operator has the corresponding squared polynomial price.

All unmasked local action terms now have upper exponent zero in every derivative through order j. Indeed they are finite sums of inner products of the above strokes and guides, their squared norms, and the plaquette discrepancy

$$\tfrac{1}{2}\big(\operatorname{chord}(g)^2 - \|\operatorname{curv}(g)\|^2\big).$$

An inner product multiplies amplitudes and adds derivative scales; this is precisely the bilinear product rule. The discrepancy has bounded derivatives because the chord squared is a fixed multiple of $\|\rho(g) - 1\|^2$, and the curvature is a fixed linear projection of the matrix displacement. At fixed j, converting a polynomial derivative scale to scale one preserves upper exponent zero.

The fine energy requires one separate observation: its clipped part need not be smooth in the fixed fine input. It is constant in the variable being differentiated. The chord is at most two and $\tau \le 1$, so the direct and linear pieces of that clipped energy are bounded by twice their fixed coefficients. If C_{path} is the fixed absolute path price, the complete fine constant is

$$F = \tfrac{1}{2}(C_{\mathrm{path}} R_G^2)^2 + 2c_{\mathrm{direct}} + 2c_{\mathrm{linear}}.$$

The remaining fine terms are, for the two curvature components, $\|G\|^2/2 - \langle G, X_f \rangle$, where G is the varying guide and X_f is fixed and bounded. The preceding product estimates apply. Thus no smoothness of the arbitrary held fine configuration is assumed.

Finally consider an extracted affine square. If a varying fluctuation w has derivatives bounded by $p(x)$ of upper exponent zero on the hot bonds in its read ball, then

$$r(w) = r(0) + Mw$$

has coordinatewise derivatives of upper exponent zero. Rows of M vanish off the hot bonds and outside radius $B/4$, so only these local bounds on w enter. Squaring, dividing by twice the square of the fixed row normalization, and summing over the at most two parent rows at an energy address preserves that exponent. This applies both to the compensated fluctuation on protected bonds and to the mean ν.

80.5 The logarithmic packet estimate

Let n be the number of seeds. If the product of protected masks is not locally zero, their closed support bounds and the compensated variation of Section 79 give derivatives of the fluctuation of upper exponent $-1/2$ on the protected $304B$ region. The mean has the same bound by the preceding argument. Thus the local action and both extracted-square costs just estimated are available at every address of the $200B$ region. There are at most $n b_{200B+1} s_L$ such addresses in each signed channel. Taking the normalized signed average preserves this bound.

Let $C_a, C_q, C_c, C_J \ge 0$ bound respectively the derivatives of the unmasked action, one extracted-square cost, the Gaussian normalization correction and the chart correction, all at scale one through order j. Their upper exponents are zero, zero, zero and $-1/2$, respectively. With ρ_L the fine-bond density, put

$$S(x) = 2h\, b_{200B+1} s_L (C_a + 2C_q) + b_{200B+2} \rho_L C_c + b_{304B+2} \rho_L C_J.$$

It has upper exponent one. The formula (426), the local bounds, and the bond census show that its nonconstant part belongs to $\mathcal{B}_j(nS, 1; y_0)$. The term $-N_A \log m_\beta$ is constant in this variation. Subtracting the value at y_0 costs at most a factor two, so

$$\Phi_A - \Phi_A(y_0) \in \mathcal{B}_j(2nS(x), 1; y_0). \tag{405}$$

This proves the first bound in (430), with the original local covariance, chart and compensated fluctuation. If the protected mask product is locally zero, the packet derivatives vanish and this bound on its logarithm is not needed.

81 Small derivatives of the remote action

The packet estimate permits a polynomial expense in its logarithm. Outside the packets we need a stronger conclusion: each local logarithmic activity, together with its derivatives, must be small. We obtain the extra power from the local quadratic comparison and the flat interpolation argument

already proved. Throughout this section the derivative order, block length, patch, forest and tags are fixed. The path bounds, flat identities and positive precision estimates retain the hypotheses of the preceding sections.

81.1 A cubic remainder in a flat chart

For a local energy address p, let $\mathcal{E}_p(U,V)$ denote its unmasked energy, with the original predictor tags. Let $I(p)$ be its set of extracted quadratic rows; $|I(p)| \le 2$. Write $f_i(V)$ for the zero-fluctuation residual and $M_i(V)u$ for the transported cut linear row. Define

$$R_p(U,V,u) = \mathcal{E}_p(U,V) - \frac{1}{2}\sum_{i\in I(p)} |f_i(V) + M_i(V)u|^2, \tag{406}$$

$$F_p(b,u) = R_p(\exp(u)\,\mathrm{replica}(\exp b), \exp b, u). \tag{407}$$

The four address types are the fine curvature, predictor penalty, coarse curvature with its two mismatch terms, and the null-row square. For the last two, the parent set is empty. These are the same unmasked expressions as in Section 72.

For every fixed k, there are a radius $r > 0$ independent of the period and a function A of upper exponent zero such that

$$F_p \in \mathcal{B}_k(A(x), 1; (b,u)), \qquad \max(\|b\|_\infty, \|u\|_\infty) \le r. \tag{408}$$

Here and below the same bound holds at every address. To verify it, use the four dressed fields and their predictor bounds in Section 80. Each cut path has length $O(R+1)$, its absolute coefficients have bounded sum, and its transport derivatives cost a polynomial in $R+1$. The local energy is a sum of squared curvature rows and inner products of the coarse stroke with linear guide rows. Its derivatives therefore cost only another polynomial in $R+1$. Each affine row in (406) is the sum of the residual already estimated there and a cut linear map applied to u. On the unit ball, the coordinate map $u \mapsto u_e$ has value and first derivative at most one and all higher derivatives zero. The product rule bounds its transported image by a polynomial in $B+1$. Squaring and summing at most two such rows proves (408). This is a bound on the unmasked formula; it makes no small-field assertion on that whole fixed ball.

Put $t = qh^{-1/2}$ with $q > 0$ fixed and sufficiently small. Every coarse field $\exp b$ with $\|b\|_\infty \le 3t$ is clean on all its plaquettes. Indeed the exponential chart gives $\mathrm{chord}(\exp b_e) \le C\|b_e\|$, and the four-link triangle inequality gives plaquette chord at most $12Cqh^{-1/2}$. Choose $q \le (12C)^{-1}$ and use $h^{-1/2}$ no larger than the coarse flag threshold. This check applies on every period.

On the radius-$3t$ ball, the fields in (407) have the exact local graft $U = \exp u\,\mathrm{replica}\,V$. The annular comparison then gives

$$|F_p(b,u)| \le E(x), \qquad E \text{ of upper exponent } -3/2. \tag{409}$$

More explicitly, the fine and predictor rows have affine error of upper exponent -1 and amplitude of upper exponent $-1/2$; the difference of their squares consequently has upper exponent $-3/2$. Their extracted rows agree with $f_i + M_i u$, since cleanliness removes the clipping on the complete read stencil. The coarse curvature remainder and null-row square have upper exponent -2. Each mismatch term is an inner product of a coarse stroke of upper exponent $-1/2$ with an error of upper exponent -1. Their sum has upper exponent $-3/2$. Taking the sum of these four nonnegative bounds gives one E for all address types. These are precisely the estimates proved in the annular argument; no inequality is differentiated in obtaining (409).

Let $Q \ge 0$ have upper exponent $-1/2$, and put $B_0 = Q + 3t$. Apply (402) to F_p, with $k = j + 3$, value bound E and fixed-ball derivative bound A. Its constants satisfy

$$W = 1000(E/t^2 + AB_0), \qquad Y = 1000E/t + 3WB_0, \qquad H = E + YB_0 + Yt + Wt^2 + At^3.$$

Their upper exponents are $-1/2, -1, -3/2$, respectively. For example E/t^2 and AB_0 have upper exponent $-1/2$, and each of YB_0, Yt, Wt^2, At^3 has upper exponent $-3/2$. Thus

$$F_p \in \mathcal{B}_j(H, t^{-1}; z) \quad (\|z\| \le Q), \qquad H \text{ of upper exponent } -3/2. \tag{410}$$

The reciprocal t^{-1} has upper exponent $1/2$ and is paid by the small derivatives of the chart coordinates in the next step.

81.2 Returning to a protected remote address

Suppose the anchor of p is farther than $200B$ from the seeds. Let $V(y)$ be a coarse variation of derivative scale with upper exponent $-1/2$, and let $u(y)$ be its coherently compensated fine increment. Hold the underlying fine configuration U_* fixed, as in Section 79. Let $\psi_p(y)$ be the product of the doubled protective cutoffs on the local read region. If ψ_p vanishes on a neighbourhood of y_0, the protected remainder has zero jets there. Otherwise the closed support argument bounds the increment on that read region by $q_0 + 4a\tau$, with q_0 of upper exponent $-1/2$.

After a signed pullback, the relative increment is Ad of this increment by the appropriate replica link. The product rule and the unitary bounds give amplitude

$$R_G^2(q_0 + 4a\tau)\{1 + (v + 1 + v)\}^j$$

for its jets through order j; this still has upper exponent $-1/2$. On tree bonds the increment is identically zero. All other bonds in the radius-B read region are active because the address is remote from the seeds. The compensated identity therefore gives, on a neighbourhood of y_0,

$$U_* = \exp u(y) \,\mathrm{replica}\, V(y) \quad \text{on the radius-}\lfloor 7B/8 \rfloor \text{ patch},$$

with the signed pullback understood on both sides.

Choose one coarse gauge from $V(y_0)$ on this patch. The coarse logarithmic coordinates $b(y)$ have jets of upper exponent $-1/2$ by the fixed-gauge construction in Section 80. Conjugate $u(y)$ by this fixed gauge and set it to zero outside the patch. This preserves its derivative bound. Denote the resulting pair of arrays by $z(y)$. For one nonnegative Q of upper exponent $-1/2$, we have $z \in \mathcal{B}_j(Q, 1; y_0)$.

All terms in (406) are determined by this patch. To check the radius, the cut guide and transported linear rows read within $B/4$ of their address; the local energy and its predicted coarse field read within $B/2$, including the fixed predictor reach. Both fit inside $7B/8$ for the previously chosen geometric threshold. The row transports, predictor tags and quadratic extraction decisions are unchanged there. Gauge covariance and the local equality of the fields consequently identify the actual remainder with $F_p(z(y))$ throughout a neighbourhood.

For a derivative of order $k \le j$, the chain rule has at most $k!$ partition terms. A term with m blocks is bounded by $Ht^{-m}Q^m$. Hence every derivative of $F_p(z(y))$ is bounded by

$$H \max(1, j!)(1 + Qt^{-1})^j.$$

The second factor has upper exponent zero. We have proved that the remote unmasked remainder has jets of upper exponent $-3/2$ whenever its protector is not locally zero. The protector has amplitude one and derivative scale bounded by a function of upper exponent zero times its support cover. Multiplying it back in preserves the amplitude exponent $-3/2$, allowing this support factor in the derivative scale.

81.3 The remote mean square and other short terms

The local mean residual m_p has remote value bound of upper exponent -1, proved in Section 75. Its jets have the same exponent. Here is why the scalar estimate suffices in this application. On a flat probe ball all sectors are clean and the actual mean agrees with its unmasked expression. The derivative estimate on the larger *shrinking* ball requires an additional check, since the intermediate probes need not satisfy the sector flag conditions. Let $B_0 = Q + 3t$ as above and write $A(z)$ for the empty-seed precision at logarithmic probe z. The same-history comparison at the identity supplies $A(0) \geq r_0 I$ for a fixed $r_0 > 0$. On $\|z\| \leq B_0$, the transported-row estimates give an entry derivative bound Cv_1, where C is fixed and v_1 has upper exponent zero. Integrating this derivative along the segment from 0 to z gives $|A(z)_{ik} - A(0)_{ik}| \leq Cv_1 B_0$. The matrices have range at most B and coordinate density at most ρ per site. Both the row and column sums of this difference are therefore at most $b_B \rho C v_1 B_0$, which tends to zero. The elementary row-column bound on the Euclidean operator norm follows by Cauchy–Schwarz: $\|Hv\|_2^2 \leq (\sup_i \sum_k |H_{ik}|)(\sup_k \sum_i |H_{ik}|)\|v\|_2^2$. Consequently $A(z) \geq (r_0/2)I$ throughout the shrinking ball for large x. Every principal restriction has the same floor. The exponentially weighted entry estimates remain uniform there; the inverse and padded-inverse jet bounds proved earlier thus apply with fixed reduced floor and width. They cost only a polynomial in B at fixed derivative order.

The empty-seed residual is the zero-fluctuation row residual minus the transported row applied to its padded-inverse response, that is $f - MPM^* f$ after the fixed row normalization. All four factors now have jet bounds of upper exponent zero on this shrinking ball. The product and row-sum estimates give the required upper exponent zero bound for the residual there. On the smaller radius-$3t$ ball every empty-seed probe is clean, so the same-history stationary comparison applies separately to each probe and gives the remote value exponent -1. No flag condition was assumed on the entire larger ball. Apply the same interpolation above with this probe value exponent -1. Then W, Y, H have upper exponents $0, -1/2, -1$. The fixed-gauge composition again costs only upper exponent zero. Finally the mean residual depends only on the radius-$24B$ patch and the unchanged windows; a point farther than $30B$ from the seeds has the same local formula as the empty-seed comparison. This gives the claimed exponent -1 for every fixed jet order, with no derivative of a gauge choice or stationary minimizer.

At most two such residuals enter a local mean-square term, so its jets have upper exponent -2. The remote logarithmic action is $-\theta\beta$ times the sum of this mean-square term and the protected remainder, where $0 \leq \theta \leq 1$ is the fixed channel weight and $0 < \beta \leq 2h$. Its amplitude therefore has upper exponent $-1/2$. This uses the same β as in the original action.

The other two short logarithmic terms are the remote normalization correction and the protected chart logarithm. Their jets have upper exponent $-1/2$ by Sections 78 and 77. Each term is set to zero on its prescribed packet neighbourhood; this indicator depends only on the frozen sector. Consequently the three short families have one common nonnegative amplitude $b(x)$ of upper exponent $-1/2$, and derivative scale

$$v_0(x)\{1 + \mathrm{cov}_B(S_p)\}, \qquad v_0 \text{ of upper exponent zero.} \tag{411}$$

Here S_p is the full enlarged read support of the term. For fixed enlargement ℓ, these supports lie in radius $(\ell + 1)B$ balls about their anchors and have uniformly bounded cover number. The number of anchors meeting a given site is at most $b_{(\ell+1)B}$ times the fixed density of the three families. Their exponentially cover-weighted site sum is therefore bounded by this polynomial count times $b(x)$ and a fixed exponential in the cover bound. It still has upper exponent $-1/2$, uniformly in the period. This supplies the short-activity part of the logarithmic price estimate.

82 Derivative prices for the Gaussian correction chains

The corrections in Section 74 are sums over ordered itineraries of physical bonds. Their operator bounds alone do not price each activity in the connected expansion. We now keep the itinerary labels and their full read supports while differentiating and summing them.

82.1 Products along a fixed itinerary

Write q for the number of colour coordinates. An itinerary of length N consists of physical bonds $e_0, \dots, e_N$; colour indices are summed within its matrix product. Suppose the entry jets of its rth matrix have amplitude $a_r \geq 0$ and common derivative scale $d \geq 0$. Repeated matrix multiplication and the product rule give

$$\mathcal{B}_j\left(q^N \prod_{r=0}^{N-1} a_r, Nd\right) \tag{412}$$

for any entry of the colour-block product. This follows by induction on N: multiplying the next entry adds d to the derivative scale, multiplies the amplitude by a_r, and introduces at most q choices for the next colour. The estimate deliberately overcounts when $N = 1$. Taking a colour trace adds a factor q. Integrating such a trace over $s \in [0, 1]$ preserves this bound when the entry estimates are uniform in s.

For the present matrices, this last differentiation under the integral is valid by the compact-glide argument of Section 78: all principal determinants are nonzero on one common parameter neighbourhood for $s \in [0, 1]$, and the finite matrix products are jointly smooth there. A closed itinerary contributes its diagonal trace; a nonclosed itinerary contributes zero to that trace.

All the matrices in these words have range at most $3B$. If some consecutive pair of bonds has all endpoint distances greater than $3B$, the corresponding matrix block vanishes identically as the coarse field varies. Thus the whole word is a zero germ. Call an itinerary admissible when every consecutive pair has endpoints at distance at most $3B$. This condition is fixed by its geometry, and is retained in every price below.

Let a be the common weighted jet amplitude of the actual precision A, let h_0 price the padded inverse of the glide, and let t_0 price its error $A_s P_s - 1$. These quantities were constructed in Section 78. The amplitudes a, h_0 have upper exponent zero; t_0 has every negative upper exponent. The first logarithmic correction has word

$$(A - 1)P_s(-E_s)^{n+1}, \qquad E_s = A_s P_s - 1.$$

It has $n + 3$ matrix factors. If d_A bounds their derivative scales, (412) and the colour trace give

$$\mathcal{B}_j\big(C_A Q_A^n, (n + 3)d_A\big), \qquad C_A = q^4(a + 1)h_0 t_0, \quad Q_A = q t_0. \tag{413}$$

The same argument applies to the flat-reference correction. Its coarse field is constant and satisfies the same entry bounds.

For the second correction put $D = AC - 1$, where C is the symmetric padded covariance. Its entry jets have amplitude w_0 with every negative upper exponent, obtained by multiplying the padded-inverse error by the full inverse envelope and then by A. If d_D is their derivative scale, the word $D(-sD)^n$ has $n + 1$ factors. Since $0 \leq s \leq 1$, its integrated trace has bound

$$\mathcal{B}_j\big(C_S Q_S^n, (n + 1)d_D\big), \qquad C_S = q^2 w_0, \quad Q_S = q w_0. \tag{414}$$

These formulas price the actual trace words, including their endpoint identifications. Discarding a factor $1/2$ or another coefficient of absolute value at most one only enlarges the bounds.

82.2 The two protected Gaussian endpoints

The inverse correction is a quadratic form with word $(-D)^{n+1}A$, of length $n+2$, and a Gaussian coordinate at each endpoint. Those coordinates are not bounded on the whole sample space. Their protective cutoffs supply the needed local bound.

Let $u(y) = \nu(y) + g_\beta z(y)$, $g_\beta = \beta^{-1/2}$, be the compensated increment on an active physical bond. If its doubled cutoff is not a zero germ, the increment jet bound has amplitude $s_0 + 4a_0\tau$, where s_0 has upper exponent $-1/2$ and $a_0 > 0$ is the fixed cutoff parameter. The mean jets have amplitude m_0 of upper exponent $-1/2$. Hence

$$z_e \in \mathcal{B}_j(p,1), \qquad p = \sqrt{\beta}\,(s_0 + 4a_0\tau + m_0), \tag{415}$$

which has upper exponent zero since $\beta \le 2h$. If either cutoff is a zero germ, the protected quadratic activity is a zero germ instead. If an endpoint is inactive, the inactive block of the completed precision and covariance is the identity; the corresponding defect block is zero. These block identities persist on the neighbourhood where the fixed principal windows remain invertible. The quadratic word therefore vanishes as a germ unless both endpoints are active. This justifies using (415) only where it applies.

Suppose the cutoff jets have amplitude one and scale d_0. The word has entry amplitude $q^{n+2}aw_0^{n+1}$. The two endpoint colour sums add q^2p^2, and the product rule adds two endpoint scales and two cutoff scales. Enlarging the resulting scale gives

$$\mathcal{B}_j\big(C_Q Q_Q^n, (n+2)d_Q\big), \qquad C_Q = p^2q^4aw_0, \quad Q_Q = qw_0, \quad d_Q = d_D + d_0 + 1, \tag{416}$$

after enlarging d_D to include the scale of A. All derivative scales here have upper exponent zero. The unprotected linear score is handled separately: its coefficient has every negative upper exponent by the mean equation, while its one protected endpoint has the bound just proved. Summing its q colours and multiplying its cutoff produces one local bond activity with every negative upper exponent, even after its fixed-radius support is counted.

82.3 Paying the derivative order before the chain length

All three itinerary families have the form

$$\mathcal{B}_j(CQ^n, (n+u)d), \qquad u \in \{3,1,2\},$$

on admissible itineraries and zero germs otherwise. To give every jet a common amplitude and unit derivative scale, multiply the amplitude by $(1+(n+u)d)^j$. The elementary inequality $n+1 \le 2^n$ implies

$$1 + (n+u)d \le 2^n\{1 + (1+u)d\}.$$

It is therefore enough to replace

$$C \longmapsto C' = C\{1+(1+u)d\}^j, \qquad Q \longmapsto Q' = 2^jQ. \tag{417}$$

No factor depending on n remains outside the geometric power. For fixed j, C' retains every negative upper exponent of C, and Q' retains every negative upper exponent of Q.

82.4 The rooted support sum

Fix a halo multiplier $m \geq 4$, also large enough to contain the previously established mean, window and compensated-variation read radii, a support exponent $a_1 \geq 0$, and a length reserve $\varepsilon > 0$. The inequality $m \geq 4$ alone is used in the count below; the additional read-radius requirement is needed when applying it to the actual activities. The support of an itinerary is the union of the radius-mB halos of the endpoint sets of all its bonds. Put

$$\rho_L = L^4 \#\{\text{axes}\}, \quad K = b_{3B+2}\rho_L, \quad s_m = b_{m+1}.$$

There are at most K choices for a next bond in an admissible step: its anchor is within $3B+2$ of the preceding anchor. Thus there are at most K^{n+u} admissible itineraries of length $n+u$ with prescribed initial bond.

Each bond halo has B-cover number at most s_m. Consecutive halos intersect because $3B+2 \leq mB$. Subadditivity along the ordered chain gives support cover at most $(n+u+1)s_m$. If this support contains a fixed site x_0, the initial anchor is within $(n+u+1)(mB+1)$ of x_0: follow the chain to a halo containing x_0 and sum the step distances. Since $n+u+1 \leq (u+1)2^n$, the number of possible initial bonds is at most

$$b_{(u+1)(mB+1)}16^n\rho_L.$$

The factor 16^n is the fourth power of 2^n; it also bounds the additive one in the lattice ball formula. Wrapping on a torus can only reduce these counts.

Multiplying the price $C'(Q')^n$, the count, the support weight and the extra length weight $e^{2\varepsilon n}$ gives the bound

$$\sum_{\lambda : x_0 \in S_\lambda} p_\lambda e^{a_1 \operatorname{cov}_B(S_\lambda)} e^{2\varepsilon n(\lambda)} \leq HC' \sum_{n \geq 0} \left[Q' 16 K e^{a_1 s_m + 2\varepsilon}\right]^n, \tag{418}$$

where

$$H = b_{(u+1)(mB+1)}\rho_L K^u e^{a_1(u+1)s_m}.$$

All sums have nonnegative terms, so grouping first by length and then by initial bond is justified without a convergence assumption. At fixed m, L, j, both K and H have upper exponent zero. The geometric ratio in (418) is eventually at most $1/2$, and its sum is at most $2HC'$. Consequently every prescribed negative upper exponent is available for the rooted price of these correction chains. This pays the actual full read supports and every itinerary label, including repeated physical bonds; it does not assume disjoint paths or independent Gaussian coordinates.

83 Prices for the inherited activities

The remaining logarithmic activities come from the incoming representation: its regular activities, cubic and quartic path terms, quadratic loop terms, and the true long-path tails on the fine and coarse levels. We convert their existing coefficient bounds into prices on the full coarse read supports. This section assumes those incoming coefficient bounds on the same representation and paths as the selected step. It proves the price conversion; maintenance of the incoming bounds along the renormalisation run is a separate construction requirement.

83.1 Coarsening a connected cover

Let π be the parent map with block length L, and let $c_M(X)$ be the connected covering number at radius M. Thus X is contained in the union of M-balls around $c_M(X)$ centres, and that union is

connected by unit lattice steps. For $B \ge 1$, $M \le 2B$ and integer $\ell \ge 0$, put

$$P_\ell = b_{\ell+6}, \qquad v_\ell = 10P_\ell/L, \qquad Y = \mathrm{halo}_{\ell B}(\pi X).$$

Then

$$c_B(Y) \le P_\ell + v_\ell c_M(X). \tag{419}$$

Here is a proof retaining the factor L^{-1}. Choose a minimal cover with centre set D. The graph joining centres at distance at most $2M+1$ is connected. Traverse a spanning tree out and back, replacing each edge by steps of max-distance at most one (diagonal steps are allowed). This gives a walk visiting all centres, of length at most $(4M+2)|D|$. Sample it every LB steps and apply π. If its length is N, the resulting centre set D' satisfies $LB|D'| \le N + LB$. The parent-distance inequality

$$L\, d(\pi x, \pi y) \le d(x,y) + L$$

shows that every projected walk point is within $2B$ of a sampled centre and that consecutive sampled centres are within $2B$. Every point of X is within $M \le 2B$ of an original centre; its parent is therefore within $3B$ of that centre's parent. Thus Y is covered by the $(\ell+5)B$-balls about D', with connected union. Tile their slightly enlarged $(\ell+6)B$-balls by B-balls with centres on a B-spaced four-dimensional grid. There are at most P_ℓ tiles per centre. These enlarged halos overlap along the sampled walk, so the resulting cover remains connected. Finally $(4M+2)|D| \le 10B|D|$ proves (419). For an empty set the inequality is immediate. Wrapping any of these walks or grids only decreases the number of distinct centres.

If $o \in X$ and $y \in Y$, the same spanning walk gives

$$d(y, \pi o) \le (\ell + 20)B\{1 + c_M(X)\}. \tag{420}$$

Indeed two points of X are at distance at most $(4M+2)c_M(X) + 2M$; apply the parent-distance inequality and then add the ℓB halo. The displayed larger constant bounds the result for $M \le 2B$, $B \ge 1$. Inspecting the fine plaquettes touching a support adds at most four fine lattice steps. Their parents lie within five coarse steps, so replacing ℓ by $\ell + 5$ pays all these reads.

83.2 From an anchor bound to a site bound

We give a general counting estimate. Suppose nonnegative weights b_i, integer costs $d_i \ge 0$, fine anchors o_i, and coarse supports Y_i satisfy

$$\sup_o \sum_{i:o_i=o} b_i e^{\sigma d_i} \le \Gamma, \qquad c_B(Y_i) \le v + v' d_i, \qquad y \in Y_i \Longrightarrow d(y, \pi o_i) \le W(1+d_i).$$

Let k, W be nonnegative integers and $0 \le g_i \le C b_i (1+d_i)^k$ with $C \ge 0$, and assume

$$a \ge 0, \quad \varepsilon > 0, \quad av' + 2\varepsilon \le \sigma. \tag{421}$$

For a fixed coarse site y, the fine anchors of labels with $d_i = n$ and $y \in Y_i$ have parents in a radius-$W(1+n)$ ball. There are at most $L^4 b_W (1+n)^4$ such anchors, because a parent has exactly L^4 residues. The sum of $g_i e^{a c_B(Y_i)}$ over these labels is at most

$$L^4 b_W \Gamma C e^{av} (1+n)^{4+k} e^{-2\varepsilon n}.$$

Use the exponential series at $\varepsilon(1+n)$ to obtain

$$(1+n)^{4+k} \le \varepsilon^{-(4+k)} (4+k)! e^{\varepsilon} e^{\varepsilon n}.$$

Summing the geometric series proves

$$\sup_y \sum_{i:y\in Y_i} g_i e^{ac_B(Y_i)} \le e^{av}\Gamma C L^4 b_W \frac{\varepsilon^{-(4+k)}(4+k)!e^{\varepsilon}}{1-e^{-\varepsilon}}. \tag{422}$$

Restricting to $d_i \ge n_0$ adds the factor $e^{-\varepsilon n_0}$. The proof works with arbitrary label multiplicities, since the input already sums all labels at an anchor. Nonnegative finite partial sums justify the estimate also for a countable alphabet.

We need an additional factor $e^{2\varepsilon d_i}$ in the output price. Apply the preceding argument to weights $b_i e^{2\varepsilon d_i}$ and exponent $\sigma-2\varepsilon$. The sufficient condition becomes $av'+4\varepsilon \le \sigma$. For inherited supports use (419) with $d_i = c_M(X_i)$ and $W = (\ell+20)B$. If $4\varepsilon < \sigma$, choose L large enough that $a10P_\ell/L+4\varepsilon \le \sigma$, before increasing x. The incoming cover radius $M = B(h)$ is eventually at most $2B(x)$: $h = x + c_L$ with c_L fixed, and the logarithmic-square radius has $B(x+c_L)/B(x) \to 1$. All factors in (422) other than ΓC then have upper exponent zero.

83.3 Path lengths, including winding paths

For a bundle of r paths based at one fine anchor, let d_i be the sum of their true lengths, including their final plaquette walks. Dividing each walk into pieces of B steps shows that its visited sites have connected B-cover number at most $1 + \text{length}/B$. All paths share the anchor, so their union with the anchor has cover at most $1 + r + d_i/B$. Applying (419) at $M = B$ gives

$$c_B(Y_i) \le P_\ell + v_\ell(1+r) + (v_\ell/B)d_i.$$

The support lies within $(\ell B + 1)(1 + d_i)$ of the parent anchor. Thus the preceding site estimate applies with $v' = v_\ell/B$. For any fixed L, this coefficient tends to zero, so the required strict exponential reserve is eventually available. No path length is replaced by its endpoint distance; in particular, winding and repeated edges retain their costs.

Suppose the path coefficients satisfy $\sum_i b_i e^{\sigma d_i} \le \Gamma$ at each translated anchor. An order-r atom evaluated with its original small-field mask has absolute value at most $b_i\tau^r$: every one of its r curvature arguments has norm at most τ, while the unhealthy branch is zero. An external multiplier bounded by $D(x)$ therefore has scalar price $D(x)\tau^r b_i$. The site conversion preserves the sum of the upper exponents of D, τ^r, and Γ, allowing any strictly larger power for the final bound. Applied to the actual families,

family	multiplier and coefficient budget	upper exponent
cubic	$2h\,\tau^3 \cdot C$	$-1/2$
quartic	$2h\,\tau^4 \cdot C$	-1
quadratic loop	$\tau^2 \cdot 2P(h)$	-1

Here C is the incoming fixed path bound, $P(h)$ is the monitor, and the channel weight has absolute value at most one. The multiplier is the original $\beta \le 2h$, wherever it occurs. The fixed number of signed orientations and plaquette axes is summed, not discarded.

For the tail of a squared path row, retain the ordered pair of path labels (i,k). With coefficients a_i, a_k and lengths l_i, l_k, its price is

$$b_{ik} = 2|a_i a_k|\, 1_{\{l_i d>R \text{ or } l_k d>R\}},$$

where $d = 1$ for the fine row and $d = L$ for the coarse row. This bounds the actual half inner product because each unmasked transported curvature has norm at most two; the kept-row indicator can

only decrease its magnitude. If $\sum_i |a_i| e^{\sigma l_i} \le C$, then

$$\sum_{i,k} b_{ik} e^{(\sigma/2)(l_i+l_k)} \le 2e^{-\sigma R/(2d)} C^2. \tag{423}$$

Indeed every retained pair has $l_i + l_k \ge R/d$; extract $e^{-\sigma R/(2d)}$ and use the product of the two weighted row sums. The genuine read support includes both full paths and the kept-row test halo. After projection that extra halo is paid by replacing ℓ by $\ell + 2$. The site bound with its length reserve is valid if $8\varepsilon < \sigma$, since its available input exponent is now $\sigma/2$. As $R = \lfloor B/100000 \rfloor$, the right side of (423) has every negative upper exponent, even after multiplication by $2h$, fixed powers of the true length and the site-conversion cost. For the coarse-level tail the parent map is the identity; its population factor is one, and the exponential still has fixed $d = L$. The argument thus covers both tails without confusing their spatial scales.

83.4 Inherited jets in the compensated variation

The incoming regular activity budget has upper exponent $-11/10$; the incoming forced activity budget has every negative upper exponent. The former bounds the original masked occurrence on its healthy domain, extended by zero, and the latter bounds the forced occurrence everywhere. Apply the anchor-to-site estimate to their original supports and masks. This preserves those exponents and all label multiplicities.

During the compensated variation the underlying fine field U_* is held fixed. Hence the regular, cubic, quartic, loop, fine-tail and forced values are constant in the coarse parameter. Their protective factors vary. On a coarse support Y, each protective bond cutoff has amplitude one and derivative scale of upper exponent zero. Covering Y by its B-cover, its full bond read set contains at most $b_B b_1 \rho_L c_B(Y)$ bonds. The product rule therefore gives the protector derivative scale $v_0(x) c_B(Y)$, with v_0 of upper exponent zero. Multiplying a constant value by this protector preserves its scalar price.

The coarse tail is the exception: its original coarse field varies. For a path of length l, the unitary product argument gives its transported curvature jets amplitude $|a_i| R_G^2$ and derivative scale $2R_G v l$. A product of two paths thus has scale $2R_G v(l_i + l_k)$, and its half inner product has amplitude at most $|a_i a_k| R_G^4$. Multiplying by the original β and channel coefficient is paid by $2hR_G^4 b_{ik}$. The same tail estimate prices these labels with every negative upper exponent, retaining the total length in the derivative scale. There is no moving coarse kept-row indicator in this term: its unmasked coarse counterpart is the smooth path-pair expression with the fixed long-pair selector.

Choose $-1/2 < t < 0$. By the strict exponent slack, the six inherited logarithmic families together have a rooted price at most h^t for sufficiently large x, with the support and 2ε extra-cost reserves just proved. For the coarse tail add its differentiated price to the scalar one; both are nonnegative and negligible, so this addition is paid by choosing an intermediate exponent between $-1/2$ and t first. The forced family has the same bound. Their common derivative scale can be taken as

$$v_1(x)\{c_B(Y_i) + d_i\}, \qquad v_1 \text{ of upper exponent zero}, \tag{424}$$

where d_i is the old support cover or the true path length, as appropriate. The bounds are uniform over the held fine configuration and preserve its original labels and parameters.

84 From a seed energy gain to packet estimates

We now supply the passage from local energy and chart estimates to the packet price used in Section 63. There are three distinct costs. Extending the energy from a seed core to its full support

introduces a small annular error. The density and normalization factors introduce a polynomial logarithmic expense. Differentiating the packet introduces a further polynomial factor. All three are paid from the same energy gain per seed.

Keep the scales h, P, d, τ, B of (268), and fix the block length, chart, shell width and all other local parameters before sending x to infinity. All constants below are independent of the lattice period. For clarity, say that $r(x)$ has upper exponent a if, for every $b > a$, $|r(x)| \le h(x)^b$ eventually. This permits subpower losses. Products add upper exponents, and a fixed finite sum preserves a common upper exponent: in each case choose a strictly smaller exponent than the one finally required and absorb fixed constants using $h \to \infty$. In particular B, d and every fixed real power of P have upper exponent zero, while τ has upper exponent $-1/2$.

84.1 Extending a core energy bound

Consider a packet with n seeds. In each signed channel let $D_2 \subseteq D_{200}$ be the sets of local energy addresses within $2B$ and $200B$ of its seeds. Their sizes satisfy

$$|D_{200}| \le n\, b_{200B+1}\, s_L,$$

where s_L bounds the number of local energy addresses per coarse site. Let e_p be the assigned energy at address p in that channel. Suppose the core energy is $\sum_{p \in D_2} e_p$ and the annular terms satisfy $e_p \ge -r(x)$ on $D_{200} \setminus D_2$, with $r(x) \ge 0$. Then

$$\sum_{p \in D_{200}} e_p \ge \sum_{p \in D_2} e_p - r(x) b_{200B+1} s_L n.$$

Average this inequality over signed channels with their normalized weights. If the averaged core energy is at least $\mu\tau^2 n$, the averaged extended energy Q_A obeys

$$Q_A \ge (\mu\tau^2 - E(x))n, \qquad E(x) = r(x) b_{200B+1} s_L. \tag{425}$$

No energy gain from one channel is multiplied by a factor from another: the average is taken on the common configuration and common packet.

The local inputs in this application are a positive seed charge μ, the displayed core bound, and an annular remainder with upper exponent $-3/2$. The latter is proved in Section 72 by comparing the fine and predictor energies with their extracted affine squares. On an address where no quadratic term was extracted, the fine and predictor energies are nonnegative; only the one-sided lower bound is needed there. On selected addresses, the local Taylor estimates bound the difference from the extracted quadratic energy in absolute value. The protected $304B$ fluctuation region and the $300B$ mask conditions allow those local estimates to be used throughout the $200B \setminus 2B$ annulus. Multiplying r by the fixed-density ball count preserves its upper exponent, so E also has upper exponent $-3/2$.

84.2 The packet density and its normalization

Let A denote the packet. Its logarithmic density has the form

$$\Phi_A = -\beta(Q_A + Q_A^{\mathrm{mean}}) + \sum_{i \in V_{200}(A)} c_i + \sum_{i \in V_{304}(A)} j_i - N_A \log m_\beta. \tag{426}$$

Here $Q_A^{\mathrm{mean}} \geq 0$ is the extracted mean-square contribution; c_i are the local Gaussian normalization corrections; j_i are the protected chart corrections; and N_A counts the free chart bonds within distance $10B$ of the packet. The bond sets and this count satisfy, with constants fixed at L,

$$|V_{200}(A)| \leq C_{200} d^4 n, \qquad |V_{304}(A)| \leq C_{304} d^4 n, \qquad N_A \leq C_{10} d^4 n.$$

Cover each seed neighbourhood by a lattice ball. There are $L^4 \#\{\text{axes}\}$ fine bonds per coarse site, so summing these bounds over seeds gives the claimed counts. Overlaps reduce the count.

The one-bond Gaussian normalization is

$$m_\beta = c_{\mathrm{chart}} (\sqrt{2\pi}\, \beta^{-1/2})^q, \qquad c_{\mathrm{chart}} > 0,$$

where q is the colour dimension. Thus

$$-\log m_\beta = -\log c_{\mathrm{chart}} - \frac{q}{2} \log(2\pi) + \frac{q}{2} \log \beta \leq C_{\mathrm{norm}} d$$

for $0 < \beta \leq 2h$ and sufficiently large x. The bound holds even when β is very small, since only an upper bound on $\log \beta$ is used. Suppose also $|c_i| \leq C_{\mathrm{cov}}$, $|j_i| \leq 1$, and that the likelihood-ticket product is bounded by $\exp(C_{\mathrm{tag}} d^4 n)$. These are uniform local analytic bounds. The other packet masks, enforcement factors and contractions take values in $[0, 1]$. Dropping the nonpositive mean-square term in (426), and using $d \geq 1$, gives

$$\Phi_A + C_{\mathrm{tag}} d^4 n \leq -\beta Q_A + K d^5 n \tag{427}$$

for a fixed $K \geq 0$.

Here the ticket bound follows directly from the normalized tag probabilities. Let N_f and N_p be the numbers of frame and predictor tags in the fixed patch. A site likelihood is N_f times its tag probability, and a bond likelihood is N_p times its tag probability. The probabilities are nonnegative and at most one: in the minimizing frame branch they are uniform on the nonempty set of minimizers, in the fallback branch they are uniform on all tags, and the predictor weights are nonnegative with sum one. Thus each likelihood is at most $N_f \leq e^{N_f}$ or $N_p \leq e^{N_p}$, respectively. The radius-r seed neighbourhood contains at most $n b_{r+1}$ sites, and the bonds incident to it number at most $n b_{r+2} \#\{\text{axes}\}$. Multiplying one factor at each such physical site and bond gives

$$\mathrm{tickets}_r \leq \exp[n\{b_{r+1} N_f + b_{r+2} \#\{\text{axes}\} N_p\}] .$$

For $r = 300B$ this is the asserted $\exp(C_{\mathrm{tag}} d^4 n)$ bound. A tag used by several local tests is still charged only once in this product. There is no independent tag resampling at each use.

A packet is set to zero if its seed and local flag conditions fail. When its value is nonzero, every retained mask is nonzero. The mask support conditions then give the local hypotheses of (425), including the control of the hot fluctuation bonds. Therefore

$$|\mathcal{P}_A| \leq \exp\{\Phi_A + C_{\mathrm{tag}} d^4 n\} \leq \exp\{((K d^5 + \beta E) - \beta \mu \tau^2) n\}.$$

Since $\beta \leq 2h$ and E has upper exponent $-3/2$, the product βE tends to zero and is eventually at most $1 \leq d^5$. We have proved

$$|\mathcal{P}_A| \leq \exp\{(C d^5 - \beta \mu \tau^2) n\}, \qquad C = K + 1. \tag{428}$$

The estimate also holds for packets which vanish, since its right-hand side is positive. This argument retains the Gaussian normalization m_β; it does not replace the original density by an unnormalized free density.

84.3 Derivatives at the edge of a mask

For a C^j map F defined near a point y in a finite-dimensional real normed space, write

$$F \in \mathcal{B}_j(U, v; y) \quad \text{if} \quad \|D^k F(y)\| \le U v^k \quad (0 \le k \le j), \qquad U, v \ge 0.$$

The zero-order bound is included. The product rule implies

$$F \in \mathcal{B}_j(U, v; y), \quad G \in \mathcal{B}_j(V, w; y) \implies FG \in \mathcal{B}_j(UV, v + w; y). \tag{429}$$

Indeed the norm of the kth derivative is bounded by $UV \sum_{r=0}^{k} \binom{k}{r} v^r w^{k-r} = UV(v+w)^k$. Repeated application gives a product bound with multiplied amplitudes and added derivative scales.

For the packet variation, the underlying fine configuration in its mask and likelihood factors is held fixed; the coarse field and its compensating fluctuation vary coherently. This matters at the boundary of a cutoff. If any of the retained mask products is identically zero on a neighbourhood of y, the packet and all its derivatives through order j vanish at y. In the complementary case, every relevant individual mask has points of nonzero value arbitrarily close to y. Continuity of its argument therefore places that argument in the *closed* support of the mask at y. For example, a cutoff which vanishes when $\|u\| > 2a\tau$ implies $\|u(y)\| \le 2a\tau$ in this case. The same argument gives the closed fine and predictor threshold inequalities. It does not claim that the mask value at y is nonzero. The scalar estimate (427) and the core estimate must hold on these closed conditions; this is the closed support version of the local input used below.

Separate the varied packet as $e^{\phi(v)} H(v)$, apart from its fixed seed-eligibility indicator. The local differentiated estimates give

$$\begin{aligned} \phi - \phi(y) &\in \mathcal{B}_j(b(x)n, 1; y), \\ H &\in \mathcal{B}_j(e^{\ell n}, c(x)n; y), \\ e^{\phi(y)+\ell n} &\le e^{(Cd^5 - g)n}, \qquad g = \beta\mu\tau^2, \end{aligned} \tag{430}$$

where $b, c \ge 0$, b has upper exponent 1 and c upper exponent zero. Here ℓ is the ticket expense per seed. The first bound is (405). To check the bound for H, first consider a coarse noise cutoff. Its argument is the squared chord of $V_e Q(U)_e^{-1}$ divided by $(W\tau)^2$, with $W > 0$ fixed and U held fixed. If the chord is strictly larger than $2W\tau$, the inner cutoff is locally zero. Otherwise the normalized matrix displacement has bounded value and derivatives of upper exponent zero: its positive derivatives are bounded using v/τ, where v has upper exponent $-1/2$ and τ^{-1} has upper exponent $1/2$. Composition with the fixed smooth cutoff on $[0, 4]$ gives derivative scale of upper exponent zero. Its value is in $[0, 1]$. The complementary cutoff $1 - \chi$ has the same property: its value is still at most one, and its positive derivatives differ only by a sign. At a boundary point use the closed chord inequality, as above. A fixed enlargement of the derivative scale absorbs the finite-order composition constants while keeping amplitude one.

The protected bond cutoffs have the corresponding bound by Section 77 and the compensated increment estimate. Fine masks, contractions and likelihoods evaluated at the held fine configuration are constants. The first two have absolute value at most one; the product of likelihoods has the stated bound $e^{\ell n}$. Apply (429) to all these factors. There are at most $nb_{300B+2}\#\{\text{axes}\}$ coarse noise cutoffs and $nb_{304B+2}\rho_L$ protected bond cutoffs, where ρ_L is the fine-bond density. Their derivative scales add, giving $c(x)n$ with upper exponent zero, as asserted. The last inequality in (430) is the closed support version just explained. Thus the differentiated local action, covariance and chart bounds proved in the preceding sections supply the required estimates.

We give the remaining derivative calculation. The repeated chain rule for e^{ϕ} has one term for each partition π of the k derivative arguments:

$$D^k(e^{\phi})(y)[v_1,\dots,v_k] = e^{\phi(y)} \sum_{\pi} \prod_{C\in\pi} D^{|C|}\phi(y)[v_i : i \in C].$$

This identity follows by induction: differentiation either adds the new argument as a singleton block or inserts it into one existing block. Each partition appears exactly once. There are at most $k!$ partitions: send each partition to the permutation whose cycles list each block in increasing order. This map is injective. For unit arguments, (430) therefore gives, for $1 \le k \le j$,

$$\|D^k(e^{\phi})(y)\| \le e^{\phi(y)} k!(1+bn)^k \le e^{\phi(y)}\{J_j(1+bn)\}^k, \qquad J_j = \max\{1, j!\}.$$

The zero-order bound is equality. Using (429) and $n \ge 1$, the packet belongs to

$$\mathcal{B}_j\big(e^{(Cd^5-g)n}, a(x)n; y\big), \qquad a(x) = J_j(1+b(x)) + c(x).$$

The function a has upper exponent 1, so $a(x) \le h^2$ eventually. As $d \ge 1$ and $d \ge \log h$ eventually, for $n \ge 1$,

$$a(x) \le e^{2d^5 n}, \qquad n \le e^{d^5 n}, \qquad (a(x)n)^j \le e^{3jd^5 n}.$$

Enlarge a to at least one if necessary. Every derivative of order at most j is now bounded by the same number:

$$\|D^k \mathcal{P}_A(y)\| \le \exp\{((C+3j)d^5 - g)n\}, \qquad 0 \le k \le j. \tag{431}$$

Thus finite differentiation changes the fixed coefficient of the logarithmic expense; it does not consume a fixed fraction of the energy gain. This argument applies separately at every point of the allowed variation, with the same constants.

84.4 Absorbing the expense and the packet count

Here is the quantitative comparison used to turn these estimates into small connected-support prices. Suppose $\mu > 0$ and $\beta \ge uh$ for some fixed $u > 0$. Define

$$M = \mu u P^2, \qquad q_* = e^{-M/2}.$$

Since $h\tau^2 = P^2$, the gain satisfies $g = \beta\mu\tau^2 \ge M$. For any fixed $D \ge 0$, integer $r \ge 0$ and $z < 2$, the monitor comparison (271) gives

$$Dd^r P^z \le M/16 \quad \text{eventually.} \tag{432}$$

Indeed apply that comparison with exponent $2 - z$ and coefficient $16D/(\mu u)$, and multiply by $\mu u P^z/16$.

Fix a packet support radius parameter ℓ and a support exponent $\alpha \ge 0$. Let H_B and D_B be the root and adjacency counts from Section 63, and let c_ℓ be its cover cost per seed. All are polynomial in B except the fixed c_ℓ. Consequently, for sufficiently large x,

$$H_B \le e^{M/16}, \qquad D_B^2 \le e^{M/16}, \qquad \alpha c_\ell \le M/16, \qquad 2 \le M/16.$$

For example, bound both counts by fixed powers of d, use (432), and then $t \le e^t$. Consider a nonnegative packet price $C_0 e^{(s-g)n}$ satisfying

$$s \le Dd^r P^z, \qquad 0 \le C_0 \le e^{Dd^r P^z}.$$

Then $C_0 \le e^{M/16}$ and the counting ratio obeys

$$D_B^2 e^{s-g+\alpha c_\ell} \le e^{M/16+M/16-M+M/16} \le q_*.$$

In particular $q_* \le 1/2$. The packet counting estimate already proved therefore bounds the site sum over packets with at least n_0 seeds by

$$H_B C_0 \frac{q_*^{\max(1,n_0)}}{1-q_*} \le \exp\{-M \max(1,n_0)/4\}.$$

To verify the last inequality, put $r_0 = \max(1, n_0)$. Since $(1-q_*)^{-1} \le 2 \le e^{M/16}$, its logarithm is bounded above by $3M/16 - Mr_0/2 \le -Mr_0/4$. This applies to (428) and (431) with $z = 0$ and $r = 5$. Hence once the stated local energy, chart and derivative inputs have been established, their full packet and support sums are small uniformly in the period. The scalar comparison selects a threshold after all fixed constants; it does not by itself construct the schedule that reaches that threshold at each renormalization step.

85 Differentiating the projected Gaussian integrals

The Gaussian used in a local activity varies with the coarse field. Differentiating its expectation therefore involves both the activity and the Gaussian density. We first reduce to the coordinates read by that activity, and then use the compensated change of variables to put all nearby integrals on one reference probability space.

85.1 Projection and the relative density

Let C be the positive covariance of Section 74, in counting colour coordinates. For a finite coordinate set I, let C_I denote its principal submatrix. Projection of the centered Gaussian with covariance C is the centered Gaussian with covariance C_I. Indeed its characteristic function at $\xi \in \mathbb{R}^I$ is $\exp(-\langle \xi, C_I \xi\rangle/2)$, by extending ξ by zero. A function depending only on these coordinates can therefore be integrated against this marginal; the other coordinates need not be independent. The marginal density is

$$p_C(z) = (2\pi)^{-N/2} \exp\{-\tfrac{1}{2}\log\det C - \tfrac{1}{2} z^T C^{-1} z\}, \qquad N = |I|, \tag{433}$$

where here and below C denotes the projected covariance when there is no ambiguity. Positivity of a principal submatrix follows by extending a nonzero vector by zero. Formula (433) follows by diagonalizing C and changing variables in the product of N one-dimensional standard Gaussian densities. It is also valid in dimension zero, with determinant and density equal to one.

Fix a base coarse parameter y_0 and write $C_0 = C(y_0)$. For fixed frozen Haar variables, the compensated transformation induces a diffeomorphism T_y of the I coordinates with $T_{y_0} z = z$ and positive Jacobian $J_y(z)$. The exact expectation identity is

$$\int F_y(w) p_{C(y)}(w)\, dw = \int J_y(z) \frac{p_{C(y)}(T_y z)}{p_{C_0}(z)} F_y(T_y z)\, d\gamma_{C_0}(z). \tag{434}$$

The positive reference density cancels, leaving the ordinary change-of-variables formula on the right. The set I includes all colours of every bond in the activity's enlarged read support. The transformation acts bond by bond, and both it and its Jacobian on I depend only on the old coordinates in I. Locality therefore identifies the transformed marginal integrand with the original activity evaluated on the transformed full field. This explains why the determinant in this formula is $\det C_I$, rather than the determinant of the covariance on the full torus.

85.2 Polynomial rank costs at the base point

There is a useful finite-order improvement over estimating a full determinant by its permutation expansion. If $A(y_0) = 0$ is an $N \times N$ matrix, multilinearity in the rows gives

$$\det(1 + A(y)) = \sum_{J \subseteq \{1,\dots,N\}} \det A_J(y). \tag{435}$$

To verify the formula, in each row choose either the row of A or the corresponding identity row. The identity rows force their columns, leaving precisely the principal minor on the chosen set. Every term of $\det A_J$ is a product of $|J|$ functions vanishing at y_0. Its derivatives of order less than $|J|$ vanish, because at least one factor receives no derivative. Thus derivatives through order j in (435) use only $|J| \le j$.

Suppose every entry belongs to $\mathcal{B}_j(v, 1; y_0)$, with $v \ge 0$. A minor of size $r \le j$ has at most $r!$ permutation terms. The product rule bounds each term by amplitude $(1 + v)^j$ and derivative scale j. There are at most $(j + 1)(1 + N)^j$ subsets of size at most j, since $\binom{N}{r} \le N^r$. Consequently

$$\det(1 + A) \in \mathcal{B}_j\big((j + 1)(1 + N)^j j!(1 + v)^j, j; y_0\big).$$

For $j = 0$ this just states the value one.

Apply this to $A(y) = C(y)C_0^{-1} - 1$. Weighted row bounds for the derivatives of C and for C_0^{-1} supply the required entry bounds by matrix multiplication. Since

$$\log\det C(y) - \log\det C_0 = \log\det(1 + A(y))$$

near y_0, composition with the logarithm near one gives a derivative bound polynomial in N, at each fixed order. The inverse-matrix estimates of Section 78 similarly control $C(y)^{-1}$. If each coordinate of $T_y z$ has derivatives bounded by W at y_0, the product rule applied to $(T_y z)^T C(y)^{-1}(T_y z)$ adds a factor at most N times the inverse row price and a fixed-order polynomial in the derivative scales and W.

Subtract the value of that quadratic form at y_0 and combine it with the relative log determinant. The exponent is zero at y_0. The exponential partition formula from Section 84 proves a relative-density bound of amplitude one and a derivative scale polynomial in N and these prices. This is a finite-rank derivative estimate at the base point. It does not yet justify taking derivatives under an integral over unbounded z.

85.3 An integrable bound on one open neighbourhood

We prove the additional assertion needed for that interchange. Let y range over a finite-dimensional real parameter space and let $C(y)$ be a C^j family of symmetric matrices near y_0, with C_0 positive definite. Suppose

$$C_0^{-1} \ge \lambda 1, \qquad \lambda > 0.$$

Let z be an arbitrary sample and let $w_z(y)$ be a family of C^j maps such that, on one neighbourhood of y_0, uniformly in z and in any auxiliary frozen variables,

$$w_z(y) - z \in \mathcal{B}_j(B, 1; y), \qquad B \ge 0. \tag{436}$$

Then there is a nonnegative function M, integrable for γ_{C_0}, which bounds every derivative through order j of $p_{C(y)}(w_z(y))/p_{C_0}(z)$ on a possibly smaller common neighbourhood.

By continuity and finite dimension, shrink the neighbourhood so that $C(y)$ is positive definite and $C(y)^{-1} \geq (\lambda/2)1$. For an explicit verification of the last inequality, make every entry of $C(y)^{-1} - C_0^{-1}$ at most $\lambda/(2(1+N))$ in absolute value. Its row and column sums are at most $\lambda/2$, so its quadratic form is bounded in absolute value by $(\lambda/2)\|z\|^2$. On a smaller closed neighbourhood, all required derivatives of C^{-1} and $\log\det C$ have finite bounds. These bounds may depend on N and the fixed period.

The derivatives of w_z have amplitude at most $B + \|z\|$. Thus the logarithm of the numerator has derivatives bounded by $K(1 + \|z\|)^2$ for a fixed K. Moreover

$$\|z\|^2 \leq 2\|w_z(y)\|^2 + 2B^2,$$

and hence

$$-\tfrac{1}{2}\log\det C(y) - \tfrac{1}{2}w_z(y)^T C(y)^{-1} w_z(y) \leq K_0 + \lambda B^2/4 - \lambda\|z\|^2/8.$$

The repeated exponential chain rule now bounds every derivative through order j of the relative density by

$$M(z) = K_1 \frac{e^{-\lambda\|z\|^2/8}(1 + \|z\|)^{2j}}{p_{C_0}(z)}. \tag{437}$$

This is integrable against γ_{C_0}: multiplication by its density cancels the denominator, and a polynomial times a strictly decaying Gaussian is integrable in finite dimension. Explicitly, $(1+t)^{2j} \leq e^{2jt}$ for $t \geq 0$, and completing the square absorbs $2jt$ into half the negative quadratic exponent. The remaining integral factors into N finite one-dimensional Gaussian integrals. The same proof works when z is a measurable function of a larger probability space with law γ_{C_0}.

85.4 Why the activity satisfies the neighbourhood hypotheses

For the compensated map of Section 79, write $u_0 = \nu_0 + \beta^{-1/2} z$ and $F_M(u) = u + \psi(M, u)$. Its displacement in Gaussian coordinates is

$$T_y z - z = \sqrt{\beta}\,\{\psi(M(y), u_0) + \nu_0 - \nu(y)\}. \tag{438}$$

Here the fixed coupling satisfies $\beta > 0$. The perturbation ψ and all its parameter derivatives are uniformly bounded for all u_0: outside the fixed radius-$2d$ ball it is zero, and inside that ball compactness applies. The unmasked mean and its derivatives are bounded on a common neighbourhood uniformly in the frozen Haar variables. Indeed the frozen cut logarithms are bounded smooth functions of the coarse variation; the finitely many source sums, row matrices and principal inverses have uniform derivative bounds there. All principal determinants stay nonzero after one common shrink. Formula (438) proves (436). The same compact argument and the finite product rule bound the Jacobian derivatives uniformly.

The remaining activity integrand also has bounded derivatives on one neighbourhood, uniformly in the Gaussian sample. The required localization is worth spelling out. Compare two fluctuation arrays $u, \widetilde{u}$ such that at each bond either they agree or both have norm greater than $4a\tau$. Every cutoff of width at most $2a\tau$ is then equal on the two arrays, and if that cutoff is nonzero the arrays agree at that bond. A masked chart logarithm therefore agrees as well. A packet's nonzero protected product forces agreement throughout its $304B$ region, which contains the read stencils of all its $200B$ action terms. Thus its logarithmic exponent and its complete value agree.

For a linear Gaussian ratio term, the score vanishes on inactive bonds and its hot endpoint carries the doubled cutoff. For a quadratic ratio term, every nonzero matrix itinerary has two hot endpoints. To check this, $AC - 1$ is zero if either coordinate is inactive, since both A and C equal

the identity on that coordinate. In a product of copies of $-(AC-1)$ followed by A, the first nonzero factor forces the initial endpoint hot, and the last factor cannot join a hot coordinate to an inactive one. Both endpoint cutoffs therefore force equality of the coordinates actually used by the quadratic term. This argument holds nearby as long as the principal inverses exist; it does not require the nearby point to satisfy the original sector inequalities.

It follows that every fixed activity factor is unchanged when large initial fluctuation coordinates are replaced by bounded ones, provided this comparison is made through the compensated map and the fine field is held fixed. Use radial truncation at radius $3d$: it leaves the radius-$2d$ ball unchanged. Outside that ball F_M is the identity; both the old and truncated coordinates then remain beyond the cutoff threshold, since $8a\tau \le d/2$. Inside it, the coordinates agree. The resulting bounded input family has uniform parameter derivatives by the compact chart construction. The frozen fine factors remain merely bounded measurable functions: their regular activities, forced activities, likelihoods and contractions use the supplied finite-volume bounds for the chosen labels. No differentiation of these frozen factors is required.

For each fixed finite product of activity factors, finite sums, products and smooth compositions on bounded sets now give a common bound on all derivatives through order j. The local action, normalization and chart functions used here are the unmasked smooth extensions already constructed. Hard selection indicators are fixed. Together with (437) and the Jacobian bound, this gives an integrable envelope for the entire right-hand side of (434), also after integrating the frozen variables against their Haar probability law.

85.5 Interchange and quantitative use

Here is the precise final differentiation argument. Suppose $G(y,\omega)$ is measurable in ω, C^j in y on an open neighbourhood, and for each $r \le j$ its rth derivative is bounded there by an integrable function $M_r(\omega)$. Then

$$D^r \int G(y,\omega)\, d\mu(\omega) = \int D_y^r G(y,\omega)\, d\mu(\omega), \qquad r \le j. \tag{439}$$

Derivative measurability follows in a fixed finite basis by taking limits of difference quotients of measurable functions. Dominated convergence first gives continuity of each derivative integral. For the induction step, the fundamental theorem of calculus on a short parameter segment writes its difference quotient as the integral of the next derivative along that segment. The bound M_{r+1} permits dominated convergence as the segment shrinks. Starting at order zero proves the derivative identity and its continuity successively through order j.

Apply this to the common reference expression above. It proves smoothness of each projected activity integral. If a sharper estimate at the base point gives $G(\cdot,\omega) \in \mathcal{B}_j(q(\omega), D; y_0)$ and $\int q\, d\mu \le Q$, then (439) and the triangle inequality give

$$\int G(\cdot,\omega)\, d\mu(\omega) \in \mathcal{B}_j(Q, D; y_0).$$

The neighbourhood envelope establishes the validity of differentiation; the sharper moment estimate supplies its period-independent price. Those moment estimates must be paid separately. In particular the period-dependent constants used to justify (439) are never used as uniform renormalization bounds.

86 Uniform moment prices after changing Gaussian coordinates

We now estimate the polynomial in the Gaussian sample introduced by differentiation of the projected density. The covariance may couple every coordinate to every other coordinate. Only its uniform diagonal bound is needed for this step.

86.1 A polynomial moment bound without independence

Let Z be a centered Gaussian vector with covariance $C \geq 0$ and $C_{ii} \leq b$. Each coordinate has exponential moment

$$\mathbb{E}e^{tZ_i} = e^{C_{ii}t^2/2} \leq e^{bt^2/2}.$$

This follows directly by completing the square in its one-dimensional Gaussian law, including the zero-variance case. For an integer $r \geq 0$,

$$(1 + |z|)^r \leq e^{r|z|} \leq e^{rz} + e^{-rz}.$$

Hence

$$\mathbb{E}(1 + |Z_i|)^r \leq 2e^{br^2/2}. \tag{440}$$

Let I be a set of physical bonds, let I_{col} contain all their colour coordinates, and put $n = |I|$, $N = |I_{\mathrm{col}}|$. Define

$$F_I(z) = 1 + n + N + \sum_{i \in I_{\mathrm{col}}} |z_i|, \qquad Y_I = 1 + n + 2N.$$

Then

$$\mathbb{E}F_I(Z)^r \leq Y_I^r \{1 + 2Ne^{br^2/2}\}. \tag{441}$$

For nonempty I_{col} choose a coordinate at which $|z_i|$ is maximal. The definition gives $F_I(z) \leq Y_I(1 + |z_i|)$, and therefore

$$F_I(z)^r \leq Y_I^r \left(1 + \sum_{i \in I_{\mathrm{col}}} (1 + |z_i|)^r\right).$$

For an empty coordinate set the same inequality is immediate. Take expectations and use (440). In particular all these moments are integrable. This argument uses one-coordinate marginals and a finite sum, without factoring a joint moment.

For the local covariance, the weighted matrix bounds of Section 74 supply a fixed b, independent of the period. Suppose an activity has nonnegative support price s, equal to the sum of its covering numbers and nonnegative length charges. A cover by radius-B balls gives

$$|\operatorname{supp}| \leq s\, b_B, \qquad |I| \leq s\, b_B \rho_L, \qquad N = q|I|,$$

where q is the colour dimension and ρ_L the fixed bond density. The first inequality follows by summing the covering balls over all factors; overlaps can only decrease the union. The second uses that every bond of I has its anchor in that union. Since b_B has upper exponent zero, (441) implies, for every fixed r,

$$\mathbb{E}F_I(Z)^r \leq A_r(x)(1 + s)^{m_r}, \tag{442}$$

for an integer m_r and a nonnegative A_r of upper exponent zero. Indeed insert the displayed bounds for n, N into (441) and expand the resulting fixed polynomial.

86.2 The derivative of the density and Jacobian

At the base parameter the compensated transformation is the identity. The coordinate derivative bound following from (438) is

$$\|D^r(T_y z)_i|_{y=y_0}\| \le \sqrt{\beta}\{u(x) + 2m(x)\} + F_I(z), \qquad 0 \le r \le j,$$

where u, m bound respectively the compensation increment and mean derivatives and have upper exponent $-1/2$. For the value, use $|z_i| \le F_I(z)$; for positive orders differentiate the compensated displacement. As $\beta \le 2h$, the first term has upper exponent zero.

The principal covariance and its inverse inherit the weighted derivative bounds proved earlier. The finite-order relative determinant argument of Section 85 therefore gives a derivative scale polynomial in N, $F_I(z)$ and functions of upper exponent zero. The Jacobian is a product of n one-bond factors. Each has value one at the base parameter and derivative scale of upper exponent zero, by Section 79. Its product has amplitude one and the sum of these scales. Since $1, n, N, |z_i| \le F_I(z)$, every term of the fixed polynomial is bounded by one common power of $F_I(z)$. We obtain an integer k and a nonnegative $D(x)$ of upper exponent zero such that the complete density ratio times Jacobian, denoted $R_y(z)$, satisfies

$$R_\cdot(z) \in \mathcal{B}_j(1, D(x)F_I(z)^k; y_0). \tag{443}$$

The exponent k depends on the derivative order and fixed construction data, not on the support size or period.

86.3 Retaining the warning gain under integration

Consider a finite product of local activity factors indexed by $\mathcal{P}$. Write $t_i \ge 0$ for their pointwise prices and s for their total support price. Suppose their proved local derivative estimates, under the coherent variation, give

$$\Phi_\cdot(z) \in \mathcal{B}_j\left(\mathbf{1}_{\mathcal{E}}(z)\prod_{i\in\mathcal{P}} t_i,\, 2r(x)s; y_0\right), \qquad r \ge 1, \tag{444}$$

where r has upper exponent zero and $\mathcal{E}$ is the joint warning event for the m warning factors. The displayed estimate implies that all derivatives through order j vanish outside $\mathcal{E}$. In this application the stronger local-vanishing property holds: an unsatisfied warning factor is zero on a neighbourhood of y_0, as in the closed-support argument for packet masks. For the other factors, their derivative scales add: each nonzero supported factor has covering number at least one, so its unit scale and length scale together cost at most twice its assigned support price. This proves the factor $2s$ in (444) from the individual bounds. The individual estimates for all species remain necessary inputs to this product step.

Multiply (444) by (443). Since $F_I \ge 1$ and $1 + s \ge 1$,

$$1 + DF_I^k + 2rs \le (1 + D + 2r)(1 + s)F_I^k.$$

Converting to derivative scale one therefore bounds all derivatives through order j of the transformed integrand by

$$\mathbf{1}_{\mathcal{E}}(z)\left(\prod_{i\in\mathcal{P}} t_i\right)(1 + D + 2r)^j(1 + s)^j F_I(z)^{kj}. \tag{445}$$

Suppose the simultaneous-warning estimate of Section 64 gives $\mathbb{P}(\mathcal{E}) \le \rho^{2m}$ with $\rho \ge 0$. By Cauchy–Schwarz and (442) at order $2kj$,

$$\mathbb{E}\big[\mathbf{1}_{\mathcal{E}} F_I^{kj}\big] \le \mathbb{P}(\mathcal{E})^{1/2} (\mathbb{E} F_I^{2kj})^{1/2} \le E(x)(1+s)^a \rho^m,$$

for a nonnegative upper-exponent-zero function E and an integer a. For example replace the square root of the moment bound by $(1 + A_{2kj})(1+s)^{m_{2kj}}$. This also proves integrability of (445). Integrating any frozen Haar variables adds no factor, since their total mass is one and the estimate is uniform in them.

Consequently the integrated derivatives obey

$$\left\| D^r \int R_y \Phi_y \, d\mu \Big|_{y=y_0} \right\| \le E(x)(1 + D + 2r(x))^j (1+s)^{a+j} \rho^m \prod_{i \in \mathcal{P}} t_i, \qquad 0 \le r \le j. \tag{446}$$

The derivative interchange is justified by the separate common neighbourhood bound of Section 85. Absorb one factor ρ into the price of each warning factor. The right-hand side then has exactly the product-price form used by the connected support summation, with one fixed polynomial in $1+s$. The support-price estimate of Section 63 allows such a fixed polynomial through its exponential reserve. Thus Gaussian differentiation preserves the available warning gain while adding only upper-exponent-zero and polynomial support costs. No independence of warning events or Gaussian coordinates has been used.

87 The common differentiated activity bound

We assemble the estimates on the actual four-part alphabet: connected seed packets, forced activities, exponentiated logarithmic activities, and warnings. All factors use the same sector, fine representation, tags, Gaussian law and compensated variation. Besides the hypotheses of the preceding local estimates, retain the deterministic safety and plateau inequalities of Section 64. Their construction is an incoming requirement at this stage. The argument below explains both their derivative use and the common price they yield.

87.1 Exponentiating the logarithmic activities

If $f \in \mathcal{B}_j(w, d; y_0)$ with $0 \le w \le 1$, then

$$e^f - 1 \in \mathcal{B}_j(C_j w, d; y_0) \tag{447}$$

for a constant depending only on j. For the value, the mean value theorem gives $|e^{f(y_0)} - 1| \le ew$. For a positive derivative order $k \le j$, use the partition formula for the exponential. A partition with $m \ge 1$ blocks contributes at most $ew^m d^k \le ewd^k$. There are at most $k!$ partitions, so $C_j = e \max(1, j!)$ suffices. This retains the small amplitude and does not require that the parameter variation stay within the small-amplitude bound on a whole neighbourhood.

Fix $-1/2 < t < 0$. The inherited logarithmic prices of Section 83, the three short prices of Section 81, and the correction chain prices of Section 82 have a common rooted bound $J(x)$ of upper exponent t, after an arbitrarily small strict exponent loss. Their supports are nonempty. Since the support exponent and extra-cost weight are nonnegative, each individual price is at most $J(x)$: choose one site in its support and retain that summand in the rooted bound. Eventually $J(x) \le 1$. Thus (447) applies to every logarithmic label with the same C_j and the same threshold,

even when the alphabet is countably infinite. Multiplying all its prices by C_j preserves their upper exponent.

The forced family needs no exponentiation. By enlarging a nonnegative function $r(x) \geq 1$ of upper exponent zero, both families have jets with derivative scale

$$r(x)\{1 + c_B(S_i) + d_i\}. \tag{448}$$

Here S_i is the full assigned support and d_i its old-cover or path-length charge; for a correction itinerary it is the correction order. The short terms have $d_i = 0$. All earlier scale bounds are dominated by this expression.

87.2 Warnings vanish as germs at a reduced threshold

Let $\xi > 0$ be the safety margin and use the smaller threshold $\xi/2$ in the Gaussian event. Suppose every coordinate within distance $2B$ of a warning anchor satisfies

$$|z_i| \leq (\xi/2)\tau\sqrt{\beta}.$$

The deterministic safety estimate at margin ξ implies that the warning factor is zero on a neighbourhood of the base parameter. We check the different dependencies, since a value equal to zero by itself would not imply vanishing derivatives.

A warning in its mandatory seed neighbourhood is identically zero by its frozen indicator. Outside that neighbourhood, the safety estimate makes every relevant fine plaquette reading strictly smaller than $\tau/4$, and every predictor chord strictly smaller than $W\tau$. The fine mask, site likelihood, bond likelihood and contraction test are evaluated at the held fine configuration U_*. Their values are consequently constant under the variation and equal to one. The contraction statement uses the fine plaquette estimates at all sites in its read set, which fits in the same $2B$ event window. The predictor chord depends continuously on the varied coarse field, with its fine predictor held fixed. Its strict inequality persists on a neighbourhood, so its cutoff also remains one there.

For a fine-bond cutoff, the reduced threshold bounds the base increment by $(C + q\xi/2)\tau$. The safety constants satisfy $C + q\xi/2 < a$, where $a\tau$ is its plateau radius. The compensated increment is continuous, so this strict inequality also persists locally. Its cutoff-minus-one is thus a zero germ. Intersecting the finitely many neighbourhoods needed by a given factor proves the assertion. In particular, every nonzero jet of a warning is supported on the reduced-margin excursion event; no assertion about differentiation of an indicator of that event is needed.

The warning amplitude is bounded by the fixed constant

$$T = 1 + \#\{\text{frame tags}\} + \#\{\text{predictor tags}\}.$$

Indeed the held likelihoods are bounded by their tag counts, all masks lie in $[0, 1]$, and the warning is a protected difference from one. Its derivative scale has the form (448) with $d_i = 0$. For a varying noise cutoff, normalize its chord by $W\tau$; the positive derivatives cost the coarse scale divided by τ, of upper exponent zero. For a bond cutoff use the corresponding normalized compensated increment. A cutoff outside its closed support is a zero germ; on the closed support the smooth scalar cutoff has bounded derivatives. Multiplying over its protective read set adds at most a function of upper exponent zero times its support cover. This proves the asserted scale in all cases, including boundary points.

For every $t < 0$, the Gaussian estimate at margin $\xi/2$ gives, after increasing the threshold in x,

$$\mathbb{P}(\text{all } m \text{ reduced-margin warnings}) \leq (h^{2t})^m. \tag{449}$$

This follows from the faster-than-every-power bound already proved, applied at an exponent strictly below $2t$. Set $\rho = h^t$. The Gaussian moment argument consumes the square root of (449), leaving the price $T\rho$ per distinct warning. Its site sum has upper exponent t: warning addresses have fixed density, each support lies in a fixed-multiple-B ball, its cover is bounded by a fixed constant, and the number of such supports through a site is polynomial in B.

87.3 Packets and the product estimate

The packet jets have price

$$p_A = \exp\{((C + 3j)d^5 - \beta\mu\tau^2)n(A)\}$$

and derivative scale one by (431). The same energy gain pays their root counts, support exponents and any prescribed fixed seed marks by the absorption argument in Section 84. Thus their available weighted site price is smaller than every fixed negative power of h. A disconnected packet has factor and price zero. All thresholds are chosen after the fixed derivative order, halo, support and length reserves.

Let $\mathcal{P}$ be a finite family of distinct labels of the four types whose assigned supports are healthy for the base coarse field: no coarse flag at the threshold $\mathrm{temp}(x)$ lies in any S_i. In particular, each packet's full enlarged support must have this property to apply its packet jet estimate. This is a separate hypothesis on the tested family; the sector's manifest flag identity alone does not imply it. Put

$$s = \sum_{i\in\mathcal{P}} \{c_B(S_i) + d_i\}.$$

Every nonzero factor has nonempty support, hence $c_B(S_i) \geq 1$. The derivative scales in (448) therefore sum to at most $2r(x)s$, also covering the unit packet scales. Let a_i denote the packet, forced or exponentiated-log price, and put $a_i = T$ for a warning. The product rule and the zero-germ property give exactly

$$\prod_{i\in\mathcal{P}} f_i \in \mathcal{B}_j\left(\mathbf{1}_{\mathcal{C}}(z)\prod_{i\in\mathcal{P}} a_i, 2r(x)s; y_0\right),$$

where $\mathcal{C}$ is the joint reduced-margin warning event. This verifies (444) for all species on the same variation. Applying the Gaussian differentiation and moment estimates yields

$$\left\| D^k \int \prod_{i\in\mathcal{P}} f_i \right\| \leq A_j(x)(1+s)^{m_j} \prod_{i\in\mathcal{P}} \widetilde{a}_i, \qquad 0 \leq k \leq j, \tag{450}$$

with A_j of upper exponent zero and $\widetilde{a}_i = a_i$ except that a warning has price $T\rho$. The integral includes the Gaussian dependence on the varied field and the frozen Haar variables, as in that argument.

The combined optional site sum has upper exponent t, and the packet site sum is negligible. For a desired output exponent $t_{\mathrm{out}} \in (-1/2, 0)$, choose $t \in (-1/2, t_{\mathrm{out}})$. Fixed upper-exponent-zero factors and the fixed polynomial in $1 + s$ are then paid by the strict exponent slack and the support reserve of Section 63. Its tree degree condition holds eventually because the geometric neighbour count has upper exponent zero whereas the vertex price has negative upper exponent. This supplies the differentiated product input for the connected-support estimates. It preserves the incoming representation and its selected history; it does not construct or reset their parameters.

88 Recovering the block integral from the activities

The preceding estimates concern a dictionary of local factors. We now verify that its absolutely convergent sum is the original block integral. This verification fixes the scalar normalization and the incidence conditions before any logarithm of a gas is taken. Throughout, fix the coarse field, forest, allocation of seeds, and tag array. The funded sector hypotheses, the chart-access conclusion and the locality statements used in the preceding sections are retained. In particular, all precisions are positive, the covariance has its specified finite range, and the same assigned supports include every variable read by their factors.

88.1 The Gaussian density and the scalar factor

Write A for the completed sector precision, A_0 for its flat reference, and C for the localized covariance. All three act on the full real coordinate space, of dimension D. Let q be the colour dimension, N the number of free forest bonds, and n the number of active free bonds. The other $N-n$ free bonds are integrated in their original Haar coordinates. Put

$$g=\beta^{-1/2},\qquad b=\sqrt{2\pi},\qquad m=c_{\mathrm{H}}(bg)^{q},$$

where $c_{\mathrm{H}}>0$ is the constant in the local probability-Haar density of Section 77. With $u=\mu+gz$, the reference Gaussian has density

$$\vartheta_C(z)=b^{-D}\exp\left(-\tfrac12\log\det C-\tfrac12 z^{T}C^{-1}z\right). \tag{451}$$

Inactive coordinates are completed by independent standard Gaussians. Their energy is $\mathscr{G}(z)=\frac12\sum_{i\ \mathrm{inactive}} z_i^2$. There are $D-nq$ such coordinates, including the tree coordinates. Consequently the change of variables in the active Haar coordinates, with the dummy Gaussian integrals inserted, gives the multiplier

$$b^{-(D-nq)}g^{nq}c_{\mathrm{H}}^{n}\exp\left(-\mathscr{G}(z)+\sum_{e\ \mathrm{active}} j_e(u)\right). \tag{452}$$

Here j_e is the localized logarithmic Haar correction; its auxiliary cutoff agrees with the actual correction wherever the smaller chart mask is nonzero. The constant in (452) is exactly

$$b^{-D}m^{N}\exp(-(N-n)\log m). \tag{453}$$

Indeed the right side is $b^{-D}m^{n}=b^{-(D-nq)}g^{nq}c_{\mathrm{H}}^{n}$. This uses $n+(N-n)=N$, not the number of active coordinates as a substitute for the number of free bonds.

The dummy insertion is exact even though the active integrand depends on the remaining Haar coordinates. Split the latter product measure into active and inactive free bonds and freeze the inactive variables. The active affine substitution has Jacobian g^{nq}. In the orthogonal coordinate decomposition into active and dummy real variables, the dummy density integrates to one. Fubini therefore proves the identity for each fixed inactive configuration and then for its Haar integral. Chart access guarantees that the nonzero original integrand is covered by this substitution; the smaller mask can be inserted there with value one. No assertion that the entire Haar group lies in one chart is used.

Let $\mathscr{L}$ denote the original positive prefactor: it is the noise normalization times $\exp(v|\Lambda_f|)$ times the coarse-field scalar extracted from the inherited exponent. The scalar retained outside all activities is

$$\mathscr{O}=\mathscr{L}\,m^{N}(\det A_0)^{-1/2}>0. \tag{454}$$

The flat reference uses the very same forest. Its determinant is not replaced by a componentwise determinant, nor is (454) charged to every component.

88.2 Restoring the masks and the exponent

We spell out the cancellations that identify the density. Let $\mathscr{E}(u)$ be the whole quadratic-plus-remainder block energy, and let $r_j(u)$ be the inherited logarithms. Let $d_p(u)$ be the individual nonlinear remainders, so the signed channel average satisfies

$$\mathscr{E}(u) = \langle \textstyle\sum_p d_p(u) \rangle + Q(u),$$

where $Q(u) = \frac{1}{2}\|Mu + b_{\mathrm{row}}\|^2$ is the physical quadratic expression. The selected row matrix M has zero inactive columns; $A = M^T M + I_{\mathrm{inactive}}$ and $\ell = M^T b_{\mathrm{row}}$. The inactive coordinates of μ and ℓ are zero. This is the rowwise square expansion used in Sections 80 and 81: each row is assigned either to its seed neighbourhood or to the remote remainder, with no omitted row. Write $c_e = \frac{1}{2}(J_e(A_0) - J_e(A))$ for the local normalization correction.

The compulsory packet exponent, summed over the packet layout, is

$$s(u) = -\beta\big(D_{\mathrm{near}}(u) + Q_{\mathrm{seed}}(\mu)\big) + \sum_{e\ \mathrm{near}} c_e + \sum_{e\ \mathrm{near}} j_e(u) - (N-n)\log m. \tag{455}$$

Here $D_{\mathrm{near}}(u)$ is the signed average of $d_p(u)$ over the posted rows at radius $200B$, and $Q_{\mathrm{seed}}(\mu)$ is the signed average of the extracted row squares at μ over those same posted rows. They are not complementary row sets. The two occurrences of "near" in the following local sums use their respective radii, $200B$ and $304B$; they are different sets. The packet layout uses disjoint halos of radius at least $500B$, so all reads through $304B$, including the extra two lattice steps, belong to a unique packet. Thus summing the packet exponents gives exactly (455). Likewise the packet products reproduce the compulsory likelihoods, flag tests and chart masks once each. A test whose location belongs to a packet has the same seed membership in that packet as in the full allocation: a seed changing the test would meet that packet's halo, and hence would lie in the same component.

Let $f_i(z)$ be the optional logarithms in the complete dictionary. On the nonzero smaller chart mask their sum is

$$\begin{aligned} \sum_i f_i(z) &= \sum_j r_j(u) + \sum_p h_p(u) + \sum_{e\ \mathrm{remote}} c_e + \sum_{e\ \mathrm{remote}} j_e(u) + \mathscr{R}(z), \\ \mathscr{R}(z) &= \tfrac{1}{2}(\log\det C + \log\det A_0) - \sum_e c_e + z^T\widehat{h} + \tfrac{1}{2}z^T(C^{-1} - A)z. \end{aligned} \tag{456}$$

The remote index sets here are the exact complements of the two near sets above. The determinant and inverse series giving $\mathscr{R}$ are those of Sections 74 and 82. Their three determinant contributions are

$$-\tfrac{1}{2}(\log\det A - \sum_e J_e(A)), \quad \tfrac{1}{2}(\log\det A_0 - \sum_e J_e(A_0)), \quad \tfrac{1}{2}(\log\det A + \log\det C).$$

Their sum is the determinant part of $\mathscr{R}$. The linear itineraries sum to $z^T\widehat{h}$ and the quadratic itineraries sum to $\frac{1}{2}z^T(C^{-1} - A)z$. Their endpoint protectors equal one on the smaller full mask. Absolute summability was proved before this recombination, so these are identities of convergent series.

The remote action terms obey

$$\sum_p h_p(u) - \beta\big(D_{\mathrm{near}}(u) + Q_{\mathrm{seed}}(\mu)\big) = -\beta\big(\langle \textstyle\sum_p d_p(u)\rangle + Q(\mu)\big). \tag{457}$$

For a near row this follows from its assigned debit and mean-square term. For a remote row it is precisely the unmasked local-square identity: its protecting cutoff equals one on the smaller full

mask. Summing first over rows and then over the finite signed channels proves (457), including rows with zero factors.

Finally, if ℓ is the linear coefficient of Q, then $\widehat{h} = -g^{-1}(A\mu + \ell)$ and $\beta g^2 = 1$ give

$$-\beta Q(u) - \mathscr{G}(z) = -\beta Q(\mu) + z^T\widehat{h} + \tfrac{1}{2}z^T(C^{-1} - A)z - \tfrac{1}{2}z^T C^{-1}z. \tag{458}$$

To verify this, expand $Q(\mu + gz)$, use that the inactive mean and linear coefficient vanish, and add the inactive identity quadratic form supplied by $\mathscr{G}$. This is an algebraic identity, valid also where the masks vanish.

Combining (455)–(458) cancels both local sums $\sum c_e$ and both full determinants. It gives

$$\begin{aligned} &-\tfrac{1}{2}\log\det A_0 + s(u) + \sum_i f_i(z) - \tfrac{1}{2}\log\det C - \tfrac{1}{2}z^T C^{-1}z \\ &\qquad = \sum_j r_j(u) - \beta\mathscr{E}(u) - \mathscr{G}(z) + \sum_{e\ \text{active}} j_e(u) - (N-n)\log m. \end{aligned} \tag{459}$$

Multiplication by $b^{-D}m^N\mathscr{L}$ and (453) now recover the exact original density and its Haar Jacobian.

There is also a finite product identity for the masks. Write a_p for a compulsory test and w_p for its optional warning. At a compulsory position $w_p = 0$ and its packet supplies a_p; at every other position the optional factor is $1 + w_p = a_p$ on the smaller full mask. The auxiliary mask warnings restore that full mask from the packet masks. If it vanishes, the entire product on both sides is zero. If it is nonzero, every doubled mask is one and the preceding identities apply literally. The complete product of physical tag likelihoods equals the number of tag choices times the original normalized likelihood. Thus a single factor $|\mathcal{T}|^{-1}$, where $\mathcal{T}$ is the tag-array space, remains outside the product. This is the uniform tag average, not a new likelihood for each use of a tag.

88.3 Absolute expansion and return to probability Haar

Fix a finite old forced-label set K and a packet layout $\mathcal{P}$. For finite sets I of logarithmic labels and J of warning labels, the expanded integrand is

$$\Phi_{K,I,J} = \prod_{P\in\mathcal{P}} P(u) \prod_{k\in K} K_k(u) \prod_{i\in I}(e^{f_i(z)} - 1) \prod_{j\in J} w_j(u). \tag{460}$$

The four alphabets form a disjoint union; coincident physical locations do not identify their labels. On the finite lattice the nonempty supports and rooted price bounds imply $\sum_i p_i < \infty$ for the whole alphabet. Separately, the all-argument logarithm estimates of Sections 81–83 give envelopes $|f_i(z)| \le a_i$ with $\sum_i a_i < \infty$. Their endpoint cutoffs are retained in this assertion. Thus $\sum_i |f_i(z)| < \infty$, and

$$\exp\left(\sum_i f_i(z)\right) = \sum_{I\ \text{finite}} \prod_{i\in I}(e^{f_i(z)} - 1).$$

For a proof, exhaust the countable alphabet by finite subsets and expand each finite product. Absolute convergence follows from $\sum_i |e^{f_i} - 1| < \infty$: outside finitely many indices, $|e^{f_i} - 1| \le 2|f_i|$. The warning alphabet is finite and its expansion is the ordinary identity for $\prod_j(1 + w_j)$.

The Gaussian moment estimate gives, with the original probability Haar variables y retained,

$$\int |\Phi_{K,I,J}(y,z)|\, d\nu_F(y)\, d\gamma_C(z) \le \prod_{v\in\mathcal{P}\sqcup K\sqcup I\sqcup J} p_v.$$

Summing the right side over all finite label sets gives at most $\prod_v (1 + p_v) \le \exp(\sum_v p_v)$. Distinct triples (K, I, J) give distinct label sets. Tonelli applied to the absolute values therefore justifies every exchange between these sums and integrals. The same argument covers the subsequent finite sum over seed allocations and tag arrays. There is no division by the value of a vacuum or flagged term.

An allocation contributes only when its flags match the coarse field, the old forced labels are admissible, and their inventories have the required incidence with the seeds. These conditions are necessary for a nonzero original integrand: the profile enforces its exact flags, and a nonzero old gas term enforces its admissibility and incidence. Set the entire integral to zero when any condition fails. On a valid allocation the density calculation above applies; on an invalid one both sides vanish. Hence the absolutely convergent sum, multiplied by $|\mathcal{T}|^{-1}\mathscr{O}$, equals the original contribution for the fixed tags and seed allocation. Summing the old forced sets, then the tags and seed allocations, recovers the probability-Haar block push with scalar factor (454).

88.4 Factorization into connected activities

It remains to show that the integral of a molecule factors over its connected components. Connectivity is defined by intersection of the full assigned supports. For fixed tags, different components read disjoint Haar coordinates. Their Gaussian coordinates may be larger than their physical read sets; the assigned supports include the prescribed covariance buffer. If two components have disjoint assigned supports, any two Gaussian coordinates read by them are farther apart than the range of C. Their cross covariance is zero. The Gaussian characteristic function consequently splits into the product of its component characteristic functions, proving independence of these coordinate projections. It follows that the Gaussian integral of the product factors. The remaining product-Haar integral factors by its disjoint read sets. All integrands are measurable and integrable by the preceding absolute bounds.

Lastly average over the tags. A component integral depends only on the site tags and on the bond tags anchored in its assigned support. These coordinate sets are disjoint for different components. The uniform measure on tag arrays is a product measure, so averaging first the bond tags and then the site tags proves the same factorization for the averaged integrals. Averages are taken after each component has been identified, using the common original tag space.

For each connected label set M, define its activity $a_M(V)$ by this averaged integral with the exact local incidence tests: its inventory must equal the flags of V inside its assigned support, and all its forced-label and packet tests must hold. The integral uses the globally flagged sector with the same local data. Locality shows that its value agrees with the restriction of the full allocation: on this support the seeds, tags, forced labels and coarse field coincide. The flags outside the support cannot change its value.

Taking connected components and taking their union are inverse on admissible label families. Indeed every label has nonempty support, so distinct components cannot share a label. Their supports are disjoint by definition. Exact global flag coverage is equivalent to exact local inventory equality in each component together with coverage by all inventories. This proves the weight-preserving factorization, including the empty family. Reindexing the absolutely convergent sum gives

$$(\mathcal{B}R)(V) = \mathscr{O}(V) \sum_{\substack{\mathcal{A} \text{ finite, compatible} \\ \bigcup_{M\in\mathcal{A}} I_M = \mathrm{Flags}(V)}} \prod_{M\in\mathcal{A}} a_M(V). \tag{461}$$

Thus the local prices bound an exact representation of the original block push. The later separation of flag-free logarithms and the iteration of the resulting bounds still use their stated construction

inputs; neither follows merely from (461).

89 Averaging the forests and separating the regular gas

The exact block gas of Section 88 was constructed with a fixed forest. To restore the lattice symmetries we average over forests before taking the logarithm of the flag-free gas. We first show that this operation leaves the scalar normalization unchanged, and then choose an average compatible with local factorization.

89.1 The flat determinant is independent of the forest

Every block forest has one outgoing edge from each nonroot vertex, directed towards the root. Distinct nonroot vertices give distinct unoriented positive-bond representatives of these edges. If two outgoing edges represented the same bond with opposite orientations, the forest height would strictly decrease in both directions, which is impossible. Thus the tree bonds are in bijection with the nonroot vertices. In particular all forests have the same number of tree bonds and the same number N of free bonds.

Let F, H be two forests and choose a bijection t between their tree bonds. We construct an integer change of basis on all bond coordinates, including the dummy tree coordinates. For a bond cochain u, let $D_F u$ set its F-tree coordinates to zero. Let $\alpha_F u(x)$ be the signed sum of u along the forest path from x to its block root, and put

$$B_F u = u + \nabla \alpha_F u.$$

Telescoping along a path proves the following four facts: $B_F u$ is zero on the F-tree bonds; $B_F u = u$ if u was already zero there; $\alpha_F u$ is zero at each root; and $B_F \nabla v = 0$ when v is zero at all roots. These assertions hold over any abelian group, since they involve only signed sums.

Define $T_{FH} u$ on the H-tree bonds by transporting the old F-tree coordinates through t, and on the other bonds by $B_H D_F u$. Then

$$D_H T_{FH} u = B_H D_F u.$$

The reverse construction uses t^{-1}. On tree coordinates it plainly recovers u. On the complementary coordinates,

$$B_F D_H T_{FH} u = B_F(D_F u + \nabla \alpha_H D_F u) = B_F D_F u = D_F u,$$

because $\alpha_H D_F u$ vanishes at the roots. Thus $T_{HF} T_{FH} = 1$; reversing the argument gives the other inverse identity. Its matrix and inverse have integer entries. Their integer determinants multiply to one, so $|\det T_{FH}| = 1$. Apply this same construction to every colour coordinate.

Let M_F, M_H be the flat selected row matrices, and let D_F^0, D_H^0 be the diagonal projections onto their tree coordinates. Flat rows annihilate gradients of root-zero functions. For a fine row this is the telescoping identity around its plaquette, preserved by each linear path coefficient. For a predictor row the coarse sampling operator sends a gradient to the coarse gradient of its root values; these values are zero. Signed coordinate changes preserve gradients, so the statement holds simultaneously in all signed channels. Hence

$$M_H T_{FH} = M_F, \qquad T_{FH}^T D_H^0 T_{FH} = D_F^0.$$

The second identity follows because the tree coordinates are simply permuted. For the completed flat precisions we obtain

$$A_F^0 = M_F^T M_F + D_F^0 = T_{FH}^T A_H^0 T_{FH}, \qquad \det A_F^0 = \det A_H^0.$$

Together with the common free-bond count, this proves that the scalar $\mathscr{O}$ in (454) is forest independent.

89.2 An independent choice at each coarse site

There are finitely many allowed forests on a finite torus: each vertex chooses one of finitely many oriented neighbouring edges and one of finitely many heights. The set is nonempty by the block-tree construction. At each coarse site x, choose independently a uniform forest F_x from this finite set. On the block with root x, use only the restriction of F_x. The assembled object is a valid forest, since each original forest edge remains in its parent block and its height decreases along that edge. Repeated global forest choices producing the same assembled forest cause no difficulty; the average is defined on the product space of choices.

For a connected component M, its integrated activity depends on these choices only at coarse sites in its full assigned support X_M. This follows from the local reads of the rows, inverse windows, tags and chart factors: agreement of the tree edges in that support gives equality of each such read and of its Gaussian marginal. To compare the Haar factors, realize both local charts as pullbacks of the same full product Haar measure; agreement of the principal Gaussian marginals then gives equality of the integrals. In particular changing a forest outside X_M does not alter the component integral. Compatible components have disjoint X_M. Independence of the site choices therefore gives

$$\mathbb{E}_F \prod_{M\in\mathcal{A}} a_M^F(V) = \prod_{M\in\mathcal{A}} \mathbb{E}_F a_M^F(V). \tag{462}$$

This is the factorization of a product measure over disjoint coordinate sets; it does not hold for an arbitrary average over one global forest.

Set $\bar{a}_M(V) = \mathbb{E}_F a_M^F(V)$. The bounds on a_M^F are uniform in F, so the same prices bound $\bar{a}_M$. The finite average commutes with the absolutely convergent gas sum, and the forest-independent $\mathscr{O}$ factors out. Thus (461) holds with $\bar{a}_M$ in place of a_M^F. A signed lattice transformation permutes the site choices and transforms every chosen forest by a bijection. Its product uniform measure is unchanged. The covariance of the individual sector formulas consequently gives the same covariance for $\bar{a}_M$, with the original transformed activity addresses.

89.3 The cost of excluding regular activities

Separate the connected labels into regular labels i with empty inventory and forced labels j with nonempty inventory. Write their values as $w_i(V)$ and $k_j(V)$, and their supports as X_i, Y_j. A nonzero regular value has no flag in its support. Fix a compatible finite forced family A whose inventories cover all flags, and put $Y_A = \bigcup_{j\in A} Y_j$. The remaining sum is the regular gas restricted to labels whose supports avoid Y_A:

$$(\mathcal{B}R)(V) = \mathscr{O}(V) \sum_{A \text{ admissible}} \prod_{j\in A} k_j(V)\, Z_{Y_A}(V). \tag{463}$$

The reindexing is exact: a compatible family splits uniquely into its regular and forced parts, and cross compatibility means precisely $X_i \cap Y_A = \varnothing$ for every retained regular label. Absolute convergence justifies summing in this order.

Assume the regular prices satisfy the connected-cluster bound of Section 62, with nonnegative support weight, rooted allowance γ and tail ratio $q_0 < 1$. Let c_Γ be the corresponding logarithmic

term, indexed by a finite ordered tuple of occurrences, with its Ursell coefficient divided by the tuple factorial. Labels may repeat in this tuple. Then

$$Z = \exp\left(\sum_{\Gamma} c_\Gamma\right) > 0.$$

For a finite set Y define

$$P_Y = \sum_{\Gamma:\, X_\Gamma \cap Y \neq \varnothing} c_\Gamma, \qquad X_\Gamma = \bigcup_{i \in \Gamma} X_i.$$

Deleting labels that meet Y replaces c_Γ by zero precisely when its union support meets Y. Otherwise it leaves the term unchanged. The same absolute majorant applies to the deleted gas. Subtracting the two absolutely convergent logarithmic series yields

$$Z_Y = Z e^{-P_Y}. \tag{464}$$

The rooted cluster estimate, summed over $x \in Y$, gives

$$\sum_{\substack{\Gamma:\, X_\Gamma \cap Y \neq \varnothing \\ |\Gamma|-1 \geq m}} |c_\Gamma| \leq |Y| \frac{\gamma q_0^m}{1 - q_0}. \tag{465}$$

Indeed each term on the left is counted by at least one site of $Y \cap X_\Gamma$; overcounting is harmless in this nonnegative sum. In particular, with $C_Y = |Y|\gamma/(1 - q_0)$,

$$Z e^{-C_Y} \leq Z_Y \leq Z e^{C_Y}.$$

The correction P_Y retains its actual sign; only its absolute estimate uses C_Y. Combining (463) and (464) gives the exact separation

$$(\mathcal{B}R)(V) = \mathscr{O}(V) Z(V) \sum_{A \text{ admissible}} \prod_{j \in A} k_j(V) e^{-P_{Y_A}(V)}. \tag{466}$$

For each fixed A, the ratio is finite and positive. The original absolute family sum implies absolute summability of this outer series: its term is exactly the signed forced product times the restricted regular gas, whose absolute value is bounded by the corresponding sum of absolute original terms. No volume-independent bound on the extensive scalar Z is asserted.

89.4 Local regular functions and their derivatives

For each regular component, the integrated construction gives a measurable local extension f_i, smooth through the purchased finite order at configurations healthy on X_i, with

$$w_i(V) = \mathbf{1}_{\{X_i \cap \mathrm{Flags}(V) = \varnothing\}} f_i(V).$$

The forest average preserves locality, gauge invariance, measurability and each uniform derivative bound, by linearity and the triangle inequality for the finite average. For a cluster tuple $\Gamma = (i_1, \ldots, i_n)$, define

$$F_\Gamma(V) = \frac{u(i_1, \ldots, i_n)}{n!} \prod_{r=1}^{n} f_{i_r}(V).$$

Then

$$c_\Gamma(V) = \mathbf{1}_{\{X_\Gamma \cap \mathrm{Flags}(V) = \varnothing\}} F_\Gamma(V). \tag{467}$$

If the union is healthy, every constituent mask is one. If it is not, a flagged site belongs to some constituent support, so that factor vanishes. This proves the equality even for repeated labels.

Suppose the local extension of the rth occurrence has derivatives through order j bounded by p_{i_r} at derivative scale one at configurations healthy on its support. Restrict throughout the following derivative estimates to configurations healthy on X_Γ; every constituent hypothesis then holds. Leibniz's formula gives product amplitude $\prod_r p_{i_r}$ at derivative scale n. Thus its derivatives through order j are bounded by

$$\frac{|u(i_1,\dots,i_n)|}{n!}(1+n)^j\prod_r p_{i_r} \le \frac{|u(i_1,\dots,i_n)|}{n!}\prod_r (2^j p_{i_r}),$$

since $1+n\le 2^n$ for $n\ge 1$. Buying the regular cluster bound with the inflated prices $2^j p_i$ therefore pays these derivatives on the same union support. A spatial derivative weight is also local: from any anchor in X_Γ, the connected-cover diameter bound places every read within a fixed multiple of $(B+1)(1+\operatorname{cover}_B X_\Gamma)$. Its polynomial cost is paid by the reserved exponential support weight as in Section 63.

Equations (466) and (467) supply the regular logarithm and the exact exclusion correction of the nonlinear step. The exclusion exponential must still be regrouped with the forced labels using its occurrence multiplicities; it cannot be dropped or replaced by an independent probability. Its quantitative allowance is the rooted price (465).

90 Regrouping the exclusion correction

The regular gas in (466) has been extracted, but the forced family still carries its exclusion factor. We now put this factor into local forced activities. The factorials and signs of the repeated logarithmic terms are retained throughout.

Fix a coarse configuration V. Continue to write k_j for a forced activity with support Y_j and nonempty inventory $I_j\subseteq Y_j$, and c_Γ for a regular logarithmic cluster term with nonempty support X_Γ. We have summable majorants

$$|k_j|\le p_j, \qquad |c_\Gamma|\le q_\Gamma, \qquad \sum_j p_j+\sum_\Gamma q_\Gamma<\infty.$$

These follow on a finite lattice from the previously purchased rooted bounds. Here q_Γ is specifically the absolute Ursell coefficient divided by the tuple factorial, multiplied by the purchased regular constituent prices. Thus the rooted cluster estimates apply to this majorant itself, not merely to $|c_\Gamma|$. All component addresses, support sets and eligibility tests are fixed before their values at V are evaluated.

For a finite admissible forced family A, the exclusion correction has the absolutely convergent expansion

$$e^{-P_{Y_A}} = \sum_{\substack{n:\ \text{finite support in the cluster alphabet}\\ n_\Gamma>0\Rightarrow X_\Gamma\cap Y_A\neq\varnothing}} \prod_{\Gamma:n_\Gamma>0}\frac{(-c_\Gamma)^{n_\Gamma}}{n_\Gamma!}, \qquad n_\Gamma\in\mathbb{N}. \tag{468}$$

For a finite cluster alphabet this is the product of the ordinary exponential series. For the whole alphabet, the sum of the absolute products is at most $\exp(\sum_\Gamma q_\Gamma)$; increasing finite subsets and dominated convergence prove the identity. Moreover the sum after multiplication by $\prod_{j\in A}k_j$ and summation over A is absolutely dominated by

$$\prod_j(1+p_j)\exp\left(\sum_\Gamma q_\Gamma\right). \tag{469}$$

The restriction that every retained cluster meet Y_A is essential: the correction deletes exactly those regular clusters.

A finite decorated family consists of a finite set of forced labels and a finite set of cluster labels, with a positive integer multiplicity attached to each cluster label. Give a forced label its support Y_j and multiplicity one, and a cluster label its support X_Γ. Its graph has one vertex per distinct label, with edges for intersecting supports. A new forced label D is a connected decorated family such that the old forced supports are pairwise disjoint and every cluster support meets the union of the old forced supports in D. Write Z_D for its union support and $I_D = \bigcup_{j\in D \text{ forced}} I_j$ for its inventory. In particular D contains a forced label and $I_D \neq \varnothing$.

Define its value by

$$\widetilde{k}_D(V) = \mathbf{1}_{\{\mathrm{Flags}(V)\cap Z_D = I_D\}} \mathbf{1}_{\{\text{the old local eligibility tests hold}\}} \prod_{j\in D \text{ forced}} k_j(V) \prod_{\Gamma\in D \text{ cluster}} \frac{(-c_\Gamma(V))^{n_\Gamma}}{n_\Gamma!}. \tag{470}$$

The cluster label itself already includes its original tuple factorial in c_Γ. The additional $n_\Gamma!$ belongs to the exponential in (468); these are two different factorials.

Consider a nonzero term in the expansion for (A, n). Split its support graph into connected components. Every retained cluster meets a forced support in A, and that forced label is in its component. Thus each component is a new forced label as just defined. Its inventory equals the flags inside its union support: a flag there belongs to some old inventory because A covers all flags, and the corresponding old forced support meets this component, so belongs to it. The reverse inclusion follows from the original inventory condition. Old eligibility tests hold for each nonzero factor. Hence the product of (470) over the components equals the original term, with every sign and factorial.

Conversely, flatten a finite compatible family of new labels whose inventories cover all flags and whose product is nonzero. Its old forced labels are pairwise compatible, both within and between the new labels. They satisfy all eligibility and exact flag-coverage conditions. A cluster label cannot occur in two new components: its nonempty support would intersect both of their supposedly disjoint union supports. Its multiplicity is therefore recovered unambiguously. Each retained cluster meets a forced support in its own component, as required in (468). The connected components of the flattened graph are exactly the original new labels. Flattening and taking components are inverse.

This weight-preserving bijection and (469) give

$$(\mathcal{B}R)(V) = \mathscr{O}(V) \exp\left(\sum_\Gamma c_\Gamma(V)\right) \sum_{\substack{\mathcal{D} \text{ finite, compatible} \\ \bigcup_{D\in\mathcal{D}} I_D = \mathrm{Flags}(V)}} \prod_{D\in\mathcal{D}} \widetilde{k}_D(V). \tag{471}$$

If there are no flags, no nonzero forced component is possible and the empty family contributes one. If there are flags, every nonzero component has a nonempty inventory. This is the required separation into a regular exponential and a forced gas.

The new values are measurable and gauge invariant: each factor and eligibility predicate is, and the products are finite. They use only the final coarse field and its flags on Z_D, by locality of the constituents and of the inventory test. Under a signed lattice transformation the old labels and the cluster occurrence tuples are permuted, with their multiplicities unchanged. Thus the value, support and inventory transform together. A common countable enumeration may then be chosen for the new labels, assigning zero to any unused addresses. Such an enumeration only reindexes the absolutely convergent gas and changes none of its values or support conditions.

For later bounds the same construction has the explicit nonnegative majorant

$$\widetilde{p}_D = \prod_{j\in D \text{ forced}} p_j \prod_{\Gamma\in D \text{ cluster}} \frac{q_\Gamma^{n_\Gamma}}{n_\Gamma!}. \tag{472}$$

Taking absolute values in (470) proves the bound by (472). We next prove the uniform rooted estimate for these majorants.

90.1 Keeping one compulsory gain through the second expansion

Let $a \ge 0$ be the desired support exponent and $\varepsilon > 0$ a reserve. Suppose the regular constituent prices obey the cluster criterion at support exponent $a+\varepsilon$, rooted allowance $u > 0$ and tail ratio $1/2$. Suppose also that the old forced prices obey

$$\sup_x \sum_{j: x\in Y_j} p_j e^{(a+\varepsilon)c_B(Y_j)} \le u e^{-M}, \qquad M \ge 0.$$

The regular cluster bound then gives

$$\sup_x \sum_{\Gamma: x\in X_\Gamma} q_\Gamma e^{(a+\varepsilon)c_B(X_\Gamma)} \le 2u.$$

Set $\eta = 20u$ and assume

$$\eta \le \tfrac{1}{2}, \qquad \frac{4b_B\eta}{\varepsilon} \le \tfrac{1}{2}, \qquad b_B = (2B+1)^4. \tag{473}$$

The second inequality is the explicit connected-tree degree bound of Section 62.

We prove that, for every site x and integer $m \ge 0$,

$$\sum_{\substack{D:\, x\in Z_D \\ |D|_{\text{copies}}\ge m}} \widetilde{p}_D e^{ac_B(Z_D)} \le 2^{-m} e^{-M}\, 2\eta. \tag{474}$$

For an upper bound forget the old compatibility tests, allow forced labels to repeat as well, and retain only connectedness and the presence of at least one forced label. On the resulting combined alphabet put $w_j = p_j$, $w_\Gamma = q_\Gamma$, and give each forced occurrence a mark M, each regular occurrence a mark zero. Inflate a price to

$$v_i = 2w_i e^{e_i}, \qquad e_j = M, \quad e_\Gamma = 0.$$

The rooted bound for $2v_i$ at exponent $a+\varepsilon$ is at most $4e^M u e^{-M} + 4(2u) = 12u \le \eta$. Thus these prices meet the distinct connected-set criterion with allowance η and tail ratio $1/2$, by (473).

To include multiplicities, first charge the desired support weight to each occurrence by setting $v_i' = v_i e^{ac_B(X_i)}$, with X_i denoting either kind of support. Since $X_i \ne \varnothing$, choose a site in X_i and use the single term of the rooted bound for $2v_i$ to obtain $v_i' \le \eta \le 1/2$. For a fixed distinct set of labels S, summing all positive multiplicities gives

$$\sum_{(n_i\ge 1)_{i\in S}} \prod_{i\in S} \frac{(v_i')^{n_i}}{n_i!} \le \prod_{i\in S} \sum_{n\ge 1} v_i' (1/2)^{n-1} = \prod_{i\in S} 2v_i'.$$

Here we used $n! \ge 1$ and $v_i' \le 1/2$. Connectedness depends only on the distinct supports, so the distinct connected-set bound now shows that the sum over all rooted connected multiplicity

families is at most $\eta/(1-1/2) = 2\eta$. Cover subadditivity on a connected set gives $c_B(\bigcup_{i\in S} X_i) \le \sum_{i\in S} c_B(X_i) \le \sum_i n_i c_B(X_i)$, justifying the charge per occurrence.

Finally the inflated product differs from the original one by $2^{\sum_i n_i} \exp(\sum_i n_i e_i)$. For a retained family these two factors are at least 2^m and e^M, respectively. Extracting them proves (474). Restricting back to the actual new labels only decreases the nonnegative sum. This argument preserves a compulsory gain even after arbitrary repeated regular corrections have been attached.

In the selected scale estimates, $u(x)$ has a fixed negative upper exponent and $B(x)$ has upper exponent zero. Both inequalities in (473), and the first regular degree bound with its own fixed positive reserve, therefore hold eventually. The compulsory mark is

$$M(x) = \frac{\mu}{4}\,\mathrm{fineMonitor}(x)^2, \qquad \mu > 0.$$

Thus (474) is precisely $40u(x)2^{-m}e^{-M(x)}$. It pays every fixed outgoing gain $\exp(-A_0 P(x)^\delta)$ with $A_0 \ge 0$ and $0 \le \delta < 2$, where $P(x) = \exp((\log\log x)^2)$ is the unshifted monitor. Indeed $P(x) \le \mathrm{fineMonitor}(x)$ eventually, the ratio $\mathrm{fineMonitor}(x)^\delta/\,\mathrm{fineMonitor}(x)^2$ tends to zero, and $40u(x)2^{-m} \le 1$ eventually. This proves the outgoing forced allowance from the incoming regular/forced prices, with the same scale and history.

91 Absorbing localized logarithms into the forced gas

The two exponential shifts in (496) arise from an exact regrouping of the forced activities with localized logarithms. We prove this regrouping, including its flag constraints and the derivative price needed for a tilted family.

All activities and logarithms in this section are real valued. Fix a coarse configuration U and let $F(U)$ be its set of flagged sites. A hard activity $K_i(U)$ has a nonempty support X_i, a nonempty inventory $I_i \subseteq X_i$, and a fixed eligibility condition. The hard factors and their eligibility predicates are measurable, gauge invariant and local on X_i, with the same final-flag convention as the logarithms below. Nonzero hard values satisfy their eligibility condition. The hard gas sums over finite families A of pairwise disjoint supports with $\bigcup_{i\in A} I_i = F(U)$:

$$G(U) = \sum_{A \text{ admissible}} \prod_{i\in A} K_i(U).$$

The empty family is allowed precisely when $F(U) = \varnothing$. Each logarithmic term $\ell_j(U)$ has nonempty support Y_j, is measurable and gauge invariant, and is local with the same threshold convention as the hard activities. Assume

$$Y_j \cap F(U) = \varnothing \implies \ell_j(U) = 0. \tag{475}$$

Suppose there are summable uniform majorants for both K_i and ℓ_j. These hypotheses will also imply absolute convergence of the new gas; it need not be assumed separately.

Use a disjoint union of the hard and logarithmic labels, assigning the factors

$$r_i = K_i, \qquad r_j = e^{\ell_j} - 1$$

and the supports X_i, Y_j, respectively. A new activity label C is a finite connected set of these labels which contains at least one hard label and whose hard supports are pairwise disjoint. Write Z_C for the union of all its supports and $J_C = \bigcup_{i\in C \text{ hard}} I_i$ for its inventory. The new activity is

$$\widehat{K}_C(U) = \mathbf{1}_{\{Z_C\cap F(U) = J_C\}} \mathbf{1}_{\{\text{all hard labels in } C \text{ are eligible at } U\}} \prod_{v\in C} r_v(U). \tag{476}$$

Its support and inventory are fixed by C before U is chosen. Its inventory is nonempty. The extra eligibility test is exactly the conjunction of the original tests, not a relaxed test. The new gas $\widehat{G}$ uses pairwise disjoint Z_C and exact coverage of $F(U)$ by the J_C.

91.1 The exact compatible-family bijection

The claimed identity is

$$\widehat{G}(U) = G(U) \exp\left(\sum_j \ell_j(U)\right). \tag{477}$$

First expand the right side as a sum over a finite admissible hard family A and a finite set B of logarithmic labels:

$$\sum_{A \text{ admissible}} \sum_{B \text{ finite}} \prod_{i\in A} K_i(U) \prod_{j\in B} (e^{\ell_j(U)} - 1). \tag{478}$$

To justify this, absolute summability of $\ell_j(U)$ implies absolute summability of $e^{\ell_j(U)} - 1$. Outside finitely many indices $|\ell_j(U)| \le 1$, and $|e^x - 1| \le 2|x|$ there. Finite products $\prod_j [1+(e^{\ell_j}-1)]$ equal the corresponding exponentials; their expansions converge absolutely, dominated by $\prod_j (1 + |e^{\ell_j} - 1|)$. Passing to the limit proves the logarithmic part of (478). The hard sum is absolutely dominated by $\prod_i (1 + |K_i|)$, so the double sum is also absolutely convergent.

Consider a nonzero term of (478) and split $A \sqcup B$ into connected components of its support intersection graph. Every component contains a hard label. In fact, if it contains a logarithmic label j, its nonzero factor implies $\ell_j(U) \neq 0$. By (475), some flagged site lies in Y_j. Exact coverage of all flags by the old hard inventories places that site in some I_i, $i \in A$. Since $I_i \subseteq X_i$, the labels i, j have overlapping supports and lie in the same component. Each component is therefore an allowed new label.

Distinct components have disjoint supports. For a component C, any flag in Z_C belongs to an old inventory I_i; its hard support then meets Z_C and so i lies in C. Conversely every I_i in C consists of flags and lies in Z_C. Thus $Z_C \cap F(U) = J_C$. Every old hard factor in the nonzero term is eligible. The product of the new activities over the components consequently equals the original term, and their inventories cover $F(U)$.

Conversely, take an admissible new family with nonzero product and flatten its labels. Within each component the old hard supports are disjoint by definition; between components they are disjoint because the whole component supports are disjoint. The old hard inventories cover $F(U)$, and the local eligibility indicators in (476) equal one. Flattening therefore gives an admissible hard family and a finite set of logarithmic labels. No label is duplicated between components, because its support is nonempty and would otherwise meet both disjoint component supports. The connected components of the flattened support graph are exactly the original new labels. This proves that flattening and taking components are inverse on nonzero terms.

The identity follows by this weight-preserving bijection and absolute convergence of (478). In particular the bijection transports its absolute sum to that of the new gas. It also includes the empty-family case: when there are no flags, every logarithm vanishes and no nonzero new component can occur.

The new activities have the required locality and symmetry. Products inherit gauge invariance; flags and all old eligibility predicates are gauge invariant, so the two indicators are also invariant. Measurability follows from measurable predicates and finite products. For locality on Z_C, agreement of the configurations and their flag data there gives agreement on every constituent support and

on every eligibility test. Finally the first indicator enforces exact local inventory equality. These arguments apply to the original addresses and predicates throughout.

91.2 Preserving the derivative allowance

Now let the hard factors be $K_i(\theta, U)$, with the logarithmic terms independent of θ. All addresses and eligibility tests remain fixed. Let $D \geq 0$, and suppose the hard factors have the joint bound of (491) with majorants p_i on $|\theta|D \leq 1$, and are independent of θ whenever their support does not contain a fixed anchor o. Suppose also that $|\ell_j(U)| \leq v_j \leq 1$, and, for an exponent $u \geq 0$,

$$\sup_x \sum_{i: x \in X_i} p_i e^{(u+1)c(X_i)} \leq b, \qquad \sup_x \sum_{j: x \in Y_j} v_j e^{(u+1)c(Y_j)} \leq w. \tag{479}$$

Choose $\kappa \geq 0$ and define inflated constituent prices

$$a_i = e^{\kappa} p_i \quad (i \text{ hard}), \qquad a_j = 2v_j \quad (j \text{ logarithmic}), \qquad \Gamma = e^{\kappa} b + 2w.$$

For a logarithmic factor the constant joint bound is $2v_j$. For a hard factor the joint bound is $e^{-\kappa} a_i$. By the product rule for joint bounds, the product in a new label C has joint bound

$$e^{-\kappa \#\{i \in C:\ i \text{ hard}\}} \prod_{v \in C} a_v \leq e^{-\kappa} \prod_{v \in C} a_v =: p_C.$$

The two indicators in (476) are independent of θ, so the same bound holds for $\widehat{K}_C$. It is independent of θ outside the anchor support as well: if $o \notin Z_C$, each hard constituent is independent of θ.

The constituent site price at exponent $u+1$ is at most Γ. If $h_B \Gamma \leq 1$, the distinct connected-set estimate (202), with reserve one, gives

$$\sup_x \sum_{C: x \in Z_C} p_C e^{uc(Z_C)} \leq e^{-\kappa}(e^{\Gamma} - 1). \tag{480}$$

Restricting to labels with nonempty hard part and separated hard supports only decreases the nonnegative sum. Thus an allowance A is sufficient whenever

$$e^{-\kappa}(e^{\Gamma} - 1) \leq A.$$

All the properties in (493) are preserved, with the same anchor and insertion debt D and with new site price A. Applying Section 94 then gives marked forced cost DA on the fused alphabet. The original derivative parameter is never rescaled when a logarithmic contribution is absorbed.

For uniform absolute convergence in this argument, the finite lattice and nonempty supports imply summability of p_i and v_j from (479). The same estimate and (480) give summability of p_C. Thus the scalar bijection, the parameter derivatives, and the site bounds refer to the same series. Applying this construction successively to the two localized logarithmic corrections proves the two gas shift identities used in (496), provided their actual corrections satisfy (475), the fixed-address conditions and the purchased bounds above. Producing those corrections from the nonlinear block integral, and paying these inequalities on the selected history, remain the analytic inputs.

92 The marked block integral on the original activity addresses

We now obtain the exact tilted identity used in Section 94. The insertion is transported through the same integral and the same connected-component decomposition as the background. There is no division by a forced activity, which may vanish, and the distinguished activity address remains fixed.

92.1 The fine inserted density

Write the fine density as

$$\rho(U) = e^{E(U)} \sum_{A \text{ admissible}} \prod_{i \in A} K_i(U).$$

Choose a bounded real measurable gauge-invariant function h, with $|h| \le w$, supported on the eligibility condition of one fixed activity address. Its dependence on the configuration is confined to that address's full inspected support. There are two possibilities. For a regular address b, its background regular activity is identically zero and h vanishes outside the healthy set for its support. For a forced address b, h vanishes whenever that forced address fails its original eligibility test. Define

$$f_h(U) = e^{E(U)} \sum_{A \text{ admissible}} B_h(A,U), \qquad B_h(A,U) = \begin{cases} h(U) \prod_{i \in A} K_i(U), & b \text{ regular}, \\ \mathbf{1}_{\{b \in A\}} h(U) \prod_{i \in A \setminus \{b\}} K_i(U), & b \text{ forced}. \end{cases} \tag{481}$$

The admissibility tests, including exact flag coverage, are those of the background. In the forced case the address is replaced even when its original value $K_b(U)$ is zero.

Here and below use the same funded sector data and all-species integrated product bounds as in Section 88. On the fixed finite lattice, choose summable uniform prices $p_i \ge |K_i(U)|$. In the regular case the absolute family sum in (481) is bounded by $w \prod_i (1 + p_i)$. In the forced case replace p_b by w and use the same product bound. Thus the series converges uniformly and absolutely. The fine exponent has a bounded measurable envelope, so f_h is bounded and measurable. Gauge invariance follows termwise from the invariance of h, the activities and the flag tests. In particular the fine Haar integral and every fixed transition slice integral exist. This argument does not require a sign for h.

92.2 One marked factor in the Gaussian expansion

Denote the elementary alphabet of the unmarked expansion by $\mathcal{I}$ and the letter assigned to b by $\star$. Its support is the same enlarged inspected support as the original address. For one sector, one tag assignment and one old Haar configuration, let $v_i(z)$ be the elementary factors of the Gaussian expansion. The marked factor is

$$H(z) = h(U(z))\, \zeta_\star(z), \qquad 0 \le \zeta_\star \le 1,$$

where $\zeta_\star$ is the larger local fluctuation cutoff on the marked support. For a finite set of letters P put

$$\Psi_h(P,z) = \mathbf{1}_{\{\star \in P\}} H(z) \prod_{i \in P \setminus \{\star\}} v_i(z). \tag{482}$$

The cutoff makes the marked factor part of the same local integral construction. On the nonzero support of the genuine common chart cutoff it equals one: the stronger common small-fluctuation bound places each marked coordinate in its plateau. If the common cutoff is zero, both sides of the chart identity are zero. Hence inserting $\zeta_\star$ changes none of the represented fine integral.

The regular and forced addresses give two slightly different expansion identities. At a blank regular address the old exponential-minus-one factor $v_\star$ is zero. Every old term containing that letter therefore vanishes. Insert $\star$ into each remaining finite letter set and replace its zero factor by H. Erasing $\star$ is the inverse map, so the resulting sum is H times the original sum. In the forced case, a family not containing the address contributes zero. For a family containing it, erase just

that forced factor and put H in its place. The optional logarithmic and warning expansions are unchanged. These two identities give precisely the two cases of (481); there is no extra multiplicity.

To justify integration of these expansions, let $r_i \geq 0$ be the summable elementary integrated prices already established for the unmarked product construction. Its bound applies to every finite set of letters, in particular to $P \setminus \{\star\}$. Since $|H| \leq w$, it gives

$$\int |\Psi_h(P,z)|\, d\nu(z) \leq \mathbf{1}_{\{\star\in P\}} w \prod_{i\in P\setminus\{\star\}} r_i, \tag{483}$$

where ν includes the old probability Haar variables and the Gaussian variables. The right side is summable over P: erasure maps the sets containing $\star$ injectively to all finite sets, whose product-price sum is at most $\exp(\sum_i r_i)$. The separate uniform envelope prices give measurability and an integrable envelope exactly as in the unmarked construction. Thus Fubini and absolute summation justify every exchange here.

For clarity, the chart and Gaussian normalization in this argument are unchanged. First split the free Haar coordinates into the active chart coordinates and their complement. On a nonzero sector the chart-support lemma puts every active bond in the original exponential chart. The product change of variables contributes $c_H^{N_{\mathrm{active}}} \prod J(u_i)$; the affine rescaling contributes $g^{qN_{\mathrm{active}}}$; the unused Gaussian coordinates integrate to one with their normalized density. The quadratic completion and logarithmic cancellation from Section 88 then apply with the extra factor $h(U)$ untouched. Its local cutoff has just been restored to one on this support. Summing the sector partition and averaging the original tags consequently recovers $\mathcal{T} f_h$, with the same scalar overhead $\mathscr{O}$ as for $\mathcal{T}\rho$.

92.3 The distinguished connected component

Split a finite elementary letter set into its connected components, using their full assigned supports. If it contains $\star$, exactly one component C contains that letter. Formula (482) factors into the marked product on C and the ordinary products on the other components. The Gaussian cross covariances between their read sets vanish; the Haar and tag read sets are disjoint. Thus the product integral factors. This is the same finite-range independence argument used in the unmarked case, now with one bounded local factor substituted.

Let $a_C^F(V)$ be an ordinary integrated component value and $b_C^F(V)$ the marked value, set to zero unless $\star \in C$. The component and flattening maps are inverse on the permitted families. The sector and eligibility masks are identical on both sides: if a component fails its local test, its corresponding ordinary or marked value is zero; on a nonzero family the union of its inventories is exactly the set of flags. Absolute convergence from (483) therefore proves

$$\mathcal{T} f_h(V) = \mathscr{O} \sum_{\mathcal{A}\ \text{admissible}} \sum_{C\in\mathcal{A}} b_C^F(V) \prod_{D\in\mathcal{A}\setminus\{C\}} a_D^F(V). \tag{484}$$

The marked value has the same locality in the forest as the ordinary one. Indeed h reads only its inspected support, and its added cutoff reads only the corresponding buffered coordinates. Forests agreeing there give equal values after pulling their free Haar variables back to the common full Haar space. Their principal Gaussian covariances agree, so integration gives equality too. We may therefore average independent forest choices at coarse sites as in Section 89. For a compatible family the marked component and all the ordinary components use disjoint sets of site choices, so the average of their product is the product of their averages. Set

$$a_C = \mathbb{E}_F a_C^F, \qquad b_C = \mathbb{E}_F b_C^F.$$

Equation (484) holds with these averaged values. The same absolute majorants apply to this finite average.

Two distinct compatible components cannot both have nonzero marked value, since both would contain $\star$. Hence, for every real θ and every compatible family,

$$\prod_{C\in\mathcal{A}}(a_C+\theta b_C)=\prod_{C\in\mathcal{A}}a_C+\theta\sum_{C\in\mathcal{A}}b_C\prod_{D\neq C}a_D.$$

The terms of degree at least two vanish individually. Together with the unmarked identity this proves the exact integral formula

$$\mathcal{T}\rho+\theta\mathcal{T}f_h=\mathscr{O}\sum_{\mathcal{A}\ \text{admissible}}\prod_{C\in\mathcal{A}}(a_C+\theta b_C). \tag{485}$$

The right side is affine despite its product notation. The masks, support assignments, inventories and distinguished fine address have never changed.

92.4 Passing the mark through the two regroupings

For θ in the interval allowed by the purchased marked prices, apply the regular cluster expansion and forced exclusion regrouping to the values $a_C+\theta b_C$. Denote their regular exponential by $a(\theta)$ and their forced gas by G_θ. The exact identities of Sections 89 and 90 give

$$\mathcal{T}\rho+\theta\mathcal{T}f_h=\mathscr{O}a(\theta)G_\theta.$$

The absolute product and cluster majorants apply uniformly on this interval. Indeed if $|a_C|\le A_C$, $|b_C|\le DB_C$ and $|\theta|D\le 1$, the affine factor has the joint value/derivative bound A_C+2B_C of Section 94. Products, cluster coefficients and exponential multiplicities retain that joint bound by the product rule proved there. This uses the marked-price purchases; the unmarked prices alone do not supply the derivative bound.

Finally let R_0,S_0 be the two fixed local logarithmic corrections absorbed into the forced gas in Section 91. That regrouping uses the same fixed background and gives $b(\theta)=G_\theta e^{R_0+S_0}$. Its background normalization is $e^{E_{\mathrm{coarse}}+R_0+S_0}=\mathscr{O}a(0)$. Substitution now proves

$$\mathcal{T}\rho+\theta\mathcal{T}f_h=e^{E_{\mathrm{coarse}}}\frac{a(\theta)}{a(0)}b(\theta),$$

which is (496) for the actual insertion (481). Differentiation is justified by the joint summable bounds, not by a formal manipulation of the series. This establishes the exact marked integral underlying that argument. The remaining quantitative task is to purchase its marked local derivative prices and principal-term estimate on the selected history; the density identity alone does not provide those estimates.

93 Prices and local derivatives of the marked activities

The marked integral permits a bounded measurable insertion. Its coarse local extension nevertheless has the derivatives needed for the projection argument. The reason is the compensated transport: it holds the fine field read by the insertion fixed. Only its smooth cutoff and the surrounding integral are differentiated.

93.1 Paying for the distinguished letter

Let X be the fine insertion support, $c = C_{B_f}(X)$ its fine cover number, s the fine support exponent, and w its uniform bound. Its fine mass is $m = e^{sc}w$. Fix the support reserve $\varepsilon > 0$, with $s > 4\varepsilon$, and let $g > 0$ be the normalization gain. Introduce

$$D = \frac{me^{-\varepsilon c}}{g}, \qquad J = ge^{-(s-\varepsilon)c}, \qquad DJ = w. \tag{486}$$

The last equality follows by cancellation of the two exponentials. It also covers $w = 0$. We give the marked letter price J in the mixed component estimate and keep the factor D outside it.

Let X' be its full enlarged coarse support. The footprint estimate of Section 83 has the form

$$C_B(X') \le C_0 + \frac{10C_0}{L}c,$$

where C_0 depends only on the fixed enlargement depth. Choose the block ratio so that $10uC_0/L \le \varepsilon$, where u is the mixed estimate's support exponent. The marked contribution to the elementary price, including the two length reserves, then obeys

$$\begin{aligned} Je^{2\varepsilon c}e^{uC_B(X')} &\le ge^{uC_0}e^{-(s-4\varepsilon)c} \\ &\le ge^{uC_0}. \end{aligned}$$

At any root there is at most this one new letter. Thus replacing the old price at that address by J increases both the optional root bound and the total root bound by at most ge^{uC_0}. All other prices and all supports remain unchanged. Applying the mixed connected-component estimate of Section 63 to this enlarged price family bounds the marked components, including any excess-cardinality, support-size and forced-count restrictions required there.

This enlargement is purchased at the same scale as the background. If its original root price and g have negative upper exponent $q < 0$, their sum has that exponent too. The geometric degree factor has upper exponent zero. It is therefore eventually at most $\beta_f^{q/2}$; this quantity tends to zero and may be used as the new tree ratio. This follows directly from the strict exponent comparison in Section 94. It makes no new choice of a history or a fine-to-coarse transition.

93.2 Holding a merely measurable insertion fixed

Fix a component P containing the marked letter, a healthy coarse configuration on its full support, and a smooth coarse probe based there. Use the compensated transport from Section 79. For fixed old Haar variables and Gaussian point z, let $U_0(z)$ be the physical fine field at the base point. On the nonzero support of the transported marked cutoff, the transport identity gives agreement with $U_0(z)$ on the inspected support of h. Since h is local, the marked factor becomes

$$h(U_0(z))\,\zeta_\star(y, z). \tag{487}$$

If the cutoff is zero, both expressions are zero. The identity holds as a germ in the coarse probe parameter y. The truncation in the compensated transport preserves this germ, because its threshold is larger than the cutoff support. This is exactly the cutoff comparison used for the unmarked transported factors.

The function $h(U_0(z))$ is constant in y and has absolute value at most w. The cutoff derivative estimate therefore gives amplitude w and derivative scale at most $J_0\sigma_\star$, where $\sigma_\star$ is the marked letter's support scale and J_0 has upper exponent zero. No derivative of h has been taken. This

observation also gives the locally uniform integrable derivative envelopes: multiply the existing smooth cutoff envelopes by the bounded fixed function $h(U_0(z))$.

Write a_i for the unmarked elementary jet amplitudes and r for their common scale factor. The sum of the support scales in the component is at most twice its geometric size W_P. Leibniz's rule applied to (487) and the unmarked factors gives amplitude

$$A_P = w \prod_{i \in P \setminus \{\star\}} a_i$$

and derivative scale at most $2(r + J_0)W_P$. The same joint warning event can be retained. Outside that event, one warning factor has the zero germ supplied by the strict small-fluctuation plateau. The marked letter is a regular or forced insertion letter, never a warning letter; erasing it leaves the warning set unchanged. Thus the zero germ still occurs among the remaining factors.

Now include the Jacobian and projected Gaussian density of Section 85. Their joint derivative scale is bounded by $U_0\mathcal{F}_I(z)^k$, with amplitude one, where $\mathcal{F}_I \geq 1$ is the Gaussian coordinate polynomial on the actual local read set I. For derivatives through order j, the full integrand has the bound

$$\mathbf{1}_{\mathcal{W}_P}(z) A_P [1 + U_0 + 2(r + J_0)]^j (1 + W_P)^j \mathcal{F}_I(z)^{kj}. \tag{488}$$

To verify the scale conversion, use

$$1 + U_0\mathcal{F}_I^k + 2(r + J_0)W_P \leq [1 + U_0 + 2(r + J_0)](1 + W_P)\mathcal{F}_I^k$$

and raise both nonnegative sides to the integer power j. The estimates concern the unmasked extension on a neighborhood of the healthy base point; they do not differentiate the flag mask.

93.3 Integrating the derivative bound

The Gaussian moment and warning estimate of Section 86 gives, for some fixed integer K and scalar F of upper exponent zero,

$$\int \mathbf{1}_{\mathcal{W}_P} \mathcal{F}_I^{kj} \, d\gamma \leq (1 + |F|)(1 + W_P)^K \rho^{N_P}.$$

Here N_P is the number of warning letters and ρ is the half-warning factor of that estimate. Explicitly, Cauchy–Schwarz uses the moment of degree $2kj$ and the square root of the joint warning probability. This explains both the doubled moment degree and the preserved half-warning gain. The old Haar measure has total mass one, so introduces no further volume factor.

Integrating (488) therefore bounds the local derivative amplitude by

$$B_* A_P \rho^{N_P} (1 + W_P)^{K+j}, \qquad B_* = (1 + |F|)[1 + U_0 + 2(r + J_0)]^j. \tag{489}$$

The finite-order differentiation-under-the-integral argument of Section 85 applies: the relative Gaussian density supplies an integrable local envelope, the transported insertion is bounded and constant in the probe, and all remaining factors have the already proved smooth envelopes. Thus this is a bound for derivatives of the actual local integral. The constants j, k, K are fixed before the coupling limit. Products, sums and fixed powers show that B_* has upper exponent zero.

Let v_P be the mixed component price with the marked elementary price replaced by J, with the warning gain and the size weight $(1 + W_P)^{K+j}$ included. The exact identity $DJ = w$ from (486) rewrites (489) as $DB_* v_P$. This proves the marked analogue of the unmarked local jet estimate from the same sector data, rather than postulating a new regularity property of h.

To make a globally measurable gauge-invariant extension, first average the local projected integral over the original finite tag choices, then average its gauge translates against product probability Haar measure. Locality is preserved because only the gauge coordinates incident to its read support matter. Haar invariance makes the result gauge invariant. At a healthy base point, the local covariance and chart bounds persist on a probe neighborhood. Change of variables in the local integral shows that it is already gauge invariant there. Hence the gauge average agrees with that local integral as a germ, and the derivative bound is unchanged. Independent forest averaging also preserves the bound. We have constructed a measurable local extension g_P satisfying

$$b_P(V) = \mathbf{1}_{\{X_P \cap \mathrm{Flags}(V) = \varnothing\}} g_P(V)$$

for regular components, with derivatives through the purchased order bounded by $DB_* v_P$ at healthy configurations. This is finite-order smoothness on the stated domain.

93.4 The logarithmic derivative and its cluster prices

Let f_P be the unmarked local extension. For a regular cluster tuple $\Gamma = (P_1, \dots, P_n)$ its unmasked marked logarithmic term is

$$h_\Gamma = \frac{u(P_1, \dots, P_n)}{n!} \sum_{r=1}^{n} g_{P_r} \prod_{s \neq r} f_{P_s}. \tag{490}$$

This is the derivative at zero of the corresponding product of affine tilted factors. All terms have the mask of the union support. Indeed on a healthy union every constituent mask is one; if the union contains a flag, each summand has a zero factor, either its marked factor or one of its ordinary factors.

Suppose f_P has local amplitude A_P^0 and g_P has amplitude DA_P^1, both at derivative scale one. Put $A_P = A_P^0 + 2A_P^1$ and inflate this price to $\widehat{A}_P = 2^{j+2} A_P$. On a healthy union each summand in (490) has amplitude at most $D \prod_r A_{P_r}$ and derivative scale n. Its derivatives through order j are consequently bounded by

$$D \frac{|u(P_1, \dots, P_n)|}{n!} n (1+n)^j \prod_r A_{P_r} \leq D \frac{|u(P_1, \dots, P_n)|}{n!} \prod_r \widehat{A}_{P_r}.$$

For the last inequality use $1 + n \leq 2^n$ and $n \leq 2^{2n}$. Thus the canonical cluster majorant at the inflated prices pays all these derivatives as well as the scalar tilt derivative. The rooted cluster estimate and its tail then give (498). The marked mixed-component estimate proved above supplies its order-zero counterpart, with the excess-cardinality restriction for nonprincipal terms. The polynomial support-weight conversion and strict exponent reserve in Section 94 therefore apply to these actual local terms.

For forced components use their absolute marked integral price from (483) and the same mixed estimate. The affine joint bounds propagate through the exclusion regrouping by the product rule, keeping the exponential multiplicity factorials. Section 90's rooted argument applies to these inflated prices and gives the summable joint forced-price family required in (493). Every nonzero marked component contains the projected insertion anchor, so the derivative vanishes at all other supports. This proves the structural and local analytic inputs to the marked differentiation. Their simultaneous scale purchases are made as follows.

93.5 One coupling threshold for the marked estimates

Fix the derivative and spatial orders, the target support exponent $a \ge 0$, and a reserve $e > 0$ with $a + 2e$ no larger than the available mixed support exponent. Let $\mathcal{O}_n$ be the unmarked rooted component price with size weight n, and C_n the corresponding marked mixed price after adjoining J. Both have upper exponent $q < 0$. Write $C_n^{(1)}$ for the marked price with excess cardinality at least one; its extra tree factor gives upper exponent $3q/2$. Put

$$t = (1 + 2^{j+2}B_*)(\mathcal{O}_n + 2C_n), \qquad v = \mathcal{O}_n + 2C_0, \qquad u = 10(t + v), \qquad \gamma = \beta_f^{q/2}.$$

These quantities are positive in the selected construction. The inflated regular prices above have rooted allowance at most t. The forced prices have allowance at most ve^{-M}, where M is the compulsory forced mark. These assertions follow by adding the unmarked and marked mixed bounds, then multiplying the regular bound by $2^{j+2}B_* \le 1 + 2^{j+2}B_*$.

The functions t, v, u have upper exponent q. Since the geometric degree factor $4(2B+1)^4/e$ has upper exponent zero, eventually

$$\frac{4(2B+1)^4 t}{e} \le \gamma, \qquad \frac{4(2B+1)^4 u}{e} \le \tfrac{1}{2}, \qquad \gamma \le \tfrac{1}{2}, \qquad u \le \tfrac{1}{2}.$$

At the same scale the shifted fine radius is at most twice the coarse radius, by the radius estimate of Section 16. There are finitely many eventual inequalities at fixed orders; choose their common threshold before choosing the finite torus or insertion. The rooted bounds are uniform in both, so this threshold works for every permitted insertion on every such torus.

For the second expansion the two rooted contributions, after inflating copies and extracting the compulsory mark, are bounded by

$$4e^M(ve^{-M}) + 4t(1-\gamma)^{-1} \le 4v + 8t \le 10(t+v) = u.$$

The connected multiplicity argument in Section 90 thus gives forced joint price at most $2ue^{-M}$. Its differentiated price is D times this quantity, as required in Section 94.

Finally, let $A_{e,r}$ be the polynomial support-weight price in that section. It has upper exponent zero at fixed orders. The nonprincipal regular tail divided by the fine mass reserve is bounded by

$$\frac{A_{e,r}}{g}\left(B_* C_n^{(1)} + t\frac{\gamma}{1-\gamma}\right).$$

Here $g = \beta_f^q$, so g^{-1} has upper exponent $-q$. Both numerator terms have upper exponent $3q/2$, since $(1-\gamma)^{-1} \le 2$. The quotient therefore has upper exponent $q/2$ and is eventually at most $\beta_f^{q/4}$. This purchases the tail and joint forced prices used by the actual marked expansion. The final principal-term and projection allowances must still be compared with the selected outgoing budget, using the same already chosen scale.

94 Differentiating the nonlinear insertion expansion

The elementary insertion is obtained by differentiating a real tilt of the same nonlinear block integral used for the background. This matters for two reasons. The background normalization must cancel exactly, and the derivative of the gas must retain the original activity addresses. We first prove the summation argument which makes this differentiation legitimate, then account for the nonprincipal terms.

94.1 A joint bound for values and derivatives

Fix a number $D \geq 0$ and an open interval I containing zero. For a differentiable real function f and a nonnegative number p, say that f has joint bound p at θ if there are $a, b \geq 0$ with

$$a + b \leq p, \qquad |f(\theta)| \leq a, \qquad |f'(\theta)| \leq Db. \tag{491}$$

The witnesses a, b may depend on θ. This is stronger than bounding the value by p and the derivative by Dp separately. Its advantage is that products have no extra factor equal to their degree. Indeed, if f_1, f_2 have witnesses $(a_1, b_1), (a_2, b_2)$, take

$$a = a_1 a_2, \qquad b = a_1 b_2 + b_1 a_2.$$

The product rule gives the required value and derivative bounds, and $a+b \leq (a_1+b_1)(a_2+b_2) \leq p_1 p_2$. Induction gives joint bound $\prod_{j \in q} p_j$ for any finite product. A constant multiplier c multiplies the bound by $|c|$. These statements include $D = 0$. For the affine tilt $f + \theta g$, if $|f| \leq a$, $|g| \leq Db$ and $|\theta| D \leq 1$, the witnesses $(a + b, b)$ give joint bound $a + 2b$.

Here is the precise infinite-sum consequence. Suppose f_i is differentiable on I and has joint bound p_i at every point of I, where $\sum_i p_i < \infty$. Both $\sum_i f_i(\theta)$ and $\sum_i f_i'(\theta)$ converge absolutely. For a closed segment joining θ and $\theta + h$ inside I, the mean value theorem gives

$$\left| \frac{f_i(\theta + h) - f_i(\theta)}{h} \right| \leq D p_i.$$

Dominated convergence on the counting measure therefore yields

$$\frac{d}{d\theta} \sum_i f_i(\theta) = \sum_i f_i'(\theta). \tag{492}$$

This proof needs no continuity of the individual derivatives. It also shows that the choice of an enumeration of the countable index set has no effect on the result.

94.2 The gas derivative and its support

Work on a fixed finite coarse lattice Λ. Let $A_j \neq \varnothing$ be the support of forced activity $j \in \mathbb{N}$. The ordinary activities, the support and inventory of each forced activity, and the admissibility test are held fixed. Only the values K_j change, along a real family $F_j(\theta, U)$, with $F_j(0, U) = K_j(U)$. Assume that, for $|\theta| D \leq 1$, these are measurable, gauge invariant forced activities with the same locality and flag conditions as K_j. Also assume

$$\begin{aligned} &F_j(\theta, \cdot) = F_j(0, \cdot) \quad \text{if } o \notin A_j, \\ &F_j(\cdot, U) \text{ has joint bound } p_j \geq 0, \\ &\sup_{x \in \Lambda} \sum_{j : x \in A_j} p_j e^{u C_B(A_j)} \leq s, \qquad u \geq 0. \end{aligned} \tag{493}$$

The middle line is required on an open neighborhood of zero contained in $\{|\theta| D \leq 1\}$, for every U. Such a neighborhood exists also when $D = 0$. Since each support is nonempty,

$$\sum_j p_j \leq |\Lambda| s < \infty.$$

For a finite family q put $p_q = \prod_{j\in q} p_j$. Expanding finite products and taking increasing finite subsets of $\mathbb{N}$ proves

$$\sum_{q\subset\mathbb{N}\ mfinite} p_q = \prod_j (1+p_j) \le \exp\Big(\sum_j p_j\Big) < \infty.$$

For fixed U, admissibility does not depend on θ. Thus each admissible product $\prod_{j\in q} F_j(\theta,U)$ has joint bound p_q, and an inadmissible term is zero. Equation (492) now proves that the gas derivative at zero is

$$\mathcal{G}^{[1]}(k)(U) = \sum_{q\ \text{admissible}} \sum_{j\in q} k_j(U) \prod_{\ell\in q\setminus\{j\}} K_\ell(U), \qquad k_j(U) = \partial_\theta F_j(\theta,U)|_0\,. \tag{494}$$

There is also absolute convergence before the marked index is summed. For joint-bound witnesses a_j, b_j at zero,

$$\sum_{j\in q} |F_j'(0,U)| \prod_{\ell\ne j} |K_\ell(U)| \le D \sum_{j\in q} b_j \prod_{\ell\ne j} a_\ell \le D \prod_{j\in q}(a_j+b_j) \le D p_q.$$

The middle inequality keeps just the terms with exactly one b in the product expansion. Summing over q proves absolute convergence of the full pointed series. This is the marked gas on the original addresses.

The marked family retains all required structural properties. If $o \notin A_j$, its derivative is zero by the first line of (493). Wherever the flag condition requires $F_j(\theta,U) = 0$, it also requires $k_j(U) = 0$. Gauge invariance and locality follow by differentiating the corresponding equalities in θ. Measurability follows by choosing any nonzero sequence $\theta_n \to 0$ in the permitted interval and writing k_j as the pointwise limit of measurable difference quotients. Finally,

$$|k_j(U)| \le D p_j \mathbf{1}_{\{o\in A_j\}}, \qquad \sum_j D p_j \mathbf{1}_{\{o\in A_j\}} e^{uC_B(A_j)} \le Ds.$$

This proves the flagged cost (249), with exactly the insertion debt D and the site bound s in (493). Complexification preserves the same majorant.

94.3 The logarithmic derivative and the exact raw identity

The localized regular terms arise from the logarithm of the positive normalization. Write this normalization as

$$a(\theta,U) = \exp\left(\sum_i \ell_i(\theta,U)\right), \qquad a_0(U) = a(0,U) > 0.$$

Suppose that each logarithmic term has a summable joint majorant, uniformly for θ near zero, and that its derivative is the localized insertion

$$\ell_i'(0,U) = \chi_{X_i}(U) h_i(U).$$

The product bound above applies, in particular, to each connected cluster term $c\prod_j f_j(\theta)/r!$: its majorant is $|c|\prod_j p_j/r!$. Thus an absolutely summable connected-cluster majorant justifies differentiating the logarithmic expansion. Equation (492) and the chain rule give

$$a'(0,U) = a_0(U) z(U), \qquad z(U) = \sum_i \chi_{X_i}(U) h_i(U). \tag{495}$$

The exact tilted block identity, after the two activity regroupings, has the form

$$\mathcal{T}\rho_{\mathrm{fine}}(U) + \theta\mathcal{T}f(U) = e^{E(U)}\frac{a(\theta,U)}{a_0(U)}\,b(\theta,U), \qquad b(0,U) = \mathcal{G}(K)(U), \tag{496}$$

where b is the gas of $F(\theta)$. All quantities use the same background and history. This identity is required for the actual block integral, for $|\theta|D \le 1$. Differentiating it, using (494) and (495), yields

$$\mathcal{T}f(U) = e^{E(U)}\mathcal{G}(K)(U)z(U) + e^{E(U)}\mathcal{G}^{[1]}(k)(U).$$

As $\rho_{\mathrm{coarse}} = e^{E}\mathcal{G}(K)$, this is exactly (246). In particular, the derivative of the normalization is multiplied by the full coarse density. Omitting a_0^{-1} before differentiation would give an incorrect factor.

The normalization cancellation in (496) is algebraic, but uses exact identities. To spell it out, let G_θ be the gas before the two regroupings, whose logarithmic shifts are R and S. Their identities read

$$b(\theta) = G_\theta e^{R+S}, \qquad e^{E+R+S} = ha_0, \qquad ha(\theta)G_\theta = \mathcal{T}\rho_{\mathrm{fine}} + \theta\mathcal{T}f.$$

Substitution proves (496). The two exact gas regroupings and their numerical majorants are inputs here, as is the exact block expansion in the last identity. Their application must retain the same support, inventory and flag data. The scalar differentiation argument does not by itself establish the additional transferred singleton-profile identity required in Section 61.

94.4 Separating the principal term from the tail

Here is the quantitative bookkeeping for the nonprincipal regular terms. It identifies precisely which connected estimates are needed. Let j be the derivative order, p the spatial weight exponent, B the cover radius, and $a \ge 0$, $e > 0$ the support exponent and its reserve. Put

$$r = pj, \qquad \Phi(C) = [\sigma(1+C)]^r, \qquad P_{e,r} = e^{-r}r!\exp(e), \qquad A_{e,r} = \sigma^r P_{e,r},$$

where $\sigma = 4(2B+1) > 0$ and e^{-r} denotes the inverse power of the reserve e. The exponential series gives $[e(1+C)]^r/r! \le \exp(e(1+C))$ for $C \ge 0$. Consequently

$$\Phi(C)e^{aC} \le A_{e,r}e^{(a+e)C}. \tag{497}$$

The localized logarithmic terms have a nonnegative integer order. An order-zero term corresponds to a single thin molecule; the higher orders are connected clusters. Terms which do not contain the insertion vanish. The remaining terms have supports containing the chosen anchor o. Write C_i for their cover numbers and $\omega_i \ge 0$ for their cluster majorants. Suppose the connected estimate gives, for every $z \ge 0$ and every site x,

$$\sum_{i:\, x\in X_i,\, \mathrm{ord}(i)\ge z} \omega_i e^{(a+e)C_i} \le t\gamma^z(1-\gamma)^{-1}, \qquad 0 \le \gamma < 1. \tag{498}$$

For positive-order terms the derivative estimate is $\|h_i\| \le D\omega_i\Phi(C_i)$. Keeping only terms containing the insertion, using $o \in X_i$, and applying (497) and (498) at $z = 1$ gives the total bound

$$\sum_{\mathrm{ord}(i)\ge 1} e^{aC_i}\|h_i\| \le DA_{e,r}\,t\gamma(1-\gamma)^{-1}. \tag{499}$$

Order zero must be treated separately, because its single-insertion molecule is the principal term. If an order-zero molecule containing the insertion is not principal, it has at least two elements.

Indeed, a singleton containing the marked element consists exactly of that element. Thus its excess cardinality is at least one. Let v_i be its nonnegative marked-molecule majorant, with the restriction that excess cardinality is at least one. The mixed marked estimate needed here is

$$\sup_x \sum_{i:\, x\in X_i} v_i e^{(a+e)C_i} \le C_1.$$

Assume the order-zero derivative bound $\|h_i\| \le DB_* v_i \Phi(C_i)$ for a nonnegative derivative price B_*. The molecule-to-order-zero-cluster map is injective, so no multiplicity is lost in the sum. Applying the reserve inequality and the site estimate at o proves

$$\sum_{\substack{\mathrm{ord}(i)=0\\ i \text{ nonprincipal}}} e^{aC_i}\|h_i\| \le DB_* A_{e,r} C_1.$$

Together with (499), this gives

$$M \le DA_{e,r}\left(B_* C_1 + \frac{t\gamma}{1-\gamma}\right). \tag{500}$$

To obtain the reserved mass of Section 61, take a at least its reserved support exponent. The same calculation with a larger a continues to apply if the displayed estimates have that reserve.

In the actual insertion normalization, $D = m\exp(-\varepsilon C_{\mathrm{fine}})/g$, where $g > 0$ is the gain and $\varepsilon > 0$ is the fine support reserve. Therefore, defining

$$T_{\mathrm{tail}} = \frac{A_{e,r}}{g}\left(B_* C_1 + \frac{t\gamma}{1-\gamma}\right),$$

we have $M \le me^{-\varepsilon C_{\mathrm{fine}}} T_{\mathrm{tail}}$. This accounts for both the support decay and the normalization gain; they are not independent assumptions on M.

94.5 The strict power reserve

For completeness, the tail comparison used in choosing the coupling can be verified directly from the price exponents. Write $H(x) = \beta_{\mathrm{fine}}(x)$, with $H(x) \to \infty$. A function has upper exponent q if for every $v > q$, eventually $|f(x)| \le H(x)^v$. Products add upper exponents and sums with the same upper exponent retain it: use half the available exponent margin in each bound and absorb the fixed factor two into the remaining power of H. Fixed powers of a function with upper exponent zero still have upper exponent zero.

Let $q < 0$ be the connected-expansion exponent. The prices used above have the following exponents, with L, derivative orders and e fixed before x is taken large:

price	$A_{e,r}$	B_*	C_1	t	γ	g^{-1}
upper exponent	0	0	$3q/2$	q	$q/2$	$-q$.

Here $\gamma = H^{q/2}$ and $g = H^q$; hence eventually $0 \le \gamma \le 1/2$ and $(1-\gamma)^{-1} \le 2$. Both terms defining T_{tail} consequently have upper exponent $q/2$. Since $q/2 < q/4$, the definition gives, eventually,

$$T_{\mathrm{tail}} \le H^{q/4}, \qquad M \le me^{-\varepsilon C_{\mathrm{fine}}} H^{q/4}. \tag{501}$$

This is a strict exponent comparison. An upper exponent zero by itself would not justify paying a price from a fixed power of $\log H$.

Sections 62 and 63 prove the connected and mixed site estimates from their vertex prices; Section 91 proves the two gas regrouping identities under their exact flag and reserve hypotheses. The remaining analytic work is to construct the actual tilted block expansion, establish its vertex prices and localized derivative estimates, and purchase all these bounds and their stated upper exponents on the selected history. With those inputs, the differentiation, flagged cost, nonprincipal tail accounting and strict-power comparison above supply the corresponding inputs to the elementary projection estimate.

95 Paying for the elementary projection on the selected scales

We now verify the principal reserve and the numerical projection allowance used in Section 61. All quantities refer to the marked expansion constructed in Sections 92 and 93. In particular the regular logarithmic terms and the differentiated forced array are those of the same background transition. We retain the incoming sector and path estimates of that construction.

95.1 The singleton identity before evaluating the gas

For each real elementary insertion M, denote by $F_j^M(\theta, U)$ the actual tilted forced coefficient after the two regroupings of Section 92. The index j includes its fixed support and flag inventory. Define one common transition map on elementary insertions by

$$\mathcal{D}(M)_j(U) = \frac{d}{d\theta} F_j^M(\theta, U)\Big|_{\theta=0}.$$

The coefficientwise differentiability and domination were proved in Sections 93 and 94. The raw forced coefficient k_j in that expansion is this very derivative. Hence

$$k_j(U) = \mathcal{D}(M)_j(U) \tag{502}$$

for every original output address and every configuration. This fixes the transition map from the constructed coefficient array itself; it is not selected from its gas evaluation.

Let $\mathcal{P}$ denote the profile map, which sums coefficients with a common support and inventory. Its absolute convergence follows from the dominated forced price. For real insertions M_1, M_2 of the same type and majorant, and $v = L^d$, the raw complex forced array is $v(\mathcal{D}(M_1) + i\mathcal{D}(M_2))$. The local splitting of Section 58 produces a coefficient field Q and a response ψ with

$$\mathcal{P}(Q_{\text{forced}}) - \text{hook}(\psi) = v\{\mathcal{P}(\mathcal{D}(M_1)) + i\mathcal{P}(\mathcal{D}(M_2))\}. \tag{503}$$

Indeed splitting adds exactly $\text{hook}(\psi)$ to the raw profile, and the profile map is linear by absolute convergence. Subtracting that hook proves the identity. Its sign agrees with the compensating response in the ordinary part of the density.

The input profile is zero for an ordinary insertion. For a flagged insertion it is its inserted value at its original support and inventory, and zero at other addresses. Thus the input ordinary part and input profile in this transition are precisely those of $M_1 + iM_2$. Their fine cost is at most $2m$. Taking $M_2 = 0$ proves the claimed singleton relation for a real insertion, with the same available cost $2m$.

These identities persist under affordable linear combinations. For a countable family with coefficients z_i and $\sum_i |z_i| m_i < \infty$, the forced-price estimate bounds the sum of absolute output coefficients by a fixed multiple of this sum. Tonelli's theorem applies to the nonnegative majorants

on the product of insertion and output indices. One may therefore interchange those sums in the profile and obtain the linear combination of (503). The permutation covariance of the actual arrays also survives: differentiate their coefficientwise relabelling identity and then apply the same absolutely convergent profile map. This verifies the marked-profile requirement of Section 61 on the selected transition.

95.2 The principal cluster already has a paid majorant

For an ordinary insertion there is exactly one principal term: the order-zero cluster whose molecule consists of the insertion letter alone. Its support is the selected enlarged insertion support X_*, and its majorant is denoted by ω_*. The distinction between this term and the order-zero tail is the excess-cardinality condition in Section 94. For a flagged insertion this cluster cannot be regular, so there is no principal term.

Write $C_* = C_B(X_*)$, and use the positive prices t, v, u and γ of Section 93. The rooted logarithmic estimate at order zero, evaluated at the principal anchor, gives

$$\omega_* e^{(a+e)C_*} \le \sum_{i:\, o\in X_i} \omega_i e^{(a+e)C_B(X_i)} \le \frac{t}{1-\gamma} \le 2t \le 1.$$

The first inequality retains one nonnegative summand. The last uses $u = 10(t+v) \le 1/2$. In the combined expansion one may instead use the common bound $4v + 8t \le u$; either bound leaves the same unit reserve. No separate smallness assumption on this individual cluster is needed.

The local derivative estimate already proved there has amplitude $D\omega_*\Phi(X_*)$, where

$$D = \frac{m e^{-\varepsilon C_{B_f}(X)}}{g} \le \frac{m}{g}, \qquad \Phi(X_*) \le V e^{C_*/2}.$$

Here m is the fine reserved mass and $g > 0$ the normalization gain. At fixed derivative order p and spatial power r, one can take

$$V = \big[4(2B+1)\big]^{rp} C_{rp}, \qquad C_{rp} = \sup_{z\ge 0} (1+z)^{rp} e^{-z/2} < \infty.$$

Finiteness follows from polynomial growth versus exponential decay. If $a \ge b + 3$ and $e \ge 0$, where b is the outgoing support exponent, then

$$e^{(b+2)C_*} \|h_*\| \le DV\omega_* e^{(b+5/2)C_*} \le DV\omega_* e^{(a+e)C_*} \le Vm/g.$$

This proves the principal estimate in (248) for the actual localized term. If $C_* \ge B$, the same proof retains $e^{-B/2}$, because $a + e - (b + 5/2) \ge 1/2$.

For a complex insertion the real and imaginary parts have the same support, type and mass majorant. Their principal terms are at the same index. The triangle inequality therefore gives total principal mass at most $2|z|Vm/g$ after multiplication by z. For a flagged input both principal contributions are zero. This verifies the principal contribution used in the complex raw expansion, including the convention of pairing a real insertion with an identically zero imaginary part.

95.3 Combining the three bills

Fix a degree d and block ratio L. Let H_d be the local splitting factor of Section 61, and put

$$c_d = \mathrm{rate}(L)/H_d > 0.$$

We first pay the two small bills in (250). Put $H = \beta_f(x)$ and

$$P(x) = e^{(\log\log x)^2}, \qquad P_f = P(H), \qquad \kappa = e^{-P(x)^{1/5}/2}, \qquad s = 2e^{-\mu P_f^2/16}, \quad \mu > 0.$$

These are the actual square root of the outgoing forced allowance and the second forced-gas price. Eventually $e < x \le H$, so $1 \le P(x) \le P_f$. If $P_f \ge 64/\mu$, then

$$\tfrac{1}{2}P(x)^{1/5} \le \tfrac{1}{2}P_f \le \mu P_f^2/32, \qquad \kappa^{-1}s \le 2e^{-\mu P_f^2/32}.$$

The last bound has every negative upper exponent in H. Indeed, with $y = \log\log H$, the logarithm of its ratio to H^{-a} is at most $\log 2 - (\mu/32)e^{2y^2} + ae^y$, which tends to $-\infty$ for every fixed $a > 0$. Since $g^{-1} = H^{-q}$ with $q < 0$,

$$2L^d g^{-1}\kappa^{-1}s \longrightarrow 0.$$

The tail bound (501) likewise gives $2L^d H^{q/4} \to 0$. The debt and tail both contain the factor $me^{-\varepsilon C_{B_f}(X)} \le m$. Choose one coupling threshold after which the two preceding scalar factors are at most c_d. Multiplying by this nonnegative mass factor proves both small-bill inequalities uniformly over all insertions, including $m = 0$; no division by m is used.

The remaining local splitting condition is paid at the same time. Since $B = \lceil(\log x)^2\rceil$, the logarithm of $(2B+1)^4\kappa$ is bounded above, for large x, by

$$C + 8\log\log x - \tfrac{1}{2}e^{(\log\log x)^2/5},$$

which tends to $-\infty$. Hence $(2B+1)^4\kappa \le 1$ eventually. This establishes all three inequalities in (250), including the hit-width bound. The thresholds depend on the fixed stage and degree, but not on the finite volume or insertion.

Define

$$E_d(x) = 2L^d V(x)/g(x), \qquad Q_d(x) = \big(c_d + |E_d(x)|\big)H_d.$$

The principal bill is at most $|E_d(x)|m$. Hence both it and the tail bill are bounded by $e_0 = (c_d + |E_d(x)|)m$. The local splitting proof gives coefficient, remainder and response bills at most $H_d e_0 = Q_d(x)m$. Its density is exactly $L^d\mathcal{T}f$, and the fine singleton expansion costs at most $2m$. This uses the separate marked-profile identity of the expansion as well as its scalar density identity.

The quantity Q_d may grow as a power of the fine coupling. That is harmless for this last projection: its permitted allowance is

$$Q_d(x) \le e^{\sqrt{\beta_f(x)}}. \tag{504}$$

We verify this inequality without changing the selected block or insertion. The price $V(x)$ has upper exponent zero, since it is a fixed power of $1 + B(x)$ times a constant. The inverse gain $g(x)^{-1} = \beta_f(x)^{-q}$ has upper exponent $-q$. Thus Q_d has some finite upper exponent, with L, d and the stage profile fixed. For any function F of finite upper exponent v, choose $w > \max(v, 0)$. Eventually $|F(x)| \le \beta_f(x)^w$. Writing $y = \sqrt{\beta_f(x)} \to \infty$, we have

$$\beta_f(x)^w e^{-\sqrt{\beta_f(x)}} = y^{2w}e^{-y} \longrightarrow 0.$$

For example, $2w\log y - y \to -\infty$, since $\log y/y \to 0$. Consequently $F(x) \le e^{\sqrt{\beta_f(x)}}$ eventually. Applying this to Q_d proves (504). The threshold is uniform in the finite volume and the insertion because Q_d depends only on the displayed stage data.

95.4 The simultaneous multiple-insertion requirements

There are a few additional numerical inequalities on this same transition. We verify them explicitly, since their order of selection matters. Fix an occupancy bound $N \in \mathbb{N}$, a support exponent $b \geq 0$, and a positive suppression constant. Write

$$H = \beta_f(x), \qquad \eta = H^{-2}, \qquad \alpha = b + \tfrac{1}{2}, \qquad s(x) = 2(1+N)\eta(Ne^{\alpha}+1) + (1+N)\tau(x),$$

where τ is the forced price from the second exclusion estimate. Its Gaussian suppression gives any prescribed negative upper exponent, in particular -2; this is the price estimate used in Section 16. Thus s has upper exponent -2. Since $B(x)$ has upper exponent zero, so does every fixed polynomial in $B(x)$. It follows that eventually

$$\begin{gathered}(2B+1)^4 s(x) \leq \tfrac{1}{2}, \qquad e^{s(x)} - 1 \leq 1, \qquad \tau(x) \leq 1, \\ \alpha\big(2 \cdot 1024B + 1\big)^4 \leq \sqrt{H}.\end{gathered} \tag{505}$$

Indeed the first and third quantities tend to zero. For the second first arrange $s(x) \leq \log 2$. For the fourth compare its upper exponent zero with the strictly larger exponent $1/2$. The expression for s is the sum of the two ordinary copied contributions and the forced contribution; in particular the factor $1 + N$ is retained in each of the three terms.

The geometric floor required for the local transport is selected earlier. If $R_{j,i}$ is its fixed support-enlargement charge for a transition from stage j to stage i, choose the integer length floor

$$n_j \geq \max_{i \leq j} \left\lceil e^{R_{j,i}} \right\rceil .$$

Every block length $L_j \geq n_j$ then satisfies $e^{R_{j,i}} \leq L_j$ for each permitted transition. Only finitely many indices occur in this maximum. This requirement is independent of the coupling and is therefore included before selecting the lengths in Section 201.

After the lengths are fixed, at stage j there are finitely many permitted predecessor stages i and degrees $d \leq i$. Choose the maximum of their thresholds for (504), (505), and the tail and forced payments. Raise the common entry threshold to this maximum. Section 201 produces one realization paying all these requirements. Every later compatible realization paying enlarged thresholds retains them.

Thus the elementary projection allowance can be taken to be $e^{\sqrt{\beta_f}}$ simultaneously for every permitted degree at each stage, on the same nonlinear transition as the ordinary and multiple-insertion estimates. With the marked-profile identity, this supplies the elementary input in Section 170; its finite linear assembly then gives the terminal price there. The underlying sector and path estimates are still those of the construction, rather than consequences of this numerical selection.

96 Comparing the principal insertion with its Gaussian reference

The principal cluster is not itself a small remainder. Its leading part is a local Gaussian reference observable. We compare these two objects on one probability space, keeping the insertion fixed while differentiating the surrounding densities. This is necessary even when the insertion is only bounded and measurable.

96.1 The common reference integral

Fix a forest, its tags, a regular insertion of amplitude $\tau \geq 0$, and a healthy base configuration. Let $y \in \mathbb{R}^j$, $j \leq n$, parameterize an ordered local variation about that configuration, with base point

0. The read coordinates form a finite set of size N_0, including every colour of each read bond. Write $C_6(y), C_9(y)$ for the actual-sector and Gaussian-reference covariances on these coordinates, and $w_6(y,z), w_9(y,z)$ for the two compensated coordinate maps. At the base point

$$C_0 = C_6(0), \qquad w_6(0,z) = z.$$

Both covariances are positive in a neighborhood of zero. Let $J(y,z)$ be the common compensated Jacobian and put

$$R_a(y,z) = J(y,z)\frac{p_{C_a(y)}(w_a(y,z))}{p_{C_0}(z)}, \qquad a = 6, 9,$$

where p_C is the Gaussian density in (433). The reference probability measure is always γ_{C_0}.

Let $h(z)$ be the original masked insertion evaluated at the fixed base configuration and its original Gaussian coordinates. It is measurable and $|h(z)| \le \tau$. The exact two integrands are

$$R_6(y,z)\chi_6(y,z)h(z), \qquad R_9(y,z)\chi_9(y,z)h(z). \tag{506}$$

Here χ_6 is the sector cutoff and χ_9 the reference cutoff, both smooth with values in $[0,1]$. The first identity follows from the compensated change of variables of Section 85: on the support of its cutoff the new configuration agrees with the fixed one on the insertion's read support. Locality therefore identifies their insertion values. For the second identity, the reference cutoff implies that the first cutoff is one and that the fixed configuration is healthy on the insertion support. The mask can then be removed, and the same local agreement identifies the unmasked reference insertion with $h(z)$. Where a cutoff vanishes both sides of its identity are zero. Thus no derivative of h is taken in either formula, and no insertion-dependent change of Gaussian measure has been made.

We state the quantitative finite-dimensional data used below explicitly. They are uniform over all the forests, tags and local variations under consideration. There are $N, Q, S \ge 1$, $\delta \ge 0$, and a fixed $M > 0$ such that:

1. $N_0 \le N$ and $C_0 \le MI$;
2. the entries of C_6, C_9 have jets through order n bounded by Q at zero, their inverses at zero have entries bounded by Q, and the jets of $C_9 - C_6$ are bounded by δ;
3. for $\|z\|_\infty \le S$, the entries of w_6, w_9 and the scalar J have jets bounded by Q, and those of $w_9 - w_6$ are bounded by δ;
4. on a common neighborhood of zero both integrands in (506) vanish for $\|z\|_\infty > S$. The cutoff jets on this ball are bounded by a fixed polynomial in N, Q, S. For every z with $\|z\|_\infty < v$, both cutoffs are identically one on some neighborhood of zero.

Here a jet bound includes the value and the operator norms of derivatives of all orders at most n. The last neighborhood may depend on z; the common compact support neighborhood does not. The covariances, coordinate maps and Jacobian are smooth on the compact coordinate ball and a neighborhood of the base parameter. These hypotheses are the specific comparison data; they are not consequences of an abstract equality between the two gas profiles.

96.2 A quantitative density difference

There is a polynomial P_n with positive coefficients, depending only on n, such that, after increasing P_n if necessary and provided

$$\delta P_n(N, Q, S) \le 1, \tag{507}$$

the following bounds hold for $\|z\|_\infty \le S$:

$$\|R_6(\cdot,z)\|_{J_0^n} \le P_n(N,Q,S), \qquad \|R_9(\cdot,z)-R_6(\cdot,z)\|_{J_0^n} \le \delta P_n(N,Q,S). \tag{508}$$

We prove the polynomial form of these estimates rather than leave the dependence on the number of read coordinates implicit.

For matrices use the maximum entry norm. Multiplication costs at most N per contracted index. Differentiating $CC^{-1}=I$ gives, for a nonempty labelled derivative set I,

$$D_I(C^{-1}) = -C^{-1}\sum_{\varnothing\neq A\subseteq I}(D_A C)D_{I\setminus A}(C^{-1}).$$

Induction in $|I|\le n$ bounds all inverse jets by a polynomial in N,Q. The exact identity

$$C_9^{-1}-C_6^{-1} = C_9^{-1}(C_6-C_9)C_6^{-1}$$

and the Leibniz rule give a factor δ in every derivative of the inverse difference. The remaining factors have the polynomial bounds just obtained.

For the logarithmic determinant, its positive-order derivatives follow by repeatedly differentiating $D\log\det C = \operatorname{tr}(C^{-1}DC)$. Telescoping each product shows that the difference of these derivatives for C_9 and C_6 is δ times a polynomial. For the zeroth order join $C_6(0)$ to $C_9(0)$ by the straight positive segment. Since the endpoint inverse operator norms are at most NQ, both endpoints are at least $(NQ)^{-1}I$; the whole segment has the same lower bound. Hence

$$|\log\det C_9(0)-\log\det C_6(0)| \le \int_0^1 N\|C_s^{-1}\|\|C_9(0)-C_6(0)\|\,ds \le N^3Q\delta.$$

The two quadratic forms in the Gaussian exponent differ by

$$\begin{aligned} w_9^TC_9^{-1}w_9 - w_6^TC_6^{-1}w_6 &= (w_9-w_6)^TC_9^{-1}w_9 \\ &\quad + w_6^TC_9^{-1}(w_9-w_6) + w_6^T(C_9^{-1}-C_6^{-1})w_6. \end{aligned}$$

Every derivative therefore has the same δ factor.

Let $\ell_a = \log[p_{C_a(y)}(w_a(y,z))/p_{C_0}(z)]$. The preceding calculations bound all jets of $\ell_9-\ell_6$ by δ times a polynomial. All positive-order jets of ℓ_6 have polynomial bounds, and $\ell_6(0,z)=0$. The labelled-partition formula

$$D_Ie^\ell = e^\ell\sum_{\pi\in\mathrm{Part}(I)}\prod_{A\in\pi}D_A\ell$$

thus gives polynomial jets of e^{ℓ_6}. For $e^{\ell_9-\ell_6}-1$, the value is bounded by $e|\ell_9-\ell_6|$ under (507). Each positive-order partition has at least one derivative factor, so all its terms retain a factor δ. The other factors can be bounded by one times an enlarged polynomial, using (507). Finally

$$R_9-R_6 = Je^{\ell_6}(e^{\ell_9-\ell_6}-1)$$

and the Leibniz rule prove (508). All operations involve at most n derivatives, so the number and degree of polynomial factors are fixed, independently of N.

96.3 The good-coordinate region and the Gaussian tail

On $\|z\|_\infty < v$, the cutoff jets equal those of the constant one. The difference of the two integrands consequently has jet norm at most $\tau\delta P_n(N,Q,S)$. On the rest of the compact ball, the product rule and (508) bound its jets by $\tau P_n(N,Q,S)$, after enlarging the polynomial once more. Outside that ball the integrands vanish on a parameter neighborhood, so their jets vanish too.

The reference Gaussian obeys the dimension-explicit tail estimate

$$\gamma_{C_0}\{\|z\|_\infty \ge v\} \le 2Ne^{-v+M/2}. \tag{509}$$

For a single coordinate, Markov's inequality applied to e^{z_i} and e^{-z_i} gives $\mathbb{P}(|z_i| \ge v) \le 2e^{-v+(C_0)_{ii}/2}$. Sum over the at most N coordinates and use $(C_0)_{ii} \le M$. No independence of these coordinates is required.

Differentiation under the integral is justified even though h need not be continuous. On a fixed compact coordinate ball the smooth density-cutoff factors and all their derivatives through order n are bounded on a smaller parameter neighborhood. Multiplication by the fixed measurable h supplies an integrable constant bound. Difference quotients are dominated using the fundamental theorem of calculus along a short parameter segment. Apply dominated convergence, successively through order n, as in Section 85. Integration of the two pointwise bounds and (509) gives

$$\left\| \int (R_6\chi_6 - R_9\chi_9) h \, d\gamma_{C_0} \right\|_{J_0^n} \le \tau P_n(N,Q,S)\big(\delta + 2Ne^{-v+M/2}\big). \tag{510}$$

Finite forest and tag averages preserve this estimate, because their weights are nonnegative and sum to one. The insertion and the reference transition remain paired before taking these averages.

96.4 How the comparison is paid

Write $\beta_f(x)$ for the fine inverse coupling, and use the upper-exponent convention of Section 200. Suppose N, Q, S have upper exponent zero, δ has upper exponent $-1/2$, and e^{-v} has every negative upper exponent. Then the multiplier of τ in (510) has upper exponent $-1/2$. Indeed a fixed polynomial in exponent-zero quantities still has upper exponent zero. Its product with δ has exponent $-1/2$, whereas its product with e^{-v} can be assigned any negative exponent. A finite sum preserves the larger exponent. The smallness requirement (507) therefore holds beyond a finite post-length threshold.

Here are the exact scales that produce the comparison data in the construction. The local support diameter is at most $D_x = B_x(4B_x + 2) + 2B_x$, and its read dimension is at most

$$N(x) = (2D_x + 1)^4 L^4 \, 4 \dim \mathfrak{g}.$$

This has upper exponent zero for fixed L. The local direction bound is $2\theta_f(x)(1 + D_x)^p$, of upper exponent $-1/2$. Let $J_0(x)$ bound the unscaled displacement of the two means, and $V_0(x)$ the covariance difference, including their jets. The construction requires the bounds

$$J_0 : \ -1, \qquad V_0 : \ -1/2, \tag{511}$$

where the numbers denote upper exponents. Here β is the actual fine inverse coupling of the incoming representation, within its prescribed window about $\beta_f(x)$. Gaussian coordinates divide the mean displacement by $g = \beta^{-1/2}$, so a common jet difference bound is

$$\delta(x) = |V_0(x)| + g^{-1}|J_0(x)|.$$

Since g^{-1} has upper exponent $1/2$, this has exponent $-1/2$. Absolute mean and coordinate bounds, covariance inverse bounds, and the fixed-order Jacobian bounds are exponent zero after this rescaling. The support radius in Gaussian coordinates is bounded by $g^{-1}(5a\theta_f + |c|\theta_f)$, also exponent zero. The product cutoffs have polynomial fixed-order jet bounds: in a derivative of order at most n, at most n factors are differentiated; the others have magnitude at most one. There are at most N^n choices of these factors. The one-factor derivatives are bounded by the fixed smooth cutoff and its rescaled aperture.

Choose the good-coordinate radius

$$v(x) = \frac{\varepsilon}{128(\dim \mathfrak{g} + 1)}\, \mathcal{M}_f(x),$$

where $\mathcal{M}_f$ is the fine monitor. The monitor estimate gives every negative upper exponent for e^{-v}. The construction's aperture comparison is $\theta_f/g \geq \mathcal{M}_f/2$. Eventually $\delta \leq \varepsilon \mathcal{M}_f/[128(\dim \mathfrak{g} + 1)]$; hence

$$g \dim \mathfrak{g}\,(v + \delta) < \varepsilon\theta_f/8.$$

Each reference coordinate then lies strictly inside its cutoff plateau when $\|z\|_\infty < v$. Continuity gives the required parameter neighborhood; the cutoff implication in the common integral gives the other plateau as well. This proves the tail and numerical purchases directly, once the primitive mean and covariance comparison (511) and their absolute jet bounds have been supplied. Those primitive estimates concern the actual sector and reference covariances; the value-only remote mean bound does not by itself prove them.

97 The mean and covariance jets in the principal comparison

We prove the extra smallness used in (511). The two means need not agree away from the flat configuration. At the flat configuration their values vanish and their first derivatives differ only by the local inverse error. This gives one more small factor for their difference than for the covariance difference.

97.1 Uniform derivatives of the two local kernels

We first record why the kernel bounds are uniform in the period. All row estimates in this subsection use the maximum of weighted row and column sums from Section 74. For a fixed derivative order n, suppose a family of precisions $A(b)$ has uniformly positive lower bound $r > 0$, uniform exponential row bound, and derivatives through order n with weighted row bounds at most $C(x)$, where C has upper exponent zero. The logarithmic fields b range over a ball of radius $v(x)$, with upper exponent $-1/2$. These hypotheses hold for the unmasked local kernels under the fixed path and positive precision bounds used in Section 80.

The inverse derivative formula, in multilinear notation, is

$$D_I A^{-1} = \sum_{(I_1,\dots,I_k)} (-1)^k A^{-1}(D_{I_1}A)A^{-1}\cdots(D_{I_k}A)A^{-1}. \tag{512}$$

The sum is over ordered partitions of the labelled set I into nonempty blocks. Differentiate $AA^{-1} = I$ and induct on $|I|$ to prove this formula: the first differentiated block is the nonempty subset falling on A, and the induction partitions the remaining labels. Submultiplicativity and the inverse decay estimate of Section 74 bound each term by a fixed power of $C(x)$ and of the inverse row bound. The number of terms depends only on n. Thus all inverse jets have upper exponent zero.

The same argument applies to every principal window, with a small reduction of the exponential weight when the windows vary. Indeed, at some fixed weight $\gamma > 0$ the ordered-partition formula bounds each inverse derivative on every window by K_n. An assembled column is one such inverse column extended by zero; thus each of its entries is at most $K_n e^{-\gamma \operatorname{dist}(a_i, a_j)}$. At weight $\gamma/2$, both row and column sums are bounded by

$$K_n m \sum_{q \geq 0} (2q+1)^4 e^{-\gamma q/2},$$

where m bounds the number of coordinates at an anchor. This bound does not require two columns to use the same window. All windows and coordinate projections remain fixed during differentiation. Symmetrizing preserves the estimate.

On the reference side the coupling is the matrix of the unmasked curvature and predictor derivatives, restricted by the fixed keep test to the active coordinates. Denote it by $D(b)$; its precision is exactly $D(b)^* D(b)$. Away from zero this coupling differs from the completed-sector coupling: it uses the actual derivatives of curved plaquettes and of the predictor, whereas the completed sector transports their flat derivatives. We bound the reference derivatives separately. A derivative of order j of a plaquette derivative is a derivative of order $j+1$ of its finite product of link matrices; the product rule bounds it by the same finite path-length moments, now through order $j+1$. The predictor contribution is treated with its fixed-ball derivative bound through that order. Purchase these bounds through $n+1$ when requesting n coupling derivatives. All coefficients are bounded by the absolute path sums already used in Section 80. Summing their exponentially weighted rows and columns, and then applying the fixed keep and active projections, gives an exponent-zero bound for each reference coupling derivative. The product rule gives the precision derivative bound, with a finite sum over subsets of the derivative labels. The flat identity proved below identifies $D(0)^* D(0)$ with the active principal block of the completed sector precision; hence Section 73 gives its lower bound r. Its perturbation on the logarithmic ball is bounded by $C(x)v(x)$, so it remains at least $r/2$ eventually. Thus the inverse argument applies to the reference matrix as well as the completed one.

The reference panels have a different geometric size from the sector windows. For $B \geq 120$ they contain every active coordinate whose parent anchor is within $\lfloor B/60 \rfloor$ of the column anchor. Each endpoint of a bond lies within one coarse step of its parent anchor. The triangle inequality therefore bounds the separation of any two such endpoints by $\lfloor B/60 \rfloor + 2$; the reference middle-panel criterion follows from $20\lfloor B/60 \rfloor + 80 \leq B$. These are the two panel inclusions required here. The inverse-error proof uses only this inner radius, not an equality of the two window systems. Accordingly the reference error costs $e^{-c\lfloor B/60 \rfloor}$ in place of e^{-cB}. Both have every negative upper exponent.

For clarity, these derivatives are taken with the forest, tags, read windows and sector indicators fixed. The unmasked matrix entries are products of path derivatives and parallel transports. Their row costs are finite sums of the absolute path prices multiplied by fixed powers of path length. The fixed-ball predictor bounds in Section 80 make those costs exponent zero. On the shrinking ball, the change from the flat precision is at most $C(x)v(x)$ in operator norm, by the fundamental theorem of calculus. Eventually this is less than $r/2$, so the same inverse estimates apply throughout the ball. This proves the required neighborhood of positivity without changing any sector indicator.

Both affine sources are products of these kernel rows, their coupling matrices and the unmasked residual vector. Their derivatives therefore have exponent-zero bounds as well. The residual and its derivatives have the fixed-ball bounds (401). A row acting on a bounded vector uses its absolute row sum, not the total number of lattice coordinates. The preceding product estimates are consequently uniform over all finite periods and all allowed forests and tags.

97.2 The exact comparison at the flat configuration

Let A denote the full flat sector precision, padded by the identity on inactive coordinates, and let U denote its active coordinate set. At a flat field all parallel transports are identities. Differentiating a plaquette gives its signed exterior difference, and differentiating the predictor noise gives the flat coarse response with the same sign in both constructions. The normalization of the signed average is the same $|\mathcal{S}|^{-1/2}$ in both matrices. Evaluating on each colour coordinate vector therefore identifies the restricted sector coupling with the reference coupling. Taking its Gram matrix gives the exact flat identity

$$A_{UU} = A_{\mathrm{ref}}. \tag{513}$$

The padding contributes zero on U. Rows discarded by the reference keep test already vanish in this coupling, so their removal does not change the product.

Since inactive rows and columns of A are identity rows and columns, A is the direct sum of A_{UU} and an identity matrix. Hence

$$(A^{-1})_{UU} = A_{UU}^{-1}.$$

This conclusion uses the actual direct-sum property; it would not follow merely by compressing an arbitrary inverse. Let P_6 be the sector window inverse and P_9 the reference window inverse at the flat point. The window-error calculation of Section 74, applied to their respective windows, gives

$$\|(P_6)_{UU} - P_9\|_0 \le t(x), \tag{514}$$

where t has every negative upper exponent. Indeed each error is bounded by its fixed inverse-decay constant times $e^{-c\lfloor B_x/60\rfloor}$, and the cutoff radius B_x dominates every fixed multiple of $\log \beta_f(x)$. Taking the sum of the two errors preserves this property. Symmetrizing (514) gives the same bound for the flat covariance difference.

Write the actual and reference means in logarithmic coordinates as

$$m_6(b) = -P_6(b)^* s_6(b), \qquad m_9(b) = -P_9(b)^* s_9(b).$$

On the actual side the formula is restricted to U after the matrix-vector product. Each source is its coupling transpose applied to the residual. The residuals vanish at $b = 0$: all curvature strokes are zero, the predictor is the identity, and all coarse guides are linear in zero strokes. Thus both sources and both means vanish there.

The source derivatives agree after restriction:

$$(Ds_6(0)h)|_U = Ds_9(0)h. \tag{515}$$

To see this, write $s = M^* a$. At zero, $a(0) = 0$, so its derivative is $M(0)^* Da(0)h$ and the derivative of M contributes zero. The two residual formulas are the same unmasked curvature and noise strokes, after restricting the kept rows. On omitted rows the coupling is zero, as in (513). This proves (515). The actual source derivative also vanishes outside U, because the corresponding coupling columns are zero.

Differentiating the mean products now gives the exact identity

$$D(m_9 - m_6)(0)h = \big((P_6(0))_{UU} - P_9(0)\big)^* Ds_9(0)h. \tag{516}$$

Derivatives of P_a multiply the zero source and disappear. By the row bound and the exponent-zero source derivative bound,

$$\|D(m_9 - m_6)(0)\| \le t(x)C(x).$$

This proves a first-derivative comparison, not merely an equality of quadratic forms.

97.3 Transferring the estimates along a local variation

We give the elementary Taylor argument used for this transfer. Let $f : E \to F$ be C^n, $n \ge 2$, on a neighborhood of the closed ball of radius $v \le 1$, and suppose

$$f(0) = 0, \qquad \|Df(0)\| \le Ct, \qquad \|D^k f(b)\| \le C \quad (2 \le k \le n,\ \|b\| \le v).$$

Let u be a C^n chart about the parameter base point with $\|u(0)\| \le v$ and $\|D^k u(0)\| \le v^k$ for $1 \le k \le n$. Taylor's formula along the segment from 0 to b gives

$$\|f(b)\| \le Ct\, v + Cv^2/2, \qquad \|Df(b)\| \le C(t + v).$$

For a derivative of $f \circ u$ of order $k \ge 1$, a partition with r blocks contributes $D^r f(u(0))$ applied to the derivatives of u of those block sizes. Their product costs v^k. When $r = 1$ this is at most $C(t + v)v$; when $r \ge 2$ it is at most Cv^2, since $k \ge 2$ and $v \le 1$. There are at most $n!2^n$ partitions for any $k \le n$. Together with the value estimate this proves

$$\|f \circ u\|_{J_0^n} \le n!2^n\{C(t + v)v + Cv^2\}. \tag{517}$$

Similarly, if $g(0)$ has norm at most t and all its positive-order derivatives on the ball are bounded by $2C$, the mean-value theorem and the same partition formula give

$$\|g \circ u\|_{J_0^n} \le n!2^n(t + 2Cv). \tag{518}$$

Apply (517) to $f = m_9 - m_6$, using (516). Apply (518) to each entry of the covariance difference, using (514). The common logarithmic chart from Sections 77 and 80 supplies the required u. For a support anchored at o, its radius is at most D_x. The chart's derivative scale can be chosen as

$$v(x) = C_n\, 2\theta_f(x)(1 + \mathrm{reach}(x))^p(1 + D_x)^p.$$

This follows by applying the weighted direction bound at o to every read site and using the uniform derivatives of the logarithm on the fixed chart. An ordered product of k local variations contributes the kth power of that scale. It has upper exponent $-1/2$ and is eventually at most one.

97.4 Returning from the local logarithm to the physical field

The preceding comparison is made with blank local sector data. We must identify its readings with the physical ones; global agreement of the fields or of their sector indicators is not available. Let Y be the allocated coarse support, disjoint from the sector's seed set. For each coordinate read by the insertion, with parent anchor p, its allocation includes

$$\mathrm{halo}_{R+4}\{p\} \subset Y, \qquad R = 45(1 + B).$$

The field is healthy on Y, and the period and chart bounds ensure that this read region has the local gauge chart of Section 77. Fix that gauge g once, at the base configuration. For every nearby configuration W define

$$b(W)_e = \begin{cases} \log((g \cdot W)_e), & \mathrm{dist}(p, e_-) \le R, \\ 0, & \mathrm{dist}(p, e_-) > R. \end{cases}$$

The finite set of logarithms stays in its common chart on a neighborhood of the base point. Thus b is smooth there, and $\exp b(W) = g \cdot W$ on the entire read halo. No assertion is made about

plaquettes crossing the boundary of its zero extension. The derivative estimates used above hold for precisely this extension: the entries outside the halo have zero derivatives.

Here is why this local agreement suffices. A seed in the read halo would belong to Y, contrary to seed separation. The local indicator tests therefore agree with the blank tests there. The forest, signed choices, widths and fixed path data are the same. Every entry used by a sector window, and every source entry used in its local mean, reads at most $17B$ from the column anchor. This fits inside R. In the reference construction the middle and end panels and their field queries are contained in the same allocated halo, with the extra four sites allowing for the bond endpoints. For the reference construction this containment can be checked numerically. The panel radius is at most $\lfloor B/20 \rfloor$. If E denotes the extent of a single path query, the coupling read radius is $E+10B+4$, and the covariance read radius is this quantity plus $2\lfloor B/20 \rfloor$. The mean adds the residual read radius, the coupling range, and one more panel radius; write the resulting radius as S. The fixed range inequalities give $S + E \leq 36B$. Even an allowed displacement of the query anchor by $7(1 + B)$ therefore costs at most $7(1 + B) + 36B \leq 45(1 + B) = R$. These inequalities cover the whole local product, including source entries in a mean row. Consequently their restricted coupling and precision entries, and their source entries, agree after the local gauge change. Equal principal matrices have equal inverses; their restricted matrix products also agree. This proves equality of the mean readings and window covariance blocks. It uses the local tests only, so sector indicators outside the halo can be arbitrary.

More explicitly, write O_j for the colour adjoint of the fixed gauge at the parent of coordinate j. For a mean read at k, and a covariance block between k and l, the preceding identification gives

$$O_k m_a^{\mathrm{phys}}(W)_k = m_a^{\mathrm{loc}}(b(W))_k, \qquad C_a^{\mathrm{phys}}(W)_{kl} = O_k^* C_a^{\mathrm{loc}}(b(W))_{kl} O_l, \qquad a = 6, 9.$$

For the sector covariance its locality initially identifies the column at k; symmetry gives the displayed row identity. The reference identity reads that row directly. Thus the same chart at k compares any requested block, including l outside the insertion's support. The mean identity on the reference side holds for every fixed fine field and auxiliary parameter: its local affine source is the same one just identified. All these equalities hold on a neighborhood, so their derivatives through the required order agree as well.

A colour adjoint is an isometry, so returning from the local mean does not change its norm. Each entry of O_j has magnitude at most one, so the covariance entry bound acquires at most $(\dim \mathfrak{g})^2$. The gauge is fixed while differentiating. Consequently the actual unscaled mean difference has upper exponent -1, whereas the covariance difference has upper exponent $-1/2$, exactly as in (511).

The argument uses the same local histories, flags and reference windows on both sides. It does not identify the two precisions on nonflat fields. The hypotheses inherited from the earlier kernel and chart sections are the common path bounds, their local smooth formulas, and the positive precision estimates; these remain part of the selected construction throughout.

98 Attenuation under the principal Gaussian reference

We apply the polynomial cancellation just proved to the reference integral in Section 96. The important feature is the order of constants: the coefficient of L^{-d-1} is chosen before L. Bounds for the nonlinear chart and the Gaussian can be chosen after L, because their additional costs tend to zero as the entrance coupling increases. Throughout, $n \geq \max\{2, d + 1\}$, the insertion has the fine regularity through order n used below, and its polynomial moments through degree d vanish. The coarse weight exponent does not exceed the fine one.

98.1 The first derivative of the reference mean

Let u be a coarse logarithmic field and $v_F(u)$ the stationary relative field in tree coordinates from Section 75. At the identity the full affine row satisfies

$$D_\infty v_F(u) + r_0'(u) = 0.$$

This is (362), with the same signed normalization on the two terms. The reference construction uses the cut row D_R and its cut residual derivative r_R'. The exponential path tail gives, uniformly in the period,

$$\|D_R v_F(u) + r_R'(u)\|_\infty \le C(1+R)^a e^{-\sigma R}\|u\|_\infty. \tag{519}$$

To see this, subtract the full identity. The row difference acts on $v_F(u)$, whose sup norm is at most $C_{\mathrm{resp}}\|u\|_\infty$. The guide difference acts on u. Each is a sum of omitted path rows; their weighted absolute tails give the displayed bound, including the finite path-length power. All inactive coordinates are omitted on both sides, and tree coordinates of v_F are zero.

Let $A_R = D_R^* D_R$, let P_R be the reference principal-window inverse, and put $E_R = A_R P_R - I$. The reference mean in active coordinates is $m(u) = -P_R(u)^* D_R(u)^* r_R(u)$. Since $r_R(0) = 0$, its derivative at zero is

$$Dm(0)u = -P_R(0)^* D_R(0)^* r_R'(u).$$

The elementary normal-equation identity therefore gives

$$Dm(0)u - v_F(u) = E_R(0)^* v_F(u) - P_R(0)^* D_R(0)^* (D_R(0) v_F(u) + r_R'(u)).$$

The window and row bounds of Section 97, together with (519), imply

$$\|Dm(0) - v_F\|_{\infty\to\infty} \le C\{e^{-c\lfloor B/60\rfloor} + (1+R)^a e^{-\sigma R}\}. \tag{520}$$

Here and below the restriction to active counting coordinates and the fixed colour reassembly contribute only fixed factors. Because $R = \lfloor B/100000\rfloor$ and B grows quadratically in the logarithm of the coupling, the right side is eventually at most the coarse small-field threshold θ_c. This proves the required first-order approximation for the selected windowed mean, without replacing it by the exact inverse mean.

98.2 The nonlinear local display

The tree-straightening potential is a linear site field $\phi(u)$. Its discrete gradient corrects the tree response to the actual installed A_0 response, denoted $\mathcal{A}u$:

$$\phi(u)(e_-) - \phi(u)(e_+) + v_F(u)_e + \overline{u}_e = (\mathcal{A}u)_e.$$

This follows by summing the tree potential along a bond: the interior terms telescope, leaving its endpoint difference. The root correction is included in the same potential, as in the response bound of Section 75.

For noise w in fine link logarithmic units define the local logarithmic display

$$\Phi(u,w)_e = \log\left(e^{\phi(u)(e_-)} e^{\widetilde{m}(u)_e + w_e} e^{\overline{u}_e} e^{-\phi(u)(e_+)}\right),$$

where $\widetilde{m}$ is the colour-reassembled mean, zero on tree links. The logarithm is taken on one common fixed ball. Its existence and uniform derivatives follow from the fixed-dimensional group chart, the mean row bounds, the tree-potential bound and the replica bound. Shrink the input ball so that

the four entries of the logarithm's argument lie in that chart. The derivative of $\log(e^a e^b e^c e^{-d})$ at zero is $a+b+c-d$. Consequently, with $T(u,w)=\mathcal{A}u+w$, there are $Q\ge 1$ and $r>0$, uniform over all periods, forests and tags, such that

$$\Phi(0)=0,\quad \|D\Phi(0)-T\|\le Q\theta_c,\qquad \|D^j\Phi(z)\|\le Q\quad (j\le n,\ \|z\|\le r). \tag{521}$$

The constants Q,r may depend on the already chosen block factor. The tangent error is (520); the other bounds follow by the finite product and composition rules and the separate reference inverse estimates already proved.

Let $a(y)=(u(y),w)$, where $u(y)$ is the fixed-gauge local logarithm along the ordered coarse variation. Write s for its common size and derivative scale, so that

$$\|a(0)\|\le s,\qquad \|D^j a(0)\|\le s^j\quad (1\le j\le n),\qquad \theta_c\le s\le \min\{1,r\}.$$

The noise is controlled separately from the health condition. In counting Gaussian coordinates z, it is $w=g\,\mathrm{tape}(z)$, up to the fixed colour isometries. We use a compactly supported C^n cutoff Γ with $|\Gamma|\le 1$ and

$$z\in\operatorname{supp}\Gamma\quad\Longrightarrow\quad g\|\mathrm{tape}(z)\|\le \varepsilon\theta_f\le s.$$

The chart bound gives $s=C\theta_c$ times a fixed polynomial in its read radius, with the local-gauge constant C purchased before L. The nonlinear display constants Q,r are chosen after L. Thus the asserted bound for a holds on the cutoff support, including its boundary. Only the coarse component varies. Integrating the first two derivative bounds in (521) gives

$$\|\Phi(z)-Tz\|\le Q(\theta_c+\|z\|)\|z\|,\qquad \|D\Phi(z)-T\|\le Q(\theta_c+\|z\|).$$

Apply the partition chain rule to $(\Phi-T)\circ a$. A term with one block costs at most $2Qs\cdot s^j\le 2Qs^2$. A term with at least two blocks costs at most $Qs^j\le Qs^2$; linear maps have zero derivatives of those orders. The value is bounded by $2Qs^2$ too. Thus for a constant c_n depending only on the derivative order,

$$\|D^j(\Phi\circ a)(0)-\mathcal{A}D^j u(0)\|\le c_nQs^2+\|w\|,\qquad j\le n. \tag{522}$$

The noise term is needed only at $j=0$ but is harmless at all orders.

Each $\mathcal{A}D^j u(0)$ has the graded decomposition of Section 60, with cost M chosen before L. Put the error in (522) entirely into its last piece. Since the fine direction width is at least θ_f, all the display derivatives have a graded decomposition at scale L^{-1} and cost

$$M+\frac{c_nQs^2+\|w\|}{\theta_f}L^{d+1}. \tag{523}$$

This operation requires no polynomial property of the error.

98.3 The common healthy segment

For a short fine support, meaning that its cover number q obeys $q^3<L$, the whole inspected support lies in the read halo of one coarse chart. Seed separation, identical local flags, and the mean-locality argument of Section 97 identify the displayed configuration with the physical Gaussian-reference configuration on that support, after two gauge transformations. One gauge is the fixed coarse chart gauge; the second is the exponential of $\phi(u)$. Gauge invariance of the insertion removes both. The equality holds as a germ in the coarse variation, so derivatives of the insertion agree. Neither the frozen fine field nor the Gaussian coordinates are resampled in this identity.

The cutoff restricts to configurations healthy on X. We also need health on the radial segment of the display. Here is the small elementary estimate that supplies it. If a plaquette has four logarithmic bonds of norm at most $h \le 1$, let $K(t)$ be the matrix chord $K(t) = (\mathrm{mat}(\mathrm{plaquette}(t)) - I)/c_G$, where c_G is the fixed colour normalization and the norm is the ambient matrix norm of the health test. The group product and this chord map have bounded second derivatives on this fixed ball, so

$$\|K(t) - tK(1)\| \le C_G t(1-t)h^2, \qquad 0 \le t \le 1.$$

Indeed subtract the affine interpolation of K at its endpoints and integrate its second derivative against the Dirichlet Green kernel on $[0,1]$; the kernel integral is $t(1-t)/2$. All four bonds in the touched-plaquette halo are included. The chord at zero is zero. Hence, if the endpoint is healthy and $C_G h^2 \le \theta_f$,

$$\|K(t)\| \le t\theta_f + t(1-t)\theta_f \le \theta_f.$$

Take $h = Qs$. The two required smallness conditions $Qs \le 1$ and $C_G(Qs)^2 \le \theta_f$ hold eventually: s is θ_c times a fixed polynomial in the read radius, and θ_c^2/θ_f has negative upper exponent. The direction width bound follows likewise from $\theta_f(1 + 6(1+B)L)^p \le 1$. These purchases occur after L. We can therefore apply the displaced cancellation bound to every partition in the derivatives of $f \circ \exp \circ \Phi \circ a$. It gives the fine-insertion price

$$C_{n,d}\tau L^{-d-1}\left(M + \frac{c_n Q s^2 + \|w\|}{\theta_f} L^{d+1}\right)^n. \tag{524}$$

The finite number of partitions is included in $C_{n,d}$.

98.4 Separate local gauges for a winding support

For a general support X, one common logarithm need not exist. We instead base the insertion chart at the actual received configuration W_0. Keep the frozen fine field, the Gaussian coordinates, and every localization tag fixed. For each tested fine bond i, choose the coarse tree gauge g_i on the read halo centered at its coarse parent. The local mean and covariance agreement proved earlier identifies this local calculation with the same received configuration. The gauges may differ with i.

Let $b_i(y)$ be the coarse logarithm in this gauge along the coarse variation, and let w_i be the adjoint transform of the same noise. Use the unstraightened tree display

$$S_i(y) = e^{\widetilde{m}(b_i(y))_i + (w_i)_i} e^{\overline{b_i(y)}_i}.$$

Its linear response is the tree response $\mathcal{W}b = v_F(b) + \bar{b}$, whose weighted bound is M. Take $\log(S_i(y)S_i(0)^{-1})$ and return it by the fixed adjoint g_i^{-1} at the initial endpoint. The variable straightening potential $\phi(b_i(y))$ is absent from this relative display. Denote the result by $Z_i(y)$; set it to zero on untested bonds. The relative logarithm is defined near $y = 0$ because both arguments lie in the fixed chart. It satisfies

$$Z(0) = 0, \qquad W(y)_i = \exp(Z_i(y))(W_0)_i \quad \text{on every tested bond.}$$

This is an identity of germs: the relative exponential cancels the base display in that bond's own gauge. Finitely many tested bonds admit one common neighborhood. Local dependence of f then gives $f(W(y)) = f(\exp(Z(y))W_0)$ there. No compatibility between different g_i is needed, since Z is an ordinary fine bond field in the original coordinates. In particular this argument does not require a logarithm of W_0 around a noncontractible loop.

Here is the bound on this field. Write $\omega(i) = \theta_f(1 + \operatorname{dist}(i_-, o))^{p_f}$ for the fine direction width. Let $C \geq 1$ be the fixed local gauge response constant, $M \geq 1$ the weighted tree-response constant just specified, and put

$$H_X = (2 + \operatorname{breadth}(X))^{np_c}, \qquad A_X = 2MC + \frac{(c_n Q + M)s^2 H_X}{\theta_f}.$$

The first derivative of b_i obeys the coarse weighted direction bound with cost C; the linear response estimate and $\theta_c \leq 2\theta_f$, $p_c \leq p_f$, therefore give $2MC\omega(i)$. For derivatives of order $j \geq 2$, the local chart bound gives

$$\|D^j b_i(0)\| \leq s^j(1 + \operatorname{dist}(\operatorname{parent}(i_-), \operatorname{parent}(o)))^{jp_c} \leq s^2 H_X.$$

The last inequality uses $s \leq 1$, $j \leq n$, and the diameter bound for a tested bond relative to $o \in X$. The linear response on this bounded field costs at most $Ms^2 H_X$.

The relative display on two copies of its inputs vanishes at zero; its derivative there is the linear response of their difference, up to $Q\theta_c$. Its first derivative away from zero differs by at most a constant times Qs, and derivatives of higher order are bounded by Q. The partition chain rule, exactly as for (522), bounds the remaining terms at order $j \geq 1$ by $c_n Q s^2 H_X$. Adjoint return preserves the norm. Since $\omega(i) \geq \theta_f$, these estimates prove

$$\|D^j Z_i(0)\| \leq A_X \omega(i), \qquad 1 \leq j \leq n.$$

The cutoff and the local health estimate make W_0 healthy. Apply the ordinary fine regularity bound to $f(\exp(Z)W_0)$ at $Z = 0$. For each partition its argument fields have the displayed fine widths, so that the bound is at most $C_n \tau A_X^n$ after summing partitions. The zeroth derivative is bounded by τ. This proves a general-support bound whose leading constant $C_n(2MC)^n$ is chosen before L. Every dependence on the later chart constant Q is accompanied by s^2/θ_f, which can be made small after L is fixed. There is no sum of separate insertion derivatives over bonds: the entire field Z is inserted into the same regularity estimate.

98.5 Integrating the density derivatives

Use the same finite Gaussian coordinates for all derivatives, with its positive covariance and the frozen fine field fixed. The moment and density differentiation arguments of Sections 85 and 86 apply to this reference as well: its precision floor, inverse jets and local affine source were proved in Section 97. The cutoff is held fixed in these coordinates. Only on its support is the unmasked regular insertion evaluated; the health conclusion in the preceding paragraph applies there.

We detail the dependence of the bound on the support. Let q be its fine cover number and let $H = (r_0 + 48)(1 + B)$, where r_0 is the fixed allocation radius. The number of read coordinates and all polynomial weights on this set are bounded by fixed powers of $H(1 + q)$, with constants allowed to depend on L. Differentiating a Gaussian density inserts sums of products of inverse-covariance entries and at most quadratic coordinate factors. At order at most n the result is a polynomial in the Gaussian coordinates of degree at most $2n$. The derivative bounds and the product rule therefore bound the price in (524), including the density factors, by

$$C_0 \tau L^{-d-1}\{1 + \rho^{2n}(1 + \|z\|)^{4n}\}. \tag{525}$$

Here C_0 depends on the pre-L graded-field cost and the regularity conversion constants, while

$$0 \leq \rho \leq K_L(\theta_f + g/\theta_f)\{H(1 + q)\}^{a_n}.$$

To check this factor, insert $w = gz$ into (523). The nonlinear remainder s^2/θ_f is bounded by θ_f times a polynomial in B because the two thresholds have bounded ratio. The noise contributes $(g/\theta_f)\|z\|$. Every positive-order density or covariance variation has at least one coarse direction factor, which is at most θ_c times such a polynomial; enlarge K_L and a_n to include it. Separate the fixed term M from these errors, and use $(a+b)^{2n} \le (2a)^{2n}(1+b^{2n})$ for $a \ge 1$, $b \ge 0$. This gives (525). For a general support, use the separate-gauge bound just proved, with its leading constant $C_n(2MC)^n$ and error $(c_nQ+M)s^2H_X/\theta_f$. Combining it with the same density factors gives $C_1\tau$ in place of $C_0\tau L^{-d-1}$, with possibly larger fixed exponents. Its pre-L constants are the ordinary local regularity and linear-response constants.

The Gaussian moment of $(1+\|z\|)^{4n}$ is bounded by $K'_L\{H(1+q)\}^{b_n}$. This follows by expanding the norm bound into coordinate moments and using the covariance upper bound, exactly as in Section 86. Choose $h_n = 2na_n + b_n$. For each fixed L, increasing the entrance coupling makes

$$[K_L(\theta_f + g/\theta_f)]^{2n} K'_L H^{h_n} \le 1.$$

Indeed θ_f times any fixed power of B tends to zero, and, uniformly over the incoming beta window, $g/\theta_f \le 2/\operatorname{monitor}(\beta_f)$. The monitor grows faster than every fixed power of B. This argument does not assign a negative power exponent to g/θ_f. The resulting integrated derivative bounds are

$$\begin{cases} C\tau L^{-d-1}(1+q)^{h_n}, & q^3 < L, \\ C'\tau(1+q)^{h'_n}, & q^3 \ge L. \end{cases} \tag{526}$$

All differentiation under the integral is justified by the same polynomial Gaussian majorants on a common parameter neighborhood.

98.6 Absorbing the enlarged support

Let $A > 0$ be the fine support exponent, $e \ge 0$ the requested coarse exponent including its reserve, and q' the coarse cover number of the allocated output support. The geometric cover estimate gives

$$Lq' \le (L + 10q)t_{r_0},$$

where t_{r_0} depends only on the fixed allocation radius. Choose L so that $20et_{r_0} \le LA$. Then

$$eq' \le et_{r_0} + \tfrac{1}{2}Aq. \tag{527}$$

For every fixed integer h, the function $(1+q)^h e^{-Aq/2}$ is bounded on $q \ge 0$: it tends to zero at infinity and is continuous on compact intervals. Thus the short-support case of (526) and (527) yield

$$e^{eq'}\,\|\text{reference integral}\|_{J^n} \le C_2 L^{-d-1}\tau e^{Aq}.$$

For the long-support case, $q^3 \ge L$ implies $1 \le ((1+q)^3/L)^{d+1}$. Insert this inequality into its bound, then absorb the larger fixed polynomial in the same half of the support exponential. We obtain the identical form with another constant C_3, also chosen before L. This is the required gain for all supports, including wrapping ones in the long-support case.

Finally, the allotted regular price has the form $L^{-d-2/5}/(2h)$, where $h > 0$ depends only on the fixed profile. Since $L^{-d-1}/L^{-d-2/5} = L^{-3/5}$, choose L large enough that both C_2L^{-d-1} and C_3L^{-d-1} are at most half that price. All nonlinear chart, moment and Gaussian-comparison thresholds are chosen afterwards. This preserves the order of constants needed for the one-step regular estimate.

99 Nonlinear singleton inheritance

A surviving singleton contributes an exponential of an old regular activity. Its first order part must be retained as a Gaussian reference observable. The error after this subtraction is smaller for two distinct reasons: the exponential remainder is quadratic, and the actual Gaussian law is close to the reference law. We prove both estimates on the same coordinates and then sum them over the inherited occurrences.

99.1 The nonlinear comparison on fixed coordinates

Use the notation and quantitative hypotheses of Section 96. In particular, R_6, R_9 are density ratios with the same compensated Jacobian, γ_{C_0} is the fixed reference probability measure, and $w(z)$ is the original masked activity. Thus w is measurable and $|w(z)| \le \tau$. Assume

$$0 \le \tau \le K \le 1.$$

Write $c(y,z)$ and $v(y,z)$ for the actual and reference cutoffs, respectively. They take values in $[0,1]$ and have the common compact support and local plateau properties specified there. The exact integrand for the singleton plus its reference observable is

$$T(y,z) = R_6(y,z)\{e^{-w(z)c(y,z)} - 1\} + R_9(y,z)v(y,z)w(z). \tag{528}$$

The plus sign in the second term cancels the negative linear term of the first. The reference guard ensures that, wherever it is nonzero, the actual cutoff is one and the frozen field is healthy on the insertion support. Consequently the same w occurs in both terms, by the compensated locality identity. No derivative of w is needed.

Let P_0, P_1, P_2 denote fixed positive polynomials in the local coordinate count and jet bounds, enlarged below when necessary. The density comparison already proved gives

$$\|R_6(\cdot,z)\|_{J_0^n} \le P_0, \qquad \|R_9(\cdot,z) - R_6(\cdot,z)\|_{J_0^n} \le \delta P_1.$$

Here and below the jet norm includes derivatives of every order from zero through n. The polynomial constants depend on n, not on the number of occurrences or the torus volume. If $Q_c \ge 1$ bounds the cutoff jets, impose also $KQ_c \le 1$. For a nonempty derivative set I, the derivative of e^{-wc} is

$$e^{-wc} \sum_{\pi \in \mathcal{P}(I)} \prod_{A \in \pi} (-wD_A c).$$

Since $|w|Q_c \le 1$, every product has size at most τQ_c, up to the fixed number of partitions. For the zero-order term, $|e^{-wc} - 1| \le e\tau$. Product differentiation in (528) therefore gives the global bound $\|T(\cdot,z)\|_{J_0^n} \le \tau P_2$ on the compact read ball.

On the smaller ball where both cutoffs have germ one at $y = 0$, the exact identity is

$$T = R_6(e^{-w} - 1 + w) + (R_9 - R_6)w.$$

For real $|w| \le 1$, the exponential series gives

$$|e^{-w} - 1 + w| \le |w|^2 \sum_{k \ge 2} \frac{1}{k!} \le |w|^2 \le \tau K.$$

The two factors involving w are constant in the differentiation parameter. Hence on this ball

$$\|T(\cdot,z)\|_{J_0^n} \le \tau(KP_0 + \delta P_1). \tag{529}$$

Let its radius be v_0 in the coordinate maximum norm and suppose $C_0 \le MI$. The coordinate Gaussian tail bound used in the principal comparison gives

$$\gamma_{C_0}\{\|z\|_\infty \ge v_0\} \le 2N \exp(-v_0 + M/2).$$

The same compact-coordinate domination justifies differentiating (528) under the integral: on a common parameter neighborhood the smooth factors and their finitely many derivatives are bounded on the fixed compact ball, and w is bounded. Splitting that integral into the good ball and its complement proves

$$\left\| \int T(\cdot, z)\, d\gamma_{C_0}(z) \right\|_{J_0^n} \le \tau\{KP_0 + \delta P_1 + 2Ne^{-v_0 + M/2} P_2\}. \tag{530}$$

This estimate includes the entire near-weight on the bad set; it does not use cancellation outside the common plateau.

In the renormalization step, K is the incoming regular budget, of upper exponent $-11/10$. The local covariance and mean comparison of Section 97 gives δ of upper exponent $-1/2$: its two terms are the covariance discrepancy and the unscaled mean discrepancy multiplied by the inverse noise scale. The local coordinate counts, cutoff bounds, and polynomials above have upper exponent zero. The plateau radius is a fixed positive multiple of the fine monitor and its Gaussian tail has every negative upper exponent. Thus the braces in (530) have upper exponent $-1/2$. In particular, all the smallness conditions in the density comparison and $KQ_c \le 1$ hold after increasing the entrance coupling, with the block size already fixed. Denote a common nonnegative bound for the braces by $b(x)$.

99.2 The retained reference term

We spell out the summation cost of the reference term, since a per-occurrence gain alone does not give the required coarse budget. Let X_i be the fine support of occurrence i, a_i its fine anchor, Y_i its allocated coarse support, and $\bar{a}_i$ the parent of a_i. Put $q_i = \mathrm{cov}_f(X_i)$ and $q_i' = \mathrm{cov}_c(Y_i)$ for their cover numbers. Suppose the incoming amplitudes satisfy

$$\sup_a \sum_{i : a_i = a} e^{Aq_i} \tau_i \le E_{\mathrm{in}}, \qquad A > 0. \tag{531}$$

All sums use the given occurrence enumeration, including repeated supports. Fix derivative order $n \ge 2$, spatial weight order p, and allocation radius r_0. Let $t \ge 0$ be the full requested coarse support exponent, including any reserve needed later.

The reference attenuation proof in Section 98 applies to each forest and tag. In its notation the pointwise derivative bound is

$$B_0 \tau_i d_L(q_i)\{1 + \rho_i^{2n}(1 + \|z\|)^{4n}\}, \qquad d_L(q) = \begin{cases} L^{-5}, & q^3 < L, \\ 1, & q^3 \ge L. \end{cases}$$

Here B_0 is fixed before L. The same local Gaussian moment estimate bounds the integral of the polynomial factor. More precisely, write $H = (r_0 + 48)(1 + B)$, where B is the scale radius. There are fixed integers a, v such that

$$\rho_i \le K_L(\theta_f + g/\theta_f)[H(1 + q_i)]^a, \quad \int (1 + \|z\|)^{4n}\, d\gamma \le M_L[H(1 + q_i)]^v.$$

The fine and coarse thresholds here have the bounded ratio required by that proof. With $h = 2na+v$, impose the explicit post-L condition

$$[K_L(\theta_f + g/\theta_f)]^{2n} M_L H^h \le 1.$$

It is available by the threshold and monitor argument proved there; no negative power rate is assigned to g/θ_f. Integration yields $B_0\tau_i d_L(q_i)\{1 + (1+q_i)^h\}$. Since

$$d_L(q) \le L^{-5}(1+q)^{15},$$

including when $q^3 \ge L$, the reference price is at most $2B_0\tau_i L^{-5}(1+q_i)^{15+h}$.

The support compression estimate gives, provided $20t\, t_{r_0} \le LA$,

$$tq_i' \le t\, t_{r_0} + \tfrac{1}{2}Aq_i.$$

Choose R so that $(1+q)^{15+h} \le Re^{Aq/2}$ for $q \ge 0$. Such an R exists by polynomial versus exponential growth and depends only on the displayed pre-L data. Thus for $C' = 1 + 2B_0 R e^{tt_{r_0}}$, the weighted reference price is bounded by

$$e^{tq_i'} s_i \le C' L^{-5} e^{Aq_i}\tau_i. \tag{532}$$

A finite average over forests and tags whose sum of absolute weights is at most one preserves this estimate. This is the actual fixed average used in the singleton construction; no new Gaussian marginal or smaller-period replacement is introduced.

Each coarse site has exactly L^4 fine parents in four dimensions. For a coarse site $\bar{a}$, summing (532) first over each fine anchor above $\bar{a}$ and then over those L^4 anchors proves

$$\sup_{\bar{a}} \sum_{i:\bar{a}_i=\bar{a}} e^{tq_i'} s_i \le (C'/L) E_{\mathrm{in}}. \tag{533}$$

The factor L^4 is essential. It is paid by the fifth power in the per-occurrence attenuation.

99.3 Inheritance of the small remainder

The corresponding nonlinear remainder is the actual singleton plus the positive reference observable. Equation (530) bounds its local ordered jets by $\tau_i b(x)$ for each fixed forest and tag. The common finite average preserves this bound.

For completeness, the coordinate-to-field step uses the same local variation as the actual singleton, not a separately chosen test field. Restrict each direction to Y_i, which leaves every derivative of a Y_i-local function unchanged. For $q_i' \le B$ the weighted admissibility bounds imply the local direction bound used in the Gaussian comparison. The support cover restriction and the coarse threshold bound by twice the fine scale threshold give the required direction norm bound directly. Indeed the anchor belongs to Y_i and the cover restriction bounds each support-point distance from it by the local radius D_x; thus the truncated direction satisfies

$$\|U\| \le \theta_c(1 + D_x)^p \le 2\theta_f(1 + D_x)^p.$$

Apply the comparison to the ordered parameter map formed from these actual restricted directions. Its mixed derivative is the prescribed ordered field derivative: evaluate the corresponding multilinear differential on the coordinate unit vectors, each of norm one. This conversion incurs no inverse rescaling factor. Consequently the estimate controls the prescribed spatially weighted derivatives through order n on healthy configurations whenever $q_i' \le B$. Regularity, gauge invariance, and

locality on every support follow from the actual singleton regularity and the reference construction. The latter is invariant under the orthogonal colour-coordinate gauge change; locality follows from agreement on Y_i. Its measurable extension outside the healthy domain has the same local germ there.

For the remainder use the weaker consequence $tq_i' \le tt_{r_0} + Aq_i$ of support compression. Combining it with (531) and again counting L^4 fine anchors above a coarse anchor gives

$$\sup_{\bar a} \sum_{i:\bar a_i=\bar a} e^{tq_i'} \tau_i b(x) \le L^4 e^{tt_{r_0}} b(x) E_{\mathrm{in}}. \tag{534}$$

For fixed L, r_0, t, the coefficient on the right retains upper exponent $-1/2$. Equations (533) and (534) are the two required inheritance bounds. The first applies to every reference support; the derivative assertion for the remainder retains the restriction $q_i' \le B$. The pre-existing full singleton regularity and the common funded path are inputs to this step. The argument does not replace their construction or the estimates for clusters containing more than one old occurrence.

100 Extra decay from the degree of a cluster

After extracting the singleton reference, many terms can be paid without a further subtraction. The relevant degree counts elementary occurrences inside every connected component, together with an extra grade for certain species. Counting only the number of components in a cluster would miss this gain. We follow the two successive connected expansions and retain their separate convergence conditions.

100.1 Grading the elementary activities

Write $\beta_+(x) = \beta_F(G, x, L)$ for the shifted fine coupling and set

$$D(x) = \beta_+(x)^{3/8}.$$

Eventually $D \ge 1$. Give the six inherited logarithmic species the following additional grades:

species	regular	cubic	quartic	quadratic loop	fine tail	coarse tail
h	1	0	1	1	2	2.

These are extra grades; a further factor D per occurrence will be inserted at the connected-component stage.

Section 83 supplies the required full-support site bounds. Fix $-1/2 < t < 0$, and choose s strictly between $-1/2$ and t. The ungraded regular, quartic and loop prices can each be bounded by $\beta_+^{-15/16}$ eventually. Multiplication by D changes this exponent to $-9/16 < s$. The cubic price has upper exponent $-1/2$ and is bounded by β_+^s. Both true path tails can be bounded by β_+^{-2}; multiplication by D^2 changes this to $-5/4 < s$. The six prices therefore sum to at most $6\beta_+^s \le \beta_+^t$ eventually. These inequalities preserve the original supports and all length charges. The forced inherited family retains every negative upper exponent and receives no extra grade.

The same argument applies to the local derivatives. Hold the original insertion fixed in the compensated coordinates as in Section 83. Its transported cutoff has jet scale bounded by an upper-exponent-zero function times the full cover plus length charge. The coarse tail has an additional varying coarse-field term; its own jet price has upper exponent -2, and so the factor D^2 again leaves exponent $-5/4$. Adding this term to its old price preserves the common bound β_+^t by first

taking the intermediate exponent s. The cutoff product rule only changes the upper-exponent-zero jet scale.

For the other optional letters, assign grade zero to the quadratic local term and grade two to the remaining small correction family. Their ungraded prices are, respectively, of upper exponent $-1/2$ and at most -2. The density and exponential jet estimates of Section 87 therefore preserve a common upper exponent t after the grades are inserted. More explicitly, if a_i is a nonnegative logarithmic amplitude and its jet scale is b_i, the full-support site bound implies $a_i \le 1$ eventually: choose a root in its nonempty support, bound this single nonnegative summand by the site sum, and drop its weights, all at least one. The zero-order exponential difference is bounded by ea_i. For a positive-order derivative, Faà di Bruno's formula is a finite sum indexed by partitions; a partition with q blocks contributes at most $ea_i^q b_i^r$ at order r. Since $a_i^q \le a_i$, this bounds every jet through order n by $C_n a_i \max(1, b_i)^n$. The scale b_i retains its full cover and length charges; only its scalar prefactor has upper exponent zero. The polynomial charge factor is paid by the reserve in the next paragraph. The previously proved warning estimates still give the same joint warning power after multiplying by these deterministic grades. Thus the grading is compatible with the actual Gaussian derivative majorants, not only their zeroth-order values.

The passage from graded letter prices to differentiated component integrals uses the common random field. For a component P, take I to be the full set of fine bonds read by its enlarged union support, and let J be its joint warning event. For the Gaussian coordinates $z_{b,c}$ indexed by bond and colour, define

$$\mathcal{F}_I(z) = 1 + |I| + |I \times \mathcal{C}| + \sum_{(b,c)\in I\times\mathcal{C}} |z_{b,c}|,$$

where $\mathcal{C}$ is the finite colour index set. The product-rule bound has the form

$$A_P \mathbf{1}_J \mathcal{F}_I^{kj}, \qquad A_P = \prod_{i\in P} \alpha_i \, V(x)(1 + S(P))^j,$$

where α_i are the nonnegative elementary packet and letter prices, $V \ge 0$ has upper exponent zero, and $S(P)$ is the full footprint charge. The read-count bound is $|I| \le (1 + S(P))b_B d_L$, with the fixed block-density factor d_L. The Gaussian moment estimate of Section 86, at the doubled order $2kj$, gives $\int \mathcal{F}_I^{2kj}\, d\mu \le [E(x)(1 + S(P))^K]^2$ for some fixed K and upper-exponent-zero $E \ge 0$. The joint warning bound is $\mu(J) \le \rho^{2b(P)}$, where $b(P)$ is the number of warning labels and $\rho \ge 0$ is the retained warning price. Cauchy–Schwarz therefore yields

$$\int A_P \mathbf{1}_J \mathcal{F}_I^{kj}\, d\mu \le A_P E(x)(1 + S(P))^K \rho^{b(P)}.$$

The remaining Haar integral has mass one. This is precisely the integrated component majorant, with its same warning power and a larger fixed footprint exponent. It uses no independence between overlapping reads or warning events. The local jet construction and its common integrable majorant justify differentiation as proved in Section 87.

100.2 Recovering the footprint weight in the first expansion

We state the needed positive-weight calculation explicitly. For an elementary label i, let X_i be its support, $u_i \ge 0$ its price, and $e_i \ge 0$ its additional length charge. For a finite connected set P put

$$C(P) = c\left(\bigcup_{i\in P} X_i\right), \qquad S(P) = \sum_{i\in P} (c(X_i) + e_i).$$

Here c is the connected cover number at the current scale. The cover subadditivity along a spanning tree gives $C(P) \le \sum_i c(X_i)$. Fix $a \ge 0$, $\epsilon > 0$ and integer $k \ge 0$. If the elementary site estimate is available with exponent $a + 2\epsilon$ on cover and 2ϵ on the length charge, the first tree estimate yields its rooted connected sum. It also pays the stronger summand

$$\mathbf{1}_{\{z \in \bigcup X_i\}} \prod_{i \in P} u_i \, e^{aC(P)} (1 + S(P))^k e^{\epsilon S(P)}.$$

Indeed

$$(1+S)^k \le e^{\epsilon} k! \epsilon^{-k} e^{\epsilon S}, \qquad S \ge 0,$$

by the exponential series at $\epsilon(1+S)$. Together with cover subadditivity this bounds the displayed summand by $e^{\epsilon} k! \epsilon^{-k}$ times the summand already paid by the elementary site estimate. The same ϵ therefore pays both the retained full-footprint exponential and the polynomial moment; we have used two copies of its reserve, not discarded either cost.

Apply this calculation to the mixed elementary alphabet of Section 63. Its mandatory packet weights already include their seed payment, while its optional letters have the grades above. For a component P, let $h(P)$ be the sum of its extra grades, $|P|$ its number of elementary occurrences, $m(P)$ its number of mandatory seeds, and $S(P)$ its full footprint charge. The strengthened component multiplier is

$$D^{h(P)+|P|} \exp\{M m(P) + \epsilon S(P)\}, \tag{535}$$

where $M \ge 0$ is the seed mark. This factor is the product of the elementary factors $D^{1+h_i} e^{M m_i}$ and the retained footprint factor. Multiplying every elementary price by D multiplies its site budget by D. Consequently the first expansion is valid if

$$\frac{4 b_B D r_0(x)}{\delta_0} \le \frac{1}{2}, \tag{536}$$

where r_0 and $\delta_0 > 0$ are the original elementary root price and its reserve, and $b_B = (2B+1)^4$. Its component root price is $D\omega(x)$, where ω is the original component price including the footprint polynomial just proved. All these statements require the graded elementary site bound on the same incoming representation.

100.3 The second expansion and its degree gain

Let j be the field derivative order, and let $B_1(x) \ge 0$ be the upper-exponent-zero constant from the component derivative estimates. A convenient second-expansion root price is

$$\sigma(x) = (1 + 2^j B_1(x)) \omega(x).$$

Choose a target support exponent $a \ge 0$ and a reserve $\eta > 0$ with $a + 2\eta$ no larger than the available component exponent. In addition to (536), impose

$$\frac{4 b_B D \sigma(x)}{\eta} \le \frac{1}{2}. \tag{537}$$

The cluster estimate of Section 62 now applies to the strengthened component weights, with root price $D\sigma$ and tail ratio $1/2$.

For a cluster $\Gamma = (P_1, \dots, P_q)$, counted with its actual occurrence multiplicities, define

$$d(\Gamma) = \sum_{r=1}^{q} \{h(P_r) + |P_r|\}, \quad m(\Gamma) = \sum_r m(P_r), \quad s(\Gamma) = \sum_r S(P_r).$$

Its strengthened weight is its old positive cluster majorant times

$$D^{d(\Gamma)}e^{Mm(\Gamma)+\epsilon s(\Gamma)}.$$

This identity follows by multiplying (535) over the cluster occurrences; repeated labels are counted at each occurrence. Suppose

$$D^3 \le e^M, \qquad D^3 \le e^{\epsilon T}, \qquad T \ge 0.$$

For any cluster satisfying at least one of

$$d(\Gamma) \ge 3, \qquad m(\Gamma) \ge 1, \qquad s(\Gamma) \ge T, \tag{538}$$

the strengthened multiplier is at least D^3. Every other factor is at least one. Its unstrengthened price is therefore bounded by D^{-3} times the strengthened price.

Spatially weighted derivatives of order at most j, with spatial power p, cost $[\ell(x)(1+C(\Gamma))]^{pj}$, where $\ell(x) = 4(2B(x)+1)$. The same exponential-series inequality pays this polynomial from the reserve η, at cost

$$H_\eta(x) = \ell(x)^{pj}e^\eta(pj)!\eta^{-pj}.$$

Using the cluster root bound at exponent $a+\eta$, then removing this reserve, proves the site estimate at exponent a

$$\sup_z \sum_{\substack{\Gamma:\ z\in\operatorname{supp}\Gamma \\ \Gamma \text{ satisfies (538)}}} \|\Phi_\Gamma\|_{j,p}e^{aC(\Gamma)} \le D^{-3}H_\eta(x)\frac{D\sigma(x)}{1-1/2}. \tag{539}$$

Here Φ_Γ denotes the cluster contribution and the norm is its local spatially weighted jet bound. Restricting a nonnegative sum to the displayed class is legitimate without analyticity of a field-dependent classification: the class depends only on labels, supports and charges. Uniform forest averaging preserves the bound.

100.4 Paying an arbitrary fraction of the forecast

We check the exponents and the order of choices. Suppose the unstrengthened site bound is

$$H_i(x)\frac{\sigma(x)}{1-1/2} \le \beta_+(x)^\varphi, \qquad \varphi < -3/8, \tag{540}$$

for some $i > 0$ with $a + 2i \le E$, where E is the available component exponent. This receipt follows from the unstrengthened component construction when its upper exponent q satisfies $q < \varphi$: B_1 and H_i have upper exponent zero, ω has upper exponent q, and therefore the entire left side has upper exponent q. Taking the reserve $i = (E-a)/2$ also makes its ordinary degree price at most $1/2$ eventually, since $q < 0$. The same graded local construction must supply the actual component majorants; the scalar exponent calculation alone does not construct them. Put $\eta = (E-a)/2$. Then $0 < i \le \eta$ and, writing $r = pj$, we have

$$\eta^{-r}r! \le i^{-r}r!, \qquad H_\eta(x) \le e^\eta H_i(x).$$

Also $H_i(x) \ge \eta^{-r}r! > 0$, since $\ell \ge 1$ and $e^i \ge 1$. Thus (540) bounds σ by a fixed constant times β_+^φ. Since b_B has upper exponent zero and $3/8 + \varphi < 0$, it implies (537) at this fixed reserve for all large x. The first condition follows in exactly the same way from the strictly smaller than $-3/8$ exponent of r_0.

The seed mark and the large-footprint threshold obey e^{-M} and $e^{-\epsilon T}$ estimates with every negative upper exponent, so the two inequalities $D^3 \le e^M, e^{\epsilon T}$ hold eventually as well. Substituting (540) into (539) gives

$$e^{\eta} D^{-2} \beta_+^{\varphi} = e^{\eta} \beta_+^{\varphi - 3/4}.$$

The incoming regular forecast is $\delta_E(x) = x^{-11/10}$. Since

$$\varphi - \frac{3}{4} + \frac{11}{10} = \varphi + \frac{7}{20} < 0,$$

the last price is at most $c\delta_E(x)$ for any prescribed $c > 0$ after increasing the entrance coupling. The comparison uses $\beta_+(x) \ge x$ and the negative exponent of δ_E.

For each stage only finitely many transitions and fixed derivative orders occur. First take the maximum of their block-size requirements; then take the maximum of their coupling thresholds and the previous entrance value. This yields one increasing collection of entrance requirements. Enlarging them preserves all the inequalities above. It does not require a uniform threshold over infinitely many derivative orders. The estimates for the remaining degree-one and degree-two classes, and the identities which attach their subtractions to the renormalized path, are separate steps.

101 Comparing the local factors at neighboring couplings

We now prove the compensated comparison used in the fixed-volume continuity argument. The important simplification is to preserve the physical fine field while changing the Gaussian coordinate. Thus the incoming measurable activity is evaluated at the same point in the two representations. Only its value difference, and not a derivative of that activity, is needed.

Fix a derivative order $j \in \mathbb{N}$ and a fine torus. For a scalar function of the coarse variation y, write

$$\|f\|_{j,y_0} = \max_{0 \le r \le j} \|D^r f(y_0)\|.$$

All derivatives below are taken on a common admissible local chart, with its fixed normalized directions. In particular, the assertions include existence and continuity of these derivatives near y_0. We use the product estimate

$$\|fh\|_{j,y_0} \le 3^j \|f\|_{j,y_0} \|h\|_{j,y_0}. \tag{541}$$

The Leibniz formula has 2^r terms at order r, so the slightly larger factor 3^j is valid also for $j = 0$.

The base representation supplies summable elementary prices $u_i \ge 0$, with $B = \sum_i u_i < \infty$, for its compensated factors w_i. All logarithms here are real. Assume explicitly a common base value bound $|\ell_i(y_0)| \le M$ for their real exponents; in the construction this follows from the summable incoming logarithmic prices and the held-field estimate. A bound on $e^{\ell_i} - 1$ alone would not control a large negative ℓ_i. We also use the following local estimates already required by the compensated construction: uniform jets of its enforcement cutoffs; a bound As for the packet action jets on every packet of size s whose cutoff is not identically zero near y_0; and, for its density compensation K_I and quadratic exponent q_I, for every subset I of read bonds,

$$\|K\|_{j,y_0} \le C_0 F_0(z)^N, \qquad \|q\|_{j,y_0} \le C_0 F_0(z)^2. \tag{542}$$

Here $N \in \mathbb{N}$, $C_0 \ge 0$, and $F_0(z) = 1 + d + \sum_{r=1}^{d} |z_r| \ge 1$ uses the full finite Gaussian coordinate space. The constants N, C_0 are common to every such I; we suppress that index in the formulas. These estimates must hold for the same base representation, forests and compensation maps. Their

uniform purchase in the scale construction is a separate requirement; the comparison below uses their actual values at the fixed stage.

Let R' be a neighboring representation with the same dictionaries, paths and masks as R, and let $t > 0$ bound its input differences as in the fixed-volume continuity argument. Put $\lambda = g(\beta)/g(\beta')$. The identity $g(\beta')\lambda = g(\beta)$ implies that the fine field at $(R', \lambda z)$ is precisely the fine field at (R, z). It also gives $\beta\lambda^2 = \beta'$ and makes the compensated noise course for R' equal to λ times the course for R.

101.1 Elementary exponential comparisons

Two elementary facts explain the logarithmic and packet estimates. First, suppose $\|e^f - 1\|_{j,y_0} \le u$, $u \le B$, and $|f(y_0)| \le M$. Then

$$\|f\|_{j,y_0} \le C_{j,M,B}u. \tag{543}$$

Indeed $f = \log(1 + (e^f - 1))$. The derivatives of $v \mapsto \log(1 + v)$ are bounded on $[e^{-M} - 1, e^M - 1]$. In each positive-order chain-rule term there is at least one derivative of $e^f - 1$. A product of r such derivatives is bounded by $u^r \le u \max(1, B)^{r-1}$. At order zero, the mean value theorem between 0 and $e^{f(y_0)} - 1$ gives the same linear bound. The chain rule is used on the open interval $v > -1$, so no assertion of a fixed neighborhood contained in that compact interval is needed.

Second, if $\|e^f - 1\|_{j,y_0} \le B$ and $\|h\|_{j,y_0} \le v \le D$, then

$$\|e^{f+h} - e^f\|_{j,y_0} \le C_{j,B,D}v. \tag{544}$$

Use $e^{f+h} - e^f = e^f(e^h - 1)$. The zero-order estimate for the second factor is $|e^{h(y_0)} - 1| \le e^D|h(y_0)|$. For positive orders, the partition formula for derivatives of e^h gives finitely many products of derivatives of h, each containing at least one factor. Thus $\|e^h - 1\|_{j,y_0} \le C_{j,D}v$, by $v^r \le v \max(1, D)^{r-1}$. Apply (541) and $\|e^f\|_{j,y_0} \le B + 1$.

101.2 The four elementary kinds of factor

We compare every neighboring factor after the noise rescaling. For a packet s, write its base compensated factor as w_s. The exact ratio identity is

$$w'_s - w_s = w_s(e^{H_s} - 1), \qquad H_s = -(\beta' - \beta)E_s - n_s\{\log m(\beta') - \log m(\beta)\}.$$

Here E_s is its compensated action including the fixed far-field term, and n_s is the number of chart normalizers in the packet. The restrictions deciding whether the packet occurs are the same in both representations. If its enforcement cutoff vanishes on a neighborhood of y_0, both germs vanish. Otherwise the packet action estimate applies. Let s_* be the maximum packet size and let m_* be the total number of fine bonds. Both are finite on this torus. Then, with $Y = 1 + As_* + m_*$,

$$\|H_s\|_{j,y_0} \le tY, \qquad \|w_s\|_{j,y_0} \le B.$$

Choose $t \le Y^{-1}$. The exponential comparison and product rule give $\|w'_s - w_s\|_{j,y_0} \le C_{\mathrm{p}}t$, uniformly over the finite packet catalogue and all its admissible auxiliary choices.

For a forced incoming factor, the held fine field is the same and the enforcement cutoff is the same. Its value difference is at most b_i, and its cutoff has a common jet bound C_{e}. Hence its factor difference has jet bound $C_{\mathrm{e}}b_i$. This differentiates only the cutoff, not the incoming forced activity.

For logarithmic factors write $w_i = e^{\ell_i} - 1$. The bounded base logarithm and the summable base factor prices give $\|\ell_i\|_{j,y_0} \le Cu_i$ by (543). The actual logarithmic catalogue has two cases.

An inherited regular activity has logarithm minus its held value times the common enforcement cutoff; its logarithm difference therefore has jet bound $C_{\mathrm{e}}a_i$. Every other catalogue entry is either unchanged by the rescaling, or is multiplied by λ^2. This follows term by term from the quadratic noise terms and $\beta\lambda^2 = \beta'$; the path and loop terms without that quadratic scaling remain unchanged. Thus

$$\|\ell_i' - \ell_i\|_{j,y_0} \le C\{\widetilde{a}_i + tu_i\},$$

where $\widetilde{a}_i$ places the regular difference amplitudes at their inherited logarithmic addresses and is zero elsewhere. The address embedding is injective, so $\sum_i \widetilde{a}_i \le t$. Since $t \le 1$ and $u_i \le B$, the right side is at most $C(1+B)$. Applying (544) gives

$$\|w_i' - w_i\|_{j,y_0} \le C_{\mathrm{l}}(\widetilde{a}_i + tu_i).$$

Finally, warning factors are exactly unchanged: their charge sets, forced supports and physical field agree. Their difference price is zero.

There are finitely many packet labels, say n_*. Combining the four cases gives nonnegative difference prices v_i satisfying

$$\|w_i' - w_i\|_{j,y_0} \le v_i, \qquad \sum_i v_i \le tK_*, \quad K_* = n_*C_{\mathrm{p}} + C_{\mathrm{e}} + C_{\mathrm{l}}(1+B). \tag{545}$$

The constants are uniform over the auxiliary choices at this fixed torus. This is stronger than continuity of each individual factor: the whole catalogue has a summable difference bound.

101.3 Products over components

Set $a_i = u_i + v_i/t$. Then $\sum_i a_i \le B + K_*$, $\|w_i\|_{j,y_0} \le a_i$, and $\|w_i' - w_i\|_{j,y_0} \le ta_i$. For every finite set of labels P, put

$$p_P = \prod_{i\in P} 3\,3^j a_i.$$

For $0 < t \le 1$, induction on $|P|$ yields

$$\left\|\prod_{i\in P} w_i'\right\|_{j,y_0} \le p_P, \qquad \left\|\prod_{i\in P} w_i' - \prod_{i\in P} w_i\right\|_{j,y_0} \le tp_P. \tag{546}$$

For the induction step use $\|w_i'\|_{j,y_0} \le 2a_i$ and

$$w_i'G' - w_iG = (w_i' - w_i)G' + w_i(G' - G).$$

Each product costs at most 3^j; the chosen factor $3\,3^j a_i$ pays both terms and the bound for the new product. The empty product has value one and difference zero.

Summing nonnegative products over finite subsets gives

$$\sum_P p_P \le \exp\{3\,3^j(B + K_*)\}. \tag{547}$$

For a finite catalogue this is $\prod_i(1 + 3\,3^j a_i)$ and $1 + x \le e^x$. Exhaustion and monotone convergence prove the countable case. Restricting to admissible molecules only decreases the sum. All supports and selection conditions agree under the common address identification.

We must identify these products with the actual component integrands. Assume both representations satisfy the incoming room and activity bounds for this same sector. The protected

compensation map holds the physical fine field fixed on a neighborhood of y_0. On that neighborhood, the definition of each component therefore equals the product of the elementary compensated factors with its fixed charge set. The neighboring compensation has the same protected patch: its noise course is multiplied by λ, and its coupling cancels that factor in the physical field. Applying the same identity to the neighboring representation proves equality of its component germ with the primed product. Intersect the two neighborhoods. Health of the union of the assigned supports implies health of every factor support, because each is contained in that union. Thus all the bounds above apply to these actual germs, and their derivatives are the derivatives of the displayed finite products.

101.4 The density compensation and the Gaussian ratio

The exact compensation identity has the anchored form

$$K'(y,\lambda z) = K(y,z)\exp\{(1-\lambda^2)(q(y,z)-q(y_0,z))\}.$$

The subtraction at y_0 is essential. Put $h(y) = (1-\lambda^2)(q(y,z)-q(y_0,z))$. Then $h(y_0)=0$, and every positive-order derivative through j is bounded by $tCF_0(z)^2$. In the partition formula for a derivative of e^h at y_0, a term with $r \geq 1$ blocks is bounded by $t^r C^r F_0(z)^{2r}$. Since $t \leq 1$ and $F_0 \geq 1$, it is at most $tC^r F_0^{2j}$. The zero-order difference $e^{h(y_0)}-1$ is zero. Consequently, also when $j=0$,

$$\|e^h - 1\|_{j,y_0} \leq tC_j F_0(z)^{2j}, \qquad \|K'-K\|_{j,y_0} \leq tC_1 F_0(z)^{N+2j}.$$

No exponential of an uncontrolled positive polynomial in z is introduced here.

Write g_P, g'_P for the component products in (546). With the fixed Gaussian density ratio $J_\lambda(z)$, which is independent of y, use the identity

$$J_\lambda K' g'_P - K g_P = J_\lambda (K'-K) g'_P + (J_\lambda - 1) K g'_P + K(g'_P - g_P).$$

The preceding estimates and the product rule give

$$\|J_\lambda K' g'_P - K g_P\|_{j,y_0} \leq tC_2 p_P F_0(z)^{N+2j}\{J_\lambda + |J_\lambda - 1|/t + 1\}. \tag{548}$$

Here C_2 is fixed and $\sum_P C_2 p_P < \infty$ by (547). This is the required local compensated comparison. The Gaussian estimates then integrate it with a common integrable dominator and preserve all jets through order j.

101.5 Packet values outside the healthy jet domain

For the forced output one also needs value estimates for packets outside the healthy domain used above. An upper bound on the packet energy itself is neither available nor needed. The actual packet has the form

$$P_\beta = m_0 \exp\{-\beta u + v_\beta\}, \qquad 0 \leq m_0 \leq M_0, \quad u \geq -d_0, \quad v_\beta \leq A_0,$$

with $d_0, M_0 \geq 0$ fixed on this torus. The packet energy lower bound supplies d_0; bounded cutoff and charge factors supply M_0. The unchanged condensation and chart terms, and the finite number of normalizers, give $|v_{\beta'} - v_\beta| \leq n_0 t$ for a finite $n_0 \geq 0$. Assume $|\beta' - \beta| \leq t \leq \min(1, \beta/2)$ and put

$$c = \beta/2 > 0, \qquad D = A_0 + n_0 + (\beta+1)d_0.$$

If $-d_0 \le u \le 0$, both exponential arguments are at most D, and their difference is at most $t(d_0+n_0)$. The mean value theorem gives a bound $tm_0e^D(d_0+n_0)$. If $u > 0$, both arguments are at most $D - cu$ and their difference is at most $t(u+n_0)$. Thus

$$|P_{\beta'} - P_\beta| \le tm_0e^D(u+n_0)e^{-cu} \le tm_0e^D(c^{-1}+n_0),$$

using $cu \le e^{cu}$ and $e^{-cu} \le 1$. Combining both cases yields

$$|P_{\beta'} - P_\beta| \le tM_0e^D(d_0+2n_0+c^{-1}). \tag{549}$$

Zero packet masks remain zero under the common data and require no estimate. This proves the packet value comparison uniformly over the unconstrained positive-energy tail. The corresponding forced product and Gaussian integration arguments use value bounds; they do not assert derivative estimates on that tail.

For completeness, the other unrestricted value factors have the same summable comparison. Let $r_i \ge 0$ bound the base logarithms and put $R_0 = \sum_i r_i < \infty$. Their differences are bounded by $d_i = tr_i + \widetilde{a}_i$, so $\sum_i d_i \le t(R_0+1)$. Both the base and new logarithms are at most $2R_0+1$ when $t \le 1$. The mean value theorem gives the exponential-factor difference price $e^{2R_0+1}d_i$. Forced incoming factors have difference price b_i, since their common cutoff lies in $[0,1]$; warnings cancel. Together with (549), the finite packet catalogue therefore has an overall elementary difference budget tC_3.

The corresponding base value prices are summable as well. Packets and warnings have finite catalogues. The logarithmic catalogue has its summable base price, while the forced incoming prices are bounded by their rooted weighted prices. Every forced support is nonempty: it contains the parent of an inspected incoming inventory site. Summing the rooted bound over the finitely many torus sites therefore bounds the unrooted sum of forced prices. The weights removed in this argument are at least one.

Apply the product argument above with $j = 0$ to these value prices. There are summable $p^{\mathrm{v}}_P \ge 0$ such that $|G'_P| \le p^{\mathrm{v}}_P$ and $|G'_P - G_P| \le tp^{\mathrm{v}}_P$ for the rescaled full component values. Before integration use

$$|J_\lambda G'_P - G_P| \le |J_\lambda - 1|p^{\mathrm{v}}_P + tp^{\mathrm{v}}_P.$$

The Gaussian ratio estimate at moment order zero bounds the integral by $tC_4p^{\mathrm{v}}_P$. Haar variables keep their normalized measure. The common support and charge data also identify all eligibility conditions. A normalized finite tag average preserves this bound; if a component is ineligible, both terms are zero. This proves the forced value comparison needed for the exponential expansion of the output array.

102 Continuity of the nonlinear output on a fixed torus

The nonlinear step must vary continuously with its incoming coupling and activity array. This assertion concerns the actual array in a fixed support dictionary. Equality of the evaluated gas would be insufficient. The constants in this section may depend on the fixed torus; uniform small prices for the scale construction are established separately.

102.1 Which data are held fixed

Consider two incoming representations R, R' on the same fine torus. They have the same regular and forced addresses, thresholds, clipping convention and path data. Write $\beta, \beta' > 0$ for their

couplings. Their regular and forced activity differences have nonnegative amplitudes a_i, b_i with

$$\sum_i a_i \le t, \qquad \sum_i b_i \le t.$$

The regular differences here are differences of the actual masked occurrences. Their supports and healthy masks agree exactly. Let $g(\beta) = \beta^{-1/2}$ and put $\lambda = g(\beta)/g(\beta') > 0$. The remaining comparison conditions are

$$|\lambda^2 - 1| \le t, \quad |\beta' - \beta| \le t, \quad |\log m(\beta') - \log m(\beta)| \le t, \qquad 0 < t \le \min(1/4, \beta/2), \tag{550}$$

where $m(\beta) > 0$ is the fixed local Haar-chart normalizing factor. All forests, tags, component labels and output address identifications are retained. In particular, their compatibility and eligibility conditions agree under the identity on elementary labels.

For a continuous incoming curve these conditions hold locally for arbitrarily small t. The activity conditions follow from its array continuity; multiplication by the fixed healthy mask preserves the value difference and makes both occurrences zero off that mask. The scalar conditions follow from continuity of the positive coupling, its inverse square root, and the logarithm of m. This observation uses the same path throughout the curve.

102.2 A Gaussian rescaling estimate

We explain why changing the coupling does not require differentiating the incoming measurable activities. On one fixed Gaussian space of dimension d, with positive precision matrix Q, put

$$p(z) = c_Q e^{-z^T Q z/2}, \qquad J_\lambda(z) = \lambda^d \frac{p(\lambda z)}{p(z)}.$$

Then for every integrable function F,

$$\int F(z)\, d\gamma_Q(z) = \int J_\lambda(z) F(\lambda z)\, d\gamma_Q(z). \tag{551}$$

This is the ordinary change of variables $w = \lambda z$. The Haar factor, when present, is left unchanged. Since $g(\beta')\lambda = g(\beta)$, the rescaled incoming field has the same physical noise; the corresponding compensation map keeps the original insertion values fixed.

For $|\lambda^2 - 1| \le t \le 1/4$, we have $\lambda \ge 1/2$ and $|\log \lambda| \le t$. Consequently

$$|\log J_\lambda(z)| \le t\{d + \tfrac{1}{2}\|Q\|\,|z|^2\}.$$

Let $F_0(z) = 1 + d + \sum_{r=1}^d |z_r|$ and choose a finite constant C_Q such that the braces are at most $C_Q F_0(z)^2$. The inequality $|u - 1| \le (1 + u)|\log u|$ for $u > 0$ follows by integrating e^s between 0 and $\log u$. Thus

$$|J_\lambda - 1| \le t C_Q (1 + J_\lambda) F_0^2.$$

All polynomial moments of F_0 are finite, by the finite Gaussian moment argument. If $M_k = \int F_0^k\, d\gamma_Q$, then (551) and $F_0(\lambda^{-1} z) \le 2F_0(z)$ give

$$\int J_\lambda F_0^k\, d\gamma_Q \le 2^k M_k, \int |J_\lambda - 1| F_0^k\, d\gamma_Q \qquad \le t C_Q (1 + 2^{k+2}) M_{k+2}. \tag{552}$$

The same estimates hold for the full finite collection of Gaussian read coordinates and the product with normalized Haar measure. No smaller marginal is substituted when the parameter changes.

Here is the precise pointwise input needed to integrate local jet differences. Fix integers $j, k \geq 0$. For every component P, suppose its compensated integrands f_P, f'_P have, on each admissible local variation, the bound

$$\|J_\lambda f'_P(\cdot, \lambda z) - f_P(\cdot, z)\|_{J^j} \leq t c_P F_0(z)^k \{J_\lambda(z) + t^{-1}|J_\lambda(z) - 1| + 1\}, \qquad \sum_P c_P \leq C. \tag{553}$$

Here $c_P \geq 0$. The constants and k are common to the neighboring representation and to the forests and tags on this fixed torus. This is the pointwise compensated comparison estimate, a stronger input than individual boundedness of the two integrands. Its role is explicit: (552) gives uniform integral bounds. For a pointwise dominator, use

$$J_\lambda(z) \leq (5/4)^{d/2} \exp(z^T Q z/8).$$

Together with $t^{-1}|J_\lambda - 1| \leq C_Q(1 + J_\lambda)F_0^2$, this bounds the integrand after division by t by a fixed polynomial times $1 + \exp(z^T Q z/8)$. Multiplying by the Gaussian density leaves a decaying exponential $\exp(-3z^T Q z/8)$, so this dominator is integrable. Integration and the derivative identities of Section 85 give

$$\|F'_P - F_P\|_{J^j} \leq t C_1 c_P, \qquad \sum_P C_1 c_P \leq C_1 C. \tag{554}$$

For example one may take $C_1 = (1 + 2^k)M_k + C_Q(1 + 2^{k+2})M_{k+2}$. The same bound survives normalized forest and tag averaging and the common local extension, since their germs agree on the healthy domain. The construction of (553) from the individual logarithmic, packet and compensation factors remains a separate local estimate; it is not inferred merely from continuity of the scalar coupling.

102.3 A summable difference through the cluster logarithm

We prove the summation step rather than interchange an infinite series with a limit without a common bound. Write $u_P \geq 0$ for component jet prices. Fix a target support exponent $a \geq 0$; the cluster comparison itself also holds for arbitrary real a. Suppose the ordinary cluster criterion holds for $2^j u_P$, with reserve $\delta > 0$, root price $\gamma > 0$ and ratio $1/2$. Suppose component differences have prices $t d_P$, where $d_P \geq 0$ and $\sum_P d_P \leq C$. On the fixed torus let

$$K_0 = \sum_{Y \subseteq \Lambda} e^{(a+\delta)c(Y)}, \qquad \kappa = \frac{\gamma}{2K_0 2^j (1 + C)} > 0.$$

The finite sum includes every possible support. For every root, the additional vertex budget from $2^j \kappa d_P$ is at most $K_0 2^j \kappa C \leq \gamma/2$. The augmented prices $2^j(u_P + \kappa d_P)$ therefore satisfy the cluster criterion with root price $3\gamma/2$ and ratio $3/4$: the degree quantity is linear in the root price. This chooses one neighborhood size κ after the torus and old criterion are fixed.

For $0 \leq t \leq \kappa$, set $\theta = t/\kappa \leq 1$. The finite product identity gives, in the joint jet bounds,

$$\left\| \prod_{r=1}^q F'_{P_r} - \prod_{r=1}^q F_{P_r} \right\|_{J^j} \leq \theta(1 + q)^j \left\{ \prod_{r=1}^q (u_{P_r} + \kappa d_{P_r}) - \prod_{r=1}^q u_{P_r} \right\}. \tag{555}$$

To verify it, expand each $F'_P = F_P + (F'_P - F_P)$. Every nonempty subset of difference factors contains at least one factor θ; its higher powers are bounded by θ. At derivative order r there are q^r assignments to the factors, bounded by $(1 + q)^j$. Summing the nonempty subsets gives exactly

the difference of the positive products on the right. Repeated component labels are still distinct positions in this calculation.

Since $1 + q \le 2^q$ for $q \ge 1$, the derivative factor is paid by 2^j per component. Multiply by the absolute Ursell coefficient and its factorial normalization, then use the augmented cluster criterion. Summing rooted bounds over all $|\Lambda|$ sites bounds the unrooted sum, because every nonzero cluster has nonempty support. Thus cluster differences have nonnegative prices te_Γ with

$$\sum_\Gamma e_\Gamma \le \kappa^{-1}|\Lambda|\frac{3\gamma/2}{1-3/4}.$$

This estimate uses the original cluster criterion with a paid perturbation margin; no new uniform small-price theorem for the neighboring representation has been assumed.

102.4 Forced products and the output array

The same product calculation applies to the forced exponential expansion. If its component values have summable bounds h_i and differences bounded by td_i, with $0 \le t \le 1$, then a finite multiplicity vector ν satisfies

$$\left|\prod_i \frac{(F_i')^{\nu_i}}{\nu_i!} - \prod_i \frac{F_i^{\nu_i}}{\nu_i!}\right| \le t\prod_i \frac{(h_i + d_i)^{\nu_i}}{\nu_i!}.$$

Indeed apply the same difference-of-products argument to repeated factors and divide by their positive factorials. Summing over all finite multiplicity vectors yields

$$\sum_\nu \prod_i \frac{(h_i + d_i)^{\nu_i}}{\nu_i!} = \exp\left(\sum_i (h_i + d_i)\right),$$

by finite products of the exponential series followed by monotone convergence. Compatibility or eligibility conditions only remove terms; they agree under the fixed address identification. This proves a summable forced difference bound.

The remaining fixed path terms are affine in the coupling. If $S_i(\beta)$ is such a term, then $S_i(\beta) - S_i(\beta') = (\beta-\beta')\{S_i(1)-S_i(0)\}$. Their two summable derivative prices therefore bound the difference by t times a summable array. A finite symmetry average preserves summability after its explicit fixed derivative factor. Any support weight $e^{Ac(Y)}$ on this fixed torus is at most $e^{A|\Lambda|}$, and moving an anchor costs at most a fixed power of $1 + \operatorname{diam}\Lambda$. Hence all these operations convert the preceding sums into the prescribed regular array budget in the jet norm and forced array budget in the value supremum norm, bounded by Kt for a finite K depending on the base representation and torus.

This also supplies the common majorant needed along a subsequence. If input parameters x_n tend to x, select an increasing subsequence n_k for which the output differences from x have summable nonnegative prices $d_{k,i}$ with $\sum_i d_{k,i} \le 2^{-k}$. If h_i majorizes the output at x, put

$$H_i = h_i + \sum_k d_{k,i}.$$

Tonelli's theorem gives $\sum_i H_i < \infty$, and the triangle inequality shows that H_i bounds both the output at x and all outputs on this subsequence, component by component in the required jet or value norm. The regular and forced arrays are treated separately in this construction. It proves the required continuity and subsequential domination in the common dictionary, conditional on the local compensated difference estimate (553) and its forced counterpart.

103 Supplying the common comparison estimates

The preceding comparison arguments use one collection of local bounds for the base representation. We show how the local estimates already proved provide that collection, and how its parameters are chosen before a running representation is inserted. Constants in a fixed-volume continuity estimate may depend on that volume. The radius of the allowed coarse variations and the large-coupling threshold used for the construction have a stronger uniformity.

Fix the branch data, a positive target support reserve, the derivative order j, the cutoff convention, and positive width and packet-energy parameters. The scale factor is taken above the common floor in the local activity and Gaussian estimates. Fix a coarse variation scale $v(x) \geq 0$ of upper exponent $-1/2$. All estimates in this section refer to the same fine representation, sector, tags and held-field compensation.

103.1 Elementary factors and conditional packet actions

The construction in Section 87 provides prices for the elementary compensated factors together with their support-dependent derivative scales. At the fixed derivative order, choose a positive spare support and extra-cost reserve δ. If s denotes the support-cover plus length charge, then $(1+s)^j \leq C_{j,\delta} e^{\delta s}$ for $s \geq 0$: the continuous function $(1+s)^j e^{-\delta s}$ is bounded on $[0,\infty)$. Apply the weighted price estimate with this reserve before removing the weights. This pays the factor $r(x)^j (1+s)^j$ from the derivative scale and gives scale-one jet prices at every order through j. The warning catalogue is finite on the fixed torus and its finite jet prices can also be summed. To pass from a rooted weighted sum to its unweighted sum on a fixed torus, sum the rooted inequality over all its sites. Every nonzero factor has nonempty assigned support and is counted at least once; all removed support and extra-cost weights are at least one. The same argument applies to the base logarithmic prices. It gives both the finite total elementary price and the finite common bound on the absolute values of the real logarithms used in Section 101. The prices need not be small after this volume-dependent summation.

There is a separate condition on the packet action: its differentiated bound is needed whenever the enforcement cutoff is not a zero germ. Write that cutoff as a finite product of its bond cutoffs. If any bond cutoff were zero on a neighborhood of the base point, the whole product would be zero there. Thus a nonzero packet germ makes every relevant bond cutoff a nonzero germ. The compensated cutoff estimates then bound the corresponding increments and their derivatives by

$$s(x) + 2a\tau(x),$$

where s has upper exponent $-1/2$ and $a > 0$ is the fixed compensation radius parameter. The mean coordinates have the same upper exponent, by Section 80. Enlarge these bounds to one nonnegative function of upper exponent zero.

The last subsection of Section 80 applies to an arbitrary held fine field. Each local packet-action summand is a held fine action, a varying coarse action, or an extracted affine square. Its derivatives are bounded by the corresponding local constants there. For a packet with s blocks, the number of such energy addresses is at most

$$s\, b(200B+1)\, n_L,$$

where $b(r)$ is the finite lattice-ball count at radius r and n_L is the fixed number of fine energy addresses per coarse block. Summing the local jet bounds, including the far-field term, gives $A(x)s$ with one finite $A(x) \geq 0$. The same bound holds on every admissible chart and for every held fine field. In particular this supplies the conditional packet-action estimate, without requiring smoothness of the fine input or an upper bound on its uncut energy.

103.2 Every read subset has the same polynomial bound

Let I be any subset of fine bonds, including the empty subset. The projected covariance contains all colours of those bonds. Write $C_I(y)$ for this covariance and $w_I(y,z)$ for the compensated Gaussian coordinate. Its inverse exists by the common positive covariance estimate. The quadratic exponent is

$$q_I(y,z) = \tfrac{1}{2} w_I(y,z)^T C_I(y)^{-1} w_I(y,z).$$

The inverse-matrix estimates and weighted matrix product estimates of Sections 78 and 85 apply to every principal read subset with the same row constants. This is where the projected covariance bounds, rather than merely positivity of the full covariance, are used.

For clarity, the finite-dimensional multiplication giving the quadratic bound can be seen directly. Suppose the coordinate jets of w_I are at most W, and the jets of the inverse matrix have row sums at most B_r at order $r \le j$. A derivative of $w_I^T C_I^{-1} w_I$ distributes its ordered directions among three factors, giving at most 3^j terms. Sum the middle matrix row first. Each term is at most $|I| \dim \mathfrak{g}$ times $W^2 \max_{r \le j} B_r$. The factor $1/2$ can be discarded in an upper bound. On the full torus the compensated-coordinate estimates give

$$W \le \left[1 + g(\beta)^{-1}\{s(x) + 2A_{\mathrm{mean}}(x)\}\right] F_0(z) = V_0 F_0(z),$$

where F_0 uses every fine bond and colour and $V_0 \ge 1$ is finite. Since $|I| \dim \mathfrak{g}$ is bounded by the full Gaussian dimension, we obtain

$$\|q_I\|_{j,y_0} \le C_* F_0(z)^2 \quad \text{for every } I. \tag{556}$$

All constants on the right are independent of the chosen subset. For $I = \varnothing$, the quadratic form is zero.

The relative density compensation K_I is the Jacobian from Section 79 times the Gaussian density ratio from Section 85. At the base point its value is one. The finite-order determinant and inverse estimates there give, for some integer k and nonnegative finite $U(x)$, a derivative scale bounded by $U(x) F_I(z)^k$, where

$$F_I(z) = 1 + |I| \dim \mathfrak{g} + \sum_{(e,a) \in I \times \mathfrak{g}} |z_{e,a}|.$$

In this expression the colour index means a fixed orthonormal basis. The bound uses only the prescribed derivative order, the local row prices and the compensation Jacobian bounds. Its polynomial degree is fixed before the subset is selected. Since $1 \le F_I \le F_0$,

$$\max_{0 \le r \le j} \{U(x) F_I(z)^k\}^r \le (1 + U(x))^j F_0(z)^{kj}.$$

Together with (556), this supplies (542) simultaneously for all I, with $N = kj$ and $C_0 = (1 + U(x))^j + C_*$. The quadratic constant C_* is allowed to depend on the torus and base coupling; no uniform volume bound is asserted for it.

The same compensated map provides the held-field and protected-patch identities. The open-neighborhood envelope in Section 85 justifies differentiation of its integrals. Thus the algebraic pointwise comparison uses the actual integrands to which that differentiation theorem applies.

103.3 Unrestricted packet values

The lower packet-energy bound in the seed estimates applies on the support of the packet mask. Adding the nonnegative far-field energy preserves this lower bound. At a fixed stage it gives $u \ge$

$-E(x)s \geq -E(x)s_*$ in the notation of (549). The product of profile, charge and enforcement factors lies in $[0,1]$. Its remaining ticket factor is bounded by $\exp(|c_{\mathrm{ticket}}|s_*)$, so this number can be taken as M_0.

Each condensation value has a finite base bound, and each local Haar-chart correction has a finite absolute bound. There are at most as many terms as fine bonds. The remaining normalization term is $-n_s \log m(\beta)$, again with n_s no larger than that finite bond count. Their sum gives a finite A_0 bounding v_β from above. Only this last normalization term changes between the compared representations; hence $|v_{\beta'} - v_\beta| \leq n_* t$. These are exactly the hypotheses of the damping estimate already proved. Positive unbounded energies are controlled by $\beta' \geq \beta/2$ and require no further cutoff.

103.4 Choosing the common variation scale and thresholds

We spell out the quantifier order needed for a running history. The common scale-factor floor is chosen from the elementary-activity and Gaussian local estimates before the derivative order, cutoff convention and positive width parameters are supplied. For a scale factor above that floor and fixed branch data, the compensation chart has a fixed positive radius. Its protected-field requirement is eventually satisfied because $\tau(x) \to 0$.

The elementary activity estimate is applied at the prescribed scale $v(x)$ and the compensated-increment scale $s(x)$, both of upper exponent $-1/2$. The packet-action, Jacobian, density and quadratic estimates just established have eventual validity at those same scales. Take the intersection of their finitely many eventual conditions. This fixes one threshold in x before the finite tori and the incoming representation are supplied. The associated constants and summable prices are then instantiated at that actual representation. In particular, a later fixed-volume comparison neighborhood may depend on its total prices and covariance, as the continuity proof permits.

To combine this with the common integrated derivative estimate, choose a fixed $Z > 0$ large enough for the normalized exponential coarse variations at derivative order j. Set

$$t_0(x) = 2\tau(x), \qquad v(x) = (j+1)Zt_0(x).$$

This function is eventually positive and has upper exponent $-1/2$. The elementary inequality $jZt_0 \leq v$ pays the ordered variation directions. Apply both the local comparison construction and the integrated Gaussian estimate at this same v. The latter supplies the ordinary component price and derivative-scale data. The cluster budget construction, with the strictly positive target support reserve, then supplies the ordinary cluster criterion with its strict spare margin. All are evaluated at the same incoming data; no alternate representation is substituted between these steps.

Finally, a schedule has only finitely many transitions at each fixed stage. Take the maximum of their scale-factor floors at that stage. After those factors are fixed, take a common upper threshold for the finitely many transition conditions, enlarge the stage entry coupling to that threshold, and preserve any earlier lower bound. Increasing entry couplings further preserves these conclusions. Consequently the comparison data and cluster margin are available uniformly for each already supplied transition satisfying the underlying sector, shell-access, protected-chart and incoming activity hypotheses. This selection does not construct those incoming data.

For an application to a running curve one must also retain its renewed price bounds, the geometric bounds on the downward map, the admissible initial action, and the inequality preExponent $/B \leq$ width $/2$. The curve has the same clipping and path data and the same regular and forced dictionaries. Its displayed output must equal the constructed nonlinear output under the chosen dictionary identifications. With these hypotheses, the fixed-volume theorem applies to that actual output, giv-

ing regular jet and forced value control and its subsequential summable majorants. The construction of an entire history satisfying all these hypotheses remains a separate step.

104 The remaining small-field budget inequalities

The nonlinear step uses several estimates at different small powers of the coupling. We verify their numerical compatibility before choosing the stage thresholds. Throughout this section the block factor and stage data are fixed. Write $z = \beta_F(x)$ for the shifted fine coupling, $B = B(x)$ for the block radius, and $\tau = \tau(x)$ for the fluctuation threshold. Thus $z \to \infty$, B has upper exponent zero, and τ has upper exponent $-1/2$. We also use the stronger growth facts $B/\log z \to \infty$ and $P_F(x)^2/\log z \to \infty$ for the actual radius and fine monitor. They follow from their prescribed logarithmic growth laws; upper exponent zero alone would not suffice for the exponential estimates. All constants may depend on the fixed derivative order, spatial weights, path prices and positive support-width reserves.

104.1 The low-degree action and its support tail

The explicit low-degree correction has four contributions: the quadratic reset term, cubic and quartic bare-action terms, and a quadratic loop term. Their rooted jet prices, with the fixed support weights already included, have the form

$$u \le C_2|\alpha|\tau^2 + C_3|\beta|\tau^3 + C_4|\beta|\tau^4 + C_{\mathrm{loop}}\Lambda\tau^2. \tag{557}$$

The constants include the finite coordinate and orientation costs, and the anchored support prices for the corresponding degrees. This is the direct sum of the quadratic and homogeneous action bounds: a degree-r term tested in directions of size τ contributes τ^r, while its coefficient contributes the displayed $|\alpha|$, $|\beta|$, or Λ. The stage fixes the derivative order, spatial power and support exponent, so these constants are unchanged when its actual threshold varies.

Assume $|\alpha| \le A$, $|\beta| \le 2z$, and $0 \le \Lambda \le \Lambda_0(x)$, where A is fixed and Λ_0 has upper exponent zero. The four terms in (557) have upper exponents respectively $-1, -1/2, -1, -1$. Hence one common nonnegative majorant for u has upper exponent $-1/2$. Since $-1/2 < -7/16$, eventually

$$u \le z^{-7/16}. \tag{558}$$

If $\sigma > 0$ is the fixed support width, then $e^{-\sigma B/4}$ has every negative upper exponent. Multiplying the same majorant by this tail gives, after increasing the threshold,

$$e^{-\sigma B/4}u \le z^{-23/20}. \tag{559}$$

Also $(a+3)/B \le \sigma/4$ eventually for any fixed target support exponent a. The thresholds here are uniform over the stated ranges of α, β, Λ and over the actual finite torus.

The reset bound itself follows from the existing coupling estimate. If the running reset satisfies $|\alpha + c_G \log L| \le A_0$, then $|\alpha| \le A_0 + |c_G \log L|$. Enlarging the right side by one makes a fixed positive bound available for this chosen block factor. It does not require selecting a different reset at each application.

104.2 The strengthened cluster remainder

Let $D = z^{3/8}$ be the extra-degree weight. Write $\mathcal{P}_\eta = C_j e^\eta \eta^{-pj}$ for the derivative price at positive reserve η, where $j, p \in \mathbb{N}$ are the fixed derivative order and spatial power; the fixed factor C_j includes

$(pj)!$ and the lattice scale factor. Suppose the ordinary component/cluster budget at reserve $i > 0$ has already given

$$\mathcal{P}_i \, S(1 - 1/2)^{-1} \le z^q, \qquad q \le -7/16.$$

Here $S \ge 0$ is the ordinary component activity price and i lies below the fixed available reserve $y > 0$. The extra-degree argument of Section 100 gives a residual bound

$$D^{-3}\mathcal{P}_y, DS(1 - 1/2)^{-1}.$$

Since $i \le y$, the inverse powers satisfy $y^{-pj} \le i^{-pj}$, while $e^i \ge 1$. Consequently $\mathcal{P}_y \le e^y \mathcal{P}_i$. The displayed remainder is at most

$$e^y D^{-2} z^q = e^y z^{q-3/4}.$$

Now $q - 3/4 \le -19/16 < -23/20$. The strict gap absorbs the fixed factor e^y and proves the common residual budget $z^{-23/20}$. This calculation consumes the same ordinary price S used in the first cluster expansion; it does not assume an independent price for the strengthened expansion.

104.3 Polynomial localization costs and the incoming allowance

The auxiliary periods in the five-period construction have side lengths bounded by a fixed polynomial in B. Their volumes, local ball counts and ordered-derivative transport costs therefore all have upper exponent zero. Multiplication by any of them preserves the upper exponent of a price.

In particular, localizing the ordinary error of size $z^{-7/16}$ produces a total auxiliary error of upper exponent $-7/16$. A further fixed support conversion, including the constant exponential of the support-cover bound, still has that exponent. Since $-7/16 < -1/8$, its final price is eventually at most $z^{-1/8}$. Similarly, localizing a price $z^{-23/20}$, including the factor $j!3^j$ from Section 110, gives a price of upper exponent $-23/20$.

Let $G(x)$ be the sum of the latter localized price and the finitely many unlocalized $z^{-23/20}$ remainders, each with its fixed coefficient. Then G also has upper exponent $-23/20$. If $c > 0$ is the fixed regular allowance constant, the strict inequality $-23/20 < -11/10$ gives

$$G(x) \le \tfrac{c}{2} z^{-11/10} \le \tfrac{c}{2} x^{-11/10}$$

eventually; the second inequality uses $z \ge x > 0$. Thus these remainders use at most half of the available regular allowance. All polynomial localization factors are paid before this final comparison.

104.4 Absorbing the forced correction at its prescribed price

A second exponential absorption must retain a prescribed positive forced allowance v, rather than replace it by a larger unspecified constant. Let $a, b \ge 0$ be its hard and soft input prices, with

$$a \le v/8, \qquad b \le z^{-1/8}.$$

Put $p = z^{1/16}$. The actual allowance has the monitor estimate $v \le C \exp(-cP_F(x)^2)$ uniformly for the allowed coupling band, with fixed $C, c > 0$. Thus $vp \le 1$ eventually. Also, uniformly for b in the stated range,

$$b \le 1, \qquad pb \le 1/8, \qquad (2B + 1)^4 \{(8p)^{-1} + 2b\} \le 1$$

eventually: the two powers in the last braces are strictly negative and the prefactor has upper exponent zero.

Choose

$$\kappa = -\log(vp) \ge 0, \qquad I = e^\kappa a + 2b.$$

Then $I \le (8p)^{-1} + 2b$, so $(2B+1)^4 I \le 1$ and in particular $0 \le I \le 1$. Using $e^I - 1 \le 2I$ gives

$$e^{-\kappa}(e^I - 1) \le 2vp\{(8p)^{-1} + 2b\} = v(1/4 + 4pb) \le 3v/4 \le v.$$

These are the exact smallness and output inequalities required by the forced-gas absorption, at the original allowance v. Its uniform monitor estimate follows from the coupling-band comparison for the gap: the normalized monitor argument throughout that band is at least half its value at the center. Squaring it only replaces the positive decay constant by one quarter. All constants and the threshold are chosen before the particular coupling in the band.

Each assertion above is eventually valid for a fixed transition. Take the finite maximum of its required thresholds at each stage, then enlarge the entry coupling accordingly. Auxiliary periods are also chosen above their fixed geometric lower bounds, and the volume growth exponent is increased to at least one. These choices are preserved by any later increase of the delays. They supply the numerical budgets; the actual local action identities, stationary comparison and funded incoming representations are still required where each budget is used.

105 Local action prices for the stationary construction

We verify the localized derivative input of Section 106 for the actual unmasked block action. The action is a sum over sites. Its positive-order coordinate derivatives have uniform local bounds, although its scalar value need not have a volume-independent bound.

Fix the block side L, the patch, the direct clip, and the kernel parameters, including its positive frame and predictor cutoffs and positive penalty coefficient. Fix also an order $n \ge 3$ and an exponential path width $\mu > 0$. The incoming path data have row, cubic and quartic budgets at most C_0 at width μ; the loop budget is at most Λ. These budgets are the sums of the absolute coefficients with their specified exponential length weights. All constants below may depend on this fixed data and on the group, but not on the auxiliary torus, its forest, or the tag choices.

105.1 From a finite read set to a localized derivative

A function based at a site v has read radius R if changing coordinates anchored farther than R from v leaves the function unchanged. A derivative containing any one such coordinate is zero: restrict to a line varying that coordinate, differentiate the constant function, and commute derivatives. If derivatives through order n have operator norm at most A, then, for $q \le n$,

$$|\partial_{i_1} \cdots \partial_{i_q} u| \le A e^{anR} \exp\left(-a \sum_{h=1}^{q} \operatorname{dist}(v, a_{i_h}) \right). \tag{560}$$

If every tested coordinate lies in the read set, the sum of their distances is at most nR; otherwise the left side vanishes. This also holds for vector-valued expressions with norms in place of absolute values.

For a path atom of total length t, the fine forest chart and a signed motion enlarge its read radius to at most $2t + 1 + L$. The path itself travels at most $2t$ in fine units; reversing an oriented bond can move its initial point by one, and the coarse replica coordinate is anchored at most L from the fine bond. This estimate remains valid for winding paths: periodic distance is bounded by lifted path length.

Choose

$$a = \frac{\mu}{4(n+1)(1+L)}.$$

The elementary inequality

$$e^{\mu t/2} e^{an(2t+1+L)} \le e^{\mu t} e^{\mu} \tag{561}$$

follows by comparing exponents: the second exponent is at most $\mu t/2 + \mu$. The holomorphic path bounds at width $\mu/2$ give a derivative price $A_n e^{\mu t/2}$ for a slot, and $A_{n,k}\,\mathrm{magnitude}(q) e^{\mu t/2}$ for a degree-k atom q. To recall why the constants are uniform, each matrix transport is a finite product; on the fixed complex neighborhood its norm is bounded exponentially in path length. The Cauchy estimate on a fixed smaller neighborhood bounds derivatives by that exponential times a constant depending only on n and the neighborhood. Composing with the fine exponential chart uses its uniform link derivatives and the finite partition formula. The fixed color projections and multilinear atom contractions contribute only the constants $A_n, A_{n,k}$. Combining this estimate with (560) and (561) pays the localized derivative by the original width-μ coefficient budget.

For an infinite row or atom family, the paid coefficient sequence is summable. It bounds the values and derivatives through order n on the common coordinate ball. Uniform convergence and the fundamental theorem of calculus along coordinate segments justify termwise differentiation, inductively in the derivative order. The underlying holomorphic series converges normally on a smaller complex neighborhood, so it is smooth to all orders there. Consequently a row based at v has localized derivative cost $C_0 A_n$, and a cubic or quartic family has cost $C_0 A_{n,3}$ or $C_0 A_{n,4}$. A loop family has cost $\Lambda A_{n,2}$. The latter dependence is linear in Λ.

105.2 Summing sites without a volume factor

Suppose a site term u_v has derivatives bounded by

$$|\partial_{i_1} \cdots \partial_{i_q} u_v| \le B \exp\left(-a \sum_h \mathrm{dist}(v, a_{i_h}) \right).$$

Fix a tested coordinate i_t and set $s = \sum_h \mathrm{dist}(v, a_{i_h})$. Then $\mathrm{dist}(v, a_{i_t}) \le s$ and

$$\sum_h \mathrm{dist}(a_{i_t}, a_{i_h}) \le (q+1)s \le (n+1)s.$$

Splitting the exponent in two gives

$$e^{-as} \le e^{-(a/2)\,\mathrm{dist}(v, a_{i_t})} \exp\left(-\frac{a}{2(n+1)} \sum_h \mathrm{dist}(a_{i_t}, a_{i_h}) \right).$$

Summing v costs only the distance-shell sum at width $a/2$. If at most h terms have the same base site, multiply this cost by h. We have obtained the bound (562), with width $a/[2(n+1)]$, for every positive derivative of the site sum. The argument needs a tested coordinate as root, so it makes no assertion at order zero. Averaging over signed motions preserves the cost: each motion permutes sites and the averaging weights sum to one.

The classical site density has three types of terms. The direct plaquette term has a fixed read radius and derivative bounds from its four unitary factors and the smooth group penalty. The squared-row term is $\|\ell(U)\|^2/2$. Its row value and derivatives are bounded by the row budget just proved; the finite product rule bounds its derivatives by a constant depending on C_0. Its localized bounds are preserved by that product rule, since the exponential factors for the split derivative slots multiply to the factor for all slots. Finally the cubic and quartic terms have the atom bounds above. Adding these terms and summing sites proves the required localized derivative estimate for the fine classical action. The classical price depends on C_0 and the direct clip, and does not depend on the loop budget Λ.

105.3 The prediction penalty on the same ball

The minimizing frame and predictor of Section 69 are smooth near the identity. The finite-read transplantation argument of Section 80 makes their fixed-order neighborhood and derivative bounds uniform over periods. We must also verify that all tags select that same branch there.

On a coordinate ball of radius r, each fine bond chord is at most $C_1 r$, by integrating its first derivative from the identity. Every path used by a frame is of bounded length for the fixed patch, so its transport chord is at most $C_2 r$. Shrink r until the strict frame test holds. The uniqueness of the normalized minimizing frame identifies all frame tags there. Its coordinates and first derivatives are uniformly bounded on a smaller ball. A connector contains the frame at each of its two endpoints and a straight fine path. Its chord is therefore at most $C_3 r$, with a common constant $C_3 > 0$.

Let Y_i be these connectors and $w_i \geq 0$ their weights, $\sum_i w_i = 1$. Comparing the prediction cost with the identity gives

$$\min_u \sum_i w_i \operatorname{price}(Y_i u^{-1}) \leq \tfrac{1}{2}(C_3 r)^2.$$

Choose $r < \sqrt{\zeta}/C_3$, where $\zeta > 0$ is the predictor cutoff. This inequality is then strictly below the cutoff. Also

$$\left\| \sum_i w_i \operatorname{mat}(Y_i) - 1 \right\| \leq c_G C_3 r,$$

where c_G converts chord to matrix norm. Shrink r further so this lies in the fixed nearest-point tube. The selected predictor is the smooth nearest-point projection of this weighted matrix average. Hence every predictor tag agrees with this same smooth expression throughout the ball. Product and composition rules bound its derivatives uniformly, using the same actual endpoint frames in both occurrences.

For each coarse bond, the smoothing penalty is the group price of the coarse field times the inverse predicted field. Its value and its derivatives through order n have a common bound on this ball. Its read set is the finite union of the patch paths at the two endpoints and the coarse bond coordinate. Apply (560) with that fixed radius, then the site-summation argument. The embedded coarse sites are distinct; there are four coarse bond directions at each such site. Thus this sum has the same localized derivative bound, with multiplicity four, multiplied by the fixed kernel penalty coefficient.

The full tested action is the sum of this smoothing penalty and the fine classical action. Taking the smaller common radius and adding their costs proves (562) for the actual action, with all tags equal there. At the identity the curvature slots, plaquette cost and prediction discrepancy vanish; their first derivatives in the full action vanish as well, since the direct penalty is stationary and every other term has degree at least two in these vanishing slots. Thus $f(0) = 0$ and $D_y f(0) = 0$. Together with the tree-gauge Hessian gap, this supplies all inputs to the uniform stationary construction. The argument fixes the block and path budgets first; it does not assert a new estimate for the evolution of those budgets along a history.

106 A uniform stationary chart and its auxiliary-volume derivatives

We construct the stationary branch on the real coordinate neighborhood used by the nonlinear step. The neighborhood must be uniform in the auxiliary torus; the subsequent derivative bounds may pay a fixed polynomial in its volume. These are different requirements.

106.1 Local derivative data and the Hessian gap

Let I and J be the coarse and free fine real-coordinate sets, and write $w = (x, y) \in \mathbb{R}^I \times \mathbb{R}^J$. Both coordinate spaces initially carry the supremum norm. Give every coordinate an anchor a_i on the fine torus, with at most K coordinates at each site. Put

$$S_a = K \sum_{r \geq 0} (2r+1)^4 e^{-ar} \qquad (a > 0).$$

The distance-shell bound implies $\sum_i e^{-a \operatorname{dist}(v, a_i)} \leq S_a$, uniformly in the torus and in v.

Fix an order $N \geq 3$. The local action estimates provide a smooth real function $f(x, y)$ on $\|w\|_\infty < r_0$ and constants $a, C, m, r_0 > 0$ such that $f(0) = 0$, $D_y f(0) = 0$, and

$$|\partial_{i_1} \cdots \partial_{i_q} f(w)| \leq C \exp\left(-a \sum_{h=1}^{q} \operatorname{dist}(a_{i_t}, a_{i_h}) \right) \quad (1 \leq q \leq N, 1 \leq t \leq q). \tag{562}$$

The vertical Hessian $H(0) = D_y^2 f(0)$ satisfies $H(0) \geq mI$ in the Euclidean quadratic-form sense. The constants and coordinate multiplicity are independent of the auxiliary torus after the block data have been fixed. The Hessian gap is the tree-gauge energy comparison in Section 138.2; the derivative estimate is the additional quantitative local-action input.

For two points v, w in the original coordinate ball, differentiate a Hessian entry along their joining segment. Applying (562) with the first Hessian coordinate as root, and summing the varied coordinate, gives

$$|H(v)_{ij} - H(w)_{ij}| \leq C S_a \|v - w\|_\infty e^{-a \operatorname{dist}(a_i, a_j)}.$$

Both row and column sums of the difference are therefore at most $C S_a^2 \|v - w\|_\infty$. Its Euclidean operator norm and its supremum operator norm have this same bound. Choose

$$r_H = \min\left\{ r_0/2, \frac{m}{2(1 + C S_a^2)} \right\}.$$

Then $H(w) \geq (m/2) I$ on $\|w\|_\infty \leq r_H$. The second-derivative case of (562) also gives uniformly bounded exponentially weighted row and column sums of $H(w)$, after reducing a to $a/2$. The inverse argument of Section 74, applied with lower bound $\min(1, m/2)$, supplies an exponential bound on $H(w)^{-1}$. Summing that bound gives a fixed constant $J_0 \geq 1$ with

$$\|H(w)^{-1}\|_{\infty \to \infty} \leq J_0. \tag{563}$$

The Euclidean bound $\|H(w)^{-1}\|_{2 \to 2} \leq 2/m$ follows directly from the positive gap. No conversion between the two vector norms is used to obtain either uniform bound.

106.2 Contraction on a common coordinate ball

Let $F(x, y) = D_y f(x, y)$, viewed as its array of real coordinates. Choose

$$r_V = \min\{ r_H, [4 J_0 (1 + C S_a^2)]^{-1} \},$$
$$r_X = \min\{ r_V, r_V / [4 J_0 (1 + C S_a)] \}.$$

For $\|x\|_\infty < r_X$, define on the closed ball $\|y\|_\infty \leq r_V$

$$T_x(y) = y - H(0)^{-1} F(x, y).$$

The Hessian difference estimate gives

$$\|D_y T_x(y)\|_{\infty\to\infty} \le J_0 C S_a^2 r_V \le \tfrac{1}{4}.$$

The second-derivative bound, summed with its vertical coordinate as root, gives

$$\|F(x,0) - F(0,0)\|_\infty \le C S_a \|x\|_\infty.$$

Consequently $\|T_x(0)\|_\infty \le r_V/4$ and

$$\|T_x(y)\|_\infty \le \tfrac{1}{4}\|y\|_\infty + r_V/4 \le r_V/2.$$

Thus T_x maps the complete closed ball into itself and contracts by $1/4$. Its iterates from zero form a Cauchy sequence: consecutive differences are bounded by a geometric sequence of ratio $1/4$. Their limit $\rho(x)$ is a fixed point by continuity. If two fixed points existed, their distance would be at most one quarter of itself, so they would coincide. We have proved

$$\|\rho(x)\|_\infty \le r_V/2, \qquad F(x,\rho(x)) = 0, \qquad \rho(0) = 0,$$

with uniqueness among all stationary points in the closed r_V-ball.

At each $(x,\rho(x))$ the vertical derivative of F is the invertible Hessian $H(x,\rho(x))$. The finite-dimensional smooth implicit-function theorem therefore supplies a local smooth stationary graph. It remains inside the open r_V-ball after restricting its neighborhood, and uniqueness identifies it with the graph just constructed. These local identifications prove smoothness of ρ throughout $\|x\|_\infty < r_X$. If several local tags give the same function on the original coordinate ball, the constructed graph is stationary for every tag and is the same graph for all of them. No extension outside the common ball is asserted.

Differentiating $F(x,\rho(x)) = 0$ once gives

$$D\rho(x) = -H(x,\rho(x))^{-1} D_x F(x,\rho(x)).$$

The rooted row sum bounds $\|D_x F\|_{\infty\to\infty}$ by CS_a. Hence $\|D\rho\|_{\infty\to\infty} \le J_0 C S_a$. Integration along a coarse line segment proves

$$\|\rho(x) - \rho(x')\|_\infty \le C_\rho \|x - x'\|_\infty, \qquad C_\rho = 1 + J_0 C S_a. \tag{564}$$

In particular the constant used to spread fine-field slack is independent of the auxiliary volume.

106.3 Fixed-order derivatives with polynomial volume cost

Let $W \ge 1$ bound $|I| + |J|$. We now allow polynomial costs in W. A scalar q-linear form with coordinate entries bounded by C has supremum operator norm at most CW^q: expand its arguments in coordinate basis vectors and bound the resulting at most W^q terms. For a vector-valued form with supremum output norm the same bound holds coordinate by coordinate. Passing that output to its Euclidean norm costs at most $\sqrt{W} \le W$.

These elementary conversions turn (562) into polynomial bounds for the ordinary derivatives of f through order N. Its value has the same kind of bound on the fixed ball, because $f(0) = 0$ and the first derivative can be integrated radially. They also give polynomial bounds for every derivative of ρ through order $N-1$. For completeness, differentiate $F \circ \Gamma = 0$, where $\Gamma(x) = (x,\rho(x))$. At order $k \ge 2$, the partition formula reads

$$0 = H(\Gamma(x)) D^k \rho(x) + \sum_{\substack{\pi \text{ a partition of } \{1,\dots,k\} \\ |\pi| \ge 2}} D^{|\pi|} F(\Gamma(x)) [D^{|B|}\Gamma(x) : B \in \pi].$$

Every block in the sum has size less than k; the single-block term is exactly the isolated Hessian term, since the coarse component of $D^k\Gamma$ vanishes. The first derivative was already bounded in (564). Inductively, if the lower derivatives have polynomial bounds $P_h(W)$, the sum is bounded by

$$C \sum_{\substack{\pi \vdash \{1,\dots,k\} \\ |\pi| \geq 2}} W^{|\pi|} \prod_{B \in \pi} (1 + P_{|B|}(W)).$$

Multiplying by J_0 bounds $D^k \rho$. This is a fixed polynomial in W at every fixed k, and uses derivatives of f only through order $k+1$. The Euclidean output version follows by the preceding factor W. Thus the coordinate conversion does not affect the uniform radius or the uniform first-derivative bound.

Put $M(x) = H(\Gamma(x))$. The chain rule gives polynomial bounds for its entry derivatives through order $N-2$. A matrix with entries bounded by B has Euclidean operator norm at most WB, so these are polynomial operator bounds as well. The inverse identity gives, for each nonempty set of derivative slots A,

$$\partial_A M^{-1} = -M^{-1} \sum_{\varnothing \neq B \subseteq A} (\partial_B M)(\partial_{A \setminus B} M^{-1}).$$

All inverse derivatives on the right have lower order. Together with $\|M^{-1}\|_{2\to 2} \leq 2/m$, this proves polynomial bounds by induction. Finally

$$D \log\det M[h] = \operatorname{tr}(M^{-1} DM[h]).$$

Repeated product differentiation and $|\operatorname{tr} A| \leq W \|A\|_{2\to 2}$ give polynomial bounds for every positive derivative of $\log\det M$ through order $N-2$. Its centered value is bounded by integrating the first derivative. Its uncentered value is bounded too: the eigenvalues lie between $m/2$ and a fixed upper row-sum bound, so the sum of their logarithms has absolute value at most a fixed constant times W. In particular all determinants used here are positive.

106.4 The loop exponent and the Gaussian comparison data

The remaining exponent is

$$\psi(x,y) = \ell(\mathcal{D}_x(y)) - \sum_{i \in J_{\mathrm{bond}}} \log J_H(y_i),$$

where J_{bond} denotes the free bonds and J_H the single-bond Haar Jacobian. The notation separates the bond set from the earlier real-coordinate set J. On the fixed combined coordinate unit ball, the actual loop coefficient budget bounds each coordinate derivative through the requested order by a fixed constant times Λ, summed over at most W sites. Its exponentially summable coefficient series converges normally there, which justifies differentiation before summation. Conversion to operator derivatives costs at most a further power of W. On the half-radius Haar chart, Section 77 bounds the derivatives of $\log J_H$. Summing over at most W free bonds and composing with the fixed Lie-coordinate maps has polynomial cost in W. Thus the ψ bounds have the form

$$P_0(W) + P_1(W)\Lambda$$

for fixed nonnegative polynomials P_0, P_1.

Apply the preceding construction at order $N = n+3$. It supplies bounds through order $n+2$ for ρ, through order $n+3$ for f, and through order $n+1$ for the relative log determinant. The

loop and Haar estimates are purchased through order $n+4$. Take coarse radius u and fluctuation parameter radius v with

$$(1+C_\rho)u+vD<\min(r_H,1),\qquad u\le r_X,\quad c_{\mathrm{Lie}}(C_\rho u+vD)\le r_H^{\mathrm{Haar}}/2.$$

Here c_{Lie} is the fixed norm of the Lie-coordinate map and $r_H^{\mathrm{Haar}}>0$ is the Haar-chart radius. On every segment $(x,\rho(x)+tbz)$, with $\|x\|_\infty<u$, $|b|<v$, $0\le t\le 1$ and $\|z\|_2\le D$, these inequalities verify the needed action, coordinate and Haar domains. The stationary Hessian is positive with inverse covariance bound $2/m$, and stationarity removes the linear vertical Taylor term. These are exactly the data required in Section 107.

For the selected auxiliary volumes, W is bounded by a polynomial in the auxiliary size parameter. All the prices obtained here are therefore polynomial in that parameter, except for the single linear loop factor. With $\Lambda\le 2P_F$ their common bound can be written $p_0+p_1P_F$, with polynomially bounded p_0,p_1. This verifies the precise price dependence used in Section 108. The remaining quantitative inputs are the local-action estimate (562), the initial Hessian gap, and the loop coefficient and Haar-chart budgets. The construction never changes the selected finite-volume action or its tags.

107 The logarithm of a truncated stationary Gaussian integral

The local small-field comparison uses a normalized integral near the actual stationary point. Its first correction has order β^{-1}. Two details matter: the cutoff is retained when the coupling parameter vanishes, and the stationary Hessian varies with the coarse field. We prove the corresponding parameter estimate with both features.

107.1 The normalized integral

Let E be the finite-dimensional space of coarse logarithmic coordinates and let $\mathbb{R}^d$ be the fine integration space, with its Euclidean norm. Let $\varphi(x,y)$ be the real action, $\psi(x,y)$ the additional real exponent, and $\rho(x)$ the stationary point. Assume

$$D_y\varphi(x,\rho(x))=0,\qquad D_y^2\varphi(x,\rho(x))=M(x)>0.$$

On the parameter domains under consideration assume smoothness of ρ, φ, ψ and $\frac12\log\det M$, with derivatives through the orders used below. More precisely, a derivative estimate through order k for the integral uses orders $k+2$ for φ and ρ, and order k for ψ and the relative log determinant, on every segment $(x,\rho(x)+tbz)$ with $0\le t\le 1$ and $|z|\le D$. Let their common bounds be p. These are bounds on the actual moving stationary branch and its Hessian, not on a separately chosen quadratic approximation.

Define the smooth divided difference

$$V(b,x,z)=\int_0^1(1-t)D_y^2\varphi(x,\rho(x)+tbz)[z,z],dt.$$

Taylor's formula and stationarity give

$$V(b,x,z)=\frac{\varphi(x,\rho(x)+bz)-\varphi(x,\rho(x))}{b^2}\quad(b\neq0),\qquad V(0,x,z)=\tfrac12 z^TM(x)z.$$

The integral formula supplies its derivatives also at $b = 0$. Put $s(x) = \frac{1}{2} \log\det M(x)$ and

$$A(b,x,z) = s(x) - s(0) + V(0,0,z) - V(b,x,z) \\ + \psi(x,\rho(x)) - \psi(x,\rho(x)+bz).$$

With γ_0 the Gaussian of covariance $M(0)^{-1}$, set

$$R(b,x) = \int_{|z|\le D} e^{A(b,x,z)}\, d\gamma_0(z), \qquad L(b,x) = \log R(b,x). \tag{565}$$

The normalization is fixed at the base point in the measure; the varying determinant is explicitly present in A.

At $b = 0$ the density identity gives exactly

$$R(0,x) = \gamma_x\{|z| \le D\}, \qquad \gamma_x = N(0, M(x)^{-1}). \tag{566}$$

Indeed the factor $e^{s(x)-s(0)}$ converts the determinant prefactor, and $V(0,0,z) - V(0,x,z)$ converts the quadratic exponent. The two ψ terms cancel. The result is the truncated Gaussian mass, which need not equal one.

There is also an exact parity identity

$$R(-b,x) = R(b,x). \tag{567}$$

Changing z to $-z$ preserves the ball and the centered Gaussian. The segment $\rho(x) + t(-b)(-z)$ equals the original segment, and the two appearances of z in the quadratic evaluation cancel their signs. Thus $A(-b,x,-z) = A(b,x,z)$, proving the identity before taking any expansion.

107.2 Uniform derivatives on the truncated ball

The chain rule in the integral defining V gives a polynomial bound $P_k(p,D)$ for all derivatives of A through order k, uniformly for $|z| \le D$. To see the polynomial nature, the map $(b,x) \mapsto (x, \rho(x) + tbz)$ has derivatives bounded by fixed polynomials in $1 + p + D$; each chain-rule term has at most k factors. Evaluating the two additional slots at z costs at most D^2. Integration against $1 - t$ adds no large constant. The ψ and log-determinant terms have the same type of bound.

Purchase $k \ge 1$ (replace a requested order zero by order one here). Suppose $|b| < v \le 1$, $\|x\| < u \le 1$, and $P_k(p,D)\max(u,v) \le 1$, enlarging P_k to include its first order bound. Since $A(0,0,z) = 0$, integration along the parameter segment from zero gives $|A(b,x,z)| \le 1$. Derivatives of e^A through order k are consequently bounded by another fixed polynomial in P_k, times e. The cutoff ball is fixed in these parameters, and the reference Gaussian has mass one. This bound is integrable and uniform on the open parameter domain, so differentiation under the integral is justified at every order through k.

If $\gamma_0\{|z| \le D\} \ge 1/2$, then

$$e^{-1}/2 \le R(b,x) \le e. \tag{568}$$

It follows by integrating $e^{-1} \le e^A \le e$ on the ball. Derivatives of log are bounded on this fixed positive interval, so the chain rule gives a common bound for all derivatives of L through order k. If p, D have upper exponent zero, this bound also has upper exponent zero. The smallness condition on u, v holds eventually whenever both have a fixed negative upper exponent.

107.3 The cutoff error when the dimension grows

The auxiliary tori have dimension at most $Q(x)$, where Q is bounded by a fixed polynomial in $1 + (\log \beta_F(x))^2$. Suppose $M(y)^{-1} \le cI$ uniformly on the coarse-field ball, with $c > 0$ fixed. Let $P_F(x)$ be the fine monitor and take $D = d_0 P_F(x)$, where $d_0 > 0$ is fixed. For the actual fine monitor, $P_F(x)/\log z \to \infty$, and every fixed polynomial in $\log z$ is eventually bounded by $P_F(x)^\epsilon$ for every $\epsilon > 0$, where $z = \beta_F(x)$. These stronger growth properties are used in the following tail estimate.

Here is a dimension-explicit tail estimate. If a centered Gaussian has covariance at most cI in dimension at most $P \ge 1$, then for $r, s \ge 0$,

$$\mathbb{P}\{|Z| > Pr\} \le 2P \exp(-sr + cs^2/2).$$

If every coordinate has absolute value less than r, then $|Z|^2 \le Pr^2 \le P^2 r^2$. A union bound and the one-coordinate exponential-moment estimate therefore prove the assertion. For $P = 1 + Q(x)$, the monitor growth gives $P \le \sqrt{P_F(x)}$ eventually. Choose $r = d_0\sqrt{P_F(x)}$ and $s = r/c$. Then $Pr \le D$ and

$$1 - \gamma_y\{|z| \le D\} \le 2(1 + Q(x)) \exp\{-d_0^2 P_F(x)/(2c)\}. \tag{569}$$

The right side has every negative upper exponent. In particular it is at most $1/2$ eventually, proving the mass condition above.

If $0 \le \epsilon \le 1/2$ and $1 - \epsilon \le r \le 1$, integration of $1/t$ on $[r, 1]$ gives $|\log r| \le 2\epsilon$. Applying this to (566) shows that $L(0, y)$ has a value smaller than every fixed negative power of β_F, uniformly on the allowed coarse ball. Derivatives require an additional argument; a small value by itself is insufficient.

107.4 Turning a small value into small jets

We give the finite-order interpolation used for that purpose. Fix $j \in \mathbb{N}$, $a > 0$, and real d, m. There is an exponent B such that, for all sufficiently large z, the following implication holds in any finite-dimensional normed spaces. If on $\overline{B}(y, z^{-a})$ a function F has derivatives through order $j + 1$ bounded by z^d, and its value is bounded by z^B, then its jets through order j at y are bounded by z^m.

The elementary one-step estimate is

$$\|D^{r+1}F(y)\| \le 2v/s + Cs$$

whenever $\|D^r F\| \le v$ and $\|D^{r+2}F\| \le C$ on the radius-s ball. Subtract the tangent map of $D^r F$ at y, integrate its derivative along a segment of length s, and use the second-derivative bound to control the tangent-map error. This gives the estimate in every unit direction, hence in operator norm, without a dimension factor.

Induct on j. The case $j = 0$ takes $B = m$. In the induction step choose

$$c = a + \max(0, d - m) + 1, \qquad b = m - 1 - c, \qquad s = z^{-c}.$$

Eventually $2s \le z^{-a}$. Apply the induction hypothesis with target exponent b on every radius-s ball whose center lies within s of y; those balls lie in the original domain. A sufficiently negative common value exponent B then bounds jets through order $j - 1$ by z^b on that inner ball. The one-step estimate bounds the order-j derivative at y by

$$2z^{b+c} + z^{d-c}.$$

Both exponents are strictly less than m, and $b < m$ also bounds all lower derivatives. Increasing the lower threshold on z proves the assertion. This is a finite induction, with all exponents chosen before the function or dimension is supplied.

Apply it to $F(y) = L(0, y)$ on a microscopic ball of radius z^{-1} inside the coarse-field domain. The derivative bound of upper exponent zero is at most z eventually. The tail estimate provides any value exponent B the induction requests. Hence, for every fixed real e,

$$\|L(0,\cdot)\|_{J^j,y} \le z^e \tag{570}$$

eventually, with the stated microscopic margin.

107.5 The even second-order gain

Purchase two additional parameter derivatives in the earlier bounds. For every $r \le j$, the function $b \mapsto D_y^r L(b, y)$ is even by (567). Its first derivative at zero vanishes. If its second derivative is bounded by $C(x)$ for $|b| \le \epsilon$, Taylor's formula gives

$$\|D_y^r L(\epsilon, y) - D_y^r L(0, y)\| \le C(x)\epsilon^2.$$

The derivative constant C has upper exponent zero. Combining with (570), assuming $\beta > 0$ and $\epsilon = \beta^{-1/2} < v$, we obtain

$$\|L(\beta^{-1/2},\cdot)\|_{J^j,y} \le C(x)\beta^{-1} + z^e. \tag{571}$$

This is the required first-order inverse-coupling price, uniform for the prescribed auxiliary volumes and coupling band, with the closed microscopic ball $\overline{B}(y, z^{-1})$ contained in the coarse-field parameter domain, provided the actual stationary branch, segment derivative and Hessian bounds stated above hold. Identifying this truncated integral with the full nonlinear push additionally requires its separate exterior estimate; that estimate is not part of the parity argument.

108 The retained stationary region and its exterior

The Gaussian comparison must describe a region of the actual fine configuration space. In particular, the fine-field cutoff must be one throughout that region. We establish this using the funded expansion to find a configuration with slack, then propagating the slack by a uniform Lipschitz estimate.

108.1 Coordinates and the normalized integral

Fix an auxiliary finite torus, a tree gauge, and the actual coarse field $V(x)$ in logarithmic coordinates $x \in X$. Let Y be the Euclidean space of free fine logarithmic coordinates, $d = \dim Y$, and N the number of free fine bonds. Its coordinate map is

$$\mathcal{D}_x(y)_i = \exp(y_i)\, V(x)_i^{\mathrm{rep}}.$$

Here y_i includes the fixed orthonormal identification with the Lie algebra, and tree bonds are restored by the gauge fixing. The space X uses the supremum norm on real color coordinates. The space Y uses its Euclidean norm for Gaussian integration; write $\|y\|_\infty$ for the supremum norm after forgetting that Euclidean norm. Thus $\|y\|_\infty \le \|y\|_2$. The combined chart norm is $\max(\|x\|_\infty, \|y\|_\infty)$. Write $\rho(x)$ for the actual stationary branch, $b = \beta^{-1/2}$, and

$$\mathcal{S}_x = \mathcal{D}_x\big(\rho(x) + b\overline{B}_Y(0, D)\big).$$

This is a compact measurable subset of the free configuration space. All Haar measures in this section are probability measures.

The local Haar density of Section 77 is $c_H J(y_i)$ on each free bond. Set

$$\psi(x,y) = \ell(\mathcal{D}_x(y)) - \sum_{i=1}^{N} \log J(y_i), \qquad M(x) = D_y^2 \phi(x, \rho(x)),$$

where ϕ is the fine classical action plus the smoothing penalty, and ℓ is the fine loop correction. Assume that the stationary chart estimates hold on the region in question, with $M(x) > 0$. Introduce

$$\Phi(x) = \beta\phi(x,\rho(x)) + \psi(x,\rho(x)) + \tfrac{1}{2}\log\det M(x), \qquad \Omega = \pi_\beta(\sqrt{2\pi}\, b)^d c_H^N,$$

where $\pi_\beta > 0$ is the smoothing-kernel prefactor. The tagged integrand of the pure push is

$$U_x(U) = \sum_k \pi_\beta \lambda_k(U) e^{-\beta[a(U)+n_k(U,V(x))]-\ell(U)} \mathbf{1}_{\{\text{fine field healthy}\}}. \tag{572}$$

The nonnegative tag weights satisfy $\sum_k \lambda_k(U) = 1$; a dead tag has zero weight. Tree-gauge integration identifies the pure push with $Z(x) = \int U_x\, dU$. The incoming density and kernel have integrable envelopes, so all restricted integrals below exist.

Suppose every point of $\mathcal{S}_x$ is fine healthy, the whole logarithmic ball is inside the Haar chart, and all live tag actions there equal ϕ. Then the product chart formula and the affine change $y = \rho(x) + bz$ give

$$\int_{\mathcal{S}_x} U_x\, dU = \Omega e^{-\Phi(x)} R(b,x), \tag{573}$$

with exactly the truncated integral R of Section 107. Indeed the Jacobian is $b^d c_H^N \prod_i J(y_i)$, the tag sum is one, and $\beta b^2 = 1$. The quadratic Gaussian factor contributes $(2\pi)^{d/2}(\det M(0))^{-1/2}$; its determinant and the relative determinant in the exponent defining R combine to $(\det M(x))^{-1/2}$. The remaining exponent is precisely $-\Phi(x)$. No tag-count factor occurs. Injectivity of the Haar chart ensures that the inverse image of $\mathcal{S}_x$ within that chart is the chosen closed ball, so this change of variables also handles its boundary.

108.2 The energy cost outside the retained region

Let $\gamma > 0$ be the stationary chart gap and put $\Delta = \gamma D^2/4$. On the common safe chart assume the fine Hessian satisfies $D_y^2\phi \geq (\gamma/2)I$. Stationarity and Taylor's integral formula then imply

$$\phi(x,y) - \phi(x,\rho(x)) \geq \frac{\gamma}{4}\|y - \rho(x)\|^2. \tag{574}$$

Here the safe chart is the inverse image of the closed combined coordinate supremum ball, while the displayed quadratic lower bound uses the Euclidean norm on Y. Both endpoints lie in that chart, and its convexity contains their entire joining segment. Consequently any point of that chart outside $\mathcal{S}_x$ pays at least Δ after multiplication by β.

We also need to recognize chart points without presupposing that the fine field is in the chart. The tree-gauge geometric estimate supplies a constant C_0 such that an alive tag, fine plaquette distances at most δ, and prediction-wire distances at most δ imply

$$\operatorname{dist}(U_i, V(x)_i^{\mathrm{rep}}) \leq (C_0 s + 1)\delta$$

for every free bond. Here s bounds the number of plaquettes in the fixed local comparison box. The estimate follows by comparing that bond with its tree-gauge replica: the local filling contributes at most $s\delta$, its geometric comparison costs C_0, and the prediction wire contributes the final δ. Choose logarithm-chart constants $r_0, C_{\log} > 0$, so that distances at most r_0 have logarithms bounded by $C_{\log}$ times the distance. For a prescribed safe coordinate radius $r_s > 0$, choose

$$\delta \le \min\left\{ \frac{r_0}{C_0 s + 1}, \frac{r_s}{C_{\log}(C_0 s + 1)} \right\}.$$

Taking the logarithm of each relative free bond now yields $U = \mathcal{D}_x(y)$ with $\|y\|_\infty \le r_s$. The coarse coordinates are already inside the safe radius. This argument uses the geometric comparison for alive tags only.

Write τ for the fine health threshold, and suppose $\tau \le \delta$. Let $W \ge 1$ bound all relevant site and coordinate counts. The path budgets on a healthy field give

$$|\ell(U)| \le W\Lambda\tau^2, \qquad a(U) \ge -WC(\tau^3 + \tau^4).$$

The second bound uses the nonnegative quadratic part of the action and the absolute cubic and quartic budgets. Define

$$B_{\mathrm{loop}} = \beta C(\tau^3 + \tau^4) + \Lambda\tau^2.$$

Let m_W bound $|\phi(x, \rho(x))|/\|x\|$ on the coarse chart, and let $p + t_W$ bound $|\Phi(x) - \beta\phi(x, \rho(x))|$. The latter consists of the loop/Haar bound p and a log-determinant bound t_W. If a prediction wire exceeds δ, its nonnegative smoothing penalty is at least $\kappa\delta^2/2$, where $\kappa > 0$ is the kernel penalty constant. Assume

$$\beta m_W \|x\| + \Delta \le \beta\kappa\delta^2/2. \tag{575}$$

For a healthy point outside $\mathcal{S}_x$, there are now two cases. If all prediction wires are small, the preceding chart-recognition argument and (574) apply. Otherwise the large-wire penalty and (575) apply. In both cases the exponent in each live tag satisfies

$$-\beta[a(U) + n_k(U, V(x))] - \ell(U) \le -\Phi(x) - \Delta + WB_{\mathrm{loop}} + p + t_W.$$

Unhealthy fields and dead tags contribute zero. Summing the tag weights therefore proves a pointwise exterior bound.

To express it in stationary units, put $m_H = c_H(\sqrt{2\pi}\, b)^{\dim \mathfrak{g}}$, so $\Omega = \pi_\beta m_H^N$, and define

$$Y_* = -N \log m_H - \Delta + WB_{\mathrm{loop}} + p + t_W. \tag{576}$$

Then

$$U_x(U) \le \Omega e^{-\Phi(x)} e^{Y_*} \quad (U \notin \mathcal{S}_x). \tag{577}$$

Since the ambient Haar measure has mass one, the same bound holds for the exterior integral.

108.3 Finding one configuration with slack

Let $Z_{\mathrm{sl}}(x)$ denote the portion of the pure push allocated to the empty coarse and fine defect sector. The sector allocation has values between zero and one. Its nonzero support requires all fine plaquette distances to be at most $\tau/2$. The exact funded expansion of this sector is

$$Z_{\mathrm{sl}}(x) = \mathcal{O}(x) \sum_I e_I(x), \qquad e_\varnothing(x) = 1,$$

where $\mathcal{O}(x) > 0$ is the same sector overhead as in the block integral, and the expansion gives $\sum_I |e_I(x)| \le e^{\varepsilon}$. These are the actual funded sector identities, with its fixed tree and allocation. They imply

$$Z_{\mathrm{sl}}(x) \ge \mathcal{O}(x)(2 - e^{\varepsilon}). \tag{578}$$

To see the factor two, separate the empty term: the absolute sum of the remaining terms is at most $e^{\varepsilon} - 1$, so their sum is at least $1 - e^{\varepsilon}$. The expansion's absolute convergence justifies this separation. On the auxiliary volume, its rooted budget gives $\varepsilon = |\Lambda_c|\, o(x)$, with a negative upper exponent for o. Since $|\Lambda_c|$ has upper exponent zero, $e^{\varepsilon} \le 3/2$ eventually.

We compare $\mathcal{O}$ to the stationary normalization before using (578). On a healthy coarse field its exact formula is

$$\mathcal{O} = \Omega \exp\{-\beta a_{\mathrm{coarse},2}(x) - \tfrac{1}{2} \log\det A_0\}.$$

Here $a_{\mathrm{coarse},2}$ is the coarse quadratic action and A_0 is the empty-sector covariance precision matrix. The funded matrix bound gives $\log\det A_0 \le W a_*$. Classical matching through degree four, with fifth-derivative cost K_5, and the coarse cubic and quartic budgets give

$$|\phi(x, \rho(x)) - a_{\mathrm{coarse}}(x)| \le W K_5 \|x\|^5, \quad |a_{\mathrm{coarse}}(x) - a_{\mathrm{coarse},2}(x)| \le W C'(\tau_c^3 + \tau_c^4).$$

Combining these inequalities with the bound $p + t_W$ yields

$$\mathcal{O} \ge \Omega e^{-\Phi - e_*}, \qquad e_* = \beta W[K_5\|x\|^5 + C'(\tau_c^3 + \tau_c^4)] + p + t_W + W a_*/2. \tag{579}$$

If

$$Y_* + e_* + 2 \le -\Delta/2, \qquad Y_* \le -\Delta/2, \tag{580}$$

then the exterior upper bound is strictly less than $\frac{1}{2}\mathcal{O} \le Z_{\mathrm{sl}}$, since $e^{-2} < 1/2$. If $\mathcal{S}_x$ contained no point with all fine plaquettes at most $\tau/2$, the allocated integrand would vanish throughout $\mathcal{S}_x$. Outside it, that integrand is bounded by U_x. Its integral would consequently be bounded by the exterior upper bound, a contradiction. Thus there exists

$$U_0 = \mathcal{D}_x(\rho(x) + b z_0), \qquad \|z_0\| \le D, \qquad \mathrm{toll}_i(U_0) \le \tau/2 \quad \text{for all } i.$$

This existence argument does not differentiate the fine cutoff.

108.4 Spreading the slack and choosing the parameters

The fine plaquette tolls are uniformly Lipschitz, with a constant $L_0 > 0$ independent of the auxiliary volume, on the combined coordinate supremum unit ball, with Lipschitz distance measured in that same norm. Indeed each plaquette contains four links; the uniform first-derivative estimate for a link, the product rule, and unitarity bound the derivative of its matrix holonomy. The reverse triangle inequality for the matrix norm then bounds the difference of tolls. Integrating that derivative along the joining segment proves the asserted Lipschitz estimate.

Let C_ρ be the supremum-coordinate Lipschitz bound for the stationary branch, with $\rho(0) = 0$. For $\|y - x\| \le r$ and $\|z\| \le D$, the supremum distance between the combined coordinates of $\mathcal{D}_y(\rho(y) + bz)$ and U_0 is at most

$$(1 + C_\rho) r + 2bD.$$

We used $\|z - z_0\|_\infty \le \|z - z_0\|_2 \le 2D$. Thus every such field is fine healthy provided

$$L_0[(1 + C_\rho) r + 2bD] \le \tau/2. \tag{581}$$

The two configurations lie in the coordinate unit ball under the radius conditions below. The same conditions place the entire region inside the Haar chart and inside the region where all tag actions coincide.

Here are the simultaneous purchases used in the construction. The action constants C, C' are fixed and nonnegative, while κ, γ are fixed and positive. The constant C_ρ is fixed at this stage independently of the auxiliary volume. The loop budget obeys $0 \le \Lambda \le 2P_F$. Write $z = \beta_F$, P_F for the fine monitor, $u = z^{-7/16}$ and $r = z^{-1}$, and take

$$\beta \in [z/2, 2z], \qquad \tau = z^{-1/2} P_F, \qquad D = d_0 P_F, \qquad d_0 = (16 L_0)^{-1}.$$

Every fixed polynomial in the auxiliary size parameter is $o(P_F^\epsilon)$ for every $\epsilon > 0$; P_F itself has upper exponent zero. These are the monitor's two distinct growth properties. Since $b \le 2z^{-1/2}$, $2L_0 bD \le \tau/4$, and $L_0(1 + C_\rho)r \le \tau/4$ eventually. This proves (581). Moreover $(1 + C_\rho)u + uD$ tends to zero. It is eventually smaller than the fixed safe radii, the action-coincidence radius and one, while its fine-coordinate component times the Lie-coordinate norm cost is at most half the Haar-chart radius. Also $b < u$ and $\tau \le \delta$ eventually.

The coefficients m_W, t_W, W, a_*, K_5 are bounded by fixed polynomials in the auxiliary size parameter. The stationary jet price has the more precise form

$$p = p_0 + p_1 P_F,$$

with both p_0, p_1 bounded by such polynomials: only the loop price contributes the linear monitor factor. Also $-N \log m_H$ is bounded above by a polynomial in that parameter, including its logarithmic coupling cost. The two action remainders $W B_{\mathrm{loop}}$ and $\beta W[K_5 u^5 + C'\tau_c^3 + C'\tau_c^4]$ tend to zero, for $\tau_c \le 2\tau$. It follows that every positive term in $Y_* + \Delta + e_* + 2$ is $o(P_F^2)$. Since $\Delta = \gamma d_0^2 P_F^2/4$, both inequalities in (580) hold eventually. Finally $m_W u \to 0$ and $\Delta/\beta \to 0$, proving (575) uniformly for $\|x\| < u$. All these thresholds can be enlarged together at a fixed stage.

We can now use (573): slack propagation has verified its cutoff condition on the entire retained region. Split the positive full integral into that region and its complement, and use (577) and (580). The result is

$$0 \le Z(y) - \Omega e^{-\Phi(y)} R(b, y) \le \Omega e^{-\Phi(y)} \exp(-\gamma D^2/8)$$

for every $\|y\| < u$ with $\|y - x\| \le r$, where the center x is coarse healthy. This is the exterior estimate required in Section 109, on its full microscopic comparison ball. Its hypotheses retain the actual funded sector expansion, the stationary chart and Hessian estimates, and the classical matching for the same coefficient data.

109 The small-field remainder of the pure nonlinear push

We combine the truncated stationary integral with the full positive push and the classical matching conditions. The full logarithm and the stationary logarithm are kept distinct until their difference has been estimated.

109.1 From an exterior bound to a logarithmic error

Let $Z(x)$ be the actual pure nonlinear push in coarse logarithmic coordinates, and let $\Omega > 0$ be its parameter-independent normalizing factor. The stationary construction supplies a real normalizing exponent $\Phi(x)$ and the positive truncated integral $R(\beta^{-1/2}, x)$ of Section 107. Suppose its exterior estimate is

$$0 \le Z(x) - \Omega e^{-\Phi(x)} R(\beta^{-1/2}, x) \le \Omega e^{-\Phi(x)} \eta. \tag{582}$$

The integral over the retained stationary region is exactly the middle term, with the same moving Hessian and Haar-chart factor. The complementary integral is nonnegative. Its upper bound in (582) is a substantive exterior estimate, not a consequence of positivity.

Assume $R \geq a_0 > 0$. Then $Z > 0$ and, for

$$E(x) = \log Z(x) - \log \Omega + \Phi(x) - \log R(\beta^{-1/2}, x),$$

we have

$$0 \leq E(x) \leq \eta / a_0. \tag{583}$$

Indeed the ratio $Z/(\Omega e^{-\Phi} R)$ lies between 1 and $1 + \eta/R$, and $0 \leq \log(1+s) \leq s$ for $s \geq 0$. In the construction $a_0 = e^{-1}/2$ and $\eta = \exp(-cP_F^2)$ with a fixed $c > 0$.

109.2 Polynomial derivative bounds for the full logarithm

The coarse field enters the positive integral through the smoothing kernel. At a fixed fine configuration and fixed tag its contribution has the form

$$U(x, \omega) = a(\omega) e^{p(x,\omega)}, \qquad a(\omega) \geq 0.$$

The fine density and tag likelihood are held in a. The exponent p is minus β times the finite sum of kernel penalties between the varied coarse bonds and the fixed predicted bonds. On a fixed logarithmic ball, smoothness of the group exponential and the penalty gives a common derivative bound

$$\|p(\cdot, \omega)\|_{J^k, x} \leq C_k |\beta| \, n_b,$$

where n_b bounds the number of coarse bonds. It is uniform in the held fine field because the group representation is unitary and that field enters only through fixed group multiplication.

Subtract $p(x, \omega)$ before differentiating the exponential. The finite partition formula gives

$$\|U(\cdot, \omega)\|_{J^k, x} \leq K_k U(x, \omega),$$

where K_k is a fixed polynomial in $1 + |\beta| n_b$. Finite tag summation preserves this relative bound. Differentiation under the Haar integral is justified by the bounded fine density and kernel on the compact configuration space, together with the uniform derivative bound on a neighborhood. Consequently

$$\|Z\|_{J^k, x} \leq K_k Z(x). \tag{584}$$

At a point where $Z(x) > 0$, normalize by that one value: $f(y) = Z(y)/Z(x)$. Then $f(x) = 1$ and its jets are bounded by K_k. The logarithm chain rule at one bounds the jets of $\log Z(y) - \log Z(x)$ by a fixed polynomial in K_k. This argument never divides an absolute derivative bound by an uncontrolled small partition function; the factor $Z(x)$ has already canceled in (584).

The other terms defining E have polynomial derivative bounds: Φ consists of the stationary action multiplied by β, the stationary extra exponent and the relative log determinant; the truncated logarithm has the bounds proved in the preceding section. For auxiliary volumes polynomial in the block radius and β in its allowed band, these give a bound z^d through any fixed positive order, for some fixed d, where $z = \beta_F(x_{\text{scale}})$. The order-zero bound for E follows directly from (583). The exponentially small value bound (583) holds on a microscopic ball of radius z^{-1} about each tested coarse point. Applying the interpolation lemma of the preceding section yields, for every prescribed e,

$$\|E\|_{J^j} \leq z^e. \tag{585}$$

This uses the exterior estimate throughout that ball. An estimate only at its center would not suffice.

109.3 The exact decomposition and the two cancellations

Write $a_{\mathrm{fine}}(x)$ for the fine action evaluated on the actual stationary branch, $a_{\mathrm{coarse}}(x)$ for the coarse classical action, and $d_{\mathrm{coarse}}(x)$ for the prescribed coarse quantum and reset contribution. Define

$$C(x) = a_{\mathrm{coarse}}(x) - a_{\mathrm{fine}}(x), \qquad W(x) = d_{\mathrm{coarse}}(x) - \psi(x, \rho(x)) - \tfrac{1}{2}\log\det M(x).$$

Thus the normalizing exponent is $\Phi = \beta a_{\mathrm{fine}} + \psi(x, \rho(x)) + \frac{1}{2}\log\det M(x)$. The actual corrected logarithm

$$G(x) = \log Z(x) + \beta a_{\mathrm{coarse}}(x) + d_{\mathrm{coarse}}(x)$$

satisfies the exact identity

$$G = \log\Omega + L + E + \beta C + W, \qquad L(x) = \log R(\beta^{-1/2}, x). \tag{586}$$

Every function in this identity uses the same stationary branch, coefficient data and finite torus.

Classical matching through degree four says $D^r C(0) = 0$ for $0 \le r \le 4$. Suppose its derivatives of orders 1 through $n \ge 5$ on the ball of radius $h \le 1$ are bounded by K_C. Taylor's formula, applied also to each derivative of order less than five, gives

$$C \in \mathcal{B}_n(K_C h^5, h^{-1}). \tag{587}$$

For orders at least five the original derivative bound suffices, because $h^r \le h^5$; for lower orders integrate the fifth derivative along the radial segment, using the zero jets at its starting point.

There is a distinct cancellation for the linear part of W. The actual corrected logarithm G is invariant under the prescribed lattice translations and signed coordinate symmetries. For each coordinate basis vector supported on the bond (b, μ), central inversion followed by translation by $2b+e_\mu$ fixes that unoriented bond and reverses its orientation. It therefore sends the single-coordinate logarithmic vector e_i to $-e_i$. Differentiating invariance at zero gives $DG(0)e_i = -DG(0)e_i$, hence $DG(0) = 0$. Since $DC(0) = 0$, (586) implies

$$DW(0) = -DL(0) - DE(0).$$

In particular, if the relevant jets of L, E are bounded by a, v, then $\|DW(0)\| \le a + v$. This cancellation uses the symmetry of the full logarithm, rather than asserting that its separate stationary terms are individually even in every coarse coordinate.

The finite-volume quadratic matching estimate supplies a further bound $\|D^2 W(0)\| \le q$. Its operator norm includes the conversion from the tested momentum waves to the actual coordinate space. If the number of coordinates is at most M, that conversion has a fixed polynomial cost in M. Thus the exponentially small finite-volume bound has the form

$$q \le P(M)\Theta\Lambda e^{-\nu\ell}, \qquad \Theta, \nu > 0,$$

where ℓ is the auxiliary torus side and P is a fixed nonnegative polynomial including the coordinate norm conversion. This quadratic matching is a separate input from classical matching through degree four.

If derivatives of W of orders three through n are bounded by K_W on the same ball, Taylor's formula now gives

$$W - W(0) \in \mathcal{B}_n\big(K_W h^3 + 3\{(a+v)h + qh^2\}, h^{-1}\big). \tag{588}$$

For orders zero, one and two, integrate the third derivative and retain the displayed first and second jets. For higher orders, $h^r \le h^3$ pays the original bound. The factor three is a common upper bound for all these finite Taylor coefficients.

109.4 The final small-field price

Take $h = z^{-7/16}$ and assume $\beta \in [z/2, 2z]$. The stationary estimate gives $a = C(x_{\mathrm{scale}})\beta^{-1} + z^{-2}$ with C of upper exponent zero; (585) gives $v = z^{-2}$. On the radial segment to the tested point, the centered stationary logarithm has price ah at scale h^{-1}, by integrating its first derivative; the higher derivative bounds give the same price since $h \le 1$. The centered error has price $2v$. Using (586)–(588) therefore gives

$$G - G(0) \in \mathcal{B}_j\big(ah + 2v + \beta K_C h^5 + K_W h^3 + 3\{(a+v)h + qh^2\}, h^{-1}\big) \qquad (1 \le j \le n). \tag{589}$$

All constants refer to the same original data.

The coefficients K_C, K_W, the coordinate bound M, and the loop price Λ have upper exponent zero; Θ, ν are fixed positive constants. The auxiliary side is at least the prescribed logarithmic-square scale, so its exponential finite-volume factor pays every fixed inverse power; in particular q may be bounded by a function of upper exponent -1. The terms in the displayed amplitude have upper exponents at most

$$-23/16, \quad -2, \quad -19/16, \quad -21/16, \quad -23/16, \quad -15/8.$$

Their maximum is $-19/16 < -23/20$. After one common increase of the stage threshold, the amplitude is at most $z^{-23/20}$. The estimate holds at every healthy coarse point with the two microscopic margins needed for the comparison ball and the radial segment. It supplies the centered chart estimate used in Section 110. Its remaining hypotheses are the actual stationary-branch bounds, classical and quadratic matching, and the exterior estimate (582); none is replaced by a comparison with a different finite-volume model.

110 Extending small-field derivative estimates to the boundary

A small-field configuration may have a plaquette exactly at the health threshold. Identities first proved where every plaquette is strictly below that threshold must therefore be extended to this boundary. We follow the actual radial argument. It uses the smallness of the logarithmic field and a second-derivative estimate; it does not assume that the healthy set is convex.

110.1 A radial retreat lemma

Let $X : [0,1] \to E$ be a twice continuously differentiable curve in a finite-dimensional normed space, with

$$\sup_{0 \le s \le 1} \|X''(s)\| \le K < t, \qquad \|X(1) - X(0)\| \le t.$$

Then $\|X(s) - X(0)\| < t$ for all $s < 1$ sufficiently close to one. If the endpoint inequality is strict, this is continuity. In the remaining case put $D = X(1) - X(0)$, so $\|D\| = t$, and $g(s) = X(s) - sD$. The fundamental theorem of calculus gives

$$\|X'(1) - D\| = \left\| \int_0^1 \{X'(1) - X'(r)\}\, dr \right\| \le K < t.$$

Since $g'(1) = X'(1) - D$, the one-sided difference quotient satisfies $\|g(s) - g(1)\| < (1-s)t$ near one. For $0 \le s < 1$ in that neighborhood,

$$X(s) - X(0) = sD + g(s) - g(1), \qquad \|X(s) - X(0)\| < st + (1-s)t = t.$$

The inequality is strict even when the original endpoint lies exactly on the boundary.

Let W be the actual logarithmic coordinate of a coarse field, and write $U_s = \mathrm{actual}(sW)$ for the field obtained by exponentiating its bond coordinates. For one plaquette let $X(s) = \rho(\mathrm{Hol}_\square U_s)$, where ρ is the fixed unitary representation. Then $X(0) = 1$ and

$$\mathrm{chord}(\mathrm{Hol}_\square U_s) = \|X(s) - 1\|/r_G,$$

with the fixed representation normalization $r_G > 0$. On a fixed small logarithmic ball the exponential has bounded first and second derivatives. If the coordinate norm is at most $w \le 1$, its bond derivative scale is at most Cw. The product rule over the four plaquette bonds gives

$$\sup_{s\in[0,1]} \|X''(s)\| \le r_G(4r_G Cw)^2.$$

Thus, if $(4r_G Cw)^2 < \tau$ and all endpoint plaquette chords are at most τ, the retreat lemma with $t = r_G\tau$ makes every plaquette strictly healthy for $s < 1$ close enough to one. There are finitely many plaquettes, so one neighborhood works for all of them.

In the construction the logarithmic radius is $h = \beta_F^{-7/16}$, while the health threshold has scale $\tau = \beta_F^{-1/2}$ times its prescribed slowly varying factor. The coordinate norm conversion has a fixed constant. Since h^2 has exponent $-7/8 < -1/2$, the displayed strict curvature inequality holds eventually, uniformly over every field in that logarithmic ball. This is the needed strict radial approximation.

110.2 Transferring chart jets to ordered group derivatives

We record the normalization used with this approximation. Let j be fixed. Compactness on a small Lie-algebra chart supplies constants $J, d > 0$ for derivatives through order j of the local logarithm. Shrink the chart so that the logarithm inverts the exponential there. Let $U = \mathrm{actual}(W)$, with $\|W\| \le h$, and suppose

$$f \circ \mathrm{actual} \in \mathcal{B}_j(T, h^{-1}; W).$$

Consider an admissible group-valued variation $V(y)$ through U, whose link derivative scale is C. The local logarithm and the inverse linear coordinate map produce a chart variation $a(y)$ with $a(y_0) = W$ and $V = \mathrm{actual} \circ a$ near y_0. The logarithm chain rule bounds its positive-order derivatives by

$$j!J\{(1 + r_G)C\}^r \qquad (1 \le r \le j).$$

Assume

$$(1 + r_G)C \le 1, \qquad j!J(1 + r_G)C \le h.$$

Then every derivative of a through order j, including its value, is at most h. In the chain rule for $f \circ \mathrm{actual} \circ a$, an outer derivative of order m costs Th^{-m} and its m inner factors cost at most h^m. They cancel. Bounding the finite partition count conservatively gives

$$\|f \circ V\|_{j,y_0} \le j!3^j T. \tag{590}$$

The same estimate applies to ordered left derivatives along any list of at most j directions of norm at most τ. Indeed use the corresponding ordered products of exponentials as the variation; the compact exponential bounds give a link scale at most $C_j j\tau$. Choose C to dominate that value. In the present construction C has upper exponent $-1/2$, strictly below the radius exponent $-7/16$, so both normalization inequalities hold eventually. The construction also proves the required local differentiability of f in the group chart.

110.3 Subtracting a summable activity array

Let u_i be regular activities with nonempty assigned supports and nonnegative jet prices k_i. Suppose their rooted support-weighted price is at most b per torus site and their derivatives in directions bounded by τ are at most k_i. Then

$$\sum_i k_i \le |\Lambda| b.$$

This follows by summing over the roots and removing weights at least one. On the open set $\mathcal{S}$ of strictly healthy configurations, every u_i is differentiable to order j. Their series can be differentiated term by term there. To check the needed domination for any fixed list of directions, scale every direction by a common positive number so that its norm is at most τ. Multilinearity of the ordered derivative then multiplies the summable majorant k_i by only one fixed power of that number. The common majorant is uniform on $\mathcal{S}$. The local differentiation theorem for normally convergent derivative series therefore applies, order by order. In particular, for $U \in \mathcal{S}$ and normalized directions,

$$\left|\sum_i u_i(U)\right| \le |\Lambda| b, \qquad \left\| D_\ell \sum_i u_i(U) \right\| \le |\Lambda| b \quad (|\ell| \le j). \tag{591}$$

Suppose the local action identity on $\mathcal{S}$ is

$$f(U) = z - H(U) - \sum_i u_i(U),$$

where z is constant. For any function q write $\widetilde{q}(U) = q(U) - q(1)$. The identity field belongs to $\mathcal{S}$, because $\tau > 0$. Subtracting its value yields

$$\widetilde{H}(U) = \sum_i u_i(1) - \sum_i u_i(U) - \widetilde{f}(U).$$

Assume now that the chart bound with amplitude T and scale h^{-1} holds for $\widetilde{f} \circ \mathrm{actual}$ at every W' such that $\|W'\| + 2d < h$ and $\mathrm{actual}(W') \in \mathcal{S}$. This is a bound on the centered function itself. Apply (590) to $\widetilde{f}$. Combining it with (591) gives

$$\|D_\ell \widetilde{H}(U)\| \le j! 3^j T + 2|\Lambda| b. \tag{592}$$

The extra factor two covers the centered value at derivative order zero; for positive orders the identity-field constant differentiates to zero.

Now let $U = \mathrm{actual}(W)$ have no flagged plaquettes, allowing equality at the threshold, and assume $\|W\| + 2d < h$ with $d > 0$. For $0 \le s \le 1$, $\|sW\| + 2d < h$ as well. The radial retreat places U_s in $\mathcal{S}$ for $s < 1$ sufficiently close to one, so the same interior estimate applies there. Assume H is regular on the full healthy domain, as supplied by the local action construction. Its ordered derivatives through order j are continuous at U in the local group chart. Since $U_s \to U$, passing to the limit in (592) gives the same bound at U. Only H needs this boundary regularity: no boundary smoothness is inferred for the original masked series or for f from its interior identity.

111 The principal regular insertion after one step

We combine the reference attenuation with the comparison to the actual principal cluster. This supplies the remaining ordinary principal term in the zero-coefficient transition of Section 61. The forced and nonprincipal terms retain the prices proved in Section 95.

Let f be a real regular fine insertion on X, of price $\tau \ge 0$, and let Y be its selected coarse enlargement. Write $q = C_{B_f}(X)$, $q' = C_{B_c}(Y)$, and $m_f = e^{a_f q}\tau$ for its fine mass. Its polynomial moments through degree d are assumed zero whenever X is nonwrapping. The order and weight requirements are

$$\max\{2, d+1\} \le n_c, \qquad d \le p_f, \qquad d+2 \le J_f,$$

where J_f is the fine kernel's available derivative level. Together with the local regularity required in the preceding sections, these permit the strip derivatives and the extra insertion derivative used in the Taylor argument. Let $e = a_c + 2$ be the exponent charged by the coarse reserved mass, and put

$$\mathfrak{m}_c(h) = e^{eq'}h, \qquad \gamma_L = \frac{L^{-2/5}}{h_d}, \qquad w_L = \frac{\gamma_L}{2L^d},$$

where $h_d \ge 1$ is the fixed projection and frame cost. We prove a coarse derivative bound of reserved mass at most $w_L m_f$ for the actual principal real insertion.

The constants in the short and long reference estimates, say A_d and A'_d, are chosen before L. Choose L so that the support compression inequality holds and

$$A_d L^{-d-1} \le w_L/2, \qquad A'_d L^{-d-1} \le w_L/2.$$

This is possible since each reduces to $4h_d A_d \le L^{3/5}$ or the analogous inequality for A'_d. The strip-to-polynomial bound is chosen before these two constants; it is uniform in all periods. Only after these choices do we increase the entrance coupling to meet the reference smallness conditions.

Denote the actual principal insertion by F and the reference Gaussian integral by H. For $q' \le B_c$, Section 96 gives, uniformly over the forests and replica choices,

$$|D^j(F - H)| \le \tau z_\beta, \qquad 0 \le j \le n_c,$$

for the admissible coarse ordered directions. Here $z_\beta \to 0$ after L is fixed. Averaging over the forests and replica choices preserves this bound: the norm of a finite average of differences is at most their common bound. The enlargement estimate gives

$$\mathfrak{m}_c(\tau z_\beta) \le e^{et_r} z_\beta m_f,$$

where t_r is the fixed allocation cost. Increase the entrance coupling so that $e^{et_r} z_\beta \le w_L/2$.

If $q^3 < L$, the short-support geometry makes X nonwrapping, so its assumed moments vanish. The short reference estimate then gives

$$\mathfrak{m}_c(\text{price of } H) \le A_d L^{-d-1} m_f.$$

If $q^3 \ge L$, use the separate-gauge reference bound. Multiplying it by $[(1+q)^3/L]^{d+1} \ge 1$ inserts L^{-d-1}; the extra support power is absorbed by the unused exponential fine mass, as proved in Section 98. Thus its reserved mass is at most $A'_d L^{-d-1} m_f$. In either case the reference price and comparison price add to at most $w_L m_f$.

It remains to consider $q' > B_c$. Here the principal reserve from Section 95 suffices directly. The large-support derivative bound has a fixed polynomial cost in q' multiplied by an exponential with at least one spare unit of support exponent. For a fixed integer k,

$$(1+q')^k e^{-q'} \le \left[\sup_{t \ge 0} (1+t)^k e^{-t/2}\right] e^{-B_c/2}.$$

The supremum is finite: differentiation of $k\log(1+t)-t/2$ puts its maximum at $t=\max\{0,2k-1\}$. The remaining multiplier also contains the covariance and chart price divided by the Gaussian scale: denote it by $E_L(\beta)$. It has finite upper power exponent in the incoming coupling, as proved in Section 95; it is not a fixed constant. The principal reserved mass is at most $E_L(\beta)e^{-B_c/2}m_f$. For some fixed a, C, eventually $E_L(\beta)\le C\beta^a$. Since B_c grows quadratically in $\log\beta_c$, and $\beta_f=\beta_c+$ a fixed shift after L is chosen, $\log(C\beta^a e^{-B_c/2})\to-\infty$. Increase the coupling until $E_L(\beta)e^{-B_c/2}\le w_L$. This estimate applies without a logarithmic chart on the whole support.

All cases give the asserted principal bound. For a complex insertion, apply the result to its real and imaginary parts. Each has the same admissible fine price, and combining their two coarse bounds costs $2w_Lm_f$. Physical normalization multiplies by L^d, giving

$$L^d(2w_Lm_f)=\gamma_Lm_f.$$

This is the principal share used by the ordinary projection estimate. It is produced for the same insertion, support, forest average and transition as the other terms; no replacement of the transition operator is involved. The case of a primary coefficient of positive grade uses moments only through one less than that grade and is handled separately below.

112 Calibration of the primary coefficient transition

The primary transition must do more than bound a regular insertion. Its coefficient must approach the prescribed linear transport on the finite coefficient frame. We prove this by the same three comparisons used in the construction: the actual principal term to its Gaussian reference, the Gaussian reference to the linear predicted field, and that field to the finite polynomial probe calculation.

Fix a frame of total degree d and a nonconstant coordinate i. Write $g_i\in\{1,\dots,d\}$ for its spatial grade and ℓ_i for its order of vanishing in the field amplitude. These are different gradings. The primary insertion h_i is the local section at fine threshold θ_f, normalized by $\theta_f^{-\ell_i}$ as in Section 59. Its regularity price is bounded by the fixed fine-frame constant T_f. All constants in this section may depend on the chosen frame and the block factor L. Periods and the incoming coupling remain uniform. Both adjacent kernel levels are at least four, and the fine level is at least d.

112.1 Removing the Gaussian fluctuation at the flat field

Let h be one real component of h_i. Denote by $H(V)$ its reference Gaussian integral and by

$$h_0(V)=h\left(\exp\bigl(\mathcal{A}\,\mathrm{crop}_R(\log V)\bigr)\right)$$

the linear prediction, with the installed response $\mathcal{A}$ and $R=45(1+B_c)$. This formula is used as a germ at $V=1$; its crop is in the identity gauge. Let k be the derivative order of a frame probe. Purchase regularity through order $n\ge k+1$, as well as $n\ge 2$. The additional derivative will control the change of the insertion's base point.

Use the fixed Gaussian measure at $V=1$ and let $R_V(z)$ be the normalized density ratio. Put $A_V(z)=h(W(V,z))$, where W is the actual received reference configuration. With the cutoff Γ of the preceding section, the exact difference of integrands is

$$\Gamma R_VA_V-h_0(V)=\Gamma R_V(A_V-h_0(V))+\Gamma(R_V-1)h_0(V)+(\Gamma-1)h_0(V). \tag{593}$$

Thus no derivative of the varying density is discarded.

Write $P(z) \ge \max\{1, \|\mathrm{tape}(z)\|\}$ for the fixed Gaussian coordinate norm augmented by one. On the cutoff support, the display comparison of Section 98 gives, for every jet through order n, an error of fine direction price

$$p(z) = C_n \frac{Qs^2 + g\|\mathrm{tape}(z)\|}{\theta_f}.$$

The linear predictor's argument jets have bounded fine direction price $W \ge 1$. The nonlinear jets have price at most $W + p(z)$. For a partition with r arguments, subtract the two chain-rule terms by first changing their base point and then their arguments one at a time. Changing the base point costs at most $C\tau p(z)(W+p(z))^r$: integrate the $(r+1)$st derivative of the insertion along the healthy radial segment proved earlier. Changing the arguments costs at most $Cr\tau p(z)(W + p(z))^{r-1}$. All these terms use $r + 1 \le n$. Summing the finitely many partitions proves

$$|D^j(A_V(z) - h_0(V))|_{V=1} \le C_n \tau p(z)(W + p(z))^n, \qquad j \le k.$$

The same argument without subtraction gives $|D^j h_0(1)| \le C_n \tau$.

The density-ratio calculation in Section 85 gives a small factor θ_f times a polynomial in the read radius and $P(z)^2$ in every positive-order jet of $R_V - 1$ at 1. Its value there is zero since $R_1 = 1$. More explicitly, differentiate $\frac{1}{2}\log\det C(V) - \frac{1}{2}\langle z, C(V)z\rangle$ with the reference precision C. Each positive-order term contains at least one admissible coarse direction, and the inverse and precision jets bound it by $K_L\theta_f(1 + B_c)^a P(z)^2$. The partition formula for its exponential multiplies such factors, retaining at least one.

The cutoff error is controlled even outside its support. It is one on the plateau $g\|\mathrm{tape}(z)_a\| \le \varepsilon\theta_f/8$ for every coordinate a. If a coordinate fails this test, then $(g/\theta_f)P(z)^2 \ge \varepsilon/8$, while $|\Gamma - 1| \le 2$. Consequently

$$|\Gamma(z) - 1| \le \frac{16}{\varepsilon}\frac{g}{\theta_f}P(z)^2.$$

This bound does not require evaluating A_V off the cutoff support: there the first two terms of (593) vanish identically.

Collect the displayed small factors into $J_\beta \ge 0$ so that

$$J_\beta \le K_L(\theta_f + g/\theta_f)(1 + B_c)^a, \qquad p(z) \le J_\beta P(z)^2.$$

The bounded threshold ratio supplies $s^2/\theta_f \le K_L\theta_f(1 + B_c)^a$. Increase the coupling until $J_\beta \le 1$. Leibniz' rule in (593) and the partition bounds then give

$$|D^j(\Gamma R_V A_V - h_0(V))|_{V=1} \le C_n \tau J_\beta P(z)^{4(n+1)}, \qquad j \le k.$$

All factors from higher-order products are absorbed by the displayed power, since $P \ge 1$ and $J_\beta \le 1$. The fixed Gaussian moments bound its integral by $K'_L(1 + B_c)^b$. Compact cutoff support and the common local smoothness justify differentiation under the integral; the polynomial bound also dominates the subtracted constant term. The monitor estimate from the preceding section implies

$$\max_{j \le k} |D^j H(1) - D^j h_0(1)| \longrightarrow 0 \quad (\beta \to \infty). \tag{594}$$

The bound is uniform in every forest, replica choice and allowed period. Averaging therefore preserves it. Combining it with the actual-to-reference comparison gives the same convergence for the actual principal insertion.

112.2 Normalized stencil functions and their jets

We give the finite-dimensional estimate needed to sample h_0 on the frame probes. Let H_i be the analytic unnormalized stencil function of coordinate i, on the finite-dimensional space of its link logs. It satisfies $D^j H_i(0) = 0$ for $j < \ell_i$. For $0 < t \le 1$ set

$$H_{i,t}(v) = t^{-\ell_i} H_i(tv).$$

On any fixed ball $\|tv\| \le R_0$, Taylor's formula and compactness give, for every prescribed order j,

$$\|D^j H_{i,t}(v)\| \le C(1 + \|v\|)^{\ell_i}. \tag{595}$$

Indeed let $a = \max\{\ell_i - j, 0\}$. The vanishing derivatives imply $\|D^j H_i(tv)\| \le C\|tv\|^a$ by integrating a times from zero. For $a = 0$, use the maximum of the continuous derivative on the ball. The scaling factor is $t^{j-\ell_i+a} \le 1$, with nonnegative exponent; and $\|v\|^a \le (1 + \|v\|)^{\ell_i}$. Taking a maximum over the finitely many required orders proves the bound with one C.

Suppose two C^k curves v, q have jets at zero bounded by $U + e, U$, respectively, and their jet differences through order k are at most $e \ge 0$. Suppose also their values, multiplied by t, lie in the fixed ball. Along the segment joining their values, use (595) through order $k + 1$. For a partition with $r \le j \le k$ blocks, the difference of the outer rth derivatives costs at most $C(1 + U + e)^{\ell_i} e$. Replacing its r arguments one at a time costs at most $Cr(1 + U + e)^{\ell_i + r} e$. Thus

$$\max_{j \le k} \|D^j(H_{i,t} \circ v)(0) - D^j(H_{i,t} \circ q)(0)\| \le C_{k,i,U}(1 + e)^{\ell_i + k} e. \tag{596}$$

This estimate is uniform as $t \downarrow 0$. The same partition calculation without subtraction bounds the jets of $H_{i,t} \circ q$ uniformly. The constant is a finite sum over partitions, not a limit at the singular normalization $t = 0$.

112.3 The cropped torus probe and the full polynomial forecast

A probe consists of a fixed polynomial potential and a derivative order $k \le d$. Its test-link curve is the holonomy curve defined in Section 59. At derivative order $j \le k$, its link-log jet on the covering lattice is a polynomial field of degree at most dj. This follows by expanding the exponential, product and logarithm to order j: every term is a product of at most j potential factors, each of degree at most d. Integration along a unit bond and finite differences do not increase that bound. The Taylor coefficients are fixed by the probe.

Normalize the coarse probe by its threshold $y = \theta_c$ and the fine link field by $t = \theta_f$, with $y/t \le 2$ and $y \le 1$. Apply the installed response to the cropped torus link-log curve and restrict to the fixed fine stencil; call the resulting normalized curve v_R. Let q be the full covering-lattice prediction obtained by applying the same response to the polynomial link-log jets. Only its k-jet is used; it is equivalently obtained by the finite polynomial jet algebra of the probe. Both curves vanish at zero. The exact correspondence between periodic and covering convolution holds because the retained kernel is periodized from its absolutely summable covering kernel.

On covering sites of distance at most R, the crop and the full probe have identical jets. The room inequality $2R < c$ ensures that these sites have their canonical lifts, so there is no seam error there. Outside this region, the probe jets have polynomial growth and the cropped jets have norm at most $K(1 + 4R)^{dk}$. The response strip bound gives

$$\max_{j \le k} \|D^j v_R(0) - D^j q(0)\| \le C\{1 + K(1 + 4R)^{dk}\} e^{-\sigma R/2} =: e_R. \tag{597}$$

To verify the tail estimate, take half the exponential weight to bound the polynomial factor uniformly, and sum the remaining weighted kernel. The finite stencil offsets and four directions supply only a fixed multiplier. The torus sum is the same sum regrouped by residue class; absolute convergence permits this regrouping. This is the cropped-versus-full comparison, including all derivative orders, for the actual retained response.

The full predicted jets have a bound U independent of the period and R. Indeed convolution of a degree-dj polynomial costs its fixed weighted kernel moment. In a nonzero chain-rule term with $r \geq 1$ blocks, the threshold factor is $y^r/t \leq y/t \leq 2$. The finitely many probe partitions therefore have a common bound. Apply (596) with $t = \theta_f$ and $R_0 = 1$: both input values are zero and hence lie in this fixed positive-radius ball. Since $e_R \to 0$, the resulting normalized stencil jet difference tends to zero. Multiplication by the fixed physical factor L^d preserves that convergence.

To identify the prescribed coefficient transport explicitly, let $\mathscr{S}_t$ be diagonal multiplication by t^{ℓ_i}, let $\mathscr{D}_L$ be diagonal multiplication by L^{g_i}, and let $\mathscr{V}_L$ be diagonal multiplication by L^{d-g_i}. If $\mathscr{R}$ is coefficient recovery of the full polynomial forecast, the construction's heading is the matrix

$$\mathscr{S}_{\theta_c}\mathscr{R}\,\mathscr{D}_L\,\mathscr{S}_{\theta_f^{-1}}, \qquad x_i = \mathscr{S}_{\theta_c}\mathscr{R}\mathscr{D}_L\mathscr{S}_{\theta_f^{-1}}\mathscr{V}_L e_i.$$

On column i, the last three diagonal factors contribute $L^{g_i}\theta_f^{-\ell_i}L^{d-g_i} = L^d\theta_f^{-\ell_i}$. The remaining coarse scaling substitutes θ_c times the probe potential into its polynomial jet. Consequently, for every frame probe p,

$$\mathrm{probe}_p(x_i) = L^d D^{k_p}(H_{i,\theta_f} \circ q_p)(0).$$

This identity is an equality of the finite jet calculation: evaluate the polynomial transport on the probe, substitute the response's polynomial moments, and use the local section's exact coefficient recovery. The normalization on each side is $L^d\theta_f^{-\ell_i}$, with the same coarse threshold in the probe. Thus (597) identifies the linear prediction's probes with those of x_i, to an error tending to zero. It does not replace the retained response by another continuum kernel.

112.4 From probes to the paid primary coefficient

Let F_i be the actual complex principal insertion, already multiplied by L^d, and let b_i be its extracted coefficient vector. The nonwrapping coarse stencil allows coefficient recovery by the fixed finite probe family. Ordered derivative convergence in (594) implies probe convergence: each probe derivative is a finite combination of ordered derivatives along its fixed test-link curve. The coefficients are uniformly bounded by its finite local price. The last subsection and the triangle inequality therefore give

$$\max_p |\,\mathrm{probe}_p(b_i - x_i)| \longrightarrow 0.$$

A finite separating family of linear probes controls the coefficient norm: on its unit sphere their maximum has a positive minimum by compactness. Equivalently use the finite recovery matrix from Section 59. Denote its bound by T_{test}. Then $\|b_i - x_i\|_1 \leq T_{\mathrm{test}}\delta_\beta$, where $\delta_\beta \to 0$ is the maximum displayed probe error.

The remaining regular rows have weighted sum tending to zero after L is fixed, by the nonprincipal estimate of Section 94. The extraction map on them costs the fixed read constant T_{read}. If their summed majorant is u_β, the total extracted coefficient differs from x_i by at most

$$p_\beta = T_{\mathrm{test}}\delta_\beta + T_{\mathrm{read}}u_\beta \longrightarrow 0.$$

The equality splitting the sum into its principal row and the other rows is justified by their absolute weighted summability.

Move this coefficient error into the local-section remainder. The local splitting identity of Section 58 makes this an exact equality of densities and of the ordinary and flagged profiles. Its additional cost is at most $C_{\mathrm{sec}}p_\beta$, with C_{sec} the fixed section and symmetry cost. Choose the probe error to spend half the desired allowance $L^{-2/5}$ and the nonprincipal error to spend the other half. The probe family and frame are finite, so one entrance threshold meets all these demands. This proves the small primary coefficient error for the prescribed transport x_i, on the actual selected transition.

For completeness, its principal remainder has the required graded bound as well. The local section for coordinate i has zero moments through degree g_i-1. Apply the short reference estimate with that degree to its fixed stencil; it gives $C_iL^{-g_i}$ times the fine mass. After L is fixed, choose the comparison error so that its reserved mass is at most L^{-g_i} times that mass. Apply this to the real and imaginary parts and multiply by L^d. The resulting principal price is at most

$$2(C_i+1)L^{d-g_i}m_i.$$

Here the constant C_i is purchased before the block factor using the weighted response derivatives through order g_i-1. The projection calculation already proved bounds the remaining cost by fixed multiples of this price and m_i. Since $L^{d-g_i}\geq 1$ and the normalized local-section mass m_i is bounded by the fixed frame credibility constant, choose the primary ledger constant E_J larger than their finite sum over the frame. This gives the graded remainder budget $E_JL^{d-g_i}$. It is the ordinary primary budget used in the calibrated coefficient-field recursion, together with the small coefficient error just proved.

113 Polynomial transport and the invertible coefficient map

The primary comparison of Section 112 uses a particular coefficient map. We prove the two-sided bound needed to iterate it, and then combine the primary, regular and forced insertions on one transition. The coefficient norm throughout is the sum of the absolute values of the coordinates.

113.1 Exact polynomial transport on the cover

Fix the positive blocking penalty $k>0$. Let L be the block side, $u=L^{-1}$, and let $j,t\geq 4$ be the adjacent fine and coarse kernel levels. Write $A_x^{ab}(z)$ for the real one-form transport row at the fine site x, with output direction a, input direction b, and coarse displacement z. The actual input site in this row is $\lfloor x/L\rfloor+z$; the floor here means the centered block quotient. The strip estimate gives constants $\sigma,C>0$, selected before L, such that

$$\sum_z |A_0^{ab}(z)|e^{\sigma|z|_1}\leq Cu.$$

The analogous rows at other residues are exponentially summable too. We first show that, for every vector-valued polynomial p of degree less than j,

$$(Ap)(x)_a=\sum_b\sum_z A_0^{ab}(z)p_b(ux+z). \tag{598}$$

This is an exact identity, including for x outside the central block.

Here are the low-frequency details. Denote the analytic alias column of Section 8 by $\mathcal{A}_l(q)$, in its rescaled one-form convention. Fourier reconstruction and $x=L\lfloor x/L\rfloor+\mathrm{rem}(x)$ give

$$e^{iq\cdot\lfloor x/L\rfloor}\widehat{A}_x^{ab}(q)=u\sum_l e^{i(q+2\pi l)\cdot x/L}\mathcal{A}_l(q)_{ab}.$$

For a nonzero alias l, both terms in the regular one-form formula have the factor

$$z_l(q)^j, \qquad z_l(q) = \operatorname{area}_L(-(q+2\pi l)/L), \qquad z_l(0) = 0.$$

Indeed the transposed filter contains the jth power of this area factor, and the nonprincipal gain contains the same factor. Removing it leaves the analytic port, rail and gain factors from the component estimates. Their denominators are nonzero at the origin. Thus $\mathcal{A}_l = z_l^j F_l$ with F_l analytic there. Every derivative of order less than j of this term, even after multiplication by the displayed exponential, vanishes at zero: in the product of j vanishing factors some factor remains undifferentiated. The principal column has $\mathcal{A}_0(0) = I$, obtained by the regular principal formula at zero. In particular

$$\sum_z A_0^{ab}(z) = u\delta_{ab}. \tag{599}$$

Restrict q to $s\xi$ for a real covector ξ, and differentiate $r < j$ times in s. Exponential summability justifies differentiation of the Fourier series: its differentiated terms are bounded by a polynomial in $|z|_1$ times the summable exponential. All nonprincipal aliases disappear. Comparing the resulting principal expression at x and at zero, after inserting $e^{is\xi(ux)}$ at zero, gives

$$\sum_z A_x^{ab}(z)\xi(\lfloor x/L \rfloor + z)^r = \sum_z A_0^{ab}(z)\xi(ux+z)^r.$$

Polarization of the symmetric r-linear forms gives the same identity for products of r possibly different covectors. Such products span the homogeneous polynomials. Summing the homogeneous parts and the finitely many colour coordinates proves (598).

Let p now have homogeneous degree $r < j$. Expanding $p(ux+z)$ multilinearly, a term containing s copies of x has a factor u^s. Its remaining $r-s$ factors are bounded by a constant times $(1+|z|_1)^{r-s}$. Consequently its coefficient tensor has norm at most

$$C_r \|p\| u^{s+1}, \qquad 0 \le s \le r.$$

The constants are independent of L: use $\sup_{v\ge 0}(1+v)^{r-s}e^{-\sigma v} < \infty$ and the row bound. Passing from diagonal polynomial values to symmetric tensor norms costs only a fixed finite-dimensional constant. This follows by compactness of the tensor unit sphere and injectivity of the diagonal map, or directly by polarization. The degree-r term is exactly $u^{r+1}p(x)$ by (599). Adding homogeneous parts proves that, if $\deg p < a \le j$, then

$$\deg(Ap) < a, \qquad \deg(Ap - u^a p) < a - 1, \tag{600}$$

and the degree-s output tensor is bounded by $C_a u^{s+1}$ times the sum of the input tensor norms. Lower-degree input terms do not affect the asserted highest-degree identity.

113.2 The two gradings of the primary matrix

Let e_i be the canonical coordinate basis of the degree-d frame, with e_0 the unit, and write (g_i, ℓ_i) for spatial grade and field-amplitude grade. Both lie between zero and d. Let R be the unscaled polynomial coefficient transport used in Section 112. We claim

$$R_{bi} = u^{g_i}\delta_{bi} \qquad (g_b \le g_i), \tag{601}$$

$$R_{bi} = 0 \qquad (\ell_b < \ell_i), \qquad |R_{bi}| \le D u^{g_i}. \tag{602}$$

Here $D \ge 1$ is chosen before L, for the fixed frame and kernel levels, and we assume $d \le j$.

To verify the first identity, differentiate the local holonomy display in directions of engineering weights $w_1, \dots, w_n$. A block of a chain-rule partition has weight $a = \sum_{r \text{ in block}} w_r$. Its unsmoothed potential is polynomial of degree less than a. By (600), the smoothed potential is u^a times that potential plus a polynomial of degree less than $a - 1$. Expand the outer multilinear derivative of the ith section in these sums. If the total weight is at most g_i, every term using one lower-degree correction has weight strictly below g_i and vanishes by the primary section's lower-grade cancellation: its multilinear derivative is zero on polynomial tuples of total engineering weight less than g_i. Indeed its local-section tests read the homogeneous label e_i, so all smaller spatial grades vanish (Section 59); the filtered gauge-coordinate argument of Section 60 then gives exactly this lower-weight assertion. No vanishing of its grade-g_i test is claimed. The remaining term has factor $u^{\sum_r w_r}$. The unsmoothed section reads exactly e_i, by the defining local-section identity. Extracting its homogeneous part of output bidegree (g_b, ℓ_b) proves (601). This also covers order zero.

For the second identity, the section has ordinary derivatives zero through order $\ell_i - 1$ at the flat field. In a derivative of total order less than ℓ_i of its composition, the outer derivative has no more arguments than that total order. It therefore vanishes. This proves the amplitude-grade restriction, regardless of spatial grade.

For the uniform bound, use the homogeneous decomposition following (600) in every inner block of the chain rule. A term of polynomial degree s in that block has weight $s+1$ and coefficient norm at most Cu^{s+1}. Expand the outer multilinear derivative. Terms whose weights sum to less than g_i vanish; every other term has a factor at most u^{g_i}, since $0 < u \le 1$. The fixed stencil, bounded derivative order and finite partition set give a constant for each polynomial test. A finite family of these tests controls the coefficient norm, as proved in Section 58. Taking the maximum over the finitely many columns and tests proves the last inequality in (602).

Let $f, c \in (0, 1]$ be the fine and coarse small-field thresholds, with $c/f \le \pi$ and $f/c \le \pi$, where $\pi \ge 1$. The actual coefficient map B has entries

$$B_{bi} = c^{\ell_b} f^{-\ell_i} L^{g_i} R_{bi}.$$

If $\ell_b \ge \ell_i$, then $c^{\ell_b}/f^{\ell_i} \le (c/f)^{\ell_i} \le \pi^d$; otherwise the entry is zero. Set $a = \pi^d D$. We have

$$a \ge 1, \qquad |B_{bi}| \le a, \qquad |B_{ii}| = (c/f)^{\ell_i} \ge a^{-1},$$

and an off-diagonal nonzero entry requires $g_i < g_b$. The unit is fixed. The actual polynomial transport commutes with signed coordinate motions and conjugation: change variables in its absolutely convergent real kernel sum, use the covariance of the local section, and then take its finite jet. The three diagonal scalings in B preserve those actions because the bidegrees do.

113.3 A bound for the inverse without a perturbative assumption

Let n be the number of coordinates and put $K = a + na^2 \ge 1$. For $y = Bv$, induction on the grade proves

$$|v_i| \le K^r \|y\|_1 \quad (g_i < r).$$

The claim is empty for $r = 0$. If $g_i < r + 1$, the other nonzero entries in row i come from grades less than g_i, hence less than r. Isolating the diagonal entry and applying the induction hypothesis gives

$$|v_i| \le a \left(|y_i| + \sum_{b \ne i} |B_{ib} v_b| \right) \le a\|y\|_1 + na^2 K^r \|y\|_1 \le K^{r+1} \|y\|_1.$$

Summing with $r = d + 1$, and bounding the forward map row by row, yields

$$\|Bv\|_1 \le C_B\|v\|_1, \qquad \|v\|_1 \le C_B\|Bv\|_1, \qquad C_B = n(a + na^2)^{d+1}. \tag{603}$$

Thus B is injective and, in this finite-dimensional space, bijective. Substituting $v = B^{-1}w$ gives the same bound for its inverse. The unit column and grade-zero row show also that B commutes with deletion of the unit coordinate, denoted by core. In particular this argument requires no smallness of the off-diagonal entries.

114 Three elementary insertions on one history transition

Fix one funded interpolating transition, its point on the interpolation interval, and its fine and coarse representations R_f, R_c. Retain the actual local slots and their support and inventory identifications. The marked derivative fixes one map $\mathcal{D}$ from real elementary insertions to coarse forced arrays, as in Section 95. The equalities identifying R_f, R_c with adjacent entries of the chosen history are part of these data. None of the selections below changes that transition or its input and output representations.

Write $\mathcal{T}$ for its density transfer, $v = L^d$ for the physical factor, $\eta = L^{-2/5}$ for its small error allowance, and κ_f, κ_c for the square roots of the fine and coarse forced allowances. They satisfy $0 < \kappa_f, \kappa_c \le 1$. For a support X and majorant $h \ge 0$, its fine mass is

$$m_f(X, h) = e^{b_f C_{B_f}(X)} h.$$

The frame's finite motion price is $C_{\mathrm{orb}} \ge 1$. Let C be the selected coefficient allowance, at least the price C_B in (603) and the graded remainder price in the primary estimate. All these choices precede the final coupling threshold.

114.1 Keeping the three exact identities

For an inserted density F, the local splitting constructs a coefficient field Q and a response ψ such that

$$\begin{aligned} \mathrm{dens}(Q) &= v\mathcal{T}F, \\ \alpha = \sum_r z_r \,\mathrm{ord}(M_r), &\qquad \omega = \sum_r z_r \,\mathrm{flag}(M_r), \\ \mathcal{P}_c(Q_{\mathrm{forced}}) - \mathrm{hook}(\psi) &= v\sum_r z_r \mathcal{P}_c(\mathcal{D}M_r). \end{aligned} \tag{604}$$

Here α is the input exponential entry, ω its input forced profile, M_r are real elementary insertions, and $\mathcal{P}_c$ groups output coefficients with the same support and inventory. The ordinary and flagged entries are separate functions; the density combines them with the background density and gas. The first identity follows from the actual integral expansion and the exact subtraction identity of Section 58. The last is the retained singleton identity (503). The middle identities simply specify the input decomposition; they are not inferred from the first.

For a complex elementary insertion use its real and imaginary parts with the same majorant, and coefficients $1, i$. Their input cost is at most twice their common mass. The response ψ is supported on sets containing the origin, is local, gauge invariant and measurable, and vanishes on healthy configurations of its support. If its reserved mass bound is a_0, the local splitting supplies remainder budgets $a_1, a_2 \ge 0$ with

$$\mathrm{rem}_{\kappa_c}(Q) \le a_1 + a_2, \qquad C_{\mathrm{turn}}a_1 + a_2 \le r, \qquad a_0 \le a_1 \le r,$$

where $C_{\mathrm{turn}} \geq 1$. More precisely the first two budgets retain the separate ordinary and forced estimates, so the same bound r holds after the allowed finite motions. These assertions are the summation and support-join bounds already proved in the local splitting; the response is the sum of its rebates. Absolute majorants justify both its grouping by support and the profile identity in (604).

114.2 The three choices and their prices

For the primary insertion with coordinate e_i, $i \neq 0$, take its local section on the canonical stencil. Its exponential entry is its masked section, its input forced profile is zero, and its density is that entry times the fine background density. The small-error and graded remainder estimates of Section 112, followed by the same local splitting, give

$$C_{\mathrm{orb}}\|Q.x - BL^{d-g}e_i\|_1 \leq \eta\|\operatorname{core} e_i\|_1, \qquad r = C\|L^{d-g}\operatorname{core} e_i\|_1.$$

The diagonal notation L^{d-g} acts coordinatewise. Let τ_f be the common fine local-section derivative bound. Its stencil is contained in one B_f-covering set once the selected geometric floor is met. Thus its input mass is at most $e^{b_f}\tau_f$. Both real components therefore fit the common input allowance

$$t_f = 2e^{b_f}\tau_f. \tag{605}$$

This allowance is independent of the coordinate.

For a regular insertion f on $X \ni 0$, assume the local regularity, gauge invariance and vanishing holonomy tests through degree d, with the prescribed derivative and spatial-weight bound h. Its input exponential entry is $\chi_X f$, its forced profile is zero, and its density is $\rho_f \chi_X f$. Put $m = m_f(X, h)$. Section 111 pays its principal term; the nonprincipal and forced payments are those of Section 95. The split field therefore satisfies

$$C_{\mathrm{orb}}\|Q.x\|_1 \leq \eta m, \qquad r = \eta m, \qquad \text{input allowance } 2m.$$

All three equalities in (604) use this same masked insertion.

For an allowed forced slot s, let $f \neq 0$ be its local, gauge-invariant measurable insertion, with the slot's support and eligibility conditions, $|f| \leq h$, and origin in its support. Put $m = m_f(X_s, h)$. Its exponential entry is zero. Its input profile equals f at the original support–inventory address and zero at every other address. It has no ordinary principal cluster: the marked slot is flagged, whereas that cluster requires an ordinary insertion. The nonprincipal and forced prices thus give the zero-leading-coefficient bound without a principal bill. The fine forced norm charges this insertion by m/κ_f. Since $0 < \kappa_f \leq 1$, enlarging the already paid bound ηm gives the required form

$$C_{\mathrm{orb}}\|Q.x\|_1 \leq \eta m/\kappa_f, \qquad r = \eta m/\kappa_f, \qquad \text{input allowance } 2m. \tag{606}$$

The input allowance remains $2m$; it is not multiplied by κ_f^{-1}. The output remainder uses κ_c through the local splitting. The two square roots serve different roles. For $f = 0$ take zero fields and zero response, so no support claim for a nonzero insertion is needed.

114.3 One finite set of insertion types for every grade

On the fixed finite fine torus, form a set containing the ordinary slot selected for every support containing the origin. Adjoin the forced slots selected by the retained slot realization for all support–inventory tags. Both indexing sets are finite, so their union is finite even if different indices select

the same slot. This set contains all three input types above. It depends on the representation and its slot realization, not on the coefficient vector later to be transported.

The map $\mathcal{D}$ was fixed before choosing the degree d. For each d with $d+3$ at most the coarse kernel level, apply the three estimates and (603) with its own frame. Its common allowance is (605), using the stage's local-section price, which bounds all those grades. The exact history identifications replace the interpolating input and output by $R_{N,m+1}$ and $R_{N,m}$ in every equality. This is substitution in fixed maps; it selects no new transition. The transition remains sound because its unmarked density identity, singleton derivative and motion identities were established for this interpolating point.

Finally the common scale payments give, with s_f the fine column allowance and K_f the next coefficient budget,

$$\eta(1+s_f)C_{\mathrm{orb}} \le L^{-3/8}, \qquad (C+\eta s_f)C_{\mathrm{orb}} \le K_f.$$

These are exactly the two purchase inequalities in (1). Together with the selected bank and room bounds there, they allow Proposition 30.2 to combine all columns into one linear coefficient family. At a fixed stage there are only finitely many permitted earlier stages and grades. Taking the maximum of their coupling thresholds supplies all these estimates simultaneously, and any subsequent increase of those thresholds preserves them.

This proves the three elementary estimates, the inverse-map bound, and their common-history use relative to the retained funded interpolation, its slot realization and the common scale payments. The construction of those nonlinear interpolation data remains the separate path-construction problem.

115 Stability in the common support dictionary

The fixed-scale contraction argument needs convergence of a one-step map on each fixed input. Its output is a countable list of local terms. Pointwise convergence of the terms and boundedness of their total price would not be enough: price could escape to later addresses. The needed extra property is uniform smallness of the tails. We prove the summation and local-subtraction argument with this property explicit. The supports, thresholds and address encodings are fixed throughout.

115.1 Uniform tails and products

For nonnegative numbers $a_{N,i}$, indexed by a countable set I, say that their tails are uniformly small if

$$\sup_N \sum_{i\in I} a_{N,i} < \infty, \qquad \forall \epsilon > 0\ \exists I_0 \subset I \text{ finite}: \quad \sup_N \sum_{i\notin I_0} a_{N,i} < \epsilon.$$

The definition is unchanged if the tail sums are replaced by sums over arbitrary finite subsets of $I \setminus I_0$. Indeed all summands are nonnegative. Domination, addition and multiplication by a fixed nonnegative constant preserve this property. Extension by zero through a fixed injection preserves it as well.

It also passes to finite-subset products. If $\sum_i a_{N,i} \le C$, set $A_{N,S} = \prod_{i\in S} a_{N,i}$ for finite subsets $S \subset I$, including the empty subset. For every finite $J \subset I$,

$$\sum_{S\subset J} A_{N,S} = \prod_{i\in J}(1+a_{N,i}) \le \exp\Big(\sum_{i\in J} a_{N,i}\Big) \le e^{C}.$$

This bounds the entire sum by monotone passage through finite J. If S is not contained in I_0, choose one of its elements outside I_0 and overcount by all such choices. Removing the chosen element gives

$$\sum_{S\not\subset I_0} A_{N,S} \le \sum_{i\notin I_0} a_{N,i} \sum_{T\subset I\setminus\{i\}} A_{N,T} \le e^C \sum_{i\notin I_0} a_{N,i}.$$

Thus the finite collection of subsets of I_0 captures all but an arbitrarily small price. Compatibility restrictions only remove nonnegative terms and preserve the conclusion. This is the tail estimate for the finite-subset products in the gas.

There is a corresponding assertion for connected clusters, where indices may repeat. Suppose the elementary prices $a_{N,i}$ have uniformly small tails and obey the rooted cluster criterion with one fixed margin $\rho < 1$. The rooted tree bound controls the total price of clusters of length at least $k+1$ by

$$V\gamma\frac{\rho^k}{1-\rho},$$

where V is the number of sites. Every cluster has a nonempty support, so summing the rooted bounds over sites bounds the unrooted sum. Choose k making this bound small. At each of the finitely many remaining lengths $r \le k$, the absolute connected-graph coefficient divided by $r!$ has a finite maximum: there are finitely many graphs on r labeled vertices. The remaining price is bounded by a constant times a product of r elementary prices. The total price of ordered tuples with some entry outside I_0 is at most

$$rC^{r-1}\sum_{i\notin I_0} a_{N,i}, \qquad C = \sup_N \sum_i a_{N,i}.$$

Choose I_0 to make these finitely many bounds small. The finite set of words of length at most k in I_0 then captures all but an arbitrarily small cluster price. This proves uniform tails for the connected expansion.

For marked clusters the margin is obtained as follows. Suppose the unmarked criterion has margin $1/2$ and rooted allowance $\gamma > 0$, and the marked elementary prices $v_{N,i}$ have uniformly small tails and total bound B. Let M bound the support weight in the rooted criterion on this finite torus. Take

$$\delta = \frac{\gamma}{2(1+MB)} > 0.$$

The additional rooted price of δv is at most $\delta MB \le \gamma/2$. Thus $a + \delta v$ satisfies the criterion with allowance $3\gamma/2$. The degree price is linear in this rooted allowance: its original bound is $\mathrm{degreePrice}(\gamma) \le 1/2$, and hence $\mathrm{degreePrice}(3\gamma/2) = (3/2)\,\mathrm{degreePrice}(\gamma) \le 3/4$. For each word, expansion of the product gives

$$\sum_{j=1}^{r} v_{N,i_j} \prod_{h\ne j} a_{N,i_h} \le \delta^{-1} \prod_{j=1}^{r} (a_{N,i_j} + \delta v_{N,i_j}).$$

The right side includes all terms with exactly one marked factor and has nonnegative remaining terms. At derivative order j, the product rule costs at most $(r+1)^j \le 2^{jr}$; absorb 2^j into each elementary price before this comparison. The paid derivative criterion supplies these enlarged prices. An unmarked price may be replaced by the smaller of its uniform-tail majorant and its rooted-criterion majorant, preserving both bounds. The cluster argument just proved then gives uniform tails for the differentiated logarithm, not only a bound on its total price.

Let $u_{N,i}, v_{N,i}$ take values in a Banach space, and suppose their norms have uniformly small tails. If $u_{N,i} - v_{N,i} \to 0$ for every fixed i, then

$$\sum_i u_{N,i} - \sum_i v_{N,i} \longrightarrow 0. \tag{\dagger}$$

To prove it, choose a common finite I_0 making each norm tail less than $\epsilon/4$. The difference of the two sums over I_0 tends to zero, and is eventually less than $\epsilon/2$. The triangle inequality proves the assertion. Absolute convergence justifies the subtraction. The same proof applies to supremum norms of bounded functions and to norms of their fixed-order derivatives.

One useful form of uniform convergence does not require a compact configuration space. If for every sequence of configurations U_N one has $f_N(U_N) - g_N(U_N) \to 0$, then

$$\sup_U |f_N(U) - g_N(U)| \longrightarrow 0.$$

Otherwise some subsequence would have supremum at least ϵ; choose on it points with difference greater than $\epsilon/2$, and complete them to a sequence arbitrarily. This contradicts the hypothesis. The same selection argument applies to derivative evaluations along normalized local charts. It is this uniform form, rather than pointwise convergence at each fixed configuration, that is needed for moving field insertions.

115.2 From entrywise convergence to a small remainder price

At the fixed scale let X_i be the regular supports and Y_i the forced supports. Write $M(X)t$ for the support price of an amplitude $t \geq 0$. Since the torus is finite, it has finitely many supports, and

$$M_* := \max_X M(X) < \infty.$$

The price is linear in the amplitude. Suppose two regular lists $f_{N,i}, g_{N,i}$ have derivative majorants with uniformly small tails, and for every fixed i the difference has arbitrarily small derivative majorant for sufficiently large N. For any $\epsilon > 0$, choose I_0 so that the sum of the two tail majorants is below $\epsilon/(2M_*)$. On I_0 use the common eventual majorant

$$\frac{\epsilon}{2M_*(|I_0| + 1)}.$$

Outside I_0 use the sum of the original two majorants. The resulting majorants bound every difference, and their total support price is at most ϵ. No supremum over infinitely many convergence thresholds has been taken.

For forced lists use the identical argument with uniform value bounds. If their contribution to the remainder norm is divided by $\kappa > 0$, make their total unnormalized support price at most $\kappa\epsilon/2$, and the regular price at most $\epsilon/2$. Their sum has remainder seminorm at most ϵ. This proves smallness in the full remainder space of Section 117. In particular it does not identify arrays merely because they have the same gas evaluation.

115.3 Local subtraction preserves this convergence

Let $f_{N,i}$ be a regular local term on X_i. Its subtraction uses the fixed linear jet projection π_i, the fixed local section σ_i, and the enlarged support $\widehat{X}_i$:

$$p_{N,i} = \pi_i f_{N,i}, \qquad r_{N,i} = f_{N,i} - \sigma_i p_{N,i}.$$

The mask identity leaves the forced rebate

$$b_{N,i} = \chi_{X_i} f_{N,i} - \sigma_i p_{N,i} - \chi_{\widehat{X}_i} r_{N,i}.$$

At this fixed scale the already established projection and local section estimates bound each of these three linear maps by a fixed constant times its input derivative majorant. Subtracting two inputs gives exactly the same maps of their difference:

$$\Delta p_i = \pi_i \Delta f_i, \quad \Delta r_i = \Delta f_i - \sigma_i \pi_i \Delta f_i, \quad \Delta b_i = \chi_{X_i} \Delta f_i - \sigma_i \pi_i \Delta f_i - \chi_{\widehat{X}_i} \Delta r_i.$$

Thus the projected coefficients converge by (†), and the residual and rebate lists retain the required small-tail bounds and entrywise convergence. Masks may be discontinuous at the threshold. They are identical on the two sides and have modulus at most one, so they do not create a continuity assumption at that threshold.

The forced correction also contains products with the background activities. On a fixed support–inventory profile it is a sum of terms

$$b_{N,i}(U)\, 1_{\mathcal{E}(i,S,U)} \prod_{j \in S} K_{N,j}(U).$$

Here $\mathcal{E}$ is the fixed compatibility and flag condition. For two backgrounds its indicator is the same: the address dictionary, supports, inventory and threshold are the same. For a fixed finite S, the identity

$$\prod_{j \in S} u_j - \prod_{j \in S} v_j = \sum_{j \in S} (u_j - v_j) \prod_{k<j} u_k \prod_{k>j} v_k$$

in any chosen ordering shows convergence of the product when each activity converges uniformly and is uniformly bounded. The product-tail estimate above and (†) justify summing over S. Multiplying by the convergent rebate, summing the finitely many profiles, and using the fixed extension of each profile to its address therefore preserves convergence. The same argument applies to the raw forced output before the rebate is added.

Consequently local subtraction, barcode grouping, addition of the background hook, and placement in the common dictionary preserve convergence of coefficients and smallness of the full remainder difference. These are exact operations on the common representation. Changing its forced dictionary midway would invalidate the product and mask comparisons.

115.4 Summing the fixed input columns

For one fixed input remainder, its regular and forced prices are summable. The selected one-step expansion has one primary column and one column for each regular and forced input entry. The primary column has a single fixed bound; the other columns have bounds proportional to these summable input prices, with the forced price divided by its fine-scale κ. Thus there is a summable sequence w_i bounding, uniformly in terminal cutoff, both the coefficient norm and remainder price of output column i.

Suppose corresponding individual output columns have the same regular supports, convergent coefficient differences, and remainder differences tending to zero in the preceding sense. Choose a finite set of columns outside which $\sum_i 2w_i$ is less than $\epsilon/2$. On that finite set take a common convergence threshold and sum the small coefficient and remainder prices. Outside it use $2w_i$. This proves convergence after all columns have been merged. The unit column is constant and requires no limiting argument. For the nonunit primary column the fixed input function is its local-section

basis element. For regular columns it is the fixed masked regular input; for forced columns it is the fixed profile of the input array. Hence each column comparison uses the same inserted function on the two histories, although its background changes.

Signed motions and complex conjugation are bounded linear or real-linear maps on these data, with the fixed motion prices already proved. The finite motion average therefore preserves the convergence. Applying the fixed coefficient pencil and the common profile capture gives convergence of the actual one-step coefficient map and its recorded remainder on every fixed coefficient vector. The coefficient space is finite dimensional; comparison on its finite basis, followed by the triangle inequality, gives operator-norm convergence. This is the local stability input used by the fixed-scale graph argument.

To apply this result to the selected histories one still needs the individual marked-column comparisons, with uniformly small tails, for their ordinary and forced expansions. Their hypotheses are more specific than a uniform total price bound: convergence of the adjacent path and Gaussian kernels, convergence of the background activities and coupling, and the common marked input are all required. The next analytic comparison must supply these on the same selected histories. The preceding proof isolates the exact summation, subtraction and normalization steps after that comparison; it does not replace it by an unproved assertion of continuity of the infinite expansion.

116 Comparing the marked local integrals

We now supply the local analytic comparison used in Section 115. Fix an adjacent pair of scales and one period. Compare two sequences of paid transitions on the same support dictionary. Assume their fine and coarse canonical path coefficients, retained Gaussian rows and couplings have common limits. The path sums have the strict-width bounds proved above. For the two backgrounds assume that each ordinary and forced activity difference tends uniformly to zero and that their activity majorants have uniformly small tails. In a marked comparison the inserted function and its address are the same on both sides and its paid size is bounded. These assumptions will later be supplied by the background trajectory construction.

The claim is convergence of the difference of each local marked integral and of every paid derivative on its healthy support. It includes uniformity over moving coarse configurations and normalized local charts. The principal issue is that the Gaussian measures in the two integrals vary. We first put them on a common physical variable; this also keeps possibly discontinuous inserted functions at exactly the same argument.

116.1 One common physical integration variable

Fix a forest, its finite coordinate choice and the support pattern of a local term. Its integral is over a compact gauge slice Y with probability Haar measure and a real Gaussian coordinate $z \in \mathbb{R}^D$. The coordinate change in the local integral has the form

$$w = \mu_N(Y) + g_N z, \qquad z = r_N(w - \mu_N(Y)), \quad r_N = g_N^{-1}.$$

Coordinate permutations belonging to the fixed support pattern are included in this notation; they have unit absolute Jacobian. The paid coupling window gives constants $0 < u \le v < \infty$ such that $u \le r_N \le v$, and the local mean estimate gives $\|\mu_N(Y)\| \le B$ independently of N, Y. If ρ_N is the centered Gaussian density, the transformed density is

$$d_N(Y, w) = r_N^D \rho_N(r_N(w - \mu_N(Y))).$$

The actual covariance bounds give fixed $C, \tau > 0$ with $\rho_N(z) \le Ce^{-\tau\|z\|^2}$. Since

$$\|w\|^2 \le 2\|w - \mu_N(Y)\|^2 + 2B^2,$$

we have

$$d_N(Y, w) \le v^D C e^{\tau u^2 B^2} e^{-\tau u^2 \|w\|^2/2}. \tag{‡}$$

Also

$$1 + \|r_N(w - \mu_N(Y))\| \le (1 + v + vB)(1 + \|w\|).$$

Thus a local integrand with bound $C_k(1 + \|z\|)^k$ becomes bounded, after multiplication by d_N, by a constant times

$$e^{-\tau u^2 \|w\|^2/2}(1 + \|w\|)^k.$$

This is integrable on $\mathbb{R}^D$. The constants may depend on the fixed scale, period, support and derivative order, but not on the terminal cutoff, the slice point, or the moving coarse configuration.

The local differentiation theorem of Section 85 writes each derivative as an integral of the differentiated compensated integrand against the Gaussian measure frozen at the base configuration. The bounds in Sections 84 and 93 give the polynomial bound just used. Therefore the change of variable is applied to that already justified derivative integral; it does not differentiate a moving measure without compensation.

116.2 Why the transformed integrand differences vanish

First hold the finite support pattern and coordinate choice fixed. At a fixed physical point (Y, w) the fine configuration read by an inserted activity is independent of the Gaussian covariance and of the coupling: the identity $\mu_N + g_N r_N(w - \mu_N) = w$ cancels their dependence exactly. Hence a common bounded marked function is evaluated at the same fine configuration on both sides. No continuity of that function is needed. An unmarked background activity has the same property; its two values differ by the assumed uniform activity difference.

The other factors in the compensated integrand are smooth finite expressions in the local path and retained-kernel data. Their continuity can be checked directly. A fixed length cutoff leaves only finitely many canonical paths: their displacement vectors lie in a finite lattice box. Each path holonomy is a finite product of link matrices; all its fixed-order derivatives are finite sums of such products. Convergence of its coefficient therefore gives convergence of these derivatives along any sequence of bounded local chart jets. Infinite canonical path sums have the same conclusion by the strict-width summability proved in Section 117.

The local Gram matrices are finite sums and products of those rows. The common positive lower bound passes to their limits. Their inverses converge by

$$A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}.$$

Repeated differentiation of $AA^{-1} = I$ proves convergence of all required inverse jets by induction. Determinants are polynomials in the entries, and the determinant stays strictly positive. Their logarithms and inverse square roots, and their fixed-order derivatives, consequently converge. This covers the selected covariance, means, Gaussian densities and density ratios, including the projected covariance in the compensation. The denominator density at a fixed w has a strictly positive limit; its reciprocal therefore causes no loss of pointwise convergence. Uniform domination is supplied separately by (‡) and the compensated derivative bounds.

Before differentiating these factors, retain the full protected support of each activity and of the marked insertion. The base coarse configuration is healthy on the enlarged union of these supports,

and the normalized chart is based there. Equation (398) holds on every active protected bond wherever the broader cutoff is nonzero near the base point; inactive bonds stay fixed by definition. Indeed, a nonzero cutoff germ gives the base fluctuation bound $4at$; continuity gives the strict $8at$ bound on a neighborhood. The compensator identity then applies on all the finitely many protected bonds on one common neighborhood. Where a cutoff has zero germ, the complete factor has zero germ. Consequently an activity or mark h_N occurs, as a germ in the chart parameter y, only as

$$h_N(U_N(Y,w))\,\zeta_N(y;Y,w),$$

with $U_N(Y,w)$ independent of y and ζ_N smooth. This is exactly the held-factor identity (487), applied to each relevant read support. All derivatives fall on ζ_N and the remaining smooth factors. Uniform convergence of activity values is therefore sufficient; convergence of their uncompensated derivatives is not being inferred.

For completeness, the elementary factors in a local product fall into four mathematical cases. A cell-action factor is the exponential of a finite local action and fixed cutoff functions; its action jets converge by the preceding path argument. A forced-activity factor is its value at the common fine configuration times a smooth local cutoff. An ordinary logarithmic factor is the exponential minus one of an ordinary activity value together with the local quadratic compensation; the first value is compared uniformly and the second by the matrix argument. A warning factor is a fixed local mask or charge times a smooth cutoff, possibly minus its constant term. The flag and compatibility decisions have the same values on both sides, because their geometric support data, threshold and physical configuration are identical. Product and chain rules now prove that the differences of the fixed-order jets of every finite product tend to zero. With one marked factor, replace the corresponding ordinary factor by the common inserted function times its cutoff; the same proof applies.

Moving configurations do not require convergence along the whole sequence. For any subsequence, take a further subsequence on which their finitely many link matrices and the bounded chart jets converge. An order-r derivative evaluation can be pulled back to r real parameters, so these finitely many jets lie in one fixed finite-dimensional bounded set. Compactness of the gauge group and of the closure of that set gives the subsequence. The support and coordinate choices range over finite sets, so take them constant on a further subsequence. The calculations above make the difference of the two transformed integrands tend to zero there. A common bounded mark need not itself converge: it multiplies a smooth-factor difference tending to zero, and its uniform bound suffices. If a difference failed to tend to zero in the original sequence, a subsequence would stay a positive distance from zero and could not have such a further subsequence. Thus the differences tend to zero without any preliminary convergence assumption on the moving configurations. Shared discontinuous indicators may also vary: for their common value $\mathbf{1}_{E_N}$ use $|\mathbf{1}_{E_N}(A_N-B_N)| \le |A_N-B_N|$. Only the finite structural support and coordinate choices are frozen; no simultaneous stabilization of indicators for every (Y,w) is needed.

116.3 Integration and local derivative norms

Let $F_N(Y,z)$ and $G_N(Y,z)$ be the two compensated derivative integrands. For fixed Y,w, their density-weighted transforms have difference tending to zero. Each is dominated by an integrable function of w independent of Y,N. Dominated convergence first in w proves convergence of the difference of their inner integrals for almost every Y. The norms of these differences are bounded by the integral of the sum of the two majorants. A second application of dominated convergence against the probability Haar measure in Y gives convergence of the complete local integrals. This

argument also justifies the two uses of Fubini, since absolute integrability was established before passing to the limit.

Apply this to every moving base configuration and normalized chart of each order up to the paid order. The selection argument of Section 115 turns it into uniform convergence of those derivative evaluations. On the fixed finite torus, the conversion from normalized chart derivatives to the local derivative price has a finite constant: all site separations are bounded, and only finitely many derivative orders occur. Therefore each individual ordinary output entry has difference tending to zero in its required derivative price. For the forced entry use order zero and arbitrary moving configurations; its uniform value difference tends to zero.

Finite averaging over forests and coordinate choices preserves both statements. The ordinary marked logarithm is a sum of finite products with one marked factor, and the forced marked polynomial is formed by finite-subset products, addition of the ordinary logarithm, and differentiation in its one source parameter. Each operation preserves the entrywise convergence just proved. The marked cluster and product estimates in Section 115 give their uniform tails. Local subtraction and the background hook then give the convergence of each fixed input column in the actual common dictionary. Summation and finite motion averaging give the adjacent-scale stability required in Section 117.

This comparison uses both backgrounds' common-limit hypotheses; it does not prove convergence of the background activity trajectory. It proves the needed continuous dependence of the selected ordinary and marked local transformations, with their masks, Gaussian measures, derivatives and countable output lists retained.

117 Convergence at a fixed scale and normalization

The full-mesh argument in Section 31 compares the final observable with one at a fixed intermediate scale. We now prove the convergence mechanism at that scale. The period and the intermediate scale are fixed throughout; the terminal scale tends to infinity. In particular the configuration space in each convergence statement below is one fixed finite torus.

117.1 Contraction of the remainder graph

At scale m, keep every regular support and forced address in one fixed dictionary. Let E_m be its vector space of admissible remainder arrays, with the seminorm obtained by minimizing their total regular mass plus κ_m^{-1} times their forced mass. The regular mass uses the prescribed derivative bounds. This seminorm retains the actual arrays; quotienting by equality of their gas evaluations is not part of its definition. The elementary sum and scaling estimates give its triangle inequality and absolute homogeneity. Evaluation of the masked ordinary sum has norm at most one, and evaluation of any forced support–inventory profile has norm at most κ_m. For example,

$$\left|\sum_a \chi_{X_a} r_a(U)\right| \le \sum_a \|r_a\|_{X_a} \le \|r\|_{E_m}, \qquad |\mathcal{P}(r_{\text{forced}})(b,U)| \le \kappa_m \|r\|_{E_m}.$$

To justify the last inequalities when the infimum defines the norm, first use any admissible majorant within ε of the infimum and let $\varepsilon \downarrow 0$. The same estimates applied to a difference give uniform Lipschitz bounds on these evaluation maps. The absolutely summable majorants justify subtracting the series.

Represent a linear family of remainders by a map $G : F_d \to E_m$ with

$$\|Gv\|_{E_m} \le s_m \|\operatorname{core} v\|_1, \qquad s_m \ge 1.$$

Two such maps are within $a \geq 0$ if

$$\|(G - H)v\|_{E_m} \leq a s_m \|v\|_1 \quad \text{for every } v.$$

This relation is symmetric, has the additive triangle property, and any two allowed graphs are within 2.

One step produces from G a coarse graph ΦG and an invertible coefficient map I_G. Before reparametrizing by the coarse coefficient, write its remainder map as $J_G = (\Phi G) I_G$. The paid regular and forced estimates of Section 114, applied to differences of insertions on the same supports, give

$$\begin{aligned} \|(I_G - I_H)v\|_1 &\leq \eta a s_{m+1} C_{\mathrm{orb}} \|v\|_1, \\ \|(J_G - J_H)v\|_{E_m} &\leq \eta a s_{m+1} C_{\mathrm{orb}} \|v\|_1 \end{aligned} \tag{607}$$

when G, H are within a. Here the same raw supports are used in the two expansions. Subtraction cancels their primary insertion. The ordinary integral and the marked singleton derivative are linear in the remaining insertion; local subtraction uses the same linear section. Thus the difference has the price of the difference of the inputs. If representatives are chosen through derivative majorants, the bound is obtained with those majorants and then their infimum. Finite motion averaging costs C_{orb}, exactly as in the weighted sum in Section 30. This explains why one must retain the common slots and supports.

The same one-step estimate gives $\|I_G^{-1} v\|_1 \leq s_{m+1} \|v\|_1$. Put $\delta = \eta s_{m+1} C_{\mathrm{orb}}$ and let $b = L^{-3/8}$. The schedule's two relevant payments are

$$\delta \leq b, \qquad 4 s_{m+1} b \leq 1.$$

The second follows from its bank inequality: if $q = L^{1/8} \geq 1$, that inequality gives $4 s_{m+1} \leq q$, while $b = q^{-3}$. For a coarse vector x, set $y = I_G^{-1} x$. Then

$$\begin{aligned} \|(\Phi G - \Phi H)x\|_{E_m} &\leq \|(J_G - J_H)y\|_{E_m} + \|\Phi H (I_H y - x)\|_{E_m} \\ &\leq (1 + s_m) \delta a \|y\|_1 \\ &\leq (1 + s_m) \delta a s_{m+1} \|x\|_1 \leq \tfrac{1}{2} a s_m \|x\|_1. \end{aligned}$$

Thus the coarse graphs are within $a/2$. This contraction acts on the complete regular and forced remainder, after reparametrizing by its actual coefficient map.

117.2 Forgetting the terminal graph

Write $G_{m,N}$ for the selected graph at scale m in the history with terminal index N. For each fixed m, let $\Phi_{m,N}$ be its one-step map. Retain the following precise local stability input: for every fixed admissible graph H, the graphs $\Phi_{m,N} H$ are Cauchy in the preceding "within" relation, and their coefficient maps $I_{m,N,H}$ are Cauchy in operator norm. Their dependence on H has the uniform bounds just proved. This is a statement about the selected adjacent-scale kernels and their marked derivatives; the contraction alone does not imply it.

First, if H_N is Cauchy, then $\Phi_{m,N} H_N$ is Cauchy. For a prescribed small error, fix a sufficiently late N_0. Compare $\Phi_{m,N} H_N$ with $\Phi_{m,N} H_{N_0}$ using contraction, compare the latter at two terminal indices using fixed-input stability, and compare back to $\Phi_{m,M} H_M$. The two outside errors are half the corresponding input errors. The identical three-term comparison with (607) proves that I_{m,N,H_N} is Cauchy.

For a fixed integer $h \geq 0$, replace $G_{m+h,N}$ by one fixed admissible graph and propagate it downward through h steps. The resulting graph $V_{m,N}^{(h)}$ is Cauchy by the preceding paragraph and finite induction. The exact history recursion and the bound 2 on any initial graph difference imply

$$G_{m,N} \text{ is within } 2^{1-h} \text{ of } V_{m,N}^{(h)}$$

for all sufficiently large N. Given $\varepsilon > 0$, choose h so that $2^{1-h} < \varepsilon/4$, and then use the Cauchy property of $V^{(h)}$. The triangle inequality proves that $G_{m,N}$ is Cauchy. Only a finite number of levels is used for each error tolerance; no infinite product of inverse maps occurs.

Now fix the label y at its reference scale r. Its coefficient at that scale is exactly y, because the terminal coefficient is normalized by the inverse of its readout at r. For each fixed $m \geq r$, let $x_{m,N}$ be the resulting coefficient. The history relation is

$$x_{m,N} = I_{m,N,G_{m+1,N}} x_{m+1,N}.$$

Suppose $x_N = S_N y_N$ is Cauchy, S_N is Cauchy in operator norm, and $\|S_N^{-1}\| \leq b$. Then y_N is bounded and

$$\|y_N - y_M\|_1 \leq b\big(\|x_N - x_M\|_1 + \|S_M - S_N\|\,\|y_M\|_1\big) \longrightarrow 0.$$

Starting with the constant reference coefficient, finite induction therefore proves that $x_{m,N}$ is Cauchy at every fixed m. Moreover

$$G_{m,N}x_{m,N} - G_{m,M}x_{m,M} = G_{m,N}(x_{m,N} - x_{m,M}) + (G_{m,N} - G_{m,M})x_{m,M}$$

is Cauchy in E_m. This proves convergence of the coefficient and Cauchyness of the full remainder data in E_m at the fixed scale, conditional only on the specified adjacent-scale stability input and its paid history realization. Their coefficient norms are bounded for all terminal indices; finitely many indices before the eventual recursion can be included by enlarging the bound.

117.3 Uniform convergence of the inserted functions

The primary exponential-entry map is linear in $x_{m,N}$ and uniformly bounded on the fixed finite torus. Indeed each of its finitely many local-section columns is continuous on the compact gauge configuration space; multiplication by its fixed mask preserves its supremum bound. Expanding in the coefficient basis therefore bounds its difference by $C\|x_{m,N} - x_{m,M}\|_1$. The ordinary remainder and each forced profile are uniformly Cauchy by their evaluation bounds in the first subsection. Adding them proves that the exponential entries and forced profiles of the inserted fields are uniformly Cauchy over all configurations. Their bounded scalar-valued function spaces are complete, so they have uniform limits; no completeness of E_m is needed. This proof permits discontinuous masks: their supports and thresholds are fixed at this scale. Translation, signed coordinate motion and conjugation preserve the conclusion. The fixed physical scaling factor does too.

For a finite set of inserted labels all these statements hold on one common tail. Their coefficient prices have finite bounds M_i. Given positive allotted prices ζ_i, take $w_i = \zeta_i/(|M_i| + 1) > 0$. Then w_i times the actual coefficient price is at most ζ_i, uniformly in terminal index. This is the positive weight choice used in the fixed-rung vault of Section 31.

117.4 Convergence of the background density

We spell out the background convergence needed for the same normalized moments. At the fixed scale retain common path stencils and activity addresses. Suppose the path coefficients converge in

the weighted sums of their differences: for one $v > 0$,

$$\sum_a |c_{N,a} - c_a| e^{v \operatorname{length}(a)} \longrightarrow 0$$

for each one-form row, and the analogous atom-budget sums tend to zero for the quadratic loop, cubic and quartic arrays. All paths have the common bounds of the construction. Suppose also that the couplings β_N are Cauchy, and the background activity arrays are Cauchy in their common norm of rooted ordinary and forced masses. These are the precise background inputs used here.

The path convergence gives uniform convergence of the action on the fixed configuration space. To see this directly, each transported curvature in a row has norm at most 2, so a row difference is at most twice its coefficient-difference budget. A degree-q atom difference, with the same stencil, is at most 2^q times its coefficient budget. The masks are identical for the two terms and hence do not add an error. The common path bounds bound all row values uniformly; consequently $|\|u\|^2 - \|w\|^2| \le (\|u\| + \|w\|)\|u - w\|$ treats the quadratic row action. There are finitely many sites, plaquette directions and signed motions. Summing their estimates, and using boundedness and convergence of β_N, proves the assertion for the full symmetrized action.

For the activity arrays, sum a rooted budget over the finitely many sites. Every ordinary term has its designated site, and every forced term has a nonempty marked support, so each term is counted at least once. If V is the number of sites and b, k are the fixed two activity prices, the resulting uniform evaluation bounds are $Vb\|A\|$ for the ordinary sum and $Vk\|A\|$ for a forced profile. Applied to differences, these give uniform convergence of the ordinary activity sum and every support–inventory profile. The fixed torus has finitely many such profiles. Its gas is a finite sum over compatible subsets of these profiles, each multiplied by the fixed flag indicator. It therefore converges uniformly as well. This finite profile description does not require the original activity list to be finite; its regrouping was justified by the absolute budgets.

Write the full density as ρ_N and its scalar term as s_N. Its scalar-stripped density

$$\widetilde{\rho}_N = e^{-s_N} \rho_N$$

is the product of the gas and the exponential of minus the path action and ordinary activity sum. All exponent arguments are uniformly bounded on a tail. Exponentiation is uniformly continuous on that bounded interval, so the stripped density is uniformly Cauchy. Its integral Z_N^0 against the probability Haar prior is Cauchy, since

$$|Z_N^0 - Z_M^0| \le \sup_U |\widetilde{\rho}_N(U) - \widetilde{\rho}_M(U)|.$$

The weighted path convergence just used follows from coefficientwise convergence at a strictly larger exponential width. Here is the useful precise implication. If $v < w$, the row coefficients at width w have a common budget B, and each coefficient converges, then

$$|c_{N,a} - c_a| e^{v \operatorname{length}(a)} \le 2B e^{-(w-v) \operatorname{length}(a)}.$$

Here the canonical row index enumerates a plaquette direction and a vector $x \in \mathbb{Z}^4$; its chosen coordinate path has length $|x|_1 + 4$. Consequently, for every $\delta > 0$,

$$\sum_a e^{-\delta \operatorname{length}(a)} = |\mathcal{A}| e^{-4\delta} \left(\frac{1 + e^{-\delta}}{1 - e^{-\delta}} \right)^4 < \infty,$$

where $\mathcal{A}$ is the finite set of plaquette directions. This is a sum over the chosen coordinate paths, not over all walks. Dominated convergence for series gives convergence of the whole row difference budget.

For an atom of degree $q = 2, 3, 4$, the canonical index enumerates q plaquette directions and $q-1$ displacement vectors. The total length is $\ell = \sum_{j=1}^{q-1} |x_j|_1 + 4q$. Its coefficients have the form $e^{-w\ell} C_N(a, \nu)$, where ν runs through the finite color multi-indices. The coefficient arrays C_N converge coordinatewise and have a common Hilbert norm bound B; finite partial sums give the same bound for their limit C. The atom difference budget at width $v < w$ is exactly

$$\sum_{a,\nu} e^{-(w-v)\ell(a)} |C_N(a, \nu) - C(a, \nu)|.$$

Each summand is at most $2Be^{-(w-v)\ell(a)}$. Summing this majorant gives a finite multiple of

$$e^{-4q(w-v)} \left(\frac{1 + e^{-(w-v)}}{1 - e^{-(w-v)}} \right)^{4(q-1)}.$$

A second application of dominated convergence proves the required atom-budget convergence. Choose v strictly between the paid width and the minimum of the larger row and three atom widths. All four conclusions then hold at this one width, on a common tail. It is essential that these are the same canonical stencils; convergence of an unrelated finite-volume matrix is insufficient.

117.5 A positive limit of the partition functions

Positivity of each Z_N^0 alone would not suffice. We need one strictly positive lower bound independent of N. Let θ be the fixed scale's small-field threshold, let $H = 2\widehat{\beta}_m$ be the common upper bound on its coupling, and let F, T be the ordinary and loop allowances there. With A the path price and $\mathcal{C}$ its quadratic clipping data, set

$$K = V\{F + \mathrm{loopsPrice}(\theta, H, A, T) + H\,\mathrm{baseBudget}(\mathcal{C}, A, \theta)\}.$$

These quantities are finite and fixed while N varies. The small-link-ball proof in Section 19 gives

$$Z_N^0 \geq a_m := \exp\{-K - |E_m^{\mathrm{bond}}| d_G \log(16/\theta)\} > 0. \tag{608}$$

Here E_m^{bond} is the finite set of positive bonds, not the remainder space, and d_G is the ambient real matrix dimension. The bound follows by taking every link in the ball of chord radius $\theta/4$: all flags vanish, the action is bounded by K, and the product Haar probability is at least $(\theta/16)^{|E_m^{\mathrm{bond}}| d_G}$. Nonnegativity of the full selected density permits restriction of its integral to this event. The scalar factor cancels in Z_N^0. Replacing β_N by H only increases the nonnegative action bound. Thus the Cauchy limit Z^0 satisfies $Z^0 \geq a_m > 0$.

Finally let $P_{N,i}(U)$ and $F_{N,i,b}(U)$ be the inserted exponential entries and forced profiles already shown uniformly Cauchy. Fix a finite census algebra, whose nonconstant generators are nilpotent. The scalar-stripped restored polynomial is

$$e^{-\text{path action}_N - \text{ordinary sum}_N} \exp_{\mathrm{nil}}\left(\sum_i t_i P_{N,i} \right) \mathcal{G}\left(\text{background profiles}_N + \sum_i t_i F_{N,i} \right).$$

The nilpotent exponential and the grouped gas are finite polynomials in their entries. Hence every mode is uniformly Cauchy and bounded. Their measurability follows from that of the local sections, masks and coefficient arrays and from the absolutely convergent groupings. Their Haar integrals are Cauchy. Dividing by $Z_N^0 \to Z^0 > 0$ proves convergence of each normalized fixed-scale mode. The scalar-stripping identity identifies this ratio with the actual restored history moment, since its numerator and partition function contain the identical factor e^{-s_N}. This proves the fixed-rung convergence required by the full-mesh comparison, under the stated selected branch and background convergence inputs. Their production must still be established for the same funded history.

118 A uniform analytic price for the bare plaquette energy

The path construction uses uniform analytic bounds for the unperturbed Wilson energy, before it compares that energy with a general path. This section records the finite-volume estimate and its proof. All constants are independent of the torus side.

Write $\mathcal{A}$ for the finite set of positive oriented plaquette types, $N_{\mathcal{A}} = |\mathcal{A}|$, and let d_G be the dimension of the fixed matrix representation. For a link field U on a torus of side m, put

$$W_m(U) = \sum_{x \in (\mathbb{Z}/m\mathbb{Z})^4} \sum_{a \in \mathcal{A}} \left(1 - \frac{\operatorname{Re}\operatorname{tr} U(\partial(x,a))}{d_G}\right).$$

Here $U(\partial(x,a))$ is the ordered four-edge holonomy. The estimate below is a coefficientwise price uniform in m; the action itself is extensive. We use Lie-algebra coordinates z at the identity and write $\widetilde{W}_m(z) = W_m(\exp z)$; every derivative below means the derivative of this coordinate function at $z = 0$. The norm on link-coordinate arrays and on their local Fourier-mode restrictions is the supremum norm.

For $t \in \mathbb{N}$, set $n = t + 1$. A coefficient picture of order t assigns a function of t independent complex momenta $q_1, \dots, q_t$ to every list of n colors and oriented-link directions; the zeroth momentum is fixed by $q_0 = -\sum_{j=1}^{t} q_j$. The width-one strip is the set where every coordinate of each independent momentum has imaginary part of absolute value at most one. Let $\mathrm{CTax}(R, 0) = 1$ and $\mathrm{CTax}(R, k+1) = \mathrm{CTax}(R,k)/(R/2^{k+1})$.

Proposition 118.1 (Uniform bare-energy coefficient price)**.** *There are constants $R_* > 0$ and $A_* < \infty$, depending only on the compact gauge representation and plaquette convention, and for each t there is a single picture $\mathcal{W}_t$ that works for every torus side m. Its components are holomorphic on the width-one momentum strip and satisfy*

$$\sup_{q \text{ in the strip}} |\mathcal{W}_t(q)| \le A_t := N_{\mathcal{A}} A_* \exp\bigl(16(t+1)^2\bigr) \frac{\mathrm{CTax}(R_*, t+1)}{(t+1)!}.$$

Moreover, let $p_0, \dots, p_t$ be dual-torus momenta, $c_0, \dots, c_t$ colors, and $b_0, \dots, b_t$ oriented-link directions. Let $e_{p,b,c}$ denote the corresponding Fourier-mode variation of the link field, and put $q = (p_1, \dots, p_t)$. Then

$$\frac{1}{n!} D^n \widetilde{W}_m(0)\bigl[e_{p_0,b_0,c_0}, \dots, e_{p_t,b_t,c_t}\bigr] = \begin{cases} |(\mathbb{Z}/m\mathbb{Z})^4|\, \mathcal{W}_t(\mathbf{c}, \mathbf{b}; q), & \sum_{j=0}^{t} p_j = 0, \\ 0, & \sum_{j=0}^{t} p_j \neq 0. \end{cases}$$

Thus the volume factor in the zero-total-momentum coefficient is exactly the torus cardinality, and every nonzero-total-momentum coefficient vanishes.

Proof. First construct a single plaquette extension. Let $\mathfrak{g}_{\mathbb{C}}$ be the complexification of the represented Lie algebra and let $\mathcal{E} : \mathfrak{g}_{\mathbb{C}} \to \mathrm{Mat}_{d_G}(\mathbb{C})$ be the linear representation map. For a complex link-coordinate array z, the two oriented link matrices are $\exp(\mathcal{E} z_e)$ and $\exp(-\mathcal{E} z_e)$. These depend holomorphically on z, since the matrix exponential is entire. Multiplication along the four edges, trace, and the average of the forward and reverse loop traces are holomorphic too. On real coordinates that average is the real trace of the holonomy, because for the unitary holonomy H one has $\operatorname{tr}(H^{-1}) = \overline{\operatorname{tr}(H)}$. Thus this extension restricts to the real plaquette energy summand.

Continuity of the exponential at zero gives $R_* > 0$ such that complex link coordinates in the radius-R_* ball give oriented matrices within $\mathrm{rad}(1)$ of their physical unitary matrices. In the source's

Frobenius norm, this implies the correct product estimate as follows. If M is within $r = \mathrm{rad}(1)$ of a unitary V, then for every matrix X,

$$\|MX\|_F \le \|VX\|_F + \|(M-V)X\|_F \le (1+r)\|X\|_F,$$

since left multiplication by V is an isometry and the operator norm of $M-V$ is at most its Frobenius norm. Iterating this inequality through four factors, starting with $X = I$, gives $\|M_1M_2M_3M_4\|_F \le (1+r)^4\|I\|_F$. The reverse loop has the same bound. The continuous pulse

$$(M,N) \longmapsto 1 - \frac{\operatorname{tr} M + \operatorname{tr} N}{2d_G}$$

is bounded on the product of the two closed Frobenius balls of radius $T = (1+r)^4\|I\|_F$ about zero, since both four-edge products lie there. Compactness in finite dimension gives a common bound A_*. Together these facts give a uniform holomorphic extension bounded by A_* on a fixed ball, at every plaquette anchor and every torus size.

Each plaquette is supported in the radius-four coordinate box about its anchor: every link in its four-edge boundary has an anchor at max-distance at most four. Hence its real local action is unchanged when the cover configuration is changed outside that box.

To define the common local symbol, place each radius-four box in the auxiliary torus $\Lambda_9 = (\mathbb{Z}/9\mathbb{Z})^4$, with $9 = 2d+1$ and $d = 4$. The box embeds in this torus without identifications. For each plaquette type, evaluate its local Taylor form there on the restricted Fourier modes $y \mapsto \mathbf{1}_{(b,c)} \exp(i\langle p, \operatorname{lift} y\rangle)$, and sum these local symbols over the plaquette types; this defines $\mathcal{W}_t$ at the corresponding independent momenta. For a period of any side m, pull a period-m field z to the auxiliary torus by

$$(P_m z)(y,b,c) = z(\mathrm{wrap}_m(\operatorname{lift} y), b, c).$$

Then the local plaquette function satisfies $\mathrm{local}_m(z) = \mathrm{local}_9(P_m z)$. The pullback of a period-m Fourier mode is exactly the restricted Fourier mode used above. This identity holds even when $m < 9$ and distinct auxiliary sites wrap to the same period-m link: evaluating local_9 on $P_m z$ retains all repeated-link contributions. Thus the local symbol is the same for every period, including the small periods. The picture variables are momenta, rather than link coordinates. To obtain the strip bound, evaluate the order-n Taylor multilinear form of the local plaquette extension on the Fourier-mode vectors e_{p_j,b_j,c_j}. The Cauchy derivative estimate on successively shrinking balls of radii $R_*/2, R_*/2^2, \dots, R_*/2^n$ bounds its order-n derivative by $A_* \,\mathrm{CTax}(R_*, n)$. The Taylor coefficient is normalized by $1/n!$. For a momentum in the width-one strip, a Fourier mode restricted to the radius-four stencil has supremum norm at most $\exp(16n)$: the dependent momentum has imaginary coordinates bounded by t, and a stencil displacement has four coordinates, each of absolute value at most four. There are n input modes in the multilinear form, so their product costs at most $\exp(16n^2)$. This proves the claimed bound for each plaquette type. Summing over the finite set $\mathcal{A}$ multiplies it by $N_{\mathcal{A}}$ and yields a single picture $\mathcal{W}_t$, independent of the torus side.

It remains to check the finite-volume coefficient identity. Expand the translated plaquette sum and apply the multilinear coefficient to an ordered momentum/color/direction tuple. Translating a local plaquette coefficient by an anchor multiplies its Fourier mode by the character at total momentum $\sum_{j=0}^{t} p_j$. The sum over all anchors is zero unless that total momentum is zero; in that case each anchor contributes the same local coefficient, so the sum is exactly $|(\mathbb{Z}/m\mathbb{Z})^4|$ times the local symbol $\mathcal{W}_t$ evaluated at the output phase. This proves the displayed identity, including its volume normalization. □

Remark 118.2 (Use in the selected path estimate). The source's all-orders summation lemma applies this same local-extension argument to a countable family of interaction atoms, with the summability condition

$$\sum_j b_j \exp\big((t+1)^2\sigma\, 4d_j\big) < \infty,$$

where d_j is the support length and b_j its absolute coefficient times its exponential length weight. For the bare action the family is finite in plaquette type, has support length four, and the preceding bound gives its summability directly. Applying the lemma at orders three and four yields the two Wilson reference pictures used in the one-step path estimate. They are uniform over torus sides; no limit in the volume is used here.

119 The weighted quadratic row estimate

The part of the one-step action quadratic in the row field has a uniform coefficient price. Its cost is the square of the exponentially weighted row budget. The proof is an exact two-index expansion followed by a summable Cauchy estimate; no finite-volume truncation of the row family is needed.

Let CTax be the iterated Cauchy derivative factor defined in Section 118. Let $\mathcal{A}$ be the finite set of plaquette types. Write d_{col} for the number of color coordinates and d_{mat} for the dimension of the matrix representation; these constants have different roles below. For each $a \in \mathcal{A}$ and $i \in \mathbb{N}$, let $s_{a,i}(U;x)$ be a color vector obtained by evaluating one path slot at the translated anchor x, and let $\ell_{a,i}$ be that slot's length. Let $c_{a,i} \in \mathbb{R}$ be its row coefficient, and assume $\ell_{a,i} \in \mathbb{N}$. The color vector takes values in the finite-dimensional real Euclidean color space on physical fields; its complex extension applies the complex-linear extension of the real color projection to half the difference between the forward slot word and its reversed word. On real fields this is the original skew-color slot. The quadratic extension is bilinear, $\sum_c s_c s_c$, not a Hermitian norm. Assume the row budget

$$\sum_i |c_{a,i}| e^{\sigma \ell_{a,i}} \le C \quad (a \in \mathcal{A}), \qquad \sigma > 0. \tag{609}$$

We use the following elementary path bound. For each $\sigma > 0$ there are a common complex coordinate radius $R_{\sigma/2} > 0$ and a constant S_G such that, on the radius-$R_{\sigma/2}$ ball about the identity, every color component of a slot of length ℓ has a holomorphic extension bounded by $S_G e^{(\sigma/2)\ell}$. More explicitly, if $H_a(z)$ is the matrix product along the slot walk and $H_a^{\mathrm{rev}}(z)$ the product along the reversed slot walk, the extension is

$$s_a^{\mathbb{C}}(z) = \Pi_{G,\mathbb{C}}\left(\frac{H_a(z) - H_a^{\mathrm{rev}}(z)}{2}\right),$$

where $\Pi_{G,\mathbb{C}}$ is the complex-linear extension of the real skew-color projection. On real links, the reversed full slot walk is the inverse walk; its string matrix is the adjoint of the forward string matrix. Their half-difference is skew-adjoint, and the physical color projection of this skew matrix is exactly the transported-curvature slot. Thus the complex extension agrees with the original real slot on physical fields. The two words each contain at most 2ℓ edge factors. Choose the radius so that each complexified edge matrix and its inverse has operator norm at most $e^{\sigma/4}$; this is possible by continuity near the compact unitary group. Each word then has norm at most $e^{(\sigma/2)\ell}$. Apply the bounded complex-linear projection; its norm and the fixed factor from the difference are absorbed in S_G. This proves the bound uniformly in location and length.

Define the quadratic row action on a torus of side m by

$$Q_m^{(2)}(U) = \frac{1}{2} \sum_{x \in (\mathbb{Z}/m\mathbb{Z})^4} \sum_{a \in \mathcal{A}} \left\| \sum_i c_{a,i} s_{a,i}(U;x) \right\|^2 .$$

The sum over i converges absolutely on the real compact configuration space by (609) and the path bound. On the common complex coordinate ball, the two-index expansion below also converges uniformly: its absolute majorant is the product of the two row sums with exponent $\sigma/2$, which is bounded by the assumed row budget. Thus the complexified action is holomorphic there before any derivatives or momentum coefficients are taken.

Proposition 119.1 (Uniform price of the quadratic row action)**.** *For each $t \in \mathbb{N}$, put $n = t+1$ and*

$$\eta_t = \min\Big(1, \frac{\sigma}{8n^2}\Big) .$$

There is one coefficient picture $\mathcal{Q}_t$ valid at every torus side, holomorphic for t independent complex momenta of width η_t, whose components are bounded by

$$\frac{|\mathcal{A}| d_{\mathrm{col}}}{2} C^2 S_G^2 \frac{\mathrm{CTax}(R_{\sigma/2}, n)}{n!}.$$

For each ordered list of n Fourier-mode variations, its coefficient is zero unless their total momentum vanishes; in the zero case it is the torus cardinality times the value of $\mathcal{Q}_t$ at the independent momenta.

Proof. For each a, i, j, expand the squared norm into the two-slot term

$$\Gamma_{a,i,j}(U;x) = \frac{1}{2} c_{a,i} c_{a,j} \sum_{c=1}^{d_{\mathrm{col}}} s_{a,i,c}(U;x) s_{a,j,c}(U;x).$$

It has support length at most $\ell_{a,i} + \ell_{a,j}$ and total absolute coefficient $d_{\mathrm{col}} |c_{a,i} c_{a,j}|/2$. Therefore its total weighted activity satisfies, by Tonelli and (609),

$$\begin{aligned} \sum_{a,i,j} \frac{d_{\mathrm{col}}}{2} |c_{a,i} c_{a,j}| e^{\sigma(\ell_{a,i}+\ell_{a,j})} &= \frac{d_{\mathrm{col}}}{2} \sum_a \left(\sum_i |c_{a,i}| e^{\sigma \ell_{a,i}} \right)^2 \\ &\le \frac{|\mathcal{A}| d_{\mathrm{col}}}{2} C^2. \end{aligned}$$

We now estimate the coefficient of each two-slot term. On the common radius-$R_{\sigma/2}$ ball, the product of its two holomorphic slot extensions is bounded by $S_G^2 e^{(\sigma/2)(\ell_{a,i}+\ell_{a,j})}$. The Cauchy derivative estimate on the nested radii $R_{\sigma/2}/2, \dots, R_{\sigma/2}/2^n$ contributes $\mathrm{CTax}(R_{\sigma/2}, n)/n!$ to the normalized order-n coefficient. Restricting a width-η_t Fourier mode to a support of length ℓ costs at most $e^{4n\eta_t \ell}$; for n inputs the total additional factor is at most $e^{4n^2 \eta_t \ell}$. Since $4n^2 \eta_t \le \sigma/2$, this factor is at most $e^{(\sigma/2)\ell}$. Consequently the slot-extension and Fourier-mode factors together are bounded by $S_G^2 e^{\sigma(\ell_{a,i}+\ell_{a,j})}$.

The weighted activity sum just computed is thus a summable majorant for the series of coefficient pictures, uniformly on the smaller momentum strip. The Weierstrass M-test gives uniform convergence on compact subsets; its sum is holomorphic, and Cauchy's integral formula permits termwise extraction of each Taylor coefficient. On the real field, absolute convergence also permits

expanding the row square and interchanging the sums, so this summed picture represents exactly $Q_m^{(2)}$.

Finally, translating each local two-slot term over all anchors multiplies its Fourier coefficient by the character of the total momentum. The character sum on the finite torus is zero at nonzero total momentum and equals the torus cardinality at zero total momentum. For each pair atom, its finite support lies in a max-distance box of radius $d = \ell_{a,i} + \ell_{a,j}$. Compute its local symbol on the auxiliary torus Λ_{2d+1}; this box has no internal identifications there. Pull any period-m field back to this auxiliary torus by $z(y) = z_m(\mathrm{wrap}_m(\mathrm{lift}\, y))$. The identity holds even when $m < 2d+1$, so repeated links are retained. Summing the local symbols over atoms gives one picture independent of m, with the asserted volume factor and price. □

Corollary 119.2 (The averaged path-kernel coefficient price)**.** *Suppose the unaveraged path kernel has the exact decomposition*

$$Q_{p,R,m} = \lambda W_m + Q_m^{(2)}, \qquad 0 \le \lambda \le Z,$$

with $Q_m^{(2)}$ as above. Then the unaveraged order-n picture has width η_t and price at most

$$A_{raw} = Z A_t + \frac{|\mathcal{A}| d_{\mathrm{col}}}{2} C^2 S_G^2 \frac{\mathrm{CTax}(R_{\sigma/2}, n)}{n!}, \qquad n = t+1.$$

Let $\mathcal{S}$ be the finite signed-coordinate group and define the average $Q^\sharp(U) = |\mathcal{S}|^{-1} \sum_{s\in\mathcal{S}} Q_{p,R,m}(\rho_s U)$, where ρ_s is the corresponding signed coordinate action on link fluctuations. For this averaged kernel, the common picture has width η_t and price at most

$$A_{avg} = \mathrm{ReflectBill}(t, \eta_t) A_{raw}, \qquad \mathrm{ReflectBill}(t,\eta) = \exp\big((t+1)^2 \eta\big).$$

Both coefficient identities hold at every torus side, with the exact zero-total-momentum volume factor.

Proof. The unaveraged assertion follows by adding the bare-action picture and the quadratic picture, after restricting the former to width $\eta_t \le 1$. For completeness, the signed action on a picture entry is

$$(\mathrm{rot}_s\, L)(x, (j_r)_{r=1}^n, p) = \left(\prod_{r=1}^{n} \mathrm{reflectFee}_s(\mathrm{plotted}(p,r), k_r) \right) L\big(x, (k_r)_{r=1}^n, \mathrm{rotatePhase}_s(p)\big),$$

Here $n = t+1$ and $j_r \in \{0,1,2,3\}$ is the coordinate axis attached to the rth slot and $k_r = s_{\mathrm{perm}}^{-1}(j_r)$ is the corresponding input axis, used both in the fee and in the original picture's axis argument. Thus the original picture is evaluated on the tuple $(k_1, \dots, k_n)$. Each factor $\mathrm{reflectFee}_s$ is either 1 or minus the Fourier character $\exp(i\langle p, -e_{s(k_r)}\rangle)$ of one unit direction. On the width-η_t strip, each independent momentum has imaginary coordinates bounded by η_t, and the plotted momentum (including the dependent $p_0 = -\sum_{r=1}^t p_r$) has imaginary coordinates bounded by $n\eta_t$. Thus each such factor has price at most $e^{n\eta_t}$. The product of the n factors therefore costs at most $e^{n^2\eta_t}$. The map $p \mapsto \mathrm{rotatePhase}_s(p)$ applies the signed coordinate action s^{-1} independently to each independent momentum; the dependent momentum transforms by the same action because it is minus their sum. This preserves the coordinate strip and incurs no additional exponential price. This is precisely the picture transformation formula: the character factors account for the translated base point in the signed coordinate action, and the remaining coefficient is the original picture evaluated on the reindexed directions and transformed momenta. Hence each rotated picture has

price at most ReflectBill$(t, \eta_t)A_{raw}$. The normalized finite signed average preserves the same bound by the triangle inequality. Linearity of the coefficient identity shows that the averaged picture represents the averaged kernel and retains the same zero-momentum volume factor. This is the exact orientation-average step needed for the input kernel. □

120 Soft response of the cubic and quartic curvature terms

This section supplies the order-two and order-three response estimates used by the selected one-loop route. The argument has two parts. First, the absolute path budgets give common analytic pictures for the cubic and quartic terms, uniformly over the volume sequence. Second, a finite gauge-variation identity removes the unwanted low-momentum coefficients. The latter step is needed: analyticity and an absolute activity bound alone do not imply the extra powers of momentum.

120.1 Summable local atoms and their coefficient pictures

An atom of degree r consists of r transported curvature slots and a finite coefficient array c_α, where α records one color coordinate for each slot. Write $d(a)$ for the sum of the slot lengths and

$$b_\sigma(a) = \sum_\alpha |c_\alpha| e^{\sigma d(a)}.$$

The corresponding local function is the polynomial in those r slot values. For a sequence a_i, its absolute activity budget is

$$\sum_i b_\sigma(a_i) \le A.$$

Gauge invariance means that simultaneous adjoint rotation of all slot arguments leaves each polynomial unchanged. These are precisely the two properties of the cubic and quartic streams used below.

Proposition 120.1 (Uniform coefficient picture for an atom stream)**.** *Fix $r, t \in \mathbb{N}$, $r \le 4$, and $\sigma > 0$. There are constants $\eta_{r,t} > 0$ and $C_{r,t} < \infty$, depending only on G, r, t, σ, such that the following holds. If $\sum_i b_\sigma(a_i) \le A$, then the translation sum of the local atom functions has an exact order-$(t+1)$ Fourier coefficient picture $P_{r,t}$. Its components are holomorphic on the independent momentum strip of width $\eta_{r,t}$, have uniform bound*

$$\|P_{r,t}\|_{\eta_{r,t}} \le A C_{r,t},$$

and represent the finite-volume Taylor coefficient on every torus. At nonzero total momentum the coefficient vanishes; at zero total momentum it is the torus cardinality times the picture value. The bound is independent of the torus side and of the number of atoms.

If the atoms are gauge invariant, the same conclusion holds for their orientation-averaged action, with the price multiplied by

$$\exp((t+1)^2 \eta_{r,t}).$$

Proof. The one-slot estimate in the proof of Proposition 119.1 gives, on one common complex ball about the physical identity, a bound

$$|s_{\nu,\alpha_\nu}(z)| \le S_G e^{(\sigma/2)\ell_\nu}.$$

It was obtained from the complexified skew-color projection of the forward slot string minus its reverse; thus it applies to the real color coordinates and their complex-linear extension. Multiplying the r slot bounds and summing the coefficient magnitudes gives

$$|a_i^{\mathbb{C}}(z)| \le S_G^r b_{\sigma/2}(a_i).$$

The complex extension is entire in the finitely many link variables in the atom's stencil. Cauchy's derivative estimate on nested balls of radii $R, R/2, \dots, R/2^{t+1}$ bounds the normalized order-$(t+1)$ derivative by

$$b_{\sigma/2}(a_i) S_G^r \frac{\mathrm{circleTax}(R, t+1)}{(t+1)!}.$$

Here $\mathrm{circleTax}(R, 0) = 1$ and $\mathrm{circleTax}(R, k+1) = \mathrm{circleTax}(R, k)/(R/2^{k+1})$; the factorial is the Taylor coefficient normalization.

The local coefficient picture inserts one Fourier character for each of the $t+1$ differentiated slots. A slot support lies in the four-dimensional box of radius at most $d(a_i)$. On a momentum strip of width η, the imaginary part of each plotted momentum is bounded coordinatewise by $(t+1)\eta$. Hence the product of the $t+1$ characters is bounded by

$$\exp\bigl(4(t+1)^2 \eta d(a_i)\bigr).$$

Choose $\eta = \sigma/(8(t+1)^2)$, so $4(t+1)^2\eta = \sigma/2$. The resulting term price is bounded by

$$b_{\sigma/2}(a_i) S_G^r \frac{\mathrm{circleTax}(R, t+1)}{(t+1)!} e^{(\sigma/2) d(a_i)} = C_{r,t} b_\sigma(a_i),$$

where $C_{r,t} = S_G^r \,\mathrm{circleTax}(R, t+1)/(t+1)!$. Thus the coefficient pictures form a normally convergent series on every smaller strip. The same majorant on the common complex link ball permits termwise differentiation of the atom series before extracting any momentum coefficient. This proves analyticity and the claimed uniform bound.

For the finite-volume identity, translate the local atom over all anchors. The sum of its characters is the finite-torus character sum: it is zero unless the total momentum is zero, and equals the number of sites when the total momentum is zero. The local picture is defined on a cover box large enough to contain the stencil, then pulled back by the wrapping map. This keeps the identity exact even on small tori where different cover points wrap to the same link. Absolute summability just proved permits exchanging the atom sum, derivative, and finite anchor sum.

For an orientation s, the picture transforms by multiplying the $t+1$ reflected-character fees and evaluating the original picture at the inverse-permuted direction labels and transformed independent momenta. Each fee has strip price at most $e^{(t+1)\eta}$, because its plotted momentum has coordinatewise imaginary parts bounded by $(t+1)\eta$. The product therefore costs at most $e^{(t+1)^2\eta}$. The normalized finite orientation average preserves this bound by the triangle inequality, and its coefficient identity is the average of the exact finite-volume identities. This proves the final assertion. □

120.2 The finite gauge-variation identity

A coefficient picture of order t records the normalized $(t+1)$st Taylor coefficient. For an external color word v, color run a, and independent momentum tuple p, its one-leg contraction at leg j is

$$(\mathcal{C}_j P)_{v,a}(p) = \sum_{\mu \in \mathcal{A}} q(p_j, \mu) P_{v, \mathrm{update}(a,j,\mu)}(p), \qquad q(p_j, \mu) = e^{i p_{j,\mu}} - 1,$$

where $\mathcal{A}$ is the finite set of spatial axes. The color word v is held fixed; the sum is over the spatial difference axis.

Lemma 120.2 (Ward partition identity)**.** *Let $m_k \to \infty$, and let F_k be real analytic, gauge invariant actions with $DF_k(0) = 0$. Suppose an order-t picture P is uniformly bounded on a momentum strip and records the $(t+1)$st coefficient of F_k exactly on every torus. Suppose further that for each $b = 2, \dots, t$ there is a uniformly bounded order-$(b-1)$ picture recording the bth coefficient, with each of its entries of ray order at least s. Then each contraction $\mathcal{C}_j P$ has ray order at least s.*

Proof. Differentiate the composition of F_k with the infinitesimal gauge change $t+1$ times. The chain rule is the finite sum over ordered set partitions of the $t+1$ slots. Gauge invariance makes this sum zero: the pure-gauge bundle has no component in the distinguished gauge coordinate. The all-singleton partition is, by the exact coefficient record, $-|\Lambda_{m_k}|(t+1)!\,\mathcal{C}_j P$. The factor $q(p_j, \mu)$ is precisely the Fourier multiplier of the one-link difference in the distinguished slot. The one-block term vanishes since $DF_k(0) = 0$.

For a partition with b blocks, $2 \le b \le t$, the lower recorded picture has order $b-1$ because it records the bth derivative. Its arguments are the sums of the external momenta in the corresponding blocks. The chain-rule contribution is a finite sum of products of this picture entry and continuous block-derivative coefficients; it therefore has ray order at least s. Thus on the finite period mesh the contraction is a finite sum of such lower terms divided by the positive factor $|\Lambda_{m_k}|(t+1)!$. Every real momentum tuple is a limit of zero-total-momentum mesh tuples: round the t independent momenta to the period mesh, and set the remaining momentum (the first, distinguished leg) to their negative sum, retaining the integer winding in the momentum lift. Exact recording gives the same factor $|\Lambda_{m_k}|$ in the top and each lower coefficient. Cancel this common volume factor in the finite-mesh identity before passing to the limit. The lower terms are continuous, so the identity passes to every real tuple. Their finite sum has ray order s, proving the claim. □

Lemma 120.3 (Recovering powers from the contracted kernel)**.** *Assume the picture entries are analytic at zero. For every fixed color word and run with at least one independent momentum variable, contractions of ray order at least two imply that each entry has ray order at least one. If the contractions have ray order at least three and there are at least two independent momentum blocks (equivalently, at least three external legs), then each entry has ray order at least two.*

Proof. The first-order expansion $q(\lambda p_j, \mu) = i\lambda p_{j,\mu} + O(\lambda^2)$ shows, after dividing a contraction of order two by λ, that

$$\sum_{\mu \in \mathcal{A}} p_{j,\mu} P_{v,\mathrm{update}(a,j,\mu)}(0) = 0.$$

Choose p supported at the unique axis which updates a_j back to itself. Then $P_{v,a}(0) = 0$. Analyticity gives ray order one.

For a contraction of order three, the next coefficient, using that constant terms vanish, is

$$\sum_{\mu \in \mathcal{A}} p_{j,\mu} DP_{v,\mathrm{update}(a,j,\mu)}(0)[p] = 0. \tag{WR}$$

Fix leg j and a vector w whose jth block is zero. Let q be the coordinate unit vector on leg j at the axis that updates a_j back to itself. Evaluate (WR) at $q + w$ and subtract its value at q. Only the selected axis remains, giving $DP_{v,a}(0)[w] = 0$. The equation at q also gives $DP_{v,a}(0)[q] = 0$. Thus the derivative vanishes on all vectors transverse to leg j. With two distinct legs, split any vector into a part supported on the first and a part with that leg zero; apply the transverse conclusion using the other leg for the first part and the first leg for the second. The full differential vanishes. Together with the vanishing constant term, this gives ray order two. □

120.3 Cubic and quartic terms

Let $A_i^{(3)}$ and $A_i^{(4)}$ be gauge-invariant atom streams of degrees three and four, each with activity budget at most A at width σ. Write T_r for their finite signed-orientation-averaged translation sums.

Proposition 120.4 (Two soft pictures for the curvature sum)**.** *If $m_k \to \infty$, there are $\eta_* > 0$ and $K_* < \infty$, depending only on G, σ, such that*

$$F_k = T_3(m_k) + T_4(m_k)$$

has exact recorded pictures P_2 of order 2 and P_3 of order 3 with uniform price at most AK_*. The first picture has vanishing order 2; the second has vanishing order 1. The functions F_k are analytic, gauge invariant, and vanish to order at least three at zero.

Proof. Apply Proposition 120.1 to each of the four choices

$$(r, t) = (3, 2), (3, 3), (4, 2), (4, 3).$$

For each fixed t, add the degree-three and degree-four pictures. Their recording identities add exactly, and their prices add. Choose the minimum of their four positive widths and the sum of their four finite price constants; this yields one η_* and one K_*, uniform over the stream and all periods.

The individual atom series vanish to at least their degree at zero, because each transported curvature slot vanishes there. Absolute convergence of the differentiated series preserves these vanishing orders, so F_k has order at least three. Gauge invariance is preserved by the finite orientation average. The zero picture records its second Taylor coefficient, and the zero first derivative is automatic. Thus the order-two picture supplies the lower-order coefficient needed for the order-three Ward argument.

Apply Lemma 120.2 with $t = 2$ to the order-two picture. The zero lower coefficient and the order-three vanishing of F_k give contractions of order three. Lemma 120.3 then upgrades the order-two picture to vanishing order two. For the order-three picture, apply Lemma 120.2 with $t = 3$; the zero second coefficient and the order-two picture just obtained supply the lower-order data at block lengths two and three. The contractions vanish to order two, and Lemma 120.3 gives vanishing order one for the order-three picture. These are the two claimed Ray orders. □

120.4 The curvature stream and the Wilson subtraction

For a path family R, write

$$C_k = \sum_{i \geq 0} \mathrm{turn}(R_i^{(3)})(m_k) + \sum_{i \geq 0} \mathrm{turn}(R_i^{(4)})(m_k), \qquad I_k = \mathrm{intended}(p, R; m_k), \quad W_k = \mathrm{Wilson}(m_k).$$

The path bounds provide the cubic and quartic activity budgets at most A and their gauge invariance. The curvature part satisfies $I_k = J_k + C_k$, where J_k is the quadratic path action.

Proposition 120.5 (Curvature stream pictures)**.** *Assume the cubic and quartic budgets and invariance just stated. There are constants $\eta_c > 0$, $K_c < \infty$ such that for $k = 1, 2, 3$ the curvature sequence C_k has a priced, exactly recorded picture of order k, vanishing order $4 - k$, with price at most AK_c.*

Proof. Proposition 120.4 supplies the order-two and order-three pictures. For order two its vanishing order is two, as required. For order three its vanishing order is one. The order-one picture is identically zero: C_k vanishes to third order, so its second Taylor coefficient is zero. The recorded error is therefore zero, its price is zero, and it has any positive vanishing order. Taking the common minimum width and the maximum of the two prices proves the assertion. □

Now impose the selected metric hypotheses: a bed B is attached to R; its metric row is bounded on a strip of width σ and price C; the metric floor bound holds at every torus side and the sworn condition holds at each selected side m_k. The following estimates are the only uses of these assumptions. The metric-slot picture has price $b_0 = \mathrm{bedCharge}(\sigma, C)$ and records the quadratic coefficient of J_k. The unit leader has price $b_1 = \mathrm{bedCharge}(\sigma, 1)$ and records the Wilson coefficient. The floor-start condition gives the zero-momentum normalization $B(0) = I$. The sworn identity and the attached-path Hessian identity identify the actual quadratic coefficient of J_k with the metric-slot picture. In Fourier variables, the difference of the two quadratic pictures has the form

$$\widehat{\mathrm{curl}}(p)^*(B(p) - I)\widehat{\mathrm{curl}}(p).$$

Each curl factor is a one-link difference and is $O(|p|)$; analyticity and $B(0) = I$ give $B(p)-I = O(|p|)$. Their product therefore has ray order three. This is the order-one picture for $J_k - W_k$, with price $b_0 + b_1$. The higher column pictures are supplied by Proposition 119.1; they record the third and fourth coefficients of J_k. The bound $\mathrm{bedCharge}(\sigma, D)$ is the product of the two face counts and the elementary exponential strip bound for the two slot factors, so it is independent of the torus side.

The row estimate of Proposition 119.1, with its signed average, supplies priced pictures for J_k at orders two and three. Proposition 118.1 supplies the corresponding Wilson pictures at both orders, with price constants X_2, X_3 and positive width one. Subtracting pictures preserves exact recording and adds their prices.

Proposition 120.6 (The selected Wilson-subtracted response)**.** *Assume that R has path bounds (τ, A, Λ). Its attached bed has metric strip bound (σ, C), the floor holds at every torus side, and the sworn condition holds at each m_k. Also suppose*

$$0 \le p.\mathrm{direct} \le Z.$$

Then there are $s > 0$ and $M < \infty$, uniform in the sequence and path, such that $I_k - W_k$ has exact recorded pictures of orders $j = 1, 2, 3$, each priced at width s and price M, with vanishing order $4 - j$. One valid price is

$$M = \mathrm{bedCharge}(\sigma, C) + \mathrm{bedCharge}(\sigma, 1) + P_2 + P_3 + X_2 + X_3 + AK_c,$$

where P_2, P_3 are the two row-action prices from Proposition 119.1, and K_c is from Proposition 120.5. One valid width is the minimum of σ, 1, the two row widths, and η_c.

Proof. Set $F_k = J_k - W_k$. Let S_1 be the metric-slot picture minus the unit leader picture. It records the second Taylor coefficient of F_k, has price $b_0 + b_1$, and has vanishing order three by the floor and zero-metric normalization. In the notation of the Ward lemma it is the lower picture of order one for F_k. Let S_2 be the row picture for J_k minus the order-two Wilson picture, and let S_3 be the order-three row picture minus the order-three Wilson picture. They exactly record the third and fourth Taylor coefficients of F_k and have prices at most $P_2 + X_2$ and $P_3 + X_3$.

The action F_k is gauge invariant and has zero first derivative. For $t = 2$, apply Lemma 120.2 to S_2, with the single lower picture S_1; the resulting contraction has vanishing order three, and

Lemma 120.3 gives vanishing order two for S_2. For $t = 3$, apply the partition lemma to S_3, using S_1 and the now improved S_2 at its two lower block counts. The contraction has vanishing order two, so the same recovery lemma gives vanishing order one for S_3. This supplies the exact three records and prices required for F_k.

The cubic and quartic terms C_k have the order-one, order-two, and order-three pictures of Proposition 120.5. Add the order-j picture of C_k to the order-j picture of F_k for each $j = 1, 2, 3$. The sum records $F_k + C_k = I_k - W_k$ exactly. Its price is bounded by the displayed M, and its vanishing order is at least $4 - j$, since the order of a sum is at least the minimum order of its summands. Taking the common minimum strip width gives s. □

120.5 From coefficient pictures to local response arrays

We state the finite conversion in its concrete two-sided form. For $j = 1, 2, 3$, a coefficient picture P_j records exactly the normalized $(j+1)$st derivative of U_m on every finite torus. It is uniformly priced at width s if each entry is holomorphic and bounded by the same constant on the independent-momentum strip of width s. It has ray order r if, on every real ray through zero, each entry is $O(|\lambda|^r)$. For an analytic entry this means every Taylor homogeneous polynomial below degree r vanishes: evaluating that polynomial on each real direction gives zero, hence the polynomial is identically zero. Thus ray order r supplies exactly the Taylor-vanishing hypothesis used in Lemma 120.7. The required input at level j is ray order $3 - j$.

For $n = 0, 1, 2$, an external color address is an $(n + 2)$-tuple of one-link color-coordinate labels, each a pair consisting of a spatial axis and a Lie-color basis index. The multilinear test fields w_r carry the corresponding link and site dependence. The stitch address set St_n is the set of ordered $(2 - n)$-tuples from $\{0, \dots, n\} \times \mathcal{A}$. The fixed rule ℓ_n routes each key (r, μ) to its assigned response leg and axis: for $n = 0$ the two keys are routed to legs zero and one; for $n = 1$ its single key (r, μ) is routed to leg $r + 1$; and for $n = 2$ there are no keys. Let Shift_n be the integer lattice on the $(n + 1)$ independent momentum-axis coordinates. A response array is a map

$$F_n : \mathsf{Color}_{n+2} \times \mathsf{St}_n \times \mathsf{Shift}_n \longrightarrow \mathbb{C}.$$

Its width at η is the supremum, over the finite color and stitch addresses, of

$$\sum_{z \in \mathsf{Shift}_n} e^{\eta |z|_1} |F_n(b, i, z)|.$$

The stitched Fourier picture is

$$\widetilde{Q}_n(b, p) = \sum_{i \in \mathsf{St}_n} \left(\prod_{r=0}^{n+1} \prod_{e \in \ell_n(i)_r} (e^{i \operatorname{plot}_r(p) \cdot e} - 1) \right) \sum_z F_n(b, i, z) e^{ip \cdot z}. \tag{SR1}$$

Here $\operatorname{plot}_r(p)$ are the $n + 2$ external momenta: the first is minus the sum of the $n + 1$ independent momenta, and the remaining ones are those independent momenta in order. For $n = 0$ the two listed differences are distributed one per leg; for $n = 1$ there is one difference, and for $n = 2$ there are none.

To state the response identity, split a link fluctuation into two species x_+, x_- and write $\mathsf{D}_m(x) = \operatorname{content}(\operatorname{input}(x_+) \operatorname{input}(x_-))$. Let v_n put species $-$ in the first n slots and species $+$ in the final two. Let $\mathcal{R}_{n,m}(U_m)$ be the $(n + 2)$-linear derivative of $U_m \circ \mathsf{D}_m$ at zero, restricted to those species in that order; it is a tensor whose components are indexed by external color addresses b. For a list

of axes $\ell = (e_1, \dots, e_r)$, let $\Delta_\ell = (T_{e_1} - I) \cdots (T_{e_r} - I)$ be the corresponding product of translated one-link differences. For a test tuple $w = (w_0, \dots, w_{n+1})$, define

$$\mathrm{Local}_{m,\ell_n(i)}(F_n(b,i,\cdot))[w] = \sum_z F_n(b,i,z) \sum_{x \in \Lambda_m} \prod_{r=0}^{n+1} (\Delta_{\ell_n(i)_r} w_r)(x + \mathrm{wrap}_m(\mathrm{lane}_r(z)), b_r).$$

Here $\mathrm{lane}_0(z) = 0$, while $\mathrm{lane}_r(z)$ for $r \geq 1$ is the shift in the $(r-1)$st independent momentum coordinate. The width bound makes the Laurent sum absolutely convergent. The response identity required below is componentwise and multilinear:

$$(\mathcal{R}_{n,m}(U_m))_b[w] = \sum_{i \in \mathsf{St}_n} \mathrm{Local}_{m,\ell_n(i)}(F_n(b,i,\cdot))[w] \qquad (b \in \mathsf{Color}_{n+2}). \tag{SR2}$$

Lemma 120.7 (Finite difference division)**.** *Let E be a finite set of phase coordinates. Suppose f is 2π-periodic in each coordinate and holomorphic, bounded by A on a strip of width $s > 0$, and its Taylor expansion at zero has no terms of degree below q. Then on a narrower strip it has a finite representation*

$$f(p) = \sum_{(a_1,\dots,a_q) \in E^q} \prod_{r=1}^{q} (e^{ip_{a_r}} - 1)\, g_{a_1,\dots,a_q}(p),$$

where the residual functions are holomorphic on width at least $s/2^q$ and uniformly bounded there by $AC(E,q,s)$. For $q = 0$ take the one-term representation $g = f$.

Proof. For $J \subseteq E$, write $p^{[J]}$ for p with the coordinates in J set to zero. For each a, periodicity implies that $f(p) - f(p^{[\{a\}]})$ vanishes whenever $p_a \in 2\pi\mathbb{Z}$. Consequently it is divisible, as a holomorphic periodic function, by $d_a(p) = e^{ip_a} - 1$. The quotient is removable at every zero. On a fundamental real period cell, away from fixed neighborhoods of these zeros the denominator is bounded below; inside those neighborhoods the quotient is bounded by Cauchy's estimate on a narrower strip. Thus

$$f(p) - f(p^{[\{a\}]}) = d_a(p) g_a(p)$$

with a uniform bound for g_a. In the telescoping identity over J, replace f at each step by the function with the previously selected coordinates staged to zero. Each corresponding quotient lowers the vanishing order by one. Induction on q applied to these residuals therefore yields the displayed finite sum. At most $|E|^q$ terms occur; at each division, Cauchy's estimate on the current strip of width r bounds the quotient on width $r/2$; after q divisions the residual strip has width $s/2^q$. Each step multiplies the bound by a constant depending only on E, q, s. The fully staged term is zero by the order-q hypothesis. This proves the result. □

For the local response array, the number of factors is $q = 2 - n$, not n. If $n = 1, 2$, apply the factorization to every picture entry with $q = 2 - n$ differences and place them on the prescribed stitch legs. For $n = 0$, two factors are required, one on each of the two response legs. The division first gives the two factors in the single independent momentum; the second is moved to the other leg by the exact identity

$$q(p,\mu) = -\,\mathrm{mode}(p, \mathrm{step}\,\mu) q(-p,\mu).$$

The phase is absorbed into the Laurent coefficient, with its strip cost included in the constant. Thus this construction yields precisely the three stitch patterns: two factors for $n = 0$, one for $n = 1$, and none for $n = 2$.

To control the Laurent width, let g be one residual, bounded by B on a strip of width η. Shifting the contour in each periodic coordinate gives $|\widehat{g}(\nu)| \le Be^{-\eta|\nu|_1}$. At width $\eta/2$,

$$\sum_{\nu} e^{(\eta/2)|\nu|_1}|\widehat{g}(\nu)| \le B\left(\sum_{k\in\mathbb{Z}} e^{-(\eta/2)|k|}\right)^{|E|} = B\left(1+\frac{2}{e^{\eta/2}-1}\right)^{|E|}.$$

Thus each residual defines a Laurent entry with a quantitative width bound; summing the at most $|E|^q$ residuals multiplies it by that finite factor. This is the coefficient-norm estimate used in the array construction.

We now verify the response identity and its price using the actual mixed-field composition. Let $D(x)$ be the local map which forms the product of the two split fields and then takes its logarithmic link coordinate. For a Boolean word v on the external slots, the response of U_m is the corresponding mixed derivative of $U_m \circ D$ at zero. At level n, the word has n entries of the first species followed by two entries of the second species; its length is $n+2$.

For a set partition π of these $n+2$ slots into b nonempty blocks, order the blocks canonically by their least slot. The chain rule gives the derivative of U_m of order b evaluated at the sums of external momenta in the blocks, multiplied by the derivatives of D on each block. The single-block term is zero because $DU_m(0) = 0$. For $2 \le b \le n+2$, the action coefficient has picture order $b-1$. Its available ray order is $3-(b-1) = 4-b$, which is at least $2-n$. Thus every term in the mixed response picture has ray order $2-n$ and is analytic on a common narrower strip.

More explicitly, for a partition π and a basis phase assignment w, the corresponding mixed coefficient is

$$b! \sum_{a_1,\dots,a_b} \left(\prod_{r=1}^{b} c_{\pi,r}(a_r; w_r)\right) P_{b-1}\big(a_1,\dots,a_b; \textstyle\sum_{j\in\pi_1} p_j, \dots, \sum_{j\in\pi_b} p_j\big),$$

where $c_{\pi,r}(a_r; w_r)$ is the coordinate coefficient of the derivative of D on block r. The factorial is the normalization between the symmetric derivative picture and the raw bth action derivative; it does not enumerate permutations of the blocks. If the input pictures are bounded by A, its absolute price is bounded by

$$Ab! \sum_{a_1,\dots,a_b} \prod_{r=1}^{b} |c_{\pi,r}(a_r; w_r)|.$$

Summing the absolute coefficient multiplier (the displayed sum without the input price A) over partitions, color basis labels, and phase basis labels defines the response constant P_n. These coefficients are uniformly finite: the map D_m is the same pointwise product-and-log map on each link. On a fixed identity neighborhood in the finite-dimensional gauge group, this map is analytic and bounded. Cauchy's estimate on a smaller ball bounds every derivative of order at most four by a constant depending only on the group and the order. The number of partitions and color/phase labels is finite, so $P_n < \infty$ independently of m. The common analytic strip for the mixed picture can be chosen as

$$s_n = \frac{s}{(n+3)^2}.$$

Every independent external momentum has imaginary coordinates at most s_n, and the missing momentum, their negative sum, has coordinates at most $(n+1)s_n$. A block sum of at most $n+2$ external momenta therefore has imaginary coordinates at most $(n+2)(n+1)s_n < s$. Thus the input pictures can be evaluated at every block sum. The mixed picture is bounded by AP_n on width s_n when the input picture prices are at most A.

For each entry of the mixed picture $\widetilde{Q}_n$, finite difference division now produces the prescribed $2-n$ stitch factors and residual functions on width $d_n = s_n/2^{2-n}$. Shifting each contour within the smaller strip of width $d_n/2$ gives Laurent coefficients bounded by $Be^{-(d_n/2)|z|_1}$ when the residual supremum is B. At array width $d_n/4$, their weighted sum is bounded by

$$B\left(1+\frac{2}{e^{d_n/4}-1}\right)^{D_n},$$

where D_n is the finite number of independent Laurent coordinates. The number of partitions and addresses is finite, so all residuals assemble into a response array of width at most AC_n for a constant C_n depending only on the finite labels and input width.

To check exactness, evaluate the factorization on a finite torus. Each Laurent term is absolutely summable by its weighted width. Its spatial translation sum is a finite character sum; orthogonality makes it zero unless total momentum is zero, in which case it equals the number of sites. The coefficient picture recording identity then identifies the sum with the mixed derivative of $U_m \circ D$. This is the exact finite-volume response identity, not only equality of Taylor polynomials.

Proposition 120.8 (Three-level response arrays)**.** *Let U_m be real analytic, with $U_m(0) = 0$ and $DU_m(0) = 0$. Suppose it has exact coefficient pictures P_j at orders $j = 1, 2, 3$, whose entries are 2π-periodic in each phase coordinate, bounded on one common strip of width $s > 0$, and have ray orders at least $3-j$. Then there are a common positive strip width η_* and a finite constant C_* such that the response arrays at levels $n = 0, 1, 2$ satisfy (SR1), the exact finite-period response identity, and width at most C_* times the sum of the three input picture prices, uniformly in m.*

Proof. For each n, first form the mixed picture $\widetilde{Q}_n$ by the ordered-partition chain rule. Every term is the input picture $P_{|b|-1}$ at the block-summed momenta, multiplied by the block coefficients. The one-block term is zero. Since $P_{|b|-1}$ has ray order $4-|b| \geq 2-n$, the mixed picture has ray order $2-n$. Apply the finite difference lemma to $\widetilde{Q}_n$ with $q = 2-n$ and assign the factors to the prescribed legs; for the two-leg split use the phase identity above. The finite character sum proves the response identity. The Laurent estimate bounds the width. For each $n = 0, 1, 2$, let C_n be the finite price multiplier from the ordered-partition and division construction, including the Laurent coefficient sum. For the common input strip s, the mixed-picture strip at level n is $s_n = s/(n+3)^2$, and the residual strip is $d_n = s_n/2^{2-n}$. Choose

$$\eta_* = \min_{0\leq n\leq 2}(d_n/4), \qquad C_* = C_0 + C_1 + C_2.$$

Then all arrays have width at most C_* times the sum of the three input picture prices. These finite constants and the positive width are independent of the period and action. □

The division and response construction above outputs an array of exactly this type. The factorization proves (SR1) entrywise. The finite-volume character identity proves (SR2); Laurent absolute convergence justifies the sum over shifts. Thus the exact response is supplied at each finite period, not merely in the limiting Taylor series.

Apply the conversion to $I_m = (I_m - W_m) + W_m$. Proposition 120.6 supplies pictures with ray orders $3, 2, 1$ at orders $1, 2, 3$. The Wilson pictures have ray orders $2, 1, 0$, so the sums have the required orders $2, 1, 0$. Their exact records and prices add. The value and first derivative of I_m vanish by the quadratic normalization and cubic vanishing of the curvature part. We obtain three uniformly priced response arrays. Let M_g denote the displayed price in Proposition 120.6, and let P_W be the sum of the leader and the two higher Wilson picture prices. The common array width is bounded by $K = 3C_*(M_g + P_W)$.

Define the mockup width property by the common Laurent bound above at each level $n \le 2$, and define its response property to be (SR2) at every such level. Then the theorem below supplies both properties, conventionally called the width and exact-response requirements for the three mockups.

Theorem 120.9 (Selected soft-response arrays)**.** *Fix $\sigma > 0$, $M, Z, A \ge 0$, $\tau > 0$, and $\varepsilon \in \mathbb{R}$. There are $\eta > 0$ and $K < \infty$ such that for every divergent period sequence m_k, clip p, path family R, attached bed B, and $\Lambda \ge 0$ satisfying*

$$p.\mathrm{direct} \le Z, \quad R.\mathrm{Bounds}(p, \tau, A, \Lambda), \quad B \text{ attached to } R, \quad B.\mathrm{metric.strip}(\sigma, M),$$

with $\mathrm{Floor}(B, p, n, \varepsilon, M)$ *for every torus side n, and the sworn metric identity at each selected side m_k. Put $U_{m_k} = \mathrm{intended}(p, R; m_k)$. Then these actions admit arrays F_0, F_1, F_2 of width at most K satisfying (SR1) and (SR2) on every period. Set $F_n = 0$ for $n > 2$. One may take $K = 3C_*(M_g + P_W)$, where M_g is the price displayed in Proposition 120.6, P_W is the finite Wilson picture price, and C_* is the finite response-array multiplier above. The constants η, K depend only on the listed scalar bounds and the fixed gauge and finite coordinate sets.*

Proof. The Wilson-subtracted proposition supplies order-j pictures of ray orders $4-j$, for $j = 1, 2, 3$. The Wilson pictures have ray orders $3 - j$. Their sums therefore have the required ray orders $3 - j$. Prices and exact recording identities add. The quadratic normalization and the cubic vanishing of the curvature part give $U_m(0) = 0$ and $DU_m(0) = 0$. Apply Proposition 120.8. Its finite constants depend only on the listed bounds and the finite color, phase, and stitch sets, so the common width and price are uniform in the path, period, and selected action. □

Scope. This section proves the uniform atom, Ward, and Wilson-subtracted coefficient estimates, together with the finite division and the exact stitched response identity. The dependence of downstream applications on the displayed metric, floor, and sworn hypotheses remains as stated.

121 From Ward identities to invariant curvature coefficients

The leading Taylor terms of a gauge-invariant action satisfy a Ward identity in every field argument. We now turn these identities into an exponentially summable expansion in curvature variables. The momenta of the arguments are constrained to sum to zero, so one cannot simply apply independent one-variable divisions to all of them. The following construction resolves that dependence before choosing the invariant, real coefficients of the path expansion.

121.1 A single exterior division with auxiliary constraints

Put $q_a(p) = e^{ip_a} - 1$. On alternating coordinate tensors in four dimensions let E_a denote exterior multiplication by the ath basis vector and let $\iota_a = E_a^{\mathsf{T}}$. Write

$$\delta(p) = \sum_{a=1}^{4} q_a(p)\iota_a.$$

Thus δ lowers exterior degree by one and $\delta^2 = 0$. Tensor factors other than the one on which this operator acts are coefficient spaces. Operators on distinct factors commute. All transposes here are algebraic transposes, without complex conjugation. This convention is appropriate for the multilinear Taylor coefficients of a scalar action.

A periodic coefficient on d complex momentum coordinates has strip bound (σ, A) if it is holomorphic on $|\operatorname{Im} p_j| < \sigma$ and has absolute value at most A there. Tensor bounds mean this entry by entry. Say that it has order s at zero if its restriction to each real ray $p = tu$ is $O(t^s)$ as $t \to 0$. For an analytic function this is equivalent to the vanishing of all homogeneous Taylor polynomials of degree less than s: restrict the Taylor expansion to each ray and use that a polynomial vanishing on every real vector is zero.

The division from Section 11 has the following strip version. For each coordinate j, there is a periodic holomorphic $D_j f$ such that

$$(e^{ip_j} - 1) D_j f(p) = f(p) - f(p|_{p_j=0}), \qquad |D_j f|_{\sigma/4} \le A\, d_\sigma, \tag{610}$$

where one may take

$$d_\sigma = \frac{4}{\sigma} \left(\frac{2(1+\sigma/4)}{\sigma/4} \right)^d .$$

Indeed, Fourier expansion gives coefficients $\widehat{f}(z)$ with

$$\sum_z e^{(\sigma/2)|z|_1} |\widehat{f}(z)| \le A \left(\frac{2(1+\sigma/4)}{\sigma/4} \right)^d .$$

Replace each monomial in coordinate j by the finite geometric sum displayed in that earlier section. It has at most $|z|_1$ terms, all with displacement norm at most $|z|_1$. On the smaller strip its absolute value is at most $|z|_1 e^{(\sigma/4)|z|_1}$, which is bounded by $(4/\sigma) e^{(\sigma/2)|z|_1}$. This proves normal convergence, the bound, and the identity, including the zeros of the divisor. If f has order s, this division has order at least $\max(s-1, 0)$. To verify this, the right side of (610) has order s, while the lowest nonzero homogeneous term of the divisor is ip_j. A nonzero lowest homogeneous term of the quotient therefore gains exactly one degree upon multiplication.

Suppose a tensor f of exterior degree r is a cycle, $\delta f = 0$. Set $f_0 = f|_{p_a=0}$ and

$$B_a = E_a D_a f, \qquad R_a = (I - E_a \iota_a) f_0 .$$

The Cartan identity $\delta E_a + E_a \delta = q_a I$ gives

$$f = \delta B_a + R_a . \tag{611}$$

For example, where $q_a \neq 0$ the cycle identities for f and f_0 imply $\delta D_a f = -\iota_a f_0$; substitution gives the displayed formula. Continuity extends it to $q_a = 0$. The residual is still a cycle, has zero entries on faces containing a, and evaluates all other entries at $p_a = 0$. Successive residual operations in the four coordinates therefore leave zero when $r > 0$. When $r = 0$, they leave the value at $p = 0$, so the same conclusion holds if that value vanishes. Adding the four lifts proves the following fact.

Lemma 121.1. *Let $r, s \in \mathbb{N}$. Let f have strip bound (σ, A), order s, and exterior degree r. Assume $\delta f = 0$ and either $r > 0$ or $f|_{p=0} = 0$. Then $f = \delta B$, where B has degree $r+1$, strip bound $(\sigma/4, 4Ad_\sigma)$, and order at least $\max(s-1, 0)$. Here other momentum coordinates may remain as parameters, and $p = 0$ in the degree-zero hypothesis means only that the four coordinates of the divided momentum vanish.*

If, in addition, for each exterior component the vector of coefficients in the other tensor factors belongs to a fixed linear subspace of that other-factor coefficient space, depending only on the other momenta, the same membership holds separately for every exterior component of B at every real momentum.

Proof. Only the last assertion remains. Away from $q_a = 0$, the quotient in (610) is a scalar multiple of the difference of two vectors in that subspace. On $q_a = 0$, vary just p_a and take the limit. The subspace is fixed during this limit and is closed, since the coefficient space is finite dimensional. Exterior multiplication, subtraction, and coordinate restriction preserve membership as well. Thus each of the four lifts and their sum preserve the constraint. Each residual retains the original strip bound and order; summing four lifts accounts for the factor four, with only one loss of Taylor order. □

121.2 Restoring the dependent momentum constraint

There are t independent momenta $p_1, \dots, p_t \in \mathbb{C}^4$, and we set $p_0 = -\sum_{j=1}^{t} p_j$. Write $\delta_j = \delta(p_j)$ acting on tensor factor j. A Ward identity on that factor is $\delta_j f = 0$. We first allow the zeroth factor to have degree $r+1$ and all independent factors to have degree one.

One can solve the zeroth-factor cycle while preserving all Ward identities except, provisionally, the first. Exchange p_1 and p_0 by the linear involution

$$(p_1, p_2, \dots, p_t) \longmapsto (-p_1 - \cdots - p_t, p_2, \dots, p_t).$$

It preserves the integer periodicity lattice and sends a sufficiently narrow strip into the original strip; width $\sigma/(t+1)$ is sufficient for $t \geq 1$. Apply Lemma 121.1 in the new first variable and exchange back. The Ward subspaces for factors $2, \dots, t$ do not depend on this variable, so they are retained. The cost is a fixed multiple of the old cost, the width remains strictly positive, and Taylor order drops by at most one. Denote this operation simply as a zeroth-factor lift. Likewise, if a coefficient has no first tensor factor and vanishes when $p_1 = 0$, the degree-zero case of the lemma adds a degree-one first factor whose contraction is the original coefficient, while retaining the remaining Ward identities.

Lemma 121.2 (Dimension shift)**.** *Fix $t, r \geq 0$ with $r + 1 + 2t > 4$ and $\sigma > 0$. There are $\eta > 0$ and $K < \infty$, depending only on these data, such that the following holds for every $A \geq 0$. If u has strip bound (σ, A), degrees $(r+1, 1, \dots, 1)$, and $\delta_j u = 0$ for $j = 0, \dots, t$, there is v of degrees $(r+2, 1, \dots, 1)$ with strip bound (η, KA) satisfying*

$$\delta_0 v = u, \qquad \delta_j v = 0 \quad (1 \leq j \leq t).$$

Proof. Induct on t, allowing r to vary. At $t = 0$ the degree $r+1$ exceeds four, so $u = 0$ and one takes $v = 0$. For $t \geq 1$, first lift the zeroth cycle, temporarily forgetting the first Ward identity. This gives X with $\delta_0 X = u$ and all remaining Ward identities. Contract its first factor:

$$V = \delta_1 X, \qquad \delta_0 V = \delta_1 u = 0.$$

A second zeroth-factor lift gives Y of zeroth degree $r+3$ with $\delta_0 Y = V$ and the remaining Ward identities. Restrict Y to $p_1 = 0$, obtaining Z. Since $q(0) = 0$, $V|_{p_1=0} = 0$, so Z is a zeroth-factor cycle in $t-1$ independent variables. Its zeroth degree is $r+3$, and

$$r + 3 + 2(t-1) = r + 1 + 2t > 4.$$

The induction hypothesis gives W of degree $r+4$ with $\delta_0 W = Z$, retaining every remaining Ward identity. Extend W constantly in p_1, writing the extension as W', and set

$$L = Y - \delta_0 W'.$$

Here δ_0 uses the full momentum $-p_1-\cdots-p_t$. Thus $\delta_0 L = V$ by $\delta_0^2 = 0$, while $L|_{p_1=0} = 0$ by the induction identity. The degree-zero first-factor lift now gives F with $\delta_1 F = L$ and all remaining Ward identities. Finally put $v = X - \delta_0 F$. Commutation on distinct factors gives

$$\delta_0 v = u, \qquad \delta_1 v = V - \delta_0 L = 0.$$

The other identities hold term by term. All operations are finite sums, fixed linear changes of variables, restrictions, exterior contractions, and the lifts already bounded above. At each stage choose a smaller positive common width. Their number is determined by t, r, and every bound remains linear in A. This gives η and K before the particular input tensor is chosen. □

Take now a tensor f with $t+1 \geq 3$ degree-one factors and all Ward identities. The dimension-shift lemma with $r = 0$ lifts factor zero while preserving the independent identities. Lift each independent factor in turn by Lemma 121.1. At each step the identities in the remaining factors are auxiliary subspaces independent of the divided momentum; the already lifted indices are just finite coefficient indices. We obtain g with all factors of degree two such that

$$f(p_1, \ldots, p_t) = \delta_0 \delta_1 \cdots \delta_t g(p_1, \ldots, p_t). \tag{612}$$

Its strip width and cost multiplier depend only on the original width and the number of factors. In particular the cubic and quartic applications have constants independent of the torus.

121.3 The quadratic case and retained moments

For two factors the dimension inequality fails. Suppose instead that $f(p)$ has both Ward identities at momenta $(-p, p)$ and order $s+2$ at zero. The preceding construction can be shortened in a way that retains precisely s orders after both lifts. Lift factor zero to obtain X of degrees $(2,1)$ and order $s+1$, with $\delta_0 X = f$. The contraction $V = \delta_1 X$ has order $s+2$, since the contraction symbol vanishes to first order. It is a zeroth-factor cycle. Lift it to obtain Y of zeroth degree three and order $s+1$ with $\delta_0 Y = V$.

There are no other momenta now. Hence $Y|_{p=0} = 0$, and the degree-zero first-factor lift gives D of degrees $(3,1)$, order s, and $\delta_1 D = Y$. Set $Z = X - \delta_0 D$. Then

$$\delta_0 Z = f, \qquad \delta_1 Z = 0, \qquad Z \text{ has order } s+1.$$

One last first-factor lift gives g of degrees $(2,2)$ and order s, with $\delta_0 \delta_1 g = f$. All bounds remain linear in the initial strip cost, with a positive width chosen before f and s. For the loop remainder we use $s = 2$: fourth-order vanishing of the potential coefficient yields second-order vanishing of its curvature coefficient.

121.4 Real coefficients and invariance at each placement

Fix a real orthonormal basis of the finite-dimensional Lie algebra. The tensors now also have one colour index for each field factor. Assume the original tensor satisfies

$$\overline{f(-p)} = f(p) \quad \text{for real independent momenta } p,$$

and that its real and imaginary colour tensors are invariant under the simultaneous adjoint action of the gauge group. The exterior lifts need not respect these properties, so we restore them explicitly.

For each entry of g, take the real part of its Fourier coefficients. The resulting real coefficient array has symbol

$$g_{\mathbb{R}}(p) = \tfrac{1}{2}\big(g(p) + \overline{g(-p)}\big).$$

Since $\overline{q(-p)} = q(p)$ for real p, contraction of $g_{\mathbb{R}}$ is still f. If g has order s, so does $g_{\mathbb{R}}$. Fourier coefficient estimates on a smaller strip give an exponentially weighted summation bound. More precisely, from strip width w and entry bound B one obtains width $w/2$ and bound

$$B\left(\frac{2(1+w/4)}{w/4}\right)^{4t}.$$

Let $\mathcal{I}$ be the subspace of invariant real colour tensors, and choose a fixed linear projection P onto $\mathcal{I}$. Apply P separately at each tuple of face directions and each tuple of lattice displacements. This gives real coefficients $R_{\mathbf{a},\mathbf{c}}(z)$ that are invariant at every such placement. Exterior contractions do not act on colour indices, so they commute with P. Since P fixes the real and imaginary parts of f, the reconstructed potential coefficient remains f. The weighted bound is multiplied by at most

$$\sum_{i,j} |P_{ij}| < \infty.$$

This is a fixed finite-dimensional projection; no assertion that its norm is one is needed. It also preserves each retained order of vanishing. Thus, for some $\eta > 0$ and $C < \infty$ fixed by width, arity, and gauge group, every face and colour entry obeys

$$\sum_{z\in\mathbb{Z}^{4t}} e^{\eta|z|_1}|R_{\mathbf{a},\mathbf{c}}(z)| \le CA. \tag{613}$$

For two factors with retained order two, these coefficients have both zero total mass and zero first moments, entry by entry:

$$\sum_z R(z) = 0, \qquad \sum_z z_a R(z) = 0 \quad (a = 1,\dots,4). \tag{614}$$

Indeed the symbol at zero is the first sum. Exponential summability implies $\sum_z |z|_1|R(z)| < \infty$, so its derivative along the ath real momentum coordinate is $i\sum_z z_a R(z)$. The symbol's value and first derivatives vanish by its retained second order. This proves both identities with absolute convergence.

121.5 The invariant path expansion

For each tuple $(\mathbf{a}, z)$ of oriented coordinate plaquettes and displacements, put the zeroth plaquette at the origin and plaquette j at $z_j \in \mathbb{Z}^4$ for $1 \le j \le t$. Use the coefficient colour tensor $R_{\mathbf{a},\cdot}(z)$ for this atom. It is invariant by the preceding construction. In the path convention of Section 138, the atom's length is exactly

$$|z|_1 + 4(t+1).$$

The four edges of every plaquette, including the anchored one, contribute to this length. Summing the weighted absolute coefficients over the finite face and colour index sets and over all displacements gives total atom budget at most

$$CA\,\#\{\text{face tuples}\}\,\#\{\text{colour tuples}\}\,e^{4\eta(t+1)}.$$

An enumeration of this countable family changes neither its sum nor its budget. In the two-factor case, filtering the enumeration by a fixed pair of face directions and a fixed colour tuple turns (614) into the required zero total loop coefficient and zero first moment of the relative displacement of the two legs. The latter displacement is $z_1 - 0$.

The potential coefficient reconstructed from these atoms is exactly (612), with the two-factor version proved separately above. Absolute convergence permits periodization and evaluation on every finite torus. This result constructs the invariant coefficients from the specified analytic Ward data. The subsequent stationary action calculation must still supply those data, their vanishing orders, and the scalar normalization of the recorded Taylor terms.

122 Weighted derivatives of the centered block frame

The penalty estimate in Section 128 requires uniformity as the block side increases. Here the averaging in the frame construction is essential. We prove the weighted derivative bound at the identity, retaining the same minimizing frame and predictor as in Section 69.

122.1 A calculus on finite probability spaces

For a probability measure ν on a finite set, define

$$\|f\|_{\nu,0} = \max |f|, \qquad \|f\|_{\nu,1} = \Big(\sum \nu |f|^2\Big)^{1/2}, \qquad \|f\|_{\nu,2} = \sum \nu |f|.$$

An input direction has charge zero, one or two. A family of smooth maps has derivative bound C through order n if every derivative of order $1 \le k \le n$, evaluated on admissible directions of charges b_i with $\sum b_i \le 2$, has norm at most C in the output norm of charge $\sum b_i$. Zeroth derivatives are specified separately. Products add charges. The only nontrivial product rule for these norms is $L^2 \cdot L^2 \subset L^1$, which is Cauchy–Schwarz. Averaging over one variable contracts each norm by the triangle inequality or Jensen's inequality. If $\nu' \le H\nu$ with $H \ge 1$, changing the measure costs at most H in all three norms.

These observations give quantitative product and composition rules. For a fixed smooth finite-dimensional map F at z, set

$$\mathcal{C}_{n,F,z}(C) = \sum_{k=1}^{n} \sum_{\pi \in \mathcal{P}_k} \|D^{|\pi|}F(z)\| C^{|\pi|},$$

where $\mathcal{P}_k$ is the set of partitions of $\{1, \dots, k\}$. If $f(0) = z$ and f has derivative bound C through order n, the partition chain rule and the preceding product inequalities bound $F \circ f$ by $\mathcal{C}_{n,F,z}(C)$. Each differentiated direction appears once, so the sum of charges is unchanged. If $DF(z) = 0$, the same chain rule bounds the derivative of order $n+1$ using derivatives of f only through order n: the excluded single-block partition is the only one that would contain $D^{n+1}f$.

122.2 Long paths with small link derivatives

Suppose each link matrix equals 1 at the origin and has derivative bound μC through a fixed target order N, where $0 < \mu \le 1$. A path contains at most K/μ links, with K fixed. Its ordered product has a bound depending on K, C, N only. The following induction proves this without an exponential loss in the number of links.

For $0 \le n < N$, assume all partial products have derivatives through order n bounded by Q_n, independently of path length. Write a new factor as $1 + v$ and a partial product as $1 + u$, where $u(0) = v(0) = 0$. In the derivative of order $n+1$ of uv, both factors have positive derivative order. Thus derivatives of u only through order n occur, while every term has a factor μ from a derivative of v. The finite Leibniz sum is bounded by $\mu R_n(Q_n, C)$, for a polynomial R_n depending only on

n and the matrix multiplication norm. The new factor alone contributes at most μC. Starting with the empty product and adding at most K/μ factors bounds the derivative of order $n+1$ by $K(C + R_n(Q_n, C))$. Taking its sum with Q_n defines a common bound Q_{n+1}. The induction begins at order zero, where every partial product has value 1.

Apply this on the enlarged block $\mathcal{B}$ of radius $(J+3)L$. For pairs of endpoints, a coordinate path has four straight pieces, each of length at most $2(J+3)L$. At a fixed position along one piece, the link read is a translate of the coordinatewise splice of the two endpoints. Swapping the corresponding endpoint coordinates is a bijection of $\mathcal{B} \times \mathcal{B}$ and preserves its uniform measure. A translate restricted to reads inside $\mathcal{B}$ is injective; reads outside the path contribute the constant identity. Splitting positive and negative directions costs at most two; inverse link matrices are adjoints and have the same derivative norms. Consequently the preceding link estimate applies on the pair space, with a constant independent of L. Multiplying the four pieces and averaging the finitely many coordinate orders gives uniform derivative bounds for E_{xy}, defined as the average of the adjoint (equivalently, inverse) of the transport along $x \to y$, with $E_{xy}(0) = 1$.

122.3 Eliminating the center of the frame

Let F_x be the matrix of the minimizing frame after recentering by the nearest group element to its weighted mean. Let ν be the uniform measure on $\mathcal{B}$, and ω the actual smoothing probability used for this recentering. The latter is the distribution of the sum of J independent uniform half-block points. Its reduction modulo L is uniform: fix all summands except the first and sum the first over its complete residue system. Hence

$$\omega(x) \le L^{-4}, \qquad \omega(x) \le H_J \nu(x),$$

where H_J is a fixed upper bound for $|\mathcal{B}|/L^4$. This proves the measure comparison needed below with no assumption that ω itself is uniform on $\mathcal{B}$.

Let $\mathcal{F}$ denote the matrix-valued nearest-group map near 1, let $\Pi = D\mathcal{F}(1)$ be the orthogonal projection onto the Lie algebra, and put $R(z) = \mathcal{F}(z) - \Pi z$. Then $DR(1) = 0$. Define

$$J_0 = \sum_y \nu(y) F_y, \qquad B_x = \sum_y \nu(y) F_y (E_{xy} - 1), \qquad T_x = J_0 + B_x.$$

The frame equation and its normalization are

$$F_x = \mathcal{F}(T_x), \qquad \sum_x \omega(x) \Pi F_x = \Pi 1. \tag{615}$$

Indeed, minimization at each frame coordinate says that it is the nearest group element to the average of the transported other frame coordinates. At the identity this average is 1, so the smooth nearest map applies in a neighborhood. Common left multiplication by the recentering group element preserves the minimizing equation. The nearest-point normal equation for that recentering element gives the second equality in (615). Existence and smoothness near the identity are those of the selected minimizer already constructed; only its uniform derivatives are at issue here.

Put $S_x = \Pi B_x + R(T_x)$. Since $\Pi^2 = \Pi$, the two equations imply the exact identity

$$F_x = \Pi 1 - \sum_y \omega(y) \Pi S_y + S_x. \tag{616}$$

To check it, first write $F_x = \Pi J_0 + S_x$. Apply Π, average with ω, and solve for the constant ΠJ_0 using the normalization. This is the step that removes the derivative of the unknown center.

We now induct on derivative order. Suppose derivatives of F through order n have uniform bound A_n. The transport E has derivatives through order $n+1$ bounded by a fixed D_{n+1}. In differentiating $F_y(E_{xy}-1)$ to order $n+1$, the term with $D^{n+1}F_y$ vanishes because $E_{xy}(0)-1=0$. Every remaining term is bounded by the product rules, and averaging in y gives a bound for B_x through order $n+1$. It follows that T_x is bounded through order n. Since $DR(1)=0$, the composition rule above bounds $R(T_x)$ through order $n+1$ using only those derivatives. Thus S_x has a bound A'_n, depending only on A_n, D_{n+1}, G, n. Equation (616) now bounds F_x through order $n+1$ by $(1+H_J)A'_n$. The induction starts at order zero, with $F_x(0)=1$. Every constant depends on G, J, n and the incoming link price, but not on L. In particular no inverse of a rooted-frame counting-norm Hessian has been inserted into this argument.

122.4 The predictor and penalty

The connector for a coarse direction consists of the centered frame at one endpoint, a straight path of length L, and the inverse centered frame at the translated endpoint. Use a padded box of radius $(J+5)L$ for the link fields. Translation from either endpoint patch into this box is injective, and the ratio of the two normalized counting measures is bounded in terms of J only. The long-path argument and the frame induction therefore give uniform derivative bounds for the connector. Changing its average to the smoothing probability costs H_J. Composing the resulting matrix average, whose value is 1, with $\mathcal{F}$ bounds the actual predictor through order n. The strict local predictor test identifies this expression with every selected tag near the identity, as in Section 105.

Next apply the fixed group exponential, multiplication, adjoint and nearest-group maps. The fine-link directions have norm at most μC on the padded box, while the finitely many coarse and gauge coordinates have norm at most C. Composition with an exponential preserves a factor μ in every positive fine-link derivative: each chain-rule term contains μ^k, $k \ge 1$, and $\mu^k \le \mu$. Thus these are precisely the link bounds used above. The penalty in one direction is the fixed smooth function

$$Z \longmapsto \frac{k}{2c_G^2}\|Z-1\|^2$$

of the coarse group element times the inverse predictor; k is the penalty coefficient and c_G the fixed color normalization. The finite composition rule now gives its derivative bound B_n. All constants were selected before L. Summing the four directions and complexifying gives (635).

122.5 Verification for the Fourier legs

We finally check the weighted stencil hypotheses just used. Write $k_l(p) = (p+2\pi l)/L$, and let $P(p)$ be the normalized block Fourier matrix, whose entries have magnitude L^{-2} at real p. The weighted norm of a fine block vector u is

$$\|u\|_p^2 = \sum_l h_l(p)^2 |(P(p)^{-1}u)_l|^2, \qquad h_l(p) = \big(|q(\operatorname{Re} k_l(p))|^2 + \mu^2\big)^{1/2}.$$

Here $q_a(k) = e^{ik_a} - 1$; color components are included in the sum. On the strip of width one, $P(p)$ and its inverse have Euclidean operator norms at most e^4: factor them into the unitary real Fourier matrix and diagonal imaginary phases. Since $h_l \ge \mu$, a unit weighted vector satisfies $\|u\|_{\ell^2} \le e^4\mu^{-1}$. Dividing by L^2 gives its normalized block L^2 bound $e^4\mu$. Also

$$|q_a(k_l(p))| \le 5h_l(p).$$

For this last inequality, split the symbol into its real-momentum value and its change in the imaginary direction. The latter is at most 4μ on this strip by $|e^z - 1| \le e^{|z|}|z|$ and $\mu \le 1$; the former is at most h_l. Fourier multiplication therefore bounds one fine difference in unnormalized ℓ^2 by $5e^4$, and in normalized L^2 by $5e^4\mu^2$. These are exactly the one-internal-leg bounds used in Section 128. The auxiliary scalar part of the internal norm is ordinary Euclidean, so its coordinates are bounded by one. Smooth bridge coordinate legs satisfy the required supremum bounds by (634), with a fixed color-dimension factor when components are combined into a vector.

To extend a block Fourier vector to the padded stencil, write $z = \bar{z} + Lr(z)$, where $\bar{z}$ is the chosen residue representative. Throughout the padded box, $|r(z)|_1 \le 4 + 4(J + 5)$. The extension is $e^{ip\cdot r(z)}u(\bar{z})$, whose phase is bounded by $\exp(4 + 4(J + 5))$. The map $z \mapsto (r(z), \bar{z})$ is injective, and $r(z)$ ranges over a fixed finite set depending only on J. Comparing normalized counting measures therefore costs another constant depending only on J. The fine field retains its factor μ in the appropriate supremum or normalized L^2 norm. Coarse stencil shifts are at most one step and cost at most e; taking real or imaginary parts only decreases the coordinate norms. This verifies every fine, coarse and gauge stencil bound used to obtain B_n and closes the weighted penalty and internal Fourier inputs of the preceding section. The quantitative smooth bridge bound (634) is the remaining independent input.

123 Uniform inverse bounds for the stationary matrices

We use the block Fourier norm of Section 122. Its weight at alias l is $h_l(p) = (|q(\operatorname{Re} k_l(p))|^2 + \mu^2)^{1/2}$. The internal space is the direct sum of the fine one-form space in this norm and the coarse scalar space in its Euclidean norm. We denote it by W_p. The dual always pairs opposite momenta. The weights are not holomorphic functions of p, so a complex inverse estimate requires more than a direct application of Cauchy's inequality to a weighted matrix.

123.1 Continuing a real lower bound to a complex strip

Use the maximum norm on the four momentum coordinates throughout. Let X_p, Y_p be two weighted norms, represented by invertible matrices $S_X(p), S_Y(p)$: $\|u\|_{X_p} = |S_X(p)^{-1}u|_2$. Suppose that, at each real momentum x, these matrices admit holomorphic replacements on the ball $|z - x| < \sigma$. The replacements agree with the actual matrices on the vertical slice $\operatorname{Re} z = x$ and compare to the actual weighted norms in both directions by a fixed factor H, and first restrict $0 < \sigma \le 1$. Let K be periodic and holomorphic, with weighted operator bound A on the strip of width σ, and suppose its weighted real matrix has least singular value at least $a > 0$. Then there are $\eta > 0$ and $F < \infty$, depending on σ, H, A, a only, such that $K^{-1} : X_p \to Y_p$ has norm at most F for p in the strip of width η.

Here is a quantitative proof. At a real anchor x, transform $K(z)$ using the holomorphic replacement matrices. The resulting matrix $Q_x(z)$ is holomorphic and has Euclidean operator norm at most H^2A on the ball of radius σ. Cauchy's estimate on complex lines, tested against unit vectors on both sides, gives

$$\|Q_x(p) - Q_x(x)\| \le k|p - x|, \qquad k = \frac{H^2A}{\sigma/2}, \quad |p - x| < \sigma/2.$$

Choose $\eta = \min\{\sigma/2, a/(2(1 + k))\}$. For $x = \operatorname{Re} p$, the actual and replacement norms coincide, and

$$|Q_x(p)v| \ge (a - k|p - x|)|v| \ge (a/2)|v|.$$

Thus $Q_x(p)$ is invertible and its inverse has norm at most $2/a$. The same is true of the weighted inverse of $K(p)$. Holomorphy of the inverse follows from the adjugate formula; periodicity follows from uniqueness of the inverse. No dimension enters this argument.

For the block Fourier weights the required replacement is explicit: freeze h_l at the real anchor x, while retaining the holomorphic Fourier matrix $P(z)$. The inequality

$$h_l(z) \le 6h_l(x), \qquad h_l(x) \le 6h_l(z) \quad (|z-x| \le 1)$$

follows from $h_l \ge \mu$, the triangle inequality and the Lipschitz bound $|q(\operatorname{Re} k_l(z)) - q(k_l(x))| \le 4\mu|z-x|$. The Fourier matrices cancel in the comparison of the two norms; the remaining matrix is diagonal with these ratios. Hence $H = 6$ works, uniformly in L. Direct sums with the ordinary scalar norm and dual norms have the same property.

123.2 The scalar and gauge-completed fine matrices

Put $\omega(k) = \sum_a q_a(-k)q_a(k)$ and define

$$s_\mu(k) = \omega(k) + \mu^2, \quad d_\mu(k) = \frac{\mu^2}{s_\mu(k)}, \quad c_\mu(k) = \mu^2 d_\mu(k)^9, \quad \gamma_\mu(k) = \omega(k) + c_\mu(k).$$

At real k, $\omega(k) = |q(k)|^2$. The sign convention in this display uses $q_a(-k)q_a(k) = |q_a(k)|^2$ on the real slice. For $x \ge 0$,

$$x + \mu^2 \left(\frac{\mu^2}{x+\mu^2}\right)^9 \ge 2^{-10}(x+\mu^2). \tag{617}$$

If $x \ge \mu^2$, its first term alone suffices. If $x < \mu^2$, the ratio is greater than $1/2$, so the second term is at least $2^{-9}\mu^2 \ge 2^{-10}(x+\mu^2)$. On the alias strip $|\operatorname{Im} k|_\infty < \mu/128$, comparison with the real symbol gives $|s_\mu(k)| \ge \frac{1}{2}(|q(\operatorname{Re} k)|^2 + \mu^2)$. Indeed the change of q has norm at most $4\mu/128$ and the change of its quadratic product is bounded by twice this times the real norm plus its square; $2ab \le a^2 + b^2$ absorbs this into less than half the real denominator. In particular all displayed rational symbols are holomorphic there, $|d_\mu| \le 2$, and $|c_\mu| \le 2^9\mu^2$.

Let $H(k)$ be the fine two-form metric, bounded by Θ on the required strip, and suppose its real kinetic form satisfies the all-period lower bound $\varepsilon|d_1(k)v|^2$. Write $K_f = d_1(-k)^T H(k) d_1(k)$ and $d_0(k)v = q(k) \wedge v$. The scalar matrix $s_\mu I$ and the scalar matrix $\gamma_\mu I$ have weighted real floors 1 and 2^{-10}. The gauge-completed one-form matrix

$$D_\mu(k) = K_f(k) + d_0(k)d_0(-k)^T + c_\mu(k)I$$

has weighted floor $2^{-10}\min\{\varepsilon, 1\}$. This follows from

$$|d_1 v|^2 + |d_0^* v|^2 = |q|^2|v|^2$$

and (617). The exterior identity is obtained by expanding the wedge and contraction in the coordinate basis. The all-period kinetic bound applies at every real momentum by density of finite-torus frequencies and continuity.

For these scalar and fine symbols restrict further to $\sigma \le 1/128$. Their complex weighted upper bounds are also uniform. Each exterior matrix costs at most a fixed color-direction constant times $|q|$, $|q(k_l)| \le 5h_l$, and $|c_\mu(k_l)| \le 2^9 h_l^2$. Thus division by h_l at both ends leaves a fixed bound depending only on G, Θ. The block Fourier conjugations cancel between opposite momenta, leaving a unitary alias permutation and these diagonal weights. The preceding continuation argument gives inverse bounds for all three matrices on one coarse strip, before L and before any higher-order interaction budget are chosen.

123.3 A real lower bound modulo gauge directions

Let $Q(p)$ be the linear block mean on one-forms and $J(p)$ its scalar counterpart, so $Qd_0 = d_{0,c}J$. Set

$$\mathcal{R}a = (-d_0 a, -Ja), \qquad N(u,r) = d_{0,c}r - Qu, \qquad \mathcal{G}(u,r) = \langle u, K_f u\rangle + \kappa |N(u,r)|^2,$$

where $\kappa > 0$ and all operators are expressed in the same block Fourier coordinates. The curl and N annihilate $\mathcal{R}a$, so $\mathcal{G}$ is unchanged when a gauge vector is subtracted. We claim that, for some $a_0 > 0$ depending only on $G, J, \kappa, \varepsilon$,

$$\text{for every } v \text{ there is } a \text{ such that } a_0 \|v - \mathcal{R}a\|_{W_p}^2 \le \mathcal{G}(v). \tag{618}$$

Choose the real coarse momentum in $[-\pi, \pi]^4$. For every nonzero alias, $|q(k_l)| \ge 2\mu$. Subtract its longitudinal component by solving $d_0(k_l)^*(u_l + d_0(k_l)a_l) = 0$; the scalar denominator is $|q(k_l)|^2 > 0$. The principal alias is left available to eliminate the coarse scalar coordinate. In normalized Fourier coordinates the scalar block mean on that alias is $\mu^2 \,\mathrm{rollMean}(k_0)$, which is nonzero. No uniform bound on the eliminating gauge parameter is needed. Here rollMean is the product of the four normalized half-block geometric sums, raised to power J. For $|k_a| \le \pi/L$, each sum has modulus at least $1/16$; this follows from the geometric-sum formula and $\sin t \ge 2t/\pi$ for $0 \le t \le \pi/2$, with the value at zero understood by continuity. Thus $|\,\mathrm{rollMean}\,| \ge c_J = (1/16)^{4J}$. We obtain a representative $(P(p)b, 0)$ with all nonprincipal b_l transverse.

Put $A_h = \sum_{l \ne 0} h_l^2 |b_l|^2$ and $B_0 = |QP(p)b|^2$. The kinetic bound and transversality give

$$\mathcal{G}(v) \ge (\varepsilon/2) A_h + \kappa B_0.$$

The principal part of $QP(p)$ has smallest singular value at least $(c_J/16)\mu$: the straight connector contributes a geometric sum of size at least $L/16$, and P contributes μ^2. The full block mean has weighted norm at most a constant C_J. To verify this upper bound directly, each fixed-position read has coefficients bounded by $C_J\mu^4$ on $O_J(L^4)$ sites, hence Euclidean row norm at most $C'_J\mu^2$; sum the L positions and use $h_l \ge \mu$. The same argument gives the scalar averaging norm $\|J\| \le C'_J\mu^2$. Separating the principal and nonprincipal aliases and squaring the triangle inequality now gives

$$((c_J/16)\mu |b_0|)^2 \le 2B_0 + 2C_J^2 A_h.$$

Finally $h_0^2 \le ((2\pi)^2 + 1)\mu^2$. It follows that $\|(Pb, 0)\|_{W_p}^2 = A_h + h_0^2 |b_0|^2$ is bounded by a fixed multiple of $(\varepsilon/2)A_h + \kappa B_0$, proving (618). This proof includes $p = 0$.

123.4 Adding the gauge-fixing square

Let F be the actual linear gauge-fixing map, periodic and holomorphic on the chosen strip, and satisfying $F(-p)^T = F(p)^*$ on real momenta. Its required identities and estimate are

$$F\mathcal{R} = \Gamma_\mu, \qquad |Fv| \le D\|v\|_{W_p}, \tag{619}$$

where the bound holds throughout that complex strip, and Γ_μ is the block scalar operator with symbol γ_μ. These are exact same-frame identities; they remain inputs to this subsection until the gauge-fixing construction is identified with these symbols. The bound on J, the definition of $\mathcal{R}$ and (617) give

$$\|\mathcal{R}a\|_{W_p} \le C\,|\Gamma_\mu a| = C|F\mathcal{R}a|,$$

with C independent of L. For the fine component this is the pointwise inequality $h_l|q_l| \le h_l^2 \le 2^{10}\gamma_\mu(k_l)$; for the coarse component use $\mu^2 \le h_l^2$ and the scalar mean bound above. Given v, take a from (618) and put $b = \|v - \mathcal{R}a\|_{W_p}$, $c = |Fv|$. Then

$$\|v\|_{W_p} \le Cc + (CD+1)b, \qquad a_0 b^2 + c^2 \le \mathcal{G}(v) + |Fv|^2.$$

Squaring the first inequality proves a uniform positive lower bound for the full quadratic form $\mathcal{G} + F^*F$; for example its inverse floor can be taken as $\max\{2C^2, 2(CD+1)^2/a_0\}$. On the complex strip its weighted upper bound is the sum of the kinetic bound, κ times the square of the bound for N, and D^2. The holomorphic opposite-momentum transpose replaces F^* off the real slice. The continuation lemma therefore gives a common inverse width and cost for this actual stationary fluctuation matrix.

Once the common matrix is identified on the Fourier grids of an unbounded family of tori, the identification extends to the entire real slice by density, and then to the strip by analytic uniqueness. Consequently any two stationary matrix presentations representing that same family inherit this one inverse estimate. This last statement does not identify arbitrary independently chosen matrices: the all-period representation and the exact identities (619) must refer to the same construction.

124 The selected bridge and gauge-fixing map

The preceding estimates must be supplied on the actual retained horizon. We verify two identifications: the fine-difference bounds of its linear stationary bridge, and the exact gauge identity of its fluctuation matrix.

124.1 The unchanged one-form transport

Let A_0 be the one-form transport of the retained tower at a fixed adjacent pair of scales. The construction in Section 11 retains this row unchanged; only the curvature completion rows are replaced. Apply Theorem 6.3 with $e = 1$. The schedule has $j \ge 4$, so every difference word u of length at most two is covered. Its actual Laurent row satisfies

$$\sum_z e^{w|z|_1} |(\nabla^u A_0)(z)| \le D L^{-1-|u|}.$$

The constants have already been chosen uniformly over the horizon. The price enlargement used in the retained construction can only increase D and narrow w.

Define $\Phi(p)$ by Fourier evaluation of A_0, with the color identity on its color factor. Fourier evaluation commutes with a cover difference, including its centered carry: this is exactly the residue-difference identity in the proof of the cited theorem. Thus, for a coordinate leg,

$$|\nabla^u \Phi(p) e_a(z)| \le D L^{-1-|u|} \qquad (p \in \mathrm{strip}(w),\ |u| \le 2).$$

Absolute exponential summability gives holomorphy and periodicity on the strip. Restriction to a smaller width preserves every bound. This proves (634) for the selected bridge and supplies the identical hypothesis in the inherited-loop estimate.

We must also identify this row with the derivative of the stationary representative used below. Write $W = A_0$ in finite-volume coordinates. In the original tree gauge chart the positive internal Hessian gives a unique local stationary branch. Its fine tangent is not W itself: straightening up the tree adds an infinitesimal gauge transformation, so it is

$$E = W + d_0 K,$$

where K is the linear tree-straightening map. Gauge this branch back by the group exponential of Kx. The derivative of the resulting fine field is $E - d_0 K = W$. At a block root the straightening identity reads $K_{\mathrm{root}} = -\operatorname{fixRoot} W$. The derivative of the rooted frame is $-\operatorname{fixRoot} W$ as well; multiplication by the inverse root gauge factor cancels it. Thus recentering the gauged-back branch gives the joint derivative $(W, 0)$. Gauge covariance preserves its stationarity. We use uniqueness only in the original tree chart: the centered merit on the full joint space has gauge null directions. Finite Fourier inversion now identifies this actual derivative on every torus grid with Φ. Hence the difference bounds above concern the selected stationary representative.

124.2 The exact gauge identity

Work first on one finite torus. Let d be the infinitesimal fine gauge action at the identity, in the sign convention of the joint action, and let $\mathcal{R}$ be its expression in centered block coordinates. Its two components are $(-d_0 a, -Ja)$ as in the preceding section. Let X be the negative right logarithmic derivative of the frame in these coordinates, and let C_μ be the positive regulator with scalar symbol c_μ. The linear gauge-fixing map is

$$F = d^* \operatorname{pr}_{\mathrm{fine}} + C_\mu X.$$

For a gauge curve through the identity, frame covariance gives its frame ratio as the inverse gauge transformation. Applying the negative right logarithm therefore returns exactly the gauge parameter near zero. Differentiating this identity gives $X\mathcal{R} = I$, after expressing the gauge tangent in the same coordinates. Also the fine component of the gauge tangent is d. Consequently

$$F\mathcal{R} = d^* d + C_\mu = \Gamma_\mu. \tag{620}$$

There is a coordinate issue in this calculation. The rooted gauge coordinates and the centered coordinates are related by a triangular change whose derivative is

$$(u, r) \longmapsto (u, r + K_0 u),$$

where K_0 is the derivative of the recentering frame. The frame covariance identity gives $K_0 d_0 a = a_{\mathrm{root}} - Ja$. It follows by substitution that this triangular derivative sends the centered gauge tangent to the rooted gauge tangent. The fine projection is unchanged. The chain rule sends X to its product with that same triangular derivative, so both sides of (620) remain identical. This verifies the identity in the centered chart used for the estimates, including its sign; it would be incorrect to combine the rooted X with a centered $\mathcal{R}$ without this change.

Fourier transformation yields the identity on every torus grid. The symbols are common to the unbounded family of periods, by the local stencil construction. They therefore agree at every real momentum by density and at every complex momentum in their common strip by analytic uniqueness. For real momentum, the Fourier symbol of each real finite-volume matrix satisfies $F(-p) = \overline{F(p)}$. This applies to the frame linearization, the fine difference, and the real regulator, hence to F itself. Density of the common finite Fourier grids gives this identity at every real momentum, and therefore $F(-p)^T = F(p)^*$ there. Thus the bilinear holomorphic continuation used below agrees with the positive Hermitian square on the real slice. The same argument identifies the constant stationary matrix with $\mathcal{G} + F(-p)^T F(p)$: the fine action contributes its attached kinetic form, the block penalty contributes $\kappa N(-p)^T N(p)$, and the gauge-fixing term contributes the displayed square. Each is the Hessian of the corresponding term of the same action at its identity.

124.3 The uniform bound for the gauge-fixing map

For a unit vector in W_p, the fine derivative term d^*u is bounded in unnormalized Euclidean norm by a constant depending only on G, by the Fourier multiplier estimate in Section 122. The frame derivative Xv has normalized block L^2 norm at most a constant, by the centered-frame induction, restriction to one residue block, and the finite coordinate operations defining its logarithm. Restriction from the padded patch to this block costs a bounded ratio of volumes. Its unnormalized Euclidean norm is therefore at most CL^2. On the fixed small coarse strip the regulator has Euclidean operator norm at most $C'\mu^2$: its alias symbol is $\mu^2(\mu^2/(\omega+\mu^2))^9$, with the denominator estimate of the preceding section, and the two Fourier conjugations have uniformly bounded norms. Hence

$$|C_\mu Xv| \le C'C\mu^2L^2 = C'C.$$

Adding the fine derivative term gives the required bound $|Fv| \le D$, with D fixed before L. The same stencil construction is holomorphic and periodic in the external momentum, since it uses only linear Fourier legs in finite real derivatives at the identity and the holomorphic regulator symbol. This proves both parts of (619) for the selected data.

The inverse estimates of Section 123 now apply to the actual common stationary matrix. At the origin the two stationary matrix presentations used for the action and its comparison have this same matrix. Their inverse bounds consequently use the same width and cost; no bound for their positive-order derivatives is asserted by this constant-matrix identification.

125 The kinetic response on the selected bridge

We supply the fine kinetic matrix bound used in Section 126, and identify its three derivative kernels on the selected finite periods. Let F_n, $0 \le n \le 2$, be the response arrays of Theorem 120.9, of common Laurent width $t_* > 0$ and price $C \ge 0$. Retain all the theorem's hypotheses on the incoming path and metric. Let Φ be the actual bridge of Section 124, with fine differences through order one bounded by $D\mu^{1+j}$. Fix $0 < \sigma \le 1$ in its strip and require $\mu\sigma \le t_*$. This last condition is imposed by increasing L after t_* and σ have been chosen.

125.1 Contracting the response arrays

For external momenta $p_1, \dots, p_n$ and coarse coordinate legs $a_1, \dots, a_n$, put $v_j = \Phi(p_j)e_{a_j}$. For an internal input momentum q the output momentum is $q' = q + \sum_j p_j$. Define a matrix $K_n(p_1, \dots, p_n; q)$ by its bilinear pairing with arbitrary internal vectors u, w:

$$u^T K_n(p_1, \dots, p_n; q)w = \mathcal{B}_n(p_1, \dots, p_n, -q', q)[v_1, \dots, v_n, u, w].$$

Here $\mathcal{B}_n$ is the block-periodized multilinear form obtained by inserting the array F_n in (SR2) and summing the base site over one fine residue block. Translation of a residue across a block boundary carries the corresponding Bloch phase. This defines the pairing on coordinate vectors, hence defines the matrix uniquely. We only use momenta for which p_j, q, q' belong to the width-σ strip.

Every stitch uses words of length at most one, and its lengths obey

$$(n+2) + \sum_{j=1}^{n+2} |\ell_j| = 4. \tag{621}$$

This is the precise derivative count: two differences for the value, one for the first background derivative, and none for the second. Set $R = D + 5e^4$. In normalized counting measure on a residue

block, the external leg with word ℓ_j has supremum norm at most $\mu^{1+|\ell_j|}R$. A unit internal leg in the weighted Fourier norm has normalized L^2 norm at most the same expression, by the multiplier estimate of Section 122. There are exactly two such internal legs. Hölder's inequality therefore bounds the normalized sum of their product and all external factors by

$$\prod_{j=1}^{n+2} \mu^{1+|\ell_j|}R.$$

Restoring the unnormalized residue sum multiplies by L^4. Equation (621) gives the cancellation

$$L^4 \prod_{j=1}^{n+2} \mu^{1+|\ell_j|}R = R^{n+2}.$$

Thus neither a count of fine sites nor a conversion of a full matrix to entrywise bounds introduces an extra power of L.

We also check complex translation factors. For a shift z of a residue leg, its Bloch extension differs from the permuted residue vector by a phase with modulus at most $\exp(\sigma(8+\mu|z|_1))$. To see the bound, write the translated site as its residue plus L times its block quotient. The difference between the two residues has coordinate sum at most $8L$; the remaining displacement contributes $|z|_1$. Multiplying the phases of the $n+2$ legs costs at most

$$e^{8\sigma(n+2)}e^{\mu\sigma|z|_1},$$

where now z is the whole array shift. The last factor is paid by its Laurent width. Summing shifts, color addresses and stitch addresses therefore gives

$$\|K_n\| \le A_n, \qquad A_n = N_n e^{8\sigma(n+2)}R^{n+2}C, \qquad 0 \le n \le 2, \tag{622}$$

where N_n is the product of the numbers of color and stitch addresses at level n. These numbers depend only on the fixed group and n. The norm is the operator norm between the weighted fine space and its bilinear dual at the two indicated momenta. Taking $A_K = A_0 + A_1 + A_2$ supplies the common bound required in the determinant comparison.

Absolute Laurent convergence, uniformly on every strictly smaller strip, also proves holomorphy of these matrices. Moving a momentum by a reciprocal lattice vector changes its residue phases and its block quotient phases by inverse factors, so the matrices are periodic. This verifies the common analytic extension directly from the arrays and the bridge.

125.2 Identification with the actual Hessian derivatives

Let I_m be the incoming action on a fine torus and put

$$\mathcal{D}(x_+, x_-) = \mathrm{content}(\mathrm{input}(x_+)\,\mathrm{input}(x_-)).$$

Near zero, $I_m(\mathcal{D}(x_+, x_-))$ is exactly the action at the background $\mathrm{input}(x_-)$ after the left increment $\mathrm{input}(x_+)$. Its Hessian in x_+ is therefore the fine kinetic Hessian in the left coordinates used for the stationary determinant. Differentiating it n times in the background $x_- = Wx$, and testing with two arbitrary internal vectors, gives precisely the mixed response with n minus-species legs followed by two plus-species legs. Analyticity permits this reordering of derivatives. This is the species order in (SR2).

On a coarse Fourier wave, the actual installed matrix W produces the block wave with residue amplitude $\Phi(p)e_a$, by the finite Fourier identity of Section 124. Insert these waves and the two internal block waves into (SR2). The sum over coarse base sites is zero unless the total coarse momentum vanishes; otherwise it is the coarse site cardinality. In the latter case the remaining residue and shift sum is exactly the defining pairing of K_n above. Fourier waves span the full finite-volume space, and testing against every outgoing wave determines a matrix. Thus K_0, K_1, K_2 are the actual value, first derivative and second derivative kernels of the same kinetic Hessian on every selected torus, including nonzero momentum transfer. They are not only limiting coefficient pictures.

Finally the local bridge legs used by the frame and penalty estimates obey their required common pointwise bound: at a residue the color Euclidean norm is at most $\sqrt{d_1}\, D\mu$, because there are d_1 one-form color coordinates. Enlarging the local price to $\sqrt{d_1}D + e^4$ also covers the fixed coordinate legs. This price is independent of L. Consequently the local estimates, the kinetic triple and all finite-volume identifications in the determinant comparison refer to one and the same selected bridge and path.

126 Uniform prices for the determinant comparison

The absolute one-loop term counts all fine variables in a block and therefore has a bound proportional to L^4. Its comparison with the gauge-completed fine determinant has a better bound. We prove that bound before extracting the quadratic coefficient. Throughout this section L is odd, $\mu = L^{-1}$, and all constants are chosen from the fixed group, patch size, bridge price and penalty coefficient, before L. The fine kinetic response is an additional input: on a positive common strip of width σ, assume its value and first two background kernels satisfy $\|K_0\|, \|K_i\|, \|K_{ij}\| \le A_K$ in the weighted fine one-form norms, at every admissible set of intermediate momenta. Section 125 constructs that response and its bound for the selected actions.

126.1 Differentiated Fourier matrices

We need the value and the first two background derivatives of a matrix. An external leg of momentum p changes an input momentum q to $q + p$. Write these three kernels as $M_0(q)$, $M_i(p; q)$ and $M_{ij}(p, r; q)$. All momenta that occur as intermediate inputs or outputs must lie in the specified strip. A bound on these three kernels always uses the weighted norms at their respective input and output momenta. It does not mean an entrywise bound. The holomorphic rear reverses the endpoints while retaining the external momenta:

$$(M^*)_0(q) = M_0(-q)^T, \quad (M^*)_i(p; q) = M_i(p; -q - p)^T, \quad (M^*)_{ij}(p, r; q) = M_{ij}(p, r; -q - p - r)^T.$$

This convention is used in all products below.

Products obey the ordinary Leibniz rule with these intermediate momenta. Thus common bounds A, B for two triples give the bound $4AB$ for their product: there is one value term, two first-derivative terms, and four second-derivative terms. If M_0^{-1} has bound F, its derivative kernels follow by differentiating $M^{-1}M = 1$:

$$\begin{aligned}(M^{-1})_i &= -M^{-1}M_iM^{-1},\\ (M^{-1})_{ij} &= M^{-1}M_iM^{-1}M_jM^{-1} + M^{-1}M_jM^{-1}M_iM^{-1}\\ &\quad - M^{-1}M_{ij}M^{-1}.\end{aligned}$$

Intermediate momenta in these formulas are determined by composition. In particular a common inverse-jet bound is $F+F^2A+2F^3A^2+F^2A$. These formulas use only the inverse at zero background; they impose no uniform neighborhood of invertibility in the full background space.

Here are the local inputs to this calculus. Let $d(U)$ be the fine covariant difference at background U. On each edge its difference from $d(1)$ is the adjoint action of the edge variable minus the identity, followed by the fixed endpoint shift. The identity

$$d(U) = d(1) + (d(U) - d(1))$$

is therefore a fixed shift operator multiplied by local smooth functions of one fine link. Positive derivatives through order two of these functions have a factor μ: every differentiated fine link has size at most μC, and products of more such factors are smaller since $\mu \le 1$. Passing from Euclidean norms to a single weighted fine norm costs at most $e^4\mu^{-1}$, by Section 122. The factor μ pays this cost. The constant term $d(1)$ has a uniform weighted bound directly from its multiplier q and $|q| \le h$. Consequently the first two background derivatives of $d(U)$ are uniformly bounded in both placements needed for d^*d and dd^*.

The derivative of the centered frame, denoted by $X(U)$, has bound CL^2 from the joint weighted space to the ordinary fine scalar space, through two background derivatives. Indeed these are at most third derivatives of the frame, with one internal direction of charge one and the remaining external directions of charge zero. The frame calculus bounds the normalized L^2 output; removing that normalization multiplies by L^2. The logarithm and the finite chart changes preserve this estimate. The block constraint derivative $Q(U)$ and its coarse penalty form $H(U)$ have uniformly bounded kernels. The actual penalty Hessian $B(U)$ has the same property: its first two background derivatives require derivatives through order four, with two internal directions of charge one. Their total charge is two, exactly the L^1 case of the weighted penalty estimate.

These estimates apply on one common smaller strip. The external directions are the actual stationary bridge of Section 124. The same finite stencils define the kernels on every torus. Thus the preceding bounds apply to their common holomorphic Fourier matrices, rather than to separately chosen extensions of finite-grid matrices.

126.2 Why the regulator has a uniform trace bound

For a map between weighted Hilbert spaces, let $\|\cdot\|_1$ denote the nuclear norm after transforming to orthonormal coordinates. We use

$$\|ABC\|_1 \le \|A\|\|B\|_1\|C\|, \qquad |\operatorname{tr} B| \le \|B\|_1.$$

The second inequality is applied to endomorphisms with the same input and output momentum. Both statements follow from a singular value decomposition; the first also follows by decomposing B as a sum of rank-one operators. They introduce no dimension factor.

Set

$$P(U) = \mu^2(d(U)^*d(U) + \mu^2 I)^{-1}, \qquad C(U) = \mu^2 P(U)^9.$$

Here and below the star means transpose with opposite Fourier momentum in the holomorphic formulas. On the real slice it is the usual adjoint. The scalar inverse estimate of Section 123, the preceding product and inverse rules, and the two weight conversions show that all three kernels of P have a common Euclidean bound independent of L. The factor μ^2 cancels the two costs $O(\mu^{-1})$. It follows that the three kernels of C have operator bound $C_0\mu^2$.

We need the stronger nuclear bound, for which the ninth power is useful. At zero background, P_0 has alias eigenvalues $d_\mu(k_l) = \mu^2/(\omega(k_l) + \mu^2)$. On the strip of width $1/128$, after choosing a real fundamental cube, the alias distance $n = \operatorname{dist}_\infty(l, 0)$ satisfies

$$|d_\mu(k_l)| \le \frac{2}{(1+n)^2}.$$

For $n = 0$ this is the scalar denominator bound. For $n > 0$, the real alias difference has size at least $2\mu n$; the same denominator comparison gives the displayed estimate. The number of residues with distance n is at most $(2n+1)^4$. Hence

$$\sum_l |d_\mu(k_l)|^3 \le 8 \sum_{n \ge 0} \frac{(2n+1)^4}{(n+1)^6} \le 128 \sum_{n \ge 0} (n+1)^{-2} < \infty.$$

The fixed color multiplicity and the bounded Fourier conjugations only multiply this constant. Thus $(P^3)_0$ has a uniform nuclear bound, although P_0 itself need not have one.

If a triple V has operator bound a and its value V_0 has nuclear bound b, the triple for V^3 has nuclear bound $20a^2b$. To see this, distribute at most two derivatives among three factors. Every term has an undifferentiated factor, which can be estimated in nuclear norm; all other factors use operator norm. The value, first derivative and second derivative have at most one, three and nine terms, respectively. The larger constant 20 also accommodates the grouped product estimates above. Apply this to $V = P^3$. We obtain

$$\|C_0\|_1,\ \|C_i\|_1,\ \|C_{ij}\|_1 \le C_1 \mu^2. \tag{623}$$

This argument needs only the value-level spectral calculation for P^3; derivatives of its eigenvectors are not used.

126.3 Cancellation of the fine kinetic form

Let b project the joint space onto its fine one-form part, and let e project it onto its coarse scalar part. Let s_a be the fixed map selecting the one-form component in direction a. Denote the fine kinetic Hessian triple by K. Define

$$X = d^* b, \qquad T = C X_{\mathrm{fr}}, \qquad F = X + T,$$

where X_{fr} is the centered frame derivative denoted by $X(U)$ above. The change of notation separates it from d^*b. The three matrix triples to be compared are

$$\begin{aligned} M_{\mathrm{act}} &= b^* K b + B + F^* F, \\ M_{\mathrm{core}} &= b^* K b + Q^* H Q + F^* F, \\ M_{\mathrm{rank}} &= b^* \left(K + dd^* + \sum_{a=1}^{4} s_a^* C s_a \right) b + e^* e. \end{aligned}$$

The first uses the full penalty Hessian; the second retains its constraint pullback. Their distinction will matter in the next comparison of low-momentum coefficients.

The weighted norm bounds above, including the bound for K, give a common operator bound A for all three triples and for $d^*d + C$. Their zero-background inverses have one common bound F_0 on a smaller strip of width η, by the previous two sections. Indeed the zero-background penalty Hessian equals its constraint pullback, and the gauge-completed fine matrix is precisely the auxiliary

matrix treated there. Take the minimum of the finitely many widths and the maximum of their costs.

The difference of the second and third triples is exactly

$$M_{\mathrm{core}} - M_{\mathrm{rank}} = Q^* H Q + T^* F + X^* T - b^* \sum_a s_a^* C s_a b - e^* e. \tag{624}$$

This is an identity at all three derivative levels. In particular the entire K triple cancels, not just its value at the origin. It follows by expanding $F = X + T$ and using $X^* X = b^* dd^* b$.

Every term on the right has a nuclear bound independent of L. For $Q^* H Q$, take the nuclear bound on the fixed-dimensional coarse space before multiplying by Q^* and Q; the estimate $\|H_j\|_1 \le \dim(\text{coarse one-forms})\|H_j\|$ works for each derivative. The same argument applies to $e^* e$. For $T = C X_{\mathrm{fr}}$, (623) cancels the L^2 frame cost because $\mu^2 L^2 = 1$. Multiplication by F or X preserves its nuclear bound. Finally, converting the regulator term to weighted fine norms costs at most $e^8 \mu^{-2}$, paid by its factor μ^2; the four component maps s_a have fixed bounds. Leibniz's rule proves the same estimates through order two. Thus there is k, independent of L and of the fine kinetic price, with

$$\|(M_{\mathrm{core}} - M_{\mathrm{rank}})_j\|_1 \le k, \qquad j = 0, 1, 2. \tag{625}$$

The inverse bound F_0 and absolute bound A may still depend on the stated kinetic hypotheses. That dependence has not been removed by this cancellation.

126.4 The second logarithmic derivative

For any of the invertible triples define its two-leg trace density by

$$\mathcal{T}_M(i, j; p, q) = \operatorname{tr}\left[M_0(q)^{-1} M_{ij}(-p, p; q) - M_0(q)^{-1} M_i(-p; q + p) M_0(q + p)^{-1} M_j(p; q) \right].$$

This is the second derivative of $\log\det M$, obtained by differentiating $D \log\det M = \operatorname{tr}(M^{-1} D M)$. It requires no choice of a global logarithm on the strip. The formula itself is holomorphic wherever the constant inverses exist. Restrict both independent momenta to width $\eta/2$, so every intermediate momentum lies in width η.

If the fiber dimension is d, the operator bounds give

$$|\mathcal{T}_M| \le d(F_0 A + F_0^2 A^2). \tag{626}$$

Here $d \le L^4(d_0 + d_1)$, with d_0, d_1 the fixed scalar and one-form color dimensions. This explains the absolute L^4 cost.

For $M = M_{\mathrm{core}}$ and $N = M_{\mathrm{rank}}$ put

$$D = F_0 k F_0, \qquad E = DA + F_0 k.$$

The resolvent identity gives $\|M_0^{-1} - N_0^{-1}\|_1 \le D$. The elementary telescoping identity $UV - U'V' = (U - U')V + U'(V - V')$ then bounds the difference of $M_0^{-1} M_{ij}$ and $N_0^{-1} N_{ij}$ by E in nuclear norm. For the four-factor term, first compare the initial two factors, at cost E; compare the next inverse, at additional cost $F_0 A D$; and finally compare the last derivative. Keeping the remaining factors in operator norm gives

$$\|M_0^{-1} M_i M_0^{-1} M_j - N_0^{-1} N_i N_0^{-1} N_j\|_1 \le (E F_0 + F_0 A D) A + F_0 A F_0 k.$$

All these products respect the displayed intermediate momenta. Taking traces proves

$$|\mathcal{T}_{M_{\mathrm{core}}} - \mathcal{T}_{M_{\mathrm{rank}}}| \le E + (E F_0 + F_0 A D) A + F_0 A F_0 k. \tag{627}$$

There is no dimension factor in this estimate.

The one-loop densities are $\frac{1}{2}\mathcal{T}_{M_{\mathrm{act}}} - \mathcal{T}_{d^* d+C}$, $\frac{1}{2}\mathcal{T}_{M_{\mathrm{core}}} - \mathcal{T}_{d^* d+C}$ and $\frac{1}{2}\mathcal{T}_{M_{\mathrm{rank}}} - \mathcal{T}_{d^* d+C}$. Their individual bounds may all be enlarged to twice the right-hand side of (626). In the difference of the latter two densities the scalar determinant cancels exactly, leaving half the difference in (627). This proves the required uniform comparison price. Identification with the thermal stationary coefficient, and the disappearance of the remaining action-versus-core quadratic difference, are separate steps below.

127 The thermal determinant and its contact term

We now relate the determinant matrices to the stationary integral and show that replacing the full penalty Hessian by its constraint pullback does not alter the leading quadratic coefficient. We retain the actual stationary branch, predictor and tree gauge from the preceding sections. The group has nonabelian simple real Lie algebra. The finite-volume stationary Hessian gap and the predictor branch hypotheses remain in force. We also retain the exact matching of the incoming and outgoing quadratic forms with the selected bridge; in particular its constraint map obeys $QW = I - \kappa^{-1} K_c$ for the selected outgoing kinetic matrix K_c. Stationarity and the gap alone do not identify an independently chosen outgoing matrix. All identities below concern germs at the trivial field.

127.1 The Haar factor in tree coordinates

Let y denote the free tree-gauge coordinates of a stationary point, and let $j(y) > 0$ be the product of their Haar density ratios, normalized by $j(0) = 1$. Define the left-increment coordinate map, componentwise, by

$$s_y(z) = \log\bigl(\exp(z)\exp(y)^{-1}\bigr), \qquad s_y(y) = 0.$$

It is a local diffeomorphism. Right invariance of probability Haar measure and the ordinary change-of-variables formula give

$$j(z) = |\det Ds_y(z)| j(s_y(z))$$

on the common coordinate neighborhood. For completeness, apply both change-of-variables formulas to any compact subset of that neighborhood. The two continuous positive densities have equal integrals over every such set. If they differed at a point, continuity would give a closed ball of positive Lebesgue measure on which one density is strictly larger, contradicting that equality. Evaluating at $z = y$ yields

$$j(y) = |\det Ds_y(y)|. \tag{628}$$

Thus the Haar density cannot be discarded when the stationary point moves away from zero.

Let H be the Hessian of the joint action in left coordinates at a stationary configuration, let R be its fine gauge-tangent matrix, and let P insert the free tree increments. Gauge invariance and stationarity give $HR = 0$; H is symmetric. The square matrix $S = [R\ P]$ has $\det(S)^2 = 1$ in the physical counting bases. To verify this last assertion, order the fine vertices by tree height, assigning to a nonroot vertex its upward tree edge and to a root its coarse scalar coordinate. On the tree edges the gauge variation is the signed difference between the vertex and its parent, because those edge variables equal the identity. At a root it is minus the root gauge parameter. The resulting gauge block is triangular, with diagonal entries ± 1. The free-increment block is the identity, and the other upper block is zero. Hence the squared determinant is one, regardless of the free edge variables.

Let L be the gauge-fixing derivative and $\Gamma = LR$ the positive scalar gauge matrix. Set $C = P^T H P$ and $B = LP$. Direct multiplication gives

$$S^T(H + L^T L)S = \begin{pmatrix} \Gamma & B \\ 0 & I \end{pmatrix}^T \begin{pmatrix} I & 0 \\ 0 & C \end{pmatrix} \begin{pmatrix} \Gamma & B \\ 0 & I \end{pmatrix}.$$

Consequently

$$\det(H + L^T L) = \det(\Gamma)^2 \det C. \tag{629}$$

The tree-chart Hessian H_{tree} is $Ds_y(y)^T C Ds_y(y)$: the additional second-derivative term in the chain rule vanishes by stationarity. Combining this with (628) proves

$$j(y)^2 \det M_{\text{act}} = \det H_{\text{tree}} \det \Gamma^2.$$

The tree gap, positive Haar density, and positive scalar determinant show that every determinant inside the logarithms is positive near the origin. Thus

$$\frac{1}{2} \log\det H_{\text{tree}} - \log j(y) = \frac{1}{2} \log\det M_{\text{act}} - \log\det \Gamma. \tag{630}$$

Adding the inherited loop gives the thermal term of the stationary Gaussian integral with exactly its original normalization.

127.2 Replacing the stationary graph by its linear tangent

Write $\zeta(x)$ for the actual centered stationary representative. Section 124 gives $\zeta(0) = 0$ and $D\zeta(0)x = (Wx, 0)$. Each scalar logarithmic determinant under consideration is invariant under simultaneous constant color rotation in the coarse, fine and scalar coordinates. Indeed these rotations act orthogonally; they conjugate the Hessian and the scalar gauge matrix. The centered-frame change is equivariant because group multiplication, the nearest-group map and the logarithm are equivariant under conjugation.

An invariant linear functional on one copy of the Lie algebra is zero. Its derivative along $\mathrm{Ad}(\exp ta)b$ annihilates every bracket $[a, b]$. The derived algebra is a nonzero ideal and therefore is the whole algebra by nonabelian simplicity. Applying this argument to each coordinate proves that the first derivative of each invariant scalar function vanishes at zero. The second-derivative chain rule therefore gives

$$D^2\big(f(x, \zeta(x))\big)_{x=0}[u, v] = D^2 f(0)[(u, Wu, 0), (v, Wv, 0)].$$

This is why the linear external legs used in the Fourier matrix calculus compute the thermal Hessian of the actual nonlinear branch. There is no assertion that the nonlinear branches themselves coincide.

Rooted and centered internal increments are related by a triangular coordinate change with determinant one. On shell the full Hessians transform by congruence; the constraint-pullback Hessian does so even off shell. Hence this change preserves their determinants in the calculations below. Gauge changes preserve the same determinants by orthogonal congruence. These observations justify working in rooted coordinates for the contact calculation and centered coordinates for the uniform estimates.

127.3 The finite-volume contact identity

Let $w_x(v)$ be the logarithmic change of the block predictor when an internal left increment v is applied at background x; $w_x(0) = 0$. Let $b(x)$ be the gradient, with respect to this predictor increment, of the coarse penalty. Write $A(x) = M_{\mathrm{core}}(x)$ in rooted coordinates and define

$$S_x(a)_{ij} = \langle a, D_v^2 w_x(0)[e_i, e_j]\rangle, \qquad Z(x) = S_x(b(x)).$$

The Hessian chain rule for the penalty says exactly

$$M_{\mathrm{act}}(x) = A(x) + Z(x).$$

It separates the pullback of the penalty Hessian from the gradient of the penalty paired with the second derivative of its constraint. At zero $b(0) = 0$, and hence $Z(0) = 0$.

Define a coarse vector $Y(x)$ by

$$\langle a, Y(x)\rangle = \mathrm{tr}\big(A(x)^{-1} S_x(a)\big).$$

It is analytic for this fixed finite volume. Gauge covariance follows directly from this formula: the internal matrices transform by the same congruence and the predictor output by the coarse adjoint action, so cyclicity of trace gives the coarse adjoint action on Y. At zero each component of $Y(0)$ is fixed by constant color rotations and is therefore zero. This also follows by differentiating the adjoint action: an invariant vector is central, and a nonabelian simple Lie algebra has zero center.

Let K_c be the coarse kinetic matrix and put $J = -K_c$. The exact linear bridge identity is

$$QW = I - \kappa^{-1} K_c.$$

The quadratic penalty at the identity is $\frac{\kappa}{2}|x-z|^2$. Its gradient in z has derivative $\kappa(z-x)$. Applying the bridge identity therefore gives

$$Db(0) = J. \tag{631}$$

Along the stationary graph this remains the same derivative, because its tangent is W. In particular $DZ(0)u = S_0(Ju)$.

The identity $\det(A + Z) = \det A \det(1 + A^{-1}Z)$ and $Z(0) = 0$ give, through second order,

$$\log\det(A+Z) - \log\det A = \mathrm{tr}(A^{-1}Z) - \tfrac{1}{2}\,\mathrm{tr}(A^{-1}ZA^{-1}Z) + O(|x|^3).$$

Since $\mathrm{tr}(A^{-1}Z) = \langle b, Y\rangle$ and $b(0) = Y(0) = 0$, differentiation gives the exact identity

$$\begin{aligned} D^2(\log\det M_{\mathrm{act}} - \log\det M_{\mathrm{core}})(0)[u,v] &= \langle Ju, DY(0)v\rangle + \langle Jv, DY(0)u\rangle \\ &\quad - \mathrm{tr}\big(A(0)^{-1}S_0(Ju)A(0)^{-1}S_0(Jv)\big). \end{aligned} \tag{632}$$

The first two terms include the derivative of the inverse matrix; it has not been dropped. The vanishing of $Y(0)$ is what removes the term involving the second derivative of b.

There is also a Ward identity

$$DY(0)d_c a = 0 \tag{633}$$

for every coarse scalar field a. To prove it on the actual branch, gauge the trivial stationary configuration by $\exp(ta)$ at coarse sites, then restore tree gauge. Gauge invariance preserves stationarity; uniqueness in the tree chart identifies the restored configuration with the selected stationary branch on this curve. The coarse field has tangent $-d_c a$. Covariance of Y and $Y(0) = 0$ show that Y vanishes along the curve. Differentiation gives (633). Replacing the branch by its linear tangent does not alter $DY(0)$, by the first-derivative chain rule. This proves the identity including the zero modes of the coarse difference.

127.4 Passing the identity to the common momentum picture

Fix L for this paragraph. The rooted frame and its change to centered coordinates are analytic finite stencils. The inverse of A_0 is holomorphic on a fixed strip. These facts construct a common holomorphic matrix kernel for $A(x)^{-1}S_x(a)$ and for its square, with two kinds of external legs: a coarse background leg and a separate coarse test leg a. Indeed express $S_x(a)$ using the second derivative of the local predictor, conjugate the centered inverse by the triangular change, and multiply. Every factor has the same finite-volume Fourier representation; products and inverse derivatives preserve it. Only a finite bound for this fixed L is required here.

Take normalized traces per coarse site. Finite Fourier inversion identifies them with averages of the common kernels over the internal momentum grid. Holomorphy on a smaller strip bounds the Fourier coefficients exponentially, so these averages converge to the torus integral, uniformly for real external momentum. The same convergence holds when an external momentum varies along grid points approaching a prescribed real momentum: subtract the limiting integral at that grid point and use its continuity. Thus both the finite-volume contact identity and its Ward identity pass to all real momenta along any divergent family of coarse periods.

Denote the resulting mixed background/test trace kernel by $M_{ij}(p)$, and the second test/test derivative kernel of the trace of the square by $B_{ij}(p)$. Explicitly, at zero background its finite-volume bilinear value on test fields a, b is

$$2\,\mathrm{tr}\big(A(0)^{-1}S_0(a)A(0)^{-1}S_0(b)\big).$$

Thus B includes both ordered cross terms. They are analytic near zero. The Ward identity says

$$\sum_i (d_c(-p)a)_i M_{ij}(p) = 0.$$

Fix a component $i = (\nu, \alpha)$ and take a in color α and $p = te_\nu$. Only that directional difference survives, giving $q_\nu(-te_\nu)M_{ij}(te_\nu) = 0$. The first factor is nonzero for all sufficiently small nonzero real t. Continuity gives $M_{ij}(0) = 0$. Analyticity then implies $M_{ij}(p) = O(|p|)$ along every ray, not just that coordinate ray.

For a fixed external component a, put $x_j(p) = (-K_c(p)e_a)_j$. Since $K_c(p) = d_1(-p)^T H_c(p) d_1(p)$ with H_c analytic, $x_j(p) = O(|p|^2)$. The infinite-volume version of (632), including the factor one half in the thermal determinant, has diagonal density

$$\frac{1}{2}\left[W(p) + W(-p) - \frac{1}{2}V(p)\right], \qquad W(p) = \sum_j x_j(p) M_{aj}(p), \quad V(p) = \sum_{i,j} x_i(-p) x_j(p) B_{ij}(p).$$

The factor $1/2$ in front of V arises because the second derivative of the trace of the square contains the two equal cross terms. Now W has ray order at least three and V has ray order at least four. Their sum therefore has zero quadratic Taylor coefficient. Passing from a Hessian density to the normalized quadratic picture introduces one more factor $1/2$ and preserves this vanishing.

We have proved that the thermal action and the core determinant have the same leading quadratic coefficient. Combining this exact cancellation with the uniform core-versus-rank bound from Section 126 bounds their remaining coefficient difference by Cauchy's inequality on the common strip. The qualitative rooted-frame constants used in the vanishing argument do not enter that uniform comparison bound.

128 Classical vertices after stationary elimination

We now construct the cubic and quartic coefficient pictures of the stationary action. Two estimates must be kept distinct. The fine action is controlled by its vanishing order in momentum; the internal contraction in the quartic term is controlled in the weighted fluctuation norm. A coordinatewise estimate for that contraction would introduce an unwanted factor equal to the number of fine variables in a block.

128.1 The weighted vertex hypotheses

Write $\mu = L^{-1}$, where L is the block side. All momenta in this section are coarse momenta. Let $\Phi(p)$ be the linear fine-field response to a coarse field. In block coordinates the full tangent vector is

$$\mathcal{L}_p h = (h, \Phi(p)h, 0).$$

The last component is the gauge coordinate. An internal vector is written $\iota v = (0, v)$, with the internal space including the fine field and the auxiliary scalar variables. Let $\|\cdot\|_p$ be its weighted fluctuation norm. The corresponding bilinear dual norm is

$$\|s\|_{p,*} = \sup_{\|v\|_{-p}\le 1} |v^T s|.$$

The sign in this definition is dictated by momentum conservation.

Fix $0 < \eta \le 1$ and $D \ge 0$. The quantitative bridge hypothesis is holomorphy and periodicity together with

$$\sup_z |\nabla^j \Phi(p) e_a(z)| \le D\mu^{1+j}, \qquad 0 \le j \le 2, \quad p \in \mathrm{strip}(\eta). \tag{634}$$

Here e_a is a unit color-direction coordinate and ∇^j is any ordered string of j fine differences, with the coarse Fourier phase included at a block crossing. Choose $D_* = 1 + D + 5e^4$. The same bound with D replaced by D_* applies to the fine component of $\mathcal{L}_p e_a$. For one internal vector of norm at most one, the weighted discrete Fourier estimate supplies the analogous bounds with one unit of integrability charge, through difference order one. More explicitly, if H_b denotes the normalized block norm $L^{2/b}$ for $b > 0$ and L^∞ for $b = 0$, the required estimates are

$$H_0(\nabla^j \mathcal{L}_p e_a) \le D_* \mu^{1+j}, \qquad H_1(\nabla^j \iota v) \le D_* \mu^{1+j} \quad (j \le 1).$$

Only the fine-field component occurs in these two inequalities. The weighted Fourier estimate and the block penalty estimate below are explicit inputs here; an ordinary bound for chart derivatives at fixed L would not imply either uniform estimate.

An n-linear vertex has price C_n if its absolute value is at most C_n on smooth coordinate legs, and also with the first leg replaced by ιv, $\|v\|_p \le 1$, and all other legs smooth. These are *unnormalized* derivatives. Their coefficient pictures therefore carry a factor $1/n!$.

For the block penalty we use the following precise receipt. For $n = 3, 4$, its real nth derivative in each of the four coarse directions is bounded by B_n, on the stencil fields induced by these legs, either all smooth or with one internal leg. The constants B_n depend on the fixed group, patch enlargement, penalty coefficient and D, but not on L or the incoming interaction budget. The stencil field bounds are to hold for both real and imaginary parts. Complex multilinearity then gives price

$$N_n = 2^n\, 4B_n. \tag{635}$$

Indeed, expand each complex leg into its real and imaginary parts. There are 2^n terms, each with a coefficient of modulus one; sum the four directional penalties. This deduction does not replace the required real weighted stencil estimate.

128.2 Why the residual vertices gain a scale factor

The coefficient pictures proved in Section 120 split the fine classical action into its Wilson part and its residual part. At degree three the Wilson picture vanishes to order one and the residual picture to order two; at degree four their respective orders are zero and one. Lemma 120.7 writes each as a finite sum of products of one-link difference factors and analytic coefficient arrays. The factors may be assigned to legs other than the first. Thus the leg that will later be tested in the internal norm carries no differences. For the four cases, the number of legs n and the number of differences q satisfy

	n	q	$n+q-4$
Wilson cubic	3	1	0
Wilson quartic	4	0	0
residual cubic	3	2	1
residual quartic	4	1	1.

To see the significance of the last column, consider one term with j_i differences on leg i. Equation (634) contributes $\prod_i \mu^{1+j_i} = \mu^{n+q}$. Summation over one fine block contributes $L^4 = \mu^{-4}$. The result is μ^{n+q-4}. This count remains valid with one internal leg: use its normalized L^2 bound and the supremum bounds on the other legs, then Hölder on the normalized block measure. The displaced arguments and analytic coefficient sums are bounded by the absolute exponentially weighted coefficient sum furnished by the difference factorization. Fourier phases are bounded on the chosen strip; these factors and the finite color and difference labels are included in the constant. The first leg has no differences, so its weaker derivative allowance is sufficient. This proves both parts of the vertex price.

Choose the coarse width η before fixing the incoming path width τ and budget A. The Wilson arrays and the penalty prices then give constants $a, b \geq 0$ independent of τ, A, L. For each τ, A the residual arrays give a fine width $\sigma' > 0$ and constants $\ell, r \geq 0$. Provided $\mu\eta \leq \sigma'$, addition of the three contributions gives holomorphic periodic vertices T_3, T_4 with prices

$$C_3 = a + \mu\ell, \qquad C_4 = b + \mu r. \tag{636}$$

These vertices represent the actual block action on every torus in the chosen family. To verify this, insert Fourier waves into each local translation sum. Summing the coarse anchor gives zero unless the total momentum is zero, and otherwise gives the coarse volume. The remaining fine-block sum is exactly the array just estimated. The germ identity between the block action and the sum of the fine action and block penalty identifies these derivatives for the actual tags. No change of stationary branch has been made.

128.3 The stationary contraction formula

Let $\mathcal{A}(x, y)$ be this real analytic block action, with coarse variable x and internal variable y. Suppose $D_y\mathcal{A}(x, y(x)) = 0$, $y(0) = 0$, and $M = D_y^2\mathcal{A}(0, 0)$ is positive definite. Put $E(x) = \mathcal{A}(x, y(x))$ and $\mathcal{L}h = (h, Dy(0)h)$. Differentiating stationarity once gives

$$D^2\mathcal{A}(0)[\mathcal{L}h, \iota v] = 0. \tag{637}$$

Define the internal covector

$$s(h, k)[v] = D^3\mathcal{A}(0)[\iota v, \mathcal{L}h, \mathcal{L}k].$$

A second differentiation of stationarity gives

$$D^2y(0)[h, k] = -M^{-1}s(h, k).$$

The resulting third and fourth derivatives are

$$D^3E(0)[h_1,h_2,h_3] = D^3\mathcal{A}(0)[\mathcal{L}h_1,\mathcal{L}h_2,\mathcal{L}h_3], \tag{638}$$

$$D^4E(0)[h_1,h_2,h_3,h_4] = D^4\mathcal{A}(0)[\mathcal{L}h_1,\mathcal{L}h_2,\mathcal{L}h_3,\mathcal{L}h_4] - \sum_{\{\{i,j\},\{k,l\}\}} s(h_i,h_j)^T M^{-1} s(h_k,h_l), \tag{639}$$

where the sum runs over the three unordered pairings of four indices.

For completeness, these identities can be checked without suppressing any chain-rule terms. Along a ray, write $y(th) = tDy(0)h + t^2u/2 + t^3v/6 + O(t^4)$. The terms involving $D_y\mathcal{A}(0)$ vanish by stationarity. The mixed Hessian terms involving $\mathcal{L}h$ vanish by (637). The cubic coefficient is therefore $D^3\mathcal{A}(0)[\mathcal{L}h]^3/6$. The fourth-order coefficient is

$$\frac{D^4\mathcal{A}(0)[\mathcal{L}h]^4}{24} + \frac{s(h,h)[u]}{4} + \frac{u^TMu}{8} = \frac{D^4\mathcal{A}(0)[\mathcal{L}h]^4}{24} - \frac{s(h,h)^TM^{-1}s(h,h)}{8}.$$

Here $u = -M^{-1}s(h,h)$. Polarization of these symmetric multilinear forms proves (638)–(639), including the coefficient and sign of all three pairings.

128.4 Common analytic pictures and their prices

Assume that the inverse of the same internal Hessian has a periodic holomorphic Fourier symbol $V(p)$, with the weighted operator estimate

$$\|V(p)s\|_p \le F\|s\|_{p,*} \tag{640}$$

on the strip of width η, where $F \ge 0$ is fixed before the incoming width and budget. This is an input about the actual inverse, not an arbitrary comparison matrix.

With smooth legs at momenta p, q, let

$$s_{ab}(p,q)[v] = T_3(-(p+q),p,q)[\iota v, \mathcal{L}_p e_a, \mathcal{L}_q e_b].$$

By the one-internal-leg estimate, $\|s_{ab}(p,q)\|_{p+q,*} \le C_3$. Consequently every contraction in (639) has modulus at most C_3FC_3. This is duality in the weighted norm; no sum of coordinatewise bounds and no dimension factor is required. The pairing uses opposite momenta on its two internal legs.

For n external legs, take the first momentum to be minus the sum of the other $n-1$. On the independent strip of width $\eta/(2n)$ all external momenta and all paired internal momenta lie in the original strip. Products, contractions and sums of the holomorphic arrays are therefore holomorphic there. Periodicity follows from that of the vertices, bridge and inverse. Equations (638)–(639), divided by the factorials, define common coefficient pictures P_3, P_4 with

$$\|P_3\|_{\eta/6} \le \frac{C_3}{6}, \qquad \|P_4\|_{\eta/8} \le \frac{C_4 + 3C_3FC_3}{24}. \tag{641}$$

They record the stationary derivatives exactly on every torus: the vertex identities already have the coarse-volume normalization, and Fourier inversion of the internal Hessian makes its contraction precisely V at the paired momentum. Thus the contraction formula preserves the same momentum-conservation rule and the same volume factor.

Finally choose L sufficiently large that $\mu\ell, \mu r \le 1$. Both pictures are then bounded on their common strip by

$$M_* = 1 + \frac{a+1}{6} + \frac{(b+1) + 3(a+1)F(a+1)}{24}.$$

The width and M_* are chosen before τ, A; only the lower threshold for L depends on the incoming path. This order of choices is what allows the stationary output construction to be iterated. The weighted bridge, weighted penalty and common inverse estimates remain the precise supplier interfaces of this section.

129 Priced local families give the three face estimates

We derive the upper-face bound, the full lower-face bound, and the lower-face-minus-halo bound used in the annulus estimate. The proof is the local price calculation in the selected construction. We then verify the scale and data choices for the two populations used by the annulus: all centered configurations and the regulator-thin shell. We derive the selected row's contact identities and the scale-weighted face prices, including the separate shell restriction required by the unregulated reference inverses.

129.1 Derivative prices

For analytic matrix paths $M_i(t)$ near $t = 0$, with values in a fixed finite-dimensional matrix space of dimension d, write

$$\mathsf{P}_r(M; C) \quad \Longleftrightarrow \quad M_i \text{ is analytic near 0 and } r_i^j \, \|M_i^{(j)}(0)\|_{\mathrm{op}} \le C \quad (j = 0, 1, 2), \tag{F1}$$

uniformly in the index i. Here $\|\cdot\|_{\mathrm{op}}$ is the Euclidean operator norm; it is submultiplicative, and $|\operatorname{tr} M| \le d\|M\|_{\mathrm{op}}$, where d is the matrix dimension. Constants may be enlarged by a fixed factor without changing the notation.

The elementary rules used below are as follows. If M, N have prices (r, C) and (r, D), then $M + N$ has price $(r, C + D)$ and MN has price $(r, 4CD)$. For the product, the zeroth and first derivatives are the ordinary product rule; the second derivative has four terms, each bounded by submultiplicativity and the corresponding two price bounds. If $a > 0$ and $C, D \ge 0$, $M(0)$ is invertible and $\|M(0)^{-1}\|_{\mathrm{op}} \le D/a$, while M has price (r, aC), differentiating $M^{-1}M = I$ once and twice gives

$$\mathsf{P}_r(M^{-1}; \mathcal{I}(C, D)/a), \qquad \mathcal{I}(C, D) = D + 2CD^2 + 2C^2D^3. \tag{F2}$$

Indeed, $(M^{-1})' = -M^{-1}M'M^{-1}$ and $(M^{-1})'' = 2M^{-1}M'M^{-1}M'M^{-1} - M^{-1}M''M^{-1}$; submultiplicativity bounds the terms by the three summands in (F2). Finally,

$$\mathsf{P}_r(M; Cr_i^{-2}) \implies |(\operatorname{tr} M_i)''(0)| \le dCr_i^{-4}, \tag{F3}$$

by the $j = 2$ case of (F1) and the trace inequality. These are the finite-dimensional price rules underlying the estimates below.

129.2 Upper, lower, and halo faces

For each configuration i, let x_i be its background, e_i its boost, $\mu_i > 0$ its regulator and $T_i > 0$ its scale. The local geometry used here is

$$\|e_i\| \le 1, \qquad \|x_i\| \le T_i \le r_*, \qquad 0 < \mu_i \le T_i. \tag{F4}$$

Let S_i denote the total stationary denominator (interacting plus reference), and let J_i and J_i^{ref} be the two two-leg arch insertions. Write $K_{0,i} = S_i(0)^{-1}$ for the inverse at the unshifted background and $K_{1,i}(t)$ for the inverse at the displaced background $x_i + te_i$. Write $D_{0,i}$ and $D_{1,i}(t)$ for the

corresponding free/reference inverses. The unshifted inverse is constant in t; the shifted inverse varies. With B_i^-, B_i^+ the displaced insertion legs, define the scalar upper and lower faces by

$$U_i(t) = \operatorname{tr}\left[K_{0,i}(J_i(t) + J_i^{\mathrm{ref}}(t))\right], \qquad L_i(t) = \operatorname{tr}\left[K_{0,i}B_i^-(t)K_{1,i}(t)B_i^+(t)\right]. \tag{F5}$$

The halo is $H_i(t) = \operatorname{tr}[D_{0,i}\widetilde{B}_i^-(t)D_{1,i}(t)\widetilde{B}_i^+(t)]$, with the four factors the corresponding free/reference values. Thus it replaces both inverses and both legs.

The selected row transfer yields the following prices, with one constant $C_0 \geq 0$ independent of i:

$$\mathsf{P}_{T_i}(K_{0,i}; C_0 T_i^{-2}), \quad \mathsf{P}_{T_i}(K_{1,i}; C_0 T_i^{-2}), \tag{642}$$

$$\mathsf{P}_{T_i}(J_i^{\mathrm{ref}}; C_0 \mu_i^2 T_i^{-2}), \qquad \mathsf{P}_1(J_i; C_0), \tag{643}$$

$$\mathsf{P}_{T_i}(B_i^{\pm}; C_0 T_i), \tag{644}$$

$$\begin{aligned}&\mathsf{P}_{T_i}(K_{0,i} - D_{0,i}; C_0\,\delta_i T_i^{-2}), \quad \mathsf{P}_{T_i}(D_{0,i}; C_0 T_i^{-2}),\\ &\mathsf{P}_{T_i}(D_{1,i}; C_0 T_i^{-2}), \quad \mathsf{P}_{T_i}(K_{1,i} - D_{1,i}; C_0\,\delta_i T_i^{-2}),\\ &\mathsf{P}_{T_i}(B_i^{\pm} - \widetilde{B}_i^{\pm}; C_0\,\delta_i T_i),\end{aligned} \tag{645}$$

where $C_0 \geq 0$ and $\delta_i = T_i + \mu_i^2/T_i^2$. The tilded factors are the reference/halo factors just described. Also $\|K_{0,i}\|_{\mathrm{op}} \leq C_0 T_i^{-2}$. The ordinary face prices (F6a)–(F6c) apply to the all-centered population. The extra differences (F6d) are required only on the shell $\mu_i \leq T_i = \|x_i\|_2$; at the center $x_i = 0$ the unregulated halo inverse need not exist. The derivation from this same selected row and the reference field is given below; in particular, the free inverses in (F6d) are used only on the shell.

Proposition 129.1 (Face estimates)**.** *Under (F4)–(F6c), there are constants $A, B, C \geq 0$, independent of i, such that*

$$|U_i''(0)| \leq \frac{A}{T_i^2} + B\frac{\mu_i^2}{T_i^6}, \qquad |L_i''(0)| \leq \frac{C}{T_i^4}, \tag{F7}$$

and, whenever the additional shell prices (F6d) hold,

$$|L_i''(0) - H_i''(0)| \leq C\frac{\delta_i}{T_i^4}. \tag{F8}$$

Proof. For the reference part of the upper face, (F6a)–(F6b) and the product rule give price $C\mu_i^2/T_i^4$ at scale T_i. By (F3), its second derivative is at most $C\mu_i^2/T_i^6$. For the interacting part, $S_i(0)^{-1}$ is constant in t and has norm at most C_0/T_i^2; combine this with the unit-scale price of J_i and apply the trace bound at unit scale. This gives A/T_i^2. Adding proves the first inequality in (F7).

The lower-face package supplies price C/T_i^2 at scale T_i for the four-factor product in (F5): each inverse has price proportional to T_i^{-2} and each displaced leg has price proportional to T_i. Applying (F3) yields C/T_i^4, proving the second inequality.

For the difference estimate, assume in addition (F6d), as is valid on the shell. Since $T_i \leq r_*$ and $\mu_i \leq T_i$, we have $\delta_i \leq r_* + 1$. The sum rule for prices and (F6c)–(F6d) therefore give each tilded leg the ordinary price $\mathsf{P}_{T_i}(\widetilde{B}_i^{\pm}; C_1 T_i)$, with C_1 depending only on C_0 and r_*. Use the exact ordered telescoping identity

$$\begin{aligned}A_1A_2A_3A_4 - B_1B_2B_3B_4 = {}&(A_1 - B_1)A_2A_3A_4 + B_1(A_2 - B_2)A_3A_4\\ &+ B_1B_2(A_3 - B_3)A_4 + B_1B_2B_3(A_4 - B_4).\end{aligned} \tag{F9}$$

Both inverse differences, at the unshifted and shifted legs, have price $C\delta_i/T_i^2$ by (F6d); the leg differences have price $C\delta_i T_i$ by (F6d). The remaining inverse factors cost C/T_i^2 and the remaining legs cost CT_i. In each of the four summands the product rule therefore gives price at most $C\delta_i/T_i^2$ at scale T_i. Apply (F3) to the trace of their sum. Its second derivative is bounded by $C\delta_i/T_i^4$, as claimed. □

129.3 The Hstar populations and the two scale choices

Let G be simple and fix the reference family, positive regulator fee, grey bound, and local row data. An Orion datum supplies a centered track x, regulator μ, color c, bed, clip and a local family l; the row condition is imposed on the combined family $l + V$. An Hstar datum is such an Orion datum together with the condition that x lies in the closed centered box. The selected population uses the axial mark $(1, c)$ and the unit axial boost e_0.

For each selected row, the path bounds give caps for the tab, the one-leg insertions, and the two-leg insertion, together with the two soft contacts. The regulated reference path supplies its own bounds. Combining the paths gives the total stationary denominator and its scale-T^2 lower bound. The derivation of (F6) below shows explicitly how these data yield the required prices. In particular, no scale-weighted estimate is inferred from analyticity alone: the rates come from the path budget, its one-leg weight T^{-1}, and its two-leg weight T^{-2}; the extra shell estimates use subtraction identities.

For a centered-box track, each coordinate has modulus at most π, so $\|x\|_2 \le \sqrt{4\pi^2} = 2\pi$; the regulator satisfies $0 < \mu \le 1$. The all-centered scale is

$$T = \|x\|_2 + \mu, \qquad 0 < T \le 1 + 2\pi =: r_*. \tag{F10}$$

Thus (F4) holds and (F7)'s upper estimate gives the first two-wave bound after combining it with the lower-face estimate. On the shell $\mu \le \|x\|_2$, reset the scale to $T = \|x\|_2$. Then $0 < \mu \le T \le r_*$, and the same package gives both the upper estimate and (F8). Use the low-face identity of Proposition 130.1 to replace the halo by the continuum pole. Since $T \le r_*$,

$$T^{-2} \le r_* T^{-3}, \qquad T^{-2} \le r_*^2 T^{-4}, \qquad \frac{\mu^2}{T^6} \le \frac{1}{T^4}, \qquad \frac{\delta}{T^4} = \frac{1}{T^3} + \frac{\mu^2}{T^6}.$$

Indeed, the response is half the difference of interacting and reference upper-minus-lower traces. The upper estimate and lower-face estimate in (F7), with $\mu \le T$, therefore give $|\kappa(x)| \le aT^{-4}$. For the shell discrepancy, the exact algebra is

$$\kappa(x) - \Pi_G(x) = \frac{1}{2}\left(\frac{1}{2}U_I'' - U_R'' + (L_R'' - H_R'') - \frac{1}{2}(L_I'' - H_I'')\right), \tag{F12}$$

where the reference/interacting halo combination is the pole identity in Proposition 130.1. The two upper terms are bounded by $C/T^2 + C\mu^2/T^6$, and the lower remainders by $C\delta/T^4$; the displayed scale inequalities give the two powers in (F11). Enlarge constants once to cover the bounded part of the centered box outside the unit ball. We obtain constants a, b independent of x and μ such that

$$|\kappa(x)| \le a(\|x\|_2 + \mu)^{-4}, \qquad |\kappa(x) - \Pi_G(x)| \le b\|x\|_2^{-3} + b\mu^2\|x\|_2^{-6} \quad (\mu \le \|x\|_2 \le 1). \tag{F11}$$

These are the uniform local estimates consumed in Section 130.

129.4 The local row bridge to the interacting family

Here are the exact row identities behind the local contact conditions. For a selected bed B_i, write $M_i(p) = B_i.\mathrm{metric}(p)$ and let $M_*(p)$ be the constant leader metric. Let $D(p)$ be the first edge symbol, with components $q_\nu(p)$, and let $A_0(p)$ be the zero-form symbol. The color-diagonal projection of the direction matrix is denoted by T. Put

$$G_i(p) = \text{the two-point row of the interacting film}, \qquad J_{i,b}(p,k) = \text{its marked two-leg row}.$$

For the selected construction, the row package has a common analytic strip and fee bound; it satisfies the denominator matching identity

$$(\text{interacting film} + \text{regulator film})_{\mathrm{tab}}(p) = \text{the bed denominator}(p), \tag{F13}$$

the all-grade floor (in particular, the grade-one normalization $M_i(0) = M_*(0)$), the local action soft order-three condition, and the vacuum matching condition (the color-diagonal vacuum row equals the finite Wilson row). These properties all concern the same selected bed and the same combined local row. The regulator satisfies $0 < \mu \le 1$ and its tab is the scalar multiple $s_\mu(p)I$ on the strip. They are properties of the selected row package, rather than consequences of analyticity alone.

Let $\omega(p) = \sum_\nu q_\nu(-p)q_\nu(p)$. The exact row and free identities use the color-diagonal extension $\mathsf{T}(A)_{(i,c),(j,d)} = \mathbf{1}_{c=d}A_{ij}$ of a direction matrix; this extension preserves matrix sums and products. They are

$$G_i(p) - |p|^2 I = \mathsf{T}\big[P_1(-p)^T(M_i(p) - M_*(p))P_1(p)\big] + (\omega(p) - |p|^2)I, \tag{F14}$$

$$J_{i,b}(p,k) - v_b(p,k) = J^{\mathrm{loc}}_{i,b}(p,k) + \big(J^{\mathrm{vac}}_{i,b}(p,k) - v_b(p,k)\big), \tag{F15}$$

and

$$\mathsf{T}\big[P_1(-p)^T M_*(p)P_1(p) + P_0(p)P_0(-p)^T\big] = \omega(p)I. \tag{F16}$$

Here v_b is the linear vector insertion determined by the first radial coefficient of the vacuum marked row.

Derivation of (F14)–(F16). Write $P_0(p)$ and $P_1(p)$ for the zero-form and first-edge symbols, so $A_0 = P_0$ and $D = P_1$. The selected denominator matching equates the tab of the interacting film plus its regulator film to the prescribed bed denominator. The two sides expand as

$$\begin{aligned}(\text{interacting} + \text{regulator})_{\mathrm{tab}}(p) &= G_i(p) + s_\mu(p)I,\\ (\text{bed denominator})(p) &= \mathsf{T}\big[P_1(-p)^T M_i(p)P_1(p) + P_0(p)P_0(-p)^T\big] + s_\mu(p)I.\end{aligned}$$

The regulator tab is scalar on the slot space, so equality cancels that term entrywise and gives, on real momenta,

$$G_i(p) = \mathsf{T}\big[P_1(-p)^T M_i(p)P_1(p) + P_0(p)P_0(-p)^T\big].$$

The first summand is the metric cross product; the second is the zero-form product. The cancellation is exact, so it introduces no regulator remainder. For the leader bed, the degree-zero exterior identity gives

$$\mathsf{T}\big[P_1(-p)^T M_*(p)P_1(p) + P_0(p)P_0(-p)^T\big] = \omega(p)I.$$

Indeed, expand the first product by direction. The longitudinal rank-one part of the zero-form product cancels the complementary part of the first-edge product, leaving $\sum_\nu q_\nu(-p)q_\nu(p)$ times the identity. The exact degree-zero exterior identity is $P_0(p)P_0(-p)^T + P_1(-p)^T M_* P_1(p) = \omega(p)I$, since $\omega(p) = \sum_\nu q_\nu(-p)q_\nu(p)$ by definition. Since M_* is the constant identity metric, this proves

(F16). Subtracting the leader identity from the row-to-film identity proves (F14), with the lattice-dispersion term $\omega(p) - |p|^2$ retained explicitly.

For the marked row, the local film is the sum of the selected local action and the fixed free quadratic film. Linearity of the marked coefficient gives (F15). The soft-3 hypothesis assigns radial order two to the one-mark layer; composing that layer with the linear two-momentum map gives the required second-order vanishing. The vacuum term is identified by grey matching with the Wilson three-leg coefficient. Write $\tau(u)_a = iu_a$ and denote the color structure coefficient by c_{bxy}. Differentiating the three plaquette placements gives its first radial coefficient

$$[-(2\tau(k)_a + \tau(p)_a)\delta_{ij} + (\tau(k)_j + 2\tau(p)_j)\delta_{ia} + (\tau(k)_i - \tau(p)_i)\delta_{ja}]c_{bxy}.$$

Each placement contributes one of the terms in the defining plaquette coefficient; summing the two orientations and using $\tau(k+p) = \tau(k) + \tau(p)$ gives this formula. The fixed free-film marked coefficient has first radial coefficient

$$[-\tau(k)_j\delta_{ia} - \tau(k+p)_i\delta_{ja}]c_{bxy}.$$

Adding these expressions and simplifying gives

$$v_{b,ij}(p,k) = [-(2\tau(k)_a + \tau(p)_a)\delta_{ij} + 2\tau(p)_j\delta_{ia} - 2\tau(p)_i\delta_{ja}]c_{bxy}.$$

This is homogeneous of degree one. Thus the vacuum marked row minus v_b has zero value and zero first radial coefficient; analyticity implies that it has radial order at least two. With the soft local term, (F15) has the required quadratic contact. The common strip bounds then give uniform compact derivative bounds by Cauchy's formula. □

Lemma 129.2 (Row data give the priced insertion family)**.** *For a bounded family of selected rows with the common strip/fee bounds, denominator matching, all-grade floor, soft order-three local action and grey vacuum matching just described, the interacting film has uniform compact derivative prices through order three. Its two-point row has a cubic radial remainder after subtraction of $|p|^2 I$, and its marked row has a quadratic radial remainder after subtraction of v_b. Consequently it supplies the compact caps and the two contact orders used in the ordinary (non-halo) part of (F6b)–(F6c).*

Proof. For each row layer and each fixed pair of matrix entries, the common strip fee and Cauchy's estimate bound derivatives of every fixed order on a compact real ball. The tab, spot and arch are fixed finite linear combinations of these entries after fixed linear maps of momentum; the chain rule therefore gives uniform compact derivative caps through the orders used here. A finite sum of these caps bounds the matrix operator norm, since the matrix dimension is fixed.

For the two-point contact, (F14) and the grade-one floor give $M_i(0) - M_*(0) = 0$. Analyticity makes this metric difference $O(|p|)$. Each edge symbol vanishes linearly, $q_\nu(p) = ip_\nu + O(|p|^2)$, so the metric term in (F14) is $O(|p|^3)$. Expanding each product $q_\nu(-p)q_\nu(p)$ shows $\omega(p) - |p|^2 = O(|p|^3)$ as well.

For the marked row, the local soft condition contributes $O((|p|+|k|)^2)$ and the vacuum-minus-v_b term has the same order by the preceding first-coefficient calculation. Thus (F15) has quadratic contact. To convert these radial orders into derivative prices, note that if an analytic scalar entry g has radial order at least m, then each homogeneous Taylor polynomial of degree $j < m$ vanishes on every real direction and hence vanishes identically. Taylor's formula with integral remainder applied to $D^n g$ on the segment from 0 to the base point a gives

$$\|D^n g(a)\| \le K\|a\|^{m-n}/(m-n)!, \qquad n < m,$$

when the mth derivative is bounded by K on that segment. For $\|a\| \le T$, this gives $T^n\|D^n g(a)\| \le KT^m$. At orders above the contact order use the compact derivative cap and the uniform upper bound $T \le r_*$. This proves the local compact and contact prices. The trajectory-price argument below derives the total-field prices in (F6a) and the shell-separation prices in (F6d) from these contacts and the reference path budgets. □

Hstar to the local row package. For every Hstar datum the associated Orion datum contains one selected bed, clip, regulator, and local action. Its denominator matching field is the identity (F13), its fee field supplies the uniform analytic strip, and its all-grade floor supplies the grade-one normalization. Its local soft field is soft order three. Applying this row argument to the single combined family $l + V$ gives the interacting row used by the annulus. The common free reference is the quadratic film. If the external vacuum V is grey and has the required fees, the finite-volume Wilson matching above identifies its marked vertex and hence its linear term. These statements establish the row contacts from the selected Hstar data. The scalar regulator term and its positive floor are then included in the total denominator; the next price calculation uses the free comparison only on the shell, where its inverse exists.

Assume $\epsilon > 0$, as in the selected floor package. The denominator lower bound is also tied to this same bed. At a real momentum p, write

$$D_i^{\mathrm{bed}}(p) = \mathsf{T}\big[P_1(-p)^T M_i(p) P_1(p) + P_0(p) P_0(-p)^T\big] + s_\mu(p) I$$

for the denominator matrix in (F13). If $\mathrm{peb}(p,\mu)$ denotes the positive quadratic weight, the cold geometry gives

$$T^2 \le 64\,\mathrm{peb}(p,\mu).$$

The bed-floor estimate for this matrix gives

$$\frac{\min(\epsilon,1)}{2^{10}}\,\mathrm{peb}(p,\mu)\|z\|^2 \le \mathrm{Re}\langle z, \mathrm{dais}(p) z\rangle.$$

Using the denominator matching to identify this matrix with the selected total denominator, we obtain

$$cT^2\|z\|^2 \le \mathrm{Re}\langle z, S_i(0) z\rangle, \qquad c = \frac{\min(\epsilon,1)}{2^{16}}.$$

The same cold lower bound and the free reference floor give the ghost denominator bound with $c = 2^{-16}$. For either square matrix S, the coercive estimate implies $cT^2\|z\|^2 \le \mathrm{Re}\langle z, Sz\rangle \le \|z\|\,\|Sz\|$. Hence $\|Sz\| \ge cT^2\|z\|$; in finite dimension this makes S invertible with $\|S^{-1}\|_{\mathrm{op}} \le (cT^2)^{-1}$.

Trajectory prices. We derive (F6) from the row contacts and the regulated reference field. At scale T_i, a path estimate with tab budget L_i consists of three component bounds. The tab derivatives satisfy $T_i^j\|M_i^{(j)}(0)\|_{\mathrm{op}} \le CL_i$ for $j = 0, 1, 2$; the one-leg slot has budget L_i/T_i, and the two-leg arch has budget L_i/T_i^2. These are separate components of the path estimate; the spot and arch bounds do not follow from the tab bound alone. For the interacting row, the compact derivative caps come from the common strip and selected fees. Along $x_i + te_i$, each derivative is a momentum derivative contracted with e_i, whose norm is at most one. If an analytic entry has radial order m, all Taylor tensors of degree below m vanish: their diagonal polynomial vanishes in every real direction, hence the polynomial itself is zero. Taylor's integral remainder applied to its jth derivative therefore bounds it by $C\|x_i\|^{m-j}$ for $j < m$; for $j = m$ use the compact derivative cap. Since $\|x_i\| \le T_i \le r_*$, this gives the path budget CT_i^m for $j \le m$. Thus the trajectory transfer with tab budget L_i gives

$$\mathsf{P}_{T_i}(\mathrm{tab}; CL_i), \qquad \mathsf{P}_{T_i}(\mathrm{spot}; CL_i/T_i), \qquad \mathsf{P}_{T_i}(\mathrm{arch}; CL_i/T_i^2). \tag{F17}$$

Thus the interacting transfer uses $L_i = T_i^2$, while the regulated reference transfer uses $L_i = \mu_i^2$.

For the interacting row, (F14) says the tab is $|p|^2 I + O(|p|^3)$; therefore its path has tab budget CT_i^2. For a marked leg, (F15) says it is the linear vector term plus a quadratic remainder. Along the path, the linear term has size $O(T_i)$, first derivative $O(1)$, and zero second derivative. The quadratic remainder has T_i^j-weighted jth derivative $O(T_i^2)$ for $j = 0, 1, 2$; since $T_i \le r_*$, this is at most $O(T_i)$. Thus the leg has budget CT_i. The uniform arch cap gives budget C at unit scale. The regulated reference field has the following three trajectory budgets, which we derive from its actual construction below:

$$C\mu_i^2, \qquad C\frac{\mu_i^2}{T_i}, \qquad C\frac{\mu_i^2}{T_i^2}.$$

These weights follow from its construction. Begin with the massive free film, the free Wilson film plus $\mu_i^2 I$. The free Wilson path has tab budget T_i^2: its tab has a quadratic zero, while its compact derivatives are uniformly bounded. Its marked leg has a linear first term and a quadratic remainder, hence budget T_i; its two-leg arch has a uniform unit-scale cap. The added constant mass has tab budget T_i^2 since $\mu_i \le T_i$, and has zero spot and arch. The massive film therefore has the three-component path budgets $T_i^2, T_i, 1$. Its scalar tab is the free dispersion plus μ_i^2; the cold geometric inequality bounds it below by $T_i^2/64$. Thus its inverse has norm at most $64T_i^{-2}$ and its inverse path has tab budget T_i^{-2}. If $C = A^{-1}$, B_i is the original one-leg slot, and D_{ij} is the original two-leg slot, the inverse film's insertion formulas, with $\widehat{B}_i$ and $\widehat{D}_{ij}$ denoting the inverse film's slots, are

$$\widehat{B}_i(p,q) = -C(q+p)B_i(p,q)C(q),$$

$$\widehat{D}_{ij}(p,q,r) = -\big[\widehat{B}_j(q,r+p)B_i(p,r) + \widehat{B}_i(p,r+q)B_j(q,r) + C(r+p+q)D_{ij}(p,q,r)\big]C(r).$$

The one-leg term has size $T_i^{-2}\, T_i\, T_i^{-2} = T_i^{-3}$. Each of the three two-leg summands has size T_i^{-4}: the first two use $T_i^{-3}\, T_i\, T_i^{-2}$, and the last uses $T_i^{-2}\, 1\, T_i^{-2}$. Thus the inverse film has tab, one-leg, and two-leg budgets T_i^{-2}, T_i^{-3}, and T_i^{-4}. The reference construction multiplies this inverse film by μ_i^2, giving budgets μ_i^2/T_i^2, μ_i^2/T_i^3, and μ_i^2/T_i^4. The same prices hold for derivatives of orders $j = 0, 1, 2$ along every front or rear path: apply the product rule to each displayed slot formula and use the corresponding Week estimate for each differentiated factor. The shifted momenta have path velocities among $e_i, -e_i$ and their sums, whose norms are bounded by 2; this changes only fixed constants. Each derivative of total order j costs T_i^{-j}, which is canceled by the T_i^j factor in the path price. Since $\mu_i^2/T_i^2 \le 1$, the inverse-reference film has the Week estimate with tab budget 1, and consequently spot and arch budgets T_i^{-1} and T_i^{-2}.

The reference insertion is formed from nine copies of this inverse-reference film and then weighted by μ_i^2. For two films with tab/spot/arch budgets $(L, L/T, L/T^2)$ and $(M, M/T, M/T^2)$, the product has tab budget LM. Its spot formula has two summands, each bounded by LM/T; its arch formula has four summands, each bounded by LM/T^2. Iterating through the nine factors changes only fixed numerical constants: the tab, spot, and arch rates remain 1, T_i^{-1}, and T_i^{-2}. The final weight gives μ_i^2, μ_i^2/T_i, and μ_i^2/T_i^2. Each directional sandwich in the free reference row has constant unit-budget factors on both sides, so it preserves these prices. Summing the four directions changes only the common constant. This proves the displayed reference budgets from the field construction.

Because $\mu_i \le T_i$, adding the local and reference tab budgets gives $C(T_i^2 + \mu_i^2) \le 2CT_i^2$. The total denominator has the same T_i^2 tab budget on both the unshifted and boosted path. The bed floor proved above gives its inverse norm at $t = 0$ at most C/T_i^2. Differentiating the inverse with (F2) gives both inverse prices in (F6a). The reference arch budget is exactly $C\mu_i^2/T_i^2$, and the interacting arch has unit-scale budget C, proving (F6b). The two total marked legs have budget at most $C(T_i + \mu_i^2/T_i) \le 2CT_i$, which proves (F6c). All constants here are uniform over the selected

Hstar family because the fees, compact caps, regulator bound, floor constant and centered-box radius are uniform.

On the shell, set $T_i = \|x_i\|_2 > 0$. The unregulated reference denominator along the path is

$$R_i(t) = \|x_i + te_i\|_2^2 I, \qquad R_i(0) = T_i^2 I.$$

Its path budget is CT_i^2 and its inverse has price C/T_i^2 by (F2). Let $S_i(t)$ be the total regulated denominator. Its difference from R_i splits into the local tab minus the free quadratic tab, which has budget CT_i^3 by (F14), and the regulator tab, which has budget $C\mu_i^2$. Thus

$$\mathsf{P}_{T_i}(S_i - R_i; C(T_i^3 + \mu_i^2)) = \mathsf{P}_{T_i}(S_i - R_i; C\delta_i T_i^2).$$

Both inverse paths have price C/T_i^2; the identity

$$S_i^{-1} - R_i^{-1} = -S_i^{-1}(S_i - R_i)R_i^{-1}$$

and the product rule therefore give the two inverse differences in (F6d) with price $C\delta_i/T_i^2$.

For the marked legs, subtract the unregulated free leg. The local marked row and the free Wilson row have the same linear vector coefficient by the explicit coefficient calculation above. Their difference has radial order two. Hence the local marked row minus the halo leg has price CT_i^2 along the shell path. The regulated reference leg has the one-leg budget $C\mu_i^2/T_i$ from its path spot component. Since

$$T_i^2 \le \delta_i T_i, \qquad \frac{\mu_i^2}{T_i} \le \delta_i T_i,$$

their sum has price $C\delta_i T_i$, proving the leg part of (F6d). This derivation requires $T_i > 0$ and is used only on the shell. At $x_i = 0$ the unregulated reference path is singular, so no halo inverse is asserted there.

129.5 Uniform path bounds from the selected records

For completeness, we describe how the path hypotheses used above are obtained uniformly for the selected family. Each selected local action has the same fee bound and positive strip width, and the external vacuum has its own common fee bound on that strip. Their sum therefore has a uniform fee bound. Cauchy's integral formula on a smaller common strip bounds every fixed momentum derivative of each matrix entry uniformly; for the tab use derivatives through order three, and for the spot and arch through order two. The momentum maps entering the three components are fixed linear maps, so the chain rule multiplies these estimates only by their fixed operator norms. The matrix dimension is fixed, hence the entrywise bounds imply operator-norm bounds. The additional fixed free film is a finite polynomial and has uniform derivative bounds on the centered box. Adding it gives the compact path caps used in (F17).

The selected action's soft hypothesis is componentwise: its unmarked layer vanishes to radial order three, its one-mark layer to order two, and its two-mark layer to order one. The local marked insertion uses the one-mark layer composed with a linear two-momentum map, so it has the second-order vanishing required for the local contact. The grey vacuum's marked row has the same linear term as the free Wilson row: the three plaquette placements give the displayed coefficient v_b, and the fixed free-film term supplies the complementary part. Their difference has order two. Thus the selected combined row has exactly the tab and marked-leg contacts used in (F14)–(F15), with uniform compact caps.

Finally, the reference field is constructed from the same regulator μ. Its trajectory tab budget is μ^2, and its two other components carry the factors T^{-1} and T^{-2}. The local field has tab budget

T^2. Since $\mu \le T$, their sum has tab budget at most $2T^2$. The selected House floor and the cold geometric bound give the total tab a coercive lower bound cT^2; the elementary inverse calculation preceding (F17) then yields the inverse estimates for both the unshifted and boosted paths. On the shell, the free comparison is positive because $T = \|x\|_2 > 0$, and the same uniform strip/cap constants apply after resetting the scale. This verifies that the family-level path bounds used in the face proof are common to all selected rows and all regulators in the stated range.

130 The annular pole and the two-wave estimate

The selected one-step construction needs two facts about one finite-volume response. Its logarithmic coefficient is an integral of the response over four-dimensional momentum space, and that response is close to the continuum quadratic pole on an annulus. This section gives the exact polar calculation and expands the finite-volume two-channel estimate that supplies the two radial bounds. The remaining local chart and weighted-face estimates are stated at their precise input interface and cross-referenced to the units that produce them.

130.1 From the finite response to a momentum integral

Let E_4 be the fundamental momentum cube in Euclidean coordinates, with normalized measure $c_{\mathrm{wt}}\,dx$ and $c_{\mathrm{wt}} = (2\pi)^{-4}$. It has finite volume and contains the unit ball. For a quadratic response picture F and color coordinate a, let $\mathcal{K}_a(x)$ be one half of the second derivative in the distinguished response direction, with the other momentum equal to x, and set $g_a(x) = \mathrm{Re}\,\mathcal{K}_a(x)$. The coefficient functional $\alpha_a(F)$ is the second Taylor coefficient of the analytic stock picture associated with F. The Fourier stock formula and differentiation under the integral give

$$\alpha_a(F) = c_{\mathrm{wt}} \int_{E_4} g_a(x)\,dx. \tag{AT1}$$

The common analytic bound on the differentiated symbol supplies the integrable majorant. Equation (AT1) is an identity for the analytically assembled response picture. The equality to a finite sampled pole-test coefficient is a separate recording statement and is not asserted by (AT1).

For $x \in \mathbb{R}^4 \setminus \{0\}$ let $r = |x|$ and define the gauge pole

$$\Pi_G(x) = 4C_G \left(\frac{x_1^2}{r^6} - 4\frac{x_1^2 x_0^2}{r^8} - \frac{1}{r^4} \right), \tag{AT2}$$

where C_G is the quadratic Casimir in the chosen representation. The spherical area of S^3 is $2\pi^2$. Rotational invariance gives

$$\int_{S^3} \theta_1^2\,d\theta = \frac{2\pi^2}{4}, \qquad \int_{S^3} \theta_1^2\theta_0^2\,d\theta = \frac{2\pi^2}{24}. \tag{AT3}$$

For the second identity, set $A = \int \theta_0^4$ and $B = \int \theta_0^2\theta_1^2$. Coordinate permutations make these independent of the distinct indices. From $\sum_i \theta_i^2 = 1$, $4A + 12B = 2\pi^2$. Rotating the 0-axis to $(e_0 + e_1)/\sqrt{2}$ gives $A = (2A + 6B)/4$, hence $A = 3B$ and $B = 2\pi^2/24$. The first identity follows by integrating $\sum_i \theta_i^2 = 1$.

Consequently the angular mean of the parenthesis in (AT2) is

$$r^{-4}\left(\frac{1}{4} - \frac{4}{24} - 1 \right) = -\frac{11}{12} r^{-4}.$$

Since the Euclidean polar Jacobian is $r^3 dr\, d\theta$,

$$\int_{\mu<|x|\le 1} |x|^{-4}\, dx = 2\pi^2 \int_\mu^1 \frac{dr}{r} = 2\pi^2 \log \mu^{-1}.$$

Using $c_{\mathrm{wt}} = (2\pi)^{-4}$, we obtain the exact pole law

$$c_{\mathrm{wt}} \int_{\mu<|x|\le 1} \Pi_G(x)\, dx = -\frac{11 C_G}{24\pi^2} \log \mu^{-1} = -\lambda_G \log \mu^{-1}, \qquad \lambda_G = \frac{11 C_G}{24\pi^2}. \tag{AT4}$$

This fixes the sign and normalization of the logarithm; in particular the pole is not defined only up to an angular constant.

130.2 The finite-volume two-channel calculation

We now state the two scale receipts used by the finite-volume construction. Let x be a centered momentum, $r = |x|$, and $0 < \mu \le 1$. Write $\mathcal{K}(x) \in \mathbb{C}$ for the normalized response coefficient and $g(x) = \mathrm{Re}\,\mathcal{K}(x)$. The six complex second-derivative traces D, Y, c, d, c_0, d_0 are defined so that D is the interacting upper trace, Y the reference upper trace, c the reference lower trace, d the interacting lower trace, and c_0, d_0 their free halo traces. The exact identities are

$$\mathcal{K} = \frac{1}{2}\left(\frac{1}{2}(D-d) - (Y-c)\right), \qquad \Pi_G = \frac{1}{2}\left(c_0 - \frac{1}{2}d_0\right), \tag{AT5a}$$

$$\mathcal{K} - \Pi_G = \frac{1}{2}\left(\frac{1}{2}D - Y\right) + \frac{1}{2}\left((c-c_0) - \frac{1}{2}(d-d_0)\right). \tag{AT5b}$$

The first identity is the second derivative of the split-field response after subtracting its reference film. The second is the free halo identity, with marked direction and color fixed. The third is their algebraic difference; the real-valued pole Π_G is embedded in $\mathbb{C}$ in these formulas. All are exact at each finite volume.

For clarity, the second identity follows directly from the two free propagator numerators. In the selected unit color direction e_b, use the normalization $\mathrm{tr}(\mathrm{ad}(e_b)^2) = -C_G$. Put $\rho = |x|^2$, $c_{\mathrm{col}} = -C_G$, and $L = -4x_1^2$. Along the marked displacement te_0, the common denominator is $\Delta(t) = \rho + 2x_0 t + t^2$. The ghost numerator is the scalar orbital factor L times the color trace, giving Lc_{col}/ρ. For the vector numerator, write $p = -2ix_1$ and let $A(t)$ have entries $A_{01}(t) = -2it$, $A_{10}(t) = 2it$, and all other entries zero. Then $p^2 = L$ and direct multiplication gives

$$\mathrm{tr}\big[(pI + A(-t))(pI + A(t))\big] = 4p^2 - 8t^2 = 4L - 8t^2.$$

Multiplying by the same color trace and scalar denominator factor gives the vector numerator $\rho^{-1}(4L - 8t^2)c_{\mathrm{col}}$. Differentiating the quotients twice at $t = 0$ gives

$$\left(\frac{Lc_{\mathrm{col}}/\rho}{\Delta}\right)''_{t=0} = -\frac{2Lc_{\mathrm{col}}}{\rho^3} + \frac{8Lc_{\mathrm{col}}x_0^2}{\rho^4}, \qquad \left(\frac{\rho^{-1}(4L-8t^2)c_{\mathrm{col}}}{\Delta}\right)''_{t=0} = -\frac{16c_{\mathrm{col}}}{\rho^2} - \frac{8Lc_{\mathrm{col}}}{\rho^3} + \frac{32Lc_{\mathrm{col}}x_0^2}{\rho^4}.$$

Thus the orbital trace minus one half of the vector trace is

$$\frac{8c_{\mathrm{col}}}{\rho^2} + \frac{2Lc_{\mathrm{col}}}{\rho^3} - \frac{8Lc_{\mathrm{col}}x_0^2}{\rho^4} = 8C_G\left(\frac{x_1^2}{r^6} - 4\frac{x_1^2x_0^2}{r^8} - \frac{1}{r^4}\right) = 2\Pi_G(x),$$

which proves the halo identity in (AT5a). The first identity is the definition of the marked split-field second response and its reference subtraction; (AT5b) is then obtained by subtracting the just-computed free identity and collecting terms.

The first family of estimates is the all-centered-record receipt, with $T_1 = r + \mu$ and $R_0 = 1 + 2\pi$:

$$0 \le r, \quad T_1 = r + \mu \le R_0, \qquad |\mathcal{K}| \le A_1 T_1^{-4}. \tag{AT6}$$

The second is the shell receipt, available only on $\mu \le r \le 1$ and using $T_2 = r$:

$$|D| \le u r^{-2} + v\mu^2 r^{-6}, \qquad |Y| \le u_{\mathrm{ref}} r^{-2} + v_{\mathrm{ref}} \mu^2 r^{-6}, \tag{AT7}$$

$$|c - c_0| \le A_0 (r + \mu^2 r^{-2}) r^{-4}, \qquad |d - d_0| \le B_0 (r + \mu^2 r^{-2}) r^{-4}. \tag{AT8}$$

All constants $A_1, u, v, u_{\mathrm{ref}}, v_{\mathrm{ref}}, A_0, B_0$ are nonnegative and uniform in the finite volume and selected record. These two receipts are kept separate: the all-centered estimate uses scale $r + \mu$, while the shell comparison uses scale r. Their source producer is the common finite family of local stationary records; its exact proof obligations are listed in the correspondence file.

Proposition 130.1 (Two wave bounds from the finite response)**.** *Suppose (AT5a)–(AT8) hold uniformly. Then*

$$|g(x)| \le \frac{A_1}{(r+\mu)^4}, \qquad |g(x) - \Pi_G(x)| \le \frac{A_2}{r^3} + \frac{A_2 \mu^2}{r^6} \quad (\mu \le r \le 1), \tag{AT9}$$

where

$$A_2 = \max\{u + u_{\mathrm{ref}} + A_0 + B_0, v + v_{\mathrm{ref}} + A_0 + B_0\}.$$

Proof. The first estimate follows from (AT6) and $|\operatorname{Re} z| \le |z|$. On the annulus, use (AT5b) and (AT7)–(AT8):

$$|\mathcal{K} - \Pi_G| \le (u + u_{\mathrm{ref}}) r^{-2} + (v + v_{\mathrm{ref}}) \mu^2 r^{-6} + (A_0 + B_0)(r^{-3} + \mu^2 r^{-6}). \tag{AT10}$$

Here $\frac{1}{2}$ factors are bounded by one, and $(r + \mu^2 r^{-2}) r^{-4} = r^{-3} + \mu^2 r^{-6}$. Since $r \le 1$, $r^{-2} \le r^{-3}$. The displayed A_2 bounds (AT10) by $A_2(r^{-3} + \mu^2 r^{-6})$. Taking real parts gives $|g - \Pi_G| \le |\mathcal{K} - \Pi_G|$, proving (AT9). □

130.3 The two applications of the finite-volume receipts

The common family of local records is used twice, with different scale parameters. Each record supplies a centered momentum x, a positive regulator $\mu \le 1$, and the same complex response coefficient $\mathcal{K}(x)$. The centered-box condition $|x_j| \le \pi$ gives $|x| \le 2\pi$.

For the all-centered estimate, use scale $T_1 = |x| + \mu$. The whole-frame bound from the two channels is

$$|D - d| \le A_{\mathrm{int}} T_1^{-4}, \qquad |Y - c| \le A_{\mathrm{ref}} T_1^{-4}.$$

The exact first identity in (AT5a) then gives $|\mathcal{K}| \le (A_{\mathrm{int}}/4 + A_{\mathrm{ref}}/2) T_1^{-4}$. This is the first radial estimate; it applies also at $x = 0$.

For the annular comparison, restrict the same records to $\mu \le |x|$ and use the second scale $T_2 = |x|$. The upper channel is split into a principal term and a tadpole term; their charged bounds are $u T_2^{-2}$ and $v\mu^2 T_2^{-6}$. The same split for the reference channel has bounds $u_{\mathrm{ref}} T_2^{-2}$ and $v_{\mathrm{ref}} \mu^2 T_2^{-6}$. The lower-channel remainders after subtracting their respective halos are bounded by $A_0(T_2 + \mu^2/T_2^2) T_2^{-4}$ and $B_0(T_2 + \mu^2/T_2^2) T_2^{-4}$. The free quotient calculation gives $\Pi_G = \frac{1}{2}(c_0 - \frac{1}{2} d_0)$, so these four estimates feed exactly into (AT5b) and yield (AT7)–(AT10).

These scale receipts arise in the source through two separate applications of the finite-family construction: the first applies the frame estimate to every centered record, while the second restricts to the subfamily $\mu \le |x|$ before applying the shell estimate. They use the same response coefficient and common constants. The annulus proof therefore does not interchange a limit with a volume-dependent integral. The equality of this analytic coefficient with the finite sampled pole-test coefficient is a separate recording step; it is not inferred from (AT1).

Source boundary. This unit proves the normalized pole integral and the algebraic conversion from the two explicit finite-volume scale receipts to the radial bounds. The receipts themselves are produced by the local stationary-family proof (‘Rally.register‘ and ‘Rally.shell‘) and remain a source-correspondence obligation until that finite-record construction is expanded in the next unit. In particular, the stationary-chart and row-price units are not claimed here to establish these upper/tadpole and lower/halo estimates. The later ‘HouseWord‘/yield argument must also prove the finite recording error decays after division by the site count, with its strip chosen before the requested blocking side. The remaining path-history, retained-cutoff, and flux-capacity suppliers are tracked in the evidence file.

131 Finite-volume recording and uniform drift

The continuum pole calculation controls a coefficient of the analytically assembled response. The finite-volume argument needs two additional steps: an exact determinant identity at each torus size, and a quantitative bound on the difference between a torus mean and its zero Fourier mode. This section proves that bound, explains the determinant calculation and positive-matrix domain, and gives the exact scale order in the final annulus output. The local stationary estimates used to feed this construction remain those listed in the preceding section and its evidence record.

Write $\mathcal{H}_\sigma(A)$ for the class of periodic functions that are holomorphic when each imaginary momentum coordinate has modulus less than σ and have supremum norm at most A there. The coefficient of a translation-invariant two-leg picture is its mixed second derivative; its finite-volume realization at a conserving pair is ℓ^4 times that coefficient. Write $\mathrm{Err}_\ell(\mathcal{Y}, \mathcal{P}; \delta)$ when every mixed two-leg coefficient of the finite test $\mathcal{P}$ differs from this finite-volume realization for $\mathcal{Y}$ by at most $\ell^4\delta$.

131.1 A torus mean differs from the zero mode by an exponential tail

Let $\ell \geq 1$ be an integer and let f be a periodic function of four real momenta, holomorphic and bounded by A on the complex strip of width $\sigma > 0$. Write its absolutely convergent Fourier series as

$$f(\theta) = \sum_{n \in \mathbb{Z}^4} c_n e^{in\cdot\theta}, \qquad |n|_1 = \sum_{j=0}^{3} |n_j|.$$

The normalized mean over the side-ℓ grid is

$$\langle f \rangle_\ell = \ell^{-4} \sum_{q \in (\mathbb{Z}/\ell\mathbb{Z})^4} f(2\pi q/\ell).$$

Character orthogonality in each coordinate says

$$\ell^{-4} \sum_{q} e^{2\pi i n\cdot q/\ell} = \mathbf{1}_{n \in \ell\mathbb{Z}^4}.$$

Consequently $\langle f \rangle_\ell - c_0$ is the sum of precisely the nonzero Fourier coefficients whose indices lie in $\ell\mathbb{Z}^4$. Every such index satisfies $|n|_1 \geq \ell$. Cauchy's integral formula, applied after shifting each coordinate contour by $\pm i(\sigma/2)$, and the geometric-series bound in four coordinates give

$$\sum_{n \in \mathbb{Z}^4} e^{(\sigma/4)|n|_1} |c_n| \leq A \left(2 \frac{1 + \sigma/4}{\sigma/4} \right)^4. \tag{RY1}$$

For completeness, the one-coordinate sum used here is bounded by

$$\sum_{k\in\mathbb{Z}} e^{-h|k|} = 1 + 2\sum_{k\ge 1} e^{-hk} \le 2\frac{1+h}{h}, \qquad h = \sigma/4;$$

the last inequality follows from $e^h \ge 1 + h$. The contour shift gives the factor $e^{-(\sigma/2)|n|_1}$ for an individual coefficient, leaving the weight $e^{(\sigma/4)|n|_1}$ in (RY1). Therefore

$$|\langle f\rangle_\ell - c_0| \le \mathfrak{r}(\sigma, A)e^{-\sigma\ell/4}, \qquad \mathfrak{r}(\sigma, A) = A\left(2\frac{1+\sigma/4}{\sigma/4}\right)^4. \tag{RY2}$$

This estimate is for the actual full grid at each ℓ; no subsequence selection is involved.

Lemma 131.1 (Recorded second-order coefficients)**.** *Let $F_{ij}(p, q)$ be two-momentum response entries in $\mathcal{H}_\sigma(A)$ jointly in p, q. Let $\mathcal{P}_\ell$ be a real finite-volume test and assume its exact local coefficient identity: for every pair of colored sites q, q', the mixed second coefficient of $\mathcal{P}_\ell$ is zero when $q+q' \ne 0$, and otherwise is*

$$\ell^4 \langle F_{ij}(\mathrm{mon}(q'), \cdot)\rangle_\ell. \tag{RY3}$$

Define the internal zero mode and limiting picture by

$$\widehat{F}_{ij,0}(p) = \int_{\mathbb{T}^4} F_{ij}(p, q)\, dq, \qquad \mathcal{Y}_{ij}(p) = \tfrac{1}{2}\,\widehat{F}_{ij,0}(p),$$

where Haar measure on $\mathbb{T}^4$ is normalized to one. We use the translation-invariant two-leg convention in which the mixed second derivative is the polarized coefficient (there is no factorial in the definition of that coefficient). Thus the finite-volume realization at a conserving pair is ℓ^4 times this coefficient, namely $\ell^4\, 2\mathcal{Y}_{ij}$. Then $\mathcal{Y}$ is interpreted as the translation-invariant two-leg picture: its mixed coefficient at external momenta p, p' is zero unless $p + p' = 0$, and in the conserving case is $2\mathcal{Y}_{ij}(p') = \widehat{F}_{ij,0}(p')$. $\mathcal{Y} \in \mathcal{H}_{\sigma/2}(A)$, and for every side ℓ,

$$\mathrm{Err}_\ell(\mathcal{Y}, \mathcal{P}_\ell; \mathfrak{r}(\sigma, A)e^{-\sigma\ell/4}). \tag{RY4}$$

Proof. The normalized Haar integral defining $\widehat{F}_{ij,0}$ is holomorphic in p on the width-σ strip: on each compact substrip its integrand and derivatives are uniformly bounded, so differentiation under the integral is valid. Its norm is at most A, hence the half-scaled picture is bounded by A on the width-$\sigma/2$ strip. For each fixed real p, apply (RY2) to the one-variable function $q \mapsto F_{ij}(p, q)$; its Fourier zero mode is exactly $\widehat{F}_{ij,0}(p)$. This gives

$$|\langle F_{ij}(p, \cdot)\rangle_\ell - \widehat{F}_{ij,0}(p)| \le \mathfrak{r}(\sigma, A)e^{-\sigma\ell/4}$$

on the real external momentum torus. If $q + q' \ne 0$, both mixed coefficients in the definition of Err_ℓ vanish by momentum conservation. If $q + q' = 0$, the second Taylor coefficient of $\mathcal{Y} = \frac{1}{2}\widehat{F}_{ij,0}$ has the factor $2! = 2$, so it equals $\widehat{F}_{ij,0}$; subtracting (RY3) and multiplying the mean estimate by the number ℓ^4 of sites gives (RY4). □

For a finite side ℓ, set

$$\mathcal{Y}_{\ell,ij}(p) = \tfrac{1}{2}\,\ell^{-4} \sum_{q\in(\mathbb{Z}/\ell\mathbb{Z})^4} F_{ij}(p, 2\pi q/\ell)$$

at each external grid momentum. By (RY3), its second finite-volume coefficient is $2\ell^4\mathcal{Y}_{\ell,ij}(p)$. The limiting picture also has the exchange and reality symmetries

$$\mathcal{Y}_{ij}(p) = \mathcal{Y}_{ji}(-p), \qquad \overline{\mathcal{Y}_{ij}(p)} = \mathcal{Y}_{ij}(-p). \tag{RY4a}$$

At each finite side, the logarithmic determinant is a real scalar function of real field coordinates. Equality of its mixed derivatives gives exchange of the two external legs, while reality of its Hessian gives conjugation under Fourier inversion. Thus the finite-volume normalized pictures have both relations at every allowed grid momentum. Estimate (RY2) says $\mathcal{Y}_{\ell,ij}(p) \to \mathcal{Y}_{ij}(p)$ uniformly over the external grid points as $\ell \to \infty$. Thus their difference from $\mathcal{Y}$ tends uniformly to zero on these grids as the side tends to infinity. The union of the full side grids is dense in the momentum torus, and $\mathcal{Y}$ is continuous by its analytic strip bound. Passing along cofinal sides and then using density proves (RY4a) everywhere. This supplies the Hessian symmetry and reality of the half-scaled zero-mode picture; it does not follow from factorial normalization alone.

The factor $1/2$ in $\mathcal{Y}$ is essential: the finite-volume mixed second derivative of $\frac{1}{2}\widehat{F}_{ij,0}$ contributes $2!\,\frac{1}{2}\widehat{F}_{ij,0} = \widehat{F}_{ij,0}$. This is why (RY3) compares the normalized grid mean to the continuum coefficient and why the error is the per-site Fourier error multiplied by the site count exactly once.

131.2 Determinant identity and positive-matrix domain

In the application, the pole test is

$$\mathcal{P}_\ell(x) = \tfrac{1}{2}\log\det P_\ell(x) - \log\det V_\ell(x),$$

where P_ℓ is the positive rank matrix and V_ℓ the positive reference matrix. The finite ticket identifies the zero-, one-, and two-leg Taylor layers of P_ℓ with those of the interacting response, and those of V_ℓ with the reference response. The floor and positive-regulator hypotheses make both matrices positive definite at $x = 0$; by continuity their real logarithmic determinants are smooth nearby. Ordinary matrix differentiation gives

$$\begin{aligned} D\log\det P[H] &= \operatorname{tr}(P^{-1}DP[H]),\\ D^2\log\det P[H,K] &= \operatorname{tr}\big(P^{-1}D^2P[H,K] - P^{-1}DP[K]P^{-1}DP[H]\big). \end{aligned} \tag{RY8}$$

and the same formulas for V. Write $A(r)$ for either matrix family, $B_i(p,r)$ for its first variation (from background momentum r to $r+p$), and $C_{ij}(p,q,r)$ for its mixed second variation (from r to $r+p+q$). The shifted fiber block of the mixed derivative of the logarithmic determinant at the ordered pair of legs is the trace of

$$A(r+p+q)^{-1}C_{ij}(p,q,r) - A(r+p+q)^{-1}B_i(p,r+q)A(r+q)^{-1}B_j(q,r). \tag{RY8a}$$

This follows by differentiating $D\log\det A[H] = \operatorname{tr}(A^{-1}DA[H])$ and using

$$D(A^{-1})[K] = -A^{-1}DA[K]A^{-1}.$$

There is one ordered product for the specified ordered pair; exchanging the legs gives the corresponding other entry, not an extra copy in (RY8a). Sum this expression over internal momentum and apply the weights in the finite ticket. Formula (RY8a) is the shifted fiber block of the Hessian. Translation invariance makes the full trace entry zero unless the external momenta conserve; in the conserving case summing the fiber blocks gives the normalized torus mean multiplied by ℓ^4. This finite-volume determinant calculation precedes the Fourier tail limit.

The Fourier estimate is entrywise. Any closed-cone passage for finite reflection matrices and the smooth-word approximation to point configurations belong to the separate reflection argument.

131.3 Order of choices in the all-period output

Here is the exact downstream implication, with its inputs exposed. Fix $\sigma, \epsilon, \tau > 0$ and $C, Z, A \geq 0$. Assume a baseline analytic family H has strip fee (k, E) with $k > 0$, $E \geq 0$, and matches the free Wilson response at every volume. Assume the response service chooses $v > 0, M \geq 0$, depending only on these external data, with the following property: for every sequence of positive integer sides $\ell_j \to \infty$, every bed B, target p, path R, and budget Λ satisfying the fixed (τ, A, Λ) path bound, attachment of B to R, metric strip bound (σ, C), direct-term bound Z, all grade floors (ϵ, C), and sworn conditions at every ℓ_j, it provides a family T with fee (v, M), whose zero-, one-, and two-leg layers vanish to orders three, two, and one respectively, and exact analytic cancellation of the target response against the Wilson term at every ℓ_j. Require also the exact finite-layer realization: for every ℓ_j, the interacting and reference matrices are real analytic on the real momentum torus, Hermitian positive definite at the origin with a positive lower eigenvalue bound uniform in j, and their Taylor layers of orders zero, one and two are the layers specified by the response ticket. The resulting shifted-fiber kernel $F_{c,\ell_j}(p, q)$ is the restriction of one common periodic kernel $F_c(p, q)$, holomorphic and bounded by some K whenever every imaginary momentum coordinate has modulus less than some $z > 0$, uniformly in j and color c. This is the common-strip and common-layer content needed for the full-grid limit; positivity at the origin alone does not imply it. Assume also the uniform two-wave output: for every combined family below, every color c, and every regulator $0 < y \leq 1$, the same constants $a, b \geq 0$ give

$$|\kappa_{y,c}(x)| \leq a(|x| + y)^{-4}, \qquad |\kappa_{y,c}(x) - \Pi_{G,c}(x)| \leq b|x|^{-3} + by^2|x|^{-6} \quad (y \leq |x| \leq 1) \tag{RY9}$$

on the centered momentum box. This states the uniform local estimate needed from the next producer; it is not deduced from the finite-volume recording argument.

Theorem 131.2 (All-period recording and drift)**.** *Under these hypotheses, for every admissible path datum, every odd integer $L \geq 1$, and every side sequence $\ell_j \to \infty$ there are a picture $\mathcal{Y}_L$, constants $z_L > 0$, $K_L \geq 0$, and numbers $e_{L,j} \geq 0$ such that*

$$\mathcal{Y}_L \in \mathcal{H}_{z_L/2}(K_L), \qquad e_{L,j} = \mathfrak{r}(z_L, K_L)e^{-z_L\ell_j/4} \longrightarrow 0, \qquad \mathrm{Err}_{\ell_j}(\mathcal{Y}_L, \mathcal{P}_{\ell_j}; e_{L,j}), \tag{RY10}$$

and, for every color c,

$$|\alpha_c(\mathcal{Y}_L) + \lambda_G \log L| \leq D_0, \quad D_0 = c_{\mathrm{wt}}\left[a\left(\frac{\pi^2}{2} + |E_4|\right) + 3\pi^2 b\right]. \tag{RY11}$$

The constant D_0 is independent of L and of the side sequence.

Proof. Choose the baseline before the response family. Put $w = \min(k, v) > 0$; restrict both fee bounds to w and add the families, obtaining fee $(w, M + E)$. Now fix L and set

$$y = L^{-1}, \qquad s = \min\{\sigma, w, y/128\} > 0. \tag{RY5}$$

Then $0 < y \leq 1$, and $s \leq y/128$ is the phase condition in the local determinant-floor estimate. At every ℓ_j, the response ticket matches the target minus the Wilson term, while the baseline ticket matches the Wilson term. Adding their zero-, one-, and two-leg Taylor identities gives the target ticket. The finite determinant formula (RY8) turns this into the exact local coefficient identity (RY3), and the floor and positive regulator bounds ensure both determinants are nonzero at zero. The all-grade floors and metric assumptions identify the regulated interacting table with the tested local determinant table; the reference table is identified by its explicit regulator formula. These

are the two tab identities required by the local positivity/floor argument at scale s. The exact finite-layer and common-strip receipt in the hypotheses supplies $z_L > 0$, $K_L \geq 0$ and the common analytic two-momentum kernel in $\mathcal{H}_{z_L}(K_L)$. Applying Lemma 131.1 yields

$$e_{L,j} = \mathfrak{r}(z_L, K_L) e^{-z_L \ell_j/4}, \qquad \mathrm{Err}_{\ell_j}(\mathcal{Y}_L, \mathcal{P}_{\ell_j}; e_{L,j}). \tag{RY12}$$

The positive z_L and $\ell_j \to \infty$ imply $e_{L,j} \to 0$ for this fixed L. No uniformity of z_L, K_L in L is required or asserted.

The same response satisfies (RY9), uniformly in color c, at regulator y. The coefficient-integral identity (AT1) identifies $\alpha_c(\mathcal{Y}_L)$ with the normalized integral of $\operatorname{Re} \kappa_{y,c}$. The radial calculation in the preceding section then gives the estimate, with the same D_0 for every color. (RY11): $\log y^{-1} = \log L$, while a, b were fixed uniformly before choosing L. Taking $L_{\min} = 1$ supplies the all-odd-side clause needed by the outer construction; oddness is not otherwise used in this estimate. □

Scope. This proves the finite Fourier tail, the exact conversion of local coefficient identities into a site-count-scaled recording error, and the downstream assembly from the stated baseline, response-service, local-positivity/floor, and uniform two-wave outputs. It does not reprove the local stationary-family producers of the all-centered and shell bounds, nor the general path-history service that supplies the response family. Those are the remaining analytic inputs recorded in the companion evidence. No finite-volume coefficient is identified with the continuum integral without first passing through (RY12).

132 The finite-volume and annulus estimates use one response

The finite-volume recording and the annulus bound must be attached to the same analytic response. This is a short compatibility step, but it prevents an invalid substitution of a different family after taking the Fourier limit. We state the join explicitly.

Fix a simple gauge group, a clip and path, a bed, and a sequence of torus sides tending to infinity. Let W be one fee-bounded analytic Wilson response with its exact Wilson tickets at those sides. Let Q be one fee-bounded correction response, with soft orders three, two, and one in its zero-, one-, and two-mark layers, whose tickets at every chosen side are the finite-action tickets minus the Wilson tickets. Assume the bed floor, metric-strip bound, attachment, and sworn bounds required by the finite determinant test hold for this same bed and path at every side. Assume the Wilson reference is grey, meaning that its Wilson tickets are the exact Taylor tickets of the finite Wilson action. Finally, assume the selected local-frame family has the common finite-volume action and frame data used in the face estimates of Section 129. Assume W and Q have a common positive response strip width η and finite fee fixed before the odd side L is chosen. This is the width used for the uniform radial constants; it is distinct from the possibly smaller strip z_L of the regulated recording kernel. For the logarithmic estimate, fix $0 < \mu \leq 1$ before applying the radial bound, with $\mu = L^{-1}$ for the requested odd side L. The Wilson fee, correction fee, floor parameter, and metric bound are common to the family; the bed, clip, and side sequence are the same in the recording and face estimates. The strip width and bound for the recorded kernel may depend on this fixed L.

Lemma 132.1 (Same-response compatibility)**.** *Under these assumptions set $F = Q + W$. Then the following conclusions concern this one F and the same selected data:*

1. *For every chosen torus side, the zero-, one-, and two-mark tickets of F are exactly the corresponding finite-action tickets. These are the three marked response layers used to identify the*

action variations in the finite test. If the regulated rank and reference matrices have their positive floors, the finite pole test is their logarithmic-determinant difference, and its conserving mixed coefficient is the site count times the normalized mesh mean of the shifted-fiber kernel of F.

2. *The analytic kernel obtained from those shifted-fiber coefficients is independent of the torus side. Define the analytic picture by*

$$\mathcal{Y}_{ij}(p) = \frac{1}{2}\int_{\mathbb{T}^4} F_{ij}(p,q)\,dq,$$

with normalized Haar measure. Its mixed two-leg coefficient is $2\mathcal{Y}_{ij}$, so the finite-volume realization at a conserving pair is $n_i^4\,2\mathcal{Y}_{ij}$. If the common kernel has bound K on a strip of width $z > 0$, this picture satisfies the recorded estimate

$$\mathrm{Err}_{n_i}(\mathcal{Y}, \mathcal{P}_{n_i}; \mathfrak{r}(z,K)e^{-zn_i/4})$$

for every chosen side n_i.

3. *For each color c, the radial bounds obtained from the same local-frame family and the same Wilson reference give*

$$|\alpha_c(\mathcal{Y}) + \lambda_G \log \mu^{-1}| \le D_0.$$

Here D_0 is fixed before the odd side parameter is chosen, while the strip constants z, K and hence the recording error may depend on L.

Proof. The fee bounds on Q and W restrict to their common smaller strip and add. The exact response identities add as well: at each chosen side the zero-, one-, and two-mark ticket of Q is the finite-action ticket minus the Wilson ticket, while the corresponding ticket of W is the Wilson ticket. Hence these three tickets of F are exactly the finite-action tickets. The order-zero remainder construction fixes the normalization; no extra factor of the number of sites is inserted here.

For each side, the positive floors make the two determinant matrices invertible at zero. Differentiating their logarithmic determinants gives one direct second variation and the single ordered product of first variations for each specified ordered pair. The ticket identity therefore identifies that finite mixed coefficient with the conserving mesh sum of the common shifted-fiber kernel. Nonconserving entries vanish by momentum conservation. The fiber trace contributes a site count exactly once. The uniform strip bound is shared by all sides because the same F is used. The full-mesh Fourier estimate of Lemma 131.1 now applies to these exact finite identities, giving the displayed error at every side, without a subsequence extraction.

For the third conclusion, apply the all-centered and shell face estimates to the same selected local-frame family. The interacting and regulated reference paths are the ones formed from F and the fixed Wilson response; the shell comparison uses the same bed and only momenta with positive unregulated scale. The resulting two radial estimates are uniform in the color and were fixed before choosing the odd side. Integrating over the inner ball, the annulus, and the part outside the unit ball gives the logarithmic estimate by Lemma 138.1. The analytic coefficient in that integral is $\alpha_c(\mathcal{Y})$ for the picture defined above: both it and the finite recorded coefficients are obtained from the same shifted-fiber kernel, with the factor $1/2$ canceled by the two-leg mixed derivative. This proves that the finite recording and the radial drift belong to one response. □

Selected construction of the common data. In the selected construction, the baseline W is the fee-bounded Wilson response. The correction Q is the response produced from the attached path and bed. Its finite tickets are the difference between the intended action and the Wilson action. Combining them before evaluating any side gives $F = Q + W$. The fee, ticket, floor, and metric conditions used above are all on this combined response and the original selected bed.

The local-frame bounds are produced from that same Wilson reference: the finite-volume vector and ghost responses, together with their one-leg and two-leg extensions, are obtained from one family of local frames. The all-centered estimate is applied at scale $|x| + \mu$; the shell estimate is applied to the restriction $\mu \le |x|$, where the unregulated reference inverse exists. The two applications give the upper face and the lower face-minus-halo price, respectively. These are precisely the hypotheses used in the preceding face calculation; no change of bed, reference, or finite-volume sequence is made between the face estimates and the ticket identity.

In source terms, this is the composition of the Wilson baseline, the path correction, ticket addition, finite determinant recording, and uniform radial evaluation. The declaration-level correspondence and the exact shared-object audit are in the companion evidence record.

133 Symmetries of the selected stationary germs

The Taylor construction of the output action uses the symmetries of three finite-volume functions: the classical stationary value, the completed determinant correction, and the inherited loop evaluated on the stationary fine field. We prove these symmetries for the selected stationary branch. In particular the determinant argument includes the moving frame and its regularization.

Fix a block step, its spanning forest, a positive regulator μ, and the analytic stationary branch $U(V)$ in the original rooted tree chart. The price gap is strictly positive. Write t for a choice of block representatives. For a coarse field V and a joint fine field and frame U, let $S_t(V, U)$ be the classical action with the blocking penalty. Let $H_t(V, U)$ be its Hessian in the joint left coordinates, and let $F_t(U)$ be the completed gauge-fixing derivative. In the notation of the determinant construction, put

$$M_t(V,U) = H_t(V,U) + F_t(U)^* F_t(U), \qquad \mathcal{B}_t(V,U) = \tfrac{1}{2}\log\det M_t(V,U) - \log\det\Gamma(U).$$

Here $\Gamma(U) = d_U^* d_U + C_\mu(U)$ is the same scalar gauge-orbit operator, with $C_\mu = \mu^2 P_\mu^9$. All matrices in this section are real. The positive price gap and positive regulator make the relevant determinants positive near the stationary origin, so these expressions define analytic real germs there.

133.1 Transport of the completed determinant

A fine gauge transformation k and coarse gauge transformation u act on the joint field and frame. Their tangent maps in joint, fine-link and scalar coordinates are orthogonal: they are products of adjoint colour rotations and coordinate permutations. Denote the joint tangent map by T and the scalar map by A. Equivariance of the block ribbon and of its frame logarithm gives

$$F_t(U')T = AF_t(U).$$

We verify why this identity includes both parts of the gauge fixing. For its frame part, the logarithm near the identity obeys $\log(khk^{-1}) = \mathrm{Ad}_k \log h$; differentiating the block-frame identity gives the intertwining of its derivative. For its link part, the covariant difference intertwines the scalar and link adjoint maps, and transposition gives the corresponding identity for the adjoint difference. The

scalar resolvent and hence its ninth power intertwine because multiplication and inversion preserve an intertwining relation. These are the two summands of F_t.

The classical action is gauge invariant. In left increment coordinates the transformed action composed with T equals the original action near the increment origin. Twice differentiating this identity yields

$$T^* H_t(V', U')T = H_t(V, U).$$

There is no derivative of T in this calculation: the base fields are fixed while differentiating the increment variable. Combining this congruence with the gauge-fixing identity and $A^*A = I$ gives

$$T^* M_t(V', U')T = M_t(V, U).$$

Similarly $\Gamma(U')$ is orthogonally conjugate to $\Gamma(U)$. Taking determinants and using $(\det T)^2 = (\det A)^2 = 1$ proves

$$\mathcal{B}_t(V', U') = \mathcal{B}_t(V, U). \tag{646}$$

The same transformation preserves S_t and the inherited loop.

Signed lattice transformations require a slightly different tangent map. For a transformed link e, let e' be the original link it reads. If its orientation is preserved, the tangent reads $X_{e'}$; if reversed, it reads $-\operatorname{Ad}_{U_{e'}^{-1}} X_{e'}$. This map is orthogonal and satisfies the exact increment identity

$$s^*(\exp(X)U) = \exp(T_U X)\, s^* U.$$

The scalar tangent is the coordinate permutation; the joint tangent also includes the corresponding frame permutation. The block representative changes from t to the transported choice r. Equivariance of the ribbon, frame logarithm and covariant difference therefore gives, with these tangent maps,

$$F_t(s^*U)T_U = AF_r(U), \qquad T_U^* H_t(s^*V, s^*U)T_U = H_r(V, U).$$

For the penalty term this uses invariance of the group price under inversion and conjugation: reversing both links in its relative product leaves its price unchanged. The same resolvent argument handles C_μ. Taking determinants as above gives

$$\mathcal{B}_t(s^*V, s^*U) = \mathcal{B}_r(V, U). \tag{647}$$

This is the completed moving determinant. It does not assert full signed covariance of the separate auxiliary fine-lattice pole symbol.

133.2 Returning a moved solution to the original chart

A transformed stationary solution need not lie in the original tree chart. Gauge it back using the tree gauge map. This map is analytic near the trivial joint field, sends that field to itself, and leaves the coarse input fixed when the coarse gauge transformation is the identity. The transformed stationarity equation is still zero: under the invertible tangent map its first derivative is the original zero first derivative. The local uniqueness in the original chart, proved by the stationary implicit-function construction, now identifies the regauged solution with the selected solution at the transformed coarse input. We use uniqueness only in that chart.

The classical value, the determinant correction by (646), and the inherited loop are unchanged by this regauging. Consequently their values along any analytic stationary probe through the trivial field equal their values on the selected branch at that probe's coarse input. The argument also applies to parameter-dependent gauge transformations: freeze the parameter when comparing

the action and matrices, and then use analyticity and the local branch uniqueness on a common neighborhood.

We must also remove the changed representative r in (647). The tested classical actions for t and r agree as germs in both the coarse and tree-chart variables, by the local branch construction of Section 105. Therefore their stationary values and their tree-chart Hessians agree. The moving Haar identity of Section 127 expresses each thermal value as that tree Hessian determinant with the same Haar factor, or equivalently as the same inherited loop plus $\mathcal{B}_t$ or $\mathcal{B}_r$. Equality of the tree Hessians and cancellation of the common inherited loop give equality of $\mathcal{B}_t$ and $\mathcal{B}_r$ along the selected branch. Thus both tested functions are independent of the representative as germs. This uses equality on a neighborhood of the moving stationary point, which follows from equality of the joint germs; equality at the stationary value alone would not identify their Hessians.

133.3 The identities used by the Taylor coefficients

Apply the preceding argument first to a signed lattice transformation, then to constant colour conjugation, and finally to a small variable coarse gauge transformation. For the last operation the transformed coarse logarithm is its actual nonlinear gauge action; transform the joint frame accordingly and gauge the resulting stationary solution back to the original chart. Each of the three analytic stationary functions is unchanged as a germ under all these operations. The inherited loop has the same invariances directly from its gauge invariant atoms and finite signed average.

Differentiating these germ identities supplies the signed and colour symmetries and the nonlinear gauge identities of the finite-volume Taylor coefficients in Section 134. When the lower positive-degree coefficients vanish, the differentiated nonlinear gauge identity reduces to its leading linear Ward identity, as proved there. All three functions here use the same stationary branch, regulator and finite torus; no symmetry claim is imported from a separately chosen stationary solution.

134 Constructing the stationary output action

We now apply the curvature lift to the Taylor coefficients of the stationary action. The construction has two separate parts. Cubic and quartic atoms match the classical action through degree four. The quadratic response determines one scalar change in the kinetic coefficient and an invariant loop remainder with vanishing moments. Both parts use the same coarse bed and the same sequence of tori.

134.1 Recovering Ward data from finite-volume coefficients

For an analytic real function F of the link coordinates, write $C_n(F) = D^n F(0)/n!$, extended complex multilinearly to complex link fields. A unit Fourier wave has a specified momentum, coordinate direction, and colour basis vector. A common coefficient picture P for functions F_k on tori Λ_k means that

$$\left| C_n(F_k)[v_1, \dots, v_n] - \mathbf{1}_{\sum p_j = 0} \, |\Lambda_k| \, P(p_1, \dots, p_{n-1}) \right| \le |\Lambda_k| \varepsilon_k \tag{648}$$

for every such tuple of unit waves; the zeroth momentum in the picture is minus the sum of the other momenta. Direction and colour indices in P are understood. We assume a common positive strip width and finite component bound, and $\varepsilon_k \to 0$. The side lengths tend to infinity. Exact recording is the case $\varepsilon_k = 0$.

Suppose F_k is invariant, as a germ at the identity, under gauge transformations and under simultaneous adjoint colour rotations. Assume its positive-degree derivatives below degree n vanish.

Differentiate the gauge-invariance identity once in the gauge parameter and $n-1$ times in the link variables. The term where all link derivatives fall on F_k gives its nth derivative with one argument equal to the linear gauge variation $d_0 h$. Every other term contains a positive-degree derivative of F_k of degree less than n and is zero. Consequently

$$C_n(F_k)[v_1, \dots, d_0 h, \dots, v_n] = 0.$$

On Fourier waves this is exactly the contraction by $q(p_j)$ in factor j. It holds in every factor by symmetry of the derivative. This is also the leading case of the finite gauge-variation identity in Section 120.

Approximate any tuple of real independent momenta by finite-torus frequencies and choose the zeroth frequency to make their sum zero. Representatives may differ by 2π multiples; both the picture and the symbols q are periodic. Divide (648) by $|\Lambda_k|$. The error tends to zero, while continuity of the common picture and of q permits passage to the limit in the finite contraction sum. Thus the picture has all continuous-momentum Ward identities. Conjugating real Taylor coefficients and reversing every wave gives $\overline{P(-p)} = P(p)$ by the same limit. Differentiating the constant colour-rotation invariance and passing to the limit makes the real and imaginary colour tensors invariant. These are precisely the hypotheses of Section 121.

For $n = 2$, only the first derivative of F_k needs to vanish. This follows from colour invariance when the gauge Lie algebra is simple: an invariant linear functional kills every bracket, and the brackets span the algebra. The constant value of F_k need not be zero.

134.2 Matching a leading Taylor coefficient

Suppose $n \geq 3$, each F_k is invariant under signed coordinate permutations as a germ, F_k vanishes to order n, and P records its nth coefficient exactly, at strip cost A. The curvature lift produces invariant real atoms of degree n with total exponential budget KA, at a positive width chosen before A and P. Let S be their translation sum and then their average over the finite group of signed coordinate permutations.

Here is the normalization check. Each transported plaquette slot has zero constant term and linear term equal to its discrete curl. The leading coefficient of the product of n slots is therefore the product of these curls. The normalized coefficient $C_n(S)$ is the symmetrization of that ordered product, namely the average over the $n!$ permutations of its arguments. On unit Fourier waves, summing the anchor over the torus gives zero unless the momenta sum to zero, and gives $|\Lambda_k|$ otherwise. The remaining absolutely convergent displacement sum is exactly the curvature symbol contracted in every factor. It is P by the lifting identity. Since $C_n(F_k)$ is symmetric, symmetrizing this identity retains it. Averaging orientations also retains it, because F_k is invariant under those orientations. Reflections may multiply a wave by a unit-modulus edge phase; these phases are included when applying the same identity to the transformed wave. Thus $C_n(S) = C_n(F_k)$ on every tuple of unit Fourier waves.

These waves span the complexified link space, by finite Fourier inversion. Multilinearity gives equality on arbitrary arguments and hence on real fields. Both functions are analytic, and each atom vanishes to order n; the exponential budget permits termwise differentiation on a common complex link ball as in Proposition 120.1. It follows that

$$F_k - S = O(|x|^{n+1}) \tag{649}$$

along every real ray. In the approximate version of (648), permutation and orientation averages are convex averages, so the same argument retains the coefficient error $|\Lambda_k|\varepsilon_k$ on each tuple of unit waves. It does not insert a factorial or a further volume factor.

134.3 The cubic and quartic construction

Let E_k be the classical stationary action in the chosen chart, and let J_k be the quadratic path action obtained from the coarse bed roots. Both are analytic, gauge invariant, and invariant under signed coordinate permutations. Their values and first derivatives at zero are zero. The stationary Hessian identity and the exact bed attachment give the same Hessian for both, namely the coarse Gaussian form. Thus

$$F_k = E_k - J_k$$

has order three. Suppose E_k has exact common coefficient pictures of degrees three and four with component bound $M \geq 1$ at width σ. The row estimates supply such pictures for J_k at positive widths and fixed bounds P_3, P_4. Choose the minimum of the three widths. Both residual pictures then have cost at most

$$M(1 + P_3 + P_4).$$

Apply (649) at $n = 3$ to obtain an invariant cubic atom action S_3 of budget $K_3 M$ and with $F_k - S_3$ of order four.

Its fourth coefficient must also be subtracted. In general an atom of degree three has nonzero higher Taylor terms. The uniform atom-picture estimate gives an exact common degree-four picture for S_3, of cost at most $K_3 M B_4$ at a positive width. Subtract it from the degree-four picture of F_k, and take the minimum width. The resulting cost is at most a fixed multiple of M. Apply the leading-removal construction at $n = 4$ to $F_k - S_3$. This produces an invariant quartic action S_4, again with budget bounded by a fixed multiple of M, and

$$E_k - J_k - S_3 - S_4 = O(|x|^5). \tag{650}$$

The two atom families and their common positive width are chosen independently of k. They consequently match the action on every torus in the sequence. Taylor expansion on rays followed by polarization proves equality of the full derivatives of orders zero through four. The chart's tag-independence is an equality of germs, so the same derivatives agree for every allowed choice of tag, without choosing a different path for each tag.

134.4 Removing the unique quadratic kinetic coefficient

We need one further consequence of the Ward and lattice symmetries. Let $f_{ab}^{uv}(p)$ be an analytic quadratic coefficient picture, where a, b are coordinate indices and u, v colour indices, with momenta $(-p, p)$. Suppose it comes from the common recording of analytic, gauge-invariant, signed-coordinate-invariant real functions as above. It then has the exchange symmetry

$$f_{ab}^{uv}(p) = f_{ba}^{vu}(-p).$$

The leading homogeneous terms transform as ordinary coordinate tensors under signed permutations: the reflected-edge phases have constant term equal to the relevant coordinate sign. Once all lower homogeneous terms vanish, the higher terms of these phases do not contribute at the order being considered.

First $f(0) = 0$. Indeed the first-order part of the Ward identity for a constant tensor is $\sum_b p_b f_{ab}^{uv}(0) = 0$ for all p, and coordinate vectors detect every entry. For its linear term write $K_{ab}^{uv}(p)$. The left and right Ward identities give

$$\sum_b p_b K_{ab}^{uv}(p) = 0, \qquad \sum_a p_a K_{ab}^{uv}(p) = 0.$$

They imply $K^{uv}_{ab}(e_b) = K^{uv}_{ab}(e_a) = 0$ and $K^{uv}_{ab}(e_j)+K^{uv}_{aj}(e_b) = 0$. If a, b, j are distinct, the coordinate permutation exchanging b and j identifies the two terms, so both vanish. When indices coincide the preceding two identities already imply the same conclusion. The linear term is zero and f has order two.

Let $Q^{uv}_{ab}(p)$ be its quadratic homogeneous term. Fix $a \neq b$ and one unit colour vector e_u. We claim that

$$Q^{uu}_{aa}(e_b) = 0 \tag{651}$$

forces $Q = 0$. At fixed a, p, the colour form $Q^{uv}_{aa}(p)$ is real, symmetric, and adjoint invariant. Reality follows from momentum reversal and even homogeneity; symmetry follows from the exchange identity. Such a form is a scalar multiple of the invariant inner product. Indeed its self-adjoint representing operator commutes with every adjoint operator; each eigenspace is an ideal, so simplicity leaves only one eigenvalue. Therefore (651) kills that whole colour form. Coordinate permutations transfer the conclusion to $Q^{uv}_{xx}(e_y)$ for all $x \neq y$. The case $x = y$ follows directly from the leading Ward identity.

Polarize Q, writing $T^{uv}_{xy}(e_k, e_l)$ for its symmetric bilinear coefficient in momentum. Reflection in coordinate r multiplies this entry by $(-1)^{\mathbf{1}_{r=k}+\mathbf{1}_{r=l}}$ on the momentum side and by $(-1)^{\mathbf{1}_{r=x}+\mathbf{1}_{r=y}}$ on the tensor side. Unless these parities agree for every r, the entry is zero. For $x = y$ this eliminates every mixed momentum coefficient; the diagonal coefficients were just shown to vanish. For $x \neq y$ it leaves only the mixed pair $\{k, l\} = \{x, y\}$. Finally insert $p = e_x + e_y$ in the Ward identity with first tensor index x. All other entries have vanished, so it kills this last mixed coefficient as well. Thus $Q = 0$ in every colour component.

The picture now has order three. Reversal of all coordinate signs acts by $+1$ on its two tensor indices but by -1 on a homogeneous cubic polynomial. Its cubic term is therefore zero. We have proved that the single test (651) gives order four.

Let $H(p)$ be the coarse metric row, with $H(0) = I$, and let b_H be the exactly recorded quadratic picture of its quadratic path action. Its quadratic homogeneous term is

$$(b_H)^{uv,[2]}_{ab}(p) = \tfrac{1}{2}\big(|p|^2\delta_{ab} - p_a p_b\big)\delta_{uv}.$$

This follows by replacing both discrete curls by their linear terms $ip\wedge$ and using $H(0) = I$; the action has coefficient $1/2$ and its two waves have opposite momenta. In particular the transverse unit test is $1/2$. Fixing distinct axes $0, 1$ and any unit colour basis vector e_u, define

$$\alpha = 2\,\mathrm{Re}\, f^{uu,[2]}_{11}(e_0).$$

Then $f - \alpha b_H$ passes the zero test. It inherits the Ward, reality, exchange, and lattice symmetries by subtracting the actual quadratic action, so it has order four. This establishes the normalization of α before any curvature lift is chosen.

134.5 The loop remainder and its real-wave estimate

Let U_k denote the local determinant and Jacobian response and V_k the inherited loop response. Suppose their common pictures are f and g, with strip bounds Φ and R on width σ, respectively. Assume f records U_k with error $|\Lambda_k|\varepsilon_k$, while g records V_k exactly and has order four. Assume also that J_k has the exact picture b_H, of bound B_H, at that width. All three functions have the invariances used above. Then

$$h = f - \alpha b_H + g$$

has order four, strip cost at most $\Phi + |\alpha| B_H + R$, and records $U_k - \alpha J_k + V_k$ with the same error. The two-factor lift, with retained order two, gives an invariant loop atom family of budget

$$D(\Phi + |\alpha| B_H + R). \tag{652}$$

It has zero total coefficient and zero first displacement moments. Let S_2 be its orientation-averaged action. The normalization argument above shows that the difference between $C_2(S_2)$ and $C_2(U_k - \alpha J_k + V_k)$ on every pair of unit complex Fourier waves is bounded by $|\Lambda_k|\varepsilon_k$.

Put $d = 4 \dim \mathfrak{g}$, the number of one-form direction and colour coordinates at a site. A complex wave with all component amplitudes bounded by one is a sum of d unit basis waves, so bilinearity bounds the coefficient difference by $d^2|\Lambda_k|\varepsilon_k$. A real wave is half the sum of that complex wave and the opposite-momentum wave with conjugate amplitude. Expanding two real waves gives four terms, each with factor $1/4$, and hence the same bound. Finally $D^2 = 2C_2$:

$$\left|D^2(S_2 - U_k + \alpha J_k - V_k)(0)[w_p, w_q]\right| \le 2d^2|\Lambda_k|\varepsilon_k. \tag{653}$$

The momenta p, q are arbitrary; no opposite-momentum restriction has been added.

134.6 One common output path

Take the row family to be the coarse bed roots and take the loop, cubic, and quartic families just constructed. Their exact row attachment is unchanged by adding the other three families. The metric identity follows from that root attachment and the coarse bed identity. Invariance and the two loop moment identities hold separately for the constructed atoms. At any positive width below the common construction width, the row and cubic/quartic budgets are satisfied whenever

$$A' \ge \max\{\mathrm{rowPrice}(w, M_H), KM\}, \qquad \Lambda' \ge D(\Phi + |\alpha| B_H + R).$$

Here M_H is the root-strip bound and M the classical Taylor bound. Width reduction only decreases each weighted absolute sum. Thus this is a path with all the required summability, metric, invariance, and moment properties, not merely a list of formal jets.

Its classical part is $J_k + S_3 + S_4$ and obeys (650). Its response part is $\alpha J_k + S_2$. The stationary determinant identity identifies the actual thermal response, as a germ, with $U_k + V_k$. Consequently (653) is the required Hessian comparison for the actual response. If

$$2d^2\varepsilon_k \le \Theta\Omega e^{-g\,\mathrm{side}(\Lambda_k)},$$

its right side is at most $|\Lambda_k|\Theta\Omega e^{-g\,\mathrm{side}(\Lambda_k)}$. The stationary gap gives positivity of the stationary fluctuation Hessian, so its determinant remains positive near zero; the analytic chart gives the required smooth response germs. These conclusions all concern the same output path. Production of the common determinant picture, the inherited picture with its fourth-order bound, and the displayed exponential error is still required when this construction is used in a transition.

135 Transporting the inherited loop response

The loop term entering a stationary step already has zero total coefficient and zero first displacement moments. These cancellations make its quadratic momentum picture vanish to fourth order. We prove that the stationary pullback retains this order and gains L^{-2} in its strip bound, where L is the blocking side. The order statement and the quantitative strip statement require different arguments.

135.1 The fine-lattice picture

Group the loop coefficients by the two plaquette directions, the two colour indices, and the relative displacement z of their bases. Denote the resulting real coefficient array by $r(z)$. Its exponential norm is bounded by the loop budget Λ: the norm of the relative displacement is no greater than the sum of the two slot lengths. Absolute summability permits grouping all atoms with the same displacement. The path's moment identities give

$$\sum_z r(z) = 0, \qquad \sum_z z_a r(z) = 0.$$

Consequently its Fourier series $\widehat{r}(p)$ vanishes to order two at zero. The quadratic coefficient of the loop action is the sum over face indices of

$$d_1(-p)^{\mathsf{T}} \widehat{r}(p) d_1(p), \tag{654}$$

with its two field arguments symmetrized and its lattice orientations averaged. Each curl vanishes to first order. Thus every component of the resulting common picture S has order four. Its strip bound is at most $C_\tau \Lambda$ at the original path width $\tau > 0$: each curl entry costs at most $4(e^\tau + 1)$, there are finitely many face pairs, and the signed orientation average has the fixed reflection cost already estimated in Proposition 120.1.

Formula (654) records the actual finite-volume quadratic coefficient. At zero total momentum, the two slot offsets appear only through their relative displacement; the anchor sum gives the number of fine sites. At nonzero total momentum it is zero. This verifies the identity before taking a large-volume limit. Invariance and the finite Ward identity give

$$S(p) d_0(p) = 0, \qquad d_0(-p)^{\mathsf{T}} S(p) = 0. \tag{655}$$

All tensors below use this unconjugated bilinear convention. On real momenta, $d_0(-p)^{\mathsf{T}} = d_0(p)^*$.

135.2 A strip bound with four finite differences

Put $\mu = L^{-1}$. Let X be the derivative of the stationary chart, represented by a common coarse-to-fine Fourier bridge. Assume that, on a fixed positive coarse strip of width σ, its fine differences of order $j \le 2$ obey the bounds

$$|\Delta^j X| \le D \mu^{1+j}. \tag{656}$$

These are the bridge bounds supplied by the stationary chart.

Apply the finite-difference division of Lemma 120.7 to the order-four picture S. It writes S as a finite sum of products of four symbols $e^{ip_a} - 1$ times Fourier series having a common exponential budget $C\Lambda$ on a positive width w depending on τ. Distribute two differences to each of the two field arguments. Moving a difference from momentum p to $-p$ uses

$$q_a(p) = -e^{ip_a} q_a(-p).$$

The extra phase is a unit lattice translation of the remaining coefficient; its cost is absorbed in C. Thus every term has exactly two differences on each bridge leg.

Pull this representation back by X. On a smaller fixed coarse strip, choose it narrow enough that all fine phases lie in the coefficient strip, in particular $\mu\sigma \le w$. The exponentially weighted coefficient sum bounds every displaced contraction uniformly. For each pair of legs, (656) contributes

$$D^2 \mu^3 \mu^3 = D^2 \mu^6.$$

Summing the fine anchor within one coarse block contributes $L^4 = \mu^{-4}$. Hence the complete common coarse picture H has, on a positive width σ' depending only on σ, component bound

$$|H|_{\sigma'} \le C'\Lambda\mu^2. \tag{657}$$

The finite numbers of colour, direction, and difference addresses are included in C', which depends on the fixed widths and D, not on L, the torus, or Λ. Normal convergence of the coefficient sums proves holomorphy. Their covariance under integer momentum shifts and that of X prove periodicity of the common picture. Expanding the same absolutely convergent sums on a finite torus gives its exact recording identity for the linear pullback of the fine loop action.

The nonlinear stationary pullback has the same second derivative. Indeed the fine loop action V has $DV(0) = 0$, and the stationary map ψ satisfies $\psi(0) = 0$ and $D\psi(0) = X$. The chain rule therefore reads

$$D^2(V \circ \psi)(0)[a, b] = D^2V(0)[Xa, Xb] + DV(0)[D^2\psi(0)[a, b]] = D^2V(0)[Xa, Xb].$$

The coordinate and tree-gauge maps identify these with the actual inherited response; changes of allowed tags agree as germs. Thus H records that response, with no additional error.

135.3 The alias formula and the coarse–fine equation

To prove the order at zero, use the real-momentum alias representation. For $l \in (\mathbb{Z}/L\mathbb{Z})^4$, let

$$k_l(p) = \frac{p + 2\pi l}{L}$$

with any fixed integer representatives; periodicity removes the choice of representatives. Let $N_l(p)$ be the alias component of X in the normalization for which a coarse unit wave pulls back into fine waves of amplitudes $N_l(p)$. With the source normalization of these amplitudes the common picture satisfies

$$H_{ab}(p) = L^4 \sum_l N_{-l}(-p)_a^{\mathsf{T}} S(k_l(p)) N_l(p)_b. \tag{658}$$

For clarity, this equality follows first at each finite coarse Fourier grid. Expand both pulled-back waves in the finite set of aliases. Fine translation invariance kills every pair except opposite aliases; the ratio of fine to coarse site counts is L^4. Approximate an arbitrary real p along grids whose side lengths tend to infinity. The bridge and both common pictures are continuous and the alias set is finite, so the identity passes to every real p. The same argument works with per-volume errors tending to zero. Here both recording identities are exact.

Write $K_f(k) = d_1(k)^* B(k) d_1(k)$ for the fine Gaussian form and $K_c(p)$ for the coarse one. The stationary Hessian equations imply, in the block Fourier coordinates,

$$K_f(k_l(p)) N_l(p)_a = A_l(p) K_c(p) e_a, \tag{659}$$

where $A_l(p)$ is the corresponding component of the adjoint block-mean map and the inverse block Fourier change of coordinates. It is analytic near zero for fixed L. To obtain (659), differentiate the stationary equation, which gives $K_f X = Q^* K_c$, and diagonalize the fine Gaussian form in its alias blocks. This identity also follows on arbitrary real momenta from its exact finite-grid versions by the preceding density argument. Since K_c contains a curl on each side,

$$K_c(p) = O(|p|^2).$$

135.4 Suppressing the nonprincipal aliases

Fix $l \neq 0$ and real p near zero. Then $k = k_l(p)$ stays away from zero modulo 2π. Let $D(k) = d_0(k)$ and $\omega(k) = \sum_a |q_a(k)|^2 > 0$. Project the bridge component off its longitudinal part:

$$N_l^{\perp}(p) = \big(I - D(k)\omega(k)^{-1}D(k)^*\big)\, N_l(p).$$

The matrix $D(k)^*D(k)$ is $\omega(k)I$ on colours, so $D(k)^*N_l^{\perp}(p) = 0$. The fine kinetic form kills pure gauge vectors. Combining this fact with (659) gives

$$\big(K_f(k) + D(k)D(k)^*\big)N_l^{\perp}(p) = A_l(p)K_c(p). \tag{660}$$

The matrix on the left is invertible at $p = 0$. Here the fine floor is required for the same bed on every finite period, with one positive lower constant. Passing its Fourier inequality to real frequencies gives

$$\langle v, K_f(k)v\rangle \geq c\|d_1(k)v\|^2, \qquad c > 0.$$

The coordinate exterior identity is

$$\|d_1(k)v\|^2 + \|D(k)^*v\|^2 = \omega(k)\|v\|^2.$$

It follows either by expanding both squares or from the anticommutation of exterior multiplication and contraction. A vector in the kernel of $K_f(k) + D(k)D(k)^*$ must make both nonnegative terms zero, and hence must be zero. Finite dimensionality gives invertibility. The matrix depends continuously on p, so its inverse is bounded near zero for this fixed l, L. Equation (660) now proves

$$N_l^{\perp}(p) = O(|p|^2).$$

This is a qualitative order statement. No uniform lower bound in L is asserted for this inverse.

The Ward identities (655) permit replacing both N factors in each alias term of (658) by their transverse projections. For $l \neq 0$, each contributes order two; $S(k_l(p))$ is bounded near $p = 0$. Every nonprincipal term therefore has order four. For the principal alias, $k_0(p) = \mu p$, and $S(\mu p)$ already has order four. Its two bridge factors are analytic and bounded near zero. This term has order four as well. There are finitely many aliases, and multiplication by the fixed factor L^4 does not change Taylor order. Hence H has order four.

Together with (657) and the exact recording identity, this supplies the inherited loop picture used in Section 134. Its strip bound is uniform in the blocking side under the stated bridge prices; its order-four property uses the all-period floor and the exact coarse–fine stationary equation. In particular it does not assume that the longitudinal alias amplitudes themselves are small.

136 Transporting the pole coefficient through the block bridge

The annulus calculation concerns a fine-lattice determinant, whereas the stationary output uses its pullback to coarse fields. We prove that their leading quadratic coefficients agree. This uses both quadratic matching equations for the selected bridge, including its normalization at zero momentum.

Let Π_f be a common quadratic picture for the fine pole functional

$$P_f(x) = \frac{1}{2}\log\det\left(K(x) + d(x)d(x)^* + \sum_a s_a^* C(x)s_a\right) - \log\det\left(d(x)^*d(x) + C(x)\right).$$

The notation is that of Section 126. Suppose its finite-volume recording error is at most the fine site cardinality times e_m, with $e_m \to 0$ along the selected periods. Let Π_c be the common picture of the same functions pulled back by the installed linear bridge W, with coarse per-site errors tending to zero. Both pictures are holomorphic on positive strips. Fix the block side L throughout the argument.

The block determinant matrix M_{rank} is the direct sum of the fine pole matrix and an identity on coarse scalar coordinates, after the coordinate permutation separating the two kinds of variables. Thus its determinant equals the fine pole determinant. The scalar determinant is the same in both presentations. The Fourier matrix identities proved above therefore identify the coarse rank picture with the picture of $P_f \circ W$; this is an equality for the actual selected finite-volume functions.

The pole function is invariant under local gauge changes, constant color rotations and unsigned coordinate permutations. For gauge changes this follows by conjugating every factor: the fine Hessian in left coordinates, the covariant difference, its resolvent regulator and the component maps all transform by the corresponding orthogonal adjoint actions. Determinants are unchanged. The same argument, with coordinate permutations whose direction signs are all positive, proves this permutation invariance. Its first derivative at zero vanishes by the simple-color argument in Section 127. Passing the finite-volume identities to the common picture, as in Section 134, shows that its bilinear matrix $S(p)$ has both Ward identities

$$S(p)d_0(p) = 0, \qquad d_0(-p)^T S(p) = 0,$$

The first-order part of the Ward identities forces $S(0) = 0$. To prove that the linear part also vanishes, fix the two color indices and write it as $S_{ab}(p) = \sum_c T_{ab,c} p_c + O(|p|^2)$. The two Ward identities, at quadratic order, give

$$T_{ab,b} = T_{ab,a} = 0, \qquad T_{ab,c} + T_{ac,b} = 0.$$

These equations also kill the case $a = b$: then $T_{aa,c} = -T_{ac,a} = 0$. If a, b, c are distinct, the unsigned permutation interchanging b, c and fixing a gives $T_{ab,c} = T_{ac,b}$; the displayed antisymmetry makes both zero. All repeated-index cases are already zero. Thus S vanishes to order at least two. This uses only unsigned permutations; no reflection symmetry of the pole functional is assumed. The finite-volume errors disappear after division by the site cardinality. No rate uniform in L is used in this step.

Write $H(p)$ for the bilinear matrix of Π_c and $N_l(p)$ for the fine alias amplitudes of W. The finite-wave expansion from Section 135 gives

$$H(p) = L^4 \sum_l N_{-l}(-p)^T S(k_l(p)) N_l(p), \qquad k_l(p) = \mu(p + 2\pi l). \tag{661}$$

We spell out the role of errors here. A pulled-back coordinate wave is a finite sum of fine alias waves with amplitudes $N_l(p)$. For fixed L, these amplitudes remain bounded as the real momentum approaches a specified value. Multilinearity bounds the error on each pair by the original per-site error times the finite sum of the products of its amplitudes. The ratio of fine to coarse site cardinalities is L^4. Hence the error still tends to zero. Opposite aliases are the only surviving pairs, by fine translation invariance; all other pairs tend to zero. Continuity and a dense sequence of external grid momenta prove (661) at every real momentum.

For $l \neq 0$, the transverse projection of $N_l(p)$ has order two at $p = 0$, by the all-period kinetic floor and the primal matching equation proved in Section 135. The two Ward identities of S remove the longitudinal parts of both amplitudes. Thus every nonprincipal term of (661) has order at least four. This argument does not require S to vanish at the nonprincipal alias.

It remains to determine the principal amplitude at zero. The dual matching equation is

$$Q(p)\Phi(p) = I - \kappa^{-1} K_c(p).$$

Since $K_c(0) = 0$, its value at zero is $Q(0)\Phi(0) = I$. Each nonprincipal alias amplitude at zero is a pure gauge vector, because its transverse projection vanishes there. The block mean commutes with the gauge difference, so it maps this vector to a coarse gradient at momentum zero, which is zero. Hence those aliases contribute nothing to $Q(0)\Phi(0)$. On the principal constant alias, the normalized transverse weights sum to one and the straight block path has L edges; therefore the mean acts as multiplication by L. We conclude

$$N_0(0) = \mu I. \tag{662}$$

In particular this normalization is a consequence of the actual two matching equations, rather than a choice of Fourier convention made after the comparison.

Analyticity gives $N_0(p) = \mu I + O(|p|)$. Since S vanishes to order two and $k_0(p) = \mu p$, expanding the two principal amplitudes yields

$$H(p) - L^4 \mu^2 S(\mu p) = O(|p|^3)$$

along every ray. The nonprincipal terms have already been shown to be of order four. If S_2 is the homogeneous quadratic term of S, the quadratic term of the expression on the right is

$$L^4 \mu^2 \mu^2 S_2(p) = S_2(p).$$

Thus H and S have identical quadratic Taylor terms, and in particular identical drift coefficients in every color component.

Apply this result with the fine pole picture supplied by the annulus estimates and the coarse rank picture supplied by the determinant trace formula. The latter's leading coefficient equals the former's. The contact cancellation and the dimension-free core-versus-rank estimate then show that the actual thermal drift differs from the fine annulus drift by a bounded amount, with the bound fixed before L. Consequently a fine estimate

$$|\alpha_f + b_G \log L| \le N$$

gives

$$|\alpha_{\mathrm{thermal}} + b_G \log L| \le N + C_{\mathrm{cmp}},$$

where C_{cmp} is the Cauchy coefficient bound from the uniform determinant comparison. The fine annulus estimate itself, including its common finite-volume recording, remains the input named at the start of this application.

137 Convergence of the free precision horizon

We prove the retained-horizon convergence used for the stationary paths. The cutoff tends to infinity while the scale and block side remain fixed. All coefficients are kept on the lattice cover. Finite-torus matrices are used to identify their limits; they are not substituted for the cover coefficients.

137.1 Compactness with an exponential row bound

A row K between finite coordinate sets I, J has coefficients $K_{ab}(z)$, $z \in \mathbb{Z}^4$. Suppose, eventually in N,

$$\sum_z e^{\sigma|z|_1}|K_{N,ab}(z)| \le C \qquad (a \in I, b \in J), \qquad \sigma > 0. \tag{663}$$

A subsequence of any given cutoff sequence converges in every coefficient: enumerate the countably many entries, extract successively in their bounded real intervals, and take the diagonal subsequence. If its limit is K, pass to the limit in each finite subsum of (663). The nonnegative sums remain bounded by C, so the full sum has the same bound.

For every $0 \le \tau < \sigma$ this coefficient convergence gives

$$\sum_z e^{\tau|z|_1}|K_{N,ab}(z) - K_{ab}(z)| \longrightarrow 0. \tag{664}$$

Indeed its summands are at most $2Ce^{-(\sigma-\tau)|z|_1}$, and

$$\sum_{z\in\mathbb{Z}^4} e^{-t|z|_1} = \left(\frac{1+e^{-t}}{1-e^{-t}}\right)^4 < \infty \qquad (t > 0).$$

First make the tail of this majorant small, and then use convergence on its finite complement. This also proves convergence of the Fourier symbols on real momenta and of the periodized matrices on every fixed torus, since their entries are sums of subfamilies of the same coefficients. The argument applies to a cross row by including its finitely many block residues in I.

Two ways of identifying the resulting cluster limits will be useful. If the Fourier symbols have a prescribed limit on a dense subset of the real momentum torus, any two cluster rows have the same symbol there. Their symbols are continuous by absolute summability, so they agree everywhere. Integrating against $e^{-ip\cdot z}$ recovers the coefficient $K_{ab}(z)$, proving equality of the rows. Alternatively, it suffices that the cluster rows have equal matrices on an unbounded sequence of torus sides. For a fixed displacement z the appropriate periodized entry is

$$\sum_{h\in\mathbb{Z}^4} K_{ab}(z + nh).$$

Its term $h = 0$ is $K_{ab}(z)$. All other terms have displacement at least $n - |z|_1$ and their sum tends to zero by absolute summability. Thus these matrices recover every cover coefficient as $n \to \infty$. In either case the cluster limit is unique. If the original sequence did not converge to it in some coordinate, a subsequence staying a fixed distance away would have a convergent further subsequence, a contradiction. Hence the full sequence converges, with (664) at every smaller width.

137.2 The primitive multipliers along the actual scale schedule

Fix a scale m, and let $v_{d,m}$ be the train consisting of the next d prescribed blocking steps, with a top train at its fine end. Its mass side $M_{d,m}$ tends to infinity, since each block side is at least three. The actual path at terminal cutoff N uses precisely $v_{N-m,m}$; for fixed m, its depth $N - m$ tends to infinity. Consequently each depth limit below also applies to the selected terminal cutoff sequence. For real momentum p and a direction a, put

$$u_{d,m,a}(p) = D_{v_{d,m},a}(p/M_{d,m})D_{v_{d,m},a}(-p/M_{d,m}),$$

using the cumulative drive of Section 7. A single step with patch P_m and side L_m gives the exact identity

$$u_{d+1,m,a}(p) = |\operatorname{pullLine}_{P_m,a}(p/L_m)|^2 u_{d,m+1,a}(p/L_m).$$

Each rail and centered plank is an average of unit complex numbers on real momenta, up to a unit phase. Its modulus is at most one. Thus the factor in front lies in $[0,1]$. Starting with $u_{0,m,a} = 1$, induction on d in the displayed recurrence shows

$$0 \le u_{d+1,m,a}(p) \le u_{d,m,a}(p) \le 1.$$

This proves convergence for every real p, also after any fixed reciprocal translation of p.

The noise list of $v_{d+1,m}$ is the list for $v_{d,m}$ with one additional train appended. This follows by induction from the list rule under one hop: both old lists undergo the same hop, with the same top element prepended. Each additional real-momentum feed is nonnegative. The accumulated noise $N_{d,m,a}(p)$ is therefore nondecreasing. The uniform noise estimate of Section 7 bounds it above, independently of d for p in the base cube. At depth one it is one. Its limit $N_{m,a}(p)$ consequently satisfies

$$1 \le N_{m,a}(p) < \infty.$$

We also need the high-alias part $H_{v_{d,m},a}(p)$. Extend its finite sum to $\mathbb{Z}^4$ by setting it to zero outside the centered representatives of the nonzero residues of $M_{d,m}$. Each fixed $z \neq 0$ is represented for all sufficiently large d. Put $q = p + 2\pi z$. The corresponding summand is then exactly

$$\frac{u_{d,m,a}(q)}{d_{M_{d,m}}(q)}.$$

The numerator has a limit by the preceding monotonicity. The identity

$$M(e^{iq_a/M} - 1) \longrightarrow iq_a$$

gives $d_M(q) \to |q|^2$. For p in the base cube and $z \neq 0$, $q \neq 0$, so its denominator has a nonzero limit. The uniform alias bounds already proved give one summable majorant

$$C \prod_{a=1}^{4} (1 + |z_a|)^{-2}$$

for these extended summands. The sum of this product is finite by the one-dimensional inverse-square series. Dominated convergence for the series proves convergence of the high-alias part. At $z = 0$ the extended summand is identically zero; no denominator at the origin is used in this argument.

For $p \neq 0$ the full directional covariance multiplier is

$$L_{d,m,a}(p) = \frac{u_{d,m,a}(p)}{d_{M_{d,m}}(p)} + H_{v_{d,m},a}(p) + k^{-1} N_{d,m,a}(p), \qquad k > 0.$$

It has a finite real limit $L_{m,a}(p) \ge k^{-1} > 0$. The pulse and dual-pulse symbols are the rational matrix expressions in these multipliers given in Section 6. Their directional denominators have just been shown to have positive limits. The remaining scalar denominator has the form

$$\sum_a \frac{|e^{ip_a} - 1|^2}{L_{m,a}(p)}.$$

It is strictly positive off the origin in the base cube: every term is nonnegative and some numerator is nonzero. Consequently every entry of both symbols converges there. At zero their normalized values are the identity for every cutoff. The uniform strip bounds of the free-kernel estimates and the first subsection now give convergence of both rows on the cover, with a retained exponential bound.

137.3 The adjacent one-form and two-form transports

Fix the block side $L = L_m$. Unlike the mass side in the preceding high-alias sum, L does not change with the terminal cutoff. For $p \neq 0$ in the base cube, all the finitely many adjacent aliases $q_l = (p + 2\pi\ell_L(l))/L$ are nonzero real momenta in that cube. Apply the directional limit above at the coarse momentum p and the fine momenta q_l. The principal and nonprincipal transport formulas of Section 8 then have convergent factors.

We spell out the additional denominator check in those formulas. The finite alias denominator is

$$\mathcal{V}_N(p) = \sum_l |m_{P_m}(q_l)|^2 \sum_a L_{N,a}^{\mathrm{scaled}}(q_l).$$

Here $L_{N,a}^{\mathrm{scaled}}$ is the fine directional multiplier with the fixed positive scaling L^{-2}. Every summand has a nonnegative real limit, each inner sum has a strictly positive limit, and the principal patch mass $m_{P_m}(p/L)$ is nonzero on the base cube. Hence $\lim_N \mathcal{V}_N(p) > 0$. All other divisions in the transport formulas are by the positive directional or scalar denominators just checked. Finite matrix products, sums and division by these nonzero limits prove convergence of every alias transport. Fourier reconstruction over the finite set of L^4 aliases proves convergence of the adjacent transport symbols. Their uniform row bounds again identify unique cover limits. Possible exceptional momenta at the origin are removed by continuity of the cluster symbols, not by taking a quotient there.

137.4 Convergence of amendments, exterior lifts and roots

Several operations preserve the cover convergence just established. Sums, transposes and scalar multiples do so directly. For products, take the minimum of two positive available widths; the convolution of rows has a uniform bound there, multiplied by the finite middle coordinate count. Its symbol is the product of the two symbols, so the first subsection identifies its unique limit. For an inverse row with a uniform bound, suppose $K_N^{-1} K_N = I$ on every torus. Every cluster limit Q of the inverse rows satisfies $QK = I$ there by finite matrix multiplication. A square matrix with a left inverse is invertible, so that inverse is unique. Recovery from all periods then identifies Q on the cover, and proves convergence.

The exterior lifts in Section 11 are finite combinations of coordinate evaluation at $p_a = 0$, fixed exterior matrices and divided differences

$$\frac{\widehat{K}_N(p) - \widehat{K}_N(p|_{p_a=0})}{e^{ip_a} - 1}.$$

Their uniform bounds at a smaller positive width were proved there. For $p_a \neq 0$ the displayed expression converges, because both numerator terms do and the denominator is fixed. Thus any two cluster symbols agree off that coordinate hyperplane, and by continuity everywhere. This proves convergence of the divided row. Repeating the argument through the finitely many axes proves convergence of the one-sided and two-sided exterior lifts, with all their fixed matrix factors.

It remains to justify the positive square roots; convergence of an arbitrary matrix factor would be false. Suppose P_N have a common exponential row bound, their torus matrices are positive

semidefinite, and $P_N^{\mathsf{T}} P_N$ converge on every torus. Two cluster limits P, Q have positive semidefinite torus matrices, since symmetry and nonnegativity of each quadratic form pass to the limit. Their squares are equal. A positive semidefinite square root is unique: if $A \geq 0$ and $A^2 = B$, then A commutes with B; on each eigenspace of B, diagonalizing the symmetric restriction of A forces all eigenvalues to be the nonnegative square root of that eigenvalue of B. Hence A is fixed on every eigenspace. Apply this to P and Q on each torus, then recover the cover coefficients. The unique-cluster-limit argument proves convergence of P_N itself.

The amended beds have exactly these properties. If B_N is the dual-pulse row, its amended inverse precision is

$$A_N = B_N + l(d_2 B_N)^{\mathsf{T}}(d_2 B_N), \qquad l \geq 0,$$

and its precision is $G_N = A_N^{-1}$. Products and the inverse argument give their limits. The root row has square $G_N - cI$, where c is the fixed direct clipping coefficient. Its uniform strip bound and positive semidefinite torus values are supplied by Section 9, so the positive-root argument applies. The original pulse, dual pulse, amended precision, its inverse and the root row therefore converge together.

For the curvature root, use the explicit amended row of Section 11:

$$R_N = D_N^{\mathsf{T}}(C_N + lI - hS_N^{\mathsf{T}} S_N) D_N, \qquad D_N = (I + l d_2 B_{c,N} d_2^{\mathsf{T}})^{-1}.$$

Here l, h are the fixed coarse and fine amendment coefficients; C_N, S_N are the two exterior lifts constructed there. Their convergence follows from the operations just proved. The positivity and uniform strip price of R_N and its positive root are already established. Since that root's square is R_N, it too converges. All of these statements concern the actual chosen positive roots; no new choice of signs or bases has been made.

137.5 Completing the retained tower

The full precision factors $J_{f,N}, J_{c,N}$ consist of the fixed direct block and the root rows, so they converge on every torus. Write $G_{f,N}, G_{c,N}$ for the corresponding precisions and $R_{f,N} = G_{f,N}^{-1}$, $R_{c,N} = G_{c,N}^{-1}$. With the two-form transport T_N and curvature root P_N, the exact tower blocks are

$$\begin{aligned} M_{F,N} &= J_{f,N} R_{f,N} T_N J_{c,N}^{\mathsf{T}}, \\ M_{N,N} &= k^{-1/2} d_{1,c}^{\mathsf{T}} J_{c,N}^{\mathsf{T}}, \\ M_{O,N} &= \begin{bmatrix} I - J_{c,N} R_{c,N} J_{c,N}^{\mathsf{T}} \\ P_N d_{2,c} R_{c,N} J_{c,N}^{\mathsf{T}} \end{bmatrix}. \end{aligned}$$

These are the same formulas as (57), on each selected physical torus of side $n_m(p)$, with its prescribed fine partner. Here $n_m(p)$ denotes the period sequence fixed at scale m by the schedule. Here p is fixed while N increases. The committed transition identities apply on precisely these tori. Every finite matrix factor has a limit there, hence so does each block. Their uniform strip bounds were purchased in the retained tower construction. The period-recovery criterion from the first subsection gives convergence of their cover coefficients. Use the selected sides above, which tend to infinity as p does; no assumption on convergence of an unrelated periodic realization is needed.

Together with the one-form predictor and two-form transport, these are precisely the five retained tower rows. Thus at every fixed scale all five bed rows and all five tower rows converge, and each family keeps some positive exponential width and finite budget. The purchases of widths and budgets are made at the fixed stages, before choosing the later terminal cutoff. The eventual normalization changes discussed in Section 169 preserve these limits. This supplies the free-horizon convergence input of that section from the actual recursively selected free kernels.

138 Attaching finite-cutoff paths on the selected stationary route

This section records the selected stationary route from its horizontal annulus estimate through finite-cutoff path attachment. It separates three different jobs: producing the annulus estimate, using it in a one-step path construction, and assembling the resulting transitions. Only the last job is a finite induction. The quantitative inputs to the one-step construction are stated at their actual interface and are not inferred from that induction.

138.1 The one-step path receipt

A path is a collection of row, loop, cubic, and quartic coefficients. For a clip p, width τ, row price A, and total price Λ, the predicate $R.\mathrm{Bounds}(p, \tau, A, \Lambda)$ is the source path budget: each exponentially weighted row family has sum at most A; the loop atom-budget family has sum at most Λ; and the cubic and quartic atom-budget families each have sum at most A. In addition, every atom is invariant, the row symbols satisfy the metric identity, row-symbol terms are summable, and the loop coefficient and its first moment have summable terms and sum to zero. The attachment predicate $\mathrm{Attached}(B, R)$ says that the path coefficients at each plaquette and cover offset sum to the bed's root entry. In particular, these are simultaneous global summability conditions, not merely pointwise bounds.

The precise annulus estimate, denoted $\mathrm{HorizCalc}(G, \sigma, \varepsilon, C, Z, \tau, A)$, has the following quantifiers. It supplies an integer $L_{\min}$ and $D_0 \geq 0$. For every odd side $L \geq L_{\min}$, every side sequence $m_i \to \infty$, every bed B, clip p, path R, and Λ, assume

$$R.\mathrm{Bounds}(p, \tau, A, \Lambda), \quad \mathrm{Attached}(B, R), \quad B.\mathrm{metric.strip}(\sigma, C), \quad p.\mathrm{direct} \leq Z,$$

along with $\mathrm{Floor}(B, p, n, \varepsilon, C)$ for every n and $\mathrm{Sworn}(B, m_i, G)$ for every i. Then there are one picture Y, nonnegative firm constants v, E, and a single error sequence $e_i \to 0$ such that

$$\mathrm{Firm}(v, E, Y_{ij}) \quad \text{for every } i, j, \qquad \mathrm{Recorded}(Y, \mathrm{PoleTest}(p, R, \mathrm{inch}(L)), \mathrm{cents}(m_i)e_i) \quad \text{for every } i,$$

and $|\alpha_j^{\mathrm{ann}}(Y) + \lambda_G \log L| \leq D_0$ for every color j. Here Recorded is the finite-mesh test identity up to its stated error; no passage to a limit in i is part of this assertion.

The selected producer of this estimate is the theorem pouring. Its proof first obtains a Wilson reference H with positive strip width and finite fee from vintageWilson. The three Wilson jets used in the construction are exactly orders one, two, and three; the order-zero remainder construction jig turns them into a single reference whose fee controls all required jets. Next, saunter constructs a second fee-controlled series V for the difference between the desired finite-volume action and the Wilson reference. It applies the supplied path bound, attachment, bed strip, floor, and sworn hypotheses to prove the analytic binding identities for that difference. The combined series $F = H + V$ therefore binds the desired action, has a common positive width $w = \min(w_H, w_V)$, and has summed fee. The exact Taylor tickets follow by adding the two binding identities and applying the fee-to-ticket estimate; the finite-mesh error is generated by the common sequence $\mathrm{cents}(m_i)e_i$ and its scalar factor tends to zero along every admissible side sequence.

For the coefficient of the logarithm, astronomy applies to the same fee-controlled, grey reference. It yields constants $A_1, A_2 \geq 0$ and, for every admissible local perturbation with scale μ, the two uniform radial estimates

$$|g(x)| \leq \frac{A_1}{(\|x\| + \mu)^4}, \qquad |g(x) - h_G(x)| \leq \frac{A_2}{\|x\|^3} + \frac{A_2\mu^2}{\|x\|^6} \quad (\mu \leq \|x\| \leq 1).$$

The first is used on the unit ball in the four-dimensional momentum space; the second compares the annular integrand with the continuum pole. These are the concrete inputs to the following estimate, rather than a general claim that the determinant varies continuously.

Lemma 138.1 (Radial logarithmic coefficient bound)**.** *Assume $0 < \mu \le 1$, $A_1, A_2 \ge 0$, and let $E_4 \subset \mathbb{R}^4$ be a measurable finite-volume region containing the unit ball. Let g be real measurable and integrable on E_4, and let h_G be integrable on the annulus $I_\mu = \{x : \mu < \|x\| \le 1\}$. Assume the radial estimates*

$$|g(x)| \le \frac{A_1}{(\|x\| + \mu)^4} \quad (x \in E_4), \qquad |g(x) - h_G(x)| \le \frac{A_2}{\|x\|^3} + \frac{A_2\mu^2}{\|x\|^6} \quad (x \in I_\mu).$$

Assume also the coefficient and pole identities

$$\alpha_j^{\mathrm{ann}}(F) = c_{\mathrm{wt}} \int_{E_4} g(x)\,dx, \qquad c_{\mathrm{wt}} \int_{I_\mu} h_G(x)\,dx = -\lambda_G \log \mu^{-1}, \quad c_{\mathrm{wt}} > 0.$$

For the productive picture F, set

$$C_{\mathrm{rad}}(A_1, A_2) = A_1 \left(\frac{\pi^2}{2} + \mathrm{vol}(E_4) \right) + 3\pi^2 A_2.$$

Then

$$|\alpha_j^{\mathrm{ann}}(F) + \lambda_G \log \mu^{-1}| \le c_{\mathrm{wt}} C_{\mathrm{rad}}(A_1, A_2).$$

Proof. Write E_4 for the finite-volume four-dimensional integration region, $B_1 = \{x : \|x\| \le 1\}$, $B_\mu = \{x : \|x\| \le \mu\}$, $I = \{x : \mu < \|x\| \le 1\}$, and $Q = E_4 \setminus B_1$. Polar coordinates give $\mathrm{vol}(B_r) = \pi^2 r^4/2$. On B_μ, the first radial estimate implies $|g(x)| \le A_1 \mu^{-4}$, so $|\int_{B_\mu} g| \le A_1 \pi^2/2$. On Q, one has $\|x\| \ge 1$, and hence $|g(x)| \le A_1$; therefore $|\int_Q g| \le A_1 \,\mathrm{vol}(E_4)$.

On I, integrate the second estimate in radial coordinates. The two radial integrals are

$$\int_I \|x\|^{-3}\,dx = 2\pi^2(1-\mu), \qquad \int_I \|x\|^{-6}\,dx = 2\pi^2 \frac{\mu^{-2} - 1}{2}.$$

Thus

$$\int_I |g - h_G| \le 2\pi^2 A_2 (1-\mu) + \pi^2 A_2 \mu^2 (\mu^{-2} - 1) \le 3\pi^2 A_2.$$

The pole integral is exactly the logarithmic term: $c_{\mathrm{wt}} \int_I h_G = -\lambda_G \log \mu^{-1}$, by the angular average of the gauge pole. The Fourier formula for α_j^{ann} is $c_{\mathrm{wt}} \int_{E_4} g$. Splitting the integral over $B_\mu \cup I \cup Q$ and using the three absolute bounds proves the claim. The angular identity and the two radial integrals are computed in the source by polar integration; no uniformity beyond the displayed radial estimates is used. □

138.2 The stationary chart receipts

The chart data consumed by the path interpolation is produced at each fixed finite volume. It is useful to give this construction separately: the positive Hessian gap, critical chart, and installed-matrix approximation enter the later path theorem as supplied data, but their source is an implicit-function argument, not an additional limit assumption.

Start with the planted comparison for a coarse site i and a fine edge $e = (x, a)$. Let L be the blocking side and write $\ell = L.\mathrm{val}$; let p_i be the planted root frame and P_i the local predictor. The planting estimates give nonnegative constants C_0, D_0 and, with

$$\alpha_0 = 2\max(1, \mathrm{predictorCut}^{-1})(2C_0 + D_0), \quad u_0 = (8\ell + 1)D_0\ell, \quad v_0 = C_0\ell^2 + \alpha_0\ell^{-2} + C_0\ell^2,$$

the finite-volume edge estimate

$$\operatorname{chord}\big(U_e(\text{replica}\,V)_e^{-1}\big) \le (u_0+v_0)\ \operatorname{port}_i(U) + \mathbf{1}_{\{i\neq i+\text{parentStep}(e)\}}\operatorname{wire}(P_i,V,e).$$

Here the first term is obtained by following the forest root walk: its length is at most the patch radius, and the planted bound charges each walk edge by $D_0\ell$ times the port. If the edge stays in its parent cell, the lifted path closes at the root, so this is the entire estimate. If it crosses to a neighboring parent, the endpoint root-to-frame chords each cost at most $C_0\ell^2$ port, while the predicted edge has chord at most $\alpha_0\ell^{-2}$ port. The triangle inequality adds these three terms, and the residual is precisely the displayed wire term. This is the two-case proof of the planted comparison; the tree walk and crossing endpoint costs are not hidden in a generic locality claim.

For a fixed step, summing the squared port terms over coarse sites uses Cauchy–Schwarz twice: first over the box offsets and plaquette types, then over the box offsets in the site sum. Injectivity of the map from a coarse site/plaquette pair to its embedded fine plaquette prevents extra multiplicity. The resulting factor is exactly the product of the box cardinality, the plaquette count, and the site count appearing in the source's curl factor. The probe wire is summed over coarse bonds and paid by the second factor, the probe factor. Combining the two with the floor bound yields

$$\sum_e \|v_e\|^2 \le \beta_\ell\,\mathcal{D}(v) + \gamma_\ell\,\mathcal{P}(v),$$

where the source coefficients are exactly

$$\beta_\ell = (2C^2)|\operatorname{Site}(L)|\,|\operatorname{Axis}|(|X|\,|\operatorname{PlaqAxes}|)\,|X|, \qquad \gamma_\ell = 2|\operatorname{Site}(L)|,$$

with X the fixed patch-offset set and C the edge-comparison constant. Thus ℓ is the blocking side cardinality, not the coarse torus side. Here $\mathcal{D}$ is the squared physical color derivative and $\mathcal{P}$ the squared predictor penalty. The finite-volume Hessian identity expresses the tested Hessian as the target metric form plus penalty $\mathcal{P}(v)$; the floor bounds the first term below by $a\mathcal{D}(v)$. Assume $a>0$, penalty >0, and $\beta_\ell,\gamma_\ell\ge 0$. Set

$$\eta = \frac{\min(a,\text{penalty})}{1+\beta_\ell+\gamma_\ell}.$$

Then $\eta>0$, $\eta\beta_\ell\le a$ and $\eta\gamma_\ell\le$ penalty, and the preceding two inequalities give the strict price gap

$$\eta\sum_e \|v_e\|^2 \le D^2\mathcal{T}(0)[v,v].$$

This is the quantitative Hessian lower bound used by the chart construction.

The remaining chart receipts follow in four steps. First, the exponential coefficient budgets in the path bound make each uncut row and atom series converge uniformly on a complex neighborhood of the physical field. Termwise analyticity and summability therefore give smoothness of the finite-volume main action. The chart compatibility hypothesis supplies the required analytic composition along every smooth field chart. Second, in a forest chart the lifted field agrees near the origin with the ordinary field coordinates; the tested action is the flat action composed with this lift. Differentiating this germ identity and using the finite-volume Hessian identity gives the prismatic second variation and the comptroller bilinear form.

Third, fix one allowed choice of local test and write $F(x,z)$ for the tested action in coarse coordinates x and forest coordinates z. At $(0,0)$, $F=0$ and $DF=0$. The price gap implies that the vertical Hessian $D_z^2F(0,0)$ is positive definite, hence invertible in the finite dimensional forest

space. The implicit-function theorem applied to $D_z F(x,z) = 0$ gives a smooth branch $z = \rho(x)$ with $\rho(0) = 0$. All local choices define the same tested-action germ on the branch, so the same ρ is critical for each of them.

Fourth, differentiate the stationarity equation at zero:

$$D^2 F(0,0)\big[(x, D\rho(0)x), (0,v)\big] = 0 \quad \text{for every forest direction } v.$$

The comptroller formula identifies this bilinear form as the sum of the bed metric term and the penalty pairing of the predictor residual. The installed tower matrix satisfies that same linear equation by the diagonal two-scale trial. Uniqueness, from invertibility of the vertical Hessian, therefore yields $D\rho(0)x = \text{relative}(W, x)$, the guessed-chart identity. Finally, the chain rule at the stationary branch removes the term involving $D_z F(0,0)$; the same comptroller formula and the diagonal tower identity give the target bed metric as the Hessian of the reduced action. This is the heat identity. These arguments prove the fixed-period stationarity, installed-tangent, and reduced-Hessian identities used by the transition; they do not assert that their neighborhoods or Hessian gaps are uniform in the period.

The exact interface for the one-step construction is the following. Fix a kernel choice, a phase J, positive ε, w, δ, and nonnegative D, Z, C, Z'. The source theorem chooses a reel R_1 satisfying $R_1.\text{Price}(s)$, nonnegative κ, and positive s. For every $\tau > 0$ and $A \geq 0$, assume $\text{HorizCalc}(G, w, \varepsilon, D, Z, \tau, A)$. It then chooses $d \geq 0$, $\gamma > 0$, a period threshold u, and functions $\beta(L)$, $\Lambda_0(L)$. For every patch with phase J and odd side at least u, every pair of side sequences m_i, n_i with $m_i \to \infty$, every path and beds/towers, the following hypotheses are required:

- an armada whose every constituent step is a larger step, and the two bed hulls (metric and root strip bounds, floors, and sworn conditions);
- a path R attached to the coarse bed with $R.\text{Bounds}(p, \tau, A, \Lambda)$, the diagonal two-scale trial, and the stated header/energy bounds;
- for each forest chart F_i, the exact receipts

  $$\text{PriceGap}(F_i, p, R, \eta), \qquad \text{Critical}(F_i, p, R, \rho_i), \qquad \text{Guessed}(F_i, W_i, \rho_i),$$

  and the heat bound;
- the local-ratio chart and the specified direct-coefficient cutoffs for both clips. The annulus picture is produced inside this proof from HorizCalc; it is not an additional input.

The remaining scalar hypotheses are $1 \leq \Lambda$, $0 < \eta$, the phase identity, the period threshold, and the two direct clip bounds. The armada must have the selected kernel and all its patches must be branched. These conditions are part of the one-step input, as are the hypotheses used to define each forest chart.

The content of these hypotheses is precise. PriceGap is the positive lower bound $\eta \sum_e \|v_e\|^2 \leq D^2\mathcal{T}(0)[v,v]$ for the tested action $\mathcal{T}$. Critical asserts that the chart ρ is smooth, vanishes at zero, and is stationary for every tested action. Guessed says that the derivative of $t \mapsto \rho(tx)$ at zero is the chart's linear prediction from the installed matrix. The preceding subsection proves these receipts at each fixed finite volume by the planted sensitivity estimate, the Hessian comparison, and the implicit-function theorem. The output ClassicalMatch is instead a conclusion: for every Taylor order $j \leq 4$ and every local choice, the tested action along ρ and the main action of the new path have equal jth derivatives at the origin.

The one-step construction gives a path Y and scalar α with

$$Y.\mathrm{row} = \mathrm{rowPath}(T), \qquad \mathrm{Attached}(T, Y),$$

and all four source components of Y.Bounds obey the requested row/loop/cubic/quartic budgets whenever $0 < \eta' \le s$, $\kappa \le A'$, $\Lambda_0(L) \le \Lambda'$ and $\Lambda/128 \le \Lambda'$. It also gives

$$|\alpha + \lambda_G \log L| + 1 \le D_1.$$

It gives the order-four classical matching identity and, whenever $\beta(L) \le \Theta$ and $c' \le \gamma$, the finite-cutoff bound with amplitude Θ and decay rate c'. The proof constructs the classical match by reducing the tested action to the new pole test and invoking fourth-order exactness of the local response; the exponential estimate comes from the funded Fourier ticket and the exponential rate comparison below. The bounded path is obtained by adding the blank path to the produced row and higher-degree coefficients, then applying the sum-budget estimate to each degree separately.

The scalar α is the local quadratic coefficient of the constructed picture. Its normalization is compared with the annular coefficient $\alpha_j^{\mathrm{ann}}(Y_\pi)$ by a finite response identity. The response identity uses two determinant pictures. The first is the half-medal response minus the reference determinant; the second is the half-rank-up response minus the same reference determinant. The source's screened response estimate bounds the difference of their quadratic coefficients by

$$\left|\alpha_j^{\mathrm{ann}}(\mathrm{medal} - \mathrm{reference}) - \alpha_j^{\mathrm{ann}}(\mathrm{rank\text{-}up} - \mathrm{reference})\right| \le B_{\mathrm{det}},$$

where B_{det} is the displayed finite reserve price $\mathrm{biteCost}(\eta/2, \mathrm{reserveCost}(\cdots))$. The rank-up picture is recorded against the pulled-back pole test at the coarse meshes; the annulus picture is recorded against the pole test itself at the cofinal fine meshes, with error tending to zero. These are different finite-volume tests, so their quadratic coefficients cannot be identified by comparing a single test value. The exact comparison is an alias argument.

Let L be the blocking ratio between the two periods, $\mu = L^{-1}$, $X(p)$ the analytic bridge between the two period tests, S the annulus picture, and H the rank-up picture. For a picture P, write $\mathcal{R}_P(p)$ for its matrix of quadratic test coefficients. The finite-mesh recording identities and period-stride approximation imply, for every real momentum p,

$$\mathcal{R}_H(p) = L^4 \sum_{\ell \in \mathbb{T}_L} \mathcal{S}_S(\mathrm{alias}_\ell(p), N_X(-p, -\ell), N_X(p, \ell)),$$

where N_X are the bridge's nose maps and $\mathcal{S}_S$ is the annulus matrix pairing. To prove this identity, choose actual mesh momenta $p_i \to p$ and stride representatives for each alias. The normalized recording error for the annulus test tends to zero; the pulled-back rank-test error also tends to zero after its period normalization. The bridge factorization expresses each coarse test as the finite sum over its fine aliases. At a fixed pair of aliases the recorded mixed coefficient converges to the corresponding entry of $\mathcal{S}_S$. If the aliases do not pair as $\ell = -k$, the two mesh centers remain distinct and the coefficient tends to zero. The paired case gives the displayed entry. There are finitely many aliases at this fixed period, so the limits pass through the sum. This is the content of the finite-mesh transfer; it uses both recorded families, not equality of their unsmeared tests.

The hypotheses used in this transfer are the ones present in the selected rank/pole comparison: X is analytic and has the required uniform strip; the same matrix maps satisfy the exact coarse-to-fine premise; both metric rows have their stated strip bounds; the coarse row has the required floor, attachment, and sworn conditions; the rank response is screened; the cofinal recorded errors tend to zero; and the period meshes are related by the source's injective replication map. These

assumptions give the normalized pulled-back recording and the noise-free alias identity. For the Taylor step below, additionally use the exact second-bed identity ($N_X(0,0) = \mu I$), the transverse primal identity, and the annulus picture's second-order profile. The nonprincipal alias estimate uses the source contractibility, firm bounds, and the estimate for the separated noncentral noses; it does not follow from the pole recordings alone.

The factor L^4 is essential. Here L is the blocking ratio between the two periods, not the coarse physical period. Let $\ell = 0$ denote the principal alias. The second bed identity fixes the zero-momentum bridge nose at $N_X(0,0) = \mu I$. The annulus picture has a second-order Taylor remainder, so its principal pairing differs from $\mu^2 \mathcal{R}_S(\mu p)$ by $O(|p|^3)$. For a nonprincipal alias, the full nose need not vanish to second order. The source high-halo estimate controls its horizontal and transverse components; contractibility then replaces the full nose by its horizontal component inside the pairing, without changing that pairing. The transverse primal identity and the floor bound make this horizontal representative vanish to second order. Only after this replacement does pairing with the annulus quadratic profile give an $O(|p|^4)$ contribution. Summing the finitely many aliases therefore gives

$$\mathcal{R}_H(p) - L^4 \mu^2 \mathcal{R}_S(\mu p) = O(|p|^3).$$

Both sides are analytic near zero. Their Taylor coefficients through order two must agree: along each ray, a nonzero coefficient of degree at most two cannot be $O(|p|^3)$; polarization recovers the full quadratic form. The second derivative of the rescaled principal term is $L^4 \mu^4 D^2 \mathcal{R}_S(0)$, and $L^4 \mu^4 = 1$. Thus

$$\alpha_j^{\mathrm{ann}}(\text{rank-up} - \text{reference}) = \alpha_j^{\mathrm{ann}}(Y_\pi).$$

Thus $|\alpha - \alpha_j^{\mathrm{ann}}(Y_\pi)| \le B_{\mathrm{det}}$. If the logarithmic coefficient bound from the annulus is D_0, then

$$|\alpha + \lambda_G \log L| + 1 \le B_{\mathrm{det}} + D_0 + 1.$$

The path construction chooses $d = B_{\mathrm{det}} + D_0 + 1$ (or a larger bound). This is the kinetic normalization step: it controls the finite-volume quadratic coefficient by comparison with the continuum logarithmic coefficient at the actual annulus scale. No unquantified limit in the cutoff is used here.

The exact order-four match is proved from the fifth-order remainder ticket. Let $f(x)$ be the tested action along the critical chart and let $g(x)$ be the main term of the new path. The response construction gives smoothness near zero and a ray estimate $f(tx) - g(tx) = O(|t|^5)$ for each direction x. Taylor's theorem in the single real variable t then forces the coefficients of t^j to agree for $j = 0, 1, 2, 3, 4$. Those coefficients are the diagonal values of the symmetric j-linear derivatives at zero. The polarization identity for symmetric multilinear forms shows that equality on all diagonals implies equality of the full derivatives. This proves ClassicalMatch at every requested order; the input critical chart and the price gap are used to obtain the smooth response and the response ticket, and are not deduced from the path identity.

For the path budget, the row component is the bed root array itself. Its exponentially weighted row sum is bounded by

$$\mathrm{rowPrice}(s', M) = |\mathrm{PlaqAxes}|\, M e^{4s'} \le \mathrm{rowPrice}(w, M), \qquad s' \le w,$$

because a row entry at offset y is charged by the bed root strip and the walk used to encode the slot has length $|y|_1 + 4$. The remaining three components are the loop, cubic, and quartic atom families constructed by the response step. Their weighted absolute budgets are separately bounded by the three output prices; invariance, zero loop jets, and exact metric normalization are preserved by the construction. The sum-budget axiom implies each family is summable, and the row-layout

identity plus the floor bound gives the metric clause. This is the explicit content behind the claimed *Y*.Bounds, not an appeal to pointwise coefficient control.

The finite-cutoff ticket is an absolute bound on the weighted remainder, with original rate k at least the requested rate s. If its unweighted amplitude is bounded by Γ, and the final amplitude budget is $\Lambda_1 \geq 1$, then for every cutoff $a \geq 0$,

$$|wR|e^{-ka} \leq \Gamma e^{-ka} \leq \Gamma e^{-sa} \leq \Gamma\Lambda_1 e^{-sa}.$$

The proof uses only $s \leq k$ and nonnegative amplitudes. This is the finite-cutoff exponential-rate comparison used in the one-step conclusion.

The source path budget is built in degree-specific steps. First, the blank row path copies the target bed roots; if those roots have strip price M, its row budget is $\mathrm{rowPrice}(w, M) = |\mathrm{PlaqAxes}|Me^{4w}$. Second, local determinant response supplies cubic and quartic coefficients. If the cubic response price is A', the quartic response price is C', and the determinant correction is paid at price D_*, the source jet estimates give the exact price bounds

$$\frac{A'}{3!}, \qquad \frac{C' + 3A'D_*A'}{4!}.$$

The factors 3 and 3!, 4! arise from the three pairings of four derivatives and the Taylor factorials. In the determinant calculation the cubic response is paired with the determinant correction in each of three channels; each product is bounded by $A'D_*A'$. These coefficients are recorded against the third- and fourth-order Taylor jets, respectively. Third, the loop correction is obtained from the Fourier coefficient of the quadratic response after subtracting its scalar quadratic term. The row path and these three atom families are assembled into one path. Its metric identity follows from the row-layout sum and the floor condition; loop normalization and invariance are preserved coefficientwise. This is why the one-step construction has a separate row price and a separate total higher-degree price, rather than charging all coefficients to a single norm.

The final budget invoice in that estimate is elementary but essential. Let $\gamma \geq 0$, $V, \Lambda \geq 0$, and suppose $\Lambda/128 \leq \Lambda'$, the terms arising before the annulus correction obey

$$2\big(C + (d + |y|)V\big)\gamma \leq \Lambda', \qquad |x + y| + 1 \leq d, \qquad B\gamma \leq \tfrac{1}{256}.$$

Then

$$\big(C + |x|V + B\Lambda\big)\gamma \leq \Lambda'. \tag{665}$$

Indeed $|x| \leq d + |y|$, by $x = (x + y) - y$, and

$$(C + |x|V + B\Lambda)\gamma \leq \big(C + (d + |y|)V\big)\gamma + B\gamma\Lambda \leq \Lambda'/2 + \Lambda'/2.$$

The final term uses $B\gamma \leq 1/256$ and $\Lambda/128 \leq \Lambda'$, so $B\gamma\Lambda \leq \Lambda'/2$. This is the only step where the auxiliary width of the horizontal picture is traded against the outer row budget.

The exponential cutoff invoice is similarly explicit. If $\Lambda_1 \geq 1$ and $|wR| \leq \Gamma$, with $s \leq k$ and $a \geq 0$, then

$$wRe^{-ka} \leq \Gamma\Lambda_1 e^{-sa}. \tag{666}$$

The bound $|wR| \leq \Gamma$ gives $wR \leq \Gamma$, while $e^{-ka} \leq e^{-sa}$; multiplying the latter by the nonnegative $\Gamma\Lambda_1$ and using $\Lambda_1 \geq 1$ proves the claim. This is the rate comparison used to preserve a finite-cutoff tail when the controlling rate is weakened.

138.3 Canonical representatives for all-period path jets

The path realization uses a closed-subspace projection to choose one representative of prescribed jets at every tested period. Let H be a real Hilbert space, $S \subset H$ a closed subspace, and for each integer $n \geq 0$ let $J_n : H \to E_n$ be a continuous linear map into a real Hilbert space E_n. Define the joint invisible subspace

$$K = S \cap \bigcap_{n \geq 0} \ker J_n, \qquad \mathcal{R} = S \cap K^{\perp}.$$

Both are closed: each kernel is closed, and intersections of closed subspaces are closed. Given $C \in S$, orthogonally project onto K and put $C_* = C - \Pi_K C$. Since $\Pi_K C \in K \subset S$, one has $C_* \in S$; projection orthogonality gives $C_* \in K^{\perp}$, and hence $C_* \in \mathcal{R}$. As $J_n(\Pi_K C) = 0$, every prescribed jet is preserved, $J_n(C_*) = J_n(C)$, and the Pythagorean identity gives $\|C_*\|^2 = \|C\|^2 - \|\Pi_K C\|^2 \leq \|C\|^2$.

This representative is unique. If $C_*, D_* \in \mathcal{R}$ have the same jets for every n, then $C_* - D_* \in S$ and all its jets vanish, so it lies in K. It also lies in $K^{\perp}$, whence $\|C_* - D_*\|^2 = 0$ and $C_* = D_*$. Apply this with the admissible correction series as S and the derivatives at the tested periods as the maps J_n. The source admissible subspace is closed, and the joint invisible subspace is exactly the common kernel used in the orthogonal-complement selection. When the tested side sequence exhausts all physical periods, this proves all-period jet injectivity on the chosen orthogonal complement. When it does not, the conclusion is exactly for the specified test sequence. This is the canonical realization argument: it uses the orthogonal complement of the common jet kernel, not an assumed injectivity of the jet map on the original space.

138.4 The arbitrary-input factor-64 envelope

The numerical factor used by the selected global caller is an allowance amplification at the input to the transition supplier. We give both layers of the finite reindexing, since the two maxima/minima use different pairs.

For each fixed output index i, the one-step theorem is first called at both neighboring input indices $j = i$ and $j = i + 1$. If the two calls return widths $w_{i,i}, w_{i+1,i}$ and row/cubic/quartic incoming-price thresholds $v_{i,i}, v_{i+1,i}$, the common parameters are

$$s_i = \min(b_i, w_{i,i}, w_{i+1,i}), \qquad A_i = \max(E_i, v_{i,i}, v_{i+1,i}).$$

Here b_i, E_i are the width and price supplied by the blank-path estimate. For each of the two calls, s_i is no larger than its width and its price threshold is no larger than A_i, so the same pair of finite-volume estimates can be used at either adjacent input scale. The flux theorem is universal in the admissible incoming path, bed, tower, armada, and chart; therefore this uniformization does not choose a favorable incoming object.

Next fix the requested positive width sequence τ_j and nonnegative incoming path-price allowance C_j. For each fixed input index j, the one-step calls indexed by the two possible output indices $i = j$ and $i = j - 1$ are unified by a second set of extrema: take the maximum of their drift bounds, period thresholds, and side-dependent cutoff amplitudes, and the minimum of their positive decay rates. At $j = 0$, the predecessor is interpreted as 0, so the single available call is used twice. These choices make one set of output data valid whether the selected transition uses the same-scale receipt or the preceding-scale receipt. This is the second neighbor reindexing in the source construction; it is distinct from the width/price extrema at fixed output index above.

In the public application, the arbitrary allowance sequence C_j is passed as $64A'_j$, where A'_j is any nonnegative sequence dominating the base path-price sequence. Thus the one-step conclusion is established for an incoming path budget $64A'_j$; the factor 64 is part of the input allowance, not an

output estimate obtained from a smaller path. At the selected level, A'_j is the profile path price, so the row, cubic, and quartic incoming budgets are $64\,\mathrm{paths}_j$. The loop halo is separately defined with budget $64\,\mathrm{loopCap}_j$. The source's scale schedule verifies the loop charges used by the transition: if L_j is the side at output level j and m its preceding level, the side-dependent threshold satisfies $W_j(L_j) \le \mathrm{loopCap}_m$, and

$$\frac{64\,\mathrm{loopCap}_{m+1}}{128} = \frac{\mathrm{loopCap}_{m+1}}{2} \le \mathrm{loopCap}_m.$$

The inequality follows from the schedule's loop-capacity condition. An incoming path with ordinary profile bounds already satisfies the halo bounds by componentwise monotonicity, since each budget is nonnegative and $1 \le 64$. That monotonicity only converts an already constructed path; the arbitrary-input transition itself comes from the pairwise flux calls and the two finite extrema operations. Its horizontal receipt and local response prices remain the explicit inputs described in the one-step section.

138.5 From adjacent one-step paths to one finite history

Let a be the scale schedule, and write $J_m = a.\mathrm{clock}.J(m)$. Its slow-step property gives

$$J_m \le J_{m+1} \le J_m + 1.$$

Suppose that for every pair (j, i) with $j = i$ or $j = i+1$ the one-step receipt is available, with width $w_{j,i} > 0$, bound $A_{j,i}$, and path $Y_{j,i}$. Define the common scale width and bound by

$$s_i = \min\{b_i, w_{i,i}, w_{i+1,i}\}, \qquad A_i = \max\{E_i, A_{i,i}, A_{i+1,i}\}.$$

Taking minima and maxima preserves positivity and nonnegativity. At either adjacent index $j \in \{i, i+1\}$, one has $s_i \le w_{j,i}$ and $A_{j,i} \le A_i$. Thus the same path package works at both ends of the transition. Its common price is the maximum of the two one-step prices, and its common cutoff rate is the minimum of the two one-step rates. This finite neighbor maximum/minimum is the entire unification step in the global construction; no limit in the scale index is taken here.

The next finite construction is where attached paths are assembled. Assume that for each terminal index N there is a branch row R_N, and for each $m < N$ there is a transition rule which, from any admissible path at $m+1$, returns a path at m, an amplitude α_m, and a stationary coefficient ρ_m, together with a bubble identity and the wing bound. Then for each fixed N one constructs rows Y_m for $0 \le m \le N$ by finite downward induction. Set $Y_N = R_N$. Given Y_{m+1}, apply the transition rule to obtain Y_m and its amplitude/stationary data. Induction preserves admissibility at the next stage because the previous step furnished exactly the attachment and bound required as input. The resulting finite table has all adjacent bubble identities and all wing bounds.

Doing this separately for each N defines a single horizon of paths: its Nth column is the finite table just constructed. The terminal rows give the branch-row property, the step outputs give the adjacent bubbles, and the one-step row identities give the global row ties and path bounds. Hence one obtains one finite-history path object satisfying all finite-cutoff path conditions, with every cutoff retained on the same horizon. This argument is not a compactness extraction and does not choose a different horizon at different cutoffs. It is the finite descending construction furnished by the transition rule.

138.6 Selected cutoff limits

The ten retained bed and tower row families must converge on the lattice cover along the actual terminal cutoff sequence. Section 137 proves this for the recursively selected free horizon, using its uniform strip bounds, positive root choices and committed transition identities. Its proof includes the high-alias and noise limits, the exterior lifts and amendments, and recovery of cover coefficients from the unbounded selected period sequence. Section 169 then applies these limits to the same stationary construction and shows that the terminal Wilson normalization preserves every fixed-scale limit.

138.7 Exact source boundary

The selected route first supplies the horizontal annulus estimate, then applies the one-step flux construction at adjacent scale pairs, constructs each bubble and wing, and assembles one finite history. A separate retained cutoff argument gives coefficientwise limits for the bed and tower rows. The present proof expands the final invoice, the neighborwise unification, the finite downward assembly, with the retained-horizon convergence proved separately in Section 137. It does not rederive the annulus producer's analytic Fourier estimates or the full multicomponent one-step flux construction from its recorded trial, price-gap, and curvature inputs. Those quantitative estimates are the exact remaining analytic input to this chapter; the evidence ledger identifies their declarations and the selected callers. No new extraction or unannounced subsequence is used in the finite-history assembly.

139 Uniqueness of the limiting background paths

The path coefficients require their own convergence argument. Uniform bounds give convergent subsequences, but do not identify their limits. The identification uses the order of the Taylor coefficients in the stationary action: first the cubic term, then the quartic term, and finally the quadratic loop correction. We explain both the finite-step comparison and the passage from subsequences to the whole sequence.

At each scale m write a path as its prescribed one-form row and three coefficient arrays

$$(a_m, b_m, c_m).$$

Here a_m represents the quadratic loop correction, b_m the cubic term, and c_m the quartic term. Each array belongs to the selected closed subspace of its real coefficient Hilbert space. The selected subspace is the orthogonal complement of the kernel of all finite-period Taylor readouts. Consequently two selected arrays with the same readouts on every prescribed period are equal. This statement concerns the arrays themselves, not just their evaluation on one torus.

For $q = 2, 3, 4$ and the fixed encoding width $w_{m,q}$, use the norm

$$\mathfrak{n}_{m,q}(C) = \sum_{a,\nu} e^{-(w_{m,q}-v_m)\ell(a)} |C(a,\nu)|, \qquad w_{m,q} > v_m.$$

Here v_m is the paid path width, and the canonical lengths and finite color index ν are as in Section 117. All weights are strictly positive. Thus this expression is a norm on the coefficient space on which it is finite; homogeneity and the triangle inequality follow term by term. Let $s_{m,q} > 0$ be its paid allowance. The ordinary outputs satisfy $\mathfrak{n}_{m,q} \le s_{m,q}$. The extended input domain permits $64 s_{m+1,q}$, and keeps the other two arrays at their ordinary allowances. The same finite-step construction is available for every input in this domain. We use this actual extended domain below; existence for one selected input alone would not suffice.

139.1 The triangular affine identities

Fix the fine and coarse one-form rows. Consider three admissible finite-step inputs and outputs, indexed by $0, 1, t$, with the same rows and the same forest, coordinates and normalization conventions. The following identities hold whenever the indicated interpolation is admissible:

$$b_t^{\mathrm{in}} = b_0^{\mathrm{in}} + t(b_1^{\mathrm{in}} - b_0^{\mathrm{in}}) \implies b_t^{\mathrm{out}} = b_0^{\mathrm{out}} + t(b_1^{\mathrm{out}} - b_0^{\mathrm{out}}), \tag{667}$$

$$c_t^{\mathrm{in}} = c_0^{\mathrm{in}} + t(c_1^{\mathrm{in}} - c_0^{\mathrm{in}}) \implies c_t^{\mathrm{out}} = c_0^{\mathrm{out}} + t(c_1^{\mathrm{out}} - c_0^{\mathrm{out}}), \tag{668}$$

$$a_t^{\mathrm{in}} = a_0^{\mathrm{in}} + t(a_1^{\mathrm{in}} - a_0^{\mathrm{in}}) \implies \begin{cases} a_t^{\mathrm{out}} = a_0^{\mathrm{out}} + t(a_1^{\mathrm{out}} - a_0^{\mathrm{out}}), \\ \alpha_t = \alpha_0 + t(\alpha_1 - \alpha_0). \end{cases} \tag{669}$$

In (668) the three input cubic arrays must agree. In (669) both their cubic and quartic arrays must agree. No equality of those arrays is needed in (667).

Here is the stationary-action argument for the first two identities. In finite-dimensional chart coordinates let $S(x, y)$ be the analytic action, $y = \psi(x)$ its stationary graph, and $S_{yy}(0, 0) = H$ its invertible Hessian in the eliminated coordinates. All these stationary graphs have $\psi(0) = 0$. Write $z(x) = (x, \psi(x))$. Differentiating $S_y(x, \psi(x)) = 0$ once determines $D\psi(0)$ from the common quadratic jet of S. Repeated differentiation determines each next derivative by an equation

$$HD^r\psi(0) = -R_r,$$

where R_r is a finite sum of contractions of derivatives of S of order at most $r+1$ and derivatives of ψ of order less than r. This follows directly by the multivariable chain rule: the only term containing $D^r\psi$ is obtained by differentiating the y argument of S_y once and the graph r times.

Suppose three actions have the same jet through order r, with $r = 2$ or 3, and their order-$(r+1)$ jets have the displayed affine relation. Their graph jets through order $r-1$ agree. In $D^{r+1}(S \circ z)(0)$, terms containing a single highest graph derivative cancel by stationarity. Terms containing $D^r z$ pair with $D^2 S$ and Dz and vanish by the derivative of stationarity, since $D^r z$ is vertical. All remaining terms involve the common lower jets, except the single term

$$D^{r+1}S(0)(Dz(0), \ldots, Dz(0)).$$

It follows that the reduced order-$(r+1)$ jet has the same affine relation. Equivalently, one may subtract the three chain-rule expansions: every common lower-jet term cancels, and the displayed term is the only possible difference.

Cubic path terms vanish through order two, and quartic terms through order three. The quadratic action depends only on the common row. Thus the preceding calculation with $r = 2$ proves the affine cubic identity for the reduced actions. The finite-step classical matching equates these jets with the coarse path jets. Subtracting their common coarse-row contribution and using the injectivity of the selected cubic readout gives (667). If the input cubic arrays agree, this identity first makes the output cubic arrays agree. The calculation with $r = 3$ then applies; subtracting the common coarse-row and cubic contributions gives (668) by the quartic readout.

For the loop identity the classical action, including its cubic and quartic terms, is the same for all three inputs. The implicit function theorem and the invertibility of H give the same stationary germ. Its Hessian determinant and the coordinate Jacobian contribution are therefore the same. The remaining one-loop expression is the loop correction evaluated on that germ; it is linear in a^{in}. Its quadratic jet is affine. The coarse one-loop decomposition has the form

$$\alpha Q_0 + Q_a + \mathcal{E},$$

where Q_0 is the common normalized quadratic row form, Q_a is the loop readout, and $\mathcal{E}$ is the finite-period error. The latter has the proved exponential bound in the period. Subtract the three decompositions. The all-period detection statement for the quadratic forms makes the remaining cover form zero. Its normalization functional takes the value one on Q_0 and zero on the selected loop subspace. Applying it gives the identity for α. The residual loop readouts then vanish on every period; injectivity on the selected loop subspace gives the identity for a^{out}. This identifies exactly where quadratic normalization and all-period detection enter the proof.

139.2 An enlarged input ball gives contraction

Choose one of the three arrays, subject to equality of its preceding arrays in the order b, c, a. Write $s' = s_{m+1,q}$ and $s = s_{m,q}$. Let x, y be two ordinary inputs and put $z = \mathfrak{n}_{m+1,q}(y - x)$. If $z = 0$, then $x = y$, and the affine identity at parameter zero gives equal selected outputs. If $z > 0$, use the real parameter

$$t = 32s'/z.$$

The replacement $x + t(y - x)$ has norm at most

$$\mathfrak{n}_{m+1,q}(x) + tz \le 33s' \le 64s'.$$

Keeping the other two arrays fixed preserves their invariant subspaces and bounds. In the loop case it also preserves the linear loop normalization constraints. Thus this is an allowed extended input. Its output and that of x both have norm at most s. By the affine identity, their difference is t times the difference of the original two outputs. Hence

$$t\mathfrak{n}_{m,q}(y^{\mathrm{out}} - x^{\mathrm{out}}) \le 2s, \qquad \frac{\mathfrak{n}_{m,q}(y^{\mathrm{out}} - x^{\mathrm{out}})}{s} \le \frac{1}{16}\frac{z}{s'} \le \frac{1}{4}\frac{z}{s'}. \tag{$*$}$$

We retain the weaker factor $1/4$ used in the history argument. The extrapolated parameter need not lie in $[0, 1]$; the extended input domain is essential for this proof.

139.3 Passing finite histories to cluster histories

Fix any sequence of terminal cutoffs tending to infinity. At each fixed scale, the three coefficient arrays have uniformly bounded Hilbert norms and the scalar corrections α are bounded. Enumerate the countably many scales and coefficient coordinates. Successive extraction in bounded real intervals, followed by the diagonal subsequence, gives convergence of all those coordinates and all scalars at once. Finite sums of coordinate squares show that each limiting array has the same Hilbert bound. A closed linear constraint is retained: for a test vector in its orthogonal complement, approximate by a finite-coordinate vector; the uniform Hilbert bound controls the omitted inner product. Thus the limits remain in the selected subspaces.

The one-form rows have their common limit from the retained kernel horizon. The width argument in Section 117 gives convergence of the path sums at every smaller width. It also gives convergence of every finite chart derivative. Indeed, on a sufficiently small complex chart ball each length-ℓ path and each fixed derivative costs a polynomial in ℓ times $e^{\epsilon\ell}$. Choose ϵ smaller than the available width difference; the remaining exponential factor sums the polynomial. Termwise differentiation is then justified by uniform absolute convergence.

The limiting eliminated Hessian obeys the same positive lower bound, by passage to the limit in its quadratic form. The analytic implicit function theorem supplies a stationary germ for the limiting action. The derivative recursion above proves convergence of every derivative of the finite

stationary germs: inversion of the finite Hessian is continuous on the matrices with this common lower bound, and each R_r is a finite expression in previously convergent derivatives. The limits also have the same classical matching through degree four. For the one-loop expression, the derivatives of $\log\det H$ are finite sums of products of H^{-1} and derivatives of H. The positive lower bound and the same recursion therefore give convergence of its quadratic jet, including the fixed coordinate Jacobian. All finite-period error inequalities pass to the limit. The row normalization does too. Thus the resulting cluster history satisfies the same local relations and paid output bounds as the finite histories.

The input at scale m equals the output at scale $m+1$ in these cluster histories: it is an exact finite-history identity on every sufficiently late cutoff, so coordinatewise passage to the limit preserves it. All periods and forests needed here have already been fixed by the schedule. The derivative recursion defines their limiting stationary germs from the same limiting path; no independent subsequence is chosen for each germ.

139.4 Comparison of two cluster histories

Let F and H be two cluster histories with the same retained row horizon. A finite-step comparison cannot simply be applied to them as if they came from the same finite cutoff. Instead fix a scale and an array type, with its preceding input arrays equal. Take finite steps tending to F, and replace just this array in each fine input by the corresponding array of H. The replacement is an ordinary admissible input: the bounds and linear invariance constraints hold separately for its three arrays. Construct its output at that very same cutoff. Extract a further subsequence for these modified outputs and their scalar corrections.

The preceding limit argument gives a valid cluster step. Its chosen input array is that of H, and its preceding arrays are also those of H. The affine identity at parameter zero shows that its chosen output array equals that of H. Apply $(*)$ before taking the subsequence limit. The weighted norms of the differences converge by the strict-width dominated convergence argument. We obtain

$$g_m \le \tfrac{1}{4} g_{m+1}, \qquad g_m := \frac{\mathfrak{n}_{m,q}(H_m - F_m)}{s_{m,q}}, \qquad 0 \le g_m \le 2.$$

Iteration gives $g_m \le 2\,4^{-h}$ for every integer $h \ge 0$; hence $g_m = 0$. Apply this first to the cubic arrays at every scale, which require no preceding equality. Then apply it to the quartic arrays, using the cubic equality just obtained. Finally apply it to the loop arrays, using both equalities. All three arrays in the two cluster histories agree.

Their rows already agree, so their full input and output paths agree. The loop identity at parameter zero identifies their scalar corrections. Uniqueness of the stationary germ identifies all its derivatives for every fixed forest and period. We have therefore proved uniqueness of the entire cluster history, including the scalar and stationary data needed in the next scale comparison.

If the original sequence failed to converge to this history in any one coordinate, scalar, or fixed chart derivative, a subsequence would stay outside some neighborhood of its asserted limit. The compactness construction applies to that subsequence and gives a further subsequence with the unique cluster limit, a contradiction. Thus the entire cutoff sequence converges. The strict-width argument upgrades its coefficientwise convergence to the weighted path convergence used for fixed-scale normalized moments. This closes that part of the fixed-scale input once the finite-step extended-domain construction, retained row horizon, and canonical all-period detection used above are supplied on the selected history.

140 Analytic prices from real coupling estimates

The parameter estimates used for the background history have a useful feature: they are obtained from real configurations. The local weights contain exponentials whose exponents need not be bounded, and bounding their absolute values before using the real estimates would lose the required smallness. We prove the interpolation argument that avoids this loss. The real local estimates to which it is applied retain their full derivative and support weights.

140.1 An algebra of parameter series

For a real power series $f(t) = \sum_{r\ge 0} f_r t^r$ and $R \ge 1$, set

$$|f|_R = \sum_{r\ge 0} |f_r| R^r.$$

Only series with finite displayed norm will be used. Their products are given by the absolutely convergent Cauchy product, and $|fg|_R \le |f|_R |g|_R$. Consequently $|\exp f|_R \le \exp(|f|_R)$. If the constant coefficient f_0 is real, it is better to keep its sign:

$$|\exp f|_R \le e^{f_0} \exp(|f - f_0|_R). \tag{670}$$

These assertions follow by summing the nonnegative absolute coefficient bounds for $\sum_{m\ge 0} f^m/m!$; absolute convergence also identifies that series with the exponential on $[-R, R]$. For a finite real test $\lambda = \sum_h a_h \delta_{t_h}$, $|t_h| \le 1$, put

$$M_R(\lambda) = \sup_{r\ge 0} \left| \sum_h a_h (t_h/R)^r \right|.$$

Termwise summation then gives

$$\left| \sum_h a_h f(t_h) \right| \le |f|_R M_R(\lambda). \tag{671}$$

Thus an absolute coefficient estimate supplies every finite-test estimate simultaneously.

Fix a real normed space E, a point $x \in E$, and an integer $j \ge 0$. For a C^j function V near x, write $D_u^i V(x) = \frac{d^i}{ds^i} V(x + su)\Big|_{s=0}$. A diagonal jet bound with amplitude C and derivative scale d means $|D_u^i V(x)| \le C d^i$ for $0 \le i \le j$ and $\|u\| \le 1$. When $i = 0$ we use $d^0 = 1$, including when $d = 0$. The parameter estimates below hold for each such derivative, with one common amplitude. Conversion to full multilinear derivative norms uses the fixed finite-order polarization constants already included in the local derivative prices.

140.2 The larger real nodes

Let $\ell(t) = a + bt$, and suppose

$$R \ge 1, \qquad 1 \le a - 6R|b|, \qquad a + 6R|b| \le 1 + \delta, \qquad \delta \le 1. \tag{672}$$

In particular $0 \le \delta \le 1$, and $1 \le \ell(t) \le 1 + \delta$ whenever $|t| \le 6R$. Set $p(t) = 1 - \ell(t)^2$. For every real q define

$$A(q) = (1 - a^2)q + 2a|bq|R + b^2|q|R^2.$$

By (670),

$$|\ell|_R \le 1+\delta, \qquad |\exp(pq)|_R \le e^{A(q)}. \tag{673}$$

The first inequality uses $a \ge 1$ and $a + R|b| \le a + 6R|b|$. No estimate on $|q|$ is required.

Choose $\sigma = -1$ if $qb \ge 0$, and $\sigma = 1$ otherwise. For $r = 0, \dots, 2j$, put

$$u_r = 3 + \frac{r}{2j+1}, \qquad \xi_r = \sigma u_r, \qquad t_r = R\xi_r.$$

These are distinct real points in $[-6R, 6R]$. At every one of them,

$$A(q) \le p(t_r)q. \tag{674}$$

Here is the sign-sensitive estimate. Put $d_* = R|b|$; then $a \ge 1 + 6d_*$ and $qbR\sigma = -|q|d_*$. For any $3 \le u \le 6$, we have

$$2ad_* + d_*^2 \le 3ad_* \le ad_*u \le 2ad_*u - d_*^2u^2.$$

The last inequality is $d_*u \le a$, multiplied by d_*u; the other two follow from $d_* \le a$ and $u \ge 3$. Expanding $p(R\sigma u)q$, and using $q \le |q|$, now gives

$$\begin{aligned} p(R\sigma u)q &= (1-a^2)q + 2a|q|d_*u - qd_*^2u^2 \\ &\ge (1-a^2)q + |q|(2ad_*u - d_*^2u^2) \ge A(q). \end{aligned}$$

This proves (674) for either sign of q.

140.3 Interpolating the local weight with its derivatives

Let $L, \Gamma : E \to \mathbb{R}$ be C^j near x, and let n be a nonnegative integer. Consider

$$J(t,y) = L(y)\ell(t)^n e^{p(t)\Gamma(y)}. \tag{675}$$

Choose the preceding nodes with $q = \Gamma(x)$. Suppose that at those $2j+1$ nodes the real local estimates give

$$|D_u^i J(t_r, x)| \le Cd^i \quad (0 \le i \le j,\ \|u\| \le 1), \qquad C, d \ge 0. \tag{676}$$

There is a constant K_j, depending only on j, such that each function $t \mapsto D_u^i J(t,x)$ has an absolutely convergent series with

$$|D_u^i J(\cdot, x)|_R \le K_j(1+\delta)^n Cd^i. \tag{677}$$

In particular this controls all finite tests by (671). The constant is independent of a, b, R, n, C, d and of the value of $\Gamma(x)$.

We prove the assertion, including the reason that only finitely many real nodes suffice. Differentiating along $x + su$ and factoring out the exponential gives

$$D_u^i J(t,x) = P_{i,u}(t)\ell(t)^n e^{p(t)\Gamma(x)}, \qquad \deg P_{i,u} \le 2i. \tag{678}$$

Indeed, for the exponential alone its ith derivative is the exponential times a sum, over partitions of $\{1, \dots, i\}$, of products $\prod_B p(t) D_u^{|B|}\Gamma(x)$. Each product has at most i factors $p(t)$, each of degree two. Leibniz's rule with L gives (678), also for $i = 0$. No division by L or by a derivative of Γ occurs.

Let $\mathcal{L}_r(z)$ be the Lagrange basis polynomials for the nodes $\xi_0, \dots, \xi_{2j}$. Polynomial interpolation gives

$$P_{i,u}(t) = \sum_{r=0}^{2j} P_{i,u}(t_r)\mathcal{L}_r(t/R).$$

Define K_j as the larger, for the two possible signs σ, of $\sum_r |\mathcal{L}_r|_1$. This is finite, since all the nodes are distinct. Also $|\mathcal{L}_r(\cdot/R)|_R = |\mathcal{L}_r|_1$. By (676), (678), and $\ell(t_r)^n \geq 1$,

$$|P_{i,u}(t_r)|e^{p(t_r)q} \leq Cd^i.$$

Multiplying each interpolation term by $\ell(t)^n e^{p(t)q}$ and using (673) and (674), its coefficient norm is at most

$$|P_{i,u}(t_r)|(1+\delta)^n e^{A(q)}|\mathcal{L}_r|_1 \leq (1+\delta)^n Cd^i|\mathcal{L}_r|_1.$$

Summing proves (677). Notice that the node sign may depend on $\Gamma(x)$. Both sets of nodes belong to the same allowed real interval, and K_j controls either choice; there is no need to differentiate the choice of sign.

140.4 The Gaussian density and the local support count

For a fixed Gaussian coordinate z, let $Q(z) \geq 0$ be its quadratic energy in the reference density. Under the rescaling $z \mapsto \ell(t)z$ in D real dimensions, the density ratio, including the Jacobian, is

$$g(t,z) = \ell(t)^D e^{(1-\ell(t)^2)Q(z)}.$$

The same argument, now with $q = Q(z) \geq 0$, yields

$$|g(\cdot,z)|_R \leq (1+\delta)^D. \tag{679}$$

Indeed (674) gives $A(q) \leq p(t_r)q \leq 0$, since $\ell(t_r) \geq 1$. The bound is uniform in z. Thus it can be multiplied into a local coefficient majorant before integration against the fixed reference Gaussian measure.

In the local application, the exponent n in (675) counts the seed coordinates actually present in the local weight, each with its color dimension. The Gaussian dimension D is likewise that of the relevant marginal. These counts are determined by the local support; replacing them by the dimension of the entire torus would destroy a bound uniform in volume. The exact local change-of-density identity is used before applying (679). The remaining support-dependent powers $(1+\delta)^{n+D}$ are included in the cover-price reserve.

140.5 Keeping the small amplitude of ordinary activities

There is a second, complementary estimate. Let $f : \mathbb{R} \times E \to \mathbb{R}$ be real-valued, and suppose that $t \mapsto f(t,y)$ is a polynomial of degree at most k on $[-1,1]$ for every y, where k is fixed. Choose distinct fixed nodes $s_0, \ldots, s_k \in [-1,1]$. Suppose that each $f(s_r,\cdot)$ is C^j near x and that

$$|D_u^i(e^{f(s_r,\cdot)} - 1)(x)| \leq vd^i, \qquad 0 \leq v \leq \tfrac{1}{2}. \tag{680}$$

There are constants $A, B \geq 1$, depending only on j, k and the nodes, such that, whenever $BR^k v \leq 1$,

$$|D_u^i(e^{f(\cdot,\cdot)} - 1)(x)|_R \leq AR^k vd^i. \tag{681}$$

In this formula the derivative is in the second variable and the series norm in the first.

To check the estimate, write $h = e^{f(s_r,\cdot)} - 1$ at a fixed node. At x we have $|h(x)| \leq v \leq 1/2$ and $f(s_r,\cdot) = \log(1+h)$ near x. The zeroth derivative is bounded by $2v$, by integrating $1/(1+s)$ between 0 and $h(x)$. For a positive-order derivative, the chain rule is a finite sum over partitions, with one derivative of $\log(1+s)$ and products of derivatives of h. The former is bounded on

$[-1/2, 1/2]$; a term with $r \geq 1$ factors of h is bounded by a constant times $v^r d^i \leq v d^i$. Hence $|D_u^i f(s_r, \cdot)(x)| \leq C_j v d^i$ at every node.

Polynomial interpolation in t is valid for every y and so also after the indicated derivatives. The coefficient norm of any fixed degree-k Lagrange basis polynomial at radius $R \geq 1$ is bounded by R^k times its norm at radius one. It follows that

$$|D_u^i f(\cdot, x)|_R \leq C_* R^k v d^i.$$

Choose $B \geq C_*$. Put $\varepsilon = C_* R^k v \leq 1$. In the Banach algebra of parameter series, the zeroth-order estimate $|e^f - 1|_R \leq e^\varepsilon - 1 \leq e\varepsilon$ follows from the exponential series. For $i > 0$, the same partition formula for derivatives of e^f gives a factor $|e^f|_R \leq e$ and products bounded by $\varepsilon^r d^i \leq \varepsilon d^i$. The finite sum gives (681), with A large enough for all $i \leq j$. This proof also covers $d = 0$: all positive derivatives then vanish.

The incoming ordinary activity in the coupling family has degree at most two in this real parameter. Thus the radius cost in this last argument is R^2, with constants fixed by the derivative order. The local stock weight uses the preceding real-node argument instead, because its exponent need not be small. Together with the local real jet estimates, the fixed Gaussian change of density, and the summable support prices, these two arguments provide the parameter estimates for the finite forest integrals. Summation over connected supports and the subsequent local projections still require their stated quantitative bounds; the interpolation proved here does not replace those bounds.

141 Summable parameter estimates for the sector expansion

We now combine the real-parameter interpolation with the local Gaussian and connected-support estimates. The important point is that one uses the same supports, labels and protected fine reads throughout the real parameter family. A separate estimate at each parameter, with an unrelated choice of labels, would not bound a finite linear combination of the resulting activities.

Fix the group, the block side L, a derivative order j, a nonnegative integer m measuring the spatial weight in the derivative budget, and all the positive safety and support reserves. Write $y = \beta_f(x)$ for the reference fine coupling, which tends to infinity with the scale parameter x. A nonnegative function has upper exponent a below if, for each $a' > a$, it is eventually at most $y^{a'}$. Finite sums, products and fixed powers have the corresponding upper-exponent bounds. Constants, local cover counts, and all the fixed-order derivative factors of upper exponent zero may therefore be absorbed using a strict exponent reserve.

The incoming sector is assumed to satisfy the local hypotheses of Sections 87 and 85, uniformly for all real points $|t| \leq 6R$ in the coupling band. The path, clipping rule, ordinary and forced support dictionaries, and held fine configurations are common. The coupling is $\beta(t) = \beta_0 \ell(t)^2$, where (672) holds. The incoming ordinary activity values and forced activity values on each fixed support are polynomials of degree at most two on $[-1, 1]$. We take

$$R = y^{1/256}, \qquad \delta = y^{-1/128}. \tag{682}$$

The geometric radius $B(x)$ and its ball count $b_B = (2B + 1)^4$ have upper exponent zero. Fix the incoming support exponent E and the desired outgoing exponent e with $E > e \geq 0$. All estimates below are eventually uniform in the allowed periods, sectors and configurations. The threshold is chosen after the fixed data just listed. Construction of those incoming sectors and of their uniform safety estimates is still required; the purpose of this section is to prove the analytic output estimates from them.

141.1 Common prices before integration

The original optional-activity rooted allowance has upper exponent $a \le -7/16$. To obtain one price valid along the parameter family, sum the nonnegative prices at the three nodes $-1, 0, 1$ for each ordinary and forced label. The sum over all such labels through a root increases by at most a factor three, and each individual price is at most three times the old rooted allowance. This last assertion uses nonempty supports and nonnegative support weights.

The complete local logarithmic factor is still polynomial of degree at most two: the old vacuum and action terms either acquire the factor $\ell(t)^2$ or remain constant, the incoming regular occurrence is quadratic by hypothesis, and the enforcement factor uses held fine reads and is parameter-independent. The forced local factor likewise is its incoming quadratic value multiplied by a fixed enforcement factor. Apply the small-activity estimate of Section 140 to the ordinary exponential-minus-one factors. Its smallness conditions are satisfied eventually: if $g(x)$ is the incoming allowance, then $3g(x) \to 0$ and $R^2 g(x)$ has negative upper exponent. For a forced factor use ordinary degree-two interpolation at the same three nodes; there is no logarithm to take. The resulting common prices cost at most a fixed constant times $1 + R^2$. Their upper exponent is at most

$$a + \frac{2}{256} \le -\frac{55}{128} < -\frac{2}{5}.$$

Thus both are paid by an enlarged target with upper exponent $-2/5$, after increasing the scale threshold. This explains the specific reserve in (682).

For seed packets, use the larger-real-node estimate with the full stock weight. If a packet contains $s \ge 1$ seeds, its real price has the form

$$\exp\{(C\mathfrak{d}^5 - \beta\mu\tau^2)s\},$$

where $\mathfrak{d} \ge 1$ is the local geometric control function. Its stock-coordinate count satisfies $n \le C_1 s\mathfrak{d}^5$. The interpolation factor is $K_j(1+\delta)^n$. Since $0 \le \delta \le 1$, $(1+\delta)^n \le e^n$, and $K_j \le e^{K_j}$ for $K_j \ge 0$. Hence increasing C to $C + C_1 + K_j$ absorbs the entire factor, because $s\mathfrak{d}^5 \ge 1$. The negative packet energy is unchanged. The absorption and root-count argument of Section 84 consequently still applies.

Warnings are constant in this parameter when evaluated with the specified held fine reads and rescaled fluctuation coordinate. Their coarse-field jets retain the common reduced-margin excursion indicator of Section 87. No derivative of that indicator is taken. Products of the four species thus have one common nonnegative coefficient majorant, with the same excursion event and the same support set as before interpolation.

There is one further factor from the local Gaussian ratio and the local stock coordinate count. Let $I(S)$ be the set of fine bonds read by the local term with coarse support S, and let d_G be the real color dimension. Its combined power is bounded by $2|I(S)|d_G$. If $c_B(S)$ is the coarse cover number, then

$$|I(S)| \le |S|\,\mathrm{dens}(L) \le c_B(S) b_B\,\mathrm{dens}(L), \qquad b_B = (2B+1)^4.$$

For any fixed $\alpha > 0$, (682) gives, eventually, $2\delta b_B\,\mathrm{dens}(L) d_G \le \alpha$. Using $1 + \delta \le e^\delta$ proves

$$(1+\delta)^{2|I(S)|d_G} \le e^{\alpha c_B(S)}. \tag{683}$$

Only the dimension of the local read set has been charged. The factor is paid by a support-exponent reserve of size α.

141.2 The moment estimate after a finite parameter test

Let A be a connected finite set of primitive labels. Denote its assigned union support by S_A, its nonnegative shape charge by s_A, and its warning set by W_A. Let $a_l \geq 0$ be the common primitive price for label l, with its warning gain included only after Gaussian integration. Put $P_A = 1 + s_A \geq 1$. The estimates just proved, the compensated Jacobian estimates, and the product rule give a coefficient bound for every derivative through order j of the local integrand. After a finite test λ, its norm is bounded by

$$M_R(\lambda)\, C_j e^{\alpha c_B(S_A)} \left(\prod_{l \in A} a_l^{\circ} \right) P_A^j\, \mathbf{1}_{\mathcal{E}_A}(z)\, F_{I(A)}(z)^{Kj}. \tag{684}$$

Here a_l° is the price before inserting the warning probability gain, $\mathcal{E}_A$ is the joint reduced-margin excursion event, $F_{I(A)} \geq 1$ is the Gaussian coordinate control function on the local read set, and K is the fixed Jacobian polynomial order. The constant C_j in this display may be replaced by a nonnegative function of upper exponent zero; this includes the compensated derivative scales. More explicitly, if those scales are r and $L' F_{I(A)}^K$, their product-rule cost is bounded by

$$\{1 + (2^K L' + 2r)\}^j P_A^j F_{I(A)}^{Kj}.$$

Indeed $F_{I(A)}, P_A \geq 1$ and $s_A \leq P_A$, so $1 + 2^K L' F_{I(A)}^K + 2r s_A$ is at most $\{1 + (2^K L' + 2r)\} P_A F_{I(A)}^K$. Fixed polarization constants are included in C_j.

For clarity, the probabilistic step in this estimate is as follows. The local Gaussian moment calculation in Section 86 supplies an integer k and a function $V(x)$ of upper exponent zero such that

$$\left(\int F_{I(A)}^{2Kj}\, d\nu \right)^{1/2} \leq (1 + |V(x)|) P_A^k.$$

The common warning estimate is $\nu(\mathcal{E}_A) \leq \rho^{2|W_A|}$. Cauchy–Schwarz therefore gives

$$\int \mathbf{1}_{\mathcal{E}_A} F_{I(A)}^{Kj}\, d\nu \leq (1 + |V(x)|) P_A^k \rho^{|W_A|}.$$

This is the source of the warning price in a_l; no independence of the warning events is assumed. Combining with (684), and then averaging the finitely many forest and tagging choices, gives the derivative price

$$p_A = \pi(x) e^{\alpha c_B(S_A)} \left(\prod_{l \in A} a_l \right) P_A^{k+j}, \tag{685}$$

where π has upper exponent zero and is independent of λ. The bound on the tested derivative is $M_R(\lambda) p_A$.

Passing derivatives through the integral is justified by the nearby integrable bounds in Section 85. There are only finitely many parameter points in a test, so finite summation commutes with derivatives and integrals. A finite average over forests preserves the same bound. On healthy supports the protected gauge average equals the unaveraged local expression as a germ: the safety plateau is open and the gauge change leaves the protected expression invariant. Thus the estimate is for the actual coarse activity, including its derivatives. An inactive support gives the zero germ. All of these comparisons use the common support dictionary, not a parameter-dependent relabeling.

The connected-set estimate for the enlarged primitive prices gives

$$\sup_z \sum_{A : z \in S_A} p_A e^{(E-\alpha) c_B(S_A)} \leq \pi(x) u_{k+j}(x). \tag{686}$$

Here E is the incoming support exponent and u_{k+j} is the connected primitive allowance with the displayed shape moment. It has upper exponent $-2/5$. To see the support bookkeeping, the left side contains $e^{\alpha c_B(S_A)} e^{(E-\alpha)c_B(S_A)} = e^{Ec_B(S_A)}$. Thus the mixed connected-set estimate is applied with its original support weight, and its shape moment is exactly $k + j$. This invokes the same connected-set and moment bound used for the non-parameter sector, with the enlarged primitive prices just constructed.

141.3 The second connected expansion

Recover an absolute parameter series for each derivative from the finite-test estimate with radius R. Section 145 gives radius $R/2$ and price $2p_A$, once $R \geq 2$. The argument can first be applied to each scalar derivative; the fixed-order polarization bound controls the full derivative. A logarithmic cluster with q occurrences is a product of these primitive integrated activities times its Ursell coefficient and $1/q!$. Derivatives through order j cost at most q^j. Since $q \leq 2^q$ for $q \geq 1$, this is paid by multiplying each vertex price by 2^j. Consequently the absolute cluster price is

$$w_\Gamma = \frac{|u(\Gamma)|}{q!} \prod_{r=1}^{q} (2^{j+1} p_{A_r}).$$

This includes repeated labels and their tuple factorial exactly as in Section 62.

Fix the desired outgoing exponent e with $0 \leq e < E$, and set $\alpha = (E - e)/4 > 0$. Put

$$Y(x) = (1 + 2^{j+1}\pi(x)) u_{k+j}(x).$$

Equation (686) pays the enlarged vertex prices at exponent $e + 2\alpha$, because $e + 2\alpha \leq E - \alpha$. The connected-tree degree criterion is

$$\frac{4b_B Y(x)}{\alpha} \leq \frac{1}{2}.$$

It holds eventually: b_B has upper exponent zero and Y has upper exponent $-2/5$. The cluster estimate then gives

$$\sup_z \sum_{\Gamma : z \in S_\Gamma} w_\Gamma e^{(e+\alpha)c_B(S_\Gamma)} \leq 2Y(x).$$

The spatially weighted derivative budget at a root costs at most $\{s_B(1 + c_B(S_\Gamma))\}^{mj}$, where $s_B = 4(2B + 1)$. To verify the geometry, connect the balls of a minimal connected cover. A path through at most $c_B(S_\Gamma)$ balls bounds the distance between two sites of S_Γ by $c_B(S_\Gamma)(4B + 2) + 2B$, which is at most $s_B(1 + c_B(S_\Gamma))$. Locality allows the field variation to be restricted to this support. The weighted direction norm on that support is therefore increased by the stated distance factor, and taking at most j derivatives gives the power mj. For every fixed integer $d \geq 0$,

$$P_{\alpha,d} := \sup_{r \geq 0} (1 + r)^d e^{-\alpha r} < \infty.$$

For example, expand $(1 + r)^d$ and bound each r^l by $l!\alpha^{-l} e^{\alpha r}$, using the exponential series. The extra spatial factor is thus paid by the remaining α reserve. Including the fixed polarization factor C'_j we obtain the final rooted derivative price

$$2C'_j s_B^{mj} P_{\alpha,mj} Y(x). \tag{687}$$

It has upper exponent $-2/5$. Since $-2/5 < -3/8$, it is eventually at most $y^{-3/8}$. This is the ordinary parameter-test allowance at radius $R/2$, on the actual connected-cluster dictionary.

141.4 The forced common majorant

For forced terms no coarse derivatives are required. The same finite-test and radius-halving argument gives ordinary constituent series prices with rooted allowance $2u_0(x)$ and forced constituent prices with rooted allowance $2u_0(x)e^{-M(x)}$. The nonnegative mark $M(x)$ is the compulsory warning gain. Form the exclusion-correction activities exactly as in Section 90. Each is a finite product of forced factors and powers of regular cluster factors, with the same multiplicity factorials. Absolute coefficient norms multiply, so (472) is also a majorant for the whole parameter series at radius $R/2$.

The remaining eligibility and flag indicators depend on the fixed coarse configuration and common supports, and are independent of the parameter. Multiplication by such an indicator only decreases the coefficient bound. This yields one majorant per final support, valid for all coarse configurations, rather than a separate majorant selected after evaluating a configuration.

Apply the forced-exclusion bound with its $u = 2u_0(x)$ and the reserve α. Its second-expansion allowance is $Z(x) = 20u = 40u_0(x)$. Both $Z \le 1/2$ and $4b_B Z/\alpha \le 1/2$ hold eventually. Hence the final common majorants k_D satisfy

$$\sup_z \sum_{D: z \in S_D} k_D e^{ec_B(S_D)} \le e^{-M(x)} \frac{Z(x)}{1 - 1/2}. \tag{688}$$

This keeps the compulsory exponential gain through arbitrarily many regular exclusion corrections. In particular, the parameter extension has not consumed that gain.

Equations (687) and (688) are the broad analytic estimates for the sector output before its finite Taylor and local projection operations. They use the uniform local sector estimates, the same connected-support sums, and the same Gaussian warning bounds as the real one-step construction. The later projection prices and the construction of the allowed coupling family are separate inputs when these estimates are applied to the selected background history.

142 Recovering a local density from five auxiliary periods

The local projection in the nonlinear step uses five small periodic boxes. Their side lengths differ by two. A fourth finite difference cancels all the contributions from folded boundary faces and retains the interior contribution. We give the construction, since both the exact identity and its derivative cost matter for the parameter estimates.

142.1 Folded fields and boundary types

For an integer $w \ge 1$, fold the cyclic set $\mathbb{Z}/(2w)\mathbb{Z}$ onto $\{0, \dots, w\}$ by

$$\pi_w(q) = \min(\bar{q}, 2w - \bar{q}), \qquad 0 \le \bar{q} < 2w.$$

Its two endpoints have one preimage each and its other points have two. Apply this map in each of the four coordinates. Adjacent sites map to adjacent sites, with direction either preserved or reversed. For a gauge field U on a larger torus and an anchor o, the folded field $\Phi_{w,o}U$ on the side-$2w$ torus reads the link of U between $o + \pi_w(q)$ and $o + \pi_w(q + e_\mu)$; a reversed link is read as its inverse. Here and below a lattice point is reduced modulo the large period when used to index U.

Fix $r \ge 1$ and $w \ge 2r$. In one coordinate, give a folded point $p \in \{0, \dots, w\}$ its boundary type:

$$\begin{cases} (\text{left}, p), & p < r, \\ (\text{right}, w - p), & w - p < r, \\ \text{interior}, & r \le p \le w - r. \end{cases}$$

The cases are disjoint. For an offset s with $|s| \le r$, the folded displacement from p is respectively

$$|p+s|-p, \qquad d-|d-s| \ (d=w-p), \qquad s. \tag{689}$$

These identities follow directly from the two affine branches of π_w. They also describe links in a local box when both endpoints are in the box. Thus, after moving the anchor to cancel the folded base point, the local field near a site depends only on its four boundary types. The displacement formulas do not depend on w. An all-interior type gives the original, unfolded field near the chosen anchor.

For a left or right type of depth $d \in \{0, \dots, r-1\}$, the number of cyclic points of that type is 1 if $d = 0$ and 2 otherwise. The interior count is

$$2(w-2r+1).$$

Indeed its folded coordinates are $r, \dots, w-r$, all strictly between the two endpoints, and each has two preimages. In four dimensions the number of sites of a prescribed type vector is the product of its four one-dimensional counts.

Let $\mathcal{T}_r$ be the finite set of four-coordinate type vectors, and let τ_* be the all-interior type. For any function $H : \mathcal{T}_r \to \mathbb{R}$, the preceding count proves

$$\sum_{i=0}^{4} c_i \sum_{q \in (\mathbb{Z}/2(w+i)\mathbb{Z})^4} H(\mathrm{type}_{w+i,r}(q)) = H(\tau_*), \qquad c_i = \frac{(-1)^i \binom{4}{i}}{2^4 \cdot 4!}. \tag{690}$$

To verify this exactly, fix a type vector with d interior coordinates. Its multiplicity at side $2(w+i)$ is a polynomial in i of degree d. The fourth difference annihilates that polynomial when $d < 4$. For the all-interior type the polynomial is $2^4(w+i-2r+1)^4$, whose fourth difference is $2^4, 4!$. Division by that factor gives one. Summing over the finitely many type vectors proves (690). In particular, no boundary term has been discarded before cancellation.

142.2 Compatible local densities

Write $f_0(o, U)$ for a density on the large torus and $f_i(q, V)$ for densities on the five auxiliary tori of sides $2(w+i)$. The compatibility conditions needed here are explicit: the auxiliary densities transform covariantly under signed lattice symmetries, their values on globally healthy fields agree whenever their centered radius-r link patterns agree, and the same agreement holds with f_0 on globally healthy fields. Finally $f_0(o, \cdot)$ is local on the radius-r neighbourhood of o. The large period is at least $4r + 16$. These are compatibility of local functions at fixed coupling and fixed path; no equality of partition functions on different periods is required.

Set $\widetilde{f}_i(q, V) = f_i(q, V) - f_i(q, \mathbf{1})$ and define $\widetilde{f}_0$ in the same way. The compatibility properties persist after this centering: the identity fields are healthy and have identical local link patterns. Define

$$g(o, U) = \sum_{i=0}^{4} c_i \sum_{q \in \Lambda_i} \widetilde{f}_i(q, \Phi_{w+i,o} U), \qquad \Lambda_i = (\mathbb{Z}/2(w+i)\mathbb{Z})^4.$$

Choose $z \ge w + 4$. The function $g(o, \cdot)$ reads only the radius-z neighbourhood of o. For each auxiliary site q set $o_{i,q} = o - \pi_{w+i}(q)$, coordinatewise, and put

$$h(o, U) = \sum_{i=0}^{4} c_i \sum_{q \in \Lambda_i} \mathbf{1}_{\{U \text{ healthy on } N_z(o_{i,q})\}} \widetilde{f}_i(q, \Phi_{w+i,o_{i,q}} U). \tag{691}$$

The shift ensures that the physical point corresponding to q is o. Here $N_z(a)$ denotes the radius-z lattice neighbourhood of a.

If U is healthy on $N_{2z}(o)$, every indicator in (691) equals one: $\operatorname{dist}(o, o_{i,q}) \le w + i \le z$ and $N_z(o_{i,q}) \subseteq N_{2z}(o)$. The folded auxiliary fields are then globally healthy. A folded plaquette is either an original plaquette with an orientation change, a conjugate of such a plaquette, or a degenerate loop with identity holonomy. The plaquette chord is invariant under conjugation and inversion; thus its bound follows from the healthy large-field bound on the read set.

Reflection covariance first moves q to its canonical folded representative. The displacement formulas (689), followed by compatibility, show that the summand depends only on its type vector. For the all-interior vector it equals $\widetilde{f}_0(o, U)$. One can make the last comparison using a globally healthy extension on the original large torus. In each cyclic coordinate of length $c \ge 4r$, take the track that increases from 0 to $2r$ in $2r$ steps, decreases back to 0 in the next $2r$ steps, and remains at 0 for the remaining $c - 4r$ steps. Place its image at $o - r$ and center its argument at the midpoint of the increasing segment. The resulting field agrees with U on the radius-r box about o and reads only that box everywhere. Each plaquette is again an original plaquette with orientation changed or an identity loop, so the extension is globally healthy. Compatibility applies to this field on the original period; locality then identifies its f_0 value with $f_0(o, U)$. Applying (690) gives

$$h(o, U) = \widetilde{f}_0(o, U) \quad \text{if } U \text{ is healthy on } N_{2z}(o). \tag{692}$$

The compatibility in the selected construction is established by passing two given periods to their product period. Replicate each field to that period; the short local density is preserved by replication. Translate both base points to zero. Their local patterns still agree there, so locality of the common-period density gives equality. The selected auxiliary representations use the same path, coupling, profile and finite cutoff. It is these equalities that permit the replication comparison. The required replication and local-generation identities are part of the short-density construction; this argument does not assert compatibility for independently chosen histories.

142.3 The exact additive decomposition

Let $F_a(U)$ be the ordinary logarithmic entries retained in a finite short-support panel. Their allocation coefficients ω_{oa} are nonnegative, sum to one in o for each a, and satisfy $S_a \subseteq N_r(o)$ whenever $\omega_{oa} \neq 0$. Put

$$E(o, U) = \sum_a \omega_{oa} \mathbf{1}_{\{U \text{ healthy on } S_a\}} F_a(U).$$

Its sum over o is exactly the masked contribution of the panel. The unmasked short density is $f_0(o, U) = \sum_a \omega_{oa} F_a(U)$. Translation covariance makes $f_0(o, \mathbf{1})$ independent of o. Define the discrepancy

$$d(o, U) = E(o, U) - f_0(o, \mathbf{1}) - h(o, U).$$

This definition, followed by the change of variable $o \mapsto o - \pi_{w+i}(q)$ in each finite site sum, proves

$$\sum_a \mathbf{1}_{\{U \text{ healthy on } S_a\}} F_a(U) = |\Lambda| f_0(0, \mathbf{1}) + \sum_o \mathbf{1}_{\{U \text{ healthy on } N_z(o)\}} g(o, U) + \sum_o d(o, U). \tag{693}$$

Indeed this change of variable identifies $\sum_o h(o, U)$ with the middle sum. It does not require that U itself be healthy. If U is healthy on $N_{2z}(o)$, the allocation supports near o are healthy, so $E(o, U) = f_0(o, U)$. Equation (692) then gives $d(o, U) = 0$. Thus the discrepancy is supported on an unhealthy enlarged neighbourhood. The regular output retains g, the constant contribution is extracted, and the forced correction uses $e^{-d} - 1$. The negative sign follows by solving (693) for the new logarithmic sum in the density convention of the nonlinear step.

142.4 The derivative cost of folding

Let the derivative order be j, the spatial weight power be m, and the small-direction threshold be $\theta > 0$. Suppose a function F on an auxiliary torus has all left derivatives through order j bounded by q when each direction has norm at most θ. On a healthy radius-z neighbourhood, the corresponding bound for $F \circ \Phi_{w+i,o}$, with directions allowed to grow as $\theta(1 + \operatorname{dist}(o, x))^m$, is

$$2^j (1 + z)^{mj} q. \tag{694}$$

The factor two per derivative is needed because a reversed link transforms a left variation into a right variation. We justify that point rather than identifying the two derivatives.

On a forward auxiliary link the induced variation is a left variation by the corresponding original direction; on a reversed link it is a right variation by the negative original direction. The two types have disjoint link coordinates, fixed by the folding map. Consequently left and right variations on different types commute. For a list of k derivatives, expand each variation into its left and right part. There are at most 2^k terms. In each term, right derivatives can be moved past all left derivatives, and at the fixed field converted to left derivatives by adjoint conjugation. Their norms are unchanged because the group metric is adjoint invariant. Explicitly, start with the empty list. For a leading mixed direction (v_L, v_R) and each converted list s for the remaining directions, the two new lists are $v_L :: s$ and $s +\!\!+ [\mathrm{Ad}_V\, v_R]$, with the adjoint chosen to turn the right action at the fixed field V into a left action. Thus the right directions occur in reverse order. The disjoint link supports ensure that this recursion introduces no derivatives of a varying conjugating field. Each resulting direction has norm at most $X\theta$, where $X = (1+z)^m$. Multilinearity and rescaling by X^{-1} bound the term by $X^k q$. Summation and $k \le j$ prove (694). The smooth chart property under composition follows since a link read, its inverse and a constant identity link are smooth maps.

It follows that the ordinary localized density has derivative price

$$\mathcal{O}_z(q_0, \dots, q_4) = \sum_{i=0}^{4} |c_i|\, 2^j (1 + z)^{mj} q_i, \tag{695}$$

when q_i bounds the centered total auxiliary density $\sum_q \widetilde{f}_i(q, \cdot)$. The support is $N_z(o)$. For the ordinary derivative budget, each entry has an assigned anchor; its budget at a root o sums only entries assigned to o, weighted by $e^{ec_B(S)}$. Thus if $c_B(N_z(o)) \le C_z$, the entry $g(o)$ costs $\mathcal{O}_z(q) e^{eC_z}$ in that anchored budget. A sum over every support containing a fixed site would instead cost the additional factor $(2z + 1)^4$. A fixed signed-orientation average and its local allocation multiply this price by the already purchased orientation cost. All these factors depend only on the fixed local geometry, and have upper exponent zero in the scale parameter.

143 Parameter estimates after the local projection

We transfer the estimates of Section 141 to the actual ordinary and forced arrays after localization. All finite tests below use the common parameter family, and every local cutoff is applied to its original fixed support. We retain the compatibility and local-generation identities specified in Section 142.

The ordinary budget below is anchored: at site o it sums the nonnegative derivative prices of entries whose assigned anchor is o, with their exponential support weights. The forced budget sums all supports containing the site. The sector bounds imply the anchored ordinary bounds by restricting their support sums to the assigned anchors.

Let P be the outgoing derivative budget, with derivative order j, spatial weight power m, radius B, threshold θ and support exponent b. Put $b_* = b + 3$. Let $y = \beta_f(x)$ and $R = y^{1/256}$, as

before. Assume the path rows have their purchased strip and coefficient bounds, their regular local readings are smooth on healthy supports, and the support reserve fits inside half the path strip. Their ordinary rooted allowance, before the finite orientation average, is at most $y^{-7/16}$. These are the path inputs to the present argument. The parameter-dependent action and quadratic rows are multiplied by $\ell(t)^2$, while the other two row types are unchanged. Since $|\ell^2|_R \le (1+\delta)^2 \le 4$, their finite-test allowance at radius $R/2$ is at most four times the original allowance.

Let $T_P > 0$ be the fixed orientation and derivative allocation cost. Combining the path terms and the sector logarithmic terms, and applying that allocation, gives the price

$$W(x) = T_P\{y^{-3/8} + 4y^{-7/16}\}. \tag{696}$$

For every finite test λ, the ordinary family has rooted derivative price $M_{R/2}(\lambda)W(x)$ at exponent b_*. A fixed finite short-support panel may be removed from this family with the same bound: keep the original nonnegative price on the retained addresses and put zero on the deleted ones. Neither the parameter dictionary nor its support sets change. This observation includes every finite truncation used in the short-density construction, with no dependence of W on the truncation order.

143.1 Allocating and testing the auxiliary densities

For the five auxiliary periods Λ_i of Section 142, form the same finite-panel local densities $f_i(q, U)$. The nonnegative allocation coefficients are bounded by one, sum to one over q, and only assign an entry to anchors in its prescribed neighbourhood. Write s for the radius of that neighbourhood and $b_s = (2s+1)^4$.

For a tested density at a fixed site q, take absolute values in its allocation formula. Each nonzero entry is assigned to an anchor within $N_s(q)$. Summing first over those anchors and then using the rooted bound gives

$$\left|\sum_h a_h f_i(t_h, q, U)\right| \le M_{R/2}(\lambda) b_s W(x) \tag{697}$$

on globally healthy U. The same estimate holds for the masked panel on an arbitrary field: outside its healthy support an entry is zero. Here the mask is independent of the parameter, so testing commutes with masking.

The total auxiliary density is exactly the sum of the panel entries, since the allocations sum to one. Summing the derivative prices over anchors therefore bounds its derivatives by

$$|\Lambda_i| W(x) M_{R/2}(\lambda).$$

Subtracting its value at the identity field costs at most twice that amount, uniformly through derivative order j; positive derivatives of the subtracted constant vanish. Thus the two probe prices needed below are

$$p_i = 2|\Lambda_i| W(x), \qquad q_i = b_s W(x). \tag{698}$$

The first bounds derivatives of the centered total density; the second bounds each uncentered local density on healthy fields. The identity field is healthy because its plaquette holonomies are all the identity and $\theta > 0$.

143.2 The ordinary output price

Let $g(t, o, U)$ be the five-period local density and $d(t, o, U)$ its discrepancy. Put

$$\mathcal{O}_z(p) = \sum_{i=0}^{4} |c_i|\, 2^j (1+z)^{mj} p_i, \qquad C_z \ge \sup_o c_B(N_z(o)).$$

All operations in the formula for g are finite linear operations on the densities. Hence a finite test commutes with that formula. Apply (694) to the tested auxiliary total densities. The result is the regular support price $M_{R/2}(\lambda)\mathcal{O}_z(p)$ at each anchor. Multiplying by the support weight gives $M_{R/2}(\lambda)\mathcal{O}_z(p)e^{b_*C_z}$. The existing complementary panel has price $M_{R/2}(\lambda)W(x)$. After the orientation allocation, the total ordinary output price is therefore

$$A(x) = W(x) + T_P\mathcal{O}_z(p)e^{b_*C_z}. \tag{699}$$

This is a bound on the full derivative budget on each healthy support, not just on the sum of the array values.

In the selected geometry, s, z and all five auxiliary volumes have upper exponent zero; the cover bound C_z is fixed by the halo and support construction. The derivative order and spatial weight power are fixed before the scale is increased. Consequently A has upper exponent $-3/8$, and

$$A(x) \le y^{-1/4} \tag{700}$$

eventually. The strict exponent gap absorbs every fixed orientation constant and every factor of upper exponent zero. This is the broad ordinary bound used in the finite-test sharpening of Section 144; its radius is $R/2 = y^{1/256}/2$.

143.3 A common series majorant for the discrepancy

A discrepancy value is the difference between the allocated masked panel, its identity-field constant, and the shifted masked auxiliary sum. Applying (697) to the first two terms gives $2b_sW(x)M_{R/2}(\lambda)$. For the last term, each centered auxiliary local value is bounded by $2q_iM_{R/2}(\lambda)$ when its mask is one, since the folded field is then globally healthy. It is zero when its mask is zero. Hence, for every field U and anchor o,

$$\left|\sum_h a_h d(t_h, o, U)\right| \le M_{R/2}(\lambda)\left\{2b_sW(x) + \sum_{i=0}^{4} |c_i||\Lambda_i|\, 2q_i\right\}.$$

Write D for the expression in braces. The finite-test functional argument gives coefficients $d_n(o,U)$ satisfying $|d_n(o,U)| \le D(2/R)^n$ for every U, with the same D. Therefore $D(2/R)^n$ is a common coefficientwise envelope; its weighted sum at radius $R/4$ is $\sum_n D2^{-n} = 2D$. Thus

$$|d(\cdot, o, U)|_{R/4} \le a(x), \qquad a(x) = 2\left\{2b_sW(x) + \sum_{i=0}^{4} |c_i||\Lambda_i|\, 2q_i\right\}. \tag{701}$$

This bound is uniform in U, and $R \ge 4$ eventually. In particular there is one coefficient majorant for the discrepancy at each fixed support. The sum over orientation copies uses the normalized finite orientation average; for values it has norm one. Assigning each copy its actual averaging weight and counting the anchors whose radius-$2z$ support contains a root gives the rooted price

$$H(x) = (4z+1)^4 a(x) e^{(b+2)C_{2z}}, \qquad C_{2z} \ge \sup_o c_B(N_{2z}(o)). \tag{702}$$

This has upper exponent $-3/8$. The signs of the discrepancy are retained in the functions; they disappear only in this nonnegative bound.

143.4 Fusing the discrepancy with the forced array

Let $\gamma(x)$ be the forced series allowance supplied by (688), and let $\phi(x)$ be the prescribed outgoing forced allowance. Use the sector allowance γ at support exponent $b+2$, weakening its available $b+3$ bound. The discrepancy allowance H is also at exponent $b+2$. The final forced bound will have exponent $b+1$, since the connected-set estimate consumes the reserve of one. Both allowances are evaluated at the same scale and the same reference band. The numerical payment chooses $\kappa \geq 0$ so that, with

$$I = e^{\kappa}\gamma + 2H,$$

the support-tree criterion at reserve one holds and

$$H \leq 1, \qquad 4b_B I \leq \tfrac{1}{2}, \qquad 2e^{-\kappa} I \leq \phi. \tag{703}$$

These are the inflation, degree and return inequalities of the exponential payment argument in Section 200. Its hypotheses hold here: H has negative upper exponent, while γ retains the compulsory exponentially small mark. The prescribed outgoing gain leaves the stated positive margin in that mark. The selected payment is uniform over the coupling band.

Here is why those inequalities pay the analytic coefficients, not only the real values. Give an old forced entry with series majorant k_l the inflated price $e^{\kappa}k_l$, and give a discrepancy entry with series price h_l the price $2h_l$. A nonempty support and the rooted bound imply $h_l \leq H \leq 1$. The coefficient algebra therefore gives

$$|e^{-d_l} - 1|_{R/4} \leq e^{h_l} - 1 \leq 2h_l.$$

The last inequality holds on $[0,1]$, for example by convexity of the exponential between its endpoint values and $e - 1 < 2$. Every final forced component contains at least one old forced entry. Its exact product of the old forced factors and the exponential-minus-one discrepancy factors consequently has series norm at most

$$e^{-\kappa} \prod_{l \in D} F_l, \qquad F_l = \begin{cases} e^{\kappa}k_l, & l \text{ old forced}, \\ 2h_l, & l \text{ discrepancy}. \end{cases}$$

Indeed each old forced factor contributes $e^{-\kappa}$; retaining just one of them only increases the bound since $\kappa \geq 0$. The eligibility predicate for a component is independent of t, so it preserves the same majorant. The connected-set estimate, with rooted allowance I and degree ratio at most $1/2$, gives the final rooted allowance $2e^{-\kappa}I \leq \phi$. This proves a common coefficient majorant at radius $R/4$ for all configurations and all entries of the actual final forced array.

The exact additive decomposition of Section 142 identifies these ordinary and forced outputs with the selected one-step source. Their dictionary is fixed by the common path and incoming supports. The auxiliary test representations may be used outside $[-1,1]$ to provide the real-node bounds, but on $[-1,1]$ they agree with the given parameter family. Thus all the finite tests and series recoveries above concern that same family. The ordinary bound (700) and the forced common majorant are precisely the two broad analytic inputs retained in Section 144. Construction of the funded path and its real-sector hypotheses remains separate.

144 Sharpening the finite-test prices

The real parameter tests of Section 145 need small prices on one fixed parameter interval. The local expansion first provides a bound on a growing interval and a better pointwise bound. A finite polynomial interpolation combines these two estimates. We give the interpolation and its numerical purchase, keeping the underlying wide-interval analytic estimate visible.

144.1 A finite interpolation estimate

Let $\mathcal{E}$ be a vector space of families on a fixed support dictionary, with a nonnegative budget $\mathcal{N}$ satisfying the triangle inequality and absolute homogeneity. A budget defined as an infimum over admissible summable majorants has these properties: add or scale the majorants and then take their infima. The argument below equally applies by retaining the majorants themselves. For a finite real signed measure λ on $[-1,1]$, set

$$M_r(\lambda)=\sup_{n\ge 0}\left|\int (t/r)^n\,d\lambda(t)\right|,\qquad r\ge 1.$$

Let $F:[-1,1]\to\mathcal{E}$. Suppose

$$\mathcal{N}\left(\int F\,d\lambda\right)\le BM_v(\lambda),\qquad \mathcal{N}(F(t))\le b\quad(-1\le t\le 1),$$

where $1\le q\le v$ and $b,B\ge 0$. The integrals here are finite linear combinations; no measurability of F is required. For each integer $k\ge 0$, a constant $U=U(k,q)\ge 0$, independent of the family, its dictionary and its volume, gives

$$\mathcal{N}\left(\int F\,d\lambda\right)\le M_q(\lambda)\left\{Ub+(1+U)(q/v)^{k+1}B\right\}. \tag{704}$$

Choose the $k+1$ distinct points $t_i=i/(k+1)$, $0\le i\le k$, and their Lagrange polynomials ℓ_i of degree at most k. Write $\ell_i(t)=\sum_{j=0}^{k}c_{ij}t^j$ and put

$$U_i=\sum_{j=0}^{k}|c_{ij}|q^j,\qquad U=\sum_{i=0}^{k}U_i.$$

For $\eta=M_q(\lambda)$, we have $|\int \ell_i\,d\lambda|\le \eta U_i$. Define

$$\Pi\lambda=\sum_{i=0}^{k}\left(\int \ell_i\,d\lambda\right)\delta_{t_i}.$$

Lagrange interpolation of each polynomial of degree at most k gives $\int t^j\,d(\lambda-\Pi\lambda)=0$ for $j\le k$. For $j>k$, use $|t_i/q|\le 1$ to get

$$\left|\int (t/q)^j\,d(\lambda-\Pi\lambda)\right|\le \eta+\sum_i\left|\int \ell_i\,d\lambda\right|\le \eta(1+U).$$

Since $q/v\le 1$, multiplication by $(q/v)^j$ proves

$$M_v(\lambda-\Pi\lambda)\le \eta(1+U)(q/v)^{k+1}.$$

The pointwise price and the triangle inequality give $\mathcal{N}(\int F\,d\Pi\lambda)\le \eta Ub$. Adding this to the wide-interval bound on $\lambda-\Pi\lambda$ proves (704). Thus the same constant works for the full regular derivative budget, including all its site sums; no count of dictionary entries enters the estimate.

144.2 The fixed-radius price at large coupling

Fix the stage and block length, and put $y = x + a$, where $a > 0$ is the fixed reference coupling increment. The broad analytic estimate to be sharpened has radius and price

$$v(x) = \tfrac{1}{2} y^{1/256}, \qquad B(x) = y^{-1/4}.$$

The local pointwise price has the form

$$b(x) = 3tC_{\mathrm{st}} x^{-11/10} + R(x),$$

where $t > 0$ is a small response parameter and $C_{\mathrm{st}} > 0$ is the fixed stage conversion constant. The remainder R has upper exponent at most $-23/20$: concretely, for every $\epsilon > 0$, $|R(x)| \le C_\epsilon y^{-23/20+\epsilon}$ eventually. This estimate comprises the local comparison and truncation errors; it is separate from the broad analytic bound.

Fix $q \ge 2$ and $s > 0$, take $k = 300$ in (704), and choose t small enough that $6Ut \le s$. The leading pointwise contribution is at most $(s/2)C_{\mathrm{st}} x^{-11/10}$. The other part of the bound is

$$UR(x) + (1+U)(2q)^{301} y^{-301/256-1/4}.$$

Its upper exponent is at most $-23/20$, because $301/256 + 1/4 = 365/256 > 23/20$. Since $23/20 > 11/10$ and $y/x \to 1$, that expression is eventually bounded above by $(s/2)C_{\mathrm{st}} x^{-11/10}$. Hence the fixed-radius test price is

$$sC_{\mathrm{st}} x^{-11/10}. \tag{705}$$

There is no requirement that U be small. It depends only on 300 and q; the response parameter is chosen after it, and the coupling threshold after all those choices. For example $t \le s/[6(1+U)]$ always suffices. Since $v(x) \to \infty$, the condition $q \le v(x)$ also holds beyond that threshold.

144.3 The reset and the final regular price

The reset and local Taylor subtraction are linear in the raw logarithmic family and in its extracted coupling coefficient. On a fixed stage, the exact regular output price has the form

$$C_0 H + C_1 |\alpha| \theta(x)^3, \tag{706}$$

where $C_0, C_1 \ge 0$ are independent of the coupling, period and input family, H is the raw test price, and $\theta(x) = x^{-1/2} \exp((\log\log x)^2)$ is the fluctuation threshold. The two constants collect respectively the reset, low-path and high-path derivative prices, and the high-path remainder. This is the same local subtraction estimate used in the nonlinear transition, applied to a finite linear combination before taking its majorant.

The stage conversion constant is selected as

$$C_{\mathrm{st}} = \frac{1}{64(1+C_0)}.$$

It is positive and satisfies both $C_{\mathrm{st}} \le 1/64$ and $C_0 C_{\mathrm{st}} \le 1/64$. Fix $d > 0$, take $0 < s \le d$ in (705), and retain the fixed bound $|\alpha| \le A$. The first term of (706) is at most $(d/64)x^{-11/10}$. The second is $o(x^{-11/10})$, since

$$\frac{\theta(x)^3}{x^{-11/10}} = x^{-2/5} \exp(3(\log\log x)^2) \longrightarrow 0.$$

Increasing the coupling threshold therefore gives regular output price at most $dx^{-11/10}$.

This remains a finite-test statement. If the moment bound of the finite test is η, the raw price is ηH, the constant path correction becomes the same finite combination of α, and its modulus is at most ηA by the zeroth moment test. The linear reset and the triangle inequality thus multiply (706) by η. Dividing by the positive regular normalization gives the regular part of (711). The scalar coupling coefficient is a bounded linear projection of the raw family, with price $H/\theta(x)^2$. Consequently its finite-test price is exactly the scalar allowance $f_m = sC_{\mathrm{st}}x^{-11/10}/\theta(x)^2$ used there.

144.4 The forced price and the remaining analytic input

A finite-test bound alone may choose its majorant after the test, so recovery of a power series for each value separately does not yet produce one common support majorant. We supply that step. Let S_i be the countable list of nonempty supports and let $e \geq 0$ be the cover exponent. Suppose that for every finite test λ there are numbers $k_\lambda(i) \geq 0$ such that

$$\left|\int F(t,i,U)\,d\lambda(t)\right| \leq k_\lambda(i) \quad \text{for every configuration } U,$$
$$\sum_{i:z\in S_i} k_\lambda(i)e^{eC(S_i)} \leq BM_q(\lambda) \quad \text{at every root } z.$$

Here $C(S_i)$ is the cover number. Nonemptiness and $e \geq 0$ imply $k_\lambda(i) \leq BM_q(\lambda)$ for each i. For each entry and configuration, the closed-span argument of Section 145 therefore extends finite-test evaluation to a bounded functional $T_{i,U}$ on the closed span of the moment vectors $v(t) = ((t/q)^n)_n$.

Fix n. Choose finite tests λ_j whose moment vectors $v(\lambda_j)$ converge in sup norm to e_n; these approximants exist by the coordinate-vector construction above. In particular $M_q(\lambda_j) \to 1$. Choose their majorants k_j as in the hypothesis. Their coordinates lie in one common bounded interval. A diagonal subsequence on the countable indices i gives $k_j(i) \to b_{n,i} \geq 0$. For each finite set of indices and root z, pass to the limit in its weighted sum. Taking the supremum over finite sets yields

$$\sum_{i:z\in S_i} b_{n,i}e^{eC(S_i)} \leq B.$$

Also $|T_{i,U}e_n| \leq b_{n,i}$, by continuity of $T_{i,U}$. This last conclusion holds for every U on the same subsequence: the chosen majorants bounded all configurations from the start, and convergence of the moment vectors is independent of U. No subsequence indexed by configurations is selected.

For $1 \leq r < q$, put

$$K_i = \sum_{n\geq 0}(r/q)^n b_{n,i}.$$

The bounds $b_{n,i} \leq B$ make each series finite. Tonelli's theorem for the nonnegative weighted series gives

$$\sum_{i:z\in S_i} K_i e^{eC(S_i)} \leq \frac{B}{1-r/q}.$$

The coefficients $(r/q)^n T_{i,U}e_n$ have total absolute sum at most K_i and reconstruct $F(t,i,U)$ as a series in t/r. Thus one common support majorant works for every configuration. Applied to the soft correction from the linear reset with $r = q/2$, this proves the required factor-two price and radius loss.

The compatible finite-subset products preserve absolute coefficient summability: their coefficients obey convolution, and the sum of coefficient norms of a product is bounded by the product of the corresponding sums. The proof is Tonelli's theorem for the nonnegative coefficient majorants.

Thus the forced absorption argument of Section 200 applies also to these parameter coefficient prices, with the same enlarged support reserve and the same scalar parameter κ. It gives a forced output price at most d times the prescribed forced normalization after increasing the entry threshold. The test inequality follows by applying each finite moment test to the absolutely summable coefficient series. This supplies the forced clause of (711) whenever the broad ordinary and forced analytic estimates have been supplied.

The broad estimates used in this section are substantive: the ordinary raw logarithmic family must have radius $y^{1/256}/2$ and price $y^{-1/4}$ in the full derivative budget, and the raw forced family must have its common coefficient majorant at the paid broad radius. The pointwise remainder R must have the stated upper exponent. The interpolation and reset estimates above do not themselves prove those statements. They explain exactly how those local analytic estimates, once obtained for the same transition and parameter line, give the small prices needed by the coupled history argument.

145 Real parameter tests and the background contraction

The small Lipschitz constant used in the background history does not follow from a small bound on the values alone. It follows from the paid bounds on finite linear combinations along each real parameter line. We prove the conversion explicitly. Only real configurations are used in these tests.

145.1 Recovering a convergent series from finite tests

Let E be a real Banach space, $p > 1$, $C \ge 0$, and $f : [-1, 1] \to E$. Suppose that, for every finite list $t_1, \dots, t_k \in [-1, 1]$ and real weights $a_1, \dots, a_k$,

$$\left\| \sum_{j=1}^{k} a_j f(t_j) \right\| \le C \sup_{n \ge 0} \left| \sum_{j=1}^{k} a_j (t_j/p)^n \right|. \tag{707}$$

No continuity of f is assumed here. There are vectors $b_n \in E$ with $\|b_n\| \le C$ such that

$$f(t) = \sum_{n=0}^{\infty} (t/p)^n b_n \qquad (-1 \le t \le 1). \tag{708}$$

To prove this, use the sup norm on sequences and let $v(t) = ((t/p)^n)_{n \ge 0}$. Its coordinates tend to zero. Define, on the span of these sequences,

$$T\left(\sum_j a_j v(t_j) \right) = \sum_j a_j f(t_j).$$

If the sequence on the left is zero, (707) makes the proposed value zero. Thus T is well defined and linear, and its norm is at most C. It extends uniquely to the closed span, because E is complete.

Every coordinate vector e_n lies in that closed span. For $n = 0$, use $v(0) = e_0$. Inductively assume this for $e_0, \dots, e_{n-1}$. For $0 < t \le 1$, put $r = t/p$. The vector

$$r^{-n} \left(v(t) - \sum_{j<n} r^j e_j \right)$$

has its first n coordinates zero, its nth coordinate one, and subsequent coordinates $r, r^2, \dots$. Its sup distance from e_n is $r \to 0$ as $t \downarrow 0$. This proves the claim. Set $b_n = Te_n$, so $\|b_n\| \le C$. The geometric series $v(t) = \sum_n (t/p)^n e_n$ converges in the sup norm. Apply T to it to obtain (708).

For $1 \le a < p$, the same identity can be written $f(t) = \sum_n (t/a)^n c_n$ with $c_n = (a/p)^n b_n$ and

$$\sum_n \|c_n\| \le \frac{C}{1 - a/p}. \tag{709}$$

This is the radius loss used when passing from finite tests to absolutely summable analytic coefficients. In particular, taking $a = p/2$ costs at most a factor two when $p \ge 2$. The argument also applies to maps into the completion of a seminormed space; no completeness of its original representatives is needed.

Two consequences will be used. First, a test with one point and weight one gives $\|f(t)\| \le C$. Second, subtracting the series at zero gives

$$\|f(t) - f(0)\| \le C \frac{|t|}{p - |t|}. \tag{710}$$

Both conclusions hold even if the initial pointwise construction of f involved arbitrary choices: the finite tests themselves force the displayed series and hence continuity on $[-1, 1]$.

145.2 Applying the tests to the actual activity coordinates

At one adjacent pair of scales, retain the actual common dictionary of regular and forced arrays. Its norm is the maximum of the normalized regular and forced budgets, with all the fixed symmetry and support constraints retained. Finite linear combinations remain in this vector space. Suppose a line of allowed inputs has normalized coupling $\lambda + t\mu$ and activity array $x + ty$, $|t| \le 1$. Let $F(t)$ be its selected output array in the completion and $Q(t)$ its scalar coupling correction.

The exact finite-test receipts used here are the following. For any finite weights a_j and points t_j, write

$$\eta_r = \sup_{n \ge 0} \left| \sum_j a_j (t_j / r)^n \right|.$$

The combined regular output array has normalized budget at most $d\eta_{q/2}$, as does the combined forced output array. The scalar correction satisfies

$$\left| \sum_j a_j Q(t_j) \right| \le f_m \eta_q.$$

The forced statement follows from its single common summable coefficient majorant; the regular statement is a bound on the actual mixed arrays, with their derivative prices, not just on values of their gas. The definition of the array norm and the isometric embedding into its completion consequently give

$$\left\| \sum_j a_j F(t_j) \right\| \le d\eta_{q/2}, \qquad \left| \sum_j a_j Q(t_j) \right| \le f_m \eta_q. \tag{711}$$

These are precisely the two instances of (707) needed below. Padding the dictionaries with zero entries, or using another representative of the same array, preserves them.

We spell out why the admissible lines are the ones used in the finite-test construction. Their activity values are affine in t on each fixed support. On a healthy configuration the regular mask is one and the value is the corresponding affine combination; on an unhealthy configuration the common mask is zero and the same equality holds. Forced values are affine without a derivative condition. The inverse coupling along the line is $\widehat{\beta}\{w(\lambda + t\mu)\}^2$, with the same reference $\widehat{\beta}$ and $w = \widehat{\beta}^{-1/32}$. Thus it is exactly the prescribed quadratic coupling line, not an independent choice of complexified gauge fields. The protected supports, normalization, path data and transition are the same for all t. The analytic estimates that pay (711) are retained as quantitative inputs here; the deduction of the contraction from those estimates is complete below.

145.3 The full-band Lipschitz estimates

Take $q \geq 4$, $d, f_m \geq 0$, $2d \leq \gamma = 1/128$, and $s_m = 2f_m$. Assume $w \leq 1/128$, as purchased by the coupling threshold. Consider two inputs $z = (\lambda, x)$ and $z' = (\lambda', x')$ with $\|x\|, \|x'\| \leq 1/8$ and $17/20 \leq w\lambda, w\lambda' \leq 23/20$. Use the product norm

$$h = \|z' - z\| = \max\{|\lambda' - \lambda|, \|x' - x\|\}.$$

The constant-line tests give $\|F(z)\|, \|F(z')\| \leq d$ and $|Q(z)|, |Q(z')| \leq f_m$. If $h > 1/4$, the triangle inequality therefore gives

$$\|F(z') - F(z)\| \leq 2d \leq 8\gamma h, \qquad |Q(z') - Q(z)| \leq 2f_m \leq 8f_m h = 4s_m h.$$

For $h = 0$, no equality of raw representatives is needed. For any $0 < \varepsilon \leq 1$, take the line $z + t(z' - z)/\varepsilon$, $|t| \leq 1$. Its coupling direction is zero and its activity direction has seminorm zero. Every point therefore has the same activity norm as z, at most $1/8$, so the line is admissible. Its value at $t = \varepsilon$ is z'. Equation (710) bounds the output differences by $d\varepsilon/(q/2 - \varepsilon)$ and $f_m\varepsilon/(q - \varepsilon)$, respectively. Letting $\varepsilon \downarrow 0$ proves equality of the output classes and of the scalar correction. This also proves that the map respects the seminorm kernel; it does not identify regular representatives on unhealthy configurations.

For $0 < h \leq 1/4$, set $v = (z' - z)/(4h)$. Its norm is $1/4$, and the full line $z + tv$, $|t| \leq 1$, is allowed. Its activity norm is at most $1/8 + 1/4 < 1$. Its normalized coupling lies in the broader interval

$$33/40 \leq w(\lambda + tv_1) \leq 47/40,$$

since $w|tv_1| \leq 1/512 < 1/40$, the margin between the narrower and broader intervals. The prescribed coupling-line construction therefore supplies (711) on this line. Apply (710) at $t = 4h \leq 1$ to get

$$\|F(z') - F(z)\| \leq \frac{4d}{q/2 - 1} h \leq 4dh \leq 8\gamma h, \qquad |Q(z') - Q(z)| \leq \frac{4f_m}{q - 1} h \leq 8f_m h.$$

This proves exactly (714) and (715). The output bounds also follow from the constant-line tests, since $d \leq 1/256 < 1/8$ and $f_m = s_m/2$.

The use of one-dimensional series here is sufficient for these Lipschitz estimates. The same real finite-test method also produces multilinear Taylor coefficients by polarization when a full analytic parameterization is required. That additional conclusion is not needed in the convergence argument of Section 146.

146 Convergence of the coupling and activity histories

We next prove convergence of the background on which the marked comparison depends. The coupling and the activities must be treated together. The activity recursion contracts towards coarser scales, whereas the coupling is fixed at the bottom scale. A finite telescoping identity joins these two directions. In particular, the proof does not assume that the activity space has compact bounded balls.

146.1 The normalized variables and their paid estimates

Fix a period. Let $\widehat{\beta}_m > 0$ be the reference coupling at scale m, and set

$$w_m = \widehat{\beta}_m^{-1/32}, \qquad B_m = \widehat{\beta}_m w_m = \widehat{\beta}_m^{31/32}, \qquad b_m(\lambda) = \widehat{\beta}_m (w_m \lambda)^2.$$

A coupling β is represented by $\lambda = w_m^{-1}\sqrt{\beta/\widehat{\beta}_m}$. On the paid band its normalized variable satisfies

$$17/20 \le w_m \lambda \le 23/20.$$

Thus, for two variables in that band,

$$B_m|\lambda - \lambda'| \le |b_m(\lambda) - b_m(\lambda')|. \tag{712}$$

Indeed the difference on the right is $B_m|\lambda-\lambda'|\,|w_m\lambda+w_m\lambda'|$, whose last factor is at least $17/10 > 1$.

Let $\mathcal{H}_m$ be the completion of the seminormed space of actual regular and forced activity arrays with the prescribed weighted derivative and value prices. The embedding preserves distances. In particular, equality of evaluated gases alone is not the equivalence relation used here. Let x_m denote the embedded array. The finite-history construction gives $\|x_m\| \le 1/8$. For a terminal index N, write

$$x_m = F_{m,N}(\lambda_{m+1}, x_{m+1}), \qquad q_m = Q_{m,N}(\lambda_{m+1}, x_{m+1}) \quad (m < N)$$

for its activity transformation and the scalar correction to the coupling. These maps are initially defined on the actual allowed arrays; the notation does not assume an extension to every element of the completion. Their output and comparison estimates are

$$\|F_{m,N}(\lambda, x)\| \le 1/8, \qquad |Q_{m,N}(\lambda, x)| \le s_m/2, \tag{713}$$

$$\|F_{m,N}(\lambda, x) - F_{m,N}(\lambda', x')\| \le c\max\{|\lambda-\lambda'|, \|x-x'\|\}, \tag{714}$$

$$|Q_{m,N}(\lambda, x) - Q_{m,N}(\lambda', x')| \le 4s_m \max\{|\lambda-\lambda'|, \|x-x'\|\}, \tag{715}$$

where $c = 8/128 = 1/16$ and $s_m \ge 0$. The numerical schedule also pays

$$\sum_{j<m} s_j \le \frac{1}{128} B_m. \tag{716}$$

The coupling recursion is

$$b_m(\lambda_m) - b_{m+1}(\lambda_{m+1}) - \alpha_{m,N} = q_m, \tag{717}$$

where $\alpha_{m,N}$ is the path correction, already convergent at each fixed m by Section 139.

We use the following precise continuity statement for the local transformation: if two sequences of terminal indices tend to infinity, their input normalized couplings have a common limit, and their actual input arrays are Cauchy with difference tending to zero in $\mathcal{H}_{m+1}$, then the differences of their output arrays and scalar corrections tend to zero. This statement concerns arbitrary allowed inputs,

including the truncated histories constructed below. Its analytic content is the common-variable comparison of Section 116 without a mark, together with the weighted head-and-tail argument of Section 115. Here is the precise tail implication. For a regular entry put $b_f(i) = \sup_U |f_i^{\mathrm{reg}}(U)|$, with its actual mask, and for a forced entry put $b_f(i) = \sup_U |f_i^{\mathrm{forced}}(U)|$. Summing the site's paid budgets gives, separately for these two families,

$$\sum_i b_f(i) \le C_m \|f\|, \qquad b_f(i) \le b_{f-g}(i) + b_g(i).$$

The constants are the finite number of sites times the corresponding positive regular or forced normalization. Fix one sufficiently late array f_{N_0} and choose a finite set outside which its b-sum is small. For every later array, its tail sum is at most that fixed tail plus $C_m\|f_N - f_{N_0}\|$. Enlarge the finite set for the finitely many earlier arrays. This gives the uniform value-price tails needed by the compensated comparison. It does not assert that arbitrary activity arrays can be truncated in their full derivative norm.

For the scalar output, use convergence of the finite projected coupling coefficient itself: its projection is bounded by the local derivative price, whose convergence is supplied by the compensated comparison. This is in addition to convergence of the output activity array. Subtracting the convergent incoming coupling and path correction gives continuity of Q by (717). The bounds (713)–(715) are the quantitative one-step hypotheses used in this section; their purchase is kept separate from the convergence proof.

146.2 First fix the normalized coupling history

Suppose two sequences of allowed normalized coupling histories $\lambda_{m,N}$ and $\lambda'_{m,N'}$ have the same limit for every fixed m. For an integer $h \ge m$, construct a truncated activity history by putting $x_h^{[h]} = 0$ and applying the same finite-history maps backwards down to m. This construction is allowed for every prescribed coupling history in the band: (713) keeps all its arrays in the ball of radius $1/8$. It does not impose the coupling recursion on this auxiliary history.

Compare it with the original activity history for the same terminal index and the same prescribed couplings. At scale h the difference is at most $1/4 \le 1$. By (714), backwards induction gives

$$\|x_m - x_m^{[h]}\| \le c^{h-m}. \tag{718}$$

For fixed h, the two truncated histories have common limits at every scale at most h. At h they are both zero. If this is known at $i+1$, their arrays are Cauchy and their difference tends to zero there; the local continuity statement and the common normalized coupling limit give output differences tending to zero at i. To get convergence, apply this comparison to arbitrary pairs of subsequences of either truncated history. If that history were not Cauchy, one could select indices $u_n, v_n \ge n$ whose distances stay above some $\varepsilon > 0$, contradicting the comparison. Completeness of $\mathcal{H}_i$ then gives a limit, common to the two histories. This completes the finite backwards induction. It also proves common limits of their scalar corrections at scales below h.

Now choose $h - m$ so large that $2c^{h-m} < \varepsilon/2$. Equation (718) and the convergence of the two truncated histories show that the original activity difference is eventually below ε. Apply the same arbitrary-subsequence argument once more. The original activity histories converge in $\mathcal{H}_m$ to the same limit. Their scalar corrections then have common limits by local continuity at the next scale. Thus convergence of all the normalized couplings implies convergence of all the activities and costs.

146.3 Identifying the possible coupling limits

At each fixed scale the band for $\lambda_{m,N}$ is a compact real interval. Successive subsequence selection at scales $0, 1, \dots$ followed by the diagonal subsequence gives a subsequence on which every normalized coupling converges. The preceding argument attaches activity and scalar-correction limits to it. Denote one such limiting history by (λ_m, x_m, q_m) and another by (λ'_m, x'_m, q'_m). Suppose the bottom couplings of the two sequences have a common limit. Passing to the limit in (717) and subtracting gives the exact finite identity

$$b_m(\lambda_m) - b_m(\lambda'_m) = -\sum_{j<m} (q_j - q'_j). \tag{719}$$

The common limits of $\alpha_{j,N}$ cancel. There is no infinite sum and no terminal boundary term in this identity.

The one-step inequalities also pass to these limits, despite being initially defined only for actual input arrays. To see this without postulating an extension of $F_{m,N}$, construct at each terminal index a new backwards activity history with the fixed normalized coupling script $(\lambda'_i)_i$ and zero terminal array. Its couplings are allowed, because the band is closed. By the previous subsection, its activities and costs have precisely the limits x'_m, q'_m. Compare it to the first history at the same terminal index using (714) and (715), then pass to the limit. Writing

$$d_m = \max\{|\lambda_m - \lambda'_m|, \|x_m - x'_m\|\},$$

we obtain

$$\|x_m - x'_m\| \le c d_{m+1}, \qquad |q_m - q'_m| \le 4 s_m d_{m+1}. \tag{720}$$

First observe that $d_m \le 1/4$ for all m. The activity bound follows from $\|x_m\|, \|x'_m\| \le 1/8$. For the normalized coupling, (712), (719), the bound $|q_j|, |q'_j| \le s_j/2$ and (716) give

$$B_m |\lambda_m - \lambda'_m| \le \sum_{j<m} s_j \le B_m/128.$$

Thus $D = \sup_m d_m$ exists and lies in $[0, 1/4]$. Using the sharper cost comparison now gives

$$B_m |\lambda_m - \lambda'_m| \le 4 \sum_{j<m} s_j d_{j+1} \le \frac{4D}{128} B_m.$$

Together with (720) this yields $d_m \le (8/128)D$ for every m. Taking the supremum, $0 \le D \le D/16$, hence $D = 0$. Both limiting normalized coupling and activity histories are identical. The costs are identical too, by their one-step comparison.

Every subsequence has a further subsequence with all normalized couplings convergent, and all such limits have just been identified. If the original normalized coupling at some scale failed to converge to this value, a subsequence would remain outside one fixed neighborhood, contradicting its convergent further subsequence. Thus the entire normalized coupling sequence converges at every scale. The first part of the proof then gives convergence of the entire activity and correction sequences. In particular this applies when the bottom coupling is fixed exactly, as in the selected public construction.

146.4 Why the cumulative payment can be imposed

The numerical inequality (716) is a finite cumulative bound, not summability of a fixed sequence of costs over all scales. We record its purchase to distinguish the two assertions. On a stage with positive reference increment a, put $B(x) = x^{31/32}$. For $x \geq a > 0$, the mean value theorem and monotonicity of $x^{-1/32}$ give

$$B(x+a) - B(x) \geq a\frac{31}{32}2^{-1/32}x^{-1/32}.$$

The corresponding cost allowance is twice the activity allowance divided by the squared fluctuation threshold. Its paid form is

$$s(x) = \frac{2vC_{\mathrm{st}}x^{-11/10}}{t(x)^2}, \qquad t(x) = x^{-1/2}\exp((\log\log x)^2),$$

where $v, C_{\mathrm{st}} > 0$ are fixed at this stage. For sufficiently large x, $t(x) \geq x^{-1/2}$, so $s(x) \leq 2vC_{\mathrm{st}}x^{-1/10}$. Since $-1/10 < -1/32$, increasing the entry threshold gives

$$s(x) \leq \frac{1}{128}\{B(x+a) - B(x)\}.$$

There are only finitely many permitted next stages at each stage, so their thresholds can be increased together after their block lengths, positive increments and constants have been fixed. The reference recursion is $\widehat{\beta}_{m+1} = \widehat{\beta}_m + a_m$. Summing the preceding inequalities over $j < m$ telescopes and gives

$$\sum_{j<m} s_j \leq \frac{1}{128}(B_m - B_0) \leq \frac{1}{128}B_m.$$

This proves the cumulative estimate required in the uniqueness argument. It preserves the chosen maps and block lengths, just as the other entry-threshold increases do.

147 One nonlinear return on a whole parameter interval

The local estimates must supply a return which varies continuously with its input. Selecting a new support dictionary independently at every parameter would not suffice. We explain how to obtain the exact integral identity, both activity bounds and continuity in one dictionary on an arbitrary compact parameter interval. The scale purchases used below are fixed before that interval and the finite volume are chosen.

Fix a finite cutoff N, adjacent levels $m+1, m$ with $m < N$, and a period. Write p_f, p_c for the two activity budgets, x_0 for the reference coarse coupling and L for the block side. The fine reference coupling is $\beta_F(x_0) = x_0 + s_G \log L$. Fix the already selected incoming and outgoing paths, their block bridge and the scalar shift $\gamma = \alpha_{N,m}$. Let $I = [a, b]$ with $a \leq b$, and let Q_x, $x \in I$, be a continuous family of fine representations in the permitted fine band. Continuity here includes the regular and forced coefficient arrays in their fixed labelled norms. Each Q_x has the selected fine path and satisfies the same history constraints.

We use the following purchased estimates on these data. The regular principal price is at most $E/16$, the complementary regular price at most $2E/16$, and the preliminary forced price at most $K(x)/8$, where

$$E = c_{\mathrm{st}}\delta_E(x_0), \qquad K(x) = K_{\mathrm{forecast}}(Q_x.\beta), \qquad \delta_E(x_0) = x_0^{-11/10}, \quad c_{\mathrm{st}} > 0.$$

These are the principal and complementary bounds of the preceding local expansion, in the actual coarse budget. The selected path has its full quadratic, cubic, quartic and loop bounds, with width $\sigma_c > 0$; its loop allowance is the selected allowance at level m. The initial action is admissible in this same budget. If A_c denotes the support exponent used before the finite symmetry average, the scale purchases also give

$$A_c = b_c + 3 \le a_{\mathrm{earn}}, \qquad A_c/B_c \le \sigma_c/4. \tag{721}$$

Here b_c is the coarse support exponent, B_c its radius and a_{earn} the earned exponent of the local expansion. In particular, these width and exponent conditions concern the same output path as the bounds, rather than a less expensive substitute.

147.1 Keeping the physical and auxiliary data fixed

At each $x \in I$, apply the local expansion to Q_x on the physical torus. Its hypotheses are supplied as follows. The block equation and both adjacent kernel bounds come from the selected horizon. Its all-period positive lower bounds apply to the two selected kernel levels; the bridge intertwining identities and the periodized rows are those of that same horizon. The input representation supplies the small-field and forced budgets and the fine band condition. The expansion is applied with the selected derivative orders, spatial weights, earned exponent, and its two error reserves. Its differentiated remainder bounds are retained with its output, as well as the undifferentiated integral identity. They cannot be recovered from that identity alone.

Use the expansion at $x = a$ as the reference dictionary. For every other x, identify elementary labels by the identity: the forests, geometric supports, sector labels and component types are unchanged. The path identity and the fixed regular and forced input dictionaries make this an admissible identification. Thus all output arrays can be read in the reference dictionary. This is precisely the comparison used in Section 102; no claim that an arbitrary choice of coordinates depends continuously on x is needed.

The auxiliary tori used for local comparisons carry the same block bridge and the same two paths. On such a torus the reference fine representation is the bare path action with the pale activity array: its density is

$$\mathbf{1}_{\{\text{all fine sites healthy}\}} \exp\{-\beta \mathcal{K}_f - \mathcal{L}_f\}.$$

Here $\mathcal{K}_f$ and $\mathcal{L}_f$ are the fine kinetic and loop terms. The pale array has zero regular cost and its stated forced cost. The fine path bounds and the clock inequality for the loop allowance therefore give the required incoming budgets. The fine threshold, radius and band are exactly those at $\beta_F(x_0)$. The all-period kernel bounds apply on every such torus, so the local expansion and its differentiated remainder bounds are available there too. The auxiliary expansions provide local comparisons; the density reconstruction below is performed on the physical torus.

147.2 The finite short density and its complement

The regular dictionary contains two kinds of entries: connected interaction clusters and path atoms. Divide them into a retained infinite part and an exhaustible short part. The selected interaction clusters have at most two polymer vertices, each formed from at most two elementary path letters of word length at most B_c. Their connected coefficient is nonzero and each polymer is admissible. These letters have empty input roster: they contain neither incoming regular insertions nor incoming forced slots. Every cluster outside this particular class is retained in the complementary array. Its regular price is at most $3E/16$ by the principal and complementary estimates. The retained path

part consists of atoms of word length greater than B_c. Let $\mathcal{P}_d$ contain the selected interaction clusters and the short path atoms whose elementary indices lie below the finite cutoff d. The panels are increasing and exhaust every nonretained entry. Leave the forced array unchanged at this intermediate operation, and set the regular entries of $\mathcal{P}_d$ to zero. Call the resulting array $T_{x,d}$.

This complement is the sum of the retained infinite part and the unselected short tail, entry by entry. For a selected short entry both sides vanish; for a retained entry only the retained summand is present; for every other entry only the tail is present. Thus the triangle inequality adds their regular budgets without changing any support. The path tail requires a separate estimate. If its original width is σ_c and its word length exceeds B_c, exponential summability and $A_c/B_c \le \sigma_c/4$ give a tail bound

$$e^{-\sigma_c B_c/4}\mathcal{C}_{\mathrm{path}} \le g, \qquad g = \beta_F(x_0)^{-23/20}.$$

Here $\mathcal{C}_{\mathrm{path}}$ is the finite path-jet price at the chosen stage. The first inequality follows by splitting the exponential row weight into the part paying the support weight and the part paying the tail; the remaining positive weight is discarded. The second is one of the scale purchases, imposed after the stage is fixed. The finite symmetry average costs C_{turn}, so the retained total has budget $3E/16 + C_{\mathrm{turn}}g$.

The unselected short tail has budget at most g for all sufficiently large d, uniformly in $x \in I$. Here is the uniformity argument in the actual jet norms. On the fixed torus changing an anchor costs at most $((1+D)^p)^j$, where D is its diameter and p, j are the spatial power and derivative order. The sum of the rooted regular majorants is finite, since there are finitely many anchors. Thus at any fixed x there is a summable majorant in the reference anchors, and exhaustion of the short labels makes its unselected sum tend to zero. Norm continuity gives a neighborhood of x where the difference array has arbitrarily small total price, so the same truncation works there. A finite subcover of I gives one truncation for the whole interval. Consequently

$$\|T^{\mathrm{reg}}_{x,d}\|_{p_c} \le h_0 := \frac{3}{16}E + C_{\mathrm{turn}}g + g. \tag{722}$$

The dependence of the tail index on the fixed torus and interval is allowed; its bound h_0 uses the purchased, volume-independent budget.

Allocate each entry F_a of the finite panel among its allowed anchors with nonnegative weights ω_{oa} summing to one. For an interaction cluster this weight is the reciprocal of the number of its allowed anchors; for a path atom it is the corresponding fixed local allocation. Average over signed lattice symmetries and translations. Define the unmasked short density

$$f_{x,d}(o,U) = \sum_{a \in \mathcal{P}_d} \omega_{oa} F_{x,a}(U).$$

This is the finite connected-cluster contribution plus the finite path-atom contribution, with the same scalar shift γ. The allocation is local: a nonzero weight implies that the entry's support lies in a fixed radius-r neighborhood of o. The weights sum to one even after the finite average. The masked panel contribution is therefore the sum of

$$E_{x,d}(o,U) = \sum_{a \in \mathcal{P}_d} \omega_{oa} \mathbf{1}_{\{U \text{ healthy on } \operatorname{supp} F_a\}} F_{x,a}(U).$$

147.3 Compatibility and the five auxiliary offers

Use the auxiliary bare inputs described above on the five tori of sides $2(w+i)$, $0 \le i \le 4$, with $w \ge 2r$. Their coupling is $Q_x.\beta$, and their path, budget, bridge, scalar shift and finite label cutoff

d are the physical ones. Let $f^{(i)}_{x,d}$ denote the corresponding short density. These are the actual auxiliary offers to the five-period construction of Section 142.

We check its compatibility condition. Under replication to a larger period, each short cluster meeting the designated anchor has a unique lift within the fixed local patch. Conversely projection gives its unique short cluster on the smaller period. The geometric margin $4r+16 \le |\Lambda|^{1/4}$ prevents wrapping in these identifications. They preserve the elementary component types and the cutoff on their labels, so they give a bijection of the finite panels at that anchor. The local connected coefficient is preserved as follows. On globally healthy fields the mask in each selected cluster equals one, so its paid masked response equals the actual connected coefficient. The graph coefficient depends only on the overlaps and equality relations of the selected polymers; the local bijection preserves both. Its factorial normalization is unchanged. The remaining coefficient is the product of the corresponding polymer integrals.

For each polymer choose matching forests on the two periods. The selected letters have empty roster, and the fields are globally healthy, so the local integrals have no imposed input insertions or flags. Let I be the finite set of fluctuation coordinates read by the larger-period integrand, and let $e : I \hookrightarrow J$ be its embedding into the smaller-period coordinate set. The period margin makes this map injective. The local windows correspond bijectively: their principal matrices have identical entries, since the local fields and paths agree. Their inverses therefore correspond, as do their extensions by zero outside the windows. The covariance is the symmetrized local inverse. Consequently, if C' and C are the two covariances, then

$$C'|_{I\times I} = C[e(I), e(I)].$$

This assertion concerns finite covariance submatrices; the full Gaussian spaces can have different dimensions.

For completeness, a centered Gaussian with covariance C has characteristic function

$$\exp(-\langle t, Ct\rangle/2).$$

Its projection to the coordinates $e(I)$ therefore has characteristic function

$$\exp(-\langle t, C[e(I), e(I)]t\rangle/2).$$

The covariance identity shows that this projected law is the same as the larger Gaussian's marginal on I. Locality expresses both integrands as the same measurable function of these coordinates: the block tags, frame tags, elementary path words, and transported mean agree. Thus their Gaussian integrals agree. This equality holds for every pair of outer Haar-slice values, because the selected, unrostered terms are independent of those slice choices after the matching local data are fixed. The inner integral is consequently a common constant; integrating against the normalized outer Haar measures preserves it. This proves equality of the local integrals, including their normalizations.

The forests used by the expansion are averaged, so equality for one matching pair must be lifted to equality of those averages. Forests are generated from finite independent local choices. Restrict these choices to the embedded smaller patch and project each local choice to its smaller-period version. The resulting map P between the finite choice sets has constant-size fibers: restriction forgets independent coordinates, and the local projection has constant-size fibers at each retained coordinate. Hence for any function F on the smaller choice set $\mathcal{R}$,

$$\frac{1}{|\mathcal{R}'|}\sum_{r'\in\mathcal{R}'} F(Pr') = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} F(r).$$

Locality lets one replace an arbitrary larger forest by the matching lift of its restriction without changing the polymer integral. Apply the displayed identity first to the block and frame tags, then to the forest choices. This proves equality of the actual averaged polymer coefficients. The normalized anchor weight is also preserved because its allowed-anchor set is in bijection. For path atoms the identity follows directly from replicating the same word of link increments. Finally the finite symmetry average commutes with replication: translation fibers all have equal cardinality, which cancels against the normalized average. Thus the short density is preserved on globally healthy replicated fields. Passing two tori to their product period and using locality there proves the required compatibility for equal local patterns. The auxiliary bare input and the physical input can be compared here because the short cluster construction uses the same path and the same healthy local generation; terms retained from the incoming regular array remain in the complementary part, rather than being identified with bare-input terms.

To price the centered auxiliary total, use its exact decomposition into the compensated interacting action, the complementary regular array and a constant. The first term has the healthy derivative bound of Section 110. For an auxiliary torus Λ_i, its complementary array has budget $C_{\mathrm{turn}}(g+g)+g$ after the same uniform-tail choice. Summing over its anchors bounds every allowed derivative of the total by $|\Lambda_i|[C_{\mathrm{turn}}(g+g)+g]$. Centering subtracts its flat value and costs at most a further copy of this bound. Consequently the centered short total has derivative price

$$q_i = \mathcal{T}_j(g) + 2|\Lambda_i|[C_{\mathrm{turn}}(g+g)+g],$$

where $\mathcal{T}_j(g)$ is the healthy-boundary derivative price of order j for the compensated action. The finite-volume continuity argument applies to the entire closed fine coupling band on each of these five fixed tori, so one d suffices for all their couplings. The comparison charts are admissible by the purchased local gauge extension and logarithmic-radius margin. These are precisely the hypotheses used in the boundary-jet estimate, and apply to the same compensated action and selected stationary path.

Let $g_{x,d}(o,U)$ be the folded counterterm made from these five centered offers. Formula (695) gives its jet bound $c' = \mathcal{O}_z(q_0,\dots,q_4)$ on the radius-z healthy neighborhood. The paid gauge extension puts every such healthy folded field into the chart just used; gauge invariance preserves the bound. Let M bound the coarse covering number of this neighborhood. Its weighted regular cost per anchor is

$$s' = c' e^{A_c M}.$$

This follows by multiplying its jet bound by its support weight, which is at most $e^{A_c M}$. It introduces no volume factor.

147.4 The discrepancy is a forced term

Use the shifted folded sum $h_{x,d}(o,U)$ of (691), and define

$$D_{x,d}(o,U) = E_{x,d}(o,U) - f_{x,d}(o,1) - h_{x,d}(o,U).$$

The exact identity (693) becomes

$$A_x(U) = A_{T_{x,d}}(U) + |\Lambda| f_{x,d}(0,1) + \sum_o \chi_{N_z(o)}(U) g_{x,d}(o,U) + \sum_o D_{x,d}(o,U). \tag{723}$$

Here A_x is the original regular sum, and $\chi_{N_z(o)}$ is its healthy indicator. The formula holds at every field. On a healthy radius-$2z$ neighborhood the allocation supports are healthy, and the folded

identity identifies h with the centered f. Hence $D_{x,d}(o,U) = 0$ there. The discrepancy is local on that enlarged neighborhood and has the gauge invariance and measurability inherited from its finitely many summands. Its value bound is obtained by the triangle inequality for the panel sum and the five folded sums; this is the local value estimate purchased along with the original expansion. Write q for this value bound, and M_2 for the covering-number bound of the doubled support. Its soft absorption cost is

$$b_{\mathrm{soft}} = \mathrm{ballCost}(2z)\, q e^{(b_c+2)M_2}.$$

The factor $\mathrm{ballCost}(2z)$ bounds the number of anchors whose discrepancy support can meet a fixed site; the extra exponent reserve is the one needed for the raised forced budget. Finite symmetry averaging preserves the corresponding normalized anchor sum. The local value estimates and logarithmic-radius geometry give $b_{\mathrm{soft}} \le \beta_F(x_0)^{-1/8}$ after increasing the entry threshold. This is a separate purchase from the regular jet cost $s' = c' e^{A_c M}$: q bounds the discrepancy on arbitrary fields, whereas c' bounds the folded counterterm's healthy jets.

Crucially, the last sum in (723) is not discarded. Put $-D_{x,d}$ into the logarithmic correction array, and absorb it into the forced gas by the exact identity $e^{-D} = 1 + (e^{-D} - 1)$ and the support-join construction of Section 91. The vanishing just proved supplies the eligibility condition of each new forced term. The old forced gas and this correction are combined with their actual support and inventory labels. Their hard input budget is at most $K(x)/8$, and their soft absorption cost is at most $\beta_F(x_0)^{-1/8}$; the purchased absorption inequalities give forced output budget $K(x)$. This operation changes the forced gas exactly by the factor $\exp(-\sum_o D_{x,d}(o,U))$.

The regular output consists of $T_{x,d}$ and the folded counterterms, followed by the fixed symmetry average. Its budget is at most $h_0 + C_{\mathrm{turn}} s'$. The scale purchase gives

$$C_{\mathrm{turn}} g + g + C_{\mathrm{turn}} c' e^{A_c M} \le E/2.$$

To justify that this purchase is available, g and c' have upper exponent $-23/20$, while E has exponent $-11/10$. Their fixed stage constants and powers of the logarithmic radius do not change this strict inequality of exponents. Thus an entry threshold makes the displayed inequality hold for every later coupling. Combining it with (722) gives the bound $11E/16 \le E$. The forced estimate above is retained separately.

147.5 The exact return and its parameter dependence

Choose one spanning tree on the physical torus, the canonical sector measure, and the bookings associated with the selected expansion. Keep them fixed on I. Let $\mathcal{X}$ be the enlarged reference dictionary, and let v_x be the array formed from the truncated local expansion, the local counterterms and the forced correction just constructed. The scalar correction is

$$\tau_x = \text{the original scalar payment at } x - \text{the constant term removed by the local subtraction.}$$

This sign follows by moving the removed constant from the action to its vacuum parameter. The exact decomposition of the local expansion and the local subtraction identity now give, for every coarse configuration, the original Gaussian return evaluated using $(\mathcal{X}, v_x, \tau_x)$. This is equality of the complete unnormalized density, with the selected γ, not merely equality of a finite jet at the identity. Absolute summability of the regular arrays justifies subtraction and regrouping: the known and subtracted regular occurrences add term by term to the original occurrences, and the discrepancy factor is included in the forced gas as in (723). The finite regrouping and forced absorption preserve the integral identity by their exact exponential and gas identities.

All regular entries are local and gauge invariant with the prescribed jets; the forced entries are local, measurable and gauge invariant. Their bounds are E and $K(x)$ respectively. The common-dictionary parameter comparison survives truncation, addition, fixed pullbacks and finite averages, each of which is continuous in these array norms. For the remaining infinite arrays, the summable comparison majorants justify passage to the limit. Thus v_x has the required continuous parameter dependence in the same dictionary $\mathcal{X}$.

We have proved the interval return needed by the scalar coupling reset: for every compact interval and every allowed input curve there is one dictionary, one tree and one sector choice, with an exact return and both local budgets throughout the interval. This statement still uses the simultaneous scale purchases listed at the start. Their selection must precede all intervals, histories and periods. The scalar reset then has a separate task: choose the new coupling and vacuum parameters without losing this density identity or its labelled estimates.

148 Reorganizing the actual return before resetting the coupling

We now turn the local return into the particular activity family used in the scalar selection. This requires an identity on every field. The difference between a local quadratic counterterm and the clipped kinetic action has a contribution on unhealthy fields, which must remain in the forced gas.

Fix the compact parameter interval, dictionary and local return of Section 147. Work first at one parameter value. Let $V = |\Lambda|$, let $0 < t \le 1$ be the coarse threshold, and write f_i for its regular entries, anchored at o_i on supports X_i. Let k_i be its forced entries. The budget has radius $B \ge 2$, support exponent $b > 0$, derivative order $q \in \mathbb{N}$, $q \ge 2$, and spatial power $p \in \mathbb{N}$, $p \ge 1$. The early regular majorants $u_i \ge 0$ obey

$$\|f_i\|_{o_i,X_i} \le u_i, \qquad \sum_{i:o_i=o} e^{(b+3)c_B(X_i)} u_i \le H \quad (o \in \Lambda), \tag{724}$$

where c_B is the covering number. The forced majorants obey the rooted hit bound at exponent $b+1$, of cost w. All entries transform under the same permutations of labels for translations and signed lattice symmetries. The selected path has width $\sigma > 0$, row and higher-atom price C, and relaxation radius r_c with

$$12 \le r_c\sigma, \qquad (b+1)/B \le \sigma/4.$$

Its clipping parameters satisfy the geometric conditions ensuring that the kept-row test is local in the radius-B halo. These are the conditions already used for the actual clipped path. The torus side is greater than 20.

148.1 The scalar coefficient extracted from the local Hessians

For each i put $Z_i = X_i \cup \mathrm{halo}_2\{o_i\}$. On a nonwrapping Z_i, let $Q_i(a,d)$ be the coefficient of the Hessian of f_i on affine fields with constant curvature in the plaquette directions a, d. The colour dependence is the invariant inner product: simplicity of the Lie algebra identifies every invariant bilinear form with a scalar multiple of it. Set $Q_i = 0$ when Z_i wraps. This convention is preserved by the label symmetries. The Hessian estimates give

$$t^2|Q_i(a,a)| \le u_i, \qquad t^2|Q_i(a,d)| \le 3u_i. \tag{725}$$

Indeed, test a unit colour vector on one affine curvature direction, scaled by t, for the first inequality. Its link size is bounded by $t\,\mathrm{dist}(e_-,o)$, hence by $t(1+\mathrm{dist}(e_-,o))^p$, so it is an admissible direction.

For the second use the direction $t(A_a + A_d)/2$. Expanding its symmetric Hessian gives four times this diagonal test minus the two single-direction tests, divided by two. The derivative bounds give $(4 + 1 + 1)u_i/2 = 3u_i$. For wrapping supports the inequalities hold because $Q_i = 0$.

The series $Q_o(a, d) = \sum_{i:o_i=o} Q_i(a, d)$ converges absolutely by (724). Reindexing the label series by a translation shows that Q_o is independent of o. Reindexing by a reflection of an axis lying in exactly one of two distinct plaquette pairs changes the sign of the corresponding off-diagonal coefficient, so that coefficient is zero. An unsigned permutation carries any plaquette pair to any other, so all diagonal coefficients agree. Consequently there is one real number Δ with

$$Q_o(a, d) = \Delta\, \delta_{ad}, \qquad t^2|\Delta| \le H. \tag{726}$$

The last bound uses the diagonal estimate in (725) at a single anchor; no volume factor occurs.

The curvature read in the counterterm is the signed clover. For an axis pair $a = (i, j)$, put

$$\phi_a(o, U) = \frac{1}{4} \sum_{\epsilon,\eta\in\{\pm1\}} \epsilon\eta\, \mathrm{curv}_G\big(U(o; \epsilon e_i, \eta e_j, -\epsilon e_i, -\eta e_j)\big),$$

where the argument is the holonomy of the indicated four-step loop based at o, and curv_G is the fixed normalized Lie-algebra projection. Each loop lies in the radius-two halo. A gauge transformation acts by the adjoint at o; a signed lattice symmetry permutes these four loops and their signs. This gives exactly the signed two-form transformation law at the common anchor. On the exponential line of an affine link field of constant curvature w, the derivative of each oriented loop's curvature at zero is $\epsilon\eta w_a$. The average therefore has derivative w_a, with no extra normalization factor. Define

$$P_i(U) = \frac{1}{2} \sum_{a,d} Q_i(a, d)\langle\phi_a(o_i, U), \phi_d(o_i, U)\rangle, \qquad r_i = f_i - f_i(1) - P_i.$$

The affine Hessian of r_i vanishes on nonwrapping supports, and $r_i(1) = 0$. Write χ_X for the healthy-set indicator on X. Define the supported discrepancy

$$d_i = f_i(1) + \chi_{\{o_i\}} P_i - \chi_{X_i} f_i + \chi_{Z_i} r_i.$$

It vanishes when Z_i is healthy. This is the exact masked splitting of Section 58, specialized to the quadratic projection and with the displayed sign convention. Put

$$c_0 = \sum_i f_i(1), \quad D = \sum_i d_i, \quad R_0 = \sum_i \chi_{Z_i} r_i, \quad h(U) = \frac{1}{2} \sum_o \chi_{\{o\}}(U) \sum_a |\phi_a(o, U)|^2.$$

All sums converge absolutely. For example each summand is bounded by a fixed multiple of u_i, and summing (724) over the finite set of anchors bounds $\sum_i u_i$ by VH. The isotropic coefficient identity now gives

$$\sum_i \chi_{X_i} f_i = c_0 + \Delta h - D + R_0. \tag{727}$$

The common singleton indicator at each anchor is essential to summing the quadratic counterterms before using isotropy.

148.2 Replacing the quadratic counterterm by the selected path

Write $\mathcal{K}$ for the actual clipped kinetic action and $\mathcal{L}$ for its loop part. The early return is

$$e^{V\tau - \beta\mathcal{K}(U) - \mathcal{L}(U) - \sum_i \chi_{X_i}(U) f_i(U) - \alpha\mathcal{Q}(U)} \mathcal{G}(U). \tag{728}$$

Here $\mathcal{G}$ is the early forced gas and $\mathcal{Q}$ is the masked low-channel presentation of the quadratic action. It includes the kept-row halo in each support. Thus it is specified before the reset; it is not the full kinetic action on arbitrary fields.

We describe the needed identity channel by channel. A low channel is either a plaquette term or an ordered pair of entries in one row of the selected path. Denote its coefficient by a_d, its unmasked quadratic field by P_d, its local constant-curvature substitute by Q_d, and its actual clipped or kept field by L_d. For a row pair,

$$a_d = c_i c_j, \qquad P_d = \tfrac{1}{2}\langle \phi_i, \phi_j \rangle, \qquad Q_d = \tfrac{1}{2}\langle \phi_{a_i}, \phi_{a_j} \rangle.$$

The latter curvature reads are at the common anchor. For a plaquette, a_d is the direct coefficient, P_d is its plaquette action and Q_d is half the squared local curvature. Let Y_d be the union of the radius-B anchor halo with the two walks when present. On its healthy set, $L_d = a_d P_d$. Define

$$\begin{aligned} F_d &= \Delta a_d (Q_d - P_d), \\ J_d &= (\alpha + \Delta) L_d - \Delta a_d \chi_{\{o\}} Q_d \quad \text{on the complement of the healthy set of } Y_d, \end{aligned}$$

and set $J_d = 0$ on that healthy set. Directly in these two cases,

$$(\alpha + \Delta) L_d + \chi_{Y_d} F_d = \alpha \chi_{Y_d} a_d P_d + \Delta a_d \chi_{\{o\}} Q_d + J_d. \tag{729}$$

This also proves that the discrepancy is supported on an unhealthy Y_d. No derivative of its indicator is needed.

Average these identities over signed symmetries and sum over translations, with normalization $1/|\mathcal{S}|$ for the signed group $\mathcal{S}$. This normalization is applied to every channel, including its discrepancy. To identify the quadratic substitute, let $S(0)$ be the actual zero-momentum row matrix. Absolute row summability permits the row-pair sum to be regrouped as

$$\sum_d a_d Q_d(w) = \tfrac{1}{2}\{c_{\mathrm{direct}} \|w\|^2 + \|S(0)w\|^2\} = \tfrac{1}{2}\|w\|^2.$$

The second equality is the selected metric normalization $c_{\mathrm{direct}} I + S(0)^T S(0) = I$. Multiplying by the singleton healthy indicator and summing anchors gives h; the signed average preserves the squared norm of the clover. Thus the averaged sum of $a_d \chi_{\{o\}} Q_d$ is exactly h. The averaged sum of $\chi_{Y_d} a_d P_d$ is $\mathcal{Q}$. For every cubic or quartic path atom A_d, add the regular correction $-(\alpha + \Delta) A_d$ on its original mask. These atoms have zero affine Hessian because their curvature factors vanish at the identity and their degrees are at least three. Denote the complete averaged regular correction by F, and the averaged sum of J_d by J. Since the low and high channels sum to $\mathcal{K}$, equation (729) becomes

$$(\alpha + \Delta)\mathcal{K} + F = \alpha \mathcal{Q} + \Delta h + J. \tag{730}$$

The low correction also has zero affine Hessian: on an affine field at the identity, transport along each path has the same first curvature derivative as its constant-curvature substitute. Thus $Q_d - P_d$ has zero second derivative there. Polarization gives the mixed assertion. This argument is only used on nonwrapping supports, where the affine field and its walks have their specified lifts.

Combining (727) and (730) in (728) gives

$$e^{V(\tau - c_0/V) - (\beta + \alpha + \Delta)\mathcal{K} - \mathcal{L} - (R_0 + F)} \mathcal{G}\, e^{D + J}. \tag{731}$$

Every sign follows by substituting the two displayed identities. This equality holds even where $\mathcal{G}$ vanishes or changes sign. Absorb $D + J$ into that gas using the supported exponential construction of Section 91. Its hypotheses hold because each discrepancy vanishes on its healthy support. This constructs the actual output at coupling $\beta + \alpha + \Delta$ and vacuum parameter $\tau - c_0/V$.

148.3 The regular and forced costs of this reorganization

We give the numerical bounds so that this identity is accompanied by estimates for its actual arrays. Let C_{pair} be the fixed ordered-derivative price for a pair of local curvatures, and put

$$C_{\mathrm{sub}} = 2 + \tfrac{3}{2}|\mathcal{A}|^2 C_{\mathrm{pair}}, \qquad C_{\mathrm{reset}} = e^{81b} C_{\mathrm{sub}},$$

where $\mathcal{A}$ is the six-element set of plaquette axes. Equation (725) and the product rule bound both the regular norm of r_i and the value of d_i by $C_{\mathrm{sub}} u_i$. The added radius-two halo increases the cover number by at most 81, whence the regular cost is at most $C_{\mathrm{reset}} H$. Its hit cost, denoted $S(H)$, is the footprint payment with population radius $4B + 2$, support increment 81, input exponent $b + 3$ and output exponent $b + 1$. This is the explicit footprint sum of Section 14.2; it is linear in H and has only polynomial dependence on B at fixed stage.

Let C_{turn} be the ordered-derivative cost of a signed pullback. For the low and high channels let L_0, L_1 be their rooted regular sums after support weighting but before this pullback cost. Let L_h be the low-channel hit sum, with the relaxation factor e^{ℓ_d / r_c} included. The channel estimates give

$$\begin{aligned} \|F_{\mathrm{low}}\| &\le C_{\mathrm{turn}} |\Delta| t^2 L_0, \\ \|F_{\mathrm{high}}\| &\le C_{\mathrm{turn}} |\alpha + \Delta| t^3 L_1, \\ \|J\|_{\mathrm{hit}} &\le \left(2t|\alpha + \Delta| + \tfrac{1}{2} C_{\mathrm{pair}} t^2 |\Delta|\right) L_h. \end{aligned} \tag{732}$$

Here is the summability behind these costs. A low support satisfies $B c_B(Y_d) \le 85B + \ell_d$ and a high support satisfies $B c_B(Y_d) \le 9B + \ell_d$. The weighted row-pair sum is the product of the two weighted row sums, hence is bounded by C^2; the direct plaquette contribution adds its fixed coefficient. Derivatives of fixed order multiply these charges by a fixed polynomial in $1 + \ell_d$. The inequality $(1 + \ell)^k \le C_{k,\epsilon} e^{\epsilon \ell}$, obtained by maximizing $(1 + x)^k e^{-\epsilon x}$ on $x \ge 0$, pays this polynomial from half the width. The remaining half pays the support weight. For the hit sum, $r_c^{-1} \le \sigma/12$ and $(b + 1)/B \le \sigma/4$ leave positive width for the rooted footprint sum. These statements hold term by term with absolute coefficients, so also for the countable channel sums.

The low regular estimate in (732) follows from the two curvature factors in P_d, Q_d. For its discrepancy, the kept-row bound is $|L_d| \le 2t e^{\ell_d / r_c} |a_d|$, and the singleton-masked substitute has size at most $C_{\mathrm{pair}} t^2 / 2$. The triangle inequality gives the third estimate. The high estimate uses $t^4 \le t^3$ and the cubic and quartic path bounds. Finally (726) and $t \le 1$ imply

$$|\alpha + \Delta| t^3 \le |\alpha| t^3 + H, \qquad 2t|\alpha + \Delta| + \tfrac{1}{2} C_{\mathrm{pair}} t^2 |\Delta| \le 2t(|\alpha| + H/t^2) + \tfrac{1}{2} C_{\mathrm{pair}} H.$$

Thus the full regular and soft costs are bounded respectively by

$$\begin{aligned} E_{\mathrm{new}} &= C_{\mathrm{reset}} H + C_{\mathrm{turn}} \{ H L_0 + (|\alpha| t^3 + H) L_1 \}, \\ S_{\mathrm{new}} &= S(H) + \{ 2t(|\alpha| + H/t^2) + \tfrac{1}{2} C_{\mathrm{pair}} H \} L_h. \end{aligned}$$

The numerical purchases require $E_{\mathrm{new}} \le \delta$ and the hard/soft absorption inequalities for the pair (w, S_{new}) and target γ. The payment proved in Section 200 gives those inequalities on the selected sufficiently late entries. The output therefore has regular cost δ and forced cost γ, with the original locality, symmetry and affine-Hessian conditions. These numerical purchases must be imposed on this same output budget.

148.4 The two healthy tests and the actual reset

Suppose the early cost obeys $H \le F/4$, where F is the scalar fund used in Section 149. On a globally healthy field all masks are one and the early gas is one. The early density is consequently

$$H_x(U) = \exp\{V\tau - \beta\mathcal{K}(U) - \alpha\mathcal{K}_2(U) - \mathcal{L}(U) - E(U)\}, \qquad |E(U)| \le VH.$$

Here $\mathcal{K}_2$ is the quadratic part and E the early regular sum. At $U = 1$, the three path terms vanish, so positivity is explicit and

$$\left|V^{-1}\log H_x(1) - \tau\right| \le H, \qquad |E(U) - E(1)| \le 2VH \le VF/2.$$

The difference of the full kinetic action and its quadratic part has size at most $2VCt^3$: each cubic and quartic atom is bounded by its row charge times t^3 or t^4, respectively, and $t^4 \le t^3$. The selected path construction bounds $|\alpha| \le A_j$, uniformly in cutoff and period, where A_j is its drift bound plus $|s_G \log L_j|$ for the fixed stage. Purchase $2A_jCt^3 \le F/2$. This is possible at each fixed stage since $t = x^{-1/2}e^{(\log\log x)^2}$, while F is a positive stage constant times $x^{-11/10}$; their ratio tends to zero. Adding the two errors proves the precise surplus bound VF at the nontrivial healthy test. The vacuum error is bounded by $F/4$.

The output arrays in (731) depend on the computed Δ, and are fixed while the trial dial varies. Their dial-motion cost is therefore zero. Their continuity in the original parameter follows from the common-dictionary comparison: the projection is linear and bounded by (725), its sum is absolutely controlled, and the remaining operations are sums, fixed pullbacks and the paid forced absorption. The scalar selection therefore applies to these arrays. Its residual is $(d - \Delta)\mathcal{K}(U_*)$, so the selected dial is exactly Δ. Its vacuum value is exactly $\tau - c_0/V$ by (731) at the identity.

Finally retain this constructed family as the witness for the history step. The recipe conditions refer to its actual supports, normalized density identity, regular and forced bounds, and bound $t^2|\Delta| \le H$. They are supplied by the construction above and the separately purchased recipe permissions. Equality of densities by itself would not supply those permissions. Once the common funding provides them, this is the continuous one-step map used in the surviving-interval induction.

149 Selecting the coupling and the vacuum constant

An exact local return contains a small correction to the kinetic coefficient. We must show that this correction lies in the purchased interval and depends continuously on the incoming representation. Two healthy configurations suffice to select the two scalar parameters. The exact density identity at other configurations comes from the local return, and must be retained separately.

149.1 A separating configuration in every volume

Assume the gauge Lie algebra is nonzero, as it is for our simple gauge group. Let $V_\Lambda = |\Lambda|$ be the number of sites in an even-sided torus. Let $a_0 > 0$ be the direct plaquette coefficient of the selected clipped kinetic action $\mathcal{K}$, and let $C \ge 0$ bound its cubic and quartic arrays. Fix a healthy-field threshold $0 < t \le 1$. There is a number $\eta > 0$, depending on the stage bounds but not the torus or t, and an all-healthy configuration U_* such that

$$\mathcal{K}(U_*) \ge V_\Lambda \eta t^2. \tag{733}$$

Here and below the trivial configuration is denoted by 1.

To construct it, first choose $\epsilon > 0$ such that every chord value in $[0, \epsilon]$ occurs in G. Indeed, take a nonzero Lie-algebra vector c. The derivative at zero of the curvature of $\exp(sc)$ is c, so its chord cannot vanish identically. Continuity and the intermediate value theorem along that path give the assertion. Set

$$r = \min\{\epsilon, 1/4, a_0/[512(1+C)]\} > 0, \qquad h = rt,$$

and choose $u \in G$ of chord h. Set the links in direction 0 equal to u on even coordinate-1 slices and to 1 on odd slices; set all other links equal to 1. Even side length makes this periodic. Every $(0,1)$ plaquette has holonomy u or u^{-1} and hence chord h. Every plaquette has chord at most $4h \le t$, by the four-link triangle inequality. Thus the configuration is healthy.

For each signed lattice symmetry the sum of the direct plaquette terms is at least $V_\Lambda a_0 h^2/2$: the symmetry permutes plaquettes, and the indicated plane already contributes that amount. The other quadratic terms are nonnegative on this healthy configuration. The absolute cubic-plus-quartic contribution per site is at most

$$C(4h)^3 + C(4h)^4 \le 128Ch^3 \le a_0 h^2/4.$$

The first inequality uses $4h \le 1$, and the second uses $512Ch \le a_0$. Averaging the signed actions preserves the lower bound $V_\Lambda a_0 h^2/4$. This proves (733) with $\eta = a_0 r^2/4$.

149.2 The scalar equation on a purchased interval

Fix the coarse path, clip and budget. Let x range in an interval I, and let $H_x(U)$ be the actual unnormalized transferred density. Assume $H_x(1), H_x(U_*) > 0$. Write

$$c(x) = \log H_x(1) - \log H_x(U_*), \qquad b(x) = \beta_f(x) + \alpha_{N,m}.$$

For a trial correction $d \in [-w, w]$, $0 < w \le 1$, write its density as

$$e^{V_\Lambda v} e^{-(b(x)+d)\mathcal{K}(U) - \mathcal{L}(U) - A_{x,d}(U)} \mathcal{G}_{x,d}(U). \tag{734}$$

Here $\mathcal{L}$ is the fixed loop action, $A_{x,d}$ the regular activity sum, and $\mathcal{G}_{x,d}$ the forced gas. The regular entries vanish at 1, and $\mathcal{K}(1) = \mathcal{L}(1) = 0$ because each path atom vanishes at the trivial configuration. The forced gas equals 1 on healthy configurations. In particular the vacuum parameter is forced to be

$$v(x) = V_\Lambda^{-1} \log H_x(1). \tag{735}$$

Put $k = \mathcal{K}(U_*) > 0$ and define the residual

$$f(x,d) = (b(x)+d)k + \mathcal{L}(U_*) + A_{x,d}(U_*) - c(x).$$

The trial matches H_x at 1 and U_* exactly when v has value (735) and $f(x,d) = 0$.

Here are the estimates required for this selection. Let $\delta \ge 0$ be the regular allowance per site and $F \ge 0$ the reset allowance per site. Assume

$$|b(x)k + \mathcal{L}(U_*) - c(x)| \le V_\Lambda F, \tag{736}$$

$$|A_{x,0}(U_*)| \le V_\Lambda \delta,$$

$$|A_{x,d'}(U_*) - A_{x,d}(U_*)| \le V_\Lambda F |d' - d|,$$

$$2(F + \delta) \le \eta t^2 w. \tag{737}$$

The arrays have fixed labels, are continuous in their regular and forced norms along parameter curves, and have locally summable comparison majorants. These conditions imply continuity of their evaluated sums along the parameter and dial: truncate a summable majorant, use continuity for the finite part, and bound the two tails by that majorant. The same argument works if domination is supplied on a further subsequence of every convergent sequence. Otherwise a sequence violating continuity would have a further dominated subsequence which still violates it. We also assume continuity of the two positive values of H_x and of $b(x)$.

From (733) and (737),

$$V_\Lambda(F+\delta) \le kw/2, \qquad V_\Lambda F \le k/2.$$

Consequently, for $d \le d'$,

$$f(x,d') - f(x,d) \ge \tfrac{1}{2}k(d'-d), \qquad |f(x,0)| \le kw/2.$$

Thus $f(x,-w) \le 0 \le f(x,w)$. Continuity in d gives a zero $d(x) \in [-w,w]$, and the strict slope bound makes it unique in that interval. It also gives the useful inverse estimate

$$|d(x) - d(y)| \le \frac{2}{k}|f(x,d(y)) - f(y,d(y))|. \tag{738}$$

For fixed y, the right side tends to zero as $x \to y$; this proves continuity of d including at endpoints. Substitution into the fixed labelled arrays preserves their continuity and their budgets. Together with (735) this produces a continuous family of representations with the purchased correction bound $|d(x)| \le w$.

149.3 Identification with the physical local correction

In the actual local construction the activity array is independent of the trial dial. The local projection and compulsory-packet absorption first give a number $\Delta(x)$ and an array S_x for which (734), at $d = \Delta(x)$ and

$$v_0(x) = \tau_x - V_\Lambda^{-1} \sum_i v_x^{\mathrm{reg}}(i,1),$$

is exactly $H_x(U)$ for every U. This is the exact local reorganization; its scalar sign is fixed by evaluating the identity at $U = 1$. Its regular arrays vanish at the flat configuration after subtraction. With $A_{x,d}$ and $\mathcal{G}_{x,d}$ independent of d, taking the two healthy evaluations of that exact identity gives

$$f(x,d) = (d - \Delta(x))k. \tag{739}$$

Therefore the selected zero is $d(x) = \Delta(x)$ and $v(x) = v_0(x)$. This identifies the selected continuous family with the actual reorganized local return, rather than merely producing a family with the same two scalar tests.

One may also express the full-field justification as a quantitative error bound. For fixed x, U, put $q = \mathcal{K}(U)$ and $r_U = H_x(U)/H_x(1)$. The trial normalized shape equals $e^{(\Delta-d)q}r_U$. Since $|d| \le 1$,

$$|e^{(\Delta-d)q}r_U - r_U| \le e^{(1+|\Delta|)|q|}|q|\,|r_U|\,|\Delta - d| = \frac{e^{(1+|\Delta|)|q|}|q|\,|r_U|}{k}|f(x,d)|.$$

This follows from $|e^z - 1| \le e^M|z|$ for real $z \le M$, $M \ge 0$. It applies also when the forced gas is nontrivial or signed. At the selected zero it proves exact equality at every configuration; the constant here may depend on x, U and the finite volume.

The remaining compatibility check uses the actual local construction. Its subtraction estimate gives $t^2|\Delta(x)| \le E(x)$, its regular and forced arrays satisfy the purchased output allowances, and its exact normalized density is the same transferred density. These are precisely the input and output conditions of the selected recipe. Using that same local construction as witness therefore preserves the retained history constraints and the next-step constraints. Equality of densities alone would not imply those constraints for arbitrary arrays or dictionaries. The forced estimate is retained separately, including its exponential monitor bound.

This proves the scalar selection and its full-field identification relative to the physical reorganization and the displayed purchases. In that application the dial-motion estimate is zero. The surplus estimate comes from the exact all-healthy local expression and the regular budget, together with the purchased cubic/quartic discrepancy between the full kinetic action and its quadratic part. The next construction must verify those purchases on the selected history; the scalar argument itself does not choose a new path or new prices.

150 Shooting through the coupling bands

A nonlinear step is available when its input coupling belongs to the prescribed band. Its output need not belong to the next band for every input parameter. We construct a surviving interval at each level; this preserves the activity continuity needed for the subsequent step.

Fix the scale sequence and all the preceding nonlinear choices. Let $s = s_G > 0$, $L_i \ge 2$ be the integer block sides, and set

$$S_0 = 0, \qquad S_{i+1} = S_i + \log L_i, \qquad \widehat{\beta}_i = \bar{\beta} + sS_i, \qquad \bar{\beta} > 0.$$

The permitted band is $[\widehat{\beta}_i/2, 2\widehat{\beta}_i]$. Suppose the purchased initial family exists for every cutoff and period, and that the continuous return of the preceding section applies on any compact interval whose inputs satisfy the history conditions and belong to the fine band. Preserve its exact density identity and its retained next-step conditions. The adjacent coupling bound has the form

$$|\beta_{i+1} - \beta_i - s\log L_i| \le \varepsilon_{i+1}, \qquad \varepsilon_{i+1} \le \tfrac{s}{4}\log L_i. \tag{740}$$

Here $\varepsilon_i \ge 0$. These are bounds on the actual returned representations.

150.1 Restricting a curve without crossing a barrier

We first record the elementary interval argument. Let f be continuous on $[x, y]$, and suppose $f(x) \le d \le u \le f(y)$. There are $z \le w$ in $[x, y]$ such that

$$f(z) = d, \quad f(w) = u, \quad f([z, w]) \subset [d, u].$$

The case $d = u$ follows by taking one point of the nonempty level set. Otherwise let w be the first point of $[x, y]$ at which $f = u$. The level set is nonempty by the intermediate value theorem, and compactness makes its minimum belong to it. Let z be the last point of $[x, w]$ at which $f = d$; this level set is likewise nonempty and compact. If $f(c) < d$ at some $c \in [z, w]$, the intermediate value theorem between c and w gives a later point with value d, contrary to the choice of z. If $f(c) > u$, the same theorem between x and c gives an earlier point with value u, contrary to the choice of w. This proves the claim without any monotonicity assumption on f.

Define inner barriers

$$\ell_i = \bar{\beta} + \tfrac{s}{2}S_i, \qquad u_i = \bar{\beta} + 2sS_i.$$

Both lie in the permitted band, since $S_i \geq 0$. If an input at level $i+1$ has coupling ℓ_{i+1}, its returned coarse coupling satisfies, by (740),

$$\beta_i \leq \ell_{i+1} - s \log L_i + \varepsilon_{i+1} \leq \ell_i - \tfrac{s}{4} \log L_i < \ell_i.$$

If the input coupling is u_{i+1}, the corresponding lower bound is

$$\beta_i \geq u_{i+1} - s \log L_i - \varepsilon_{i+1} \geq u_i + \tfrac{3s}{4} \log L_i > u_i.$$

Thus the two endpoints cross the next two barriers in the required order, even though the intermediate output curve may oscillate.

150.2 Descending through a finite history

At cutoff N restrict the initial bare-action family to $[\ell_N, u_N]$. Its coupling is its parameter, it is continuous in the labelled activity norms, and it satisfies the prescribed initial conditions. Suppose a compact parameter interval already carries a family of histories from N down to $i+1$, with every represented coupling in its band, and with the endpoint couplings at level $i+1$ equal to ℓ_{i+1}, u_{i+1}. Apply the continuous nonlinear step to the level-i+1 curve on this interval. The endpoint inequalities just proved and the interval lemma give a subinterval on which the output coupling stays between ℓ_i, u_i, attaining these values at the two ends.

Restrict every earlier representation in the history to this subinterval and append the newly constructed level-i representation. Restriction preserves the fixed dictionaries, their norm continuity, all activity bounds and every previously established adjacent identity. The new step supplies the same properties at level i. Induction therefore reaches level zero. Since $\ell_0 = u_0 = \bar{\beta}$, either endpoint supplies a history with exactly the desired coarse coupling. No representation was extended across an interval on which its input band condition failed.

There is also one fixed neighborhood of coarse targets that uses the same selected data. Put

$$h = \min\{\bar{\beta}/2, s \log L_0/4\} > 0.$$

For any $|b - \bar{\beta}| \leq h$, descend only to level 1 and apply the last nonlinear step there. Its lower endpoint returns at most $\bar{\beta} - s \log L_0/4 \leq b$, and its upper endpoint returns at least $\bar{\beta} + 3s \log L_0/4 \geq b$. The intermediate value theorem selects an input parameter with output coupling b. That output lies in the level-zero band because $|b - \bar{\beta}| \leq \bar{\beta}/2$. For $N = 0$, use the initial family directly at parameter b. This construction asserts existence, not uniqueness of a shooting history or continuity of its choice as b varies.

For every cutoff and period select one such history. The selected paths depend on cutoff and level, and were fixed before the period; therefore their identity across periods is preserved. Each selected history starts at the bare Wilson density with vacuum parameter zero, satisfies all its local activity bounds, and has the exact full-density transfer at each step. The stronger retained conditions are preserved by taking them as part of the next-step requirement throughout the interval induction. The induction selects no new bridges, paths or support exponents.

150.3 The running-coupling estimate

Let $E_0 = 0$ and $E_i = \sum_{j<i} \varepsilon_{j+1}$. Telescoping (740) gives, uniformly over every selected cutoff $N \geq i$ and period,

$$|\beta_i - \widehat{\beta}_i| \leq E_i + |b - \bar{\beta}|. \tag{741}$$

The later stage purchases give the more precise ratio bound

$$\frac{\varepsilon_{i+1}}{\log L_i} \le J_{i+1}^{-2}, \qquad J_i \longrightarrow \infty.$$

Fix $\epsilon > 0$ and choose j so that $J_{i+1}^{-2} \le \epsilon/2$ for $i \ge j$. Then

$$E_i \le E_j + \tfrac{\epsilon}{2}(S_i - S_j), \qquad i \ge j.$$

Since $S_i \to \infty$, eventually $E_j + |b - \bar{\beta}| \le \epsilon S_i/2$. Substitution into (741) yields

$$|\beta_i - \widehat{\beta}_i| \le \epsilon S_i$$

uniformly in cutoff and period. In particular the error is $o(S_i)$. There is no assertion that the sum of the step errors is uniformly bounded in the depth. The estimate uses their small ratio to the accumulated logarithmic scale.

151 Installing the terminal Wilson normalization

The preceding startup construction uses a particularly simple terminal path. We must install it in the common horizon while preserving the stationary estimates used at the next step. Agreement through fourth order suffices for the required classical and one-loop coefficients; it does not identify the full nonlinear stationary maps.

151.1 Transporting the terminal quadratic columns

Write the old terminal full column, at any period, as

$$A = \begin{pmatrix} \sqrt{c_0} I \\ R \end{pmatrix}, \qquad E = \begin{pmatrix} I \\ 0 \end{pmatrix}, \qquad Z = A + E.$$

The terminal metric identity says $A^* A\, d = d$, where d is the lattice exterior derivative on one-forms. Since $E^* A = A^* E = \sqrt{c_0} I$, it implies

$$Z^* A d = Z^* E d.$$

Moreover

$$Z^* Z = (1 + \sqrt{c_0})^2 I + R^* R$$

is positive definite. Put

$$P = Z (Z^* Z)^{-1} Z^*, \qquad U = 2P - I.$$

Direct multiplication gives $P^* = P$, $P^2 = P$ and $U^* U = I$. Also $P(Ad + Ed) = Ad + Ed$ and $PAd = PEd$, whence

$$UAd = Ed.$$

Construct the analogous reflection U_0 for the desired terminal column $A_0 = (\sqrt{c_0} I, \sqrt{1 - c_0} I)^{\mathsf{T}}$. Then $X = U_0^* U$ is an isometry and $XAd = A_0 d$.

These are periodizations of common exponentially decaying kernels. To check the uniform estimate, suppose the old row kernel has an exponentially weighted absolute coefficient bound at width $\eta > 0$. Its Fourier symbol and reflected transpose are holomorphic on a smaller strip. On the real torus the Gram matrix is bounded below by $(1 + \sqrt{c_0})^2 I$. The coefficient bound controls uniformly the change of its symbol under a small imaginary displacement. Choose a smaller positive

width so that this change has operator norm at most half that lower bound. The Neumann series then constructs a holomorphic inverse on the whole smaller strip, bounded by $2/(1+\sqrt{c_0})^2$. Fourier coefficient estimates on a further smaller strip give an exponentially weighted absolute sum: shift the contour in each coordinate in the sign of the Fourier index and sum the resulting geometric series. Matrix dimension is fixed by the row types, so these constants are independent of period and block length. Multiplication and transposition give the same kind of bound for U, U_0, X. This is the strip inverse argument used for the free kernels.

Modify only the finest transfer column m_f, replacing it by Xm_f. Since $X^*X = I$, its Gram matrix is unchanged. Its mixed identity with the terminal curvature column is unchanged after replacing that column by A_0, because $XAd = A_0 d$. The other transfer columns, including the mean column and the transport matrix, are unchanged. Thus all quadratic transport and orthogonality identities remain valid. The exponential bounds for the modified column follow by convolution of its old bound with the bound for X. There is only this one modification at the terminal step, so its cost does not accumulate along the history.

At each stage choose the new strip width below both the original width and the width just obtained; choose the new upper price above one, the old price and this convolution price; and choose the new lower price below one and the old lower price. These choices depend only on the stage profiles and old kernel prices. They precede the block lengths, delays and realization. The resulting horizon agrees with the old one below its terminal level, its towers agree when $m+1 < N$, and their mean and transport columns agree also at $m+1 = N$. Its terminal direct metric is the identity. This proves the revised-horizon assertions used in Section 153.

151.2 Finite jets of stationary values

We record the analytic calculation controlling the change of path. Let $F, G : X \times Y \to \mathbb{R}$ be analytic germs at zero on finite-dimensional real spaces. Suppose their derivatives through order $k+1$ agree. Let $u, v : X \to Y$ be analytic germs with $u(0) = v(0) = 0$ and

$$D_Y F(x, u(x)) = 0, \qquad D_Y G(x, v(x)) = 0.$$

Assume the vertical Hessian of F at zero is invertible. Then the graphs $(x, u(x))$ and $(x, v(x))$ agree through order k, and the stationary values $F(x, u(x))$, $G(x, v(x))$ agree through order $k+1$.

To prove the graph assertion, use the analytic local diffeomorphism

$$\Psi_F(x, y) = (x, D_Y F(x, y)).$$

Its derivative is block triangular with invertible diagonal blocks I and $D_Y^2 F(0)$. The corresponding map Ψ_G has the same jets through order k. On the two stationary graphs these maps both equal $(x, 0)$. Composing their jet identities with the local inverse of Ψ_F proves equality of graph jets through order k. On a stationary graph the chain rule reduces to

$$D\big[F(x, u(x))\big] = D_X F(x, u(x)),$$

because the vertical derivative vanishes. The right sides for F, u and G, v agree through order k, using the graph identity and the derivative jets of F, G. Their values at zero also agree. This proves equality of stationary-value jets through order $k+1$.

In the application $k = 3$. The old terminal action and the new startup action both match the Wilson action through order four, by their terminal normalization and Section 152. This matching holds after substitution of the real fine-field chart on each forest slice. Indeed that chart is analytic

and sends zero to the identity field; compose the matched real jets with its local logarithm. The noise term is identical in the two tested actions, so adding it preserves the matching.

The vertical Hessians at zero are equal, hence the old positive gap is preserved exactly. The analytic implicit function theorem constructs the new critical graph. Near zero the tested-action germs are independent of the branch tag under the chosen kernel's branch hypothesis; thus this one graph solves every tested critical equation. Its first derivative agrees with the old graph, so it has the same linear mean. The preceding calculation also preserves the classical stationary-value jets through order four. This is the precise classical matching required by the outgoing coarse path, which is left unchanged.

151.3 The one-loop coefficient and retained limits

The vertical Hessian matrices of the two tested actions have matching jets through order two, since their actions match through order four. The determinant is a finite polynomial in matrix entries, so the determinants have matching two-jets. Composing with the stationary graphs preserves this matching: the graphs agree through order three, more than is needed here. Their positive determinant at zero allows the real logarithm.

The Haar correction is a finite product of the same local chart Jacobian J along the vertical graph coordinates. Since J is analytic near zero and $J(0) = 1$, these products and their logarithms also have matching two-jets. Both terminal loop potentials are zero. It follows that the two thermal functions

$$\frac{1}{2}\log\det D_Y^2 F(x, u(x)) - \log\prod_b J(u(x)_b)$$

and its counterpart for G, v have identical Hessians at zero. The finite-period wave estimate depends on these Hessians and the unchanged outgoing classical data. Its bound is therefore preserved with exactly the old scalar correction, gap and tolerance. Positivity near the new stationary graph follows from continuity and its positive definite Hessian at zero.

Apply this construction only at $m + 1 = N$; at every lower step keep the old graph, path and scalar correction. The new terminal path is the startup path, while

$$R_{N,m}^{\mathrm{new}} = R_{N,m}^{\mathrm{old}} \quad (m < N), \qquad \alpha_{N,m}^{\mathrm{new}} = \alpha_{N,m}^{\mathrm{old}}, \qquad u_{N,m}^{\mathrm{new}} = u_{N,m}^{\mathrm{old}} \quad (m + 1 < N).$$

The matching argument above supplies all stationary conditions at the changed step. The revised quadratic horizon supplies its same linear guess and all-period kernel identities. For every fixed level, the changed data consequently agree with the old data on a tail of cutoffs. Their fixed-level limits and eventual extended-domain bounds persist by this exact tail equality. The new terminal graph need not have the old graph's complete analytic expansion, and no such assertion is used.

152 The terminal Wilson family

The shooting argument needs a continuous initial family on the entire terminal coupling band, with one fixed local dictionary and common summable bounds. We construct it directly from the Wilson density. This also explains why its terminal path can be chosen before any trajectory is selected.

Fix a compact gauge group as in the construction, a clipping profile with direct coefficient $0 < c_0 \le 1$, linear coefficient $0 < c_1 \le 1/4$, and relaxation length $\lambda > 0$. Write $q = 1 - c_0$. For a plaquette holonomy u, put

$$r(u) = \mathrm{chord}(u), \qquad k(u) = \mathrm{curv}(u), \qquad W(u) = \tfrac{1}{2} r(u)^2.$$

Thus $0 \le W \le 2$ and $\|k(u)\| \le r(u)$. The gauge normalization gives $k(\exp X) = X + O(\|X\|^2)$ near zero.

152.1 A curvature-coordinate remainder

The analytic inverse function theorem applied to $X \mapsto k(\exp X)$ gives a local inverse v near zero. Set $u(x) = \exp v(x)$ and $w(x) = W(u(x))$. Conjugation covariance of curvature and uniqueness of the local inverse show $w(\mathrm{Ad}_g x) = w(x)$ as germs at zero. The identities $k(u^{-1}) = -k(u)$ and $W(u^{-1}) = W(u)$ give $w(-x) = w(x)$. The first and second derivatives at zero are respectively zero and the Euclidean inner product. Thus the fourth Taylor polynomial is

$$T_4 w(x) = \tfrac{1}{2}\|x\|^2 + Q_4(x), \qquad Q_4(x) = \frac{1}{4!} D^4 w(0)[x,x,x,x].$$

The odd terms vanish by evenness. The quartic form Q_4 is adjoint-invariant, by differentiation of the conjugation identity. Define the global smooth residual

$$h(u) = W(u) - \tfrac{1}{2}\|k(u)\|^2 - Q_4(k(u)).$$

For every smooth family through the identity its composition with h has vanishing derivatives through order four. Indeed near the identity $u = u(k(u))$, so this is the Taylor remainder $(w - T_4 w)(k(u))$; the chain rule preserves the stated vanishing.

We need more than a Taylor estimate at the all-identity link field. For every derivative order K and spatial power P there are $C > 0$ and $t_0 \in (0,1]$, independent of the torus, such that

$$\left|\partial_{a_1} \cdots \partial_{a_l} h(U_0 U_1 U_2^{-1} U_3^{-1})\right| \le C t^5, \qquad 0 \le l \le K, \tag{742}$$

when the plaquette chord is at most $t \le t_0$ and the four-link directions have norm at most $t5^P$. Derivatives here are successive left increments on the links.

Here is a uniform proof. For fixed links U_1, U_2, U_3 and a small face logarithm x, reconstruct the first link as

$$U_0(x) = \exp(x) U_3 U_2 U_1^{-1}.$$

The face holonomy is then $\exp x$. After l left differentiations the resulting smooth function of x vanishes to order $(5-l)_+ = \max(5-l, 0)$ at zero. Each differentiation can lower the vanishing order by at most one: introduce its increment parameter as one further variable, differentiate the vanishing Taylor polynomial, and restrict that parameter to zero. All remaining links range over a compact group and the directions, after division by t, range over a fixed compact ball. Taylor's integral remainder therefore gives a common bound $C_l \|x\|^{(5-l)_+}$ for the unscaled derivatives. A fixed local logarithm satisfies $\|x\| \le C_{\log} r(u)$ when $r(u)$ is sufficiently small. Restoring the l factors of t gives $C_l' C_{\log}^{(5-l)_+} t^{l+(5-l)_+} \le C_l'' t^5$, since $t \le 1$. Take the largest constant and smallest threshold over $0 \le l \le K$. For a real lattice plaquette, the four links may coincide because of wrapping; substitution into the four-link identity remains valid. Each lies within distance four of any chosen plaquette corner, so the normalized spatially weighted directions obey the required bound $t5^P$. This proves (742) on every healthy plaquette, including equality in the health threshold.

152.2 The principal path and the nonnegative debt

Choose the terminal path to have one row $\sqrt{q}\, k(u)$ at each elementary plaquette, quartic term qQ_4, and zero cubic and loop terms. Its zero-momentum metric is $c_0 I + qI = I$. Its path supports have length four, and its only nonzero quartic coefficients form a finite list. Hence at any fixed positive

strip width their weighted path sums are finite. Choosing the stage path allowance above their maximum, and the column allowance above the prescribed factorial, supplies the startup profile before choosing block lengths. Its uncut principal action differs from the Wilson action by $\sum qh(u)$, so the two actions have identical jets through order four at the identity.

Fix an integer support scale $B \geq 2$ and threshold $t > 0$ satisfying

$$2 \leq te^{(B-2)/\lambda}. \tag{743}$$

At a plaquette with base site x, the row is kept when every plaquette based at y has chord at most $te^{d(x,y)/\lambda}$. Its base action is

$$b(u) = \begin{cases} c_0 r(u)^2/2, & r(u) \leq t, \\ c_1 t r(u), & r(u) > t, \end{cases} + \mathbf{1}_{\text{kept}}\, q\|k(u)\|^2/2.$$

The quartic term is masked by health of the full four-corner plaquette support. Define the debt at that plaquette by

$$d(U) = \begin{cases} W(u) - c_1 t r(u), & r(u) > t, \\ \mathbf{1}_{\text{healthy}}\, q\|k(u)\|^2/2 + \mathbf{1}_{\text{not healthy}}\, qW(u) - \mathbf{1}_{\text{kept}}\, q\|k(u)\|^2/2, & r(u) \leq t. \end{cases}$$

A direct split into the healthy, large-chord and remaining cases gives

$$W(u) = b(u) + q\mathbf{1}_{\text{healthy}} Q_4(k(u)) + q\mathbf{1}_{\text{healthy}} h(u) + d(U). \tag{744}$$

The debt satisfies $0 \leq d(U) \leq 2$, since $q\|k(u)\|^2/2 \leq qW(u) \leq 2$. On a large-chord plaquette,

$$d(U) \geq r(u)^2/2 - tr(u)/4 \geq t^2/4.$$

If the base site is farther than B from every flag, the plaquette is healthy and the row is kept. For nearby plaquettes this follows from absence of flags; for the remaining plaquettes (743) and the global chord bound $r \leq 2$ make the kept test automatic. Thus its debt vanishes there. The same argument shows that debt at a base site can be determined from the field in its radius-B neighborhood, with the prescribed boundary flag information.

Average (744) over signed coordinate changes. Let $s_* = 1/|\mathcal{S}|$, where $\mathcal{S}$ is that finite symmetry group, and index the oriented plaquettes by $i = (s, x, b)$. Their supports are the actual transformed four-corner supports Y_i, with native site x, residual h_i and debt d_i. The signed average is exact, because the Wilson energy is invariant:

$$\beta E_W(U) = A_\beta^{\text{top}}(U) + \beta s_* q \sum_i \mathbf{1}_{\text{healthy}(Y_i)} h_i(U) + \beta s_* \sum_i d_i(U). \tag{745}$$

This uses the transformed supports and native sites in every channel.

152.3 Forced packets and the exact Wilson density

Let $\mathcal{D}(U)$ be the flag set. Join two flags when their closed radius-$2B$ balls overlap, and let D run over the connected components for this relation. Put

$$P(D) = \bigcup_{x \in D} B_{2B}(x), \qquad C_D(U) = s_* \sum_{i:\, x_i \in B_B(D)} d_i(U).$$

The regions $P(D)$ of distinct components are disjoint and $P(D) \cap \mathcal{D}(U) = D$. Every nonzero debt lies in exactly one of these sums: it lies within B of a flag, while membership in two such neighborhoods would join the two flag components. Consequently

$$\sum_D C_D(U) = s_* \sum_i d_i(U).$$

The energy price is

$$C_D(U) \geq |D| s_* t^2/16. \tag{746}$$

To prove it, every flag is a corner of a large-chord plaquette. Such a plaquette has at most four corners and its base is within distance two of the flag, hence in $B_B(D)$. If $\mathcal{A}$ is the set of these plaquettes in the identity channel, then $|D| \leq 4|\mathcal{A}|$. Each contributes at least $t^2/4$; all other debts are nonnegative. Multiplication by s_* proves (746).

In a fixed dictionary include every nonempty connected flag set D with address $(P(D), D)$, and put

$$K_D(\beta, U) = \mathbf{1}_{\{P(D) \cap \mathcal{D}(U) = D\}} e^{-\beta C_D(U)}.$$

Other addresses have zero coefficient. The forced gas sums products over disjoint supports with total flag inventory $\mathcal{D}(U)$. Exactly one family can have nonzero product: the components of $\mathcal{D}(U)$ just constructed. Indeed disjoint connected regions covering that inventory give its component decomposition. Thus the gas equals $e^{-\beta s_* \sum_i d_i(U)}$, including the empty product when there are no flags. Put the regular residuals $\beta s_* q h_i$ in their healthy supports and take every extra logarithmic attachment to be zero. Equation (745) now proves that the full density is exactly $e^{-\beta E_W(U)}$ at every field.

All coefficients are gauge-invariant and measurable; the local residuals are smooth in left increments. Their value and Hessian at the identity vanish. Permuting the signed channels and lattice sites gives the required translation and signed-coordinate covariance. The packet charge reads only its support with the given flags, since the radius-B neighborhood of a site in $B_B(D)$ lies in $P(D)$. These are the locality conditions of the startup family.

152.4 Uniform prices on the whole coupling band

Let $a > 0$ be the support exponent. A plaquette support has cover number at most 5^4. At a fixed native site there are $|\mathcal{S}|$ times the number $n_\square$ of plaquette directions regular entries. By (742) their regular budget for $0 \leq \beta \leq v$ is at most

$$|\mathcal{S}| n_\square e^{a 5^4} v C t^5. \tag{747}$$

The constants depend on the stage, not on volume or cutoff.

Choose positive numbers σ, κ, γ and a lower band endpoint l such that

$$\kappa + \sigma \leq l\, s_* t^2/16, \tag{748}$$

$$(4B+1)^4 e^{(a+2)5^4 - \sigma} \leq \gamma, \qquad (2B+1)^4 \gamma \leq 1, \tag{749}$$

$$(2B+1)^4 (e^\gamma - 1) \leq 1. \tag{750}$$

For $\beta \geq l$, (746) implies

$$|K_D(\beta, U)| \leq e^{-\kappa} e^{-\sigma |D|}.$$

The elementary ball supports $B_{2B}(x)$ have cover number at most 5^4, and at most $(4B+1)^4$ of them contain a specified site. Thus (749) bounds their rooted support sum with exponent $a+2$ by γ.

Apply the connected-support estimate (202), with spare exponent one, to these ball supports. It gives the packet majorant

$$\sup_x \sum_{D:\, x\in P(D)} e^{-\sigma|D|} e^{(a+1)\operatorname{cov}_B(P(D))} \le e^{\gamma}-1.$$

This is one common summable majorant for the entire band $\beta \ge l$. The regular and forced budgets, and the extra one-unit support reserve, have now been proved. After the finite gas grouping the forced cost is at most $e^{-\kappa}(e^{e^{\gamma}-1}-1)$, by the same connected-support estimate with (750).

We verify that the actual schedule pays these inequalities. Write $s = \log \widehat{\beta}$ and set

$$M = e^{(\log s)^2}, \quad t = e^{-s/2}M, \quad B = \lceil s^2 \rceil, \quad \gamma = e^{-M/2}, \quad \delta_E = e^{-11s/10}, \quad \delta_K = e^{-M^{1/5}}.$$

At a fixed stage take $\sigma = M$, $l = e^s/2$ and $\kappa = M$. Then $lt^2 s_*/16 = s_* M^2/32 \ge 2M$ eventually, giving (748). Since B grows polynomially in s while M grows faster than every such polynomial, (749) and (750) hold eventually. More explicitly $(4B+1)^4 \le M/4$ and $(a+2)5^4 \le M/4$ eventually, so their product with e^{-M} is at most $e^{-M/2}$; also $e^{\gamma}-1 \le 2\gamma$ once $\gamma \le 1$. Since $\gamma \to 0$, we may also require $e^{e^{\gamma}-1}-1 \le 1$. The grouped forced cost is then at most e^{-M} and hence at most δ_K.

The logarithm of (747), with $v = 2e^s$, equals

$$C - \frac{3}{2}s + 5(\log s)^2,$$

where C is a stage constant. It is at most $-11s/10$ for large s, because $(\log s)^2 = o(s)$. Also $t \to 0$, $B \to \infty$, and $\log(te^{(B-2)/\lambda}) = -s/2 + (\log s)^2 + (B-2)/\lambda \to \infty$. Thus the residual threshold and (743) are paid at the same entry delay.

The stronger startup reserve needed by the later common services is obtained without changing this family. For any fixed additional support exponent $\eta \ge 0$, replace a by $a+\eta$ in the preceding prices and delay the stage further. Take instead $l = e^s/4$ and $\kappa = s_* M^2/128$. Eventually $M \le \kappa$, and

$$\kappa + M \le 2\kappa = s_* M^2/64 = lt^2 s_*/16.$$

The same packet bounds then give a forced allowance at most $e^{-s_* M^2/128}$ before grouping, with a still smaller grouped cost after the reserve is paid. These choices depend on the stage and its fixed extra requests, and work for all periods and cutoffs realizing that stage. Taking their entry thresholds together with the other already chosen thresholds therefore preserves the order of choices in the common construction.

152.5 Continuity and the initial history

The dictionary and the masks above do not depend on β. Regular coefficients depend linearly on β, so their normalized derivative differences have the bound $C|\beta - \beta'|t^5$. For a packet, $0 \le C_D(U) \le 2s_*|\mathcal{S}|\,|\Lambda| n_\square$, and, for $\beta, \beta' \ge 0$,

$$|e^{-\beta C_D(U)} - e^{-\beta' C_D(U)}| \le C_D(U)|\beta - \beta'|.$$

There are finitely many nonzero addresses on a fixed torus. Their weighted difference sums consequently tend to zero with $\beta' - \beta$, uniformly in U. Together with the band-wide majorants above this proves the required curve property and invariant-atom continuity. No continuity of a health indicator as a function of U is asserted.

Pack the dictionary into the fixed terminal dictionary by its coupling-independent injection. Zero unused entries contribute neither cost nor logarithm, and every nonzero coefficient, support, inventory, and density is preserved. The finite gas grouping and its continuous summable bounds therefore give a continuous family on $[\widehat{\beta}/2, 2\widehat{\beta}]$ with exact Wilson density, regular allowance δ_E and grouped forced allowance δ_K. Its scalar constant is zero and its coupling is the input parameter β.

The revised common horizon has exactly the terminal path constructed above. For its default recipe there is no additional initial permission condition. The exact density and paid ingredients thus start the retained history at level N and give its starter relation. This proves the initial-family assertion used in Section 153 for every N, p, including $N = 0$.

153 Choosing the nonlinear and stationary data together

The stationary construction, local expansion and scalar reset must use one set of paths and one scale schedule. We now describe the simultaneous choice and the way its identities are retained. The analytic inputs are the path-attachment estimates, their extended-domain convergence bounds, and the local nonlinear estimates proved in the preceding sections. This section concerns their common application; a numerical choice of large scales cannot replace an analytic estimate.

153.1 The order of the choices

Prescribe the finite derivative orders, spatial powers and support exponents at every stage. They may tend to infinity. At a fixed stage j, the adjacent target stage is either j or its immediate predecessor; thus every maximum below involves finitely many requirements. First choose the free-kernel prices and block floors. The stationary construction gives a positive upper width σ_j and a lower path allowance A_j. The startup expansion gives another lower path allowance. Choose the profile's path allowance above their maximum, and its width positive and at most σ_j. Keep its direct coefficient below the purchased free-kernel bound. The profile construction retains the prescribed orders and support exponents. These choices precede the actual block lengths.

There are two levels of kernel price: the free construction's price and the price required after replacing the terminal action by the Wilson action. Choose the latter from the former now, at the fixed profiles. Its validity is uniform over all later enlargements of the block floors. Consequently the local nonlinear estimates can request this final price without choosing it from a horizon that has not yet been built.

Fix the startup and multiple-insertion requirements. Enlarge each block floor by the maximum of the startup floor, its scalar comparison floor and the multiple-insertion floor. Further enlarge it by the floors for the nonlinear geometry and the auxiliary local comparisons. The local targets have the same gauge chart, halo 500, earned exponent $b_i + 3$, error exponent $-7/16$ and degree zero at target stage i. The relevant comparison radius exceeds the halo by at least five. These are the targets used to prove the interval return, including the finite short-panel localization.

Choose any allowed block sequence above these floors. Only now select the wage bounds and local price tables; their constants can depend on the actual block lengths. An arbitrary larger wage family is still allowed. Choose positive postal rates below any prescribed positive upper rates, together with their forecast bounds. The rate agreement needed by the scalar reset can be arranged explicitly. If ρ_{ji} is the selected postal rate, take the post-step forcing rate g_j no larger than

$$\min\{\rho_{jj}/4, \rho_{j,j-1}/4\}.$$

At stage zero the repeated predecessor is harmless. Set the reset cost for the transition to i equal

to four times the local credit constant at i. Thus

$$g_j \le \rho_{ji}/4, \qquad c_i \le q_{ji}/4 \tag{751}$$

at every permitted transition. These are positive quantities. The post-step gap is supplied by the stationary estimate for this actual block sequence; take the auxiliary multiplier equal to one and the remaining post-step rate no larger than its stationary cap.

Next choose protection parameters below any prescribed positive upper bounds and below one. The local expansion records can then be selected with their smallness parameter at most $1/4$. The order is significant: the interval return uses these records on every input curve, so their selection precedes all parameter intervals and finite periods.

Finally fix the insertion menus and auxiliary finite lists. Given any previous entry delays, enlarge them successively to pay the following already established estimates: the sector integrals and their active Gaussian prefactors; the marked and unmarked local bounds; the two forced absorptions; the small-field localization and parameter comparison; and the scalar reset and startup prices. Impose both the fixed local reserve $1/16$ and any requested positive smaller regular reserve τ, with forced reserve $1/8$. The support and path conditions are identical in these two applications.

Here is a precise reason these successive enlargements are compatible. Each estimate supplies a stage-indexed lower threshold $d_j^{(r)}$ and a conclusion valid whenever the entry coupling exceeds that threshold. Replace the current threshold at stage j by the finite maximum of it and all the $d_j^{(r)}$ for that stage, and then by its running maximum up to j. Shell lower bounds are treated the same way. The conclusions persist at every later delay. Requirements on a finite auxiliary list are imposed after that list is fixed, before the final delay is selected. This proves the quantifier order

profiles ; block lengths ; rates and gaps ; protection ; menus and lists ; delays ; scale realization.

The last realization is arbitrary among those paying these delays. In particular no threshold is selected from a later finite-horizon solution. Section 201 constructs a realization, with periods having arbitrarily late multiples of every integer. This is the period divisibility needed for the five auxiliary comparisons of the nonlinear return.

153.2 Keeping the stationary and free data on that realization

Let a be one such realization. Build its free horizon M^0 with the initial kernel prices, and its retained horizon M^1 with the next prices. The retention identities keep their periodized covariance, row and tower data in the precise combinations used by the stationary construction. In particular the full-period lower bounds are supplied for M^1, not for a separately selected quadratic form. Apply the stationary construction to M^1 with the purchased gap and shell bounds. It supplies simultaneously, for every cutoff N, paths $R^1_{N,m}$, scalar corrections $\alpha_{N,m}$ and stationary maps at every allowed period and forest. Its stronger version supplies the extended-domain estimates used in Section 169. The free convergence of Section 137 concerns these same retained towers.

Replace only the terminal normalization, using the revised horizon M and paths $R_{N,m}$. The construction gives the exact identities

$$\begin{aligned} R_{N,N} &= R_N^{\mathrm{Wilson}}, & R_{N,m} &= R^1_{N,m} \quad (m < N),\\ \alpha^{\mathrm{revised}}_{N,m} &= \alpha_{N,m}, & \rho^{\mathrm{revised}}_{N,m} &= \rho^1_{N,m} \quad (m+1 < N). \end{aligned}$$

The entire tower agrees with that of M^1 when $m+1 < N$, and its transport matrix agrees already when $m < N$. The new horizon satisfies the final nonlinear kernel price chosen before the block

lengths. At the terminal level its canonical direct coefficient is at most one, which gives the required Wilson positivity bound. At lower levels the retained all-period bounds apply.

For any fixed m, all the displayed identities hold on the tail $N > m+1$. The limits and eventual larger-width bounds in Section 169 are therefore preserved: remove finitely many terms from each sequence and use equality of the remaining terms. This argument applies to each stationary derivative as well as each path and tower coefficient. It also preserves the extended-domain statement, because the one-step data coincide on this tail. Thus the final nonlinear construction carries both the stationary convergence data and the free-horizon convergence data. The terminal normalization introduces no independently selected background.

153.3 The same local return on every input interval

On this final horizon the sector construction supplies its exact Gaussian expression, including the active determinant, Haar factors, sieve and coupling prefactor. The local price tables selected above apply to this expression. The parameter comparison uses a single reference dictionary and the identity identification of its elementary labels, as in Section 147. The auxiliary locality estimates are retained when the delays are increased: they are stated beyond their entry thresholds, with all block lengths, rates and local dictionaries fixed.

Consequently, for every cutoff, every adjacent pair of levels and period, and every allowed continuous input curve on a compact interval, that section supplies the exact return in one dictionary, with early regular cost

$$H = c_i \delta_E(\widehat{\beta}_m)$$

and the forecast forced cost. Both the $1/16$ and the τ reserve estimates refer to this same constructed horizon and local price data. The selected tree and sector choice can depend on the interval and period, as permitted by the scalar reset, but are fixed throughout that interval.

We check the remaining recipe conditions rather than assuming that they follow from the density identity. Write

$$\delta = \delta_E(\widehat{\beta}_m)/32, \qquad \gamma = \exp\{-\mathfrak{k}(\rho_{ji}, \widehat{\beta}_{m+1}, \beta)\}$$

for the actual regular and forced prices after reorganization, where $\mathfrak{k}$ is the purchased postal decay exponent and β is the incoming coupling. We verify the comparison with the recipe prices explicitly. Put $x = \widehat{\beta}_m$, $y = x + c = \widehat{\beta}_{m+1}$, where $c = s_G \log L \geq 0$, and write $g(\beta) = \beta^{-1/2}$ and $t(x) = g(x)P(x)$. Let $C > 1$ be the exchange constant and let $H_{\mathrm{post}} \geq 1$, $D \in \mathbb{N}$, $\rho > 0$ be the selected postal constants. With $\ell_\theta(y) = P(y)^\theta$, define

$$\begin{aligned} M(x,y) &= H_{\mathrm{post}}(1 + B(x) + P(y) + 2\ell_\theta(y))^D, \\ R(x,y,\beta) &= M(x,y)\left[g(\beta)^3 + g(\beta)^{-D}\left(e^{-\rho B(x)} + e^{-\rho P(y)^2/(1+B(x))^D}\right)\right], \\ T(x,y,\beta) &= C\left[\left(\frac{C}{L} + \frac{1}{16C}\right)\delta_E(y) + R(x,y,\beta)\right]. \end{aligned}$$

All terms in M and R are nonnegative. The recipe's portion and charges are precisely

$$\text{portion} = T, \qquad \text{charge}_E = T + M(x,y)t(x)^3, \qquad \text{charge}_K = H_{\mathrm{post}} e^{-\mathfrak{k}}, \quad \mathfrak{k} = \rho\left(\frac{t(y)}{g(\beta)}\right)^2.$$

In particular $T \geq \delta_E(y)/16$. Since c is fixed after L is chosen, the elementary shifted-gap comparison gives, for all sufficiently large x, $\delta_E(x) \leq 2\delta_E(y)$. The reset constants of Section 148 have the form

$$c_i = \frac{1}{64(1 + B_i^{\mathrm{reset}})}, \qquad B_i^{\mathrm{reset}} \geq 0, \qquad B_i^{\mathrm{reset}} c_i \leq \frac{1}{64}, \qquad c_i \leq \frac{1}{64}.$$

Here B_i^{reset} is the coefficient of H in the regular reset price; write $v_i \ge 0$ for the coefficient of $|\alpha|t(x)^3$ there. It follows that

$$H = c_i\delta_E(x) \le \frac{\delta_E(y)}{32} \le T, \qquad \delta = \frac{\delta_E(x)}{32} \le \frac{\delta_E(y)}{16} \le T \le \mathrm{charge}_E\,.$$

Moreover the uniform stage bound $|\alpha| \le A_j$ and the delayed cubic estimate $64v_iA_jt(x)^3 \le \delta_E(x)$ give

$$\mathrm{regularNeed} = B_i^{\mathrm{reset}}H + v_i|\alpha|t(x)^3 \le \frac{\delta_E(x)}{64} + \frac{\delta_E(x)}{64} = \delta.$$

Thus the actual local cost is paid, as well as the recipe's larger allowance. The soft reset price is bounded by a stage constant times $(1+B(x))^4t(x)$; the late estimate for this expression in the reset section bounds it by the forecast's mild reserve. Its remaining hard reserve is exactly the purchased forecast comparison at this incoming band. These two reserves are retained in the forced absorption of the local return.

Since $H_{\mathrm{post}} \ge 1$, $\gamma = e^{-\mathfrak{k}} \le \mathrm{charge}_K$. This proves

$$H \le \mathrm{portion}, \qquad \delta \le \mathrm{charge}_E, \qquad \gamma \le \mathrm{charge}_K\,. \tag{752}$$

The incoming band lies in $[y/4, 4y]$, so $g(y)/2 \le g(\beta) \le 2g(y)$. Hence $\mathfrak{k} \ge \rho P(y)^2/4 \ge g_jP(y)^2$, using (751). Since $P(x+c)/P(x) \to 1$ and $P(x) \to \infty$, eventually $g_jP(x+c)^2 \ge P(x)^{1/5}$. The coarse forced allowance is $e^{-P(x)^{1/5}}$, so

$$\gamma \le e^{-g_jP(y)^2} \le e^{-P(x)^{1/5}}. \tag{753}$$

The coarse regular allowance dominates δ by its factor 32. Thus the output satisfies the costs of the next call as well as those of the recipe batch.

For clarity, the corresponding implication on the actual arrays is purely monotone. If u_i and v_i are their nonnegative regular and forced majorants at costs δ, γ, the same majorants work at the larger charges in (752); no supports, functions or labels change. Section 148 gives the exact normalized density identity for the same batch and $t^2|\Delta| \le H \le \mathrm{portion}$. The permission predicate used in this construction imposes no additional batch restriction. Together these facts are exactly the batch relation of the chosen recipe. Appending this batch to the incoming finite history supplies the retained-history predicate at level m and the next-step predicate. This proves those predicates for the output that was actually constructed.

The terminal startup construction, treated separately below, must supply one continuous family on the full terminal band. Its regular and forced bounds, common summable majorants and invariant jets survive the fixed packing into the terminal dictionary. Its exact Wilson density starts the retained history. This is the required initial-family assertion even when $N = 0$.

Once that startup assertion is proved, we can apply the actual scalar reset at every input interval and then Section 150. The result is a finite history at every cutoff and period, with the prescribed level-zero coupling, the exact transferred densities and the same selected paths $R_{N,m}$. Its per-edge coupling error is the purchased reset width, and its forced estimate includes (753). No assertion of uniqueness of these shooting histories is required. Their subsequent convergence and the additional observable comparisons use the retained common data and the estimates of their respective sections.

154 Paying the observable estimates on the same history

The nonlinear return and the observable estimates have to be available before a finite-volume history is selected. We now make the additional choices in that order. The resulting statements are uniform

over every allowed representation on the purchased horizon. In particular, enlarging the finite volume does not require a new choice of block lengths.

Retain the profiles, horizons and targets of Section 153. At a transition from stage j to stage i, write b_i for the outgoing support exponent, e_{ji} for the target's earned exponent, and q_{ji} for its error exponent. The purchases give

$$e_{ji} > b_i + 3, \qquad q_{ji} \le -7/16.$$

These are two different reserves: the first pays support enlargement, whereas the second gives decay with the coupling. Fix a nondecreasing integer occupancy schedule $N_j \ge 1$. At each stage we need only the two possible target stages and the finitely many grades through j.

154.1 The broad parameter estimate and its localized price

Put $y = x + s_G \log L$, $B = \lceil (\log x)^2 \rceil$ and $r = y^{1/256}$. Apply Section 141 with outgoing exponent $e = b_i + 3$, the actual common support dictionary, and the fixed derivative order and spatial power at stage i. Its positive support reserve is $(e_{ji} - e)/4$. The input allowance has upper exponent at most $-7/16$, so the integrated constituent price has upper exponent $-2/5$ and the rooted derivative price is eventually at most $y^{-3/8}$. For each finite real test λ on the parameter interval, its ordinary logarithmic family F therefore satisfies

$$\mathcal{N}_e \left(\int F(t)\, d\lambda(t) \right) \le y^{-3/8} M_{r/2}(\lambda).$$

Here $\mathcal{N}_e$ is the rooted derivative budget at support exponent e, and $M_{r/2}$ is the moment-test norm of Section 144. The forced family has one common absolute coefficient majorant at radius $r/2$, with the compulsory warning gain retained. The construction proves this for all compatible input representations and all real parameter lines; the majorant is not selected after the finite test.

The hypotheses on the real line deserve mention. Its coupling is the same quadratic line used in Section 140. Its activity coordinates are affine, hence of degree at most two. The path, cutoff, charts and input dictionaries are fixed on the line. The enlarged real interval remains in the purchased band, by the radius choice there. The compensated derivative and packet-action estimates are supplied by Section 103. Thus all the local hypotheses of the sector estimate hold on these actual lines, including their held fine fields and their cutoffs.

We check that the finite localization does not destroy the useful power. Let C_i be the signed-pullback derivative cost and define

$$w(x) = C_i \{ y^{-3/8} + 4y^{-7/16} \}.$$

This has upper exponent $-3/8$. The five auxiliary tori in Section 142 have sides bounded by a fixed multiple of $1+B$. Their numbers of sites, local ball counts, and weighted changes of anchor are fixed powers of $1 + B$. The interpolation weights are fixed. For example, a contribution from auxiliary torus Λ_ν has price at most $2|\Lambda_\nu| w(x)$; its weighted change of anchor costs at most $2^k (1 + cB)^{pk}$ at derivative order k and spatial power p. The local allocation costs at most another fixed ball count and $\exp((b_i + 2) C_{\mathrm{cover}})$, with C_{cover} fixed. Consequently both the localized raw derivative price and its ordinary rooted price have upper exponent $-3/8$. The latter is eventually at most $y^{-1/4}$, and the former at most $y^{-1/8}$. These comparisons use strict exponent margins and are uniform over the period and input representation.

The forced price survives the same localization with its forecast normalization. The enlarged spatial factors have upper exponent zero. The compulsory Gaussian warning is $\exp(-\mu P(y)^2/c)$

with $\mu > 0$ and fixed $c > 0$; the permitted forecast spends a strictly smaller part of that exponent. The ratio of the spent price to the forecast tends to zero faster than any fixed power of y, as in Section 200. Thus the ordinary price $y^{-1/8}$ and the soft correction can be paid simultaneously in the forced absorption inequalities. This uses the chosen positive forecast rate below its permitted fraction of μ, already selected before the delays. More precisely, the forecast hard rate is at most $\mu/16$. Its soft allowance is twice the zero-order compulsory activity price, $2A_{\mathrm{comp}}(x)$, and the resizing estimate is applied at fraction $1/16$ before the final absorption. This keeps both the factor two and the stated rate reserve. No new rate is selected for an individual input.

154.2 The fixed parameter radius and all derivatives

Choose a fixed $q \geq 4$ and a desired normalized output allowance $0 < s \leq 1/256$. The finite interpolation in Section 144, with degree 300, has a constant $U = U(300, q)$. Choose the small response parameter $t > 0$ so that $6Ut \leq s$. This choice precedes the nonlinear purchases; Section 153 permits any positive such t. The pointwise localization estimate is

$$3tC_{\mathrm{st}}x^{-11/10} + R(x), \qquad R \text{ has upper exponent } -23/20.$$

Combining it with the just-proved broad bound gives the fixed-radius price $sC_{\mathrm{st}}x^{-11/10}$. Indeed the discarded parameter tail is at most

$$(1+U)(2q)^{301}y^{-365/256},$$

which is $o(x^{-11/10})$, as is $UR(x)$. The linear reset and the common forced coefficient majorant give the two normalized output prices at radius $q/2$ from that section. The scalar projection has price $sC_{\mathrm{st}}x^{-11/10}/\theta(x)^2$. Thus the finite-test hypotheses of Section 145 are available for the same return as its exact density identity.

The differentiated output also requires one common local collection of constants before an input is supplied. This follows by making the choices in Sections 87 and 103 at the largest requested order k. For the chart derivatives use a coarse radius

$$q_k(x) = Z_k\, 2\theta_f(x), \qquad v(x) = (k+1)q_k(x),$$

where $Z_k > 0$ is the fixed angular chart constant and $\theta_f > 0$ is the fine fluctuation threshold. Then $v(x) > 0$, $kq_k(x) \leq v(x)$, and both radii have upper exponent $-1/2$. The comparison estimates allow any such radius. Fix the cutoff convention before this collection of constants is chosen. The first integrated polymer is the same unmasked local function used in Section 87. On a healthy support its mask is identically one in a neighborhood, so its ordered derivatives are the derivatives of that unmasked function. The angular-to-chart conversion at radius q_k uses $kq_k \leq v$ and supplies all orders through k. This identifies the pre-mask derivative estimate as well as the masked activity; the former is required by later differentiated insertions. The Gaussian normalization estimates apply to every read subset with the same row bounds, including the empty subset. The enforcement and warning derivatives have respectively the bounds proportional to the cover number and to one plus the warning-support cover number; these are exactly the cutoff terms retained in the differentiated activity construction.

Reserve a positive part of the target support exponent before summing the differentiated factors. At every order through k, its exponential weight absorbs the polynomial cover and length factors. For completeness, choose a common positive support reserve $\Delta = e_{ji}/2$ at outgoing exponent zero.

If $A_k(x) \geq 0$ is the common integrated derivative constant of upper exponent zero and $u_k(x) > 0$ the constituent allowance, its differentiated allowance is

$$s_k(x) = (1 + 2^k A_k(x))u_k(x).$$

It has the same negative upper exponent as u_k. The connected-tree criterion $4(2B+1)^4 s_k/\Delta \leq 1/2$ therefore holds eventually. The ordinary and forced unweighted constituent allowances likewise become at most $1/2$, and their tree criteria at reserve Δ hold at the same threshold. Since $0 + \Delta + \Delta = e_{ji}$, these are exactly the two reserves required in the renewed first and second expansions. The unstrengthened construction in Section 100 proves their majorants and all polynomial cover moments; we use it here at exponent zero, with the same Δ in both expansions. Thus renewal does not introduce an unrelated set of constituent prices.

The resulting connected activity majorants and nearby integrable domination are those of Sections 87, 85 and 102. Enlarge the finitely many constants to common nonnegative ones and intersect their large-coupling intervals. All those constants depend on the stage, block length, chart and purchased prices, but not on the finite volume or input representation. The subsequent unweighted total sums may depend on the finite volume, as the continuity proof allows. This distinguishes the uniform local derivative estimates from the fixed-volume domination used to pass limits under sums.

154.3 Insertion estimates and their common delays

The primary, regular and forced one-insertion estimates use the same local expansion. Their common heading and inverse bound were proved in Section 113. The primary calibration uses the extra derivative available when $d + 3 \leq i$; the regular case uses its vanishing leading coefficient, and the forced case retains its square-root forced-allowance normalization. The auxiliary local integral and its comparison are those of Sections 96 and 97. The finite-support conditions are imposed by increasing the shell and entry bounds, after their fixed support radii have been selected.

For multiple insertions, the raised forced estimate and the ordinary estimate have distinct prices. Use both, as in Section 17. The tariff has ordinary allowance y^{-1}, single-insertion allowance y^{-2}, and forced allowance $2e^{-\mu P(y)^2/16}$. The occupancy and projection inequalities of Section 95 apply with $N = N_j$. In particular the local projection bill is at most $e^{\sqrt{y}}$, while its two error bills fit the fixed local splitting allowance. The required geometric floor is imposed before L is selected; the occupancy and error inequalities are imposed afterwards.

Here is the common choice explicitly. At stage j take the maximum of the pre-length floors for the finitely many allowed target stages, the primary comparisons, and the projected insertion geometry. Choose L_j above it. Having chosen wages, rates and protection, choose the primary records with smallness at most $1/4$. For each allowed target stage, take the finitely many eventual thresholds for all grades through j, the broad and fixed-radius parameter bounds, the differentiated output, the ordinary and raised menu bounds, and the projection inequalities. Include all previous thresholds. Replace their maximum at j by its running maximum up to j. Enlarge the shell bound by the required fixed comparison radii and its growth lower bound by one.

Every conclusion used above is valid beyond its threshold, so all are preserved by any further enlargement. The quantifiers are therefore

$$\text{profiles and block floors;} \quad \text{block lengths;}$$
$$\text{prices and protection;} \quad \text{delays;}$$
$$\text{all compatible local inputs.}$$

The scale realization and the selected finite-volume history come last. On that history each of the estimates applies to its actual held transition; the endpoint representation, its density identity, its singleton derivative and its marked profile all retain their original common dictionary. In particular the observable estimates do not choose another solution of the scalar shooting problem.

This supplies the additional local purchases needed to apply the one-history transfer constructions. The passage from those local purchases to the complete centered service and the ultraviolet field family still requires their simultaneous global assembly.

155 Producing and symmetrizing the marked campaign

This section supplies the campaign used by the fixed-stencil construction. It has two parts. First, the equation ideal is read by microscopic link observables whose Taylor errors lie strictly above the requested grade. Second, finite-dimensional linear algebra combines these reads into one campaign, and finite group averages impose the signed and real symmetries. The filtration throughout counts factors that are color coordinates of based curvature slots; it is not a count of raw link entries.

155.1 The microscopic filtration and its Taylor data

Fix a gauge group, a holomorphic link action, and a cutoff degree d. A slot S is one of the based, transported plaquette-curvature readouts on the cover lattice. For a real basis vector e_a of the color space, write

$$x_{S,a}(U) = e_a \cdot S(U).$$

Let $\mathcal{P}_{\geq r}$ be the complex span of finite products of at least r such coordinates, with the empty product allowed when $r = 0$. Thus a single slot coordinate has filtration degree one even though, as a function of link matrices, it is a nonlinear polynomial or analytic expression.

For a label p of grade $m = (r, s)$, a marked read consists of a field function f and a polynomial q with three properties. The function is gauge invariant, has at least s slot factors in the span sense above, and its link-action derivatives through order d agree with those of q. The correction $q - p$ has strictly larger grade stamp

$$\mathrm{st}(r, s) = (s, r)$$

in lexicographic order. When p lies in the Maxwell equation ideal, the read f is additionally microscopic: its moment against every sufficiently large-volume spectator which is local away from a fixed anchor vanishes. These are precisely the properties needed for later stencil pricing and for the exact equation-ideal payment.

The elementary analytic input is a first-order germ for each slot. Put Snap_d for the finite-dimensional space of radial link jets through degree d. For every slot S, the function

$$J \longmapsto S\big(U_d(J)\big), \qquad J \in \mathrm{Snap}_d,$$

is real analytic near zero, transforms by the adjoint action under a constant gauge transformation at the base point, vanishes at zero, and has derivative equal to the linearized plaquette curvature. To see the analyticity directly, on one link the connection coefficient is a linear function $A_J(t)$ of J. On the Banach space $C([0,1], \mathfrak{A})$ of continuous paths in the finite-dimensional matrix algebra $\mathfrak{A}$ containing the group representation, the Volterra operator

$$(T_J u)(t) = \int_0^t u(s) A_J(s)\, ds$$

has norm at most $\sup_s \|A_J(s)\|$. In a neighborhood of $J = 0$ this is less than one, so the unique solution of the link equation is

$$u_J = (I - T_J)^{-1}\mathbf{1} = \sum_{n\geq 0} T_J^n \mathbf{1}.$$

The series converges locally uniformly in the finite-dimensional parameter J, and hence is analytic there. Uniqueness identifies it with the given link transport. A slot is obtained by a finite product of these transports and their inverses, followed by the plaquette curvature map; it is therefore analytic as well. Constant-gauge covariance follows by conjugating the ODE, and differentiating at $J = 0$ gives the signed sum of the edge tangents around the plaquette, which is the linearized curvature. This proves the required slot germ without imposing any regularity on a configuration space of infinite link fields.

155.2 Avoiding one edge while spanning Taylor tests

The next finite-dimensional fact is the source of the locality needed for products. Regard Snap_d itself as the real vector space $\mathcal{T}_d$ of radial jets. For a slot S, let

$$B_S = D_0\big(J \mapsto S(U_d(J))\big) : \mathrm{Snap}_d \to \mathfrak{g}.$$

If e is a prescribed cover edge, call a slot S safe for e if its spatial support avoids e along the translated path used to define S.

Lemma 155.1 (Safe Taylor tests span)**.** *For each edge e, the real linear span of the component functionals of B_S, over slots S safe for e, is all of $\mathcal{T}_d^*$.*

Proof. For $d < 2$ the radial-jet space has no Taylor components and the claim is immediate. Assume $d \geq 2$. Take a jet $A \in \mathrm{Snap}_d$ annihilated by every listed functional. For every pair of axes $i < j$, the safe-slot geometry excludes only a finite set of plaquette base points. At each other base point it supplies a detour transport path avoiding e. The hypothesis says the resulting slot derivative on A is zero. The slot-germ identity therefore gives the full sampled curl equation below. Here $C_{ij}(A_\ell)(x,t)$ denotes the axes-(i,j) finite-mesh curl component of the degree-ℓ Taylor field at position x and mesh t; it is homogeneous under simultaneous scaling of x and t.

$$\sum_{\ell=0}^{d-2} C_{ij}(A_\ell)(x_y, 1) = 0$$

at every lattice point y outside the finite exceptional set. We now induct downwards: once shells above k have been shown to vanish, this same equation reduces to its sum over $0 \leq \ell \leq k$. For $t > 0$ define

$$F_k(x,t) = \sum_{\ell=0}^{k} t^{k-\ell} C_{ij}(A_\ell)(x,t).$$

Homogeneity of the degree-ℓ curl component gives

$$F_k(tx_y, t) = t^k \sum_{\ell=0}^{k} C_{ij}(A_\ell)(x_y, 1) = 0.$$

Choose lattice sites y_t with $tx_{y_t} \to x$ as $t \downarrow 0$. For $x \neq 0$ the finite exceptional set is avoided for all sufficiently small t. Continuity of F_k gives $F_k(x,0) = C_{ij}(A_k)(x,0) = 0$, since every lower-shell

term has a positive power of t at $t = 0$. Continuity extends this identity to $x = 0$. Thus the top coefficient has zero curl. For a homogeneous radial-gauge potential of degree $k + 1$, this forces the potential to vanish directly. Differentiate its radial-gauge identity $\sum_j x_j A_j(x) = 0$ in coordinate i and use curl-freeness and Euler's identity:

$$0 = A_i + \sum_j x_j \partial_i A_j = A_i + \sum_j x_j \partial_j A_i = (k+2)A_i.$$

Hence $A_i = 0$ for every coordinate, so the top shell vanishes. Remove it from the sampled equation and repeat with the next lower index. Descending induction proves $A = 0$. In a finite-dimensional space a family of functionals spans the dual exactly when its common kernel is zero, which proves the claim. □

The slot linearization in this proof is concrete. For a based plaquette readout it is the oriented sum of the four link tangents, transported to the base point; the ODE calculation above proves this formula for the actual link action. Only finitely many plaquette base points are excluded; at each remaining point the detour path is chosen separately. The radial argument therefore applies to every Taylor test, not just to tests supported on a generic set.

Lemma 155.2 (Equation-factor and spectator reads)**.** *Let a be a homogeneous polynomial in the equation ideal, of grade $m = (r, s)$ with $r \le d$. Then a is a finite linear combination of terms*

$$\ell \prod_{j=1}^{k} z_j,$$

where ℓ is a homogeneous linear label functional annihilating the Maxwell solution-potential subspace and each z_j is a homogeneous linear label atom. Their field reads are assembled from slot-coordinate observables. Each term has a raw read f and polynomial q with $q - a$ of strictly larger grade stamp, $f \in \mathcal{P}_{\ge s}$, and f microscopic. Individual terms need not be gauge invariant.

Proof. Write $\mathcal{E}_j$ for a projection onto the space of radial potentials obtained from on-shell Maxwell curvatures at order j, and let $\mathcal{E}$ act coefficientwise. The equation ideal is the kernel of $p \mapsto \mathrm{changed}(\mathcal{E})p$. If p belongs to that ideal, then

$$p = p - \mathrm{changed}(\mathcal{E})p.$$

Expand p into monomials before applying the substitution. For a literal slot atom u, split $u = (u - \mathcal{E}u) + \mathcal{E}u$. The first summand annihilates every Maxwell solution potential. For a product $u_1 \cdots u_k$, use the telescoping identity

$$\prod_{j=1}^{k} u_j - \prod_{j=1}^{k} \mathcal{E}u_j = \sum_{i=1}^{k} (u_i - \mathcal{E}u_i)\Big(\prod_{j<i} \mathcal{E}u_j\Big)\Big(\prod_{j>i} u_j\Big).$$

The projection preserves the radial grade. Thus every summand has one solution-annihilating linear factor and a finite spectator product, with the original total grade. This proves the displayed factorization directly from the equation-ideal definition; no invariant-factor assumption is made.

Fix one generator $\ell \prod_j z_j$. The functional ℓ annihilates the Maxwell solution-potential subspace. The required duality statement concerns only the top Taylor layer. For a homogeneous potential A_ℓ, define its transverse plaquette difference by

$$T_\ell(p, b; x) = N_\ell(p, b; x, 1) - N_\ell(p, b; x - e_b, 1),$$

where N_ℓ is the sampled curl on the plaquette based at p. Apply the plaquette incidence sign in direction ν to T_ℓ to obtain the transverse force density. The corresponding line term $L_\ell(\nu, p; x, t)$ is the same incidence sum of the oriented arc; each arc is the difference of the two ordered triple averages of $\partial_b(\partial_a A_c - \partial_c A_a)$ along the plaquette edges. The source's scaling and zero-mesh identities are

$$tL_\ell(\nu, p; tx, t) = t^\ell T_\ell(\nu, p; x), \qquad \sum_p L_\ell(\nu, p; x, 0) = -\sum_b \partial_b \operatorname{curl}(A_\ell)_{b\nu}(x).$$

If every positive path-force test vanishes through order $j+2$, then for every permitted lattice site the finite force sum vanishes. Its line form is

$$F_j(x,t) = \sum_{\ell=0}^{j} t^{j-\ell} \sum_p L_\ell(\nu, p; x, t).$$

The scaling identity shows that $tF_j(tx_y, t)$ is t^j times the vanishing finite force sum. Since $t > 0$, this implies $F_j(tx_y, t) = 0$. For $x \neq 0$, choose lattice sites y_t with $tx_{y_t} \to x$; they eventually avoid the finite exceptional set. Let $t \downarrow 0$ and use continuity. Only the top coefficient remains, and the zero-mesh identity gives $\sum_b \partial_b \operatorname{curl}(A_j)_{b\nu} = 0$. The curl is also closed by the Bianchi identity, hence is an on-shell Maxwell curvature. The exact potential sequence represents this curvature as the curl of a radial potential; uniqueness in radial gauge identifies that potential with A_j. Thus A_j lies in the image of the Maxwell potential map. Continuity extends the assertion to $x = 0$.

Now let a linear functional on the top Taylor layer annihilate that solution-potential image. It vanishes on A_j whenever the positive force tests vanish on A. Thus its composition with projection onto the top Taylor layer annihilates the common kernel of the force family. Finite-dimensional duality puts this composed functional in the span of the positive path forces. This is the force representation used to construct the equation-factor read; no statement about the full force-kernel in all Taylor layers is needed.

Write the equation-factor atom as one of grade $(j+2, 1)$, and let the spectator word have grade equal to the sum of the grades of its factors. At the intermediate cutoff $j+2$, express the linearized equation factor as a finite sum of transported link forces. Replacing each linearized curvature by its actual slot read gives the forced factor with error in slot filtration at least two. The spectator word is read by induction on its length: for each factor, the safe-span lemma supplies slots avoiding the force edge. The product rule then puts every error term at least one slot above the spectator's own factor count. These reads have the desired minimum slot count and preserve the one-link microscopic identity.

For the requested cutoff $d \geq j+2$, extend the intermediate Taylor functional to the full jet and multiply by the unchanged spectator word. The extension has two distinct errors. First, the discrepancy between the constructed read and its own lifted Taylor polynomial has one more slot factor than the target's second grade. Second, the lifted linear functional agrees with the original equation atom on the intermediate jet; its remaining Taylor tail has strictly higher first grade while retaining the same slot count. Multiplying this tail by the spectator word preserves the strict first-grade increase. Thus, writing w for the lifted polynomial and v for the target generator, the required comparison is the sum

$$w - v = \big(w - \operatorname{Taylor}_{\leq d}(\widetilde{\ell}\, a)\big) + \big(\operatorname{Taylor}_{\leq d}(\widetilde{\ell}\, a) - v\big).$$

The first term is strictly later in the slot-count coordinate of the grade stamp; the second is strictly later in the first-grade coordinate. Both are therefore strictly above the target stamp. This two-part comparison, together with the forced edge read, gives the required positioned read for the generator. Finite sums and scalar multiples extend it to the full homogeneous ideal element.

The reads in this lemma are raw reads; gauge averaging is done only after the terms for the invariant label have been assembled. □

The conditional moment calculation in this argument is the one-link Haar integration-by-parts identity. For a walk w from the anchor to a source edge e, the transported force is

$$\mathrm{Ad}_{\mathrm{travel}(U,0,w)}\,\mathrm{field}(U,e).$$

Move the transport to the other slot using invariance of the color inner product, then expand in a real orthonormal color basis (E_a). Every term has the form

$$\langle E_a, \mathrm{field}(U,e)\rangle\, S_a(U),$$

where S_a is the spectator times a coefficient depending only on the walk transport. For the translated test at anchor o, choose the Micro radius larger than the support radius of the walk and its source edge, and choose the side threshold so these cover sites do not wrap. The test spectator is then local outside that radius and is unchanged when only the translated source link is varied. Haar integration by parts on that link gives

$$\int (\partial_{E_a} S_a - \beta\langle E_a, \mathrm{field}(U,e)\rangle S_a)\, d\mu_\beta(U) = 0.$$

The derivative term vanishes; the derivative of the bare density is the displayed force term. The envelope bound supplies an integrable majorant for differentiating under the integral, so the identity also holds for the bounded measurable spectators permitted in the Micro definition. Summing over a gives the force moment zero. This identity is valid for either sign of β; the nonzero condition in Micro is used only when dividing out the scalar factor in the moment normalization.

Scope of the factor lemma. The source's equation-ideal identity is the kernel statement used above. The force calculation used is only the top-component implication proved from the radial current limit: simultaneous vanishing of the positive path-force tests puts the highest radial coefficient in the image of the Maxwell potential map. The conditional one-link Haar identity is the moment input defining the equation leg. The argument above proves the monomial factorization, edge-avoiding spectator assembly, and two-part filtration comparison from these facts.

155.3 The all-grade campaign

The previous lemmas give the exact pointwise producer needed for the finite-dimensional assembly.

Proposition 155.3 (Positioned equation-ideal read)**.** *For every pure-grade label p in the Maxwell equation ideal, of grade $m = (r,s)$ with $r \le d$, there are a gauge-invariant function f and a polynomial q such that f has slot degree at least s, the action derivatives through order d of f and q agree, $q - p$ has grade strictly above m, and f is microscopic.*

Proof. Use the exact ideal factorization to write p as a finite sum of gap atoms times spectator monomials. Apply the factor-read lemma to those generators and add the resulting raw reads. Their polynomial targets sum to p; their errors remain in the strictly higher-stamp subspace, and the slot-degree and microscopic conditions pass to finite sums. The resulting raw function need not yet be gauge invariant.

Now average the pair over the compact gauge group at the anchor. A constant gauge transformation rotates the color components of each of the finitely many fixed slot addresses among a

finite basis. Thus the orbit of this particular finite sum lies in a finite-dimensional span, and its Haar integral is an ordinary finite-dimensional vector integral. The label p is gauge invariant, so the target of every rotated read is still p; its Taylor correction remains in the same strictly higher stamp subspace. Forms, slot filtration, and the microscopic moment condition are linear and preserved by the integral. A based transported slot under an arbitrary local gauge transformation changes only by the adjoint action of that gauge transformation at its base point: the interior transports cancel in pairs. Averaging over the anchor group therefore makes the read invariant under the full local gauge action. This produces the required f and q. □

The proposition applies to the actual action, rather than to a formal substitute: the slot germ is obtained from the link ODE above. Its one-factor linearization is the oriented plaquette sum. The finite-generation formula for the ideal is obtained degree by degree by expanding each polynomial into monomials, applying the projection onto the Maxwell solution-potential subspace, and splitting one literal equation factor from the remaining slot coordinates. The safe-span annihilator proof handles the edge avoidance needed to multiply the spectators. These are the mathematical steps packaged by the positioned read in the source.

Theorem 155.4 (Campaign and imprint at fixed cutoff)**.** *Assume at cutoff d there is a complex-linear map S from labels to link-field functions with unit value one, gauge-invariant values, and exact action Taylor forms through order d. In addition, for each pure-grade basis label of grade (r, s), assume its value has a finite representation as a linear combination of products of based curvature-slot color coordinates, each product having at least s factors. This is the powered slot-degree condition used below. Then there is a complex-linear campaign F whose grade-m value has a read with slot degree at least the second grade, whose polynomial correction has strictly larger stamp than the input label, and whose equation-ideal values are microscopic. The campaign may be chosen to commute with signed coordinate changes and the real structure.*

Proof. We first construct the unsymmetrized campaign. The sound section itself provides a linear map X with

$$Xp = (p, S(p)), \qquad X1 = (1, 1).$$

It gives the required action forms, gauge invariance, and slot degree for each pure grade. It need not be microscopic on the equation ideal.

For a fixed grade m, restrict the input p to $\mathrm{Grade}(m)$ and let

$$P_m = \mathrm{Grade}(m) \cap \mathcal{I}_d,$$

where $\mathcal{I}_d$ is the finite-dimensional Maxwell equation ideal. Choose a basis (b_i) of P_m. For each b_i, apply Proposition 155.3 and write its read as (q_i, f_i). Extend $b_i \mapsto (q_i, f_i)$ linearly on P_m. The result is a payment map on the ideal part of this grade: forms, strict-stamp correction, powered membership, and the Micro conclusion are all linear conditions and hence pass through the basis expansion.

Choose a linear projection L_m from the full label space onto P_m which restricts to the identity on P_m. If X_m is the map just constructed on P_m and S_m is the sound-section map, set

$$F_m(p) = X_m(L_m p) + S_m(p - L_m p).$$

Both summands have the required grade-m payment, so their sum does too. If this grade-m input p lies in the equation ideal, grade preservation of the ideal gives $L_m p \in P_m$ and $p - L_m p \in P_m$. Since L_m is the identity on P_m, applying L_m to $p - L_m p$ gives zero and also $p - L_m p$; therefore

$p - L_m p = 0$. Thus for ideal inputs the entire value is the microscopic marked value $X_m(p)$. If $m = (0,0)$, the unit is not in the equation ideal: the ideal has no nonzero constant component. Hence $L_m 1 = 0$ and the sound map supplies $F_m(1) = (1,1)$.

There are only finitely many relevant grades,

$$\{0,\dots,d\} \times \{0,\dots,d\}.$$

Decompose $p = \sum_m p_m$ by its homogeneous grade and define $F(p) = \sum_m F_m(p_m)$. On an input of pure grade m, every other grade component vanishes, so the corresponding payment remains exact. On the unit only grade $(0,0)$ contributes, so the start condition holds. This is the claimed campaign.

Now symmetrize. Let $\mathcal{S}$ be the finite signed coordinate permutation group, θ_s its action on labels, and ϕ_s its joint action on a label/function pair. Define

$$F^{\mathrm{sgn}} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \phi_s F \theta_{s^{-1}}.$$

Signed changes preserve grade, the strictly higher-stamp subspace, the slot-product filtration, the action-form identity, and the equation ideal. They preserve the moment condition by changing variables in the finite volume measure. Thus every summand remains a campaign and the average is one. Reindexing by $s \mapsto ts$ shows $F^{\mathrm{sgn}} \theta_t = \phi_t F^{\mathrm{sgn}}$.

Let c be the joint real-conjugation involution on labels and label/function pairs. It is conjugate-linear, commutes with the signed action, and preserves each campaign condition. Set

$$F^{\mathrm{imp}} = \tfrac{1}{2}\big(F^{\mathrm{sgn}} + cF^{\mathrm{sgn}}c\big).$$

The midpoint is complex-linear because the two conjugate-linear actions on the source and target cancel. It remains a campaign; $c^2 = 1$ gives real equivariance, while commutation of c with the signed action retains signed equivariance. Both averages fix the unit. This is the required imprint. □

The exact algebraic assembly above uses no limit in volume and no compactness argument. Its analytic input is the positioned read. The source proves that input from the radial ODE germ, the finite Taylor annihilator argument, the equation-ideal generation formula, and the conditional-Haar microscopic identity described above. The section input is supplied by the integrated section producer, as the following corollary records.

Corollary 155.5 (Actual fixed-cutoff imprint)**.** *For every gauge group, holomorphic link action H, and cutoff d, the source's actual section construction yields a campaign and an imprint at that cutoff.*

Proof. Use the local-sections construction of Section 59. For cutoff d it gives the section $T_d F$ and pose polynomials P_α with exact action forms through order d, unit, and gauge covariance. Its pose certificate says that every monomial in the pose polynomial of a grade-(r,s) basis label has total factor degree at least s. Expanding the finite pose polynomial writes each such basis value as a finite sum of slot-coordinate products with at least s factors. Thus the powered slot-degree hypothesis of the theorem holds. Applying the theorem gives a campaign; its finite signed and real averages give the imprint. □

156 A fixed marked section and its calibration budgets

The marked finite-volume route needs a single local section at each label degree. That section must be chosen before the volume, cutoff, or derivative budget is known. It must also price every later derivative request at that degree. We construct that section first, then show how the schedule selects its budgets and how the approximate field change is made invertible.

156.1 One section for every later derivative budget

Let H be the fixed holomorphic gauge action. At a fixed degree d, let E_d be a finite-dimensional complex label space with a real homogeneous basis $(e_j)_{j\in J_d}$. The basis contains the unit and has grades (r_j, s_j) with $r_j \le d$. The algebra used here is not filtered by raw link-entry count. Let $\mathfrak{S}$ be the collection of based and transported curvature slots on the cover lattice, let I be the real color-coordinate basis, and put $\mathrm{Chip} = \mathfrak{S} \times I$. For $c = (S, a) \in \mathrm{Chip}$ let $x_c(U)$ be color coordinate a of curvature slot S at configuration U. Define $\mathcal{P}_{\ge s}$ as the complex span of the finite products $\prod_{\nu=1}^{m} x_{c_\nu}$ with $m \ge s$. Thus one factor means one curvature-coordinate slot, even though it is itself polynomial in link entries. This is the source filtration called Powered.

The input to the section construction is an imprint: linear maps $A_d : E_d \to E_d$ and $B_d : E_d \to \mathcal{P}_{\ge 0}$ with unit normalization $A_d 1 = 1$, $B_d 1 = 1$, gauge invariance of $B_d p$, $\mathrm{Forms}(H, d, B_d p, A_d p)$, and $B_d p \in \mathrm{Micro}$ whenever p belongs to the equation ideal, together with the campaign's exact-action identity. They also have the following filtration clauses. If p has grade at least (r, s), then $A_d p - p$ has strictly larger source stamp

$$\mathrm{st}(r, s) = (s, r)$$

in lexicographic order, and $B_d p \in \mathcal{P}_{\ge s}$. The order is lexicographic in the displayed order, so increasing the stamp never lowers the factor coordinate s. In addition, A_d and B_d each commute with the signed-coordinate action and the real structure; equivariance of only B_d would not suffice to pass it through A_d^{-1}. A campaign without these two equivariances is turned into such an imprint by the finite group and conjugation averages described in the source; the campaign producer itself is a separate input to this local theorem.

For clarity, the symmetrization just mentioned is finite-dimensional. If Φ_s is the joint signed action on labels and fields, and $\mathfrak{c}$ is the joint real-conjugation involution, put

$$F_0 = \frac{1}{|\mathcal{S}|} \sum_{s\in\mathcal{S}} \Phi_s \, F \, \theta_{s^{-1}}, \qquad F_1 = \frac{1}{2}\big(F_0 + \mathfrak{c} F_0 \mathfrak{c}\big),$$

where $\mathcal{S}$ is the signed-coordinate group. Each summand is still a campaign: the grade is preserved, the correction remains strictly above its source stamp, the field remains in $\mathcal{P}_{\ge s}$, and the unit and exact action identities are preserved. A finite average preserves these linear conditions. Reindexing the first sum by left multiplication shows that F_0 commutes with every signed action; the involution identity $\mathfrak{c}^2 = 1$ shows F_1 commutes with conjugation, and the two actions commute, so both equivariances hold for F_1. Both averages fix the unit. This is the mathematical content of passing from a campaign to an imprint; the existence of the unsymmetrized campaign from the action is not reproved in this section.

Theorem 156.1 (Fixed-stencil marked portfolio from an imprint)**.** *Given an imprint on E_d, there are a finite centered stencil S_d, a linear section*

$$T_d : E_d \longrightarrow \mathbb{C}[\textit{link entries on } S_d],$$

and an invertible linear regrading $R_d : E_d \to E_d$ such that

1. *$0 \in S_d$ and S_d is invariant under signed coordinate permutations;*
2. *evaluating T_d at the coordinates of $R_d p$ on every link field gives $B_d p$ exactly;*
3. *T_d is gauge invariant, has the correct unit, is compatible with signed pullbacks and conjugation, and has the prescribed action projection;*
4. *each basis entry has a finite pose supported in S_d whose degree is at least s_j;*
5. *for every derivative order k and power P, there is a finite number $C_{d,k,P}$ such that each basis entry has source rate $C_{d,k,P}$ at factor degree s_j. This is a bound on the healthy part of the translated stencil, for every budget point of order k and power P, torus, anchor, and tested direction list allowed by that budget point.*

The section and stencil depend on d and the fixed action/frame only; they do not depend on k, P, torus size, or cutoff.

Proof. First invert the triangular regrading. On the finite grade decomposition, the condition that $A_d p - p$ has strictly higher grade makes A_d an upper-triangular block matrix with identity diagonal. Back substitution solves $A_d x = y$ grade by grade: at the lowest grade, x has the same component as y; after subtracting that component, the next grade is solved in the same way. Finiteness of the grade set makes this process terminate and gives a unique x. Thus $R_d = A_d$ is invertible, and the same induction shows that its inverse preserves the lower grade bound.

For each basis vector set $f_j = B_d(R_d^{-1} e_j)$. By the filtration property, f_j is a finite linear combination of monomials, each with at least s_j curvature-coordinate factors. Each monomial uses only a finite list of slots. A product is represented by concatenating its two lists of slots. A sum is represented by taking the disjoint union of the lists and adding the corresponding polynomials. The empty list represents constants. Induction over the finite monomial expansion therefore gives, for each j, a finite slot list and a pose polynomial of degree at least s_j realizing f_j.

There are only finitely many basis vectors. The union of their slot stencils is finite, so choose a centered box containing it. Enlarge the box symmetrically if needed; this preserves containment and makes the stencil invariant under signed coordinate permutations. On this one box, the polynomial realization is constructed for every basis vector. Extend these values linearly from the basis to define T_d. Linearity gives the exact evaluation identity for every p, not merely for basis labels.

The evaluation identities also transfer all structure. Gauge invariance, the unit normalization, signed covariance, conjugation reality, and the action-projection condition are the corresponding properties of B_d. Because evaluation agrees on the entire basis, the section has these properties on all labels. The basis poses give the common-support and degree witnesses.

It remains to show the derivative budgets for all later queries. The source rate statement is localized: for a stencil S, order k, power P, side n, and anchor o, it controls derivatives after expansion at o on $\mathrm{Healthy}(\theta, S + o)$, for direction lists obeying the budget point's direction bound at o. The extra derivative-rate parameter here is zero, so the threshold factor remains exactly θ^s at every tested derivative order. Each color-coordinate curvature slot has rate degree one by the local slot estimate. For a binary product, the product rule gives at most 2^k terms at order k. Iterating over a product of m factors gives at most m^k terms; in each pose the number m is fixed, so this changes only its finite constant. Factor degrees add under multiplication. Hence a monomial of at least s_j chips has rate degree at least s_j. Finite sums add their constants, and lowering a rate from a larger degree to s_j is valid because $0 < \theta \leq 1$. Applying this to the finite pose expansion of each basis field gives one finite constant $c_j(k, P)$ for the exact localized rate above. Summing basis estimates with coefficient moduli gives the linear price for arbitrary labels. No uniformity in d, k, or P is claimed. □

The last item is the currency furnished by the finite section. In the application, the profile at stage i requests the particular order k_i and power P_i appearing in that profile. For each $d \le i$, choose one available amount $a_{di} \ge 0$ for this request. It remains a valid credit after replacing a negative amount by zero, since all budgets are nonnegative. The section is not rebuilt when these requests change.

156.2 Buying all stage and degree conditions

For a stage i, let c_i be its column count, let q_d be the stencil size of the chosen section together with the fixed local section, and let s_i be its support exponent. The source sphere price is

$$\mathsf{S}_i(q_d, a) = 2 + e^{81 s_i}(a + c_i + 2q_d).$$

For each fixed pair $d \le i$, the chosen credit a_{di}, the sphere price, and the coefficient $2(1 + c_i)$ are finite. The stencil's orbit cost is finite, so its sliver

$$\delta_d = \frac{1}{2\,\mathcal{O}_d^2}$$

is strictly positive. Since the volume purse is $\mathsf{P}(L) = L^{1/8} \to \infty$, choose an integer n_{di} such that

$$2(1 + c_i)\mathsf{S}_i(q_d, a_{di}) \le \mathsf{P}(L)\delta_d \qquad (L \ge n_{di}).$$

The credit clause and nonnegativity hold at every such L. Separately, for each j choose v_j so that

$$(1 + 2c_j)\,\mathsf{D}(L) \le \overline{\mathsf{P}}(L) \qquad (L \ge v_j),$$

where the source door toll and purse are $\mathsf{D}(L) = \mathrm{rate}(L) = L^{-2/5}$ and $\overline{\mathsf{P}}(L) = \mathrm{barunit}(L) = L^{-3/8}$. Their ratio is $L^{-1/40}$, so the inequality holds once $L^{1/40} \ge 1 + 2c_j$. For a fixed stage i, take the maximum of the finitely many n_{di} with $d \le i$ and v_j with $j \le i$. This single threshold n_i proves all clauses required at stage i simultaneously. The resulting nonnegative table a_{di} and thresholds n_i are exactly the same section's all-stage purchase data.

The later delayed budget has a similar finite-threshold argument. Write $\beta_F(x, L) = x + b_L$, where b_L is the fixed logarithmic scale shift, and write w_d for the positive power in the reciprocal tail $\mathsf{f}_d(x) = x^{-w_d}$. For fixed L, d, and all sufficiently large x, one has $0 \le b_L \le x$ and hence $y = x + b_L \le 2x$. The tail comparison needed by the delay is equivalent, after multiplying by the positive denominators, to

$$L^{d+1} y^{w_d} \le y x^{w_d}.$$

The left side is at most $L^{d+1} 2^{w_d} x^{w_d}$, which is at most the right side once $x \ge L^{d+1} 2^{w_d}$. The remaining conditions at a fixed stage-profile pair are just that x exceed the fixed profile cost and the finitely many credits requested at degrees d below that profile. They therefore hold eventually as well. For a fixed stage, only that stage and its adjacent predecessor are tested; the conjunction is finite. Choose a threshold for every stage and replace these thresholds by their prefix maxima, also taking the maximum with the original entrance delay. The new entrance delay is monotone, dominates the old one, and satisfies every adjacent-stage inequality whenever the actual delay exceeds it. This proves the delay calibration rather than assuming a single threshold works for all stages at once.

At the selected history caller, these are the three eventual conditions needed by the marked budget package. For each fixed degree d, the section credit is queried at the profile of the actual stage $J(N)$. The strict reference-depth margin implies $d \le J(N)$ for all sufficiently large N. The

purchased stage table supplies the credit and flight inequality at that stage, evaluated at its preceding block side. The delay purchase supplies the allowance comparison: evaluate its adjacent-stage estimate at the selected entrance value and use the schedule identity

$$\widehat{\beta}_{i+1} = \beta_F(\widehat{\beta}_i, L_i) = \widehat{\beta}_i + G.\mathrm{slopeConstant} \log L_i.$$

The resulting amount is nonnegative and no larger than the new running allowance. Thus, at every sufficiently late stage for this fixed degree, there are simultaneously a valid derivative credit, the allowance bound, and the sphere/flight bound. These are exactly the three fields of the marked budget package used by the selected field-calibration call. It uses the same portfolio section throughout: the delay changes available budget, not the section or its stencil.

156.3 A small coordinate error gives an invertible calibration

The section is now fixed. At a given cutoff, the field calculation produces, for each coefficient basis vector e_j, a proposed correction u_j satisfying

$$\|e_j - u_j\|_1 \le \delta_d,$$

where $\|x\|_1$ is the sum of the absolute values of the finitely many coordinates and $\delta_d = (2\mathcal{O}_d^2)^{-1}$. The required linear calibration is not inferred from columnwise closeness alone; the factor $\mathcal{O}_d$ accounts for the finite symmetry and the change from the coordinate basis to the section's coefficient coordinates.

Lemma 156.2 (Robust finite-dimensional calibration)**.** *Let E be the coefficient space of the chosen frame, with its ℓ^1 norm. Let $\mathcal{O} \ge 1$ bound both the sum of the chip coefficients and the ℓ^1 norm of each symmetry action on the frame. For columns u_j, define the source repainting map by*

$$Dx = \sum_{(g,j)} \chi_{g,j}(x)\, \rho_g u_j, \qquad x = \sum_{(g,j)} \chi_{g,j}(x)\, \rho_g e_j,$$

where g ranges over the finite signed/conjugation symmetry list, ρ_g is its action on the coefficient frame, and the fixed complex- linear chip coefficients obey $\sum_{g,j} |\chi_{g,j}(x)| \le \mathcal{O}\|x\|_1$. If $\|e_j - u_j\|_1 \le (2\mathcal{O}^2)^{-1}$ for every basis vector, then D is invertible and $\|D^{-1}\|_{1\to 1} \le 2$.

Proof. Subtracting the two repainting formulas gives $(I - D)x = \sum_{g,j} \chi_{g,j}(x)\rho_g(e_j - u_j)$. The sum of chip moduli is at most $\mathcal{O}\|x\|_1$, and applying the symmetry to a column error costs at most another factor $\mathcal{O}$. Consequently

$$\|(I - D)x\|_1 \le \mathcal{O}^2\delta_d\|x\|_1 = \tfrac{1}{2}\|x\|_1.$$

The triangle inequality gives $\|Dx\|_1 \ge \frac{1}{2}\|x\|_1$; therefore D is injective. Since E is finite dimensional, it is surjective and invertible. The same lower bound with $x = D^{-1}y$ gives $\|D^{-1}y\|_1 \le 2\|y\|_1$. □

156.4 Application to the selected field calibration

For a cutoff and mark, let V_d be the fixed section above. The selected field package supplies an eventual nonnegative credit b_N for the profile's derivative order and power, with b_N no larger than the available running allowance. It also supplies the flight inequality

$$2(1 + c_N)\mathsf{S}_N(q_d, b_N) \le \mathsf{P}(L_{N-1})\delta_d,$$

and the local spread/insertion data. These exact conditions are the preconditions for the finite field-calibration step. That step first constructs one approximate column u_j for each frame basis vector with the sliver error above, then applies Lemma 156.2 to the whole coefficient map. Its inverse norm is at most 2 and its repainted columns agree exactly with the selected field change. The coefficient error is thus removed by one invertible linear regrading, shared by all labels at this cutoff.

The price attached to each constructed column is the explicit schedule amount

$$E_N = 4(1+c_N)\mathsf{S}_N(q_d,b_N) + e^{81(s_N+2)}\mathsf{Z}_N\,\zeta_{d,N}\,u_N^{-1}, \qquad \mathsf{Z}_N = \big(\mathsf{C}(V_d) + 2\mathsf{C}(V_d^{\mathrm{loc}})\big)\mathsf{N}_{d,N},$$

where C denotes the fixed derivative price of a local section and $\mathsf{N}_{d,N}$ the number of active sites in the selected insertion region. The first summand pays the local stencil and sphere terms; the second pays the activity term after the reserve factor $\zeta_{d,N}/u_N$. The same bound controls the marked matter estimate for all earlier opened scales and all signed motions, because those motions are included in the orbit cost and the local price.

For a fixed degree, the slow-growth assertion uses all the source components, not just the flight inequality. The running credit is bounded eventually by $\widehat{\beta}_N$ by the Braze hymn; the cutoff profile and the schedule give a slow bound for $\widehat{\beta}_N$ and for the column count. The scale estimate for the profile gives a slow bound for $\exp(81(s_N+2))$. The number of insertion sites is bounded by the finite seat count, whose slow bound follows from the tab schedule. Finally the Fridge estimates give the required slow bounds for $\zeta_{d,N}$ and u_N^{-1}. Since the class of slow sequences is closed under finite sums and products, each summand in E_N is slow and hence so is E_N. Omitting either of the Fridge bounds would not justify this conclusion.

The section theorem and stage/delay arguments above prove the fixed-stencil portfolio, all later credit requests, and the schedule's three eventual field-budget inputs. The local invitation and field-spread estimates used after that point remain separate substantive suppliers; they do not follow from existence of the section or from the inverse lemma. Their source declarations, actual selected caller, and remaining estimate boundary are listed in the evidence file.

157 The marked observable at the terminal level

The terminal vacuum construction must also carry each local field and its forced profile. We give this construction before propagating the field down the selected history. All activities, supports and predicates in this section belong to that history.

157.1 Reserved addresses and the missing part of the field

Fix a grade d, a cutoff N and a period p. Write R for the terminal representation, F for its grade-d coefficient frame, S for its sound local section, $o = 0$ for its origin, Y for its footprint and $0 < \theta \le 1$ for its threshold. The forced activities are K_i, with supports X_i and nonempty inventories $I_i \subseteq X_i$. Their eligibility requires both the prescribed flags on X_i and the original additional predicate. A compatible collection has pairwise disjoint supports, and its inventories cover exactly the flagged sites.

The activity dictionary used in the startup and return constructions has, for every pair $I \subseteq X$ with $I \neq \varnothing$, an infinite sequence of reserved addresses. The address indexed by (X, I, j) has support X, inventory I, and zero activity. Its additional predicate is precisely the condition that the flagged sites in X equal I. Here is why these properties survive the construction. The dictionary embeds the old hard labels in one set of coordinates and reserves a disjoint set for every shape (X, I). The

new connected objects encode finite sets of these coordinates. A reserved singleton cannot be an image of an old object: that would put its reserved coordinate in the old coordinate embedding. Its coefficient therefore remains zero. Reading a singleton recovers exactly its support and inventory; its eligibility test is the original test for that one shape. This is the same operation in startup and every return. In particular it supplies the reserved addresses in the actual selected terminal representation.

We use these addresses to encode a finite compatible nonempty family b of old forced labels whose supports all meet Y. Put

$$T_b = Y \cup \bigcup_{i \in b} X_i, \qquad I_b = \bigcup_{i \in b} I_i.$$

Choose the reserved address $s(b)$ for shape (T_b, I_b), with serial number encoding (d, b). This is injective in b. No old activity is overwritten: $K_{s(b)} = 0$ before the marked coefficient is installed.

For $v \in F$ let P_v be the primary, masked part of the section, and let

$$f_v(U) = S(\theta^{-1} \cdot_F v)(o, U) - P_v(U).$$

Here $\cdot_F$ is the graded dilation: on a coordinate of degree r it multiplies by θ^{-r}. The primary part retains the vacuum coordinate everywhere and the scaled nonvacuum part on healthy configurations. Hence $f_v = 0$ when Y is healthy; otherwise it equals the scaled nonvacuum section. The map $v \mapsto f_v$ is complex-linear, measurable, gauge invariant, and depends only on the field and flags in Y. These assertions follow directly from the corresponding properties of the sound section and the mask. If q is the maximum coordinate degree and each axis section has modulus at most $C_d \geq 1$, then

$$|f_v(U)| \leq C_d \theta^{-q} \| \operatorname{core} v \|_1. \tag{T1}$$

Indeed expand the nonvacuum coefficient in the finite axis basis and use $0 < \theta \leq 1$, so $\theta^{-r} \leq \theta^{-q}$ for every axis.

Define the new forced coefficient at $s(b)$ by

$$J_v(s(b), U) = \mathbf{1}_{\{T_b \cap \operatorname{flags}(U) = I_b\}} f_v(U) \prod_{i \in b} K_i(U), \tag{T2}$$

and set it to zero at all other addresses. Its support and inventory are exactly T_b, I_b. When it is nonzero, the indicator ensures the original additional predicate of the reserved address. Old factors K_i retain all their own predicates. The dependence, invariance and measurability requirements thus hold on the original dictionary. The marked coefficient has lead v, zero regular correction, and forced part J_v.

157.2 The exact density and forced profile

Consider an old compatible collection c at a configuration where $f_v(U) \neq 0$. Split it uniquely into the labels meeting Y and the labels disjoint from Y:

$$b = \{i \in c : X_i \cap Y \neq \varnothing\}, \qquad r = c \setminus b.$$

The set b is nonempty. Indeed Y is then not healthy, and a flagged site of Y lies in some inventory of c, hence in its support. Replace b by $s(b)$ and leave r unchanged. The result is compatible: its old remaining supports are disjoint from Y and from all supports in b, hence from T_b. Inventories still cover exactly the same flags. Conversely, from a compatible marked collection containing $s(b)$,

replace that address by b. Its other supports are disjoint from T_b, so this inverse restores precisely the original collection. These two operations are inverse before any summation.

For each paired term, (T2) times its remaining old activity factors is $f_v(U)$ times the original product. The following absolute estimate justifies summing these identities. Suppose $k_i \geq 0$ majorizes $|K_i|$ and has the actual weighted site bound

$$\sum_{i:x\in X_i} M(X_i)k_i \leq w, \qquad M(X) = \exp\{E\ \mathrm{cover}_B(X)\}, \tag{T3}$$

where $E \geq 0$ is the purchased support exponent. Because every X_i in b meets Y, the connected covering inequality gives

$$M(T_b)\prod_{i\in b} k_i \leq M(Y)\prod_{i\in b} M(X_i)k_i.$$

Moreover the sum of $M(X_i)k_i$ over labels meeting Y is at most $|Y|w$: sum (T3) over $x \in Y$. For every finite set of such labels, the sum of products over its nonempty subsets is $\prod_i(1+M(X_i)k_i)-1 \leq e^{|Y|w}-1$. Increasing the finite sets and using nonnegativity proves

$$\sum_b M(T_b)\sup_U |J_v(s(b),U)| \leq C_d\theta^{-q}\|\operatorname{core} v\|_1 M(Y)(e^{|Y|w}-1). \tag{T4}$$

Dropping the compatibility restrictions only enlarges this bound. It supplies a summable majorant for the new coefficients. The lawful old gas bounds supply the remaining dressing majorants, so the regrouping above is absolutely convergent. Its sum is exactly f_v times the old gas. Adding the primary contribution proves the pointwise identity

$$\mathrm{density}(v)(U) = R(U)S(\theta^{-1}\cdot_F v)(0,U) = a_N^d R(U)\,\mathrm{initial}_{N,p}(v)(U). \tag{T5}$$

The last equality is the definition of the physical normalization of initial.

There is also an identity before multiplication by the unmarked gas. Group the forced coefficients according to their support and inventory. The address $s(b)$ contributes exactly the expression in (T2), and all addresses outside its image contribute zero. The bijection between b and the finite compatible families meeting Y identifies this profile with the attachment of the singleton response f_v on Y. This is the forced-profile identity needed by the adjacent transfer construction. It follows from equality of the individual products and the summable majorant (T4), not from the density identity (T5).

The footprint, graded dilation and healthy mask commute with signed lattice motions. The sound section also commutes with the independent complex-conjugation action. Therefore f_v transforms into $f_{g\cdot v}$, with the same scalar conjugation when appropriate. Reindexing the compatible collections preserves their supports and inventories and proves the same covariance for the attached profile. No equivariant choice of the serial numbers is needed: the profile sums over addresses. The density, coefficient index and forced profile thus have the common motion covariance required at terminal birth.

157.3 Paying the birth cost before choosing the history

It remains to pay the factor in (T4), uniformly in the period. The axis bound C_d comes from the fixed finite local section. Its stencil has cardinality at most a fixed R_d, and, once the cover radius contains that stencil, $\mathrm{cover}_B(Y) \leq 1$. Thus the fee is at most

$$C_d\theta^{-q}e^E(e^{R_d w}-1). \tag{T6}$$

On a fixed profile stage, E, C_d, R_d, q are fixed. For the hatted coupling x the chosen threshold satisfies $\theta^{-1} \le x$ once $x \ge 1$, and the forced allowance is

$$w(x) = \exp\{-\exp((\log\log x)^2/5)\}, \qquad 0 < w(x) \le 1.$$

For each fixed integer n, $\sqrt{w(x)}x^n \to 0$. To see this, put $u = \log\log x$. The logarithm of that product is $-\frac{1}{2}e^{u^2/5} + ne^u$, which tends to $-\infty$ because $e^{u^2/5-u} \to \infty$. Also $e^z - 1 \le ze^z$ for $z \ge 0$, by integration of e^t over $[0, z]$. Consequently (T6) is at most

$$K_d x^q w(x), \qquad K_d = C_d e^E R_d e^{R_d}.$$

Choose $x \ge \max(1, K_d)$ and $\sqrt{w(x)}x^{q+1} \le 1$. Then this last expression is at most $\sqrt{w(x)}$. This proves the terminal remainder bound with charge $\|\operatorname{core} v\|_1$, uniformly in N, p at this profile stage.

At stage j there are only finitely many grades and earlier profiles through j. Take a common maximum of their fee thresholds and stencil fit thresholds, then take running maxima in j. Include this delay among the purchases of Section 154 before applying the common scale selection. At level m, both its grade $d \le J(m)$ and its actual profile index $J(m)$ lie in the finite list purchased at $J(m+1)$. The fee therefore fits at every admitted terminal level, on the same selected scale sequence. With (T1)–(T5), this supplies the terminal marked data and all of their bounds for the backward transfer construction.

158 The local reserve and marked one-body invitation

This unit closes two finite estimates used when a marked label is inserted at the terminal scale of a selected history. The first is the smallness of the forced reserve relative to a fixed section and its finite frame. The second converts a paid local section into the exact coefficient and remainder data called a *Cream*. It keeps the actual one-body insertion, its normalization, and its exponential support price visible. The later recentering of this Cream through all earlier scales is a separate step and is not claimed here.

158.1 The forced reserve beats every fixed frame price

Let F_d be a finite frame with grades (r_j, s_j), $j \in J_d$, and set

$$M_d = |J_d|, \qquad k_d = \max_{j \in J_d} s_j.$$

For a budget threshold $\vartheta_N \in (0, 1]$, define the finite seat price

$$\mathrm{Seat}_{d,N} = \sum_{j \in J_d} \vartheta_N^{-s_j}.$$

Assume the scale parameter $\beta_N \to \infty$ and $\vartheta_N^{-1} \le \beta_N$. Define

$$\mathrm{stain}(x) = \exp\left(-\exp\left(\frac{(\log\log x)^2}{5}\right)\right) \quad (x > 1), \qquad \mathfrak{f}_N = \mathrm{stain}(\beta_N),$$

and, for fixed $x, y \ge 0$,

$$\mathrm{Storm}_{d,N}(x, y) = (x + 2y)\,\mathrm{Seat}_{d,N}\,.$$

Lemma 158.1 (Reserve versus a fixed section)**.** *For every fixed d and $x, y \geq 0$,*

$$\sqrt{\mathfrak{f}_N}\ \mathrm{Storm}_{d,N}(x,y) \leq 1$$

for all sufficiently large N. For every fixed finite stencil X on the cover lattice,

$$|\,\mathrm{regional}(X,0)|\,\mathfrak{f}_N \leq 1$$

for all sufficiently large N as well.

Proof. Write $M = M_d$, $k = k_d$, and $A = x + 2y$. Since each $s_j \leq k$ and $\vartheta_N^{-1} \leq \beta_N$,

$$\mathrm{Seat}_{d,N} \leq M\beta_N^k.$$

Choose N large enough that $\beta_N \geq \max\{1, AM\}$. Then $AM\beta_N^k \leq \beta_N^{k+1}$, so it suffices that $\sqrt{\mathrm{stain}(\beta_N)}\,\beta_N^{k+1} \leq 1$. For completeness, put $v = \log\log x$. The logarithm of the left side is

$$(k+1)e^v - \tfrac{1}{2}e^{v^2/5}.$$

As $v \to \infty$, $v^2/5 - v \to \infty$, hence $e^{v^2/5} \geq 2(k+1)e^v$ eventually. This proves the required super-polynomial decay and the first assertion.

Let $m_X = |X|$. Regionalization is an image, so $|\,\mathrm{regional}(X,0)| \leq m_X$. Eventually $\beta_N \geq \max\{1, m_X\}$. The just-proved estimate with exponent one implies $\sqrt{\mathfrak{f}_N}\,\beta_N \leq 1$, while $0 < \mathfrak{f}_N \leq 1$; therefore

$$|\,\mathrm{regional}(X,0)|\mathfrak{f}_N \leq m_X\sqrt{\mathfrak{f}_N} \leq \beta_N\sqrt{\mathfrak{f}_N} \leq 1.$$

This proves the finite local allowance estimate. □

The volume-independent geometric cover estimate is also elementary. Let B_N be the radius in the budget's support norm and assume $B_N \to \infty$. For fixed X, every point of X is eventually in the radius-B_N cube. Consequently $\mathrm{regional}(X,0)$ lies in the halo of the origin. The cube-tiling cover of that halo uses at most 81 support blocks, so

$$\mathrm{coverNumber}(B_N, \mathrm{regional}(X,0)) \leq 81$$

eventually. In the source schedule $B_N \to \infty$ follows from $\beta_N \to \infty$ and the radius estimate in the finite-volume budget. The three local conditions used below are therefore obtained with the same terminal threshold: the regional forced mass is at most one, its support cover number is at most 81, and the square-root reserve times the fixed storms price is at most one.

158.2 From a paid section to a Cream

We record the exact one-body statement. Let C be a selected finite-volume family, N its terminal level, p the period, and d a label degree. Write $F = F_d$, T for the fixed local section, V for the purchased section, $\theta_N \in (0,1]$ for the terminal budget threshold, $w_N = \mathfrak{f}_N$ for the forced allowance, $u_N > 0$ for the startup allowance, and $\zeta_{d,N} \geq 0$ for the mark budget. Let $R_N = C.\mathrm{path.rep}(N,p,N)$ be the terminal representative, α_N its support exponent, $a_N > 0$ the physical lattice spacing, and B_N the support radius in the terminal budget point. Put

$$\begin{aligned} X &= \mathrm{stencil}(V) \cup \mathrm{stencil}(T), \\ D &= \mathrm{regional}(X,0), \qquad c_N = \text{column count}, \\ s_N &= \mathrm{Storm}_{d,N}\big(\mathrm{price}(V), \mathrm{price}(T)\big). \end{aligned}$$

where the two prices are the global value-norm constants of V and T; they are distinct from the derivative credit b_N. Assume $\alpha_N \geq 0$ and the following exact local inputs:

1. C is a selected family which makes the action and $C.\mathrm{ref}(d) \le N$. The section V is sound, and its Credit holds at the terminal order and power with a nonnegative price b_N.

2. Every $z \in X$ satisfies $20(|z_j|+1) \le \mathrm{side}(N,p)$ for each coordinate j. The local regional bounds are $0 < w_N$, $|D|w_N \le 1$, and $\mathrm{coverNumber}(B_N, D) \le 81$.

3. The selected terminal representative is lawful and has a terminal Keeper. Also $u_N > 0$, $\zeta_{d,N} \ge 0$, and $\sqrt{w_N}s_N \le 1$.

4. At the terminal budget point, every placed basis column of T is regular on its footprint and has derivative bound $c_N\theta_N^{s_j}$ for grade s_j; this is the selected-family price clause.

For $v \in F$, define the normalized cylinder of the purchased section by

$$\mathcal{O}_{N,p}(v)(U) = (a_N^d)^{-1} V.\mathrm{places}(F.\mathrm{scaled}(\theta_N^{-1})v, 0, U).$$

For a basis vector e_j of F, define

$$\begin{aligned} Q_N &= 2 + e^{81\alpha_N}\big(b_N + c_N + 2|X|\big), \\ E_N &= 4(1+c_N)Q_N + e^{81(\alpha_N+2)}s_N\frac{\zeta_{d,N}}{u_N}. \end{aligned}$$

Proposition 158.2 (One-body marked invitation)**.** *Under these inputs, for each basis vector e_j there is a coefficient record Z_j and a regular price $r_N = b_N + c_N$ such that, with*

$$\begin{aligned} t_N &= e^{81\alpha_N}r_N + 2|D|\sqrt{w_N}\,e^{81\alpha_N}s_N, \\ \overline{A}_N &= e^{81(\alpha_N+2)}s_N. \end{aligned}$$

The record fields are

$$\begin{aligned} Z_j.x &= e_j, \\ &\quad Z_j.\mathrm{Format}(0, C.\mathrm{params}, R_N), \\ &\quad Z_j.\mathrm{remBound}(0, R_N, \sqrt{w_N}, t_N), \\ F.\mathrm{coeffSize}(Z_j.x) + t_N &\le Q_N. \end{aligned}$$

and $0 < Q_N$. There are an attenuation S_j and an actual reserved mass

$$A_j = e^{(\alpha_N+2)\,\mathrm{coverNumber}(B_N,D)}s_N$$

with $\mathrm{Attainment}(R_N.p, 0, S_j, A_j)$ *and $A_j\zeta_{d,N} \le u_N E_N$, and*

$$\mathrm{profile}(R_N.\mathrm{activities}, Z_j.\mathrm{forced}) = \mathrm{hook}(R_N.\mathrm{activities}, R_N.p.\mathrm{threshold}, S_j).$$

For every configuration U, its density and continuity clauses are

$$\begin{aligned} Z_j.\mathrm{density}(0, T, R_N, U) &= (a_N^d)R_N.\mathrm{eval}(U)b_j(U), \\ b_j(U) &= (a_N^d)^{-1}V.\mathrm{places}\big(F.\mathrm{scaled}(\theta_N^{-1})e_j, 0, U\big), \\ &\quad b_j \text{ is continuous.} \end{aligned}$$

The reserved-mass bound is

$$\overline{A}_N\zeta_{d,N} \le u_N E_N, \qquad 4(1+c_N)Q_N \le E_N.$$

Equivalently, these are the Cream *fields used by the selected field calibration.*

Proof. Fix j. The coordinate vector

$$a = F.\mathrm{scaled}(\theta_N^{-1})e_j, \qquad z = F.\mathrm{coords}^{-1}(a).$$

By the inverse-coordinate identity,

$$F.\mathrm{scaled}(\theta_N)\, F.\mathrm{coords}(z) = e_j.$$

Set $y = V.\mathrm{places}(a, 0, \cdot)$. The pointwise section $V.\mathrm{places}(a, 0, \cdot)$ reads only the translated stencil of V; by the stencil inclusion, this is contained in D. The margin assumption implies that the placement is nonwrapping: each coordinate of every stencil point has absolute value strictly smaller than one twentieth of the torus side, so the translated stencil has a unique representative in the centered box. Thus the source's read relation for z holds on D. The same statements apply to T.

The Credit estimate for V applies to each basis column on its footprint. For the rescaled axis $a = F.\mathrm{scaled}(\theta_N^{-1})e_j$, only column j is nonzero; its coefficient has modulus $\theta_N^{-s_j}$. The terminal Credit gives every ordered derivative test admitted by the terminal budget, on its healthy translated footprint, the bound $b_N\theta_N^{s_j}$ for the unscaled column. Scaling cancels these factors, so $y = V.\mathrm{places}(a, 0, \cdot)$ has the same regularity and derivative bound b_N. The selected-family price clause gives the identical conclusion for $g = T.\mathrm{places}(a, 0, \cdot)$ with bound c_N. Regularity and derivative sup bounds are closed under subtraction, so $r = y - g$ is regular on D and every such derivative is bounded by $r_N = b_N + c_N$. This is the bound used in the regular part of the remainder. The global value-norm constants in s_N are different: the definition of $\mathrm{price}(V)$ and finite-coordinate expansion give, for every coefficient vector v,

$$|V.\mathrm{places}(v, 0, U)| \le \mathrm{price}(V) F.\mathrm{coeffSize}(v),$$

and likewise for T. The scaled-coefficient estimate gives the explicit envelopes

$$\|V.\mathrm{places}(a, 0, U)\| \le \mathrm{price}(V)\, \mathrm{Seat}_{d,N}, \qquad \|T.\mathrm{places}(a, 0, U)\| \le \mathrm{price}(T)\, \mathrm{Seat}_{d,N}\,.$$

These are the uniform pointwise bounds used in the residual estimate. The local insertion construction uses this read relation and the terminal Keeper to split the target into its unit coordinate, the core of the local section, and a residual which vanishes at the identity. With $g = T.\mathrm{places}(a, 0, \cdot)$ and residual $r = y - g$, define

$$\begin{aligned} b_0 &= r - \mathrm{mask}_D(r) + \xi - \mathrm{mask}_A(\xi), \\ \xi &= T.\mathrm{places}\big(F.\mathrm{scaled}(\theta_N^{-1})F.\mathrm{core}(e_j), 0, \cdot\big). \end{aligned}$$

where A is the local-section footprint and each mask is the indicator of the corresponding healthy set. The read identity, unit normalization, and the projection condition in section soundness give the null-value and gauge-vanishing conditions needed for this residual. On the healthy set for D, $\mathrm{mask}_D(r) = r$ and, since $A \subseteq D$, $\mathrm{mask}_A(\xi) = \xi$; hence $b_0 = 0$ there. The read and projection identities give the required null-value and gauge-vanishing conditions for r. The triangle inequality and the two section prices then give the pointwise bound

$$|b_0(U)| \le \mathrm{price}(V)\, \mathrm{Seat}_{d,N} + 2\,\mathrm{price}(T)\, \mathrm{Seat}_{d,N} = s_N.$$

This is the exact "storms" majorant; the two local-section terms are charged separately.

The target entering the coefficient density is

$$b_j(U) = (a_N^d)^{-1} y(U), \qquad y(U) = V.\mathrm{places}(F.\mathrm{scaled}(\theta_N^{-1})e_j, 0, U).$$

Thus $b_j = \mathcal{O}_{N,p}(e_j)$.

For clarity, the record is built by the source's finite forcing construction, not by an unspecified existence argument. Start with the unpriced terminal record and its coefficient vector e_j. Its regular part is the section remainder on D; the forced correction is supported on D. To price the correction, let

$$\mathrm{mergePrice}(D,h) = e^{\alpha_N \,\mathrm{coverNumber}(B_N,D)} h\big(e^{|D|w_N} - 1\big).$$

When $|D|w_N \le 1$, the elementary inequality $e^t - 1 \le 2t$ for $0 \le t \le 1$ yields

$$\mathrm{mergePrice}(D,h) \le \sqrt{w_N}\, 2|D|\sqrt{w_N} e^{\alpha_N \,\mathrm{coverNumber}(B_N,D)} h.$$

Apply activity garnishing to the singleton family consisting of support D and correction b_0, using the canonical support, inventory, and admissibility dictionary supplied by the terminal Keeper constructed in Section 157. In the construction, the temporary enlarged eligibility dictionary is transported back to the original one by the canonical crossing map. The inputs are the origin support, attachment, nonnegative bound s_N, and the merge-price budget above. It returns a forced family whose gas-mark sum is $b_0(U)$ times the activity gas density and whose profile is the prescribed hook. Add this family to the regular coefficient produced by the support split. Pointwise, the exponential entry contributes the unit coordinate plus the masked residual, while the gas-mark sum contributes b_0; their sum is the target. The regular remainder's price is its ordinary mass plus the merge price. This construction also gives an attainment with reserved mass

$$A_j = e^{(\alpha_N+2)\,\mathrm{coverNumber}(B_N,D)} s_N \le e^{81(\alpha_N+2)} s_N.$$

The regular part of the remainder is bounded by $e^{81\alpha_N} r_N$. The merge-price estimate, in the $\sqrt{w_N}$-weighted remainder budget, contributes at most $2|D|\sqrt{w_N} e^{81\alpha_N} s_N$. Their sum is t_N. Since $r_N = b_N + c_N$, $|D| \le |X|$, and $\sqrt{w_N} s_N \le 1$,

$$t_N \le e^{81\alpha_N}(b_N + c_N + 2|X|), \qquad 1 + t_N \le 2 + e^{81\alpha_N}(b_N + c_N + 2|X|) = Q_N.$$

Here the extra 1 is the coefficient size of e_j; this proves the Cream charge clause. The actual reserved mass is

$$A_j = e^{(\alpha_N+2)\,\mathrm{coverNumber}(B_N,D)} s_N,$$

and the cover bound gives $A_j \le \overline{A}_N$. Multiplying the latter envelope by $\zeta_{d,N}/u_N$ gives the reserved-mass clause from the second summand of E_N; the first summand gives $4(1 + c_N)Q_N \le E_N$. This proves all the stated record fields. □

The proposition closes the one-body invitation consumed by `Parterre.inviting` once the exact terminal margins, Keeper, Credit, and local regional bounds are in place. The remaining selected field calibration step is `Figment.figment`: it chooses a single recentered label for all earlier-scale root tests. Its source proof uses the selected Music's ordered adjacent-scale transitions, Antilles' reverse induction, the paid height/Caroling estimates, and the finite cutoff remainder transfer. Those are not consequences of the local reserve and invitation arguments above; their exact inputs and proof status are tracked in the evidence ledger.

159 The common history and its centered service

We now apply the local constructions to a single family. The input is the simultaneous purchase of Sections 153 and 154, including the stationary and free convergence estimates on their common horizon. We retain the prescribed positive blocking penalty throughout.

Choose increasing profile orders and spatial powers, both at least the stage index, and support exponents at least 100. The particular choices $6 + j$, $2 + j$ and $100 + j$ suffice for the elementary construction; further finite-stage requests may enlarge them before the profiles are fixed. Choose the fixed parameter radius and the small response reserve as in Section 154. Make the block-length choices before their numerical costs. Apply Section 201 to all the resulting requests, with a nondecreasing occupancy schedule. This gives one sequence of levels, couplings and periods, with $J(m) \to \infty$.

159.1 Choosing the history before its observable comparisons

For every cutoff N and period p, the terminal family is the exact Wilson family of Section 152. The interval return and scalar reset are applied with the strengthened local conditions of Section 154. Thus every member of the resulting continuous interval family carries those conditions. To prove that its bottom couplings cover the required interval, forget only these additional conditions and apply the barrier and covering argument of Section 150. This leaves the interval, its endpoints and every replayed history unchanged. Choose a parameter in that very interval whose bottom coupling is the prescribed common value, and retain its original strengthened certificate. Retain that member, its actual representations $R_{N,p,m}$ and its adjacent transitions. All of them use the already purchased paths, kernels and clipping data. Choosing the members for the countably many pairs (N, p) introduces no new scale request.

The original certificate at the chosen parameter supplies the local parameter estimates on these histories. Thus the comparisons and fixed-level limits in Sections 146, 169 and 117 apply to this family. The uniform local derivative bank is that of Section 154. In particular, the fixed-level convergence is available for the actual selected activity arrays, not just for their stationary paths.

Choose a reference level t_d for each grade d such that

$$d + 3 \le J(t_d).$$

These levels exist by divergence of J. They may be chosen nondecreasing by taking finite prefix maxima. At each adjacent step choose the wing containing the origin principal modes and the keeper keys for all relevant support/inventory shapes. These keys form a finite set: the torus is finite, so its support powerset and the pairs $I \subseteq X$ with $I \neq \varnothing$ are finite. The wing uses these shapes, not the countably many reserved serial addresses. Hence it covers every forced coefficient while remaining finite. Take the finite union required for the admitted grades. Section 113 supplies one heading and one local transfer map for that entire wing. Consequently the same map works for all those grades. Enlarging the wing preserves its finite interpolation proof and the exact endpoint representation.

Apply the finite backward construction of Section 20, starting at the terminal marked section constructed in Section 157. This gives the finite transfer data at every grade, cutoff and period. The three exact identities in (604) identify the coefficient, exponential entry and forced profile throughout the construction. In particular, its terminal density is $a_N^d R_{N,p,N}(U)\,\mathrm{initial}_{N,p}(v)(U)$; the physical observable $\mathrm{initial}_{N,p}$ already contains the factor a_N^{-d}. The local maps used in this construction are the adjacent maps already chosen above; the downward induction does not select another background history. The unit, gauge invariance and signed-motion identities are preserved by the exact local section and by the equivariant finite assembly. Thus these transfer data form one family on the retained history.

159.2 The service at every reference level

Fix a grade d, a period p, a cutoff N, and $t_d \le k \le N$. We verify the inputs of Theorem 24.1 for this family.

At every step above k, the ordinary and raised menu estimates of Section 17 apply to the selected transition. Its input and output are exactly $R_{N,p,m+1}$ and $R_{N,p,m}$. Its three tariff inequalities give the local duty bound and the same single-marker derivative. The terminal clause is supplied by Section 157: the reserved slots, marked density and forced-profile identity are constructed on the exact startup representation, with their separate entrance fee paid before scale selection. This proves retention all the way to the terminal level.

The room and bank constants are those chosen for the local coefficient maps. The scale inequalities in Section 23 give their common marker prices

$$\zeta_{d,m} = \widehat{\beta}_m^{-(\deg F_d+4)}, \qquad \phi_m = \widehat{\beta}_{m+1}^{-1}, \qquad \delta_m = \widehat{\beta}_{m+1}^{-2}, \qquad \eta_m = \tfrac{1}{4}\widehat{\beta}_m^{-1/2}.$$

Here F_d is the finite coefficient frame. Their strict positivity, local coefficient bounds, adjacent-price growth and support inequalities are proved there. Since $d \le J(k) \le J(m)$, the finite-grade inequalities purchased at stage $J(m)$ apply at every $m \ge k$. The occupancy bound at level m is $N_{J(m+1)}$, which is nondecreasing because both maps are nondecreasing.

The same scale estimates give

$$B_m \ge 8, \qquad \delta_m \phi_m e^{B_m/128} \ge 1, \qquad B_{m+1}^2 \le L_m B_m^2.$$

The stain/contact bound pays the simultaneous forced support requirement. These inequalities continue to hold after restriction to any reference level k; no constant is selected from N. For the local-section norm use its fixed camera constant C_d. The fitted stencil has cover at most one and its values are bounded by C_d, uniformly in the cutoff and period. Its paid coefficient price is precisely the bank price in the scale estimates. The nested-support and footprint bounds are the ones proved in Section 21 on this same level sequence.

The backward transfer data supply the remaining finite source record. Its coefficient and activity bounds are the two separate ledgers in Section 20; its adjacent identities use the selected maps, and its terminal forced-profile identity is the third identity in (604). That identity is essential: the scalar integral identity alone would not give the terminal marked polynomial. All these conclusions hold for every (d, k, N, p) in the stated range, with the already fixed reference levels. They are exactly the service hypotheses, now supplied by the local purchases and their finite backward construction.

We may therefore apply Theorem 24.1 and the terminal birth calculation. At a reference level k, a finite label list with paid weights and separated origins has normalized moment bound

$$\exp\{|\Lambda_k|\mathcal{T}_k + 6|\mathcal{V}|\}.$$

The replica geometry of Section 21 gives the aligned site identity $|\Lambda_k| = M(a_m/a_k)^4$ for M replicated labels. The normalized Mth root is consequently bounded by

$$\exp\{(a_m/a_k)^4\mathcal{T}_k + 6\}/w.$$

This is the centered estimate for the same family, simultaneously at all suitable periods. The fixed-level convergence just retained and the removal of higher marker terms in Section 25 then give convergence along the full mesh sequence at every fixed suitable period. The finite-difference bounds of Section 28 apply to the same transfer data, so its limiting correlations have the smoothness proved in Section 27.

The further calibration of an arbitrary marked section and the rank and ultraviolet comparisons must still be made on this family. They will use the same reference levels and transfers; the centered construction above is already fixed before those comparisons or a volume subsequence are chosen.

160 A fixed-coupling history for the rank comparison

The rank tests compare the interacting fields with a free field over a short interval of scales. They require an auxiliary integration history. The coupling is held fixed on that interval, and the local additive remainder is controlled in value. We give the density construction and its bounds here. The calibrated history constructed earlier supplies the endpoint and the kernels; it is still the history defining the fields.

160.1 The additive remainder and its exact density factor

On a finite torus Λ, write a representation's density as

$$R(U) = \exp\{v|\Lambda| - A(U) - E(U)\}\, G(U), \qquad E(U) = \sum_{\alpha} E_\alpha(U).$$

Here A is its path action, G its forced gas with the prescribed supports, inventories and eligibility tests, and E_α are real, measurable, gauge-invariant local functions: their values agree when the local links and the relevant final flag data agree. Each E_α vanishes unless its support X_α is healthy at the representation's threshold. Give X_α an anchor $a_\alpha \in X_\alpha$. Suppose $b_\alpha \geq 0$ majorizes $|E_\alpha|$ and, for the representation's support weight $M(X) \geq 1$,

$$\sum_{\alpha : a_\alpha = x} M(X_\alpha) b_\alpha \leq q \quad (x \in \Lambda). \tag{754}$$

The sums in this inequality are assumed summable. Since the anchor set is finite, summing it gives

$$\sum_{\alpha} |E_\alpha(U)| \leq |\Lambda| q.$$

Thus the sum defining E converges absolutely, is measurable and gauge invariant, and e^{-E} has the finite-volume bound $e^{|\Lambda|q}$. Let R^0 be obtained by deleting just the additive terms. Its paths, coupling, scalar, forced activities and their tests are unchanged. Then

$$R = e^{-E} R^0. \tag{755}$$

For any finite family of source variables, the same identity holds for its polynomial-valued density. Consequently the integrability of that density follows from the corresponding envelope for R^0. The estimate $e^{|\Lambda|q}$ is used only for finite-volume integrability; it is not used as a uniform bound in the volume limit.

160.2 A local multiplier keeps the expansion on the actual density

Here is the local operation behind the construction. For a masked additive term E_α, insert the letter

$$h_\alpha = e^{-E_\alpha} - 1.$$

The identity $\prod_\alpha (1 + h_\alpha) = e^{-E}$ is exact. Its infinite version follows from absolute summability, since $|e^{-t} - 1| \leq e^{|t|}|t|$ and $\sum_\alpha b_\alpha < \infty$. If the weighted footprint estimate for these letters has allowance ω, attach them to the same component alphabet as the fluctuation integral. The component estimate of Section 63 and the reassembly identity of Section 88 apply with this enlarged alphabet. The old forced labels and their eligibility conditions are retained. In particular the resulting identity integrates $e^{-E} R^0 = R$. This construction cannot be replaced by integrating R^0 alone.

For later use, denote by δ the weighted allowance of the unmarked coarse letters, including the multiplier letters. The regular-gas estimate of Section 89, at the purchased support exponent, supplies a masked logarithmic family ℓ_γ with anchored cost at most 16δ, and

$$G_{\mathrm{reg}}(U) = \exp\{-\sum_\gamma \ell_\gamma(U)\}. \tag{756}$$

The elementary numerical conditions used at this step include

$$16\delta \le 1, \qquad e^{2\delta} - 1 \le 1 - e^{-16\delta}, \qquad 16\delta\,\mathcal{R}(B) \le 1.$$

Here $\mathcal{R}(B)$ is the connected-support counting constant in the regular logarithm bound, and B is the coarse support radius. They are imposed on the same letter prices, not on a different auxiliary normalization. If the integrand has holes around marked sites, the forced-exclusion identity gives the corresponding localized logarithms with those holes. The scalar identity without holes and these identities with holes therefore have the same normalization.

Choose the new coarse scalar from the positive fluctuation prefactor. If the prefactor is $C > 0$, its contribution to the scalar per coarse site is $|\Lambda_c|^{-1} \log C$. Positivity follows from the positive coupling, the positive chart ratio, and the positive determinant of the flat precision. In particular this logarithm does not choose a branch. Let A_c be the retained coarse path action and βB_c the reference action used in the fluctuation formula. Its local correction is $\beta B_c - A_c$. Add its masked local pieces to ℓ_γ. If the former have anchored cost ϵ, the combined family has cost

$$\epsilon + 16\delta. \tag{757}$$

Indeed the two nonnegative majorant sums add at every anchor. The identity for its exponential is exactly

$$e^{v_c|\Lambda_c| - \beta B_c}\, G_{\mathrm{reg}} = e^{v_c|\Lambda_c| - A_c - E_c}.$$

The regular/forced decomposition of the same expansion then gives

$$R_c = \mathcal{K}_\beta R_f \tag{758}$$

on every coarse configuration, including configurations with flags. To check this last assertion, first keep the source polynomial in the finite expansion. Its coefficient of the empty inventory equals the unmarked scalar density. The reassembly and exclusion identities hold coefficientwise and preserve exact flag coverage. Setting the source family empty therefore identifies their scalar coefficients on every configuration. There is no division by a density or by a forced activity in this argument.

160.3 Symmetrizing the additive family

The newly obtained sum E_c is invariant under signed lattice motions, but an individual choice of anchored pieces need not have a symmetric indexing. Let H be the finite group of these motions on the torus. Replace a piece f_α by the family

$$|H|^{-1} f_\alpha(gU), \qquad (g, \alpha) \in H \times \mathcal{I},$$

with the correspondingly transported support and anchor. For fixed g, the anchored bound is the old bound at the inverse image of the anchor. Multiplication by $|H|^{-1}$ and summation over g preserve that bound. Absolute summability permits interchange of the finite group sum and the piece sum. Since the total was invariant, the new total is $|H|^{-1} \sum_g E_c(gU) = E_c(U)$. Left multiplication in H permutes the new indices, giving the required covariance of individual addresses. Reindex this countable family and, if necessary, add zero terms. The gas and all forced addresses are unchanged. This produces the locality, measurability and motion properties together with (758).

160.4 The cost over a short interval

Fix a block ratio $L > 1$ and a width parameter $b \geq 0$. For large s put

$$y = \log\log s, \quad A_s = e^{y^3}, \quad q_n = s^{-1/200} A_s^{4n}, \quad \omega_n = q_n A_s, \quad \delta_n = q_n A_s^3.$$

The two parts of the comparison interval have lengths h_1, h_2. Their definitions are the ceilings

$$h_1 = \left\lceil \frac{(b+2)y^2}{\log L} \right\rceil, \qquad h_2 = \left\lceil \frac{\log(M_s^{10})}{\log L} \right\rceil, \qquad h = h_1 + h_2,$$

where $M_s = \lceil (\log s)^2 \rceil$; the support wall is M_s^2. The support-radius estimates give $\log(M_s^{10}) \leq y^2$ eventually. Thus $h \leq C_{L,b} y^2 + 2 \leq y^4$ eventually. For every fixed integer $r \geq 0$, uniformly for $0 \leq n \leq h$,

$$q_n A_s^r \leq s^{-1/400} \tag{759}$$

for all sufficiently large s. In fact its logarithm is at most $-(\log s)/200 + (4y^4 + r)y^3$. For $y \geq 1$ the positive term is at most $(4+r)y^7$. Since $y^7 e^{-y} \to 0$ and $\log s = e^y$, it is at most $(\log s)/400$ eventually, proving the claim.

The footprint price of an additive input with cost q_n is at most

$$C_{L,a,t}(42B+1)^4 q_n,$$

where the support reserve t is strictly positive, a is the fixed support exponent, and $B \leq M_s^2$. All factors except the displayed radius and q_n are fixed before s is increased. The wall estimates imply $C_{L,a,t}(42M_s^2+1)^4 \leq A_s$ eventually. Hence the multiplier letters cost at most ω_n. The remaining unmarked letters have their purchased local bounds; their combined logarithm is allowed cost $16\delta_n$. The path correction costs at most $s^{-1/200}$. These bounds close the recurrence because, as soon as $A_s \geq 17$,

$$s^{-1/200} + 16\delta_n \leq q_{n+1}. \tag{760}$$

Indeed $s^{-1/200} \leq q_n \leq \delta_n$ and $q_{n+1} = A_s \delta_n$.

For clarity, the small exponential conditions also hold uniformly through this interval. If $0 \leq t \leq 1$, then $e^t - 1 \leq 2t$. Consequently $e^{2\delta_n} - 1 \leq 4\delta_n$. If $0 \leq \delta_n \leq 1/16$, the inequalities $e^{16\delta_n} \geq 1 + 16\delta_n$ and $(1+16\delta_n)^{-1} \leq 1 - 4\delta_n$ give

$$e^{2\delta_n} - 1 \leq 1 - e^{-16\delta_n}.$$

The second rational inequality follows after multiplying by its positive denominator: $12\delta_n - 64\delta_n^2 \geq 0$. The factors $\mathcal{R}(B)$, the insertion inflation and any fixed comparison polynomial have at most the purchased powers of e^{y^2}; multiplying (759) by such factors still gives zero in the limit. The forced-gas allowance is smaller than every negative power of s, uniformly for the local scale parameter in $[s/2, s]$. It therefore pays the support-counting factors in the normalization-with-holes estimates. All these are eventual conditions after L, b and the finite menu have been fixed.

160.5 Joining to the endpoint

Choose a comparison depth k and cutoff $N \geq k$. Above k, retain the original history. At k, use its actual representation in the chosen local presentation. Below k, set the coupling identically $\beta_{N,p,k}$ and apply the preceding construction down to the lower edge of the comparison interval. The local scale parameter lies in $[s/2, s]$, with s the clock coupling at k, and the endpoint bound $|\beta_{N,p,k} - s| \leq s/8$ places this constant coupling in the purchased local range throughout the interval.

Inductively, the input at a step has cost q_n, the output has cost at most q_{n+1} by (760), and their unnormalized densities satisfy (758) at that very same coupling. Only the new entry is installed at each backward step. Thus all entries already constructed, including the endpoint and its marked labels, are retained. Composition proves the density identities for the entire auxiliary history. The comparison with one or two marked sites uses the same component alphabet and the same hole normalization. Its leading and remainder estimates are separate from this unmarked construction; they must be verified on these actual densities before the rank tests can be applied.

161 Several marked insertions on the restarted history

We keep the actual restarted density from Section 160. Its additive remainder is removed only in the auxiliary unmarked normalization. The marked integral itself must retain that remainder. We first prove the finite algebra that identifies all the selected insertion coefficients, then explain this mixed normalization and the extra local price it requires.

161.1 Selecting a cell in the marker algebra

For a finite set V, let

$$\mathscr{A}_V = \mathbb{C}[z_v : v \in V]/(z_v^2 : v \in V).$$

Its basis consists of $z_A = \prod_{v\in A} z_v$, $A \subseteq V$. Products satisfy $z_A z_B = z_{A\cup B}$ for disjoint sets and are zero otherwise. If F has zero constant coefficient, define $\exp_* F = \sum_{j=0}^{|V|} F^j/j!$. Every product of more than $|V|$ nonconstant monomials has a repeated variable, so it vanishes. Hence this finite exponential satisfies the usual addition formula for commuting elements of zero constant coefficient, by the binomial theorem. In particular

$$\prod_\alpha (1 + y_\alpha z_{A_\alpha}) = \exp_*\left(\sum_\alpha y_\alpha z_{A_\alpha}\right) \quad (A_\alpha \neq \varnothing). \tag{761}$$

There is no disjointness assumption on the A_α: their overlapping products vanish on both sides.

For $T \subseteq V$, define π_T by setting $z_v = 0$ for $v \notin T$. Thus $\pi_T(\sum_A c_A z_A) = \sum_{A\subseteq T} c_A z_A$. It is an algebra homomorphism, and therefore commutes with $\exp_*$. For a finite cell set $\mathcal{C}$ and mark set J, put $V = \mathcal{C} \times J$. The cell embedding $\iota_c : \mathscr{A}_J \to \mathscr{A}_V$ sends z_j to $z_{(c,j)}$; it too is an algebra homomorphism.

Fix one coarse cell c and a selected set $T \subseteq \{c\} \times J$. The incoming insertion pieces are indexed by pairs (c', α). Each piece has a nonempty marker support $A_{c',\alpha}$ lying in $\{c'\} \times J$. Its value in a particular sector, forest, tag and Gaussian coordinate is $y_{c',\alpha}$. The spatial geometry selects exactly those pieces for which $A_{c,\alpha} \subseteq T$; write $\mathcal{I}_T$ for this selected index set and write $B_\alpha \subseteq J$ for their marker supports with the cell coordinate removed. Define the planted polynomial

$$P_T = \sum_{\alpha\in\mathcal{I}_T} y_{c,\alpha} z_{B_\alpha}.$$

Applying π_T to (761) proves

$$\pi_T \prod_{c',\alpha} (1 + y_{c',\alpha} z_{A_{c',\alpha}}) = \pi_T \iota_c(\exp_* P_T). \tag{762}$$

To check the selection explicitly, a piece from $c' \neq c$ has at least one marker outside T and is killed. A piece from c is retained exactly when its entire marker support is contained in T. Because π_T is

multiplicative, a product containing a killed factor cannot contribute through a different coefficient. This proves the identity at all orders simultaneously.

The spatially planted value is the same masked field value as in the left side. Write it as $\Gamma_\alpha(\xi)h_\alpha(U(\xi))$, where $0 \le \Gamma_\alpha \le 1$ is its local cutoff. On the region where $\Gamma_\alpha > 0$, the chart's healthy-field condition makes the original mask equal to one on the full inspected support. Where $\Gamma_\alpha = 0$, both products are zero. Consequently this formula agrees everywhere with $\Gamma_\alpha \,\mathrm{masked}(h_\alpha)(U(\xi))$. The sector and tag are fixed during this verification. They are not chosen separately for the different readers.

All linear maps on $\mathscr{A}_V$ are continuous because this space is finite dimensional. Hence an integrable polynomial-valued function can be integrated coefficientwise, and its integral commutes with π_T and ι_c. If ν_t are the finitely many sector/tag measures and λ_t their scalar weights, (762) gives

$$\sum_t \lambda_t \int \pi_T \prod_{c',\alpha} (1 + y_{c',\alpha,t} z_{A_{c',\alpha}})\, d\nu_t = \pi_T \iota_c \left(\sum_t \lambda_t \int \exp_*\, P_{T,t}\, d\nu_t \right). \tag{763}$$

Here integrability is provided by the common positive marker weights and the paid product envelopes: the absolute value of each coefficient is bounded by its weighted polynomial norm divided by its positive monomial weight. The Gaussian/Haar envelope bounds that norm by an integrable function. Thus no interchange of a merely formal infinite series is used in (763).

161.2 The marked density and its three kinds of factors

Let P denote the fine representation and Q the coarse one. Write $\mathcal{K}$ for a finite family of lifted background packets, with full spatial supports X_b and fine flag inventories F_b. Its unmarked coefficients reproduce the activities of P at each activity address. This addresswise identity is stronger than equality of their summed gases. For regular insertion pieces h_α with nonempty marker supports, the exact polynomial density is

$$D_P(U) = e^{\mathrm{exponent}_P(U)} \exp_* \left(\sum_\alpha z_{A_\alpha} \,\mathrm{masked}(h_\alpha)(U) \right) \sum_{B\ madmissible} \prod_{b \in B} K_b(U). \tag{764}$$

Admissibility requires pairwise disjoint full supports and the exact identity $\bigcup_{b\in B} F_b = \mathrm{flags}_P(U)$. It is not enough to require just disjoint inventories.

Fix the actual coarse configuration, positioned chart, forest, tag and Gaussian coordinate. The unmarked chart expansion, including its Haar Jacobian and Gaussian normalization, has the following sliced form. For each fine flag set F,

$$\mathbf{1}_{\{\mathrm{flags}_P(U)=F\}} J(U) e^{\mathrm{exponent}_P(U)} = \Lambda \sum_{A\ mcompatible(F)} \prod_{a\in A} p_a. \tag{765}$$

Here J denotes the full scalar chart factor, and Λ is the same positive loading used by the unmarked construction. Each p_a has a fixed coarse support, a fine inventory and a coarse inventory. Compatibility means disjoint coarse supports, fine inventory union F, coarse inventory union equal to the coarse flags, and the sector key specified by those same packets. These are precisely the data of the unmarked sector expansion of Section 88.

To obtain the marked identity, first fix a background packet family B. If its full supports overlap its contribution is zero. Otherwise multiply (765), with $F = \bigcup_{b\in B} F_b$, by $\prod_{b\in B} K_b$ and by the insertion product (761). The indicator on the left imposes exactly the background flag test in (764). Expand the insertion product as a sum over its subsets. On the right we now have

three independent kinds of factors: scalar body packets p_a, background packets K_b, and marked insertion pieces. Index their union by the disjoint union of these three alphabets. This is a bijection between the three subset choices and the subsets of the union alphabet. The flag and sector tests are unchanged. If a marker is used twice, the corresponding product is zero in $\mathscr{A}_V$, exactly as on the expanded left side. Summing over B proves the full marked chart identity at every coefficient.

Integrate with the common Gaussian/Haar product envelopes, sum the sector partition and average the tags. The coefficientwise argument of Section 14 then applies to this same three-alphabet expansion. It gives the pushed density $\mathcal{T}D_P$ represented by the resulting coarse polynomial activities and their regular exponential. In particular this step does not follow by repeating the one-address affine argument of Section 92; that argument alone has no mixed higher coefficients.

161.3 Restoring the additive remainder before normalization

Put $P^0 = \mathrm{white}(P)$, so that its additive remainder has been deleted while its coupling, path, masks and forced activities are unchanged. In the notation of Section 160, $P = e^{-E}P^0$. On the actual chart the additional exponent is a sum $\sum_i f_i = -E(U)$ over local coarse supports. Suppose its support prices $b_i \geq 0$ satisfy $|f_i| \leq b_i$ and have anchored weighted sum at most $d \leq 1$. The functions f_i are real-valued. Let M_i be the real-valued unperturbed local exponents, with

$$|e^{M_i} - 1| \leq w_i, \qquad w_i \geq 0, \qquad \text{anchored weighted sum of } w_i \leq e \leq 1.$$

The weights are at least one, so individually $b_i \leq d$ and $w_i \leq e$. Hence $e^{M_i} \leq 2$ and $|e^{f_i} - 1| \leq 2|f_i| \leq 2b_i$. The latter elementary inequality holds for real $|f_i| \leq 1$, by convexity of the exponential on $[0, 1]$ and $e - 1 < 2$, and by $1 - e^{-t} \leq t$ for $t \geq 0$. It follows that

$$|e^{M_i+f_i} - 1| \leq |e^{M_i} - 1| + e^{M_i}|e^{f_i} - 1| \leq w_i + 4b_i.$$

Thus the perturbed auxiliary price is at most $e + 4d$. The base Gaussian quadratic form and the scalar loading have not changed. This proves (765) for P itself from the neutral chart expansion: add f_i to the same local exponents and use their exact sum $-E(U)$. The full-support locality of these terms was established in the restarted-history construction, including the final flag tests.

For the actual scale purchases, write ω for the permitted perturbation price. The neutral auxiliary price is at most ω and $d \leq \omega$, so the preceding bound is at most 5ω and in particular at most the reserved 6ω. The two other elementary letter families have anchored prices at most $u^{-1/2}$ each, where u is the retimed coupling scale. Since $u^{-1/2} \leq \omega$, their combined price is at most 8ω. The background large-field letters retain the decay $\exp(-\alpha \,\mathrm{hot}^2 \,\mathrm{shares})$; the separate purchase $2a_{\mathrm{coast}} \leq \alpha/4$ pays their forced inflation. With the reserved line multiplier L_0, the integrated cluster bound is therefore purchased through

$$L_0(e^{8\omega} - 1) \leq \delta_{\mathrm{step}}.$$

This is the local unmarked price used by the restarted recurrence. The fixed stage constants and Gaussian envelopes in this calculation are those of the neutral compiler at the same physical parameters. The perturbation estimate does not assert derivative bounds for E.

There are now two distinct, compatible uses of the representations. The body identity (765) is for P, including E. The unmarked auxiliary alphabet consists of these *same corrected* body packets p_a, together with the archived background packets of P^0. It does not use the former neutral body packets. Its loading is the same Λ as in (765).

The normalization needed here has two precise parts. Write Z_j for its zero-marker regular components. For every deleted spatial support D and every coarse configuration V, require the identity

$$\Lambda(V)\,\mathcal{G}_{Z;D}(V) = e^{\mathrm{exponent}_Q(V)} \prod_a (1 + p_a^{\mathrm{pay}}(V)) \prod_{b:\, X_b^{\mathrm{hole}} \cap D \neq \varnothing} (1 + h_b(V)). \tag{766}$$

Here $\mathcal{G}_{Z;D}$ is the compatible gas of the Z_j whose full supports are disjoint from D. The factors p_a^{pay} and h_b are the fixed boundary and hole corrections of the unmarked regrouping, with their original masks. Secondly, the unmarked coarse forced family must embed into the forced components of this alphabet, with identical values, supports and inventories, and must exhaust its zero-marker forced components. The forced family, together with these same hole and boundary correction factors, must reproduce the archived activity of Q at each activity address, with its full support and flag inventory. This addresswise identity is required before applying the exclusion regrouping. These are the all-deleted-support identity and the forced-family embedding supplied by the unmarked forest and exclusion construction. They must be established for the corrected p_a and this actual coarse Q; equality of the two archived fine backgrounds alone does not establish them.

We verify that these two statements, once available, pass to the full marked alphabet. First coalesce background packets with the same address and marker mode. Their zero-marker values are exactly the archived activities at that address, by the addresswise background identity. This embeds the entire unmarked alphabet into the full alphabet, preserving values, full supports, inventories and sector keys. Every remaining elementary insertion has a nonempty marker support. If a full connected component has empty marker support, every elementary factor in it must therefore have empty marker support: its marker support is their union. Hence it comes from the unmarked alphabet. Conversely every unmarked component has the same connectedness and eligibility tests after the embedding. This is a bijection of zero-marker components preserving their full supports. It restricts to the components disjoint from any given D. Thus $\mathcal{G}_{Z;D}$ is unchanged, and (766) holds for the full alphabet at every deleted support. The same component bijection transports the forced-family embedding and its exhaustion property.

Finally fix a compatible marked forced family and let D be the union of its full supports. Its residual zero-marker gas is precisely $\mathcal{G}_{Z;D}$. Substitute (766). A hole factor occurs exactly when its support meets at least one member of that forced family, since D is their union. Expanding these factors and regrouping by their incident forced components is the weight-preserving exclusion bijection of Section 90. The resulting polynomial has exponent $\mathrm{exponent}_Q$ and the actual coarse archived background, by the transported forced-family embedding. Summing over the marked forced families therefore represents $\mathcal{T}D_P$ with this coarse vacuum normalization. This uses (765) for P and (766) for the corrected auxiliary alphabet; it never substitutes $\mathcal{T}D_{P^0}$ for the marked integral.

This proves the finite algebra and the additive-remainder correction needed by the all-order compiler. Its quantitative marked use still requires the common Gaussian reader estimates and their cumulant approximation, the day and night marker prices, and the selected spatial support bounds. In particular the cell selection (763) must be applied with the same readers and same sector measures as the body expansion, at the purchased slice radius. Those analytic and geometric estimates are separate from the polynomial identities proved here.

162 The Fourier window of the actual curvature covariance

The Gaussian comparison requires estimates for the covariance selected by the construction, uniformly in its ancestry and physical period. We derive these estimates before considering the cumulative transport. Fix the positive kernel penalty and a smoothing order $J \geq 4$. All constants in

this section may depend on these two choices and on the finite derivative order, but not on block lengths or ancestry.

162.1 A uniformly elliptic symbol

Let $H(p)$ be the two-form pulse symbol of the chosen ancestry, in radian frequency $p \in \mathbb{R}^4$. The component and positive-row constructions give constants $\alpha, \eta > 0$ and $\Omega < \infty$ such that its Laurent coefficients have strip cost Ω at width α, its action on every finite period has quadratic lower bound η, and $H(0) = I$. These are properties of the original pulse, before any replacement of the curvature completion. Exponential summability implies, for every integer $m \geq 0$,

$$\sup_p \|D^m H(p)\| \leq C_m.$$

Indeed each derivative multiplies the coefficient at z by at most $|z|^m$, and $|z|^m e^{-\alpha|z|_1}$ is bounded. The finite-period lower bound also holds at every real frequency. First extend the quadratic bound of the real finite-period operator to complex vectors by splitting them into real and imaginary parts. Then test the period-n matrix on the vector $e^{ip\cdot x}v$ for a grid frequency p. Its quadratic form and squared norm both acquire the same factor n^4, which cancels. Approximate an arbitrary real frequency by grid frequencies and use continuity. Thus

$$\mathrm{Re}\langle H(p)v, v\rangle \geq \eta|v|^2.$$

Put $q_a(p) = e^{ip_a} - 1$, and let $X(p)$ and $U(p)$ be exterior multiplication by $q(p)$ from one-forms to two-forms and from scalars to one-forms, respectively. Define

$$F(p) = X(p)^* H(p) X(p) + U(p)U(p)^*, \qquad P(p) = X(p)F(p)^{-1}X(p)^* \quad (p \neq 0),$$

and set $P(0) = 0$. On the cube $|p_a| < 3\pi/2$, the elementary identity for exterior multiplication and contraction gives

$$\|Xv\|^2 + \|U^*v\|^2 = |q|^2\|v\|^2, \qquad \mathrm{Re}\langle Fv, v\rangle \geq \min(\eta, 1)|q|^2\|v\|^2.$$

Here $|q| \geq c|p|$: the continuous function $|e^{it} - 1|/|t|$ has a positive lower bound on $[-3\pi/2, 3\pi/2]$, with its value at zero defined to be one. Consequently $F(p)$ is invertible for $p \neq 0$ in this cube.

We record the derivative estimates with their exact gain. A function has order s if, through the finite number of derivatives under consideration, its mth derivative is bounded by $C_m|p|^{s-m}$. Products add orders by Leibniz' rule. Taylor's formula gives $q - ip$ order two and $H - I$ order one; q itself has order one. The exterior identity at $q = ip$ gives

$$F(p) = |p|^2 I + E(p), \qquad E \text{ has order 3.}$$

Thus $T = |p|^{-2}F$ has order zero, $T - I$ order one, and $\|T^{-1}\|$ is uniformly bounded by coercivity. Differentiating $TT^{-1} = I$ proves inductively that $D^m T^{-1} = O(|p|^{-m})$: the term containing $D^m T^{-1}$ is isolated and every remaining term is a product of derivatives whose orders total m. Furthermore $T^{-1} - I = T^{-1}(I - T)$ has order one. It follows that

$$P \text{ has order 0}, \qquad P - P_0 \text{ has order 1}, \quad P_0(p) = X_0(p)|p|^{-2}X_0(p)^*, \tag{767}$$

where X_0 is exterior multiplication by ip. These estimates are uniform over all the selected ancestries.

We verify that P is the actual curvature multiplier. Let D be the original one-form covariance and $R = X^*HX$. The transport identities already proved give $RDX^* = X^*$ and $XU = 0$. Write $B = UU^*$. Then $XB = 0$, $BX^* = 0$, and $RB = BR = 0$. Hence $F = R + B$ commutes with B, as does its inverse, and $XF^{-1}B = 0$. Multiplying $RDX^* = X^*$ on the left by XF^{-1} yields

$$XF^{-1}X^* = XF^{-1}RDX^* = XDX^*.$$

At zero both curvature multipliers vanish. Thus no change of the finite-period covariance has been made.

162.2 A smooth chart without changing the finite Fourier sum

Choose a smooth nondecreasing function g that is zero on $(-\infty, -5\pi/4]$ and one on $[-3\pi/4, \infty)$, and put

$$h(p) = \prod_{a=1}^{4} \big(g(p_a) - g(p_a - 2\pi)\big).$$

It is supported in $|p_a| \le 5\pi/4$, equals one near zero, and satisfies

$$\sum_{z \in \mathbb{Z}^4} h(p + 2\pi z) = 1.$$

The last identity is the product of four telescoping sums; only finitely many terms are nonzero. Define the unit-frequency chart $K(\xi) = h(2\pi\xi)P(2\pi\xi)$, entry by entry and with the usual Kronecker delta on colour indices. Its support lies in $|\xi| \le 2$. For an integer displacement x, periodicity and the partition identity show that the exact period-n covariance is

$$\kappa_n(x) = n^{-4} \sum_{z \in \mathbb{Z}^4} e^{2\pi i x \cdot z/n} K(z/n). \tag{768}$$

The sum is finite. Grouping z by its residue modulo n proves the formula, including even n and the zero frequency. At zero we retain the actual value $P(0) = 0$, rather than assigning a continuous value to a direction-dependent limit.

For two lists of coordinate directions, of lengths a, b, let $u = a + b$. Differencing the first endpoint multiplies $K(\xi)$ by $e^{2\pi i \xi_c} - 1$; differencing the second endpoint uses $e^{-2\pi i \xi_c} - 1$. Denote their product by $m(\xi)$, and the corresponding product of $2\pi i \xi_c$ and $-2\pi i \xi_c$ by $m_0(\xi)$. Repeatedly using $e^{it} - 1 - it = O(t^2)$, with its differentiated Taylor bounds, shows that $m - m_0$ has order $u + 1$ and m, m_0 have order u. By (767), the remainder

$$G(\xi) = m(\xi)K(\xi) - m_0(\xi)e^{-(2\pi)^2|\xi|^2} P_0(2\pi\xi)$$

satisfies, for $0 \le v \le u + 6$,

$$\begin{aligned} &\|D^v G(\xi)\| \le C|\xi|^{u+1-v} && (0 < |\xi| \le 1), \\ &\|D^v G(\xi)\| \le C|\xi|^{-5} && (|\xi| \ge 1). \end{aligned} \tag{769}$$

For the first estimate, subtract and add $m_0 K$ and use the product rule. The extra Gaussian differs from one to order two. Outside the unit ball the compactly supported term has bounded derivatives, and every derivative of the Gaussian term decays faster than any power. Finite maxima give the same C for all face pairs and direction lists of bounded length.

162.3 Fourier decay and quadrature at the pole

Write $\mathcal{F}_+ f(x) = \int e^{2\pi i x\cdot\xi} f(\xi)\,d\xi$. We first show directly that (769) implies

$$|\mathcal{F}_+ G(x)| \le C'|x|^{-u-5} \quad (|x| \ge 1). \tag{770}$$

Set $r = |x|^{-1}$ and choose a smooth radial cutoff equal to zero on the ball of radius $r/2$ and one outside the ball of radius r. The omitted inner term has integral at most $C\int_0^r t^{u+1}t^3\,dt \le C'r^{u+5}$. For the outer term integrate by parts $u+6$ times in the unit direction $x/|x|$. Each term where a derivative hits the cutoff is supported on $r/2 \le |\xi| \le r$ and has integral $O(Cr^{-1})$. For the remaining terms, derivatives of order $u+6$ have integral bounded by

$$C'\left(\int_r^1 t^{-5}t^3\,dt + \int_1^\infty t^{-5}t^3\,dt\right) \le C''/r.$$

Boundary terms vanish by first inserting a distant smooth cutoff and then sending its radius to infinity; the displayed bounds control its derivatives as well. Multiplication by $|x|^{-u-6}$ proves (770).

Next let $f = mK$. It is compactly supported, bounded by C, and satisfies $\|Df(\xi)\| \le C/|\xi|$ off zero. Partition frequency space into cubes of side $1/n$, and use their grid corner as the sample point. Its distance from any point of that cube is at most $2/n$. If this distance is less than $|\xi|/2$, the mean value bound gives

$$|f(\xi) - f(z/n)| \le C'/(n|\xi|).$$

Otherwise the same estimate follows from the bound $2C$ on the difference, after increasing C'. This also covers a sampled corner at zero. The phase difference is at most $2\pi|x|\,|\xi - z/n|$. Integration over a fixed enlarged support ball is legitimate because $|\xi|^{-1}$ is locally integrable in four dimensions. We obtain

$$\left|\mathcal{F}_+ f(x) - n^{-4}\sum_z e^{2\pi i x\cdot z/n} f(z/n)\right| \le C'\frac{1+|x|}{n}. \tag{771}$$

This estimate treats the singular grid point without removing it.

Finally the Gaussian-regularized continuum term has the required Maxwell limit. In the present Fourier convention its scalar potential multiplier, for $t > 0$ (here $t = (2\pi)^2$), is $(2\pi)^{-2}e^{-t|\xi|^2}/|\xi|^2$. The identity

$$\frac{e^{-t|\xi|^2}}{|\xi|^2} = \int_t^\infty e^{-s|\xi|^2}\,ds$$

and the product Gaussian integral give, at $x \ne 0$,

$$\mathcal{F}_+\left((2\pi)^{-2}\frac{e^{-t|\xi|^2}}{|\xi|^2}\right)(x) = \frac{1}{4\pi^2|x|^2}\left(1 - e^{-\pi^2|x|^2/t}\right).$$

Fubini is justified by $\int_t^\infty (\pi/s)^2\,ds < \infty$. All polynomial frequency moments are integrable, so differentiation under the integral supplies both curls and both endpoint derivative lists, with the opposite sign at the second endpoint. The difference from the derivatives of $1/(4\pi^2|x|^2)$ is a derivative of a Gaussian times that potential. It is bounded by any prescribed inverse power for $|x| \ge 1$.

Combining this fact with (768), (770) and (771) proves

$$\left|\Delta_x^a\Delta_y^b\kappa_n(x,y) - \mathcal{K}_{a,b}(x-y)\right| \le C_{a,b}\left(|x-y|^{-a-b-5} + \frac{1+|x-y|}{n}\right), \qquad |x-y| \ge 1. \tag{772}$$

Here each superscript denotes its specified direction list, and $\mathcal{K}_{a,b}$ is the corresponding continuum Maxwell curvature covariance. The constants are uniform in ancestry and period. The covariance

has closed rows and columns because its construction is dDd^* and $d^2 = 0$. Its entries are real by conjugating the finite Fourier sum and replacing the frequency by its negative. No positivity of a later normal-ordered covariance is inferred.

For a fixed comparison order $d \geq 1$, only $a, b \leq d-1$ are needed. The zero-order comparison can use $d = 1$ and discard the extra data. The construction reserves their finitely many chart constants at smoothing order J before choosing the block length. At the chosen comparison level the reserved chart cost is paid by $H^{1/4}$, where H is its monitor. The finite constants from the multiplier and Fourier estimates are then absorbed successively into $H^{1/2}$ and H. This yields (772) with $C_{a,b}$ replaced by H, on that same chosen history. The quantitative clearance at the chosen level remains part of the common scale selection; merely taking an unrelated level large would not establish it.

163 Cumulative transport and the actual Gaussian comparison

We prove the low-frequency transport bounds on the same atlas as Section 162. Write κ_i for its original real curvature covariance and Ψ_i for the installed map from level i to level $i+1$, with the identity on colour coordinates. Installation sums all periodic images of the cover row. Fix a comparison order $d \geq 1$ and a plateau of smoothing order J with $d - 1 \leq J - 2$.

163.1 Moving differences through an actual step

For a word w of $j \leq d-1$ coordinate directions, the selected transport has a real quotient row Q_w satisfying

$$\Delta_w \Psi_i = Q_w \Delta_w, \qquad \sup_x \sum_y |Q_w(x,y)| \leq R L_i^{-j-2}, \tag{773}$$

where $L_i = a_i/a_{i+1}$ and $R \geq 1$ depends only on the kernel penalty and adjacent smoothing orders. The quotient needs proof; a bound on the left side alone would not suffice.

Theorem 6.3, applied to the actual pulse-multiplied curvature column, gives its differentiated row strip cost CL_i^{-j-2}. The original pulse's strip cost simply multiplies the established column cost. To move coarse differences to the right, divide the symbol successively by $e^{ip_a} - 1$. The division is regular on the real principal cube before summing aliases. On the principal alias use

$$e^{ip_a/L_i} - 1 = \frac{e^{ip_a} - 1}{L_i \operatorname{rail}_{L_i}(p_a/L_i)}.$$

The denominator has a positive lower bound by the principal rail estimate. On a nonprincipal alias $q = (p + 2\pi l)/L_i$, the column contains the box filter $A(q)^J$. The geometric-series identity is

$$(e^{iq_a} - 1)A(q) = (e^{ip_a} - 1)s_a(q),$$

where s_a is a product of the other coordinate box filters and a phase, divided by L_i. For j directions use j box factors. The remaining power A^{J-j} has nonnegative degree. The remaining multiplier is continuous by the nonprincipal denominator bounds already proved. Thus the full differentiated symbol has a continuous quotient by the product of all coarse differences.

A coefficientwise division also preserves a narrower strip bound. In one coordinate use the exact finite identity

$$\frac{e^{inp_a} - 1}{e^{ip_a} - 1} = \begin{cases} \sum_{h=0}^{n-1} e^{ihp_a}, & n > 0, \\ -\sum_{h=n}^{-1} e^{ihp_a}, & n < 0, \\ 0, & n = 0. \end{cases}$$

There are $|n|$ terms with no greater displacement. Reducing width σ to $\sigma/2$ costs at most $2/\sigma$, since $|n|e^{\sigma|n|/2} \le (2/\sigma)e^{\sigma|n|}$. This divides the symbol minus its value on $p_a = 0$; that value vanishes by the preceding divisibility. Equality off the coordinate hyperplane extends to it by continuity. Repeat along w; divisibility by the remaining factors survives at each step. There are finitely many words of length at most $d-1$, so one narrower strip and one cost work for all. The scalar factor L_i^{-j-2} is unchanged. Periodization and the triangle inequality turn this cover bound into an absolute row bound; the sum over the six two-form types contributes their cardinality. This proves (773) with a cost chosen before L_i.

Closedness is preserved on all cochains, including harmonic ones. Write the actual step as $B_f T H_c$, using reciprocal pulse B and pulse H. Its exterior transport C satisfies

$$C d_c B_c = d_f B_f T, \qquad B_c H_c = I.$$

Multiplying by H_c gives $d_f \Psi_i = C d_c$. No global exactness of the input cochain is required. Reality follows from the real cover coefficients.

163.2 Absolute rows of a product

For $\ell \le k$ put $m = k - \ell$, $\mathcal{L} = \Psi_{k-1} \cdots \Psi_\ell$ and $S = a_\ell / a_k = \prod_{i=\ell}^{k-1} L_i$. If every jth difference of f is bounded by v, iteration gives

$$\|\Delta_w \mathcal{L} f\|_\infty \le R^m S^{-j-2} v.$$

This holds for arbitrary real f. For a specified output row, take $f(y)$ to be the sign of $\Delta_w \mathcal{L}(x, y)$, using zero when that entry vanishes. Then $|f| \le 1$ and every jth difference has absolute value at most 2^j. Consequently

$$S^{j+2} \sum_y |\Delta_w \mathcal{L}(x, y)| \le R^m 2^d. \tag{774}$$

The sign test need not be closed. Reality and preservation of closedness pass separately to the product.

Fix $b \ge 6$ and $a > 0$. After R, choose a plateau block length $L > 1$ so that $4(b+2)\log R \le a \log L$. With $m = \lceil (b+2)\log H / \log L \rceil$ we have

$$R^m \le R H^{a/4} \le H^{a/2}, \qquad R^m 2^d \le H^a$$

eventually. The second inequality absorbs fixed 2^d into $H^{a/2}$. Also $S = L^m \ge H^{b+2}$. The original covariance entries have a fixed bound C_J uniform in period, by summing the bounded curvature multiplier over its normalized finite Fourier grid. Paying $C_J \le H^a$, and writing $P = H^a$, gives

$$S^{j+2} \sum_y |\Delta_w \mathcal{L}(x, y)| \le P, \qquad |\kappa_\ell(y, z)| \le P. \tag{775}$$

One finite table of costs is selected before the block length, then its clearance threshold. On the allowed plateau the prescribed number of steps is exactly m. All estimates therefore concern the same chosen level and history, uniformly in period.

163.3 Mean and covariance after normal ordering

For a real closed coarse cochain u with $\|u\|_\infty \le P$, set $\mu = \mathcal{L} u$ and $B = \mathcal{L} \kappa_\ell \mathcal{L}^{\mathsf{T}}$. Finite summation gives

$$\begin{aligned} S^{j+2} |\Delta_w \mu(x)| &\le P^2, \\ S^{j+2} S^{h+2} |\Delta_w^x \Delta_v^y B(x, y)| &\le P^3. \end{aligned} \tag{776}$$

For the second inequality bound each covariance entry by P; the remaining double sum factors into the absolute row sums of (775). The mean and both covariance arguments are closed, by the original closedness and its preservation.

Use $r = H^b$, $a = \lambda/4$, $0 < \lambda \le 1/4$, and the purchased condition $S \ge rH^2$. For $j \ge 0$, $r^{j+2}H^4 \le S^{j+2}$. Hence the weighted means are at most $H^{-4}P^2 \le H^{-3}$. Retaining only one factor H^{-4} for the covariance gives the convenient bound

$$r^{j+h+4}|\Delta_w^x \Delta_v^y B(x,y)| \le H^{-4+3\lambda/4}. \tag{777}$$

Fix distinct continuum positions x_i and their selected rounded lattice centres z_i at magnification r. Required offsets have subpower size in H, so normalized pair displacements stay in compact sets away from zero. The free covariance and the finitely many derivatives used here are bounded on these sets. For an off-diagonal pair, multiply (772) by r^{j+h+4}. The singular error is at most $CH/r = CH^{1-b} \le H^{-4}/2$ eventually, since $b \ge 6$. The period error is bounded by a fixed power of H divided by the lattice side. The selected mesh estimates make it at most $H^{-4}/2$ eventually, uniformly in physical period. Thus the scaled error is at most H^{-4}, and the scaled covariance is bounded by $H^{\lambda/8}$ eventually.

Let K_{ij}^0 be the polynomial test-cochain covariance built from the free jet covariance at z_i/r. For both derivative orders $j, h \le d-2$, at the origins its weighted differences differ from the corresponding free curvature derivatives by at most H^{-4}. Indeed a difference of order j kills test cochains of degree less than j, and the degree-j term recovers its homogeneous curvature tensor exactly. Terms of degree $q > j$ carry $r^{j-q} \le r^{-1}$. Expanding the finite covariance matrix in the jet basis bounds the total error by $C/r \le H^{-4}$ eventually. The coefficient constants are fixed and the free entries are bounded on the compact configuration set. If one endpoint has order $d-1$, its derivative of K_{ij}^0 is zero, since all test cochains have degree at most $d-2$. Set $K_{ii}^0 = 0$, in accordance with free normal ordering.

The actual normal-ordered covariance is

$$\Gamma_{ij}(s,t) = \mathbf{1}_{i \ne j}\kappa_k(z_i + s, z_j + t) - B(z_i + s, z_j + t). \tag{778}$$

Its diagonal is $-B$, generally nonzero. At origins with both $j, h \le d-2$, the two comparison errors and (777) give

$$r^{j+h+4}|\Delta_w^s \Delta_v^t(\Gamma_{ij} - K_{ij}^0)(0,0)| \le 2H^{-4} + H^{-4+3\lambda/4} \le H^{-3}$$

eventually; only the last term is needed on the diagonal. For $d = 1$ this range of origin derivatives is empty. At a top-order endpoint the polynomial comparison vanishes and

$$H^{\lambda/8} + H^{-4+3\lambda/4} \le H^{\lambda/4} = P.$$

These, with the mean and closedness bounds, are precisely the finite-reading estimates with $P = H^{\lambda/4}$ and $E = H^{-3}$ used in Section 164. They hold uniformly over the admitted closed real coarse backgrounds. Positive semidefiniteness of Γ is not assumed.

Application must retain the purchased plateau and clearance conditions and the identification of this atlas with the actual reader law. These belong to the common history and reader construction; an arbitrary symmetric model cannot replace that identification.

164 One coordinate system for the finite Gaussian comparison

The moment estimate in Section 165 requires a common space containing both the physical jet coordinates and the discarded higher-order information. We construct that space from finite lattice readings. The covariance may be indefinite after normal ordering; only its symmetry is used in this construction.

164.1 Recovering polynomial curvature coordinates from differences

Write $\Delta_a f(x) = f(x+e_a) - f(x)$ on $\mathbb{Z}^4$. A real curvature cochain is specified on positively oriented plaquettes and colour coordinates and extended antisymmetrically to all ordered pairs of axes. We call it closed when

$$\Delta_a f_{bc} + \Delta_b f_{ca} + \Delta_c f_{ab} = 0.$$

For an integer $d \geq 0$, let K_d index a basis of the radial polynomial connection jets whose curvature degree j satisfies $j + 2 \leq d$. Its basis element has grade $g_\alpha = j + 2$. Let t_α be its lattice plaquette cochain, obtained by integrating the polynomial connection along each edge and taking its discrete curl. These are the polynomial test cochains of the local section construction. They are closed. Differences of order greater than j vanish, while order-j differences at zero are the components of $j!$ times the homogeneous curvature tensor. Edge integration does not change this highest homogeneous term.

For clarity, the finite sampling argument needs only closedness, not the divergence equation. Given a closed cochain, its order-j differences at zero form a symmetric derivative tensor. Applying the closedness equation after any $j-1$ differences gives the cyclic identity for this tensor. It therefore defines a homogeneous closed polynomial two-form. For $j = 0$ this assertion just says that the tensor is antisymmetric. The radial primitive is obtained by contracting this two-form with the position vector and dividing by $j + 2$, as in the local jet construction. Its curl is the given two-form. Thus these highest-order readings are those of a radial polynomial connection jet.

The map from the span of the t_α of curvature degree at least q to all origin differences of orders $q, \ldots, d-2$ is injective. Indeed, take the largest degree with a nonzero coefficient. The differences of that order kill all smaller degrees and recover its full curvature tensor. Vanishing of those readings forces that radial connection jet, hence its basis coefficients, to be zero. Descend through the finite list of degrees. The same descent, now subtracting the recovered top-degree test cochain, shows that the origin readings through order $d-2$ of every closed cochain are matched by a unique linear combination of the t_α.

Choose a linear left inverse of each of these finite injective sampling maps. For a key α of curvature degree j, use the left inverse involving only differences of orders at least j. This gives a real functional ℓ_α which is a finite linear combination of origin differences, with

$$\ell_\alpha(t_\beta) = \delta_{\alpha\beta}, \qquad \Pi_d f = \sum_{\alpha \in K_d} \ell_\alpha(f) t_\alpha. \tag{779}$$

For closed f, $\Pi_d f$ has exactly its origin differences through order $d-2$. Compatibility of the different left inverses follows from uniqueness of the full expansion just proved: restriction to orders at least j deletes all lower-degree test cochains, and the left inverse therefore recovers the same remaining coefficients. The functionals are defined on arbitrary cochains; the matching assertion is only used for closed ones.

Let c_α be the sum of the absolute values of the coefficients of ℓ_α in its difference representation, and choose $C_d \geq 1$ with $c_\alpha \leq C_d$ for every key. This is a finite constant, selected before any comparison scale or stencil radius.

164.2 Residual coordinates and their bounds

Fix a finite set S of addresses, all with lattice distance $|x|_1 \leq M$, where M is a nonnegative integer, and a radius $r \geq 1$. For a cochain f define

$$L_\alpha f = r^{g_\alpha} \ell_\alpha(f), \qquad \alpha \in K_d,$$
$$L_z f = r^{d+1} (f - \Pi_d f)(z), \qquad z \in S.$$

The coordinate set is the disjoint union $K_d \sqcup S$. Every functional L uses only finitely many values, since a finite difference does so. For every $z \in S$ there is the exact identity

$$f(z) = \sum_{\alpha \in K_d} r^{-g_\alpha} t_\alpha(z) L_\alpha f + r^{-(d+1)} L_z f. \tag{780}$$

It is simply the definition of $f - \Pi_d f$ and holds even when f is not closed.

Put $q = \max(d-1, 0)$. Suppose a closed f has the weighted origin bounds

$$r^{j+2} |\Delta_w f(0)| \le P \quad (j = |w| \le d-2)$$

and the weighted top-difference bounds

$$r^{q+2} |\Delta_w f(x)| \le P \quad (|w| = q,\ |x|_1 \le M).$$

All plaquette and colour components are included. Assume $P \ge 0$. For physical coordinates, $g_\alpha \le j+2$ for every difference entering ℓ_α, so $|L_\alpha f| \le P c_\alpha$.

For the residual bound set $h = f - \Pi_d f$. Its origin differences below order q vanish, and its order-q differences equal those of f, since $\Pi_d f$ has degree at most $d-2$. If $d = 0$ or 1, the key set is empty and the last assertion just means $h = f$. An elementary discrete remainder estimate gives

$$|h(z)| \le M^q \sup_{|w|=q,\ |x|_1 \le M} |\Delta_w h(x)|.$$

To prove it, descend from order q to zero. At each lower order the relevant difference is zero at the origin. Sum its first differences along a shortest signed coordinate path to z. The path has at most M edges and stays inside the ball. A negative step uses the forward difference at the new endpoint. Each descent therefore multiplies the bound by at most M. At $q = 0$ no descent is required, with $M^0 = 1$. Since $d + 1 \le q + 2$ and $r \ge 1$, we obtain

$$|L_\alpha f| \le P C_d, \qquad |L_z f| \le P M^q. \tag{781}$$

If the weighted origin differences instead have bound $E \ge 0$, the physical coordinates satisfy the sharper bound $E C_d$.

Apply the same argument at both ends of a kernel $K(z, w)$ whose rows and columns are closed. If its mixed weighted readings of the preceding types are bounded by P, and the origin–origin ones by E, then, for $T \ge \max(C_d, M^q)$,

$$|(L_a \otimes L_b) K| \le P T^2, \qquad |(L_\alpha \otimes L_\beta) K| \le E T^2. \tag{782}$$

Indeed fix one difference at the first endpoint, divide its bound by its positive weight, and apply (781) at the second endpoint. Apply it again at the first. The endpoint operations commute because both are finite linear combinations of evaluations. Closedness is preserved under these finite linear combinations. This justifies both applications, including residual coordinates.

164.3 The exact free block and the common covariance

Fix a separated configuration of n insertion points. Let $F_{i\alpha, j\beta}$ be its free covariance in the radial jet coordinates. This is a real symmetric matrix: the underlying free kernel and the change to these coordinates are real. Normal ordering sets its same-insertion block to zero. Define the reference cochain kernel

$$K^0_{ij}(z, w) = \sum_{\alpha, \beta \in K_d} F_{i\alpha, j\beta}\, r^{-g_\alpha} t_\alpha(z)\, r^{-g_\beta} t_\beta(w).$$

Its rows and columns are closed. Equation (779) shows that its matrix under the L coordinates is exactly F on the physical block and zero whenever either coordinate is residual. This is an exact identity, not a small-error statement.

Let Γ_{ij} be the actual symmetric cochain kernel and μ_i its actual mean. Suppose the means obey the two bounds of the previous subsection with constants P, E, and $\Gamma_{ij} - K^0_{ij}$ obeys their two-endpoint versions. Define one enlarged mean and covariance by

$$m_{ia} = L_a \mu_i, \qquad Q_{ia,jb} = (L_a \otimes L_b)\Gamma_{ij}.$$

If $T \geq \max(C_d, M^q)$, their entries satisfy

$$|m_{ia}| \leq PT, \qquad |m_{i\alpha}| \leq ET,$$
$$|Q_{ia,jb} - F^{\mathrm{ext}}_{ia,jb}| \leq PT^2, \qquad |Q_{i\alpha,j\beta} - F_{i\alpha,j\beta}| \leq ET^2,$$

where F^{ext} has zero residual rows and columns. Consequently, if

$$B \geq 1, \quad PT \leq B, \quad PT^2 \leq B, \quad |F_{i\alpha,j\beta}| + PT^2 \leq B, \tag{783}$$
$$ET \leq \varepsilon, \qquad ET^2 \leq \varepsilon, \tag{784}$$

all enlarged entries are bounded by B, while the physical block and mean are within ε of $(F, 0)$. Applying (780) at one and at both endpoints proves that the actual stencil mean and covariance are the linear and bilinear pullbacks of (m, Q). This gives the common pullback required for every slot and every subset in the moment comparison.

Here are the purchases used there. Let $H \to \infty$ be its monitor, $0 < \lambda \leq 1/4$, and set

$$T = P = H^{\lambda/4}, \quad E = H^{-3}, \quad B = H^{\lambda}, \quad \varepsilon = H^{-2}.$$

The stencil radius has subpower growth, so $M^q \leq T$ eventually; also $C_d \leq T$. Then $PT^2 = H^{3\lambda/4} \leq B/2$ eventually, while $PT \leq PT^2$ since $T \geq 1$. The exponents $-3 + \lambda/4$ and $-3 + \lambda/2$ are at most -2, which proves the two error inequalities. On a compact separated configuration set the finitely many free entries are bounded by continuity of the differentiated kernel; the same bound holds for the rounded configurations eventually. Thus $|F| \leq B/2$ eventually and all of (783) holds.

164.4 The calibrated polynomial price

The actual polynomial map on a fixed grade space has a coefficient bound of the form

$$|\mathcal{H}u|_1 \leq H \,|\,\mathrm{awake}(u)|_1, \qquad |\,\mathrm{awake}(u)|_1 \leq 2|Tu|_1.$$

Here T is its tested coefficient map, and awake is the coefficient vector in the retained common normalization. For a target label v of grade at most d, the calibration equation is

$$D_a T u = \mathrm{weight}_r\, v, \qquad a > 0, \quad a^{-1} \leq 2H, \quad r = H^b \geq 1.$$

D_a multiplies a basis coordinate by a to its connection degree; weight_r multiplies it by r to its grade. Let q_* be the maximum connection degree in the fixed frame. Coordinatewise inversion gives

$$|Tu|_1 \leq (2H)^{q_*} r^d |v|_1, \qquad |\mathcal{H}u|_1 \leq 2^{q_*+1}|v|_1 H^{bd+q_*+1} \leq H^{bd+q_*+2}$$

eventually. The last threshold depends on the fixed target label, and can be taken uniformly on a fixed bounded set of labels. It is uniform in the cutoff and period when the displayed input bounds are uniform. The finite loss $q_* + 2$ is chosen before the amplification exponent b.

Finally the weighted norm of the reconstruction coordinate at $z \in S$ is exactly

$$\sum_{\alpha \in K_d} |t_\alpha(z)| + 1.$$

The powers r^{-g_α} and $r^{-(d+1)}$ cancel their variable weights. Each t_α is a polynomial of fixed degree at most $d-2$, so this expression is at most a fixed multiple of $(1+M)^d+1$. It has subpower growth in H. Hence it is eventually at most H^λ. If the exact calibrated low-grade identity and the fixed polynomial degree bound k hold, the weighted tail estimate of Section 165 applies with these very coordinates and this coefficient price. Choosing $b \geq (q_* + 2) + 3$ leaves its required factor H^{-2}.

The construction above proves the coordinate and price conversion. Its application still requires the actual closed-kernel and mean difference estimates on the selected atlas, and the exact low-grade identity for its polynomial map. Those data must refer to the same readers as the joint Gaussian model. The coordinate conversion does not obtain them from covariance symmetry alone.

165 From calibrated Gaussian moments to cumulants

The marked comparison uses all subsets of a finite insertion word. Its Gaussian calculation must therefore control products before taking the finite logarithm. We give the degree-dependent estimates with the powers used in the scale purchases. The covariance and calibration hypotheses are stated explicitly; their common-atlas construction is needed when this calculation is applied to the selected history.

165.1 A dimension-independent polynomial estimate

For a polynomial p in finitely many commuting variables, write $|p|_1$ for the sum of the absolute values of its coefficients. It is submultiplicative: expand a product, apply the triangle inequality to each coefficient, and then sum over the two original monomial indices. Let $\mathcal{W}_{Q,\mu}$ be the Wick evaluation with symmetric covariance Q and mean μ. No positivity is needed in this algebraic calculation. The defining recurrence, for a product of j coordinate variables, is

$$\mathcal{W}_{Q,\mu}(X_{i_1} \cdots X_{i_j}) = \mu_{i_1} \mathcal{W}_{Q,\mu}(X_{i_2} \cdots X_{i_j}) + \sum_{\ell=2}^{j} Q_{i_1 i_\ell} \mathcal{W}_{Q,\mu}\left(\prod_{a \neq 1,\ell} X_{i_a} \right), \tag{785}$$

with value one on the empty product. Repeated coordinate indices are allowed: their positions in this recurrence remain distinct.

Assume $B \geq 1$, $|\mu_i| \leq B$ and $|Q_{ij}| \leq B$ for every coordinate used. Define $C_d(B) = [2B(d+1)]^d$, including $C_0(B) = 1$. Then every polynomial of degree at most d satisfies

$$|\mathcal{W}_{Q,\mu}(p)| \leq |p|_1 C_d(B). \tag{786}$$

If a second pair (Q', μ') obeys the same bounds and $|Q_{ij} - Q'_{ij}|, |\mu_i - \mu'_i| \leq \varepsilon$, then

$$|\mathcal{W}_{Q,\mu}(p) - \mathcal{W}_{Q',\mu'}(p)| \leq |p|_1 \varepsilon C_d(B). \tag{787}$$

Here $\varepsilon \geq 0$ can be arbitrary.

We prove both estimates on monomials by simultaneous induction on the degree ceiling d. At degree zero their assertions are respectively $1 \leq 1$ and $0 \leq \varepsilon$. For a nonempty monomial of degree at most $d+1$, the recurrence has one mean term and at most d covariance terms. Every remaining

product has degree at most d. Thus the first bound is at most $(d+1)BC_d(B) \le C_{d+1}(B)$. For a difference of terms use $ab - a'b' = a(b-b') + (a-a')b'$. The induction bounds its absolute value by $B\varepsilon C_d(B) + \varepsilon C_d(B)$, at most $2B\varepsilon C_d(B)$. Summing at most $d+1$ such terms gives $2B(d+1)\varepsilon C_d(B) \le \varepsilon C_{d+1}(B)$. The inequalities follow from $C_d(B) \le [2B(d+2)]^d$. The empty monomial satisfies both bounds at every degree ceiling. Finally expand p in monomials and sum their absolute coefficients. This proves (786) and (787) without a factor depending on the number of coordinate variables.

165.2 Calibrated vertices and their products

Fix n insertion slots, a common polynomial degree bound k, and possibly different grades g_i for the slots. Here n, k, g_i are nonnegative integers, and all error parameters ξ, ε are nonnegative. After substituting the actual linear reader map, write H_i for the polynomial of slot i. Let F_i be its intended free polynomial, embedded in the physical coordinates of the same enlarged variable space. Suppose the grade truncation of H_i through grade g_i is exactly F_i. Every variable has a nonnegative integer grade. For $r \ge 1$, write

$$|p|_{1,r} = \sum_{\alpha} |p_\alpha| r^{\mathrm{grade}(\alpha)}.$$

Every monomial of $H_i - F_i$ has grade at least $g_i + 1$, whence

$$r^{g_i+1} |H_i - F_i|_1 \le |H_i|_{1,r}. \tag{788}$$

This uses the exact low-grade identity, so no difference of two large low-grade coefficients is being estimated.

Suppose the incoming polynomial has coefficient norm at most a_i and degree at most k, and every substituted linear coordinate has weighted norm at most $b \ge 1$. Expanding each monomial and using submultiplicativity gives $|H_i|_{1,r} \le a_i b^k$. Thus the purchase

$$a_i b^k \le \xi r^{g_i+1}$$

implies $|H_i - F_i|_1 \le \xi$. Also assume $|F_i|_1 \le A$, where $A \ge 1$. Then $|H_i|_1 \le A + \xi$. For a subset I of slots, telescope the product one factor at a time. Every summand contains one difference $H_i - F_i$ and $|I| - 1$ factors whose norm is at most $A + \xi$. Therefore

$$\left| \prod_{i \in I} H_i - \prod_{i \in I} F_i \right|_1 \le |I| \xi (A+\xi)^{|I|-1} \le n \xi (A+\xi)^n. \tag{789}$$

For $I = \varnothing$ the difference is zero. Polynomial degrees are at most nk, both before and after substitution and truncation.

The actual reader covariance and mean must be the pullbacks of a common pair (Q, μ) on the enlarged variable space. Substitution then commutes with Wick evaluation. To verify this, apply (785) to products of substituted linear forms. Their means are the linear pullbacks of μ and their pairings are the bilinear pullbacks of Q. Induction on the number of factors gives the identity on monomials; linearity gives it on all polynomials. This establishes the needed identity without changing Gaussian variables separately for each slot.

Assume all entries of (Q, μ) are bounded by $B \ge 1$, and its restriction to the physical coordinates is within ε of the free covariance with zero mean, whose entries are also bounded by B. The F_i

involve only these physical coordinates. Insert and subtract the Wick evaluation of $\prod_{i\in I} F_i$ at (Q,μ). Equations (786)–(789) give, for every subset I,

$$\left|\mathcal{W}_{Q,\mu}\left(\prod_{i\in I} H_i\right) - \mathcal{W}_{\mathrm{free}}\left(\prod_{i\in I} F_i\right)\right| \le \{n\xi(A+\xi)^n + A^n\varepsilon\}\, C_{nk}(B). \tag{790}$$

The unused residual coordinates do not enter the second term. They do enter the first term, and their bounds in B cannot be omitted.

165.3 The scale powers that pay for the estimate

Put $H = \exp((\log\log s)^2)$ for the diverging monitor along the comparison scales, and choose

$$\lambda = \frac{1}{4(nk+k+1)}, \qquad B = H^{\lambda}, \qquad \xi = \varepsilon = H^{-2}.$$

The parameter s is taken sufficiently large that the logarithms are defined and $H \ge 1$. Since $\lambda nk \le 1/4$,

$$C_{nk}(B) \le [2(nk+1)]^{nk} H^{1/4}.$$

The right side of (790) is therefore at most

$$[2(nk+1)]^{nk}\{n(A+1)^n + A^n\}H^{-7/4} \le H^{-3/2}$$

for all sufficiently large s. The last threshold depends on the fixed word and degree bound. It is uniform over the finite volume and replicas once the preceding covariance and vertex hypotheses hold uniformly.

For completeness, the grade-tail purchase follows from the same powers. If $r = H^b$, $a_i \le H^{bg_i+E}$, the linear-coordinate substitution cost is at most H^{λ}, and $b \ge 0$, $b \ge E+3$, then

$$a_i(H^{\lambda})^k \le H^{bg_i+E+1/4} \le H^{-2}H^{b(g_i+1)}.$$

The first inequality uses $\lambda k \le 1/4$; the second follows from $E + 1/4 \le b - 2$. Thus every grade is treated at its own truncation level while the arity and polynomial degree determine the common Gaussian growth allowance.

165.4 Taking the finite logarithm

Use the squarefree algebra $\mathscr{A}_{\{1,\dots,n\}}$ from Section 161. For a polynomial q with constant coefficient one, define

$$\log_* q = \sum_{j=1}^{n} \frac{(-1)^{j+1}}{j}(q-1)^j.$$

Formal power-series identities truncated after degree n give $\exp_*(\log_* q) = q$ and $\log_*(\exp_* P) = P$ for P with zero constant coefficient. These identities can also be checked coefficientwise: all powers above n vanish, so the usual one-variable formal identities in the nilpotent variable $q-1$ are finite identities here. In particular the cumulant polynomial is uniquely determined by the moment polynomial.

We give an explicit local stability bound. Use the coefficient ℓ^1 norm on this algebra. If $|q-1|_1, |q'-1|_1 \le M$, the identity $u^j - v^j = \sum_{a=0}^{j-1} u^a(u-v)v^{j-1-a}$ yields

$$|\log_* q - \log_* q'|_1 \le \left(\sum_{j=1}^{n} M^{j-1}\right)|q-q'|_1.$$

If the coefficientwise moment error is at most $H^{-3/2}$, then $|q - q'|_1 \le 2^n H^{-3/2}$. The free moment polynomial q' varies continuously on the compact comparison window, which is contained in the distinct-point configuration domain, so its coefficient norm is bounded. The moment error then bounds $|q - 1|_1$ as well. Consequently there is a constant C for this fixed word and window such that

$$|\log_* q - \log_* q'|_1 + H^{-3/2} \le (C+1)H^{-3/2} \le H^{-31/32} \tag{791}$$

for all sufficiently large s. The source uses the coefficient maximum norm; the same conclusion follows from its being bounded by the ℓ^1 norm. The finite factor 2^n is absorbed only after n is fixed.

On the actual intermediate Gaussian history, let $\mathcal{S}_{l,k}$ be the successive linear substitution and Gaussian integration operator from level k to level l. Define the moment polynomial by

$$q(z) = \sum_{I \subseteq \{1,\dots,n\}} z_I \, \mathcal{S}_{l,k} \left(\prod_{i \in I} P_i \right) (y).$$

Its constant coefficient is one, since every substitution and normalized Gaussian integration preserves one. The connected polynomial obtained by the finite partition formula is $\log_* q$: expanding the displayed finite logarithm gives that partition formula directly, with coefficient $(-1)^{j-1}(j-1)!$ for a partition into j nonempty blocks. Evaluation at the fixed intermediate field y commutes with this finite polynomial formula. This is the exact cumulant used in the planted comparison. Therefore the moment estimate for the same reader covariance and mean implies (791) for it.

If the planted local coefficient differs from this cumulant by at most $H^{-3/2}$, its sum with the compensating negative free cumulant is bounded by $H^{-31/32}$, by (791) and the triangle inequality. Signed copies and complex conjugation preserve these absolute coefficient bounds. This is the small initial remainder needed for the subsequent marked propagation. Its use still requires the local planted-coefficient error, the exact low-grade calibration, and the common reader covariance estimates at the displayed powers; they are not consequences of finite-dimensional logarithm continuity.

166 Realigning and propagating the marked comparison

The finite polynomial comparison must survive each block integration. At an intermediate level the marked expansion contains the desired principal insertion, a small regular remainder, and a forced remainder. We explain how its leading pieces are reassigned to the prescribed supports and how the resulting errors accumulate. The identities use the actual representation at that level.

166.1 Why a connected principal component stays in one cell

Let $\rho \ge 0$ be an elementary slice radius. Each elementary principal piece has a nonempty set of marked reader sites and its full support is within distance ρ of every one of those readers. Assume distinct replica cells have reader separation strictly greater than dB, and

$$(3 + 2n)\rho \le dB \le bB.$$

Consider a connected component in the graph joining elementary pieces whose full supports intersect. Two marked readers on the same piece are at distance at most ρ, since a reader lies in that piece's support. Readers on two adjacent pieces have distance at most 2ρ, by using a point in the intersection. Both bounds are at most dB, so the readers belong to the same replica cell. Following a path of intersecting pieces shows that all marked readers of the component belong to one cell.

There are at most n distinct marker labels in that cell. Make a graph on the labels used by the component, joining labels appearing on the same or on adjacent pieces. The preceding support path gives a walk between any two of these labels. Erase cycles from the walk. The remaining simple path uses at most $n-1$ edges, each of geometric length at most 2ρ. Thus the distance between any two marked readers is at most $2n\rho$. Every point in the component's full support is at distance at most ρ from one of its readers, so it is at distance at most $(2n+1)\rho \le dB$ from every marked reader of the component. Its marker diameter is also at most dB. This proves the required enlarged-support bounds independently of how many elementary pieces the component contains.

For the selected construction take the integer slice radius supplied by the local expansion, bounded by $aB/(3+2n)$, where a is the chosen horizon. The purchase $6n+3 \le a/(3+2n)$ is the separate local room condition needed to produce these elementary slices. Once they are available, the argument above uses $(3+2n)\rho \le aB$ and the strict separation of the replica cells. Subsequent regrouping of pure principal components preserves their full supports and marker unions, so the same argument applies to the principal connected logarithm.

166.2 Marker prices and spatial prices

Fix $n \ge 1$ marks in each replica cell. Every principal piece has a nonempty marker subset $I \subseteq \{1,\dots,n\}$ and a prescribed spatial support $X_{c,I}$ in cell c. The geometric hypotheses are the following: pieces with the same marker mode have a unique prescribed support; each actual principal support is contained in that support; and two prescribed supports containing the same spatial site belong to the same replica cell. These statements are made for full inspected supports, including the masks. Let

$$\operatorname{chg}_e(X) = \exp(e \operatorname{cov}_B(X)), \qquad \operatorname{chg}_e(X_{c,I}) \le R_e,$$

where $e \ge 0$ is the support exponent, B the current radius and cov_B its covering number. Use the constant marker weight $0 < r \le 1$. The weight of a mode I is $r^{|I|} \le 1$.

If each prescribed piece has weighted coefficient price at most Z, its total incidence price at a fixed mark (c,j) is at most $2^n Z$. Indeed only subsets I of that same cell can contain (c,j), and there are at most 2^n subsets. This estimate is independent of the number of replica cells.

A second estimate converts incidence price to spatial price. Suppose actual pieces s_a have nonnegative prices p_a and nonempty marker supports, and the total price of pieces containing any fixed mark is at most N. Assign piece a to its prescribed support $X_{c(a),I(a)}$. At a spatial site v, if any such support contains v, all such supports lie in one cell c. Charge each summand to one of its nonempty marks and then overcount by summing over every mark in that cell. It follows that

$$\sum_{a:\, v \in X_{c(a),I(a)}} p_a \operatorname{chg}_e(X_{c(a),I(a)}) \le nNR_e. \tag{792}$$

This is why the spatial charge is proportional to n, while the number of principal subset types is bounded by 2^n.

166.3 Replacing the leading pieces on their prescribed supports

Write the current regular source as a sum of remainder pieces H_a and leading pieces S_b. Let $F_{c,I}$ be the desired principal source at mode (c,I), with support $X_{c,I}$. Group the S_b by this marker mode and define on its healthy region

$$L_{c,I}(U) = \sum_{b:\, \operatorname{mode}(S_b)=(c,I)} S_b(U) - F_{c,I}(U).$$

Assume $|L_{c,I}(U)| \le y$ there. Attach the full support $X_{c,I}$ to $L_{c,I}$ and apply its healthy mask. The incidence price of these new regular remainders is at most

$$2^n R_e y. \tag{793}$$

Indeed each individual weighted price is at most $R_e y$, because $r^{|I|} \le 1$, and the preceding subset count applies.

The mask change also has a forced part. A leading piece may be healthy on its smaller support while its prescribed larger support is not healthy. Split its original masked value into its value on the larger healthy region and the complementary part. The latter is supported where the enlarged support contains a flag. Insert these complementary pieces into the forced alphabet, keeping their original marker modes. The identity is an identity of functions on every field: on the larger healthy region it is the defining equality for $L_{c,I}$; on its complement the difference has just been assigned to the forced part. Apply the exact regular/forced regrouping of Section 91 to the enlarged supports. No division by an activity value is involved.

Here is its quantitative form. Suppose the previous forced spatial price is s, the leading incidence price is Z, and the new forced inflation exponent is $g \ge 0$. If

$$e^g s + nZR_{e+1} \le \delta, \qquad (2B+1)^4 \delta \le 1, \tag{794}$$

then the regrouped forced price is at most

$$e^{-g}(e^{\delta} - 1). \tag{795}$$

To apply the canonical regrouping estimate, the old forced pieces contribute $e^g s$ and the newly moved pieces contribute at most nZR_{e+1} by (792). These are the two terms of its local inflation budget. Its support-counting factor is exactly $(2B+1)^4$, so (794) is its smallness condition. Its exponential product bound gives (795). The same regrouping bijection preserves the archived background coefficient at each address. Thus a regular remainder price z becomes at most $z + 2^n R_e y$, while the forced price changes as displayed, and the polynomial density is unchanged.

166.4 The recurrence along the comparison interval

Fix a top comparison depth k, an intermediate level l, a larger arity allowance $u \ge \max(2, n)$, and a monitor $H > 1$ fixed by k. Here k, l, u, n are natural numbers with $l \le k$, and all scalar price bounds are nonnegative. Let the tolerance exponent be $q \ge 6$ and the regular step multiplier be $\alpha \ge 0$. Set

$$d_0 = H^{-q}, \qquad d_j = (\alpha + 2)^j d_0.$$

The principal sources are the connected polynomials of the same Gaussian history at the current level, with the same cell and marker labels. At the terminal comparison level their higher connected coefficients vanish: the formal logarithm of $\prod_i (1 + z_i P_i)$ is $\sum_i z_i P_i$. Its singleton terms are exactly the launched calibrated insertions. Consequently the initial marked representation has regular remainder price d_0, after the specified terminal-source purchase, which must also bound its initial forced price by G_k.

Suppose the actual one-step marked compiler at each level has the following quantitative outputs when its incoming remainder price is $z \ge d_0$. Before realignment it supplies remainder price at most $(\alpha + 1)z$, principal incidence price at most H^{-2}, and forced price at most G_i^2 at the coarse level.

Here $G_i = e^{-g_i} \le 1$ is the required final forced price. Its projected principal polynomial differs from the next prescribed connected polynomial by at most y on every required healthy region. Purchase

$$2^n R_e y \le d_0, \qquad G_i + nH^{-2}R_{e+1} \le H^{-1}, \tag{796}$$

$$(2B_i + 1)^4 H^{-1} \le 1, \qquad e^{H^{-1}} - 1 \le 1. \tag{797}$$

The forced inequality is (794) with $s = G_i^2$, $g = g_i$ and $\delta = H^{-1}$. Its output is at most G_i. The regular price becomes at most

$$(\alpha + 1)z + d_0 \le (\alpha + 2)z.$$

Induction therefore bounds the remainder at level i by d_{k-i}, with forced price G_i and the exact prescribed principal polynomial. At the induction step the density is the push of the preceding marked density along the same transition kernel; the realignment changes only its representation. Hence this is an estimate on the original launched observable, not a sequence of separately chosen approximants.

The total number of steps is purchased so that $(\alpha + 2)^{k-l} \le H$. It follows that $d_{k-l} \le H^{1-q}$. The marker weight throughout this interval is $r = H^{1/(8u)-U}$, with the fixed cost exponent U chosen so that $0 < r \le 1$. The same radius is used at all replicas and finite periods. If the later propagation requires an extra factor H^{uV+2} in the regular price, purchase $1 - q \le -uV - 2$. Then $H^{1-q} \le H^{-uV-2}$ supplies that margin. These two inequalities are separate purchases: a small initial error does not alone control arbitrarily many steps or an arbitrary later marker-weight change.

166.5 The scalar envelope used in the replica estimate

We record the final elementary step linking a paid marked representation to Section 167. Suppose a polynomial density on a torus of q_0 sites has the form

$$D(U) = e^{\mathrm{exponent}_P(U)} \exp_*\left(\sum_a z_{I_a} h_a(U)\right) \sum_{B \text{ admissible}} \prod_{b \in B} K_b(U).$$

The I_a are nonempty, and each background packet has nonempty full spatial support. Let $\|\cdot\|_{\boldsymbol{r}}$ denote the coefficient ℓ^1 norm weighted by positive marker weights. Assume regular incidence price at most ℓ per mark and forced spatial price at most L per site, with nonnegative support exponent. If there are M_0 marks in total, double counting gives

$$\sum_a \|z_{I_a} h_a\|_{\boldsymbol{r}} \le M_0 \ell, \qquad \sum_b \|K_b\|_{\boldsymbol{r}} \le q_0 L.$$

A nonempty support contributes at least once to the corresponding incidence sum. Dropping the support weight is legitimate because it is at least one. Masks can only lower the regular value bounds. Submultiplicativity gives

$$\left\| \exp_*\left(\sum_a z_{I_a} h_a\right) \right\|_{\boldsymbol{r}} \le e^{M_0 \ell}.$$

Dropping the admissibility restrictions in the nonnegative majorant bounds the forced sum by $\prod_b (1 + \|K_b\|_{\boldsymbol{r}}) \le e^{q_0 L}$. Therefore

$$\|D(U)\|_{\boldsymbol{r}} \le \exp\{\mathrm{exponent}_P(U) + q_0 L + M_0 \ell\}. \tag{798}$$

For the replica estimate, first complete any later propagation to the selected output torus whose site count is exactly the number q_0 of replica cells. This equality is a property of that selected output level, not of an arbitrary intermediate torus. At this output there are $M_0 = q_0 n$ marks. Thus $L, \ell \le 1$ give per-site expense at most $1+n$, exactly the arity dependence in the determinant ticket. The weight of the coefficient carrying a chosen nonempty subset I in each of the q_0 cells is the q_0th power of its single-cell weight. Apply the normalization estimate and then take the q_0th root. The denominator is the partition function of the same pushed density, preserved by the unmarked coefficient identity at every step.

The recurrence above states its one-step hypotheses quantitatively. The joint Gaussian reader estimate supplies the discrepancy y and the principal source bounds; the mixed compiler supplies the exact marked push and the pre-realignment prices. The selected scale construction must buy all of these on the common history, including the later propagation below l. Neither the recurrence nor the scalar envelope asserts those upstream estimates without their corresponding purchases.

167 The volume cost of the rank tests

The replicated rank tests require the correct power of the coupling in both the numerator and the partition function. We prove the finite-torus estimate with that power. Write $D = \dim_{\mathbb{R}} \mathfrak{g}$, μ_G for probability Haar measure, and $\chi(u)$ for the normalized chord of $u \in G$ used in the action. Thus $\chi(1) = 0$, it is unchanged by inversion and conjugation, and $\chi(uv) \le \chi(u) + \chi(v)$.

167.1 Haar bounds in the physical dimension

There are constants $0 < c_- \le 1 \le c_+ < \infty$, depending only on G, such that

$$\mu_G\{\chi(u) \le r/4\} \ge c_- r^D \qquad (0 < r \le 1), \tag{799}$$

$$\int_G e^{-\chi(u)/t}\, d\mu_G(u) \le c_+ t^D \qquad (t > 0). \tag{800}$$

To see this, use the actual Haar chart of Section 77. Its density is bounded above and below by positive constants near zero. Differentiability of the represented exponential, together with the normalization of the colour norm, gives, after shrinking the chart,

$$\tfrac{1}{2}\|z\| \le \chi(\exp z) \le 2\|z\|.$$

A sufficiently small chord ball is contained in that chart: the faithful matrix embedding induces the topology of the compact group. Comparison with Euclidean balls of radii $r/2$ and $3r$ therefore gives $a r^D \le \mu_G\{\chi \le r\} \le b r^D$ for $0 < r \le r_0$. For $r \in [r_0, 1]$ the lower bound extends by monotonicity after decreasing a. The upper bound extends to all $r > 0$ by the total mass bound one, after increasing b.

For $k \ge 0$, on $kt \le \chi(u) < (k+1)t$ the integrand in (800) is at most e^{-k}. Summing over these measurable shells bounds the integral by

$$b t^D \sum_{k=0}^{\infty} e^{-k} (k+1)^D.$$

The series converges, since its successive-term ratio tends to e^{-1}. This proves the upper bound; rescaling the lower bound by $1/4$ proves (799). The proof also covers $D = 0$: powers of positive radii are then one. No connectedness of G is needed.

167.2 An axial tree and the four unintegrated links

Let $\Lambda = (\mathbb{Z}/L\mathbb{Z})^4$, $L > 1$, and $V = |\Lambda|$. Represent each coordinate by $0, \dots, L-1$. For $x \neq 0$ let $p(x)$ be its largest nonzero coordinate index. Join x to $x - e_{p(x)}$. The sum of the coordinate representatives strictly decreases on every such edge, so these edges form a tree rooted at zero. The tree has $V-1$ positive links; there are $4V$ positive links altogether. Thus there are $3V+1$ free links after fixing the tree links to one.

For completeness, tree gauge fixing preserves the integral of a gauge-invariant integrable function. Integrate successively from the leaves of the tree. At a leaf, use its tree link as the gauge variable at that vertex and multiply all its other incident links by that variable or its inverse. Conditional on the other links, these changes preserve their Haar measures. The tree link becomes one, and the remaining gauge variable integrates to one. Induction removes all $V-1$ tree variables. This proves the identity with product probability Haar measure on the free links, including for a disconnected group.

Keep also the four links $(-e_i, i)$, one wrap link in each direction. They are free, distinct, and are not in the tree. We now associate a different plaquette to each of the other $3V-3$ free links so that their integrals can be performed successively. Here is an explicit ordering, which also deals with wrap links. Set $p(0) = 0$, and for a positive link $e = (x, i)$ put

$$t(e) = \max(i, p(x)), \qquad \pi(e) = \big(t(e),\, \mathbf{1}_{\{i=t(e)\}},\, \textstyle\sum_j x_j\big),$$

ordered lexicographically.

If $i < t(e) = t$, choose the plaquette based at $x - e_t$ in directions (i, t). Its i-link at x is e. Its two t-links are tree links, because $x_t \neq 0$ and every coordinate above t vanishes. Its remaining i-link has smaller tier or the same tier and smaller coordinate sum. It therefore has smaller priority whenever it is free.

If $i = t(e)$ and e is free, then $x_i = L-1$ and all coordinates above i vanish: otherwise it is a tree link. Put $w = x + e_i$. Since e is not the retained wrap link, $w \neq 0$. Let $j = p(w) < i$ and choose the plaquette based at $x - e_j$ in directions (j, i). Its i-link at x is e. The j-link based at $w - e_j$ is a tree link. The other j-link has smaller tier or has the same tier with second priority coordinate zero. The remaining i-link has the same tier and second coordinate but smaller coordinate sum, since $x_j \neq 0$. Thus again every other free link on this plaquette has smaller priority.

The chosen plaquettes are distinct. If two links shared a chosen plaquette, each would have smaller priority than the other. Moreover, a maximal-priority remaining link occurs in no plaquette chosen for another remaining link. With all other links fixed, its own plaquette holonomy has the form aub or $au^{-1}b$, so its distribution is Haar. Fubini and descending induction therefore show that for continuous $F : G \to [0, 1]$,

$$\int \prod_{p \text{ plaquette}} F(U(\partial p))\, d\mu(U) \leq \left(\int_G F\, d\mu_G \right)^{3V-3}. \tag{801}$$

Indeed discard all unchosen factors, which are at most one, and then integrate the chosen factors in that order. The four edges of a plaquette are distinct for $L > 1$, including $L = 2$; thus each integration uses its selected variable exactly once, possibly inverted.

167.3 The normalized estimate

Consider a lawful representation R on this torus, with gauge-invariant path action and activities, and with nonnegative density, coupling β, threshold $\theta \leq 1$, scalar exponent σ, and clip constants

satisfying $0 < c_{\mathrm{lin}} < c_{\mathrm{dir}}/16$. The path and activity bounds below are the bounds of that same representation. These assumptions ensure that its density and the action bounds used in tree gauge fixing are gauge invariant. Put

$$g = \beta^{-1/2}, \quad 0 < g \le \theta, \quad c = \min(1, c_{\mathrm{lin}}), \quad b(g) = N_{\mathcal{A}}\big(c_{\mathrm{dir}} g^2/2 + (Ag)^2/2\big).$$

Here A bounds the linear curvature rows, and $N_{\mathcal{A}}$ is the number of positive plaquette types. Let r be the per-site bound for the additive remainder and the cubic, quartic and loop terms. Those bounds, with their actual support weights, give

$$\mathrm{exponent}_R(U) \ge \sigma - V\{r + \beta b(g)\} \qquad \text{if every link has chord at most } g/4, \tag{802}$$

$$\mathrm{exponent}_R(U) \le \sigma + V(r + N_{\mathcal{A}}) - \sum_p \frac{\chi(U(\partial p))}{g/c}. \tag{803}$$

Here is the local verification of these two uses of the path bounds. On the small-link set, every plaquette has chord at most g, all flags vanish, and each linear curvature row has norm at most Ag by its absolute row bound. Summing its squared contribution and the direct plaquette contribution gives $Vb(g)$. The other terms have absolute sum at most Vr. For the upper estimate put $a = \chi(U(\partial p))$. The clipped direct term obeys

$$ca/g - 1 \le \beta \ \mathrm{clipped}(a).$$

In the quadratic region this follows from $c(a/g-1)^2 \ge 0$, $c \le 1$, and $c \le c_{\mathrm{dir}}$. In the linear region use $c \le c_{\mathrm{lin}}$, $g \le \theta$ and $\beta g^2 = 1$. The other quadratic row term is nonnegative. The sum over plaquettes is unchanged by signed motions; hence averaging over them preserves this lower bound on the path action. Subtracting the remaining action and additive terms at their absolute bound gives (803).

Gauge-fix the axial tree and restrict its $3V+1$ free links to $\{\chi \le g/4\}$. The tree links are one, so (802) applies. All flags vanish there, and the forced gas equals one. Nonnegativity on the complement and (799) yield the strictly positive bound

$$Z_R := \int R \, d\mu \ge e^{\sigma - V(r+\beta b(g))} (c_- g^D)^{3V+1}. \tag{804}$$

Now let f be an integrable complex function, $z \ge 0$, and suppose

$$z|f(U)| \le e^{\mathrm{exponent}_R(U) + VX}.$$

Using (803), then (801) and (800), gives

$$z\left|\int f \, d\mu\right| \le e^{\sigma + V(r + N_{\mathcal{A}} + X)} \{c_+ (g/c)^D\}^{3V-3}.$$

Define $h = \log c_+ - D \log c \ge 0$ and

$$P_R(X) = 2r + N_{\mathcal{A}} + \beta b(g) + X + 3h - 4 \log c_-.$$

Division by (804) gives

$$\frac{z|\int f \, d\mu|}{Z_R} \le \exp\{V P_R(X) - 4D \log g\}. \tag{805}$$

To check the constants, the powers of g^D differ by exactly four. Also $(3V-3)h \le 3Vh$ and $-(3V+1)\log c_- \le -4V\log c_-$, since $V \ge 1$ and $\log c_- \le 0$. These are the only enlargements in this division.

The same estimate holds for the auxiliary history of Section 160. If its additive terms have value bound δ per site, delete them temporarily to obtain R^0. Then $e^{-V\delta}R^0 \le R \le e^{V\delta}R^0$ because $R \ge 0$, and $R^0 = e^E R \ge 0$. Apply the preceding estimate to R^0 with numerator expense $X+\delta$, and use $Z_{R^0} \le e^{V\delta}Z_R$. The extra numerator and denominator expenses total $2V\delta$. This is exactly the contribution of δ to $2r$ in P_R. Derivative estimates on those auxiliary additive terms are not needed.

167.4 Taking the replica root

Let a terminal tiling have q cells, and push its full marked density to a torus with exactly q sites along the auxiliary history. The unnormalized scalar integral remains the original partition function Z. Suppose each coefficient of the pushed marked density has the local estimate needed above with coefficient price $a^{q|I|}$ and per-site expense $X \le 1+M$, where M is the maximum number of markers per cell and $I \ne \varnothing$ is the chosen marker subset. The signed orientation copies use conjugation on the reversed cells. Thus the coefficient carrying I in every cell is exactly Z times the replicated Wilson expectation. Equation (805) gives

$$a^{q|I|}|\mathbb{E}(\text{replicated coefficient})| \le e^{qP_R(X)-4D\log g}.$$

Taking the positive qth root proves

$$\|\text{coefficient } I\|_{\text{root}} \le \frac{\exp\{P_R(X) - 4D\log g/q\}}{a^{|I|}}. \tag{806}$$

This retains the finite-volume harmonic correction $-4D\log g/q$.

The clock buys this last expense uniformly in the period. At an output level m, $\beta \le 4\widehat{\beta}_m$, so $-4D\log g = 2D\log\beta \le 8D\widehat{\beta}_m$. The clock coupling grows linearly in its logarithmic spatial scale, whereas the minimal torus site count grows at least exponentially in that scale. Consequently $8D\widehat{\beta}_m/q \le 1$ eventually, even at the smallest period. For the remaining terms, use the same clip and row bound A as the clock stage, and suppose

$$\theta_R \le \theta_{\text{clock}}, \qquad \Lambda \le \Lambda_{\text{clock}}, \qquad \delta \le 1.$$

The late-scale purchases are

$$\widehat{\beta}_m\theta_{\text{clock}}^3 \le 1, \qquad \widehat{\beta}_m\theta_{\text{clock}}^4 \le 1, \qquad \Lambda_{\text{clock}}\theta_{\text{clock}}^2 \le 1.$$

Consequently the actual remainder bound satisfies

$$r = \delta + \beta(A\theta_R^3 + A\theta_R^4) + \Lambda\theta_R^2 \le 1 + 8A + 1, \qquad \beta b(g) = \frac{N_{\mathcal{A}}}{2}(c_{\text{dir}} + A^2).$$

With $X \le 1+M$, a fixed stage ticket bounding $P_R(X)+1$ is therefore

$$T_M = 2(1+8A+1) + N_{\mathcal{A}} + \frac{N_{\mathcal{A}}}{2}(c_{\text{dir}} + A^2) + (1+M) + 3h - 4\log c_- + 1.$$

The final 1 pays for the harmonic correction, rather than enlarging an already spent ticket. Choose the comparison depth k on a constant-profile interval and purchase

$$T_M \le I_k = \frac{\log \mathcal{M}_k}{16M},$$

where $\mathcal{M}_k$ is the diverging clock monitor. This is a post-block-length scale delay. Set $a = e^{2I_k}$. Since $|I| \geq 1$, (806) is at most $e^{I_k}/a = e^{-I_k} \to 0$. For the rank detectors $M = 2$. These estimates apply to the one-point and corrected two-point coefficients themselves. The local marked construction must supply their displayed coefficient price on the same auxiliary density; no estimate for a different denominator is used here.

168 Three comparison scales and weak convergence

The comparison estimate is initially available at grid points whose coordinate projections are distinct. A test function may cross the excluded hyperplanes. We use three scales to pass from the former estimate to the latter test: one for the numerical comparison, one for a bound everywhere on a compact noncoincident set, and one for a derivative bound away from thin strips. Their different costs are essential to the argument.

168.1 The geometric and analytic estimates

Fix an arity $m \geq 1$ and use the Euclidean norm on $E_4^m = (\mathbb{R}^4)^m$, retaining the maximum norm for pair separation. For $R > 0$ set

$$K_R = \{x : \|x\|_2 \leq R, \quad \|x_i - x_j\|_\infty \geq R^{-1} \ (i \neq j)\}.$$

This is a compact subset of the noncoincident configuration space. Every compact subset of that space is contained in some K_R. Define

$$\Omega_\sigma = \{x : |x_{i,\nu} - x_{j,\nu}| > 8\sigma \text{ for every } i \neq j,\ 1 \leq \nu \leq 4\}.$$

Thus the complement of Ω_σ is a union of coordinate strips, not a neighborhood of just the actual coincidence set.

Let S be the smooth interacting table already extracted along the ordered mesh and volume filter $\mathcal{F}$, and let $Q(x)$ be the free correlation for a fixed word of labels. Fix a comparison index k. Its physical radius is $r_k > 0$; its normalized grid spacing is $h_k > 0$; its strip width is $\sigma_k > 0$. The calibrated insertion maps may still depend on the finite cutoff pair. Suppose they are eventually bounded in their finite-dimensional operator space. On any convergent subfilter of these maps write Y for the limiting tuple and

$$U_{k,Y}(x) = S(r_k x; Y).$$

All the following estimates are to hold for every such Y, with constants independent of that subfilter. The three required estimates are

$$|U_{k,Y}(y) - Q(y)| \leq e_k \qquad \text{at generic grid points } y \in K_{2R}, \tag{807}$$

$$|U_{k,Y}(x)| \leq A_k^m \qquad (x \in K_R), \tag{808}$$

$$\|DU_{k,Y}(x)\| \leq 4mB_k^m/\sigma_k \qquad \big(|x_{i,\nu} - x_{j,\nu}| > 4\sigma_k\big). \tag{809}$$

The last estimate is needed only in the noncoincident domain. The quantities e_k, A_k, B_k are nonnegative. Assume

$$e_k \longrightarrow 0, \qquad \sigma_k \longrightarrow 0, \qquad 4mh_k \leq \sigma_k \quad \text{eventually}, \tag{810}$$

$$A_k^m \sigma_k \longrightarrow 0, \qquad B_k^m h_k/\sigma_k \longrightarrow 0. \tag{811}$$

We first prove the weak comparison from these estimates, then explain the three scale choices that give exactly these powers.

Choose the coordinatewise lattice address y of x, so $\|y-x\| \le 4mh_k$. If $x \in \Omega_{\sigma_k}$, the closed ball of radius σ_k about x has every coordinate difference larger than $4\sigma_k$: moving its two endpoints changes a difference by at most $2\sigma_k$. In particular the segment from x to y remains in the region of (809). The fundamental theorem of calculus on that segment gives

$$|U_{k,Y}(x) - U_{k,Y}(y)| \le 16m^2 B_k^m h_k/\sigma_k.$$

For all sufficiently large k, this y belongs to K_{2R} whenever $x \in K_R$. Indeed the compact set K_R lies in the interior of K_{2R}, and $\|y-x\| \le \sigma_k \to 0$. It is also generic by the strict coordinate separation just proved. The free function Q is uniformly continuous on K_{2R}. If $\omega_Q(t)$ is its modulus of continuity there, the three-term triangle inequality gives

$$\sup_{x \in K_R \cap \Omega_{\sigma_k}} |U_{k,Y}(x) - Q(x)| \le \alpha_k := 16m^2 B_k^m h_k/\sigma_k + e_k + \omega_Q(\sigma_k) \longrightarrow 0. \tag{812}$$

This convergence is uniform over the limiting tuples Y under consideration.

There is an explicit linear bound for the volume of the omitted strips:

$$|K_R \setminus \Omega_\sigma| \le C_{m,R}\sigma, \qquad C_{m,R} = 64m(m-1)(2R)^{4m-1}. \tag{813}$$

For a fixed ordered pair $i \ne j$ and coordinate ν, make the linear change replacing $x_{j,\nu}$ by $x_{j,\nu} - x_{i,\nu}$ and leaving the other coordinates unchanged. Its determinant is one. The restricted coordinate has interval length 16σ and each of the remaining $4m-1$ coordinates lies in an interval of length $2R$. This bounds the volume of that strip intersected with K_R by $16\sigma(2R)^{4m-1}$. There are $4m(m-1)$ choices. Subadditivity proves (813), including intersecting strips. For $m = 1$ the union is empty and the constant is zero.

Let f be a smooth test supported in K_R, and put $C_Q = \sup_{K_R} |Q| < \infty$. Split its integral into the clear region and its complement. Equations (808)–(813) give the quantitative bound

$$\left| \int f(x)\{U_{k,Y}(x) - Q(x)\}\, dx \right| \le \|f\|_\infty \left\{ |K_R|\alpha_k + C_{m,R}(A_k^m + C_Q)\sigma_k \right\}. \tag{814}$$

The right side tends to zero. Both integrands are continuous on a neighborhood of the support, so these are ordinary integrals; no extension across coincident configurations is involved.

168.2 Returning to cutoff-dependent insertions

It is necessary to apply this estimate to the actual insertion maps, whose values depend on the cutoff. Fix k. Their eventual boundedness gives a compact closure in their finite-dimensional operator space. We use the ordered point and smeared limits of the same table S; these limits also hold on subfilters. If insertion maps converge to a map T along a subfilter, multilinearity in a fixed finite basis proves that their point and smeared correlations converge to those with the limiting labels. For example, expanding an m-tuple gives finitely many terms

$$\sum_{a_1,\dots,a_m} t_{1,a_1}(N,p) \cdots t_{m,a_m}(N,p) U_{N,p}(e_{a_1}, \dots, e_{a_m}).$$

Every scalar coefficient converges and every fixed basis correlation has the specified limit. Finite sums and products therefore give the claimed convergence for varying insertions. For the smeared

expression, use the fixed dilated test $r_k^{-4m} f(\,\cdot\,/r_k)$; a change of variables identifies its limiting integral with $\int f(x) U_{k,Y}(x)\,dx$.

Denote the right side of (814) by $E_k(f)$. We claim that the cutoff smeared comparison error is eventually at most $E_k(f)+\eta$, for every $\eta > 0$. If not, the set where it exceeds this number is frequent in $\mathcal{F}$. Restricting to that set gives a nontrivial filter. Compactness of the insertion maps supplies a further convergent subfilter. Along it the smeared error converges, by the preceding finite expansion, to a number at most $E_k(f)$. But its values are all larger than $E_k(f) + \eta$, so their limit is at least that latter number. This is a contradiction. Consequently the actual cutoff errors have the same asymptotic upper bound, and that bound tends to zero as $k \to \infty$. This compactness argument is performed separately at each fixed k and does not replace the original ordered extraction by a new continuum table.

168.3 The three scale inequalities

We describe precisely the estimates needed from the marked construction. Fix an upper arity $n \geq 2$, a maximum grade D, and a word of length $m \leq n$ in grades at most D. Let $H_k > 1$ be the diverging comparison monitor. Choose three exponents $a, b, e > 0$ and a strip exponent $c < 0$, and set

$$r_k = a_k H_k^a, \qquad h_k = H_k^{-a}, \qquad \sigma_k = H_k^c.$$

At a fixed comparison depth, all three panels must use the same calibrated readout map T. The lift at exponent a and the lift at exponent b then differ by the grade weight $H_k^{(a-b)j}$ on grade j. With a common finite basis, the sum of the absolute expansion coefficients of a fixed label of grade at most D is bounded by a constant times $H_k^{(a-b)D}$. The constant depends on the label, not on k. This uses the same map T on the two sides; two unrelated calibrations would not give the stated identity.

The marked comparison at the a-panel supplies (807) with $e_k = C\varepsilon_k \to 0$. The centered one-point bounds at the b-panel, expanded in that common basis and in the finitely many signed and conjugated orientations, supply the bulk coefficient bound with

$$A_k = C_b H_k^{(a-b)D}.$$

The analogous centered difference bound at the e-panel supplies

$$B_k = C_e H_k^{(a-e)D}.$$

Here the product and difference estimates are the centered cell estimates of Section 18. Their use requires actual cell widths, not just these coefficient prices. If w_b, w_e are the output levels of the latter two panels, the schedule estimates are

$$a_{w_b - 1}/a_k \leq H_k^{b+6}, \qquad 6m a_{w_b} < a_{w_b - 1}, \tag{815}$$

$$a_{w_e}/a_k \leq H_k^{e+5}. \tag{816}$$

The first gives a bulk cell strictly smaller than $r_k/(2R)$ once $b + 6 < a$; the second gives a difference cell smaller than $r_k \sigma_k$ once $e + 5 < a + c$. Applying the centered bounds and their continuum limits then gives (808) and the directional bound B_k^m/σ_k. Summing over the $4m$ coordinate directions gives exactly (809).

Here is the elementary rounding calculation underlying the width expenses 5 and 6. On the relevant constant-block interval let its block factor be $L \leq H_k$ and let V be its cover parameter,

with $V^{10} \le H_k$. The panel height is the least h with $L^h \ge H_k^{b+2}$, and its surplus is the least s with $L^s \ge V^{10}$. Minimality gives $L^h \le LH_k^{b+2}$ and $L^s \le LV^{10}$. Thus

$$a_{w_b}/a_k = L^{h+s} \le H_k^{b+5}.$$

One earlier block costs at most another H_k, giving the first inequality in (815). The block factors at these output levels tend to infinity; hence they eventually exceed $6m$, proving its second inequality. The identical calculation with e gives (816). The allowance of each panel must place all these levels in the same constant-block interval.

It remains to choose the exponents so that both integral errors decay. The exact sufficient inequalities are

$$\begin{aligned} c < 0, \qquad b + 6 < a, \qquad e + 5 < a + c, \\ (a - b)Dn + c < 0, \qquad (a - e)Dn - a - c < 0. \end{aligned} \tag{817}$$

They imply $-a < c$ and hence $4mh_k \le \sigma_k$ eventually. They also give

$$A_k^m \sigma_k = C_b^m H_k^{(a-b)Dm+c} \to 0, \qquad B_k^m h_k/\sigma_k = C_e^m H_k^{(a-e)Dm-a-c} \to 0.$$

Indeed $a > b, e$, so replacing m by n increases the exponents. For a concrete choice, write $q = Dn$, $x = 7q + 1$, $y = 6 + x$, and take

$$a = 10 + D + x + yq + y + 7, \qquad b = a - 7, \qquad e = a - y, \qquad c = -x.$$

Then $b + 6 = a - 1$, $e + 5 = a - x - 1$, and the last two exponents in (817) are respectively -1 and $-17 - D - y$. Also $a, b, e > D + 6$, so the additional lower bounds on panel exponents are respected. These choices depend only on the finite grade and arity bounds. They make no assertion of a uniform estimate over all words.

Finally, if these common-panel data are available at arbitrarily large comparison depths, choose an increasing sequence of those depths. All the eventually valid numerical inequalities remain valid along it. There are only finitely many basis words under fixed grade and arity bounds; take the maximum of their finitely many thresholds. The preceding proof applies to each word on this one sequence, and multilinearity extends it to arbitrary labels within these bounds. The empty word is the constant one and needs no strip estimate. Unit, conjugation and signed-symmetry properties of the insertion maps pass through this construction whenever they hold eventually at the finite cutoffs. Producing the common marked panel data on the selected Wilson history is a separate upstream task; the argument here proves their exact weak-convergence consequence.

169 Applying the convergence estimates to the selected histories

The preceding convergence arguments concern particular arrays, paths and stationary branches. We now identify these objects along one construction. This identification is needed because equality of the actions at one period would not identify their coefficient arrays, and because a cutoff-dependent change of the top boundary need not preserve a limit unless it eventually leaves each fixed scale unchanged.

169.1 One simultaneous stationary construction

Fix the scale schedule, clipping data and coordinate conventions. Denote the terminal cutoff by N and the scale by $m \le N$. The finite stationary construction of Section 138 gives paths $R_{N,m}$, scalar

corrections $\alpha_{N,m}$ and stationary maps $\rho_{N,m}^{p,F}$, where p is a period and F an allowed forest. All these objects are chosen together. Their one-step identities use $R_{N,m+1}$ as input and $R_{N,m}$ as output. The path bounds hold at the prescribed scale widths and allowances.

For the convergence assertion we use the strengthened version of this construction. At scale m it fixes three coefficient Hilbert spaces, with encoding widths $\delta_{m,2}, \delta_{m,3}, \delta_{m,4}$ strictly larger than the paid width w_m. The nonlinear path terms are encoded in their canonical selected subspaces. It also supplies the following uniformity: every admissible replacement of the input path with the same attached row and with all path and loop allowances enlarged by 64 has an output in these same selected spaces, at the ordinary output allowances. The output has the same classical and thermal matching identities on every period and forest. Finally, the underlying free horizon converges at each scale in all of its retained components, and the root rows have an eventual bound at some width $v_m > w_m$. These are the extended-domain, canonical-readout and retained-horizon hypotheses used in Section 139. They belong to the one-step analytic construction; neither finite backward induction nor the argument below supplies them.

Here is why its coefficient representation is the one needed in that section. Select the three coefficient arrays of each path once, at (N, m), and use that same selection in every occurrence of the path. For $m + 2 < N$, the input arrays of the step at m are then exactly the output arrays of the step at $m + 1$. The row at either occurrence is also the same attached root row. Thus the input/output equality holds in the coefficient spaces, before taking any limit. The one-step identities and bounds are available for all these chosen arrays. For a replacement of one array the other arrays remain fixed; their closed linear constraints and their separate budgets are preserved. The enlarged input domain therefore supplies exactly the additional steps used in the contraction argument.

Apply Section 139 to these data. At every fixed scale there are one limiting path R_m and one scalar α_m such that every coefficient of $R_{N,m}$ converges to the corresponding coefficient of R_m, and $\alpha_{N,m} \to \alpha_m$. The limiting path has the original paid bounds. For every fixed period, forest and derivative order r,

$$D^r \rho_{N,m}^{p,F}(0) \longrightarrow D^r \rho_m^{p,F}(0). \tag{818}$$

The limiting stationary germ is determined by R_{m+1}, its retained quadratic data and its forest. Its Hessian is positive with the same lower bound. Consequently no choice of a new subsequence for a forest or derivative order is involved in (818). The scalar limit is the one obtained from these same paths and thermal identities.

169.2 A single width for all four coefficient families

The common dictionaries are part of the conclusion just obtained. Write $r_{N,m}(a)$ and $r_m(a)$ for row coefficients on the canonical row dictionary, and $c_{N,m,q}(a, \nu)$ and $c_{m,q}(a, \nu)$ for the actual, exponentially encoded coefficients of the atoms of degree $q = 2, 3, 4$. Their supports, lengths $\ell(a)$ and color multi-indices ν are independent of N on a tail. Choose

$$w_m < \widetilde{w}_m < \min\{v_m, \delta_{m,2}, \delta_{m,3}, \delta_{m,4}\}.$$

Then, for every $\varepsilon > 0$, all sufficiently large N satisfy

$$\sum_a |r_{N,m}(a) - r_m(a)| e^{\widetilde{w}_m \ell(a)} \le \varepsilon, \tag{819}$$

$$\sum_{a,\nu} |c_{N,m,q}(a, \nu) - c_{m,q}(a, \nu)| e^{\widetilde{w}_m \ell(a)} \le \varepsilon \qquad (q = 2, 3, 4). \tag{820}$$

For the row statement one may either sum the finitely many plaquette directions or impose the estimate separately on each; shrinking the requested ε by their number gives the displayed version.

For completeness, the two tail bounds that permit this simultaneous choice are different. If the row budget at v_m is B_m, its coefficient difference at the smaller width is bounded by

$$2B_m e^{-(v_m-\widetilde{w}_m)\ell(a)}.$$

For an atom, remove its encoding factor $e^{-\delta_{m,q}\ell(a)}$. The resulting coefficient arrays have a common Hilbert bound $C_{m,q}$, so their individual differences have absolute value at most $2C_{m,q}$. Their weighted majorant is therefore $2C_{m,q}e^{-(\delta_{m,q}-\widetilde{w}_m)\ell(a)}$. Both majorants are summable on the canonical dictionaries by the geometric sums in Section 117. The coordinate limits and a finite-head, summable-tail argument prove (819)–(820). There are only four families and finitely many row directions, so their eventual estimates hold on one tail. This gives one path limit with one strictly larger width, as required in the local integral comparison.

169.3 Changing the terminal normalization

The final Wilson normalization replaces the top path at $m = N$. Let $\widetilde{R}_{N,m}$, $\widetilde{\alpha}_{N,m}$ and $\widetilde{\rho}^{p,F}_{N,m}$ denote the resulting data. The change of coordinates retains the exact identities

$$\begin{aligned}
\widetilde{R}_{N,m} &= R_{N,m} && (m < N),\\
\widetilde{\alpha}_{N,m} &= \alpha_{N,m} && \text{for all } N, m,\\
\widetilde{\rho}^{p,F}_{N,m} &= \rho^{p,F}_{N,m} && (m+1 < N).
\end{aligned}$$

It also leaves the free bed unchanged for $m < N$ and the entire associated tower unchanged for $m + 1 < N$; its transport row is unchanged already for $m < N$. These are identities of the maps and coefficient data, not merely equalities of their evaluated actions.

For fixed m, all the identities hold once $N > m + 1$. Deleting this finite initial segment changes none of the limits above, none of the eventual larger-width bounds and none of the retained free horizon limits. Equality of the stationary maps gives equality of every derivative at zero. Thus the final normalized construction has exactly the same R_m, α_m and stationary jets. In particular, replacing the terminal path has not introduced a second background for the activity estimates.

169.4 The activity and coupling limits on this background

Fix a period p. The regular supports, forced inventories and norms at scale m depend on the fixed scale and period, not on N. Embed their actual arrays into the one completion $\mathcal{H}_{m,p}$ of Section 146. The representative of each finite history has exactly its prescribed activities; the embedding is the same one used in the finite-test estimates. Its norm is at most $1/8$. At every retained scale its path is $\widetilde{R}_{N,m}$, by the construction's input/output identities. The scalar in its coupling recursion is $\widetilde{\alpha}_{N,m}$, not a newly chosen correction.

The finite real-test estimates and their ray consequence in Section 145 give the two one-step comparison bounds in (714)–(715). Their constants are $8/128$ for the activity and $4s_m$ for the scalar correction. The entry-threshold calculation at the end of Section 146 gives $\sum_{j<m} s_j \le B_m/128$ for these same allowances. The larger-width path limit above, the convergent stationary jets and the retained free horizon supply the background data in the common-variable comparison. The compensated parameter estimates of Sections 101 and 102, with the suppliers of Section 103, then give its local continuity, including the projected scalar output. Each application retains the funded transition and output identities stated there. This continuity applies also to the admissible

auxiliary histories with a prescribed coupling script and zero terminal activity, since the same one-step construction is available for every input in its allowed ball.

All hypotheses of Section 146 are therefore on one and the same family. For any terminal sequence $N \to \infty$ and any corresponding finite trajectories whose bottom couplings tend to b, there are, at every fixed m, p, limits

$$\beta_{N,m} \longrightarrow \beta_m, \qquad x_{N,m} \longrightarrow x_m \text{ in } \mathcal{H}_{m,p}, \qquad q_{N,m} \longrightarrow q_m.$$

Here $q_{N,m}$ is the scalar activity correction, distinguished from the path correction $\alpha_{N,m}$. They obey

$$\beta_m - \beta_{m+1} - \alpha_m = q_m.$$

Two sequences with the same bottom limit b give the same limits: interleave them, whose terminal indices still tend to infinity, and apply the same uniqueness proof. No compactness of a ball in $\mathcal{H}_{m,p}$ is used. The norm convergence and the positive partition lower bound in Section 117 now apply to the actual normalized densities at that scale.

This application retains the strengthened stationary construction, its convergent free horizon and the funded nonlinear transition. The argument establishes compatibility of their consequences; the analytic production of these inputs remains the responsibility of the corresponding construction estimates.

170 The coefficient field at the last grouping scale

We connect the local-field representation to the degree-bounded coefficient field in the cutoff estimate. This step uses a projection estimate for one elementary insertion. We state its content explicitly and prove the extension to the whole grouped observable, including the normalization that determines its final price.

Let $m+1$ and m be adjacent fine and coarse scales, with mesh sizes $a_{m+1} = a_m/L$ and $L \geq 1$. Fix a degree $d \geq 0$. For every fine elementary insertion M anchored at the origin, assume there is a coarse coefficient field Q_M of degree at most d with the prescribed locality and covariance format, satisfying

$$\operatorname{dens}(Q_M) = L^d \mathcal{T} \operatorname{dens}(M), \qquad |Q_M|_{\mathrm{coeff}} \leq x \operatorname{cost}(M), \qquad \operatorname{rem}(Q_M) \leq x \operatorname{cost}(M), \tag{821}$$

where $x \geq 0$. Here the remainder bound is the weighted sum of the local regular remainder and the dominated forced remainder at this scale. Assume, in addition, the singleton transition expansion of Section 57: the fine expansion cost for this insertion is at most $2 \operatorname{cost}(M)$, and its coarse response cost is at most $x \operatorname{cost}(M)$. The relation for the quoted profile has multiplier L^d, consistently with the first identity in (821). These statements concern the actual background transition and its actual coefficient fields; they are not an abstract polynomial approximation assumption. Their analytic proof remains an input here.

170.1 From one insertion to a grouped field

Suppose a grouped fine field has zero separately recorded constant and an elementary representation

$$(g, q) = \sum_j z_j \big(\operatorname{ord}(M_j), \operatorname{flag}(M_j)\big), \qquad \sum_j |z_j| \operatorname{cost}(M_j) \leq A,$$

where the list is finite and every insertion is anchored at the origin. Choose Q_j by (821), and let $Q = \sum_j z_j Q_j$. The sum means addition of coefficient vectors and of forced arrays, and concatenation

of the regular remainder lists. Thus all locality and covariance conditions are preserved. The triangle inequality gives

$$|Q|_{\mathrm{coeff}} \le xA, \qquad \mathrm{rem}(Q) \le xA. \tag{822}$$

For the regular remainder, multiply each majorant by $|z_j|$ and sum its weighted costs. For the forced remainder, do the same for its uniformly summable majorants. These constructions prove the asserted bound without any cancellation of signed terms. Finite linearity of integration gives

$$\mathrm{dens}(Q) = L^d \mathcal{T}\, \mathrm{coin}_R(g, q).$$

Indeed the linear density of (g, q) is the sum of the elementary densities with coefficients z_j, by the singleton formula (212). Each elementary density is required to be integrable under the fine Haar prior and in each transition slice, as in Section 57; the lawful representation and transition-kernel properties supply this in the application. Thus finite summation commutes with the transition integral. The empty list gives $Q = 0$ and the same conclusion.

The singleton transition expansions concatenate in precisely the same way. Their fine cost is at most $2A$ and their coarse response cost at most xA. Dividing the output field and its profile by L^d changes the latter cost to $L^{-d}xA$, while the fine cost stays $2A$. The multiplier in the quoted-profile identity is now one. This is the normalization needed by the one-scale transport lemma.

Require also that the whole marked input density is integrable under the fine prior and the transition slices; this is the hypothesis needed below for taking polynomial coefficients and regrouping through integration. Suppose the other grouped fields are translated, coordinate transformed or conjugated copies of (g, q), with origins that are embeddings of their coarse origins. The transition commutes with these motions, and the coefficient-field format is stable under them. Apply the same motion to $L^{-d}Q$ in each cell. The quoted-profile expansions transform by relabelling the finite set of slots; their cost bounds remain $2A$ and $L^{-d}xA$. The singleton coefficient of the integrated marked density is the linear density of the corresponding transformed copy of $L^{-d}Q$: taking a coefficient commutes with integration, and the preceding density identity commutes with the motion. Thus the singleton-identification hypothesis of Section 57 holds for every copied cell, with one and the same coefficient field Q.

If the incoming mark weights I_v satisfy

$$2I_v(2A) \le 1, \qquad I_v L^{-d} xA \le 1, \tag{823}$$

and the support, occupancy, numerical and new-weight hypotheses of that transport lemma hold, it follows that the integrated marked density has the required local representation with these coefficient fields and higher-array price one. This conclusion retains the entire marked density, not only its singleton coefficients.

170.2 The terminal price and physical normalization

At the terminal grouping stage, let $z > 0$ be any lower bound for the common grouped weight supplied by the grouping induction. Section 56 gives $A = 4/z$. Take $x \ge 1$ and reduce the incoming weights to

$$I_v = \frac{z}{16x}.$$

They do not exceed the available grouped weight. Reducing positive weights decreases every higher coefficient price. Moreover

$$2I_v(2A) = \frac{1}{x} \le 1, \qquad I_v L^{-d} xA = \frac{L^{-d}}{4} \le 1.$$

This verifies (823). If the outgoing weights obey $0 < J_v \le \delta_m I_v$, where δ_m is the first reduction factor in the transport lemma, its final reweighting condition follows term by term: the product of the J_v is at most the product of $\delta_m I_v$, and the allowed support exponential is at least one.

We have obtained a coarse field Q with both bounds at most $4x/z$, appearing in the density with prefactor L^{-d}. Define the physically normalized coefficient field

$$Q' = a_{m+1}^d Q.$$

Scaling multiplies its coefficient and remainder costs by the positive number a_{m+1}^d. Since

$$a_m^{-d} a_{m+1}^d = L^{-d},$$

this changes neither the ordinary singleton nor its flagged profile in the density. It only rewrites them with the standard physical prefactor a_m^{-d}. We obtain

$$|Q'|_{\mathrm{coeff}},\ \mathrm{rem}(Q') \le a_{m+1}^d \frac{4x}{z}. \tag{824}$$

This is the terminal projection price used in the cutoff accounting: take z equal to its available grouping budget and x equal to the elementary projection allowance.

Finally, regrouping and backwards integration are compatible. The regrouping map reads the coefficient of the complete old group at each new mark; it is a linear map between finite-dimensional polynomial spaces. It therefore commutes with every integrable transition. At the finest scale, reading a complete group in the marked birth density gives the product of its constituent observables. An induction over the intervening transitions shows that the terminal reaped density is the backwards-integrated density of that product. Consequently the coefficient field in (824) represents the intended cluster product through the same sequence of scales. This identifies the observable to which the cutoff price applies; the elementary estimate (821) and the nonlinear transition estimate still require their own proofs.

171 Decay of the matched reference error

The reference comparison uses a second cone argument. Here the coefficient vector is small compared with the remainder cost. The initial condition is imposed at the coarse comparison scale; the resulting cone propagates towards finer scales and bounds the error in the opposite direction.

Let μ be the coefficient ℓ^1 norm, including the unit coordinate. At scale i write v_i for the coefficient vector of a signed comparison and $r_i \ge 0$ for its remainder cost. These are the data of the actual difference, not the sum of the costs of its two summands. On an interval $l \le i \le k$, assume $v_l = 0$ and a one-step comparison of the following form. Put

$$q_i = L_i^{1/8}, \qquad \varepsilon_i = q_i^{-3}, \qquad Y_i = \mu(D_{L_i} v_{i+1}),$$

where D_L multiplies the coordinate of degree $j \le d$ by L^{d-j}. An invertible coefficient map B_i and its inverse have norm at most $K_i \ge 1$, and the transition satisfies

$$\begin{aligned} \mu(v_i - B_i D_{L_i} v_{i+1}) &\le \varepsilon_i \{\mu(\mathrm{core}\, v_{i+1}) + r_{i+1}\}, \\ r_i &\le K_i \mu(\mathrm{core}\, D_{L_i} v_{i+1}) + \varepsilon_i r_{i+1}. \end{aligned} \tag{825}$$

Here core deletes the unit coordinate. Let $C_i \ge 1$ be the column allowances. The schedule is required to satisfy

$$64 K_i^2 \le C_i, \qquad q_i \ge 100000 M_i^6 (1 + C_{i+1})^2, \qquad M_i \ge K_i \ge 1. \tag{826}$$

Finally suppose the finest-scale cost obeys

$$r_k \le w + C_k\{w + \mu(v_k)\}, \qquad w \ge 0. \tag{827}$$

We prove

$$e^{(s_k - s_i)/8}\{r_i + \mu(v_i)\} \le 4(1 + C_k)w \qquad (l \le i \le k), \tag{828}$$

where $s_{i+1} - s_i = \log L_i$.

First, deleting the unit coordinate cannot increase μ, and $L_i \ge 1$ gives $\mu(v_{i+1}) \le Y_i$. Apply B_i^{-1} to (825) and use the triangle inequality:

$$Y_i \le K_i\{\mu(v_i) + \varepsilon_i(Y_i + r_{i+1})\}, \qquad r_i \le K_i Y_i + \varepsilon_i r_{i+1}. \tag{829}$$

Set $A_i = 32(1 + C_i)^2$. The schedule implies

$$A_i \ge 4K_i^2, \quad 4K_i\varepsilon_i \le 1, \quad 4K_i A_{i+1}\varepsilon_i \le 1, \quad (4K_i^2 + 1)q_i\varepsilon_i \le 1. \tag{830}$$

Indeed (826) gives $q_i \ge 128K_i^2(1 + C_{i+1})^2 \ge 1$. Since $\varepsilon_i \le q_i^{-1}$ and $q_i\varepsilon_i \le q_i^{-1}$, the last three claims follow from $4K_i \le q_i$, $128K_i(1 + C_{i+1})^2 \le q_i$, and $4K_i^2 + 1 \le q_i$, respectively. The first follows from $C_i \ge 64K_i^2$ and $C_i \ge 1$.

Suppose the cone inequality $A_i\mu(v_i) \le r_i$ holds. For this one step abbreviate $K = K_i$, $\varepsilon = \varepsilon_i$, $u = \mu(v_i)$, $Y = Y_i$ and $t = r_{i+1}$. The second inequality in (829) yields

$$4Ku \le Y + \varepsilon t,$$

because $4K^2u \le r_i \le KY + \varepsilon t$ and $K \ge 1$. Substitute this bound for u in the first inequality:

$$Y \le \tfrac{1}{4}(Y + \varepsilon t) + K\varepsilon Y + K\varepsilon t \le \tfrac{1}{2}Y + (K + \tfrac{1}{4})\varepsilon t.$$

Thus $Y \le (2K + 1/2)\varepsilon t \le 4K\varepsilon t$. Using (830), we obtain

$$A_{i+1}\mu(v_{i+1}) \le A_{i+1}Y \le t, \qquad q_i r_i \le q_i(4K^2 + 1)\varepsilon t \le t.$$

The first statement propagates the cone. The second gives the gain in remainder cost. At $i = l$ the cone holds because $v_l = 0$; induction therefore proves both inequalities at every step. Multiplying the latter gives

$$e^{(s_k - s_i)/8} r_i \le r_k.$$

At the finest scale the cone also implies $2C_k\mu(v_k) \le r_k$. Insert this in (827):

$$r_k \le (1 + C_k)w + \tfrac{1}{2}r_k, \qquad r_k \le 2(1 + C_k)w.$$

Since $\mu(v_i) \le r_i$ everywhere in the cone, this proves (828).

171.1 Keeping the error affordable along the interval

The gain estimate is combined with a sequence of mark weights. Let $\eta_i > 0$, put $b_l = b > 0$ and $b_{i+1} = \eta_i b_i$. Let $\rho_i > 0$ satisfy

$$\eta_i L_i^{d+1}\rho_i \le \rho_{i+1}.$$

If $E \ge 0$ and

$$b_l a_l^{-d}\frac{E}{e^{(s_k - s_l)/8}} \le \rho_l,$$

then the same estimate holds at every $i \ge l$ up to k:

$$b_i a_i^{-d} \frac{E}{e^{(s_k - s_i)/8}} \le \rho_i. \tag{831}$$

In fact, the expression on the left is multiplied in one step by $\eta_i L_i^d q_i \le \eta_i L_i^{d+1}$. This proves the assertion by induction. Applied with $E = 4(1 + C_k)w$, it bounds the weighted coefficient-plus-remainder cost of the matched difference at all intermediate scales. The construction of that signed difference and the proof of its one-step comparison (825) remain separate inputs: this numerical argument does not obtain them by subtracting two unrelated error estimates.

172 Constructing the signed reference difference

The cone in Section 171 applies only to a comparison produced on a common background history. We describe that construction. Its principal feature is to remove the coefficient vector after each backward step, retaining the removed part in a reference label. This provides a family of exact comparisons, from which a single label is chosen at the coarsest scale.

Fix the degree d and scales $l \le k$. Assume the bank and column inequalities (826), with $K_i, C_i \ge 1$, as in Section 171. At each scale i, let $\mathcal{P}_i(b)$ be the calibrated reference coefficient field with bare label b in the fixed coefficient space. Its coefficient vector is $I_i b$, where I_i is a linear isomorphism. Assume its density, ordinary field and flagged profile depend linearly on b, and its remainder has a nonnegative subadditive bound $h_i(b)$ with

$$h_i(b) \le C_i \mu(\operatorname{core} I_i b) \le C_i \mu(I_i b). \tag{832}$$

The reference densities satisfy the exact degree-d transition identity. Their coefficient and remainder data satisfy (825), with a common comparison map for each reference label, as supplied by the calibrated construction.

We also require a small transition estimate for a coefficient field S whose coefficient vector is zero. If its remainder is bounded by $q \ge 0$, its backward transition has a representation U such that

$$\operatorname{dens}(U) = L_i^d \mathcal{T}_i \operatorname{dens}(S), \quad \mu(U.x) \le \eta_i q, \quad \operatorname{rem}(U) \le \eta_i q, \quad (1 + 2C_i)\eta_i \le \varepsilon_i. \tag{833}$$

Here $\eta_i \ge 0$ and $\varepsilon_i = L_i^{-3/8}$. The fields and their sums retain the coefficient-field format and the integrability needed for the displayed transition identities. These are analytic inputs for the same history, including its calibrated reference fields.

172.1 Removing the coefficient after each step

Start with a coefficient field Q at scale k, with remainder bound $t \ge 0$ and $\mu(Q.x) + t \le w$. Put

$$c_k = I_k^{-1} Q.x, \qquad S_k = Q - \mathcal{P}_k(c_k), \qquad q_k = t + h_k(-c_k).$$

The minus sign denotes a sum with the field $\mathcal{P}_k(-c_k)$; its regular remainders are concatenated with those of Q. Thus $S_k.x = 0$ and its remainder is at most q_k. For every bare label v, define

$$D_k(v) = S_k + \mathcal{P}_k(c_k + v), \qquad r_k(v) = q_k + h_k(c_k + v).$$

Linearity of the reference density and vector gives

$$D_k(v).x = Q.x + I_k v, \qquad \operatorname{dens}(D_k(v)) = \operatorname{dens}(Q) + \operatorname{dens}(\mathcal{P}_k(v)). \tag{834}$$

The cost, despite possible cancellation in this identity, is kept as the sum of the displayed nonnegative remainder bounds. Since $t \le w$ and $\mu(Q.x) \le w$,

$$r_k(v) \le w + C_k\{w + \mu(D_k(v).x)\}.$$

This is exactly the finest-scale condition (827).

Suppose $c_{i+1}, S_{i+1}, q_{i+1}$ have been constructed, with $S_{i+1}.x = 0$. Apply (833) to obtain U_i, put $e_i = \eta_i q_{i+1}$, and set

$$\begin{aligned} b_i &= I_i^{-1} U_i.x, & c_i &= c_{i+1} + b_i, \\ S_i &= U_i - \mathcal{P}_i(b_i), & q_i &= e_i + h_i(-b_i). \end{aligned}$$

Then $S_i.x = 0$, and q_i bounds its remainder. Define $D_i(v) = S_i + \mathcal{P}_i(c_i + v)$ and $r_i(v) = q_i + h_i(c_i + v)$. Their densities satisfy

$$\operatorname{dens}(D_i(v)) = \operatorname{dens}(U_i) + \operatorname{dens}(\mathcal{P}_i(c_{i+1} + v)) = L_i^d \mathcal{T}_i \operatorname{dens}(D_{i+1}(v)).$$

The first equality cancels the two reference terms with labels $-b_i$ and b_i by linearity. The second uses both transition identities and linearity of the integrable transition. The same cancellation holds separately for ordinary fields and flagged profiles; it is not merely an equality after taking expectation.

We verify the signed one-step estimates. Fix v and write $y = c_{i+1} + v$. Then

$$D_{i+1}(v).x = I_{i+1} y, \qquad D_i(v).x = I_i y + U_i.x.$$

Use the comparison map belonging to the primary label y. Its coefficient error is at most

$$\varepsilon_i\{\mu(\operatorname{core} I_{i+1} y) + h_{i+1}(y)\}.$$

The added vector $U_i.x$ costs at most $\eta_i q_{i+1} \le \varepsilon_i q_{i+1}$. Thus the coefficient inequality in (825) holds for the actual signed vectors, with cost $r_{i+1}(v)$.

For the remainder, (832) gives $h_i(\pm b_i) \le C_i e_i$. Subadditivity yields

$$\begin{aligned} r_i(v) &= e_i + h_i(-b_i) + h_i(y + b_i) \\ &\le h_i(y) + (1 + 2C_i) e_i \\ &\le K_i \mu(\operatorname{core} D_{L_i} I_{i+1} y) + \varepsilon_i\{h_{i+1}(y) + q_{i+1}\}. \end{aligned}$$

This is the other inequality of (825). In particular it is a bound for the very same presentation that supplies the exact density identity. No subtraction of unsigned remainder bounds is used.

Now choose the single label $v = -c_l$ and write $b = c_l$. The coefficient vector of $D_l(-b)$ vanishes. The construction therefore meets every hypothesis of Section 171, including its terminal condition. Writing $Z_i = D_i(-b)$ and $r_i = r_i(-b)$ gives

$$\operatorname{dens}(Z_k) = \operatorname{dens}(Q) - \operatorname{dens}(\mathcal{P}_k(b)), \qquad r_i + \mu(Z_i.x) \le \frac{4(1 + C_k)w}{e^{(s_k - s_i)/8}}. \tag{835}$$

If $E \ge 4(1 + C_k)w$, then $w \le E$ and at $i = k$

$$\mu(I_k b) = \mu(Q.x - Z_k.x) \le w + E \le 2E.$$

This is the quota bound for the matched reference label.

172.2 The singleton transition expansion is retained

To use this difference inside a marked density, more than its coefficient and remainder estimates is needed. Suppose the zero-vector transition in (833) comes with the singleton expansion of Section 57, of fine cost $2q_{i+1}$ and coarse response cost e_i. Suppose the primary transition for label y has fine cost at most

$$\operatorname{orb}(d)\mu(I_{i+1}y)(\psi_i + 2C_{i+1}),$$

and coarse response cost $h_i(y)$. Both expansions must use the same quoted-profile map; their finite sets of slots can then be united. Their sum has the ordinary and flagged profiles of $D_{i+1}(v)$ and the coarse profile of $D_i(v)$, by the cancellations just proved. Its fine cost is at most

$$\operatorname{orb}(d)\mu(D_{i+1}(v).x)(\psi_i + 2C_{i+1}) + 2r_{i+1}(v).$$

Its response cost $e_i + h_i(y)$ is at most $r_i(v)$, because

$$h_i(y) = h_i\big(-b_i + (y + b_i)\big) \le h_i(-b_i) + h_i(y + b_i).$$

Thus the same signed difference carries the singleton transition expansions needed for the marked transport argument. Combining (835) with (831) makes these expansions affordable once the prescribed one-step allowances are imposed. The production of the small zero-vector transition and the calibrated primary transition bounds remains an analytic dependency of this construction.

173 From a coefficient field to a finite-cutoff reference observable

The signed comparison must control the original finite-cutoff observable, not only its coefficient vector. This requires a simultaneous marked argument. We explain the finite-dimensional matching and the exact replicated difference used in that argument.

Fix a degree d and scales $\operatorname{ref}(d) \le m \le k < N$. All paths below use the same finest scale N and the same history; m is the opening scale and k the matching scale. Let Q represent a continuous finite-cutoff observable f at scale k through the full marked local representation. Suppose its coefficient-plus-remainder bound is at most $c > 0$, and put

$$E \ge 4(1 + C_k)c, \qquad G = e^{(s_k - s_m)/8}, \qquad w > 0.$$

Assume the signed transition hypotheses of Section 172, the one-scale marked transport hypotheses of Section 57, and the necessary cell separation and chamber locality at each scale. The common affordability condition is

$$32(1 + C_k)wa_m^{-d}E/G \le \theta_m/2. \tag{836}$$

Here θ_i are the single-source allowances; their growth condition is the one used in (831). The coefficient paths must also satisfy the corresponding one-step singleton cost conditions there and in the marked transport lemma. No availability of these analytic hypotheses is inferred from (836) alone.

173.1 An approximate coordinate basis

Apply the signed reference construction first to Q. It gives a label b with $\mu(I_k b) \le 2E$ and a path representing $Q - \mathcal{P}_k(b)$. Its opening price is $2w$; the required initial affordability follows from (836) because $2 \le 32(1 + C_k)$.

For each coefficient coordinate j, apply the same construction to the primary field with vector e_j and zero remainder. Use opening price $8Ew$ and coefficient allowance $4(1+C_k)$. Its affordability expression is exactly the left side of (836). Obtain labels b_j. The cone estimate at k, now with terminal bound 1, gives

$$32(1+C_k)^2\mu(e_j - I_k b_j) \le 2(1+C_k), \qquad \mu(e_j - I_k b_j) \le \tfrac{1}{4}.$$

Define $Az = \sum_j z_j I_k b_j$. Since μ is the coefficient ℓ^1 norm,

$$\mu(z - Az) \le \sum_j |z_j| \mu(e_j - I_k b_j) \le \tfrac{1}{4}\mu(z).$$

It follows that $\mu(z) \le 2\mu(Az)$. Thus A is injective and, in this finite-dimensional space, surjective. Solve $As = I_k b$. Injectivity of I_k gives

$$b = \sum_j s_j b_j, \qquad \sum_j |s_j| \le 2\mu(I_k b) \le 4E. \tag{837}$$

No uniform condition number of an arbitrarily chosen reference basis is being assumed; the bound comes from the explicitly constructed approximate coordinate vectors.

Let ℓ_j be the physically normalized primary field for e_j at scale k, with its local mask, and set

$$\ell = -\sum_j s_j \ell_j, r = \mathcal{B}_{k,N}\ell,$$

where $\mathcal{B}_{k,N}$ is the backward observable transfer to the finest lattice. All ℓ_j lie in the same fixed stencil window. Hence ℓ does too, and the room condition places r inside the centered chamber. This common window is essential for the replicated transfer identity below.

173.2 The full replicated difference

We spell out that identity, keeping every mixed term. Write $\mathcal{T}_{N,k}$ for the forward density transfer, adjoint to $\mathcal{B}_{k,N}$ under the Haar priors. For copied fine observables O_i and copied coarse observables F_i, the coefficient of a mark set S after introducing the coarse sources is

$$\sum_{A \subseteq S} \mathcal{T}_{N,k} \left(\rho_N \prod_{i \in S \setminus A} (-O_i) \right) \prod_{i \in A} F_i. \tag{838}$$

Here ρ_N is the unnormalized Wilson density. The same forward transfer is used for the whole fine product in each summand. It has not been replaced by a product of transferred factors.

For coarse functions in the common local window, the room condition ensures that backward transfer respects their copied product: $\mathcal{B}_{k,N}(\prod_{i\in A} F_i)$ is the product of the corresponding copies of $\mathcal{B}_{k,N}F$. Using adjointness in each summand of (838), its integral is therefore

$$\int \rho_N \prod_{i \in S} (\mathcal{B}_{k,N} F_i - O_i)\, dU.$$

This follows by the finite subset expansion of the product. The interchange is justified by bounded local observables, integrable fine densities and the transition-slice bounds. Further forward transfers leave this integral unchanged. Thus the coefficient of the complete mark set is the Wilson partition function times the replicated difference moment.

The singleton bounds used here follow from the priced coefficient basis and the remainder bounds. Explicitly, require the masked basis evaluation bound with column constant $C_m \geq 1$ and remainder divisor at most one. If a coefficient field has vector q and remainder cost t, the triangle inequality gives $|g| \leq (1 + C_m)\mu(q) + t$ before physical normalization. The sum of the norms of its flagged profiles is at most t: sum the absolutely summable forced majorants and use that every support weight is at least one. The same argument bounds the ordinary remainder by t; the primary term is bounded by $(1+C_m)\mu(q)$ using the priced basis. If $va_m^{-d}(t+\mu(q)) \leq \theta_m$ and $C_m\theta_m \leq 1$, these bounds give normalized singleton allowances 2 and 1. Translations and conjugation preserve them. This is the sharper bound for the matched coefficient paths; it is separate from the general centered-field estimate.

Suppose the marked transport has reached scale m with common mark weight $v > 0$, higher-array price one, and singleton bounds

$$v|g_i(U)| \leq 2, \qquad v \sum_X |q_i(X, U)| \leq 1.$$

The integrated marked coefficient estimate (103), with singleton allowances 2 and 1, then bounds the complete moment by

$$v^M |\,\text{moment}\,| \leq \exp\bigl(|\Lambda_m|\mathcal{V}_m + 5M\bigr).$$

There are M copied cells. Write $S_0 = \exp(|\Lambda_m|\mathcal{V}_m/M + 5)$, the cell price. Taking the Mth root gives the centered root bound S_0/v. This is an estimate for the complete replicated difference; no estimate on its individual mixed summands is substituted.

173.3 Combining the comparison paths

The one-scale transport is stable under affordable finite linear combinations of its single-source paths. Explicitly, if path j has positive opening price v_j and coefficient a_j, the condition

$$\sum_j w|a_j|/v_j \leq 1 \tag{839}$$

preserves every common normalized cost bound. All paths use the same multiplicative weight evolution: at scale i the opening prices v_j and w are multiplied by the same positive factor. Thus the fractions in (839) are unchanged at each intermediate scale. To verify the cost bound, if $v_j \sum_X |F_{j,X}| \leq C$, then

$$w \sum_X \left| \sum_j a_j F_{j,X} \right| \leq \sum_j \frac{w|a_j|}{v_j} v_j \sum_X |F_{j,X}| \leq C.$$

The same calculation applies to singleton values, elementary transition costs and response majorants. Sum the dominated indexed arrays using their summed majorants. The transition identities are linear, so remain exact. Applying the marked transport successively proves the claim for the whole interval.

Use the base difference path with coefficient 1 and price $2w$, and the coordinate paths with coefficients $-s_j$ and price $8Ew$. Equation (837) gives

$$\frac{w}{2w} + \sum_j \frac{w|s_j|}{8Ew} \leq 1.$$

At scale k, the reference terms cancel separately in the ordinary and flagged profiles, because $-b+\sum_j(-s_j)(-b_j)=0$. For a conjugating copy, use the conjugate of every coefficient a_j in the portfolio; for an ordinary copy use a_j itself. Their absolute values coincide, so the cost proof is unchanged. The cancellation in a conjugating copy is the conjugate of the displayed identity. Thus cancellation holds for every copied profile. The resulting ordinary field is the one for Q plus ℓ; the flagged profile is exactly the one for Q. Introducing ℓ into the marked representation of f is multiplication by its squarefree source exponential. The identity (838), with $O=-f$ and $F=\ell$, therefore identifies the complete coefficient with the replicated moment of $f+r$. The transport and complete-moment estimate give

$$\mathcal{R}(f+r)\le S_0/w,$$

where $\mathcal{R}$ is the centered root seminorm.

For each coordinate path the same argument, starting from the terminal reference field and inserting ℓ_j at scale k, gives

$$\mathcal{R}\big(\mathcal{B}_{k,N}\ell_j-\mathcal{O}(b_j)\big)\le S_0/(8Ew).$$

Here $\mathcal{O}(b_j)$ denotes the fine observable attached to that bare label by the calibrated terminal identification. This application requires the terminal reference representation and its singleton profile identity on the same history. They remain among the analytic inputs to the matched construction. The chamber seminorm inequality, backward-transfer linearity and (837) imply

$$\mathcal{R}\big(-r-\mathcal{O}(b)\big)\le 4E\frac{S_0}{8Ew}=\frac{S_0}{2w}.$$

Finally $f-\mathcal{O}(b)=(f+r)+(-r-\mathcal{O}(b))$ lies in the common chamber class, so its centered root is at most $2S_0/w$. The matched reference observable has quota at most $2E$, as proved above. These are exactly the two bounds used in the cutoff matching argument.

174 Capturing a cluster and paying for a cutoff match

The local match used in Section 175 is constructed before any continuum limit. One follows a product of local observables down through the scales, merging groups whenever their parent sites cease to be separated. The bookkeeping must be independent of the number of replicated cells. Otherwise it would be unusable in the centered root estimate. We give the geometric selection and the scale accounting explicitly, and state where the remaining local RG estimates enter.

174.1 Regrouping without a growing arity cost

Let a cluster have $h\ge 1$ marked sites and distinguish one of its marks. At a given scale, call sites separated if their periodic maximum-coordinate distance is at least B^2, where $B\ge 1$ is the purchased support radius. Among the subsets containing the distinguished mark choose one of maximal cardinality with separated sites. Every omitted mark lies at distance less than B^2 from a chosen mark: otherwise it could be added. Assign each omitted mark to one such representative, and assign each chosen mark to itself. This defines a partition with a chosen representative in every part. If the original sites were not separated, the number of parts strictly decreases.

Copies in different centered cells have the same relative geometry, up to translations and signed coordinate transformations, which preserve distance. Use the same partition in each copy. Cross-cell separation and separation of the chosen representatives in the home cell then give separation of all representatives. In particular the number of regroupings is at most $h-1$, regardless of how many cells are copied.

The cost of connecting a mark to its representative can be charged one scale before their parents meet. To check the geometry, let the block ratio be L, let K be the parent's support radius and B the child's, and let x, y be two child sites whose parents have distance less than K^2. Each child is at distance less than $L/2$ from the embedded parent. The triangle inequality gives

$$\operatorname{dist}(x, y) < L(K^2 + 1) \le L(K + B + 1)^2 =: c.$$

A pair at distance at most cB can be covered by at most $(2c+1)^4$ B-blocks: both points lie in the radius-cB halo of one point, which is tiled by at most $2c + 1$ blocks per coordinate. Since $B \ge 1$, the preceding pair therefore has covering number at most

$$(2c + 1)^4 \le 16(1 + c)^4. \tag{840}$$

This is the halo charged to the individual grouping operation. It does not assert that the entire cluster lies in a single small halo at every intervening scale.

Here are the two local RG estimates used in the induction. A group has a positive weight W measuring the allowed size of its marked representation. If its parents stay separated, descent through a block of ratio L changes its weight to W/L^4. If p groups merge into $q < p$ groups at a scale with connecting-cover bound ℓ, their new weights can be taken to be

$$\widetilde{W}_A = \frac{1}{8} \prod_{r \in A} (\lambda W_r), \qquad \lambda = \frac{\delta^2}{h} e^{-(\nu+1/2)\ell}. \tag{841}$$

Here ν is the purchased support exponent and $\delta > 0$ the local smallness parameter. These statements also retain the exact marked polynomial after regrouping, its covariance between copies, its integrability, and its permitted local representation. The analytic proofs of these two local representation estimates remain construction inputs. In particular their weight rules alone are not a proof that a marked representation exists.

Choose one $0 < \alpha \le 1$ no larger than every grouping factor λ that will be needed. If j is the initial scale, i the current scale and $s \le h$ the current number of groups, put

$$C(i, s) = \left(\frac{a_j}{a_i}\right)^{4h} (\alpha/8)^{h(h-s)} \prod_{r=1}^{h} W_r^{\text{initial}}.$$

We claim that the product of the current group weights is at least $C(i, s)$. This is equality initially. For an ordinary descent,

$$C(i, s) = L_i^{-4h} C(i + 1, s) \le L_i^{-4s} \prod_r W_r,$$

since $s \le h$ and $L_i \ge 1$. For a genuine grouping, $q < p \le h$ and $\alpha/8 \le 1$ imply

$$C(i, q) \le C(i, p)(\alpha/8)^h.$$

Thus

$$8^q C(i, q) \le \alpha^h C(i, p) \le \lambda^p \prod_r W_r.$$

We used $8^q \le 8^h$ and $\alpha^h \le \alpha^p \le \lambda^p$. The last expression is exactly the product of the new numerators in (841), because the groups partition the old marks. This proves the induction after the subsequent descent as well. The regrouping of marked polynomials commutes with descent: it is a fixed linear map on a finite coefficient space, so it commutes with integration against the

transition kernel. Its composition law is the composition law of the partitions. These observations keep the induction on the original exact product.

The initial single-field estimates give weights $W_r^{\mathrm{initial}} = \delta_j \eta_j p_r(j)/8$, where $p_r(j) > 0$ is the paid weight of the rth field. At the final parent scale $m+1$ merge all remaining groups. Their number s is at most h, and the last grouping factor is also at least α. The terminal merge uses the weight $\lambda^s \prod_r W_r$ without the factor $1/8$ reserved at intermediate regroupings; it supplies the final local projection directly. Since $\alpha^h \le \lambda^s$ and $C(m+1,0) \le C(m+1,s)$, the resulting weight is at least

$$X = \alpha^h \left(\frac{a_j}{a_{m+1}}\right)^{4h} (\alpha/8)^{h^2} (\delta_j \eta_j/8)^h \prod_{r=1}^{h} p_r(j). \tag{842}$$

The final local projection estimate produces a degree-q coefficient whose coefficient-size and remainder norm are each bounded by

$$P = a_{m+1}^q \, \mathcal{B}_m \frac{4}{X}. \tag{843}$$

Here $\mathcal{B}_m$ is the purchased local projection bound. This projection estimate, together with the marked-representation estimates above, is still an analytic input. The preceding argument proves the precise scale cost it must be given; no factor depends on the number of copies.

174.2 A single subpower allowance for all intermediate levels

The halo estimate ensures that each grouping factor has zero logarithmic rate. Explicitly the purchased bound is $(\nu_{i+1}+1)\ell_i \le 16\sqrt{\widehat{\beta}_i}$, while $\sqrt{\widehat{\beta}_i}/s_i \to 0$ and $\log \delta_{i+1} = o(s_i)$. Taking logarithms of (841) therefore gives $\log \lambda_i = o(s_i)$ for fixed h. These are schedule and running-coupling estimates; their production is a separate input.

This pointwise-in-scale fact also gives a uniform allowance for all previous scales. If $u_i > 0$ and $\log u_i = o(s_i)$ for an increasing $s_i \to \infty$, set $\alpha_j = \min(1, u_0, \dots, u_j)$. Then $\alpha_j \le 1$, $\alpha_j \le u_i$ for $i \le j$, and $\log \alpha_j = o(s_j)$. Indeed, for any $\epsilon > 0$, eventually $u_i \ge e^{-\epsilon s_i} \ge e^{-\epsilon s_j}$ for $i \le j$ in the tail. The positive minimum of the finite initial segment is also eventually at least $e^{-\epsilon s_j}$. Hence $e^{-\epsilon s_j} \le \alpha_j \le 1$. Apply this to the grouping factors. Arbitrarily many intermediate scales thus require only one subpower allowance for the fixed cluster arity.

174.3 The two output prices

Take sequences of levels k, m, j of logarithmic rates b, a, c relative to $t_n \to \infty$, with $0 < b < a < c$. Here k sets the centered test scale, m the matching scale, and j the insertion grid. All level sequences tend to infinity. Let the field degrees be d_r, put $w = \sum_r d_r$, and define

$$\tau = cw + 4h(c-a).$$

The calibrated single-field weight has the form $p_r(j) = a_j^{d_r} \mathcal{W}_r(j)$, with $\log \mathcal{W}_r(j) = o(s_j)$. All factors in (842) other than the displayed mesh powers have zero logarithmic rate. The gentle schedule gives rate a also at $m+1$. Consequently X has rate $-\tau$.

Set

$$E = 4(1+\zeta_m)(2P), \qquad V = 2E, \qquad z = \exp\big(((q+\tfrac{1}{16})(a-b) - \tau)t_n\big). \tag{844}$$

Equations (843) and (844) give rate $\tau - qa$ for both E, V. The final comparison of the local coefficient with the normalized reference fields has the following quantitative form. Provided its

positive matching allowance z satisfies

$$\mathcal{M}(z;k,m) \le \delta_m X/(16\mathcal{B}_m), \tag{845}$$

$$32(1+\zeta_m) z\, a_k^{-q} E/G(m,k) \le \vartheta_q(k)/2, \tag{846}$$

it produces a reference label $Q_{N,p}$ with quota at most V and centered discrepancy price at most

$$A = 2\mathcal{S}(k)/z. \tag{847}$$

Here $G(m,k) = \exp((s_m - s_k)/8)$, ϑ_q is positive with zero logarithmic rate, and $\mathcal{M}$ is the allowance transported from the centered scale to m. Its bound is

$$\mathcal{M}(z;k,m) \le z(a_m/a_k)^{q+1} \frac{\vartheta_q(m)}{\vartheta_q(k)}.$$

If the centered scale has first been opened to a finer admissible level, use that opened level in these formulas. The opening retains rate b, as checked below. The actual comparison producing (847), including its representation and reference-field identification, remains to be proved in the construction part; these displayed inequalities specify exactly the budget it needs.

Both payment conditions hold eventually by a strict comparison of rates. The right side of (845) has rate $-\tau$, whereas the upper bound for its left side has rate

$$(q+\tfrac{1}{16})(a-b) - \tau - (q+1)(a-b) = -\tau - \tfrac{15}{16}(a-b) < -\tau.$$

The left side of (846) has rate

$$(q+\tfrac{1}{16})(a-b) - \tau + qb + (\tau - qa) - \tfrac{1}{8}(a-b) = -\tfrac{1}{16}(a-b) < 0,$$

while its positive right side has rate zero. If positive sequences have unequal logarithmic rates, division and taking logarithms shows that the one with the smaller rate is eventually smaller. This justifies both comparisons, including all their fixed constants. Since $\mathcal{S}$ has upper rate zero, (847) has upper rate $\tau - (q+1/16)(a-b)$, as required in Input 175.1.

Here is the opening check. Write $K(k) \ge k$ for the opened level, $u = a_k/a_{K(k)} \ge 1$, and $B_{K(k)} \ge 1$ for its positive vacuum budget. The schedule ensures, for every $\epsilon > 0$,

$$u^4 B_{K(k)} \le \epsilon s_{K(k)} \quad \text{eventually in } k.$$

Since $s_{K(k)} - s_k = \log u \le u \le u^4 B_{K(k)}$, we have $s_k/s_{K(k)} \to 1$. Opening a sequence of rate b therefore retains that rate. The actual centered factor is

$$\mathcal{S}(k) = \exp\bigl(u^4 t_{K(k)} + 5\bigr),$$

where $t_{K(k)} \le B_{K(k)}$ eventually. Its logarithm divided by t_n is at most $\epsilon s_{K(k)}/t_n + 5/t_n$. Taking the upper limit and then $\epsilon \downarrow 0$ proves its upper rate zero.

174.4 Keeping the match in the continuum limit

For fixed intermediate levels and a fixed cluster on the j-grid, require

$$8 \max_r |x_r| < a_{m+1}, \qquad |x_r - x_s| > 2a_j B_j^2 \quad (r \ne s).$$

Cross-cell separation persists during every intervening descent. The initial room condition is $(B_{m+1}^2+1)a_{m+1} \le a_k$, which follows eventually from subpower B_{m+1} and $a > b$. At an intermediate level $i \ge m+1$, put $L = a_{m+1}/a_i$. Distinct parent cell centers at level $m+1$ have lattice separation at least $a_k/a_{m+1} \ge B_{m+1}^2 + 1$. Their embeddings at level i therefore have separation at least $L(B_{m+1}^2+1)$. The children each deviate from their embedded parent by less than $L/2$. The triangle inequality leaves separation greater than LB_{m+1}^2. The purchased support propagation bound $B_i^2 \le LB_{m+1}^2$ then gives the required separation at level i. This propagation bound is one of the local RG inputs retained above. The argument applies simultaneously to every pair of marks belonging to different copied cells.

For sufficiently large periods there is no wrapping in this finite configuration. Its points have the required periodic separation and have common parent zero at level $m+1$: each is inside half the parent block by the first bound and the maximum-coordinate estimate. For sufficiently fine cutoff N, the fixed stencils of the cluster fields and of every degree-q field fit in the centered positive chamber. The cutoff comparison can consequently be applied on the ordered tail, first in period and then in N.

Choose its labels $Q_{N,p}$ on that tail and set them to zero elsewhere. Their quotas are bounded by the same V. Equation (196) bounds their reference norms by a constant depending on the fixed matching level, so finite-dimensional compactness gives a convergent family on a finer nontrivial filter. Every eventual discrepancy and quota bound persists on that same filter. Section 52 converts the discrepancy bound to all character phases, and Section 53 gives the physical low-degree distances and the whole-label bound. These are all statements about the same limiting label Q.

Finally the geometric requirements hold for the grid clusters of Section 175. If their scale is x with $a < x < c$, their radius has rate $-x < -a$, so they fit strictly inside the $m+1$ block. Their pairwise separation is eventually larger than $e^{-(x+\epsilon)t_n}$, while $a_j B_j^2$ has rate $-c$ because B_j is subpower. Choosing $x + \epsilon < c$ proves the separation condition. The fixed support-radius room at the centered scale is also automatic: a fixed radius plus a subpower truncation radius, multiplied by a_m, has rate $-a < -b$. It therefore fits inside the centered scale a_k. This completes the accounting and extraction of the local matching input, conditional on the explicit analytic representation and comparison estimates above.

175 From a matched local field to the raw OPE

The raw comparison in Input 176.1 is obtained in three steps. At a fine lattice configuration the product is replaced by a single local field of a sufficiently high degree. The finite tests defining the prescription then identify its low-degree coefficients. Finally, derivatives with respect to the cluster positions extend the estimate from the lattice to every separated configuration. The last step uses the filtration: taking a derivative increases the field degree by one.

We use upper rates as in Section 176, with any sequence $t_n \to \infty$. A cluster $X_n = (x_{i,n})_{i \in I}$ is at scale $x > 0$ if $\max_i |x_{i,n}|$ has upper rate $-x$ and, for every $\delta > 0$,

$$|x_{i,n} - x_{j,n}| > e^{-(x+\delta)t_n} \quad (i \ne j) \quad \text{eventually.}$$

These conditions tolerate factors with zero logarithmic rate. In particular, they include every cluster of radius Mr_n and separation ηr_n, where $t_n = -\log r_n$. Spectators in this section remain a fixed positive distance from the origin and one another, although their positions and phases can vary with n.

175.1 Intermediate scales and generic grids

Write a_k for the mesh and $s_k = \log(1/a_k)$ for its logarithmic scale. The following elementary consequence of the mesh schedule will be useful. Suppose s_k is strictly increasing, tends to infinity, and $s_{k+1}/s_k \to 1$. Then for every $d > 0$ and every $t_n \to \infty$ there are integers $j_n \to \infty$ such that $s_{j_n}/t_n \to d$. Indeed choose the first j_n with $s_{j_n} > dt_n$. Eventually $j_n > 0$, and

$$s_{j_n-1} \le dt_n < s_{j_n}, \qquad 1 < \frac{s_{j_n}}{dt_n} \le \frac{s_{j_n}}{s_{j_n-1}} \longrightarrow 1.$$

The zero-rate variant follows by replacing dt_n with $\sqrt{t_n}$. The gentle-mesh condition used in the construction gives exactly the ratio property, since $s_{k+1} - s_k = \log L_k = o(s_k)$. The proof here uses that condition; its production for the chosen trajectory remains part of the construction estimates.

A configuration is *coordinate-generic* if, for each of the four coordinates separately, the coordinates of its distinct points are pairwise different. Such configurations can approximate arbitrary ones at the mesh scale with a bound independent of their location. To see this, enumerate I by $0, \dots, N-1$, set $K = N + 1$, and for mesh $h > 0$ put

$$y_{i,\mu} = h\left(K\left\lfloor \frac{x_{i,\mu}}{hK} \right\rfloor + i\right), \qquad \mu = 1, \dots, 4.$$

Different point indices have different residues modulo K, so each coordinate is injective. The floor inequality and $0 \le i < K$ give $|y_{i,\mu} - x_{i,\mu}| \le hK$, hence

$$\max_i |y_i - x_i| \le 2Kh. \tag{848}$$

This is the empty-exterior version of the grid construction; no claim is made that it preserves coordinate distances from fixed exterior points. Only ordinary Euclidean separation of the full configuration is needed below.

If $h_n = e^{-dt_n + o(t_n)}$ with $d > x$, the rounded cluster and every configuration in the closed ball

$$B_n = \{Z : \|Z - X_n\| \le \|Y_n - X_n\|\}$$

remain at scale x, using the maximum of the Euclidean norms on the configuration space. Here is the separation check. Given $\delta > 0$, choose $0 < \epsilon < \delta$ with $x + \epsilon < d$ and use the original separation at exponent $x + \epsilon$. The displacement has strictly smaller exponential order, so eventually twice its norm is at most half that separation. Thus the perturbed separation is at least $\frac{1}{2}e^{-(x+\epsilon)t_n} > e^{-(x+\delta)t_n}$. The radius bound follows from the triangle inequality. Fixed exterior clearances also persist, because these clusters tend to the origin.

175.2 Reading a high-degree match

We state the still needed quantitative local matching input precisely. All fields and tests in it belong to the same extracted hierarchy and the prescription already constructed in Section 50. For a cluster label list p, write $w(p) = \sum_i \deg p_i$ and

$$\tau(p; a, d) = d\, w(p) + 4|I|(d - a).$$

The second term is the four-dimensional volume cost of the local matching estimate.

Input 175.1 (Local matching estimates)**.** *Fix $0 < b < a < x < d$, a nonempty cluster label list p, a coordinate-generic configuration sequence X_n on meshes of logarithmic rate d, at cluster scale*

x, and an integer $q \geq 0$. There are degree-q local labels Q_n, and nonnegative budgets A_n, V_n, with upper rates

$$A_n : \quad \tau(p; a, d) - (q + \tfrac{1}{16})(a - b), \qquad V_n : \quad \tau(p; a, d) - qa,$$

with the following properties. The discrepancy between the cluster product and Q_n inserted at the origin, tested against any fixed spectator label list at fixed positive clearance, has upper rate at most that of A_n, uniformly along every choice of phase and exterior configuration. The same is true for the mixed state and the finite tests of the prescription. The labels obey

$$\|Q_n\| \leq J_n V_n, \qquad J_n \text{ has upper rate } qa. \tag{849}$$

For each $1 \leq m \leq q$ there are $z_{m,n}$ in the label space of degree q whose physical classes lie in the included space E_{m-1}, and

$$\|Q_n - z_{m,n}\| \leq F_{m,n} V_n, \qquad F_{m,n} \text{ has upper rate } (q - m)a. \tag{850}$$

All norms here are fixed finite-dimensional label norms. The original labels Q_n, before passing to their physical classes, satisfy these bounds. The estimates hold for every fixed cluster label list, including lists formed by differentiating its fields.

In the source the same Q_n arises as a finite-dimensional sublimit of cutoff matches, retaining their local discrepancy and quota bounds. The individual channel estimate below is essential: a bound on the whole degree-q label would lose the powers that distinguish low and high channels. Input 175.1 has not yet been derived here from the finite Wilson law.

Let $f_{\alpha,n}$ denote the raw coefficient of the product and let $v_{\alpha,n} = b_\alpha([Q_n])$ be the coefficient of its matched field, where b_α is the fixed functional of the prescription and $m = |\alpha| \leq q$. Then

$$v_{\alpha,n} \text{ has upper rate } \tau(p; a, d) - ma. \tag{851}$$

For $m = 0$, apply the fixed linear functional to (849). For $m > 0$, coherence of the prescription implies $b_\alpha([z_{m,n}]) = 0$: this detector kills every field of degree less than m. Hence

$$|v_{\alpha,n}| = |b_\alpha([Q_n - z_{m,n}])| \leq \|b_\alpha \circ \pi_q\| \, \|Q_n - z_{m,n}\|,$$

where π_q is the quotient map. Combining (850) with the rate of V_n proves (851).

Each raw coefficient is a finite linear combination of exterior tests. Applying the discrepancy part of Input 175.1 to those tests therefore gives

$$f_{\alpha,n} - v_{\alpha,n} \text{ has upper rate } \tau(p; a, d) - (q + \tfrac{1}{16})(a - b). \tag{852}$$

For a specified output degree m, choose $q \geq m$ so large that $(q + \frac{1}{16})(a - b) \geq (m + 1)a$. Then (851) and (852) show that $f_{\alpha,n}$ has upper rate $\tau(p; a, d) - ma$.

The analogous remainder argument retains the entire product correlation. Let ℓ_n be the functional obtained by inserting a field at the origin beside the chosen spectators. Assume, as furnished by the separated correlation estimate, that $\ell_n(u)$ has upper rate zero for every fixed field u and every phase sequence. To truncate at D, choose $q \geq D$ with $(q + \frac{1}{16})(a - b) \geq (D + 1)a$. Exact reconstruction of $[Q_n]$ by its prescription coefficients gives

$$\begin{aligned} S_{\chi_n}(p, z; X_n, Y_n) - \ell_n(H_D(X_n, p; 0)) = {} & \big[S_{\chi_n}(p, z; X_n, Y_n) - \ell_n([Q_n])\big] \\ & + \sum_{|\alpha| \leq D} (v_{\alpha,n} - f_{\alpha,n}) \ell_n(u_\alpha) \\ & + \sum_{D < |\alpha| \leq q} v_{\alpha,n} \ell_n(u_\alpha). \end{aligned} \tag{853}$$

The first term and each low-channel difference have upper rate at most $\tau(p;a,d)-(D+1)a$. Each high channel has that rate as well, by (851), because $a>0$ and $|\alpha|\ge D+1$. The sums are finite. This proves the raw grid remainder estimate with upper rate $\tau(p;a,d)-(D+1)a$. The matching label may depend on q; the raw coefficients and the truncation on the left do not.

175.3 Interpolation by descending through the grading

We give the interpolation argument explicitly. Suppose functions $h_n(p,X)$ are indexed by finite field lists of the same length, with integer total degree $w(p)$. Assume that for every scale-x sequence:

1. $h_n(p,X_n)$ has a rough upper rate $xw(p)+c$;
2. on every coordinate-generic grid sequence of logarithmic mesh rate $d>x$, it has upper rate $dw(p)+e$;
3. it is eventually differentiable along each such sequence, with identity

$$\partial_{i,\mu}h_n(p,X_n)=h_n(p',X_n).$$

Here p' is obtained by differentiating the corresponding field, so $w(p')=w(p)+1$.

The constants c,e are independent of p; they may depend on fixed truncation and exterior data. Then at every scale-x sequence the improved upper rate is $dw(p)+e$.

Choose an integer $N\ge 0$ with $(d-x)N\ge c-e$. We prove by induction on $k\ge 0$ that the improved rate holds whenever $N\le w(p)+k$. For $k=0$ the rough rate is already no larger than the improved one. For the induction step round X_n to Y_n by (848). Every choice $Z_n\in B_n$ stays at scale x. The induction hypothesis applied to the differentiated lists shows that each coordinate derivative at Z_n has upper rate $d(w(p)+1)+e$. In the fixed finite-dimensional configuration space,

$$\|Dh_n(p,Z_n)\|\le\sum_{i,\mu}|\partial_{i,\mu}h_n(p,Z_n)|,$$

so the derivative norm has that rate too.

These bounds and differentiability hold eventually uniformly on B_n. For any specified positive slack, failure would give infinitely many indices with a bad point; choosing a bad point there and the center elsewhere produces a sequence $Z_n\in B_n$ contradicting the corresponding eventual assertions. The mean-value inequality on the convex ball thus gives

$$|h_n(p,Y_n)-h_n(p,X_n)|\le\sup_{Z\in B_n}\|Dh_n(p,Z)\|\,\|Y_n-X_n\|.$$

The displacement has upper rate $-d$, so the difference has upper rate $dw(p)+e$. The grid value has the same rate, which proves the induction step. Taking $k=N$ covers every field list. Only finitely many derivative levels are used for any one application, although the assumptions must be available at every finite degree.

For clarity, the mean-value inequality here needs no continuity of the derivative. On each segment in B_n, differentiability implies continuity, and the scalar mean-value theorem applied to real projections of the complex-valued function bounds its increment by the derivative bound times the segment length. The finite-dimensional operator estimate above follows by expanding a vector in its coordinate basis and applying the triangle inequality.

175.4 Removing the intermediate-scale losses

The differentiability identities needed for this interpolation are exactly the derivative-field identities of the hierarchy. They persist under finite linear combinations of detector tests and under subtraction of the finite raw truncation. At a separated configuration its neighborhood remains separated, so the totalized values away from that neighborhood play no role in differentiation.

Suppose the rough separated correlation estimates give upper rate $xw(p)$ for the cluster with fixed-clearance spectators, uniformly in the phase. They give the same rate for each raw coefficient, since its detector list is fixed and finite. A finite truncation also has rough rate $xw(p)$: its coefficient has that rate and each fixed single-field spectator insertion has rate zero. Therefore both the raw coefficient and the raw discrepancy satisfy the rough interpolation bound with $c = 0$.

For a coefficient of degree m, the grid bound just proved is

$$dw(p) + 4|I|(d - a) - ma.$$

Use interpolation with $e = 4|I|(d - a) - ma$ to obtain this bound at all scale-x clusters. For a remainder through degree D, use instead $e = 4|I|(d - a) - (D + 1)a$. These offsets do not change when a cluster field is differentiated, as required by the induction.

Now set $a = x - \epsilon$, $d = x + \epsilon$, and $b = a/2$, with $0 < \epsilon < x$. The mesh schedule supplies all three intermediate levels. The resulting coefficient upper rate is

$$(w(p) - m)x + \epsilon\big(w(p) + 8|I| + m\big),$$

and the remainder upper rate is

$$(w(p) - D - 1)x + \epsilon\big(w(p) + 8|I| + D + 1\big).$$

Given any desired positive slack above the limiting exponent, choose ϵ small enough that the displayed excess is less than half the slack, then use the remaining half in the upper-rate bound. This proves rates $(w(p) - m)x$ and $(w(p) - D - 1)x$, respectively. No uniformity of the intermediate constructions as $\epsilon \to 0$ is needed.

Translating the cluster center to the origin preserves all distances and the complete correlation. The raw coefficients use relative positions, so the same estimates hold at arbitrary centers. If the cluster is empty, its product is the identity. Exact reconstruction of that degree-zero field makes every raw truncation exact; positive-degree detectors annihilate it, and degree-zero coefficients are constant. Thus this case also satisfies the stated rates. Taking $x = 1$ proves Input 176.1, conditional on Input 175.1, the rough separated correlation estimates, and the derivative identities already described. The remaining task is to produce those matching and correlation estimates for the actual Wilson trajectory.

176 Transferring the OPE estimates to homogeneous fields

The raw prescription and its covariant homogeneous version need not agree exactly on a product of fields. Their difference must therefore be estimated before the homogeneous coefficients can be used in the expansion. We give that transfer, retaining the complete correlations, including all unit and disconnected terms. The quantitative estimate for the raw prescription remains the construction input identified below.

Fix cluster fields $\boldsymbol{v} = (v_i)_{i \in I}$, with $w = \sum_i \deg v_i$. For a sequence of positive numbers $r_n \to 0$, set $t_n = -\log r_n \to \infty$. Say that a sequence f_n has upper rate a if for every $\delta > 0$,

$$\|f_n\| \le e^{(a+\delta)t_n} = r_n^{-a-\delta} \quad \text{for all sufficiently large } n.$$

This notation permits zero values. Finite sums preserve a common upper rate, products add rates, and fixed linear maps preserve rates. For sums, first use half the desired slack and then absorb the finite number of terms into $e^{\delta t_n/2}$. The same device absorbs a fixed operator norm. Products follow by dividing the slack between the factors. In particular any bounded sequence has upper rate zero.

If $\ell_n : E \to \mathbb{C}$ is a sequence of linear functionals on a fixed finite-dimensional space, and $\ell_n(e)$ has upper rate b for every fixed e, then applying it to vectors u_n of upper rate a gives upper rate $a+b$. To prove this, expand u_n in a fixed basis. Each coordinate has rate at most a, each basis evaluation has rate at most b, and the finite sum has rate at most $a+b$. The identical finite-basis argument works for multilinear forms with several varying inputs. We will use it for bounded rotated detector labels.

Input 176.1 (Raw product comparison)**.** *For every fixed cluster label list, every $M \geq 0$, $\eta > 0$, every positive sequence $r_n \to 0$ and configurations with*

$$|x_{i,n} - o_n| \leq M r_n, \qquad |x_{i,n} - x_{j,n}| \geq \eta r_n \quad (i \neq j),$$

the raw coefficients of (173) *have upper rate $w - |\alpha|$. For any fixed spectator label list $\boldsymbol{z}$, any $c > 0$, any positions satisfying*

$$|y_{j,n} - o_n| \geq c, \qquad |y_{j,n} - y_{k,n}| \geq c \quad (j \neq k),$$

and any sequence of phases or mixed-state choices χ_n, the following assertions hold. For every D, the raw discrepancy

$$S_{\chi_n}(\boldsymbol{v}, \boldsymbol{z}; \boldsymbol{x}_n, \boldsymbol{y}_n) - \ell_{\chi_n, D, o_n, \boldsymbol{y}_n, \boldsymbol{z}}(H_D(\boldsymbol{x}_n, \boldsymbol{v}; o_n)) \tag{854}$$

has upper rate $w - D - 1$. For every fixed $u \in E_j$, the separated insertion $\ell_{\chi_n, j, o_n, \boldsymbol{y}_n, \boldsymbol{z}}(u)$ has upper rate zero. These statements hold for every finite spectator list, including empty lists, and for conjugated cluster fields as well. The same underlying hierarchy and physical inclusions are used throughout.

In the source these are the fixed-exterior specialization of the raw coefficient and full-discrepancy estimates; the stronger statements allow several positive logarithmic scales. This input includes uniform control along arbitrarily moving configurations and phases. A bound for each fixed configuration with a threshold depending on that configuration would be insufficient.

176.1 The averaged readout error

Fix a raw truncation degree D and an output degree m, which may differ. Abbreviate $h_n = H_D(\boldsymbol{x}_n, \boldsymbol{v}; o_n)$. We claim that

$$C_m(\boldsymbol{x}_n, \boldsymbol{v}; o_n) - L_{mD} h_n \quad \text{has upper rate } w - D - 1. \tag{855}$$

First consider only the rotation average. Let

$$Z_m(K; \boldsymbol{x}, \boldsymbol{v}; o) = R_m(K) H_m(K^{-1}\boldsymbol{x}, R(K^{-1})\boldsymbol{v}; K^{-1}o)$$

be its integrand. On the common separated region, expanding the finite tests gives the exact identity

$$\begin{aligned} &Z_m(K; \boldsymbol{x}, \boldsymbol{v}; o) - Z_m(K; (o), (h); o) \\ &= \sum_{|\alpha| \leq m} \sum_{\lambda} b_{\alpha,\lambda} \big[S_{\mathrm{mix}}(\boldsymbol{v}, R(K)\boldsymbol{w}_\lambda; \boldsymbol{x}, o + K\boldsymbol{z}_\lambda) - S_{\mathrm{mix}}(h, R(K)\boldsymbol{w}_\lambda; o, o + K\boldsymbol{z}_\lambda) \big] R_m(K) \iota_{|\alpha|,m} u_\alpha. \end{aligned} \tag{856}$$

Here each sum over λ is restricted to the finite support of b_α. To verify the identity, write each raw coefficient as its exterior correlation at relative origin. Rotate the entire correlation by K and translate by o. The cluster positions and labels become $\boldsymbol{x}, \boldsymbol{v}$, and the detector positions and labels become exactly those displayed. Subtract the same evaluation with the single field h at o, and use linearity in that slot.

Take $h = h_n$ and any sequence $K_n \in O(4)$. All the rotated detector positions have a common positive separation from o_n and from one another: these are the fixed finite nonzero detector distances, preserved by rotations. All rotated detector labels and output vectors are bounded, by continuity of the representations on the compact group. Apply Input 176.1 to each fixed basis choice of detector labels in (856), then use the multilinear finite-basis argument above to allow their bounded variation with K_n. Every summand and hence their finite sum has upper rate $w - D - 1$.

This statement for every rotation sequence implies a uniform estimate in K at each sufficiently large n. Indeed, for a fixed slack, if there were a violating rotation at infinitely many indices, choose one at each such index and any rotation at the remaining indices. The resulting sequence would contradict the rate estimate just proved. Thus the norm of the integrand difference is eventually bounded uniformly by $e^{(w-D-1+\delta)t_n}$. Both integrands are continuous on the compact group once the shrinking cluster lies in $\mathcal{U}_m$. Integrating against probability Haar measure gives the same bound for the difference of their averages.

Apply the same argument to the conjugated part. Its single comparison field is $J_D h_n$, rather than the raw truncation recomputed from the conjugated cluster. This distinction causes no extra assumption. Conjugating a complete correlation conjugates all its labels, and the raw discrepancy is linear in each spectator slot. Hence the discrepancy with cluster $J\boldsymbol{v}$, comparison field $J_D h_n$, and arbitrary spectator labels is the complex conjugate of the original discrepancy with conjugated spectator labels. That original discrepancy is covered by the input and the bounded-label argument. It has the same norm and upper rate. Output conjugation is a fixed real-linear map on a finite-dimensional space and preserves the rate. Adding the two averages and dividing by two proves (855); translation invariance identifies its centered singleton readout with $L_{mD} h_n$. Applying the fixed projection K_m yields

$$F_m(\boldsymbol{x}_n, \boldsymbol{v}; o_n) - P_{mD} h_n \quad \text{has upper rate } w - D - 1. \tag{857}$$

176.2 Coefficient and remainder bounds

Set $D = m$ in (857). Applying P_{mm} to the raw shadow H_m kills every term of degree less than m, by inclusion compatibility and (180). The remaining finitely many terms consist of degree-m raw coefficients, of upper rate $w - m$, times fixed vectors. Their sum has that rate. The difference in (857) has the smaller rate $w - m - 1$, so F_m has upper rate $w - m$. Taking its fixed channel coordinates gives the same rate for each coefficient in (182).

For the remainder, retain the raw truncation $h_n = H_D$ and abbreviate the spectator functional at degree j by $\ell_{n,j}$. The exact reconstruction identity in (179) and compatibility of spectator insertions give

$$\ell_{n,D}(h_n) = \sum_{j \le D} \ell_{n,j}(P_{jD} h_n).$$

Consequently the final homogeneous discrepancy is exactly the raw discrepancy (854) plus

$$\sum_{j \le D} \ell_{n,j}\big(P_{jD} h_n - F_j(\boldsymbol{x}_n, \boldsymbol{v}; o_n)\big). \tag{858}$$

Each vector difference has upper rate $w - D - 1$ by (857). The insertion functional has rate zero on every fixed vector by the input. The finite-dimensional evaluation argument therefore gives rate $w - D - 1$ for every term of (858), and for their finite sum. Adding the raw discrepancy proves the same rate for the final discrepancy, along any prescribed configuration and phase sequence.

Theorem 176.2. *Under Input 176.1, the coefficients of* (182) *satisfy the public separated OPE bounds. Specifically, for every $\epsilon > 0$, fixed channel (m, α) and fixed $M \geq 0$, $\eta > 0$, at all sufficiently small $r > 0$,*

$$|c_{\boldsymbol{v}}^{m,\alpha}(\boldsymbol{x}; o)| \leq r^{m-w-\epsilon}$$

uniformly over $|x_i - o| \leq Mr$ and $|x_i - x_j| \geq \eta r$. For each finite D, fixed spectator labels and $c > 0$, one also has

$$\left| S_\chi(\boldsymbol{v}, \boldsymbol{z}; \boldsymbol{x}, \boldsymbol{y}) - \sum_{j \leq D} \sum_{\alpha} c_{\boldsymbol{v}}^{j,\alpha}(\boldsymbol{x}; o) S_\chi(e_{j,\alpha}, \boldsymbol{z}; o, \boldsymbol{y}) \right| \leq r^{D+1-w-\epsilon},$$

uniformly over the same clusters, all spectators with center and mutual clearance at least c, and all phases and the mixed state.

Proof. The preceding arguments prove these inequalities eventually along every admissible sequence, by taking slack ϵ in the rate definition. If a uniform threshold for the coefficient failed, choose for each n a violating radius $0 < r_n < 1/n$ and an admissible cluster at that radius. The resulting sequence contradicts its eventual rate bound. If the uniform remainder threshold failed, choose in addition a violating spectator configuration and a violating phase at each radius. The rate bound was proved for arbitrary such choices, so gives the same contradiction. A finite initial segment where a cluster might meet a spectator is immaterial: for sufficiently small r, $Mr < c/2$ makes the combined configuration separated, while the stated lower bounds separate all points within either group. The center and spectators are already separated by c. □

The coefficients used in this theorem were chosen before D, the phase and the spectators. Only the small-radius threshold may depend on those finite demands. The theorem transfers the quantitative raw comparison; it does not produce that comparison from the Wilson measure.

177 Free-reference rank from finite point probes

This section proves the finite-dimensional reference-rank facts used in the trace calibration. The calculation uses only four-dimensional Euclidean Green-function derivatives, the signed coordinate symmetries, and simplicity of the compact gauge Lie algebra. All spaces below are real, even though polynomial coefficients are temporarily complexified to encode real observables.

177.1 Low-grade invariant labels

Write $\mathfrak{g}$ for the compact simple real gauge Lie algebra, with its invariant inner product, and e_0, e_1, e_2, e_3 for the coordinate basis of $\mathbb{R}^4$. A grade-d label is a polynomial in all radial jets, with jet order r having engineering weight $r + 2$. In particular, grades four and five may include linear terms in higher jets; those terms are removed below by gauge simplicity. After that reduction, their nonlinear terms depend only on the curvature F and, at grade five, its first derivative J. A signed permutation s acts on a coordinate tensor by permuting its indices and multiplying by the product of the corresponding coordinate signs. The raw grade-four space consists of symmetric two-index arrays of grade-four labels, real under complex conjugation and covariant under this signed action.

The raw grade-five space consists of one-index arrays of grade-five labels with the same reality and covariance conditions.

The following elementary color fact is used repeatedly. If a real linear functional ℓ on $\mathfrak{g}$ is invariant under the adjoint action, then differentiating $\ell(\mathrm{Ad}_{\exp(tx)}\, y) = \ell(y)$ at $t = 0$ gives $\ell([x, y]) = 0$. Simplicity and nonabelianity imply $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, hence $\ell = 0$. For an invariant real bilinear form B, define T by $B(x, y) = \langle Tx, y\rangle$. The invariance of B and of the positive inner product implies $T[a, x] = [a, Tx]$ for every a, x. Therefore its self-adjoint and skew-adjoint parts $S = (T + T^*)/2$ and $A = (T - T^*)/2$ both commute with the adjoint action. Every eigenspace of S is an ideal; the spectral theorem and simplicity show $S = \lambda I$. Also $-A^2$ is positive self-adjoint and commutes with the adjoint action, so the same argument gives $-A^2 = cI$ for some $c \geq 0$. If $c > 0$, then $J = A/\sqrt{c}$ is a skew-adjoint complex structure commuting with every ad_a. This implies $\mathrm{ad}_{Jx} = J\,\mathrm{ad}_x$. The right side is self-adjoint, because it is a product of commuting skew-adjoint operators, whereas the left side is skew-adjoint by invariance of the inner product. It must vanish; invertibility of J would then make every ad_x vanish, contradicting nonabelianity. Thus $c = 0$ and $A = 0$, so $T = \lambda I$. Every invariant bilinear form is consequently a scalar multiple of the inner product. Applying this argument to real and imaginary parts covers complex-valued coefficients too. Consequently the color dependence of an invariant quadratic term is just the inner product. The engineering weights are 2 for F and 3 for its first derivative J; a jet of order r has weight $r + 2$. At grade four or five, a monomial can therefore contain at most two factors. The possible terms are a constant, one linear functional of each jet of weight at most the grade, a symmetric quadratic form in F, and (only at grade five) a mixed bilinear form in (F, J). Every linear jet term vanishes by the preceding color argument, including the higher-derivative linear terms allowed by these two grades. Under simultaneous reversal of all four coordinate signs, F is even while the output vector index and J are odd. Hence constants and quadratic F terms vanish in a covariant vector. Only the mixed term remains in grade five.

For precision, the source represents the radial degree-zero and degree-one potentials by the following spanning families, for $c \in \mathfrak{g}$:

$$(\ell^0_{ij} c)_a(z) = \tfrac{1}{2}(\delta_{aj} z_i - \delta_{ai} z_j)c,$$
$$(\ell^1_{rkl} c)_a(z) = \tfrac{1}{3}(\delta_{al} z_k - \delta_{ak} z_l) z_r c.$$

The first family is antisymmetric in i, j; the second is antisymmetric in k, l and satisfies

$$\ell^1_{rkl} + \ell^1_{klr} + \ell^1_{lrk} = 0.$$

This identity follows by adding the three displayed quadratic monomials: each term cancels its opposite. The families span the actual radial jet spaces: for every degree-zero potential A_0 and degree-one potential A_1, respectively,

$$A_0 = \tfrac{1}{2} \sum_{i,j} \ell^0_{ij}(\mathrm{curl}^{ij}_0\, A_0), \qquad A_1 = \tfrac{1}{2} \sum_{r,k,l} \ell^1_{rkl}(\mathrm{curl}^{rkl}_1\, A_1).$$

Indeed, applying the corresponding curl to the right sides gives the identity on each curl coordinate, and the displayed formulas for the legs then give equality of the potential coefficients. Thus a bilinear form in these jet slots is determined by its values on pairs of legs. Signed covariance and pair antisymmetry follow directly from the same formulas. Put $f_{ij} = \mathrm{curl}^{ij}_0\, A_0$ and $j_{rkl} = \mathrm{curl}^{rkl}_1\, A_1$; the cycle identity is precisely the Bianchi relation $j_{rkl} + j_{klr} + j_{lrk} = 0$.

For a grade-five vector label, let M_p denote the mixed bilinear term in its low-grade decomposition. Choose a unit color vector v and define

$$A_{p;r;ij;kl} = M_p(\ell^0_{ij}(v), \ell^1_{rkl}(v)).$$

Reality makes these coefficients real. The adjoint-invariant bilinear-form calculation shows that this unit color vector records the full color coefficient. The leg spanning identities show that the coefficients determine every M_p. The cycle relation for ℓ^1 gives the Bianchi relation for A. Hence $A_{p;r;ij;kl}$ is antisymmetric in i,j and in k,l, and obeys

$$A_{p;r;ij;kl} = s_p s_r s_i s_j s_k s_l \, A_{s(p);s(r);s(i)s(j);s(k)s(l)} \tag{1}$$

for every signed permutation, where the signs $s_a \in \{\pm 1\}$ are those of the coordinate reflection. The finite signed-coordinate orbit count in dimension four has the following explicit form: if a coefficient satisfying (1) is nonzero, then its two antisymmetric pairs have distinct entries, and either (i) $p = r$ and the two pairs are parallel, or (ii) $p \neq r$ and there is a third coordinate u, distinct from p, r, such that the two pairs are $(p,u),(r,u)$ in either order. Permuting coordinates and reversing an antisymmetric pair reduces these cases to the four rows

$$(p,r;ij;kl) = (0,0;01;01),\ (0,0;12;12),\ (0,2;01;21),\ (0,2;21;01). \tag{2}$$

To check the count directly, a reflection in coordinate a multiplies the coefficient by -1 exactly when a belongs to an odd number of the six indices p, r, i, j, k, l; a nonzero entry requires even parity at every coordinate. If $p = r$, this forces the supports of the two antisymmetric pairs to agree, giving case (i). If $p \neq r$, parity forces the pairs to share exactly one coordinate, distinct from both p, r, giving case (ii). Coordinate permutations put the resulting supports in the forms in (2), and antisymmetry fixes the order signs. This proves the stated orbit count.

Lemma 177.1 (Three probes determine every covariant vector label)**.** *Let $A_{p;r;ij;kl}$ be a real array satisfying the antisymmetries, Bianchi relation, and signed covariance above. If*

$$A_{0;0;01;01} = A_{0;2;01;21} = A_{0;2;21;01} = 0,$$

then every coefficient of A vanishes.

Proof. By the orbit count, it suffices to check the four rows in (2). Three are the displayed hypotheses. For the remaining row, use Bianchi in the differentiated pair with derivative index 0 and pair 12:

$$A_{0;0;12;12} + A_{0;1;12;20} + A_{0;2;12;01} = 0.$$

The third term is minus the third displayed probe by reversing the first pair 12 to 21. Interchanging 1 and 2 carries the second term to the same third probe, up to the sign from reversing its first or differentiated pair. Both terms therefore vanish, so $A_{0;0;12;12} = 0$. Thus all four representatives vanish; signed covariance and antisymmetry propagate zero to every entry. □

177.2 The raw spaces have dimensions at most four and three

For a raw grade-four tensor $Q = (Q_{ab})$, choose a unit color vector v. The four real linear probes are: the real constant coefficient of Q_{00}; and the coefficients of $\langle F_{01}, F_{01}\rangle$ and $\langle F_{12}, F_{12}\rangle$ in Q_{00} and $\langle F_{01}, F_{21}\rangle$ in Q_{02}. If all four vanish, the constant vanishes. Polarization in the two F slots extracts the symmetric quadratic coefficient. For the quadratic coefficient array, signed parity says that if $a = b$, the two antisymmetric field pairs have the same support. Relative to the fixed output axis a, their common support either contains a or avoids a; coordinate permutations fixing a reduce these to the probes Q_{00} on $(01,01)$ and $(12,12)$. If $a \neq b$, the two field-pair supports have symmetric difference $\{a,b\}$, so they share a third coordinate distinct from a, b. Coordinate permutation reduces this to Q_{02} on $(01,21)$. Pair reversals change only signs; exchanging the output indices gives the

same coefficient by $Q_{ab} = Q_{ba}$. These are exactly the three quadratic probes listed above. The invariant-color argument makes vanishing on the chosen unit color vector sufficient: every color coefficient is its inner product multiple. The linear term is zero by color simplicity. Thus these four probes are injective, and

$$\dim \mathcal{R}_4 \le 4,$$

where $\mathcal{R}_4$ denotes the raw symmetric grade-four table space.

For a raw grade-five vector $Q = (Q_p)$, polarize the mixed term by evaluating the grade-zero and grade-one slots at (F, J), $(F, 0)$, $(0, J)$, and $(0, 0)$ and taking the alternating sum. This cancels the constant and pure-degree terms and leaves exactly the mixed bilinear coefficient. The three probes are the three entries in Lemma 177.1. The Bianchi identity supplies the fourth orbit representative, while the color argument and the all-coordinate reflection remove the other grades as shown above. Thus the probe map is injective into $\mathbb{R}^3$ and

$$\dim \mathcal{R}_5 \le 3.$$

This proves the two raw-dimension bounds without assuming a tensor classification beyond the finite parity count just given.

177.3 The normalized free stress basis

Let $\nu = (4\pi^2)^{-1}$ and $d_G = \dim \mathfrak{g} > 0$. The normalized four-dimensional massless Green function is $G(x) = \nu|x|^{-2}$. At the unit vector e_0, direct differentiation gives

$$\partial_i\partial_j G(e_0) = -2\nu\delta_{ij} + 8\nu\delta_{i0}\delta_{j0}, \tag{859}$$

$$\partial_k\partial_i\partial_j G(e_0) = 8\nu(\delta_{ij}\delta_{k0} + \delta_{ki}\delta_{j0} + \delta_{i0}\delta_{kj}) - 48\nu\delta_{i0}\delta_{j0}\delta_{k0}. \tag{860}$$

For example, differentiating G once gives $\partial_i G(x) = -2\nu x_i|x|^{-4}$. Differentiating again gives

$$-2\nu\delta_{ij}|x|^{-4} + 8\nu x_i x_j|x|^{-6}.$$

One more derivative and $x = e_0$ give the displayed formulas.

For antisymmetric index pairs define the free curl kernel

$$C_{ab;km}(x) = \delta_{kb}\partial_a\partial_m G(x) - \delta_{mb}\partial_a\partial_k G(x) - \delta_{ka}\partial_b\partial_m G(x) + \delta_{ma}\partial_b\partial_k G(x),$$

and the two-contraction chain

$$\mathcal{C}_{ab,cd;ij,kl}(x) = C_{ab;ij}(x)C_{cd;kl}(x) + C_{ab;kl}(x)C_{cd;ij}(x).$$

The Wick pairing of two separated point insertions with coefficient vector $w = (w_0, w_1, w_2, w_3)$ is d_G times

$$\mathcal{D}_w^{ab}(H) = \begin{cases} w_1 \sum_i H_{ai,ai} + w_2 \sum_{i,j} H_{ij,ij}, & a = b, \\ w_3 \sum_i H_{ai,bi}, & a \ne b, \end{cases}$$

For fixed field pairs (ij, kl) set $H_{pqrs}(x) = C_{pq;ij}(x)C_{rs;kl}(x) + C_{pq;kl}(x)C_{rs;ij}(x)$. The probe is $S_w(ij, kl; ab; x) = d_G\mathcal{D}_w^{ab}(H(x))$; the vacuum contribution to a one-point 00 entry is w_0. Substitution of (859) into the curl factors gives

$$S_w(01, 01; 00; e_0) = d_G\nu^2(32w_1 + 64w_2), \tag{861}$$

$$S_w(12, 12; 00; e_0) = d_G\nu^2(64w_2), \tag{862}$$

$$S_w(02, 12; 01; e_0) = d_G\nu^2(-16w_3). \tag{863}$$

After factoring out $d_G\nu^2$, the coefficient rows for these three probes, in columns (w_1, w_2, w_3), are

$$(32, 64, 0), \qquad (0, 64, 0), \qquad (0, 0, -16).$$

These finite sums are evaluated by inserting the displayed Kronecker-delta Hessian into C and then into H and $\mathcal{D}_w$. The second equation recovers w_2, the first then recovers w_1, and the third recovers w_3, since $d_G\nu^2 > 0$. Together with the one-point vacuum coefficient w_0, these four probes prove injectivity of the free stress coefficient map into the free stress table. The raw bound $\dim \mathcal{R}_4 \le 4$ and the surjective averaging projection from raw arrays to the free table show that the free table has dimension at most four. Thus the coefficient map is an isomorphism.

Lemma 177.2 (Free stress coefficient map is an isomorphism)**.** *The four coefficient tensors consisting of the vacuum scalar, the two independent diagonal curvature contractions, and the off-diagonal shear contraction form a basis of the free symmetric grade-four stress table.*

Proof. The four probes above prove injectivity from a four-dimensional space; the raw upper bound proves equality of dimensions, hence bijectivity. □

177.4 The two conserved free directions

Let $\mathcal{T}_4$ be the free symmetric grade-four stress table and $K_0 = \ker(\mathrm{skim} : \mathcal{T}_4 \to \mathbb{R})$. Let $\mathcal{V}_5$ be the grade-five vector table after imposing the free Maxwell equation, and let $D_0 : K_0 \to \mathcal{V}_5$ be the free divergence restricted to this zero-skim subspace. Write π for the quotient projection from the raw vector table onto $\mathcal{V}_5$. The projection is onto by the equivariant section obtained by averaging any real section over the finite signed permutation group. The equation-of-motion array is

$$E_b = \sum_{i,a} \langle F_{bi}, \partial_a F_{ai} \rangle.$$

It is a nonzero raw covariant vector: take a nonzero color vector v, set $F_{01} = v$ and $\partial_0 F_{01} = v$ at one jet, with all other independent jet coordinates zero. The defining sum then evaluates its $b = 0$ component to $\langle v, v \rangle > 0$. The Maxwell equation makes E vanish after the quotient. Thus the kernel of π is nonzero. Rank–nullity and $\dim \mathcal{R}_5 \le 3$ give $\dim \mathcal{V}_5 \le 2$.

Let ϕ_1, ϕ_2 be the two linear functionals on $\mathcal{V}_5$ obtained by the point-divergence probes $(01, 01)$ and $(12, 12)$ at e_0. For w as above, let $S_w(ij, kl; ab; x)$ denote the response with field-pair insertion (ij, kl), output component (a, b), and separation x. Differentiating the chain product gives

$$\partial_u \mathcal{C} = (\partial_u C_{ab;ij}) C_{cd;kl} + C_{ab;ij} (\partial_u C_{cd;kl}) + (\partial_u C_{ab;kl}) C_{cd;ij} + C_{ab;kl} (\partial_u C_{cd;ij}).$$

Inserting (859)– (860) and summing $u = 0, 1, 2, 3$ gives

$$\sum_u \partial_u S_w(01, 01; u0; e_0) = d_G\nu^2(-256w_1 - 512w_2 + 128w_3), \tag{864}$$

$$\sum_u \partial_u S_w(12, 12; u0; e_0) = d_G\nu^2(-512w_2 - 128w_3). \tag{865}$$

The two coefficient rows in (w_1, w_2, w_3) are linearly independent. Consequently $\phi_1 \circ D_0$ and $\phi_2 \circ D_0$ are independent, so ϕ_1, ϕ_2 are independent on $\mathcal{V}_5$. Together with the upper bound $\dim \mathcal{V}_5 \le 2$, this gives $\dim \mathcal{V}_5 = 2$, and $\Phi = (\phi_1, \phi_2) : \mathcal{V}_5 \to \mathbb{R}^2$ is an isomorphism. The two probe equations have common kernel $\mathbb{R}(1, -1/4, 1)$ on (w_1, w_2, w_3): the second gives $w_2 = -w_3/4$, and then the first gives $w_1 = w_3$. The scalar coefficient w_0 is removed by zero skim. Thus the kernel of $(\mathrm{skim}, \Phi \circ D_0)$ is

exactly the Weyl line $(0, 1, -1/4, 1)$. Since Φ is an isomorphism, this also proves the Weyl tensor has zero divergence. The zero-skim domain has dimension three, and the restricted divergence has one-dimensional kernel, so it has rank two and is onto the two-dimensional target. Its kernel is precisely the Weyl line.

Proposition 177.3 (Free rank and closure dimensions)**.** *The free symmetric grade-four stress table has dimension four. The Maxwell-reduced grade-five vector table has dimension two. The free zero-skim divergence map is onto that vector table and its kernel is the one-dimensional Weyl line.*

Proof. The first assertion is Lemma 177.2. The previous paragraph proves the second and third assertions. The scalar skim is exactly the real one-point average of the 00 entry; it is not the tensor trace. The statement uses only the explicit probe formulas and the finite-dimensional quotient argument just given. □

178 A positive free detector at every jet order

This section proves the all-grade reference-field separation needed in Section 181 and for the finite point scanner. The point is that positivity must hold for a mixture of derivative orders, not just separately at each order. We establish it by an explicit triangular change of probes. All vector spaces in the one-leg calculation are real; ordinary complex conjugation is added only when passing to the Wick-polynomial pairing.

178.1 The Kelvin response and a triangular family of probes

Let $\mathscr{S}_j$ be the finite-dimensional real space of homogeneous degree-j Maxwell field strengths in four-dimensional Euclidean space, with the fixed gauge-color representation. Thus $P \in \mathscr{S}_j$ is an antisymmetric two-tensor-valued homogeneous polynomial satisfying the Maxwell differential and algebraic identities. Let $\langle \cdot, \cdot \rangle_F$ denote the positive Frobenius inner product, including the positive color inner product. For a nonzero point x, write

$$r_x = |x|^{-2}, \qquad I(x) = r_x x, \qquad \vartheta_x = I - 2r_x\, x \otimes x.$$

The matrix ϑ_x is the orthogonal reflection that reverses the line through x and fixes its orthogonal complement. Its induced action on two-tensors is denoted $\vartheta_x^{(2)}$.

For a symmetric coefficient tensor A of order j with two antisymmetric field indices, write componentwise

$$(\mathcal{D}_A f)_{im}(x) = \sum_{(a_1,\dots,a_j)\in\{0,1,2,3\}^j} \frac{1}{j!} D^j f(x)[e_{a_1}, \dots, e_{a_j}]\, A_{im}(e_{a_1}, \dots, e_{a_j}).$$

The sum is over ordered derivative tuples; later the two field indices are summed in both orders in the Frobenius pairing. This is the normalized Taylor coefficient convention. The colour pairing is the fixed positive real inner product. Write $c_j = (-2)^j/(4\pi^2)$, the coefficient of the four-dimensional Green function.

We first establish the differentiated Green identity used below:

$$(\mathcal{D}_A G)_{im}(x) = c_j |x|^{-2(j+1)} A_{im}(x), \qquad x \neq 0, \quad A \in \mathscr{S}_j, \qquad G(x) = \frac{1}{4\pi^2|x|^2}. \tag{866}$$

For $j = 0$ this is evaluation of the Green function. Suppose it holds at order j and let A have order $j + 1$. For each coordinate direction e_a, freeze the first tensor slot to get $A_a \in \mathscr{S}_j$. The normalized Taylor coefficient satisfies

$$(\mathcal{D}_A f)_{im} = \frac{1}{j+1} \sum_a \partial_a (\mathcal{D}_{A_a} f)_{im}.$$

This follows by expanding the ordered derivative tuples: each of the $j + 1$ possible first slots gives the same symmetric coefficient. Applying the induction hypothesis and differentiating gives

$$(\mathcal{D}_A G)_{im} = \frac{c_j}{j+1} \left(-2(j+1)|x|^{-2(j+2)} \sum_a x_a (A_a)(x) + |x|^{-2(j+1)} \sum_a \partial_a (A_a)(x) \right).$$

For completeness, the trace identity here is harmonicity in the derivative slots, not the Maxwell divergence equation alone. Write $F_{im}(x)$ for the homogeneous field components represented by A. The Maxwell equations give $\partial_a F_{ai} = 0$ and $\partial_a F_{im} + \partial_i F_{ma} + \partial_m F_{ai} = 0$. Apply ∂_a and sum over a in the second identity. The two terms involving the divergence vanish, so $\Delta F_{im} = 0$. By the normalized coefficient convention, $(A_a)_{im} = (j+1)^{-1} \partial_a F_{im}$; hence

$$\sum_a \partial_a (A_a)_{im} = (j+1)^{-1} \Delta F_{im} = 0.$$

Euler's homogeneous identity, or direct contraction of the first slot, gives

$$\sum_a x_a (A_a)_{im}(x) = F_{im}(x) = A_{im}(x).$$

Since $c_{j+1} = -2c_j$, this proves (866). It is an off-origin identity; no distributional contact term is involved.

The Euclidean Green kernel and its two curls give a response field associated with P. The Kelvin calculation needed here is the following identity, with the normalization fixed by the Green function:

$$\mathcal{R}(P)(y) = \text{strength}(j) |y|^{-4} \vartheta_y^{(2)} P(I(y)), \qquad y \neq 0, \quad \text{strength}(j) = 4(j+1)(j+2)(-2)^j \, \text{normal}. \tag{867}$$

Here normal $= 1/(4\pi^2) > 0$ is the scalar in (866). In particular,

$$(-1)^j \, \text{strength}(j) > 0. \tag{868}$$

The sign follows directly from the displayed formula: the factors $4(j+1)(j+2)$ and normal are positive and $(-1)^j (-2)^j = 2^j$.

We recall the differential reduction giving this normalization. Put $R_i(x) = \sum_a x_a P_{ai}(x)$ and $c_j = (-2)^j$ normal. The differentiated Green potential is $c_j |x|^{-2(j+1)} P_{ai}(x)$. The divergence-free equation removes the derivatives falling on P in its divergence, so the resulting vector is

$$D_i(x) = -2(j+1) c_j |x|^{-2(j+2)} R_i(x).$$

The curl of the field response is $2(\partial_m D_i - \partial_i D_m)$. Closedness, the Bianchi identity, and Euler homogeneity give

$$\partial_i R_m(x) - \partial_m R_i(x) = (j+2) P_{im}(x).$$

Also $\partial_i |x|^{-2(j+2)} = -2(j+2) x_i |x|^{-2(j+3)}$. Substitution yields

$$4(j+1)(j+2) c_j |x|^{-2(j+2)} \left[P_{im}(x) - 2|x|^{-2} \big(x_i R_m(x) - x_m R_i(x) \big) \right].$$

The bracket is the two-index reflection of $P(x)$. Homogeneity gives

$$|x|^{-2(j+2)}\vartheta_x^{(2)}P(x) = |x|^{-4}\vartheta_x^{(2)}P(I(x)).$$

This proves (867) off the origin and identifies its coefficient; no contact term is used.

For an orthogonal reflection s, define its full action on a solution by $(s\cdot Q)(v) = s^{(2)}Q(sv)$: it acts on both field indices and every derivative slot. For $x \neq 0$, let $\mathrm{Leg}_x(P,Q)$ be the one-leg kernel: contract the second derivatives of the scalar Green kernel at separation x with the dual probes of P and Q, and apply the two curls. The one-leg reflected pairing is

$$B_x^{j,k}(P,Q) = \mathrm{Leg}_x(P, \vartheta_x\cdot Q).$$

Orthogonal covariance and exchange symmetry of the Green kernel imply

$$B_x^{j,k}(P,Q) = B_x^{k,j}(Q,P).$$

The use of the reflection in the second slot is essential: this is the one-leg reflected form, not the unreflected covariance.

To triangularize all grades at once, for $0 \le k \le j$ let

$$T_{j,k}^x P = \frac{\mathrm{strength}(j)}{\mathrm{strength}(k)}\,\mathrm{Taylor}_k\big(P(\,\cdot - I(x))\big), \qquad P \in \mathscr{S}_j.$$

The Taylor coefficient is taken at the origin, and the translated coefficient is again a homogeneous degree-k Maxwell solution. Set

$$\Sigma_x P = \sum_{k=0}^{j} T_{j,k}^x P \in \mathscr{S}_{<j+1}, \qquad P \in \mathscr{S}_j.$$

The sum is finite. By the exact Taylor formula for a degree-j polynomial, for every y,

$$\sum_{k=0}^{j} \mathrm{strength}(k)(T_{j,k}^x P)(y) = \mathrm{strength}(j)P(y - I(x)). \tag{869}$$

Apply (867) term by term and use (869). The resulting response is $\mathcal{R}_x(\Sigma_x P) := \sum_{k=0}^{j}\mathcal{R}(T_{j,k}^x P)$. For y near x, inversion gives

$$I(x+tv) - I(x) = t\,V_{x,v}(t), \qquad V_{x,v}(t) = |x+tv|^{-2}\big(\vartheta_x v - t|v|^2 I(x)\big),$$

where $V_{x,v}$ is smooth near $t=0$. Since P is homogeneous of degree j, the shifted response along this line is t^j times a smooth function. Consequently its Taylor coefficients through order $j-1$ vanish, while its order-j coefficient is

$$\mathrm{strength}(j)|x|^{-2(j+2)}(\vartheta_x\cdot P)(v),$$

where the reflected solution is defined by $(\vartheta_x\cdot P)(v) = \vartheta_x^{(2)}P(\vartheta_x v)$. To connect the response computation to the one-leg pairing, the two-curl Green identity gives

$$B_x(\Sigma_x P, Q_k) = (-1)^k\,\big\langle \vartheta_x\cdot Q_k, \mathrm{Taylor}_k\big(\mathcal{R}_x(\Sigma_x P)\big)(x)\big\rangle_F\,.$$

Here $Q_k \in \mathscr{S}_k$, and the normalized Taylor tensor includes all ordered derivative tuples. The sign $(-1)^k$ is the sign from differentiating the second insertion. The vanishing lower Taylor coefficients imply $B_x(\Sigma_x P, Q_k) = 0$ when $k < j$; at $k = j$ the order-j coefficient gives $B_x(\Sigma_x P, Q_j) =$

$\lambda_j(x)\langle P, Q_j\rangle_F$, because $(\vartheta_x \cdot Q_j, \vartheta_x \cdot P)$ have the same Frobenius product as (Q_j, P). When the second input is also transformed, its components have grades at most its original grade, and its top component is unchanged because $T^x_{k,k}$ is the identity. Thus these same two identities give the triangular identities for the full one-leg kernel. Extend B_x by bilinearity to the finite tower. If $\mathscr{S}_{<m} = \bigoplus_{k<m} \mathscr{S}_k$, then the formulas

$$B_x(\Sigma_x P, \Sigma_x Q) = 0 \qquad (P \in \mathscr{S}_j,\ Q \in \mathscr{S}_k,\ k < j), \tag{870}$$

$$B_x(\Sigma_x P, \Sigma_x Q) = \lambda_j(x)\langle P, Q\rangle_F \qquad (P, Q \in \mathscr{S}_j), \tag{871}$$

where

$$\lambda_j(x) = (-1)^j \,\mathrm{strength}(j)|x|^{-2(j+2)} > 0. \tag{872}$$

The factor $(-1)^k$ in the one-leg kernel comes from differentiating the second insertion; at equal grade it combines with the Taylor coefficient to give precisely (872).

Now fix a finite cutoff n and form the tower

$$\mathscr{S}_{<n} = \bigoplus_{j=0}^{n-1} \mathscr{S}_j.$$

On this tower the reflected Gram form is

$$B_x(P, Q) = \sum_{j,k<n} B_x^{j,k}(P_j, Q_k).$$

Apply the triangular transform Σ_x separately in each input grade. The cross-grade identities (870) and their transposes remove every off-diagonal block, while (871) gives

$$B_x(\Sigma_x P, \Sigma_x Q) = \sum_{j<n} \lambda_j(x)\langle P_j, Q_j\rangle_F. \tag{873}$$

Every coefficient on the right is positive. Since each Frobenius inner product is positive definite, the right side is positive for every nonzero tower. Moreover, $T^x_{j,j}$ is the identity: the top-degree Taylor coefficient of $P(\,\cdot - I(x))$ is P. Therefore the matrix of Σ_x on the ordered grades is lower triangular with identity diagonal. Its strictly grade-lowering part $N_x = \Sigma_x - I$ satisfies $N_x^n = 0$, so

$$\Sigma_x^{-1} = I - N_x + N_x^2 - \cdots + (-1)^{n-1} N_x^{n-1}.$$

Thus the diagonalization is an invertible change of probes, and the full mixed-grade Gram form is positive definite. This is a simultaneous statement about the entire tower; positivity of the separate diagonal blocks alone would not imply it.

178.2 Wick polynomials and strict reflected positivity

Let $\mathscr{A}$ be the complex polynomial algebra generated by the formal field-strength jet coordinates at all derivative orders. Evaluation on homogeneous Maxwell solutions restricts each polynomial to the on-shell jet space, imposing both the differential Maxwell equations and the algebraic Bianchi identities. The filtered shell sector $\mathscr{Q}_{\le d}$ is the image of this restriction map through grade d, hence is canonically the quotient by its kernel. Full reflection preserves each grade and this filtered sector. Each linear coordinate restricted to $\mathscr{S}_j$ is a linear functional on a finite-dimensional inner-product space, so the finite-dimensional Riesz representation theorem writes it as pairing with a vector of $\mathscr{S}_j$. Consequently every polynomial in finitely many jet coordinates is a polynomial in the coordinates

of one sufficiently large finite tower. A homogeneous solution $P \in \mathscr{S}_j$ defines its linear Wick symbol $\mathsf{d}(P)$. Products of these symbols generate the polynomial Wick algebra.

For two distinct points x, y, put $z = x - y$. Let $\mathcal{W}_{x,y}(A, B)$ denote the free two-point moment of the polynomial insertions A and B. The reflected pairing is linear in its first argument and conjugate-linear in its second:

$$\langle A, B\rangle^{\mathrm{ref}}_{x,y} = \mathcal{W}_{x,y}\big(A, \vartheta_z \cdot \overline{B}\big),$$

where ϑ_z acts on every tensor index of the second field insertion. Complex conjugation of polynomial coefficients is a separate operation in this formula. The free moment is the normal-ordered cross-site Gaussian matching moment: only pairings joining the two distinct insertion sites occur. Removing one linear field factor from the first insertion therefore gives the sum of its contractions with the factors in the second insertion. This follows by separating the unique partner of the removed factor in the finite matching formula for a free moment. With two insertion points there is only one possible partner site. In orthogonal coordinates the identity reads

$$\langle X_i p, q\rangle^{\mathrm{ref}}_{x,y} = d_i \langle p, \partial_i q\rangle^{\mathrm{ref}}_{x,y}, \qquad \langle 1, X_i q\rangle^{\mathrm{ref}}_{x,y} = 0, \qquad \langle 1, 1\rangle^{\mathrm{ref}}_{x,y} = 1. \tag{874}$$

The first identity is the differentiated removal rule, and the second is the same rule with no factor available to contract on the first side.

Choose a real orthogonal basis $e_1, \dots, e_M$ for the tower Gram form (873), with squared norms $d_i > 0$. Such a basis exists by finite-dimensional Gram–Schmidt: at each step, the orthogonal complement of the span already chosen is positive definite, so a nonzero vector in it has strictly positive squared norm; normalize or retain its positive norm. In the polynomial coordinates $X_1, \dots, X_M$ associated with e_i, the reflected pairing satisfies

$$\langle X^\alpha, X^\beta\rangle^{\mathrm{ref}}_{x,y} = \begin{cases} \displaystyle\prod_{i=1}^{M} \alpha_i! d_i^{\alpha_i}, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

For completeness, this formula follows by induction on $|\alpha|$. When $\alpha = 0$, the last two identities in (874) give 1 if $\beta = 0$ and 0 otherwise. If $\alpha_i > 0$, apply the first identity with $p = X^{\alpha - e_i}$ and $q = X^\beta$:

$$\langle X^\alpha, X^\beta\rangle = \beta_i d_i \langle X^{\alpha - e_i}, X^{\beta - e_i}\rangle,$$

where the right side is zero when $\beta_i = 0$. Induction shows that the pairing vanishes unless $\alpha = \beta$. When they agree, removing each occurrence of coordinate i contributes successively $d_i\alpha_i, d_i(\alpha_i - 1), \dots, d_i$, so its total factor is $\alpha_i! d_i^{\alpha_i}$. Multiplication over coordinates gives the formula. For arbitrary polynomials $p = \sum_\alpha p_\alpha X^\alpha$ and $q = \sum_\alpha q_\alpha X^\alpha$, it follows that

$$\langle p, q\rangle^{\mathrm{ref}}_{x,y} = \sum_\alpha p_\alpha \overline{q_\alpha} \prod_{i=1}^{M} \alpha_i! d_i^{\alpha_i}, \qquad \langle p, p\rangle^{\mathrm{ref}}_{x,y} = \sum_\alpha |p_\alpha|^2 \prod_{i=1}^{M} \alpha_i! d_i^{\alpha_i}. \tag{875}$$

Each weight is strictly positive, including the empty multi-index, whose weight is 1. Thus the reflected norm is strictly positive for every nonzero polynomial. Every polynomial uses only finitely many jet orders; choose a tower cutoff containing those orders, and the argument applies. If two representatives define the same shell polynomial, their difference restricts to zero on every solution and has zero pairing with every insertion. The formula is therefore well defined on the shell quotient and strictly positive on each nonzero class.

Theorem 178.1 (Strict all-grade free detector)**.** *For any two distinct Euclidean points, the reflected free two-point form on the polynomial algebra of gauge-invariant field-strength jets, after quotienting by the Maxwell equations and the Bianchi identities, is strictly positive: if P is a nonzero quotient class, then*

$$\langle P, P\rangle^{\mathrm{ref}}_{x,y} > 0.$$

Consequently, the family of separated two-point evaluations at any fixed pair detects every field in each finite-dimensional grade.

Proof. The Kelvin calculation gives the one-leg triangular identities and the strictly positive tower pivots in (870)–(872). Equation (873) therefore diagonalizes the entire finite tower with positive diagonal entries. The removal-rule calculation gives (875), whose summands are nonnegative and whose summand for any nonzero coefficient is positive. Every polynomial lies in a finite tower, proving strict positivity for each nonzero shell class.

For the detector statement, fix a grade d and a nonzero class v in its free field space. Choose any separated pair. The reflected evaluation of v against its reflected conjugate is the strictly positive number just proved. Hence at least one vector in any spanning basis of the grade has a nonzero pairing with v; otherwise linearity would make its pairing with every grade-d vector, including the reflected conjugate of v, vanish. Thus the full family of basis-column evaluations is injective. □

178.3 The finite point scanner

We spell out the map used by the construction. Fix a reference frame and a grade d. Choose a basis $b_0, \dots, b_m$ of its free field space with $b_0 = 1$, the scalar unit; the remaining basis elements are images of vectors in that same reference frame. At a separated pair (x, y) define the row scanner

$$\mathcal{R}_{x,y}(v) = \big(\mathrm{Use}_{x,y}(v, b_j)\big)_{j=0}^{m}, \qquad \mathrm{Face}_{x,y}(w) = \vartheta_{x-y}\big(\overline{w}\big).$$

Here the bar is ordinary coefficient conjugation, and the reflection acts on all tensor indices. By the definitions of the free-field quotient and the reflected polynomial pairing,

$$\mathrm{Use}_{x,y}(v, \mathrm{Face}_{x,y}(w)) = \langle v, w\rangle^{\mathrm{ref}}_{x,y}. \tag{876}$$

If $\mathcal{R}_{x,y}(v) = 0$, then the bilinear form $w \mapsto \mathrm{Use}_{x,y}(v, w)$ vanishes on the chosen basis and therefore on the whole grade. In particular, take $w = \mathrm{Face}_{x,y}(v)$. Equation (876) and strict positivity imply $v = 0$. So the scanner is injective. Notice that it is the conjugate and the reflection together that place the second entry in the reflected positive form; reflection alone or conjugation alone is not the required operation.

The scalar unit and source-frame basis make this an admissible finite scanner for the later construction. The unit is retained as the first column, and the other columns use the same frame from which the free table was built. This proves the all-grade reference detector only. It makes no positivity claim about the interacting measure, and the eventual interacting comparison and replication bounds remain separate inputs.

179 Finite projections and transport of the equation kernel

Free separation does not by itself identify the interacting null space. We prove the finite-dimensional endpoint, keeping the actual marked correlation estimates and local equation estimates as explicit inputs. All spaces below are finite-dimensional over $\mathbb{C}$.

179.1 Uniform projections and a common extraction

Let V have a norm, an isometric conjugate-linear involution J, and a representation ρ of a finite group Γ commuting with J. Fix an isomorphism $Q : \mathbb{C}^m \to V$ from Euclidean space. For any subspace $U \subset V$, let Π_U be the orthogonal projection onto $Q^{-1}U$. Then

$$A_U^0 = I - Q\Pi_U Q^{-1}, \qquad \|A_U^0\| \le C_0 := \|I\| + \|Q\|\,\|Q^{-1}\|$$

kills U and satisfies $x - A_U^0 x \in U$. Any linear map A having these two properties satisfies $A^2 = A$ and $\ker A = U$: apply A to $x - Ax$ for the first assertion, and use $x = x - Ax$ when $Ax = 0$ for the second.

When U is invariant under the symmetries, put

$$B_U = \frac{1}{|\Gamma|}\sum_{g\in\Gamma} \rho(g) A_U^0 \rho(g)^{-1}, \qquad A_U = \tfrac{1}{2}(B_U + JB_UJ).$$

Each average still kills U and differs from the identity by a map into U. It is therefore an idempotent with kernel U. Reindexing by left multiplication proves that B_U commutes with $\rho(g)$; commutation of J and $\rho(g)$ proves the same for A_U, and $JA_UJ = A_U$. The composition JB_UJ is complex-linear. Moreover

$$\|A_U\| \le \frac{C_0}{|\Gamma|}\sum_{g\in\Gamma} \|\rho(g)\|\,\|\rho(g)^{-1}\|.$$

This bound is independent of U. If invariance is known only at late cutoffs, use A_U^0 at earlier ones and take the maximum of the two bounds. No isometry of the given signed representation is needed.

For every grade d, apply this construction to its coefficient space V_d and finite equation spaces $U_{d,N,p}$. The bounds may depend on d, but not on mesh cutoff N or period index p. There is a common refinement with mesh limits for every fixed (d,p) and then period limits for every d. Indeed, enumerate the countably many pairs (d,p), take successive subsequences in their compact matrix balls, and take a diagonal subsequence. Choose its jth term from the jth nested subsequence, with index beyond the jth index of each of the first j subsequences and beyond the preceding chosen index. Its tail then belongs to every fixed earlier subsequence, so every fixed pair retains its limit. Repeat in the period variable for the countably many grades. Bounded lifted vectors indexed by a comparison scale and finitely many columns can be included in the same enumeration. Previously existing limits survive these refinements.

Write $A_{d,p}$ for the inner limits and T_d for the outer limits. Continuity of multiplication gives $A_{d,p}^2 = A_{d,p}$ and $T_d^2 = T_d$. Below the index $c = (N,p)$ follows this ordered filter: first every sufficiently large selected period, then every sufficiently large selected mesh cutoff, whose threshold may depend on the period. The two-stage limits give $A_{d,c} \to T_d$ along this filter. There is no claim of uniform convergence in p.

179.2 The all-test null space and the local equation input

Fix a grade, write $V = V_d$, $A_c = A_{d,c}$ and $T = T_d$, and let Θ consist of all smooth compactly supported spectator tests on separated configuration domains with one specified slot of this grade. Let $f_{\theta,c} \in V^*$ be its finite-cutoff correlation functional and assume $f_{\theta,c} \to f_\theta$ in operator norm. Convergence on a fixed basis suffices, since for its dual basis e_j^*

$$\|f_{\theta,c} - f_\theta\| \le \sum_j |(f_{\theta,c} - f_\theta)(e_j)|\,\|e_j^*\|.$$

Define

$$M = \bigcap_{\theta \in \Theta} \ker f_\theta.$$

This includes every arity and spectator word, not only the finite pair measurements used below to bound dimension.

The local equation input is as follows. For every family $v_c \in \ker A_c$ with subpower inverse-calibrated norm, there is a comparison insertion Z_c whose separated spectator moments vanish exactly, and whose difference from the physical insertion of v_c has centered discrepancy price tending to zero on this same filter. Subpower means that for every $\epsilon > 0$ the inverse-calibrated norm is eventually at most $\exp(\epsilon s_N)$, with $s_N \to \infty$. The insertions are measurable and bounded at each cutoff, and their products with the fixed spectator word are integrable.

Section 53.3 proves that bounded coefficient families have subpower inverse-calibrated norm. Its smeared-discrepancy estimate sends the tested difference from Z_c to zero. The compact test support is eventually separated on the lattice, so the tested sum for Z_c is zero. Subtraction of the finite-cutoff integrals is legitimate by integrability. Consequently

$$v_c \in \ker A_c, \quad (v_c) \text{ bounded} \quad \Longrightarrow \quad f_{\theta,c}(v_c) \longrightarrow 0. \tag{877}$$

For $v \in \ker T$, the family $v_c = (I - A_c)v$ is bounded, is killed by A_c, and tends to v. Hence (877) and norm convergence of the functionals imply

$$\ker T \subseteq M. \tag{878}$$

The local equation input is substantive; it does not follow from free positivity.

It remains applicable after further refinement. Extend a vector family from the refined grid of mesh and period indices by zero outside that grid. The kernel condition is preserved. Subpower bounds on the original ordered filter hold conditionally on membership in the refined grid: monotonicity of the two index sequences translates the refined thresholds into original thresholds. Outside that grid the bound is automatic. Apply the original local estimate to this extension, then restrict back. Thus no control of the original family at unused cutoff pairs is required.

179.3 Purchased measurements and exact dimension

Let E be the free field space, $r = \dim E$, and $b_1, \dots, b_r$ a basis. By Section 178, choose finitely many free measurements $\ell_a \in E^*$ for which $Lx = (\ell_a(x))_{a=1}^h$ is injective. The additional interacting comparison consists of vectors $u_{k,j} \in V$ and linear measurements $g_{k,a} \in V^*$ satisfying

$$g_{k,a}|_M = 0, \qquad g_{k,a}(u_{k,j}) \longrightarrow \ell_a(b_j). \tag{879}$$

Here k tends to infinity after taking the mesh and period limits for each fixed comparison scale. Neither $u_{k,j}$ nor $g_{k,a}$ is required to converge as $k \to \infty$.

For the application, $g_{k,a}$ is a generic-point correlation linear in its first insertion, with its other insertions fixed at that k. It vanishes on M because the corresponding correlation is continuous on the open generic configuration domain and has zero integral against every compactly supported smooth test there. A nonzero value could be rotated by a constant complex phase to have positive real part on a small neighborhood; a nonnegative smooth bump supported there would then have nonzero integral. This contradiction proves pointwise vanishing. It uses no point evaluation at a coordinate tie.

The convergence in (879) is the paid comparison input. Its precise usable form, after the cutoff limits, is

$$|g_{k,a}(u_{k,j}) - \ell_{a,k}(b_j)| \le P_{k,a,j}\varepsilon_k, \qquad P_{k,a,j}\varepsilon_k \to 0, \qquad \ell_{a,k}(b_j) \to \ell_a(b_j).$$

The factors $P_{k,a,j}$ are coefficient norms of the common free moment profile at the rescaled separated points. Continuity of that finite profile bounds them. The lifted vectors for fixed k must be bounded uniformly in both cutoff indices, so the preceding common extraction retains them. These requirements are not consequences of the finite matrix argument.

The matrix $(\ell_a(b_j))$ has rank r. Choose an r-row minor with nonzero determinant. Its counterpart in $(g_{k,a}(u_{k,j}))$ is nonzero for all sufficiently large k, by entrywise convergence and continuity of the determinant. A linear relation between the classes $[u_{k,j}]$ in V/M, evaluated by these rows, has every coefficient zero. Thus

$$r \le \dim(V/M). \tag{880}$$

For $r = 0$ this inequality is automatic and no minor is needed.

Assume also the finite equation count

$$\dim V \le \dim \ker A_c + r \quad \text{frequently along the same filter.} \tag{881}$$

In the application, a label space W, equation ideal $\mathcal{E}$, and actual calibrated isomorphisms $L_c : W \to V$ satisfy $\dim(W/\mathcal{E}) = r$ and $\ker A_c = L_c\mathcal{E}$. Rank-nullity then gives equality in (881). The free detector identifies $W/\mathcal{E}$ with the free space; it does not establish this separate identity for the actual finite kernel.

If A, T are idempotents with $\|A - T\| < 1$, the map $\ker A \to \ker T$, $v \mapsto (I - T)v$, is injective. Indeed a vector in its kernel has $Tv = v$ and $Av = 0$, so $\|v\| = \|(A - T)v\| \le \|A - T\|\|v\|$, forcing $v = 0$. Interchange A, T to obtain equal kernel dimensions. Applying this eventually to $A_c \to T$, and intersecting with the frequent set in (881), gives $\dim \ker T \ge \dim V - r$. Combining this with (878) and (880) proves

$$M = \ker T, \qquad \dim(V/M) = r. \tag{882}$$

The purchased classes are therefore eventually a basis of V/M. This supplies the kernel hypothesis in the normalized compact test argument of Section 53, provided its projections and tests are these same ones. That argument then applies to its unbounded approximating vectors without asserting convergence of moving physical subspaces.

179.4 Exact transport and quotient symmetries

Suppose $\ker A_c = L_c\mathcal{E}$ as above, with L_c intertwining the real involutions and signed actions at all sufficiently late cutoffs. The subspace $\mathcal{E}$ is invariant, and the selected A_c commute with the actions on V. Their limit T commutes too, by passage to the limit on each fixed vector. Define

$$Y_c = TA_c + (I - T)(I - A_c). \tag{883}$$

Direct multiplication, using both idempotence identities, gives

$$Y_c - I = (2T - I)(A_c - T), \qquad Y_cA_c = TA_c = TY_c.$$

Thus $Y_c \to I$. Choose one late cutoff satisfying all the symmetry conditions and $\|Y_c - I\| < 1$. Its inverse is the norm-convergent series $\sum_{n\ge0}(I - Y_c)^n$: multiplication of the partial sums leaves the remainder $(I - Y_c)^{n+1}$, which tends to zero. In particular $\|Y_c^{-1}\| \le (1 - \|Y_c - I\|)^{-1}$. The intertwining identity and invertibility give $Y_c(\ker A_c) = \ker T$.

Since Y_c is a real polynomial expression in T, A_c, it commutes with both symmetries. Therefore $\mathcal{L} = Y_c L_c$ is an isomorphism with $\mathcal{L}\mathcal{E} = M$, and descends to

$$W/\mathcal{E} \simeq V/M \tag{884}$$

intertwining the induced conjugations and signed actions. Only one sufficiently late L_c is used; convergence of L_c is unnecessary.

If a smooth representative of the limiting correlation table has the same tested correlations, its physical null space equals M. Physical nullity implies tested nullity by integration, while the converse follows by the smooth bump argument for every separated spectator word. Hence (884) identifies that physical quotient with the free quotient. Construction of the smooth representative and its derivative hierarchy, the actual calibrated maps L_c, the local equation estimate, and the paid comparisons remain separate inputs. This section proves the finite projection, dimension and symmetry transport on their common family and extraction.

180 The trace calibration on the selected construction

The trace asymptotic uses a one-dimensional conserved grade-four line and a calibrated limiting sequence of such tensors. Here we prove the finite- dimensional rank argument conditional on the interacting-center and scale-dependent comparison receipts stated below. The free reference rank is proved in Section 177. Detailed boundary-measure and all-radii arguments remain in Sections 194 and 196.

180.1 The conserved tensor space has dimension one

Let K_0 be the real finite-dimensional space of reference grade-four stress tensors and V_0 its divergence space. Let

$$D_0 : K_0 \longrightarrow V_0$$

be the reference divergence. Let K be the interacting spin-kernel space, $D : K \to V$ its fixed divergence, and, at scale i,

$$O_i : K \longrightarrow V_0$$

the comparison map into the reference two-component table. Only O_i depends on scale. The interacting-center dimension equivalence and the scale-dependent comparison are source inputs stated below; the elementary rank conclusion is proved conditional on them. The free reference statement is not assumed: it is proved in Section 177. Concretely, K_0 is the subspace of reference symmetric grade-four tensors with zero scalar skim, and V_0 is the reference grade-five vector table. The interacting K is the centered grade-four kernel space, and D is the divergence rank map restricted to it. The optical map takes values in the corresponding reference vector table.

1. D_0 is surjective and $\dim \ker D_0 = 1$;
2. $\dim K = \dim K_0$;
3. eventually $\ker O_i = \ker D$;
4. eventually there is a linear map $S_i : \mathbb{R}^2 \to K$ such that $O_i S_i : \mathbb{R}^2 \to V_0$ is bijective.

Item 2 is the selected source's grade-four center-equivalence receipt. Its coherent coordinate maps and centering projections are recorded in the evidence file; this chapter does not infer it from the free reference rank.

Lemma 180.1 (Rank-one conservation law)**.** *Under these four properties,* $\dim\ker D = 1$.

Proof. Choose i large enough that properties 3–4 hold. Since O_iS_i is surjective, O_i is surjective. Rank–nullity and the kernel identity give

$$\dim\ker D = \dim\ker O_i = \dim K - \dim V_0.$$

The reference surjection D_0 gives

$$\dim K_0 = \dim V_0 + \dim\ker D_0 = \dim V_0 + 1.$$

Property 2 identifies $\dim K = \dim K_0$, so subtraction yields $\dim\ker D = 1$. □

The free reference computation is proved in Section 177. In the notation there, the scalar skim is the real one-point average of the 00 tensor entry, the free symmetric grade-four table has dimension four, and the Maxwell-reduced grade-five vector table has dimension two. The zero-skim divergence map is onto and its kernel is the Weyl line. In particular, this is a proved finite-dimensional input to item 1, not a free-rank hypothesis. The source map T from the four coefficient parameters is identified with that chapter's free stress coefficient map. Its point-probe identities and the low-grade signed-covariance classification are proved there.

To prove item 2, let S be the reference table and T the interacting table. The coherent field equivalence $U : S.Field_4 \to T.Field_4$ preserves conjugation and signed action and maps the reference unit to the interacting unit. For a stress array B, define entrywise

$$(\mathsf{U}B)_{ab} = U(B_{ab}), \qquad (\mathsf{C}_T B)_{ab} = B_{ab} - \mathbb{E}_T(B_{ab})\,1_T.$$

Coherence makes U preserve symmetry, reality, and signed covariance. Centering then maps these arrays into the interacting centered kernel space. Thus

$$\mathsf{S}(B) = \mathsf{C}_T(\mathsf{U}B)$$

is a linear map from the reference centered kernel to the interacting one. Define the reverse map $\mathsf{R}(B) = \mathsf{C}_S(\mathsf{U}^{-1}B)$. For a centered reference array B, write $\lambda_{ab} = \mathbb{E}_T(U(B_{ab}))$. Since $U(1_S) = 1_T$,

$$\mathsf{U}^{-1}(\mathsf{C}_T(\mathsf{U}B))_{ab} = B_{ab} - \lambda_{ab}1_S.$$

The reference entry is centered, so $\mathbb{E}_S(B_{ab}) = 0$; also $\mathbb{E}_S(1_S) = 1$. Applying C_S gives $B_{ab} - \lambda_{ab}1_S - (-\lambda_{ab})1_S = B_{ab}$. This proves $\mathsf{RS} = I$. The same calculation with S, T reversed proves $\mathsf{SR} = I$. Hence the maps are inverse linear equivalences, which establishes item 2. This proof uses the source coherence, unit, and centering identities; the evidence crosswalk gives their exact declarations.

Property 3 follows from the actual optical map. At grade five the source formula is

$$(O_iT)(a) = b_i\,U_{i,5}^{-1}\big((DT)(a)\big), \qquad b_i > 0,$$

where $U_{i,5}$ is an isomorphism and $b_i = \mathrm{crown}(Z, J, \mathrm{Stern}(V), 5, i) > 0$ is the positive scalar normalization. The optical map is this scalar times $U_{i,5}^{-1}$ composed with the divergence rank map. Therefore $O_iT = 0$ exactly when $DT = 0$. Positivity and invertibility hold on a cofinal set of scales. Property 4, by contrast, also requires the scale-dependent two-direction limits and stable inversion established below; the free reference calculation does not prove it.

In particular, for every tensor T at all sufficiently large scales,

$$O_iT = 0 \quad\Longleftrightarrow\quad DT = 0. \tag{885}$$

Property 4 follows once the actual source maps S_i have the stated columns u_i^0, u_i^1 and $\mathcal{B}_i = O_i \circ S_i$ converges to an isomorphism. The stable-inversion argument is written next; the scale- dependent convergence estimates remain source inputs.

180.2 Solving the two-direction correction

At scale i, let $u_i^0, u_i^1 \in K$ be the two correction tensors and define $S_i : \mathbb{R}^2 \to K$ by $S_i(e_0) = u_i^0$ and $S_i(e_1) = u_i^1$. Set $\mathcal{B}_i = O_i \circ S_i : \mathbb{R}^2 \to V_0$. Let $T_i^0 \in K$ be the uncorrected scaled tensor and $X_i = O_i(T_i^0) \in V_0$. Thus exact cancellation $\mathcal{B}_i c_i + X_i = 0$ is precisely the optical conservation condition $O_i(T_i^0 + S_i c_i) = 0$. The source's limiting comparison matrix $\mathcal{B}_\infty$ is an isomorphism. The following elementary lemma isolates the only finite-dimensional inverse argument.

Lemma 180.2 (Stable inversion in two directions)**.** *Let E, F be real normed spaces of dimension two. Suppose linear maps $\mathcal{B}_i : E \to F$ converge in operator norm to a bijection $\mathcal{B}_\infty$. Then there are i_0 and $C_0 < \infty$ such that for $i \geq i_0$, $\mathcal{B}_i$ is bijective and*

$$\|\mathcal{B}_i^{-1} y\| \leq C_0 \|y\| \qquad (y \in F).$$

If $\|X_i\| \leq C_1 A_i$ with $A_i \geq 0$, the unique solution $c_i = -\mathcal{B}_i^{-1} X_i$ satisfies

$$\mathcal{B}_i c_i + X_i = 0, \qquad \|c_i\| \leq C_0 C_1 A_i.$$

Proof. The unit sphere of E is compact. Since $\mathcal{B}_\infty$ is injective, the continuous function $u \mapsto \|\mathcal{B}_\infty u\|$ has a positive minimum m on that sphere. For large i, $\|\mathcal{B}_i - \mathcal{B}_\infty\| < m/2$, and hence $\|\mathcal{B}_i u\| \geq (m/2)\|u\|$ for every $u \in E$. Thus $\mathcal{B}_i$ is injective, and equality of the finite dimensions makes it surjective. The same lower bound gives $\|\mathcal{B}_i^{-1} y\| \leq (2/m)\|y\|$. Taking $C_0 = 2/m$ proves the claims, including the equation and bound for c_i. □

The receiver argument is the already integrated Lemma 197.1 in Section 197; it is not duplicated here.

In the construction, E is the real vector space of grade-five coefficient arrays satisfying reality and signed covariance, not the full unconstrained label space. The space H is the reference vector table. Its canonical field map is the quotient projection applied to the grade-five coefficient array; its kernel is the line generated by the nonzero equation-of-motion array x. The scalar functional ρ is the free divergence price after that same quotient projection. The exceptional vector is the medium

$$w_i = \beta_i^{-1} \sigma_i^X - \sigma_i^L,$$

where σ_i^X is the transported chosen response, σ_i^L is the transported reference response at the same scale, and β_i is the positive scale parameter. The source proves $w_i \to x$, weighted decay of its field image, and weighted decay under ρ. The finite chart gives the eventual right inverse u_i, while the limiting chart gives the section U. The lemma shows how these receipts produce a uniformly bounded receiver with small weighted error; it does not prove the response, price, or field-decay estimates themselves.

In the trace construction, $E = \mathbb{R}^2$ labels the two correction sectors, $F = V_0$ is the reference table space, and A_i is the reciprocal scale price. The limiting matrix is invertible because the two reference directions are linearly independent. Its convergence is pointwise on a finite basis, hence in operator norm. The uncorrected divergence has norm at most $C_1 A_i$. Solving the system therefore produces coefficients with the same A_i bound and exact cancellation. Applying the flux functional to this identity gives the coefficient normalization. The precise weighted remainder estimates and the coefficient bound below show that the false-sector term vanishes, while the true-sector term is $-4c_i^1 + o(1)$. Since the uncorrected term tends to $-\kappa$, the true-sector coefficient tends to $-\kappa/4$. In the selected construction κ is the slope constant. The source's limiting physical tensor is therefore fixed by its Ward flux, not by an arbitrary rescaling.

The coefficient argument uses the following estimates from the actual selected source, stated here in mathematical form. For the two correction directions u_i^0, u_i^1, a linear flux functional ℓ_i, correction size A_i, and positive weight a_i:

$$\|X_i\| \le C_1 A_i, \qquad \|c_i\| \le C_0 C_1 A_i, \qquad A_i \le a_i, \tag{886}$$

$$\ell_i(X_i) \longrightarrow -\kappa, \qquad a_i \ell_i(u_i^0) \longrightarrow 0, \qquad a_i\big(\ell_i(u_i^1) + 4\big) \longrightarrow 0, \tag{887}$$

$$\|c_i\| \le C_3 a_i \quad \text{eventually.} \tag{888}$$

The exact cancellation identity is

$$X_i + c_i^0 u_i^0 + c_i^1 u_i^1 = 0.$$

By (888) and the weighted limits, $c_i^0 \ell_i(u_i^0) \to 0$ and $c_i^1(\ell_i(u_i^1)+4) \to 0$. Hence $c_i^1 \ell_i(u_i^1) = -4c_i^1 + o(1)$. Apply ℓ_i to the exact cancellation identity. Since $\ell_i(X_i) \to -\kappa$, it gives $-\kappa - 4c_i^1 + o(1) = 0$, so $c_i^1 \to -\kappa/4$. The selected local estimates establish the displayed bounds; the correspondence table records the exact producer declarations. This argument is the finite-dimensional calibration used by the source's corrected stress sequence.

For clarity, the two limits used after solving the correction system follow from the following explicit scale-dependent receipts. Let $B_i > 0$, let L_i, Ω_i, Δ_i be tensors, and define $\mathcal{Y}_i(T) = B_i^{-1}\mathcal{P}_i(T)$ for a linear scaled tensor map $\mathcal{P}_i$. Put $U_{L,i} = \mathcal{P}_i(L_i)$, $U_{\Omega,i} = \mathcal{P}_i(\Omega_i)$, and $U_{\Delta,i} = \mathcal{P}_i(\Delta_i)$. Assume

$$U_{L,i} \longrightarrow R_0, \qquad U_{\Omega,i} \longrightarrow R_\Omega, \qquad U_{\Delta,i} \longrightarrow R_\Delta, \qquad \frac{RQ_i}{B_i} \longrightarrow 0, \tag{889}$$

and $T_i = B_i L_i + c_i^0 \Omega_i + c_i^1 \Delta_i$ with $|c_i^0| + |c_i^1| \le C R Q_i$. For the trace limit, assume in addition

$$B_i \operatorname{tr} U_{L,i} \longrightarrow 0, \qquad h_i \operatorname{tr} U_{\Omega,i} \longrightarrow 0, \qquad |c_i^0| \le C' h_i, \qquad \operatorname{tr} U_{\Delta,i} \longrightarrow 4O, \qquad c_i^1 \longrightarrow -\kappa/4. \tag{890}$$

Here $h_i \ge 0$ is the weight in the false-sector trace estimate. These hypotheses keep the inverse-normalization scale B_i distinct from that weight; the source uses distinct quantities for them. Here O denotes the reference trace field, so the limit $4O$ is field-valued.

Lemma 180.3 (Normalized and trace limits from the correction receipts)**.** *Under* (889) *and* (890)*,*

$$\mathcal{Y}_i(T_i) \longrightarrow R_0, \qquad B_i \operatorname{tr} \mathcal{Y}_i(T_i) \longrightarrow -\kappa O.$$

Proof. By linearity and the definition of $\mathcal{Y}_i$,

$$\mathcal{Y}_i(T_i) = U_{L,i} + \frac{c_i^0}{B_i} U_{\Omega,i} + \frac{c_i^1}{B_i} U_{\Delta,i}.$$

The coefficient bound and $RQ_i/B_i \to 0$ imply $c_i^a/B_i \to 0$ for $a = 0, 1$. Each U-sequence converges and is therefore bounded. Both correction terms vanish, proving the first limit. Taking traces before normalization gives

$$B_i \operatorname{tr} \mathcal{Y}_i(T_i) = B_i \operatorname{tr} U_{L,i} + c_i^0 \operatorname{tr} U_{\Omega,i} + c_i^1 \operatorname{tr} U_{\Delta,i}.$$

The first term tends to zero. The second is bounded by $C' h_i |\operatorname{tr} U_{\Omega,i}|$, which tends to zero. The last tends to $(-\kappa/4)(4O) = -\kappa O$. □

In the source application, $\mathcal{P}_i$ is the scaled comparison map, B_i is the trace amplitude, and $\mathcal{Y}_i$ is the normalized year map. The source's base and correction beam limits, washout ratio, base trace vanishing, false-sector weighted trace vanishing, and true-sector trace limit instantiate the displayed receipts. The lemma proves the normalized tensor and trace conclusions once these estimates are installed; it does not identify the two weights or infer one from the other.

180.3 The calibrated stress sequence

Write $r_i > 0$ for the selected scale and U_i for its grade-four chart. For a tensor T, set

$$Y_i(T) = r_i^4 U_i^{-1} T.$$

The corrected sequence T_i is conserved by the comparison identity (885). The finite-dimensional correction above gives the explicit limits

$$Y_i(T_i) \longrightarrow R_0, \qquad B_i \operatorname{tr} Y_i(T_i) \longrightarrow -\kappa O, \tag{891}$$

where R_0 is the reference stress tensor and O the reference trace field. For the physical centered tensor R, the same separated shell-flux test has limits $\ell_i(R) \to y \neq 0$ and $\ell_i(T_i) \to y$. Since the conserved space has dimension one, eventually $T_i = t_i R$. Linearity gives $t_i = \ell_i(T_i)/\ell_i(R) \to 1$, so dividing (891) by t_i proves

$$Y_i(R) \longrightarrow R_0, \qquad B_i \operatorname{tr} Y_i(R) \longrightarrow -\kappa O.$$

This is precisely the normalization argument used in Lemma 196.1. The source supplies the two flux limits by its physical and reference translation Ward identities and the separated-jet comparison. The argument above makes explicit why the common nonzero Ward value forces the multiplicative constant to be one.

180.4 Every vanishing positive scale has a strict subsequence

The scale part of the source proves a slightly stronger assertion than a single convenient scaling. Fix any sequence $s_n > 0$ with $s_n \to 0$. At each integer level d, the finite-volume growth and funding estimates give the following selection property: for every lower index N and every $\theta > 0$ there are $n > N$, a level $j \geq d$, and a finite-level chart $V_{d,j}$ such that

$$s_n < \theta, \qquad V_{d,j} \text{ has cutoff-}d \text{ matching receipts at location } j, \qquad r(V_{d,j}) = s_n. \tag{892}$$

The matching receipts cover grades through d, and their accuracy is the one requested at cutoff d. The estimate is uniform in the requested lower index: its only use of N is to ask the scale sequence to pass below a positive threshold after N.

Lemma 180.4 (Recursive strict scale selection)**.** *Suppose the one-stage selection property* (892) *holds at every level. Then there is a strictly increasing sequence of indices n_d and matching charts V_{d,j_d} at levels $j_d \geq d$ such that s_{n_d} is strictly decreasing to zero and $r(V_{d,j_d}) = s_{n_d}$.*

Proof. Choose an admissible triple (n_0, j_0, V_{0,j_0}) at level zero, using the one-stage property with lower index zero and any positive threshold. Having chosen (n_d, j_d, V_{d,j_d}), apply the property at level $d+1$ with lower index n_d and threshold $s_{n_d} > 0$. It gives $(n_{d+1}, j_{d+1}, V_{d+1,j_{d+1}})$ with $n_{d+1} > n_d$ and $s_{n_{d+1}} < s_{n_d}$. Induction gives strict monotonicity of the indices and strict decrease of the selected scales. Since $n_d \to \infty$ and $s_n \to 0$, the subsequence s_{n_d} tends to zero. The selected chart receipts hold at each level by construction. □

The selected construction obtains the one-stage property from its funded finite-volume selection estimate. It finds a scale below θ past any prescribed index, then chooses a chart with scale-location index at least d and with the accuracy requested at cutoff d. The recursive construction above is the complete subsequence argument; it does not assume that a preselected scaling already follows the requested sequence.

180.5 Corrected lifts and compatibility across grades

The gradewise maps obtained by finite-volume extraction and projection need not preserve the filtration inclusions. In the source, extraction and projection are performed first; the correction is then made directly in the selected table fields. Write E_d for the reference field space in grade d and S_d for that selected table field space. Let

$$\iota_d^E : E_d \longrightarrow E_{d+1}, \qquad \iota_d^S : S_d \longrightarrow S_{d+1}$$

be their inclusion maps. For $d < e$, write $\iota_{d,e}^E$ and $\iota_{d,e}^S$ for their composites, and use the identity maps when $d = e$. They obey the transitivity identities. The reference inclusion is injective. Choose a linear left inverse $R_d : E_{d+1} \to E_d$, so $R_d \iota_d^E = 1$. When fields carry the finite signed symmetry and real structure, this retraction can be chosen to preserve them: begin with any left inverse and average its conjugates over the finite signed group. Then average that equivariant map with its conjugate-linear reflection. The signed actions commute with the real involution, so this second averaging preserves signed equivariance. Each conjugate remains a left inverse because the inclusion intertwines both actions. This is the finite-dimensional splitting used below.

For each finite-volume index k, let $U_{k,d} : E_d \to S_d$ be the raw chart maps. Define the corrected maps recursively by

$$C_{k,0} = U_{k,0}, \tag{893}$$

$$C_{k,d+1} = \iota_d^S C_{k,d} R_d + U_{k,d+1}(1 - \iota_d^E R_d). \tag{894}$$

The first summand corrects the part of a field already represented at the previous grade; the second retains the new complement. Directly using $R_d \iota_d^E = 1$ gives

$$\begin{aligned} C_{k,d+1}\iota_d^E &= \iota_d^S C_{k,d} R_d \iota_d^E + U_{k,d+1}(\iota_d^E - \iota_d^E R_d \iota_d^E) \\ &= \iota_d^S C_{k,d}. \end{aligned}$$

Induction and transitivity therefore give, for every pair $d \le e$,

$$C_{k,e}\iota_{d,e}^E = \iota_{d,e}^S C_{k,d}. \tag{895}$$

This is an exact identity at each index k, before the $k \to \infty$ limit. It is the needed balanced inclusion relation; unrelated isomorphisms at early indices are not used to obtain it.

The same correction preserves the weak symmetry conditions by induction, since R_d, both inclusions, and $U_{k,d}$ intertwine the signed and real structures. The unit in each grade is the iterated inclusion of the constant unit in grade zero. Thus the exact grade-zero unit identity, together with the balanced relation, gives the unit identity at every grade. It also preserves the tested approximation. To see the exact finite-slot identity, let I be a subset of the tuple's slots already using C, and let $\mathcal{W}_k^I(v)$ be the tested chart correlation with C in those slots and U in the others. For $i \notin I$ with $v_i \in E_{d+1}$, put $p_i = R_d v_i$ and $q_i = (1 - \iota_d^E R_d)v_i$. Linearity in that slot, formula (894), and the table inclusion identity give the exact equality

$$\begin{aligned} \mathcal{W}_k^{I\cup\{i\}}(v) - \mathcal{W}_0(v) = {} & \big(\mathcal{W}_k^{I\cup\{i\}}(v[i \mapsto p_i]) - \mathcal{W}_0(v[i \mapsto p_i])\big) \\ & + \big(\mathcal{W}_k^{I}(v[i \mapsto q_i]) - \mathcal{W}_0(v[i \mapsto q_i])\big). \end{aligned}$$

Here $v[i \mapsto p_i]$ has grade d in slot i, and $v[i \mapsto q_i]$ has grade $d + 1$. Induction on the number of corrected slots, using this grade reduction for a corrected slot, expresses the fully corrected discrepancy for any fixed tuple as a finite sum of raw tested discrepancies at fixed labels. Every

summand tends to zero by raw approach, so corrected correlations approach the reference. The finite sum's size and label norms may depend on the tuple; this qualitative argument does not retain the raw numerical rate without tracking those factors.

The raw maps $U_{k,d}$ below are formed by the finite-dimensional ultrafilter limits and grade projections. Apply the correction after this projection, exactly as in the source. The balanced identity (895) is then exact for each fixed pair of grades. Once tested-correlation convergence makes the corrected maps $C_{k,d}$ isomorphisms eventually at each fixed grade, the selected isomorphisms obey the inclusion identity for each fixed pair on a tail of indices. The exceptional tail may depend on the pair of grades; no single tail uniform in all grades is asserted. This is the compatible gradewise scaling conclusion used by the source.

Here are the finite-dimensional gluing details. At cutoff k, let

$$\mathscr{L}_k = \prod_{d=0}^{k} \operatorname{Hom}_{\mathbb{C}}(E_d, A_d),$$

where E_d is the free reference grade-d field space and A_d is the finite family coefficient space at grade d. Write S_d for the selected table field space. This is finite-dimensional. The finite-volume lifts into $\mathscr{L}_k$ are bounded along the one fixed ultrafilter: the source has a uniform lift bound on its late set. The closure of a bounded set in a finite-dimensional space is compact, so every ultrafilter on that late set has a limit $Y_k \in \mathscr{L}_k$. This is done for each integer k using the same ultrafilter. Let $P_d : A_d \to S_d$ be the selected table's projection. Put

$$U_{k,d} = P_d \circ Y_k(d) : E_d \longrightarrow S_d \qquad (d \le k).$$

The exact finite-volume reality, signed-reflection, and unit identities hold on a set belonging to the ultrafilter. Their maps are continuous, so they pass to Y_k and then to $U_{k,d}$.

For fixed k, the calibrated-smearing argument also passes each finite-volume short-test bound to the limit. Let $A_{k,v}(t)$ denote the tested correlation of a fixed tuple $v = (v_1, \dots, v_n)$ after applying U_{k,d_a} and scaling by s_{n_k}; let $A_{0,v}(t)$ denote its reference correlation. For a tuple of arity $n \le k$, labels of grades at most k, and any smooth compactly supported test t inside the cutoff window, with $\sup |t| \le M$, the bound is

$$|A_{k,v}(t) - A_{0,v}(t)| \le \frac{1}{k+1}\, M \prod_{a=1}^{n} \|v_a\|. \tag{896}$$

The change-of-variables identity for the dilated test converts the finite-scale integral to the displayed tested correlation. For any $\eta > 0$, the upper-bound receipt makes the finite-scale discrepancy at most the right side plus η on a late set. Since the fixed ultrafilter lies below that late filter, the limit obeys the same bound after letting $\eta \downarrow 0$. The cutoff window radius tends to infinity with k, so it eventually contains the support of each fixed test. Since $(k+1)^{-1} \to 0$, the tested correlations converge to their reference values. This is the approach property needed for the scaling chart.

Fix a grade d. The reference smoothness and test-function uniqueness imply that the linear functionals obtained by inserting a grade-d field into smooth tested correlations separate points of E_d. Since E_d is finite-dimensional, choose finitely many such tests $\lambda_1, \dots, \lambda_m$ spanning its dual. The raw approach estimate gives convergence of their values for $U_{k,d}$, and the finite-sum correction argument gives the same convergence for $C_{k,d}$. On a finite basis the combined maps $\Lambda_k = (\lambda_{j,k})$ therefore converge in operator norm to the injective reference map $\Lambda = (\lambda_j)$. The minimum of $\|\Lambda v\|$ on the unit sphere is positive. Hence Λ_k is injective for all sufficiently large k, and since each functional factors through $C_{k,d}$, so is $C_{k,d}$. The dimension receipt gives $\dim S_d \le \dim E_d$;

injectivity gives the reverse inequality. Thus $C_{k,d}$ is an isomorphism eventually for each fixed d. The exceptional tail may depend on d.

Theorem 180.5 (Conditional gradewise ultralimit charts)**.** *Assume the scale comparison matrices $\mathcal{B}_i$ converge in operator norm to an isomorphism and $\|X_i\| \le CA_i$, so stable inversion supplies property 4 in the rank argument; property 3 follows from* (885)*. Assume also the eventual correction cancellation and estimates in* (886)*–* (890)*, together with the one-stage selection property* (892)*. For physical stress normalization, assume separated shell-flux limits $\ell_i(R) \to y \neq 0$ and $\ell_i(T_i) \to y$. Finally, for each cutoff k, assume the finite-volume lift bound, exact unit and covariance identities, the dimension receipt, and the short-test bound for every tuple of arity at most k described above. Then for every positive sequence $s_n \to 0$, there is a strictly increasing sequence n_k such that s_{n_k} strictly decreases to zero, and there are corrected maps $C_{k,d} : E_d \to S_d$ for $d \le k$ with the following properties: for each fixed d, $C_{k,d}$ is an isomorphism for all sufficiently large k; the exact unit, reality, and signed-covariance identities hold at those indices; and for each fixed tuple and compactly supported test, the tested correlation converges to its reference value as $k \to \infty$. Moreover, for each fixed pair $d \le e$, the corrected maps obey $C_{k,e}\iota^E_{d,e} = \iota^S_{d,e}C_{k,d}$ for all sufficiently large k. The tail may depend on (d, e).*

Proof. Apply the recursive scale-selection lemma to choose n_k with strictly decreasing s_{n_k}. At each cutoff k, the common-ultrafilter finite dimensional limit gives the raw projected maps $U_{k,d}$ and their tested correlation estimates. The corrected-lift calculation above produces $C_{k,d}$; the finite slot argument transfers tested convergence, reality, and signed covariance to these maps. The finite separating-test argument then proves eventual injectivity grade by grade, and the dimension receipt makes each fixed-grade map eventually bijective. The exact balanced identity holds for every corrected map by (895); for each fixed pair of grades, intersecting the two eventual bijectivity tails shows the selected maps satisfy the claimed inclusion relation. □

The corrected-lift construction proves the pairwise eventual inclusion identities for the corrected maps; it does not claim one tail uniform over all grades. The actual source theorem producing the full gallery also requires several substantive analytic estimates: the one-stage finite-volume chart selection; the scale-dependent two-direction limits; the base and correction beam limits; the weighted false-sector flux washout and true-sector flux convergence; the separated-jet comparison; and the finite chart lift, unit, covariance, short-test, and dimension receipts used in the gluing. The source crosswalk gives exact producers and the selected public route, but this unit has not independently rederived those estimates. Thus this chapter proves the finite-dimensional rank calculation, the stable inversion and coefficient extraction, the scalar subsequence recursion, and the finite-dimensional gluing conditional on the listed source receipts; it does not claim a stand-alone proof of the full gallery producer.

181 From funded chart receipts to a trace gallery

This section records the actual selected-run route from one-scale chart data to the scale-dependent charts used by the trace calibration. The analytic chart receipts are provided by the funded construction and are identified below. We prove the subsequent scale selection and finite-dimensional limit steps explicitly. The correction and Ward-flux calculations are in Section 180.

181.1 The funded chart family and its one-step output

Fix a grade d, a binding, a gauge-invariant chart budget, and a scale parameter $z \geq 0$. The funded chart receipt consists of three panels $P(t, i)$, $i = 0, 1, 2$, for $0 \leq t \leq 1$, with arc coordinates $b_i + t$. Their currency and angular parameters are constant in t, and the three base arcs obey the finite separation inequalities for the probe count at grade d. At every sufficiently large level k and every $t \in [0, 1]$, the receipt supplies a finite-volume chart witness at that level and offset. The other fields of this witness are the chart, invoice, legacy-cost, actuation, and stamp data used by the quantitative chart estimate. The single offset changes only the three arc coordinates.

The funded run constructs these panels from its paid multi-scale request. The request gives charts with a common added offset and fixed currency and angle. Forward chart estimates supply legacy costs at all large levels; the staging estimate at the common angle supplies simultaneous actuation and stamp receipts. Clipping the offset to $[0, 1]$ preserves the arc lower bound, chart factor, currency, and angle. The resulting finite-level data are the hypotheses of the one-step chart theorem.

That theorem also requires a simple gauge group, the funded chart-control estimate, a bound on the generator through grade d, an index l with $l \leq \mathrm{trays}(d)$, a positive sequence tending to zero, and the selected extraction order, smooth table, point limits, and reference hierarchy used by the station construction. It returns a strictly increasing level curve and, at a selected point, data (k, i, b, c, R) such that the requested scale equals the exact clock gap at level i and base arc b. Its chart has $5 < b$ and b no larger than the permitted angular bound, covers all reference grades through d, and supplies the finite-volume charts and invoices whenever $i < N$. It also supplies a pairwise comparison receipt at cost $\mathrm{costHint}(l)$ and a station whose lift has a finite norm bound, preserves the unit, real structure, and signed action, and obeys the short-test estimate for arity at most l, tests in the stated window, and error $\mathrm{costHint}(l)$. The proof of this quantitative upgrade is the separate three-panel chart result; this section uses that exact output and proves the scalar gap interpolation and recursion. In the gallery construction one takes $l = d$, so the chart cutoff, arity cap, and finite grade cutoff agree. The clock gives, for each fixed base arc $b \geq 0$, a positive gap

$$g_b(k) = a_k m_k^b, \tag{897}$$

where $a_k > 0$, $m_k \geq 1$, $g_b(k) \to 0$, and eventually $g_b(k) \leq g_{b+1}(k+1)$. These scale-gap facts are the input to the selection argument.

Lemma 181.1 (Clock-gap interpolation)**.** *Fix $b \geq 0$. Suppose $g_b(k) > 0$, $g_b(k) \to 0$, and $g_b(k) \leq g_{b+1}(k+1)$ for all $k \geq k_0$. For every $J \geq k_0$ there is $r > 0$ such that for each $0 < s \leq r$ there are $k > J$ and $t \in [0, 1]$ with $s = g_{b+t}(k)$.*

Proof. Take $r = g_b(J) > 0$. Given $0 < s \leq r$, choose the least $k > J$ with $g_b(k) \leq s$; it exists by convergence to zero. Minimality gives $s \leq g_b(k-1)$, and the one-step comparison at $k-1$ gives

$$g_b(k) \leq s \leq g_b(k-1) \leq g_{b+1}(k).$$

The function $t \mapsto g_{b+t}(k) = a_k m_k^{b+t}$ is continuous on $[0, 1]$. The intermediate value theorem supplies the desired t. □

181.2 Choosing scales and chart levels

Lemma 181.2 (Strict scale selection)**.** *Assume the one-step chart producer has this output at every grade d: for every previous index N and every $\theta > 0$, it supplies an index $n > N$, a level $i \geq d$, and*

a grade-d chart with exact radius $s_n < \theta$ and the finite-level receipts above. Let $s_n > 0$ and $s_n \to 0$. Then there are strictly increasing indices n_d, levels $i_d \geq d$, and admissible charts at grade d and level i_d, such that

$$s_{n_0} < 1, \qquad s_{n_{d+1}} < s_{n_d}, \qquad i_d \geq d.$$

In particular $i_d \to \infty$ and the selected chart radius is exactly s_{n_d}.

Proof. For $d = 0$, apply the one-step output with previous index 0 and threshold 1. For $d > 0$, apply it with previous index n_{d-1} and threshold $s_{n_{d-1}}$. It gives $n_0 > 0$, $s_{n_0} < 1$, and thereafter $n_d > n_{d-1}$, $s_{n_d} < s_{n_{d-1}}$, with a grade-d chart at level $i_d \geq d$ and exact radius s_{n_d}. Induction proves the claims; $i_d \geq d$ implies $i_d \to \infty$. □

The one-step chart theorem obtains this output by combining the clock-gap interpolation with the funded chart family. Its quantitative comparison and station estimates are the three-panel weak-comparison result in Section 168; we use its conclusion here without duplicating those estimates.

Here is the finite-level extraction used to pass from the charts to the limiting fields. Let F be an extraction order, let $\mathcal{U}$ be an ultrafilter on period-phase pairs finer than the fused filter of F, and fix a cutoff D. Suppose $X_{N,p}$ is a tuple of linear maps, one for each grade $0, \ldots, D$, from the corresponding reference field into the common finite-period coefficient space. Assume $A \geq 0$ and, on an F-late set,

$$\|X_{N,p}\| \leq A. \tag{1}$$

The finite product of these map spaces is finite-dimensional. The closed ball of radius A is compact. Compactness gives a proper subfilter limit of the bounded tuple of maps. This subfilter is finer than $\mathcal{U}$; ultrafilter maximality identifies it with $\mathcal{U}$. Thus the tuple converges along $\mathcal{U}$ to $Y = (Y_0, \ldots, Y_D)$. This is the only compactness step: it is finite-dimensional and is performed at each fixed cutoff.

If the finite-level maps preserve the real involution and signed actions on the late set, continuity of those actions shows that each Y_d preserves them. If they send the reference unit to the scalar unit eventually, the same limit identity gives $Y_d(1) = 1$. Compose with the quotient projection to define the table map

$$U_d = \mathsf{P}_d Y_d, \qquad 0 \leq d \leq D.$$

This order matters: the finite maps converge in the coefficient space, and the table-valued maps are obtained by applying the continuous projection after taking the limit.

For a fixed tuple of n labels of grades at most D, a fixed smooth test function supported in the allowed window, and n no greater than the available arity cap, the finite-level receipt bounds the paid test error by ϵ times the label charge and the test supremum. To pass this estimate to the quotient table, use the ordered-moment limit rather than continuity of an integral whose finite-volume measure varies. For each fixed tuple of coefficient vectors, the ordered read receipt gives convergence of the corresponding finite-volume moment to the table moment. The moment is multilinear in its n label vectors, and each label space is finite-dimensional. Expanding each moving vector in a fixed basis expresses the moment as a finite sum of products of convergent coordinates and convergent fixed-basis moments. Hence the moving-label moment converges to the table moment. This is the finite-basis multilinear limit argument; it uses the common ordered extraction and the fixed compactly supported test. The inequality passes to the limit because a closed interval is closed under limits. More explicitly, for each $\eta > 0$ the late-set estimate is at most the claimed bound plus η; taking the limit and then letting $\eta \downarrow 0$ removes the slack. This proves the cutoff-level short-test estimate with no exchange of an infinite sum and a limit. It is the same moment-passage mechanism as the calibrated-smearing lemma: fixed-vector convergence is supplied by the ordered table, and finite dimensionality handles the moving vectors.

Apply this extraction separately at cutoff $D = d$ to the chart chosen at grade d and level i_d. The chart receipt supplies a bound on its finite lifts, exact reality and signed covariance, the exact unit, and a short-test error for arity $n \le d$ with coefficient $1/(d+1)$. Projecting the limits gives panels U_d. For every fixed finite label tuple, its grades are at most d eventually along the sequence of cutoffs; the support of a fixed compact test fits the expanding chart window eventually; and $1/(d+1) \to 0$. Write $\mathcal{T}_{r,U}(v;t)$ for the tested integral of the tuple v under radius r, panel family U, and test function t. Write $\mathcal{T}_0(v;t)$ for the corresponding reference test value. Hence

$$\mathcal{T}_{s_{n_d},U_d}(v;t) - \mathcal{T}_0(v;t) \longrightarrow 0$$

for every fixed tuple v and test t. This is the tested approach property. The argument is for each fixed test; it does not assert a uniform tail over all tests or all label tuples.

The final chart isomorphisms follow grade by grade. For a fixed grade d, the reference tests separate its finite-dimensional field: every nonzero vector has a nonzero separated two-point evaluation. To obtain a smooth test, keep the spectator insertion fixed and choose a separated pair where the complex value is nonzero. Multiply the spectator by a unit complex phase so that the real part of the value is positive. Continuity away from the diagonal gives a small neighborhood on which that real part remains positive. Integrating against a nonnegative smooth bump of integral one, supported in that neighborhood, gives a smooth smeared test with nonzero value. Thus smooth smeared reference tests also separate. Normalize to the unit sphere. For each point of that compact sphere, continuity gives a neighborhood on which one such test has absolute value at least half its value at the point. A finite subcover gives tests $t_1, \dots, t_m$ and a constant $\alpha > 0$ such that every unit vector has $\max_j |\mathcal{T}_0(v;t_j)| \ge \alpha$. The fixed finite set of label charges and test suprema is bounded on that sphere. The tested approach estimate therefore makes all corresponding errors less than $\alpha/2$ for sufficiently large k. If the projected grade map killed a unit vector, all its tested values would vanish, contradicting this lower bound. Thus it is injective. The dimension bound from the selected table gives the reverse dimension inequality; therefore the two dimensions agree and the map is bijective. This tail may depend on d: there is no single finite index after which every grade is simultaneously bijective. For the finitely many indices before the grade-d tail, choose any isomorphism using the same dimension equality. This defines a scaling at every index. For each fixed grade, the unit, reality, and signed symmetry identities hold on its eventual isomorphism tail; the arbitrary isomorphisms used earlier need not preserve those identities. The reference test-separation input here is the all-grade free-detector result of Section 178; the separate low-grade trace calculation is not enough to justify separation in every grade. The reviewed corrected-lift argument in Section 180 provides the compatible panel maps when required by the trace family.

181.3 The gallery handoff

The preceding argument proves the scale-selection, finite-cutoff extraction, tested-approach, and fixed-grade eventual-isomorphism clauses. It does not by itself prove the complete gallery statement. That handoff also needs the physical-array receipts in the selected-run transfer. For the same ultrafilter, hierarchy, and scaling these include convergence of the recentered physical stress to its prescribed material tensor; for every comparison table, an eventual decomposition of its rescaled transformed energy tensor into the scalar unit and canonical stress direction, with the two scalar coefficients converging respectively to the prescribed stress coefficient and to zero; and convergence of the corresponding zoomed scalar observable to the prescribed multiple of the unit. The arbitrary-null- sequence case needs these same receipts after the selected scaling has been constructed above. They are additional physical and field-array estimates, not consequences of finite-dimensional compactness or of the rank-one solve alone. The correction, one-dimensional kernel, stable inverse, flux

normalization, and physical/reference shell comparison in Section 180 remain conditional on their displayed analytic receipts.

In the selected public construction, the one-scale chart family comes from the paid multi-scale request and its subsequent chart staging. This section proves the scale-selection and tested-chart clauses of that assembly; the physical-array, stress, and flux receipts needed for the full gallery remain separate inputs.

181.4 A priced mission forces the weighted response to vanish

This subsection records the quantitative argument behind the weighted field-image estimate used by the trace receiver. It exposes the local price, scale, and detector inputs. It does not derive the mission's local price bound from the finite-volume estimates.

Let S be the selected interacting table, and let $\mathcal{U}_i^{(d)} : E_d \to S_d$ be its scale chart from the reference grade space E_d into the selected grade space S_d. Let $\mathcal{V}$ be the finite-dimensional real vector space of signed-equivariant grade-five response arrays (the source's one-point response table), and let s_i be the real-linear normalized response map into $\mathcal{V}$. Write j_i for the selected frame level and k_i for its associated scale index. The source uses two distinct positive scales:

$$z_i = \widehat{\beta}_{k_i}, \qquad \zeta_i = Z.\beta(k_i).$$

The first, z_i, appears in the polynomial weight and rate estimate; the second, ζ_i, is the Arch scale in the medium and cancels from the response identity below. Let a_i be the field dilation at level j_i, b_i the frame scale, m_i the frame cap, and r_i the physical dilation. The selected scale relations are

$$r_i = a_i b_i, \qquad 0 < b_i \le m_i, \qquad m_i = \mathrm{tiltcap}(j_i).$$

Let v_i be the selected medium and x_i its core field. Its exact core representation is

$$v_i(\alpha) = \zeta_i^{-1} \mathcal{E}_{k_i}(x_i(\alpha)) \qquad (\alpha \in \mathrm{Axis}), \tag{898}$$

where $\mathcal{E}_k$ is the source's core-to-strip map. The response normalization is

$$\mathcal{U}_i^{(5)}((s_i(v))_\alpha) = \mathcal{V}_i(v_\alpha), \tag{899}$$

where $\mathcal{V}_i = \mathrm{Vault}(S, Z, 5, k_i, c_i)$ is the grade-five strip evaluation at scale index k_i with crown normalization c_i. For x odd under central signed inversion, meaning $F(\mathrm{archV})x = -x$, the exact scaling identity is

$$\mathcal{V}_i(\mathcal{E}_{k_i}(x)) = c_i P_5 x, \qquad c_i = \zeta_i (a_i b_i)^5, \tag{900}$$

where $P_5 = S.\,\mathrm{project}_5$ maps raw grade-five labels into S_5. These are exact source identities, not consequences of the estimate below.

For a reference grade-four field $w \in E_4$, let $\lambda_{i,w}(Y)$ be the tested two-point pairing of $\mathcal{U}_i^{(5)}(Y_0)$ with $\mathcal{U}_i^{(4)} w$, at the shrinking point $r_i \zeta_i^{\mathrm{pt}}$. Denote this pairing on explicitly supplied grade-five and grade-four entries by $\mathfrak{p}_{i,w}$. The local price input is the bound

$$\left| \mathfrak{p}_{i,w}(a_i^5 P_5 x_i(0), \mathcal{U}_i^{(4)} w) \right| \le \Theta_i, \quad \text{eventually in } i \text{ for each fixed } w, \tag{901}$$

where Θ_i is the positive local price allowance. The source proves this receipt from the priced mission and its finite-volume mileage estimate; it remains an upstream input here.

The selected chart hierarchy and its derivative/orientation identities supply a finite detector. There are fixed reference fields $w_1, \dots, w_m \in E_4$ such that the limiting pairings define an injective linear map

$$\Lambda(y) = (\lambda_{w_r}(y))_{r=1}^m : \mathcal{V} \longrightarrow \mathbb{C}^m.$$

The scale-dependent maps $\Lambda_i(y) = (\lambda_{i,w_r}(y))_{r=1}^m$ converge to Λ in operator norm. The full family of grade-four insertions separates the signed-equivariant response arrays; choose a finite separating subfamily and use the selected chart and shrinking-point convergence for those tests. If $\mathcal{V} \neq \{0\}$, injectivity gives a positive minimum norm on its unit sphere. Thus some $c > 0$ satisfies

$$\|\Lambda_i y\| \geq c\|y\| \qquad \text{for all sufficiently large } i \text{ and every } y \in \mathcal{V}. \tag{902}$$

For $\mathcal{V} = \{0\}$ the desired conclusion is immediate. The separation and convergence receipts, not finite dimensionality alone, are the detector inputs.

The scalar rate input is

$$z_i^p m_i^5 \Theta_i \longrightarrow 0 \qquad \text{for each fixed integer } p \geq 0. \tag{903}$$

In the selected source, this follows by multiplying three rate tickets: the rate of z_i is bounded by the zero-rate ticket for the scale at the summit of j_i; m_i has a zero-rate fifth-power ticket; and Θ_i has rate -100δ, with the fixed source constant $\delta = 10^{-4}$. Ticket products add rates, so for fixed p the product has negative rate $-100\delta < 0$ and tends to zero. The stage comparison bounds z_i by the summit scale. These scalar receipts are listed with their exact source declarations in the evidence file.

Lemma 181.3 (Weighted field decay from the priced mission)**.** *Assume* (898)–(903)*, the night condition for* $x_i(0)$*, the response identity* (899)*, the price bound* (901)*, and the detector bound* (902)*. Then, for each fixed integer* $p \geq 0$*,*

$$z_i^p s_i(v_i) \longrightarrow 0 \quad \text{in } \mathcal{V}.$$

Proof. Set $Y_i = z_i^p s_i(v_i) \in \mathcal{V}$. By the medium representation, response normalization, and night scaling,

$$\mathcal{U}_i^{(5)}((Y_i)_0) = z_i^p \zeta_i^{-1} c_i P_5 x_i(0) = z_i^p (a_i b_i)^5 P_5 x_i(0).$$

The Arch scale ζ_i cancels; it is distinct from the polynomial weight z_i. Linearity in the grade-five entry gives, for each detector field w_r,

$$\lambda_{i,w_r}(Y_i) = z_i^p b_i^5 \mathfrak{p}_{i,w_r}(a_i^5 P_5 x_i(0), \mathcal{U}_i^{(4)} w_r).$$

The local price bound and $b_i \leq m_i$ imply

$$\|\Lambda_i Y_i\| \leq \sqrt{m}\, z_i^p m_i^5 \Theta_i \longrightarrow 0.$$

The detector lower bound gives $\|Y_i\| \leq c^{-1}\|\Lambda_i Y_i\| \to 0$. This proves the desired limit in the full finite-dimensional response space. □

The source applies this lemma to its selected medium. The proof identifies the mechanism: the powered local price controls finitely many separating coordinates, and the negative rate forces each coordinate to zero. The uniform local price receipt, mission construction, and detector convergence still depend on the selected finite-volume funding and chart construction.

182 The finite-volume Crown buffer

We record the buffer argument used to pass from a local Crown approximation to the selected two-point observable. It is useful to separate its three ingredients: a rooted product estimate for the good approximation, a finite-volume estimate for the exceptional set, and the exact product identification across the coarse-to-fine map.

Fix a scale index $i > 0$, a finite-volume index N, and a period p with $i \le k = \mathrm{summit}(i) \le N$. Put $k_- = k - 1$. Write

$$\begin{aligned} T &= \mathrm{tiles}(i,k,N,p), & F_i(q) &= \exp\Big(\frac{q}{10^4}\, G_i\Big), \\ G_i &= \mathrm{Gate}(\widehat{\beta}_i), & \Pi_k &= \mathrm{prism}(\widehat{\beta}_k), \\ \beta &= \beta_{N,p}, & \beta_k &= \beta_{N,p}(k), \qquad A_j = C.S.a(j). \end{aligned}$$

For a tiling T with cell set I, its rooted functional is $\mathsf{R}_T^{\beta}(h) = |\mathbb{E}_\beta[\prod_{q\in I} T.\mathrm{copy}(q,h)]|^{1/|I|}$, with the normalized-root convention of the source. Write $\mathcal{B}_{N,k}$ for the local extension from scale k to volume N. For distinct coarse sites a, b at scale i, let $\alpha_{i,k}(a)$ and $\alpha_{i,k}(b)$ denote their scale-k anchors, and let $\alpha_{i,N}(a), \alpha_{i,N}(b)$ denote their volume-N anchors. For scale-k fields X, Y, define the complex pairing

$$\begin{aligned} \mathcal{Q}(X,Y) &= \mathbb{E}_\beta\Big[((\mathcal{B}_{N,k}X)\circ \mathrm{translate}_{\alpha_{i,N}(a)})((\mathcal{B}_{N,k}Y)\circ \mathrm{translate}_{\alpha_{i,N}(b)})\Big], \\ X_a &= X\circ \mathrm{pull}_{\alpha_{i,k}(a)}, \qquad Y_b = Y\circ \mathrm{pull}_{\alpha_{i,k}(b)}. \end{aligned}$$

The exact product identity used below is $\mathcal{Q}(X,Y) = \mathbb{E}_\beta[\mathcal{B}_{N,k}(X_aY_b)]$ when both fields are interior to a common window and that window fits the tiling. The scale receipts give, eventually in i,

$$e^{-2\Pi_k} \le F_i(-1{,}200{,}000), \qquad 3 \le F_i(1). \tag{904}$$

For the first inequality, use $i \le k$, monotonicity of the gate, and the selected estimates

$$100{,}000\, G_k\ \mathrm{Lyra}(k) \le \Pi_k/32, \qquad G_i \le G_k.$$

Since $\mathrm{Lyra}(k) \ge 1$ eventually, this gives $120G_i \le 2\Pi_k$, which is the exponent inequality in (904). The second inequality follows from $G_i \to \infty$.

Suppose f is the field to be tested and a Crown witness provides a field w, a support P, and a window radius l such that both fields are interior to that window, the scale has room for l, and

$$|P| \le F_i(1000), \qquad \sup_U |f(U) - w(U)| \le F_i(500{,}000), \tag{905}$$

$$|f(U) - w(U)| \le F_i(-500{,}000) \quad \text{if } U \text{ is healthy for } P \text{ at threshold } \vartheta_k/16, \tag{906}$$

$$\mathsf{R}_T^{\beta}(\mathcal{B}_{N,k}w) \le F_i(-1{,}200{,}000). \tag{907}$$

Let g be the second field in the desired probe, with $\sup_U |g(U)| \le F_i(150{,}000)$ and with the required window interiority. Let $a \ne b$ be distinct coarse sites. Define the pulled-back factors

$$f_a(U) = f(\mathrm{pull}_{\alpha_{i,k}(a)}\, U), \quad w_a(U) = w(\mathrm{pull}_{\alpha_{i,k}(a)}\, U), \quad g_b(U) = g(\mathrm{pull}_{\alpha_{i,k}(b)}\, U), \quad D_a = f_a - w_a.$$

Here ϑ_k is the clock threshold at scale k; the pull maps use the scale-k anchors defined above.

We make explicit the finite-volume input used for the correction. Let ρ_k be the nonnegative density of the effective scale-k representation of the volume-N Wilson measure, and put $Z = \int \rho_k = \mathrm{Partition}(N,p)$. We use the normalized transfer identity

$$\mathbb{E}_{\beta_{N,p}}[\mathcal{B}_{N,k}h] = Z^{-1}\int \rho_k(U)h(U)\,d\mu_k(U). \tag{908}$$

Let $I_x(U)$ indicate that the local flag at site x exceeds threshold b. The one-site estimate is an incoming local-action receipt:

$$\int \rho_k(U)I_x(U)\,d\mu_k(U) \le Z\exp\left(\left(\frac{A_{k_-}}{A_k}\right)^4 \mathrm{Treasury}(k) - \beta_k\frac{\mathrm{clip}_k}{5}b^2\right). \tag{909}$$

It applies when $2b \le \vartheta_k$ and the scale has room for the radius-eight indicator. Given this receipt, obtain the corresponding estimate for a support P' by splitting configurations into those healthy for all sites of P' and the complement. On the healthy set, use the bound U_0 for the field. On the complement, there is at least one $x \in P'$ with $I_x = 1$, so the pointwise absolute integrand is at most $V_0\rho_k\sum_{x\in P'} I_x$. Integration, the triangle inequality, and (909) give

$$|\mathbb{E}_{\beta_{N,p}}[\mathcal{B}_{N,k}h]| \le U_0 + V_0|P'|\exp\left(\left(\frac{A_{k_-}}{A_k}\right)^4 \mathrm{Treasury}(k) - \beta_k\frac{\mathrm{clip}_k}{5}b^2\right). \tag{910}$$

The exponent in (910) is eventually at most $-2\Pi_k$ by the scale estimate

$$\left(\frac{A_{k_-}}{A_k}\right)^4 \mathrm{Treasury}(k) - \beta_k\frac{\mathrm{clip}_k}{5}b^2 \le -2\Pi_k, \tag{911}$$

for $b = \vartheta_k/16$. This is the exact threshold used in the witness bound. Here is the scalar calculation behind that estimate. Put $x = \widehat{\beta}_k$, $y = \mathrm{Lyra}(k)$, $q = \Pi_k$, $C = \mathrm{clip}_k$, and $m = \mathrm{monitor}(k)$. These satisfy $x, y, C > 0$ and $q \ge 0$. The selected scale receipts give

$$\left(\frac{A_{k_-}}{A_k}\right)^4 \mathrm{Treasury}(k) \le y^6, \quad 10\cdot 16^2 y(y^6+2q) \le m^2, \quad yC \ge 1, \quad x\vartheta_k^2 = m^2, \quad \beta_k \ge x/2.$$

Since $b = \vartheta_k/16$, the second and fourth inequalities imply $y^6 + 2q \le m^2/(2560y)$. The third gives $1/y \le C$, so

$$y^6 + 2q \le \frac{Cm^2}{2560} = \frac{x}{2}\frac{C}{5}\left(\frac{\vartheta_k}{16}\right)^2 \le \beta_k\frac{C}{5}b^2.$$

Subtracting from the first bound yields the exponent in (911). Thus the exceptional-set estimate is obtained from the one-site flag bound and the displayed scale receipts, not from locality by itself.

Lemma 182.1 (Good Crown term)**.** *Under the seam condition for T and the window and room hypotheses above,*

$$|\mathcal{Q}(w,g)| \le F_i(-1{,}050{,}000).$$

Proof. The translated planted inequality gives

$$|\mathcal{Q}(w,g)| \le \mathsf{R}^{\beta}_T(\mathcal{B}_{N,k}w)\,\mathsf{R}^{\beta}_T(\mathcal{B}_{N,k}g).$$

Its proof represents each anchored factor by a tiling copy of an oriented, possibly conjugated chamber-interior field; this preserves the rooted functional. The Crown bound controls the first factor by $F_i(-1{,}200{,}000)$. For the second, the rooted sup estimate says that a field bounded pointwise by $R \ge 0$ has root at most R; hence it is at most $F_i(150{,}000)$. Multiplying and using $F_i(s)F_i(t) = F_i(s+t)$ gives the claim. □

Lemma 182.2 (Exceptional-set buffer)**.** *The correction term satisfies*

$$|\mathcal{Q}(f-w,g)| \le F_i(-350{,}000) + F_i(-549{,}000).$$

Proof. By the exact product identity, the correction is the finite-volume Wilson expectation of the local field $D_a g_b$. This field has envelope obtained by multiplying the two local envelopes. Its global bound is $F_i(500{,}000)F_i(150{,}000) = F_i(650{,}000)$; on the healthy set for the translated support, the bound improves to $F_i(-500{,}000)F_i(150{,}000) = F_i(-350{,}000)$. Translation preserves the number of support sites, so the support cardinality is at most $F_i(1000)$. Apply (910) with $U_0 = F_i(-350{,}000)$, $V_0 = F_i(650{,}000)$, P' the translate of P, and $b = \vartheta_k/16$. By (911), this bounds $|\mathbb{E}_\beta[\mathcal{B}_{N,k}(D_a g_b)]|$ by

$$F_i(-350{,}000) + F_i(650{,}000)F_i(1000)e^{-2\Pi_k}.$$

Apply (904) to the last factor and combine the exponents: $650{,}000 + 1000 - 1{,}200{,}000 = -549{,}000$. Thus

$$F_i(-350{,}000) + F_i(-549{,}000). \tag{912}$$

□

Proposition 182.3 (Crown buffer for the two-point probe)**.** *Under the preceding hypotheses, if also the coarse-to-fine room condition holds, then*

$$|\mathcal{Q}(f,g)| \le F_i(-30{,}000)$$

for all sufficiently large i, uniformly in the allowed finite volumes, periods, fields, and distinct coarse sites.

Proof. Set $D_a = f_a - w_a$. By the exact product identity for the coarse pairing,

$$\mathcal{Q}(f,g) = \mathcal{Q}(w,g) + \mathcal{Q}(f-w,g).$$

The first term is bounded by Lemma 182.1 and the second by (912). Each exponent is at most $-31{,}000$, so the sum is at most $3F_i(-31{,}000)$. Since eventually $3 \le F_i(1)$, this is at most

$$F_i(-31{,}000)F_i(1) = F_i(-30{,}999) \le F_i(-30{,}000).$$

□

The proof uses the stated Crown witness and the finite-volume buffer estimate. The source's witness construction is a separate input to this transfer proposition.

183 The finite price recurrence

We give the scale induction that turns the paid direction and the local one-step response estimate into the endpoint inequality used in Section 190. The one-step response estimate is derived below from the finite-volume local insertion, local uniqueness, and block-size bounds. The exact source declarations and their remaining scale inputs are recorded in the accompanying evidence file.

Fix a reference degree d, a selected scale history H, and integers $m \le k$. Write

$$\Gamma_j = \exp\left(\frac{S_k - S_j}{8}\right), \qquad \kappa_j = \sqrt{\mathrm{monitor}(j)}, \qquad I_j = \frac{1}{\Gamma_j \kappa_j} \quad (m \le j \le k).$$

Here S_j is the scale coordinate of the history. For $m \le j < k$, let L_j be the block side length at step j, let

$$P_j = C(J(j+1)), \qquad b_j = \frac{1}{10K(J(j+1))}, \qquad \rho_j = \mathrm{rate}(L_j), \qquad s_j = \frac{b_j}{\kappa_{j+1}},$$

where $J(j)$ is the stage at rung j, C is the selected coefficient-cost profile, K is the fixed positive worth profile, and $\rho_j > 0$ is the block remainder-cost factor. Write c_k for the number of coefficient columns at the terminal rung, matching the notation of the Coffer estimate. Put $p_j = L_j^{1/8}$, and define

$$H_j = \Gamma_j^{-1} + \frac{P_j L_j^d + \rho_j}{\Gamma_{j+1}}, \qquad \varepsilon_j = \frac{I_{j+1}}{10 P_j H_j}. \tag{913}$$

All quantities in the denominators are positive.

The scale and budget choices give, at every step,

$$\kappa_{j+1} \le 2\kappa_j, \qquad 1 \le P_j, \qquad 4P_j \le p_j, \qquad 2P_j b_j \le \frac{1}{3}, \qquad \Gamma_j = \Gamma_{j+1} p_j. \tag{914}$$

Consequently

$$P_j I_j \le \frac{1}{2} I_{j+1}. \tag{915}$$

Indeed, the ratio of the left side to I_{j+1} is $P_j \kappa_{j+1}/(p_j \kappa_j)$, which is at most $2P_j/p_j \le 1/2$.

For a coefficient x at rung j, write x° for its projection to nonconstant coordinates, and let $\mathcal{L}_j(x)$ be the coefficient produced by the local block insertion. The diagonal weight map $\mathcal{V}_{L_j}$ multiplies a grade-r coordinate by L_j^{d-r}. Thus $\|\mathcal{V}_{L_j}(x^\circ)\| \le L_j^d \|x^\circ\|$. The one-step local response estimate, for a paid comparison with adjacent coefficients x_j, x_{j+1} and nonnegative remainder-cost bounds r_j, r_{j+1}, is

$$\begin{aligned} \|x_j - \mathcal{L}_j(x_{j+1})\| \le{}& s_j\big(\|x_{j+1}^\circ\| + r_{j+1}\big) \\ &+ \varepsilon_j\Big(r_j + P_j \|\mathcal{V}_{L_j}(x_{j+1}^\circ)\| + \rho_j r_{j+1}\Big). \end{aligned} \tag{916}$$

It holds uniformly for all sufficiently large volume indices, provided the adjacent fields have the required format, remainder bounds, exact density transfer, and their marked sites are related by the scale map. The separate local insertion estimate is also

$$\|x_{j+1}\| \le P_j \|\mathcal{L}_j(x_{j+1})\|. \tag{917}$$

We explain the finite-volume comparison producing the second line of (916). Let $\mathcal{Q}_d$ be the finite family of coefficient-recovery probes, with probe prices $\pi(\tau)$ for $\tau \in \mathcal{Q}_d$. Let $M_d \ge 1$ be the maximum of the operator-norm products of the local gauge variations that occur in these probes, over the finitely many ordered variation patterns of order at most d and over the compact local gauge chart. Let T_d be the finite coefficient reconstruction constant for this probe family. Thus for any coefficient difference z,

$$\|z\| \le T_d \sum_{\tau \in \mathcal{Q}_d} |R_\tau(z)|.$$

A probe of order at most d annihilates a regular source term whose support is nonwrapping and whose format has the required vanishing of its first d local variations. Let B_0 be the support radius and $\sigma > 0$ the support exponent of the local representation, and define

$$M_{B_0}(X, W) = e^{\sigma\, \mathrm{cover}_{B_0}(X)} W.$$

For each of the two coefficients, the regular part has a countable presentation by regular terms with origin in their supports and with total M_{B_0}-mass bounded by its remainder budget. Each term has a common healthy chart. If its purchased sup-bound is u_a, then on this chart every ordered directional derivative of orders $q = 0, \dots, d$ has norm at most u_a, after the local variation prices have been factored out. The chart is large enough for each recovery probe and its local stencil to fit, and the admissible local presentation has the required order-d vanishing on each nonwrapping term. These are the finite recovery and regularity hypotheses used below; a value bound alone would not imply the probe estimate. Their signed union is still absolutely summable, even when some supports overlap.

Let $\mathcal{I}_0$ be the indices of nonwrapping terms, and take a finite subset $D \subset \mathcal{I}_0$. The probe applied to the presented coefficient equals the corresponding derivative of its local presentation; this is the coefficient-readout identity. On the common quiet chart, the forced contributions vanish, and the prescribed masks replace the local masks. Thus each density is the same nonzero exponential action factor times its presented coefficient plus its regular series. Equality of the two densities lets us cancel this common factor and gives the signed difference of the presented coefficients as the sum of all signed regular terms. The finite sum over D has zero probe: every term has the vanishing property just stated. First apply the finite-order local chain rule to the finite truncation and its omitted tail, not to the infinite series. The local comparison estimate bounds the difference between the full presented field probe and the zero probe of this truncation by the sup-bound tail of the omitted terms. This estimate is uniform in finite D, because the local section has order-d regularity and the source terms have summable sup-bound mass. If the probe has order r, the chain rule expands its local link variation into ordered partitions of $\{1, \dots, r\}$. Let $\lambda_{\tau,q}$ be the price of a local logarithmic variation of order q; the finite sum of the products $\prod_{A \in \mathcal{P}} \lambda_{\tau,|A|}$ over these partitions, with the zeroth-order term included, is $\pi(\tau)$. Supremum bounds on the omitted terms and the triangle inequality therefore give

$$|R_\tau(P.x - Q.x)| \le \pi(\tau) M_d \left(\sum_{a \in \mathcal{I}_0 \setminus D} u_a + \sum_{a \notin \mathcal{I}_0} u_a \right),$$

where u_a is the sup-bound of the corresponding signed regular term. The remainder certificates give $\sum_a M_{B_0}(X_a, u_a) \le a + b$, so $\sum_a u_a$ is finite as well. For the wrapping terms, the elementary support inequality

$$10\big(h(4B_0 + 2) + 2B_0\big) < n, \quad 0 \in X, \quad X \text{ wrapping}, \quad W \ge 0 \quad \Longrightarrow \quad W \le e^{-\sigma h} M_{B_0}(X, W)$$

follows from the lower bound on the cover radius of a wrapping support. Absolute summability justifies the tail limit (without differentiating an infinite sum term by term) and permits the finite sets D to increase through all of $\mathcal{I}_0$. The nonwrapping sup-bound tail tends to zero, and the wrapping sup-bound sum is at most $e^{-\sigma h}(a + b)$; taking this limit gives the probe estimate

$$|R_\tau(P.x - Q.x)| \le \pi(\tau) M_d e^{-\sigma h}(a + b).$$

This is the truncation-and-tail argument: one does not apply a probe term by term to the infinite collection of wrapping terms.

The coefficient recovery inequality is

$$\|z\| \le T_d \sum_{\tau \in \mathcal{Q}_d} |R_\tau(z)|.$$

It follows from finite-dimensionality: the probes jointly separate coefficients, so their maximum is a norm on the coefficient space; compactness of its unit sphere gives a finite comparison constant, and a maximum is bounded by the sum. Summing the probe estimate over the finite family gives

$$\|P.x - Q.x\| \le \mathfrak{C}_d e^{-\sigma h}(a+b), \qquad \mathfrak{C}_d = T_d M_d \sum_{\tau \in \mathcal{Q}_d} \pi(\tau). \tag{918}$$

Take the torus side length n so that $10(h(4B_0+2)+2B_0) < n$. Then every probe fits in the region of nonwrapping supports. This proves the required local uniqueness estimate from coefficient recovery, local vanishing, the exponential support tail, and absolute summation.

For the insertion itself, first decompose the next-level density into its primary term, its countable regular terms, and its countable forced terms. The primary term has only finitely many coefficient coordinates. The regular series is absolutely summable by its sup-bound mass budget. The forced series is absolutely summable after division by the positive factor $y = \sqrt{\text{forcedAllowance}_{j+1}}$, using the forced part of the same remainder certificate. Write the regular mass sum as a and the forced mass sum as yb, where $a, b \ge 0$ and $a + b \le r_{j+1}$ by the remainder certificate. For a regular term of volume mass m_a, the paid atom costs are $s_j m_a$ and $\rho_j m_a$; for a forced term of mass m_a, they are $s_j m_a/y$ and $\rho_j m_a/y$. Thus the combined regular and forced costs are bounded by $s_j r_{j+1}$ and $\rho_j r_{j+1}$, respectively. The primary coefficient atom contributes the finite coordinate costs $s_j\|x^\circ_{j+1}\|$ and $P_j\|\mathcal{V}_{L_j}(x^\circ_{j+1})\|$. Summing first over the finite coefficient basis, then over the countable regular and forced lists, is legitimate because their total costs and coefficient norms are summable. At fixed volume N, write A_N for the supremum of the normalized action and vacuum factor occurring in the density integral. There are finitely many compact link variables, so the action is continuous and $A_N < \infty$. The absolute integrand of the countable atomic sum is bounded by A_N times its summable mass majorant. This is an integrable bound on the normalized finite-volume Haar space, so Fubini preserves the density identity. Thus the countable atomic sum has the pushed density. Denote its actual coefficient by $\widetilde{x}_j$. The coordinate leads sum to $\mathcal{L}_j(x_{j+1})$, and the paid insertion estimate gives $\|\widetilde{x}_j - \mathcal{L}_j(x_{j+1})\| \le s_j(\|x^\circ_{j+1}\| + r_{j+1})$. Its coarse remainder budget is at most $P_j\|\mathcal{V}_{L_j}(x^\circ_{j+1})\| + \rho_j r_{j+1}$. This is the content of the countable summation argument: dominatedness and absolute summability justify both the density identity and the interchange of summation with integration.

Compare this intermediate coefficient with the actual coefficient at rung j using the local uniqueness estimate (918). Their local densities agree by the paid comparison's density identity and the insertion's pushed-density identity. Choose h so that $\mathfrak{C}_d e^{-\sigma h} \le \varepsilon_j$, then choose the torus side long enough for the displayed support inequality. The triangle inequality gives exactly (916), uniformly over the allowed fields.

For (917), coefficient coordinates have grades $r = 0, 1, \dots, d$, so each weight $L_j^{d-r} \ge 1$ when $L_j \ge 1$. Consequently the weighted coordinate norm dominates the original norm. The legal block cap then controls that weighted norm by P_j times the norm after insertion. This is the size estimate for the admissible leading map; it is not merely the inequality $L_j \ge 1$.

Lemma 183.1 (One-step propagation)**.** *Suppose the adjacent data satisfy the local hypotheses above and, for some $t \ge 0$,*

$$r_j + \|x_j\| \le \frac{t}{\Gamma_j}, \qquad r_{j+1} + \|x_{j+1}\| \le \frac{t}{\Gamma_{j+1}}, \qquad \|x_j\| \le tI_j.$$

Then

$$\|x_{j+1}\| \le tI_{j+1}.$$

Proof. The projection to nonconstant coordinates is contractive, so the paid-cone bound gives $\|x^{\circ}_{j+1}\| + r_{j+1} \le \|x_{j+1}\| + r_{j+1} \le t/\Gamma_{j+1}$. It also gives

$$r_j + P_j\|\mathcal{V}_{L_j}(x^{\circ}_{j+1})\| + \rho_j r_{j+1} \le t\left(\Gamma_j^{-1} + \frac{P_j L_j^d + \rho_j}{\Gamma_{j+1}}\right) = tH_j.$$

Apply (916) and then (917). The triangle inequality yields

$$\|x_{j+1}\| \le P_j\left(\|x_j\| + s_j\frac{t}{\Gamma_{j+1}} + \varepsilon_j t H_j\right).$$

Using $\|x_j\| \le tI_j$, $s_j = b_j/\kappa_{j+1}$, and the definition of I_{j+1}, the right side is

$$tI_{j+1}\left(\frac{P_jI_j}{I_{j+1}} + P_jb_j + \frac{1}{10}\right) \le tI_{j+1}\left(\frac{1}{2} + \frac{1}{6} + \frac{1}{10}\right) = \frac{23}{30}tI_{j+1} \le tI_{j+1}.$$

Here $P_jb_j \le 1/6$ follows from the fourth inequality in (914). This proves the claim, including the degenerate case $t = 0$. □

Proposition 183.2 (Finite endpoint price estimate)**.** *Assume the selected history satisfies the scale inequalities* (914) *for every $m \le j < k$. Let a paid comparison have coefficient vectors x_j, remainder costs r_j, and marked sites at the spatial origin for $m \le j \le k$. Assume its left coefficient is zero, $x_m = 0$, and that its adjacent pairs satisfy the local estimates* (916) *and* (917)*. Assume also the paid direction conditions of Section 171, the room and bank inequalities* (826)*, and the terminal spread bound* (827) *with endpoint budget $w \ge 0$. Then, for all sufficiently large volume indices for which the local one-step estimates hold,*

$$\kappa_k\|x_k\| \le 4(1 + c_k)w. \tag{919}$$

The threshold in the volume index is uniform over the comparison data and w covered by the local estimate.

Proof. We first verify that the comparison satisfies the discounted paid-cone bound used in the recurrence. In the canonical cone estimate, choose at rung j the worth coefficient $K_j = K(J(j+1))$, the column allowance equal to the actual column count c_j, and the bank comparison coefficient $M_j = K(J(j+1)+1)$. The scale-slow condition gives $K_j \le K(J(j)+1)$, so the room inequality yields $64K_j^2 \le c_j$. The bank condition is exactly $100000M_j^6(1+c_{j+1})^2 \le p_j$, and monotonicity gives $M_j \ge K_j \ge 1$. The paid direction in that cone is the adjacent coefficient comparison with diagonal weights L_j^{d-r} and loss p_j^{-3}; this is the direction hypothesis assumed here. The zero left coefficient and terminal spread are precisely its boundary conditions. Therefore equation (828) gives

$$\Gamma_j(r_j + \|x_j\|) \le 4(1 + c_k)w \qquad (m \le j \le k). \tag{920}$$

Put $t = 4(1 + c_k)w$. Since $x_m = 0$, we have $\|x_m\| = 0 \le tI_m$. Suppose inductively that $\|x_j\| \le tI_j$. The paid-cone bound gives $r_j + \|x_j\| \le t/\Gamma_j$ and $r_{j+1} + \|x_{j+1}\| \le t/\Gamma_{j+1}$. In particular the local remainder allowances satisfy the two bounds in Lemma 183.1. That lemma gives $\|x_{j+1}\| \le tI_{j+1}$. Finite induction reaches $j = k$, where $\Gamma_k = 1$, so $I_k = 1/\kappa_k$. Multiplying by κ_k gives (919). The finite intersection of the volume thresholds for $j = m, \dots, k-1$ is still an eventual set. Each adjacent estimate is uniform over the specified comparison data, so one threshold works for all of them. □

Thus the endpoint estimate combines the already proved paid-cone argument of Section 171 with the one-step quantitative price recurrence proved here. The only new induction is the finite propagation from the zero left coefficient to the terminal rung.

184 Choosing the local paid budgets

The one-step insertion uses two kinds of local input: a finite list of coefficient-recovery probes, and positive budgets for the basis, regular, and forced atoms. The first choice is elementary once the probe list is finite. The eventual threshold can then be made monotone in the scale and uniform over all admissible adjacent levels. This section proves those two selection steps. The local atom estimates themselves are recorded in the source audit; they are the remaining inputs to the one-step recurrence of Section 183.

Fix a finite set $\mathcal{T}$ of probes and nonnegative constants O, H, W, G_d, a, c, where O is the motion-average norm factor, H the local readout factor, W the test factor, G_d the gauge-variation factor, a the block weight, and c the tail-decay factor. For each probe τ, let $\pi_\tau \ge 0$ be its variation price and define its local coefficient

$$q_\tau = 4aG_d\pi_\tau + 2a.$$

Given a target price $y > 0$, the required atom budget inequalities are

$$q_\tau v \le u \qquad (\tau \in \mathcal{T}), \tag{921}$$

$$O\big(Wu + 2Hav\big) \le y, \tag{922}$$

$$4OHav \le y, \tag{923}$$

$$O\big(Wv + 2Havc\big) \le y. \tag{924}$$

The two outputs u, v are the probe and summability budgets. In the Wilson application, q_τ bounds the local variation coefficient, while the other three inequalities pay respectively for regular terms, large-support terms, and the primary atom.

Lemma 184.1 (Positive local budget choice)**.** *For every finite probe family and every $y > 0$, there are $u, v > 0$ satisfying* (921)–(924)*.*

Proof. Set

$$K = \sum_{\tau\in\mathcal{T}} q_\tau, \qquad B = \max\{1,\ O[W(1+K) + 2Ha],\ 4OHa,\ O[W + 2Hac]\}.$$

These are finite nonnegative numbers. Choose

$$v = \frac{y}{2B}, \qquad u = (1+K)v.$$

Both are strictly positive. Since $q_\tau \le K \le 1 + K$, $q_\tau v \le u$. The left side of the regular-term inequality is $vO[W(1+K) + 2Ha] \le vB = y/2$. The large-support and primary left sides are respectively $4OHav \le Bv = y/2$ and $vO[W + 2Hac] \le Bv = y/2$. Each is therefore at most y, as required. □

We next record the uniform threshold selection used to turn fixed-parameter estimates into a single scale profile. At a scale index j, the allowed adjacent indices are the same-level index and its immediate predecessor; there are at most two. The coefficient grades to be checked form the finite set $\{0, \dots, j-1\}$. Thus only finitely many eventual requirements occur at each fixed j.

Lemma 184.2 (Monotone threshold envelope)**.** *Let $b_j \in \mathbb{R}$ be any prescribed lower threshold. For each j, let S_j be a finite set and, for each $s \in S_j$, let $E_{j,s} \in \mathbb{R}$ have the property that a desired conclusion $\mathcal{P}_{j,s}(x)$ holds whenever $x \ge E_{j,s}$. Then there is a nondecreasing sequence B_j such that $B_j \ge b_j$ and, for every $s \in S_j$, $\mathcal{P}_{j,s}(x)$ holds whenever $x \ge B_j$.*

Proof. Define

$$e_j = \max\big(\{b_j\} \cup \{E_{j,s} : s \in S_j\}\big), \qquad B_j = \max_{0 \le r \le j} e_r.$$

The maxima exist because the sets are finite and nonempty after adjoining b_j. The prefix maximum B_j is nondecreasing and dominates b_j. If $s \in S_j$, then $E_{j,s} \le e_j \le B_j$, so every $x \ge B_j$ satisfies the stipulated conclusion. □

Apply this lemma at each scale to the finitely many pairs consisting of an admissible adjacent level and a coefficient grade below that scale. The atom-budget lemma supplies positive budgets for each fixed pair; the finite-volume local estimates then supply an eventual threshold for the corresponding primary, regular, and forced atom conclusions. Taking the finite maximum and then the prefix maximum gives one monotone threshold profile that works simultaneously for every adjacent transition and every permitted coefficient grade. This is the uniformization step needed before the scale induction. It does not prove the local atom estimates: their mathematical inputs and exact source route are itemized in the accompanying evidence file.

185 Rate conversion and the local scale budget

This section records the quantitative selection after the Fourier-tail and jet comparison in Section 188. It proves the finite-rate summation used to select the probe estimate and the scalar domination used to choose the local chart budget. The strip, growth, weighted-vanishing, chart, and support receipts themselves remain the inputs identified in that section; the evidence record supplies the source crosswalk.

Fix the block side L. Write

$$H(x) = \mathrm{fineMonitor}(x), \quad \chi(x) = \sqrt{H(x)}, \quad \beta(x) = \beta_F(x),$$
$$R(x) = \mathrm{radiusD}(x), \quad \rho(x) = \mathrm{reach}(x), \quad b(x) = \mathrm{radiusB}(x).$$

For fixed L, these are

$$\beta(x) = x + c_L, \quad b(x) = \lceil (\log x)^2 \rceil,$$
$$R(x) = b(x)(4b(x) + 2) + 2b(x), \quad \rho(x) = 45(1 + b(x)),$$
$$\mathrm{depthW}_L(R) = L(R + 2),$$

where c_L is fixed. Here $H(x) = \exp((\log\log \beta(x))^2)$, so $H \ge 1$, $H \to \infty$, and χ is the monitor factor in the priced conclusion. For a function f, write $f \in \mathcal{R}_a$ if for all $v > a$ one has eventually $|f(x)| \le \beta(x)^v$. This is the source's strict-power rate convention. Finite sums preserve a common rate, products add rates, and $\chi \in \mathcal{R}_0$.

Lemma 185.1 (The exponential tail pays the finite probe)**.** *Let $d \in \mathbb{N}$, $\sigma > 0$, and let K, M and the priced local derivative bound Φ be fixed nonnegative constants. Let $\delta_k(x)$ and $C_1(x)$ be the quantities of (9)–(10), with $b = \rho(x)$ and $m = R(x)$. Set*

$$E(x) = \sum_{k=0}^{d} \delta_k(x), \qquad Z_d(x) = \sum_{j=0}^{d} \#\,\mathrm{Part}^{\mathrm{ord}}(j)\, 2^j E(x) C_1(x)^j\, \Phi. \tag{12}$$

Then $Z_d \in \mathcal{R}_{-1}$, and consequently

$$\chi(x) Z_d(x) \longrightarrow 0. \tag{13}$$

In particular, for every $y > 0$, eventually $Z_d(x) \le y/\chi(x)$. If the local receipts of Proposition 188.4 hold and $E(x) \le C_1(x)$, every derivative order $j \le d$ of the selected probe is bounded by $Z_d(x)t$, where t is the test amplitude and the price factor Φ has already been included in Z_d.

Proof. By the polynomial geometry receipts, R and the lift-size polynomial $(1+4L(R+2))^{dk}$ have rate zero. The width cost and all other factors in δ_k, apart from $\exp(-\sigma\rho(x)/2)$, are fixed constants or rate-zero polynomial factors. The reach cutoff has every negative rate: for each $a>0$, eventually $\exp(-\sigma\rho(x)/2)\le\beta^{-a}$. Thus each δ_k, and their finite sum E, have rate -1 (the stronger arbitrary negative rate is more than needed). The finite sum in C_1 has rate zero. By the product and finite-sum rules, $Z_d\in\mathcal{R}_{-1}$.

Since $\chi\in\mathcal{R}_0$, their product has rate -1. The strict-power definition then implies convergence to zero: choose any exponent v strictly between -1 and 0; eventually $|\chi Z_d|\le\beta^v$, and $\beta\to\infty$. This proves (13) and the stated eventual price bound. The final assertion follows by applying Proposition 188.4 at each order and bounding its finite ordered-partition sum by (12). □

Actual finite-rate receipts. In the construction the polynomial direction-growth constants are

$$h_k(x)=4KM\,C_{dk}(\sigma)\big(1+4L(R(x)+2)\big)^{dk},\qquad C_1(x)=1+\sum_{k=0}^{d}h_k(x).$$

The error at order k is

$$\delta_k(x)=8KMe^{-\sigma\rho(x)/2}(1+4R(x))^{dk}C_{dk}(\sigma/2).$$

These formulas retain the source's factor four from the four edge components and the additional factor two from the two input arrays. The coefficient $E(x)$ is required to be at most $C_1(x)$ before applying the finite multilinear comparison. This is a separate eventual estimate; it follows because $E\in\mathcal{R}_{-1}$ and $C_1\ge1$.

Proposition 185.2 (The selected local budget)**.** *Fix natural numbers n,p,r, a fixed positive integer block side L, and the admissible data C,D of the selected source chart. For x large, define the covering and response quantities*

$$\begin{aligned}
P_*(x)&=(2\ \mathrm{depthW}_L(R(x))+1)^4,\\
B_*(x)&=(r+48)(1+b(x))(1+P_*(x)),\\
A_*(x)&=\mathrm{boost}_{C,D,L}\ B_*(x)^{\,8+2p+2pn}+\frac{16}{D.\varepsilon},\\
S_*(x)&=\mathrm{clinicFee}(n,D.Q)\big(2C.C(1+R(x))^p\big)^2,\\
U_*(x)&=(A_*(x)+S_*(x))\ \mathrm{fiery}(x)+\frac{2A_*(x)+2}{H(x)},\\
K_*(x)&=1+\mathrm{guests}_{G,D,K}(n+1,L)B_*(x)^{\,4(1+4(n+1))}.
\end{aligned}$$

These abbreviations use the exact source coefficients

$$\begin{aligned}
s_L&=1+16\,\#\,\mathrm{Co}(L,\mathrm{Axis}\times\mathrm{Color}),\\
b_0&=(\mathrm{clinicFee}(n,D.Q)+C.M)(2C.C)^2,\\
\mathrm{boost}_{C,D,L}&=b_0+(b_0+1)L^3+2ns_L^2(D.Z+1)D.Z,\\
\mathrm{guests}_{G,D,K}(q,L)&=2s_L^{1+4q}\,\mathrm{inflate}_{4q}(D.K^{-1}/8),\\
\mathrm{meter}(h,D.K,m)&=2(1+h)^{m+1}\,\mathrm{inflate}_m(D.K^{-1}/8).
\end{aligned}\tag{14a}$$

Here h is the number of heads and $\mathrm{inflate}_m(v)=(1+m/v)^m$ *is the Gaussian moment inflation factor. Assume $D.\varepsilon>0$, $D.K>0$, nonnegative fee and moment parameters, and $C.C\ge1$. The*

scalar cost terms below are nonnegative by admissibility. The source clock is linear for $t \geq 0$; write $c_{\mathrm{clock}} = \mathrm{clocks}(1) \geq 0$. Assume the fixed record and chart parameters are admissible, and let $\beta \geq \beta(x)/2$. Suppose the support and budget satisfy

$$\begin{aligned} B.\mathrm{threshold} = \mathrm{fiery}(x), \quad 0 \leq P.\mathrm{threshold} \leq 2B.\mathrm{threshold}, \\ B.B \leq 2b(x), \quad P.\mathrm{power} = p, \quad \mathrm{coverNumber}(B.B, X) \leq P_*(x). \end{aligned} \tag{14}$$

Put $F_0 = B.\mathrm{threshold}$, $u = \beta^{-1/2}$, $v_\rho = C.C\,P.\mathrm{threshold}(1+\rho(x))^p$, $v_R = C.C\,P.\mathrm{threshold}(1+R(x))^p$, $h = \mathrm{heads}(B, X)$, and $\mathrm{cup} = D.Z(1+\mathrm{scope}(B, X, r, x))^{pn}$. Define the displayed scalar cost by

$$\begin{aligned} F(x) = {} & (\mathrm{clinicFee}(n, D.Q) + C.M)v_\rho^2(2+\mathrm{breadth}(B, X))^{pn}/F_0 \\ & + \frac{\mathrm{clinicFee}(n, D.Q)v_\rho^2 + u}{F_0} L^3 \\ & + n(2F_0)h^2(\mathrm{cup}+1)\,\mathrm{cup} \\ & + \frac{16}{D.\varepsilon}(F_0 + u/F_0) \\ & + \frac{\mathrm{clinicFee}(n, D.Q)v_R^2 + u}{F_0}. \end{aligned} \tag{14b}$$

This is the response-plus-paint cost used by the local chart construction. It satisfies, eventually in x,

$$0 \leq F(x) \leq U_*(x), \qquad \mathrm{meter}(\mathrm{heads}(B, X), D.K, 4(n+1)) \leq K_*(x), \tag{15}$$

and

$$\chi(x)U_*(x)K_*(x) \longrightarrow 0. \tag{16}$$

Consequently, for every $y > 0$, eventually and uniformly in all data satisfying (14),

$$F(x) \leq 1, \qquad \mathrm{clocks}(t)F(x)\,\mathrm{meter}(\mathrm{heads}(B, X), D.K, 4(n+1)) \leq \frac{y}{\chi(x)}t \quad (t \geq 0). \tag{17}$$

This is the scalar estimate consumed by the local regular-source branch.

Proof. We give the domination and limiting arguments separately. First, the cover formula gives

$$1 + \rho(x) \leq B_*(x), \quad 2 + \mathrm{breadth}(B, X) \leq B_*(x), \quad 1 + \mathrm{scope}(B, X, r, x) \leq B_*(x). \tag{18}$$

Write $a = b(x)$ and $q = \mathrm{coverNumber}(B.B, X)$. The definitions give

$$\mathrm{breadth}(B, X) = q(4B.B + 2) + 2B.B + 8 \leq q(8a+2) + 4a + 8, \quad \mathrm{scope} = ra + \mathrm{breadth} + 1.$$

With $S_X = (r+48)(1+a)(1+q)$, direct expansion, using $a, q, r \geq 0$, gives $1 + 45(1+a) \leq S_X$, $2 + q(8a+2) + 4a + 8 \leq S_X$, and $ra + q(8a+2) + 4a + 10 \leq S_X$. These are (18) with S_X in place of B_*. Since $q \leq P_*$ and the bulk formula is increasing in its cover-count argument, $S_X \leq B_*$; hence (18) follows.

Put $F_0 = B.\mathrm{threshold}$, $u = gStep(\beta) = \beta^{-1/2}$, and $S_X = \mathrm{bulk}(\mathrm{coverNumber}(B.B, X), r, x)$. Since $F_0 = \beta(x)^{-1/2}H(x)$, (14) yields

$$\frac{u}{F_0} = \frac{1}{H(x)}\sqrt{\frac{\beta(x)}{\beta}} \leq \frac{\sqrt{2}}{H(x)} \leq \frac{2}{H(x)}. \tag{19}$$

The cover inequalities (18) and $P.\mathrm{threshold} \le 2F_0$ give

$$\mathrm{loom}(P, \rho(x), C.C) \le 2C.C F_0 S_X^p. \tag{20}$$

Put $t = F_0$ and $v = \mathrm{loom}(P, \rho(x), C.C)$. The loom is $C.C\, P.\mathrm{threshold}(1 + \rho(x))^p$; (18) and $P.\mathrm{threshold} \le 2t$ give $v \le 2C.CtS_X^p$. The segment term is $(\mathrm{clinicFee} + C.M)v^2\,\mathrm{heft}\,/t$, where $\mathrm{heft} = (2 + \mathrm{breadth})^{pn} \le S_X^{pn}$. Thus, with $b_0 = (\mathrm{clinicFee} + C.M)(2C.C)^2$,

$$\mathrm{segment} \le b_0 t S_X^{2p+pn} \le b_0 \zeta(B, u) S_X^{8+2p+2pn}.$$

The tranche term is $((\mathrm{clinicFee}\, v^2 + u)/t)L^3$. Its first numerator part is bounded by $b_0 t S_X^{2p}$; its second is u/t. Since $t \le \zeta(B, u) = t + u/t$, $S_X \ge 1$, and $2p \le 8 + 2p + 2pn$, this gives tranche $\le (b_0 + 1)L^3\zeta(B, u)S_X^{8+2p+2pn}$.

For the scratch term, its exact factors are $n(2t)\,\mathrm{heads}^2(\mathrm{cup}+1)\,\mathrm{cup}$, where $\mathrm{cup} = D.Z(1 + \mathrm{scope})^{pn}$. The head count is $(2\,\mathrm{scope}+1)^4 \#\,\mathrm{Co}(L, \mathrm{Axis} \times \mathrm{Color})$. From (18), $1 + \mathrm{heads} \le s_L S_X^4$, with $s_L = 1 + 16 \#\,\mathrm{Co}(L, \mathrm{Axis} \times \mathrm{Color})$. Also $\mathrm{cup} \le D.ZS_X^{pn}$ and $\mathrm{cup} + 1 \le (D.Z + 1)S_X^{pn}$. Replacing t by $\zeta(B, u)$ therefore gives

$$\mathrm{scratch} \le 2ns_L^2(D.Z + 1)D.Z\,\zeta(B, u)S_X^{8+2pn}.$$

Adding these three bounds gives the coefficient $\mathrm{boost} = b_0 + (b_0 + 1)L^3 + 2ns_L^2(D.Z + 1)D.Z$. Since $\mathrm{opal} = \mathrm{rust} + (16/D.\varepsilon)\zeta$, and $S_X \ge 1$,

$$\mathrm{opal} \le A_*(x)(F_0 + u/F_0). \tag{21}$$

The paint term is evaluated with the loom at $R(x)$, not at the reach. Its direction is $v_R = C.C\, P.\mathrm{threshold}(1 + R(x))^p \le 2C.CF_0(1 + R(x))^p$. Squaring this inequality and dividing by $F_0 > 0$ gives $\mathrm{clinicFee}(n, D.Q)v_R^2/F_0 \le S_*(x)F_0$. The remaining paint term is u/F_0. Thus the paint contribution is bounded by

$$\mathrm{paintW} \le S_*(x)F_0 + u/F_0. \tag{22}$$

Equations (19)–(22) give $F \le U_*$.

The head-count moment estimate follows by inserting $1 + \mathrm{heads} \le s_L S_X^4$ into the explicit moment formula (14a): for $m = 4(n + 1)$,

$$\mathrm{meter}(\mathrm{heads}(B, X), D.K, 4(n + 1)) \le \mathrm{guests}_{G,D,K}(n + 1, L)S_X^{4(1+4(n+1))} \le K_*(x). \tag{23}$$

For the limit, B_*, A_*, S_*, K_* are fixed polynomials in $1 + \mathrm{radiusB}(x)$, hence have rate zero and are subpower relative to H. Indeed $\mathrm{radiusB}(x) \le 1 + (\log x)^2$, while $\beta(x) = x + O(1)$ for fixed L, and $\log H = (\log\log\beta)^2$; the logarithm of any such fixed polynomial is $O(\log\log\beta) = o(\log H)$. Thus for each $\eta > 0$, eventually each nonnegative polynomial factor Q here satisfies $Q \le H^\eta$. The first term in $\chi U_* K_*$ has rate $-1/2$, since $\mathrm{fiery} = H/\sqrt{\beta}$ and all other factors have rate zero. For the second term choose the subpower bounds $A_* + 1 \le 2H^{1/8}$, $K_* \le H^{1/8}$; then

$$\chi\,\frac{(2A_* + 2)K_*}{H} \le 4H^{1/2+1/8+1/8-1} = 4H^{-1/4} \longrightarrow 0.$$

This proves (16). Also $H \ge 1$, $K_* \ge 1$, and $F \le U_*$, so $F \le \chi U_* K_*$; (16) makes this at most one eventually. Finally $\mathrm{clocks}(t) = t\,\mathrm{clocks}(1)$ for $t \ge 0$. Choose $d_* = 1 + \mathrm{clocks}(1)$. From (16), eventually $d_*\chi U_* K_* \le y$, and (15) proves the second inequality in (17). □

The bound is conditional only on the selected local chart and support inequalities (14); the source crosswalk identifies the already-proved support-cover, rust, and Gaussian-moment estimates whose algebra is expanded above. It does not produce the separate strip, field-growth, weighted vanishing, same-history, or healthy-chart receipts required to enter this local branch.

186 Three local scalar price bounds

This section proves three scalar price conversions used by the selected trace construction. The arguments are independent of the construction of the local response and chart records that consume these bounds.

Fix a block side L, and retain $\beta(x)$, $H(x)$, and $\chi(x) = H(x)^{1/2}$ from Section 185. For a fixed price $a > 0$, write

$$\mathsf{s}_a(x) = a/\chi(x).$$

A negative strict-power rate has the following elementary consequence: if $f \in \mathcal{R}_\alpha$ with $\alpha < 0$, then $\chi(x)f(x) \to 0$. Indeed $\chi \in \mathcal{R}_0$, so the product has rate α; choose $v \in (\alpha, 0)$, and use $\beta(x)^v \to 0$.

Lemma 186.1 (Cloud price)**.** *Let Q be a target with rate exponent $q < 0$. Suppose its poster coefficient $p_Q(x)$ has strict-power rate $q/4$, and define the cloud response for a cloud Z on a representation R by*

$$\mathcal{M}_Q(Z;x) = m_R(Z)\exp(-\epsilon_Q N_R(Z))p_Q(x), \qquad m_R(Z) \ge 0, \quad N_R(Z) \ge 0, \quad \epsilon_Q > 0.$$

Then for every $a > 0$, eventually, uniformly over all clouds,

$$|\mathcal{M}_Q(Z;x)| \le \mathsf{s}_a(x) m_R(Z). \tag{1}$$

Proof. Since $q < 0$, choose v with $q/4 < v < 0$. The rate hypothesis implies $|p_Q(x)| \le a/\chi(x)$ eventually: equivalently $\chi|p_Q| \to 0$, and then make it at most a. The attenuation obeys $0 < \exp(-\epsilon_Q N_R(Z)) \le 1$, independently of the cloud. Multiply these inequalities by the nonnegative mass $m_R(Z)$. □

Lemma 186.2 (Forced gas price)**.** *Let $g(x) = \exp(-H(x)^{1/5}/2)$. For every $a > 0$, eventually*

$$\mathrm{DoorsRate}(L)g(x) \le \mathsf{s}_a(x). \tag{2}$$

Proof. Set $t = \log\log\beta(x)$, so $\log H = t^2$ and $\log\beta = e^t$. Then

$$\log\big(\chi(x)g(x)\big) = \tfrac{1}{2}t^2 - \tfrac{1}{2}e^{t^2/5} \longrightarrow -\infty,$$

because $e^{t^2/5}$ grows faster than t^2. Thus $\chi g \to 0$; multiplying by the fixed positive $\mathrm{DoorsRate}(L)$ and making the product at most a proves (2). □

Lemma 186.3 (Tuition price)**.** *For a target with goal $\beta(x)^q$, define its tuition at fixed integers $s, n \ge 0$ by*

$$\mathcal{T}_Q(x;s,n) = A(x)^{sn}\, P_{1/2}(sn)\, \beta(x)^{-q} \exp(-b(x)/2), \tag{3}$$

where A and the polynomial price $P_{1/2}(sn)$ have rate zero and $b(x) = \lceil(\log x)^2\rceil$. Then $\mathcal{T}_Q(\cdot;s,n) \in \mathcal{R}_z$ for every real z. In particular, for every $a > 0$, eventually

$$\mathcal{T}_Q(x;s,n) \le \mathsf{s}_a(x). \tag{4}$$

Proof. The first two factors in (3) have rate zero, and β^{-q} has rate $-q$. The last factor has every strict-power rate: for every fixed $c > 0$, $b(x)/2 \ge c\log\beta(x)$ eventually, since $b(x) \asymp (\log x)^2$ while $\log\beta(x) \asymp \log x$. Thus $e^{-b/2} \le \beta^{-c}$ eventually. Choose $c > \max(0, -q - z)$; the product rate $-q - c$ is then less than z, so the strict-power definition gives the claimed rate. Taking $z = -1$ and using the negative-rate consequence above proves (4). □

187 Construction of the finite local response package

We give the quantitative construction behind the local response package used by the later trace estimates. The output contains its probe error, chart error, price, and cloud allowance; these are built together because the final cloud billing uses the same price budget as the probe estimate.

Fix the scale data of the trace construction and write

$$\chi(x) = \sqrt{H(x)}, \qquad e = \mathrm{title}(P_1), \quad a = \mathrm{title}(P_0), \quad h = \mathrm{credibility}(P_1), \quad \nu = \mathrm{DoorsRate}(L).$$

Here e, a, h, ν are positive, and $e, a \geq 1$. Let F be the finite family of probes at the chosen grade. For $p \in F$, let w_p be its coefficient price, and write Λ for the common normalization factor in the probe estimate. Put

$$Z = \mathrm{testPrice}(F), \qquad T = \mathrm{readPrice}(F), \qquad c = \mathrm{orbitCost}(F)\,\mathrm{errorPrice}(F)\,\mathrm{corn}(d)\,a.$$

The source normalizations give $Z, T \geq 1$ and $c > 0$. For a positive target price ζ, define

$$\kappa = \min\left\{\frac{\nu}{2cZ}, \zeta\right\} > 0, \qquad g_p(s) = w_p \Lambda L^d\, 2(es + s) + s. \tag{1}$$

The source's finite probe family is essential here: each g_p is continuous, $g_p(0) = 0$, and therefore there is one $b > 0$ such that $g_p(s) \leq \kappa$ for all $p \in F$ and $0 \leq s \leq b$. Set $Q = b/2$.

Theorem 187.1 (Finite-probe and cloud billing)**.** *Let $d \in \mathbb{N}$, and assume $e, a, h, \nu, c, Z, T, \zeta > 0$, $e, a, Z, T \geq 1$, and a finite probe family F with weights $w_p \geq 0$ and normalization $\Lambda \geq 0$. Suppose the selected response satisfies $u \in \mathcal{R}_{-1/2}$ and $\chi \in \mathcal{R}_0$, with $\chi(x) \geq 1$ eventually. For each probe, the charged scalar is the continuous function* (1)*. Set*

$$\kappa = \min\left\{\frac{\nu}{2cZ}, \zeta\right\}, \qquad \delta_x = Q/\chi(x), \qquad C_x = \kappa/\chi(x).$$

Assume the following selected-scale receipts hold on one common eventual tail, uniformly for every admissible period, forest and tag choice, every leaf of cover at most B, and every normalized direction list in the fixed grade/block. The bounds and cutoff-tail estimates are independent of the leaf weight τ. For the chosen threshold θ and fine threshold Θ,

$$\tfrac{1}{2} \leq \frac{\Theta}{\theta} \leq 2. \tag{2}$$

For every selected probe and allowed local chart, the probe-to-run difference has norm at most $Q/\chi(x)$. For every allowed support and budget, the scale cap O_x and local moment M_x obey

$$0 \leq O_x \leq 1, \qquad \mathrm{clocks}(e)\, O_x\, M_x \leq \frac{Q}{\chi(x)}. \tag{3}$$

For every cloud Z_0 of mass $m_0 \leq h$, put $\mathcal{M}_x(Z_0) = \mathrm{marvel}_x(Z_0) \geq 0$. Its cloud receipt is

$$L^d\, 2\mathcal{M}_x(Z_0) \leq S m_0, \qquad S = \frac{\nu}{2cTh}. \tag{4}$$

Then for all sufficiently large x:

1. *the two selected thresholds satisfy the factor-two comparison;*
2. *the response amplitude $s_x = |u(x)|$ obeys $s_x \leq \delta_x$ and $s_x \leq \zeta/\chi(x)$;*

3. *the scale cap is at most one, and its clock-weighted moment is at most* δ_x*;*

4. *every selected probe has approximation error at most* δ_x*;*

5. *for each probe* p*, the explicit charge* $w_p \Lambda L^d \, 2(e s_x + \delta_x) + \delta_x$ *is at most* C_x*;*

6. *for every cloud* Z_0 *of mass at most* h*, the combined charge*

$$c\left[ZC_x + TL^d \, 2\mathcal{M}_x(Z_0)\right] \le \nu.$$

In particular $C_x \le \zeta/\chi(x)$*.*

Proof. The product rate gives $\chi u \in \mathcal{R}_{-1/2}$. Choose $q \in (-1/2, 0)$; the strict-power definition implies $\chi(x)|u(x)| \le \beta(x)^q \to 0$. Thus, eventually,

$$|u(x)| \le Q/\chi(x), \qquad |u(x)| \le \zeta/\chi(x), \qquad \chi(x) \ge 1. \tag{925}$$

The first two give the asserted response amplitude bound and the desired price bound for the amplitude. The first selected geometry estimate gives the factor-two threshold comparison.

The selected scale-cap estimate is called at price Q/e. Its price bound is linear in the allowed scalar multiplier t; evaluating it at $t = e$ gives exactly the second inequality in (3), while its companion bound gives $O_x \le 1$. The probe approximation estimate is called at tolerance Q, which gives the bound stated above directly.

For a probe p, the amplitude bound (925) gives

$$2(e|u(x)| + \delta_x) \le \frac{2(eQ + Q)}{\chi(x)}. \tag{926}$$

Consequently its charged expression is bounded by

$$\frac{w_p \Lambda L^d \, 2(eQ + Q) + Q}{\chi(x)} = \frac{g_p(Q)}{\chi(x)} \le \frac{\kappa}{\chi(x)} = C_x.$$

This is the simultaneous finite-probe choice made above; it is not a pointwise choice depending on the probe.

Finally, let a cloud have mass $m_0 \le h$. The cloud estimate (4) gives

$$TL^d(2\mathcal{M}_x(Z_0)) \le TSh = \frac{\nu}{2c}.$$

The probe part costs at most $ZC_x \le Z\kappa/\chi(x) \le Z\kappa$. By the definition of κ, $cZ\kappa \le \nu/2$. Adding the cloud and probe charges therefore gives at most ν. This proves all six clauses. □

187.1 Matching the finite-volume response estimate

We now identify the local insertion receipt in the actual response construction. Fix a leaf insertion of weight τ, and hold its masked ingredient fixed while varying the local field. The two functions compared in the source are the actual-sector and Gaussian-reference integrals

$$\Phi_6(y) = \int R_6(y,z)\chi_6(y,z)h(z)\, d\gamma_{C_0}(z), \qquad \Phi_9(y) = \int R_9(y,z)\chi_9(y,z)h(z)\, d\gamma_{C_0}(z), \tag{5}$$

where h is the fixed measurable masked insertion and $|h| \le \tau$. The density comparison in Section 96 proves, under its displayed finite-dimensional covariance, mean, cutoff, and chart hypotheses,

$$\|\Phi_6 - \Phi_9\|_{J_0^n} \le \tau P_n(N, Q, S)\big(\delta + 2Ne^{-v+M/2}\big). \tag{6}$$

The primitive mean and covariance estimates supplying δ are proved in Section 97. The fixed insertion is not differentiated. The local chart directions are the same ordered variations used in the response test, with $j \le n$ and their prescribed directional and canopy bounds.

In the exact construction, the fixed leaf is held together with its regular masked ingredient L'. For each fixed grade and block, the bounds $N(x), Q(x), S(x), \delta(x)$ and the cutoff-tail bounds are common to every admissible period, forest and tag choice, every leaf whose cover is at most B, and every normalized direction list allowed in the comparison. They are independent of the leaf weight τ. Thus the one finite-probe choice and selected response $u(x)$ work uniformly over all these inputs. The local read set, forest, tags, insertion mask, and chart are fixed during each individual differentiation. The two local observables are

$$\mathcal{O}_{\mathrm{act}}(y) = \Phi_6(y), \qquad \mathcal{O}_{\mathrm{ref}}(y) = \Phi_9(y) \tag{7}$$

and these identities follow by writing each side with its own normalized Gaussian density and using the protected-support agreement on the insertion's read set. Thus the response difference is exactly $\Phi_6 - \Phi_9$, not a comparison of different insertions or different Gaussian spaces. The local finite price cutoff makes the integrands vanish outside a common compact coordinate ball; the cutoffs are one on the protected smaller ball. Differentiation under the integral is justified with the bounded measurable h fixed, by the uniform derivative bounds on that compact ball and dominated convergence.

The selected rate budgets choose the scalar envelope

$$B_*(x) = P_n(N(x), Q(x), S(x))\big(\delta(x) + 2N(x)e^{-v(x)+M(x)/2}\big).$$

The covariance discrepancy has strict-power rate $-1/2$, the mean discrepancy has rate -1 before conversion to Gaussian coordinates, and the good-coordinate tail has every negative rate. The fixed polynomial factor has rate zero. Therefore $B_* \in \mathcal{R}_{-1/2}$. The response construction packages the concrete derivative, cutoff, and tail budgets into one scalar envelope. The component bounds are assembled from the finite product and sum rules for the displayed rates. With this envelope, for every admissible leaf and every ordered list of at most n normalized chart directions, the derivative of the actual-minus-reference observable at the chart origin is bounded by $\tau B_*(x)$, uniformly over the choices above. This is the concrete uniform derivative inequality used by the response construction.

If the branch patch required by the uniform derivative inequality does not exist, the response is zero and that inequality is vacuous. If the patch exists, the source's finite-volume estimate supplies a function f of rate $-1/2$ and the derivative bound with price $2|f|$. Choosing the response to be $u = 2|f|$ preserves that rate and bound. Thus the remaining assembly in this section is exactly the finite-test choice and scalar billing proved above; the local insertion estimate is supplied by the two cited comparison sections under their explicit hypotheses.

188 Fourier tails and local jet estimates

This section proves two quantitative ingredients of the small-support regular-source argument: a weighted convolution bound with an exponentially small tail, and the corresponding multilinear estimate for a local coefficient probe. The remaining hypotheses are named mathematically: a

uniform Fourier strip bound, a coefficient-growth receipt, and the local regularity and vanishing-jet receipts. In the construction these come from the selected finite-range kernel, the budget's order bounds, and the regularity of the masked source.

For each natural number n, write $\mathbb{Z}^4$ for the covering lattice, with $|x|_1 = \sum_{\nu=1}^{4} |x_\nu|$, and put

$$W_n(x) = (1 + |x|_1)^n, \qquad C_n(\sigma) = \left(1 + \frac{n+1}{\sigma}\right)^n \quad (\sigma > 0).$$

The elementary inequality

$$(1 + r)^n \le C_n(\sigma) e^{\sigma r} \qquad (r \ge 0) \tag{1}$$

follows by setting $s = \sigma/(n+1)$. Indeed

$$1 + r \le \left(1 + \frac{n+1}{\sigma}\right)(1 + sr) \le \left(1 + \frac{n+1}{\sigma}\right) e^{sr},$$

and taking the n-th power gives (1), since $ns \le \sigma$. Also $W_n(x + z) \le W_n(x) W_n(z)$, by the triangle inequality and $1 + a + b \le (1 + a)(1 + b)$.

Lemma 188.1 (Polynomially weighted convolution)**.** *Suppose a scalar kernel $q : \mathbb{Z}^4 \to \mathbb{R}$ and constants $\sigma > 0$, $C \ge 0$ satisfy*

$$\sum_{z \in \mathbb{Z}^4} e^{\sigma |z|_1} |q(z)| \le C \tag{2}$$

and a Banach-valued array b satisfies $\|b(x)\| \le M W_n(x)$, where $M \ge 0$. Then the convolution

$$(q * b)(x) = \sum_z q(z) b(x + z)$$

is absolutely convergent and

$$\|(q * b)(x)\| \le M C_n(\sigma) C W_n(x). \tag{3}$$

For a 4-by-4 matrix kernel whose sixteen entries each satisfy (2) with the same C, the corresponding vector-row operator obeys

$$\|(\mathsf{A} b)(x)\| \le 4 M C_n(\sigma) C W_n(x), \tag{4}$$

when the output norm is the maximum of its four component norms.

Proof. For each z,

$$|q(z)| \, \|b(x + z)\| \le M W_n(x) |q(z)| W_n(z) \le M W_n(x) C_n(\sigma) e^{\sigma |z|_1} |q(z)|,$$

where (1) was applied to $|z|_1$. Summing proves absolute convergence and (3). For a matrix row, apply (3) to each of its four entries and sum the four resulting component bounds; this gives (4). □

Lemma 188.2 (Agreement on a ball gives an exponential tail)**.** *Assume (2). Let $A_1, A_2 : \mathbb{Z}^4 \to E$, where E is a Banach space, satisfy*

$$\|A_1(y)\| \le d_1 W_n(y), \qquad \|A_2(y)\| \le d_2 W_n(y),$$

with $d_1, d_2 \ge 0$, and suppose $A_1(x + z) = A_2(x + z)$ whenever $|z|_1 < b$. Then

$$\left\| \sum_z q(z) \big(A_1(x + z) - A_2(x + z) \big) \right\| \le e^{-\sigma b/2} (d_1 + d_2) C_n(\sigma/2) C W_n(x). \tag{5}$$

Proof. The difference vanishes for $|z|_1 < b$, so the triangle inequality and submultiplicativity of W_n give an upper bound

$$(d_1 + d_2)W_n(x) \sum_{|z|_1 \geq b} |q(z)|W_n(z).$$

Using (1) with $\sigma/2$, each summand is at most

$$C_n(\sigma/2)e^{\sigma|z|_1/2}|q(z)| \leq C_n(\sigma/2)e^{-\sigma b/2}e^{\sigma|z|_1}|q(z)|.$$

Summation and (2) prove (5). □

Lemma 188.3 (Multilinear jet comparison)**.** *Here is the passage from that tail estimate to a local coefficient measurement. Let P be a linear screen onto the allowed edge directions; it is applied only to the direction arrays below. Let F be a C^N real-valued local observable. Here* $\mathrm{inc}_1(U)$ *is the configuration obtained by exponentiating the tangent field U from the identity. Write $\mathcal{J}_a = D^a(F \circ \mathrm{inc}_1)(0)$ for its symmetric a-linear derivative at the identity configuration. Assume the following three estimates for $a \leq N$:*

1. *for all admissible screened directions $v_1, \ldots, v_a$, and all nonnegative λ_i with $\|v_i\| \leq \lambda_i$,*

$$|\mathcal{J}_a(v_1, \ldots, v_a)| \leq \Phi w \prod_{i=1}^{a} \lambda_i; \tag{6}$$

2. *the already-screened reference directions $A_i = P(\theta y_i)$ satisfy $\|A_i\| \leq K$, with $K \geq 1$, and the screened reference jet vanishes:*

$$\mathcal{J}_a(P(\theta y_1), \ldots, P(\theta y_a)) = 0; \tag{7}$$

3. *directions x_i approximate them with $\|P(\theta x_i) - P(\theta y_i)\| \leq \varepsilon$, where $0 \leq \varepsilon \leq K$.*

Then

$$|\mathcal{J}_a(P(\theta x_1), \ldots, P(\theta x_a))| \leq 2^a \varepsilon K^a \Phi w. \tag{8}$$

Proof. Set $U_i = P(\theta x_i)$, $A_i = P(\theta y_i)$, and $E_i = U_i - A_i$. By multilinearity,

$$\mathcal{J}_a(U_1, \ldots, U_a) - \mathcal{J}_a(A_1, \ldots, A_a) = \sum_{\varnothing \neq S \subseteq \{1,\ldots,a\}} \mathcal{J}_a(v_1^S, \ldots, v_a^S),$$

where $v_i^S = E_i$ if $i \in S$, and $v_i^S = A_i$ otherwise. The empty-set term is zero by (7). In every remaining term at least one factor has norm at most ε; all other factors have norm at most K, by the stated bounds on the screened reference directions. Applying (6) to each term bounds it by $\varepsilon K^{a-1}\Phi w \leq \varepsilon K^a \Phi w$. There are $2^a - 1 \leq 2^a$ such terms, proving (8). □

Proposition 188.4 (Local probe bound from a strip and a jet gap)**.** *Fix natural numbers d, N with $d \leq N$, a support radius b, a chart radius m, and $\sigma > 0$, and let $L \geq 1$ be the integer block side. For each $k \leq d$, suppose the four-by-four row kernel $(q_{\nu j})_{\nu,j=1}^4$ has entries satisfying (2) with constant M. Suppose the two arrays $A_j^{(0)}, A_j^{(1)}$ satisfy*

$$\|A_j^{(h)}(y)\| \leq K W_{dk}(y) \quad (h = 0, 1), \qquad A_j^{(0)}(u + z) = A_j^{(1)}(u + z) \text{ if } |z|_1 < b, \tag{9a}$$

where $K \ge 1$, $M \ge 0$, and each kernel base u has $|u|_1 \le 4m$. Define the order-k reference and actual directions by applying this matrix row to $A^{(0)}$ and $A^{(1)}$, respectively. The screen is coordinate restriction to allowed edges, and the normalization is the same edge norm used in (6); thus no additional screen norm is inserted between the row estimate and the priced jet. Then their difference is at most

$$\delta_k = 8Ke^{-\sigma b/2}(1+4m)^{dk}C_{dk}(\sigma/2)M \tag{9}$$

in the normalized edge norm. The reference and actual directions satisfy

$$\|y_k\|,\ \|x_k\| \le 4KC_{dk}(\sigma)M(1+4L(m+2))^{dk} \le C_1,$$

where

$$C_1 = 1 + \sum_{k=0}^{d} 4KC_{dk}(\sigma)M\big(1+4L(m+2)\big)^{dk}, \tag{10}$$

Assume the local jet satisfies the price bound (6), and also that $\varepsilon_d \le C_1$. Let $\mathcal{V}_k$ denote the allowed direction subspace of degree k, and suppose $x_k, y_k \in \mathcal{V}_k$ and the weighted vanishing receipt holds:

$$\mathcal{J}_a(y_{q_1}, \dots, y_{q_a}) = 0 \quad \text{whenever } q_i \ge 1, \quad \sum_{i=1}^{a} q_i \le d. \tag{10a}$$

Assume also the order-zero receipt $F(\mathrm{id}) = 0$. Let $\gamma : \mathbb{R} \to \mathcal{X}$ be the actual C^d probe path in the tangent configuration space, with $\gamma(0) = 0$ and $\gamma^{(k)}(0) = x_k$ for $1 \le k \le d$. The observable is evaluated on $\mathrm{inc}_1(\gamma(s))$, so its base point is the identity configuration. Then the probe of total derivative order $\ell \le d$ obeys

$$\left| \frac{d^\ell}{ds^\ell} F(\mathrm{inc}_1(\gamma(s))) \Big|_{s=0} \right| \le |\operatorname{Part}^{\mathrm{ord}}(\ell)|\, 2^\ell \varepsilon_d C_1^\ell \Phi w, \qquad \varepsilon_d = \sum_{k=0}^{d} \delta_k. \tag{11}$$

Proof. For each of the four entries in an output row, (5) with $d_1 = d_2 = K$ bounds the tail by

$$2Ke^{-\sigma b/2}C_{dk}(\sigma/2)MW_{dk}(u).$$

There are four input components, so the row difference is at most $8Ke^{-\sigma b/2}C_{dk}(\sigma/2)MW_{dk}(u)$. Since $|u|_1 \le 4m$, this is (9). Summing over $k \le d$ gives a common error bound ε_d. For either direction, (4) and the growth bound on its input array give $4KC_{dk}(\sigma)MW_{dk}(u)$. In the chart the lifted base index has size at most $4L(m+2)$, giving the order-k bound above; the extra 1 ensures $C_1 \ge 1$.

Here $\operatorname{Part}^{\mathrm{ord}}(\ell)$ is the finite index set of ordered finite partitions of the ℓ differentiation slots into nonempty blocks, in the canonical ordering used by the chain rule; it is not a sum over arbitrary permutations of each set partition. For (11), use the repeated chain rule in that ordered-finpartition form. When $\ell = 0$, the derivative is $F(\mathrm{id}) = 0$ by the separate order-zero receipt. For each nonempty-order partition π of the ℓ probe derivatives, the chain-rule term is the $|\pi|$-linear jet $\mathcal{J}_{|\pi|}$ applied to the actual direction $x_{|B|}$ for each block $B \in \pi$. The all-reference term, with $x_{|B|}$ replaced by $y_{|B|}$, vanishes: the block degrees are positive and sum to $\ell \le d$, so the weighted vanishing hypothesis applies. Each reference direction is bounded by C_1, and each actual direction differs from it by at most ε_d. Since $\varepsilon_d \le C_1$, the multilinear comparison lemma (8), with $K = C_1$, bounds this chain-rule term by $2^{|\pi|}\varepsilon_d C_1^{|\pi|}\Phi w$, which is at most $2^\ell \varepsilon_d C_1^\ell \Phi w$. Summing over the finite ordered finpartition index set proves (11). □

In the selected construction, the identity behind the periodization is that the routed test path equals the local observable composed with the lifted direction. The reference direction is the finite Laurent polynomial obtained from the probe's truncated mode, while the actual input array is the analytic periodized row applied to the patched test field. These input arrays agree on the ball of radius b; their convolved directions need not agree there. The strip bound supplies (2), while the coefficient-growth receipt supplies KW_{dk}. Thus the difference is not estimated term by term without cancellation: all terms inside the agreement ball cancel, and only the exponentially weighted exterior tail is paid.

The argument above proves the Fourier-tail and multilinear comparison once the three receipts (strip, growth, and vanishing priced jet) are available. It does not prove those upstream receipts, nor does it by itself prove the source-placement or healthy-chart hypotheses in the small-support branch. Those are separate construction inputs.

189 A paid estimate for one regular source

We spell out how a single regular source term is paid in the selected one-block transition. The source is split into its real and imaginary channels. When the transported support fits inside the receiving budget radius, local coefficient probes control the inserted coefficient. When it does not fit, a direct supremum estimate controls it. The two cases use different fields of the local budget, which is why both the regular and large-support inequalities were selected in Lemma 184.1. The principal insertion attenuation needed in the local estimate is established in Section 111; the argument below keeps the regular source and both support regimes explicit.

Fix a one-step transition of block length L, a coefficient frame of maximum grade d, and a regular source f on fine support b, with pointwise bound $c \geq 0$. Let $\alpha_f > 0$ be the fine support exponent and $q_f \geq 0$ its fine cover number, and define the charged mass

$$m_0 = e^{\alpha_f q_f} c.$$

Let $\beta > 0$ be the selected coupling, let $\mathcal{M}_f(G, L, \beta) > 0$ be the fine monitor, and put $\chi = \sqrt{\mathcal{M}_f(G, L, \beta)}$. Thus a raw budget b_0 gives the actual channel price b_0/χ. Set $a = L^d$. Let $O, H, W \geq 0$ be respectively the motion-average norm, local readout, and local test factors. Let raw budgets $u, v, \xi > 0$ satisfy

$$O(Wu + 2Hav) \leq \xi, \tag{927}$$

$$4OHav \leq \xi. \tag{928}$$

These are the regular and large-support paid budgets. In the Wilson application, m_0 is the mass of the source support with its sup-bound; u, v are chosen by Section 184.

The local estimates needed from the one-step construction have the following precise form. The real and imaginary channels have transported cloud charges C_1, C_2 satisfying

$$0 \leq C_1, C_2 \leq \frac{v}{\chi} m_0. \tag{929}$$

If the transported support is at most the receiving budget radius, the local probe reconstruction error, after both channels are combined, is at most

$$O\big(Wu\chi^{-1} m_0 + Ha(C_1 + C_2)\big). \tag{930}$$

If the transported support exceeds that radius, each channel admits a uniform chart bound $S_i \le \frac{v}{\chi} m_0$ for $i = 1, 2$; the resulting two-channel coefficient error is at most

$$OHa(S_1 + S_2 + C_1 + C_2). \tag{931}$$

In either case the local construction also supplies a coefficient in the coarse format, its exact pushed-density identity, and a remainder budget at most ρm_0, where ρ is the one-step remainder rate. These are the actual local chart and transport conclusions used below, not consequences of the scalar budget inequalities alone.

Proposition 189.1 (Regular source atom)**.** *Assume the local estimates* (929)– (931) *and the coarse-format, density, and remainder conclusions just stated. If* (927)– (928) *hold, then the source term produces a one-step coefficient with coefficient error at most* $\xi\chi^{-1}m_0$*, exact pushed density, and remainder at most* ρm_0*. In the terminology of the finite price recurrence, this is the regular-source estimate with cost pair* $(\xi\chi^{-1}m_0, \rho m_0)$*.*

Proof. First suppose the transported support is within the receiving radius. By (929),

$$O\big(Wu\chi^{-1}m_0 + Ha(C_1 + C_2)\big) \le O(Wu + 2Hav)\chi^{-1}m_0 \le \xi\chi^{-1}m_0,$$

where the last inequality is (927). The local probe reconstruction estimate (930) therefore gives the claimed coefficient error.

Now suppose the transported support exceeds the receiving radius. The uniform chart estimate and the two channel bounds give

$$OHa(S_1 + S_2 + C_1 + C_2) \le OHa(4v\chi^{-1}m_0) = 4OHav\chi^{-1}m_0 \le \xi\chi^{-1}m_0,$$

by (928). Thus the large-support estimate gives the same coefficient error. The two cases exhaust the support sizes. In both cases, the local construction supplies the exact pushed-density identity, coarse format, and remainder bound ρm_0; these components are unchanged by the numerical comparison. This proves the assertion. □

Here is the source realization of the local hypotheses. The source term is turned into its real and imaginary masked channels. The single distinguished channel is the only one carrying the root label; the other channel labels have zero assigned support cost. The per-channel reserved-mass estimate is provided by the regular insertion result of Section 111. In the small-support regime, the finite local probes fit inside the chart, and the local reconstruction bound is the test-price term $Wu\chi^{-1}m_0$ plus the two readout terms HaC_1 and HaC_2. In the large-support regime, the direct chart supremum bound replaces the probes, leaving two chart and two cloud readout contributions; this is exactly the factor four in (928). The cloud inequality controls each C_i, and the insertion construction carries the same source through the block with its density identity. The finite price recurrence then sums these single-source costs over the countable regular family using its absolute mass budget.

More explicitly, let q_τ be the coefficient in Section 184. In the small-support case, the chart estimate gives, for each recovery probe τ, a contribution at most $q_\tau v\chi^{-1}m_0$. The cycle budget gives $q_\tau v \le u$; finite coefficient recovery therefore bounds the probe part by $Wu\chi^{-1}m_0$. The two real/imaginary cloud contributions are bounded by $Ha(C_1 + C_2)$, and the regular budget in (927) pays their sum. The chart construction has enough room that the support is nonwrapping and every probe stencil lies in the healthy region; the ordered derivative bounds through order d are exactly the ones used by the finite recovery map. This is the local probe conclusion of the small-support construction, not an inference from the value bound alone.

For the large-support case, the local field is built by its support reservation. The reservation includes the factor $\exp(\alpha_c q_i) \geq 1$, where $\alpha_c > 0$ is the receiving support exponent and $q_i \geq 0$ the channel's coarse cover number, so its unweighted supremum cost is no larger than its reserved mass. The construction bounds that reserved mass using the one-site talent estimate times the source mass. That estimate is at most v/χ, hence each channel has $S_i \leq (v/\chi)m_0$. The cloud bounds likewise give $C_i \leq (v/\chi)m_0$. Applying the local readout bound to the two chart bounds and the two cloud terms yields $OHa(S_1 + S_2 + C_1 + C_2) \leq 4OHavm_0/\chi$. The large-support budget pays this quantity. In both cases the chosen support labels retain the same fine source and the transition preserves its density, while the one-site reservation and turn-price inequalities give the remainder ρm_0. The local chart statements used in these paragraphs are the exact outputs of the small-support and large-support constructions; their hypotheses and producer declarations are recorded in the accompanying evidence file.

In the small-support construction, the image of the source support is first enlarged by the transport halo. The geometric placement puts this enlarged set in a parent patch and bounds every point of that patch's support by the chart radius. The condition

$$20(r_d + R_d + 1) < n_0$$

then makes both the transported source support and the parent patch nonwrapping; r_d is the chart radius, R_d the transport reach, and n_0 the parent side length. The local construction gives the actual field U_i and its smooth reference V_i on one healthy chart. For each channel weight w_i, the ordered derivatives at the identity through order d satisfy

$$\|D_0^\ell U_i - D_0^\ell V_i\| \leq 2(v/\chi)w_i,$$

For every coefficient-recovery probe $\tau \in \mathcal{T}$, evaluated at parameter zero along its test path, one also has

$$|\partial_0^\tau V_i| \leq (v/\chi)w_i.$$

The probe stencil lies in the healthy chart. The first bound controls the local variation correction in the reconstruction, and the second is the probe-price input. The finite recovery map and its probe-price sum therefore give the $Wu\chi^{-1}m_0$ contribution; applying the same row decomposition to the two real channels gives the separate $Ha(C_1 + C_2)$ contribution. This supplies the stated small-support estimate with the correct one-block normalization.

In the large-support construction, the transported cover number is larger than the receiving radius. For each real channel, the support-reservation argument constructs a regular field with supremum bound S_i and a reserved mass

$$R_i = e^{\alpha_c q_i} S_i, \qquad R_i \leq (v/\chi)m_0,$$

where $\alpha_c > 0$ is the receiving support exponent and $q_i \geq 0$ is the channel's coarse cover number. The exponential factor is at least one, so $S_i \leq R_i \leq (v/\chi)m_0$. Independently, the cloud estimate gives $C_i \leq (v/\chi)m_0$. Thus the two chart bounds and two cloud charges together cost at most

$$OHa(S_1 + S_2 + C_1 + C_2) \leq 4OHav\chi^{-1}m_0,$$

which is paid by the large-support inequality. This is the large-support estimate used in the proposition; it keeps the chart and cloud contributions separate because they come from different source bounds.

190 Atomic responses and the finite-volume root price

This section records the actual transfer used by the public trace supplier. There are two distinct operations. An atomic response first becomes a finite-volume response with a root-price estimate. The price estimate is then converted into the two components of a Bill. We give the conversions and the quantitative endpoint of the Coffer argument, while keeping the upstream response, paid-price, and price-duality receipts explicit.

190.1 From an atomic response to the Coffer input

Let X be an order-zero coefficient at scale k, g its local field, and $Z = \widehat{\beta}_k$. Write x_X for its coordinate, X^{f} for its forced part, and D_X for its density. The atomic certificate supplies a nonnegative remainder allowance s, a response L with attainment cost e, and the identities

$$\|x_X\|_{\mathrm{coeff}} + s \le q, \tag{932}$$

$$\mathrm{profile}(X^{\mathrm{f}}) = \mathrm{hook}(L), \tag{933}$$

$$D_X(U) = \mathrm{eval}(U)\, g(U). \tag{934}$$

The remainder bound is measured at scale k, at order zero and with tolerance $\sqrt{\mathrm{forcedAllowance}_k}$; the response has $\mathrm{Attainment}(L) = e$. Set $u = \mathrm{alto}_{k,d}$, the inverse field dilation, and $G' = ug$. Multiplying the density identity by u gives

$$uD_X(U) = \mathrm{eval}(U)G'(U).$$

Thus the atomic response becomes the Coffer input, with unchanged format, remainder and forced-profile data.

The selected atomic charge budget satisfies

$$4(1 + c_k)q \le Z^3, \tag{935}$$

where c_k is the number of coefficient columns at scale k. Hence

$$4(1 + c_k)(\|x_X\|_{\mathrm{coeff}} + s) \le Z^3.$$

The second needed comparison is $Z\,\mathrm{fly}_d(k) \le Z^2$, using $\mathrm{fly}_d(k) \le Z$ and $Z \ge 1$. The price construction chooses the atomic charge allowance so both comparisons hold.

The endpoint price-duality estimate used below has the following exact meaning. A response duet P runs from a level ℓ to $k \le N$ in the same selected history. At each intermediate level its coefficient has admissible order-zero format and remainder bound; its marked site is the spatial origin and these sites are related by the one-step scale map. The coefficient and cost changes satisfy the paid direction inequality, and the densities satisfy the exact one-step transfer identity. Its left endpoint coefficient vector is zero, and its terminal cost obeys the spread estimate

$$\mathrm{cost}_k \le w + c_k\big(w + \|P_k.x\|_{\mathrm{coeff}}\big), \qquad w \ge 0.$$

Write $\kappa_k > 0$ for the square root of the level-k monitor. For every such paid comparison the endpoint estimate is

$$\kappa_k \, \|P_k.x\|_{\mathrm{coeff}} \le 4(1 + c_k)w. \tag{936}$$

The zero in the left-end condition is a coefficient vector; the marked sites are at the spatial origin, not zero-valued observations. The finite recurrence and adjacent-level estimate are given in Section 183. The paid-cone estimate is supplied by the canonical argument in Section 171, under its explicit direction, room, bank, and endpoint-spread assumptions. None of these bounds follows from the atomic response alone.

190.2 The finite Coffer transfer

We keep the source parameters distinct. Write c_k for the number of coefficient columns at level k. Let X be the input coefficient, S its remainder budget, E the paid response and e its attainment cost, and $\zeta_i > 0$ its payment allowance. Let $a \in (0,1]$ be the mask fraction, $v' \geq 0$ the forced-allowance budget, and $q, B > 0$ two coefficient budgets. The actual Coffer hypotheses include

$$\|x_X\|_{\mathrm{coeff}} + S \leq q, \qquad 4(1+c_k)q \leq B, \qquad e\zeta_k \leq v'B. \tag{937}$$

They also include exact coefficient format, remainder, payment, density and forced-profile identities; scale growth and one-step response receipts; and the melting condition. If B_k is the spatial-spacing parameter of the scale budget and $R_k = \mathrm{siteFactor}(B_k)$, the latter requires support exponent at least four and

$$t = \mathrm{forcedAllowance}_k + R_k\sqrt{\mathrm{forcedAllowance}_k} + a + R_k v', \qquad (2B_k+1)^4 t \leq 3, \qquad e^t - 1 \leq 1.$$

Here B_k is not the coefficient budget B in (937).

For $J \leq m < k < N$, let $\mathrm{gain}(k,m) > 0$ denote the scale gain from m to k. The output weight is

$$w_m = a\,\frac{\zeta_m/2}{\|\,\mathrm{alto}_{m,d}\,\|(B/\,\mathrm{gain}(k,m))}. \tag{938}$$

For clarity, if $\mathcal{T}_J$ is the set of reflected scale-J cells in the volume-N tiling and $K = |\mathcal{T}_J|$, with $\mu_{N,p}$ the normalized Wilson measure, the root functional is

$$\mathrm{root}_{J,N,p}(F) = \left\| \int \prod_{x \in \mathcal{T}_J} \mathrm{copy}_x(F)(U)\, d\mu_{N,p}(U) \right\|^{1/K}.$$

Here $\mathrm{copy}_x(F)$ evaluates F on the translated reflection half of the configuration; when that reflection reverses orientation it also complex-conjugates the value. This is the reflection-positive tiling functional, with its Kth-root normalization.

Assume $N > k$, the reference grade is no larger than m, and the scale-J-to-m window is open. The response data are an admissible sequence of coefficient voices from the reference level through N, an admissible top coefficient, an exact paid response identity between each adjacent pair, and the terminal forced-profile identity. The input field has the format, remainder, attainment, and density identities stated above. Assume also the melting and payment inequalities, the endpoint estimate (936), and the room and interiority conditions from J to k. Put

$$\widetilde{G}' = \mathrm{back}_{k\to N}\, G'.$$

Under these hypotheses the finite construction returns a label b with two estimates:

$$b'\,\|x_X - \mathcal{C}_k(b)\|_{\mathrm{coeff}} \leq 4(1+c_k)q, \tag{939}$$

$$\mathrm{root}_{J,N,p}(\widetilde{G}' - \mathcal{O}_{N,p}b) \leq \frac{\mathrm{stakes}(J,m)}{w_m}. \tag{940}$$

Here b' is the endpoint price-duality constant, $\mathcal{C}_k(b)$ is the scale-k response column of b, and $\mathrm{root}_{J,N,p}$ is the reflection-positive root functional on the volume-N field, defined using the scale-J tiling. In this application the endpoint constant b' is κ_k.

For the column estimate, apply the signed reference-difference construction of Section 172 to the Coffer coefficient field X at terminal level k, with its remainder bound S and total charge q.

The construction assumes one fixed background history and calibrated primary fields $\mathcal{P}_i(v)$ whose coefficient is $I_i v$, together with the bank and column inequalities (826) and the exact degree-d density transition. At each backward step it first transports the current zero-coefficient remainder by the zero-coordinate transition (833); it then subtracts the primary field with the resulting coefficient, so that the new remainder again has coefficient zero. Linearity cancels the two primary densities exactly, and the zero-coordinate transition supplies the density of the transported remainder. The paid direction and remainder estimates (825) apply to the resulting signed difference, not to an unsigned estimate obtained by subtracting costs.

The top family has terminal charge at most q, since $\|x_X\|_{\mathrm{coeff}} + S \le q$. Choose its single reference label by setting the left endpoint coefficient to zero. The output is precisely (835): for the chosen label b, the terminal coefficient is $x_X - \mathcal{C}_k(b)$, the terminal density is $D_X - D_{\mathcal{P}_k(b)}$, and the left endpoint coefficient vector vanishes. Every intermediate field and the selected reference field are evaluated against the same finest volume and background history; this is what permits the exact density cancellation. At the terminal level the reference construction also retains its spread bound $r_k \le q + c_k(q + \|Z_k.x\|_{\mathrm{coeff}})$, the precise spread condition in the Woke duet. Thus it applies SharpRead with budget $w = q$. The stronger full terminal estimate in (835), with $C_k = c_k$, gives $r_k + \|Z_k.x\|_{\mathrm{coeff}} \le 4(1+c_k)q$; this is the coarse transport budget used by the subsequent construction. The signed dualization then gives the zero-site paid-response duet required by SharpRead, and its endpoint inequality is exactly (939). Here zero refers to the left coefficient vector and to the duet's marked spatial site being the origin. It does not assert that the observed field values vanish. For the rooted estimate, the same dual chain is planted once at each site of the finite scale-J census. The construction couples each translated negative endpoint to the local lens response, then propagates the resulting polynomial to level m. Its site-supported and box-supported parts have weighted absolute sums bounded by 2 and 1, respectively; each box term carries one unit of support cost. The planted coefficient is the moment in the root functional: integrating it against the level-m reference measure gives the volume-N Wilson partition function times the normalized moment. This identity follows at each transfer from normalization of the transition kernel and the exact response-density identity, and at the terminal level from the definition of the marked coefficient.

The polynomial-price calculation behind the finite-gas estimate is as follows. For a finite census polynomial Q, write

$$\mathcal{P}_w(Q) = \sum_s \mathrm{toll}(w,s)\|Q_s\|$$

for its weighted absolute coefficient sum. The price is submultiplicative. For box terms K_p, the finite sum over disjoint-support patterns obeys

$$\mathcal{P}_w(\mathrm{field}(K)) \le \prod_p (1 + \mathcal{P}_w(K_p)) \le \exp\left(\sum_p \mathcal{P}_w(K_p)\right),$$

because each pattern's product price is at most the product of its factors and $1 + x \le e^x$. The same submultiplicativity, applied term by term to the finite truncated exponential, gives $\mathcal{P}_w(\mathrm{pass}(Q)) \le e^{\mathcal{P}_w(Q)}$. The site and box absolute-sum receipts contribute $2K$ and K to the exponent. The remaining term is the $2fK$ contribution in the planted price estimate at $f = 1$.

By construction, the integral of the selected coefficient equals the volume-N Wilson partition function times the normalized moment M. The partition function is positive. Dividing by it and using the finite volume denominator bound yields

$$|M|\ \mathrm{toll}(w,\mathfrak{s}) \le \exp(|\Lambda_m|\,\mathrm{Treasury}(m) + K(2+1+2))\,. \tag{941}$$

This is an absolute-value estimate and does not use cancellation. Its hypotheses include the response construction's support-cost and moment identity receipts.

In the Coffer application all site weights equal w_m, so $\mathrm{toll}(w, \mathfrak{s}) = w_m^K$. The tiling identity is

$$|\Lambda_m| = K \left(\frac{A_J}{A_m} \right)^4.$$

Taking the positive K-th root of (941) yields

$$|M|^{1/K} \le \frac{\exp((A_J/A_m)^4 \operatorname{Treasury}(m) + 5)}{w_m} = \frac{\operatorname{stakes}(J, m)}{w_m},$$

which is (940). The moment identity is needed to identify this M with the desired rooted observable.

190.3 Using the output in the Bill

The Bill is evaluated on the inflated field $G' = ug$. Its root error is $\widetilde{G}' - \mathcal{O}_{N,p} b$, where $\widetilde{G}' = \mathrm{back}_{k \to N} G'$. In the Bill specialization, $a = Z^{-1}$, $v' = Z^{-1}$, $B = Z^3$, and $e = Z$, while $\zeta_i = \mathrm{fly}_d(i)$ at each scale. Thus the payment condition is $e\zeta_k = Z \,\mathrm{fly}_d(k) \le Z^2 = v'B$. The gas weight from (938) is

$$w_{\mathrm{gas}} = Z^{-1} \frac{\mathrm{fly}_d(m)/2}{\| \operatorname{alto}_{m,d} \| (Z^3 / \operatorname{gain}(k, m))}.$$

The factor Z^{-1} is the mask fraction a in the Coffer input, so this is exactly the native Bill denominator. Its norm entry follows from the atomic charge and the unit scale allowance. Its bias entry combines (939) with the separately calibrated response-column estimate; the paid-response price estimate in Section 191 bounds the resulting quantity by $\mathrm{threshold}(k)^2 \,\mathrm{AgeK}_k(-1/4)$.

The outstanding suppliers are the atomic response, the endpoint price-duality recurrence, and the local action and occupancy estimates which construct the site/box expansion and its absolute-value envelope. The argument above spells out the numerical transfer once those receipts are supplied; it does not claim to have proved those producers.

191 The paid-response price bound

We compute the quantitative root-price bound attached to a paid finite-volume response. This is the numerical input used for the two component errors in Section 192. The computation has two parts: a coefficient-size estimate from the paid label and its approximation receipt, and a negative-rate bound on the native root price.

For each frame index i, set

$$J_i = \mathrm{basin}(i), \qquad K_i = \mathrm{summit}(i), \qquad t_i = \widehat{\beta}_{K_i}, \qquad v_i = \left(\frac{A_i}{A_{J_i}} \right)^d, \qquad f_i = \mathrm{fly}_d(J_i),$$

where A_j is the field dilation at level j. Let s_i be the local stake price and γ_i the gain from level K_i down to J_i. The source's native root allowance for a grade-d response has the exact form

$$n_{d,i} = \frac{s_i}{t_i^{-1} \left(\dfrac{f_i/2}{A_{J_i}^{-d} (t_i^3/\gamma_i)} \right)}. \tag{942}$$

The norm of the inverse grade-d dilation is $A_{J_i}^{-d}$, which is the factor appearing in this expression. Simplifying gives

$$A_i^d n_{d,i} = \frac{2 s_i v_i t_i^4}{f_i \gamma_i}. \tag{943}$$

Indeed, the denominator in (942) is $f_i \gamma_i A_{J_i}^d / (2t_i^4)$, and $A_i^d / A_{J_i}^d = v_i$. Throughout this calculation the scale receipts ensure $A_i, A_{J_i}, t_i, f_i, \gamma_i > 0$ and $G_i \to \infty$.

The scalar rate receipts are expressed by tickets. For a positive sequence F_i, a rate-zero ticket means that for every $\epsilon > 0$, eventually $0 \le F_i \le \exp(\epsilon G_i)$, where $G_i = \mathrm{Gate}(\widehat{\beta}_i)$. If two sequences have tickets of rates α and α', their product has rate $\alpha + \alpha'$: choose the two ticket bounds at exponent $\alpha + \eta$ and $\alpha' + \eta$, multiply them, and then let 2η be smaller than the requested excess over $\alpha + \alpha'$. Nonnegative sums of equal-rate sequences retain that rate by the same argument and the elementary bound $u + v \le 2 \max(u, v)$.

The scale hypotheses supply the following rate-zero tickets:

$$s_i, \qquad v_i, \qquad t_i^4, \qquad f_i^{-1}.$$

The first is the stake receipt. The second is the dilation-ratio receipt at the selected basin. The third is the scale-at-summit receipt. For the fourth, $J_i \le K_i$ and the inverse-flux receipt applies to the flight scale at J_i. Multiplication, including the constant factor 2, therefore gives a rate-zero ticket for

$$B_i = \frac{2 s_i v_i t_i^4}{f_i}.$$

The gain estimate supplies eventually $\tau_i(500) \le \gamma_i$, where

$$\tau_i(q) = \exp\Big(\frac{q}{10^4} G_i \Big).$$

The ticket bound at exponent $70/10^4$ gives $B_i \le \tau_i(70)$ eventually. Dividing by the gain and using multiplicativity yields

$$A_i^d n_{d,i} = B_i / \gamma_i \le \tau_i(70) / \tau_i(500) = \tau_i(-430). \tag{944}$$

This proves the exact negative-rate price needed below for all sufficiently large i.

Lemma 191.1 (Column size from a paid label)**.** *Suppose a grade-d paid label y satisfies*

$$\|y\|_{\mathrm{coeff}} \le 1, \qquad \|y - \mathrm{Col}(x)\|_{\mathrm{coeff}} \le \vartheta_K^2 \ \mathrm{Age}_K(-1/4),$$

where the selected threshold obeys $0 < \vartheta_K \le 1$ and $\mathrm{Age}_K(-1/4) \le 1$. Then

$$\|\,\mathrm{Col}(x)\|_{\mathrm{coeff}} \le 2.$$

Proof. The triangle inequality gives

$$\|\,\mathrm{Col}(x)\|_{\mathrm{coeff}} \le \|y\|_{\mathrm{coeff}} + \|\,\mathrm{Col}(x) - y\|_{\mathrm{coeff}} \le 1 + \vartheta_K^2 \,\mathrm{Age}_K(-1/4) \le 2.$$

□

Proposition 191.2 (Paid label yields the scaled component price)**.** *Let g be an admissible finite-volume observable and let y, x be a paid grade-d label pair satisfying the hypotheses of Lemma 191.1. Assume its paid root receipt states, for every allowed window and whenever g is interior, that*

$$\mathcal{R}_{i,p}\Big(\mathcal{B}\big(A_{K_i}^{-d} g\big) - \mathcal{O}(x) \Big) \le n_{d,i}.$$

Then the paid column has size at most 2. *For* $q_i = (A_i/A_{K_i})^d$, *linearity and homogeneity give the scaled error bound*

$$\mathcal{R}_{i,p}\Big(\mathcal{B}(q_i g) - \mathcal{O}(A_i^d x)\Big) \le \tau_i(-430).$$

For all sufficiently large i, *this is the paid-component error used in the probe assembly, uniformly over the allowed windows and observables.*

Proof. The coefficient conclusion is Lemma 191.1. The inverse dilation satisfies $q_i g = A_i^d(A_{K_i}^{-d} g)$. By real-linearity of the finite-volume observation and pullback,

$$\mathcal{B}(q_i g) - \mathcal{O}(A_i^d x) = A_i^d \left(\mathcal{B}(A_{K_i}^{-d} g) - \mathcal{O}(x)\right).$$

Since $A_i > 0$, homogeneity of the root seminorm and the assumed paid receipt bound this by $A_i^d n_{d,i}$. Equation (944) then gives the stated $\tau_i(-430)$ bound, uniformly over all allowed windows and observables. □

The label-producing receipt is obtained in the source from a funded atomic activity and a scale-dependent resource transfer. The mathematical inputs are not just a coefficient bound: the atomic activity carries a bounded charge, a paid profile, and a density identity; its finite-volume conversion also uses the price-duality estimate against every allowed test. The source construction of those atomic and dual-price inputs remains part of the full trace supplier. This section discharges the coefficient and rate arithmetic once they are available.

192 Assembly of the finite-volume response probe

We spell out the finite algebra that turns two separately paid local components and a buffered chamber approximation into the probe used in the mission. The quantitative assumptions below are exactly the estimates the assembly consumes. In particular, the proof establishes every component of the probe conclusion and does not regard the existence of a record with those components as a proof.

Fix a frame level j, its volume level k, and a finite volume $N > k$. Write

$$t = \widehat{\beta}_k, \qquad a > 0, \qquad \beta > 0, \qquad q = (a/a_k)^5,$$

where a_k is the field scale at the volume level. Let τ_j be the error scale

$$\tau_j(s) = \exp\Big(\frac{s}{10^4}(\log \widehat{\beta}_j)^{48}\Big), \qquad E = \tau_j(-430).$$

Assume the scale bounds

$$t \ge 1, \quad a \ge 1, \quad 0 < \beta \le 2t, \quad 1 + 2t \le \tau_j(230), \quad 6c \le t, \tag{945}$$

where c is the coefficient cost of the finite symmetry average defined below. The error scales obey $\tau_j(s)\tau_j(s') = \tau_j(s + s')$.

Let $\mathfrak{A} = \mathrm{Signed}(4) \times \{0, 1\}$ be the finite motion set; the second factor records whether complex conjugation is applied. For $\mu = (s, e)$, write ρ_μ for the real-linear action on grade-five coefficients and σ_μ for the corresponding real-linear action on observables. For $\mu = (s, e)$ and axis d, define $\mathrm{prev}(\mu, d) = s.\mathrm{perm}^{-1}(d)$ and $\epsilon(\mu, d) = \mathrm{sign}(s, \mathrm{prev}(\mu, d))$; the sign does not depend on the

conjugation bit e. Put $R = |\mathfrak{A}|^{-1}$. The finite averages have the form

$$(\mathscr{S}_5 V)_d = R \sum_{\mu \in \mathfrak{A}} \epsilon(\mu, d)\, \rho_\mu V_{\mathrm{prev}(\mu,d)}, \qquad R \sum_{\mu \in \mathfrak{A}} |\epsilon(\mu, d)| = 1, \tag{946}$$

$$(\mathscr{S}_0 f)_d = R \sum_{\mu \in \mathfrak{A}} \epsilon(\mu, d)\, \sigma_\mu f_{\mathrm{prev}(\mu,d)}. \tag{947}$$

This is a real-linear Reynolds average. Reindexing by $\mu \mapsto g\mu$ for each signed motion g and using the defining previous-axis and sign formulas gives the exact covariance $\rho_g Z_d = \mathrm{sign}(g, d) Z_{g.\,\mathrm{perm}(d)}$. Toggling the second motion coordinate reindexes the conjugation action and proves the reality condition. In particular, central inversion fixes every axis and has sign -1, so $\rho_{\mathrm{archV}} Z_d = -Z_d$ for each d. Oddness is a consequence of this vector-valued Reynolds projection, not a claim about arbitrary grade-five labels.

Let Col be the finite-volume column map from grade-five labels to coefficient arrays. It is real-linear and commutes with each motion by finite-volume covariance. The paid labels obey $\|\operatorname{Col}(X_d)\|_1, \|\operatorname{Col}(Y_d)\|_1 \le 2$; this is the transported column bound, including the norm of the observation map, rather than a bound on the raw label alone. Define its averaged coefficient cost by

$$c = R \sup_d \sum_{\mu \in \mathfrak{A}} \|\rho_\mu\|_{\mathrm{col}}.$$

The triangle inequality yields $\|\operatorname{Col}(\mathscr{S}_5 V)_d\|_1 \le c \sup_e \|\operatorname{Col}(V_e)\|_1$. This is the precise finite column estimate used below; the signed or conjugate motion is real-linear, not assumed complex-linear or isometric in the column norm. The observable average has total variation one, so, on chamber-interior observables, the reflection seam makes it preserve a common root-seminorm bound.

Let x_d, y_d be two arrays of finite-volume observables, each interior to a fixed locality window. Let X_d, Y_d be their grade-five paid coefficients. The paid receipts and the finite-volume column map give the transported column bounds $\|\operatorname{Col}(X_d)\|_1 \le 2$ and $\|\operatorname{Col}(Y_d)\|_1 \le 2$. Write $\mathcal{O}$ for the linear finite-volume observation into a one-point observable, $\mathcal{B}$ for pullback from the coarse volume to the reflection chamber at level N, and $\mathcal{R}$ for the chamber root seminorm. Assume each paid residual, each pulled-back local field, and each observed label below is interior to the same reflection chamber, with the locality window and room conditions required for the finite-volume seam. On this interior class, assume the seam gives root-seminorm subadditivity and invariance under each motion. The exact covariance identities are

$$\mathcal{O}\,\mathscr{S}_5 = \mathscr{S}_0\,\mathcal{O}, \qquad \mathcal{B}\,\mathscr{S}_0 = \mathscr{S}_0\,\mathcal{B}. \tag{948}$$

These follow term by term from linearity and covariance of the finite-volume observation and pullback under each signed motion.

The paid-component estimates used here are the following explicit inequalities, for every axis d:

$$\mathcal{R}\big(\mathcal{B}(q x_d) - \mathcal{O}(a^5 X_d)\big) \le E, \qquad \mathcal{R}\big(\mathcal{B}(q y_d) - \mathcal{O}(a^5 Y_d)\big) \le E. \tag{949}$$

The buffered chamber estimate is this: if

$$J_d = q\big(x_d - \beta y_d\big)$$

and g is any admissible coarse observable with supremum norm at most $\tau_j(150000)$, interior to the same locality window, with that window fitting in the available room, then at any two distinct coarse sites,

$$|\mathbb{E}\,[(\mathcal{B}\mathscr{S}_0 J)_d(x)\,(\mathcal{B}g)(z)]| \le \tau_j(-30000). \tag{950}$$

The estimate is supplied by the finite-volume chamber buffer; its proof is upstream of this assembly. The scale assumptions and the paid-component bounds are supplied by the construction's budget and response receipts.

Proposition 192.1 (Probe from paid components and a chamber buffer)**.** *Under the assumptions above, define*

$$Z_d = (\mathscr{S}_5(X - \beta Y))_d, \qquad J_d = q(x_d - \beta y_d), \qquad P_d = \mathcal{O}(a^5 Z_d), \qquad U_d = (\mathcal{B}\mathscr{S}_0 J)_d.$$

Then Z is signed-covariant, with each component satisfying $\rho_{\mathrm{archV}} Z_d = -Z_d$. Moreover, for every axis d,

1. $\| \operatorname{Col}(a^5 Z_d)\|_1 \le a^5 t^4$*;*
2. *U_d is chamber-interior and $\mathcal{R}(U_d - P_d) \le \tau_j(-200)$;*
3. *for every admissible bounded local test in* (950)*, the separated-site pairing of U_d with its pullback is at most $\tau_j(-200)$ in absolute value.*

Thus P_d has exactly the coefficient, parity, root-error, and local-price parts required of the finite-volume probe.

Proof. Real-linearity of the column map and the transported column bounds give $\| \operatorname{Col}(X_d - \beta Y_d)\|_1 \le 2 + 2\beta$. Covariance lets the finite-average bound apply to Z, so

$$\| \operatorname{Col}(Z_d)\|_1 \le c(2 + 2\beta) \le c(2 + 4t) \le 6ct \le t^2.$$

Because $a, t \ge 1$, this yields $\| \operatorname{Col}(a^5 Z_d)\|_1 \le a^5 t^4$. The finite signed-motion average has the covariance just established; central inversion fixes each axis with sign -1, giving the oddness assertion for every component.

By linearity and the covariance identities,

$$U_d - P_d = \big(\mathscr{S}_0\big(\mathcal{B}(qx) - \mathcal{O}(a^5 X)\big)\big)_d - \beta\big(\mathscr{S}_0\big(\mathcal{B}(qy) - \mathcal{O}(a^5 Y)\big)\big)_d.$$

The root seminorm is invariant under each signed motion, is subadditive, and the averaging weights have total absolute value one. Hence

$$\mathcal{R}(U_d - P_d) \le (1 + \beta)E \le \tau_j(230)\tau_j(-430) = \tau_j(-200),$$

where $1 + \beta \le 1 + 2t \le \tau_j(230)$ was used. The interiority of U_d follows because each motion preserves the fixed locality window and finite averaging preserves the interior domain.

Finally, the buffered estimate gives a separated-site price at most $\tau_j(-30000)$, which is at most $\tau_j(-200)$ by monotonicity of τ_j. This proves all three conclusions. □

The proposition isolates the actual finite averaging and error arithmetic in the selected probe producer. The source's remaining inputs to this local argument are the two paid-component estimates (949), the finite-volume covariance identities, and the buffered chamber estimate (950); their exact producers and hypotheses are recorded in the evidence file. In particular, this section does not claim that arbitrary local components or an arbitrary chamber record satisfy those paid estimates.

193 From finite-volume mileage to the selected chart price

This section proves the passage from the selected finite-volume mission to its local price in the limiting table. The finite-volume separation estimate of Proposition 198.1 supplies the numerical bound used by the mileage argument. The mission's funded probe, its local approximations, and the selected chart convergence are stated explicitly below; this section does not claim to produce those analytic receipts.

Write i for the scale parameter, j_i for its selected frame level, and $k_i = \mathrm{summit}(j_i)$ for the associated volume scale. Let $a_i > 0$ be the field scale, $b_i > 0$ the zoom factor, and $r_i = a_i b_i$ the physical insertion scale. Let $\mathcal{L}$ be the ultrafilter on finite-volume indices $p = (N, s)$ used by the construction, and define

$$\tau_i(q) = \exp\Big(\frac{q}{10^4}(\log \widehat{\beta}_{j_i})^{48}\Big).$$

A mission consists of arrays $t_i(p)$ and x_i such that, for all sufficiently large i, $t_i(p) \to x_i$ along $\mathcal{L}$. Moreover, there is one fixed integer m such that for all sufficiently large i and for $\mathcal{L}$-almost every $p = (N, s)$, a probe of complexity at most m, based at level j_i, has marked grade-five field $a_i^5 t_i(p)$. For such a probe, its measured marker P has coefficient size at most $a_i^5 \widehat{\beta}_{k_i}^m$ and is odd under the central signed inversion. Its toll receipt supplies a bounded local observable U whose pullback to the finite-volume chamber is interior and whose root error relative to P is at most $\tau_i(-200)$. For every pair of distinct coarse sites x, z, it also supplies the bound

$$|\mathbb{E}_{i,p}[\widetilde{U}(x)\widetilde{g}(z)]| \le \tau_i(-200),$$

where tildes denote pullback to the finite-volume chamber. Here g is any coarse test whose pointwise absolute value is at most $\tau_i(150000)$, which is interior to the fixed grade-four locality window, and whose pullback is chamber-interior, provided that the window fits in the available room at frame level j_i inside the volume at scale k_i. For the selected grade-four insertion, a localization receipt supplies a local approximation V with root error at most $\tau_i(-300)$ and bounds for the two measured root costs. These are the quantitative inputs consumed by the finite-volume mileage estimate; their construction remains an upstream obligation.

Fix a reference grade-four field w. Let $H_{i,p}(w)$ be its finite-height insertion approximant, which is a raw grade-four label. The selected hierarchy receipt asserts $H_{i,p}(w) \to y_i(w)$ along $\mathcal{L}$, where $y_i(w)$ is the limiting raw field and $\pi_4 y_i(w) = U_i^{(4)} w$. It also ensures, on an $\mathcal{L}$-large set, that the finite transfer is admissible, the marker and insertion lie in the same reflection chamber, and the two coarse sites are distinct. Let ξ_i be the cover vector with first component $\lceil b_i \rceil$ and every other component 1. Set $z_i = b_i^{-1}\xi_i$ componentwise. The scale relation $r_i = a_i b_i$ gives $r_i z_i = a_i \xi_i$. Thus the physical measurement point has the two cover coordinates

$$\ell_i = (\xi_i, 0).$$

The separation receipt for these coordinates supplies the distinct-site hypothesis.

Set $\Theta_i = \tau_i(-100)$.

Proposition 193.1 (Mission-to-chart price transfer)**.** *Fix w. Assume the mission properties above. Assume that for all sufficiently large i and $\mathcal{L}$-almost every finite-volume index p, mileage applies to the probe and insertion and gives*

$$\big|\mathbb{E}_{i,p}\big[A_{i,p}(a_i^5 t_i(p); x_i^*)\, B_{i,p}(H_{i,p}(w); y_i^*)\big]\big| \le \Theta_i. \tag{951}$$

Here $\mathbb{E}_{i,p}$ is the normalized finite-volume expectation, $A_{i,p}$ and $B_{i,p}$ are the measured grade-five and grade-four observables at the two distinct sites x_i^*, y_i^* determined by ℓ_i. Assume the selected-table limit theorem: whenever $u_p \to u$ and $v_p \to v$ along $\mathcal{L}$,

$$\mathbb{E}_{i,p}[A_{i,p}(u_p; x_i^*) B_{i,p}(v_p; y_i^*)] \longrightarrow \mathcal{M}_i(\pi_5 u, \pi_4 v), \tag{952}$$

where π_d is the limiting projection in grade d and $\mathcal{M}_i$ is the selected table's two-entry pairing at the scale-r_i point. Then, for all sufficiently large i (with threshold allowed to depend on w),

$$\left| \mathcal{M}_i\Big(a_i^5 \pi_5 x_i,\ U_i^{(4)} w \Big) \right| \le \Theta_i.$$

Proof. Fix a sufficiently large i for which the mission, mileage, and chart receipts hold. The set on which the probe and finite-volume insertion conditions hold belongs to $\mathcal{L}$. On this set, (951) places every finite expectation in the closed disk of radius Θ_i. Mission convergence gives $a_i^5 t_i(p) \to a_i^5 x_i$, and the chart receipt gives $H_{i,p}(w) \to y_i(w)$ with $\pi_4 y_i(w) = U_i^{(4)} w$. Applying (952) shows that the finite expectations converge to $\mathcal{M}_i(a_i^5 \pi_5 x_i, U_i^{(4)} w)$; linearity of π_5 was used to take the fixed scalar a_i^5 outside. Since the disk is closed, the limit satisfies the same bound. This proves the proposition. □

193.1 The finite-volume mileage receipt

The error scale is multiplicative: $\tau_i(q)\tau_i(q') = \tau_i(q + q')$, and it is increasing in q. In the finite-volume estimate, both measured markers have root cost at most $\tau_i(5)$. The probe approximation contributes root error $\tau_i(-200)$ and separated-site price $\tau_i(-200)$; the local insertion approximation contributes root error $\tau_i(-300)$. The reflection-positive product bound controls the two cross terms by products of root costs. Bilinearity gives the exact decomposition

$$\langle P, Q \rangle = \langle U, V \rangle + \langle P, Q - V \rangle + \langle P - U, V \rangle.$$

The three terms are bounded respectively by

$$\tau_i(-200), \qquad \tau_i(5)\tau_i(-300) = \tau_i(-295), \qquad \tau_i(-200)(\tau_i(5) + \tau_i(-300)) = \tau_i(-195) + \tau_i(-500).$$

Thus the sum is at most $4\tau_i(-195)$. The selected growth receipt gives $4 \le \tau_i(1)$ eventually, hence $4\tau_i(-195) \le \tau_i(-194) \le \tau_i(-100)$. Proposition 198.1 proves this three-term estimate from the explicit root and reflection-positive bounds. The remaining hard content is construction of the probe, local approximation, and root-cost receipts at the selected scales.

In the source route, the finite-cover construction produces the mission's fixed-complexity probe, while convergence of the selected core field produces its limiting value. The limit transfer above is complete once those receipts are supplied. Producing the funded cover, the probe witness, and the quantitative localization receipts remains part of the larger construction.

194 The spectral measure of a single field

The joint translation measure acts on vectors in Hilbert space. A point field applied to the vacuum need not be such a vector. To obtain the spectral measure of its two-point function, we first keep a positive heat height, take the measure of the resulting vector, and then remove the exponential damping. The resulting measure can have infinite total mass.

194.1 Exact damping of vector measures

We first record a uniqueness fact for bounded holomorphic functions that will identify heat evolution inside the spectral integral. Suppose h is bounded and continuous on the closed upper half-plane, holomorphic in its interior, and zero on the real axis. Fix $x \in \mathbb{R}$, $s \geq 0$ and apply the Gaussian contour argument from Lemma 48.1 to $h(z+x)$ with $b = -2\varepsilon s$. The bottom integral vanishes. On the top edge,

$$e^{-ib(t+is)-\varepsilon(t+is)^2} = e^{-\varepsilon s^2} e^{-\varepsilon t^2}.$$

The nonzero constant factor can be cancelled, giving $\int e^{-\varepsilon t^2} h(x+t+is)dt = 0$. Set $\varepsilon = R^{-2}$ and $t = Ru$. After cancelling $R > 0$, let $R \downarrow 0$. Boundedness and the Gaussian majorant justify dominated convergence and give $\sqrt{\pi} h(x+is) = 0$. Thus $h = 0$. In particular, two such bounded holomorphic functions with equal continuous boundary values agree throughout the half-plane.

Let ν_v be the finite joint spectral measure of a vector $v \in \mathcal{H}_E$, and retain T_E and E_t from the earlier construction. Equation (166) gives $p_0 \geq 0$ almost everywhere for ν_v. For fixed $a \in \mathbb{R}^4$, the function

$$G(z) = \int e^{ip\cdot a} e^{izp_0}\, d\nu_v(p)$$

is bounded and continuous on the closed upper half-plane. On a neighborhood where $\operatorname{Im} z \geq \eta > 0$, its derivative is dominated by $p_0 e^{-\eta p_0} \leq \eta^{-1}$. Since ν_v is finite, dominated differentiation proves holomorphy. The bounded holomorphic function

$$F(z) = \langle v, T_E(a) E(-iz) v\rangle$$

in the open half-plane has the continuous boundary $\langle v, T_E(a) V(t) v\rangle$. By the translation spectral formula these boundary values equal $G(t)$. The uniqueness just proved therefore identifies $F = G$.

The operators E_t are self-adjoint and commute with translations. For $t \geq 0$, the preceding identity at $z = 2it$ gives

$$\langle E_t v, T_E(a) E_t v\rangle = \langle v, T_E(a) E_{2t} v\rangle = \int e^{ip\cdot a} e^{-2tp_0}\, d\nu_v(p).$$

Both measures here are finite, because $p_0 \geq 0$. Uniqueness of finite measures from their characteristic functions, proved in Section 48, yields

$$d\nu_{E_t v}(p) = e^{-2tp_0}\, d\nu_v(p). \tag{953}$$

This proof uses the continuous bounded half-plane boundary of a vector matrix coefficient, not the generally distributional boundary of a point correlation.

194.2 Point vectors and removal of damping

Fix a label v and, for $s > 0$, define

$$q_s(v) = Q_{p,(s)}(0; v) = E_{s-p} B_p(v) \Omega_E, \qquad 0 < p < 1, \quad 4p < s.$$

The vector is independent of the admissible width p. Indeed, the mixed Gram identity gives the same number for the scalar product of either of two such vectors with either one of them. Expanding the squared norm of their difference therefore gives zero. If $t \geq 0$, a width admissible at s is also admissible at $s + t$. Using that common width and the semigroup law gives

$$q_{s+t}(v) = E_t q_s(v). \tag{954}$$

Let ν_s be the finite spectral measure of $q_s(v)$. Define a positive Borel measure by

$$d\mu_v(p) = e^{2p_0}\, d\nu_1(p). \tag{955}$$

This is a well-defined sigma-finite measure: the density is finite and positive, and is bounded on each set $\{|p_0| \le n\}$. Multiplication of densities and (953)–(954) imply

$$e^{2(s+t)p_0}\, d\nu_{s+t} = e^{2sp_0}\, d\nu_s.$$

Compare s and 1 through their common larger height $s+1$. It follows that for every $s > 0$,

$$d\mu_v = e^{2sp_0}\, d\nu_s, \qquad e^{-2sp_0}\, d\mu_v = d\nu_s. \tag{956}$$

In particular μ_v is supported in the time half-space $p_0 \ge 0$, and every positive exponential damping has finite mass.

The single-insertion case of the mixed Gram identity and translation covariance of the point vectors now gives

$$F_2(x_0^0 - is, x_1^0 + is; \mathbf{x}_0, \mathbf{x}_1; Jv, v) = \int e^{-2sp_0} e^{ip\cdot(x_1 - x_0)}\, d\mu_v(p). \tag{957}$$

To see the relative sign explicitly, the two vectors are $T_E(x_0)q_s(v)$ and $T_E(x_1)q_s(v)$. Unitarity moves the first translation to the second, producing $T_E(x_1 - x_0)$; the spectral formula and (956) give the integral above.

For a self-adjoint Euclidean scalar label v, taking both real positions equal to zero identifies the left side with its Euclidean two-point function at separation $2s$. Thus

$$T_v(t) = \int e^{-tp_0}\, d\mu_v(p) \quad (t > 0). \tag{958}$$

It follows already that T_v is real, nonnegative and decreasing. The stronger forward-cone conclusion below is obtained after identifying the measure with the actual reconstructed fields.

194.3 Polynomial growth and the full boundary distribution

The fixed-label gap bound (148), applied at heights $(-s, s)$ and zero real positions, gives constants $C \ge 0$ and $k \in \mathbb{N}$ such that

$$\nu_s(\mathbb{R}^4) = \|q_s(v)\|^2 \le Cs^{-k}, \qquad 0 < s \le 1.$$

For $R \ge 1$ use $s = R^{-1}$ in (956). On $\{p : p_0 \le R\}$ one has $e^{-2sp_0} \ge e^{-2}$, and hence

$$\mu_v\{p : p_0 \le R\} \le e^2 C R^k. \tag{959}$$

This controls an entire energy sublevel set, not just a Euclidean ball. In particular $\mu_v(\{|p| \le R\}) \le AR^k$ with $A = e^2 C$. The measure is locally finite and therefore is a Radon measure on $\mathbb{R}^4$.

Here is a quantitative rapid-weight consequence. Cover $\mathbb{R}^4$ by the closed shells $2^j \le 1 + |p| \le 2^{j+1}$, $j \ge 0$. On the jth shell the weight $(1 + |p|)^{-k-1}$ is at most $2^{-j(k+1)}$, and its mass is at most $A2^{(j+1)k}$. Summing, with overlaps causing only an upper bound, gives

$$\int (1 + |p|)^{-k-1}\, d\mu_v(p) \le A2^{k+1} < \infty. \tag{960}$$

Thus every Schwartz function is absolutely integrable against μ_v, and integration is a continuous functional on Schwartz space, bounded by a weighted supremum seminorm times the finite integral in (960).

Use the unnormalized plus Fourier convention

$$f_+(p) = \int e^{ip\cdot x} f(x)\,dx, \qquad \mathcal{W}\Phi(p) = \int e^{ip\cdot(x_1-x_0)}\Phi(x_0,x_1)\,dx_0dx_1.$$

Both are Schwartz functions when their arguments are Schwartz. For the second assertion take the joint plus Fourier transform and restrict it to the injective linear subspace $p \mapsto (-p,p)$. The chain rule controls derivatives of the restriction, and $|(-p,p)| = \sqrt{2}|p|$ controls its weighted supremum seminorms.

At each height $s > 0$, Fubini in (957) is justified by $\nu_s(\mathbb{R}^4)\|\Phi\|_{L^1} < \infty$. It identifies the joint tested correlation with $\int e^{-2sp_0}\mathcal{W}\Phi(p)d\mu_v(p)$. As $s \downarrow 0$, the exponential has absolute value at most one almost everywhere and tends to one. The integrable majorant $|\mathcal{W}\Phi|$ and the joint boundary theorem give

$$W_2(Jv,v)(\Phi) = \int \mathcal{W}\Phi(p)\,d\mu_v(p). \tag{961}$$

This is an identity on all joint Schwartz tests, including those meeting the coincidence set or the null cone. No restriction of a distribution to such a set has been used.

194.4 Identification with field vectors and the forward cone

Let $\xi_v(f) = \phi(v,f)\Omega$ be the one-field vector in the reconstructed Hilbert space. With $\Phi(x_0,x_1) = \overline{f(x_0)}g(x_1)$, Fubini in the ordinary Fourier integral gives $\mathcal{W}\Phi = \overline{f_+}g_+$. The word Gram formula and (961) therefore give

$$\langle \xi_v(f), \xi_v(g)\rangle = \int \overline{f_+(p)}g_+(p)\,d\mu_v(p). \tag{962}$$

The product is integrable by the rapid-weight bound. Translation covariance gives $U(a)\xi_v(f) = \xi_v(f(\cdot - a))$, and changing variables in the Fourier integral gives $(f(\cdot-a))_+(p) = e^{ip\cdot a}f_+(p)$. It follows from (962) that the characteristic function of the finite PVM measure of $\xi_v(f)$ equals that of $|f_+|^2\mu_v$. The latter is finite by (960). Finite-measure uniqueness gives

$$d\mu^{\mathrm{field}}_{\xi_v(f)} = |f_+|^2\,d\mu_v. \tag{963}$$

The weights in this identity determine μ_v uniquely. Given a closed ball of radius n, choose a smooth compactly supported function h equal to one on that ball and take a Schwartz function f with $f_+ = h$. Such an f exists by Fourier inversion, with the factor $(2\pi)^{-4}$ in the inverse transform. The weight $|f_+|^2$ is then exactly one on the ball. Equality of all the weighted measures implies equality on every measurable subset of each ball; increasing the radius proves equality on all Borel sets by continuity from below.

By Theorem 48.2, every measure on the left of (963) vanishes outside the forward cone. Apply the preceding weights on each ball to conclude

$$\mu_v(\mathbb{R}^4 \setminus \overline{V}_+) = 0. \tag{964}$$

Thus the forward-cone statement is inherited from the reconstructed field representation, while the initial undamping argument required only positivity of the time coordinate in the Euclidean chamber.

For completeness the same identification gives Lorentz covariance of the point measure. Write $A = \Lambda^{-T}$ for the dual momentum action and $g_\Lambda(x) = g(\Lambda^{-1}x)$. Since $\det \Lambda = 1$, $(g_\Lambda)_+(Ap) = g_+(p)$. Covariance of the field vectors and of their PVM measures then gives

$$|(g_\Lambda)_+|^2 \mu_{\rho(\Lambda)v} = A_*\big(|g_+|^2 \mu_v\big) = |(g_\Lambda)_+|^2 A_* \mu_v.$$

The transformed tests run through Schwartz space, so uniqueness by the weights proves $\mu_{\rho(\Lambda)v} = A_*\mu_v$. In particular a Lorentz scalar label has a Lorentz-invariant point measure, with the same dual convention as the joint translation measure.

194.5 Application to the centered trace

Let T_{ab} be the real symmetric Euclidean stress tensor supplied by the construction, transforming as a rank-two tensor, and put $\tau = \sum_a T_{aa}$. Orthogonality gives

$$R(K)\tau = \sum_{a,i,j} K_{ia} K_{ja} T_{ij} = \sum_{i,j} \delta_{ij} T_{ij} = \tau.$$

Its one-point value $c = \langle \tau \rangle_\chi$ is independent of position by translation invariance, and real by conjugation symmetry. The centered label $\Theta = \tau - c\mathbf{1}$ is consequently real, invariant under all Euclidean orthogonal transformations, and has zero one-point function. Since the adjoint label operation is conjugation followed by time reflection, $J\Theta = \Theta$.

It is also a Lorentz scalar. Differentiating its Euclidean invariance shows that every Euclidean rotation generator annihilates it. The corresponding boost generator is $-i$ times that generator, so also annihilates it. The constant curve Θ and its boost orbit solve the same finite-dimensional constant-coefficient differential equation with the same initial value; uniqueness makes the orbit constant. Spatial rotations already fix it, and generation by boosts and spatial rotations proves full Lorentz invariance.

The one-point holomorphic continuation of Θ is zero: on positive imaginary times it is the zero Euclidean one-point function, and the one-variable identity theorem extends that equality to the half-plane for each spatial position. Its tested boundary is therefore zero. The word formula gives $\langle \Omega, \phi(\Theta, f)\Omega \rangle = 0$ for every Schwartz f. Finally (958) supplies the positive measure and Laplace formula for the very same centered trace used in Section 196. No additional positive-measure input is needed there once the earlier reconstruction hypotheses and this stress tensor are available.

195 Excluding finite mass-shell profiles

The centered trace has a short-distance behavior that can distinguish it from finite polynomial expressions in free fields. The relevant distinction is the negative power of the logarithm. We prove the exclusion directly for the mass-shell profiles appearing in the statement. The proof proceeds through finite power–logarithm approximations to Laplace integrals; it does not assume a convergent asymptotic series.

195.1 An algebra of finite asymptotic approximations

Let $\mathcal{P}$ be the real vector space of finite sums of the functions $s^a(\log s)^k$, where $a \in \mathbb{Z}$, $k \in \mathbb{N}$ and $s > 0$. It is an algebra containing the constants. Say that f has an approximation of order $b \in \mathbb{Z}$ if

$$f(s) = P(s) + O(s^b) \quad (s \downarrow 0), \qquad P \in \mathcal{P}.$$

Say that f has power–logarithm approximations if it has an approximation of every integer order. Different orders may use different finite sums. An approximation of order b is also one of any order $a \le b$, since $s^b \le s^a$ for $0 < s < 1$.

Lemma 195.1. *Functions with power–logarithm approximations form a real algebra. Every such function, and every element of $\mathcal{P}$, is $O(s^u)$ for some integer u. Multiplication by s^a shifts the order of an approximation by a.*

Proof. For $0 < s < 1$, $|\log s| \le s^{-1}$. Thus each monomial is bounded by s^{a-k}, and a finite sum is $O(s^u)$ for the smallest of the finitely many exponents $a-k$. An order-zero approximation then gives the growth claim for f, after replacing u by $\min(u, 0)$. The assertions about addition, real scalar multiplication, and shifts are immediate from the definition and closure of $\mathcal{P}$ under these operations.

For multiplication fix a desired order b. Choose ℓ with $g = O(s^\ell)$ and an approximation $f = P + O(s^{b-\ell})$. Choose u with $P = O(s^u)$ and then $g = Q + O(s^{b-u})$. The exact identity

$$fg - PQ = (f - P)g + P(g - Q)$$

shows that both terms are $O(s^b)$. Since $PQ \in \mathcal{P}$, this proves the required approximation. In particular it proves closure under every finite product, including the empty product 1. □

Lemma 195.2 (Integrating an approximation)**.** *Suppose F is differentiable on $(0, \infty)$, $F' = g$, and g has an approximation of integer order $b \ne -1$. Then F has an approximation of order $b + 1$.*

Proof. Every monomial has an antiderivative in $\mathcal{P}$. For $a = -1$ it is $(\log s)^{k+1}/(k+1)$. For $a \ne -1$, integration by parts gives the recursion

$$I_{a,k}(s) = \frac{s^{a+1}(\log s)^k}{a+1} - \frac{k}{a+1} I_{a,k-1}(s), \qquad I_{a,0}(s) = \frac{s^{a+1}}{a+1}.$$

These formulas are verified by differentiation, so no integrability at zero is being assumed. If $g = P + O(s^b)$, subtract a primitive of P from F. It remains to treat a differentiable R with $|R'(s)| \le Cs^b$ on $(0, \delta)$, where $C \ge 0$.

Put $d = b + 1 \ne 0$ and $A = C/d$. The derivatives of $As^d + R(s)$ and $As^d - R(s)$ are nonnegative. The mean value theorem therefore makes both functions increasing. Consequently, for $0 < x < y < \delta$,

$$|R(y) - R(x)| \le A(y^d - x^d). \tag{965}$$

The right-hand side is nonnegative also when $d < 0$. In that case fix $y \in (0, \delta)$; the inequality bounds $|R(x)|$ by a constant plus $|A|x^d$. Since $x^d \ge 1$ for small x, we obtain $R = O(s^d)$. If $d > 0$, the same inequality shows that $R(x)$ is Cauchy as $x \downarrow 0$: for $x, y < \eta$, its oscillation is at most $A\eta^d$. Let its limit be c. Sending $x \downarrow 0$ in (965) gives $|R(y) - c| \le Ay^d$. The constant c belongs to $\mathcal{P}$. Adding back the subtracted primitive proves the result in both cases. □

The exceptional order -1 is excluded only from this estimate for an arbitrary remainder. The logarithmic monomial itself has already been integrated explicitly. In the argument below the exceptional remainder order is handled by weakening it to order -2 before integrating.

195.2 Radial Laplace integrals and their Euler equation

For $m \geq 0$ and $j, k \in \mathbb{N}$, set

$$\omega(r) = \sqrt{m^2 + r^2}, \qquad M_{j,k}(s) = \int_0^\infty r^{j+2} \omega(r)^{k-1} e^{-s\omega(r)}\, dr, \quad s > 0.$$

The value of the integrand at $r = 0$ is immaterial. For $r > 0$, $\omega(r) \geq r > 0$. The integrands for $k = 0$ and $k = 1$ are respectively bounded by $r^{j+1}e^{-sr}$ and $r^{j+2}e^{-sr}$. These are integrable: scaling and repeated integration by parts give

$$\int_0^\infty r^q e^{-sr}\, dr = q! s^{-q-1} \quad (q \in \mathbb{N}).$$

The boundary terms vanish at infinity by exponential decay and at zero when $q > 0$; the initial case is $\int_0^\infty e^{-sr} dr = s^{-1}$. The identity $\omega^2 = m^2 + r^2$ now proves integrability for every k, by induction in steps of two, and gives

$$M_{j,k+2} = m^2 M_{j,k} + M_{j+2,k}. \tag{966}$$

For a fixed $s > 0$, differentiation of the integrand in a neighborhood where the parameter is at least $s/2$ is dominated in absolute value by the integrand for $M_{j,k+1}(s/2)$. Dominated differentiation gives

$$M'_{j,k}(s) = -M_{j,k+1}(s). \tag{967}$$

There is a second relation which improves the approximation order. For $r > 0$, differentiation in the radial variable gives

$$\frac{d}{dr}\big(r^{j+3} e^{-s\omega(r)}\big) = (j+3) r^{j+2} e^{-s\omega(r)} - s \frac{r^{j+4}}{\omega(r)} e^{-s\omega(r)}.$$

The two terms are integrable by the bounds just proved. The primitive vanishes at zero and infinity, including when $m = 0$. Integration therefore yields the Euler identity

$$s M_{j+2,0}(s) = (j+3) M_{j,1}(s). \tag{968}$$

Combining (966)–(968) gives

$$\frac{d}{ds}\big(s^{j+3} M_{j,1}(s)\big) = -m^2 s^{j+3} M_{j,0}(s). \tag{969}$$

Indeed the derivative on the left is $(j+3)s^{j+2}M_{j,1} - s^{j+3}(m^2 M_{j,0} + M_{j+2,0})$, and the Euler identity cancels the first and last terms.

Proposition 195.3. *Every $M_{j,k}$ has power–logarithm approximations.*

Proof. The initial bound is $0 \leq M_{j,0}(s) \leq (j+1)! s^{-j-2}$, so the zero polynomial gives an approximation of order $b_0 = -j - 2$. Suppose $M_{j,0}$ has an approximation of order $b \geq b_0$. The right-hand side of (969) then has an approximation of order $j + 3 + b \geq 1$. Lemma 195.2, followed by division by s^{j+3}, gives an approximation of order $b + 1$ to $M_{j,1}$. Since $M'_{j,0} = -M_{j,1}$, another application of the same lemma gives an approximation to $M_{j,0}$ of order $b + 2$, hence of order $b + 1$, provided $b + 1 \neq -1$. In the remaining case $b = -2$, weaken the approximation to $M_{j,1}$ from order -1 to order -2. Integration then gives precisely the required order -1 for $M_{j,0}$.

Induction from b_0 proves all larger orders, and weakening covers every smaller integer order. The first half of the argument now gives every order for $M_{j,1}$ as well: choose $b \geq b_0$ as large as needed. Finally (966) and closure under addition and scalar multiplication prove the assertion for all k, by induction in steps of two, simultaneously for every j. □

195.3 From modes to the logarithmic obstruction

A mode consists of a mass $m \geq 0$, integers $j, k \geq 0$, and a continuous real function P on $\mathbb{R}^3$ homogeneous of degree j under positive scalings. Its Euclidean profile is

$$K_{m,P,k}(s) = \int_{\mathbb{R}^3} \frac{\omega_m(p)^k P(p)}{2\omega_m(p)} e^{-s\omega_m(p)}\, dp, \qquad \omega_m(p) = \sqrt{m^2 + |p|^2}. \tag{970}$$

Let $d\sigma$ denote ordinary surface area measure on the unit sphere. The maximum of $|P|$ on that sphere is finite. In polar coordinates, $|P(r\theta)| \leq Cr^j$, so absolute integrability follows from that of $M_{j,k}$. Fubini's theorem and homogeneity therefore give the exact normalization

$$K_{m,P,k}(s) = \frac{1}{2}\left(\int_{S^2} P(\theta)\, d\sigma(\theta)\right) M_{j,k}(s). \tag{971}$$

A track is a finite product of mode profiles; the empty track is 1. By Proposition 195.3 and Lemma 195.1, every track has power–logarithm approximations. The same is true of the real part of a finite complex linear combination of tracks, because all the individual tracks are real.

Lemma 195.4 (No inverse logarithmic leading term)**.** *If a real function f has an approximation of order zero and*

$$\lim_{s\downarrow 0} s^8 \big(\log s^{-1}\big)^2 f(s) = A \in \mathbb{R},$$

then $A = 0$.

Proof. Write $f = P + O(1)$ with $P \in \mathcal{P}$. Since $s^8(\log s^{-1})^2 \to 0$, the same weighted limit holds for P. Restrict to $s_n = 2^{-n}$. A monomial $c_{a,k} s^a (\log s)^k$ contributes

$$c_{a,k}\big(\log(1/2)\big)^{k+2}\big(2^{-(8+a)}\big)^n n^{k+2}.$$

Combine equal pairs (a, k) and discard zero coefficients. The remaining finite sum has distinct pairs $(\lambda, q) = (2^{-(8+a)}, k+2)$, with $\lambda > 0$ and $q \geq 2$. Order them lexicographically and choose the largest pair (λ_*, q_*). If the sum is empty, its limit is zero. If $\lambda_* < 1$, every term tends to zero: writing $\lambda = e^{-c}$ with $c > 0$ reduces this to $n^q e^{-cn} \to 0$. For completeness, $e^{cn} \geq (cn)^{q+1}/(q+1)!$ proves that limit.

If $\lambda_* \geq 1$, then $D_n = \lambda_*^n n^{q_*} \to \infty$. Each smaller term divided by D_n tends to zero. With a smaller base, exponential decay dominates any remaining power of n by the same estimate; with equal base, the ratio is a negative power of n. Thus the finite sum divided by D_n tends to its nonzero leading coefficient. But the sum has the finite limit A, and division by $D_n \to \infty$ makes its limit zero, a contradiction. Only the first case is possible, and it has $A = 0$. □

Theorem 195.5. *Let $T : (0, \infty) \to \mathbb{C}$ satisfy*

$$\lim_{s\downarrow 0} s^8 \big(\log s^{-1}\big)^2 T(s) = A, \qquad A > 0.$$

Then T cannot agree on any interval $(0, \delta)$ with a finite complex linear combination of tracks built from (970)*.*

Proof. If such agreement held, the real part of that combination would have an order-zero approximation. Taking real parts in the displayed limit and using the agreement would contradict Lemma 195.4. □

This proves the exclusion once the positive trace asymptotic has been established for the constructed theory. It imposes no restriction on an infinite sum or a continuous distribution of masses, and no such exclusion is asserted here. The construction of T and its positive coefficient A is a separate quantitative step.

196 The trace coefficient and the passage to all radii

Two different normalizations enter the trace asymptotic. Translation Ward identities fix the amplitude of the stress tensor, while the running Wilson coefficient fixes the logarithmic scale. After those normalizations, the numerical coefficient comes from a free curvature contraction. Finally one must remove the rounded positions and the chosen sequence of scales. We separate these operations because none can be replaced by choosing a convenient normalization of the trace after the fact.

196.1 The free scalar contraction

Put $N = \dim_{\mathbb{R}} \mathfrak{g}$ and use an orthonormal color basis. The reference potential covariance in four Euclidean dimensions is $\delta^{AB}\delta_{ab}D(x)$, where

$$D(x) = \frac{1}{4\pi^2|x|^2}, \qquad x \neq 0.$$

Indices a, b, i, j range over the four coordinate directions and A, B over the N colors. The curvature covariance obtained by differentiating this potential is $\delta^{AB}C_{abij}(x)$. Explicitly,

$$\begin{aligned} C_{abij}(x) = {} & -\delta_{bj}\partial_a\partial_i D(x) + \delta_{bi}\partial_a\partial_j D(x) \\ & + \delta_{aj}\partial_b\partial_i D(x) - \delta_{ai}\partial_b\partial_j D(x). \end{aligned}$$

Write $r = |x|$ and $H_{ai} = \delta_{ai} - 2x_a x_i/r^2$. Differentiating D gives

$$\partial_a\partial_i D(x) = \frac{1}{4\pi^2}(-2\delta_{ai}r^{-4} + 8x_a x_i r^{-6}), \qquad C_{abij}(x) = \frac{H_{ai}H_{bj} - H_{aj}H_{bi}}{\pi^2 r^4}.$$

For the second identity expand the products of H. Their quartic terms cancel, leaving exactly the four second-derivative terms above. Moreover $H = I - 2uu^{\mathsf{T}}$ for the unit vector $u = x/r$, hence $HH^{\mathsf{T}} = I$. It follows that

$$\begin{aligned} \sum_{i,j}(H_{ai}H_{bj} - H_{aj}H_{bi})^2 &= 2\Big(\sum_i H_{ai}^2\Big)\Big(\sum_j H_{bj}^2\Big) - 2\Big(\sum_i H_{ai}H_{bi}\Big)^2 \\ &= 2 - 2\delta_{ab}. \end{aligned}$$

Summing over the sixteen ordered pairs (a, b) gives 24.

The reference scalar is

$$O(x) = \frac{1}{4}\sum_{A,a,b} :F_{ab}^A(x)F_{ab}^A(x):,$$

where the sum is over ordered coordinate pairs. Here the colons mean the quadratic Wick field: in its two-point function only contractions between the two different positions are included. There are exactly two such pairings. Color contraction gives $\sum_{A,B}(\delta^{AB})^2 = N$, and therefore

$$\langle O(x)O(0)\rangle_0 = \frac{2N}{16}\sum_{a,b,i,j} C_{abij}(x)^2 = \frac{3N}{\pi^4|x|^8}. \tag{972}$$

This computation uses the stated reference covariance and Wick rule; the identification of the constructed reference hierarchy with those data is an earlier construction requirement. The factor $1/4$ and the use of ordered pairs are essential to the coefficient.

196.2 Fixing the physical stress amplitude

Here is the finite-dimensional step that prevents an arbitrary scalar multiple of the stress tensor from entering the answer. All vector spaces in this paragraph are real, including the real spaces of self-adjoint tensor fields. Let $\mathcal{K}$ be the space of centered symmetric tensors with the required rotation symmetry, let $\mathcal{V}$ be the space in which their divergences lie, and let $D : \mathcal{K} \to \mathcal{V}$ be the divergence map. The reference theory has corresponding spaces $\mathcal{K}_0, \mathcal{V}_0$ and divergence D_0. The following precise finite-dimensional data are supplied by the stress comparison estimates: D_0 is onto, $\dim \ker D_0 = 1$, $\dim \mathcal{K} = \dim \mathcal{K}_0$, $\dim \mathcal{V} = \dim \mathcal{V}_0$, and there is a surjective map $Q : \mathcal{K} \to \mathcal{V}_0$ with $\ker Q = \ker D$. Rank–nullity then gives

$$\dim \ker D = \dim \mathcal{K} - \dim \mathcal{V}_0 = \dim \mathcal{K}_0 - \dim \mathcal{V}_0 = 1. \tag{973}$$

In the construction the surjectivity of Q is obtained from a surjective finite-dimensional comparison composed with a test map into $\mathcal{K}$; a surjective composition is sufficient to make Q itself surjective. The asserted comparison and kernel equality remain quantitative inputs here.

Let $R \in \ker D$ be the physical centered stress tensor. At scales $r_i > 0$ let U_i be the calibrated isomorphism from reference grade-four fields to interacting grade-four fields. On tensors set

$$Y_i(T)_{ab} = r_i^4 U_i^{-1}(T_{ab}).$$

Suppose the comparison construction supplies eventually conserved tensors $T_i \in \mathcal{K}$ and positive numbers B_i with

$$Y_i(T_i) \longrightarrow R_0, \qquad B_i \operatorname{tr} Y_i(T_i) \longrightarrow 4d_2 O. \tag{974}$$

The normalization test is a shell flux functional ℓ_i on tensors. For example, fixing a separated spectator configuration x, a smooth cutoff χ equal to one near that configuration, a direction b, and one distinguished field v, its form is

$$\ell_i(T) = -r_i \sum_a \int (\partial_a \chi_i)(z) S_{T_{ab}, v_i, \boldsymbol{q}_i}(z, r_i x)\, dz, \qquad \chi_i(z) = \chi(z/r_i).$$

The derivatives of χ_i are supported away from the spectators, so the integrals involve separated correlations. The translation Ward identity for the physical tensor identifies this with a derivative insertion. The needed convergence of that insertion and of the comparison shell integral has the following exact consequence:

$$\ell_i(R) \longrightarrow y \neq 0, \qquad \ell_i(T_i) \longrightarrow y. \tag{975}$$

The same nonzero y occurs because R_0 has the reference translation Ward identity. The existence of a detecting reference derivative field, and the jet estimates justifying these two limits, are part of the stress comparison input, rather than consequences of rank–nullity.

Lemma 196.1. *Equations* (973)–(975) *imply*

$$Y_i(R) \longrightarrow R_0, \qquad B_i \operatorname{tr} Y_i(R) \longrightarrow 4d_2 O.$$

Proof. The first limit in (975) implies $R \neq 0$. Since the conserved space is one-dimensional, eventually $T_i = c_i R$ for a real scalar c_i. Linearity of ℓ_i gives $c_i \ell_i(R) = \ell_i(T_i)$. For large i the denominator $\ell_i(R)$ is nonzero, and the two limits give $c_i \to y/y = 1$. In particular c_i is eventually nonzero and $c_i^{-1} \to 1$. Multiply both limits in (974) by c_i^{-1} and use linearity of Y_i and of the trace. This yields exactly the asserted limits. □

Centering the trace in the chosen phase can differ from tracing the mixed-state-centered tensor R by a fixed multiple $c\mathbf{1}$ of the unit. If $U_i\mathbf{1}_0=\mathbf{1}$ eventually and $B_i r_i^4\to 0$, then

$$U_i^{-1}(B_i r_i^4 c\mathbf{1}) = B_i r_i^4 c\mathbf{1}_0 \longrightarrow 0.$$

Consequently the selected centered trace Θ has the same limit

$$a_i := U_i^{-1}(B_i r_i^4\Theta) \longrightarrow 4d_2 O. \tag{976}$$

The identities above explain how the physical Ward normalization, rather than a later rescaling of Θ, determines this coefficient.

196.3 Rounded correlations and moving labels

We next state exactly the remaining approximation data used to pass from (976) to the trace asymptotic. Write $S_\chi(v,w;x_0,x_1)$ for the selected two-point function and $S_0(v,w;x_0,x_1)$ for the reference one. Both are bilinear in the field labels. The selected centered trace is a scalar under Euclidean rotations, and translations preserve its correlations. Thus its two-point function has the form

$$S_\chi(\Theta,\Theta;x_0,x_1) = T(|x_0-x_1|) \quad (x_0\neq x_1). \tag{977}$$

To justify this reduction, an orthogonal map sends the nonzero difference x_0-x_1 to $-|x_0-x_1|e_0$; equal-length vectors are related by a reflection, with the identity covering equality. A translation then sends the two points to $-te_0/2, te_0/2$, where $t=|x_0-x_1|$. Covariance and the scalar transformation law of Θ give (977).

Input 196.2 (Trace comparison along scales)**.** *For every sequence $s_n>0$ tending to zero there is a subsequence $r_i=s_{n_i}$, with n_i strictly increasing, and comparison data with the following properties. The maps U_i and numbers $B_i>0$ satisfy* (976) *and*

$$\frac{\log(r_i^{-1})}{B_i}\longrightarrow\frac{1}{2b_0},\qquad b_0=\frac{11C_{\mathfrak g}}{48\pi^2}>0,\qquad d_2=-\frac{b_0}{2},$$

where $C_{\mathfrak g}>0$ is the adjoint Casimir in the fixed normalization. There are rounded configurations $q_i(x)\to x$ and errors $\varepsilon_i\to 0$. Let E_0 be the reference grade-four field space and fix its finite basis (e_α). At every coordinatewise separated configuration x in the interior of the comparison window, eventually

$$\begin{aligned}&\big|S_\chi(U_ie_\alpha,U_ie_\beta;r_iq_i(x))-S_0(e_\alpha,e_\beta;q_i(x))\big|\\&\qquad\le\varepsilon_i\big(1+M_{\alpha\beta}(q_i(x))\big),\end{aligned}\tag{978}$$

where $M_{\alpha\beta}$ is continuous near x. The reference correlations are continuous there. The comparison window includes a neighborhood of $x(t)=(0,(t/2,t/2,t/2,t/2))$ for each $1/2<t<2$.

This input is the ultraviolet and stress comparison problem. The positive trace spectral measure and its Laplace formula are supplied by Section 194 for the same reconstructed centered trace. We have not inferred the comparison estimates from compactness.

Lemma 196.3. *Under Input 196.2, for each eligible fixed x,*

$$r_i^8\log^2(r_i^{-1})T(r_i|q_i(x)_0-q_i(x)_1|)\longrightarrow\frac{A}{|x_0-x_1|^8},\qquad A=\frac{3N}{\pi^4}.$$

Proof. The right side of (978) tends to zero, since $M_{\alpha\beta}(q_i(x))$ converges and is therefore bounded. Continuity of S_0 gives convergence of the left correlation to $S_0(e_\alpha, e_\beta; x)$. This also holds for convergent, varying labels: expand each of them in the fixed finite basis. Every coordinate converges, and bilinearity expresses the correlation as a finite sum of products of two coordinates and one of the convergent basis correlations. In particular, (976) gives

$$S_\chi(U_i a_i, U_i a_i; r_i q_i(x)) \longrightarrow (4d_2)^2 S_0(O, O; x).$$

The definition of a_i, bilinearity and (977) identify the expression on the left with $B_i^2 r_i^8 T(r_i|q_i(x)_0 - q_i(x)_1|)$. Multiply by $(\log(r_i^{-1})/B_i)^2$ and use (972). The limiting coefficient is

$$\frac{(4d_2)^2}{(2b_0)^2}\frac{3N}{\pi^4} = \frac{12 d_2^2 N}{b_0^2 \pi^4} = \frac{3N}{\pi^4}.$$

All divisions are legitimate since $b_0 > 0$ and $B_i > 0$. □

196.4 Removing rounding and subsequences

The rounded separation need not equal the intended radius. Pointwise convergence at rounded arguments alone does not identify $T(r_i)$. The positive spectral representation resolves this: T is real, nonnegative and decreasing, because e^{-tp_0} has those properties for each $p_0 \geq 0$. Integrability at the smaller positive time permits the comparison under the integral.

Theorem 196.4. *Input 196.2, with the reference normalization above, implies*

$$\lim_{r \downarrow 0} r^8 \log^2(r^{-1}) T(r) = \frac{3N}{\pi^4} > 0.$$

Proof. First fix any one comparison subsequence r_i. Let $c_i = r_i^8 \log^2(r_i^{-1}) \geq 0$. For $1/2 < t < 2$ take the configuration $x(t)$ in the input. Its Euclidean separation is exactly t, all four coordinate separations are nonzero, and $w_i(t) := |q_i(x(t))_0 - q_i(x(t))_1| \to t$. Lemma 196.3 gives

$$c_i T(r_i w_i(t)) \longrightarrow A/t^8.$$

If $t > 1$, eventually $w_i(t) > 1$, so monotonicity yields $c_i T(r_i) \geq c_i T(r_i w_i(t))$. If $t < 1$, eventually $0 < w_i(t) < 1$ and the reverse inequality holds. Given $\epsilon > 0$, choose $t_+ > 1$ and $t_- < 1$ sufficiently close to one that $|A/t_\pm^8 - A| < \epsilon/2$. For all sufficiently large i, the two comparison values are within $\epsilon/2$ of their limits. They bound $c_i T(r_i)$ between $A - \epsilon$ and $A + \epsilon$. Thus $c_i T(r_i) \to A$.

To obtain the limit at all positive radii, suppose it failed. There would be $\epsilon > 0$ and a sequence $s_n \in (0, 1/n)$, $n \geq 1$, for which $|s_n^8 \log^2(s_n^{-1}) T(s_n) - A| \geq \epsilon$ for every n. Input 196.2 supplies a comparison subsequence of this same sequence. The preceding argument makes its weighted values converge to A, contradicting that inequality. Finally $N > 0$ for the nonabelian simple Lie algebra under consideration, so $A > 0$. □

Combining this theorem with Theorem 195.5 proves the finite-profile exclusion from the same trace comparison input. The rank, Ward, approximation and scale-selection estimates identified above still need to be supplied by the construction; the algebraic and limiting passages do not replace them.

197 Weighted price control for the trace receiver

This section proves the scalar combination of the response estimates that feeds the rank-one receiver lemma 197.1. It isolates the field decay, unweighted price convergence, and weighted price convergence that the finite-volume producer must furnish. It does not assume the receiver itself or suppress those three estimates into a single unnamed bound.

Lemma 197.1 (Weighted price from a rank-one approximate kernel)**.** *Let E and H be finite-dimensional real normed spaces. Let $f : E \to H$ and $U : H \to E$ be linear, with $fU = \mathrm{id}_H$ and $\ker f = \mathbb{R}x$ for some $x \neq 0$. Suppose linear maps $v_i : E \to H$ and $u_i : H \to E$ satisfy*

$$v_i \longrightarrow f, \qquad u_i \longrightarrow U, \qquad v_i u_i = \mathrm{id}_H \quad \textit{eventually.}$$

Let $w_i \to x$, let $e_i \in \mathbb{R}$, and let $\rho : E \to \mathbb{R}$ be a fixed linear functional. If

$$e_i v_i(w_i) \longrightarrow 0 \quad \textit{in } H, \qquad e_i \rho(w_i) \longrightarrow 0,$$

then there are linear functionals $\mu_i : H \to \mathbb{R}$ with uniformly bounded operator norms and

$$\big\| e_i(\mu_i v_i - \rho) \big\|_{E^*} \longrightarrow 0.$$

Proof. Choose a linear functional $\lambda : E \to \mathbb{R}$ with $\lambda(x) = 1$, by extending x to a basis. Define

$$A(y) = \lambda\big(y - Uf(y)\big).$$

Since $f(y - Uf(y)) = 0$, the kernel assumption gives $y - Uf(y) = ax$ for some scalar a. Applying λ shows $a = A(y)$, and therefore

$$y = Uf(y) + A(y)x. \tag{*}$$

Set $V_i = u_i f + A \otimes w_i$, where $(A \otimes w_i)(y) = A(y)w_i$. The convergence of u_i and w_i implies $V_i - \mathrm{id}_E = (u_i - U)f + A \otimes (w_i - x)$, by (*). The right side tends to zero in operator norm. Once $\|V_i - \mathrm{id}_E\| < 1$, the Neumann series gives an inverse and

$$\|V_i^{-1} - \mathrm{id}_E\| \leq \frac{\|V_i - \mathrm{id}_E\|}{1 - \|V_i - \mathrm{id}_E\|},$$

so $V_i^{-1} \to \mathrm{id}_E$.

Put $\mu_i = \rho u_i$. These functionals have bounded norms because $u_i \to U$. For $y \in E$, the exact identity $v_i u_i = \mathrm{id}_H$ gives

$$\begin{aligned} e_i(\mu_i v_i - \rho)(V_i y) &= e_i \rho\big(u_i v_i(u_i f(y) + A(y)w_i) - u_i f(y) - A(y)w_i\big) \\ &= A(y)\, e_i\big(\rho(u_i v_i(w_i)) - \rho(w_i)\big). \end{aligned}$$

The first term inside the final difference vanishes after weighting, because

$$e_i \rho(u_i v_i(w_i)) = \rho(u_i(e_i v_i(w_i)))$$

and ρu_i is uniformly bounded. The second term vanishes by hypothesis. Thus the left side tends to zero for every fixed y. In finite dimension, convergence on a basis implies operator-norm convergence, so $e_i(\mu_i v_i - \rho)V_i \to 0$. Multiplication by $V_i^{-1} \to \mathrm{id}_E$ gives $e_i(\mu_i v_i - \rho) \to 0$ in operator norm. □

Let E be a finite-dimensional real normed space, H a finite-dimensional real normed space, $s_i : E \to H$ linear maps, and $\rho : E \to \mathbb{R}$ a fixed linear functional. Let $x \in E$ satisfy $\rho(x) = 0$. For each i, let $b_i \geq 1$ with $b_i \to \infty$, let $q_i, r_i \in E$, and let $e_i \geq 0$. Define

$$m_i = b_i^{-1} q_i - r_i.$$

Assume

$$b_i^{-1} q_i \longrightarrow 0, \qquad \sup_i |\rho(q_i)| < \infty, \qquad r_i \longrightarrow -x, \tag{979}$$

$$e_i / b_i \longrightarrow 0, \qquad e_i \rho(r_i + x) \longrightarrow 0, \qquad b_i s_i(m_i) \longrightarrow 0 \quad \text{in } H. \tag{980}$$

Lemma 197.2 (The medium has both weighted null limits)**.** *Under* (979)– (980)*,*

$$m_i \longrightarrow x, \qquad e_i \rho(m_i) \longrightarrow 0, \qquad e_i s_i(m_i) \longrightarrow 0.$$

Proof. Since $b_i^{-1} q_i \to 0$ and $r_i \to -x$, $m_i = b_i^{-1} q_i - r_i \to x$. Since $\rho(x) = 0$,

$$e_i \rho(m_i) = (e_i / b_i) \rho(q_i) - e_i \rho(r_i + x).$$

The first term tends to zero by boundedness of $\rho(q_i)$ and $e_i / b_i \to 0$, and the second by hypothesis. Also $e_i / b_i \leq 1$ eventually, so

$$\| e_i s_i(m_i) \| = (e_i / b_i) \| b_i s_i(m_i) \| \leq \| b_i s_i(m_i) \| \longrightarrow 0.$$

□

All sequences in the source application below are already reindexed by a selected level $k_i \to \infty$: b_i is the arch scale at level k_i, q_i and r_i are the chosen and reference responses there, and s_i is the chart at index i. This reindexing matters: the source does not assert the field-decay estimate at every unreindexed level. Write $p_i = \widehat{\beta}_{k_i}$ and $A_a(i) = \exp(a(\log\log p_i)^2)$. The source-specific weight is $e_i = A_{1/5}(i)$ at this selected level. The next subsection proves $e_i / b_i \to 0$ and $e_i \leq p_i$. The field input is supplied with weight p_i; since $b_i \leq 2 p_i$, this gives $b_i s_i(m_i) \to 0$ as required above.

In the selected construction, E is the real signed-equivariant space of grade-five coefficient arrays, x is its nonzero equation-of-motion direction, and ρ is the free scalar price. The vectors q_i and r_i are respectively the transported chosen response and the transported reference response. Their combination $m_i = b_i^{-1} q_i - r_i$ is the medium used to build a response functional. The scale b_i is the positive arch scale along the selected chart. Boundedness of $\rho(q_i)$ follows from the unweighted Ward-price limit; the weighted estimate on $\rho(r_i + x)$ is the scale-resolved reference-price comparison; and the last estimate is the field decay of the medium.

Combining Lemma 197.2 with Lemma 197.1, under the chart hypotheses $s_i \to f$, $u_i \to U$, $fU = \mathrm{id}$, $\ker f = \mathbb{R}x$ with $x \neq 0$, and $s_i u_i = \mathrm{id}$ eventually, gives bounded linear functionals $\mu_i : H \to \mathbb{R}$ and

$$\sup_i \|\mu_i\| < \infty, \qquad \big\| e_i(\mu_i s_i - \rho) \big\|_{E^*} \longrightarrow 0.$$

This is the response-price estimate used by the two-direction trace correction. The displayed deduction is complete; its chart and response estimates are the separate producer obligations listed in the audit.

197.1 The source rate and medium calculation

For the already selected levels $k_i \to \infty$, write $p_i = \widehat{\beta}_{k_i}$ for the running positive rate and b_i for the arch scale at k_i. The scale comparison used by the construction is

$$\frac{1}{2}p_i \le b_i \le 2p_i, \qquad p_i \longrightarrow \infty. \tag{981}$$

The age weights are $A_a(i) = \exp(a(\log\log p_i)^2)$. They satisfy

$$A_a(i)A_c(i) = A_{a+c}(i), \qquad A_a(i) \le p_i^{\delta} \quad \text{eventually for every fixed } a \in \mathbb{R} \text{ and every } \delta > 0. \tag{982}$$

For the second relation, take logarithms. Since $(\log\log p)^2 = o(\log p)$, eventually $|a|(\log\log p_i)^2 \le \delta \log p_i$. Exponentiation gives the bound. Also $A_a(i) \to 0$ directly for $a < 0$, since $(\log\log p_i)^2 \to \infty$.

In particular, for $a \in \{1/6, 1/5\}$,

$$0 \le \frac{A_a(i)}{b_i} \le 2p_i^{a-1} \longrightarrow 0. \tag{983}$$

The same argument gives $A_a(i) \le p_i$ eventually, so an estimate with weight p_i implies the corresponding estimate with weight $A_{1/5}(i)$.

Here is the scalar deduction from the response and reference receipts. Let q_i be the transported chosen response and w_i the transported reference response. The local response estimate and the separated flux limit have the form

$$\|q_i\| \le CA_{1/6}(i), \qquad \rho(q_i) \longrightarrow -\kappa, \tag{984}$$

while the reference chart and the reference-price comparison give

$$w_i \longrightarrow -x, \qquad \rho(x) = 0, \qquad A_{1/5}(i)\rho(w_i) \longrightarrow 0. \tag{985}$$

Define $m_i = b_i^{-1}q_i - w_i$. Then

$$\|b_i^{-1}q_i\| \le 2Cp_i^{-5/6} \longrightarrow 0,$$

so $m_i \to x$. Since $\rho(q_i)$ converges it is bounded, and

$$A_{1/5}(i)\rho(m_i) = \frac{A_{1/5}(i)}{b_i}\rho(q_i) - A_{1/5}(i)\rho(w_i) \longrightarrow 0$$

by (983) and (985). Finally, if the local field estimate is $p_i\, s_i(m_i) \to 0$, then $A_{1/5}(i) \le p_i$ gives $A_{1/5}(i)s_i(m_i) \to 0$. These are exactly the three limits consumed by the rank-one approximate-kernel lemma; the remaining price-receiver construction is its finite-dimensional argument, not another analytic estimate.

The source establishes the arch comparison from positivity of the arch scale and the two-sided scale control. It obtains the local response estimate from a priced local charge inequality, transfers it through the finite-volume separation limit, and gets the reference-price estimate from the calibrated reference chart. Those three analytic inputs are not proved by the scalar calculation above. Their declarations and proof boundaries are listed in the adjacent evidence record.

197.2 How the two weighted receipts are assembled

The weighted price receipt for the medium uses only the response-price limit, the reference-price cancellation, and the age-rate comparison just proved. It is therefore useful to distinguish it from the stronger source estimate that supplies the weighted field image. If $\rho(q_i) \to -\kappa$, then $|\rho(q_i)| \le M$ eventually for some M; thus

$$\left| \frac{A_{1/5}(i)}{b_i} \rho(q_i) \right| \le 2M p_i^{-4/5} \longrightarrow 0.$$

No weighted convergence rate for the response flux itself is needed here. The separate reference-chart estimate is the cancellation

$$A_{1/5}(i)\rho(w_i) \to 0.$$

This identifies the precise roles of the two source limits and avoids conflating a field-size estimate with a price estimate.

For the field image, the construction proves the stronger statement $p_i s_i(m_i) \to 0$ in the finite-dimensional chart norm. The transfer to the weaker age weight is the monotonicity estimate $A_{1/5}(i) \le p_i$; there is no loss from summing over a basis because the chart is a fixed finite-dimensional operator. The bound $\|q_i\| = O(A_{1/6}(i))$ alone proves only $m_i \to x$ and does not imply this field estimate. The latter is supplied by the local pricing argument and its finite-volume passage.

197.3 Source route and exact remaining estimates

The response-size and response-flux limits above follow from a concrete separation argument. Let $\mathcal{T}$ be the linear map from the transported finite-volume coefficient field to the grade-five response, and let λ be the flux functional on the transported fields. On the limiting field, their compatibility is $\rho(\mathcal{T} v_i) = \lambda(v_i)$. At fixed level i, write $v_i(p)$ for the finite-volume transported field indexed by the configuration parameter p. The local charged-configuration estimate gives, on a set in the separation ultrafilter,

$$\|\mathcal{T} v_i(p)\| \le C_T A_{1/6}(i), \tag{986}$$

$$|\lambda(v_i(p)) + \kappa| \le C_1 A_{-1/4}(i) + C_2 A_{-1}(i) + \frac{p_i^{-30} + p_i^{-100}}{12}. \tag{987}$$

Here C_T, C_1, C_2 are fixed constants from the finite-dimensional coefficient, response, and flux maps. The first line is the field-cost part of the estimate; the second is the flux-price part. The transported field converges along that ultrafilter to the infinite volume field v_i. Since $\mathcal{T}$ and λ are continuous linear maps, the inequalities pass to their limits. Therefore $q_i = \mathcal{T} v_i$ satisfies the response norm bound and $\rho(q_i) = \lambda(v_i) \to -\kappa$. Every term on the right tends to zero by (982), proving (984).

In the source, the finite-volume inequalities in (987) are obtained by applying the local charged-configuration bound to each typical configuration and using the transported-field error estimate. The latter is bounded by $A_{-1/4}$; the local flux estimate contributes $4A_{-1}$ and the two inverse-power costs. Thus the ultrafilter passage is an ordinary continuity argument; all quantitative work lies in the local charge bound and its field-transport error estimate. Their exact source declarations are recorded in the evidence table.

For the weighted reference-price estimate, let $c_i(p)$ be the calibrated finite-volume probe and $v_i(p)$ the transported field. The calibration gives $\rho(\mathcal{T} c_i(p)) = \rho(-x) = 0$, while the transport estimate gives $\|v_i(p) - c_i(p)\| \le C A_{-1/4}(i)$ on an ultrafilter-large set. Continuity therefore implies

$$|\rho(\mathcal{T} v_i(p))| \le \|\rho\| \, \|\mathcal{T}\| C A_{-1/4}(i).$$

Passing to the ultrafilter limit gives the same bound for $|\rho(w_i)|$. Multiplication by $A_{1/5}$ gives a quantity tending to zero because $A_{1/5}A_{-1/4} = A_{-1/20} \to 0$. This uses only a rate for the difference between the transported field and the calibrated probe; it does not assert that the probe itself converges at that rate.

The remaining chart-field input required by the general lemma is the exact limit

$$p_i\, s_i(m_i) \longrightarrow 0. \tag{988}$$

It is independent of the response-size and flux estimates above. To produce it, the source constructs finite-volume labels whose one-point expectations converge to the target core field, obtains local price bounds for each fixed grade-four test, and uses derivative and orientation compatibility to pass the bounds to a finite basis of the chart target. Finite dimensionality then converts those basis estimates to the displayed chart-norm limit. The finite-volume label existence, local price bounds, chart scaling identity, and compatibility identities are analytic inputs to this step; no claim that they follow from the scalar rate calculation is made here. Their exact declarations and the source's application route are recorded only in the evidence file.

198 A finite-volume separation estimate

We prove the separation estimate that converts a localized probe into a small two-site correlation. The proof is a three-term bilinear expansion; all source-specific quantitative inputs are written as precise hypotheses and audited separately.

Fix a finite periodic gauge configuration space Ω with normalized Wilson expectation $\mathbb{E}_\beta$, and a reflection-positive cell tiling at selected level i with $n \geq 1$ cells. Its admissible observables are bounded measurable gauge-invariant functions in the chamber-local domain. For each cell a, its copy $C_a F$ is the translated pullback of F under that cell's reflection; if the reflection has odd sign parity, the copy is complex-conjugated. Define the root bound by

$$\mathcal{R}_i(F) = \left| \mathbb{E}_\beta \prod_{a=1}^{n} C_a F \right|^{1/n}.$$

The local positivity estimate supplies the following inequalities for observables in the chamber-local domain:

$$|\mathbb{E}_\beta[F(\tau_{a(x)}\omega)G(\tau_{a(y)}\omega)]| \leq \mathcal{R}_i(F)\mathcal{R}_i(G) \quad (x \neq y), \tag{989}$$

$$\mathcal{R}_i(F+G) \leq \mathcal{R}_i(F) + \mathcal{R}_i(G). \tag{990}$$

Here x, y are coarse sites in the tiling, $a(x), a(y)$ are their fine anchor sites, and $\omega \in \Omega$ is a gauge configuration. The anchor map is injective on the two sites under consideration. The first estimate is the reflected-cell product bound; the second is the root-bound triangle inequality. We use them only when the observables and all indicated sums and differences remain in the bounded measurable, gauge-invariant, chamber-local domain. The local error scale at selected level i is

$$t_i(a) = \exp\Big(\frac{a}{10^4}(\log p_i)^{48}\Big), \qquad p_i \longrightarrow \infty.$$

It satisfies $t_i(a)t_i(c) = t_i(a+c)$ and is increasing in a once $p_i > 1$. The source growth estimate gives $t_i(1) \geq 4$ for all sufficiently large i.

Proposition 198.1 (Finite-volume two-site error)**.** *Let P, Q, U, V be observables in the chamber-local domain at selected level i, and let $x \neq y$ be coarse sites whose fine anchor sites are distinct. Suppose*

$$\mathcal{R}_i(P) \le t_i(5), \qquad \mathcal{R}_i(Q) \le t_i(5), \tag{991}$$

$$\mathcal{R}_i(Q - V) \le t_i(-300), \qquad \mathcal{R}_i(P - U) \le t_i(-200), \tag{992}$$

$$|\mathbb{E}_\beta[U(\tau_{a(x)}\omega)V(\tau_{a(y)}\omega)]| \le t_i(-200). \tag{993}$$

Then

$$|\mathbb{E}_\beta[P(\tau_{a(x)}\omega)Q(\tau_{a(y)}\omega)]| \le t_i(-100)$$

for all sufficiently large i.

Proof. Write $\langle F, G\rangle_{x,y} = \mathbb{E}_\beta[F(\tau_{a(x)}\omega)G(\tau_{a(y)}\omega)]$. By (990) and the assumptions,

$$\mathcal{R}_i(V) \le \mathcal{R}_i(Q) + \mathcal{R}_i(V - Q) \le t_i(5) + t_i(-300).$$

The definition gives $\mathcal{R}_i(-F) = \mathcal{R}_i(F)$, since negating F changes the product of its n cell copies only by the overall sign $(-1)^n$. Hence the assumed bound for $Q - V$ also bounds $V - Q$. Bilinearity gives the exact identity

$$\langle P, Q\rangle_{x,y} = \langle U, V\rangle_{x,y} + \langle P, Q - V\rangle_{x,y} + \langle P - U, V\rangle_{x,y}. \tag{994}$$

Indeed, the right hand side equals $\langle U, V\rangle + \langle P, Q\rangle - \langle P, V\rangle + \langle P, V\rangle - \langle U, V\rangle$. Using (989) on the last two terms and (991)–(993) therefore yields

$$\begin{aligned} |\langle P, Q\rangle_{x,y}| &\le t_i(-200) + t_i(5)t_i(-300) + t_i(-200)\big(t_i(5) + t_i(-300)\big) \\ &= t_i(-200) + t_i(-295) + t_i(-195) + t_i(-500) \\ &\le 4t_i(-195). \end{aligned}$$

The last inequality uses monotonicity in the exponent. Since $t_i(1) \ge 4$,

$$4t_i(-195) \le t_i(1)t_i(-195) = t_i(-194) \le t_i(-100),$$

again by monotonicity. This proves the conclusion. □

198.1 Application to a selected response insertion

For the selected construction, the proposition applies with P equal to the measured grade-five marker, Q the marker after the grade-four insertion, U the localized probe supplied by the marker receipt, and V the bounded local approximation to the inserted marker. The exact source hypotheses have the following mathematical form: both measured markers have root bound at most $t_i(5)$; the local approximation satisfies $\mathcal{R}_i(Q - V) \le t_i(-300)$; the probe-to-marker difference satisfies $\mathcal{R}_i(P - U) \le t_i(-200)$; and the distant-site probe correlation satisfies $|\langle U, V\rangle_{x,y}| \le t_i(-200)$. With these four receipts, the proposition proves the required two-site price bound $t_i(-100)$.

The proof above discharges the finite-volume assembly estimate, including its exponent arithmetic. It does not prove production of the four receipts. In the selected source, their producers are the coefficient-to-root estimate for the two measured markers, the local approximation and window bound for the insertion, the probe witness with its distant-site price estimate, and the finite-volume reflection-positivity seam. The evidence table records these exact source statements and the remaining producer boundaries. No continuum or infinite-volume limit is used in this proposition.

199 The local charged-configuration estimate

We isolate the finite-volume algebra that turns a charged configuration into a bounded field and a priced response. This is the local step used before the separation-limit passage. Its quantitative premises are stated as component bounds, so the proof does not hide the estimate inside a single named receipt.

For $t > e$, put

$$W_a(t) = \exp\bigl(a(\log\log t)^2\bigr), \qquad c(t) = t^{-30}.$$

Then $W_a(t)W_b(t) = W_{a+b}(t)$, $W_{-1}(t) \to 0$, and $c(t) \to 0$ as $t \to \infty$.

Let I be a finite set of lattice directions, Y a finite-dimensional real normed coefficient space, and let $\ell : Y^I \to \mathbb{R}$ be a continuous linear flux functional. Let $s : Y^I \to [0,\infty)$ be the source coefficient size, satisfying $\|A\|_\infty \le s(A)$ for every coefficient array A. Write $\sigma = \theta^{-2}$ with $\theta > 0$ for the local normalization. The nonlinear local coefficient map has already been evaluated in the arrays $B', K', D' \in Y^I$; these are outputs of that local construction, not images under the core linear projection. Let $X, W, E, L \in Y^I$. Assume the exact charged-configuration decomposition

$$X = W + B' - K' + D' + E + L, \tag{995}$$

and the following local coefficient and flux receipts at scale t:

$$s(X) \le \theta^2 W_{1/6}(t), \tag{996}$$

$$\ell(\sigma K') = 0, \qquad \ell(\sigma B') = h/12, \qquad |h - m| \le c(t), \tag{997}$$

$$s(W) \le \theta^2 W_{-1}(t), \qquad s(D') \le \theta^2 W_{-1}(t), \tag{998}$$

$$s(E) \le \theta^2 W_{-1}(t), \qquad s(L) \le \theta^2 W_{-1}(t). \tag{999}$$

Here h is the finite-volume charge extracted from the distinguished component, and m is its target price. These are coefficient-size receipts after the local map has been evaluated. No norm contraction of the nonlinear map is assumed.

Proposition 199.1 (Local field and flux bounds)**.** *Under the displayed assumptions,*

$$\|\sigma X\|_\infty \le W_{1/6}(t), \qquad \left|\ell(\sigma X) - \frac{m}{12}\right| \le 4\|\ell\| W_{-1}(t) + \frac{c(t)}{12}.$$

Proof. The field bound follows from the size domination and (996):

$$\|\sigma X\|_\infty \le \sigma s(X) = \theta^{-2} s(X) \le W_{1/6}(t).$$

For the price, apply ℓ to the decomposition in (995). The K' term vanishes and the B' term is $h/12$. Therefore

$$\ell(\sigma X) - \frac{m}{12} = \ell(\sigma W) + \ell(\sigma D') + \ell(\sigma E) + \ell(\sigma L) + \frac{h-m}{12}.$$

For any array A, continuity gives $|\ell(\sigma A)| \le \|\ell\|\sigma\|A\|_\infty \le \|\ell\|\theta^{-2} s(A)$. The triangle inequality and (997)– (999) give the claimed bound. There are four error terms, so the coefficient is exactly four; no direction-count factor is introduced because s bounds the full array norm. □

In the selected trace response, $m/12$ is the target flux $-\kappa$. Consequently the local bound proves a response-size estimate tending to zero after division by the arch scale and an unweighted flux limit, provided $t \to \infty$ along the selected configurations and the displayed component receipts hold. The remaining passage from typical finite-volume configurations to their separated limit is an ordinary continuity argument, as recorded in the weighted-estimate section. The component receipts themselves are separate finite-volume producer statements; this proposition does not prove them from the charged-configuration hypotheses.

200 Purchasing the nonlinear estimates on one scale sequence

The nonlinear estimates above are uniform in the finite torus once the block size, derivative orders and stage parameters have been fixed. We now choose coupling thresholds that pay them together. In particular, the marked and unmarked constructions can use the same transition. The two logarithmic corrections absorbed into the forced gas require two successive payments; both are included.

200.1 An elementary exponential payment

Let $\beta_f(x) = x + c_L$, with fixed $c_L > 0$, let $B(x) = \lceil (\log x)^2 \rceil$, and let $P_f(x)$ be the shifted fine monitor. For $C > 0$ put

$$\tau_C(x) = \exp\{-CP_f(x)^2/2\}.$$

Fix $D_0 > 0$ and a function f of negative upper exponent. For all sufficiently large x, the following assertion holds simultaneously for every pair $a, b \geq 0$ such that

$$a \leq D_0 e^{-CP_f(x)^2}, \qquad b \leq f(x).$$

There is $\kappa \geq 0$ with

$$\begin{gathered} b \leq 1, \qquad (2B+1)^4 I \leq 1, \qquad e^{-\kappa}(e^I - 1) \leq 2\tau_C(x), \\ I = e^{\kappa} a + 2b. \end{gathered} \tag{1000}$$

Take $\kappa = CP_f(x)^2/2$. Then $e^{\kappa} a \leq D_0 \tau_C(x)$, so

$$(2B+1)^4 I \leq (2B+1)^4 (D_0 \tau_C + 2f).$$

The right side tends to zero. Indeed the geometric factor has upper exponent zero, f has negative upper exponent, and τ_C has every negative upper exponent by the monitor estimate. The same choice of threshold gives $b \leq 1$ and the middle inequality of (1000). Since $(2B+1)^4 \geq 1$, also $0 \leq I \leq 1$. Convexity of the exponential on $[0,1]$ gives $e^I - 1 \leq (e-1)I \leq 2I \leq 2$. Multiplication by $e^{-\kappa} = \tau_C$ proves the last inequality. These are exactly the smallness and output-price conditions for the forced-gas absorption in Section 91. The threshold depends on C, D_0, f, L, but not on a, b.

200.2 The two soft correction prices

We verify the negative power needed in the second payment. Fix a stage, including its derivative order, spatial weight, support exponent, path widths and path prices. Let $t \in (0,1]$ be its threshold and B its radius. Let α be the coupling reset, with $|\alpha| \leq A$, where $A \geq 0$ is fixed after the block length has been chosen. If the regular remainder allowance is $0 \leq H \leq t^2$, the reset's boundary-hit coefficient is

$$h_{\mathrm{hit}} = 2t\left(|\alpha| + H/t^2\right) + \tfrac{1}{2} H C_{\mathrm{pair}}.$$

Here $C_{\mathrm{pair}} \geq 0$ depends only on the fixed derivative orders. Since $H \leq t^2 \leq t$,

$$h_{\mathrm{hit}} \leq [2(A+1) + C_{\mathrm{pair}}/2]t.$$

There are two correction families, with the support reserve used for the first absorption and the larger reserve used for the second. For either family let $R_0, K_0 \geq 0$ be respectively its reset and path prices computed at radius one. The rooted footprint sum gives

$$R(B,H) \leq R_0 (1+B)^4 H, \qquad K(B) \leq K_0 (1+B)^4.$$

To see the radius dependence explicitly, its footprint population is a fixed nonnegative multiple of either $(2B+1)^4$ or $(8B+5)^4$. The inequalities $2B+1 \le 3(1+B)$ and $8B+5 \le 13(1+B)$ compare them with their radius-one values. All remaining factors are fixed: exponential support charges, inverse positive width powers and the convergent geometric series for the unused width reserve. The same calculation works with the larger reserve, using its smaller positive width. Thus the total soft price satisfies

$$R(B,H) + h_{\mathrm{hit}} K(B) \le C_A (1+B)^4 t, \qquad C_A = R_0 + [2(A+1) + C_{\mathrm{pair}}/2] K_0. \tag{1001}$$

These constants are selected separately for the two families.

At coupling x, take

$$t(x) = x^{-1/2} \exp((\log\log x)^2), \qquad H(x) = c_{\mathrm{st}} x^{-11/10}, \qquad 0 < c_{\mathrm{st}} \le 1/64.$$

For all sufficiently large x, we have $t(x) \le 1$. For $x > 1$ we also have $H \le x^{-1} \le t^2$, so the preceding calculation applies eventually. Moreover, for any fixed constant C_A,

$$C_A (1 + B(x))^4 t(x) \le \beta_f(x)^{-1/4}$$

eventually. Divide the left side by $x^{-1/4}$ and take logarithms: the variable terms are $4\log(1 + \lceil (\log x)^2 \rceil) + (\log\log x)^2 - \frac{1}{4}\log x$, which tend to minus infinity. Finally $\beta_f(x)/x \to 1$, so the same assertion holds with the shifted scale. This proves the required soft bound uniformly for $|\alpha| \le A$, both derivative budgets and both finite-volume families.

The first hard price of the nonlinear expansion is at most $\exp(-\mu P_f^2/4)$, with $\mu > 0$. Its accompanying soft price is at most $\beta_f^{-1/8}$. In the multiple-insertion version it is at most $C_{\mathrm{cov}} \beta_f^{-1/8}$, where the positive constant C_{cov} depends only on the fixed enlargement depth. Both have negative upper exponent. Apply (1000) with $C = \mu/4$, $D_0 = 1$ to obtain the first output bound

$$2e^{-\mu P_f^2/8}.$$

Use this as the hard price for the second correction, whose soft price is bounded by (1001). A second application with $C = \mu/8$, $D_0 = 2$ gives

$$2e^{-\mu P_f^2/8} \quad \longmapsto \quad 2e^{-\mu P_f^2/16}. \tag{1002}$$

These are the two successive forced allowances used in the nonlinear transition, including the larger-reserve correction. All inequalities use the same value of x.

200.3 Choosing thresholds without changing the transition

The marked construction first chooses a sufficiently large block size to pay its footprint conversion. For that fixed block and fixed stage orders, Sections 93 and 94 give a threshold beyond which the local marked derivatives, the two connected expansions and their tail estimates hold. The chart transport radius is chosen before this threshold; the microscopic fluctuation threshold tends to zero and eventually fits inside it. The Gaussian and all-species estimates are uniform in the finite torus and in the insertion, so these choices do not depend on either of them.

There are finitely many permitted target stages at each current stage j. For each such pair (j,i), choose the block-size floor given by the marked construction and take their finite maximum. Choose the actual block size L_j above that floor. Only after L_j is fixed do we choose the coupling

thresholds needed for (1000), (1001) and the fixed-order marked estimates. This order permits their constants to depend on the actual block length and reset bound.

More explicitly, let d_j be a previously chosen entry threshold, and let $r_{j,i}$ be the threshold for all the requirements at transition (j, i). Replace it by

$$d'_j = \max\left(d_j, \max_{i \text{ permitted at } j} r_{j,i}\right). \tag{1003}$$

For any later thresholds $d''_j \geq d'_j$ and every $x \geq d''_j$, all the requirements still hold. This is immediate from $x \geq d'_j \geq r_{j,i}$; the other selected data, including the block sizes and the transition maps, are unchanged. Applying (1003) successively to the ordinary and the larger-reserve multiple-insertion requirements gives one set of thresholds satisfying both. No uniform maximum over all orders or all stages is needed.

Consequently the numerical arrival conditions previously required by constructions with one or several insertions and by the centered construction can be imposed simultaneously on the same selected scale sequence. This is a purchase of their actual displayed estimates. The construction of the underlying funded Gaussian data and the subsequent coefficient and normalization identities remain the respective analytic and algebraic steps already separated in those arguments.

201 One scale sequence for all the estimates

The estimates are indexed by finite stages, while the final theory requires arbitrarily high orders. We choose a slowly advancing stage map so that every fixed stage eventually appears, with enough time to satisfy its comparison estimates. The same choice also pays the numerical requests made after the block lengths have been selected.

201.1 A slowly advancing sequence with long constant intervals

Let b_j, c_j be arbitrary nonnegative integers. Define recursively

$$A_0 = 0, \qquad k_j = c_j + 2A_j + 1, \qquad A_{j+1} = A_j + b_{j+1} + 2k_j + 1.$$

The A_j are strictly increasing and tend to infinity. Define $J_0(m) = j$ when $A_j \leq m < A_{j+1}$. These half-open intervals partition the nonnegative integers. Then J_0 is nondecreasing, $J_0(0) = J_0(1) = 0$, and

$$J_0(m+1) \leq J_0(m) + 1, \qquad J_0(m) \longrightarrow \infty.$$

The increment bound holds because an integer step can cross at most one of the distinct integer endpoints. The divergence follows from $m \geq A_j \Rightarrow J_0(m) \geq j$.

If $J_0(m+1) = j > 0$, then $A_j \leq m+1$ and $A_j \geq b_j + 1$, so $b_j \leq m$. Also $k_j \geq c_j$, and the whole interval

$$k_j - \lfloor k_j/2 \rfloor \leq m \leq k_j + \lfloor k_j/2 \rfloor \tag{1004}$$

lies in $[A_j, A_{j+1})$: its lower endpoint is at least A_j because $k_j \geq 2A_j + 1$, and its upper endpoint is at most $2k_j < A_{j+1}$. Thus $J_0 = j$ throughout this interval. This explicit recursion simultaneously postpones entry to every stage and gives that stage a prescribed long comparison interval.

201.2 Physical scales, entry requirements and periods

Let $L_j \ge 3$ be the selected nondecreasing odd block lengths, and let j_* be the first stage. Put

$$J(m) = j_* + J_0(m), \qquad s_0 = 0, \qquad s_{m+1} = s_m + \log L_{J(m+1)}, \qquad a_m = e^{-s_m}.$$

Induction and monotonicity give

$$m \log 3 \le s_m \le m \log L_{J(m)}. \tag{1005}$$

In particular $a_m \downarrow 0$, and e^{s_m} is the integer product of the first m block lengths.

Choose an initial coupling $\beta_* > 1$ large enough to meet the first-stage entry requirement and to ensure $t(x) \le 1$ for $x \ge \beta_*$. This is possible since $t(x) = x^{-1/2}\exp((\log\log x)^2) \to 0$. Set

$$\widehat{\beta}_m = \beta_* + s_G s_m, \qquad s_G > 0.$$

For each fixed stage $j > j_*$ there are only finitely many entry and comparison bounds. Require, at sufficiently large s,

$$\begin{gathered} d_j \le \beta_* + s_G s, \qquad e_j(\beta_*) \le s, \qquad s > 0, \\ \log L_j \le s/j, \qquad V_j(s) \le e^s. \end{gathered} \tag{1006}$$

Here d_j is the coupling entry, e_j the scale entry, and the finite-volume requirement is

$$V_j(s) = h_j(1 + B(s) + B_j^+(s))^{r_j} + 100(B(s)+1) + 100((j+3)L_j + 1),$$

with nonnegative integers h_j, r_j and

$$B(s) = \left\lceil \log^2(\beta_* + s_G s) \right\rceil, \qquad B_j^+(s) = \left\lceil \log^2(\beta_* + s_G(s + \log L_j)) \right\rceil.$$

All inequalities in (1006) hold eventually. The first four follow from positive linear growth in s; for the last, both ceiling functions are $O(s)$, so $V_j(s)$ is bounded by a fixed polynomial and hence is $o(e^s)$. Select b_{j-j_*} so that they hold whenever $s \ge b_{j-j_*}\log 3$. The entry property of J_0 and the lower bound in (1005) now give all these requirements at every step with $J(m+1) > j_*$.

For the comparison interval at stage j, suppose its required window $w_j(s)$ is nonnegative and integer valued and $w_j(s)/s \to 0$, and its minimum scale is r_j^0. Choose c_{j-j_*} so large that

$$r_j^0 \le s, \qquad w_j(s) \le \lfloor k/2 \rfloor$$

whenever $k \ge c_{j-j_*}$ and $k\log 3 \le s \le k \log L_j$. Such a choice exists because eventually $w_j(s)/s \le (2\log L_j)^{-1}$. At the center $k = k_{j-j_*}$ of (1004), (1005) gives precisely these scale bounds. Thus the whole requested window around k stays at stage j.

Only finitely many mesh steps remain in the first plateau. Choose an integer M at least their finitely many values $V_{j_*}(s_m)$ and at least the required minimum physical period. Use periods

$$P_p = 4(M + p + 1).$$

The lattice side at level m is $P_p e^{s_m}$. On the first plateau it pays $V_{j_*}(s_m)$ by the choice of M. At every later step it pays $V_{J(m+1)}(s_m)$ because that quantity is at most e^{s_m}. The three summands of V_j pay respectively the specified volume, coarse-cover and fine-patch requirements.

The periods increase strictly and have arbitrarily late multiples of every positive integer c. Indeed, given z, take $p = c(M+z+1) - (M+1)$. Then $p \ge z$ and $P_p = 4c(M+z+1)$. Finally

$$0 \le \frac{\log L_{J(m+1)}}{s_m} \le \frac{1}{J(m+1)} \longrightarrow 0$$

after the first plateau, by (1006). Thus the relative logarithmic mesh increment tends to zero as required. The stage profiles, block lengths and kernel have all been retained; the new choices are their scheduling and periods.

201.3 Countably many requests on that same sequence

Let $F_{n,i} \in \mathbb{R}$ be a countable family of numerical requests selected before constructing the sequence. Define

$$E_j = 1 + \sum_{n=0}^{j} \sum_{i=0}^{j} |F_{n,i}|.$$

Raise d_j to at least $\exp(\exp E_j)$ before applying the preceding construction. Since $d_{J(m+1)} \le \widehat{\beta}_m$, monotonicity of the logarithm twice gives

$$E_{J(m+1)} \le \log\log \widehat{\beta}_m.$$

For fixed n, the divergence of J supplies a first index after which $n \le J(m+1)$. At every such m and every $i \le J(m+1)$,

$$|F_{n,i}| \le E_{J(m+1)} \le 1 + \log\log \widehat{\beta}_m. \tag{1007}$$

The first index depends on n. We have not required that a finite stage pay infinitely many independent requests.

There is also an all-stage form. For a prescribed real sequence H_j, replace d_j by $\max(d_j, H_j)$. Then

$$H_{J(m+1)} \le \widehat{\beta}_m \tag{1008}$$

for every m, including zero. Both conclusions persist on every realization paying enlarged entry thresholds, because their proofs use only a lower bound on the threshold actually paid. Combining finitely many such purchases before choosing the realization therefore retains every earlier conclusion.

201.4 Selecting the lengths before their numerical costs

The ordering of choices is essential. First fix the stage profiles, the original kernel, the row-price families and their secondary widths. Their analytic suppliers give integer block-size floors. Take the pointwise maximum of the original floor, the flat and dual support floors, the curvature floor, the comparison floors, and the treasury and period floors. A nondecreasing odd sequence L_j above these floors exists: choose each L_j above the current finite maximum and above L_{j-1}. These choices precede any numerical request that depends on L_j.

With the lengths now fixed, form the actual post-length numerical costs. If two constructions require $H_j^{(1)}$ and $H_j^{(2)}$, pay $H_j = \max(H_j^{(1)}, H_j^{(2)})$ by (1008). Their countable refinements, as well as the previously selected inverse-width requests, are paid by (1007). Add the nonlinear entry requirements of Section 200 and every other selected finite-stage threshold before constructing J and $\widehat{\beta}$. Every realization paying these enlarged thresholds preserves all the earlier purchases by the preceding monotonicity argument.

Every step then uses the original length $L_{J(m+1)}$, so its actual finite patch is exactly the patch for which the post-length cost was computed. In particular a bound of the form $e^{H_j} \le \widehat{\beta}_m$ yields $H_j \le \log \widehat{\beta}_m$ on that patch; no estimate for a different block can be substituted. A maximum floor gives each of its component floors, and a maximum post-length cost gives each component cost. Applying the native analytic suppliers to these inequalities therefore supplies their conclusions on the same original levels and quadratic horizon.

This proves the scheduling and numerical selection step. The analytic hypotheses of those suppliers, and the sublinear window bounds used above, are still the specific estimates established in their own sections. Numerical selection does not replace them.

202 Additional scale estimates for the operator-product expansion

The operator-product construction uses the same block lengths and hatted couplings as the centered estimates. Its additional numerical requirements can all be imposed after the block lengths are chosen and before the scale sequence is realized. We verify them explicitly.

For the final construction choose the profile order and spatial power at stage j to be at least $6+j$ and $2+j$, respectively, and choose the support exponent to be exactly 100 at every stage. Constant support exponents satisfy the monotonicity requirement. Increasing the order and spatial power does not require increasing this exponent: the finite profile prices are selected afresh at the chosen orders, and their later floors and delays are allowed to depend on them. Thus the common funding and observable purchases apply with this choice. The constant exponent will also be retained when intermediate clusters are regrouped.

202.1 The halo and block-length inequalities

At a transition write L for the already selected integer block side, x for the hatted coupling and $y = x + c_G \log L$ for the next hatted coupling. Set $B(x) = \lceil (\log x)^2 \rceil$ and define

$$H_L(x,y) = \big(1 + L(B(x) + B(y) + 1)^2\big)^4.$$

For fixed L and any fixed $A \geq 0$, there is an entrance threshold such that

$$AH_L(x,y) \leq \sqrt{x} \tag{O1}$$

above that threshold. Indeed $y/x \to 1$, so eventually $B(y) \leq 2B(x)$. Hence $B(x) + B(y) + 1 \leq 4(1 + B(x))$ and $H_L(x,y) = O((1 + \log x)^{16})$. The logarithm of $(1 + \log x)^{16}/\sqrt{x}$ tends to $-\infty$, proving (O1). Apply this with $A = 101$, the fine support exponent plus one. The power four is the spatial dimension; the factor L is needed because two child anchors which first meet after blocking can be separated by a block side as well as the two cover radii.

The stronger block-length condition

$$L \leq \log x \tag{O2}$$

is obtained by requiring $x \geq e^L$. It is a condition at a fixed profile stage, not a uniform bound over unchosen block lengths. Let C_i denote the column count at stage i. At a transition with fine stage j the coarse stage is either j or $j-1$. Imposing

$$x \geq \max(L_j, C_j, C_{j-1}) \tag{O3}$$

therefore gives both adjacent column bounds and the block bound required by the logarithmic estimates. For $j = 0$ use C_0 in place of C_{j-1}.

202.2 Uniform logarithmic bounds for the fixed-stage data

Write h_i for the path price at stage i, $\sigma_i > 0$ for its width, $\lambda_i > 0$ for its linear clipping parameter, r_i for its relaxation parameter, and d_i for its direct clipping parameter. These are fixed finite nonnegative numbers, with the stated strictly positive denominators. Let $D_G = \dim_{\mathbb{R}} M_{r_G}(\mathbb{C})$, and let $n_\square$ be the number of oriented plaquette-axis types used in the action. The elementary vacuum bound has the form

$$V_i(x) = 2(1 + (4h_i + 1)) + 1 + n_\square(d_i + h_i^2) + 4D_G(\log 16 + \tfrac{1}{2}\log x).$$

Thus $V_i(x) + n_\square = B_i + 2D_G \log x$ for a fixed constant B_i. At each fixed stage we can require simultaneously

$$L_j \le \log x, \qquad 10/\sigma_i \le \log x, \qquad 10/\lambda_i \le \log x, \tag{O4}$$

$$V_i(x) + n_\square \le (\log x)^2, \tag{O5}$$

$$10^5(i+1) \le \log x, \qquad 10^5(1 + h_i + 100 + r_i + \sigma_i + C_i + d_i + \lambda_i) \le \log x. \tag{O6}$$

For (O5), take $z = \log x \ge \max(1, B_i, 2D_G + 1)$; then $B_i + 2D_G z \le z + (z-1)z = z^2$. The remaining inequalities hold because $\log x \to \infty$. To allow the stage indexing at both ends of a transition, purchase (O4)–(O6) for every $i \le j$ with the actual block side L_j. This is a finite list at each j.

Take the maximum of these thresholds and those for (O1)–(O3), then enlarge the existing entrance delay by their running maxima. Every estimate holds for all larger x, so this preserves earlier purchases. At level m take $j = J(m+1)$ and $i = J(m)$. The common scale construction ensures the entrance condition at this j, and the identities for the actual clock give

$$L = L_{J(m+1)}, \qquad x = \widehat{\beta}_m, \qquad y = \widehat{\beta}_{m+1}.$$

Thus (O1)–(O6) hold on the selected family, at all steps and periods. They require no subsequent change of its history.

202.3 Two consequences used in choosing intermediate scales

The physical scale satisfies $\widehat{\beta}_m = \beta_* + c_G \log a_m^{-1}$, with $c_G > 0$. For every fixed integer $q \ge 0$,

$$a_m \widehat{\beta}_m^q \longrightarrow 0. \tag{O7}$$

Indeed, setting $s = \log a_m^{-1} \to \infty$ gives an exponential e^{-s} times a polynomial of degree q. Also, for every $c, \epsilon > 0$ and integer $q \ge 0$, eventually

$$c(\log x)^q \le \epsilon \exp\{(\log\log x)^2\}. \tag{O8}$$

The logarithm of the ratio of the left side to the right side is $\log(c/\epsilon) + qu - u^2$ with $u = \log\log x \to \infty$, so the ratio tends to zero.

Finally, if $r > 1$ and $ra_i < 1$, there is $j < i$ such that

$$a_{j+1} \le ra_i < a_j. \tag{O9}$$

Choose the first index k with $a_k \le ra_i$. It exists since i qualifies; it is positive since $a_0 = 1 > ra_i$. Set $j = k - 1$. Minimality gives the strict upper inequality and $k \le i$ gives $j < i$. These are the intermediate-scale choices used with the common clock. The analytic OPE estimates remain those of the preceding local construction; (O1)–(O9) supply its additional scale inequalities.

203 The operator-product estimates on the selected history

The local projection and the several-insertion expansion must use the same adjacent transition. We establish this compatibility on the family of Section 159. Throughout this section the profiles have support exponent 100, as selected in Section 202. The orders and spatial powers still increase. No background representation is chosen again.

203.1 Finite elementary representations of transferred fields

Fix a finite torus, a representation R and an anchor o. For a support X and a nonnegative bound h, write

$$m_R(X,h) = e^{bC_B(X)}h,$$

where $b \geq 0$ and B are its support exponent and covering radius. The reserved regular slot on each nonempty support and the reserved forced slot at each admissible support–inventory pair are retained from the marked construction. A finite elementary representation of an ordinary part g and a forced profile K means identities

$$g = p + \sum_{j\in I} z_j \operatorname{ord}(M_j), \qquad K = \sum_{j\in I} z_j \operatorname{flag}(M_j), \qquad \sum_{j\in I} |z_j| \operatorname{cost}(M_j) \leq A. \tag{1009}$$

Here I is finite, all insertion supports contain o, and $p \in \mathbb{C}$ is the separately recorded constant. In particular K is the actual profile, not merely an array with the same integrated density.

Suppose first that $g = \sum_i f_i$, where the sum is countable, each f_i is measurable and gauge invariant, depends only on its support X_i and the flags there, vanishes unless X_i is healthy, and $o \in X_i$. Assume

$$|f_i(U)| \leq h_i, \qquad \sum_i m_R(X_i,h_i) \leq A.$$

Since $b \geq 0$, $\sum_i h_i < \infty$. Group by the finitely many supports of the torus and put

$$f_X = \sum_{i:X_i=X} f_i, \qquad h_X = \sum_{i:X_i=X} h_i.$$

These series converge uniformly and absolutely. Their sums preserve measurability, gauge invariance, local dependence and the healthy-set condition. The real and imaginary parts of f_X can be placed in the reserved regular slot of X, each with bound h_X. Their coefficients are 1 and i. The resulting elementary representation has ordinary part f_X, zero forced profile and price at most $2m_R(X,h_X)$. Zero functions may be omitted. Summing over the finite support set gives (1009) with $p = 0$, $K = 0$ and price $2A$. The equality of the price sums follows by grouping an absolutely convergent nonnegative series; its support weight is constant within each group.

The same argument applies to a dominated forced array. Group its coefficients by their support–inventory pair (X,J). Suppose the grouped coefficient vanishes unless the flags in X are exactly J, where J is nonempty, and every nonzero group has $o \in X$. Use the reserved forced slot at (X,J) and split the grouped function into real and imaginary parts. An array with total weighted majorant A therefore has a finite elementary representation of its profile, with ordinary part zero and price at most $2A$. The exact inventory condition is preserved under grouping. This construction uses only finitely many pairs even when the original array has infinitely many serial addresses.

We apply this compression to an elementary insertion M at a fine scale, with mass $m = \operatorname{cost}(M) \geq 0$. Denote its block transfer by $\mathcal{T}$ and the forced derivative fixed in Section 95 by $\mathcal{D}M$. The exact differentiated expansion gives an ordinary sum and this forced derivative. Its nonprincipal ordinary mass is at most p_M; its possible principal term has mass at most E_*m; and its forced mass is at most q_M. The local derivative and support estimates give

$$2p_M \leq m, \qquad 2q_M \leq m, \qquad E_* \leq L. \tag{1010}$$

Here E_* is the fixed exponential charge for enlarging the inherited support. The last inequality is the geometric block-length floor in Section 95. More precisely, p_M is the nonprincipal regular bill

of that construction, and q_M is its forced derivative bill before division by the clipping gain. Put $\kappa = \sqrt{\text{coarse forced allowance}}$ and let H_0 be the degree-zero splitting factor. Its explicit form is

$$H_0 = 1 + 2|v_0|(1 + C_{\mathrm{orb}} + |f_0|) \geq 1, \qquad c_0 = L^{-2/5}/H_0 \leq 1,$$

where v_0, f_0 are the fixed finite local-splitting constants and $C_{\mathrm{orb}} \geq 0$ is its motion price. The two degree-zero payments are exactly

$$2p_M \leq c_0 m, \qquad 2\kappa^{-1} q_M \leq c_0 m, \qquad 0 < \kappa \leq 1.$$

They imply the first two inequalities in (1010). This uses the same unscaled forced derivative as the profile identity. The principal contribution is absent for a forced input. For an ordinary input its masked value is at most its original height: the single-insertion, order-zero cluster has no additional insertion factor. Enlarging its support therefore costs precisely E_* times its original mass. All remaining clusters are charged to p_M.

The compression just proved gives a finite elementary representation of the transferred ordinary part and the profile of $\mathcal{D}M$ with price at most

$$2(p_M + E_* m) + 2q_M \leq (2 + 2L)m \leq 6Lm,$$

where $L \geq 1$. Its inserted density is exactly $\mathcal{T}$ applied to the density of M, by the differentiated block identity. Thus the profile as well as the density is retained.

This conclusion extends to every finite representation (1009). Apply it to each M_j and multiply its output coefficients by z_j. Concatenating the finite lists gives total price at most $6LA$. The constant term remains p because $\mathcal{T}\rho_f = \rho_c$. Finite sums commute with integration: each elementary density is integrable under the lawful representation, and the transition is the already constructed kernel. The output profile is exactly $\sum_j z_j \mathcal{P}(\mathcal{D}M_j)$. This proves the transport bound needed when a grouped observable is carried through intermediate scales.

203.2 Several insertions and their common allowances

At adjacent levels $m+1, m$ set

$$x = \widehat{\beta}_m, \quad y = \widehat{\beta}_{m+1}, \quad L = L_{J(m+1)}, \quad N = N_{J(m+1)}, \quad \delta_m = y^{-2}, \quad \sigma_m = y^{-1}.$$

Let χ_m be the forced-gas price from the two successive absorptions, with its actual positive suppression constant and this block length. It is the price denoted by t in Section 17. The single-insertion and arbitrary finite-menu estimates of that section use this same $\mathcal{D}$, with input support exponent 96, ordinary output exponent $100 + 1/2$, and forced output exponent 101. Their output allowances are respectively 1 and χ_m.

To apply them to several slots in each replica cell, take the finite marker set to be the product of the cell set and its slot set. The compiler of the finite-menu construction already permits every finite marker set. The ordinary coefficient is indexed by individual markers and the forced coefficients by their subsets. Grouping by the cell coordinate is exactly the substitution used in Section 55. The quiet coefficient stays the actual background profile. A singleton coefficient stays the sum of the derivatives of its original insertions through $\mathcal{D}$. The motion identities follow by applying the same permutation to the cell coordinate and leaving the slot coordinate fixed; equal weights on corresponding slots preserve every weighted estimate. Thus increasing the number of slots does not change the background transition or its insertion transport.

Write $B = B_m$ and $\alpha = 100 + 1/2$. The enlarged-support bound of Section 54 gives the local price

$$s_m = 2(1+N)\delta_m(Ne^{\alpha}+1) + (1+N)\chi_m. \tag{1011}$$

The two copies in its first term are the healthy and unhealthy pieces of the ordinary grouped entry. The last term is the forced entry. The purchases in (505) give, simultaneously,

$$0 < \delta_m \le 1, \qquad \chi_m \le 1, \qquad (2B_m+1)^4 s_m \le \tfrac{1}{2}, \qquad e^{s_m} - 1 \le 1, \tag{1012}$$

$$\alpha(2048B_m+1)^4 \le \sqrt{\widehat{\beta}_{m+1}}. \tag{1013}$$

The allowance for the elementary degree-d projection is $X_m = \exp(\sqrt{\widehat{\beta}_{m+1}}) \ge 1$, by (504). Both that projection and the $6L$ elementary transport use the same $\mathcal{D}$ just identified.

203.3 Uniformity above the reference level

For each grade d retain the reference level t_d with $d + 3 \le J(t_d)$. If $m \ge t_d$, monotonicity gives $d \le J(m)$. The numerical purchases were made for every degree up to the predecessor stage, and for every permitted adjacent pair of stages, before the scale sequence was realized. Hence the degree-d projection is available at every such m, for every cutoff $N_0 > m$ and every period. The constants in (1012)–(1013) depend only on m, not on N_0 or the period.

The adjacent transition retained by the selected history is the one at the chosen point of its interval family. Substituting the equalities identifying its fine and coarse representations with the corresponding history entries changes none of the formulas above. Consequently the ordinary menu estimates, the several-slot estimates, the finite elementary transport, and the final local projection hold together on that history. These supply the local hypotheses used in Sections 170 and 54. The later continuum limiting and cutoff arguments use their previously established estimates; they require no new choice of background history.

204 The final handoffs and the remaining construction theorem

The final assembly must use one hierarchy throughout. An operator-product expansion for one subsequential limit and a reconstruction for another would not establish the desired result. We first state the remaining two properties of that common hierarchy, then isolate precisely what is needed to assemble it.

204.1 The operator-product and trace requirements

Let $\boldsymbol{A} = (A_i)_{i\in I}$ and $\boldsymbol{B} = (B_j)_{j\in J}$ be finite lists of graded fields and write $w(\boldsymbol{A}) = \sum_{i\in I} \deg A_i$. For each grade ℓ, let $(E_{\ell,\alpha})_{\alpha\in\mathcal{C}_\ell}$ be a finite collection of fields. The coefficients $c^{\ell,\alpha}_{\boldsymbol{A}}(\boldsymbol{x}; o)$ do not depend on the phase. The cluster geometry at scale $r > 0$ is

$$|x_i - o| \le Mr, \qquad |x_i - x_{i'}| \ge \eta r \quad (i \ne i'),$$

where $M \ge 0$ and $\eta > 0$. Spectators satisfy $|y_j - o| \ge c$ and $|y_j - y_{j'}| \ge c$ for $j \ne j'$, where $c > 0$. For every $\epsilon > 0$, each fixed coefficient obeys

$$|c^{\ell,\alpha}_{\boldsymbol{A}}(\boldsymbol{x}; o)| \le r^{\ell - w(\boldsymbol{A}) - \epsilon} \tag{1014}$$

at sufficiently small r, uniformly in the indicated cluster geometry. For each fixed truncation degree D, the expansion requires

$$\left| S^{s}_{\boldsymbol{A},\boldsymbol{B}}(\boldsymbol{x},\boldsymbol{y}) - \sum_{\ell=0}^{D} \sum_{\alpha\in\mathcal{C}_\ell} c^{\ell,\alpha}_{\boldsymbol{A}}(\boldsymbol{x};o) S^{s}_{E_{\ell,\alpha},\boldsymbol{B}}(o,\boldsymbol{y}) \right| \le r^{D+1-w(\boldsymbol{A})-\epsilon}. \tag{1015}$$

Here s can be any character or the mixed state. The small-radius threshold for the remainder is uniform over these choices and over the allowed positions, but may depend on the fields, D, M, η, c, ϵ. There is no assertion of one threshold working for all truncation degrees.

The trace requirement concerns a centered grade-four field Θ. Set

$$T(r) = S^{\chi_*}_{\Theta,\Theta}((\mathbf{0},-r/2),(\mathbf{0},r/2)).$$

Its vacuum one-point functional vanishes, and the required asymptotic is

$$\lim_{r\downarrow 0} r^{8}\log^{2}(r^{-1})T(r) = \frac{3\dim_{\mathbb{R}}\mathfrak{g}}{\pi^{4}} > 0. \tag{1016}$$

Theorem 195.5 shows that this asymptotic implies the required exclusion of finite complex linear combinations of mass-shell tracks, including empty products. The class of profiles and their exact normalization are given in Section 195. Theorem 196.4 supplies the limiting argument from the trace comparison estimates; producing those estimates remains necessary.

204.2 A precise conditional assembly

An internal extraction order is a pair (u,p) of strictly increasing maps $\mathbb{N}\to\mathbb{N}$. Refining by (v,q) gives $(u\circ v, p\circ q)$. The full mesh order is $u(N)=N$. The need for a full mesh limit at the public endpoint is stronger than convergence just along $u\circ v$.

Lemma 204.1 (Identifying the full mesh limit)**.** *Fix a test and its insertion labels. Suppose that at every suitable period p the full sequence $W_{N,p}$ converges to some ℓ_p. Suppose also that an ordered extraction supplies numbers w_p such that $W_{u(j),p}\to w_p$, where u is strictly increasing, and $w_{p(q(k))}\to s$. Then $\ell_p = w_p$ at every suitable period, and the same volume limit s holds with the full mesh limits.*

Proof. Strict increase implies $u(j)\ge j$ by induction, so $u(j)\to\infty$. The full mesh convergence therefore implies $W_{u(j),p}\to\ell_p$. To see uniqueness explicitly, for any $\epsilon>0$, take j sufficiently large that this term is within $\epsilon/2$ of each proposed limit. The triangle inequality gives $|\ell_p - w_p|<\epsilon$, hence equality. Suitability holds eventually on the volume sequence, so the two sequences of row limits eventually agree and have the same limit. □

This lemma explains how a full ultraviolet convergence result can be combined with a hierarchy constructed by subsequence extraction. It does not obtain full ultraviolet convergence from subsequential convergence: that is its additional hypothesis.

Input 204.2 (Joint construction input)**.** *For the fixed group G and each chosen positive blocking penalty b, there is one lattice family with the following properties. Its insertions are continuous, gauge-invariant and uniformly local, and its bare coefficients satisfy* (1)*. For every extraction order (u,p), there is a refinement (v,q) and one set of graded fields, phase data, correlations and selected relativistic theory as in Definition 1.1, for which all of the following hold simultaneously:*

1. *The mixed correlations satisfy* (4) *along the* full *mesh sequence at every suitable period and* (5) *along $p\circ q$.*

2. *The phase correlations have the stated smoothness, symmetry, continuity and mixture identity* (3).

3. *The fields and coefficients satisfy* (1014)–(1015) *and the centered field satisfies* (1016).

Every assertion refers to the same lattice family and the same selected hierarchy on that refinement.

Proposition 204.3 (Assembly from the joint input)**.** *Input 204.2 implies the full existence target stated in Section 1, including the operator-product and trace properties above.*

Proof. Fix G and choose $b = 1$. Select the family supplied by Input 204.2 once. Its regularity is part of that input and is unaffected by any subsequent choice of volume sequence. Now let $p : \mathbb{N} \to \mathbb{N}$ be an arbitrary strictly increasing sequence. Apply the input to the extraction order (id, p). It yields (v, q) and one hierarchy with all three simultaneous properties. Take q as the public volume subsequence and retain this hierarchy. Its ordered Wilson convergence uses the full mesh sequence by item 1, so no assertion about a mesh subsequence has been substituted for (4). Item 2 gives the phase and mixture conditions; the selected theory is part of the same data. Item 3 supplies the remaining two properties, with finite-profile exclusion following from Theorem 195.5. The construction of the family preceded the choice of p, while q and the hierarchy were chosen after p, exactly as required. □

The proposition is only the final logical assembly. The main outstanding work is to establish Input 204.2 from the Wilson measure. In particular, the finite-dimensional normalization proof does not supply the estimates uniform in spacing and volume; a convergent mesh subsequence does not establish the full mesh limit; the Euclidean Hilbert space of Section 32 needs the subsequent slab and phase estimates to support the phase decomposition and the relativistic reconstruction proved in the later sections; and separated-correlation compactness does not supply the operator-product remainder or the trace asymptotic. Until those inputs have detailed proofs or checked applicable references, this article does not contain a complete existence proof.

References

[1] K. Osterwalder and R. Schrader, *Axioms for Euclidean Green's functions II*, Communications in Mathematical Physics **42** (1975), 281–305, especially §IV.1, pp. 287–288.

[2] D. P. Williams, *A (Very) Short Course on C^*-Algebras*, lecture notes, 14 February 2024, Theorem 2.13, pp. 29–31. `https://math.dartmouth.edu/~dana/bookspapers/cstar.pdf`.

[3] E. T. Sawyer, *Lecture Notes in Advanced Real Analysis*, McMaster University, Chapter 6, Theorem 32, p. 78. `https://math.mcmaster.ca/~sawyer/Publications/Advanced_Real_Analysis.pdf`.

[4] R. B. Melrose, *Introduction to Microlocal Analysis*, Chapter 1, "Tempered distributions and the Fourier transform," §1.7, Theorem 1.3, pp.21–22. `https://math.mit.edu/~rbm/iml/Chapter1.pdf`.

[5] B. Bekka, P. de la Harpe and A. Valette, *Kazhdan's Property (T)*, author manuscript, 23 February 2007, Appendix D.3, Theorem D.3.1, p.397. `https://perso.univ-rennes1.fr/bachir.bekka/KazhdanTotal.pdf`.